Algebraic Topology

***EYNTKA* Series**

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Introduction

Algebraic topology is the study of topological spaces by assigning to them an algebraic object (integers, vector spaces, a sequence of groups, etc.) in a "functorial" manner:

1. topological information of a space is transformed into algebraic information that can be carried by continuous maps (and as shall be shown soon, most properties are preserved under the more general notion of *homotopic maps*).
2. If two topological spaces are isomorphic (more generally, homotopy equivalent), then they shall share the same algebraic information[1].

a word on how after extracting algebraic information, we also came up with ways of constructing spaces that will satisfy the algebraic information, so in sense the theory also grew the amount of spaces we work with.

By assigning algebraic invariants, we can measure and capture properties of their underlying shapes. Modern algebraic topology has a multitude of ways of assigning algebraic information each creating its own theory, including but not limited to:

1. Homotopy theory: different ways n-spheres can map into a space
2. Homology and Cohomology: famously said to formalize the notion of holes in n-dimensions. A different intuition that is arguably more practical is that it measures the failure of a function or section on a space X that can be defined on an (open) subset $U \subseteq X$ to be extended to the entire space X[2].
3. K-theory: measuring how a vector bundle fails to be trivial
4. Cobordism: measuring the failure of an n-manifold to be the boundary of an $(n+1)$-dimensional manifold

[1] The converse is only solved for some special cases, and is in general undecidable (see cite:HERE).

[2] There are many nuances on what extension means, how independent of a choice from the original set U it is or the choice of simplicial/singular homology, what choice of coefficients we choose, all of which is explored in chapter 3 and chapter 4

In this book, our primary focus will be on *homotopy*, *homology*, and *cohomology*. K-theory and cobordism can each be considered *generalized cohomology theories* (see Part III). Each of these theories associates an algebraic object $A(X)$ to a given topological space X. Crucially, this assignment is strictly functorial: a continuous map $f : X \to Y$ between spaces induces a corresponding algebraic homomorphism $A(f) : A(X) \to A(Y)$. In fact, all of these theories will be *homotopy invariant* (homotopy is defined in chapter 1), namely if $f, g : X \to Y$ are continuous and $f \simeq g$ are homotopic then $A(f) = A(g)$. Through the power of functoriality, we will see that many difficult geometric questions can be systematically reduced to problems in algebra. Some celebrated examples of theorems we will prove using these methods include:

1. Invariance of Dimension: $\mathbb{R}^n \ncong \mathbb{R}^m$ for $n \neq m$.
2. Brouwer Fixed Point Theorem: Any continuous map $f : D^n \to D^n$ from the closed unit disk to itself must have a fixed point.
3. Fundamental Theorem of Algebra: Every non-constant complex polynomial has a root.
4. Borsuk-Ulam Theorem: Any continuous map from the sphere S^n to $\mathbb{R}^n$ must map some pair of antipodal points to the same value.
5. The Hairy Ball Theorem: A continuous tangent vector field on an even-dimensional sphere (such as S^2) must vanish at some point. (Colloquially: "you cannot comb a hairy ball flat without creating a cowlick").
6. Generalized Jordan Curve Theorem: Any continuous embedding of a sphere S^{n-1} into $\mathbb{R}^n$ separates the space into exactly two connected components, an "inside" and an "outside".
7. The Ham Sandwich Theorem: Given any n measurable "objects" in $\mathbb{R}^n$, there exists a single $(n-1)$-dimensional hyperplane that divides the volume of each object exactly in half.
8. Nielsen-Schreier Theorem: Every subgroup of a free group is itself a free group. The proof in theorem 2.3.13 realizes the subgroup as the fundamental group of a covering space of a wedge of circles, and a covering space of a graph is again a graph.
9. Lefschetz Fixed Point Theorem: A vast generalization of the Brouwer fixed point theorem which states that if a certain homological invariant (the Lefschetz number) of a map $f : X \to X$ is non-zero, then f must have a fixed point.
10. H-Space Structures on Spheres: The only spheres that can be equipped with a continuous multiplication operation (making them an H-space) are S^0, S^1, S^3, and S^7, which corresponds to the existence of the real numbers, complex numbers, quaternions, and octonions.

By translating geometric settings into an algebraic framework, we gain access to computational methods that are otherwise unavailable in pure point-set topology or geometry.

A core strategy in computing these invariants is breaking down complicated spaces into simpler, well-understood building blocks. We will extensively study spaces constructed by gluing simple geometric objects, such as simplices (generalized triangles) or topological disks, together along their boundaries. Familiar spaces like the torus, spheres, the Klein bottle, and projective spaces all admit such cell structure decompositions. Working with these structures gives our algebraic calculations a geometric content. A central theme of the subject is proving that our algebraic invariants depend only on the intrinsic topology of the space itself, and are independent of any specific choice of

decomposition into simpler objects. Many early hands-on computations choose a *representation* of the space, while the theory proves these results are *independent* of choice of representation. The independence of choice of representation shows it is a property *intrinsic* to the underlying space and not the choice of representation of the space.

A natural question arises: to what extent are these algebraic invariants preserved? At a minimum we ask that $A(X)$ be a topological invariant: if $X \cong Y$ are homeomorphic, then $A(X) \cong A(Y)$. Homeomorphism is a rigid notion of sameness, namely a homeomorphism is a *bijection* that is continuous in both directions. It is the precise form of the common geometric explanation that two spaces are the same if we may bend or stretch them into each other's shapes but never tear, glue, or change dimension (like the famous coffee mug to doughnut example).

The algebraic invariants we shall build will be invariant under a much weaker condition: the coarser relation of *homotopy equivalence*, which no longer requires the map to be a bijection, so we may now also *collapse* and *thicken* our shape. For instance, continuously retracting a solid disk to its center, retracting the plane minus a point onto a circle, and deforming all of $\mathbb{R}^n$ to a single point are all examples of homotopy equivalences. In this setting, a disk and a point are homotopy equivalent even though they are not homeomorphic: homotopy equivalence has forgotten dimension and local detail, remembering only the coarser "hole structure" of a space. Since every homeomorphism is a special case of a homotopy equivalence, this really is a weakening, namely more spaces become equivalent. That flexibility lets us replace a complicated space by a homotopy-equivalent simple one, compute the invariants of the simple model, and read off the answer for the original.

0.0.1 For the Expert There is a structural reason for homotopy invariance: each invariant is computed from a chain complex, and chain complexes are only well-defined *up to chain homotopy*. A topological homotopy $f \simeq g$ induces a chain homotopy between the maps they induce, so f and g act identically on homology. In the extreme case where a space contracts to a point, this becomes a *contracting homotopy* s with $\partial s + s\partial = \mathrm{id}$, exhibiting the augmented chain complex as split exact.

0.0.2 Images in Book A lot of these images are from either Hatcher's *Algebraic Topology* ([21]) or Munkres's *Topology* ([30]). In the first edition, I shall create my own images. These were put there from when I was learning the material and wanted a quick reference.

0.0.3 Reading the numbering A star marks *bonus material*: a chapter, section or subsection that belongs in the book but that nothing later depends on. Skipping it costs nothing downstream. The star sits on the number, as in §2.7$\star$; in divisions written before this convention it sits at the front of the title instead, as in **Uniform Spaces*. The two mean the same thing.

Part I

Classical Invariants

Topology [16] attaches a group to a space. The fundamental group of the circle is $\mathbb{Z}$, computed through the covering $\mathbb{R} \to S^1$, and Brouwer's fixed point theorem for the disk follows: a retraction of the disk onto its boundary circle would factor the identity of $\pi_1(S^1) \cong \mathbb{Z}$ through the trivial group $\pi_1(D^2)$. The fundamental group sees only loops, and every loop on the sphere S^2 contracts. So π_1 cannot tell S^2 from a point, and the same argument says nothing about a retraction of the ball in $\mathbb{R}^3$ onto its boundary sphere, or about whether $\mathbb{R}^3$ and $\mathbb{R}^4$ are homeomorphic, the complement of a point being simply connected in both.

Poincaré's *Analysis Situs* of 1895 introduced the fundamental group. It also redefined, through Poincaré's homologies, the counts of independent cycles in each dimension that Betti had introduced in 1871, named them Betti numbers after him, and stated the symmetry of the Betti numbers of a closed orientable manifold now called Poincaré duality. In the late 1920s, following Noether, the Betti numbers became the ranks of homology groups. Cohomology and its cup product followed in the 1930s, and Eilenberg's singular homology of 1944 gave the definition used here, which needs no triangulation of the space.

The four chapters take these invariants in the same order and compute each from a cell structure. Chapter 1 sets up cell complexes, spaces built by attaching disks along their boundaries, and the homotopy relation between maps. Chapter 2 computes the fundamental group of a cell complex from its 2-skeleton by van Kampen's theorem and classifies covering spaces by subgroups of the fundamental group. Chapter 3 defines singular homology, proves its homotopy invariance and excision, and computes it from the cells, which settles the questions above in every dimension. Chapter 4 dualizes homology to cohomology, which the cup product makes a ring, proves Poincaré duality for closed manifolds orientable over the coefficient ring and its version for manifolds with boundary, and develops cohomology with local coefficients.

1

Combinatorial Topology and Homotopy

Let X be a topological space. Combinatorial topology aims to create a new structure on X that will be thought as a way of interpreting X as built up from simpler spaces (most commonly circles and triangles, along with their higher dimensional counterparts). How they are glued and how they are chosen to map into our space X will determine the "type" of combinatorial object we are working with. In this chapter, we shall explore the common *CW complex* that will have many important properties in the rest of this book.

We shall also explore a generalization of the notion of a continuous map known as a *homotopy*. As we shall see, a homotopy will be able to distort a shape much more drastically than a continuous map, but as an added benefit will be more flexible and give many new maps between spaces that previously were unconnected. These spaces will share many properties that are invariant under the stronger deformation given by homotopies. In the following chapters, we shall see that all the algebraic structure we shall explore shall be invariant under homotopies, showing that they are the right notion of morphism in algebraic topology.

1.1 Homotopy and Homotopy Type

For notational simplicity, we will take the convention that

$$I = [0, 1]$$

Definition 1.1.1: Homotopy

Let X, Y be topological spaces, and suppose we have two maps f, f' which are continuous maps from X to Y. We say that f is *homotopic* to f' if there is a continuous map $F : X \times I \to Y$ such that

$$F(x,0) = f(x), \qquad F(x,1) = f'(x)$$

F is called a homotopy between f and f', and we write this as $f \simeq f'$

Thinking of the graphs of functions as shapes, if we can continuously deform one shape into another, the two shapes are homotopic. Note that the topology from which we define continuity ($X \times I$) has *both* the topology of X and the Euclidean topology to define it. The "continuous deformation" is more reliant on the Euclidean portion, meaning we can use our usual intuition for the most part of deformations. In terms of I being a closed interval, we can think of this as having "one minute of time to change it, and we must do it in that time" (due to the compactness of the interval). To formalize this intuition, homotopy can be interpreted another way: give $Y^{[0,1]}$ the compact-open topology. Since $[0,1]$ is locally compact Hausdorff, the exponential law [16, The Exponential Law] shows that a homotopy is equivalently a continuous map

$$f : X \to Y^{[0,1]}$$

which sends each point x to a path in Y. Hence, a homotopy between two maps f, g can be thought of as saying there is a continuous path from every point $f(x) \in Y$ to a point $g(x) \in Y$. Notice the importance of the domain in this definition: even if $f(X) = \{y\}$, a homotopy will assign a path for each $x \in X$, hence we shall find that two images of different dimensions are homotopic.

Example 1.1.2: Homotopies

1. Let $f, g : \mathbb{R} \to \mathbb{R}^2$, $f(x) = (x, x^3)$ and $g(x) = (x, e^x)$. Then the map $F : \mathbb{R} \times I \to \mathbb{R}^2$,

   $$F(x,t) = (x, (1-t)x^3 + te^x)$$

   is a homotopy: both coordinates of F are continuous functions of (x,t), and $F(x,0) = (x, x^3)$, $F(x,1) = (x, e^x)$. Thus

   $$f \simeq g$$

2. Take $f : I \to I$ $x \mapsto x^2$. Then

   $$F(x,t) = (1-t)x^2 + tx$$

   is a homotopy from f to id_I. More generally, x^2 can be replaced by any polynomial p with $p(I) \subseteq I$ (indeed by any continuous p: $I \to I$), and this homotopy will work, since $(1-t)p(x) + tx$ is a convex combination of two points of I.

3. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function, and $g : \mathbb{R} \to \mathbb{R}$, $g(x) = 0$. Then:

   $$F : \{\mathbb{R}\} \times I \to \mathbb{R} \qquad F(x,t) = (1-t)f(x)$$

   is a continuous function, with $F(x,0) = f(x)$ and $F(x,1) = 0$. Thus, every continuous function on $\mathbb{R}$ is homotopic to the constant function.

   In terms of graphs, the graph of any continuous function on $\mathbb{R}$, with the Euclidean topology, can be continuously deformed onto the graph of the constant function 0.

More generally, if F is a homotopy whose construction is

$$F(x,t) = ((1-t)f(x) + tg(x))$$

then it is called a *straight-line homotopy.*

4. Let $f, g : \mathbb{R}^2 \setminus \{0\} \to \mathbb{R}^2 \setminus \{0\}$ where f is the identity map and $g(x) = \frac{x}{\|x\|}$. The straight-line homotopy $F(x,t) = (1-t)x + t\frac{x}{\|x\|} = \left(1 - t + \frac{t}{\|x\|}\right)x$ stays in $\mathbb{R}^2 \setminus \{0\}$, since the scalar in front of x is positive, so $f \simeq g$. We shall later show that f is not homotopic to a constant map. If the domain and codomain were $\mathbb{R}^2$, then f *would* be homotopic to a constant map, by the straight-line homotopy to 0. If the domain is $\mathbb{R}^2$ and the codomain is $\mathbb{R}^2 \setminus \{0\}$, every map h, surjective or not, is homotopic to a constant map, by $(x,t) \mapsto h((1-t)x)$, because the domain contracts.

This last example shows how the holes of a space govern which maps into it are homotopic.

Homotopy is an equivalence relation on the continuous maps between two spaces (exercise 1.3.1 (1)). Hence, it is well-defined to take maps up to homotopy.

The invariants of this book assign a group to each space and a homomorphism to each continuous map, compatibly with composition. Categories and functors are the language for such assignments. The covariant definitions below restate those of *Core Algebra* [10, Category, [Covariant] Functor, Natural Transformation and Natural Isomorphism, Equivalence of Categories], the contravariant forms are added here, and *Category Theory* [9] develops the subject.

Definition 1.1.3: Categories and Functors

A *category* C consists of a class of *objects*, a set $\mathrm{Hom}_{\mathsf{C}}(X,Y)$ of *morphisms* $f\colon X \to Y$ for each pair of objects, an *identity* $\mathrm{id}_X \in \mathrm{Hom}_{\mathsf{C}}(X,X)$ for each object, and a *composition* $\mathrm{Hom}_{\mathsf{C}}(Y,Z) \times \mathrm{Hom}_{\mathsf{C}}(X,Y) \to \mathrm{Hom}_{\mathsf{C}}(X,Z)$, $(g,f) \mapsto g \circ f$, which is associative and has the identities as two-sided units. A morphism $f\colon X \to Y$ is an *isomorphism* if there is a morphism $g\colon Y \to X$ with $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$.
A *functor* (or *covariant functor*) $F\colon \mathsf{C} \to \mathsf{D}$ assigns to each object X of C an object $F(X)$ of D and to each morphism $f\colon X \to Y$ a morphism $F(f)\colon F(X) \to F(Y)$, with $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$ and $F(g \circ f) = F(g) \circ F(f)$. A *contravariant functor* reverses morphisms: it assigns to $f\colon X \to Y$ a morphism $F(f)\colon F(Y) \to F(X)$, with $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$ and $F(g \circ f) = F(f) \circ F(g)$.

A functor of either variance carries isomorphisms to isomorphisms: if g is inverse to f, then $F(g) \circ F(f) = F(g \circ f) = F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$ in the covariant case, and the other composite and the contravariant case are the same computation.

Example 1.1.4: Categories of Spaces and Groups

1. Top has topological spaces as objects and continuous maps as morphisms, Set has sets and functions, Grp has groups and homomorphisms, and Ab has abelian groups and homomorphisms. In each, composition is composition of functions. The isomorphisms of Top are the homeomorphisms.

2. Top_* has as objects the *pointed spaces* (X, x_0), a space X with a *basepoint* $x_0 \in X$, and as morphisms the *based maps* $f\colon (X,x_0) \to (Y,y_0)$, the continuous maps with $f(x_0) = y_0$. A

composite of based maps is based.

3. Forgetting the basepoint is a functor $\mathsf{Top}_* \to \mathsf{Top}$, and regarding an abelian group as a group is a functor $\mathsf{Ab} \to \mathsf{Grp}$.

4. For an abelian group G, the assignment $A \mapsto \mathrm{Hom}(A, G)$, $f \mapsto f^*$ with $f^*(\varphi) = \varphi \circ f$, is a contravariant functor $\mathsf{Ab} \to \mathsf{Ab}$, since $(g \circ f)^*(\varphi) = \varphi \circ g \circ f = f^*(g^*(\varphi))$ and $\mathrm{id}^*(\varphi) = \varphi$.

Definition 1.1.5: Natural Transformations and Equivalences of Categories

Let $F, G \colon \mathsf{C} \to \mathsf{D}$ be functors. A *natural transformation* $\eta \colon F \Rightarrow G$ assigns to each object X of C a morphism $\eta_X \colon F(X) \to G(X)$ of D such that for every morphism $f \colon X \to Y$ the square

$$\begin{array}{ccc} F(X) & \xrightarrow{\eta_X} & G(X) \\ {\scriptstyle F(f)}\downarrow & & \downarrow{\scriptstyle G(f)} \\ F(Y) & \xrightarrow[\eta_Y]{} & G(Y) \end{array}$$

commutes, that is $G(f) \circ \eta_X = \eta_Y \circ F(f)$. For contravariant F and G the condition reads $G(f) \circ \eta_Y = \eta_X \circ F(f)$. It is a *natural isomorphism* if every η_X is an isomorphism. An *equivalence of categories* is a functor $F \colon \mathsf{C} \to \mathsf{D}$ together with a functor $G \colon \mathsf{D} \to \mathsf{C}$ and natural isomorphisms $G \circ F \Rightarrow \mathrm{id}_{\mathsf{C}}$ and $F \circ G \Rightarrow \mathrm{id}_{\mathsf{D}}$.

Lemma 1.1.6: Homotopy Is Compatible With Composition

Let $f, f' \colon X \to Y$ and $g, g' \colon Y \to Z$ be continuous maps with $f \simeq f'$ and $g \simeq g'$. Then $g \circ f \simeq g' \circ f'$. If the spaces carry basepoints x_0, y_0, z_0 and the given homotopies F and G satisfy $F(x_0, t) = y_0$ and $G(y_0, t) = z_0$ for all t, then $g \circ f \simeq g' \circ f'$ through a homotopy H with $H(x_0, t) = z_0$ for all t.

***Proof*:**
Let $F \colon X \times I \to Y$ be a homotopy from f to f' and $G \colon Y \times I \to Z$ one from g to g'. The map $H(x, t) = G(F(x, t), t)$ is the composite of the continuous map $(x, t) \mapsto (F(x, t), t)$ with G, so it is continuous, and $H(x, 0) = g(f(x))$, $H(x, 1) = g'(f'(x))$. Under the basepoint hypotheses $H(x_0, t) = G(y_0, t) = z_0$, completing the proof.

Definition 1.1.7: The Homotopy Category hTop

Let hTop denote the category whose objects are topological spaces and whose morphisms $X \to Y$ are the homotopy classes $[f]$ of continuous maps $f \colon X \to Y$, with composition $[g] \circ [f] = [g \circ f]$ and identities $[\mathrm{id}_X]$. The *pointed homotopy category* hTop_* is formed in the same way from Top_*: its morphisms are the classes of based maps under *based homotopy*, a homotopy F of based maps with $F(x_0, t) = y_0$ for all t.

Composition is well defined by lemma 1.1.6, and it is associative and unital because composition of maps is. Sending each map to its class is a functor $\mathsf{Top} \to \mathsf{hTop}$. A functor F on Top with $F(f) = F(f')$ whenever $f \simeq f'$ factors through it: $\bar{F}([f]) = F(f)$ is well defined by hypothesis and is

a functor because F is. The same holds for $\mathsf{Top}_* \to \mathsf{hTop}_*$, and section 8.1 develops the structure of hTop_* that the axioms of chapter 8 use.

Definition 1.1.8: Nullhomotopic

If f is homotopic to a constant map, we say that f is *nullhomotopic*. A space X is called *contractible* if its identity map id_X is nullhomotopic.

Equivalently, X is contractible if and only if it is homotopy equivalent to a point (definition 1.1.14). If $f\colon X \to \{*\}$ and $g\colon \{*\} \to X$ satisfy $g \circ f \simeq \mathrm{id}_X$, then $g \circ f$ is a constant map, and the converse is the first item of example 1.1.15.

The idea of nullhomotopy is that a map which deforms continuously to a constant map can be thought of as homotopically trivial.

Example 1.1.9: Nullhomotopic

Take any convex set $U \subseteq \mathbb{R}^n$ (say, a ball) and a point $x_0 \in U$. The straight-line homotopy $(x,t) \mapsto (1-t)x + tx_0$ stays in U by convexity, and deforms the identity map of U to the constant map at x_0. Thus, all such shapes are contractible, and so is every space homeomorphic to one of them, by transporting this homotopy along the homeomorphism.

Deformation retractions are a first step towards isomorphisms in hTop, as well as an important notion in and of themselves.

Definition 1.1.10: Deformation Retraction

Let $f : X \to X$ be the identity map and $A \subseteq X$ a subspace. If there exists a homotopy $f_t : X \times I \to X$ such that:

1. $f_0 = \mathrm{id}$
2. $f_1(X) = A$
3. $f_t|_A = \mathrm{id}$ for all $t \in I$

then f_t is said to be a *deformation retraction* of the space X to A. If $f_t : X \times I \to Y$ is a homotopy where for some subspace $A \subseteq X$, $f_t|_A$ is independent of t, then we call the homotopy *homotopy relative* to A, or *homotopy rel* A.

Think about taking bold-face letters or higher dimensional objects like S^n punctured at the poles and retracting them to their thin counterparts or lower dimensional counterpart. Note that being retractable to a point is slightly stronger than being contractible since the map must restrict to the identity on the point. Every deformation retraction is homotopy relative to a subspace A.

The end map f_1 of a deformation retraction is a retraction:

Definition 1.1.11: Retraction

Let $f : X \to X$ be a continuous map. Then f is said to be a *retraction* on A if $f(X) = A$ and $f|_A = \mathrm{id}$.

The definition of a retraction involves no homotopy. In particular, fewer conditions need to hold for a retraction.

Example 1.1.12: Retraction

1. If $X = Y \times Z$, and $A = Y \times \{z_0\}$, then $f(y, z) = (y, z_0)$ is a retraction.
2. If $X = \left\{x \in \mathbb{R}^2 \;\middle|\; 1 \leq \|x\| \leq 2\right\}$ is an annulus and $A = \left\{x \in \mathbb{R}^2 \;\middle|\; \|x\| = 1\right\}$ is its inner circle, then pinching everything radially to A, $f(x) = x/\|x\|$, is a retraction.
3. $X = \mathbb{R}$, $A = \mathbb{R}_{\geq 0}$, then $\sigma(t) = |t|$ is a retraction

Thus, each deformation retraction is homotopic to a retraction map. Note that for each $x \in X$, we have a retraction map to $\{x\}$. However, since not all spaces are simply connected, not all spaces have a deformation retraction to a retraction (as we shall shortly see).

Example 1.1.13: Deformation Retraction

1. The following are three different deformation retractions of $\mathbb{R}^2$ with two punctured holes:

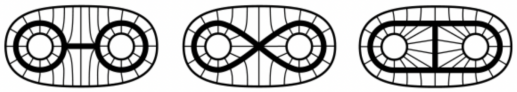


Figure 1.1: $\mathbb{R}^2 \setminus \{a, b\}$ retracted to different curves

2. There is a nice way of visualizing a deformation retraction. For $f_t : X \to Y$, take

$$M_f := (X \times I) \sqcup Y / [((x, 1) \sim y \text{ iff } f_1(x) = y)]$$

which we may visualize as the following:

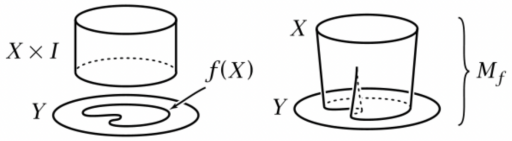


Figure 1.2: Mapping Cylinder of f

Notice that M_f also deformation retracts onto Y, namely by the homotopy $((x, t), s) \mapsto (x, (1-s)t+s)$ on the cylinder, fixing Y, which at $s = 1$ carries each (x, t) along $\{x\} \times I \subseteq M_f$ to $f(x)$.

3. Suppose X deformation retracts onto a point $x \in X$, by a homotopy $F \colon X \times I \to X$ fixing x at all times. Then every open neighborhood U of x contains an open neighborhood V of x with $V \hookrightarrow U$ nullhomotopic. Since $F(\{x\} \times I) = \{x\} \subseteq U$, compactness of I (the tube

lemma) gives an open $V \ni x$ with $F(V \times I) \subseteq U$, and F restricted to $V \times I$ is a homotopy in U from the inclusion to the constant map at x.

Consider the *comb*

$$C = ([0,1] \times \{0\}) \cup \bigcup_{r \in \mathbb{Q} \cap [0,1]} (\{r\} \times [0, 1-r])$$

It deformation retracts onto each point $(a, 0)$ of its base: first slide every tooth straight down onto the base, then contract the base to $(a, 0)$ along itself. It deformation retracts onto no point off the base. Take $p = (r_0, y)$ with $y > 0$, a small open ball B around p missing the base, and the neighborhood $U = B \cap C$ of p in C. Then U is a union of pairwise disjoint vertical segments on distinct teeth, and its path components are these segments, since the x-coordinate of a path in U is a continuous map into $\mathbb{Q}$, hence constant. Every neighborhood $V \subseteq U$ of p in C meets teeth other than the one through p: the points $(r, \min(y, 1-r))$ lie on the tooth at r and converge to p as $r \to r_0$ through the rationals. A nullhomotopy of $V \hookrightarrow U$ would join every point of V by a path in U to the constant point, putting all of V in one path component of U. That contradicts the first part. Hence, a space like so:



Figure 1.3: Contractible, but not Deformation Retractable

is contractible, but not deformation retractable[a].

[a] Though, there is a *weak deformation retract* from the zigzag space above to the darker line

Retractions are useful since they generalize the notion of invertibility, leading to the isomorphisms of hTop:

Definition 1.1.14: Homotopy Equivalence

Let $f : X \to Y$ be a continuous map. Then f is called a *homotopy equivalence* if there exists a continuous g such that $f \circ g \simeq \mathrm{id}$ and $g \circ f \simeq \mathrm{id}$. In this case, we shall write $X \simeq Y$ and call them *homotopy equivalent* or that they have the same *homotopy type*. In hTop, homotopy equivalences are the isomorphisms.

Example 1.1.15: Homotopy Equivalent

1. Suppose the shape X is contractible, so that the identity $F : X \to X$ has a homotopy $F_t : X \times I \to X$ to $F_1(X) = \{x\}$. Consider the map $f : X \to \{*\}$ where $f(X) = *$ and homotopy inverse $g : \{*\} \to X$ where $g(*)$ maps to $\{x\}$. Then the map

$$g \circ f : X \to X \qquad g \circ f(X) = \{x\}$$

is homotopic to $\mathrm{id} : X \to X$ by taking $(g \circ f)_t = F_{1-t}$.

2. A non-trivial example would be that the 3 shapes in fig. 1.1 are homotopy equivalent. This also shows that homotopy equivalence need not preserve cardinality, or even dimension, and hence is weaker than a homeomorphism[a].

3. Let $f : X \to X$ be a deformation retract onto A. Then $A \hookrightarrow X$ is a homotopy equivalence (this can be thought of as the generalization of the contractible argument). A slightly weaker condition still works: if $f : X \to X$ has a homotopy such that $f_0 = \mathrm{id}$, $f_1(X) \subseteq A$ and $f_t(A) \subseteq A$, then f is called a *weak deformation retract* onto A, and $A \hookrightarrow X$ is still a homotopy equivalence. With $i\colon A \hookrightarrow X$ and $r = f_1\colon X \to A$, the composite $i \circ r = f_1$ is homotopic to $f_0 = \mathrm{id}_X$ through the f_t, and $r \circ i = f_1|_A$ is homotopic to id_A through the restrictions $f_t|_A$, which land in A because $f_t(A) \subseteq A$.

[a]The simplest example that only requires a notion of connectivity is showing that $\mathbb{R} \ncong \mathbb{R}^2$

A large portion of this book will be examining what properties are invariant under homotopy equivalence. As pointed out, cardinality and dimension are not preserved, so in a sense they preserve more "core" topological features. We shall see that the number of "holes" will be preserved. Though this might seem like a triviality at first glance, this shall have major implications for finding when certain derivatives exist[1], and more generally for an important rough classification of topological spaces via homotopy and homology/cohomology groups.

1.1.16 Homotopies between Compact Surfaces Two *closed* surfaces (compact, without boundary, of dimension 2) that are homotopy-equivalent are in fact homeomorphic, so that homotopy adds no extra flexibility when working with closed surfaces. Boundary matters here: the annulus and the Möbius band are homotopy equivalent (both deformation retract to a circle) but not homeomorphic (see [21][a]). This is not the case in general.

[a]This comes down to the classification of compact surfaces. Genus and orientation characterize closed surfaces and are preserved by homotopy equivalences, and hence two homotopy equivalent closed surfaces must be homeomorphic.

Showing that a space is contractible may be easier than showing that it deformation retracts to a point.

Example 1.1.17: House With Two Rooms
Consider the following space:



Figure 1.4: House With Two Rooms, Hatcher

The following argument is a picture-level illustration. Its two geometric claims are read off the figure and not proved here, and the contractibility of X is proved rigorously in example 5.3.8. If X represents this space, let X_ϵ be this space with each point thickened with a filled in ϵ-ball, for ϵ small enough that X_ϵ deformation retracts to X (in fact, is homotopy equivalent to X: if $r : X_\epsilon \to X$ is the retraction and $\iota : X \to X_\epsilon$ is the inclusion, $r\iota \simeq \mathrm{id}$ and $\iota r \simeq \mathrm{id}$). Then X_ϵ is homeomorphic to the filled in closed ball $D^3 \subseteq \mathbb{R}^3$: it can be formed from a ball of clay by pushing in the two tunnels and hollowing out the two chambers. Since D^3 is contractible to a point $\{*\}$ by

[1]for those who know differential forms, when a closed form is exact

the straight-line homotopy,

$$X \simeq X_\epsilon \simeq D^3 \simeq \{*\}$$

Hence by transitivity X is contractible. In fact, X deformation retracts to a point $x_0 \in X$: if f_t is a deformation retraction of $X_\epsilon \cong D^3$ to x_0, then $r \circ f_t$ restricted to X starts at $r|_X = \mathrm{id}_X$, ends at the constant map at x_0, and fixes x_0 throughout. Seeing what this deformation retraction looks like is much harder.

1.2 Cell-Complex

As is often the case in mathematics, we understand well some simple examples, and would like to create more complicated examples using the simpler family of objects. There are many simple geometric shapes which we may choose that have very simple combinatorial properties. The disk and the triangle are examples. We restrict to the disk and its higher dimensional counterparts D^n. The rigid shape of D^n is not important, but its topological properties are. Thus, define an n-cell to be a topological space homeomorphic to D^n. Note that a 1-cell is homeomorphic to a closed interval $[a, b]$.

Definition 1.2.1: CW Complex

A *CW complex* or a *cell complex* is a space constructed in the following form:

1. Start with a discrete set X^0 whose points are regarded as 0-cells
2. Inductively, form the n-skeleton X^n from X^{n-1} by attaching n-cells e^n_α via gluing maps

   $$\varphi_\alpha : \partial e^n_\alpha \to X^{n-1}$$

   where $\partial e^n_\alpha = S^{n-1}$. That is, X^n is the quotient space of the disjoint union $X^{n-1} \sqcup_\alpha D^n_\alpha$ of X^{n-1} with the collection of n-disks D^n_α under the identifications $x \sim \varphi_\alpha(x)$ for $x \in \partial D^n_\alpha$. As a set, $X^n = X^{n-1} \sqcup_\alpha e^n_\alpha / \sim$ where each e^n_α is an open n-disk.
3. The induction stops at a finite step, at which point we set $X = X^n$, or we continue this process indefinitely and set $X = \bigcup_n X^n$ in which case X is given the weak topology via inclusion ($A \subseteq X$ is open if and only if $A \cap X^n$ is open in X^n for all n). If $X = X^n$, then X is said to be finite dimensional with dimension n.

To break down the definition, it is perhaps more intuitive to see that we have

$$X^0 \hookrightarrow X^1 \hookrightarrow \cdots$$

where X_k is obtained by gluing copies of k-cells $(e^k_\alpha)_\alpha$ to X_{k-1} via gluing maps $g^k_\alpha : \partial e^k_\alpha \to X_{k-1}$. Then X is the colimit of this diagram with these maps

$$\varinjlim_i X_i = X$$

The term *CW* stands for "Closure-finite Weak topology".

Each cell of a cell complex comes with a natural map into the complex.

Definition 1.2.2: Characteristic Map

Let X be a cell complex and e^n_α associated to $\varphi_\alpha : \partial e^n_\alpha \to X^{n-1}$ be a cell forming this complex. Then there exists a map $\Phi_\alpha : e^n_\alpha \to X$ extending the map φ_α to be a homeomorphism from the interior of D^n_α to e^n_α, in particular $\Phi_\alpha : e^n_\alpha \hookrightarrow X^{n-1} \sqcup_\alpha D^n_\alpha \to X^n \hookrightarrow X$.

The name characteristic is due to its link to *Euler's characteristic* (see section 3.5.3).

definition 1.2.1 prescribes a construction and a topology, and says nothing about separation. The property one needs first is the weakest one, and it is what lets a compact subset of a cell complex be treated as closed.

Theorem 1.2.3: CW Complexes Are Hausdorff

Every CW complex is Hausdorff.

***Proof*:**

Write X^n for the n-skeleton, $\varphi_\alpha \colon S^{n-1} \to X^{n-1}$ for the attaching maps of the n-cells and $\Phi_\alpha \colon D^n_\alpha \to X^n$ for the characteristic maps of definition 1.2.2. Throughout, X^n carries the quotient topology from $q \colon X^{n-1} \sqcup_\alpha D^n_\alpha \to X^n$, so a subset of X^n is open exactly when its preimage meets X^{n-1} and each D^n_α in an open set.

Step 1: disjoint open sets extend one dimension upward. Let U, V be disjoint open subsets of X^{n-1}. For each n-cell put

$$\hat{U}_\alpha = \left\{ rs \in D^n_\alpha \;\middle|\; s \in \varphi_\alpha^{-1}(U),\ \tfrac{1}{2} < r \le 1 \right\}$$

and define $\hat{V}_\alpha$ from V in the same way, rs being scalar multiplication in $D^n_\alpha \subseteq \mathbb{R}^n$. Each $\hat{U}_\alpha$ is open in D^n_α, being the intersection of D^n_α with the open set $\left\{ x \in \mathbb{R}^n \;\middle|\; \tfrac{1}{2} < |x| < 2,\ x/|x| \in \varphi_\alpha^{-1}(U) \right\}$. The set $U \sqcup_\alpha \hat{U}_\alpha$ is saturated for q: the only identifications are $s \sim \varphi_\alpha(s)$ for $s \in S^{n-1}$, and $s \in \hat{U}_\alpha$ holds exactly when $\varphi_\alpha(s) \in U$. So it descends to a set U' open in X^n with $U' \cap X^{n-1} = U$, and likewise V'. They are disjoint, since $U \cap V = \emptyset$ and $\varphi_\alpha^{-1}(U) \cap \varphi_\alpha^{-1}(V) = \emptyset$ forces $\hat{U}_\alpha \cap \hat{V}_\alpha = \emptyset$.

Step 2: every skeleton is Hausdorff, by induction on n. The set X^0 is discrete. Assume X^{n-1} Hausdorff and take $p \neq q$ in X^n. Each open cell $e^n_\alpha = \Phi_\alpha(D^n_\alpha \setminus S^{n-1})$ is open in X^n, its preimage being $D^n_\alpha \setminus S^{n-1}$ in that disk and empty elsewhere, and Φ_α restricts to a homeomorphism onto it.

1. If p and q lie in distinct open n-cells, those two cells separate them.
2. If they lie in the same open cell, that cell is homeomorphic to $D^n \setminus S^{n-1}$, which is Hausdorff, and the separating sets are open in X^n because the cell is.
3. If $q = \Phi_\alpha(x)$ lies in an open n-cell and $p \in X^{n-1}$, choose ρ with $|x| < \rho < 1$. Then $B = \Phi_\alpha(\{z \mid |z| < \rho\})$ is open in X^n and $C = \Phi_\alpha(\{z \mid |z| \le \rho\})$ is closed in X^n, since the preimage of each meets D^n_α in an open, respectively closed, ball and meets every other disk and X^{n-1} in the empty set. So B and $X^n \setminus C$ separate q and p.
4. If both lie in X^{n-1}, the inductive hypothesis separates them there and Step 1 pushes the separation into X^n.

Step 3: the whole complex. If the construction stops at a finite stage, Step 2 is the claim. Otherwise let $x \neq y$ in $X = \bigcup_n X^n$ and pick N with $x, y \in X^N$. Step 2 gives disjoint open $U_N, V_N \subseteq X^N$ separating them, and Step 1 extends these upward to disjoint open $U_n, V_n \subseteq X^n$ for every $n \geq N$ with $U_{n+1} \cap X^n = U_n$ and $V_{n+1} \cap X^n = V_n$. Put $U = \bigcup_{n \geq N} U_n$ and $V = \bigcup_{n \geq N} V_n$. These are disjoint, and each is open for the weak topology: for $n \geq N$ one has $U \cap X^n = U_n$, and for $n < N$ one has $U \cap X^n = U_N \cap X^n$, open in X^n as a subspace of X^N. Hence U and V separate x and y, completing the proof.

The three cases of Step 2 are the only ways two points of a skeleton can sit relative to the cells, and only the last needs the inductive hypothesis. The same extension device of Step 1 is what makes normality provable by the same induction, with Urysohn functions in place of the radial collars.

Lemma 1.2.4: Compact Subsets Meet Finitely Many Cells

Let X be a CW complex and $K \subseteq X$ a compact subset. Then K meets only finitely many open cells of X, and $K \subseteq X^N$ for some N.

***Proof*:**
Suppose K meets infinitely many open cells. Choose distinct open cells $e_1, e_2, \dots$ meeting K and points $x_i \in K \cap e_i$, and let $S = \{x_1, x_2, \dots\}$. The open cells are pairwise disjoint, so S is infinite and meets each open cell in at most one point.

Step 1: every subset $T \subseteq S$ is closed in X. By the weak topology of definition 1.2.1 it suffices to show that $T \cap X^n$ is closed in X^n for every n, by induction on n. The skeleton X^0 is discrete, so $T \cap X^0$ is closed in it. Suppose $T \cap X^{n-1}$ is closed in X^{n-1}. The skeleton X^n carries the quotient topology from $X^{n-1} \sqcup_\alpha D^n_\alpha$, and $(T \cap X^n) \cap X^{n-1} = T \cap X^{n-1}$ is closed in X^{n-1}, so $T \cap X^n$ is closed in X^n once $\Phi_\alpha^{-1}(T \cap X^n)$ is closed in D^n_α for every n-cell, Φ_α being the characteristic map of definition 1.2.2. On ∂D^n_α the map Φ_α is the attaching map φ_α, with values in X^{n-1}, so

$$\Phi_\alpha^{-1}(T \cap X^n) = \varphi_\alpha^{-1}(T \cap X^{n-1}) \cup \left(\Phi_\alpha^{-1}(T) \cap (D^n_\alpha \setminus \partial D^n_\alpha)\right)$$

The first set is closed in ∂D^n_α, by continuity of φ_α and the inductive hypothesis, hence closed in D^n_α. The second has at most one point, because Φ_α maps the open disk injectively onto the open cell e^n_α and T meets that cell in at most one point. A union of a closed set and a finite set is closed in D^n_α, which completes the induction.

Step 2: the contradiction. By Step 1 the set S is closed in X, hence closed in the compact set K, hence compact. Each singleton $\{x_i\}$ is open in S, since its complement in S is closed by Step 1. The cover of S by its singletons is then an open cover of a compact space by infinitely many sets, no finitely many of which cover S, which is impossible. Hence K meets only finitely many open cells.

As a set, X is the disjoint union of its open cells (definition 1.2.1), and each open cell lies in the skeleton of its dimension. So K lies in the union of the finitely many open cells it meets, and $K \subseteq X^N$ for N the largest of their dimensions ($N = 0$ if K is empty), as claimed.

Example 1.2.5: Examples of CW Complexes

1. Let X^0 be some collection of points. Then we may form a graph by attaching 1-cells (i.e. spaces homeomorphic to closed intervals) to points in X^0. If we so choose, we may form loops in our graphs.

2. $\mathbb{R}$ can be seen as having a CW-structure (i.e. homeomorphic to a CW-complex) by taking the 0-skeleton $\mathbb{Z}$ and attaching the 1-cells $[n, n+1]$. This process can be generalized to give $\mathbb{R}^n$ a CW-structure.

3. The house with two rooms pictured earlier is a CW complex. One such structure has twenty-nine 0-cells, fifty-one 1-cells, and twenty-three 2-cells.

4. One can get the sphere S^n by attaching an n-cell to a point e^0 along the constant map $\partial D^n \to e^0$ (there are no k-cells for $0 < k < n$, meaning $X^0 = X^1 = \cdots = X^{n-1}$). The result is $D^n/\partial D^n$, and $x \mapsto \left(2\sqrt{1 - \|x\|^2}\, x,\ 2\|x\|^2 - 1\right)$ maps D^n onto $S^n \subseteq \mathbb{R}^{n+1}$, sends ∂D^n to the point $(0, 1)$ and is injective on the open disk, so it induces a continuous bijection from the compact space $D^n/\partial D^n$ to the Hausdorff space S^n, which is a homeomorphism. Alternatively, S^n is obtained from S^{n-1} by attaching two n-cells, the upper and lower hemispheres, each along the identity of S^{n-1}. Starting from S^0, this gives a structure with two cells in each dimension $0, \ldots, n$ (try this for S^2).

5. Take $X^0 = \{*\}$ to be a single point. Then we may attach two 1-cells. Then, take one 2-cell, and identify two of its sides with one of the circles, and two of its sides with the other circle (to do this, make the map a $2 - 1$ map onto the two 1-cells). If we straighten out the circle so that it looks like squares, we would get the following visual:

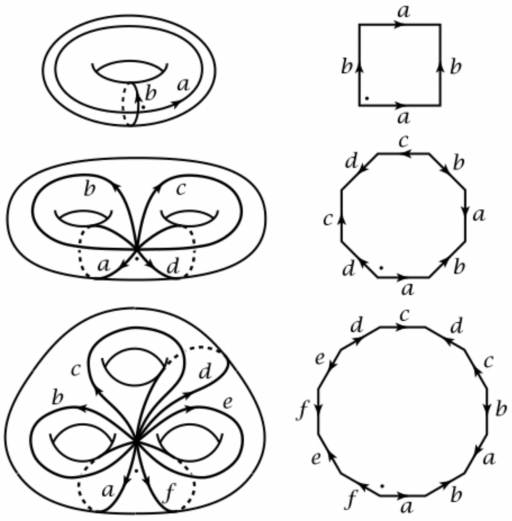


Figure 1.5: n-torus as CW complex, Hatcher

The other examples are when we are attaching a 2-cell to 4 or 6 circles respectively.

6. $\mathbb{R}\mathrm{P}^n$ is a CW-complex. We can see this by interpreting $\mathbb{R}\mathrm{P}^n$ as $S^n/(v \sim -v)$. In this case, $\mathbb{R}\mathrm{P}^n$ is $\mathbb{R}\mathrm{P}^{n-1}$ with the gluing $S^{n-1} \to \mathbb{R}\mathrm{P}^{n-1}$ gluing antipodal points together. This inductive process is well-defined as $\mathbb{R}\mathrm{P}^1 = S^1$. The trick is to glue the boundary at "twice the speed" around S^1. Hence $\mathbb{R}\mathrm{P}^n$ has one cell per dimension.

   From this, we may define $\mathbb{R}\mathrm{P}^\infty = \bigcup_n \mathbb{R}\mathrm{P}^n$ to also be a cell-complex with one cell per dimension. The space $\mathbb{R}\mathrm{P}^\infty$ can be viewed as the space of lines through the origin in $\mathbb{R}^\infty = \bigcup_n \mathbb{R}^n$.

7. $\mathbb{C}\mathrm{P}^n$ is a CW-complex. We shall show how we can create $\mathbb{C}\mathrm{P}^n$ as the quotient of D^{2n} under the relation $v \sim \lambda v$ for $v \in \partial D^{2n}$. Take $S^{2n+1} \subseteq \mathbb{C}^{n+1}$ with last coordinates real and non-negative. Then they are vectors of the form:

$$(w, \sqrt{1 - |w|^2}) \in \mathbb{C}^n \times \mathbb{C} \qquad |w| \leq 1$$

   These vectors are the graph of the function $w \mapsto \sqrt{1 - |w|^2}$ and form a disk $D^{2n}_+ \subseteq S^{2n+1}$ whose boundary S^{2n-1} consists of vectors of the form $(w, 0) \in \mathbb{C}^n \times \mathbb{C}$ with $|w| = 1$. Then putting the relation $v \sim \lambda v$ on S^{2n+1} can be identified with vectors in D^{2n}_+, and the identification is unique if the last coordinate is nonzero. If the last coordinate is zero, we take $v \sim \lambda v$ for $v \in S^{2n-1}$.

   Thus, we have $\mathbb{C}\mathrm{P}^n$ described as the quotient of D^{2n}_+ through $v \sim \lambda v$ for $v \in S^{2n-1}$. Finally, $\mathbb{C}\mathrm{P}^n$ is obtained from $\mathbb{C}\mathrm{P}^{n-1}$ by attaching a $2n$-cell e^{2n} via $S^{2n-1} \to \mathbb{C}\mathrm{P}^{n-1}$. Hence,

$$\mathbb{C}\mathrm{P}^n = e^0 \cup e^2 \cup \cdots \cup e^{2n}$$

   We may define $\mathbb{C}\mathrm{P}^\infty$ similarly to $\mathbb{R}\mathrm{P}^\infty$.

Many later constructions operate on subcomplexes.

Definition 1.2.6: Subcomplex

Let X be a cell complex. Then a *subcomplex* is a closed subspace $A \subseteq X$ that is a union of cells of X. A pair (X, A) consisting of a cell complex X and a subcomplex A is called a *CW pair*.

Since A is closed, the characteristic map of each cell in A has image contained in A, and the image of the attaching maps of A is contained in A, so A is itself a cell-complex.

Note that the closure of each cell of a cell-complex (or the closure of a collection of cells) need not be a subcomplex (though they often will be). For example, take S^1 with the usual one 0-cell and one 1-cell. Attach a 2-cell to S^1 by gluing the boundary of e^2 to a *proper subset* (or sub-arc) of S^1. Then the closure of the 2-cell is not a subcomplex since it contains only a *part* of the 1-cell.

Example 1.2.7: Subcomplexes

1. Each skeleton X^n of a cell-complex X is a subcomplex.
2. Each $\mathbb{R}\mathrm{P}^k \subseteq \mathbb{R}\mathrm{P}^n$ and $\mathbb{C}\mathrm{P}^k \subseteq \mathbb{C}\mathrm{P}^n$ is a subcomplex for $k \leq n$. In fact, these are the only subcomplexes of $\mathbb{R}\mathrm{P}^n$ and $\mathbb{C}\mathrm{P}^n$
3. The natural inclusions $S^0 \subseteq S^1 \subseteq \cdots \subseteq S^n$ are subcomplexes given the right CW structure

of S^n, namely the structure of example 1.2.5(4) in which each S^k is obtained from S^{k-1} by attaching two hemispheres, instead of the structure with one 0-cell and one n-cell.

Cell complexes shall be a fundamental tool we use. Many invariants will be defined by showing the CW structure is invariant up to homotopy, and taking advantage of the combinatorial information of the CW structure. There are many theorems that characterize when a topological space X admits a CW structure. Milnor proved that every manifold has the homotopy type of a CW complex, similar results hold for algebraic varieties, and the Lundell-Weingram Theorem characterizes some technical requirements. The general rule of thumb should be: if the object is "pretty geometric", it will admit a CW structure. So many "pathological" examples or "weird infinite dimensional" examples shall not have CW structures, but that is acceptable as it is already interesting that we can use this structure on essentially all geometric objects.

1.2.1 Operations on Cell Complex

One of the ways we shall take advantage of CW structures is by defining new ways of constructing CW structures given already known ones. Most of these constructions work more generally in the category Top. Any appropriate adjustments to the definitions can be made.

Definition 1.2.8: Product

Let X and Y be CW complexes with characteristic maps Φ_α and Ψ_β. The *product cell structure* on $X \times Y$ has as cells the products $e^m_\alpha \times e^n_\beta$, with characteristic maps $\Phi_\alpha \times \Psi_\beta \colon D^m \times D^n \cong D^{m+n} \to X \times Y$.

In general the topology that the cells determine is finer than the product topology, and the two can differ. They agree when X or Y is locally compact, for instance a finite complex, and when both have countably many cells (lemma 1.2.12).

Example 1.2.9: Product Of Cells

Let S^1 have the standard cell structure (attach I to a single point identifying the endpoints). Then the torus $T^2 = S^1 \times S^1$ has the product cell structure, with one 0-cell, two 1-cells and one 2-cell, as in fig. 1.5.

Definition 1.2.10: Quotient

Let (X, A) be a CW pair. Then X/A has a cell-structure respecting the universal property of quotients given as follows: The cells of X/A are the cells of $X - A$ plus a new 0-cell, the image of A in X/A.

Example 1.2.11: Quotient Of Cells

1. Take S^{n-1} with any cell structure, build D^n from S^{n-1} by attaching an n-cell in the natural way, and consider (D^n, S^{n-1}). Then D^n/S^{n-1} is S^n with the usual cell-structure.

For infinite complexes the product cell structure needs its own topology, and smash products and suspensions combine it with the quotients just defined.

Lemma 1.2.12: Products of CW Complexes and the Cell Topology

Let X and Y be CW complexes with the product cell structure of definition 1.2.8, and let $X \times_c Y$ denote $X \times Y$ with the *cell topology*: a set is closed exactly when its intersection with $C \times D$ is closed in the product topology of $C \times D$ for every pair of finite subcomplexes $C \subseteq X$, $D \subseteq Y$.

1. $X \times_c Y$ is a CW complex with the product cells, and each $C \times D$ as above is a finite subcomplex of it whose subspace topology is the product topology.
2. The identity map $X \times_c Y \to X \times Y$ is continuous. Every compact subset of $X \times_c Y$ and every compact subset of $X \times Y$ lies in some $C \times D$, on which the two topologies agree. Consequently the two spaces have the same maps from compact Hausdorff domains, the identity induces an isomorphism of singular chain complexes, and $H_\bullet$ and $H^\bullet$ of the two spaces agree with every coefficient group. Moreover a map $Z \to X \times_c Y$ from a CW complex Z is continuous exactly when it is continuous into the product topology.
3. If X or Y is locally compact, for instance finite, then $X \times_c Y = X \times Y$.
4. If $q\colon X \to X'$ and $r\colon Y \to Y'$ are the projections collapsing subcomplexes as in definition 1.2.10, then $q \times r\colon X \times_c Y \to X' \times_c Y'$ is a quotient map.
5. If X and Y both have countably many cells, then $X \times_c Y = X \times Y$.

***Proof*:**

Step 1: Finite factors. Let C and D be finite CW complexes. The disjoint unions of the closed cells give continuous surjections $\Phi\colon \bigsqcup_\alpha D^{m_\alpha} \to C$ and $\Psi\colon \bigsqcup_\beta D^{n_\beta} \to D$, and a set is closed in a CW complex exactly when its preimage under each characteristic map is closed, by induction up the (finitely many) skeleta of definition 1.2.1, since each skeleton carries the quotient topology of the previous one with the attached disks. The map $\Phi \times \Psi$ is a continuous surjection from a compact space to $C \times D$, which is Hausdorff by theorem 1.2.3, so it is a closed map and hence a quotient map. Its source decomposes into the products $D^{m_\alpha} \times D^{n_\beta} \cong D^{m_\alpha + n_\beta}$, which are the characteristic domains of the product cells. So a subset of $C \times D$ is closed in the product topology exactly when its preimage under every product characteristic map is closed, and the latter condition is equivalent to having closed intersection with every closed product cell, since each characteristic map is itself a closed surjection onto its image cell. The product topology on $C \times D$ is therefore the topology of the product cell structure, which proves the finite case of (1) and, since finite complexes are compact, shows the cell topology of the general $X \times_c Y$ restricts on each $C \times D$ to the product topology.

Step 2: The general cell structure. Each point of $X \times Y$ lies in a unique product open cell. A closed product cell $\bar{e} \times \bar{f}$ lies in a product $C \times D$ of finite subcomplexes, because a closed cell meets finitely many open cells and, by induction on dimension through that finiteness, lies in a finite subcomplex. Each characteristic map $\Phi_\alpha \times \Psi_\beta\colon D^{m+n} \to X \times_c Y$ is continuous, since it is continuous into the compact $C \times D$ with its product topology, which by Step 1 is its subspace topology inside $X \times_c Y$. Closure-finiteness holds because $\bar{e} \times \bar{f}$ lies in a finite subcomplex. Finally the cell topology is the weak topology determined by the closed product cells: closedness tested on every $C \times D$ is equivalent to closedness tested on every $\bar{e} \times \bar{f}$, since each closed product cell lies in some $C \times D$ and each $C \times D$ is a finite union of closed product cells carrying, by Step 1,

the topology those cells determine. Running the skeletal construction of definition 1.2.1 on these cells and attaching maps therefore reproduces exactly the topology of $X \times_c Y$, which is (1).

Step 3: Comparison of the topologies. A set closed in the product topology has closed intersection with each $C \times D$, so the identity $X \times_c Y \to X \times Y$ is continuous. A compact subset of $X \times Y$ has compact projections, each of which lies in a finite subcomplex by lemma 1.2.4 and the closed-cell argument of Step 2, so the compact set lies in some $C \times D$. A compact subset of $X \times_c Y$ lies in a finite subcomplex of it, again by lemma 1.2.4, hence in a finite union of closed product cells, hence in some $C \times D$. On $C \times D$ the topologies agree by Step 1. A continuous map from a compact Hausdorff space into either topology therefore lands in a common $C \times D$ and is continuous into the other, and in particular the singular simplices of the two spaces coincide, giving the asserted identification of singular chains and of (co)homology. For the last clause of (2), a map $g\colon Z \to X \times Y$ continuous into the product topology restricts on each closed cell of Z to a map from a compact Hausdorff space, which is then continuous into $X \times_c Y$, and a map out of a CW complex is continuous when its restriction to every closed cell is.

Step 4: A locally compact factor. Suppose X is locally compact, and let $p\colon \bigsqcup_\alpha D^{m_\alpha} \to X$ and $r\colon \bigsqcup_\beta D^{n_\beta} \to Y$ be the characteristic-map surjections, which are quotient maps as in Step 1 (the skeletal construction makes closedness in a CW complex equivalent to closedness of all characteristic preimages, in general and not only in the finite case). The source of p is locally compact, being a disjoint union of disks, so by [16, Products of Quotient Maps] applied twice,

$$p \times r = (\mathrm{id}_X \times r) \circ (p \times \mathrm{id})\colon \Big(\bigsqcup_\alpha D^{m_\alpha}\Big) \times \Big(\bigsqcup_\beta D^{n_\beta}\Big) \longrightarrow X \times Y$$

is a composition of quotient maps, hence a quotient map, where $p \times \mathrm{id}$ is a quotient map because its fixed factor is locally compact and $\mathrm{id}_X \times r$ is one because X is. The quotient topology induced by $p \times r$ is the final topology with respect to the product characteristic maps, which as in Step 2 is the cell topology. Hence the product topology equals the cell topology, which is (3).

Step 5: Products of collapses. Let q, r be as in (4) and let $g\colon X' \times_c Y' \to T$ have $g \circ (q \times r)$ continuous. It suffices to show g is continuous on $C' \times D'$ for all finite subcomplexes $C' \subseteq X'$, $D' \subseteq Y'$. Each cell of C' other than the collapse point is a cell of X, and a finite subcomplex of X containing those closed cells, enlarged by one 0-cell of the collapsed subcomplex when C' contains the collapse point, is a finite subcomplex $C \subseteq X$ with $C' \subseteq q(C)$. Choose $D \subseteq Y$ for D' in the same way. The restriction $(q \times r)|\colon C \times D \to q(C) \times r(D)$ is a continuous surjection from a compact space to a Hausdorff space, hence a quotient map, and $g \circ (q \times r)|$ is continuous, so g is continuous on $q(C) \times r(D)$ and therefore on the closed subspace $C' \times D'$.

Step 6: Countably many cells. Suppose X and Y have countably many cells. By Step 3 it suffices to show that a set W closed in $X \times_c Y$ is closed in $X \times Y$. Each closed cell lies in a finite subcomplex (Step 2), so an enumeration of the cells gives finite subcomplexes $X_1 \subseteq X_2 \subseteq \cdots$ such that every finite subcomplex of X lies in some X_k, and likewise $Y_1 \subseteq Y_2 \subseteq \cdots$. A subset $O \subseteq X$ whose intersection with every X_k is open in X_k is open in X, since it then meets every closed cell in a relatively open set. Fix $(x, y) \notin W$ and k_0 with $x \in X_{k_0}$ and $y \in Y_{k_0}$, and put $P_{k_0} = \{x\}$ and $Q_{k_0} = \{y\}$. Suppose compact sets $P_k \subseteq X_k$ and $Q_k \subseteq Y_k$ with $(P_k \times Q_k) \cap W = \emptyset$ are given. The set $W \cap (X_{k+1} \times Y_{k+1})$ is closed in the compact Hausdorff space $X_{k+1} \times Y_{k+1}$, by the definition of the cell topology and Step 1. For each $p \in P_k$, finitely many open boxes of $X_{k+1} \times Y_{k+1}$ missing W cover $\{p\} \times Q_k$, and the intersection of their first factors and the union of their second factors are open sets $U_p \ni p$ and $V_p \supseteq Q_k$ with $(U_p \times V_p) \cap W = \emptyset$. Finitely many U_p cover P_k, and their union U and the intersection V of the corresponding V_p satisfy

$(U \times V) \cap W = \emptyset$. Compact Hausdorff spaces are regular, so there are open sets U' and V' with $P_k \subseteq U' \subseteq \overline{U'} \subseteq U$ and $Q_k \subseteq V' \subseteq \overline{V'} \subseteq V$. Put $P_{k+1} := \overline{U'}$ and $Q_{k+1} := \overline{V'}$. These are compact, $(P_{k+1} \times Q_{k+1}) \cap W = \emptyset$, and P_k and Q_k lie in the interiors $\mathrm{int}_{X_{k+1}} P_{k+1}$ and $\mathrm{int}_{Y_{k+1}} Q_{k+1}$. Let O be the union of the sets $\mathrm{int}_{X_k} P_k$ for $k > k_0$, and O' the analogous union for the Q_k. Each term of O with $k < j$ lies in the term with index j, so $O \cap X_j$ is the union of the sets $\mathrm{int}_{X_k} P_k \cap X_j$ with $k \geq \max(j, k_0 + 1)$, each open in X_j. Hence O is open in X, and likewise O' is open in Y, with $(x, y) \in O \times O'$. A point of $O \times O'$ lies in $P_k \times Q_k$ for k large, since both sequences increase, so $O \times O'$ misses W. The complement of W is therefore open in the product topology, which is (5), completing the proof.

Clause (2) is what makes the cell topology harmless. Singular homology, cohomology and homotopy groups are computed from maps of compact spaces, so they cannot tell $X \times_c Y$ from $X \times Y$. Clauses (3) and (5) say when the two topologies coincide, which covers every product with a finite factor and every product of two countable complexes, such as $\mathbb{R}\mathrm{P}^\infty \times \mathbb{R}\mathrm{P}^\infty$.

Definition 1.2.13: Suspension

Let X be a cell complex. Then the *suspension* of X, denoted SX, is the quotient of $X \times I$ obtained by collapsing $X \times \{0\}$ to a point and $X \times \{1\}$ to another point.

Visually, a suspension can be seen as so:

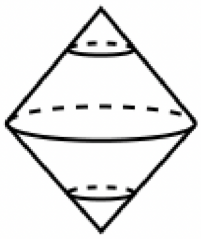


Figure 1.6: Suspension

The prototypical example of a suspension is $S(S^n) = S^{n+1}$. Since suspensions are created via quotienting and products, we see that we can create a cell structure on the cone of X:

$$CX = (X \times I)/(X \times \{0\})$$

the cylinder with one end collapsed to a point, so that SX is the union of two copies of CX glued along their common base $X \times \{1\}$.

The presentation of SX as two cones meeting along X underlies every later use of the suspension, and it holds for spaces far more general than cell complexes.

Proposition 1.2.14: Unreduced Suspension as a Union of Two Cones

Let X be a topological space, with cone $CX = (X \times I)/(X \times \{0\})$ and suspension $SX = (X \times I)/(X \times \{0\}, X \times \{1\})$ (the construction of definition 1.2.13, here applied to an arbitrary space X). Let C_-X and C_+X be the images in SX of the closed halves $X \times [0, \frac{1}{2}]$ and $X \times [\frac{1}{2}, 1]$. Then each of C_-X and C_+X is a cone on X, and

$$SX = C_-X \cup C_+X, \qquad C_-X \cap C_+X = X \times \{\tfrac{1}{2}\} \cong X$$

Each cone is contractible. If X is compact Hausdorff, then CX and SX are compact Hausdorff and the two cones $C_\pm X$ are closed subsets of SX whose union is SX.

***Proof*:**

Write $q\colon X \times I \to SX$ for the quotient map. On $X \times [0, \frac{1}{2}]$ the only identification imposed by q is the collapse of $X \times \{0\}$ to a point, so $C_-X = q(X \times [0, \frac{1}{2}])$ is a copy of the cone CX, and symmetrically $C_+X = q(X \times [\frac{1}{2}, 1])$ is a cone with cone point $q(X \times \{1\})$. Their union is $q(X \times I) = SX$, and q is injective on $X \times \{\frac{1}{2}\}$, where no collapse occurs, so $C_-X \cap C_+X$ is a copy of X. Each cone deformation retracts to its cone point by sliding along the I-coordinate, the retraction of C_-X being $\big(q(x, s), t\big) \mapsto q(x, (1-t)s)$ and that of C_+X its reflection, so both cones are contractible.

Suppose X is compact Hausdorff. Then $X \times I$ is compact Hausdorff, hence normal, and $SX = q(X \times I)$ is compact as a continuous image. Collapsing a closed subspace of a normal space to a point yields a Hausdorff quotient, and SX is obtained from $X \times I$ by collapsing the two disjoint closed subsets $X \times \{0\}$ and $X \times \{1\}$ in turn, their images staying disjoint and closed, so SX is Hausdorff. The same argument applied to the single closed subset $X \times \{0\}$ shows CX is compact Hausdorff. Finally $q^{-1}(C_-X) = X \times [0, \frac{1}{2}]$ and $q^{-1}(C_+X) = X \times [\frac{1}{2}, 1]$ are closed and saturated, so $C_\pm X$ are closed in SX, completing the proof.

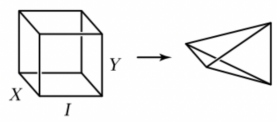
Though this might not seem useful, as we progress this construction will become increasingly important. Of particular importance is that maps can be suspended: if $f : X \to Y$, then we can suspend it to $Sf : SX \to SY$, which is induced by $f \times \mathrm{id} : X \times I \to Y \times I$ on the quotients. Often, the suspension of a "non-simply connected" object shall be simply connected, and the suspension of a map shall make it nullhomotopic.

In a suspension, all the points of X are joined to a single point. We can generalize this by joining every point in X to the points of a space Y (and vice-versa):

Definition 1.2.15: Join

Let X, Y be topological spaces. Then the *join*, denoted $X * Y$, is the quotient of the space $X \times Y \times I$ by $(x, y_1, 0) \sim (x, y_2, 0)$ and $(x_1, y, 1) \sim (x_2, y, 1)$. Thus $X \times Y \times \{0\}$ collapses to X while $X \times Y \times \{1\}$ collapses to Y.

If X, Y are CW complexes, then there is a natural CW structure on $X * Y$ with the subspaces X and Y as subcomplexes, and the remaining cells being the product cells of $X \times Y \times (0, 1)$. This process of the join operation can be visualized like so:



In fact, if $X = Y = [0, 1]$, then a join would produce the above shape. The join $X * Y$ contains a copy of X and Y, and every point $(x, y, t) \in X * Y$ is on a unique line segment joining the points $x \in X \subseteq X * Y$ and $y \in Y \subseteq X * Y$. Hence, we may think of every point in $X * Y$ as being written in the form

$$t_1 x + t_2 y$$

where $0 \leq t_i \leq 1$, $t_1 + t_2 = 1$ and $0x + 1y = y$ and $1x + 0y = x$. These expressions are all possible *convex combinations*. If we iterate and write $X_1 * \cdots * X_n$, we see that this is independent of bracketing and we can write every element in the form:

$$t_1 x_1 + \cdots + t_n x_n \qquad 0 \leq t_i \leq 1, t_1 + \cdots + t_n = 1$$

that is, we get every formal convex linear combination of the elements. A special but important case is when each X_i is a singleton. In this case, the join of two points is a line segment, the join of 3 is a triangle, the join of 4 is a tetrahedron, and so forth. In general, the join of n points is a convex polyhedron of dimension $n - 1$ known as a *simplex*.

Example 1.2.16: Join

Let X_i be two points, and say they are placed in $\mathbb{R}^n$ on a coordinate axis at unit lengths. Then the join $X_1 * \cdots * X_n$ is the union of 2^n copies of $n - 1$ dimensional simplices. The radial projection from the origin gives a homeomorphism from $X_1 * \cdots * X_n$ to S^{n-1}.

Definition 1.2.17: Wedge Sum

Let X, Y be topological spaces. Then given points $x \in X$ and $y \in Y$, the *wedge sum* (with respect to x, y) is

$$X \vee Y = X \sqcup Y / x \sim y$$

If X, Y are pointed spaces, so that by definition there is a distinguished point as part of the definition, then the wedge sum is naturally defined (no need to say "with respect to"). If X, Y are cell complexes, $X \vee Y$ is a cell complex by taking the cell complex $X \sqcup Y$ and collapsing a subcomplex to a point. A simple example of a wedge sum would be for any cell complex X, the quotient X^n / X^{n-1} is a wedge sum of the n-spheres $\vee_\alpha S^n_\alpha$ with one sphere for each n-cell of X.

Definition 1.2.18: Smash Product

Let X, Y be topological spaces. Then given $x \in X$ and $y \in Y$, the *smash product* (with respect to x and y) is

$$X \wedge Y := X \times Y / X \vee Y$$

If X, Y are cell complexes pointed by 0-cells, then giving $X \times Y$ the cell-complex topology makes the smash product a cell complex.

1.3 Homotopy Equivalence and Homotopy Extension

In this section, we shall prove some important theorems to find homotopy equivalent spaces. The first is a way of collapsing some parts of our space to a point to simplify the problem, and the second varies the way we put subspaces together.

In general, if we take X and X/A, these two spaces may have different homotopy types. For example, take the torus T^2 and $T^2/(S^1 \vee S^1) \cong S^2$. The homotopy type changes because $S^1 \vee S^1$ is not contractible. To ensure an invariance in the homotopy type we must require that the subspace we are contracting by is contractible.

Definition 1.3.1: Homotopy Extension Property

Let $f_0 : X \to Y$ be a map and $f_t|_A : A \times I \to Y$ be a homotopy on $f_0|_A$. Then if we can extend this to a homotopy on X, $f_t : X \times I \to Y$, we say that (X, A) has the *homotopy extension property*.

Note how the pair (X, A) did not include information about the codomain. Thus, for any Y if $X \times \{0\} \to Y$ is continuous and $A \times I \to Y$ is a homotopy that agrees with $A \times \{0\}$, then it may be extended to $X \times I \to Y$. If this were only valid for certain codomains, we would say that it had the homotopy extension property with respect to Y.

Proposition 1.3.2: Condition For Homotopy Extension

If a pair (X, A) has the homotopy extension property, then $(X \times \{0\}) \cup (A \times I)$ is a retract of $X \times I$. Conversely, if $(X \times \{0\}) \cup (A \times I)$ is a retract of $X \times I$ and A is closed in X or X is Hausdorff, then (X, A) has the homotopy extension property.

***Proof*:**
Let (X, A) have the homotopy extension property, and put $W = X \times \{0\} \cup A \times I$. The map $f_0 \colon X \to W$, $x \mapsto (x, 0)$, together with the homotopy $A \times I \to W$, $(a, t) \mapsto (a, t)$, of its restriction to A, extends to a map $X \times I \to W$. This map is the identity on W, so W is a retract of $X \times I$.

Conversely, suppose $X \times \{0\} \cup A \times I$ is a retract of $X \times I$. If X is Hausdorff, then A is closed, so it suffices to treat the case where A is closed. For if $r : X \times I \to X \times I$ is a retraction map onto $X \times \{0\} \cup A \times I$, then the image of r is the set of points $z \in X \times I$ with $r(z) = z$, i.e. a closed set if X is Hausdorff, so $X \times \{0\} \cup A \times I$ is closed in $X \times I$ and hence A is closed in X.

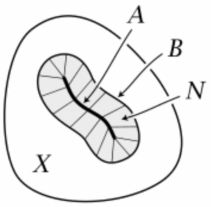
Now let A be closed. If we have two continuous maps $X \times \{0\} \to Y$ and $A \times I \to Y$ that agree on $A \times \{0\}$, we can combine them to get a map

$$X \times \{0\} \cup A \times I \to Y$$

which is continuous since it is continuous on the closed sets $X \times \{0\}$ and $A \times I$. By assumption, we have a retraction map and so composing to get $X \times I \to X \times \{0\} \cup A \times I \to Y$, we get an extension $X \times I \to Y$, hence (X, A) has the homotopy extension property.

Example 1.3.3: Pairs With the Homotopy Extension Property

1. If A has a mapping cylinder neighbourhood in X, then (X, A) has the homotopy extension property. A mapping cylinder neighbourhood is a closed neighbourhood N containing a subspace B (B is the "boundary" of N), with $N \setminus B$ an open neighbourhood of A such that there exists a map $f : B \to A$ and a homeomorphism $h : M_f \to N$ with $h|_{A\cup B} = \mathrm{id}$.



   To see that (X, A) has this property, first note that $I \times I$ retracts onto $I \times \{0\} \cup (\partial I) \times I$, and so $B \times I \times I$ retracts onto $B \times I \times \{0\} \cup B \times (\partial I) \times I$, and hence we induce a retraction of $M_f \times I$ onto $M_f \times \{0\} \cup (A \cup B) \times I$. Thus $(M_f, A \cup B)$ has the homotopy extension property by proposition 1.3.2, as $A \cup B$, the two ends of the cylinder, is closed in M_f. Thus, the homeomorphic pair $(N, A \cup B)$ does as well.

   Given a continuous map $X \to Y$ and a homotopy of its restriction to A, we can take the constant homotopy on $X - (N - B)$ and then extend over N by applying the homotopy extension property for $(N, A \cup B)$ to the given homotopy on A and the constant homotopy on B.

2. For an example of a pair without the homotopy extension property take (I, A) where $A = \{0, 1, 1/2, 1/3, ...\}$. Then there is no continuous retraction $r\colon I\times I \to W = I\times\{0\}\cup A\times I$. Indeed, $W \setminus (I \times \{0\}) = A \times (0, 1]$, and a path in it has first coordinate a continuous map into the countable set A, hence constant. So a path in W from $(1/(n + 1), 1)$ to $(1/n, 1)$ meets $I \times \{0\}$. Applying r to the segment $[1/(n + 1), 1/n] \times \{1\}$ gives such a path, so there is a point p_n of that segment with $r(p_n) \in I \times \{0\}$. Then $p_n \to (0, 1)$, while $r(p_n)$ stays in the closed set $I \times \{0\}$, which does not contain $r(0, 1) = (0, 1)$, contradicting continuity of r.

Proposition 1.3.4: Homotopy Extension For CW Pair

Let (X, A) be a CW pair. Then $X \times \{0\} \cup A \times I$ is a deformation retract of $X \times I$, hence (X, A) has the homotopy extension property.

***Proof*:**
The argument works first with the cells D^n and then with the space X. First, there is a retraction

$$r : D^n \times I \to (D^n \times \{0\}) \cup (\partial D^n \times I)$$

for instance the radial projection from the point $(0, 2) \in D^n \times \mathbb{R}$. From this retraction, we may construct a deformation retraction from $D^n \times I$ onto $(D^n \times \{0\}) \cup (\partial D^n \times I)$ by setting

$$r_t = tr + (1 - t)\mathrm{id}$$

Since $X^n \times I$ is obtained from $(X^n \times \{0\}) \cup (X^{n-1} \cup A^n) \times I$ by attaching copies of $D^n \times I$ along $(D^n \times \{0\}) \cup (\partial D^n \times I)$, we get a deformation retraction of $X^n \times I$ onto $(X^n \times \{0\}) \cup (X^{n-1} \cup A^n) \times I$. If we perform this deformation retraction during the t-interval $[1/2^{n+1}, 1/2^n]$, then the infinite concatenation of these homotopies is a deformation retraction of $X \times I$ onto $(X \times \{0\}) \cup (A \times I)$. There is no problem with continuity of this deformation retraction at $t = 0$, since it is continuous on $X^n \times I$ (being stationary in $[0, 1/2^{n+1}]$), and the CW complexes have the weak topology with respect to their skeleta, so the map is continuous if and only if its restriction to each skeleton is continuous. But then we have the retraction.

Theorem 1.3.5: Homotopy Equivalence Of Quotient

Let (X, A) satisfy the homotopy extension property and let A be contractible. Then the quotient map $\pi : X \to X/A$ is a homotopy equivalence.

***Proof*:**
Let $f_t : X \to X$ be a homotopy extending the contraction of A where:

1. $f_0 = \mathrm{id}$
2. $f_t(A) \subseteq A$ for all $t \in I$

Then $\pi \circ f_t : X \to X/A$ sends A to a point, hence there is a well-defined map $\overline{f}_t : X/A \to X/A$, notably the following diagram commutes for all t:

$$\begin{array}{ccc} X & \xrightarrow{f_t} & X \\ \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \pi} \\ X/A & \xrightarrow{\overline{f}_t} & X/A \end{array}$$

In particular $\pi \circ f_t = \overline{f}_t \circ \pi$. When $t = 1$, we have that $f_1(A)$ is a point, hence f_1 induces the map $g : X/A \to X$ in the following diagram:

$$\begin{array}{ccc} X & \xrightarrow{f_1} & X \\ \downarrow{\scriptstyle \pi} & \overset{g}{\nearrow} & \downarrow{\scriptstyle \pi} \\ X/A & \xrightarrow{\overline{f}_1} & X/A \end{array}$$

The upper triangle commutes by the definition of g, and the lower one, $\pi \circ g = \overline{f}_1$, since

$$\pi \circ g(\overline{x}) = \pi \circ g \circ \pi(x) = \pi \circ f_1(x) = \overline{f}_1 \circ \pi(x) = \overline{f}_1(\overline{x})$$

Hence, the maps g and π are inverse homotopy equivalences since

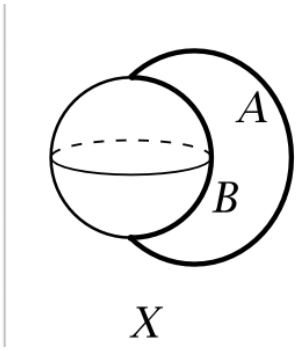
$$g \circ \pi = f_1 \simeq f_0 = \mathrm{id} \qquad \pi \circ g = \overline{f}_1 \simeq \overline{f}_0 = \mathrm{id}$$

via the homotopies f_t and $\overline{f}_t$, giving us the desired result.

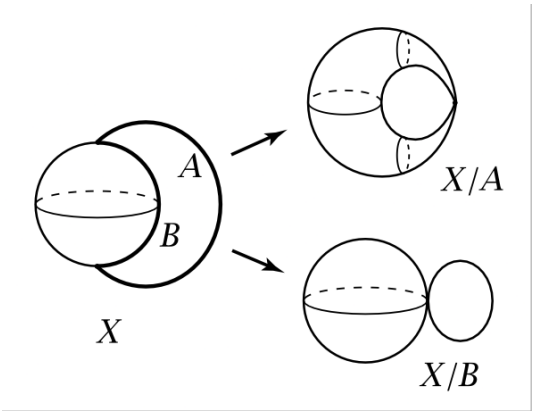
Example 1.3.6: Homotopy Equivalence Using Quotients

1. (Graphs) Let X be any graph with finitely many vertices and edges. If two endpoints of any edge of X are distinct, we can collapse the edge to a point, giving a homotopy equivalent graph with one fewer edge (since edges are contractible). This can be repeated until we get a graph X where each component has 1 vertex connected by loops (wedge sum of circles). Two wedge sums of finitely many circles are homotopy equivalent if and only if they have the same number of circles. If the numbers agree, the wedges are homeomorphic. Conversely, the fundamental group of a wedge of n circles is free of rank n (proposition 2.2.8), so its abelianization is $\mathbb{Z}^n$. Homotopy equivalent spaces have isomorphic fundamental groups (theorem 2.1.45), hence isomorphic abelianizations, and $\mathbb{Z}^n \cong \mathbb{Z}^m$ forces $n = m$, since finitely generated free $\mathbb{Z}$-modules have an invariant basis number.

2. Let X be the space S^2 along with an interval I attached at two distinct points. Let the interval I be labeled A, and let B be the arc on the sphere connecting the endpoints of A:



Give X a CW structure with the two endpoints of A and B being 0-cells, the interiors of A and B being 1-cells, and the rest of S^2 a 2-cell. Since A and B are contractible subcomplexes, proposition 1.3.4 and theorem 1.3.5 give that X/A and X/B are homotopy equivalent to X:



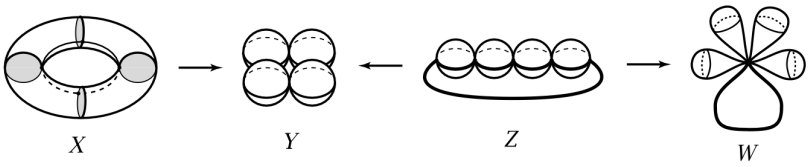
The space X/A is S^2/S^0, that is the sphere with two points identified, while X/B is $S^1 \vee S^2$.

Thus,

$$S^2/S^0 \simeq S^1 \vee S^2$$

even though at first glance they may seem quite different.

3. Let X be the union of a torus T with n meridional disks. To make a CW complex on X, take a longitudinal circle in the torus, hence intersecting each of the meridional disks at one point. These points of intersection are the 0-cells, the 1-cells are the rest of the longitudinal circle and the boundary of the meridional disks, and the 2-cells are the remaining regions. We may collapse the meridional disks to a point to yield a homotopy equivalent space Y consisting of n 2-spheres, each tangent to its neighbors (we may think of this as a necklace with n beads). From this intuition, we may imagine a space Z which is a line of n 2-spheres each touching each other tangentially (a "strand of n-beads") with a line connecting the ends of the first and last 2-sphere. Then we can collapse the line to get our space Y, showing the two are homotopy equivalent. Finally, we may take the union of arcs on the 2-spheres to form a path starting from the endpoint of the first sphere and ending on the endpoint of the last sphere, which we may then collapse to get a shape that resembles a "bouquet of flowers" with a handle:



4. (reduced suspension) Let X be a CW complex and $x \in X$ a 0-cell. Then in the suspension SX, there is a line segment $\{x\} \times I$ which we can collapse to produce the homotopy-equivalent space ΣX called the *reduced suspension*. For example, if $X = S^1 \vee S^1$ with x the point of intersection, then SX is the union of two spheres intersecting along the arc $\{x\} \times I$, so the reduced suspension is $S^2 \vee S^2$ (which we can think of as a simpler space than the suspension). In general,

$$\Sigma(X \vee Y) = \Sigma X \vee \Sigma Y$$

for arbitrary CW complexes X, Y wedged at 0-cells x and y. Indeed, $(X \sqcup Y) \times I \to (X \vee Y) \times I$ is a quotient map by [16, Products of Quotient Maps], I being locally compact Hausdorff, so both sides are quotients of $(X \times I) \sqcup (Y \times I)$. In both, the identifications collapse $(X \times \partial I) \cup (\{x\} \times I) \cup (Y \times \partial I) \cup (\{y\} \times I)$ to a single point and do nothing else, and two quotients of one space by the same relation are homeomorphic. The reduced suspension also has a simpler CW structure: in SX there are two 0-cells (the two suspension points), and an $(n+1)$-cell $e^n \times (0,1)$ for each n-cell of X other than the 0-cell x. Then the reduced suspension ΣX is the same as the smash product $X \wedge S^1$, since both spaces are the quotient of $X \times I$ with $(X \times \partial I) \cup \{x\} \times I$ collapsed to a point.

Another application of the extension property is another way of gluing shapes while keeping the homotopy type invariant:

Theorem 1.3.7: Homotopy Equivalence Via Gluing

Let (X_1, A) be a CW pair and let $f, g : A \to X_0$ be attaching maps that are homotopic. Then

$$X_0 \sqcup_f X_1 \simeq X_0 \sqcup_g X_1$$

relative to X_0

When we say that the two spaces are homotopy equivalent relative to X_0, we mean that there are maps between the two spaces that restrict to the identity on X_0 and are inverse homotopy equivalences via homotopies that restrict to the identity on X_0.

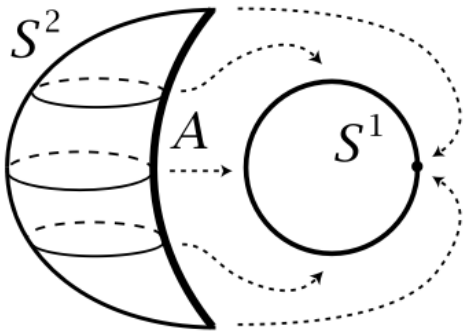
***Proof*:**
Let $F : A \times I \to X_0$ be a homotopy from f to g, and consider the space $X_0 \sqcup_F (X_1 \times I)$. This space contains $X_0 \sqcup_f X_1$ and $X_0 \sqcup_g X_1$ as the subspaces over $X_1 \times \{0\}$ and $X_1 \times \{1\}$. Then given a deformation retraction of $X_1 \times I$ onto $(X_1 \times \{0\}) \cup A \times I$ as given in proposition 1.3.4, we get an induced deformation retraction of $X_0 \sqcup_F (X_1 \times I)$ onto $X_0 \sqcup_f X_1$. Similarly, $X_0 \sqcup_F (X_1 \times I)$ deformation retracts onto $X_0 \sqcup_g X_1$. Both of these deformation retractions restrict to the identity on X_0, so together they give a homotopy equivalence

$$X_0 \sqcup_f X_1 \simeq X_0 \sqcup_g X_1$$

relative to X_0, as we sought to show.

Example 1.3.8: Attaching Spaces

1. Let X be a sphere with two points identified. We rederive that this is homotopy equivalent to $S^1 \vee S^2$. Take S^2 and S^1. Take an arc A on S^2, and attach it to all of S^1. Since A is contractible, this attaching map is homotopic to a constant map, and attaching S^2 to S^1 along A via a constant map gives $S^1 \vee S^2$:



   The result now follows from theorem 1.3.7, since (S^2, A) is a CW pair, S^2 being obtained from A by attaching a 2-cell.

2. We may do the same with the necklace of beads as seen in the previous example. The

necklace can be obtained from a circle by attaching n 2-spheres along arcs, so the necklace is homotopy-equivalent to the space obtained by attaching n 2-spheres to a circle at points, then sliding these attaching points around the circle until they all coincide, giving us the wedge-sum.

3. Let $CA = (A \times I)/(A \times \{0\})$ be the cone on A. Then if (X, A) is a CW pair,

$$X/A \simeq X \cup CA$$

For, since CA is a contractible subcomplex of $X \cup CA$, proposition 1.3.4 and theorem 1.3.5 give:

$$X/A = (X \cup CA)/CA \simeq X \cup CA$$

4. Let (X, A) be a CW pair with A contractible in X (the inclusion map $A \hookrightarrow X$ is homotopic to a constant map). Then

$$X/A \simeq X \vee SA$$

By the previous example, we have $X/A \simeq X \cup CA$. Now, since A is contractible in X, the mapping cone of the inclusion $A \hookrightarrow X$ $(X \cup CA)$ is homotopy equivalent to the mapping cone of a constant map by theorem 1.3.7, applied to the CW pair (CA, A), and that mapping cone is $X \vee SA$. As an example, $S^n/S^i \simeq S^n \vee S^{i+1}$ for $i < n$, since S^i is contractible in S^n if $i < n$. As a particular use-case, we get

$$S^2/S^0 \simeq S^2 \vee S^1$$

which we saw earlier, showing us we have two techniques to find homotopy-equivalences.

Proposition 1.3.9: Homotopy Extension and Homotopy Equivalence

Let (X, A) and (Y, A) satisfy the homotopy extension property and $f : X \to Y$ be a homotopy equivalence with $f|_A = \mathrm{id}$. Then f is a homotopy equivalence relative to A

***Proof*:**
Recall that a homotopy equivalence *relative to* A means a map $f\colon X \to Y$ for which there is $g\colon Y \to X$ with $g|_A = \mathrm{id}$ and homotopies $gf \simeq \mathrm{id}_X$ and $fg \simeq \mathrm{id}_Y$ that are stationary on A. The hypothesis $f|_A = \mathrm{id}$ already fixes f on A. What must be improved is the homotopy inverse g, which a priori need not fix A, and the two round-trip homotopies, which a priori move points of A. The homotopy extension property is exactly the tool that performs both repairs. Fix a homotopy inverse $g\colon Y \to X$, so that $gf \simeq \mathrm{id}_X$ and $fg \simeq \mathrm{id}_Y$.

Step 1: Adjusting g so that $g|_A = \mathrm{id}$. Let $h_t\colon X \to X$ be a homotopy from $gf = h_0$ to $\mathrm{id}_X = h_1$. Restricting to A and using $f|_A = \mathrm{id}$, the family $h_t|_A$ is a homotopy of maps $A \to X$ running from $(gf)|_A = g|_A$ (at $t = 0$) to id_A (at $t = 1$). Thus $h_t|_A$ is a homotopy of $g|_A$, viewed as a map $A \to X$. Since (Y, A) satisfies the homotopy extension property (definition 1.3.1), this homotopy of $g|_A$ extends over Y: there is a homotopy $g_t\colon Y \to X$ with $g_0 = g$ and $g_t|_A = h_t|_A$ for all t. Set g_1 to be the end of this homotopy. Then $g_1|_A = h_1|_A = \mathrm{id}_A$, so g_1 restricts to the identity on A, and $g_1 \simeq g$, so g_1 is again a homotopy inverse of f. Replacing the original inverse by g_1, we have arranged $g_1|_A = \mathrm{id}$ while keeping $g_1 f \simeq \mathrm{id}_X$ and $fg_1 \simeq \mathrm{id}_Y$.

Step 2: Upgrading $g_1 f \simeq \mathrm{id}_X$ to a homotopy rel A. A homotopy from $g_1 f$ to id_X is furnished by

splicing the reverse of $g_t f$ to the homotopy h_t. Concretely, define $k_t \colon X \to X$ by

$$k_t = \begin{cases} g_{1-2t}\, f, & 0 \le t \le \frac{1}{2}, \\ h_{2t-1}, & \frac{1}{2} \le t \le 1 \end{cases}$$

The two clauses agree at $t = \frac{1}{2}$, where both read $g_0 f = gf = h_0$, so k_t is a well-defined homotopy. It runs from $k_0 = g_1 f$ to $k_1 = h_1 = \mathrm{id}_X$. The point of this particular splicing is its behaviour on A. Since $f|_A = \mathrm{id}$ and $g_t|_A = h_t|_A$, on A the first clause becomes $g_{1-2t}|_A = h_{1-2t}|_A$ and the second is $h_{2t-1}|_A$, so

$$k_t|_A = \begin{cases} h_{1-2t}|_A, & 0 \le t \le \frac{1}{2}, \\ h_{2t-1}|_A, & \frac{1}{2} \le t \le 1 \end{cases}$$

which starts and ends at $h_1|_A = \mathrm{id}_A$ and whose second half retraces its first half in reverse: $k_t|_A = k_{1-t}|_A$ for all t. A homotopy with this palindromic restriction can be deformed, rel its two ends, into one that is stationary on A, as follows.

Build a homotopy of homotopies $k_{t,u} \colon A \to X$ over the parameter square $I \times I$ with coordinates (t, u), the t-axis horizontal and the u-axis vertical. Picture the square cut by the "Λ" whose apex sits at the midpoint $(\frac{1}{2}, 0)$ of the bottom edge and whose two legs run upward to the top corners $(0, 1)$ and $(1, 1)$. At height u the legs sit at $t = \frac{1-u}{2}$ and $t = \frac{1+u}{2}$, so the Λ opens out to span the entire top edge while pinching to a point on the bottom. On the bottom edge set $k_{t,0} = k_t|_A$. Outside the Λ, in the two lower regions flanking the apex, declare $k_{t,u}$ independent of u, equal to its value on the bottom edge directly below. Inside the Λ, the upper triangle containing the whole top edge, declare $k_{t,u}$ independent of t, equal to the common value carried up the legs. Concretely this reads

$$k_{t,u}|_A = \begin{cases} k_t|_A, & t \le \frac{1-u}{2} \quad \text{or} \quad t \ge \frac{1+u}{2}, \\ k_{\frac{1-u}{2}}|_A, & \frac{1-u}{2} \le t \le \frac{1+u}{2} \end{cases}$$

The two prescriptions agree across the left leg automatically, since the outside value at $t = \frac{1-u}{2}$ is $k_{\frac{1-u}{2}}|_A$, the same as the inside value. Across the right leg they agree precisely because $k_t|_A = k_{1-t}|_A$, which forces the outside value $k_{\frac{1+u}{2}}|_A$ at $t = \frac{1+u}{2}$ to equal the inside value $k_{\frac{1-u}{2}}|_A = k_{1-\frac{1+u}{2}}|_A$. So $k_{t,u}$ is well defined and continuous. Now read off the three remaining edges. The left edge $t = 0$ and the right edge $t = 1$ lie outside the Λ for every $u < 1$, hence carry the constant bottom-corner values $k_0|_A = h_1|_A = \mathrm{id}_A$ and $k_1|_A = h_1|_A = \mathrm{id}_A$. The top edge $u = 1$ lies inside the Λ, carrying the value $k_{\frac{1-u}{2}}|_A$ at $u = 1$, namely $k_0|_A = \mathrm{id}_A$. Thus $k_{t,u}$ is a homotopy of the homotopy $k_t|_A$ (its bottom edge) which is the constant id_A on the other three edges.

Now extend $k_{t,u}$ from A to all of X. Write I_t for the factor carrying the variable t of the homotopy k_t and I_u for the new factor carrying the deformation variable u. The pair to which the extension applies is $(X \times I_t, A \times I_t)$, deformed in the direction u. This pair satisfies the homotopy extension property, by the exponential law [16, The Exponential Law], I_t being locally compact Hausdorff. For any space Z, a map $X \times I_t \to Z$ is the same as a continuous map $X \to C(I_t, Z)$, and a homotopy $A \times I_t \times I_u \to Z$ in the variable u is the same as a homotopy $A \times I_u \to C(I_t, Z)$. The homotopy extension property of (X, A), applied with codomain $C(I_t, Z)$, extends the second over $X \times I_u$ starting from the first, and transposing back gives the required extension over $X \times I_t \times I_u$. View $k_{t,u}$ as a homotopy in the variable u of the map $k_t|_A$, the latter

regarded as a single map $A \times I_t \to X$ in the variable t. By the homotopy extension property for $(X \times I_t, A \times I_t)$ it extends to a homotopy of the full map $k_t \colon X \to X$, producing $k_{t,u} \colon X \to X$ with bottom edge $k_{t,0} = k_t$ the homotopy of Step 2. Reading off the left, top, and right edges of the (t,u)-square gives a homotopy from $k_0 = g_1 f$ to $k_1 = \mathrm{id}_X$ whose restriction to A is constant equal to id_A throughout (those three edges carry the constant value computed above). Hence

$$g_1 f \simeq \mathrm{id}_X \quad \text{rel } A$$

Step 3: Upgrading $fg_1 \simeq \mathrm{id}_Y$ to a homotopy rel A. It remains to make the other round-trip homotopy stationary on A as well. Since $g_1 \simeq g$, we have $fg_1 \simeq fg \simeq \mathrm{id}_Y$, so f and g_1 are homotopy inverses of each other on both sides. This symmetry lets us run Steps 1 and 2 once more with the two maps interchanged, applying them to the pair g_1, f in place of f, g. The map being shown an equivalence is now $g_1 \colon Y \to X$ with inverse $f \colon X \to Y$, and g_1 already restricts to id on A. The output is an adjusted inverse $f_1 \colon X \to Y$ with $f_1|_A = \mathrm{id}$ together with the relative homotopy produced by Step 2, which here reads

$$f_1 g_1 \simeq \mathrm{id}_Y \quad \text{rel } A$$

Two relative homotopies are now in hand: $g_1 f \simeq \mathrm{id}_X$ rel A from Step 2, and $f_1 g_1 \simeq \mathrm{id}_Y$ rel A just obtained. Working throughout with homotopies stationary on A, compute

$$f_1 = f_1 \,\mathrm{id}_X \simeq f_1 \,(g_1 f) = (f_1 g_1)\, f \simeq \mathrm{id}_Y \, f = f \quad \text{rel } A$$

where the first relation postcomposes $g_1 f \simeq \mathrm{id}_X$ rel A by f_1, and the second precomposes $f_1 g_1 \simeq \mathrm{id}_Y$ rel A by f. Hence $f_1 \simeq f$ rel A. Precomposing this relative homotopy with $g_1 \colon Y \to X$, which carries A to A by the identity so that the homotopy stays stationary on A, gives

$$f g_1 \simeq f_1 g_1 \simeq \mathrm{id}_Y \quad \text{rel } A$$

Therefore g_1 is a two-sided homotopy inverse of f with $g_1|_A = \mathrm{id}$, and both homotopies $g_1 f \simeq \mathrm{id}_X$ and $f g_1 \simeq \mathrm{id}_Y$ are stationary on A. This is precisely the statement that f is a homotopy equivalence relative to A, completing the proof.

Corollary 1.3.10: homotopy extension and deformation Retraction

Let (X, A) satisfy the homotopy extension property and let the inclusion $A \hookrightarrow X$ be a homotopy equivalence. Then A is a deformation retract of X.

***Proof*:**
Apply proposition 1.3.9 to the inclusion $i \colon A \hookrightarrow X$, with the pairs (A, A), which has the homotopy extension property because a homotopy on A is its own extension, and (X, A). It gives $g \colon X \to A$ with $g|_A = \mathrm{id}$ and $ig \simeq \mathrm{id}_X$ rel A. Reversing this homotopy gives $f_t \colon X \to X$ with $f_0 = \mathrm{id}_X$, $f_1 = ig$, so $f_1(X) = A$ because $g|_A = \mathrm{id}$, and $f_t|_A = \mathrm{id}$ for all t. This is a deformation retraction of X onto A (definition 1.1.10).

Corollary 1.3.11: Homotopy Equivalence via Deformation Retracts

The map $f : X \to Y$ is a homotopy equivalence if and only if X is a deformation retract of the mapping cylinder M_f. Hence, two spaces X and Y are homotopy equivalent if and only if there is a third space containing both X and Y as deformation retracts.

***Proof*:**

The argument runs through the mapping cylinder $M_f = (X \times I) \sqcup Y/[(x,1) \sim f(x)]$ introduced in example 1.1.13. Identify X with the bottom end $X \times \{0\} \subseteq M_f$ and Y with its image under the gluing, so that both sit inside M_f as subspaces. The map f factors through this cylinder: writing $\iota\colon X \hookrightarrow M_f$ for the inclusion of the bottom end and $r\colon M_f \to Y$ for the map sending $(x,t) \mapsto f(x)$ and fixing Y, the composite $r \circ \iota$ is exactly f. The map r is a deformation retraction of M_f onto Y: the homotopy $r_s(x,t) = (x, (1-s)t + s)$ on the cylinder coordinates, with r_s the identity on Y, slides each point (x,t) along its segment $\{x\} \times I$ up to $(x,1) = f(x) \in Y$, fixing Y throughout. In particular ι and f differ by the homotopy equivalence r, so ι is a homotopy equivalence if and only if f is.

The pair (M_f, X) satisfies the homotopy extension property. The subspace X is closed in M_f, its preimage in $(X \times I) \sqcup Y$ being $X \times \{0\}$, so by proposition 1.3.2 it suffices to produce a retraction of $M_f \times I$ onto $(M_f \times \{0\}) \cup (X \times I)$. Write the cylinder coordinates as (x,t,u), where t is the mapping-cylinder coordinate ($X = X \times \{0\}_t$ at the bottom, the glued copy of Y at the top $t = 1$) and u is the homotopy parameter coming from the extra factor of I in $M_f \times I$. On the cylinder portion the target subspace is

$$\big(X \times I_t \times \{0\}_u\big) \cup \big(X \times \{0\}_t \times I_u\big)$$

the first piece being $M_f \times \{0\}$ and the second being $X \times I$, since X sits at $t = 0$. In the square $I_t \times I_u$ this is the L-shape $(I_t \times \{0\}_u) \cup (\{0\}_t \times I_u)$ of the bottom edge together with the $t = 0$ edge.

Retract $I_t \times I_u$ onto this L-shape by radial projection from the apex $(1,2)$ sitting above the top-right corner: each point is pushed along the ray emanating from $(1,2)$ until it meets the L-shape. This projection fixes the L-shape pointwise, and it sends the entire right edge $\{1\}_t \times I_u$ to the single corner $(1,0)$, since the ray from $(1,2)$ through any $(1,u)$ runs straight down the line $t = 1$ to $(1,0)$. Performing this retraction on the (t,u)-coordinates of the cylinder gives a retraction of $X \times I_t \times I_u$ onto the L-shape target above, under which the top edge $X \times \{1\}_t \times I_u$ is collapsed by $(x,1,u) \mapsto (x,1,0)$. On the glued copy of Y the target subspace is $Y \times \{0\}_u$, and the retraction collapsing the homotopy parameter, $(y,u) \mapsto (y,0)$, retracts $Y \times I_u$ onto it. These two retractions agree on the overlap. Under the gluing $(x,1) \sim f(x)$, a point $(x,1,u)$ of the top edge is identified with $(f(x),u) \in Y \times I_u$. The cylinder retraction sends it to $(x,1,0)$, identified with $(f(x),0)$, while the Y-side retraction sends $(f(x),u)$ to $(f(x),0)$, the same point. Both maps are continuous on their closed domains and agree on the closed overlap, so by the pasting lemma they assemble into a continuous retraction of $M_f \times I$ onto $(M_f \times \{0\}) \cup (X \times I)$. Thus (M_f, X) has the homotopy extension property by proposition 1.3.2.

Suppose first that f is a homotopy equivalence. Then $\iota\colon X \hookrightarrow M_f$ is a homotopy equivalence, by the factorization $f = r \circ \iota$ with r a homotopy equivalence. Since (M_f, X) satisfies the homotopy extension property and the inclusion $X \hookrightarrow M_f$ is a homotopy equivalence, corollary 1.3.10 applies and shows that X is a deformation retract of M_f.

Conversely, suppose X is a deformation retract of M_f. A deformation retraction of M_f onto X exhibits the inclusion $\iota\colon X \hookrightarrow M_f$ as a homotopy equivalence (definition 1.1.14): the retraction

$\rho\colon M_f \to X$ satisfies $\rho \circ \iota = \mathrm{id}_X$ and $\iota \circ \rho \simeq \mathrm{id}_{M_f}$. Since $f = r \circ \iota$ with r a homotopy equivalence and ι a homotopy equivalence, f is a composite of homotopy equivalences, hence itself a homotopy equivalence. This establishes the stated equivalence: f is a homotopy equivalence if and only if X is a deformation retract of M_f.
The final assertion follows. If $f\colon X \to Y$ is a homotopy equivalence, then $Z = M_f$ is a space containing both X and Y as deformation retracts, X by the equivalence just proved and Y by the retraction r above. Conversely, if a single space Z contains X and Y as deformation retracts, then the inclusions $X \hookrightarrow Z$ and $Y \hookrightarrow Z$ are homotopy equivalences, and composing one with a homotopy inverse of the other produces a homotopy equivalence $X \simeq Y$, completing the proof.

This last corollary alludes to a more general notion known as *cobordism* which says that two manifolds are *cobordant* if their union is the boundary of a manifold of one higher dimension. Cobordism returns in section 7.6 and section 8.5, and *Differential Topology* [32] treats it from the smooth side.

Exercise 1.3.1

1. Show that homotopies form an equivalence relation.
2. Show that $S^n \wedge S^m$ is homeomorphic to S^{n+m} for all $n, m \geq 0$, beginning with $S^1 \wedge S^1 \cong S^2$.
3. Let X and Y be CW complexes pointed by 0-cells x_0 and y_0, with $X \times Y$ given the cell-complex topology. Show that $X * Y \simeq S(X \wedge Y)$, where S is the suspension of definition 1.2.13. (Hint: the subcomplex $A = \{x_0\} * Y \cup X * \{y_0\}$ of $X * Y$ is two cones glued along a segment, and collapsing it gives the reduced suspension of $X \wedge Y$ from example 1.3.6. Use proposition 1.3.4 and theorem 1.3.5, first to see that A is contractible and then for each collapse.)

2

Fundamental Group

As we saw, homotopies allow us to study spaces up to continuous deformation along paths. It is thus natural to study the paths on a space. In this section, we shall see that paths that are closed (that form a loop) shall naturally form a group structure, and this group structure shall be invariant under homotopy giving our first algebraic invariants on topological spaces. This group shall be called the *fundamental group* or *homotopy group*.

After having defined the fundamental group, we shall give many ways of computing it, how we can piece it together from the homotopy group of smaller spaces, and how homotopy groups of CW complexes can be calculated. Along the way, we shall see that most spaces X shall have a *universal cover*, say $U(X)$ that shall be simply connected (all paths are nullhomotopic) and will be a geometric representation of the fundamental group. We shall see that there are many intermediary spaces between X and $U(X)$, and the correspondence between the intermediary spaces and the fundamental group of X shall mirror that of the famous *Galois correspondence* between the intermediary fields and the subgroups of the Galois group.

2.1 Path Homotopy and Fundamental Group

Much of the subject studies homotopies, which eventually lead to homotopy groups, homology, exact sequences of homotopy groups and the Hopf fibration of the sphere. An important class of homotopies worth singling out is homotopies which are "1-dimensional". Working with one-dimensional ones does not mean we will not be working with higher-dimensional objects, but the path in the space we use that will be homotopic will be one dimensional.

Definition 2.1.1: Path-Homotopic

Two paths f, f' in X are *path-homotopic* if

1. they have the same initial and final points: $f(0) = f'(0)$, $f(1) = f'(1)$
2. There exists a continuous function $F : I \times I \to X$ such that

$$F(s,0) = f(s), \quad F(s,1) = f'(s), \quad F(0,t) = x_0, \quad F(1,t) = x_1$$

where $f(0) = f'(0) = x_0$ and $x_1 = f(1) = f'(1)$. We write

$$f \simeq_p f'$$

F is called a *path-homotopy*

Example 2.1.2: Path Homotopy

1. If we take all functions $I \to I$ of the form $x \mapsto x^n$, then all of these are path-homotopic. In fact, any path such that $f(0) = g(0) = 0$, $f(1) = g(1) = 1$ will be path-homotopic by the straight-line homotopy
2. More generally, if C is a convex set, then any two paths with the same initial and ending points are path-homotopic by the straight-line homotopy
3. There are path homotopies other than the straight-line homotopy. Among the examples of chapter 1, look for a path homotopy that is not a straight line.
4. Some paths with the same endpoints are not path-homotopic. Take $X = \mathbb{R}^2 - \{0\}$ and

$$f(s) = (\cos(\pi s), \sin(\pi s)) \qquad g(s) = (\cos(\pi s), 2\sin(\pi s))$$

Then these two functions are path homotopic using the straight-line homotopy, as the figure shows:

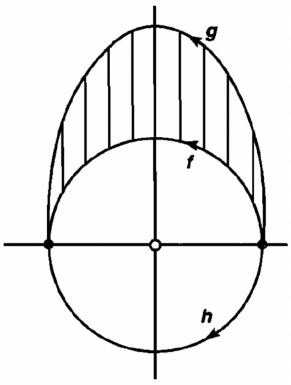


Figure 2.1: Same initial and end points, but one goes under point

In general, if the point 0 were not missing, we could have:

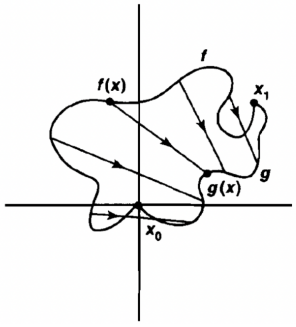


Figure 2.2: Convex space

However, the path $h(s) = (\cos(\pi s), -\sin(\pi s))$ is *not* path-homotopic to f. In the visual, this is the h path. It should perhaps seem intuitive that if we try to pass a continuous path *through* the point 0, we will introduce a discontinuity. The proof uses the computation of $\pi_1(S^1)$ in theorem 2.1.40. If $f \simeq_p h$, then $f * \tilde{h} \simeq_p h * \tilde{h} \simeq_p e_{(1,0)}$ by lemma 2.1.6 and theorem 2.1.7, where $\tilde{h}(s) = h(1-s)$. Composing this path homotopy with the retraction $x \mapsto x/\|x\|$ of $\mathbb{R}^2 - \{0\}$ onto S^1 makes the loop $f * \tilde{h}$, which runs once around S^1, trivial in $\pi_1(S^1, (1,0))$. But its lift to $\mathbb{R}$ starting at 0 ends at 1, so the isomorphism of theorem 2.1.40 sends it to $1 \neq 0$, a contradiction.

5. The polynomials $x \mapsto x^n$ from earlier are all in the same equivalence class

$$[f] = [g] = [\mathrm{id}_I]$$

the class of the path id_I from 0 to 1. They are not path-homotopic to the constant path at 0, since a path homotopy fixes both endpoints and the constant path ends at 0, not 1. As maps, however, they are *nullhomotopic*: $F'(x,t) = (1-t)x$ is a homotopy from id_I to the constant map at 0. It moves the endpoint 1, so it is not a path homotopy.

As may be expected from objects in a category, the isomorphisms form a groupoid, and the automorphisms of an object form a group. This group is rather complicated and is the study of more recent mathematics[1], and so we shall construct a new, much simpler algebraic invariant based on the paths in a space.

Definition 2.1.3: identity path

If $x \in X$, denote e_x to be the constant path $e_x : I \to X$ such that $e_x(t) = x$ for $t \in I$

The constant path at 0 in the previous example is then e_0.

The natural candidate for an operation between paths is to extend one path to the next. However, we cannot extend *arbitrary* paths from one to the next, since that will make discontinuous paths (because the end point of one path might not be the initial point of another). Thus, to combine paths we must impose this restriction:

[1] See for example the study of $\mathrm{Diff}(M)$ for a manifold M

Definition 2.1.4: combining paths Operators: $*$

If f, g are paths in X and $f(1) = g(0)$, define $f * g$ by

$$(f * g)(t) = \begin{cases} f(2t) & t \in [0, \frac{1}{2}] \\ g(2t-1) & t \in [\frac{1}{2}, 1] \end{cases}$$

Example 2.1.5: Combining Paths

Try thinking of any two paths in $\mathbb{R}^2$ and combining them into 1.

Lemma 2.1.6: Well-defined: $*$ Preserves Equivalence Relation

If $f \simeq_p f'$, $g \simeq_p g'$, then

$$(f * g) \simeq_p (f' * g')$$

***Proof*:**

Let F and G be path homotopies from f to f' and from g to g', and define $H : I \times I \to X$ by

$$H(s,t) = \begin{cases} F(2s,t) & s \in [0, \frac{1}{2}] \\ G(2s-1,t) & s \in [\frac{1}{2}, 1] \end{cases}$$

The two formulas agree at $s = \frac{1}{2}$, since $F(1,t) = f(1) = g(0) = G(0,t)$ for every t, so H is continuous by [16, Pasting Lemma]. It satisfies $H(s,0) = (f * g)(s)$, $H(s,1) = (f' * g')(s)$, $H(0,t) = f(0)$ and $H(1,t) = g(1)$, so H is a path homotopy from $f * g$ to $f' * g'$, as claimed.

Because of this, we can define $*$ on equivalence classes:

$$[f] * [g] = [f * g]$$

The operation $*$ is only defined on paths whose end and initial points match, so the group axioms hold in the following partial form.

Theorem 2.1.7: Properties Of $*$

1. (Associativity) Assume f, g, h are paths in X, $f(1) = g(0)$, $g(1) = h(0)$. Then
$$([f] * [g]) * [h] = [f] * ([g] * [h])$$
2. (identity) If f is a path in X, $f(0) = x_0$, $f(1) = x_1$, then
$$[f] * [e_{x_1}] = [f] = [e_{x_0}] * [f]$$
3. (invertible) Given a path f in X, define $\tilde{f}(t) = f(1-t)$. Then
$$[f] * [\tilde{f}] = [e_{f(0)}]$$
$$[\tilde{f}] * [f] = [e_{f(1)}]$$

***Proof*:**
First, a reparametrization does not change the class of a path. Let $\varphi\colon I \to I$ be continuous with $\varphi(0) = 0$ and $\varphi(1) = 1$, and let f be a path. Since I is convex, $F(s,t) = f\big((1-t)\varphi(s) + ts\big)$ is defined and continuous, with $F(s,0) = (f \circ \varphi)(s)$, $F(s,1) = f(s)$, $F(0,t) = f(0)$ and $F(1,t) = f(1)$. So $f \circ \varphi \simeq_p f$. By lemma 2.1.6, each class below may be computed from representatives.

1. The path $(f * g) * h$ runs through f, g and h on $[0, \frac{1}{4}]$, $[\frac{1}{4}, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$, and $f * (g * h)$ runs through them on $[0, \frac{1}{2}]$, $[\frac{1}{2}, \frac{3}{4}]$ and $[\frac{3}{4}, 1]$. So $(f * g) * h = (f * (g * h)) \circ \varphi$, where φ is linear on each of $[0, \frac{1}{4}]$, $[\frac{1}{4}, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$ with $\varphi(0) = 0$, $\varphi(\frac{1}{4}) = \frac{1}{2}$, $\varphi(\frac{1}{2}) = \frac{3}{4}$ and $\varphi(1) = 1$. The reparametrization fact gives (1).
2. We have $f * e_{x_1} = f \circ \varphi$ for $\varphi(s) = \min(2s, 1)$, and $e_{x_0} * f = f \circ \psi$ for $\psi(s) = \max(2s-1, 0)$. The reparametrization fact gives (2).
3. Define $H\colon I \times I \to X$ by $H(s,t) = f\big(2s(1-t)\big)$ for $s \in [0, \frac{1}{2}]$ and $H(s,t) = f\big((2-2s)(1-t)\big)$ for $s \in [\frac{1}{2}, 1]$. Both formulas give $f(1-t)$ at $s = \frac{1}{2}$, so H is continuous by [16, Pasting Lemma]. Since $\tilde{f}(2s-1) = f(2-2s)$, we have $H(s,0) = (f * \tilde{f})(s)$, and $H(s,1) = H(0,t) = H(1,t) = f(0)$. So $f * \tilde{f} \simeq_p e_{f(0)}$. Applying this to $\tilde{f}$, whose reverse is f, gives $\tilde{f} * f \simeq_p e_{\tilde{f}(0)} = e_{f(1)}$.

This proves the three properties, as claimed.

If $k\colon X \to Y$ is continuous and F is a path homotopy from f to f', then $k \circ F$ is a path homotopy from $k \circ f$ to $k \circ f'$, since it is continuous and inherits the four boundary conditions. And $k \circ (f * g) = (k \circ f) * (k \circ g)$, as both sides are given by the same two formulas of definition 2.1.4.

The identity in theorem 2.1.7 is not unique: it depends on the endpoints of the path. The operation $*$ just falls short of being a binary operation: it is not a function but a *partial function*. Two arbitrary paths cannot be combined. Their endpoints must match, as noted when defining $*$. If we restrict to paths that are always concatenable, we will indeed have a binary function, but we usually want to think of $*$ being defined on our entire space.

A "group" whose operation is a partial function (and hence has multiple identities, depending on how the partial function is restricted to get a function) has its own name:

Definition 2.1.8: Groupoid

A *groupoid* is a set (ex. G) with a *partial function* $* : G \times G \to G$ *and* a unary function $^{-1}: G \to G$ which then satisfy the group axioms (associativity, inverse, identity). The difference from groups is that we now have:

1. (inverse) $a^{-1} * a$ and $a * a^{-1}$ are always defined
2. if $a * b$ is defined, then $a * b * b^{-1} = a$ and $a^{-1} * a * b = b$ are also defined

How exactly the partial function $*$ is "partial" depends on the context. There is no one way of defining this partial function. Furthermore, we choose a unary function for the inverse because the "identity" is now no longer so "unique", as was shown in theorem 2.1.7.

Example 2.1.9: Groupoid Example

Let X be a space. The path-homotopy classes $[f]$ of paths in X, with $[f] * [g] = [f * g]$ defined when $f(1) = g(0)$ and $[f]^{-1} = [\tilde{f}]$ for the reverse path $\tilde{f}(t) = f(1-t)$, form a groupoid: the operation is well defined by lemma 2.1.6, and the groupoid axioms are theorem 2.1.7.

Definition 2.1.10: Path Space

Let $P(X) = \{\text{equivalence classes of paths } [f]\}$. We will call this the *path space*

There is a natural map $P(X) \to X^2$, $[f] \mapsto (f(0), f(1))$. This map is surjective if and only if X is path connected. The groupoid from earlier is

$$(P(X), *, ^{-1})$$

As mentioned, we still do not have a "unique" identity nor a total function, so we are yet to totally define a group. The limiting factor is that we cannot combine arbitrary paths. If the initial and end points were the same, this would solve our problem. We therefore restrict our attention once again, this time to paths that *loop*:

Definition 2.1.11: Loop

Given $x_0 \in X$ a *loop based at* x_0 is a path which begins and ends at x_0, that is

$$f : I \to X, \qquad f(0) = f(1) = x_0$$

Restricting to loops has an advantage. In the plane with a point missing, a loop around that point cannot be contracted to the constant map (as in example 2.1.2, where the loop $f * \tilde{h}$ appeared). This failure of loops to contract will be our key concept. With this definition, we can consider the group structure on the subset of $P(X)$ of classes of loops at a point.

Definition 2.1.12: Fundamental Group [at x_0]

Let
$$\pi_1(X, x_0) = \{[f] | f(0) = f(1) = x_0\}$$
be the path space with loops at x_0. Then $\pi_1(X, x_0)$ is a group under $*$, called *fundamental group* of X *based at* x_0, or the *first homotopy group*

2.1.13 Note There *can* be more than one element in $\pi_1(X, x_0)$. We will soon explore spaces where this is the case, and also explore cases where there is only one element.

Sometimes $\pi(X)$ is used to represent the groupoid of the space X. Given a 1st homotopy group, one naturally asks whether there is a 2nd, or an nth. There is, and it is denoted $\pi_n(X, x_0)$. The higher values of n are where the "dimensions" mentioned earlier enter. These are the subject of homotopy theory, studied in chapter 5.

Example 2.1.14: Fundamental Group

Consider $\mathbb{R}^n$ with the Euclidean topology and some $x_0 \in \mathbb{R}^n$ to define $\pi_1(\mathbb{R}^n, x_0)$. Then by the straight-line homotopy, we can collapse every loop to the constant map, so there is only one element:
$$\pi_1(\mathbb{R}^n, x_0) = \{[e_{x_0}]\}$$
More generally, a convex subset of $\mathbb{R}^n$ has trivial fundamental group (by the straight-line homotopy).

Note that this group is *not* necessarily abelian. We shall show that $S^1 \vee S^1$ has fundamental group as the free product of two copies of the integers, $\mathbb{Z} * \mathbb{Z}$.

So how many fundamental groups are there on a space? Does every point have a distinct type of group or are some of them isomorphic, or better yet, all of them isomorphic on the same space? The answers to this give us the topological property we are looking for in groups. That is, when thinking about a space, we usually think about the fundamental groups *up to isomorphism* and then the topologically invariant property are the *types* of fundamental groups. That means that the fundamental group does not rely on the point x_0 on which it was originally defined to get its structure. Moreover, it will also not rely on the *entire* space X, just the component spaces, so the interesting objects to study are the path-connected components.

Proposition 2.1.15: Group Isomorphism Between Fundamental Groups

Let α be a path in X with $\alpha(0) = x_0$ and $\alpha(1) = x_1$, and let $\bar{\alpha}(t) = \alpha(1 - t)$ be its reverse. Then
$$\hat{\alpha} \colon \pi_1(X, x_0) \to \pi_1(X, x_1), \qquad \hat{\alpha}([f]) = [\bar{\alpha}] * [f] * [\alpha]$$
is a group isomorphism, with inverse $\hat{\bar{\alpha}} \colon \pi_1(X, x_1) \to \pi_1(X, x_0)$, $[g] \mapsto [\alpha] * [g] * [\bar{\alpha}]$.

***Proof*:**

This is [16, Group Isomorphism Between Fundamental Groups], stated there for the same direction and formula, whose proof shows that $\hat{\alpha}$ and $\hat{\bar{\alpha}}$ are mutually inverse homomorphisms.

Corollary 2.1.16: Path-Connectedness And Isomorphism

If X is path-connected, $\pi_1(X, x_0)$ is isomorphic to $\pi_1(X, x_1)$ for any $x_0, x_1 \in X$.

All points of one path component therefore have isomorphic fundamental groups, and it costs nothing to assume X path-connected. The isomorphism of proposition 2.1.15 depends on the path, and the next corollary shows that it depends only on the endpoints exactly when the fundamental group is abelian.

Corollary 2.1.17: Fundamental Group Abelian

Let X be a path-connected space. Then $\pi_1(X)$ is abelian if and only if, for all points $b_0, b_1 \in X$, the isomorphism $\hat{h}\colon \pi_1(X, b_0) \to \pi_1(X, b_1)$ of proposition 2.1.15 is the same for every path h from b_0 to b_1.

***Proof*:**
Suppose $\pi_1(X)$ is abelian (abelianness does not depend on the basepoint, by corollary 2.1.16), and let h_1, h_2 be paths from b_0 to b_1. The class $c = [\bar{h}_1 * h_2]$ is a loop class at b_1 with inverse $[\bar{h}_2 * h_1]$, so for a loop f at b_0

$$\hat{h}_1([f]) = [\bar{h}_1 * f * h_1] = [\bar{h}_1 * h_2] * [\bar{h}_2 * f * h_2] * [\bar{h}_2 * h_1] = c\,\hat{h}_2([f])\,c^{-1} = \hat{h}_2([f])$$

the last step because $\pi_1(X, b_1)$ is abelian.

Conversely, suppose $\pi_1(X, b_0)$ is not abelian, so there are $[f], [g] \in \pi_1(X, b_0)$ with $[f]*[g] \neq [g]*[f]$, that is, $[\bar{g}] * [f] * [g] \neq [f]$. Pick a path h from b_0 to b_1. Then $g * h$ is also a path from b_0 to b_1, and since $\hat{h}$ is injective,

$$\widehat{g * h}([f]) = [\bar{h} * \bar{g} * f * g * h] = \hat{h}([\bar{g}] * [f] * [g]) \neq \hat{h}([f])$$

So $\hat{h}$ and $\widehat{g * h}$ differ although h and $g * h$ have the same endpoints, completing the proof.

The fundamental group of a path-connected topological group is always abelian, by the Eckmann-Hilton argument of exercise 5.1.1 (1), so the fundamental group of a connected Lie group is abelian.

Overall, if two path-connected spaces have different fundamental groups, they are topologically different. However, in general, the same fundamental group does not mean the same space (for example, $\mathbb{R}$ and $\mathbb{R}^2$ have the same fundamental group). The fundamental group that is simplest to compute is the trivial one we have seen often.

Definition 2.1.18: Simply-Connected

X is *simply-connected* if X is path-connected, and $\pi_1(X, x_0)$ is the trivial group for some (and hence all) $x_0 \in X$

Note that a space with $\pi_1(X, x_0) = 0$ need *not* be contractible (take for example the Warsaw circle of example 5.3.9, which is the graph of $\sin(1/x)$ closed up into a circle). For CW complexes, Whitehead's theorem (theorem 5.3.6) shows that a CW complex whose homotopy groups are all zero is contractible, since the map to a point is then a weak homotopy equivalence.

Lemma 2.1.19: simply connected simplification

If X is simply connected, $P(X) \cong X^2$ (i.e. a bijection), that is, a path class in X is only determined by its endpoints.

***Proof*:**
First, all paths with the same endpoints are path-homotopic. Let α, β be paths from x_0 to x_1. Then

$$\begin{aligned} [\alpha] * [\tilde{\beta}] &\in \pi_1(X, x_0) \\ [\alpha] * [\tilde{\beta}] &= [e_{x_0}] \\ [\alpha] * [\tilde{\beta}] * [\beta] &= [e_{x_0}] * [\beta] \\ [\alpha] &= [\beta] \end{aligned}$$

With this, we can show injectivity. If $\pi([\alpha]) = \pi([\beta])$, then

$$\alpha(0) = \beta(0), \qquad \alpha(1) = \beta(1)$$

then $[\alpha] = [\beta]$ by the argument above.

For surjectivity, let $(x_0, x_1) \in X^2$. Since X is path-connected (definition 2.1.18), there exists a path α with $\alpha(0) = x_0$, $\alpha(1) = x_1$, so that

$$\pi([\alpha]) = (x_0, x_1)$$

completing the proof. The map π is well defined because a path homotopy fixes both endpoints.

2.1.1 Functorial Properties

Given another topological space Y and a continuous function $f : X \to Y$, the group structure we just built up is carried along: fundamental groups are topological in nature, and continuous maps *preserve* them by inducing *homomorphisms* (the group analogue of continuous functions). This reflects the view that topology studies the properties of objects that do not change under continuous functions.

Definition 2.1.20: induced homomorphic function

Suppose $h : X \to Y$ is continuous, with $h(x_0) = y_0$ (i.e. $h\colon (X, x_0) \to (Y, y_0)$). Then define

$$h_* : \pi_1(X, x_0) \to \pi_1(Y, y_0) \qquad h_*([\alpha]) = [h \circ \alpha]$$

2.1.21 Trivia Notice that h is a continuous map which also sends the chosen point to the chosen point, so it is a morphism of the category Top_* of pointed spaces and based maps (example 1.1.4).

Proposition 2.1.22: $(-)_*$ is a Functor

1. h_* is a group homomorphism
2. If $i : (X, x_0) \to (X, x_0)$ is the identity, then i_* is the identity
3. This map is functorial, that is, if $f : (X, x_0) \to (Y, y_0)$ and $g \colon (Y, y_0) \to (Z, z_0)$ then

$$(g \circ f)_* = g_* \circ f_*$$

Furthermore, if $f : (X, x_0) \to (Y, y_0)$ is a homeomorphism, then f_* is an isomorphism.

Thus $(X, x_0) \mapsto \pi_1(X, x_0)$, $h \mapsto h_*$ is a functor $\mathsf{Top}_* \to \mathsf{Grp}$ (definition 1.1.3).

***Proof*:**
The map h_* is well defined because h composed with a path homotopy is a path homotopy.

1.

$$\begin{aligned} h_*([\alpha][\beta]) &= h_*([\alpha * \beta]) \\ &= [h \circ (\alpha * \beta)] \\ &= [(h \circ \alpha) * (h \circ \beta)] \\ &= [(h \circ \alpha)] * [(h \circ \beta)] \\ &= h_*([\alpha]) * h_*([\beta]) \end{aligned}$$

2. $i_*([\alpha]) = [i \circ \alpha] = [\alpha]$, hence it is the identity

3.

$$\begin{aligned} (g \circ f)_*[\alpha] &= [(g \circ f) \circ \alpha] \\ &= [g \circ (f \circ \alpha)] \\ &= g_*([f \circ \alpha]) \\ &= g_*(f_*([\alpha])) \end{aligned}$$

If f is a homeomorphism with inverse f^{-1}, then by (2) and the composition rule, $(f^{-1})_* \circ f_* = (f^{-1} \circ f)_* = \mathrm{id}$ and $f_* \circ (f^{-1})_* = \mathrm{id}$, so f_* is an isomorphism.

Thus, the fundamental group is invariant under homeomorphisms, so we can study it as a topological property of our space. The first task is to calculate fundamental groups of spaces.

2.1.2 Covering Spaces and Lifting Property

The way we shall seek out fundamental groups is by associating with them a new universal object that shall geometrically represent the loops of a space (hence, geometrically represent the algebraic properties of the fundamental group). The main idea is that each space X can be "covered" by a large space E which will be easier to analyze (usually simply connected). Then we may ask for a

"universal covering space" that will be a simply connected object that will somehow represent all possible "untangling" of loops, and will exist for almost every space we shall work with, the exception usually being pathological spaces such as the Hawaiian earrings (which shall soon be defined).

2.1.23 Where covering-space theory lives The point-set theory of covering spaces, evenly covered neighbourhoods, path and homotopy lifting, the lifting criterion, the universal cover, the classification by subgroups, and deck transformations, is proved in *Topology* [16], a prerequisite. Results below whose proof is not given here cite it there. Van Kampen's theorem, the homology of covering spaces, orientation coverings, and classifying spaces need more than the point-set theory and are proved in full.

Definition 2.1.24: Evenly Covered and Slices

Let $p : E \to B$ be a continuous surjective map. The open set U of B is said to be *evenly covered* by p if the inverse image $p^{-1}(U)$ can be written as the union of disjoint open sets V_α in E such that for each α, the restriction of p to V_α is a homeomorphism of V_α onto U. The collection $\{V_\alpha\}$ will be called the partition of $p^{-1}(U)$ into *slices*

Definition 2.1.25: Covering Map

Let $p : E \to B$ be continuous and surjective. If every point b of B has a neighborhood U that is evenly covered by p, then p is called a *covering map*, and E is said to be a *covering space* of B.

Example 2.1.26: Covering Maps

1. The identity map $i : X \to X$ is a covering map: every open U is evenly covered by the single slice U
2. The next most trivial cover is if we have many copies of the object. If $X \times \{1, ..., n\}$ is n disjoint copies of X, then $p(x, i) = x$ for all i is again a covering map, since all the spaces are disjoint and the restriction to any copy is a homeomorphism onto X. This function gives a picture of covering spaces in general:

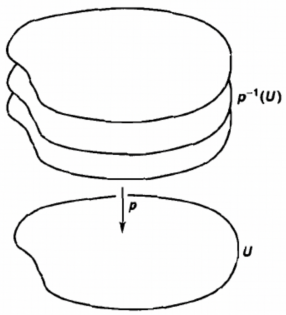


Figure 2.3: Covering space example

3. The map $p : \mathbb{R} \to S^1$ given by the equation

$$p(x) = (\cos(2\pi x), \sin(2\pi x))$$

is a covering map, by [16, The Real Line Covers The Circle]: the open half-circle $x > 0$ has preimage the disjoint union of the intervals $(n - \frac{1}{4}, n + \frac{1}{4})$, each carried homeomorphically onto it, and likewise for the half-circles $x < 0$, $y > 0$, $y < 0$. Any interval $[n, n + 1] \subseteq \mathbb{R}$ wraps itself around the circle completely. Notice that $\mathbb{R}$ is simply connected, while we shall show that S^1 is not simply connected.

4. Another covering space for the circle is $p : S^1 \to S^1$ given by $z \mapsto z^n$ treating $S^1 \subseteq \mathbb{C}$.

5. It may be tempting to say that being a local homeomorphism implies p is a covering map, but this is not the case. Consider:

$$p : \mathbb{R}_{\geq 0} \to S^1 \qquad p(x) = (\cos(2\pi x), \sin(2\pi x))$$

This function is still surjective, and a local homeomorphism (any point $x \in \mathbb{R}_{\geq 0}$ has a neighborhood homeomorphic to its image), and even is a covering map for *most* points. However, consider the point $b_0 = (1, 0)$ and some open neighborhood U around that point. Then the pre-image of U would be the same as in the third example, except for V_0 which would be $(0, \epsilon)$ for some ϵ depending on the open neighbourhood U. If we restrict p to $(0, \epsilon)$, then we do not get a homeomorphism, but an *embedding*, that is, a function for which if we restrict the codomain to its image, it becomes a homeomorphism. Restricting to $(0, \epsilon)$ gives the open neighborhood starting at 0 and ending at ϵ (relative to where sin and cos send ϵ).

Another example would be:

$$E = [0, 1] \times \{0\} \cup (0, 1) \times \{1\}, B = [0, 1]$$

which forces surjectivity on the function $(0, 1) \to [0, 1]$. Then there is not a neighborhood around $0 \in [0, 1]$, which is evenly covered. Again, this comes down to it being an embedding based on the domain rather than the co-domain.

6. The map $\varphi : \mathbb{C} \to \mathbb{C}^\times$ mapping $z \mapsto e^z$ is a covering map. Under the homeomorphisms $\mathbb{C} \cong \mathbb{R} \times \mathbb{R}$, $x + iy \mapsto (x, y/2\pi)$, and $\mathbb{C}^\times \cong \mathbb{R}_{>0} \times S^1$, $w \mapsto (|w|, w/|w|)$, the map φ becomes $(x, t) \mapsto (e^x, p(t))$, the product of the homeomorphism $x \mapsto e^x$ and the covering p of the third example, which is a covering map by proposition 2.1.27(4). It is in fact an ∞ to 1 covering map since $z + 2k\pi i \mapsto e^z$. Note that $\mathbb{C}$ is simply connected.

Proposition 2.1.27: Properties of Covering Maps

Let $p : E \to B$ be a covering map.

1. For every $y \in B$, the subspace $p^{-1}(y)$ has the discrete topology
2. p is an open map
3. If B_0 is a subspace of B, and if $E_0 = p^{-1}(B_0)$, then the map $p_0 : E_0 \to B_0$ obtained by restricting p is a covering map
4. If $p' : E' \to B'$ is also a covering map, then
$$p \times p' : E \times E' \to B \times B'$$
is a covering map

***Proof*:**
Proved in *Topology*: items 1 and 2 are [16, properties of Covering Maps] (a slice meets a fibre in exactly one point, so singletons are open in $p^{-1}(y)$, and for $A \subseteq E$ open, each point of $p(A)$ lies in $p(A \cap V_\beta)$ for some slice V_β, which is open in B because $p|_{V_\beta}$ is a homeomorphism onto the open set U), item 3 is [16, Subspace Of Covering Space] (the slices over $U \cap B_0$ are the sets $V_\alpha \cap E_0$), and item 4 is [16, Product Of Covering Maps] (the products $V_\alpha \times V'_\beta$ of slices are slices over $U \times U'$).

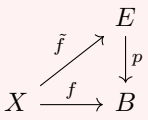
Example 2.1.28: Product of Covering Map

1. Take
$$p \times p : \mathbb{R} \times \mathbb{R} \to S^1 \times S^1$$
where p is the same covering map of the circle we introduced earlier. Then this function is a covering map for the torus, by proposition 2.1.27(4).
2. Notice that in the torus we have a homeomorphic copy of the figure 8 ($S^1 \vee S^1$). We thus have a covering map for it by restricting the covering of the torus, proposition 2.1.27(3). There are multiple ways of covering this space. One can keep one circle and unwrap the other, or unwrap both. This space will also be a space that will have a universal cover.

As mentioned, the key property of covering spaces for us is how they represent fundamental groups, through "lifting" paths from our base space up to our covering space in a way that simplifies calculations.

Definition 2.1.29: Lifting

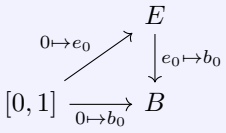
Let $p : E \to B$ be an [arbitrary] continuous function. Let $f : X \to B$ be a continuous map. A *lifting* of f is a continuous map $\tilde{f} : X \to E$ such that $f = p \circ \tilde{f}$. Equivalently, the following diagram commutes



In the theory of projective modules, homomorphisms out of a projective module can always be lifted. Key to us when dealing with homotopies is that paths lift through covering maps.

Lemma 2.1.30: Paths Lifted Through Covering Map

Let $p : E \to B$ be a covering map such that $p(e_0) = b_0$ and let $f : [0, 1] \to B$ be a continuous path with initial point $f(0) = b_0$. Then it has a unique lifting to a path $\tilde{f}$ in E with initial point $\tilde{f}(0) = e_0$, that is the following diagram commutes:



***Proof*:**
Proved in *Topology* [16, paths can be lifted]: subdivide $[0, 1]$ finely enough that each piece lands in an evenly covered set, using the Lebesgue number of the pulled-back cover, then lift piece by piece into the slice determined by the previous endpoint and glue by the pasting lemma. Uniqueness follows because each piece is connected and the slices are disjoint.

Lemma 2.1.31: Homotopies can be lifted

Let $p : E \to B$ be a covering map, and let $p(e_0) = b_0$. Let $F : I \times I \to B$ be a continuous function such that $F(0, 0) = b_0$. Then there exists a unique lift $\tilde{F} : I \times I \to E$ such that $\tilde{F}(0, 0) = e_0$. If F is a path homotopy, then $\tilde{F}$ is a path homotopy.

***Proof*:**
Proved in *Topology* [16, Homotopies Can Be Lifted, Path Homotopies can be lifted], by the same subdivision argument as the previous lemma run in two variables over the square, together with the observation that the lift of a path homotopy is again one because the fibres are discrete and the edges are connected.

Note that though paths lift, loops *need not lift to loops*. In particular, it is often the case that a loop is lifted to a path that is not closed. However, the lifts of two path-homotopic loops have the same endpoints:

Lemma 2.1.32: Path Homotopies can be lifted

Let f, g be paths starting and ending at b_0 (i.e. path-homotopic loops). If $f \simeq_p g$, then $\tilde{f}(1) = \tilde{g}(1)$

***Proof*:**
The path homotopy F between f, g lifts to a path homotopy between $\tilde{f}, \tilde{g}$. In particular, the end points of homotopic paths agree, therefore $\tilde{f}(1) = \tilde{g}(1)$.

Using this, we see that lifting continuous maps is in fact directly tied with the fundamental group:

Theorem 2.1.33: Conditions for Maps being Lifted

Let $p : (\tilde{X}, e_0) \to (X, x_0)$ be a covering map and $f : (Y, y_0) \to (X, x_0)$ where Y is path connected and locally path connected[a]. Then the lift $\tilde{f}$ exists if and only if $f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, e_0))$

[a] for each $y \in Y$, there exists an open neighbourhood that is path connected. Think of a comb-space for a connected, not locally path connected space.

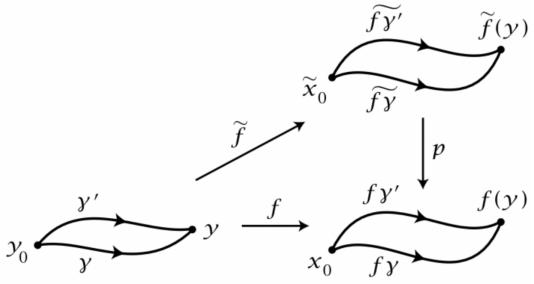
***Proof*:**
If $\tilde{f}$ exists, then by functoriality $f_* = p_* \circ \tilde{f}_*$, hence $f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, e_0))$.

Conversely, suppose $f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, e_0))$. Choose $y \in Y$, and let γ be a path in Y from y_0 to y. Then $f\gamma$ is a path in X starting at x_0 and going to $f(y)$. Then by lemma 2.1.30, $f\gamma$ lifts to a path $\tilde{f\gamma}$ starting at e_0.

Now, define

$$\tilde{f}(y) = \tilde{f\gamma}(1)$$

This map is well-defined: if γ' is another path from y_0 to y, then $h_0 = (f\gamma') * (\overline{f\gamma})$ is a loop at x_0. By assumption $[h_0] \in f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, e_0))$. The argument follows this picture:



Since $[h_0] \in p_*(\pi_1(\tilde{X}, e_0))$, there is a homotopy h_t of h_0 to a loop h_1 in X that lifts to a loop $\tilde{h}_1$ in $\tilde{X}$ at basepoint e_0. The homotopy h_t can be lifted to $\tilde{h}_t$ by lemma 2.1.31. Then since $\tilde{h}_1$ is a loop at e_0, so is $\tilde{h}_0$. By the uniqueness of the lifted paths, the first half of $\tilde{h}_0$ is $\tilde{f\gamma'}$ and the second half is $\tilde{f\gamma}$ traversed backwards, with common midpoint $\tilde{f\gamma}(1) = \tilde{f\gamma'}(1)$. Hence, $\tilde{f}$ is well-defined.

It remains to show that $\tilde{f}$ is continuous. Let $U \subseteq X$ be an evenly covered open neighbourhood of $f(y)$. Then choose $\tilde{U} \subseteq \tilde{X}$ containing $\tilde{f}(y)$ such that $p : \tilde{U} \to U$ is a homeomorphism. By local path-connectedness, choose a path-connected open neighbourhood V of y such that $f(V) \subseteq U$. For paths from y_0 to points $y' \in V$, we can take a fixed path γ from y_0 to y followed by paths η in V from y to the points y'. Then the paths $(f\gamma) * (f\eta)$ in X have lifts $(\tilde{f\gamma}) * (\tilde{f\eta})$ where $\tilde{f\eta} = p^{-1} f\eta$ and $p^{-1} : U \to \tilde{U}$ is the inverse of $p|_{\tilde{U}}$. Thus, $\tilde{f}(V) \subseteq \tilde{U}$, and $\tilde{f}|_V = p^{-1} f$, hence $\tilde{f}$ is continuous at any y, completing the proof.

Corollary 2.1.34: Unique Lifting Property

Let $p : \tilde{X} \to X$ be a covering map and $f : Y \to X$ be a continuous map. Then if $\tilde{f}_1, \tilde{f}_2 : Y \to \tilde{X}$ are two lifts of f that agree at one point of Y and Y is connected, then $\tilde{f}_1$ and $\tilde{f}_2$ agree on all of Y, that is

$$\tilde{f}_1 = \tilde{f}_2$$

***Proof*:**
The strategy is the one that recurs throughout the lifting theory: the locus where the two lifts agree is simultaneously open and closed, so connectedness of Y forces it to be all of Y. Set

$$S = \left\{ y \in Y \;\middle|\; \tilde{f}_1(y) = \tilde{f}_2(y) \right\}$$

By hypothesis S is nonempty, since the two lifts agree at some point. It suffices to show that S is both open and closed in Y, for then connectedness of Y leaves $S = Y$ as the only nonempty option, which is exactly the asserted equality $\tilde{f}_1 = \tilde{f}_2$.

Fix $y \in Y$ and write $x = f(y) \in X$. Since $p\colon \tilde{X} \to X$ is a covering map, choose an evenly covered open neighbourhood U of x, so that $p^{-1}(U) = \coprod_\alpha V_\alpha$ is a disjoint union of slices, each mapped homeomorphically onto U by p (definition 2.1.24). The two points $\tilde{f}_1(y)$ and $\tilde{f}_2(y)$ both lie in $p^{-1}(U)$, because $p\tilde{f}_i(y) = f(y) = x \in U$. Let V_1 and V_2 be the slices containing them, respectively. By continuity of the lifts there is an open neighbourhood N of y with $\tilde{f}_1(N) \subseteq V_1$ and $\tilde{f}_2(N) \subseteq V_2$. Shrinking N if necessary, N may also be taken to satisfy $f(N) \subseteq U$.

Suppose first that $y \in S$, so $\tilde{f}_1(y) = \tilde{f}_2(y)$. This common point lies in both V_1 and V_2, and distinct slices are disjoint, so $V_1 = V_2$. Call this common slice V. On N both lifts land in V, and on V the projection p is injective. For any $z \in N$ the relation $p\tilde{f}_1(z) = f(z) = p\tilde{f}_2(z)$ together with this injectivity gives $\tilde{f}_1(z) = \tilde{f}_2(z)$, so $N \subseteq S$. Hence every point of S has a neighbourhood contained in S, and S is open.

Suppose instead that $y \notin S$, so $\tilde{f}_1(y) \neq \tilde{f}_2(y)$. These two points cannot lie in the same slice, for p is injective on each slice while $p\tilde{f}_1(y) = p\tilde{f}_2(y)$. Thus $V_1 \neq V_2$, and being distinct slices they are disjoint. For $z \in N$ we then have $\tilde{f}_1(z) \in V_1$ and $\tilde{f}_2(z) \in V_2$ with $V_1 \cap V_2 = \emptyset$, so $\tilde{f}_1(z) \neq \tilde{f}_2(z)$ for every $z \in N$. Thus N is a neighbourhood of y disjoint from S, which shows that the complement $Y \setminus S$ is open, that is, S is closed.

The set S is therefore a nonempty subset of Y that is both open and closed. As Y is connected, $S = Y$, so $\tilde{f}_1(y) = \tilde{f}_2(y)$ for every $y \in Y$ and consequently $\tilde{f}_1 = \tilde{f}_2$, as we sought to show.

As mentioned, the lifts are *not necessarily* loops. When projected using the covering space, they will become loops. The key is that all the paths in B at b_0 lift to paths that start at $e_0 \in p^{-1}(b_0)$ and end at $e_k \in p^{-1}(b_0)$. We may imagine the lifting correspondence as somehow "untangling" the paths. Lifts of paths are unique once the starting point is fixed, and path-homotopic loops have path-homotopic lifts, so the endpoint of the lift of a loop at b_0 starting at e_0 depends only on the class of that loop in $\pi_1(B, b_0)$:

Definition 2.1.35: Lifting Correspondence

Let $p : E \to B$ be a covering map. Let $p(e_0) = b_0$. Given $[f] \in \pi_1(B, b_0)$, let $\tilde{f}$ be a lifting of f to E such that $\tilde{f}(0) = e_0$. Then we can define:

$$\varphi : \pi_1(B, b_0) \to p^{-1}(b_0) \qquad \varphi([f]) = \tilde{f}(1)$$

and φ is called the *lifting correspondence* derived from the covering map p, e_0.

2.1.36 Note It is important to choose a point e_0, since the fibre $p^{-1}(b_0)$ usually has many points, each with a neighbourhood mapped homeomorphically onto its image, so a choice must be made. The choice, however, does not matter much, since a neighbourhood of any such e_0 is homeomorphic to its image.

Recall that the inverse image of a point under a covering map is a discrete space (proposition 2.1.27), so this construction lands in a discrete set.

Proposition 2.1.37: Lifting Correspondence and Path-Connected Spaces

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$. If E is path-connected, then the lifting correspondence is surjective. If E is also simply-connected, then the lifting correspondence is bijective

***Proof*:**
Assume E is path-connected and let $e_1 \in p^{-1}(b_0)$. To show surjectivity, choose a path f_E in E such that $f_E(0) = e_0$, $f_E(1) = e_1$. Let $f = p \circ f_E$. Then f_E lifts f, and

$$f(0) = p(f_E(0)) = p(e_0) = b_0 \tag{2.1}$$
$$f(1) = p(f_E(1)) = p(e_1) = b_0 \tag{2.2}$$

Thus, $\varphi([f]) = f_E(1) = e_1$, and so φ is surjective, as we sought to show.

Now assume E is simply connected and say $\varphi([f]) = \varphi([g])$ so that $\tilde{f}(1) = \tilde{g}(1)$. The lifts $\tilde{f}$ and $\tilde{g}$ have the same endpoints, so $\tilde{f} \simeq_p \tilde{g}$ by lemma 2.1.19. Composing this path homotopy with p gives a path homotopy $p \circ \tilde{f} \simeq_p p \circ \tilde{g}$, that is, $f \simeq_p g$, so $[f] = [g]$, showing injectivity.

What we have essentially proven is that we can deduce information of the fundamental group of a space given the covering map, and we can at least deduce all its elements if the covering space is simply connected. At minimum, if we have a simply connected covering space, then the cardinality of $\pi_1(X, b_0)$ is equal to that of $p^{-1}(b_0)$.

Example 2.1.38: Lifts

1. Let $B = E = S^1$, and take $p : E \to B$ $z \mapsto z^2$. Notice that E is path-connected, hence the fundamental group surjects onto the pre-image of a point of the covering map. Let $b_0 = e_0 = (1,0)$. Then $\varphi : \pi_1(S^1, b_0) \to p^{-1}(b_0) = \{(1,0), (-1,0)\}$.

   We exhibit this map explicitly. Take $e_{b_0} : [0,1] \to S^1$, $e_{b_0}(t) = b_0 = (1,0)$. Then the lift of e_{b_0} starting at e_0 is the constant path $\tilde{e_{b_0}} = e_{e_0}$: that path is a lift, and two lifts agreeing at a point of the connected space I coincide (corollary 2.1.34). Then:

   $$\varphi([e_{b_0}]) = e_{e_0}(1) = e_0 = (1,0)$$

   Now let

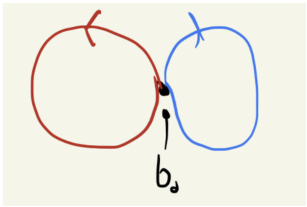
   $$f : I \to S^1, f(t) = (\cos(2\pi t), \sin(2\pi t))$$

   which is the function which will go around the circle once. Then the lift starting at e_0 is

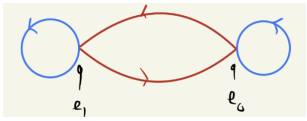
   $$\tilde{f} : I \to S^1, \tilde{f}(t) = (\cos(\pi t), \sin(\pi t))$$

   So $\tilde{f}(1) = (-1,0)$ so $\varphi([f]) = (-1,0)$ (which also verifies surjectivity). So for the first time, we showed that we have a *non-contractible loop* since the two homotopy classes are different. That is $[f] \neq [e_{b_0}]$.

2. Let X be the figure 8 as we described in a previous example (from the torus). Let the intersection of the circles be b_0. Take one loop going around the left circle counter-clockwise, and one for the right circle, clockwise, so that $[f], [g] \in \pi_1(X, b_0)$. Let E represent the following lifting space:

   (a) Figure 8

   (b) Particular Lifting Space of Figure 8

   and take e_0 as in the image. Then

   $$\begin{aligned} \varphi([f]) &= e_1 \\ \varphi([g]) &= e_0 \\ \varphi([f] * [g]) &= e_1 \\ \varphi([g] * [f]) &= e_1 \end{aligned}$$

   This tells us that $[f]$ and $[g]$ are different paths, but it seems that their product is the same. Since φ is surjective but not injective, it gives only *partial* information on how the elements of this fundamental group interact. A larger covering space reveals more.

3. Same X, same b_0, but this time, our lifting space E is different.

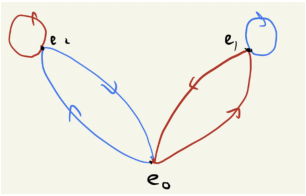


Figure 2.5: Different Lifting Space

Now,

$$\varphi([f] * [g]) = e_1 \tag{2.3}$$
$$\varphi([g] * [f]) = e_2 \tag{2.4}$$

Therefore, $[f] * [g] \neq [g] * [f]$, i.e. $\pi_1(X, b_0)$ is *not* abelian, as we claimed when $\pi_1(X, x_0)$ was introduced.

Proposition 2.1.39: Lifting Correspondence And Homomorphisms

Let $p : E \to B$ be a covering map and $p(e_0) = b_0$. Then:

1. The induced homomorphism $p_* : \pi_1(E, e_0) \to \pi_1(B, b_0)$ is injective
2. Let $H = p_*(\pi_1(E, e_0))$. Then the lifting correspondence φ induces an injective map:

$$\Phi : \pi_1(B, b_0)/H \to p^{-1}(b_0)$$

   where $\pi_1(B, b_0)/H$ is the collection of right cosets of H. The map is bijective if E is path connected
3. If f is a loop in B based at b_0, then $[f] \in H$ if and only if f lifts to a loop in E based at e_0

***Proof*:**

1. Let $\tilde{h}$ be a loop in E at e_0 where $p_*([\tilde{h}])$ is the identity element. Since it is the identity, there exists a path homotopy F between $p \circ \tilde{h}$ and the constant loop. Then let $\tilde{F}$ be the lifting of F to E such that $\tilde{F}(0, 0) = e_0$. But then $\tilde{F}$ is the path homotopy between $\tilde{h}$ and the constant loop at e_0
2. Let f, g be loops in B based at b_0 with corresponding $\tilde{f}$ and $\tilde{g}$ in E that begin at e_0. Then $\varphi([f]) = \tilde{f}(1)$ and $\varphi([g]) = \tilde{g}(1)$. We will show that $\varphi([f]) = \varphi([g])$ if and only if $[f] \in H * [g]$. First, if $[f] \in H * [g]$, then $[f] = [h * g]$ where $h = p \circ \tilde{h}$ for some loop $\tilde{h}$ in E based at e_0. Then the product $\tilde{h} * \tilde{g}$ is defined and is the lifting of $h * g$. Since $[f] = [h * g]$, the liftings $\tilde{f}$ and $\tilde{h} * \tilde{g}$ which begin at e_0 must end at the same point in E. But then $\tilde{f}$ and $\tilde{g}$ end at the same point of E so $\varphi([f]) = \varphi([g])$.
   Conversely, suppose $\varphi([f]) = \varphi([g])$ so that $\tilde{f}$ and $\tilde{g}$ end at the same point in E. Then

the product $\tilde{h}$ of $\tilde{f}$ and the reverse of $\tilde{g}$ is defined, and it is a loop in E based at e_0. By theorem 2.1.7, $[\tilde{h} * \tilde{g}] = [\tilde{f}]$. If $\tilde{F}$ is a path homotopy in E between $\tilde{h} * \tilde{g}$ and $\tilde{f}$, $p \circ \tilde{F}$ is a path homotopy in B between $h * g$ and f where $h = p \circ \tilde{h}$. But then $[f] \in H * [g]$, as we sought to show.

Furthermore, if E is path connected, then φ is surjective (proposition 2.1.37), meaning Φ is also surjective.

3. Injectivity of Φ means $\varphi([f]) = \varphi([g])$ if and only if $[f] \in H * [g]$. Applying this with g the constant loop, we get that $\varphi([f]) = e_0$ if and only if $[f] \in H$. But $\varphi([f]) = e_0$ exactly when the lift of f that begins at e_0 ends at e_0.

Finally, we compute the fundamental group of the circle.

Theorem 2.1.40: Fundamental Groups Of The Circle

The fundamental group of S^1 is isomorphic to $(\mathbb{Z}, +)$

***Proof*:**
Let $p : \mathbb{R} \to S^1$, and $p(t) = (\cos(2\pi t), \sin(2\pi t))$. Then p is a covering map (example 2.1.26) such that $p(r) = p(s) \Leftrightarrow r - s \in \mathbb{Z}$. Let $b_0 = (1, 0) \in S^1$. Then $p^{-1}(b_0) = \mathbb{Z}$. Let our starting point in E be $e_0 = 0$. Then the lifting correspondence gives a map $\varphi : \pi_1(S^1, b_0) \to \mathbb{Z}$. Moreover, since $\mathbb{R}$ is path-connected and simply connected (example 2.1.14), φ is a bijection by proposition 2.1.37. So, it remains to show that φ is a homomorphism:

$$\varphi([f] * [g]) = \varphi([f]) + \varphi([g])$$

Let $\tilde{f}, \tilde{g}$ be lifts of f, g starting at 0, so $\varphi([f]) = \tilde{f}(1) \in \mathbb{Z} = p^{-1}(b_0)$. Define $\tilde{g}_2(t) = \tilde{g}(t) + \tilde{f}(1)$. Then

$$p(\tilde{g}_2(t)) = p(\tilde{g}(t) + \tilde{f}(1)) = p(\tilde{g}(t) + k) = p(\tilde{g}(t)) = g(t)$$

since $k = \tilde{f}(1) \in \mathbb{Z}$, so the covering map projects this path onto g, meaning it is also a lift of g, just shifted over by $\tilde{f}(1)$ to where the path starts. We can see this by $\tilde{g}_2(0) = \tilde{f}(1)$. Thus, we can combine these two paths, $\tilde{f} * \tilde{g}_2$, and by what we have just shown, this is a lift of $f * g$. Thus

$$\begin{aligned} \varphi([f] * [g]) &= (\tilde{f} * \tilde{g}_2)(1) \\ &= \tilde{g}_2(1) \\ &= \tilde{g}(1) + \tilde{f}(1) = \varphi([f]) + \varphi([g]) \end{aligned}$$

completing the proof.

This also means that the fundamental group of the circle is abelian, which matches the picture of loops drawn on a circle: their order can be reversed. On the figure 8 it cannot (example 2.1.38).

2.1.3 Fundamental Group Invariant Under Homotopy

If f is a homeomorphism, then f_* is an isomorphism. However, this is a rather rigid condition. If $f \simeq g$, then $f_* = g_*$, showing that the fundamental group is invariant under the much looser

condition of homotopy. If $f_t : (X, x_0) \to (Y, y_0)$ is a homotopy such that $f_t(x_0) = y_0$, we shall call it a basepoint-preserving homotopy.

Proposition 2.1.41: Invariance Of Induced Homomorphisms Under Homotopy

Let $f_t : (X, x_0) \to (Y, y_0)$ be a basepoint-preserving homotopy. Then $(f_0)_* = (f_1)_*$ as homomorphisms $\pi_1(X, x_0) \to \pi_1(Y, y_0)$.

***Proof*:**
For a loop φ at x_0, the map $(s, t) \mapsto f_t(\varphi(s))$ is a path homotopy from $f_0 \circ \varphi$ to $f_1 \circ \varphi$, since $f_t(x_0) = y_0$ for all t. Then $(f_0)_*([\varphi]) = [f_0 \circ \varphi] = [f_1 \circ \varphi] = (f_1)_*([\varphi])$.

Corollary 2.1.42: The Fundamental Group Is a Functor on hTop_*

The assignment $(X, x_0) \mapsto \pi_1(X, x_0)$, $[h] \mapsto h_*$ is a well-defined functor $\mathsf{hTop}_* \to \mathsf{Grp}$. In particular, a based map that is an isomorphism of hTop_* induces an isomorphism of fundamental groups.

***Proof*:**
A based homotopy is a basepoint-preserving homotopy, so by proposition 2.1.41 the homomorphism h_* depends only on the based homotopy class of h. The functor $\pi_1 \colon \mathsf{Top}_* \to \mathsf{Grp}$ of proposition 2.1.22 therefore factors through hTop_*, as described after definition 1.1.7, and a functor carries isomorphisms to isomorphisms, completing the proof.

When working with homotopy equivalence, we may in fact drop the basepoint information.

Lemma 2.1.43: Induced Maps, Retractions, and Deformation Retraction

Let A be a retract of X. Then for $a_0 \in A$, the induced map $\iota_* : \pi_1(A, a_0) \to \pi_1(X, a_0)$ is injective, where $\iota : A \to X$ is the inclusion. If A is a deformation retract of X, then ι_* is also surjective, hence an isomorphism.

This means that X should have *at least* as many elements in its fundamental group as A.

***Proof*:**
Let $r : X \to A$ be a retraction. Then $r \circ \iota = \mathrm{id}_A$. Thus, $(r \circ \iota)_* = (id_A)_* = id|_{\pi_1(A,a_0)}$. Thus

$$[f] = (r_* \circ \iota_*)([f]) = r_*(\iota_*([f]))$$

showing r_* is a left inverse, thus ι_* is injective.

If $r_t : X \to X$ is a deformation retraction of X onto A so that $r_0 = \mathrm{id}$ and $r_t|_A = \mathrm{id}$ and $r_1(X) \subseteq A$, then for any loop $f : I \to X$ based at $x_0 \in A$, the composition $r_t \circ f$ gives a path homotopy (as $r_t(x_0) = x_0$) of f to a loop in A, showing ι_* is surjective, and hence an isomorphism.

The group-theoretic equivalent statement is that a homomorphism $\varphi : G \to H$ restricts to the identity on H. If H is a normal subgroup, then G is the direct product of H and $\ker(\varphi)$, and if H is not normal then we replace direct product with semi-direct product.

Lemma 2.1.44: Basepoint Path In Homotopy

Let $\varphi_t \colon X \to Y$ be a homotopy, $x_0 \in X$, and h the path $h(t) = \varphi_t(x_0)$ from $\varphi_0(x_0)$ to $\varphi_1(x_0)$. Then $\varphi_{1*} = \hat{h} \circ \varphi_{0*}$, where $\hat{h} \colon \pi_1(Y, \varphi_0(x_0)) \to \pi_1(Y, \varphi_1(x_0))$ is the isomorphism of proposition 2.1.15. That is, the following diagram commutes:

$$\begin{array}{ccc} \pi_1(X, x_0) & \xrightarrow{\varphi_{0*}} & \pi_1(Y, \varphi_0(x_0)) \\ & \searrow^{\varphi_{1*}} & \downarrow \hat{h} \\ & & \pi_1(Y, \varphi_1(x_0)) \end{array}$$

***Proof*:**
For $t \in [0,1]$ let $h_t(s) = h(ts)$, the restriction of h to $[0,t]$ reparametrized over $[0,1]$, a path from $\varphi_0(x_0)$ to $\varphi_t(x_0)$, and let $\bar{h}_t$ be its reverse. For a loop f at x_0, the loops $h_t * (\varphi_t \circ f) * \bar{h}_t$ are based at $\varphi_0(x_0)$ and depend continuously on (s,t), so they form a homotopy of loops at $\varphi_0(x_0)$:

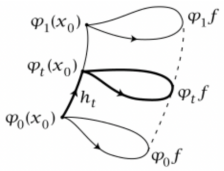


Figure 2.6: Visualization

At $t = 0$ the path h_0 is constant, so the loop is homotopic to $\varphi_0 \circ f$, and at $t = 1$ it is $h * (\varphi_1 \circ f) * \bar{h}$. Hence $[\varphi_0 \circ f] = [h] * [\varphi_1 \circ f] * [\bar{h}]$, and multiplying by $[\bar{h}]$ on the left and $[h]$ on the right gives

$$\hat{h}(\varphi_{0*}([f])) = [\bar{h}] * [\varphi_0 \circ f] * [h] = [\varphi_1 \circ f] = \varphi_{1*}([f])$$

completing the proof.

Theorem 2.1.45: Homotopy Equivalence, Then Basepoint Invariant

Let $\varphi \colon X \to Y$ be a homotopy equivalence. Then the induced homomorphism $\varphi_* \colon \pi_1(X, x_0) \to \pi_1(Y, \varphi(x_0))$ is an isomorphism for all $x_0 \in X$.

***Proof*:**
Let $\psi \colon Y \to X$ be a homotopy inverse, so that $\varphi \circ \psi \simeq \mathrm{id}_Y$ and $\psi \circ \varphi \simeq \mathrm{id}_X$, and consider

$$\pi_1(X, x_0) \xrightarrow{\varphi_*} \pi_1(Y, \varphi(x_0)) \xrightarrow{\psi_*} \pi_1(X, \psi\varphi(x_0)) \xrightarrow{\varphi_*} \pi_1(Y, \varphi\psi\varphi(x_0))$$

By lemma 2.1.44, applied to a homotopy from id_X to $\psi \circ \varphi$, the composite $\psi_* \circ \varphi_*$ of the first

two maps is $\hat{h}$ for a path h from x_0 to $\psi\varphi(x_0)$, hence an isomorphism by proposition 2.1.15. Applied to a homotopy from id_Y to $\varphi \circ \psi$, the same lemma makes the composite of the second and third maps an isomorphism. The first composite being bijective, ψ_* is surjective, and the second being bijective, ψ_* is injective. So ψ_* is a bijection, and $\varphi_* = \psi_*^{-1} \circ (\psi_* \circ \varphi_*)$ is an isomorphism, completing the proof.

Example 2.1.46: Fundamental Group Does Not Characterize Spaces

Define the *pseudocircle* to be the set $X = \{a, b, c, d\}$ with the topology:

$$\{\{a, b, c, d\}, \{a, b, c\}, \{a, b, d\}, \{a, b\}, \{a\}, \{b\}, \emptyset\}$$

which can be thought of as the topology induced by the partial order $a < c$, $b < c$, $a < d$, $b < d$ where the open sets are the downward-closed sets. Then we may define a continuous map $f : S^1 \to X$ by

$$f(x, y) = \begin{cases} a & x < 0 \\ b & x > 0 \\ c & (x, y) = (0, 1) \\ d & (x, y) = (0, -1) \end{cases}$$

This map is a *weak homotopy equivalence*: it induces an isomorphism on all homotopy groups. We do not prove this here. It is a case of McCord's theorem on finite spaces [26]. Granting it, f also induces isomorphisms on singular homology and cohomology, which are introduced later in this book (proposition 5.3.15). See this link for more details:

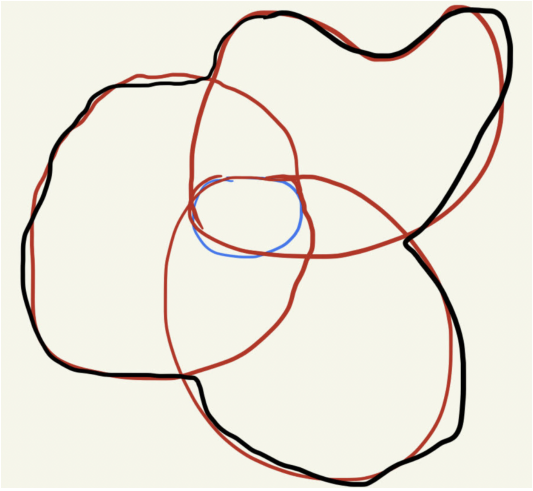
https://en.wikipedia.org/wiki/Pseudocircle

A space is often assembled from open pieces whose fundamental groups are known, and a loop that wanders through several pieces can be cut at points of the overlaps.

Proposition 2.1.47: Decomposing Spaces to find Homotopy

Let X be a topological space such that it is the union of a collection of path-connected open sets A_α each containing the basepoint $x_0 \in X$ and if each intersection $A_\alpha \cap A_\beta$ is path connected, then every loop in X at x_0 is homotopic to a product of loops each of which is contained in a single A_α.

To visualize:



and a loop f at x_0 is homotopic to $f \simeq (f_1 * f_2 * \cdots * f_n)$ where each $f_i \in A_i$, hence:

$$[f] = [f_1] * [f_2] * \cdots * [f_n]$$

breaking up the problem into one of finding the fundamental group of each A_i.

***Proof*:**
Let $f : I \to X$ be a loop with basepoint x_0. First, we may partition I so that each subinterval $[s_{i-1}, s_i]$ is mapped by f to a single A_α, namely since f is continuous each $s \in I$ has an open neighbourhood $V_s \subseteq I$ that gets mapped to A_α by f. We may take these intervals to be sufficiently small so that the closure is in A_α. Then by compactness finitely many V_s cover I, giving us our desired partition.

Let $f_i = f|_{[s_{i-1},s_i]}$ and A_i be the A_α containing the image of f_i. Then $f = f_1 * f_2 * \cdots * f_n$. Since $A_i \cap A_{i+1}$ is path connected for each i, there exists a g_i in $A_i \cap A_{i+1}$ from x_0 to $f(s_i) \in A_i \cap A_{i+1}$. Then consider

$$(f_1 * \overline{g}_1) * (g_1 * f_2 * \overline{g}_2) * \cdots * (g_{n-1} * f_n)$$

which is path-homotopic to f, since each inserted $\overline{g}_i * g_i$ is path-homotopic to a constant path (theorem 2.1.7), and each factor is a loop at x_0 in a single A_i. This is the desired composition, as we sought to show.

The relations among loops from different pieces are the content of van Kampen's theorem, theorem 2.2.4. When every piece is simply connected (with path-connected pairwise intersections, as in proposition 2.1.47), each loop in a single piece is null-homotopic and $\pi_1(X)$ is trivial.

Example 2.1.48: Fundamental Group of n-sphere
Write S^n as the union of A_1 and A_2 each homeomorphic to $\mathbb{R}^n$ and $A_1 \cap A_2$ is homeomorphic to $S^{n-1} \times \mathbb{R}$ (take A_1, A_2 to be the complements of the two antipodal points in S^n). Then choosing a basepoint $x_0 \in A_1 \cap A_2$, if $n \geq 2$ then $A_1 \cap A_2$ is path connected and so by proposition 2.1.47 every loop in S^n based at x_0 is homotopic to a product of loops in A_1 or A_2. Both $\pi_1(A_1)$ and $\pi_1(A_2)$ are zero, and hence S^n is simply connected for $n \geq 2$.

2.1.4 Interesting Consequences

Theorem 2.1.49: Cannot make a hole

Let B^2 be the unit disc in $\mathbb{R}^2$. There is no retraction from B^2 to S^1. Informally, one cannot make a hole.

***Proof*:**
The fundamental group of the circle is infinite (theorem 2.1.40), but the fundamental group of B^2 has 1 element (because it is convex, example 2.1.14). A retraction would make the map $\pi_1(S^1) \to \pi_1(B^2)$ induced by the inclusion injective (lemma 2.1.43), so there is *no* retraction.

Theorem 2.1.50: Brouwer Fixed Point Theorem

If $f : B^2 \to B^2$ is continuous, then there exists $x \in B^2$ such that $f(x) = x$

***Proof*:**
Recall that $B^2 := \{(x, y) | x^2 + y^2 \leq 1\}$ and thus is $\approx I^2$.

Assume $f : B^2 \to B^2$ is continuous. For the sake of contradiction, suppose that for every $x \in B^2$, $f(x) \neq x$. We will use f to construct a retraction of B^2 to its boundary S^1.

The key idea is that since $f(x) \neq x$, and both are in B^2, we can draw a straight line between x and $f(x)$. We can continue this line to the boundary. If we give a direction (say from $f(x)$ to x), we can then make a function $g(x)$ which will take our point x, and given this line, it will bring it to the boundary.

The key idea here is that if we had that $f(x) = x$, then there would be no ray to draw. For example, if $f(x) = \frac{x}{\|x\|}$ then every point will have a clear ray except $x = 0$. This unclearness of the ray is translated as f being discontinuous (in the case of the concrete f, we divided by 0).

More formally, we will define $G : B^2 \times \mathbb{R}_{\geq 0} \to \mathbb{R}^2$.

$$G(x, r) = x + r(x - f(x))$$

since everything is in $\mathbb{R}^2$, multiplying, adding and subtracting are allowed. Note that

$$\|G(x, r)\|^2 = \|x\|^2 + r^2 \|x - f(x)\|^2 + 2rx \cdot (x - f(x))$$

so for $\|G(x, r)\| = 1$, we use the quadratic formula to solve r:

$$r = \frac{-2x \cdot (x - f(x)) + \sqrt{4(x \cdot (x - f(x)))^2 + 4(1 - \|x\|^2) \|x - f(x)\|^2}}{2 \|x - f(x)\|^2}$$

The denominator is positive because f has no fixed points, and the discriminant is nonnegative because $\|x\| \leq 1$, so the right hand side, which we call $r(x)$, is a continuous function of x with $\|G(x, r(x))\| = 1$. It is the larger of the two roots, whose product is $(\|x\|^2 - 1)/\|x - f(x)\|^2 \leq 0$, so $r(x) \geq 0$.

Thus, we can define

$$g(x) = G(x, r(x)) \in S^1$$

and g is continuous. On the circle, $\|x\| = 1$ and $x \cdot (x - f(x)) = 1 - x \cdot f(x) \geq 0$, so $r(x) = 0$ and $g(x) = x$. Hence g is a retraction of B^2 onto S^1, contradicting theorem 2.1.49.

Theorem 2.1.51: Fundamental Theorem Of Algebra

Let $p(z)$ be a nonconstant polynomial with coefficients in $\mathbb{C}$. Then $p(z)$ has a root in $\mathbb{C}$

***Proof*:**
Without loss of generality let $p(z)$ be monic (we may scale the result at the end). Assume that $p(z)$ has no roots in $\mathbb{C}$. Then for each $r \in \mathbb{R}_{\geq 0}$, define

$$f_r(s) : S^1 \to S^1 \qquad f_r(s) = \frac{p(re^{2\pi is})/p(r)}{|p(re^{2\pi is})/p(r)|}$$

which is a loop based at 1. As we let r vary, we get loops f_r that are homotopic based at 1. Since f_0 is the trivial loop, we get that $[f_r] \in \pi_1(S^1)$ is the nullhomotopic element.

Now, fix r larger than $\max(|a_1| + \cdots + |a_n|, 1)$. Then for $|z| = r$, we get

$$|z^n| > (|a_1| + \cdots + |a_n|)|z^{n-1}| \geq |a_1 z^{n-1}| + \cdots + |a_n| \geq |a_1 z^{n-1} + \cdots + a_n|$$

Thus, it follows that

$$p_t(z) = z^n + t(a_1 z^{n-1} + \cdots + a_n)$$

has no roots on $|z| = r$ and $0 \leq t \leq 1$. Now, replacing p by p_t in the formula of f_r and letting $t \to 0$ we get

$$f_{r,0}(s) = \omega_n(s) = e^{2\pi ins}$$

Now, $[\omega_n] \in \pi_1(S^1)$ maps to n in the isomorphism presented in the proof of theorem 2.1.40, since its lift starting at 0 is $s \mapsto ns$. Thus, we have:

$$0 = \varphi([f_r]) = \varphi([\omega_n]) = n$$

and hence $n = 0$, showing that only constant functions do not have roots.

Theorem 2.1.52: Borsuk-Ulam Theorem In 2 Dimensions

For every continuous map $f : S^2 \to \mathbb{R}^2$, there exists a pair of antipodal points $x, -x$ such that $f(x) = f(-x)$

An interesting consequence of this is that S^2 is not homeomorphic to a subspace of $\mathbb{R}^2$, an intuitive but not obvious fact to prove directly.

***Proof*:**
For the sake of contradiction, suppose there is a continuous map $f : S^2 \to \mathbb{R}^2$ with no equal

antipodal images. Then define the map $g : S^2 \to S^1$ by

$$g(x) = \frac{f(x) - f(-x)}{\|f(x) - f(-x)\|}$$

Now, imagining the sphere S^2 embedded in $\mathbb{R}^3$, $S^2 \subseteq \mathbb{R}^3$ centered at the origin, take η to be the loop given by the equator

$$\eta(t) = (\cos 2\pi t, \sin 2\pi t, 0)$$

Let $h = g \circ \eta$. Then, $g(-x) = -g(x)$, so

$$h(s + 1/2) = -h(s)$$

for all $s \in [0, 1/2]$. Now, $h : I \to S^1$ can be lifted to a path $\tilde{h} : I \to \mathbb{R}$ (lemma 2.1.30). The equation $h(s + 1/2) = -h(s)$ then implies

$$\tilde{h}(s + 1/2) = \tilde{h}(s) + q/2 \qquad q \in 2\mathbb{Z} + 1$$

This q is independent of s, since by solving the equation $\tilde{h}(s + 1/2) = \tilde{h}(s) + q/2$, we see that q depends continuously on $s \in [0, 1/2]$, so q must be a constant since it is constrained to integer values. In particular

$$\tilde{h}(1) = \tilde{h}(1/2) + q/2 = \tilde{h}(0) + q$$

Thus, h represents q times the generator of $\pi_1(S^1)$. Since q is odd, we conclude that h is *not* nullhomotopic. But h is the composition $g \circ \eta : I \to S^2 \to S^1$, and η is nullhomotopic in S^2 because S^2 is simply connected (example 2.1.48), so $g \circ \eta$ is nullhomotopic in S^1 by composing the nullhomotopy of η with g. This is a contradiction. Thus, it cannot be that $f(x) \neq f(-x)$ for all x, as we sought to show.

Corollary 2.1.53: Union Of 3 Sets of Sphere contains Antipodal Points

Whenever S^2 is expressed as the union of three closed sets A_1, A_2, A_3, then at least one of these sets must contain a pair of antipodal points $\{x, -x\}$.

***Proof*:**
Let $d_i : S^2 \to \mathbb{R}$, $d_i(x) = \inf_{y \in A_i} |x - y|$ be the distance from x to A_i. This is a continuous function, since the triangle inequality gives $|d_i(x) - d_i(x')| \leq |x - x'|$, so we may apply the Borsuk-Ulam theorem (theorem 2.1.52) to

$$S^2 \to \mathbb{R}^2, \quad x \mapsto (d_1(x), d_2(x))$$

namely, we can find a pair of points $x, -x$ such that $d_1(x) = d_1(-x)$ and $d_2(x) = d_2(-x)$. If either of these two distances is zero, then $x, -x$ both lie in the set A_1 or A_2, since these are closed sets. On the other hand, if the distance from x and $-x$ to A_1 and A_2 are both strictly positive, then x and $-x$ must lie in neither of these sets, but then they must lie in A_3.

This theorem fails when considering 4 sets: radially project the four faces of a regular tetrahedron inscribed in S^2. A face lies in a plane $\{x : x \cdot c = 1/3\}$ for a unit vector c and inside the unit ball, so each point of its image is at angular distance at most $\arccos(1/3) < \pi/2$ from c. Any two points

of the image are then at angular distance less than π, so none of the four images contains a pair of antipodal points. Given the higher-dimensional analogue of Borsuk-Ulam's theorem, the same theorem works for S^n being covered by $n+1$ sets (while being able to cover it with $n+2$ sets none containing antipodal points). The above corollary even works for S^1: if it is covered by two closed sets, then one must contain antipodal points (closedness is required, since S^1 can be covered by two half-open sets).

Further consequences of these methods, omitted here, include extending maps from S^1 over B^2, results on vector fields, and the fact that a 3×3 matrix with positive real entries has a positive real eigenvalue (see [30], p. 349).

The homotopy structure also is dependent on information beyond paths, for example $\pi_n(S^2)$ is nonzero for infinitely many values of n.

	π_1	π_2	π_3	π_4	π_5	π_6	π_7	π_8	π_9	π_{10}	π_{11}	π_{12}	π_{13}	π_{14}	π_{15}
S^0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S^1	$\mathbb{Z}$	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S^2	0	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{12}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_{15}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_{12} \times \mathbb{Z}_2$	$\mathbb{Z}_{84} \times \mathbb{Z}_2^2$	$\mathbb{Z}_2^2$
S^3	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{12}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_{15}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_{12} \times \mathbb{Z}_2$	$\mathbb{Z}_{84} \times \mathbb{Z}_2^2$	$\mathbb{Z}_2^2$
S^4	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z} \times \mathbb{Z}_{12}$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_{24} \times \mathbb{Z}_3$	$\mathbb{Z}_{15}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_{120 \times Z_{12} \times Z_2}$	$\mathbb{Z}_{84} \times \mathbb{Z}_2^5$
S^5	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{30}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_{72} \times \mathbb{Z}_2$
S^6	0	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_{60}$	$\mathbb{Z}_{24} \times \mathbb{Z}_2$	$\mathbb{Z}_2^3$
S^7	0	0	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	0	0	$\mathbb{Z}_2$	$\mathbb{Z}_{120}$	$\mathbb{Z}_2^3$
S^8	0	0	0	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	0	0	$\mathbb{Z}_2$	$\mathbb{Z} \times \mathbb{Z}_{120}$

Figure 2.7: known Higher order Homotopy groups of n-spheres

One last result that shall be stated, but not proven due to its complexity, is the characterization of all simply connected closed 3-manifolds. The simplest such manifold would be S^3. Up to homeomorphism, are there any others? This was known as the *Poincaré conjecture*, and it was proven that this was *not* the case. All closed simply connected 3-manifolds are homeomorphic to S^3.

2.2 Products, Fibred Products, and Van Kampen's Theorem

We have covered what happens if we have a continuous map, homeomorphism, and homotopy equivalence between two topological spaces. Some usual constructions of topological spaces affect the fundamental group in computable ways. Most of the products we have seen will require a bit more theory, namely the *van Kampen theorem*.

Proposition 2.2.1: Fundamental Group And Product Spaces

Let X, Y be path connected spaces. Then $\pi_1(X \times Y) \cong \pi_1(X) \times \pi_1(Y)$

***Proof*:**
This follows from the universal property of products: if $f : [0,1] \to X \times Y$ is a loop, then there exist continuous maps g, h such that $f(t) = (g(t), h(t))$. These are loops in X and Y respectively, and similarly a path homotopy between two loops f, f' induces path homotopies between g, g' and h, h' (its two coordinates). Conversely, any two loops g, h combine to form a loop (g, h) in $X \times Y$, and two path homotopies combine in the same way. Hence, we have a bijection:

$$\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0), \qquad [f] \mapsto ([g], [h])$$

This map is $((p_X)_*, (p_Y)_*)$ for the projections p_X, p_Y, so it is a homomorphism by proposition 2.1.22.

Example 2.2.2: Fundamental Group Of Torus
Since $T^2 = S^1 \times S^1$, $\pi_1(T^2) = \mathbb{Z} \times \mathbb{Z}$. In particular, any loop on T^2 is a loop around the minor axis combined with a loop from the major axis. Interestingly, these loops may knot:



Figure 2.8: Torus Loop Knot

These give interesting non-trivial paths.

An important consequence that is usually rather difficult to prove classically becomes much easier:

Corollary 2.2.3: Plane Not Homeomorphic To $\mathbb{R}^n$

$\mathbb{R}^2$ is not homeomorphic to $\mathbb{R}^n$ for $n \neq 2$

***Proof*:**
The case of $n = 1$ is quickly disposed of since $\mathbb{R}^2 - \{0\}$ is path connected, while $\mathbb{R} - \{0\}$ is not. For $n \geq 2$, $\mathbb{R}^n - \{x\}$ is homeomorphic to $S^{n-1} \times \mathbb{R}$, hence by proposition 2.2.1

$$\pi_1(\mathbb{R}^n - \{x\}) \cong \pi_1(S^{n-1}) \times \pi_1(\mathbb{R}) \cong \pi_1(S^{n-1})$$

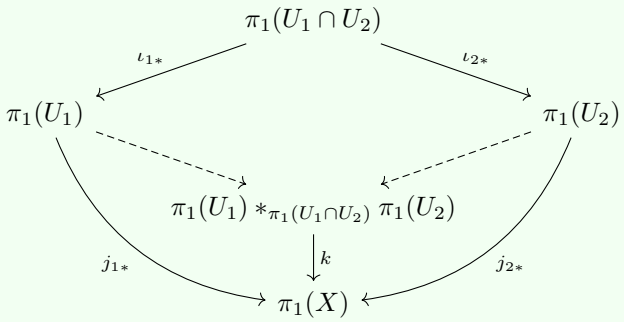
Hence, if $n = 2$ we get $\pi_1(\mathbb{R}^n - \{x\}) \cong \mathbb{Z}$ (theorem 2.1.40), and for $n > 2$ we get the trivial group (example 2.1.48, an application of proposition 2.1.47). A homeomorphism $h \colon \mathbb{R}^2 \to \mathbb{R}^n$ would restrict to a homeomorphism $\mathbb{R}^2 - \{x\} \to \mathbb{R}^n - \{h(x)\}$, which induces an isomorphism of fundamental groups (proposition 2.1.22). So $n = 2$.

For $\mathbb{R}^m \ncong \mathbb{R}^n$ if $n \neq m$, we can use higher homotopy groups, but there will be an easier proof of this using homology in chapter 3.

Proposition 2.1.47 gave us the paths of a space that can be decomposed into spaces A_α whose intersection is path-connected. However, we did not present what type of group these paths can produce, besides some trivial cases. Van Kampen's theorem gives the group structure: the fundamental group of the union is the free product of the fundamental groups of the pieces, amalgamated along that of the intersection.

Theorem 2.2.4: Van Kampen's Theorem

Let X be a topological space which is the union of two open and path connected subspaces U_1, U_2. Suppose that $U_1 \cap U_2$ is path connected and nonempty, and let $x_0 \in U_1 \cap U_2$. Write $\iota_k \colon U_1 \cap U_2 \hookrightarrow U_k$ and $j_k \colon U_k \hookrightarrow X$ for the inclusions, and let $\pi_1(U_1) *_{\pi_1(U_1 \cap U_2)} \pi_1(U_2)$ be the *amalgamated free product*, the quotient of the free product $\pi_1(U_1) * \pi_1(U_2)$ by the normal subgroup generated by the elements $\iota_{1*}(\omega)\, \iota_{2*}(\omega)^{-1}$ for $\omega \in \pi_1(U_1 \cap U_2, x_0)$. Then the homomorphism k induced by j_{1*} and j_{2*} in the commutative diagram



is an isomorphism. More generally, let $X = \bigcup_\alpha A_\alpha$ be a union of open path-connected sets, each containing x_0, and write $i_{\alpha\beta} \colon \pi_1(A_\alpha \cap A_\beta) \to \pi_1(A_\alpha)$ for the map induced by inclusion. If every $A_\alpha \cap A_\beta$ is path-connected, then the homomorphism $\varphi \colon *_\alpha \pi_1(A_\alpha, x_0) \to \pi_1(X, x_0)$ induced by the inclusions is surjective. If moreover every $A_\alpha \cap A_\beta \cap A_\gamma$ is path-connected, then $\ker \varphi$ is the normal subgroup N generated by the elements $i_{\alpha\beta}(\omega)\, i_{\beta\alpha}(\omega)^{-1}$ for $\omega \in \pi_1(A_\alpha \cap A_\beta, x_0)$, so that φ induces an isomorphism $*_\alpha \pi_1(A_\alpha)/N \cong \pi_1(X)$.

***Proof*:**

The homomorphism $\pi_1(U_1) * \pi_1(U_2) \to \pi_1(X)$ restricting to j_{k*} on the factor $\pi_1(U_k)$ kills each generator $\iota_{1*}(\omega)\, \iota_{2*}(\omega)^{-1}$, since $j_1 \circ \iota_1 = j_2 \circ \iota_2$ is the inclusion of $U_1 \cap U_2$, so it descends to k and the diagram commutes. The content of the theorem is that k is bijective. It is cleanest to prove the general statement over an arbitrary cover, since the two-set case is the special instance with index set $\{1, 2\}$. So let $X = \bigcup_\alpha A_\alpha$ be a union of open path-connected sets, each containing the basepoint x_0, with each pairwise intersection $A_\alpha \cap A_\beta$ path-connected, and consider

$$\varphi : \ast_\alpha \pi_1(A_\alpha, x_0) \longrightarrow \pi_1(X, x_0)$$

the homomorphism out of the free product whose restriction to the factor $\pi_1(A_\alpha)$ is the map $j_{\alpha*}$ induced by the inclusion $A_\alpha \hookrightarrow X$. For the cover $\{U_1, U_2\}$ the map φ is the composite of

the quotient map onto the amalgamated product with k, so surjectivity of φ gives surjectivity of k. Every triple intersection of members of $\{U_1, U_2\}$ is U_1, U_2 or $U_1 \cap U_2$, all path-connected, so the triple hypothesis holds, and the description of $\ker\varphi$ identifies the amalgamated product with $*_\alpha \pi_1(A_\alpha)/\ker\varphi$, making k injective. We prove surjectivity first, and then compute the kernel under the triple-intersection hypothesis.

Step 1: Surjectivity by subdivision against a Lebesgue number. Let $f : I \to X$ be a loop based at x_0. The sets $f^{-1}(A_\alpha)$ form an open cover of the compact metric space I, so by the Lebesgue number lemma there is a $\delta > 0$ such that every subinterval of I of length less than δ is carried by f into a single A_α. Choose a partition $0 = s_0 < s_1 < \cdots < s_n = 1$ with each $s_i - s_{i-1} < \delta$, and for each i pick an index α_i with $f([s_{i-1}, s_i]) \subseteq A_{\alpha_i}$. Writing $f_i = f|_{[s_{i-1},s_i]}$ (reparametrized to I), we have a factorization $f = f_1 * f_2 * \cdots * f_n$ in which the consecutive endpoint $f(s_i)$ lies in $A_{\alpha_i} \cap A_{\alpha_{i+1}}$.

This is precisely the situation of proposition 2.1.47, and the construction there converts f into a product of loops at x_0. Concretely, since each intersection $A_{\alpha_i} \cap A_{\alpha_{i+1}}$ is path-connected and contains both x_0 and the point $f(s_i)$, choose a path g_i in $A_{\alpha_i} \cap A_{\alpha_{i+1}}$ from x_0 to $f(s_i)$, taking g_0 and g_n to be the constant path at x_0. Inserting $\bar{g}_i * g_i$ (homotopically trivial by theorem 2.1.7) at each cut point gives

$$f \simeq (f_1 * \bar{g}_1) * (g_1 * f_2 * \bar{g}_2) * \cdots * (g_{n-1} * f_n)$$

rel endpoints. Each parenthesized factor is now a loop based at x_0, and the i-th factor has image inside the single set A_{α_i} (the path g_{i-1}, the subpath f_i, and the path $\bar{g}_i$ all lie in A_{α_i}, using that g_i was chosen inside $A_{\alpha_i} \cap A_{\alpha_{i+1}} \subseteq A_{\alpha_i}$). Hence $[f]$ is the product in $\pi_1(X)$ of the classes $j_{\alpha_i *}[i\text{-th factor}]$, each of which lies in the image of φ. As $[f]$ was an arbitrary loop class, φ is surjective.

A word in the free product is a finite sequence $[\gamma_1][\gamma_2]\cdots[\gamma_m]$ in which each γ_k is a loop at x_0 lying in some single set A_{α_k}. The surjectivity argument shows every element of $\pi_1(X)$ arises from such a word, which we call a *factorization* of the loop $\gamma_1 * \gamma_2 * \cdots * \gamma_m$. The element of the free product determined by the word lies in $\ker\varphi$ exactly when this product loop is nullhomotopic in X. We must show that any two factorizations of homotopic loops are connected by a sequence of moves, each of which preserves the image in $\pi_1(X)$ and changes the word only within the normal subgroup N generated by the relators $i_{\alpha\beta}(\omega)\, i_{\beta\alpha}(\omega)^{-1}$.

Here $i_{\alpha\beta} : \pi_1(A_\alpha \cap A_\beta) \to \pi_1(A_\alpha)$ denotes the map induced by the inclusion $A_\alpha \cap A_\beta \hookrightarrow A_\alpha$. For a loop ω in the intersection, $i_{\alpha\beta}(\omega)$ is the class of ω in $\pi_1(A_\alpha)$ and $i_{\beta\alpha}(\omega)$ the class of the same loop in $\pi_1(A_\beta)$. The relator records that these two classes become equal once mapped to $\pi_1(X)$. Two elementary moves leave the word unchanged modulo N:

1. If a loop γ_k lies in $A_\alpha \cap A_\beta$, the factor $[\gamma_k]$ may be reread from the factor $\pi_1(A_\alpha)$ to the factor $\pi_1(A_\beta)$ of the free product (replacing $i_{\alpha\beta}(\gamma_k)$ by $i_{\beta\alpha}(\gamma_k)$), since the two differ by exactly a generator of N.

2. Adjacent factors $[\gamma_k][\gamma_{k+1}]$ lying in the same A_α may be combined into the single factor $[\gamma_k * \gamma_{k+1}]$, and conversely a factor may be split, since this is just multiplication inside the group $\pi_1(A_\alpha)$ and leaves the free-product element unchanged.

Call two words *equivalent* if one is obtained from the other by a finite sequence of these two moves. The moves preserve the element of $*_\alpha \pi_1(A_\alpha)$ modulo N, so to prove $\ker\varphi = N$ it suffices to show: *if two factorizations represent loops that are homotopic in X, then the two*

words are equivalent. Granting this, a word lying in $\ker\varphi$ represents a nullhomotopic loop, hence is equivalent to the empty word, hence lies in N. The inclusion $N \subseteq \ker\varphi$ holds because $j_{\alpha*} \circ i_{\alpha\beta} = j_{\beta*} \circ i_{\beta\alpha}$ (both are the inclusion $A_\alpha \cap A_\beta \hookrightarrow X$ on π_1), so each relator is killed by φ. The remainder of the proof establishes the displayed implication.

Step 2: A staggered Lebesgue grid on the homotopy square. Suppose f and f' are loops at x_0 with chosen factorizations, and let $F : I \times I \to X$ be a homotopy rel endpoints from $f = F(\cdot, 0)$ to $f' = F(\cdot, 1)$, so F carries the left and right edges $\{0\} \times I$ and $\{1\} \times I$ to the basepoint x_0. The sets $F^{-1}(A_\alpha)$ cover the compact square $I \times I$, so the Lebesgue number lemma gives a $\delta > 0$ such that any subset of diameter less than δ maps into a single A_α. Fix first a partition $0 = t_0 < t_1 < \cdots < t_q = 1$ of the vertical coordinate with each $t_j - t_{j-1} < \delta/2$, cutting the square into the horizontal rows $I \times [t_{j-1}, t_j]$, and on each row impose its *own* vertical partition, independently of the other rows: choose $0 = s_0^{(j)} < s_1^{(j)} < \cdots < s_{p_j}^{(j)} = 1$ with each $s_i^{(j)} - s_{i-1}^{(j)} < \delta/2$, producing the closed rectangles

$$R_{ij} = \left[s_{i-1}^{(j)}, s_i^{(j)}\right] \times [t_{j-1}, t_j] \qquad (1 \le i \le p_j,\ 1 \le j \le q)$$

each of diameter below δ, so $F(R_{ij}) \subseteq A_{ij}$ for some index, written A_{ij}. The kernel computation turns on one structural choice, the only departure from a naive product grid. *Stagger the vertical cuts of adjacent rows*: having chosen the interior cut points $s_i^{(j)}$ $(0 < i < p_j)$ of each row, perturb them by amounts smaller than the minimum gap so that no interior cut point of row j coincides with any interior cut point of row $j+1$, and so that no cut point lands on the boundary in $I \times I$ between two of the assigned sets. The first condition is achievable because the cut points of two rows form two finite subsets of the open interval $(0, 1)$, and a generic small shift of one makes them disjoint. The perturbations are small enough that each rectangle still has diameter below δ, so the assignment $R_{ij} \mapsto A_{ij}$ survives. The effect of staggering is the inequality that drives Step 3: with the interior vertical cuts of consecutive rows disjoint, *no point of $I \times I$ lies in more than three rectangles R_{ij}*, equivalently every vertex of the resulting (non-product) grid is a corner of at most three rectangles. This is Hatcher's perturbation ([21, Theorem 1.20]). The top and bottom rows are exempt from any further constraint, so we may additionally require the bottom partition $\{s_i^{(1)}\}$ to refine the partition of the chosen factorization of f and the top partition $\{s_i^{(q)}\}$ to refine that of f'. The splitting move (move (ii)) makes this harmless, replacing each chosen factorization by an equivalent one adapted to the grid.

The deformation proceeds one row at a time, and within a row one rectangle at a time, reading off at each stage a word associated to a path that climbs through the grid and showing that crossing a single rectangle changes the word only by equivalences.

Step 3: Auxiliary paths land in double or triple intersections. The vertices of the staggered grid are the points $(s_i^{(j)}, t_{j-1})$ and $(s_i^{(j)}, t_j)$, the corners of the rectangles R_{ij}. For each vertex v the point $F(v)$ lies in every assigned set A_{kl} for which v is a corner of R_{kl}: the closed rectangle R_{kl} maps into A_{kl} and contains v, so $F(v) \in A_{kl}$, and the intersection of these assigned sets is therefore nonempty (it contains $F(v)$). The point of staggering is that this intersection involves at most three sets. Classify v by its location:

- If v lies on the left or right edge $\{0\} \times I$ or $\{1\} \times I$, then $F(v) = x_0$ and we take γ_v to be the constant path.
- If v lies on the bottom edge $t = 0$ or the top edge $t = 1$, it is a corner of at most the two rectangles of that single row sharing the vertical cut at v, with assigned sets A, A', and the intersection in play is the double intersection $A \cap A'$.

- If v lies on an interior horizontal line $t = t_j$ $(0 < j < q)$, it is a corner of rectangles drawn from the two rows j and $j+1$ adjacent to that line. Because the interior vertical cuts of rows j and $j+1$ are staggered to be disjoint, v is an interior cut point of at most one of the two rows. If v is an interior cut point of row j (so a shared vertical-edge endpoint of the two row-j rectangles $R_{i,j}, R_{i+1,j}$ with sets A, A'), then v is *not* a cut point of row $j+1$, hence lies in the interior of the bottom edge of a single row-$(j+1)$ rectangle $R_{m,j+1}$ with set A''. The three sets A, A', A'' are all in play and no fourth set occurs. The symmetric case (where v is an interior cut point of row $j+1$ but not of row j) likewise yields three sets. If v is a cut point of neither row (possible only at $v = (0, t_j)$ or $(1, t_j)$, already covered by the first bullet), or of exactly one row at a position interior to a single rectangle of the other, the count is three or fewer.

Thus at every interior vertex at most three assigned sets are in play. Choose, for each vertex v, a path γ_v in X from x_0 to $F(v)$ lying in the intersection of the assigned sets in play at v: when exactly two sets A, A' occur, take γ_v in $A \cap A'$, which is path-connected by hypothesis and contains both x_0 and $F(v)$. When three sets A, A', A'' occur, take γ_v in $A \cap A' \cap A''$, path-connected by the triple-intersection hypothesis. This is the sole point at which the triple hypothesis enters, and the staggering is exactly what keeps the count at three: without it an interior vertex of a product grid would be a corner of four rectangles carrying four genuinely distinct sets, whose common intersection is a quadruple intersection not assumed path-connected, and no single γ_v serving both adjacent rows could be produced. The failure of the triple hypothesis (as in the suspension of three points) breaks the kernel computation at precisely these vertices while leaving surjectivity (Step 1, which used only double intersections) intact.

Step 4: Reading words off horizontal paths and crossing one rectangle. For $0 \le j \le q$ let h_j denote the horizontal path along the grid line $t = t_j$, that is the loop $s \mapsto F(s, t_j)$ based at x_0 (it is a loop because the endpoints $s = 0, 1$ map to x_0). Then $h_0 = f$ and $h_q = f'$. A single row is now crossed using only that row's own vertical partition, which is a genuine one-dimensional grid, so within a fixed row no four-valent vertex arises and the crossing is a chain of moves across individual rectangles. Fix a row j and write $s_r = s_r^{(j)}$ for its cut points $(0 \le r \le p_j)$. For $0 \le r \le p_j$ let h_j^r be the path in X that follows the line $t = t_{j-1}$ from $s = 0$ up to $s = s_r$, then the vertical segment from (s_r, t_{j-1}) to (s_r, t_j), then the line $t = t_j$ from $s = s_r$ to $s = 1$. Hence $h_j^{p_j}$ traverses the bottom edge $t = t_{j-1}$ of row j and h_j^0 traverses the top edge $t = t_j$, that is $h_j^0 = h_j$ and $h_j^{p_j} = h_{j-1}$. Each such staircase path is cut by the row's grid into subpaths, each lying in a single assigned set A_{kl} (a subpath on $t = t_{j-1}$ lies on the bottom edge of a row-j rectangle, a subpath on $t = t_j$ on a top edge, and the vertical segment on a shared vertical edge, all of row j). Inserting the chosen $\bar{\gamma}_v * \gamma_v$ at every grid vertex v it passes through (exactly as in Step 1) turns it into a product of loops at x_0, each loop confined to a single A_{kl}. This product is a word in the free product. Call it $w(h_j^r)$.

The passage from h_j^r down across the single rectangle R_{rj} to h_j^{r-1} replaces the lower-left corner path of R_{rj} by its upper-right corner path, the two corner paths agreeing at the two diagonal vertices (s_{r-1}, t_{j-1}) and (s_r, t_j). Since $F(R_{rj}) \subseteq A_{rj}$, the four edges of R_{rj} and the constant-corner loops all map into the single set A_{rj}, and inside A_{rj} the boundary of the square is nullhomotopic (it bounds the disk $F|_{R_{rj}}$). Hence the lower-left and upper-right corner loops are homotopic *within* A_{rj}. Reading both through the same factor $\pi_1(A_{rj})$, the corresponding subwords are equal in $\pi_1(A_{rj})$, so they are related by move (ii) (combine/split inside one group) followed by replacing one in-group expression by the equal one. The only subtlety is that the two corner paths attach to the rest of the staircase through the auxiliary paths γ_v at the two diagonal

vertices v, and these γ_v were chosen in intersections of several assigned sets. Rereading γ_v from the set used by the incoming subpath to the set used by the outgoing subpath is exactly move (i), legitimate because γ_v lies in the relevant double or triple intersection by Step 3 (both the incoming and outgoing sets are among the at-most-three sets in play at v, so the loop recording the change of reading lies in their pairwise intersection inside that triple). Thus

$$w(h_j^{r-1}) \text{ is equivalent to } w(h_j^r)$$

for each rectangle, the equivalence using only moves (i) and (ii).

Step 5: Assembling the rows. Concatenating Step 4 across the rectangles of a single row j from $r = p_j$ down to $r = 0$ gives a chain of equivalences

$$w(h_{j-1}) = w(h_j^{p_j}) \sim w(h_j^{p_j-1}) \sim \cdots \sim w(h_j^0) = w(h_j)$$

so $w(h_{j-1})$ and $w(h_j)$ are equivalent. The auxiliary paths γ_v at the vertices on the line $t = t_j$ are shared between row j (where they close up $h_j = h_j^0$) and row $j+1$ (where they close up $h_j = h_{j+1}^{p_{j+1}}$): a single choice of γ_v per vertex serves both, which is legitimate precisely because Step 3 placed γ_v in the intersection of *all* the at-most-three sets in play at v, those coming from row j below and from row $j+1$ above together. Hence the word $w(h_j)$ read at the top of row j is literally the same word read at the bottom of row $j+1$, and running j from 1 to q chains the row equivalences together:

$$w(f) = w(h_0) \sim w(h_1) \sim \cdots \sim w(h_q) = w(f')$$

It remains to connect the word $w(f) = w(h_0)$ produced by the grid to the originally chosen factorization of f. But the bottom partition $\{s_i^{(1)}\}$ was arranged (Step 2) to refine the partition of the chosen factorization, and the chosen auxiliary paths γ_v at the bottom vertices may be taken to agree with the splitting paths of that factorization (both connect x_0 to the same cut points inside the same double intersections, and any two such choices differ by a loop in a double intersection, absorbed by moves (i) and (ii)). Hence $w(h_0)$ is equivalent to the chosen factorization word of f, and likewise $w(h_q)$ to that of f'. Chaining all equivalences shows the chosen factorizations of f and f' are equivalent, which is the displayed implication of Step 1.

Step 6: Conclusion. Apply the implication to a word $W \in \ker\varphi$: it factors a loop f that is nullhomotopic in X, so f is homotopic to the constant loop, whose factorization is the empty word. By Steps 2 to 5 the word W is equivalent to the empty word, hence $W \in N$ since each equivalence move alters the free-product element only by a generator of N. Combined with $N \subseteq \ker\varphi$ from Step 1, this gives $\ker\varphi = N$, so φ descends to an isomorphism

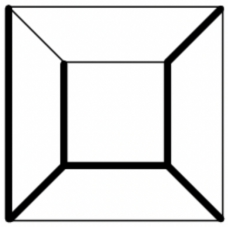
$$\ast_\alpha \pi_1(A_\alpha)/N \xrightarrow{\cong} \pi_1(X)$$

Specializing to the cover $\{U_1, U_2\}$, the group N is generated by $\iota_{1*}(\omega)\,\iota_{2*}(\omega)^{-1}$ for $\omega \in \pi_1(U_1 \cap U_2)$, so the quotient is the amalgamated free product $\pi_1(U_1) *_{\pi_1(U_1\cap U_2)} \pi_1(U_2)$ and the induced map k to $\pi_1(X)$ is an isomorphism, completing the proof.

Example 2.2.5: Use Of Van-Kampen

1. The simplest example would be a wedge-sum. Let $X = \bigvee_\alpha X_\alpha$ be a wedge of path-connected spaces whose basepoints $x_\alpha \in X_\alpha$ are all identified to a single point x_0, and suppose each x_α has an open neighbourhood $U_\alpha \subseteq X_\alpha$ that deformation retracts onto x_α. Then $*_\alpha \pi_1(X_\alpha) \cong \pi_1(X)$. Indeed, the open sets $A_\alpha = X_\alpha \vee \bigvee_{\beta \neq \alpha} U_\beta$ are path-connected, each deformation retracts onto X_α, and every intersection of two or more of them is $\bigvee_\beta U_\beta$, which deformation retracts onto x_0. So theorem 2.2.4, in its general form over a cover, applies with trivial relations. The hypothesis holds whenever each X_α is a CW complex and x_α is a 0-cell, by lemma 3.4.17 applied to the subcomplex $\{x_\alpha\}$. This covers the wedges of circles and spheres, and the wedge with a torus, used below.

2. Let X be the graph:



and T be the heavily shaded regions (T is a maximal spanning tree). Notice that T is contractible, and that there are 5 non-shaded edges. Then $\pi_1(X) = *_{i=1}^5 \mathbb{Z}$, which we deduce using van Kampen's theorem (theorem 2.2.4). For each edge $e_\alpha \in X - T$, choose an open neighbourhood of $T \cup e_\alpha$ in X that retracts onto $T \cup e_\alpha$. Call it A_α. Then the intersection $A_\alpha \cap A_\beta$ deformation retracts to T, which is contractible and path connected. Hence, we get

$$\pi_1(X) \cong *_\alpha \pi_1(A_\alpha)$$

Since each A_α deformation retracts onto a circle, we get that $\pi_1(X)$ is the free product $\mathbb{Z} * \mathbb{Z} * \mathbb{Z} * \mathbb{Z} * \mathbb{Z}$.

3. This item is an illustration that nothing later depends on. The three deformation retractions in it are read off the figures and not proved here. The fundamental groups then follow from the wedge rule of the first item. Let $X = \mathbb{R}^3 - S^1$. First, notice that this deformation retracts to the shape:



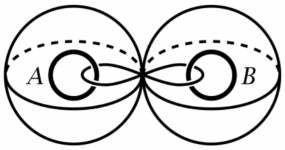
Then one can imagine that we modify this deformation retraction so that we instead have $S^1 \vee S^2$ like so:



Then since we have a deformation retraction (lemma 2.1.43), we have that

$$\pi_1(X) \cong \pi_1(S^1 \vee S^2) \cong \mathbb{Z}$$

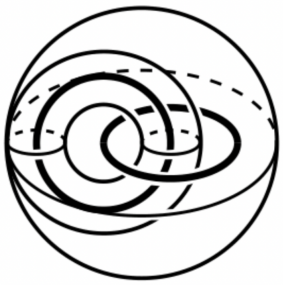
since $\pi_1(S^2) \cong 0$ (example 2.1.48). Similarly, if $X = \mathbb{R}^3 - (A \cup B)$ where A, B are two unlinked circles, then we may deformation retract to $S^1 \vee S^1 \vee S^2 \vee S^2$ to look like so:



We thus, get that

$$\pi_1(X) \cong \mathbb{Z} * \mathbb{Z}$$

On the other hand, if the two circles were linked, then we get a deformation retraction to $S^2 \vee (S^1 \times S^1)$:



which gives us, by proposition 2.2.1:

$$\pi_1(X) \cong \mathbb{Z} \times \mathbb{Z}$$

As we see, the complement of a shape can give us some "knotting" information. This is the first step into knot theory and knot invariants.

4. (torus knots) For each $m, n \in \mathbb{Z}$ where $\gcd(n, m) = 1$, define $K_{m,n} \subseteq \mathbb{R}^3$ to be the image of $f : S^1 \to S^1 \times S^1 \subseteq \mathbb{R}^3$ given by
$$f(z) = (z^m, z^n)$$
The knot $K_{m,n}$ winds around the torus m times in the longitudinal direction and n times in the meridional direction. The relatively prime criterion ensures f is injective. Consider $\pi_1(\mathbb{R}^3 - K_{m,n})$. Write $S^3 - K_{m,n}$, for the one-point compactification S^3 of $\mathbb{R}^3$, as the union of $\mathbb{R}^3 - K_{m,n}$ and an open ball about the point at infinity missing $K_{m,n}$, whose intersection with $\mathbb{R}^3 - K_{m,n}$ is homeomorphic to $S^2 \times \mathbb{R}$. Van Kampen's theorem (theorem 2.2.4) shows that the inclusion induces an isomorphism $\pi_1(\mathbb{R}^3 - K_{m,n}) \cong \pi_1(S^3 - K_{m,n})$.

   This group is $G_{m,n} = \langle a, b \big| a^m = b^n \rangle$. For $|m|, |n| \geq 2$ its center is the cyclic subgroup generated by a^m, and the quotient by it is the free product $\mathbb{Z}/m * \mathbb{Z}/n$ [21, Example 1.24, pp. 47 and 48].

5. (Hawaiian earrings) Let $X \subseteq \mathbb{R}^2$ be the union of circles C_n of radius $1/n$ with center $(1/n, 0)$

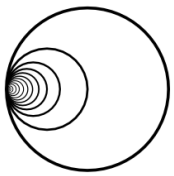


Figure 2.9: Hawaiian Earrings

   We may think that the fundamental group of this space is an infinite free product of $\mathbb{Z}$, but it is much larger. The retractions of X onto the circles C_n induce a surjection of $\pi_1(X)$ onto the uncountable product $\prod_n \mathbb{Z}$, while the fundamental group of a wedge of countably many circles is countable [21, Example 1.25, p. 49].

An important application of van Kampen's theorem is when used in tandem with cell-complexes, in particular 2-complexes, for they "add relations" by adding homotopies. Let X be a 1-skeleton to which we attach a collection of 2-cells e_α via the maps $\varphi_\alpha : S^1 \to X$. Let the resulting space be labeled Y. These maps define a loop in X. If $s_0 \in S^1$ is a basepoint, then $\varphi_\alpha(s_0)$ is the basepoint of the image of φ_α. For varying α, the basepoints may differ, but we may take a path γ_α from x_0 to $\varphi_\alpha(s_0) \in X$ that makes each loop a loop at x_0: $\tilde{\gamma}_\alpha \varphi_\alpha \gamma_\alpha$. These loops may not be nullhomotopic, but after attaching the 2-cells, they will become nullhomotopic. This means that the kernel of the map $\pi_1(X, x_0) \to \pi_1(Y, x_0)$ induced by the inclusion $X \hookrightarrow Y$ contains all $\tilde{\gamma}_\alpha \varphi_\alpha \gamma_\alpha$[2]. Since the kernel represents the added relation, we see that how we attach the cells represents new relations added to Y. Let N be the normal subgroup of $\pi_1(X, x_0)$ generated by the loops $\tilde{\gamma}_\alpha \varphi_\alpha \gamma_\alpha$. The kernel is exactly N, so the attached 2-cells determine the group exactly:

[2]Note that it is path-independent: if η_α was a different path, then $\eta_\alpha \varphi_\alpha \tilde{\eta}_\alpha = (\eta_\alpha \tilde{\gamma}_\alpha)\gamma_\alpha \varphi_\alpha \tilde{\gamma}_\alpha (\gamma_\alpha \tilde{\eta}_\alpha)$

Proposition 2.2.6: Van-Kampen and Cells

1. If Y is obtained from X by attaching 2-cells as described above, then the inclusion $X \hookrightarrow Y$ induces a surjection $\pi_1(X, x_0) \to \pi_1(Y, x_0)$ whose kernel is N, hence
$$\pi_1(Y) = \pi_1(X)/N$$
2. If Y is obtained from X by attaching n-cells for fixed $n \geq 3$, then the inclusion $X \hookrightarrow Y$ induces an isomorphism
$$\pi_1(Y, x_0) \cong \pi_1(X, x_0)$$
3. For a path-connected cell complex X, the inclusion of the 2-skeleton $X^2 \hookrightarrow X$ induces an isomorphism $\pi_1(X^2, x_0) \cong \pi_1(X, x_0)$

This shows that the fundamental group of a CW complex is determined by how we attach 2-cells.

***Proof*:**
All three parts rest on a single device: although Y is built by attaching closed cells, an open cover to which theorem 2.2.4 applies is obtained by fattening each attached cell with a collar so that the pieces are open in Y and their pairwise and triple intersections are path-connected.

1. *Attaching* 2*-cells.* Write $Y = X \cup \bigcup_\alpha e^2_\alpha$, where e^2_α is attached along $\varphi_\alpha \colon S^1 \to X$. The space Y itself does not split as a union of open sets one of which is each e^2_α, since the cells are closed and their boundaries lie in X. The remedy is to enlarge the picture. Pick a point y_α in the interior of each cell e^2_α, and set
$$B = Y - \{y_\alpha : \alpha\} \qquad \text{and} \qquad A_\alpha = e^2_\alpha \cup (\text{a collar of } e^2_\alpha \text{ inside } X)$$
made precise as follows. Each A_α is the union of the open cell, a small open collar in Y of the attaching circle, and the open cell of e^2_α itself, chosen so that A_α is open in Y and deformation retracts onto the closed cell e^2_α by pushing the collar inward. Thus A_α is contractible (it retracts to a single closed 2-cell, which is a disk), in particular $\pi_1(A_\alpha) = 0$. The set B is open and deformation retracts onto X: each punctured open cell $e^2_\alpha - \{y_\alpha\}$ deformation retracts radially onto its boundary circle in X, and these retractions, being the identity on X, glue to a deformation retraction of B onto X. Hence the inclusion $X \hookrightarrow B$ is a homotopy equivalence and $\pi_1(B, x_0) \cong \pi_1(X, x_0)$.
The intersection $A_\alpha \cap B$ is the collar with y_α removed, an open annular neighbourhood of the attaching circle, which deformation retracts onto that circle and is therefore path-connected with $\pi_1(A_\alpha \cap B) \cong \mathbb{Z}$, generated by the loop φ_α. The triple intersections $A_\alpha \cap A_\beta \cap B = A_\alpha \cap A_\beta$ are empty for $\alpha \neq \beta$ (distinct open cells are disjoint and the collars may be taken disjoint), hence vacuously path-connected. Choosing the path γ_α in X from x_0 to the basepoint $\varphi_\alpha(s_0)$ of the α-th attaching circle realizes the image of the generator of $\pi_1(A_\alpha \cap B, x_0)$ under the inclusion into B as the loop $\gamma_\alpha^{-1}\varphi_\alpha\gamma_\alpha$, that is $\tilde{\gamma}_\alpha\varphi_\alpha\gamma_\alpha$ in the notation preceding the statement.
Apply theorem 2.2.4 to the cover $\{B\} \cup \{A_\alpha\}_\alpha$ of Y, all members containing x_0 after the basepoint is joined into each A_α by a whisker inside its collar (this changes none of the homotopy types just computed). The theorem gives
$$\pi_1(Y, x_0) \;\cong\; \Big(\pi_1(B) * \mathop{\Large *}_\alpha \pi_1(A_\alpha)\Big)/M$$

where M is the normal subgroup generated by the elements $i_{B\alpha}(\omega)\, i_{\alpha B}(\omega)^{-1}$ for $\omega \in \pi_1(A_\alpha \cap B)$. Since $\pi_1(A_\alpha) = 0$, the free factor $*_\alpha\, \pi_1(A_\alpha)$ is trivial and the free product collapses to $\pi_1(B) \cong \pi_1(X, x_0)$. For each α the group $\pi_1(A_\alpha \cap B)$ is infinite cyclic generated by ω_α, the loop around the collar. Reading ω_α into A_α gives $i_{\alpha B}(\omega_\alpha) = 0$ since $\pi_1(A_\alpha) = 0$, while reading it into $B \simeq X$ gives $i_{B\alpha}(\omega_\alpha) = \tilde{\gamma}_\alpha \varphi_\alpha \gamma_\alpha \in \pi_1(X, x_0)$. Thus the defining relator of M is exactly $\tilde{\gamma}_\alpha \varphi_\alpha \gamma_\alpha$, and M, transported along the isomorphism $\pi_1(B) \cong \pi_1(X)$, is the normal subgroup N generated by the loops $\tilde{\gamma}_\alpha \varphi_\alpha \gamma_\alpha$. Consequently

$$\pi_1(Y, x_0) \;\cong\; \pi_1(X, x_0)/N$$

and under this isomorphism the quotient map is induced by the inclusion $X \hookrightarrow B \hookrightarrow Y$, since the composite $\pi_1(X) \cong \pi_1(B) \to \pi_1(Y)$ is the inclusion-induced map and is surjective with kernel N. This is the first assertion.

2. *Attaching cells of dimension $n \geq 3$.* The construction of part (1) goes through verbatim with e^2_α replaced by an n-cell e^n_α attached along $\varphi_\alpha \colon S^{n-1} \to X$: set A_α to be the open n-cell together with an open collar of its boundary sphere, so that A_α deformation retracts onto a closed n-disk and is contractible, and let $B = Y - \{y_\alpha\}$ deformation retract onto X exactly as before. The single change is in the intersection: $A_\alpha \cap B$ now deformation retracts onto the attaching sphere S^{n-1} rather than S^1. For $n \geq 3$ the sphere S^{n-1} is simply connected, so
$$\pi_1(A_\alpha \cap B, x_0) \cong \pi_1(S^{n-1}) = 0$$
by the computation of $\pi_1(S^{n-1})$ for $n - 1 \geq 2$ (example 2.1.48). Feeding this into theorem 2.2.4 with the cover $\{B\} \cup \{A_\alpha\}_\alpha$, every group $\pi_1(A_\alpha \cap B)$ is now trivial, so the normal subgroup M generated by the relators $i_{B\alpha}(\omega)\, i_{\alpha B}(\omega)^{-1}$ is itself trivial. Each $\pi_1(A_\alpha) = 0$, so the free product again collapses to $\pi_1(B) \cong \pi_1(X, x_0)$. Therefore
$$\pi_1(Y, x_0) \cong \pi_1(X, x_0)$$
the isomorphism being induced by the inclusion $X \hookrightarrow Y$. The two excluded ranges behave differently: for $n = 2$ the attaching spheres are circles and the intersections have fundamental group $\mathbb{Z}$, which is precisely part (1) and adds the relations N. For $n = 1$ attaching arcs alters the homotopy type of the 1-skeleton and the present argument does not apply. Thus the inclusion induces an isomorphism on π_1 exactly in the stable range $n \geq 3$, which is the case the third assertion invokes.

3. *The 2-skeleton suffices.* Let X be a path-connected cell complex with skeleta $X^0 \subseteq X^1 \subseteq X^2 \subseteq \cdots$, and fix the basepoint $x_0 \in X^2$ (a 0-cell, after a path-connectedness adjustment). The inclusion $X^2 \hookrightarrow X$ factors through the skeletal filtration, so it suffices to control the induced map at each stage and then pass to the union. By part (2), for every $n \geq 3$ the inclusion $X^{n-1} \hookrightarrow X^n$, which attaches the n-cells of X, induces an isomorphism $\pi_1(X^{n-1}, x_0) \cong \pi_1(X^n, x_0)$. Composing these isomorphisms from $n = 3$ upward gives, for every finite $N \geq 2$,
$$\pi_1(X^2, x_0) \xrightarrow{\;\cong\;} \pi_1(X^N, x_0)$$
each map induced by inclusion.
It remains to pass from the finite skeleta to X itself. The point is that the fundamental group sees only compact data. A loop $f \colon I \to X$ based at x_0 has compact image, and a compact subset of a cell complex meets only finitely many cells, hence lies in some finite skeleton X^N (lemma 1.2.4). Thus f is a loop in X^N, and by the isomorphism above its

class comes from $\pi_1(X^2)$. This shows the inclusion $X^2 \hookrightarrow X$ induces a surjection on π_1. For injectivity, suppose a loop f in X^2 is nullhomotopic in X, witnessed by $H\colon I \times I \to X$. The image $H(I \times I)$ is compact, hence contained in some finite skeleton X^N with $N \geq 2$ (lemma 1.2.4), so f is already nullhomotopic in X^N. Since $\pi_1(X^2) \cong \pi_1(X^N)$ via inclusion, f is nullhomotopic in X^2. Therefore $\pi_1(X^2, x_0) \to \pi_1(X, x_0)$ is injective as well, and so an isomorphism, completing the proof.

The proposition computes what attaching 2-cells does to a fundamental group, which leaves open the group it starts from. For a cell complex that group is π_1 of the 1-skeleton, a graph, and the fundamental group of a graph is the other half of the presentation recipe, used in example 2.2.9.

Definition 2.2.7: Graph and Tree

A *graph* is a CW complex of dimension at most 1. Its 0-cells are its *vertices* and its 1-cells its *edges*. A *tree* is a contractible graph. A *spanning tree* of a graph Γ is a subcomplex $T \subseteq \Gamma$ that is a tree and contains every vertex of Γ.

Proposition 2.2.8: The Fundamental Group of a Graph

Let Γ be a connected graph and x_0 a vertex.

1. Γ has a spanning tree.
2. For any spanning tree T, the quotient map $\pi\colon \Gamma \to \Gamma/T$ is a homotopy equivalence, and Γ/T is a wedge of circles with one circle for each edge of Γ not in T.
3. $\pi_1(\Gamma, x_0)$ is a free group with one generator for each edge of Γ not in T. If Γ has finitely many vertices V and edges E, that rank is $E - V + 1$.

***Proof*:**

Step 1: a spanning tree exists. Write $\Phi_\alpha\colon D^1 = [-1, 1] \to \Gamma$ for the characteristic maps of the edges (definition 1.2.2), and let $\Gamma_1 \subseteq \Gamma$ be a subcomplex. (a) A subset $C \subseteq \Gamma_1$ is closed in Γ exactly when $\Phi_\alpha^{-1}(C)$ is closed for every edge e_α of Γ_1, because the 0-skeleton is discrete and the preimage under the characteristic map of any other edge is a set of endpoints of $[-1, 1]$. In particular Γ_1 is closed, and it carries the quotient topology from its vertices and the disks of its edges. (b) By [16, Products of Quotient Maps], applied with the locally compact Hausdorff space I, a map $\Gamma_1 \times I \to \Gamma$ is continuous once its composite with $\Phi_\alpha \times \mathrm{id}_I$ is continuous for every edge e_α of Γ_1 and its restriction to $\{v\} \times I$ is continuous for every vertex v of Γ_1.

An *edge path* from x_0 to a vertex v is a finite sequence of edges in which consecutive edges share an endpoint, the first having x_0 and the last v as an endpoint. Let $d(x_0) = 0$, and for $v \neq x_0$ let $d(v)$ be the least number of edges in an edge path from x_0 to v, or ∞ if there is none. If one endpoint of an edge has finite d, so does the other. Hence the vertices of finite d together with the edges at them form a subcomplex Γ', the remaining vertices and edges form a subcomplex Γ'', and $\Gamma = \Gamma' \sqcup \Gamma''$. Both are closed by (a), so both are open, and $x_0 \in \Gamma'$, so connectedness of Γ forces $\Gamma'' = \emptyset$. Every vertex therefore has finite d.

For each vertex $v \neq x_0$ choose an edge e_v whose endpoints are v and a vertex w_v with $d(w_v) = d(v) - 1$, the last edge of a shortest edge path, and let Φ_v be its characteristic map and

$c_v \in \{-1, 1\}$ the point with $\Phi_v(c_v) = w_v$. Let T_k be the subcomplex formed by the vertices v with $d(v) \leq k$ and the edges e_v with $1 \leq d(v) \leq k$, and let $T = \bigcup_k T_k$, a subcomplex containing every vertex. If $d(v) = k+1$, then the only edge of T_{k+1} with endpoint v is e_v, as the endpoints of $e_{v'}$ are v' and $w_{v'}$, with $d(w_{v'}) \leq k$. So T_{k+1} is T_k with the arcs $\Phi_v(D^1)$, $d(v) = k+1$, each attached at the single point w_v, and

$$h^k(y, s) = y \quad (y \in T_k) \qquad h^k\big(\Phi_v(r), s\big) = \Phi_v\big((1-s)r + sc_v\big) \quad (d(v) = k+1)$$

is a well-defined homotopy $T_{k+1} \times I \to T_{k+1}$, continuous by (b), from the identity to a retraction $r_k \colon T_{k+1} \to T_k$, fixing T_k throughout. For $x \in T_m$ put $R_k(x) = r_k r_{k+1} \cdots r_{m-1}(x)$ if $k < m$ and $R_k(x) = x$ if $k \geq m$. This does not depend on m, since r_j fixes T_j, and it defines retractions $R_k \colon T \to T_k$ with $r_k R_{k+1} = R_k$, continuous by (a). Define $H \colon T \times I \to T$ by $H(x, 0) = x$ and

$$H(x, t) = h^k\big(R_{k+1}(x),\, 2^{k+1}t - 1\big) \qquad 2^{-k-1} \leq t \leq 2^{-k}$$

The two formulas at $t = 2^{-k}$ both give $R_k(x)$, and $H(x, 1) = R_0(x) = x_0$. A closed edge or a vertex of T lies in some T_m, and there $H(x, t) = x$ for $t \leq 2^{-m}$, because R_{k+1} and h^k fix T_m when $k \geq m$. So on each such cell H is given by finitely many continuous formulas on closed time intervals, and H is continuous by (b). Hence T is contractible, and it is a spanning tree.

Step 2: collapsing the tree. The pair (Γ, T) is a CW pair, so it has the homotopy extension property by proposition 1.3.4, and T is contractible, so theorem 1.3.5 makes $\pi \colon \Gamma \to \Gamma/T$ a homotopy equivalence. As T contains every vertex, Γ/T has a single 0-cell, and its 1-cells are the edges of Γ outside T, each now attached with both endpoints at that vertex. A 1-cell so attached is a circle, so Γ/T is the wedge sum (definition 1.2.17) of one circle per edge outside T.

Step 3: the fundamental group. A homotopy equivalence induces an isomorphism on fundamental groups, so $\pi_1(\Gamma, x_0) \cong \pi_1(\Gamma/T)$. Applying theorem 2.2.4 to a wedge of circles, covered by open sets each thickening one circle by a short arc of its neighbours, gives the free product of the groups $\pi_1(S^1) \cong \mathbb{Z}$, one factor per circle: every pairwise and triple intersection deformation retracts to the wedge point and is therefore simply connected. A free product of copies of $\mathbb{Z}$ is the free group on the corresponding set. If Γ is finite, every spanning tree T has $V - 1$ edges. Step 1, run inside T, produces a spanning tree $T' \subseteq T$ with one edge e_v for each vertex $v \neq x_0$, and (3), applied to the connected graph T, whose fundamental group is trivial, leaves no edge of T outside T'. So $E - (V - 1) = E - V + 1$ edges lie outside T, completing the proof.

The count $E - V + 1$ is the first Betti number of Γ. Step 2 is why a presentation of the fundamental group of a cell complex takes its generators from the 1-cells: a spanning tree contributes nothing, and each remaining edge contributes one free generator.

Example 2.2.9: Fundamental Group Of M_g

Recall in example 1.2.5(5) that we took the cell complex M_g constructed with one 0-cell, $2g$ 1-cells and one 2-cell. Note that $M_1 = T$ is the torus. We get that the 1-skeleton is the wedge sum of $2g$ circles, giving a fundamental group being the free-product of $2g$ copies of $\mathbb{Z}$. Then the 2-cell attached along the loops gives the commutators, namely $[a_1, b_1], ..., [a_g, b_g]$, as relations (proposition 2.2.6). Hence:

$$\pi_1(M_g) \cong \langle a_1, b_1, ..., a_g, b_g \mid [a_1, b_1], ..., [a_g, b_g] \rangle$$

Using this, we see that $M_g \not\simeq M_h$ if $g \neq h$ (hence $M_g \ncong M_h$). A homotopy equivalence would give $\pi_1(M_g) \cong \pi_1(M_h)$ (theorem 2.1.45), hence an isomorphism of abelianizations. The abelianization

of $\pi_1(M_g)$ is $\mathbb{Z}^{2g}$, since every relator $[a_i, b_i]$ becomes trivial, and finitely generated free $\mathbb{Z}$-modules have an invariant basis number, so $2g = 2h$.

Corollary 2.2.10: Every Group Appears As A Fundamental Group

Let G be a group. Then there exists a 2-dimensional cell complex X_G such that

$$\pi_1(X_G) \cong G$$

***Proof*:**
Let $G = \langle g_\alpha \mid r_\beta \rangle$ where g_α are the generators and r_β are the relations. Then take the 1-skeleton to be $\bigvee_\alpha S^1$ and attach the 2-cells e^2_β specified by the words r_β. The fundamental group of $\bigvee_\alpha S^1$ is the free group on the g_α (proposition 2.2.8), and by proposition 2.2.6 attaching the 2-cells quotients it by the normal subgroup generated by the r_β, which gives G.

Example 2.2.11: More Fundamental Groups

1. (non-orientable surfaces) Let N_g, for $g \geq 1$, be the cell complex with one 0-cell, g 1-cells $a_1, \ldots, a_g$ and one 2-cell attached along the word $a_1^2 a_2^2 \cdots a_g^2$. This is the non-orientable surface of genus g. Note that $N_1 = \mathbb{R}\mathrm{P}^2$ since it is D^2 with the antipodal points of ∂D^2 identified, and N_2 is the Klein bottle (the left image is the usual construction of the Klein bottle):

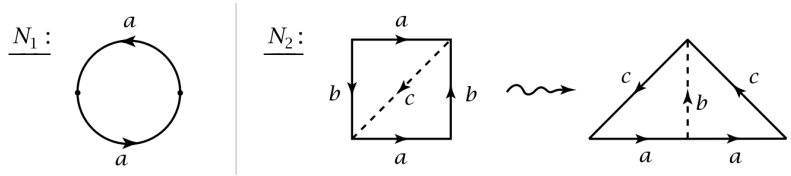


Figure 2.10: $\mathbb{R}\mathrm{P}^2$ and Klein Bottle

   Then, by proposition 2.2.8 and proposition 2.2.6,

$$\pi_1(N_g) \cong \langle a_1, \ldots, a_g \mid a_1^2 a_2^2 \cdots a_g^2 \rangle$$

   Its abelianization is $\mathbb{Z}^g$ modulo the relation $2(a_1 + \cdots + a_g) = 0$, written additively. In the basis $a_1, \ldots, a_{g-1}, a_1 + \cdots + a_g$ of $\mathbb{Z}^g$ this relation involves only the last basis vector, so the abelianization is $\mathbb{Z}^{g-1} \oplus \mathbb{Z}/2$.

   Hence N_g is not homotopy equivalent to N_h if $g \neq h$, and is not homotopy equivalent to any M_h. A homotopy equivalence would induce an isomorphism of fundamental groups (theorem 2.1.45), hence of abelianizations. The torsion subgroup of $\mathbb{Z}^{g-1} \oplus \mathbb{Z}/2$ is $\mathbb{Z}/2$, while the abelianization $\mathbb{Z}^{2h}$ of $\pi_1(M_h)$ is torsion-free (example 2.2.9), which separates N_g from every M_h. The quotient by the torsion subgroup is $\mathbb{Z}^{g-1}$, and an invariant basis number for finitely generated free $\mathbb{Z}$-modules separates N_g from N_h.

2. Let $G = \langle a \mid a^n \rangle = \mathbb{Z}_n$. Then X_G is S^1 with the 2-cell e^2 attached via the map $z \mapsto z^n$. If $n = 2$, then $X_G = \mathbb{R}\mathrm{P}^2$. The rest of this item is an illustration, which we do not prove and

which nothing later depends on. Since the attaching map wraps n times around the circle, n sheets of e^2 meet along each point of the circle, so for $n \geq 3$ the space X_G is not a surface (a 2-manifold). For example, if $n = 3$, then a neighbourhood N of the circle in X_G is the product of the tripod (a vertex with three intervals attached to it) with the interval I, with the two ends of the product identified by a one-third twist:

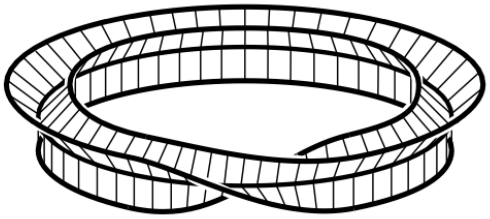


Figure 2.11: visual of X_3

2.3 Classification of Covering Spaces

Let X be a path-connected, locally path-connected space. The covering spaces of X have a theory parallel to Galois theory. In particular, every covering map shall be associated with a subgroup of $\pi_1(X, x_0)$ (for any given basepoint $x_0 \in X$), where the partial order given by the covering maps is inverse to the ordering of the groups, that is we have a contravariant functor between the partial orders (giving the Galois correspondence).

Recall that if $p : E \to B$ is a covering space where E is simply connected, then $\pi_1(B, x_0)$ is in bijective correspondence with $p^{-1}(x_0)$. Are there conditions on B under which there exists a covering map where E is simply connected? There are:

Definition 2.3.1: Semilocally Simply-Connected

Let X be a topological space. If for each $x \in X$ there exists a neighbourhood $U \subseteq X$ such that the map induced by the inclusion map $\iota_* : \pi_1(U, x) \to \pi_1(X, x)$ is trivial, then X is called *semilocally simply-connected*.

To see why this is a necessary condition, suppose $p : \tilde{X} \to X$ is a covering space with $\tilde{X}$ simply connected. Then every point $x \in X$ has a neighbourhood U having a lift $\tilde{U} \subseteq \tilde{X}$ projecting homeomorphically to U by p. Each loop in U lifts to a loop in $\tilde{U}$, and the lifted loop is nullhomotopic in $\tilde{X}$ since $\tilde{X}$ is simply connected. Hence, composing this nullhomotopy with p, the original loop in U is nullhomotopic in X.

All locally simply-connected spaces are semilocally simply-connected. CW complexes are locally contractible (lemma 2.3.2), so every connected CW complex has a universal cover.

Lemma 2.3.2: CW Complexes Are Locally Contractible

Let X be a CW complex, $x \in X$ and $W \subseteq X$ an open set containing x. Then there are an open set N with $x \in N \subseteq W$ and a homotopy $H\colon N \times I \to N$ from id_N to the constant map at x with $H(x,t) = x$ for all t. Consequently X is locally path-connected and semilocally simply connected.

***Proof*:**

Write $\varphi_\alpha\colon S^m \to X^m$ and $\Phi_\alpha\colon D^{m+1}_\alpha \to X^{m+1}$ for the attaching and characteristic maps of the $(m+1)$-cells (definition 1.2.2). A subset of X^{m+1} is open, or closed, exactly when it meets X^m in an open, or closed, set and has open, or closed, preimage under every Φ_α (definition 1.2.1). Let x lie in the open k-cell e_β, say $x = \Phi_\beta(y)$ with $|y| < 1$.

Step 1: The neighbourhood. We build open sets $N_m \subseteq X^m$ with $N_{m+1} \cap X^m = N_m$ and with closure $\overline{N_m}$ in X^m contained in W. For $m < k$ put $N_m = \emptyset$. The set $\Phi_\beta^{-1}(W)$ is open in D^k and contains y, so some closed ball $\bar{B}(y,\rho)$ lies in it and in the open disk. Let $N_k = \Phi_\beta(B(y,\rho))$, which is $\{x\}$ when $k = 0$. Its preimage is $B(y,\rho)$ under Φ_β and empty under the other characteristic maps of k-cells, so N_k is open in X^k. The compact set $\Phi_\beta(\bar{B}(y,\rho))$ is closed in the Hausdorff space X^k (theorem 1.2.3), so $\overline{N_k} \subseteq \Phi_\beta(\bar{B}(y,\rho)) \subseteq W$.

Given N_m with $m \geq k$, fix an $(m+1)$-cell α and put $A_\alpha = \varphi_\alpha^{-1}(N_m)$, open in S^m, and $L_\alpha = \varphi_\alpha^{-1}(\overline{N_m})$, compact. The open set $\Phi_\alpha^{-1}(W)$ contains L_α, since $\Phi_\alpha = \varphi_\alpha$ on S^m. Its preimage under $S^m \times [\tfrac{1}{2}, 1] \to D^{m+1}$, $(s,r) \mapsto rs$, is open and contains $L_\alpha \times \{1\}$, so the tube lemma [30, §26] gives $\varepsilon_\alpha \in (0, \tfrac{1}{2})$ with $K_\alpha = \{rs \mid s \in L_\alpha,\ 1 - \varepsilon_\alpha \leq r \leq 1\}$ contained in $\Phi_\alpha^{-1}(W)$. Put $C_\alpha = \{rs \mid s \in A_\alpha,\ 1 - \varepsilon_\alpha < r \leq 1\}$ and

$$N_{m+1} = N_m \cup \bigcup_\alpha \Phi_\alpha(C_\alpha)$$

The map Φ_α is injective on the open disk, the open cells are disjoint, and $\Phi_\alpha = \varphi_\alpha$ on S^m, so $N_{m+1} \cap X^m = N_m$ and $\Phi_\alpha^{-1}(N_{m+1}) = C_\alpha$. The set C_α is D^{m+1} intersected with the open set $\big\{z \in \mathbb{R}^{m+1} \;\big|\; 1 - \varepsilon_\alpha < |z| < 2,\ z/|z| \in A_\alpha\big\}$, so N_{m+1} is open in X^{m+1}. In the same way $K = \overline{N_m} \cup \bigcup_\alpha \Phi_\alpha(K_\alpha)$ has $K \cap X^m = \overline{N_m}$ and $\Phi_\alpha^{-1}(K) = K_\alpha$, so K is closed, and $N_{m+1} \subseteq K \subseteq W$ gives $\overline{N_{m+1}} \subseteq W$.

Let $N = \bigcup_m N_m$. Then $N \cap X^m = N_m$ for every m, so N is open in X, and $x \in N \subseteq W$. A subset of N meeting every N_m in an open set meets every X^m in an open set, so it is open.

Step 2: Retracting each stage onto the previous one. For $m \geq k$ define $R_{m+1}\colon N_{m+1} \times I \to N_{m+1}$ by $R_{m+1}(z,t) = z$ for $z \in N_m$ and

$$R_{m+1}\big(\Phi_\alpha(rs), t\big) = \Phi_\alpha\big(((1-t)r + t)s\big) \qquad (rs \in C_\alpha)$$

The two prescriptions agree where $r = 1$, the radius $(1-t)r + t$ stays between r and 1, and at $t = 1$ the point $\Phi_\alpha(s) = \varphi_\alpha(s)$ lies in N_m. Let π be the quotient map from $X^m \sqcup_\alpha D^{m+1}_\alpha$ onto X^{m+1}. Its restriction to the open saturated set $\pi^{-1}(N_{m+1}) = N_m \sqcup_\alpha C_\alpha$ is a quotient map onto N_{m+1}: a set whose preimage is open in $\pi^{-1}(N_{m+1})$ has open preimage under π. Its product with id_I is again a quotient map [16, Products of Quotient Maps], and composed with it R_{m+1} is the projection on $N_m \times I$ and $(rs,t) \mapsto \Phi_\alpha(((1-t)r+t)s)$ on $C_\alpha \times I$. So R_{m+1} is continuous. Write

$\varrho_{m+1} = R_{m+1}(-,1)\colon N_{m+1} \to N_m$. On N_k put $G(\Phi_\beta(z),t) = \Phi_\beta((1-t)z+ty)$ for $z \in B(y,\rho)$, a homotopy in N_k from the identity to the constant map at x that fixes x. It is continuous because Φ_β restricts to a homeomorphism from the open disk onto the open cell e_β, and so from $B(y,\rho)$ onto N_k.

Step 3: The contraction. For $z \in N_M$, $M \geq k$, let $t \mapsto H(z,t)$ be constant at z on $[0, 2^{k-M-1}]$, follow $R_{m+1}(\varrho_{m+2}\cdots\varrho_M(z),-)$ on $[2^{k-m-2}, 2^{k-m-1}]$ for $m = M-1,\dots,k$, and follow $G(\varrho_{k+1}\cdots\varrho_M(z),-)$ on $[\frac{1}{2},1]$, each reparametrized linearly. Consecutive pieces agree at the shared endpoints, so H is continuous on $N_M \times I$. For $z \in N_M$ the recipe read in N_{M+1} gives the same path, because R_{M+1} fixes N_M, so H is well defined on $N \times I$. By Step 1 the map from $\bigsqcup_M N_M$ to N is a quotient map, hence so is its product with id_I [16, Products of Quotient Maps], and H is continuous. It starts at the identity, ends at the constant map at x, and fixes x, which lies in N_k, where every R_{m+1} is stationary and G fixes x.

Each $z \in N$ is joined to x by the path $t \mapsto H(z,t)$ in N, so N is path-connected, and as W was arbitrary, X is locally path-connected. A loop γ in N at x is nullhomotopic in N through the loops $s \mapsto H(\gamma(s),t)$, which are based at x because H fixes x. Taking $W = X$, the map $\pi_1(N,x) \to \pi_1(X,x)$ is trivial, so X is semilocally simply connected (definition 2.3.1), as claimed.

Example 2.3.3: Not Semilocally Connected

Take $X \subseteq \mathbb{R}^2$ consisting of circles C_n of radius $1/n$ centered at $(1/n, 0)$ (i.e. the Hawaiian earrings). It is not semilocally simply-connected: every neighbourhood of the origin contains some C_n, and the loop around C_n is not nullhomotopic in X, because C_n is a retract of X (collapse the other circles to the origin) and lemma 2.1.43 applies. On the other hand, the cone $CX = (X \times I)/(X \times \{0\})$ on the shrinking wedge of circles is semilocally simply-connected since it is contractible, but is not locally simply-connected.

Lemma 2.3.4: Universal Covering Map

Let X be a path connected, locally path connected, semilocally simply connected space. Then there exists a simply connected space $\tilde{X}$ such that $p : \tilde{X} \to X$ is a covering map. Furthermore, any two simply connected covering spaces of X are isomorphic over X.

***Proof*:**

Fix a basepoint $x_0 \in X$ and let

$$\tilde{X} = \{[\gamma] \mid \gamma \text{ is a path in } X \text{ starting at } x_0\}$$

where $[\gamma]$ is the homotopy class of γ relative to its endpoints. Define $p\colon \tilde{X} \to X$ by $p([\gamma]) = \gamma(1)$, which is well defined since a homotopy relative to the endpoints fixes $\gamma(1)$. Since X is path-connected, each point $x \in X$ is the endpoint of a path starting at x_0, and hence p is surjective.

Step 1: A basis of X. Let $\mathcal{U}$ be the collection of path-connected open sets $U \subseteq X$ such that the map $\pi_1(U,x) \to \pi_1(X,x)$ induced by the inclusion is trivial for some $x \in U$. It is then trivial for every $x \in U$, since a path in U between two points induces compatible isomorphisms of the fundamental groups of U and of X at those points (proposition 2.1.15). The collection $\mathcal{U}$ is a basis of X. Given an open W and a point $x \in W$, semilocal simple connectivity gives a neighbourhood V of x with $\pi_1(V,x) \to \pi_1(X,x)$ trivial, and the path component of x in the

open set $W \cap \text{int}V$ is open, X being locally path-connected. It lies in $\mathcal{U}$, since its map on π_1 factors through $\pi_1(V, x)$.

Step 2: The sets $U_{[\gamma]}$. For $U \in \mathcal{U}$ and a path γ from x_0 to a point of U, let

$$U_{[\gamma]} = \{[\gamma * \eta] \mid \eta \text{ is a path in } U \text{ starting at } \gamma(1)\}$$

If $[\gamma] = [\gamma']$ then $[\gamma * \eta] = [\gamma' * \eta]$ for every such η, so $U_{[\gamma]}$ depends only on the class of γ, justifying the notation. The map $p\colon U_{[\gamma]} \to U$ is surjective since U is path-connected. It is injective, because two paths η, η' in U from $\gamma(1)$ to the same point form a loop $\eta * \overline{\eta'}$ in U, which is nullhomotopic in X, so that $[\gamma * \eta] = [\gamma * \eta']$. If $[\gamma'] \in U_{[\gamma]}$, say $[\gamma'] = [\gamma * \eta]$, then $U_{[\gamma']} = U_{[\gamma]}$: an element of $U_{[\gamma']}$ has the form $[\gamma * \eta * \mu]$, which lies in $U_{[\gamma]}$, and conversely an element of $U_{[\gamma]}$ has the form

$$[\gamma * \mu] = [\gamma * \eta * \overline{\eta} * \mu] = [\gamma' * \overline{\eta} * \mu]$$

which lies in $U_{[\gamma']}$. In particular two sets $U_{[\gamma]}$ and $U_{[\gamma']}$ for the same U are either disjoint or equal.

Step 3: The topology. The sets $U_{[\gamma]}$ cover $\tilde{X}$, since $[\gamma] \in U_{[\gamma]}$ for any $U \in \mathcal{U}$ containing $\gamma(1)$. If $[\gamma''] \in U_{[\gamma]} \cap V_{[\gamma']}$, then $U_{[\gamma]} = U_{[\gamma'']}$ and $V_{[\gamma']} = V_{[\gamma'']}$ by Step 2, and for $W \in \mathcal{U}$ with $\gamma''(1) \in W \subseteq U \cap V$ the set $W_{[\gamma'']}$ contains $[\gamma'']$ and satisfies

$$W_{[\gamma'']} \subseteq U_{[\gamma'']} \cap V_{[\gamma'']}$$

So the sets $U_{[\gamma]}$ form a basis for a topology on $\tilde{X}$. Each bijection $p\colon U_{[\gamma]} \to U$ is a homeomorphism: the basic open sets contained in $U_{[\gamma]}$ are the sets $V_{[\gamma']}$ with $V \in \mathcal{U}$, $V \subseteq U$ and $[\gamma'] \in U_{[\gamma]}$, and p carries $V_{[\gamma']}$ onto V. Conversely, for $V \in \mathcal{U}$ with $V \subseteq U$, the preimage of V in $U_{[\gamma]}$ is $V_{[\gamma']}$ for any $[\gamma'] \in U_{[\gamma]}$ with $\gamma'(1) \in V$, by injectivity. So p gives a bijection between bases of the two topologies, and p is continuous.

Step 4: p is a covering map. Let $U \in \mathcal{U}$. A point $[\gamma'] \in p^{-1}(U)$ lies in $U_{[\gamma']}$, so $p^{-1}(U)$ is the union of the sets $U_{[\gamma]}$ with $\gamma(1) \in U$. By Step 2 any two of these sets are disjoint or equal, and by Step 3 each is open and is carried homeomorphically onto U by p. So U is evenly covered (definition 2.1.24). The sets of $\mathcal{U}$ cover X, so p is a covering map (definition 2.1.25).

Step 5: $\tilde{X}$ is path-connected. For $[\gamma] \in \tilde{X}$ and $t \in [0, 1]$ let γ_t be the path that equals γ on $[0, t]$ and is constant at $\gamma(t)$ on $[t, 1]$. The map $t \mapsto [\gamma_t]$ is continuous. Given a basic neighbourhood $U_{[\gamma_t]}$ of $[\gamma_t]$, for t' near t the path γ maps the interval between t and t' into U, and $[\gamma_{t'}] = [\gamma_t * \eta]$ for the path η in U that runs along γ from $\gamma(t)$ to $\gamma(t')$, so $[\gamma_{t'}] \in U_{[\gamma_t]}$. This is a path in $\tilde{X}$ lifting γ from $[x_0]$, the class of the constant path, to $[\gamma_1] = [\gamma]$. Since $[\gamma]$ was arbitrary, $\tilde{X}$ is path-connected.

Step 6: $\tilde{X}$ is simply connected, and uniqueness. To show $\pi_1(\tilde{X}, [x_0]) = 0$, it suffices to show that the image of this group under p_* is trivial, since p_* is injective (proposition 2.1.39(1)). An element of the image of p_* is represented by a loop γ at x_0 that lifts to a loop in $\tilde{X}$ at $[x_0]$. The path $t \mapsto [\gamma_t]$ of Step 5 lifts γ starting at $[x_0]$, so by the uniqueness of lifts (lemma 2.1.30) it is that loop, and $[\gamma_1] = [x_0]$. Since $\gamma_1 = \gamma$, this says $[\gamma] = [x_0]$, so γ is nullhomotopic and the image of p_* is trivial. Hence $\tilde{X}$ is simply connected. That any two simply connected covering spaces of X are isomorphic over X is [16, Uniqueness of the Universal Cover], which applies since X is path-connected and locally path-connected, completing the proof.

This construction shows that a space with these conditions always has a simply connected covering

space. However, most of the time there shall be simpler means of constructing covering spaces. For example, $\mathbb{R}$ is a simply connected covering space of S^1, and the product T^n has covering space $\mathbb{R}^n$ by proposition 2.1.27(4).

Theorem 2.3.5: Galois Correspondence Of Fundamental Groups

Let X be a path-connected, locally path-connected, semilocally simply-connected space. Then there is a bijection between the set of basepoint-preserving isomorphism classes of path-connected covering spaces $p : (\tilde{X}, e_0) \to (X, b_0)$ and the set of subgroups of $\pi_1(X, b_0)$, obtained by associating the subgroup $p_*(\pi_1(\tilde{X}, e_0))$.

***Proof*:**
This is the classification of coverings, proved in *Topology* [16, Classification of Coverings] for a path-connected, locally path-connected and semilocally simply connected base. Its two halves are the existence of a covering realising a prescribed subgroup, which goes through the universal cover, and the injectivity of $p \mapsto p_*\pi_1(\tilde{X}, e_0)$, which is the lifting criterion applied in both directions.

Under the same hypotheses the classification has a categorical form: the fibre over b_0, with the right action of $\pi_1(X, b_0)$ by lifting loops, is an equivalence of categories from the coverings of X to the nonempty right $\pi_1(X, b_0)$-sets [16, The Monodromy Equivalence]. (That theorem admits the empty covering, which the surjectivity required in definition 2.1.25 excludes, and matches it with the empty π_1-set.)

Theorem 2.3.6: Uniqueness Of Covering Space

Let X be path-connected and locally path-connected. Then two path-connected covering spaces $p_1 : \tilde{X}_1 \to X$ and $p_2 : \tilde{X}_2 \to X$ are basepoint-isomorphic if and only if $\mathrm{im}p_{1_*} = \mathrm{im}p_{2_*}$.

An isomorphism here is a homeomorphism $f\colon \tilde{X}_1 \to \tilde{X}_2$ with $p_2 \circ f = p_1$, sending basepoint to basepoint.

***Proof*:**
This is the injective half of the classification of coverings, also proved in *Topology* [16, Classification of Coverings]. If there is an isomorphism f between $(\tilde{X}_1, \tilde{x}_1)$ and $(\tilde{X}_2, \tilde{x}_2)$, then we get $p_1 = p_2 f$ and $p_2 = p_1 f^{-1}$, so by functoriality (proposition 2.1.22) the groups are equal. Conversely, suppose the groups are equal. Each $\tilde{X}_i$ is path-connected by hypothesis and locally path-connected, as a covering space of a locally path-connected space [16, Coverings Inherit Local Path-Connectedness]. So by the lifting criterion (theorem 2.1.33), we may lift p_1 to a map $\tilde{p}_1 : (\tilde{X}_1, \tilde{x}_1) \to (\tilde{X}_2, \tilde{x}_2)$ with $p_2\tilde{p}_1 = p_1$. We can do the same the other way to get $p_1\tilde{p}_2 = p_2$. Then by uniqueness of lifting (corollary 2.1.34), $\tilde{p}_1\tilde{p}_2 = \mathrm{id}$ and $\tilde{p}_2\tilde{p}_1 = \mathrm{id}$, since each composition is a lift of p_2, respectively p_1, fixing the basepoint, as is the identity. These maps are continuous, and they are each other's inverses, and hence each is an isomorphism, as we sought to show.

Proposition 2.3.7: Base-Point Independence

If ignoring the basepoint, the correspondence gives a bijection between isomorphism class of path-connected covering spaces $p : \tilde{X} \to X$ and conjugacy classes of subgroups of $\pi_1(X, x_0)$

***Proof*:**

The basepoint-preserving correspondence of theorem 2.3.5 sends a covering $p\colon (\tilde{X}, e_0) \to (X, b_0)$ to the subgroup $H = p_*(\pi_1(\tilde{X}, e_0)) \subseteq \pi_1(X, b_0)$, and theorem 2.3.6 shows that this assignment is injective on basepoint-preserving isomorphism classes. Forgetting the basepoint amounts to allowing e_0 to range over the fiber $p^{-1}(b_0)$ and identifying two coverings whenever some such reassignment makes them basepoint-isomorphic.

Step 1: Moving the basepoint conjugates the subgroup. Fix the covering $p\colon \tilde{X} \to X$ with X path-connected and locally path-connected, and let $e_0, e_1 \in p^{-1}(b_0)$ be two points of the fiber over b_0. Write $H_0 = p_*(\pi_1(\tilde{X}, e_0))$ and $H_1 = p_*(\pi_1(\tilde{X}, e_1))$. Since $\tilde{X}$ is path-connected, choose a path $\tilde{\eta}$ in $\tilde{X}$ from e_0 to e_1, and let $\eta = p \circ \tilde{\eta}$, a loop in X based at b_0. Set $g = [\eta] \in \pi_1(X, b_0)$. The claim is that

$$H_1 = g^{-1} H_0 \, g$$

By proposition 2.1.15, the reverse path $\overline{\tilde{\eta}}$, from e_1 to e_0, induces an isomorphism $\widehat{\overline{\tilde{\eta}}}\colon \pi_1(\tilde{X}, e_1) \to \pi_1(\tilde{X}, e_0)$ given by $[\tilde{f}] \mapsto [\tilde{\eta}] * [\tilde{f}] * [\overline{\tilde{\eta}}]$. This is correctly typed against definition 2.1.4: for a loop $\tilde{f}$ at e_1, the concatenation $\tilde{\eta} * \tilde{f}$ is defined because $\tilde{\eta}(1) = e_1 = \tilde{f}(0)$, and the further concatenation with $\overline{\tilde{\eta}}$ is defined because $\tilde{f}(1) = e_1 = \overline{\tilde{\eta}}(0)$, so $\tilde{\eta} * \tilde{f} * \overline{\tilde{\eta}}$ traverses $\tilde{\eta}$ from e_0 to e_1, runs $\tilde{f}$, then returns along $\overline{\tilde{\eta}}$ to e_0, giving a loop at e_0. Applying p_* and using functoriality (proposition 2.1.22) together with $p \circ \tilde{\eta} = \eta$, a loop $\tilde{f}$ at e_1 is carried to

$$p_*\big([\tilde{\eta}] * [\tilde{f}] * [\overline{\tilde{\eta}}]\big) = [\eta] * p_*([\tilde{f}]) * [\overline{\eta}] = g \, p_*([\tilde{f}]) \, g^{-1}$$

As $[\tilde{f}]$ ranges over $\pi_1(\tilde{X}, e_1)$, the image $p_*([\tilde{f}])$ ranges over H_1. On the other hand $\widehat{\overline{\tilde{\eta}}}$ is a bijection onto $\pi_1(\tilde{X}, e_0)$, so the classes $[\tilde{\eta}] * [\tilde{f}] * [\overline{\tilde{\eta}}]$ range over all of $\pi_1(\tilde{X}, e_0)$ and their images under p_* range over all of H_0. Comparing the two descriptions of the same set gives $H_0 = g \, H_1 \, g^{-1}$, that is $H_1 = g^{-1} H_0 \, g$. Thus, as e_1 ranges over $p^{-1}(b_0)$, the subgroup $p_*(\pi_1(\tilde{X}, e_1))$ ranges over conjugates of H_0.

Conversely, every conjugate of H_0 arises this way. Given $g = [\eta] \in \pi_1(X, b_0)$, lift the loop η to the path $\tilde{\eta}$ in $\tilde{X}$ starting at e_0, which exists by lemma 2.1.30, and let $e_1 = \tilde{\eta}(1) \in p^{-1}(b_0)$. Then $\tilde{\eta}$ is a path from e_0 to e_1 projecting to η, so the computation above applies and yields $H_1 = p_*(\pi_1(\tilde{X}, e_1)) = g^{-1} H_0 \, g$. As g ranges over $\pi_1(X, b_0)$ the conjugate $g^{-1} H_0 \, g$ ranges over the entire conjugacy class of H_0. Therefore the set of subgroups realized by the various choices of basepoint in the fiber over b_0 is precisely the conjugacy class of H_0 in $\pi_1(X, b_0)$.

Step 2: Conjugacy classes classify the unbased coverings. Define a map from basepoint-free isomorphism classes of path-connected coverings to conjugacy classes of subgroups of $\pi_1(X, b_0)$ by sending a covering $p\colon \tilde{X} \to X$ to the conjugacy class of $H_0 = p_*(\pi_1(\tilde{X}, e_0))$ for any chosen $e_0 \in p^{-1}(b_0)$. By Step 1 this conjugacy class does not depend on the choice of e_0, so the map is well-defined on a fixed covering. It is also constant on basepoint-free isomorphism classes: if $f\colon \tilde{X}_1 \to \tilde{X}_2$ is a homeomorphism with $p_2 \circ f = p_1$ (no constraint on basepoints), pick $e_0 \in p_1^{-1}(b_0)$ and set $e_0' = f(e_0) \in p_2^{-1}(b_0)$. Then f restricts to a basepoint-preserving isomorphism $(\tilde{X}_1, e_0) \to (\tilde{X}_2, e_0')$, so by functoriality (proposition 2.1.22) $p_{1*}(\pi_1(\tilde{X}_1, e_0)) = p_{2*}(\pi_1(\tilde{X}_2, e_0'))$ as

actual subgroups, and a fortiori the two conjugacy classes agree. Hence the assignment descends to basepoint-free isomorphism classes.

The map is surjective: given any subgroup $K \subseteq \pi_1(X, b_0)$, theorem 2.3.5 produces a path-connected covering $p\colon (\tilde{X}, e_0) \to (X, b_0)$ with $p_*(\pi_1(\tilde{X}, e_0)) = K$, and this covering maps to the conjugacy class of K. Since every conjugacy class contains a subgroup, the map hits every conjugacy class.

For injectivity, suppose two path-connected coverings $p_1\colon \tilde{X}_1 \to X$ and $p_2\colon \tilde{X}_2 \to X$ have $H_1 = p_{1*}(\pi_1(\tilde{X}_1, e_1))$ and $H_2 = p_{2*}(\pi_1(\tilde{X}_2, e_2))$ in the same conjugacy class, where $e_1 \in p_1^{-1}(b_0)$ and $e_2 \in p_2^{-1}(b_0)$ are chosen basepoints. By Step 1, moving the basepoint of $\tilde{X}_2$ realizes every conjugate of H_2. In particular there is a point $e_2' \in p_2^{-1}(b_0)$ with $p_{2*}(\pi_1(\tilde{X}_2, e_2')) = H_1$ as subgroups of $\pi_1(X, b_0)$. The two coverings $(\tilde{X}_1, e_1)$ and $(\tilde{X}_2, e_2')$ now have equal images under the induced maps, so theorem 2.3.6 provides a basepoint-preserving isomorphism between them. Forgetting the basepoint, p_1 and p_2 are isomorphic as coverings. Hence coverings with conjugate subgroups lie in the same basepoint-free isomorphism class.

The assignment is therefore a well-defined surjection that is injective on basepoint-free isomorphism classes, hence a bijection between isomorphism classes of path-connected coverings $p\colon \tilde{X} \to X$ and conjugacy classes of subgroups of $\pi_1(X, b_0)$, completing the proof.

Hence, a simply-connected covering space $\tilde{X}$ is a *universal cover* since it is unique *up to isomorphism*.

2.3.1 Deck Transformations

The isomorphisms between covering spaces in theorem 2.3.6, those making the diagram

$$\begin{array}{ccc} E & \xrightarrow{\sim} & E \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle p} & \\ & B & \end{array}$$

commute, have the same properties as morphisms of fiber bundles. There is a relation between fiber bundles, covering maps, and fundamental groups. Since the study of these fiber-bundle morphisms on covering spaces predates categorical nomenclature, they have their own widely accepted name:

Definition 2.3.8: Deck Transformation

Let $p : E \to B$ be a covering map. Then an isomorphism:

$$\begin{array}{ccc} E & \xrightarrow{\sim} & E \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle p} & \\ & B & \end{array}$$

is called a *Deck transformation*. The collection of deck transformations is denoted $\mathrm{Deck}\,(p)$ or $\mathrm{Aut}(p)$. If we want to specify a particular group for a covering space, we will write $\mathrm{Aut}(p, E)$.

Since the spaces we are working with have unique universal covering maps up to isomorphism (ex. CW complexes), the notation $\mathrm{Deck}\,(p)$ is well-defined given we are always taking universal covers.

Example 2.3.9: Deck Transformation

1. Let $p : \mathbb{R} \to S^1$. Then the deck transformations will be the vertical translations, so
$$\text{Deck}\,(p) = \mathbb{Z}$$
Indeed, each translation $x \mapsto x+n$, $n \in \mathbb{Z}$, is a deck transformation, and a deck transformation is determined by the image of 0, which lies in $p^{-1}(1,0) = \mathbb{Z}$ (corollary 2.1.34).
2. Let $p : S^1 \to S^1$ be given by $z \mapsto z^n$. Then $\text{Aut}(p, S^1) = \mathbb{Z}/n\mathbb{Z}$, by the same argument with the rotations $z \mapsto \zeta z$ by the nth roots of unity ζ in place of the translations.

By the unique lifting property (corollary 2.1.34), a deck transformation is *completely* determined by where it sends a single point (assuming E is path connected). Thus, the only deck transformation that fixes a point of E is the identity.

Definition 2.3.10: Normal Covering

A covering space $p : E \to B$ is called *normal* if for each $x \in B$, and each pair of lifts $\tilde{x}, \tilde{x}'$ of x, there exists a deck transformation taking $\tilde{x}$ to $\tilde{x}'$.

For example, the covering space $\mathbb{R} \to S^1$ and the n-sheeted covering space $S^1 \to S^1$ are normal. Intuitively, a normal covering space is one with "maximal symmetry".

Proposition 2.3.11: Properties Of Normal Covering Spaces

Let $p : (E, e_0) \to (B, b_0)$ be a path-connected covering space of a path-connected, locally path-connected space B, and let H be the subgroup $H = p_*(\pi_1(E, e_0)) \subseteq \pi_1(B, b_0)$. Then:

1. The covering map is normal if and only if $H \trianglelefteq \pi_1(B, b_0)$
2. $\text{Aut}(p, E)$ is isomorphic to the quotient $N(H)/H$ ($N(H)$ the normalizer of H).

In particular, $\text{Aut}(p, E)$ is isomorphic to $\pi_1(B, b_0)/H$ if E is a normal covering. Hence the universal cover $p : E \to B$ has $\text{Aut}(p) \cong \pi_1(B)$

***Proof*:**
Both clauses are point-set covering-space theory, established in *Topology* [16, The Deck Group as a Quotient of a Normalizer, The Deck Group of the Universal Cover]: the equivalence of normality of the covering with normality of the subgroup H, and the isomorphism $\text{Aut}(p, E) \cong N(H)/H$, are proved there in the chapter on the fundamental group. The specialisation to the universal cover, where H is trivial and the deck group is all of $\pi_1(X, x_0)$, is the corollary recorded there immediately afterwards. They are quoted here in the form the homology of covering spaces uses, as we sought to show.

2.3.2 Subgroups of Free Groups

A subgroup of the fundamental group of a graph is the image of the fundamental group of a covering space (theorem 2.3.5), and that covering space is again a graph, so proposition 2.2.8 applies to it [21, Section 1.A, p. 85].

Lemma 2.3.12: Covering Spaces of Graphs

Let Γ be a graph with vertex set V and characteristic maps $\Phi_\alpha\colon D^1 \to \Gamma$ for its edges, and let $p\colon \tilde{\Gamma} \to \Gamma$ be a covering map. Then $\tilde{\Gamma}$ is a graph with vertex set $p^{-1}(V)$ and with one edge for each lift $\tilde{\Phi}\colon D^1 \to \tilde{\Gamma}$ of each Φ_α, the lift $\tilde{\Phi}$ being its characteristic map.

***Proof*:**

Step 1: lifts of edges. Let $s \in D^1$ and $\tilde{x} \in \tilde{\Gamma}$ with $p(\tilde{x}) = \Phi_\alpha(s)$. The interval D^1 is path-connected, locally path-connected and simply connected, so theorem 2.1.33 gives a lift $\tilde{\Phi}$ of Φ_α with $\tilde{\Phi}(s) = \tilde{x}$, and it is the only one by corollary 2.1.34. Let $\tilde{\Phi}$ and $\tilde{\Psi}$ be lifts of Φ_α and Φ_β with $\tilde{\Phi}(r) = \tilde{\Psi}(r')$ for some $r, r' \in (-1,1)$. Then $\Phi_\alpha(r) = \Phi_\beta(r')$ lies on an open edge, so $\alpha = \beta$, and $r = r'$ because Φ_α is injective on $(-1,1)$. The two lifts agree at r, so $\tilde{\Phi} = \tilde{\Psi}$ by corollary 2.1.34. Hence distinct lifts have disjoint images of $(-1,1)$.

Step 2: the cells of $\tilde{\Gamma}$. Each lift $\tilde{\Phi}$ of Φ_α is injective on $(-1,1)$, since $p\tilde{\Phi} = \Phi_\alpha$ is, and $p\tilde{\Phi}(\pm 1) = \Phi_\alpha(\pm 1) \in V$, so $\tilde{\Phi}$ maps ∂D^1 into $p^{-1}(V)$. A point $\tilde{x} \notin p^{-1}(V)$ has $p(\tilde{x}) = \Phi_\alpha(s)$ for a unique edge and a unique $s \in (-1,1)$, as Γ is the disjoint union of V and its open edges, and Step 1 gives a lift $\tilde{\Phi}$ with $\tilde{\Phi}(s) = \tilde{x}$. Together with Step 1, $\tilde{\Gamma}$ is the disjoint union of $p^{-1}(V)$ and the sets $\tilde{\Phi}((-1,1))$, one for each lift, each a bijective image of $(-1,1)$.

Step 3: the topology. Give $p^{-1}(V)$ the discrete topology, and let $\tilde{q}$ be the map onto $\tilde{\Gamma}$ from the disjoint union of $p^{-1}(V)$ and one copy of D^1 for each lift $\tilde{\Phi}$, given by the inclusion and the lifts. It is continuous and, by Step 2, surjective, and the only points it identifies are an endpoint c of the copy of D^1 for $\tilde{\Phi}$ and the vertex $\tilde{\Phi}(c)$. These are the identifications of definition 1.2.1 for the attaching maps $\tilde{\Phi}|_{\partial D^1}\colon \partial D^1 \to p^{-1}(V)$, so it remains to show that $\tilde{q}$ is a quotient map, that is, that a set $C \subseteq \tilde{\Gamma}$ with $\tilde{\Phi}^{-1}(C)$ closed for every lift $\tilde{\Phi}$ is closed.

Let $\tilde{x} \notin C$. Choose an open set $U \ni p(\tilde{x})$ evenly covered by p (definition 2.1.25), let $\tilde{U}$ be the slice containing $\tilde{x}$, so that $p|_{\tilde{U}}\colon \tilde{U} \to U$ is a homeomorphism, and put $D = p(C \cap \tilde{U})$. Fix an edge e_α and a connected component J of the open set $\Phi_\alpha^{-1}(U) \subseteq [-1,1]$, an interval open in $[-1,1]$. The map $\sigma = (p|_{\tilde{U}})^{-1} \circ \Phi_\alpha|_J$ lifts $\Phi_\alpha|_J$, and for $t_0 \in J$ the lift $\tilde{\Phi}$ of Φ_α with $\tilde{\Phi}(t_0) = \sigma(t_0)$, from Step 1, agrees with σ on the connected set J by corollary 2.1.34. Since $p|_{\tilde{U}}$ is a bijection onto U,

$$\Phi_\alpha^{-1}(D) \cap J = \{t \in J \mid \sigma(t) \in C\} = J \cap \tilde{\Phi}^{-1}(C)$$

which is closed in J. The components J are open and cover $\Phi_\alpha^{-1}(U)$, so $\Phi_\alpha^{-1}(D)$ is closed in $\Phi_\alpha^{-1}(U)$, and $\Phi_\alpha^{-1}(U \setminus D) = \Phi_\alpha^{-1}(U) \setminus \Phi_\alpha^{-1}(D)$ is open in $[-1,1]$. This holds for every edge and Γ^0 is discrete, so $U \setminus D$ is open in Γ (definition 1.2.1). Then $(p|_{\tilde{U}})^{-1}(U \setminus D) = \tilde{U} \setminus C$ is an open set containing $\tilde{x}$ and disjoint from C. So C is closed, $\tilde{q}$ is a quotient map, and $\tilde{\Gamma}$ is the graph described, as we sought to show.

Theorem 2.3.13: Nielsen-Schreier Theorem

Every subgroup of a free group is free.

***Proof*:**
Let F be a free group on a set S and $H \subseteq F$ a subgroup. Let Γ be the graph with a single vertex x_0 and one edge for each $s \in S$, both ends of each edge attached at x_0, a wedge of circles. Each point of Γ is joined to x_0 along its edge, so Γ is path-connected, and $\{x_0\}$ is a spanning tree. By proposition 2.2.8, $\pi_1(\Gamma, x_0)$ is free on a set in bijection with S, so there is an isomorphism $\pi_1(\Gamma, x_0) \cong F$, under which H corresponds to a subgroup $H' \subseteq \pi_1(\Gamma, x_0)$ isomorphic to H.

By lemma 2.3.2, the graph Γ is locally path-connected and semilocally simply connected, so theorem 2.3.5 provides a path-connected covering space $p\colon (\tilde{\Gamma}, \tilde{x}_0) \to (\Gamma, x_0)$ with $p_*\pi_1(\tilde{\Gamma}, \tilde{x}_0) = H'$. The homomorphism p_* is injective by proposition 2.1.39(1), so $\pi_1(\tilde{\Gamma}, \tilde{x}_0) \cong H' \cong H$.

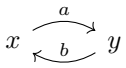
By lemma 2.3.12, $\tilde{\Gamma}$ is a graph with vertex set $p^{-1}(x_0)$, so $\tilde{x}_0$ is a vertex, and $\tilde{\Gamma}$ is connected, being path-connected. By proposition 2.2.8, applied to $\tilde{\Gamma}$ and $\tilde{x}_0$, the group $\pi_1(\tilde{\Gamma}, \tilde{x}_0)$ is free, with one generator for each edge of $\tilde{\Gamma}$ outside a spanning tree. Hence H is free, as we sought to show.

3

Homology

When working with the fundamental group $\pi_1(X)$, we must keep track of a basepoint. The composition of two loops is defined by tracing one path and then the other, a process anchored to their common starting point. This geometric restriction has an algebraic consequence: path concatenation is generally non-commutative, rendering $\pi_1(X)$ a non-abelian group[1]. As we have seen, this non-abelian nature makes computations and group classifications difficult.

Homology abandons the basepoint. Its basic objects are *cycles*, which, unlike the loops of homotopy, have no basepoint. Algebraically, instead of concatenating paths, we will consider formal sums of geometric pieces. To see how this simplifies the algebra, consider two vertices x and y, with two directed edges a and b traveling between them:



In homotopy, the concatenated loop $a \cdot b$ (based at x) and the loop $b \cdot a$ (based at y) are distinct objects living in different groups. In homology, however, we no longer think of traveling along paths in sequence. Instead, we treat a and b as independent geometric building blocks and take their formal commutative sum: $a + b = b + a$. This shift "abelianizes" our theory. All homology groups are abelian, so the methods of homological algebra apply to them.

Without basepoints and continuous functions as our primary tools, we need a new mechanism to generate our algebraic invariants. Homology achieves this by focusing on the relationship between *cycles* (closed shapes with no boundary, like a circle or a hollow sphere) and *boundaries* (the outer edges of higher-dimensional solid shapes). Homology measures to what extent cycles fail to be boundaries. A cycle represents a "hole" in our space if and only if it is *not* the boundary of some higher-dimensional piece of the space.

The chapter formalizes this in two ways:

[1] Every finitely generated group can be represented as $\pi_1(BG)$

1. *Simplicial Homology:* We will first introduce a rigid, combinatorial structure known as a Δ-complex (a generalization of simplicial complexes). By building spaces out of discrete n-dimensional triangles (simplices) glued along their faces, we can reduce the topological concepts of "cycles" and "boundaries" to finite matrices and linear algebra. This makes explicit computations possible.
2. *Singular Homology:* While simplicial homology is highly computable, it relies entirely on a specific geometric construction. To guarantee that our invariants depend only on the topological space X itself, we will introduce singular homology. This theory allows continuous maps of simplices directly into X, yielding a functorial theory that cannot be computed directly from its definition.

The chapter combines the two approaches by proving that for spaces that can be triangulated, the singular and simplicial homology groups are isomorphic, so the singular groups can be computed with the simplicial method.

Homology originally consisted of numbers, the Betti numbers and the torsion coefficients, and this is what it was when Poincaré wrote about homology at the beginning of the 20th century. The transition to the modern viewpoint with groups took place gradually over about 40 years, and was settled in 1952 by Eilenberg and Steenrod.

An informal discussion of the alternating sum in the boundary operator is this Mathematics Stack Exchange thread.

3.1 Δ-Complex

Simplicial homology is defined from a Δ-complex structure on a space. Its building block is the simplex.

Definition 3.1.1: n-Simplex

An *n-simplex* is the smallest convex set in $\mathbb{R}^m$ containing $n+1$ points $v_0, v_1, ..., v_n$ that do not lie in an affine subspace of dimension less than n, together with an ordering of the vertices. Equivalently, $v_0, v_1, ..., v_n$ are *affinely independent*: the vectors $v_1 - v_0, \ldots, v_n - v_0$ are linearly independent. The points v_i are the vertices of the simplex. If it is important to represent the vertices and their order, we denote it as $(v_0, ..., v_n)$. Each point of the simplex can be written as $\sum_i t_i v_i$ with $t_i \geq 0$ and $\sum_i t_i = 1$, uniquely by affine independence, and the numbers $t_0, \ldots, t_n$ are its *barycentric coordinates*.

Note that the ordering is part of the definition, which is why the vertices are in an $(n+1)$-tuple notation. The ordering orients the edges. The idea behind the ordering is due to the fact that rotations only produce "half" of the possible rigid-body isomorphisms of a shape. One must reflect to get the other half. In linear algebra, this is represented as the determinant being positive or negative. This is important when we are integrating and must choose a direction to integrate, and for manifolds when we must determine the Stokes orientation of the boundary. We shall also see that there are surfaces with no orientation such as the Klein bottle, and these shall be created by gluing edges at opposite orientations. For example, two 2-simplices can be glued along a common edge, and then the other edges could be glued in such a way to produce a torus T^2, a real projective plane $\mathbb{R}\mathrm{P}^2$, or a Klein bottle:

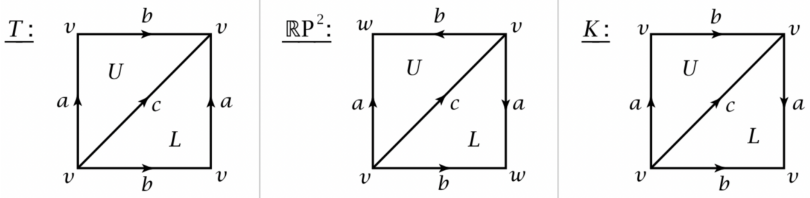


Figure 3.1: Different ways to glue a square

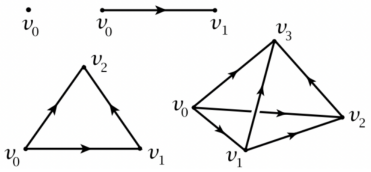
If we delete a vertex from an n-simplex, we get an $(n-1)$-simplex called the *face* of the simplex. If $(v_0, v_1, ..., v_n)$ is an n-simplex, we shall represent a face via

$$(v_0, v_1, ..., \hat{v}_i, ..., v_n)$$

where $\hat{v}_i$ indicates the vertex is removed. By convention, the vertices of a face (or any sub-simplex) will always be ordered according to their ordering of the larger simplex.

Example 3.1.2: n-simplex

1. The 0-, 1-, 2- and 3-simplices:



2. The standard n-simplex is defined as:

$$\Delta^n = \left\{ (t_0, t_1, ..., t_n) \in \mathbb{R}^{n+1} \;\middle|\; \sum_i t_i = 1,\ t_i \geq 0,\ 0 \leq i \leq n \right\}$$

whose vertices are the unit vectors along the coordinate axes. For example, Δ^2 is:

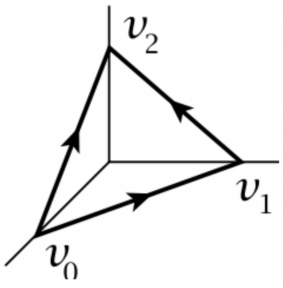


Figure 3.2: Δ^2

Definition 3.1.3: Boundary

Let Δ^n be the standard simplex. Then the union of all the faces of Δ^n is called the *boundary* of Δ^n and is written $\partial\Delta^n$. The *open simplex* $\text{int}\Delta^n$ is the interior of Δ^n.

Definition 3.1.4: Δ-Complex

Let X be a topological space. Then a Δ-*complex structure*[a] on X is a collection of continuous maps $\sigma_\alpha : \Delta^{n_\alpha} \to X$ with n_α depending on the index α such that

1. The restriction $\sigma_\alpha|_{\text{int}\Delta^{n_\alpha}}$ is injective, and each point $x \in X$ is in the image of exactly one restriction $\sigma_\alpha|_{\text{int}\Delta^{n_\alpha}}$. We label $\sigma_\alpha|_{\text{int}\Delta^{n_\alpha}}$ as e^n_α.
2. The restriction of each σ_α to a face of Δ^{n_α}, identified with $\Delta^{n_\alpha-1}$ by the affine map preserving the order of the vertices, is one of the maps $\sigma_\beta : \Delta^{n_\alpha-1} \to X$.
3. A set $A \subseteq X$ is open if and only if $\sigma_\alpha^{-1}(A)$ is open in Δ^{n_α} for each σ_α (the topology of X is the final, or quotient, topology determined by the maps σ_α).

[a] Not to be confused with a *Simplicial-complex structure*, see [21, p.107]

Figure 3.1 of (a covering space of) a torus, $\mathbb{R}\text{P}^2$, and a Klein bottle are examples of Δ-complexes. The torus and the Klein bottle each have six maps (one 0-simplex, three 1-simplices, two 2-simplices, each labeled on the diagram), while $\mathbb{R}\text{P}^2$ has two 0-simplices (v, w in the covering space). We see here the importance of the orientation of the edges. We see that X can be represented as a quotient of disjoint simplices $\Delta^{n_\alpha}_\alpha$ with $\Delta^{n_\alpha-1}_\beta$ corresponding to the restriction of σ_β of σ_α to the particular face. Hence, any Δ-complexes can be described "combinatorially" as collections of n-simplices together with functions associating each face of an n-simplex to an $(n-1)$-simplex (similar to a cell-complex).

For a geometric n-simplex $\sigma = [w_0, \dots, w_n]$ in some $\mathbb{R}^N$ (definition 3.1.1), write $t_0, \dots, t_n$ for its barycentric coordinates, so that $y = \sum_i t_i(y)\, w_i$ with $\sum_i t_i(y) = 1$ and all $t_i(y) \geq 0$, and write $\partial\sigma$ for the union of the proper faces, the set of points with some $t_i(y) = 0$.

Lemma 3.1.5: Simplices Are Disks

Let σ be a geometric n-simplex. Then there is a homeomorphism $(D^n, S^{n-1}) \to (\sigma, \partial\sigma)$ of pairs.

***Proof*:**
For $n = 0$ both pairs are a point and the empty set, so let $n \geq 1$. Let b be the barycentre, with all $t_i(b) = \frac{1}{n+1}$, and let V be the span of the vectors $w_i - b$, a vector space of dimension n since the w_i are affinely independent. Fix a linear isometry $V \cong \mathbb{R}^n$, giving V the norm $|\cdot|$ and unit ball D^n. The coordinates t_i extend to affine functions on $b + V$ with $\sum_i t_i = 1$, and for $v \in V$ set

$$\mu(v) = 1 - (n+1)\min_i t_i(b+v)$$

Each $t_i(b + \lambda v) = \frac{1}{n+1} + \lambda\big(t_i(b+v) - \frac{1}{n+1}\big)$ is affine in λ, so for $\lambda \geq 0$ the minimum over i transforms in the same way, and $\mu(\lambda v) = \lambda\mu(v)$. The function μ is continuous, and $\mu(v) > 0$ for $v \neq 0$, since the coordinates of $b + v \neq b$ are not all equal to $\frac{1}{n+1}$ and sum to 1. Moreover

$b+v \in \sigma$ exactly when $\min_i t_i(b+v) \geq 0$, that is $\mu(v) \leq 1$, and $b+v \in \partial\sigma$ exactly when $\mu(v) = 1$. Define

$$\Psi(x) = b + \frac{|x|}{\mu(x)}\, x \qquad \Phi(y) = \frac{\mu(y-b)}{|y-b|}\,(y-b)$$

for $x \in D^n - \{0\}$ and $y \in \sigma - \{b\}$, with $\Psi(0) = b$ and $\Phi(b) = 0$. By homogeneity $\mu(\Psi(x)-b) = |x|$ and $|\Phi(y)| = \mu(y-b)$, so Ψ maps D^n into σ and S^{n-1} into $\partial\sigma$, and Φ maps σ into D^n and $\partial\sigma$ into S^{n-1}, and the same homogeneity gives $\Psi\Phi = \mathrm{id}$ and $\Phi\Psi = \mathrm{id}$. Both maps are continuous away from 0 and b. With $c > 0$ the minimum of μ on the compact unit sphere of V, $|\Psi(x) - b| = |x|^2/\mu(x) \leq |x|/c$ and $|\Phi(y)| = \mu(y-b)$ tend to 0 as $x \to 0$ and $y \to b$, so Ψ is a homeomorphism of pairs, as claimed.

Proposition 3.1.6: Δ-Complexes Are CW Complexes

Let X be a Δ-complex, and let X^n be the union of the images of the σ_α with $n_\alpha \leq n$. Then X is a CW complex with n-skeleton X^n, one n-cell for each σ_α with $n_\alpha = n$, and characteristic maps $\sigma_\alpha \circ \Psi$, where $\Psi\colon (D^n, S^{n-1}) \to (\Delta^n, \partial\Delta^n)$ is a homeomorphism of pairs (lemma 3.1.5). In particular X is Hausdorff (theorem 1.2.3).

***Proof*:**
Write $e_\alpha = \sigma_\alpha(\mathrm{int}\Delta^{n_\alpha})$ for the open simplices, which partition X by condition (1).

Step 1: Preimages of skeleta. Applying condition (2) repeatedly, the restriction of σ_β to a k-dimensional face F of Δ^{n_β} is $\sigma_\gamma \circ \iota_F^{-1}$ for some γ with $n_\gamma = k$, where $\iota_F\colon \Delta^k \to F$ is the order-preserving affine homeomorphism, so σ_β maps the open face $\mathrm{int}F$ onto e_γ. The open faces of Δ^{n_β} partition it, so the image of σ_β is a union of open simplices of dimension at most n_β, and X^n is the union of the open simplices of dimension at most n. Since the open simplices are pairwise disjoint, $\sigma_\beta^{-1}(X^n)$ is the union of the faces of Δ^{n_β} of dimension at most n, a closed set. Hence X^n is closed in X by condition (3), and $\sigma_\beta(\partial\Delta^{n_\beta}) \subseteq X^{n_\beta - 1}$.

Step 2: The topology of X^n. We show that $A \subseteq X^n$ is closed in X^n if and only if $\sigma_\gamma^{-1}(A)$ is closed for every γ with $n_\gamma \leq n$. If A is closed in X^n, it is closed in X, since X^n is closed, so every $\sigma_\gamma^{-1}(A)$ is closed. Conversely, for any β the set $\sigma_\beta^{-1}(A)$ lies in the union of the finitely many faces F of Δ^{n_β} of dimension at most n, and its intersection with F is $\iota_F(\sigma_\gamma^{-1}(A))$ for the γ of Step 1. This is closed, as the image of a closed subset of the compact space Δ^k under the embedding ι_F. So $\sigma_\beta^{-1}(A)$ is closed for every β, and A is closed in X by condition (3), hence in X^n.

Step 3: Attaching cells. By Step 2 every subset of X^0 is closed, so X^0 is discrete. Let $n \geq 1$, and let $q\colon X^{n-1} \sqcup_{n_\alpha = n} \Delta_\alpha^n \to X^n$ be the inclusion on X^{n-1} and σ_α on Δ_α^n. It is continuous and surjective. By condition (1), σ_α is injective on $\mathrm{int}\Delta_\alpha^n$ with image e_α, disjoint from X^{n-1} and from the other e_β, and by Step 1 $\sigma_\alpha(\partial\Delta_\alpha^n) \subseteq X^{n-1}$. So q identifies exactly each $x \in \partial\Delta_\alpha^n$ with $\sigma_\alpha(x)$. If $q^{-1}(A)$ is closed, then $\sigma_\alpha^{-1}(A)$ is closed for $n_\alpha = n$, and for $n_\gamma < n$ the set $\sigma_\gamma^{-1}(A) = \sigma_\gamma^{-1}(A \cap X^{n-1})$ is closed since $A \cap X^{n-1}$ is closed in X^{n-1}, so A is closed in X^n by Step 2. Hence q is a quotient map, and composing with Ψ shows that X^n is obtained from X^{n-1} by attaching n-cells along the maps $\sigma_\alpha \circ \Psi|_{S^{n-1}}$, as in definition 1.2.1.

Step 4: The weak topology. Every point lies in some open simplex, so $X = \bigcup_n X^n$. If $A \cap X^n$ is closed in X^n for every n, then $\sigma_\alpha^{-1}(A) = \sigma_\alpha^{-1}(A \cap X^{n_\alpha})$ is closed for every α, so A is closed by condition (3), as we sought to show.

Now let X be a topological space with a Δ-complex structure. The structure records the simplices in each dimension:

Definition 3.1.7: Group Of n-Chains

Let X be a Δ-complex. Then $\Delta_n(X)$ is the free abelian group generated by the open n-simplices e^n_α of X. Elements of $\Delta_n(X)$ are called *n-chains*, and can be written as

$$\sum_\alpha n_\alpha e^n_\alpha$$

or similarly $\Sigma_\alpha n_\alpha \sigma_\alpha$ where $\sigma_\alpha : \Delta^n \to X$ is the characteristic map of e^n_α.

These groups generalize the notion of a path (or surface in higher dimension) without a basepoint. A loop (or closed surface) is a chain whose boundary "cancels itself out". Each face in the boundary carries an orientation, recorded by a sign.

Let $X = \Delta^n$ be the standard Δ-complex so that it can be written as $(v_0, v_1, ..., v_n)$. If $n = 1$, then we shall let:

$$\partial[v_0, v_1] = [v_1] - [v_0]$$

which can be thought of as a direction of integration. Another way of looking at it is to take the determinant of

$$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

placing the coordinates v_0 at $(1, 0)$ and v_1 at $(0, 1)$. If we swap them, then:

$$\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$

and hence the orientation is flipped. If $n = 2$, then

$$\partial[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1]$$

We can visually imagine this as standing inside the triangle and looking at the edges that are left after removing a vertex: two of them go right to left, and one will go left to right. As a determinant, we get:

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \cdot \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \cdot 1$$

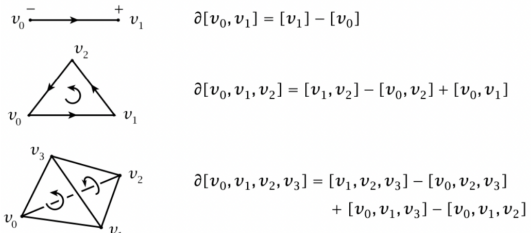
We can also think of it as the orientation that is needed to make a *cycle*, in that the resulting shape is now the path that needs to be taken to form a closed loop. Similarly, it is a consistent orientation of the boundary of the inside. Similarly, for $n = 3$, we do the same alternating sum:

$$\partial[v_0, v_1, v_2, v_3] = [v_1, v_2, v_3] - [v_0, v_2, v_3] + [v_0, v_1, v_3] - [v_0, v_1, v_2]$$

Visually, we can imagine being inside the tetrahedron and looking at the triangular faces. Then the 3 vertices of each face, in order, go either clockwise or anticlockwise. The same determinant trick will again give us a similar result: if we flip any two positions, we would get a negative. We cannot

directly have the same path intuition, but the equivalent in dimension 3 is that each face now has an outwards facing orientation so that all the faces have the same orientation and the interior of Δ^3 now has consistently oriented boundary.

The following figure illustrates this:



More generally, we see the following emerging pattern:

$$\partial(v_0, v_1, ..., v_n) = \sum_i (-1)^i (v_0, ..., \hat{v}_i, ..., v_n)$$

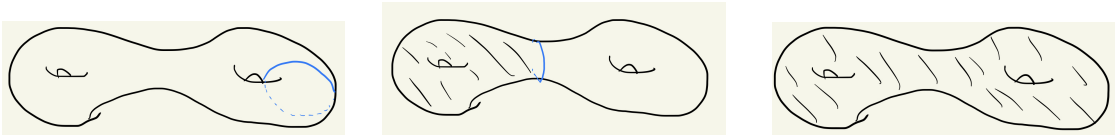
Taking the Δ-complex structure of a space X, we see that if the boundary is zero (for example, for a closed loop), then we have a *cycle*. Defining ∂ on the generators and extending linearly makes it a homomorphism:

Definition 3.1.8: Boundary Homomorphism

Let X be a Δ-complex. Then the *boundary homomorphism* $\partial_n : \Delta_n(X) \to \Delta_{n-1}(X)$ is defined by:

$$\partial_n(\sigma_\alpha) = \sum_i (-1)^i \, \sigma_\alpha|_{(v_0,...,\hat{v}_i,...,v_n)}$$

The boundary map may or may not be zero, and not all cycles of lower dimension are the boundary of a higher dimensional chain. The following figure is drawn for CW complexes, whose boundaries are easier to picture (we shall see that the homology computed from a Δ-complex agrees with the homology computed from a CW structure):



(a) Cannot be in image of ∂_2 (b) In image of ∂_2 (c) Image in ∂_2 is Zero

Figure 3.3: Cycles and their Relations

Because of the signs, the boundary of a boundary is zero:

Lemma 3.1.9: Complex Condition

The composition

$$\Delta_n(X) \xrightarrow{\partial_n} \Delta_{n-1}(X) \xrightarrow{\partial_{n-1}} \Delta_{n-2}(X)$$

produces the zero map, $\partial_{n-1} \circ \partial_n = 0$

***Proof*:**
Plugging in, we get:

$$\begin{aligned}\partial_{n-1}(\partial_n(\sigma)) = &\sum_{j<i}(-1)^i(-1)^j \ \sigma|_{(v_0,\dots,\hat{v}_j,\dots,\hat{v}_i,\dots,v_n)} \\ &+\sum_{j>i}(-1)^i(-1)^{j-1} \ \sigma|_{(v_0,\dots,\hat{v}_i,\dots,\hat{v}_j,\dots,v_n)}\end{aligned}$$

where the exponent $j-1$ in the second sum appears because, once v_i is removed with $i<j$, the vertex v_j sits in position $j-1$. Switching i and j in the second sum turns it into the negative of the first, so the two sums on the right hand side cancel.

The property in lemma 3.1.9 can be studied independently of the geometry from which it started, and it leads to many results in algebraic geometry and homological algebra. Its purely algebraic form is the following:

Definition 3.1.10: Chain Complex and Exact Sequence

A *chain complex* $C_\bullet$ is a family of abelian groups C_n, $n \in \mathbb{Z}$, with homomorphisms $\partial_n \colon C_n \to C_{n-1}$ such that $\partial_n \partial_{n+1} = 0$ for all n, in other words $\mathrm{im}\partial_{n+1} \subseteq \ker \partial_n$. If equality holds for every n, the sequence is called an *exact sequence*. In most complexes of this book $C_n = 0$ for $n < 0$, and the complex is written

$$\cdots \to C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \to \cdots \to C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} 0$$

Not every cycle is the boundary of a chain of one dimension higher. The following group measures to what degree cycles "fail" to be boundaries of higher chains:

Definition 3.1.11: nth [Simplicial] Homology Group

Given a chain complex, the *nth homology group* is defined to be

$$H_n = \frac{\ker(\partial_n)}{\mathrm{im}\partial_{n+1}}$$

Elements of $\ker \partial_n$ are called *cycles*, elements of $\mathrm{im}\partial_{n+1}$ are called *boundaries*, elements of H_n are called *homology classes*, and if two elements are in the same coset, they are called *homologous* (meaning their difference is a boundary).
In the case where $C_n = \Delta_n(X)$, the homology group will be denoted $H_n^{\Delta}(X)$ and is called the *nth Simplicial homology group* of X

The homology group measures to what degree cycles are *not* the boundary of a higher dimensional chain. If all cycles are boundaries, then the homology is trivial, and if not the homology group is a way of quantifying the obstruction. Since most homology groups in this book will be finitely generated, we shall use the free part and the torsion part of a finitely generated abelian group to interpret this obstruction.

To compare the homology of two complexes one needs maps between them. A family of homomorphisms induces a map on homology when it carries cycles to cycles and boundaries to boundaries, and commuting with the boundary maps guarantees both.

Definition 3.1.12: Chain Map

Let $C_\bullet$ and $D_\bullet$ be chain complexes (definition 3.1.10). A *chain map* $f\colon C_\bullet \to D_\bullet$ is a family of homomorphisms $f_n\colon C_n \to D_n$ with $\partial f_n = f_{n-1}\partial$ for every n, that is, such that the diagram

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & C_{n+1} & \xrightarrow{\partial} & C_n & \xrightarrow{\partial} & C_{n-1} & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle f_{n+1}} & & \downarrow{\scriptstyle f_n} & & \downarrow{\scriptstyle f_{n-1}} & & \\
\cdots & \longrightarrow & D_{n+1} & \xrightarrow{\partial} & D_n & \xrightarrow{\partial} & D_{n-1} & \longrightarrow & \cdots
\end{array}$$

commutes. Chain complexes of abelian groups and chain maps, composed degreewise, form a category $\mathsf{Ch}(\mathbb{Z})$ (definition 1.1.3).

Proposition 3.1.13: Chain Maps Induce Maps on Homology

A chain map $f\colon C_\bullet \to D_\bullet$ carries cycles to cycles and boundaries to boundaries, so it induces homomorphisms $f_*\colon H_n(C) \to H_n(D)$, $[z] \mapsto [f(z)]$. They satisfy $(g \circ f)_* = g_* \circ f_*$ and $(\mathrm{id})_* = \mathrm{id}$, so each H_n is a functor $\mathsf{Ch}(\mathbb{Z}) \to \mathsf{Ab}$, and so is the *graded homology* $H_\bullet(C) = (H_n(C))_n$, with values in graded abelian groups and degree-preserving homomorphisms.

***Proof*:**
If $\partial z = 0$, then $\partial f(z) = f(\partial z) = 0$, and $f(\partial b) = \partial f(b)$ for every b, so f maps $\ker \partial_n$ into $\ker \partial_n$ and $\mathrm{im}\partial_{n+1}$ into $\mathrm{im}\partial_{n+1}$ and induces f_* on the quotients. The identities hold on representatives, $(g \circ f)_*[z] = [g(f(z))] = g_* f_*[z]$ and $\mathrm{id}_*[z] = [z]$, and the graded statement is the collection of these over all n, completing the proof.

Example 3.1.14: Simplicial Homology Group

For reference, here are Δ-complexes for the torus, $\mathbb{R}\mathrm{P}^2$, and Klein bottle:

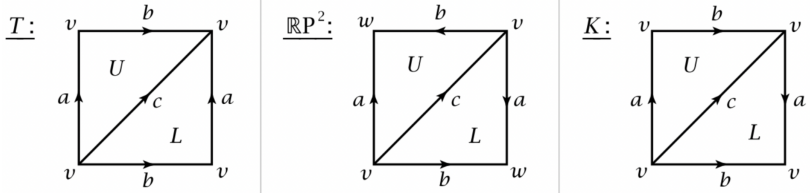


Figure 3.4: Reference Figure for Torus, Projective space, and Klein Bottle

1. Let $X = S^1$, with one vertex v and one edge e. Then $\Delta_0(S^1)$ and $\Delta_1(S^1)$ are both $\mathbb{Z}$. The map $\partial_1 : \Delta_1(S^1) \to \Delta_0(S^1)$ is $\partial_1 \equiv 0$ since

$$\partial e = v - v$$

Hence $\mathrm{im}\partial_1 = 0$. On the other hand, we always have $\ker \partial_0 = \mathbb{Z}$, and so $H_0^\Delta(S^1) = \mathbb{Z}$. Since $\partial_1 = 0$, $\ker \partial_1 = \Delta_1(S^1) = \mathbb{Z}$. For $n \geq 2$, $\Delta_n(S^1) = 0$ since there are no simplices in these dimensions. Hence:

$$H_n^\Delta(S^1) = \begin{cases} \mathbb{Z} & n \in \{0, 1\} \\ 0 & n \geq 2 \end{cases}$$

Notably, if the boundary maps are all zero, the homology groups of the complex are isomorphic to the chain groups $\Delta_n(X)$.

2. Let $X = T$ be the torus given the Δ-complex in fig. 3.4, namely it has one vertex, three edges a, b, c, and two 2-simplices U, L, giving us $\Delta_0(T) = \mathbb{Z}$, $\Delta_1(T) = \mathbb{Z}^3$, and $\Delta_2(T) = \mathbb{Z}^2$. Like before, $\mathrm{im}\partial_1 = 0$, so $H_0^\Delta(T) = \Delta_0(T) = \mathbb{Z}$. Next, $\ker \partial_1 = \mathbb{Z}^3$, and since $\partial_2 U = \partial_2 L = a + b - c$, writing $\Delta_1(T) = \ker \partial_1$ with basis $\{a, b, a + b - c\}$ we have:

$$H_1^\Delta(T) = \mathbb{Z} \oplus \mathbb{Z}$$

with basis $\{[a], [b]\}$. Since there are no 3-simplices, we have that $H_2^\Delta(T) = \ker \partial_2$, which is generated by $U - L$ since

$$\partial(pU + qL) = (p + q)(a + b - c) = 0 \qquad \iff \qquad p = -q$$

Hence, we overall have:

$$H_n^\Delta(T) = \begin{cases} \mathbb{Z} & n = 0 \\ \mathbb{Z} \oplus \mathbb{Z} & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & n \geq 3 \end{cases}$$

3. Let $X = \mathbb{R}\mathrm{P}^2$ with the Δ-complex structure given by fig. 3.4, with two vertices v, w, three edges a, b, c and two 2-simplices U, L. Then $\mathrm{im}\partial_1$ is generated by $w - v$, so

$$H_0^\Delta(\mathbb{R}\mathrm{P}^2) \cong \mathbb{Z}$$

Since $\partial_2 U = -a+b+c$ and $\partial_2 L = a-b+c$, we get that ∂_2 is injective meaning its kernel is zero and so:

$$H_2^\Delta(\mathbb{R}\mathrm{P}^2) = 0$$

Next, ∂_1 maps $\Delta_1(\mathbb{R}\mathrm{P}^2) \cong \mathbb{Z}^3$ onto $\mathrm{im}\partial_1 \cong \mathbb{Z}$, so $\ker\partial_1 = \mathbb{Z}\oplus\mathbb{Z}$ (the paths that, under the covering map, are closed). From the figure, its basis is $a-b$ and c. The quotient $\ker\partial_1/\mathrm{im}\partial_2$ is a finite group since the two groups have the same rank. To find the quotient, take as basis for $\ker\partial_1$ $(a-b+c, c)$. Then $\mathrm{im}\partial_2$ is generated by $a-b+c$ and $(a-b+c)+(-a+b+c) = 2c$, so the quotient has order 2, thus:

$$H_1^\Delta(\mathbb{R}\mathrm{P}^2) = \mathbb{Z}/2\mathbb{Z}$$

Giving us:

$$H_n(\mathbb{R}\mathrm{P}^2) = \begin{cases} \mathbb{Z} & n=0 \\ \mathbb{Z}/2\mathbb{Z} & n=1 \\ 0 & n\geq 2 \end{cases}$$

The torsion element can be thought of as reflecting the "non-orientability" of $\mathbb{R}\mathrm{P}^2$, the "twists" in our complex. We shall calculate $H_k(\mathbb{R}\mathrm{P}^n)$ using cellular homology in section 3.5.

4. For a moment assume homeomorphisms preserve homology groups. Let $X = S^n$ with the Δ-complex structure on S^n given by taking two copies of Δ^n and identifying their boundaries via identity maps, giving us a space homeomorphic to S^n. Then to compute $H_n^\Delta(S^n)$, we see that $\ker\partial_n$ is an infinite cyclic group generated by $U-L$ where U, L are the two n-simplices, since $\partial_n U = \partial_n L$ and this boundary is nonzero, its $n+1$ faces being distinct simplices. There are no $(n+1)$-simplices, so the image of ∂_{n+1} is trivial and

$$H_n^\Delta(S^n) \cong \mathbb{Z}$$

The other groups need more tools, and corollary 3.4.15 computes them all. For $n=2$ the answer is

$$H_k(S^2) = \begin{cases} \mathbb{Z} & k=0 \\ 0 & k=1 \\ \mathbb{Z} & k=2 \\ 0 & k\geq 3 \end{cases}$$

5. (Klein Bottle)

As the simplicial homology groups of a finite Δ-complex are finitely generated, they are characterized by the free-rank and the torsion. Historically, these were given the following name:

Definition 3.1.15: Betti Number And Torsion Coefficients

The number of $\mathbb{Z}$ summands of $H_n(X)$ is called the nth Betti number of X, and the integers specifying the orders of the finite cyclic summands are called the *torsion coefficients*.

In the early development of homology theory in the 1900s, it was not the groups $H_n(X)$ that were studied but the *Betti numbers* and *torsion coefficients*. It was not until the 1920s that Emmy Noether brought about the more algebraic point of view.

3.2 Singular Homology

Two questions about simplicial homology are hard to answer from a given Δ-complex: whether the groups are invariant under homeomorphism, and whether they are invariant under homotopy equivalence. The *singular homology groups* are easier to work with theoretically, though harder to visualize, and are defined for more general spaces. We shall show that they agree with the simplicial homology groups.

Definition 3.2.1: Singular n-Simplex

Let X be a space. Then a *singular n-simplex* in X is a continuous map $\sigma\colon \Delta^n \to X$, that is an element of the set

$$\mathrm{Hom}_{\mathsf{Top}}(\Delta^n, X)$$

The free abelian group $C_n(X)$ with basis this set is the group of *[singular] n-chains* in X.

The term singular expresses how the map σ need only be continuous. There may be "singularities" where the image does not look like a simplex. This generality also means that each space X has many more singular n-simplices, often infinitely many, and the boundary relies on the domain of σ rather than a geometric intuition in the codomain. The elements of $C_n(X)$ are the finite formal sums $\sum_i n_i\sigma_i$. Just as before, we define a boundary map:

$$\partial_n(\sigma) = \sum_i (-1)^i \, \sigma|_{(v_0,\ldots,\hat{v}_i,\ldots,v_n)}$$

There is implicitly an identification of $(v_0, ..., \hat{v}_i, ..., v_n)$ with Δ^{n-1} preserving the ordering of vertices so that we can think of the map $\sigma|_{(v_0,\ldots,\hat{v}_i,\ldots,v_n)}$ as being from $\Delta^{n-1} \to X$ (i.e. of a singular $(n-1)$-simplex). When not ambiguous, we shall write the boundary map $\partial_n : C_n(X) \to C_{n-1}(X)$ simply as ∂. Then we may concisely write the condition $\partial_n\partial_{n+1} = 0$ as $\partial^2 = 0$, and again elements of the image of ∂ are called boundaries, while elements of the kernel are called cycles. Thus, we again have the following definition:

Definition 3.2.2: Singular Homology Group

Let X be a topological space with singular n-simplices. Then the nth *singular homology group* is:

$$H_n(X) = \frac{\ker \partial_n}{\mathrm{im}\partial_{n+1}}$$

Unlike simplicial homology groups, singular homology groups are preserved under homeomorphisms directly from the definition: a homeomorphism $\varphi\colon X \to Y$ gives a bijection $C_n(X) \to C_n(Y)$, $f \mapsto \varphi \circ f$, compatible with the boundary maps, since it maps singular simplices of each dimension to singular simplices of the same dimension (proposition 3.3.3). Note that $C_n(X)$ is usually uncountable and may seem hard to work with, but as we shall see this flexibility gives us many more easily provable results. In fact, the singular homology group can be identified with the simplicial homology group by a special construction on a space X (labeled $S(X)$), by taking for each singular n-simplex σ an n-simplex Δ^n_σ, with each face of Δ^n_σ attached to the $(n-1)$-simplex of $S(X)$ given by the restriction of σ to that face. Then $\Delta_n(S(X)) = C_n(X)$ with the same boundary maps, so $H^\Delta_n(S(X)) = H_n(X)$ for all n, though $S(X)$ is usually huge.

In the case where X is a smooth manifold, a rough idea of an element of $H_n(X)$ is that it can be interpreted as the collection of all possible oriented sub-manifolds. The orientation information comes from the boundary map which "preserves" orientation information. Then elements of $H_1(X)$ are represented by oriented loops, elements of $H_2(X)$ are represented by maps of closed oriented surfaces into X. The term represented here can be seen as saying that the domain when mapping into X can be "thought of" as a loop, surface, etc. An oriented 1-cycle $\coprod_\alpha S^1_\alpha \to X$ is zero in $H_1(X)$ if and only if it extends to a map of a compact oriented surface with boundary $\coprod_\alpha S^1_\alpha$ into X, and the analogous statement holds for 2-cycles [21, §2.1, p. 109]. It was hoped in the early days of homology theory that this connection with boundaries of manifolds would continue in all higher dimensions, but it does not. For example, the map from $\mathbb{C}\mathrm{P}^2$ to a point is an oriented 4-cycle that is zero in $H_4(\mathrm{pt}) = 0$, but it does not extend over any compact smooth 5-manifold W with $\partial W = \mathbb{C}\mathrm{P}^2$. Chapter 7 proves this. The space $\mathbb{C}\mathrm{P}^2$ has one cell in each of the dimensions 0, 2 and 4 (example 1.2.5), so $\chi(\mathbb{C}\mathrm{P}^2) = 3$, and that chapter shows that the Stiefel-Whitney number $w_4[\mathbb{C}\mathrm{P}^2]$ is $\chi(\mathbb{C}\mathrm{P}^2)$ reduced mod 2 (section 7.4), hence 1, while every Stiefel-Whitney number of a boundary vanishes (section 7.6). These notions are explored further in a theory known as *cobordism* (section 8.5).

Proposition 3.2.3: Homology group and path-connected components

Corresponding to the decomposition of a space X into its path-components X_α,

$$H_n(X) \cong \bigoplus_\alpha H_n(X_\alpha)$$

***Proof*:**
Since a simplex is path-connected, the image of a singular simplex is path-connected and so lies in a single path-component. Hence $C_n(X)$ splits as the direct sum of subgroups $C_n(X_\alpha)$. Then the boundary maps ∂_n preserve this direct sum decomposition, taking $C_n(X_\alpha)$ to $C_{n-1}(X_\alpha)$, hence $\ker \partial_n$ and $\mathrm{im}\partial_{n+1}$ split similarly as direct sums, hence the homology group splits, that is:

$$H_n(X) \cong \bigoplus_\alpha H_n(X_\alpha)$$

as we sought to show.

We shall later see that $\tilde{H}_n(\bigvee_\alpha X_\alpha) \cong \bigoplus_\alpha \tilde{H}_n(X_\alpha)$ for good pairs $(X_\alpha, \{x_\alpha\})$ (corollary 3.4.19). The homology of a product $X \times Y$ is more complicated, and is computed in section 4.3.

Proposition 3.2.4: 0th Singular Homology Group for Path-Connected Space

If X is nonempty and path-connected, then $H_0(X) \cong \mathbb{Z}$. Hence, for any space X, $H_0(X)$ is a direct sum of $\mathbb{Z}$ for each path component.

***Proof*:**
By definition, $H_0(X) = C_0(X)/\mathrm{im}\partial_1$ (since $\partial_0 = 0$). Define the homomorphism, surjective since X is nonempty,

$$\epsilon : C_0(X) \to \mathbb{Z} \qquad \sum_i n_i \sigma_i \mapsto \sum_i n_i$$

and show that $\ker\epsilon = \text{im}\partial_1$, giving us the structure of $H_0(X)$.

($\text{im}\partial_1 \subseteq \ker\epsilon$) Note that each singular 1-simplex $\sigma : \Delta^1 \to X$ has

$$\epsilon(\partial_1(\sigma)) = \epsilon(\sigma|_{(v_1)} - \sigma|_{(v_0)}) = 1 - 1 = 0$$

hence, they are contained in the kernel.

($\text{im}\partial_1 \supseteq \ker\epsilon$) Suppose $\epsilon\left(\sum_i n_i\sigma_i\right) = \sum_i n_i = 0$. If we can find an element in $C_1(X)$ for which $\sum_i n_i\sigma_i$ is the boundary, then we are done. Since the σ_i are singular 0-simplices, their image is a point in X. Fix $x_0 \in X$. Since X is path-connected, for each i choose a path $\tau_i : I \to X$ where $\tau_i(0) = x_0$ and $\tau_i(1) = \sigma_i(v_0)$, and let σ_0 be the singular 0-simplex with image x_0. Then τ_i can be seen as a singular 1-simplex, namely $\tau_i : [v_0, v_1] \to X$, and

$$\partial\tau_i = \sigma_i - \sigma_0$$

Thus:

$$\partial\left(\sum_i n_i\tau_i\right) = \sum_i n_i\sigma_i - \sum_i n_i\sigma_0 = \sum_i n_i\sigma_i$$

So $\sum_i n_i\sigma_i$ is a boundary, giving $\ker\epsilon \subseteq \text{im}\partial_1$, as we sought to show.

So ϵ induces $H_0(X) \cong \mathbb{Z}$. The statement for an arbitrary space follows by applying this to each path component and using proposition 3.2.3.

Proposition 3.2.5: Homology Group of a Point

If X is a point, then

$$H_0(X) \cong \mathbb{Z} \qquad H_n(X) = 0,\ \forall n > 0$$

***Proof*:**

There is a unique singular n-simplex σ_n for each n, and $\partial(\sigma_n) = \sum_i (-1)^i\sigma_{n-1}$ is a sum of $n+1$ terms, and thus is 0 when n is odd and σ_{n-1} when n is even ($n \neq 0$). Thus, we get

$$\cdots \to \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0$$

with the boundary maps alternating between being an isomorphism and the trivial map except at the last $\mathbb{Z}$. In each degree $n > 0$, either the outgoing map is an isomorphism, so the cycles are 0, or the incoming map is an isomorphism, so every cycle is a boundary. Hence the homology groups of this complex are all trivial except for $H_0 \cong \mathbb{Z}$, as $\partial_1 = 0$.

It is sometimes useful to define a modified version of homology for which the homology of a point is trivial in all dimensions (including 0).

Definition 3.2.6: Reduced Homology Group

Define the *reduced homology groups* $\tilde{H}_n(X)$ to be the homology groups of the augmented chain complex:

$$\cdots \to C_2(X) \xrightarrow{\partial_2} C_1(X) \xrightarrow{\partial_1} C_0(X) \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where

$$\epsilon\left(\sum_i n_i\sigma_i\right) = \sum_i n_i$$

We require X to be nonempty, or else we would have a nontrivial homology group in dimension -1. Since $\epsilon \circ \partial_1 = 0$, ϵ vanishes on $\text{im}(\partial_1)$, and hence induces a map $H_0(X) \to \mathbb{Z}$ with kernel $\tilde{H}_0(X)$. This map is surjective because X is nonempty, and the sequence $0 \to \tilde{H}_0(X) \to H_0(X) \to \mathbb{Z} \to 0$ splits because $\mathbb{Z}$ is free. In degrees $n > 0$ the two chain complexes agree. Thus:

$$H_0(X) \cong \tilde{H}_0(X) \oplus \mathbb{Z} \qquad H_n(X) \cong \tilde{H}_n(X),\ n > 0$$

We can think of the extra $\mathbb{Z}$ in the augmented chain as representing the unique map $(\emptyset) \to X$ where $\emptyset$ is the empty simplex with no vertices. Then the augmentation map ϵ is the boundary map since:

$$\partial(v_0) = (\hat{v}_0) = (\emptyset)$$

$H_1(X)$ is the abelianization of $\pi_1(X)$ whenever X is path connected. For a proof see [21, 2.A].

3.3 Homotopy Invariance

Homotopy equivalent spaces have isomorphic homology groups. More precisely, a continuous map $f : X \to Y$ induces a homomorphism $f_* : H_n(X) \to H_n(Y)$ for each n, functorially in f, and f_* is an isomorphism if f is a homotopy equivalence.

Definition 3.3.1: Pushforward

Let $f : X \to Y$ be a continuous map. Then f induces a homomorphism $f_\sharp : C_n(X) \to C_n(Y)$ by mapping

$$\sigma \mapsto f \circ \sigma$$

which then is extended linearly:

$$f_\sharp\left(\sum_i n_i\sigma_i\right) = \sum_i n_i f_\sharp(\sigma_i) = \sum_i n_i f \circ \sigma_i$$

Lemma 3.3.2: The Pushforward Is a Chain Map

Let $f\colon X \to Y$ and $g\colon Y \to Z$ be continuous maps.

1. $f_\sharp\colon C_\bullet(X) \to C_\bullet(Y)$ is a chain map, that is $f_\sharp \partial = \partial f_\sharp$.
2. $(g \circ f)_\sharp = g_\sharp \circ f_\sharp$ and $(\mathrm{id}_X)_\sharp = \mathrm{id}$.
3. If $A \subseteq X$ and $B \subseteq Y$ satisfy $f(A) \subseteq B$, then $f_\sharp$ carries $C_\bullet(A)$ into $C_\bullet(B)$.
4. $\epsilon \circ f_\sharp = \epsilon$ on $C_0(X)$, for the augmentations ϵ of definition 3.2.6.

***Proof*:**
Both sides of each identity are homomorphisms out of the free abelian group on the singular simplices, so it suffices to check them on a singular simplex $\sigma\colon \Delta^n \to X$.

1. Composing each face of σ with f gives the corresponding face of $f \circ \sigma$, so
$$f_\sharp(\partial\sigma) = \sum_i (-1)^i f \circ \left(\sigma|_{(v_0,\ldots,\hat{v}_i,\ldots,v_n)}\right) = \sum_i (-1)^i \left(f \circ \sigma\right)|_{(v_0,\ldots,\hat{v}_i,\ldots,v_n)} = \partial(f_\sharp\sigma)$$
2. $(g \circ f) \circ \sigma = g \circ (f \circ \sigma)$ and $\mathrm{id}_X \circ \sigma = \sigma$.
3. If $\sigma(\Delta^n) \subseteq A$, then $(f \circ \sigma)(\Delta^n) \subseteq f(A) \subseteq B$.
4. For $n = 0$, $f \circ \sigma$ is a single singular 0-simplex, so $\epsilon(f_\sharp\sigma) = 1 = \epsilon(\sigma)$.

This completes the proof.

Proposition 3.3.3: Singular Chains and Homology Are Functors

The assignment $X \mapsto C_\bullet(X)$, $f \mapsto f_\sharp$ is a functor $C_\bullet\colon \mathsf{Top} \to \mathsf{Ch}(\mathbb{Z})$, and singular homology is its composite with the homology functors of proposition 3.1.13, $H_n = H_n \circ C_\bullet\colon \mathsf{Top} \to \mathsf{Ab}$, and likewise for the graded homology $H_\bullet$. Explicitly, for continuous maps $g\colon X \to Y$ and $f\colon Y \to Z$ and every n:

1. $(fg)_* = f_* g_*$ as maps $H_n(X) \to H_n(Z)$
2. $(\mathrm{id}_X)_*$ is the identity map of $H_n(X)$
3. if f is a homeomorphism, then $f_*\colon H_n(Y) \to H_n(Z)$ is an isomorphism

The same holds for reduced homology, with $f_\sharp$ extended to the augmented complexes of definition 3.2.6 by the identity of $\mathbb{Z}$ in degree -1.

***Proof*:**
By lemma 3.3.2(1) and (2), $C_\bullet$ is a functor, and composing it with the functor H_n of proposition 3.1.13 gives a functor, which is (1) and (2). A homeomorphism is an isomorphism of Top, and a functor carries isomorphisms to isomorphisms, which is (3). By lemma 3.3.2(4), $f_\sharp$

together with $\mathrm{id}_{\mathbb{Z}}$ in degree -1 is a chain map of augmented complexes, and these extended maps compose and have identities as in (2), so the same argument applies to reduced homology, as we sought to show.

Two chain maps can induce the same map on homology without being equal. It suffices that their difference be a boundary in a way that is uniform across all chains.

Definition 3.3.4: Chain Homotopy

Let $f, g\colon C_\bullet \to D_\bullet$ be chain maps. A *chain homotopy* from f to g is a family of homomorphisms $P\colon C_n \to D_{n+1}$ satisfying

$$\partial P + P\partial = g - f$$

and f and g are *chain homotopic*, written $f \simeq g$, if one exists. A chain map $f\colon C_\bullet \to D_\bullet$ is a *chain homotopy equivalence* if there is a chain map $h\colon D_\bullet \to C_\bullet$ with $h \circ f \simeq \mathrm{id}_C$ and $f \circ h \simeq \mathrm{id}_D$.

Proposition 3.3.5: chain homotopic chain maps

Chain homotopic chain maps induce the same homomorphism on homology. Consequently a chain homotopy equivalence induces isomorphisms on homology.

***Proof*:**
Let P be a chain homotopy from f to g and let z be a cycle. Then $g(z) - f(z) = \partial P(z) + P(\partial z) = \partial P(z)$ is a boundary, so $[g(z)] = [f(z)]$. If f is a chain homotopy equivalence and h is as in definition 3.3.4, then $h_* f_* = (h \circ f)_* = \mathrm{id}$ and $f_* h_* = (f \circ h)_* = \mathrm{id}$ by the first part and proposition 3.1.13, as we sought to show.

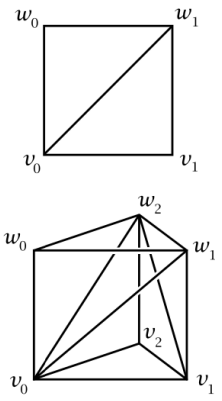
Theorem 3.3.6: Homotopy, Then Isomorphism

Let $F\colon X \times I \to Y$ be a homotopy from f to g. Then there are homomorphisms $P\colon C_n(X) \to C_{n+1}(Y)$ forming a chain homotopy from $f_\sharp$ to $g_\sharp$, and P carries each singular simplex σ to a combination of singular simplices with image in $F(\sigma(\Delta^n) \times I)$. Hence homotopic maps induce the same homomorphisms

$$f_* = g_*\colon H_n(X) \to H_n(Y)$$

for every n.

***Proof*:**
Step 1: The prism operator. Write $e_0, \dots, e_n$ for the vertices of Δ^n, and in $\Delta^n \times I$ write $v_i = (e_i, 0)$ and $w_i = (e_i, 1)$. For points $u_0, \dots, u_{n+1}$ of the convex set $\Delta^n \times I$, write $[u_0, \dots, u_{n+1}]\colon \Delta^{n+1} \to \Delta^n \times I$ for the affine map sending the k-th vertex of Δ^{n+1} to u_k. The $(n+1)$-simplices $[v_0, \dots, v_i, w_i, \dots, w_n]$, $0 \le i \le n$, decompose the prism $\Delta^n \times I$, as the figure shows for $n = 1, 2$, and only their faces enter the computation below.



Let $F\colon X \times I \to Y$ be a homotopy from f to g. For a singular simplex $\sigma\colon \Delta^n \to X$ define the *prism operator*

$$P(\sigma) = \sum_{i=0}^{n} (-1)^i \, F \circ (\sigma \times \mathrm{id}_I) \circ [v_0, \ldots, v_i, w_i, \ldots, w_n]$$

and extend linearly to $P\colon C_n(X) \to C_{n+1}(Y)$. Each simplex occurring in $P(\sigma)$ has image in $F(\sigma(\Delta^n) \times I)$.

Step 2: The boundary of a prism. Fix σ and abbreviate $F \circ (\sigma \times \mathrm{id}_I) \circ [u_0, \ldots, u_{n+1}]$ to $[u_0, \ldots, u_{n+1}]$. In $[v_0, \ldots, v_i, w_i, \ldots, w_n]$ the vertex v_j sits in position j and w_j in position $j+1$, so

$$\partial P(\sigma) = \sum_{j \le i} (-1)^{i+j} [v_0, \ldots, \hat{v}_j, \ldots, v_i, w_i, \ldots, w_n] + \sum_{j \ge i} (-1)^{i+j+1} [v_0, \ldots, v_i, w_i, \ldots, \hat{w}_j, \ldots, w_n]$$

For $1 \le k \le n$, the term $i = j = k$ of the first sum is $[v_0, \ldots, v_{k-1}, w_k, \ldots, w_n]$ and the term $i = j = k-1$ of the second sum is its negative, so these cancel. The two remaining terms with $i = j$ are $[w_0, \ldots, w_n]$, from $i = j = 0$ in the first sum, and $-[v_0, \ldots, v_n]$, from $i = j = n$ in the second. The affine map $[w_0, \ldots, w_n]$ is $x \mapsto (x, 1)$ and $F(x, 1) = g(x)$, so the first of these is $g \circ \sigma = g_\sharp(\sigma)$, and likewise the second is $-f_\sharp(\sigma)$.

The j-th face of σ is $\sigma \circ \varepsilon_j$ for the affine inclusion $\varepsilon_j\colon \Delta^{n-1} \to \Delta^n$ omitting the j-th vertex, and $(\sigma \circ \varepsilon_j) \times \mathrm{id}_I = (\sigma \times \mathrm{id}_I) \circ (\varepsilon_j \times \mathrm{id}_I)$, where $\varepsilon_j \times \mathrm{id}_I$ is affine and carries the vertices of $\Delta^{n-1} \times I$ in order to the v_k and the w_k with $k \ne j$. Hence

$$P(\partial\sigma) = \sum_{i<j} (-1)^{i+j} [v_0, \ldots, v_i, w_i, \ldots, \hat{w}_j, \ldots, w_n] + \sum_{i>j} (-1)^{i-1+j} [v_0, \ldots, \hat{v}_j, \ldots, v_i, w_i, \ldots, w_n]$$

the exponent $i-1$ recording that v_i sits in position $i-1$ of the face once v_j with $j < i$ is deleted. Each term of $\partial P(\sigma)$ with $i \ne j$ is therefore the negative of the term of $P(\partial\sigma)$ with the same simplex. Altogether $\partial P(\sigma) = g_\sharp(\sigma) - f_\sharp(\sigma) - P(\partial\sigma)$, and by linearity $\partial P + P\partial = g_\sharp - f_\sharp$ on $C_n(X)$, with $P = 0$ on $C_{-1}(X) = 0$.

Step 3: Conclusion. By Step 2, P is a chain homotopy from $f_\sharp$ to $g_\sharp$ (definition 3.3.4), and Step 1 gives the support property. By proposition 3.3.5, $f_* = g_*$, as we sought to show.

Corollary 3.3.7: Homotopy Equivalence, Then Isomorphism

Let $f : X \to Y$ be a homotopy equivalence. Then $f_\sharp \colon C_\bullet(X) \to C_\bullet(Y)$ is a chain homotopy equivalence, and $f_* : H_n(X) \to H_n(Y)$ are isomorphisms for all $n \in \mathbb{N}$, and so are the induced maps $f_* \colon \tilde{H}_n(X) \to \tilde{H}_n(Y)$ on reduced homology.

***Proof*:**
Let $g \colon Y \to X$ be a homotopy inverse, so that $gf \simeq \mathrm{id}_X$ and $fg \simeq \mathrm{id}_Y$. By lemma 3.3.2(2) and theorem 3.3.6,

$$g_\sharp f_\sharp = (gf)_\sharp \simeq (\mathrm{id}_X)_\sharp = \mathrm{id} \qquad f_\sharp g_\sharp = (fg)_\sharp \simeq (\mathrm{id}_Y)_\sharp = \mathrm{id}$$

so $f_\sharp$ is a chain homotopy equivalence with inverse $g_\sharp$, and $f_* \colon H_n(X) \to H_n(Y)$ is an isomorphism with inverse g_* by proposition 3.3.5. For reduced homology, $\tilde{H}_n = H_n$ when $n \geq 1$, since the augmented complex of definition 3.2.6 differs from $C_\bullet$ only by the group $\mathbb{Z}$ in degree -1 and the boundary ϵ into it, and $\tilde{H}_0(X) = \ker \epsilon / \mathrm{im} \partial_1$ is the kernel of the map $H_0(X) = C_0(X)/\mathrm{im}\partial_1 \to \mathbb{Z}$ induced by ϵ. By lemma 3.3.2(4), $\epsilon \circ h_\sharp = \epsilon$ on C_0 for every continuous map h, so h_* carries $\tilde{H}_0$ into $\tilde{H}_0$. Hence f_* and g_* restrict to mutually inverse maps between $\tilde{H}_0(X)$ and $\tilde{H}_0(Y)$, completing the proof.

Corollary 3.3.8: Homology Is a Functor on hTop

For each n, the assignment $X \mapsto H_n(X)$, $[f] \mapsto f_*$ is a well-defined functor $\mathsf{hTop} \to \mathsf{Ab}$, and so is $X \mapsto H_\bullet(X)$ with values in graded abelian groups. It carries the isomorphisms of hTop, the homotopy equivalences, to isomorphisms.

***Proof*:**
By theorem 3.3.6, f_* depends only on the homotopy class $[f]$, so the functor H_n of proposition 3.3.3 factors through the functor $\mathsf{Top} \to \mathsf{hTop}$ described after definition 1.1.7, and likewise $H_\bullet$. A functor carries isomorphisms to isomorphisms (the remark after definition 1.1.3), completing the proof.

Corollary 3.3.9: Homology Group Of Contractible Spaces

Let X be a space homotopy equivalent to a point. Then:

$$H_n(X) = \begin{cases} \mathbb{Z} & n = 0 \\ 0 & n \geq 1 \end{cases}$$

and the reduced homology $\tilde{H}_n(X)$ vanishes for every n.

***Proof*:**
By corollary 3.3.7 the homology and the reduced homology of X in degrees $n \geq 0$ are those of a point. For a point, proposition 3.2.5 gives $H_0 \cong \mathbb{Z}$ and $H_n = 0$ for $n \geq 1$. The point has exactly one singular 0-simplex, so the augmentation $\epsilon \colon C_0 \cong \mathbb{Z} \to \mathbb{Z}$ is an isomorphism, and

the augmented complex has $\tilde{H}_0 = \ker\epsilon/\mathrm{im}\partial_1 = 0$, while $\tilde{H}_n = H_n = 0$ for $n \geq 1$. The space X is nonempty, a homotopy equivalence with a point giving a map from the point to X, so its augmentation is onto and $\tilde{H}_{-1}(X) = 0$, as we sought to show.

3.4 Relative Homology: Quotient Complexes

In geometry, we often construct more complicated shapes from simpler ones. Often, we may decompose information about a shape X into:

1. information on $A \subseteq X$ and X/A
2. information on $A, B \subseteq X$ where $A \cup B = X$

In the right cases, the homology groups $H_n(A)$ and $H_n(X/A)$ can be used to compute $H_n(X)$, and similarly for $A, B \subseteq X$. The two main results, the second an application of the first, are:

1. Snake Lemma Sequences
2. Mayer-Vietoris Sequences

A subspace $A \subseteq X$ gives the quotient $C_n(X)/C_n(A)$, in which all chains in A are trivial:

Definition 3.4.1: Relative Chain Groups

Let X be a topological space and $A \subseteq X$ a subspace. Then

$$C_n(X, A) := C_n(X)/C_n(A)$$

is called the *relative chain group* of X with respect to A.

Notice that the boundary map $\partial : C_n(X) \to C_{n-1}(X)$ takes $C_n(A)$ to $C_{n-1}(A)$, hence we have an induced map $\partial_n : C_n(X, A) \to C_{n-1}(X, A)$, which satisfies $\partial^2 = 0$ since the boundary map of $C_\bullet(X)$ does. Thus, we have the following:

Definition 3.4.2: Relative Homology Groups

Define

$$H_n(X, A) = \frac{\ker \partial_n}{\mathrm{im}\partial_{n+1}}$$

to be the *relative homology group* where ∂_n and ∂_{n+1} are defined as above. Elements of $H_n(X, A)$ are represented by *relative cycles* (n-chains $\alpha \in C_n(X)$ such that $\partial\alpha \in C_{n-1}(A)$), and a relative cycle α is trivial in $H_n(X, A)$ if and only if it is a *relative boundary* $\alpha = \partial\beta + \gamma$, where $\beta \in C_{n+1}(X)$ and $\gamma \in C_n(A)$.

Note that the quotient $C_n(X)/C_n(A)$ *can* be viewed as a subgroup of $C_n(X)$ by considering it as having basis the singular n-simplices $\sigma : \Delta^n \to X$ whose image is *not* contained in A. However, the boundary map does not take this subgroup of $C_n(X)$ to the corresponding subgroup of $C_{n-1}(X)$, so we do not use this description.

The groups $H_n(A)$, $H_n(X)$ and $H_n(X, A)$ are related through a short exact sequence of complexes:

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
\cdots & \longrightarrow & A_{n+2} & \longrightarrow & A_{n+1} & \longrightarrow & A_n & \longrightarrow & A_{n-1} & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle i} & & \downarrow{\scriptstyle i} & & \downarrow{\scriptstyle i} & & \downarrow{\scriptstyle i} & & \\
\cdots & \longrightarrow & B_{n+2} & \longrightarrow & B_{n+1} & \longrightarrow & B_n & \longrightarrow & B_{n-1} & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle j} & & \downarrow{\scriptstyle j} & & \downarrow{\scriptstyle j} & & \downarrow{\scriptstyle j} & & \\
\cdots & \longrightarrow & C_{n+2} & \longrightarrow & C_{n+1} & \longrightarrow & C_n & \longrightarrow & C_{n-1} & \longrightarrow & \cdots \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & & 0 & &
\end{array} \tag{3.1}$$

Here each row is a chain complex with boundary maps ∂, each column $0 \to A_k \to B_k \to C_k \to 0$ is exact, and the squares commute, so that i and j are chain maps. The motivating case is $A_k = C_k(A) \subseteq B_k = C_k(X)$ with $C_k = C_k(X)/C_k(A)$.

Theorem 3.4.3: Algebraic Snake Lemma

Let $0 \to A_\bullet \xrightarrow{i} B_\bullet \xrightarrow{j} C_\bullet \to 0$ be a short exact sequence of chain complexes as in (3.1). Then there are homomorphisms $\partial\colon H_n(C) \to H_{n-1}(A)$, the *connecting homomorphisms*, such that the sequence

$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(B) \xrightarrow{j_*} H_n(C) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(B) \xrightarrow{j_*} H_{n-1}(C) \to \cdots$$

is exact. The connecting homomorphism sends the class of a cycle $c \in C_n$ to the class of the cycle $a \in A_{n-1}$ determined by $i(a) = \partial b$, where $b \in B_n$ is any element with $j(b) = c$.

The long exact sequence runs down each column of the following diagram and then along a diagonal arrow to the top of the next column:

$$\begin{array}{ccccc}
H_{n+1}(A) & & H_n(A) & & H_{n-1}(A) \\
\downarrow{\scriptstyle i_*} & & \downarrow{\scriptstyle i_*} & & \downarrow{\scriptstyle i_*} \\
H_{n+1}(B) & \overset{\partial}{\nearrow} & H_n(B) & \overset{\partial}{\nearrow} & H_{n-1}(B) \\
\downarrow{\scriptstyle j_*} & & \downarrow{\scriptstyle j_*} & & \downarrow{\scriptstyle j_*} \\
H_{n+1}(C) & & H_n(C) & & H_{n-1}(C)
\end{array}$$

This theorem holds in the general context of homological algebra, since its hypotheses concern only an exact sequence of complexes.

***Proof*:**
The chain maps i and j carry cycles to cycles and boundaries to boundaries, so they induce the homomorphisms i_* and j_*, and $j_* i_* = (ji)_* = 0$ because $ji = 0$. The definition of ∂ follows the

diagram

$$\begin{array}{ccc} & & A_{n-1} \\ & & \downarrow{\scriptstyle i} \\ B_n & \xrightarrow{\partial} & B_{n-1} \\ \downarrow{\scriptstyle j} & & \\ C_n & & \end{array}$$

Let $c \in C_n$ be a cycle. Since j is surjective, $c = j(b)$ for some $b \in B_n$. By commutativity $j(\partial b) = \partial(j(b)) = \partial c = 0$, so $\partial b \in \ker j = \mathrm{im} i$, and $\partial b = i(a)$ for some $a \in A_{n-1}$. Since $i(\partial a) = \partial i(a) = \partial\partial b = 0$ and i is injective, $\partial a = 0$. Define $\partial[c] = [a]$. This is well defined:

1. The element a is uniquely determined by ∂b, since i is injective.
2. If b' is another element with $j(b') = c$, then $b' - b \in \ker j = \mathrm{im} i$, say $b' = b + i(a')$. Then
$$i(a + \partial a') = \partial b + \partial i(a') = \partial(b + i(a')) = \partial b'$$
so b' produces $a + \partial a'$, which has the same class as a.
3. If c is replaced by $c + \partial c'$ with $c' \in C_{n+1}$, write $c' = j(b')$. Then $c + \partial c' = j(b + \partial b')$, and the lift $b + \partial b'$ has the same boundary ∂b, so a is unchanged.

The map ∂ is a homomorphism: if lifts b_1, b_2 of cycles c_1, c_2 produce a_1, a_2, then $b_1 + b_2$ lifts $c_1 + c_2$ and $i(a_1 + a_2) = \partial b_1 + \partial b_2 = \partial(b_1 + b_2)$, so $\partial([c_1] + [c_2]) = [a_1] + [a_2]$.

1. $\mathrm{im} i_* \subseteq \ker j_*$, since $j_* i_* = 0$.
2. $\mathrm{im} j_* \subseteq \ker \partial$. For a cycle $b \in B_n$, the lift of $j(b)$ may be taken to be b itself, and $\partial b = 0 = i(0)$, so $\partial j_*[b] = 0$.
3. $\mathrm{im}\partial \subseteq \ker i_*$. With c, b, a as in the definition, $i_*\partial[c] = [i(a)] = [\partial b] = 0$.
4. $\ker j_* \subseteq \mathrm{im} i_*$. Let $b \in B_n$ be a cycle with $j_*[b] = 0$, so $j(b) = \partial c'$ for some $c' \in C_{n+1}$. Write $c' = j(b')$. Then $j(b - \partial b') = j(b) - \partial j(b') = 0$, so $b - \partial b' = i(a)$ for some $a \in A_n$, and a is a cycle since $i(\partial a) = \partial(b - \partial b') = 0$ and i is injective. Hence $i_*[a] = [b - \partial b'] = [b]$.
5. $\ker \partial \subseteq \mathrm{im} j_*$. If $\partial[c] = 0$, then $a = \partial a'$ for some $a' \in A_n$. The element $b - i(a')$ is a cycle, since $\partial(b - i(a')) = \partial b - i(\partial a') = i(a) - i(a) = 0$, and $j(b - i(a')) = j(b) = c$, so $j_*[b - i(a')] = [c]$.
6. $\ker i_* \subseteq \mathrm{im}\partial$. Let $a \in A_{n-1}$ be a cycle with $i(a) = \partial b$ for some $b \in B_n$. Then $j(b)$ is a cycle, since $\partial j(b) = j(\partial b) = ji(a) = 0$, and b is a lift of it with $\partial b = i(a)$, so $\partial[j(b)] = [a]$.

This proves exactness at every term, completing the proof.

The construction of the connecting homomorphism involved choices, a lift b of the cycle c among them, and the proof of theorem 3.4.3 verified that the resulting class depends on none of them. That independence gives more than well-definedness: it makes the connecting homomorphism compatible with maps of short exact sequences, so that entire long exact sequences can be compared term by

term.

Proposition 3.4.4: Naturality of the Connecting Homomorphism

Let

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A_\bullet & \xrightarrow{i} & B_\bullet & \xrightarrow{j} & C_\bullet & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \\ 0 & \longrightarrow & A'_\bullet & \xrightarrow{i'} & B'_\bullet & \xrightarrow{j'} & C'_\bullet & \longrightarrow & 0 \end{array}$$

be a commutative diagram of chain maps whose rows are short exact sequences of chain complexes. Then the long exact sequences of theorem 3.4.3 associated to the two rows form a commuting ladder

$$\begin{array}{ccccccccccc} \cdots & \longrightarrow & H_n(A) & \xrightarrow{i_*} & H_n(B) & \xrightarrow{j_*} & H_n(C) & \xrightarrow{\partial} & H_{n-1}(A) & \xrightarrow{i_*} & \cdots \\ & & \downarrow{\scriptstyle \alpha_*} & & \downarrow{\scriptstyle \beta_*} & & \downarrow{\scriptstyle \gamma_*} & & \downarrow{\scriptstyle \alpha_*} & & \\ \cdots & \longrightarrow & H_n(A') & \xrightarrow{i'_*} & H_n(B') & \xrightarrow{j'_*} & H_n(C') & \xrightarrow{\partial'} & H_{n-1}(A') & \xrightarrow{i'_*} & \cdots \end{array}$$

***Proof*:**

There are three squares to check in each period of the ladder, and only the one containing the connecting homomorphisms has content.

For the square with i_* and i'_*, let $a \in A_n$ be a cycle. Then

$$i'_*\alpha_*[a] = [i'\alpha(a)] = [\beta i(a)] = \beta_* i_*[a]$$

using the chain-level commutativity $i'\alpha = \beta i$ in the middle equality. The square with j_* and j'_* commutes by the same computation, with $j'\beta = \gamma j$ in place of $i'\alpha = \beta i$.

For the connecting square, the claim is $\alpha_*\partial = \partial'\gamma_*$. Let $[c] \in H_n(C)$ be represented by the cycle $c \in C_n$, and unwind the definition of ∂ from the proof of theorem 3.4.3: choose $b \in B_n$ with $j(b) = c$, observe that $\partial b \in \ker j = \mathrm{im} i$, write $\partial b = i(a)$ for a unique cycle $a \in A_{n-1}$, and set $\partial[c] = [a]$. Now push the entire chase through the vertical maps. The element $\beta(b) \in B'_n$ satisfies

$$j'(\beta(b)) = \gamma(j(b)) = \gamma(c)$$

so $\beta(b)$ is a legitimate lift of $\gamma(c)$, which is a cycle since γ is a chain map. Its boundary is

$$\partial(\beta(b)) = \beta(\partial b) = \beta(i(a)) = i'(\alpha(a))$$

using first that β is a chain map and then the commutativity $\beta i = i'\alpha$. Thus the element of A'_{n-1} produced by the chase for $\gamma(c)$, run with the lift $\beta(b)$, is $\alpha(a)$. Since the proof of theorem 3.4.3 showed the resulting homology class to be independent of the choice of lift,

$$\partial'\gamma_*[c] = \partial'[\gamma(c)] = [\alpha(a)] = \alpha_*[a] = \alpha_*\partial[c]$$

which is the claimed commutativity, completing the proof.

Theorem 3.4.5: The Long Exact Sequence of a Pair in Homology

Let $A \subseteq X$, and write $i\colon A \hookrightarrow X$ for the inclusion and $j\colon C_\bullet(X) \to C_\bullet(X, A)$ for the quotient chain map.

1. The sequence
$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(X) \to \cdots \to H_0(X, A) \to 0$$
is exact, where ∂ sends the class of a relative cycle α to the class of the cycle $\partial\alpha$ of A.
2. If $A \neq \emptyset$, the sequence of (1) with $H_n(A)$ and $H_n(X)$ replaced by $\tilde{H}_n(A)$ and $\tilde{H}_n(X)$ is exact.
3. For subspaces $B \subseteq A \subseteq X$ the sequence
$$\cdots \to H_n(A, B) \to H_n(X, B) \to H_n(X, A) \xrightarrow{\partial} H_{n-1}(A, B) \to \cdots$$
is exact, the first two maps being induced by the inclusions of chain complexes.
4. A *map of pairs* $f\colon (X, A) \to (Y, B)$, a continuous map $f\colon X \to Y$ with $f(A) \subseteq B$, induces a chain map $f_\sharp\colon C_\bullet(X, A) \to C_\bullet(Y, B)$ and homomorphisms $f_*\colon H_n(X, A) \to H_n(Y, B)$, with $(g \circ f)_* = g_* \circ f_*$ and $(\mathrm{id})_* = \mathrm{id}$. The sequences of (1) for (X, A) and (Y, B) form a commuting ladder
$$\begin{array}{ccccccccc} \cdots & \longrightarrow & H_n(A) & \longrightarrow & H_n(X) & \longrightarrow & H_n(X, A) & \xrightarrow{\partial} & H_{n-1}(A) & \longrightarrow & \cdots \\ & & \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle f_*} & & \\ \cdots & \longrightarrow & H_n(B) & \longrightarrow & H_n(Y) & \longrightarrow & H_n(Y, B) & \xrightarrow{\partial} & H_{n-1}(B) & \longrightarrow & \cdots \end{array}$$
and the same holds for the sequences of (2), and of (3) for a map of triples, a continuous map carrying each of the three spaces into the corresponding one.

***Proof*:**

1. By lemma 3.3.2(3) applied to the inclusion, $C_\bullet(A)$ is a subcomplex of $C_\bullet(X)$, so $0 \to C_\bullet(A) \xrightarrow{i_\sharp} C_\bullet(X) \xrightarrow{j} C_\bullet(X, A) \to 0$ is a short exact sequence of chain complexes. Theorem 3.4.3 gives the exact sequence. For a relative cycle α the lift may be taken to be α itself, so $\partial[\alpha]$ is the class of $\partial\alpha \in C_{n-1}(A)$. The sequence ends at $H_0(X, A)$ because $H_{-1}(A) = 0$.
2. Extending $i_\sharp$ by $\mathrm{id}_\mathbb{Z}$ in degree -1 gives an injective chain map $\tilde{C}_\bullet(A) \to \tilde{C}_\bullet(X)$ of the augmented complexes of definition 3.2.6, by lemma 3.3.2(4). Its cokernel is $C_n(X, A)$ in degrees $n \geq 0$ and $\mathbb{Z}/\mathbb{Z} = 0$ in degree -1, so $0 \to \tilde{C}_\bullet(A) \to \tilde{C}_\bullet(X) \to C_\bullet(X, A) \to 0$ is a short exact sequence of chain complexes, and theorem 3.4.3 applies to it. The resulting sequence continues past $H_0(X, A)$ with $\tilde{H}_{-1}(A)$, the cokernel of the augmentation $C_0(A) \to \mathbb{Z}$, which vanishes because the augmentation sends any point of $A \neq \emptyset$ to 1, so

the sequence ends at $H_0(X, A) \to 0$ as stated.

3. The inclusions give a short exact sequence $0 \to C_\bullet(A, B) \to C_\bullet(X, B) \to C_\bullet(X, A) \to 0$, exact because $C_n(X, A) = C_n(X)/C_n(A) \cong (C_n(X)/C_n(B))/(C_n(A)/C_n(B))$, and theorem 3.4.3 applies to it.

4. By lemma 3.3.2(3), $f_\sharp$ carries $C_\bullet(A)$ into $C_\bullet(B)$, so it descends to a homomorphism $f_\sharp \colon C_\bullet(X, A) \to C_\bullet(Y, B)$ of quotients, which commutes with ∂ because $f_\sharp$ does, and the identities $(g \circ f)_\sharp = g_\sharp f_\sharp$ and $(\mathrm{id})_\sharp = \mathrm{id}$ of lemma 3.3.2(2) pass to the quotients. With proposition 3.1.13 this gives f_* and its functoriality. The chain maps $f_\sharp$ on $C_\bullet(A)$, $C_\bullet(X)$ and $C_\bullet(X, A)$ form a commutative diagram from the short exact sequence of (X, A) to that of (Y, B): the left square commutes because the first map is the restriction of the second, and the right square because the third is induced on the quotients. Proposition 3.4.4 gives the ladder. For (2) the vertical maps are extended by $\mathrm{id}_\mathbb{Z}$ in degree -1, which is a chain map of augmented complexes by lemma 3.3.2(4), and for (3) a map of triples maps each of the three relative complexes to the corresponding one, compatibly with the inclusions and quotients, and the same proposition applies.

This completes the proof.

The long exact sequence measures the difference between $H_n(X)$ and $H_n(A)$: the inclusion $A \hookrightarrow X$ induces isomorphisms $H_n(A) \cong H_n(X)$ for all n exactly when $H_n(X, A) = 0$ for all n. If the relative groups vanish, exactness of $H_{n+1}(X, A) \to H_n(A) \to H_n(X) \to H_n(X, A)$ makes each i_* an isomorphism. If every i_* is an isomorphism, exactness makes both j_* and ∂ zero, so $H_n(X, A)$, which is the kernel of ∂ and the image of j_*, is zero. The quotient in the augmented sequence of (2) is the relative complex unchanged, so the reduced relative groups are the relative groups, $\tilde{H}_n(X, A) = H_n(X, A)$.

Lemma 3.4.6: The Five Lemma

Given the following commutative diagram:

$$\begin{array}{ccccccccc} A & \xrightarrow{f_1} & B & \xrightarrow{f_2} & C & \xrightarrow{f_3} & D & \xrightarrow{f_4} & E \\ \downarrow{\scriptstyle\alpha} & & \downarrow{\scriptstyle\beta} & & \downarrow{\scriptstyle\gamma} & & \downarrow{\scriptstyle\delta} & & \downarrow{\scriptstyle\epsilon} \\ A' & \xrightarrow{g_1} & B' & \xrightarrow{g_2} & C' & \xrightarrow{g_3} & D' & \xrightarrow{g_4} & E' \end{array}$$

if the two rows are exact, and α, β, δ, and ϵ are isomorphisms, then so is γ.

***Proof*:**
The argument is a diagram chase, and it is the prototype for all the comparison arguments that follow. Throughout, "the rows are exact" means ker = im at each interior node (B, C, D on top and B', C', D' on the bottom), and "the diagram commutes" means each of the four squares commutes, so that $g_i \circ (\text{vertical}) = (\text{vertical}) \circ f_i$ for each i. The horizontal maps are named f_i and g_i so that they cannot be confused with the elements a, b, c, d the chase produces.

Step 1: Surjectivity of γ (the four-lemma using β, δ surjective and ϵ injective). Let $c' \in C'$ be arbitrary. The goal is to produce $c \in C$ with $\gamma(c) = c'$. Push c' forward along the bottom row to $d' := g_3(c') \in D'$. Since δ is surjective, choose $d \in D$ with $\delta(d) = g_3(c')$. Apply the bottom

map g_4 and use commutativity of the rightmost square:

$$\epsilon(f_4(d)) = g_4(\delta(d)) = g_4(g_3(c')) = 0$$

the last equality because $\mathrm{im} g_3 = \ker g_4$ by exactness of the bottom row at D'. As ϵ is injective, $f_4(d) = 0$, so $d \in \ker f_4 = \mathrm{im} f_3$ by exactness of the top row at D. Pick $\tilde{c} \in C$ with $f_3(\tilde{c}) = d$. The element $\tilde{c}$ need not map to c', but it is off by something killed by g_3. To see this, compute, using commutativity of the third square,

$$g_3(\gamma(\tilde{c})) = \delta(f_3(\tilde{c})) = \delta(d) = g_3(c')$$

so $g_3(\gamma(\tilde{c}) - c') = 0$, that is, $\gamma(\tilde{c}) - c' \in \ker g_3 = \mathrm{im} g_2$ by exactness of the bottom row at C'. Choose $b' \in B'$ with $g_2(b') = \gamma(\tilde{c}) - c'$. Since β is surjective, choose $b \in B$ with $\beta(b) = b'$, and set

$$c := \tilde{c} - f_2(b) \in C$$

Then, using commutativity of the second square ($\gamma \circ f_2 = g_2 \circ \beta$),

$$\gamma(c) = \gamma(\tilde{c}) - \gamma(f_2(b)) = \gamma(\tilde{c}) - g_2(\beta(b)) = \gamma(\tilde{c}) - g_2(b') = \gamma(\tilde{c}) - (\gamma(\tilde{c}) - c') = c'$$

Hence c' lies in the image of γ. As c' was arbitrary, γ is surjective.

Step 2: Injectivity of γ (the four-lemma using β, δ injective and α surjective). Let $c \in C$ satisfy $\gamma(c) = 0$. The goal is $c = 0$. Push c along the top row to $d := f_3(c) \in D$ and use commutativity of the third square:

$$\delta(f_3(c)) = g_3(\gamma(c)) = g_3(0) = 0$$

Since δ is injective, $f_3(c) = 0$, so $c \in \ker f_3 = \mathrm{im} f_2$ by exactness of the top row at C. Choose $b \in B$ with $f_2(b) = c$. Its image $\beta(b)$ is not yet zero, but it is killed by g_2: by commutativity of the second square,

$$g_2(\beta(b)) = \gamma(f_2(b)) = \gamma(c) = 0$$

so $\beta(b) \in \ker g_2 = \mathrm{im} g_1$ by exactness of the bottom row at B'. Choose $a' \in A'$ with $g_1(a') = \beta(b)$. Since α is surjective, choose $a \in A$ with $\alpha(a) = a'$. Then, by commutativity of the leftmost square ($\beta \circ f_1 = g_1 \circ \alpha$),

$$\beta(f_1(a)) = g_1(\alpha(a)) = g_1(a') = \beta(b)$$

and since β is injective this forces $f_1(a) = b$. Therefore

$$c = f_2(b) = f_2(f_1(a)) = 0$$

because $f_2 \circ f_1 = 0$ by exactness of the top row at B (the composite of two consecutive maps in an exact sequence vanishes, $\mathrm{im} f_1 = \ker f_2$). Hence $\ker \gamma = 0$ and γ is injective.

Combining the two steps, γ is a bijective homomorphism, hence an isomorphism. Surjectivity used β, δ surjective and ϵ injective, while injectivity used β, δ injective and α surjective. The full strength of the hypotheses (all four maps isomorphisms) is more than is strictly needed, and the two "four-lemma" halves recorded here are exactly the forms invoked when comparing two long exact sequences term by term, completing the proof.

Example 3.4.7: Application of Snake Lemma

1. Consider $(D^n, \partial D^n)$ and its long exact sequence in reduced homology (theorem 3.4.5(2)). Then the ∂ maps

$$\cdots \to \tilde{H}_i(D^n) \to H_i(D^n, \partial D^n) \xrightarrow{\partial} \tilde{H}_{i-1}(S^{n-1}) \to \tilde{H}_{i-1}(D^n) \to \cdots$$

   are isomorphisms for all $i > 0$ since the remaining terms $\tilde{H}_i(D^n)$ are all zero for all i (corollary 3.3.9). Thus, using $\tilde{H}_{n-1}(S^{n-1}) \cong \mathbb{Z}$ and $\tilde{H}_k(S^{n-1}) = 0$ for $k \neq n-1$ (corollary 3.4.15):

$$H_i(D^n, \partial D^n) \cong \begin{cases} \mathbb{Z} & i = n \\ 0 & i \neq n \end{cases}$$

2. Consider (X, x_0) for $x_0 \in X$ and its long exact sequence of reduced homology groups (theorem 3.4.5(2)). Then

$$H_n(X, x_0) \cong \tilde{H}_n(X)$$

   for all n since $\tilde{H}_n(x_0) = 0$ for all $n \in \mathbb{N}$ (corollary 3.3.9).

Proposition 3.4.8: Homotopy Between Relative Maps

Let $f, g\colon (X, A) \to (Y, B)$ be maps of pairs that are homotopic through maps of pairs $f_t\colon (X, A) \to (Y, B)$. Then the chain maps $f_\sharp, g_\sharp\colon C_\bullet(X, A) \to C_\bullet(Y, B)$ are chain homotopic, and $f_* = g_*\colon H_n(X, A) \to H_n(Y, B)$ for every n.

***Proof*:**
Let $F(x, t) = f_t(x)$, a homotopy $X \times I \to Y$ from f to g with $F(A \times I) \subseteq B$, and let $P\colon C_n(X) \to C_{n+1}(Y)$ be its prism operator from theorem 3.3.6, so that $\partial P + P\partial = g_\sharp - f_\sharp$. For a singular simplex σ with image in A, every simplex of $P(\sigma)$ has image in $F(\sigma(\Delta^n) \times I) \subseteq F(A \times I) \subseteq B$, so P carries $C_n(A)$ into $C_{n+1}(B)$ and induces homomorphisms $P\colon C_n(X, A) \to C_{n+1}(Y, B)$ on the quotients. The identity $\partial P + P\partial = g_\sharp - f_\sharp$ holds on $C_\bullet(X)$, so it holds for the induced maps on the quotients, and P is a chain homotopy from $f_\sharp$ to $g_\sharp$ on relative chains (definition 3.3.4). By proposition 3.3.5 the two chain maps induce the same homomorphisms on relative homology, as we sought to show.

Excision says when we can delete, or excise, a part of the space and keep the same relative homology groups:

Theorem 3.4.9: Excision Theorem

Let $Z \subseteq A \subseteq X$ such that the closure of Z is contained in the interior of A ($\overline{Z} \subseteq \text{int}A$). Then the inclusion $(X - Z, A - Z) \hookrightarrow (X, A)$ induces isomorphisms

$$H_n(X - Z, A - Z) \cong H_n(X, A) \qquad \forall n \in \mathbb{N}$$

Equivalently, for subspaces $A, B \subseteq X$ whose interiors cover X ($\text{int}A \cup \text{int}B = X$), the inclusion $(B, A \cap B) \hookrightarrow (X, A)$ induces isomorphisms

$$H_n(B, A \cap B) \cong H_n(X, A) = H_n(A \cup B, A) \qquad \forall n \in \mathbb{N}$$

The equivalence is obtained by setting $B = X - Z$ and $Z = X - B$ so that $A \cap B = A - Z$ and $\bar{Z} \subseteq \text{int}A$ is equivalent to $X = \text{int}A \cup \text{int}B$ since $X - \text{int}B = \bar{Z}$.

For points $w_0, \ldots, w_n$ of some $\mathbb{R}^N$ and a nonempty set $I \subseteq \{0, \ldots, n\}$, write $b_I = \frac{1}{|I|} \sum_{i \in I} w_i$ for the *barycentre* of the face of $[w_0, \ldots, w_n]$ spanned by the w_i with $i \in I$, and diam for the diameter of a subset of $\mathbb{R}^N$.

Lemma 3.4.10: Barycentric Simplices Are Smaller

Let $w_0, \ldots, w_n \in \mathbb{R}^N$ have convex hull λ, and let $I_0 \subsetneq I_1 \subsetneq \cdots \subsetneq I_k$ be nonempty subsets of $\{0, \ldots, n\}$. Then the convex hull of $b_{I_0}, \ldots, b_{I_k}$ has diameter at most $\frac{n}{n+1} \operatorname{diam} \lambda$.

***Proof*:**

For a finite set $F \subseteq \mathbb{R}^N$ and a point y, the function $x \mapsto |x - y|$ is convex, so its maximum over the convex hull of F is attained at a point of F. Applying this twice, the diameter of the convex hull of F is the largest distance between two points of F, so it suffices to bound $|b_{I_p} - b_{I_q}|$ for $p < q$. The point b_{I_p} lies in the convex hull of the w_i with $i \in I_q$, since $I_p \subseteq I_q$, so by the same convexity $|b_{I_q} - b_{I_p}| \leq \max_{i \in I_q} |b_{I_q} - w_i|$. For $i \in I_q$,

$$|b_{I_q} - w_i| = \Big| \frac{1}{|I_q|} \sum_{l \in I_q} (w_l - w_i) \Big| \leq \frac{|I_q| - 1}{|I_q|} \operatorname{diam} \lambda$$

because the term $l = i$ vanishes and each of the other $|I_q| - 1$ terms has length at most $\operatorname{diam} \lambda$. Since $|I_q| \leq n + 1$, the factor is at most $\frac{n}{n+1}$, as claimed.

Next, let X be a topological space and U a collection of subsets of X whose interiors cover X. Let $C_n^U(X) \leqslant C_n(X)$ be chains $\sum_i n_i \sigma_i$ such that $\text{im}\sigma_i \subseteq U_j \in U$ for some element in U. Then the boundary map takes $C_n^U(X)$ to $C_{n-1}^U(X)$, since each face of σ_i has image in the same U_j, hence $C_n^U(X)$ forms a chain complex. Let the respective homology group be denoted $H_n^U(X)$.

Lemma 3.4.11: Homology Of Covered Space

The inclusion map $\iota : C_n^U(X) \hookrightarrow C_n(X)$ is a chain homotopy equivalence, in particular there exists a map $\rho : C_n(X) \to C_n^U(X)$ such that $\iota \circ \rho$ and $\rho \circ \iota$ are chain homotopic to the identity. Thus, ι induces an isomorphism:

$$H_n^U(X) \cong H_n(X) \qquad \forall n \in \mathbb{N}$$

The proof uses the method of *small simplices*: every singular chain can be cut into pieces small enough that each piece lands inside a single member of the cover, and the cutting is performed by an operator (barycentric subdivision) that is chain homotopic to the identity, so it changes nothing on homology. The argument has four stages. The first defines subdivision on the affine chains of a single convex set by coning from the barycentre, and computes its effect on boundaries. The second transports subdivision to arbitrary singular chains and produces an explicit chain homotopy between the subdivision operator and the identity. The third shows that iterating subdivision drives the geometric size of the pieces to zero, so that after finitely many iterations every piece is U-small. The fourth assembles these into the operator ρ and the required chain homotopies.

***Proof*:**

Throughout, an *affine simplex* in a convex set $Y \subseteq \mathbb{R}^N$ is written $[w_0, \dots, w_n]$, the singular simplex $\Delta^n \to Y$ sending the ith vertex of Δ^n to w_i and extending affinely on barycentric coordinates. The free abelian group on the affine n-simplices of Y is denoted $LC_n(Y)$, the *linear chains*, a subcomplex of $C_n(Y)$ on which the boundary map acts by the usual face formula $\partial[w_0, \dots, w_n] = \sum_i (-1)^i [w_0, \dots, \hat{w_i}, \dots, w_n]$. We augment this complex by a copy of $\mathbb{Z} = LC_{-1}(Y)$ generated by the empty simplex $[\,\varnothing\,]$, with $\partial[w_0] = [\,\varnothing\,]$ for every 0-simplex, so that the augmented linear chain complex $LC_\bullet(Y)$ runs down to degree -1.

Step 1: The cone operator and subdivision of linear chains. Fix a point $b \in Y$. The *cone* operator $b \cdot (-) \colon LC_n(Y) \to LC_{n+1}(Y)$ prepends b as a new initial vertex,

$$b \cdot [w_0, \dots, w_n] = [b, w_0, \dots, w_n]$$

and is extended linearly. On the empty simplex it gives $b \cdot [\,\varnothing\,] = [b]$. A direct expansion of the face formula, separating the term that drops the apex b from the terms that drop one of the w_i, yields the cone identity

$$\partial(b \cdot \lambda) = \lambda - b \cdot (\partial\lambda) \qquad (\lambda \in LC_n(Y)) \tag{3.2}$$

valid for all $n \geq 0$, where the augmentation makes the identity hold in degree 0 as well (there $\partial(b \cdot [w_0]) = \partial[b, w_0] = [w_0] - [b]$ and $b \cdot \partial[w_0] = b \cdot [\,\varnothing\,] = [b]$). The identity says geometrically that the boundary of a cone is the base minus the cone on the boundary. It is the algebraic form of the fact that a cone is contractible, and every computation below uses it.

Define the *barycentric subdivision* operator $S \colon LC_n(Y) \to LC_n(Y)$ recursively. On the augmentation set $S[\,\varnothing\,] = [\,\varnothing\,]$. For an affine simplex $\lambda = [w_0, \dots, w_n]$ with barycentre $b_\lambda = \frac{1}{n+1} \sum_i w_i$, set

$$S\lambda = b_\lambda \cdot (S\partial\lambda) \tag{3.3}$$

and extend linearly. Unwinding the recursion, $S\lambda$ is the signed sum of the $(n+1)!$ affine simplices whose vertices are the barycentres of an increasing chain of faces of λ. This sum is the classical barycentric subdivision of the geometric simplex λ, oriented coherently. We claim S is

a chain map, $\partial S = S\partial$. This is an induction on n. In degree 0 subdivision is the identity, since $S[w_0] = w_0 \cdot S\partial[w_0] = w_0 \cdot [\varnothing] = [w_0]$, so $\partial S = S\partial$ there. For $n \geq 1$, using (3.3), then (3.2), then the inductive hypothesis $\partial S\partial\lambda = S\partial\partial\lambda = 0$,

$$\partial S\lambda = \partial(b_\lambda \cdot S\partial\lambda) = S\partial\lambda - b_\lambda \cdot (\partial S\partial\lambda) = S\partial\lambda - b_\lambda \cdot (S\partial\partial\lambda) = S\partial\lambda$$

so S commutes with ∂.

Step 2: A chain homotopy from S to the identity on linear chains, then on singular chains. We construct an operator $T\colon LC_n(Y) \to LC_{n+1}(Y)$ with

$$\partial T + T\partial = \mathrm{id} - S \tag{3.4}$$

again by induction and again by coning. Set $T = 0$ on $LC_{-1}(Y)$ and on $LC_0(Y)$ (where $S = \mathrm{id}$ already, so the right side vanishes). For an affine n-simplex λ with barycentre b_λ, define

$$T\lambda = b_\lambda \cdot (\lambda - T\partial\lambda) \tag{3.5}$$

Applying (3.2) to (3.5),

$$\partial T\lambda = (\lambda - T\partial\lambda) - b_\lambda \cdot \partial(\lambda - T\partial\lambda) = \lambda - T\partial\lambda - b_\lambda \cdot (\partial\lambda - \partial T\partial\lambda)$$

By the inductive form of (3.4) applied to the $(n-1)$-chain $\partial\lambda$, namely $\partial T\partial\lambda = \partial\lambda - S\partial\lambda - T\partial\partial\lambda = \partial\lambda - S\partial\lambda$, the parenthesised term becomes $\partial\lambda - (\partial\lambda - S\partial\lambda) = S\partial\lambda$. Substituting and recalling $S\lambda = b_\lambda \cdot S\partial\lambda$ from (3.3),

$$\partial T\lambda = \lambda - T\partial\lambda - b_\lambda \cdot S\partial\lambda = \lambda - T\partial\lambda - S\lambda$$

which rearranges to $\partial T\lambda + T\partial\lambda = \lambda - S\lambda$, that is (3.4). (The explicit closed form of T as a signed sum over flags, together with the same sign accounting, is in [21, pp. 119-124]. The recursion above determines it uniquely.)

The operators S and T now transfer from linear chains to singular chains of an arbitrary space X by naturality, the standard device of evaluating on the universal model. The identity map $\mathrm{id}_{\Delta^n} \in LC_n(\Delta^n)$ is itself an affine n-simplex (the canonical one). For any singular simplex $\sigma\colon \Delta^n \to X$, define

$$S\sigma = \sigma_\sharp(S\,\mathrm{id}_{\Delta^n}), \qquad T\sigma = \sigma_\sharp(T\,\mathrm{id}_{\Delta^n})$$

where $\sigma_\sharp\colon C_\bullet(\Delta^n) \to C_\bullet(X)$ is the chain map induced by σ, and extend linearly to $C_n(X)$. Because $\sigma_\sharp$ commutes with ∂ and the construction on Δ^n is the linear one above, the relations established for linear chains pass through: $S\colon C_\bullet(X) \to C_\bullet(X)$ is a chain map, and

$$\partial T + T\partial = \mathrm{id} - S \qquad \text{on } C_\bullet(X) \tag{3.6}$$

Naturality also gives $f_\sharp S = S f_\sharp$ and $f_\sharp T = T f_\sharp$ for any map $f\colon X \to Y$, since both sides are determined by their value on id_{Δ^n}. This naturality is what will let subdivision respect the subspaces appearing in the cover.

Step 3: Iterated subdivision makes chains U-small. The mesh of a linear chain is controlled by a single geometric estimate on the standard simplex. If λ is an affine n-simplex, then every simplex appearing in $S\lambda$ has diameter at most $\frac{n}{n+1}\operatorname{diam}\lambda$, the diameter of an affine simplex meaning that of its image, because by the unwinding of Step 1 the vertices of such a simplex are the barycentres of an increasing chain of faces of λ, so lemma 3.4.10 applies.

Iterating, the m-fold subdivision $S^m\lambda$ consists of simplices of diameter at most $\left(\frac{n}{n+1}\right)^m \operatorname{diam}\lambda$, which tends to 0 as $m \to \infty$ because $\frac{n}{n+1} < 1$. Now fix a singular simplex $\sigma\colon \Delta^n \to X$. The open sets $\{\sigma^{-1}(\mathrm{int}U_j) : U_j \in U\}$ cover the compact metric space Δ^n, so the Lebesgue number lemma furnishes $\delta > 0$ such that every subset of Δ^n of diameter less than δ lies in a single $\sigma^{-1}(\mathrm{int}U_j)$. Choose m with $\left(\frac{n}{n+1}\right)^m \operatorname{diam}\Delta^n < \delta$. Then every simplex of $S^m\mathrm{id}_{\Delta^n}$ has image (in Δ^n) of diameter less than δ, hence contained in some $\sigma^{-1}(\mathrm{int}U_j)$. Applying $\sigma_\sharp$, every simplex of $S^m\sigma = \sigma_\sharp(S^m\mathrm{id}_{\Delta^n})$ has image inside some U_j. Thus $S^m\sigma \in C_n^U(X)$: a high enough iterate of subdivision carries σ into the U-small subcomplex. The integer $m = m(\sigma)$ depends on σ, which is the point requiring care in the next step.

Step 4: Assembling ρ and the chain homotopies. The iterates of the homotopy T telescope into a homotopy from id to S^m. Define, for each $m \geq 1$,

$$D_m = \sum_{0 \leq i < m} TS^i\colon C_n(X) \to C_{n+1}(X)$$

Since S is a chain map and T satisfies (3.6), summing the relations $\partial TS^i + TS^i\partial = S^i - S^{i+1}$ over $0 \leq i < m$ telescopes to

$$\partial D_m + D_m\partial = \mathrm{id} - S^m \tag{3.7}$$

so D_m is a chain homotopy from id to S^m for each fixed m. The difficulty is that no single m works for all simplices at once. The remedy is to let m vary simplex by simplex and absorb the variation into a well-defined operator.

For each singular n-simplex σ let $m(\sigma)$ be the least integer for which $S^{m(\sigma)}\sigma \in C_n^U(X)$, finite by Step 3, and set

$$D\sigma = D_{m(\sigma)}\sigma = \sum_{0 \leq i < m(\sigma)} TS^i\sigma$$

extended linearly to $C_n(X)$. This D is a well-defined natural operator $C_n(X) \to C_{n+1}(X)$ since $m(\sigma)$ is determined by σ. Define the candidate retraction by

$$\rho = \mathrm{id} - \partial D - D\partial\colon C_n(X) \to C_n(X) \tag{3.8}$$

Being id plus a boundary-type correction, ρ is automatically a chain map ($\partial\rho = \partial - \partial D\partial = \rho\partial$). Two facts remain: that ρ lands in $C_n^U(X)$, and that ι and ρ are mutually inverse up to chain homotopy.

ρ lands in $C_n^U(X)$. For a single simplex σ we compute $\rho\sigma$ directly. Because D agrees with $D_{m(\sigma)}$ on σ but $D\partial\sigma$ uses the (smaller) values $m(\sigma')$ on the faces σ' of σ, expand

$$\rho\sigma = \sigma - \partial D\sigma - D\partial\sigma = \big(\sigma - \partial D_{m(\sigma)}\sigma - D_{m(\sigma)}\partial\sigma\big) + \big(D_{m(\sigma)}\partial\sigma - D\partial\sigma\big)$$

By (3.7) the first bracket is $S^{m(\sigma)}\sigma$, which lies in $C_n^U(X)$ by the choice of $m(\sigma)$. The second bracket is $\sum_{\sigma'} \pm(D_{m(\sigma)} - D_{m(\sigma')})\sigma'$ summed over the faces σ' of σ. Each difference $D_{m(\sigma)} - D_{m(\sigma')} = \sum_{m(\sigma') \leq i < m(\sigma)} TS^i$ is a sum of terms $TS^i\sigma'$ with $i \geq m(\sigma')$, and $S^i\sigma'$ is U-small for $i \geq m(\sigma')$, hence so is $TS^i\sigma'$ since T is natural and sends a U-small simplex (one with image in a single U_j) to chains with image in the same U_j. Therefore $\rho\sigma \in C_n^U(X)$, and ρ corestricts to a chain map $\rho\colon C_n(X) \to C_n^U(X)$.

ι and ρ are chain homotopy inverse. By construction (3.8), D is a chain homotopy between $\iota \circ \rho$ and $\mathrm{id}_{C_\bullet(X)}$: indeed $\iota\rho = \rho = \mathrm{id} - (\partial D + D\partial)$ as operators into $C_\bullet(X)$, which is exactly

the statement $\partial D + D\partial = \mathrm{id} - \iota\rho$. For the other composite, note first that ρ restricts to the identity-up-to-homotopy already at the level of $C_n^U(X)$: if σ is itself U-small then $m(\sigma) = 0$, so $D\sigma = 0$, while $D\partial\sigma$ involves only faces of σ, which are also U-small with $m = 0$, hence $D\partial\sigma = 0$. Thus $\rho\sigma = \sigma$ for U-small σ. Consequently $\rho \circ \iota = \mathrm{id}_{C_n^U(X)}$ on the nose. Restricting the homotopy D to $C_n^U(X)$ (where it vanishes) shows $\iota \circ \rho$ is chain homotopic to the identity on $C_\bullet(X)$ via D, and $\rho \circ \iota$ equals the identity on $C_\bullet^U(X)$. Hence ι is a chain homotopy equivalence with homotopy inverse ρ.

A chain homotopy equivalence induces isomorphisms on homology (proposition 3.3.5 applied to both composites), so

$$\iota_* \colon H_n^U(X) \xrightarrow{\cong} H_n(X) \qquad \forall n \in \mathbb{N}$$

completing the proof.

With the small-simplices lemma in hand, the excision theorem follows by applying it to a two-element cover.

***Proof Of the Excision Theorem*:**
By the equivalence recorded after the statement, it suffices to prove the second form: for $A, B \subseteq X$ with $\mathrm{int}A \cup \mathrm{int}B = X$, the inclusion induces isomorphisms $H_n(B, A \cap B) \cong H_n(X, A)$ for all n. Take the cover $U = \{A, B\}$, whose interiors cover X by hypothesis.

Step 1: Comparison of relative complexes. Form the U-small chains $C_n^U(X) = C_n(A) + C_n(B)$, the subgroup of chains expressible as a sum of a chain in A and a chain in B, and pass to the relative complex $C_n^U(X)/C_n(A)$. The composite

$$C_n(B) \hookrightarrow C_n^U(X) \twoheadrightarrow C_n^U(X)/C_n(A)$$

has kernel exactly $C_n(B) \cap C_n(A) = C_n(A \cap B)$, since a singular simplex with image in B that also lies in $C_n(A)$ is a simplex with image in $A \cap B$ (the basis of $C_n(X)$ being the singular simplices, an intersection of free subgroups generated by simplices is generated by the common simplices). The map is surjective because every generator of $C_n^U(X)/C_n(A)$ is represented by a simplex in B. Therefore it descends to an isomorphism of chain complexes

$$C_n(B)/C_n(A \cap B) \xrightarrow{\cong} C_n^U(X)/C_n(A) \tag{3.9}$$

natural in n and commuting with the boundary maps, hence an isomorphism $H_n(B, A \cap B) \cong H_n\big(C_\bullet^U(X)/C_\bullet(A)\big)$ on homology.

Step 2: The small-simplices lemma kills the difference between C^U and C. Consider the short exact sequences of chain complexes

$$0 \to C_\bullet(A) \to C_\bullet^U(X) \to C_\bullet^U(X)/C_\bullet(A) \to 0, \qquad 0 \to C_\bullet(A) \to C_\bullet(X) \to C_\bullet(X)/C_\bullet(A) \to 0$$

The inclusion $C_\bullet^U(X) \hookrightarrow C_\bullet(X)$ is the identity on the common subcomplex $C_\bullet(A)$ and is the map ι of lemma 3.4.11, which is a chain homotopy equivalence inducing isomorphisms $H_n^U(X) \cong H_n(X)$. The induced maps on the quotients fit into a morphism of the two short exact sequences, hence into a map of their long exact sequences, commuting with all connecting homomorphisms (proposition 3.4.4). In that ladder the maps on the $H_n(A)$ terms are identities and the maps on the $H_n^U(X) \to H_n(X)$ terms are the lemma's isomorphisms. By the five lemma (lemma 3.4.6) the induced map on the relative homology of the quotient complexes is an isomorphism,

$$H_n\big(C_\bullet^U(X)/C_\bullet(A)\big) \xrightarrow{\cong} H_n\big(C_\bullet(X)/C_\bullet(A)\big) = H_n(X, A) \tag{3.10}$$

Step 3: Conclusion. Composing the isomorphism (3.9) on homology with (3.10),

$$H_n(B, A\cap B) \cong H_n\big(C_\bullet^U(X)/C_\bullet(A)\big) \cong H_n(X, A) \qquad \forall n \in \mathbb{N}$$

and tracing the maps shows the composite is induced by the inclusion $(B, A\cap B) \hookrightarrow (X, A)$, since at the chain level it is exactly the inclusion $C_\bullet(B) \hookrightarrow C_\bullet(X)$ read modulo $C_\bullet(A)$. Translating back through $B = X - Z$, $A \cap B = A - Z$ gives the first form $H_n(X - Z, A - Z) \cong H_n(X, A)$, as we sought to show.

When A is a closed deformation retract of a neighbourhood in X, the relative homology groups are the reduced homology groups of the quotient:

Corollary 3.4.12: Snake Lemma With Absolute Homology Groups

Let (X, A) be a pair where A is a nonempty closed subspace that is a deformation retract of some neighbourhood of A in X, $\pi : (X, A) \to (X/A, A/A)$ the natural quotient map, and $\pi_* : H_n(X, A) \to H_n(X/A, A/A) \cong \tilde{H}_n(X/A)$ the induced map. Then π_* is an isomorphism:

$$H_n(X, A) \cong H_n(X/A, A/A) \cong \tilde{H}_n(X/A)$$

***Proof*:**
Let V be a neighbourhood of A in X that deformation retracts onto A, which exists by the hypothesis that (X, A) is a good pair. The proof assembles three isomorphisms: one provided by the deformation retraction, one by excision (theorem 3.4.9), and one by passing to the quotient, where the same two devices apply because $(X/A, A/A)$ inherits a good neighbourhood from V. Throughout, all maps between relative homology groups are those induced by inclusions or by the quotient map $\pi\colon (X, A) \to (X/A, A/A)$, and the diagrams below commute because they commute already at the level of singular chains.

Step 1: The deformation retraction collapses V to A. Consider the triple $A \subseteq V \subseteq X$ and the long exact sequence of the triple (theorem 3.4.3 applied to the short exact sequence of chain complexes $0 \to C_\bullet(V, A) \to C_\bullet(X, A) \to C_\bullet(X, V) \to 0$),

$$\cdots \to H_n(V, A) \to H_n(X, A) \to H_n(X, V) \xrightarrow{\partial} H_{n-1}(V, A) \to \cdots$$

Let $r_t\colon V \to V$ be the deformation retraction, so $r_0 = \mathrm{id}_V$, $r_1(V) = A$, and $r_t|_A = \mathrm{id}_A$ for all t. Each r_t is a map of pairs $(V, A) \to (V, A)$, and $r_0 = \mathrm{id}$ is homotopic through such maps to the composite $(V, A) \xrightarrow{r_1} (A, A) \hookrightarrow (V, A)$. By the homotopy invariance of relative homology (proposition 3.4.8) the identity on $H_n(V, A)$ therefore factors through $H_n(A, A) = 0$, so $H_n(V, A) = 0$ for every n. Exactness of the displayed sequence then forces the inclusion-induced map

$$H_n(X, A) \xrightarrow{\cong} H_n(X, V) \qquad \forall n \in \mathbb{N}$$

to be an isomorphism.

Step 2: Excise the subspace A. The set $Z = A$ satisfies $\overline{Z} = A \subseteq \mathrm{int}V$, since V is a neighbourhood of A and A is closed. Excision (theorem 3.4.9) applied to $Z \subseteq V \subseteq X$ gives that the inclusion $(X - A, V - A) \hookrightarrow (X, V)$ induces an isomorphism

$$H_n(X - A, V - A) \xrightarrow{\cong} H_n(X, V) \qquad \forall n \in \mathbb{N}$$

Step 3: Transport the chain to the quotient. Write $q\colon X \to X/A$ for the quotient map and set $\bar{A} = A/A$, the single point onto which A collapses, and $\bar{V} = V/A = q(V)$. Because A is closed and V deformation retracts onto A, the pair $(X/A, \bar{A})$ is again a good pair: composing r_t with q and using that $r_t|_A = \mathrm{id}_A$ shows r_t descends to a deformation retraction $\bar{r}_t\colon \bar{V} \to \bar{V}$ of the neighbourhood $\bar{V}$ of $\bar{A}$ onto $\bar{A}$. Steps 1 and 2, applied verbatim to $(X/A, \bar{A})$ with the neighbourhood $\bar{V}$, yield isomorphisms

$$H_n(X/A, \bar{A}) \xleftarrow{\cong} H_n\big((X/A) - \bar{A},\ \bar{V} - \bar{A}\big) \cong H_n(X/A, \bar{V})$$

the left arrow induced by inclusion and excision, the right by the deformation retraction.

Step 4: Identify the two complements and conclude. The quotient map q restricts to a homeomorphism of pairs

$$q\colon (X - A,\ V - A) \xrightarrow{\cong} \big((X/A) - \bar{A},\ \bar{V} - \bar{A}\big)$$

since q is a bijection away from A, is continuous, and is open there: a set $W \subseteq X - A$ is open in $X - A$ if and only if $q(W)$ is open in $(X/A) - \bar{A}$, as $q^{-1}(q(W)) = W$ for W disjoint from A. A homeomorphism of pairs induces an isomorphism on relative homology, so the induced map $q_*\colon H_n(X - A, V - A) \to H_n\big((X/A) - \bar{A},\ \bar{V} - \bar{A}\big)$ is an isomorphism.

Name the inclusion-induced isomorphisms of Steps 1 and 2 and their quotient counterparts. In the source, $a\colon H_n(X, A) \to H_n(X, V)$ comes from the deformation retraction of Step 1 and $e\colon H_n(X - A, V - A) \to H_n(X, V)$ from the excision of Step 2. In the quotient, $\bar{a}\colon H_n(X/A, \bar{A}) \to H_n(X/A, \bar{V})$ and $\bar{e}\colon H_n\big((X/A) - \bar{A},\ \bar{V} - \bar{A}\big) \to H_n(X/A, \bar{V})$ are the corresponding maps of Step 3. Each of $a, e, \bar{a}, \bar{e}$ is an isomorphism. Write $\pi'_*\colon H_n(X, V) \to H_n(X/A, \bar{V})$ for the map induced by the quotient map of pairs $(X, V) \to (X/A, \bar{V})$. These maps assemble into the diagram

$$\begin{array}{ccccc}
H_n(X, A) & \xrightarrow[\cong]{a} & H_n(X, V) & \xleftarrow[\cong]{e} & H_n(X - A, V - A) \\
{\scriptstyle \pi_*}\downarrow & & \downarrow{\scriptstyle \pi'_*} & & \cong\downarrow{\scriptstyle q_*} \\
H_n(X/A, \bar{A}) & \xrightarrow[\cong]{\bar{a}} & H_n(X/A, \bar{V}) & \xleftarrow[\cong]{\bar{e}} & H_n\big((X/A) - \bar{A},\ \bar{V} - \bar{A}\big)
\end{array}$$

in which every horizontal edge is inclusion-induced: the left-hand pair $a, \bar{a}$ are the deformation-retraction isomorphisms of Step 1, the right-hand pair $e, \bar{e}$ the excision isomorphisms of Step 2, while the three vertical maps π_*, π'_*, q_* are induced by the quotient. Both squares commute by naturality: each is the image under q of a commuting square of inclusions of pairs, so q-compatibility of the inclusions gives $\bar{a} \circ \pi_* = \pi'_* \circ a$ on the left and $\bar{e} \circ q_* = \pi'_* \circ e$ on the right. From the right square, $\pi'_* = \bar{e} \circ q_* \circ e^{-1}$ is a composite of three isomorphisms, hence an isomorphism. From the left square, $\bar{a} \circ \pi_* = \pi'_* \circ a$, and since $a, \bar{a}, \pi'_*$ are isomorphisms,

$$\pi_* = \bar{a}^{-1} \circ \pi'_* \circ a$$

is an isomorphism as well. This is exactly the map on relative homology induced by π, since at the chain level all of the inclusions and the quotient compose to the chain map of $\pi\colon (X, A) \to (X/A, \bar{A})$. Thus

$$\pi_*\colon H_n(X, A) \xrightarrow{\cong} H_n(X/A, \bar{A}) = H_n(X/A, A/A) \qquad \forall n \in \mathbb{N}$$

is an isomorphism.

Finally, $\bar{A}$ is a single point, so the long exact sequence of the pair $(X/A, \bar{A})$ in reduced homology (theorem 3.4.3, applied to reduced chains since $\bar{A} \neq \emptyset$) has the terms $\tilde{H}_n(\bar{A}) = 0$, whence

its connecting maps give $H_n(X/A, A/A) \cong \tilde{H}_n(X/A)$ for all n by the identification of relative homology with a point against reduced homology (definition 3.2.6). Chaining the isomorphisms,

$$H_n(X, A) \cong H_n(X/A, A/A) \cong \tilde{H}_n(X/A) \qquad \forall n \in \mathbb{N}$$

with the first induced by π_*, as we sought to show.

This motivates the following definition:

Definition 3.4.13: Good Pair

Let $A \subseteq X$ be a nonempty closed subspace that is a deformation retract of some neighbourhood of A in X. Then (X, A) is called a *good pair*.

For a good pair the relative groups are the reduced homology of the quotient, so the long exact sequence of the pair can be written in reduced absolute groups alone.

Theorem 3.4.14: Topological Snake Lemma

Let (X, A) be a good pair. Then there is an exact sequence:

$$\cdots \to \tilde{H}_n(A) \xrightarrow{i_*} \tilde{H}_n(X) \xrightarrow{\pi_*} \tilde{H}_n(X/A) \xrightarrow{\partial} \tilde{H}_{n-1}(A) \xrightarrow{i_*} \tilde{H}_{n-1}(X) \xrightarrow{\pi_*} \tilde{H}_{n-1}(X/A) \to \cdots$$

where i is the inclusion and $\pi\colon X \to X/A$ the quotient map. The sequence is natural for maps of good pairs: a map of pairs $f\colon (X, A) \to (Y, B)$ induces $\bar{f}\colon X/A \to Y/B$ with $\bar{f} \circ \pi = \pi \circ f$, and the maps $(f|_A)_*$, f_* and $\bar{f}_*$ carry the sequence of (X, A) to that of (Y, B) in a commuting ladder.

***Proof*:**

Step 1: The long exact sequence of the pair in reduced homology. Write $\tilde{C}_\bullet$ for the augmented chain complexes of definition 3.2.6 and $j\colon \tilde{C}_\bullet(X) \to C_\bullet(X, A)$ for the quotient map. The pair is good, so $A \neq \emptyset$, and by theorem 3.4.5(2) the sequence

$$\cdots \to \tilde{H}_n(A) \xrightarrow{i_*} \tilde{H}_n(X) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\partial} \tilde{H}_{n-1}(A) \xrightarrow{i_*} \tilde{H}_{n-1}(X) \to \cdots$$

is exact.

Step 2: Replacing the relative group by the quotient. Write $\bar{A} = A/A$ for the point onto which A collapses. By corollary 3.4.12, which applies because (X, A) is a good pair, so that A is nonempty, the quotient map induces isomorphisms $\pi_*\colon H_n(X, A) \to H_n(X/A, \bar{A})$. Step 1 applied to the pair $(X/A, \bar{A})$ gives an exact sequence

$$\tilde{H}_n(\bar{A}) \to \tilde{H}_n(X/A) \xrightarrow{j'_*} H_n(X/A, \bar{A}) \to \tilde{H}_{n-1}(\bar{A})$$

whose outer terms vanish by corollary 3.3.9, so j'_* is an isomorphism. The composite $\theta = (j'_*)^{-1}\pi_*$ is then an isomorphism from $H_n(X, A)$ to $\tilde{H}_n(X/A)$.

Step 3: Identifying the maps. The square of chain maps

$$\begin{array}{ccc} \tilde{C}_\bullet(X) & \xrightarrow{j} & C_\bullet(X,A) \\ {\scriptstyle \pi_\sharp}\downarrow & & \downarrow{\scriptstyle \pi_\sharp} \\ \tilde{C}_\bullet(X/A) & \xrightarrow{j'} & C_\bullet(X/A,\bar{A}) \end{array}$$

commutes, both composites sending a singular simplex σ of X to the class of $\pi \circ \sigma$, and the left map is a chain map of augmented complexes because $\epsilon \circ \pi_\sharp = \epsilon$. On homology this gives $\pi_* j_* = j'_* \pi_*$, with the relative map π_* on the left and the map $\pi_* \colon \tilde{H}_n(X) \to \tilde{H}_n(X/A)$ of the statement on the right, that is $\theta \circ j_* = \pi_*$. Replacing $H_n(X,A)$ by $\tilde{H}_n(X/A)$ along the isomorphism θ in the sequence of Step 1, with j_* replaced by $\theta j_* = \pi_*$ and ∂ by $\partial\theta^{-1}$, preserves exactness. The result is the sequence of the statement, with connecting map $\partial\theta^{-1}$.

Step 4: Naturality. Since $f(A) \subseteq B$, the map $\pi \circ f$ is constant on each fibre of $\pi \colon X \to X/A$, so it descends to a continuous $\bar{f}$ with $\bar{f} \circ \pi = \pi \circ f$ by the universal property of a quotient map [16, Universal Property Of Quotient Maps], and $\bar{f}$ carries $\bar{A}$ to $\bar{B}$. The squares containing i_* and π_* commute by proposition 3.3.3 for reduced homology, since $f \circ i = i \circ f|_A$ and $\bar{f} \circ \pi = \pi \circ f$. By theorem 3.4.5(4) for reduced sequences, $f_* \partial_X = \partial_Y f_*$ on $H_n(X,A)$, and the relative maps satisfy $\pi_* f_* = \bar{f}_* \pi_*$ and $\bar{f}_* j'_* = j'_* \bar{f}_*$, by functoriality for the maps of pairs $\pi \circ f = \bar{f} \circ \pi$ and by naturality of j_*. Hence $\theta_Y f_* = (j'_*)^{-1} \pi_* f_* = (j'_*)^{-1} \bar{f}_* \pi_* = \bar{f}_* \theta_X$, and the connecting square commutes: $f_* \partial_X \theta_X^{-1} = \partial_Y f_* \theta_X^{-1} = \partial_Y \theta_Y^{-1} \bar{f}_*$, as we sought to show.

Corollary 3.4.15: Homology Groups of Spheres

$$\tilde{H}_n(S^n) \cong \mathbb{Z} \qquad \tilde{H}_i(S^n) = 0 \quad (i \neq n)$$

***Proof*:**
For $n > 0$, take $(X,A) = (D^n, S^{n-1})$ so that $X/A = S^n$. This is a good pair, since S^{n-1} is a deformation retract of $D^n - \{0\}$. Then the terms $\tilde{H}_i(D^n)$ in the long exact sequence of theorem 3.4.14 for this pair are zero since D^n is contractible (corollary 3.3.9). Exactness of the sequence implies the maps

$$\tilde{H}_i(S^n) \xrightarrow{\partial} \tilde{H}_{i-1}(S^{n-1})$$

are isomorphisms for $i > 0$. Iterating, $\tilde{H}_k(S^n) \cong \tilde{H}_{k-n}(S^0)$ for $k \geq n$ and $\tilde{H}_k(S^n) \cong \tilde{H}_0(S^{n-k})$ for $k < n$. The space S^0 is two points, so $\tilde{H}_0(S^0) \cong \mathbb{Z}$ and $\tilde{H}_j(S^0) = 0$ for $j > 0$ (propositions 3.2.3 and 3.2.5), and $\tilde{H}_0(S^{n-k}) = 0$ for $k < n$ since S^{n-k} is path-connected (proposition 3.2.4). Hence the homology groups of S^n are:

$$\tilde{H}_k(S^n) = \begin{cases} \mathbb{Z} & k = n \\ 0 & k \neq n \end{cases}$$

as we sought to show.

Corollary 3.4.16: No Retract For Border Of Spheres

The border of D^n does not retract, that is ∂D^n is not a retract of D^n. Hence, every map $f : D^n \to D^n$ has a fixed point

***Proof*:**
If $r : D^n \to \partial D^n$ is a retraction, then $r \circ \iota = \mathrm{id}$, ι the inclusion map. Then the composition

$$\tilde{H}_{n-1}(\partial D^n) \xrightarrow{\iota_*} \tilde{H}_{n-1}(D^n) \xrightarrow{r_*} \tilde{H}_{n-1}(\partial D^n)$$

is the identity map on $\tilde{H}_{n-1}(\partial D^n) \cong \mathbb{Z}$. But ι_* and r_* are zero since $\tilde{H}_{n-1}(D^n) = 0$, a contradiction. For the fixed point statement, run the argument of theorem 2.1.50 in dimension n: if $f(x) \neq x$ for every x, let $r(x) = x + t(x)(x - f(x))$ be the point where the ray from $f(x)$ through x meets ∂D^n, with $t(x) \geq 0$ the larger root of the quadratic $|x + t(x - f(x))|^2 = 1$. The root depends continuously on x, and for $x \in \partial D^n$ it is 0 because $x \cdot (x - f(x)) = 1 - x \cdot f(x) \geq 0$. So r is a retraction of D^n onto ∂D^n, which does not exist.

For a nonempty subspace $A \subseteq X$ such that (X, A) is not a good pair, the cone $CA = (A \times I)/(A \times \{0\})$, attached to X along $A \times \{1\} = A$, takes the place of the quotient X/A. There are isomorphisms $H_n(X, A) \cong \tilde{H}_n(X \cup CA)$ for all n, obtained from the long exact sequence of the pair $(X \cup CA, CA)$, the excision of the cone point, and a deformation retraction onto A of the cone minus its cone point [21, Section 2.1, p. 125].

Lemma 3.4.17: CW Pairs Are Good Pairs

Let X be a CW complex and $A \subseteq X$ a nonempty subcomplex. Then A is a deformation retract of an open neighbourhood of A in X, so (X, A) is a good pair.

***Proof*:**
Write $A^n = A \cap X^n$, and $q_n \colon X^{n-1} \sqcup_\alpha D^n_\alpha \to X^n$ for the quotient maps of definition 1.2.1, with attaching maps $\varphi_\alpha \colon S^{n-1} \to X^{n-1}$. Since A is a closed union of cells, the attaching map of each cell of A has image in A.

Step 1: Neighbourhoods skeleton by skeleton. We construct open sets $N^n \subseteq X^n$ with $N^n \cap X^{n-1} = N^{n-1}$ and $A^n \subseteq N^n$, and homotopies $G^n \colon N^n \times I \to N^n$ from the identity that fix $N^{n-1} \cup A^n$ pointwise and satisfy $G^n_1(N^n) \subseteq N^{n-1} \cup A^n$. Put $N^0 = A^0$, which is open in the discrete space X^0, and let G^0 be the constant homotopy. Given N^{n-1}, let $U_\alpha = \varphi_\alpha^{-1}(N^{n-1})$, open in S^{n-1}. For an n-cell of A put $W_\alpha = D^n_\alpha$, and for any other n-cell put

$$W_\alpha = \left\{ y \in D^n_\alpha \;\middle|\; |y| > \tfrac{1}{2},\ y/|y| \in U_\alpha \right\}$$

an open subset of D^n_α. In both cases $W_\alpha \cap S^{n-1} = \varphi_\alpha^{-1}(N^{n-1})$, using for a cell of A that $\varphi_\alpha(S^{n-1}) \subseteq A^{n-1} \subseteq N^{n-1}$. Hence $N^{n-1} \sqcup_\alpha W_\alpha$ is open and saturated for q_n, so $N^n = q_n(N^{n-1} \sqcup_\alpha W_\alpha)$ is open in X^n, with $N^n \cap X^{n-1} = N^{n-1}$ and $A^n \subseteq N^n$. On W_α for a cell not in A let $g_t(y) = (1 - t + t/|y|)\, y$. It keeps the direction $y/|y|$, has $|g_t(y)| = (1-t)|y| + t \in (\frac{1}{2}, 1]$, fixes S^{n-1}, and sends W_α into U_α at $t = 1$. Let g_t be the identity on N^{n-1} and on the W_α of cells of A. These maps respect the identifications made by q_n, so they define G^n_t on N^n. By [16, Subspaces And Quotient Maps] the restriction of q_n to the open saturated set $N^{n-1} \sqcup_\alpha W_\alpha$ is a

quotient map onto N^n, and by [16, Products of Quotient Maps] so is its product with id_I, so G^n is continuous. Since $\varphi_\alpha(U_\alpha) \subseteq N^{n-1}$, the map G_1^n carries N^n into $N^{n-1} \cup A^n$.

Step 2: The neighbourhood and its deformation. Let $N = \bigcup_n N^n$. Then $N \cap X^n = N^n$ is open in X^n for every n, so N is open in X, which carries the weak topology, and $A \subseteq N$. Extend G^k to $N^k \cup A$ by the identity on A. The two definitions agree on $N^k \cap A = A^k$, and $N^k = (N^k \cup A) \cap X^k$ and A are closed in $N^k \cup A$, so the extension $\hat{G}^k$ is continuous by [16, Pasting Lemma]. For $x \in N^n$ define $H(x,t) = x$ for $t \leq 2^{-n}$, and on the interval $[2^{-k}, 2^{-k+1}]$, for $k = n, n-1, \ldots, 1$ in turn, let $H(x,-)$ run through $\hat{G}^k$, reparametrized over that interval and starting from the point $H(x, 2^{-k})$. This is well defined, since after $\hat{G}^n, \ldots, \hat{G}^{k+1}$ the point lies in $N^k \cup A$, and the definitions for $x \in N^n \subseteq N^{n+1}$ agree, since $\hat{G}^{n+1}$ fixes N^n. On each $N^n \times I$ the map H is a finite concatenation of continuous homotopies, hence continuous. The weak topology makes the map $\bigsqcup_n X^n \to X$ a quotient map, so by the two results cited in Step 1 the map $\bigsqcup_n N^n \times I \to N \times I$ is a quotient map, and H is continuous. Now $H_0 = \mathrm{id}$, every H_t fixes A pointwise and maps N into N, and $H_1(N) \subseteq N^0 \cup A = A$. So A is a deformation retract of the open neighbourhood N. As A is closed and nonempty, (X, A) is a good pair, as we sought to show.

Corollary 3.4.18: Excisions For CW Complex

Let X be a CW complex that is the union of subcomplexes A and B. Then the inclusion $(B, A \cap B) \hookrightarrow (X, A)$ induces an isomorphism

$$H_n(B, A \cap B) \cong H_n(X, A) \qquad \forall n \in \mathbb{N}$$

***Proof*:**
Suppose first that $A \cap B = \emptyset$. Then A and B are disjoint closed sets covering X, hence both open, so a singular simplex of X, having connected image, lies in A or in B, and $C_\bullet(X) = C_\bullet(A) \oplus C_\bullet(B)$. The inclusion therefore induces an isomorphism of chain complexes $C_\bullet(B, \emptyset) = C_\bullet(B) \to C_\bullet(X)/C_\bullet(A)$, hence on homology.

Now let $A \cap B \neq \emptyset$. The set $A \cap B$ is a closed union of cells, hence a nonempty subcomplex of the CW complex B, so by lemma 3.4.17 the pairs $(B, A \cap B)$ and (X, A) are good pairs. The inclusion $B \hookrightarrow X$ induces a continuous map $k\colon B/(A \cap B) \to X/A$, which is bijective, since $B - (A \cap B) = X - A$ and each quotient adds a single point. It is closed: a closed set of $B/(A \cap B)$ has as preimage a closed saturated set $F \subseteq B$, and the preimage in X of its image under k is F if F misses $A \cap B$ and $F \cup A$ otherwise, both closed in X because B and A are. So k is a homeomorphism, and it is a map of pairs from $(B/(A \cap B), *)$ to $(X/A, *)$ whose inverse induces the inverse map on homology. The square

$$\begin{array}{ccc} H_n(B, A \cap B) & \longrightarrow & H_n(X, A) \\ \downarrow & & \downarrow \\ H_n\big(B/(A \cap B), *\big) & \xrightarrow{k_*} & H_n(X/A, *) \end{array}$$

with vertical maps induced by the quotient maps commutes, because it commutes at the level of maps of pairs. The vertical maps are isomorphisms by corollary 3.4.12 and k_* is an isomorphism,

so the inclusion-induced map $H_n(B, A \cap B) \to H_n(X, A)$ is an isomorphism, as we sought to show.

Corollary 3.4.19: Wedge Sum Homology Group

Let (X_α, x_α), for α in a nonempty index set, be spaces with points such that each pair $(X_\alpha, \{x_\alpha\})$ is good, and let $\bigvee_\alpha X_\alpha$ be the wedge sum, the quotient of $\bigsqcup_\alpha X_\alpha$ identifying all the points x_α (definition 1.2.17), with inclusions $\iota_\alpha \colon X_\alpha \to \bigvee_\alpha X_\alpha$. Then for every n the map

$$\bigoplus_\alpha (\iota_\alpha)_* \colon \bigoplus_\alpha \tilde{H}_n(X_\alpha) \to \tilde{H}_n\Big(\bigvee_\alpha X_\alpha\Big)$$

is an isomorphism.

***Proof*:**
Let $X = \bigsqcup_\alpha X_\alpha$, let $A \subseteq X$ be the set of the points x_α, and let $p \colon X \to X/A = \bigvee_\alpha X_\alpha$ be the quotient map, so that ι_α is p restricted to X_α.

Step 1: Reduced and relative groups. For a space Y with a point y, the augmented complex of the point y is a subcomplex of $\tilde{C}_\bullet(Y)$ (definition 3.2.6) with quotient $C_\bullet(Y, y)$, and it has zero homology by corollary 3.3.9. By theorem 3.4.3 the quotient chain map induces isomorphisms $\tilde{H}_n(Y) \to H_n(Y, y)$, and these commute with the maps induced by maps of pointed spaces, since they are defined on chains.

Step 2: The disjoint union. The sets X_α are disjoint, open and cover X, so every singular simplex of X, having connected image, lies in exactly one X_α. Hence $C_\bullet(X) = \bigoplus_\alpha C_\bullet(X_\alpha)$ and $C_\bullet(A) = \bigoplus_\alpha C_\bullet(\{x_\alpha\})$, so $C_\bullet(X, A) = \bigoplus_\alpha C_\bullet(X_\alpha, x_\alpha)$ as chain complexes. Kernels and images of a direct sum of maps are computed summandwise, so the inclusions induce an isomorphism $\bigoplus_\alpha H_n(X_\alpha, x_\alpha) \to H_n(X, A)$.

Step 3: The quotient. The pair (X, A) is good: A is nonempty and closed, each point x_α being closed in X_α, and if V_α is a neighbourhood of x_α in X_α deformation retracting onto x_α, then $\bigsqcup_\alpha V_\alpha$ is a neighbourhood of A deformation retracting onto A, the deformation being continuous on each of the disjoint open sets $V_\alpha \times I$. By corollary 3.4.12 the quotient map induces isomorphisms $p_* \colon H_n(X, A) \to H_n(X/A, *)$.

Step 4: Assembly. For a singular simplex σ of X_α, the class of $p \circ \sigma = \iota_\alpha \circ \sigma$ is reached from σ along both routes of the square

$$\begin{array}{ccccc} \tilde{H}_n(X_\alpha) & \longrightarrow & H_n(X_\alpha, x_\alpha) & \longrightarrow & H_n(X, A) \\ {\scriptstyle (\iota_\alpha)_*}\downarrow & & & & \downarrow{\scriptstyle p_*} \\ \tilde{H}_n(X/A) & & \longrightarrow & & H_n(X/A, *) \end{array}$$

whose horizontal maps are those of Steps 1 and 2. So the square commutes, and $\bigoplus_\alpha (\iota_\alpha)_*$ is the composite of the isomorphisms of Steps 1, 2 and 3 with the inverse of the isomorphism of Step 1 for X/A, as we sought to show.

Theorem 3.4.20: Invariance Of Dimension

Let $U \subseteq \mathbb{R}^n$ and $V \subseteq \mathbb{R}^m$ be nonempty open subsets. If $U \cong V$, then $m = n$.

***Proof*:**
Take $x \in U$, and let $A = \mathbb{R}^n - \{x\}$, $B = U$, so that $A \cup B = \mathbb{R}^n$ and $A \cap B = U - \{x\}$. Since A and B are open with $A \cup B = \mathbb{R}^n$, the second form of excision (theorem 3.4.9) gives

$$H_k(U, U - \{x\}) \cong H_k(\mathbb{R}^n, \mathbb{R}^n - \{x\})$$

By the long exact sequence of reduced homology for the pair $(\mathbb{R}^n, \mathbb{R}^n - \{x\})$, in which the terms $\tilde{H}_k(\mathbb{R}^n)$ vanish since $\mathbb{R}^n$ is contractible (corollary 3.3.9), we get

$$H_k(\mathbb{R}^n, \mathbb{R}^n - \{x\}) \cong \tilde{H}_{k-1}(\mathbb{R}^n - \{x\})$$

Since $\mathbb{R}^n - \{x\}$ deformation retracts onto a sphere S^{n-1} centred at x, corollaries 3.3.7 and 3.4.15 give

$$H_k(U, U - \{x\}) \cong \begin{cases} \mathbb{Z} & k = n \\ 0 & k \neq n \end{cases}$$

Thus, by the same reasoning:

$$H_k(V, V - \{y\}) \cong \begin{cases} \mathbb{Z} & k = m \\ 0 & k \neq m \end{cases}$$

Since a homeomorphism $\varphi : U \to V$ induces an isomorphism $\varphi_* : H_k(U, U - \{x\}) \to H_k(V, V - \{\varphi(x)\})$, taking $y = \varphi(x)$ gives $n = m$, as we sought to show.

The statement is hard to prove with the methods of point-set topology alone, and homology reduces it to the computation of the local groups $H_k(U, U - \{x\})$ above.

Note that the converse is not true for general open sets (a punctured $\text{int}D^n$ and $\text{int}D^n$ have the same dimension but different homology groups). Limiting to simply connected sets, for $n = 1$ a nonempty simply connected open subset of $\mathbb{R}$ is connected, hence an open interval, and so homeomorphic to $\mathbb{R}$. For $n = 2$, a nonempty simply connected open subset of $\mathbb{R}^2 = \mathbb{C}$ is either $\mathbb{C}$ itself or, by the Riemann mapping theorem of *Complex Analysis* [31, Riemann Mapping Theorem], biholomorphic to the open unit disk, which is homeomorphic to $\mathbb{R}^2$ by $x \mapsto x/(1 - |x|)$. For $n \geq 3$ simple connectivity is not enough. A nonempty open subset of $\mathbb{R}^n$ is homeomorphic to $\mathbb{R}^n$ if and only if it is contractible and simply connected at infinity, by a theorem assembled from work of Stallings, Siebenmann, Freedman and Perelman [1, §1.1, Theorem 2]. The condition at infinity is necessary: the Whitehead manifold is a contractible open subset of $\mathbb{R}^3$ that is not homeomorphic to $\mathbb{R}^3$ [1, §1.1].

The idea of the above theorem can be generalized to find when two spaces are not locally homeomorphic:

Definition 3.4.21: Local Homology Groups

Let X be a topological space and $x \in X$ be a [closed] point. Then

$$H_n(X, X - \{x\})$$

is called the *local homology group* of the space X at x.

If U is an open neighbourhood of x in X, then excision (theorem 3.4.9) of the closed set $X - U$, which lies in the open set $X - \{x\}$ because $x \in U$ and $\{x\}$ is closed, gives

$$H_n(X, X - \{x\}) \cong H_n(U, U - \{x\})$$

meaning the local homology groups only depend on the local topology of X near x. Then a homeomorphism $f\colon X \to Y$ induces:

$$H_n(X, X - \{x\}) \cong H_n(Y, Y - \{f(x)\})$$

for all x and n. Hence, we see that the local homology group can be used to study spaces locally, and determine whether a space is locally homeomorphic at certain points.

3.4.1 Mayer-Vietoris Sequences

The long exact sequence of a pair relates absolute and relative homology groups. The Mayer-Vietoris sequence relates the homology of a union to the homology of the pieces, and is in many cases easier to compute with.

Theorem 3.4.22: Mayer-Vietoris Sequence

Let $A, B \subseteq X$ such that X is the union of the interiors of A and B. Then we have the following exact sequence

$$\cdots \to H_n(A \cap B) \xrightarrow{\varphi} H_n(A) \oplus H_n(B) \xrightarrow{\psi} H_n(X) \xrightarrow{\partial} H_{n-1}(A \cap B) \xrightarrow{\varphi} H_{n-1}(A) \oplus H_{n-1}(B) \to \cdots \to H_0(X) \to 0$$

called the *Mayer-Vietoris sequence*. Here $\varphi(x) = (i_* x, -j_* x)$ and $\psi(a, b) = k_* a + l_* b$ for the inclusions $i\colon A \cap B \to A$, $j\colon A \cap B \to B$, $k\colon A \to X$ and $l\colon B \to X$.

The idea behind this sequence is that we often use it for induction proofs, where we know information about A, B and $A \cap B$, and build up to show information about the homology of $A \cup B$.

***Proof*:**
The whole sequence is the long exact sequence of homology attached, via the algebraic snake lemma (theorem 3.4.3), to a short exact sequence of chain complexes. The work is to produce that short exact sequence and to identify its outer terms with $H_n(A \cap B)$ and $H_n(A) \oplus H_n(B)$ and its middle term, after one further isomorphism, with $H_n(X)$.

Stage 1: small chains compute $H_n(X)$. Let $C_n(A + B) \subseteq C_n(X)$ be the subgroup of chains $\sum_i n_i \sigma_i$ each of whose singular simplices σ_i has image contained in A or in B. Equivalently, $C_n(A + B) = C_n(A) + C_n(B)$ as subgroups of $C_n(X)$, the sum of the images of the inclusions of A and B. The boundary $\partial\colon C_n(X) \to C_{n-1}(X)$ carries each σ_i to an alternating sum of its faces,

and a face of a simplex with image in A (respectively B) again has image in A (respectively B). Hence ∂ maps $C_n(A+B)$ into $C_{n-1}(A+B)$, so the groups $C_\bullet(A+B)$ form a subcomplex of $C_\bullet(X)$. Write $H_n(A+B)$ for its homology.

Apply lemma 3.4.11 to the collection $U = \{A, B\}$. Its interiors $\mathrm{int}A$ and $\mathrm{int}B$ cover X by hypothesis, and the small-chain complex $C_\bullet^U(X)$ of that lemma is by construction $C_\bullet(A+B)$. The lemma provides a chain map $\rho\colon C_\bullet(X) \to C_\bullet(A+B)$ with $\iota\rho$ and $\rho\iota$ chain homotopic to the identity, where $\iota\colon C_\bullet(A+B) \hookrightarrow C_\bullet(X)$ is the inclusion. Consequently ι induces an isomorphism

$$\iota_*\colon H_n(A+B) \xrightarrow{\sim} H_n(X) \qquad \forall n \in \mathbb{N}$$

This is the single point at which the covering hypothesis $X = \mathrm{int}A \cup \mathrm{int}B$ enters. Everything else is formal.

Stage 2: the short exact sequence of complexes. For each n define homomorphisms

$$0 \longrightarrow C_n(A\cap B) \xrightarrow{\varphi} C_n(A) \oplus C_n(B) \xrightarrow{\psi} C_n(A+B) \longrightarrow 0$$

by $\varphi(x) = (x, -x)$ and $\psi(a, b) = a + b$, where each chain is regarded inside $C_n(X)$ through the relevant inclusion. Both are restrictions of maps built from the inclusion-induced chain maps $C_\bullet(A\cap B) \to C_\bullet(A)$, $C_\bullet(A\cap B) \to C_\bullet(B)$, $C_\bullet(A) \to C_\bullet(X)$ and $C_\bullet(B) \to C_\bullet(X)$, each of which commutes with ∂. Hence φ and ψ are chain maps. Exactness is checked degreewise.

1. *φ is injective.* If $\varphi(x) = (x, -x) = 0$ then $x = 0$ already in the first coordinate, so $\ker\varphi = 0$.
2. *ψ is surjective.* An element of $C_n(A+B)$ is by definition a sum $a + b$ with $a \in C_n(A)$ and $b \in C_n(B)$, and $\psi(a, b) = a + b$. Hence ψ hits every generator.
3. $\ker\psi = \mathrm{im}\varphi$. The composite $\psi\varphi(x) = x + (-x) = 0$, so $\mathrm{im}\varphi \subseteq \ker\psi$. Conversely, suppose $\psi(a, b) = a + b = 0$ as a chain in $C_n(X)$, with $a \in C_n(A)$ and $b \in C_n(B)$. Then $a = -b$ as elements of $C_n(X)$, an equality of formal $\mathbb{Z}$-linear combinations of singular simplices. Two such combinations are equal precisely when they assign the same coefficient to each singular simplex of X, so every simplex occurring in a with nonzero coefficient occurs in $-b$ with the same coefficient. Such a simplex has image in A (because it occurs in a) and in B (because it occurs in b), hence in $A \cap B$. Therefore $a \in C_n(A\cap B)$, and $(a, b) = (a, -a) = \varphi(a) \in \mathrm{im}\varphi$. This proves $\ker\psi \subseteq \mathrm{im}\varphi$.

Since φ and ψ are chain maps and the displayed sequence is exact in every degree, the diagram is a short exact sequence of chain complexes.

Stage 3: assembly. Feed the short exact sequence of Stage 2 into the algebraic snake lemma (theorem 3.4.3). It produces the long exact sequence

$$\cdots \to H_n(A\cap B) \xrightarrow{\varphi_*} H_n(A) \oplus H_n(B) \xrightarrow{\psi_*} H_n(A+B) \xrightarrow{\partial} H_{n-1}(A\cap B) \to \cdots$$

in which the homology of the middle complex $C_\bullet(A) \oplus C_\bullet(B)$ is $H_n(A) \oplus H_n(B)$, because homology commutes with the direct sum of complexes. Now replace $H_n(A+B)$ throughout by $H_n(X)$ using the isomorphism ι_* of Stage 1. Because ι is an inclusion of subcomplexes of $C_\bullet(X)$, the map ψ followed by ι is exactly the chain map $(a, b) \mapsto a + b \in C_n(X)$, so after this substitution the map $H_n(A) \oplus H_n(B) \to H_n(X)$ is the one induced by $(a, b) \mapsto a + b$, that is, the sum of the two inclusion-induced maps $H_n(A) \to H_n(X)$ and $H_n(B) \to H_n(X)$. Likewise

φ_* becomes $x \mapsto (x, -x)$ on homology, the pair of inclusion-induced maps $H_n(A \cap B) \to H_n(A)$ and $H_n(A \cap B) \to H_n(B)$. Substituting gives the exact sequence

$$\cdots \to H_n(A\cap B) \xrightarrow{\varphi} H_n(A)\oplus H_n(B) \xrightarrow{\psi} H_n(X) \xrightarrow{\partial} H_{n-1}(A\cap B) \xrightarrow{\varphi} H_{n-1}(A)\oplus H_{n-1}(B) \to \cdots$$

The sequence terminates on the right in $\cdots \to H_0(X) \to 0$, since $H_{-1}(A \cap B) = 0$ forces the snake lemma's tail to end at H_0. This is the Mayer-Vietoris sequence, completing the proof.

When $A \cap B \neq \emptyset$, augmenting the previous short exact sequence of complexes by the augmentation maps gives a formally identical Mayer-Vietoris sequence for reduced homology groups:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & C_0(A\cap B) & \longrightarrow & C_0(A)\oplus C_0(B) & \longrightarrow & C_0(A+B) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \epsilon} & & \downarrow{\scriptstyle \epsilon\oplus\epsilon} & & \downarrow{\scriptstyle \epsilon} & & \\ 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z}\oplus\mathbb{Z} & \longrightarrow & \mathbb{Z} & \longrightarrow & 0 \end{array}$$

If $A \cap B$ is path-connected, then we may think of Mayer-Vietoris sequences as an analog of van Kampen's theorem. In particular, the reduced Mayer-Vietoris sequence gives an isomorphism:

$$H_1(X) \cong \frac{H_1(A) \oplus H_1(B)}{\mathrm{im}\varphi}$$

which is exactly the abelianization of the van Kampen theorem, where H_1 is the abelianization of π_1 for path connected spaces (theorem 5.4.5).

Corollary 3.4.23: Naturality of the Mayer-Vietoris Sequence in Homology

The sequence of theorem 3.4.22 is natural in the triple (X, A, B): if $f\colon X \to Y$ is continuous with $f(A) \subseteq A'$ and $f(B) \subseteq B'$, where $X = \mathrm{int}A \cup \mathrm{int}B$ and $Y = \mathrm{int}A' \cup \mathrm{int}B'$, then the diagram

$$\begin{array}{ccccccccc} \cdots \to & H_n(A\cap B) & \longrightarrow & H_n(A)\oplus H_n(B) & \to & H_n(X) & \xrightarrow{\partial} & H_{n-1}(A\cap B) & \to \cdots \\ & \downarrow & & \downarrow & & \downarrow{\scriptstyle f_*} & & \downarrow & \\ \cdots \to & H_n(A'\cap B') & \to & H_n(A')\oplus H_n(B') & \to & H_n(Y) & \xrightarrow{\partial} & H_{n-1}(A'\cap B') & \to \cdots \end{array}$$

commutes, the unlabelled vertical maps being induced by the restrictions of f. The same holds for the reduced sequences when $A \cap B$ and $A' \cap B'$ are nonempty.

***Proof*:**
By lemma 3.3.2(3) the chain map $f_\sharp$ carries $C_\bullet(A)$ into $C_\bullet(A')$, $C_\bullet(B)$ into $C_\bullet(B')$, $C_\bullet(A \cap B)$ into $C_\bullet(A' \cap B')$, and hence $C_\bullet(A + B)$ into $C_\bullet(A' + B')$. These restrictions map the short exact sequence of Stage 2 of the proof of theorem 3.4.22 for (X, A, B) to the one for (Y, A', B'), and both squares commute at the level of chains: $(f_\sharp x, -f_\sharp x) = \varphi(f_\sharp x)$ and $f_\sharp a + f_\sharp b = f_\sharp(a + b)$. Proposition 3.4.4 gives a commuting ladder between the long exact sequences with $H_n(A+B)$ and $H_n(A' + B')$ in the third place. The isomorphisms $\iota_*\colon H_n(A + B) \to H_n(X)$ and $H_n(A' + B') \to H_n(Y)$ of Stage 1 satisfy $f_*\iota_* = \iota_* f_*$, since $f_\sharp \iota = \iota f_\sharp$ on small chains, both being $f_\sharp$ applied to a chain of X. Transporting the ladder along these isomorphisms, every square remains a composite of commuting squares, including those containing the connecting maps $\partial\iota_*^{-1}$. For the

reduced sequences the same argument runs on the augmented complexes, $f_\sharp$ being extended by $\mathrm{id}_{\mathbb{Z}}$ in degree -1 (lemma 3.3.2(4)), completing the proof.

Proposition 3.4.24: Relative Mayer-Vietoris Sequence

Let $X = A \cup B$ be a space, let $C \subseteq A$ and $D \subseteq B$, and put $Y = C \cup D$. Write $C_n(A+B) = C_n(A) + C_n(B) \subseteq C_n(X)$, $C_n(C+D) = C_n(C) + C_n(D) \subseteq C_n(Y)$ and $C_n(A+B, C+D) = C_n(A+B)/C_n(C+D)$.

1. The maps $\varphi(x) = (x, -x)$ and $\psi(a, b) = a + b$ induce a short exact sequence of chain complexes

$$0 \to C_\bullet(A \cap B, C \cap D) \xrightarrow{\varphi} C_\bullet(A, C) \oplus C_\bullet(B, D) \xrightarrow{\psi} C_\bullet(A + B, C + D) \to 0$$

   in which every group is free abelian.

2. If $X = \operatorname{int} A \cup \operatorname{int} B$ and Y is the union of the interiors of C and D in Y, then the chain map $\iota\colon C_\bullet(A + B, C + D) \to C_\bullet(X, Y)$ induced by the inclusions induces isomorphisms on homology.

3. Under the hypotheses of (2) there is a long exact sequence

$$\cdots \to H_n(A\cap B, C\cap D) \xrightarrow{\varphi} H_n(A,C) \oplus H_n(B,D) \xrightarrow{\psi} H_n(X,Y) \xrightarrow{\partial} H_{n-1}(A\cap B, C\cap D) \to \cdots$$

   with $\varphi(x) = (i_* x, -j_* x)$ and $\psi(a, b) = k_* a + l_* b$ for the inclusions of pairs $i\colon (A \cap B, C \cap D) \to (A, C)$, $j\colon (A \cap B, C \cap D) \to (B, D)$, $k\colon (A, C) \to (X, Y)$ and $l\colon (B, D) \to (X, Y)$.

For $C = D = \emptyset$ the sequence of (3) is that of theorem 3.4.22.

***Proof*:**

For a subspace $Z \subseteq X$, the group $C_n(Z)$ is the subgroup of $C_n(X)$ spanned by the singular simplices with image in Z, a subset of the basis of $C_n(X)$. A quotient of the group spanned by a set of simplices by the subgroup spanned by a subset of them is free on the remaining simplices.

1. The group $C_n(A+B)$ is spanned by the simplices with image in A or in B, and $C_n(C+D)$ by those with image in C or in D, a subset of these since $C \subseteq A$ and $D \subseteq B$. So $C_n(A + B, C + D)$ is free, and so are $C_n(A \cap B, C \cap D)$, $C_n(A, C)$ and $C_n(B, D)$. The map $x \mapsto (x, -x)$ carries $C_n(C \cap D)$ into $C_n(C) \oplus C_n(D)$, and $(a, b) \mapsto a + b$ carries $C_n(C) \oplus C_n(D)$ into $C_n(C + D)$, so both descend to the quotients, and they commute with ∂ because the inclusions are chain maps preserving all the subcomplexes involved.

   The map ψ is surjective, every element of $C_n(A + B)$ being a sum $a + b$, and $\psi\varphi = 0$. If $\varphi[x] = 0$ for $x \in C_n(A \cap B)$, then $x \in C_n(C) \cap C_n(D)$, so every simplex occurring in x has image in $C \cap D$, and $[x] = 0$. If $\psi([a], [b]) = 0$, then $a + b = c + d$ with $c \in C_n(C)$ and $d \in C_n(D)$, so $(a - c) + (b - d) = 0$ in $C_n(X)$, with $a - c \in C_n(A)$ and $b - d \in C_n(B)$. Each simplex occurring in $a - c$ then occurs in $b - d$ with the opposite coefficient, so it has image in $A \cap B$. Hence $x = a - c \in C_n(A \cap B)$, $b - d = -x$, and $([a], [b]) = ([x + c], [-x + d]) = ([x], [-x]) = \varphi[x]$.

2. The inclusions form a map of short exact sequences of chain complexes

$$\begin{array}{ccccccccc} 0 & \longrightarrow & C_\bullet(C+D) & \longrightarrow & C_\bullet(A+B) & \longrightarrow & C_\bullet(A+B,C+D) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow{\scriptstyle \iota} & & \\ 0 & \longrightarrow & C_\bullet(Y) & \longrightarrow & C_\bullet(X) & \longrightarrow & C_\bullet(X,Y) & \longrightarrow & 0 \end{array}$$

   The left vertical map induces isomorphisms on homology by lemma 3.4.11, applied to the space Y and the collection $\{C, D\}$, whose interiors in Y cover Y, and the middle one by the same lemma applied to X and $\{A, B\}$. By proposition 3.4.4 the long exact sequences of the two rows form a commuting ladder. In the window $H_n(C+D) \to H_n(A+B) \to H_n(A+B, C+D) \to H_{n-1}(C+D) \to H_{n-1}(A+B)$ the four outer vertical maps are isomorphisms, so the five lemma (lemma 3.4.6) makes ι_* an isomorphism.

3. Theorem 3.4.3 applied to the sequence of (1) gives a long exact sequence whose middle terms are $H_n(A, C) \oplus H_n(B, D)$, homology commuting with direct sums of complexes. Replacing $H_n(A+B, C+D)$ by $H_n(X, Y)$ along the isomorphism ι_* of (2) preserves exactness. The composite $\iota\psi$ is the chain map $([a], [b]) \mapsto [a+b] \in C_n(X, Y)$, which induces $k_* + l_*$, and φ induces $(i_*, -j_*)$.

For $C = D = \emptyset$ all groups are absolute, and the sequence of (1) and the map ι are those of the proof of theorem 3.4.22, so (3) is that sequence, completing the proof.

Corollary 3.4.25: Mayer-Vietoris for Deformation Retracts of Neighbourhoods

Let $X = A \cup B$, and suppose there are $U \supseteq A$ and $V \supseteq B$ with $X = \mathrm{int}U \cup \mathrm{int}V$ such that the inclusions $A \hookrightarrow U$, $B \hookrightarrow V$ and $A \cap B \hookrightarrow U \cap V$ are homotopy equivalences, for instance deformation retractions. Then the exact sequence of theorem 3.4.22 holds for A and B.

***Proof*:**
Take $C = D = \emptyset$ in proposition 3.4.24. The inclusions map the short exact sequence of part (1) for A, B to the one for U, V, and on the homology of the first two terms, $H_n(A \cap B) \to H_n(U \cap V)$ and $H_n(A) \oplus H_n(B) \to H_n(U) \oplus H_n(V)$, they induce isomorphisms by corollary 3.3.7. By proposition 3.4.4 and the five lemma (lemma 3.4.6), the map $H_n(A+B) \to H_n(U+V)$ is an isomorphism, and $H_n(U+V) \to H_n(X)$ is one by part (2). The long exact sequence of the short exact sequence of part (1) for A, B, with $H_n(A+B)$ replaced by $H_n(X)$ along the composite isomorphism, is the asserted sequence, its maps identified as in part (3), completing the proof.

Lemma 3.4.26: Stereographic Projection

Let $m \geq 0$ and let $a \in S^m \subseteq \mathbb{R}^{m+1}$. The map

$$s_a \colon S^m - \{a\} \to a^\perp \qquad s_a(z) = \frac{z - (z \cdot a)\, a}{1 - z \cdot a}$$

is a homeomorphism onto the hyperplane $a^\perp = \{w \in \mathbb{R}^{m+1} \mid w \cdot a = 0\} \cong \mathbb{R}^m$, with inverse $w \mapsto \big(2w + (|w|^2 - 1)\, a\big)/(|w|^2 + 1)$, and $|s_a(z)|^2 = (1 + z \cdot a)/(1 - z \cdot a)$.

***Proof*:**
For $z \in S^m - \{a\}$ we have $z \cdot a < 1$, so s_a is continuous, and $s_a(z) \cdot a = 0$. Since $|z| = |a| = 1$,

$$\big|z - (z \cdot a)\, a\big|^2 = 1 - (z \cdot a)^2 = (1 - z \cdot a)(1 + z \cdot a)$$

which gives the formula for $|s_a(z)|^2$, and with it $|s_a(z)|^2 + 1 = 2/(1 - z \cdot a)$ and $|s_a(z)|^2 - 1 = 2(z \cdot a)/(1 - z \cdot a)$. Write $t(w) = \big(2w + (|w|^2 - 1)\, a\big)/(|w|^2 + 1)$ for $w \in a^\perp$, a continuous map. Substituting the last two identities with $w = s_a(z)$,

$$t\big(s_a(z)\big) = (1 - z \cdot a)\Big(\frac{z - (z \cdot a)\, a}{1 - z \cdot a} + \frac{z \cdot a}{1 - z \cdot a}\, a\Big) = z$$

Conversely, for $w \in a^\perp$ orthogonality gives $|2w + (|w|^2 - 1)a|^2 = 4|w|^2 + (|w|^2 - 1)^2 = (|w|^2 + 1)^2$, so $t(w) \in S^m$, and $t(w) \cdot a = (|w|^2 - 1)/(|w|^2 + 1) < 1$, so $t(w) \neq a$. Then $1 - t(w) \cdot a = 2/(|w|^2 + 1)$ and $t(w) - \big(t(w) \cdot a\big)a = 2w/(|w|^2 + 1)$, so $s_a(t(w)) = w$. Hence s_a and t are mutually inverse continuous maps, as claimed.

Example 3.4.27: Mayer-Vietoris Sequence

1. Let $n \geq 1$, let $N = e_{n+1}$ and $S = -e_{n+1}$ be the poles of S^n, and let $A = S^n \setminus \{S\}$ and $B = S^n \setminus \{N\}$, open sets covering S^n. By lemma 3.4.26, s_S and s_N carry A and B homeomorphically onto copies of $\mathbb{R}^n$, so $\tilde{H}_i(A) = \tilde{H}_i(B) = 0$ for all i (corollary 3.3.9). Since $s_N(S) = 0$, the map s_N carries $A \cap B = S^n \setminus \{N, S\}$ homeomorphically onto $N^\perp \setminus \{0\} \cong \mathbb{R}^n \setminus \{0\}$, which deformation retracts onto the unit sphere S^{n-1} through $(w, t) \mapsto (1-t)w + tw/|w|$. So $\tilde{H}_i(A \cap B) \cong \tilde{H}_i(S^{n-1})$ by corollary 3.3.7, and the reduced Mayer-Vietoris sequence, which applies as $A \cap B \neq \emptyset$, contains the exact segment

$$0 = \tilde{H}_i(A) \oplus \tilde{H}_i(B) \to \tilde{H}_i(S^n) \xrightarrow{\partial} \tilde{H}_{i-1}(A \cap B) \to \tilde{H}_{i-1}(A) \oplus \tilde{H}_{i-1}(B) = 0$$

   Hence $\tilde{H}_i(S^n) \cong \tilde{H}_{i-1}(S^{n-1})$ for every i, which computes the homology of spheres by induction on n.

2. Let K be the Klein bottle, and see it as the union of two Möbius bands A and B glued at their boundary by a homeomorphism. Then A, B, and $A \cap B$ are homotopy equivalent to circles. The band A is a deformation retract of the open set U obtained by adding to it an open collar of the circle $A \cap B$ in B, likewise B of an open set V, and $U \cap V$, an open annulus, deformation retracts onto $A \cap B$. So corollary 3.4.25 applies, and the important

part of the Mayer-Vietoris sequence for the decomposition $K = A \cup B$ is

$$0 \to H_2(K) \to H_1(A \cap B) \xrightarrow{\varphi} H_1(A) \oplus H_1(B) \to H_1(K) \to 0$$

The map $\varphi : \mathbb{Z} \to \mathbb{Z} \oplus \mathbb{Z}$ is given by $1 \mapsto (2, -2)$ for suitable generators, because the inclusion of the boundary circle of a Möbius band induces multiplication by 2 on $H_1 \cong \mathbb{Z}$. To see this, write the band as $M = [0,1] \times [-1,1]/\sim$ with $(0,t) \sim (1,-t)$. Its core circle is $C = [0,1] \times \{0\}/\sim$, and its boundary circle ∂M is traced once by the loop $\gamma\colon [0,2] \to M$ with $\gamma(s) = (s,1)$ for $s \leq 1$ and $\gamma(s) = (s-1,-1)$ for $s \geq 1$, which closes up because $(1,1) \sim (0,-1)$ and $(1,-1) \sim (0,1)$. The maps $r_u(s,t) = (s,(1-u)t)$, $u \in I$, respect $\sim$ and descend from a continuous map on $[0,1] \times [-1,1] \times I$, so they form a deformation retraction of M onto C, and $(r_1)_*\colon H_1(M) \to H_1(C)$ is an isomorphism (corollary 3.3.7). Identify C with S^1 by $(s,0) \mapsto e^{2\pi i s}$ and ∂M with S^1 by $\gamma(s) \mapsto e^{\pi i s}$. Then $r_1(\gamma(s)) = e^{2\pi i s}$, so $r_1|_{\partial M}$ is the map $z \mapsto z^2$, of degree 2 (example 3.5.8). Hence the inclusion $\partial M \hookrightarrow M$ induces multiplication by 2, after these identifications. Applying this to A and to B, whose common boundary circle is $A \cap B$, and recalling the sign in $\varphi(x) = (i_*x, -j_*x)$, gives $\varphi(1) = (2,-2)$. Since φ is injective, we get

$$H_2(K) = 0$$

Furthermore, $H_1(K) = \mathbb{Z} \oplus \mathbb{Z}_2$, since we can choose $(1,0)$ and $(1,-1)$ as a basis for $\mathbb{Z} \oplus \mathbb{Z}$. The higher homology groups vanish by exactness, since $H_n(A \cap B)$, $H_n(A)$ and $H_n(B)$ are homology groups of circles and vanish for $n \geq 2$ (corollaries 3.3.7 and 3.4.15). Hence:

$$\tilde{H}_n(K) = \begin{cases} 0 & n = 0 \\ \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} & n = 1 \\ 0 & n \geq 2 \end{cases}$$

Example 3.4.28: Product With Sphere
Let X be a space and $n \geq 0$. Then for every k

$$H_k(X \times S^n) \cong H_k(X) \oplus H_{k-n}(X)$$

where $H_j = 0$ for $j < 0$. Regard $S^m \subseteq S^{m+1}$ as the equator $x_{m+2} = 0$, fix $e = (1,0,\ldots,0)$, which lies in every S^m, and let $p\colon X \times S^n \to X$ be the projection and $s\colon X \to X \times S^n$ the map $x \mapsto (x,e)$. Since $ps = \mathrm{id}_X$, proposition 3.3.3 gives $p_*s_* = \mathrm{id}$. Every class c is $s_*p_*c + (c - s_*p_*c)$ with $c - s_*p_*c \in \ker p_*$, and if $s_*x \in \ker p_*$ then $x = p_*s_*x = 0$. So every class is uniquely the sum of an element of $s_*H_k(X) \cong H_k(X)$ and an element of $\ker p_*$, and the claim follows once $\ker p_* \cong H_{k-n}(X)$, which is proved by induction on n.

For $n = 0$, the space $X \times S^0$ is the disjoint union of the open sets $X \times \{e\}$ and $X \times \{-e\}$, each mapped homeomorphically onto X by p. As in Step 2 of the proof of corollary 3.4.19, the two inclusions identify $H_k(X \times S^0)$ with $H_k(X) \oplus H_k(X)$, and p_* becomes $(a,b) \mapsto a+b$, whose kernel $\{(a,-a) \mid a \in H_k(X)\}$ is isomorphic to $H_k(X)$.

Let $n \geq 1$ and let $N = (0,\ldots,0,1) \in S^n$. The open sets $U = S^n - \{N\}$ and $V = S^n - \{-N\}$ cover S^n and contain e. By lemma 3.4.26, s_N maps U homeomorphically onto the hyperplane $N^\perp \cong \mathbb{R}^n$, with $s_N(e) = e$, so the straight-line homotopy to e in $N^\perp$ contracts U to e, and likewise V. Moreover s_N maps $U \cap V$ onto $N^\perp - \{0\}$, since $s_N(-N) = 0$, and it is the identity

on the equator $S^{n-1} = S^n \cap N^\perp$. The homotopy $w \mapsto (1-t)w + tw/|w|$ deformation retracts $N^\perp - \{0\}$ onto its unit sphere, which is that equator, so the inclusion $S^{n-1} \hookrightarrow U \cap V$ is a homotopy equivalence. Taking products with id_X, the maps $X \to X \times U$ and $X \to X \times V$ given by $x \mapsto (x, e)$ are homotopy equivalences with homotopy inverses the projections, and the inclusion $J\colon X \times S^{n-1} \to X \times (U \cap V)$ is a homotopy equivalence.

Apply theorem 3.4.22 to the open sets $A = X \times U$ and $B = X \times V$ of $X \times S^n$, with $A \cap B = X \times (U \cap V)$. Write p_A, p_B, p_C for the projections of A, B, $A \cap B$ to X and s_A, s_B, s_C for the maps $x \mapsto (x, e)$ into them. By theorem 3.3.6 and proposition 3.3.3, $(p_A)_*$ and $(p_B)_*$ are isomorphisms inverse to $(s_A)_*$ and $(s_B)_*$, while $(p_C)_*(s_C)_* = \mathrm{id}$. Since p_A and p_B restrict to p_C on $A \cap B$, the map φ followed by $(p_A)_* \oplus (p_B)_*$ is $c \mapsto ((p_C)_*c, -(p_C)_*c)$, so $\ker\varphi = \ker(p_C)_*$. Every $(a, b) \in H_k(A) \oplus H_k(B)$ equals $((s_A)_*a', (s_B)_*b')$ with $a' = (p_A)_*a$ and $b' = (p_B)_*b$, and the inclusions compose with s_A and s_B to s, so $\psi(a, b) = s_*(a' + b')$. Hence $\mathrm{im}\psi = s_*H_k(X)$. By exactness, ∂ vanishes exactly on $s_*H_k(X)$ and maps $H_k(X \times S^n)$ onto $\ker\varphi = \ker(p_C)_*$ in degree $k-1$. Restricted to $\ker p_*$, which meets $s_*H_k(X)$ only in 0 because $p_*s_* = \mathrm{id}$, the map ∂ is therefore an isomorphism onto $\ker\big((p_C)_*\colon H_{k-1}(A \cap B) \to H_{k-1}(X)\big)$.

Finally $p_C \circ J$ is the projection $p'\colon X \times S^{n-1} \to X$, and J_* is an isomorphism, so J_* carries $\ker p'_*$ onto $\ker(p_C)_*$. By the inductive hypothesis $\ker p'_* \cong H_{(k-1)-(n-1)}(X) = H_{k-n}(X)$ in degree $k-1$. Hence $\ker p_* \cong H_{k-n}(X)$ and $H_k(X \times S^n) \cong H_k(X) \oplus H_{k-n}(X)$.

3.4.2 Equivalence of Simplicial and Singular Homology

The theory of homology was developed through singular homology, and the computational techniques through simplicial homology. The two homology groups are isomorphic, and the isomorphism extends to relative groups. If X is a Δ-complex with $A \subseteq X$ a subcomplex (a Δ-complex formed by a union of simplices of X), the relative groups $H_n^\Delta(X, A)$ are the homology groups of $\Delta_n(X, A) = \Delta_n(X)/\Delta_n(A)$. The *canonical chain map* $\theta\colon \Delta_\bullet(X, A) \to C_\bullet(X, A)$ sends each n-simplex of X to its characteristic map $\sigma\colon \Delta^n \to X$, and induces a canonical homomorphism $H_n^\Delta(X, A) \to H_n(X, A)$. If $A = \emptyset$, the relative homology is the absolute homology.

Definition 3.4.29: Quasi-Isomorphism

A chain map $f\colon C_\bullet \to D_\bullet$ is a *quasi-isomorphism* if $f_*\colon H_n(C) \to H_n(D)$ is an isomorphism for every n.

A chain homotopy equivalence is a quasi-isomorphism, by proposition 3.3.5.

Lemma 3.4.30: The Identity Simplex Generates

For every $n \geq 0$, the identity map $\iota_n\colon \Delta^n \to \Delta^n$, viewed as a singular n-simplex, is a relative cycle whose class generates $H_n(\Delta^n, \partial\Delta^n) \cong \mathbb{Z}$ (with $\partial\Delta^0 = \emptyset$).

***Proof*:**
Each face of ι_n has image in $\partial\Delta^n$, so $\partial\iota_n \in C_{n-1}(\partial\Delta^n)$ and ι_n is a relative cycle. We induct on n. For $n = 0$, Δ^0 is a point, whose unique singular 0-simplex is ι_0, and $H_0(\Delta^0) = C_0(\Delta^0) \cong \mathbb{Z}$ is generated by its class (proposition 3.2.5).

Let $n \geq 1$, write $v_0, \ldots, v_n$ for the vertices of Δ^n, let $F_0 = [v_1, \ldots, v_n]$ be the face opposite v_0, and let Λ be the union of the other n faces of dimension $n-1$, those containing v_0. Then $\partial\Delta^n = F_0 \cup \Lambda$ and $F_0 \cap \Lambda = \partial F_0$. Let $j\colon \Delta^{n-1} \to \partial\Delta^n$ be the order-preserving affine map onto F_0. We show that the two maps

$$H_n(\Delta^n, \partial\Delta^n) \xrightarrow{\partial} H_{n-1}(\partial\Delta^n, \Lambda) \xleftarrow{j_*} H_{n-1}(\Delta^{n-1}, \partial\Delta^{n-1})$$

are isomorphisms, where ∂ is the connecting map of the long exact sequence of the triple $(\Delta^n, \partial\Delta^n, \Lambda)$.

Step 1: ∂ is an isomorphism. The set Λ is nonempty, and it is contractible: each of its faces is convex and contains v_0, so the straight-line homotopy to v_0 stays in Λ. The simplex Δ^n is contractible as well, so in the long exact sequence of the pair (Δ^n, Λ) in reduced homology (Step 1 of the proof of theorem 3.4.14) the absolute terms vanish (corollary 3.3.9), and $H_i(\Delta^n, \Lambda) = 0$ for every i. These groups flank ∂ in the long exact sequence of the triple

$$H_n(\Delta^n, \Lambda) \to H_n(\Delta^n, \partial\Delta^n) \xrightarrow{\partial} H_{n-1}(\partial\Delta^n, \Lambda) \to H_{n-1}(\Delta^n, \Lambda)$$

so ∂ is an isomorphism. By the construction of the connecting homomorphism in theorem 3.4.3, with ι_n as its own lift to $C_n(\Delta^n, \Lambda)$, it sends $[\iota_n]$ to the class of $\partial\iota_n = \sum_i (-1)^i \iota_n|_{[v_0,\ldots,\hat{v}_i,\ldots,v_n]}$. Every term with $i \geq 1$ lies in $C_{n-1}(\Lambda)$, so $\partial[\iota_n]$ is the class of $\iota_n|_{F_0} = j = j \circ \iota_{n-1}$, that is, $\partial[\iota_n] = j_*[\iota_{n-1}]$.

Step 2: j_* is an isomorphism. If $n = 1$, then Δ^0 is a point with $\partial\Delta^0 = \emptyset$, $\partial\Delta^1 = \{v_0, v_1\}$ and $\Lambda = \{v_0\}$. A singular simplex of the discrete space $\partial\Delta^1$ lies at v_0 or at v_1, so $C_\bullet(\partial\Delta^1, \Lambda) \cong C_\bullet(\{v_1\})$, and j, which sends the point to v_1, induces an isomorphism of chain complexes $C_\bullet(\Delta^0) \to C_\bullet(\partial\Delta^1, \Lambda)$. Let $n \geq 2$. The pair $(\Delta^{n-1}, \partial\Delta^{n-1})$ is homeomorphic to (D^{n-1}, S^{n-2}) by lemma 3.1.5, so it is a good pair, S^{n-2} being a deformation retract of $D^{n-1} - \{0\}$. The pair $(\partial\Delta^n, \Lambda)$ is good too: Λ is closed and nonempty, and with $\Psi\colon (D^{n-1}, S^{n-2}) \to (F_0, \partial F_0)$ a homeomorphism of pairs from the same lemma and $b = \Psi(0)$, the radial deformation retraction of $D^{n-1} - \{0\}$ onto S^{n-2}, transported by Ψ, is a deformation retraction of $F_0 - \{b\}$ onto ∂F_0. Extended by the identity on Λ, it is a deformation retraction of the neighbourhood $\partial\Delta^n - \{b\}$ onto Λ, continuous by [16, Pasting Lemma] since $F_0 - \{b\}$ and Λ are closed in $\partial\Delta^n - \{b\}$ and meet in ∂F_0, where both maps are the identity. The map j carries $\partial\Delta^{n-1}$ into Λ and $\mathrm{int}\Delta^{n-1}$ bijectively onto $F_0 - \partial F_0 = \partial\Delta^n - \Lambda$, so it induces a continuous bijection $\bar{j}\colon \Delta^{n-1}/\partial\Delta^{n-1} \to \partial\Delta^n/\Lambda$. Its source is compact, and its target is Hausdorff: two points outside Λ have disjoint open neighbourhoods missing Λ, and since a compact metric space is normal, a point outside Λ and the closed set Λ have disjoint open neighbourhoods, which are saturated. Hence $\bar{j}$ is a homeomorphism ([16, making continuous function homeomorphisms]). By corollary 3.4.12, the quotient maps induce isomorphisms from the two relative groups to the relative homology of the quotients with their base points, and these isomorphisms carry j_* to $\bar{j}_*$, since $\bar{j}$ composed with one quotient map is the other quotient map composed with j. As $\bar{j}_*$ is an isomorphism, so is j_*.

Step 3: Conclusion. By induction, $[\iota_{n-1}]$ generates $H_{n-1}(\Delta^{n-1}, \partial\Delta^{n-1}) \cong \mathbb{Z}$. By Steps 1 and 2, $H_n(\Delta^n, \partial\Delta^n) \cong \mathbb{Z}$, and $[\iota_n] = \partial^{-1} j_*[\iota_{n-1}]$ generates it, as we sought to show.

Lemma 3.4.31: Relative Homology of Consecutive Skeleta

Let X be a CW complex and $n \geq 0$. Then $H_k(X^n, X^{n-1}) = 0$ for $k \neq n$, and $H_n(X^n, X^{n-1})$ is free abelian with basis the classes $(\Phi_\alpha)_*[D^n]$, one for each n-cell e^n_α, where $\Phi_\alpha \colon (D^n, \partial D^n) \to (X^n, X^{n-1})$ is the characteristic map of definition 1.2.2 and $[D^n]$ is a fixed generator of $H_n(D^n, \partial D^n) \cong \mathbb{Z}$ (for $n = 0$, $\partial D^0 = X^{-1} = \emptyset$ and $[D^0]$ generates $H_0(D^0) \cong \mathbb{Z}$).

***Proof*:**
For $n = 0$ the pair is $(X^0, \emptyset)$, so $H_k(X^0, X^{-1}) = H_k(X^0)$, and X^0 is a discrete set with one point for each 0-cell, so proposition 3.2.3 and proposition 3.2.5 give the claim, the classes $(\Phi_\alpha)_*[D^0]$ being the classes of the points. Let $n \geq 1$. If X^{n-1} is empty, no n-cell can be attached, so X^n is empty and the claim holds with no n-cells. Assume $X^{n-1} \neq \emptyset$. The pair (X^n, X^{n-1}) is a good pair by lemma 3.4.17, X^{n-1} being a nonempty subcomplex of the CW complex X^n. The quotient X^n/X^{n-1} is the wedge sum $\bigvee_\alpha S^n_\alpha$ of the spheres $S^n_\alpha = D^n_\alpha/\partial D^n_\alpha$, one for each n-cell, since collapsing X^{n-1} collapses the attaching image of every n-cell to the common basepoint. With $q \colon X^n \to X^n/X^{n-1}$ the quotient map, $q \circ \Phi_\alpha$ is the quotient map $q_\alpha \colon D^n \to S^n_\alpha$ followed by the inclusion ι_α of the α-th summand, so the square

$$\begin{array}{ccc} H_k(D^n, \partial D^n) & \xrightarrow{(\Phi_\alpha)_*} & H_k(X^n, X^{n-1}) \\ {\scriptstyle (q_\alpha)_*}\downarrow & & \downarrow{\scriptstyle q_*} \\ H_k(S^n_\alpha, *) & \xrightarrow{(\iota_\alpha)_*} & H_k\left(\bigvee_\beta S^n_\beta, *\right) \end{array}$$

commutes, and both vertical maps are isomorphisms in every degree by corollary 3.4.12, applied to the good pairs (X^n, X^{n-1}) and $(D^n, \partial D^n)$, the sphere ∂D^n being a deformation retract of $D^n - \{0\}$ by radial projection. For a space Y with a point y, the augmented complex of the point y is a subcomplex of $\tilde{C}_\bullet(Y)$ with quotient $C_\bullet(Y, y)$, and it has zero homology by corollary 3.3.9, so by theorem 3.4.3 the quotient chain map induces isomorphisms $\tilde{H}_k(Y) \to H_k(Y, y)$, compatible with maps of pointed spaces since it is defined on chains. Under these isomorphisms, corollary 3.4.19, applied to the good pairs $(S^n_\beta, *)$, says that the maps $(\iota_\beta)_*$ assemble into an isomorphism $\bigoplus_\beta H_k(S^n_\beta, *) \to H_k(\bigvee_\beta S^n_\beta, *)$, and corollary 3.4.15 says that $H_k(S^n_\beta, *) \cong \tilde{H}_k(S^n_\beta)$ is 0 for $k \neq n$ and infinite cyclic for $k = n$, generated by $(q_\beta)_*[D^n]$. Hence $H_k(X^n, X^{n-1}) = 0$ for $k \neq n$, and $H_n(X^n, X^{n-1})$ is free abelian with basis the elements $q_*^{-1}\big((\iota_\beta)_*(q_\beta)_*[D^n]\big) = (\Phi_\beta)_*[D^n]$, completing the proof.

Theorem 3.4.32: Singular And Simplicial Homology

For every Δ-complex pair (X, A) the canonical chain map $\theta\colon \Delta_\bullet(X, A) \to C_\bullet(X, A)$ is a quasi-isomorphism, that is, $H_n^\Delta(X, A) \to H_n(X, A)$ is an isomorphism for all $n \in \mathbb{N}$. It is natural for inclusions of subcomplex pairs: if $X' \subseteq X$ and $A' \subseteq A$ are subcomplexes with $A' \subseteq X'$, then θ for (X', A') followed by the map induced by the inclusion is the map induced by the inclusion followed by θ for (X, A).

***Proof*:**
The strategy is to prove the absolute case ($A = \emptyset$) for finite-dimensional X by induction on the dimension, comparing the skeletal long exact sequences of simplicial and singular homology through the canonical chain map and invoking the five lemma (lemma 3.4.6) on the resulting ladder. The base of the induction is the computation of the relative groups $H_n(X^k, X^{k-1})$, which turn out to be free abelian on the k-simplices in *both* theories. Once this is in hand the five lemma propagates the isomorphism up the skeleta. Two further reductions, a direct-limit argument for infinite-dimensional X and a second five-lemma comparison for the relative case, complete the theorem.

Throughout, write X^k for the k-skeleton of X, the union of all simplices of dimension $\leq k$. The canonical map of the statement is induced by the chain map $\Delta_n(X) \to C_n(X)$ sending each open n-simplex e_α^n to its characteristic map $\sigma_\alpha\colon \Delta^n \to X$, a singular n-simplex. This chain map commutes with the boundary operators because the simplicial boundary $\partial e_\alpha^n = \sum_i (-1)^i e_\alpha^n|_{[v_0,\ldots,\hat{v}_i,\ldots,v_n]}$ is precisely the singular boundary of σ_α under the identification of the i-th face of σ_α with the characteristic map of the corresponding $(n-1)$-simplex. It therefore descends to homology and, by passing to quotients, to the relative groups. The naturality clause holds already on chains: a simplex of X' has the same characteristic map, regarded in X, whether it is viewed as a simplex of X' or of X.

Step 1: The relative groups of a skeletal pair, simplicially. Fix $k \geq 0$. By construction $\Delta_n(X^k, X^{k-1}) = \Delta_n(X^k)/\Delta_n(X^{k-1})$ is the free abelian group on the open k-simplices when $n = k$ and is zero for $n \neq k$, since the only simplices of X^k not already in X^{k-1} are the k-simplices, and a Δ-complex has simplices in dimension n generating Δ_n. The relative simplicial boundary maps therefore vanish (their source or target is 0), so the relative simplicial homology is the relative chain group itself:

$$H_n^\Delta(X^k, X^{k-1}) = \begin{cases} \Delta_k(X^k) & n = k \\ 0 & n \neq k \end{cases}$$

free abelian on the k-simplices of X.

Step 2: The relative groups of a skeletal pair, singularly. The same groups are now computed in singular homology, for $k \geq 1$, the only case used in Step 3, where X^0 is treated directly. By proposition 3.1.6, X^k is a CW complex with $(k-1)$-skeleton X^{k-1}, one k-cell for each k-simplex e_α^k, and characteristic maps $\sigma_\alpha \circ \Psi$, where $\Psi\colon (D^k, S^{k-1}) \to (\Delta^k, \partial\Delta^k)$ is a homeomorphism of pairs. A homeomorphism of pairs induces an isomorphism on relative homology, so the generator $[D^k]$ of $H_k(D^k, \partial D^k) \cong \mathbb{Z}$ fixed in lemma 3.4.31 may be chosen with $\Psi_*[D^k] = [\iota_k]$, the generator of lemma 3.4.30. Then lemma 3.4.31 gives $H_n(X^k, X^{k-1}) = 0$ for $n \neq k$, and $H_k(X^k, X^{k-1})$ is free abelian with basis the classes

$$(\sigma_\alpha \circ \Psi)_*[D^k] = (\sigma_\alpha)_*[\iota_k] = [\sigma_\alpha \circ \iota_k] = [\sigma_\alpha]$$

one for each k-simplex. Comparing with Step 1, the canonical map sends the basis element e^k_α of $H^\Delta_k(X^k, X^{k-1})$ to the basis element $[\sigma_\alpha]$. The canonical map

$$H^\Delta_n(X^k, X^{k-1}) \longrightarrow H_n(X^k, X^{k-1})$$

carries a basis bijectively to a basis, hence *is an isomorphism for every n and every $k \geq 1$.*

Step 3: The induction on skeleta for finite-dimensional X. Assume first that X is finite-dimensional and that $X = X^k$ for some k. Argue by induction on k that $H^\Delta_n(X^k) \to H_n(X^k)$ is an isomorphism for all n. The base case $k = 0$ is immediate: X^0 is a discrete set of points, and both $H^\Delta_0(X^0)$ and $H_0(X^0)$ are free abelian on those points with the canonical map the identity on generators, while all higher groups vanish. For the inductive step, the inclusion of pairs gives a commuting ladder relating the long exact sequence of the simplicial pair (X^k, X^{k-1}) (the algebraic long exact sequence of theorem 3.4.3 applied to $0 \to \Delta_\bullet(X^{k-1}) \to \Delta_\bullet(X^k) \to \Delta_\bullet(X^k, X^{k-1}) \to 0$) to that of the singular pair (definition 3.4.2):

$$\begin{array}{ccccccccc}
H^\Delta_{n+1}(X^k, X^{k-1}) & \longrightarrow & H^\Delta_n(X^{k-1}) & \longrightarrow & H^\Delta_n(X^k) & \longrightarrow & H^\Delta_n(X^k, X^{k-1}) & \longrightarrow & H^\Delta_{n-1}(X^{k-1}) \\
\downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
H_{n+1}(X^k, X^{k-1}) & \longrightarrow & H_n(X^{k-1}) & \longrightarrow & H_n(X^k) & \longrightarrow & H_n(X^k, X^{k-1}) & \longrightarrow & H_{n-1}(X^{k-1})
\end{array}$$

The squares commute by proposition 3.4.4: the canonical chain map defines a morphism between the two short exact sequences of chain complexes (it is natural for inclusions of pairs and intertwines the simplicial and singular boundaries, as noted at the outset), and the proposition gives the ladder, including the connecting squares. The two outer vertical maps adjacent to the centre, those on the relative groups, are isomorphisms by Step 2, and the two vertical maps on $H^\Delta_*(X^{k-1}) \to H_*(X^{k-1})$ are isomorphisms by the inductive hypothesis. The five lemma (lemma 3.4.6) applied to the five-term window centred at $H^\Delta_n(X^k) \to H_n(X^k)$, with the four flanking maps isomorphisms, forces the central map $H^\Delta_n(X^k) \to H_n(X^k)$ to be an isomorphism. Since n was arbitrary, the inductive step is complete, and the theorem holds for all finite-dimensional X (in the absolute case).

Step 4: The passage to infinite-dimensional X. Suppose now X is arbitrary. The point is that homology, simplicial and singular alike, is detected on finite-dimensional skeleta, because both simplicial and singular chains are *compactly supported*: a chain is a finite sum of (characteristic maps of) simplices, and the image of each simplex is a compact subset meeting only finitely many open simplices of X, hence contained in some skeleton X^k, by lemma 1.2.4, X being a CW complex with these skeleta by proposition 3.1.6. Concretely, a class in $H^\Delta_n(X)$ or $H_n(X)$ is represented by a cycle whose finitely many simplices all lie in X^k for k large enough, and a boundary witnessing triviality likewise lies in some X^m. Therefore the natural maps

$$\varinjlim_k H^\Delta_n(X^k) \xrightarrow{\cong} H^\Delta_n(X), \qquad \varinjlim_k H_n(X^k) \xrightarrow{\cong} H_n(X)$$

over the skeletal filtration are isomorphisms in both theories: surjectivity because every cycle is supported in some X^k, and injectivity because every bounding chain is supported in some X^m. (In the singular case this is the statement that singular homology commutes with the directed union $X = \bigcup_k X^k$, the directed system having the weak topology with respect to its skeleta.) The canonical map is compatible with the filtration, so it induces a map of directed systems. By Step 3 each map $H^\Delta_n(X^k) \to H_n(X^k)$ is an isomorphism, and the colimit of isomorphisms is an

isomorphism. Passing to the colimit gives that $H_n^\Delta(X) \to H_n(X)$ is an isomorphism for every n, removing the finite-dimensionality hypothesis.

Step 5: The relative case. Finally, let (X, A) be an arbitrary Δ-complex pair, $A \subseteq X$ a subcomplex. The absolute case just proved applies to both X and A. The simplicial relative chain complex $\Delta_\bullet(X, A) = \Delta_\bullet(X)/\Delta_\bullet(A)$ sits in the short exact sequence $0 \to \Delta_\bullet(A) \to \Delta_\bullet(X) \to \Delta_\bullet(X, A) \to 0$, and the singular one in $0 \to C_\bullet(A) \to C_\bullet(X) \to C_\bullet(X, A) \to 0$. The canonical chain map is a map between these short exact sequences. Their associated long exact sequences (theorem 3.4.3) form a commuting ladder (proposition 3.4.4)

$$\begin{array}{ccccccccc} H_n^\Delta(A) & \longrightarrow & H_n^\Delta(X) & \longrightarrow & H_n^\Delta(X, A) & \longrightarrow & H_{n-1}^\Delta(A) & \longrightarrow & H_{n-1}^\Delta(X) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ H_n(A) & \longrightarrow & H_n(X) & \longrightarrow & H_n(X, A) & \longrightarrow & H_{n-1}(A) & \longrightarrow & H_{n-1}(X) \end{array}$$

in which the four vertical maps flanking the central one, namely those on $H_*^\Delta(A) \to H_*(A)$ and $H_*^\Delta(X) \to H_*(X)$, are isomorphisms by Steps 3 and 4. The five lemma (lemma 3.4.6) then forces the central map $H_n^\Delta(X, A) \to H_n(X, A)$ to be an isomorphism for every n. As (X, A) was an arbitrary Δ-complex pair, this establishes the theorem in full, completing the proof.

A subgroup of a finitely generated abelian group is finitely generated[2]. Hence if $\Delta_n(X)$ is finitely generated, so is $H_n^\Delta(X)$, a quotient of a subgroup of $\Delta_n(X)$, and so is $H_n(X)$ by theorem 3.4.32.

Example 3.4.33: Finding Explicit Generators of $H_n(S^n)$

Let $n \geq 1$. By lemma 3.4.30, the identity $\iota_n \colon \Delta^n \to \Delta^n$, read as a singular simplex, is a relative cycle whose class generates $H_n(\Delta^n, \partial\Delta^n) \cong \mathbb{Z}$. In the language of theorem 3.4.32, give Δ^n the Δ-complex structure whose simplices are its faces, with the affine face inclusions as characteristic maps. Then $H_n^\Delta(\Delta^n, \partial\Delta^n) \cong \mathbb{Z}$ is generated by the top simplex, and the canonical map sends it to $[\iota_n]$.

By lemma 3.1.5 the pair $(\Delta^n, \partial\Delta^n)$ is homeomorphic to $(D^n, \partial D^n)$, a good pair, so $\Delta^n/\partial\Delta^n$ is a sphere S^n with a point x_0, the image of $\partial\Delta^n$. Let $q \colon \Delta^n \to S^n$ be the quotient map. By corollary 3.4.12 the class of the singular simplex $q \circ \iota_n = q$ generates $H_n(S^n, x_0)$. The faces of q are all the constant simplex c_{n-1} at x_0, so $\partial q = \sum_i (-1)^i c_{n-1}$, which is 0 for n odd and $c_{n-1} = \partial c_n$ for n even, c_n being the constant n-simplex at x_0. The map $H_n(S^n) \to H_n(S^n, x_0)$ is an isomorphism for $n \geq 1$, by the long exact sequence of the pair (theorem 3.4.3), the homology of a point (proposition 3.2.5), and, for $n = 1$, the injectivity of $H_0(x_0) \to H_0(S^1)$, the class of a point having augmentation 1 (proposition 3.2.4). Moreover c_n is a chain in x_0. Hence $H_n(S^n) \cong \mathbb{Z}$ is generated by the class of the cycle q when n is odd and of the cycle $q - c_n$ when n is even.

3.4.3 Retractions

A retraction $r \colon X \to A$ splits the long exact sequence of the pair (X, A), by the following lemma.

[2]This need not be the case for non-abelian groups.

Lemma 3.4.34: Splitting Lemma

Let $0 \to A \xrightarrow{i} B \xrightarrow{j} C \to 0$ be a short exact sequence of abelian groups. Then the following are equivalent:

1. there is a homomorphism $p\colon B \to A$ with $p \circ i = \mathrm{id}_A$ (a retraction of i)
2. there is a homomorphism $s\colon C \to B$ with $j \circ s = \mathrm{id}_C$ (a section of j)
3. there is an isomorphism $\varphi\colon A \oplus C \to B$ with $\varphi(a,0) = i(a)$ and $j(\varphi(a,c)) = c$ for all $a \in A$ and $c \in C$

If any of these are satisfied, the exact sequence is said to split, and in particular $B \cong A \oplus C$. For a section s as in (2), the map $\varphi(a,c) = i(a) + s(c)$ is an isomorphism as in (3), so that B is the internal direct sum of $i(A)$ and $s(C)$.

***Proof*:**
Abelian groups are $\mathbb{Z}$-modules, so [10, Splitting Criteria] applies with $R = \mathbb{Z}$. It states that, for a short exact sequence of modules, a retraction of i exists if and only if a section of j exists, and that both are equivalent to $B = i(A) \oplus C'$ for a submodule $C' \subseteq B$.

1. The equivalence of (1) and (2) is the cited equivalence of a retraction of i with a section of j.
2. Given a section s as in (2), set $\varphi(a,c) = i(a) + s(c)$. Then $\varphi(a,0) = i(a)$ and $j(\varphi(a,c)) = j(i(a)) + j(s(c)) = c$, since $j \circ i = 0$. If $\varphi(a,c) = 0$, applying j gives $c = 0$, and then $i(a) = 0$ forces $a = 0$ because i is injective. For $b \in B$ the element $b - s(j(b))$ lies in $\ker j = \mathrm{im} i$, say $b - s(j(b)) = i(a)$, so $b = \varphi(a, j(b))$. Hence φ is an isomorphism as in (3).
3. Given φ as in (3), set $s(c) = \varphi(0,c)$. Then $j(s(c)) = c$, so s is a section as in (2).

The isomorphism φ of (3) gives $B \cong A \oplus C$, and the computation in item 2 shows that for any section s the map $(a,c) \mapsto i(a) + s(c)$ is such an isomorphism, so every $b \in B$ is uniquely $i(a) + s(c)$, completing the proof.

Proposition 3.4.35: Retraction And Split Sequence

Let $A \subseteq X$ and let $r\colon X \to A$ be a retraction, that is a continuous map with $r \circ i = \mathrm{id}_A$ for the inclusion $i\colon A \hookrightarrow X$. Then for every n:

$$H_n(X) \cong H_n(A) \oplus H_n(X, A)$$

***Proof*:**
By proposition 3.3.3(1) and (2), $r_* i_* = (ri)_* = (\mathrm{id}_A)_* = \mathrm{id}$ on $H_n(A)$ for every n, so each $i_*\colon H_n(A) \to H_n(X)$ is injective, with retraction r_*. The long exact sequence of the pair, which is theorem 3.4.3 applied to $0 \to C_\bullet(A) \to C_\bullet(X) \to C_\bullet(X, A) \to 0$, contains

$$H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(X)$$

with j the quotient chain map. Exactness at $H_{n-1}(A)$ gives $\mathrm{im}\partial = \ker i_* = 0$ (for $n = 0$ the target $H_{-1}(A)$ is zero), and exactness at $H_n(X, A)$ then gives $\mathrm{im} j_* = \ker \partial = H_n(X, A)$. With exactness at $H_n(X)$ and the injectivity of i_*, this shows that

$$0 \to H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{j_*} H_n(X, A) \to 0$$

is a short exact sequence. The map r_* is a retraction of i_*, so the sequence splits by lemma 3.4.34, and $H_n(X) \cong H_n(A) \oplus H_n(X, A)$, as we sought to show.

3.5 Application and Computation of Homology

This section computes the homology of important spaces and develops methods for computing the homology of CW complexes. The *degree* of a map of spheres generalizes the winding number. Cellular homology of a CW complex is isomorphic to singular (and hence simplicial) homology, and gives the Euler characteristic. The applications include the theorem of *invariance of domain* and the Lefschetz fixed point theorem.

3.5.1 Degree

In complex analysis, the winding number of a function gives information about the number of roots as well as the values of path integrals. The following is a generalization of this concept:

Definition 3.5.1: Degree

Let $n \geq 0$. A continuous map $f\colon S^n \to S^n$ induces an endomorphism f_* of the infinite cyclic group $\tilde{H}_n(S^n)$ (corollary 3.4.15), the chain map $f_\sharp$ commuting with the augmentation, so $f_*(\alpha) = d\alpha$ for a unique $d \in \mathbb{Z}$. Then d is called the *degree* of f, and is denoted $\deg f$. For $n \geq 1$ the reduced and unreduced groups agree in degree n, so $\deg f$ is equally the integer by which f_* multiplies on $H_n(S^n)$.

Proposition 3.5.2: Properties Of Degree

Let $n \geq 1$ and let $f, g\colon S^n \to S^n$ be maps.

1. $\deg \mathrm{id} = 1$
2. If f is not surjective, then $\deg f = 0$. Note the converse is not true
3. If $f \simeq g$, then $\deg f = \deg g$ (the converse is more difficult to prove, see [21, Corollary 4.25][a])
4. $\deg fg = \deg f \deg g$. Hence if f is a homotopy equivalence, then $\deg f = \pm 1$
5. $\deg f = -1$ if f is a reflection of S^n in a hyperplane through the origin, fixing the points of a subsphere S^{n-1} and interchanging the two complementary hemispheres
6. The antipodal map $-\mathrm{id}\colon S^n \to S^n$ given by $x \mapsto -x$ has degree $(-1)^{n+1}$
7. If f has no fixed points, then f is homotopic to the antipodal map, and hence $\deg f = (-1)^{n+1}$

[a] This was first proved by Hopf in 1925

***Proof*:**

1. Immediate since $\mathrm{id}_* = \mathrm{id}$ (proposition 3.3.3(2)).
2. If $x_0 \in S^n - f(S^n)$, then f can be factored as
$$S^n \to (S^n - \{x_0\}) \hookrightarrow S^n$$
and $H_n(S^n - \{x_0\}) = 0$, since $S^n - \{x_0\}$ is homeomorphic to $\mathbb{R}^n$ (lemma 3.4.26), which is contractible (corollary 3.3.9). Hence $f_* = 0$ by proposition 3.3.3(1).
3. If $f \simeq g$, then $f_* = g_*$ by theorem 3.3.6, so $\deg f = \deg g$.
4. By proposition 3.3.3(1), $(fg)_* = f_* g_*$ is multiplication by $\deg f \deg g$. If g is a homotopy inverse of f, then $fg \simeq \mathrm{id}$, so $\deg f \deg g = \deg \mathrm{id} = 1$ by (3) and (1), and the integer $\deg f$ is ± 1.
5. First let $r(x_0, x_1, \ldots, x_n) = (-x_0, x_1, \ldots, x_n)$. Let $\Phi\colon \Delta^n \to D^n$ be a homeomorphism of pairs onto (D^n, S^{n-1}), the inverse of the map of lemma 3.1.5, and put $\sigma_\pm(y) = \big(\pm\sqrt{1 - |\Phi(y)|^2}, \Phi(y)\big)$. These map Δ^n homeomorphically onto the closed hemispheres $\pm x_0 \geq 0$, agree on $\partial\Delta^n$, where they parametrize the equator, and satisfy $r \circ \sigma_+ = \sigma_-$. Together with the restrictions of σ_+ to the faces of Δ^n they form a Δ-complex structure on S^n (definition 3.1.4): the open hemispheres and the images of the open faces of Δ^n partition S^n, the faces of $\sigma_\pm$ are simplices of the structure, and the topology is that of a finite union of closed sets. Write U and L for the two n-simplices. Their boundaries agree and are nonzero, the $n+1$ faces being distinct simplices, so the simplicial cycles in degree n are the multiples of $U - L$, and there are no $(n+1)$-simplices. Hence $H_n^\Delta(S^n) \cong \mathbb{Z}$ is generated by $U - L$, and by theorem 3.4.32 the singular cycle $\sigma_+ - \sigma_-$ generates $H_n(S^n)$. Since $r_\sharp(\sigma_+ - \sigma_-) = \sigma_- - \sigma_+$, the map r_* is multiplication by -1. A reflection in any

hyperplane through the origin is hrh^{-1} for an orthogonal map h carrying the hyperplane $x_0 = 0$ to the given one, and by (4) its degree is $\deg h \deg r \deg h^{-1} = \deg r = -1$, since $\deg h \deg h^{-1} = \deg \mathrm{id} = 1$.

6. The map $-\mathrm{id}$ is the composite of the $n+1$ reflections changing the sign of one coordinate of $\mathbb{R}^{n+1}$, each of degree -1 by (5), so its degree is $(-1)^{n+1}$ by (4).

7. If $f(x) \neq x$, the segment $t \mapsto (1-t)f(x) - tx$ from $f(x)$ to $-x$ does not pass through the origin: its endpoints are unit vectors, and a zero at some $t \in (0,1)$ would make $f(x)$ a positive multiple of x, hence equal to x. Hence if f has no fixed points, the formula

$$f_t(x) = \frac{(1-t)f(x) - tx}{|(1-t)f(x) - tx|}$$

defines a homotopy from f to the antipodal map, and $\deg f = (-1)^{n+1}$ by (3) and (6).

This completes the proof.

The converse of (2) fails. The map $g\colon S^1 \to S^1$, $g(e^{2\pi it}) = e^{4\pi i \min(t,1-t)}$ for $t \in [0,1]$, winds once around the circle and back, so it is surjective. It factors through the contractible interval $[0,1/2]$ via $e^{2\pi it} \mapsto \min(t, 1-t)$, so $g_* = 0$ on $H_1(S^1)$ (corollary 3.3.9) and $\deg g = 0$.

Theorem 3.5.3: Hairy ball theorem

Let $X = S^n$. Then X has a continuous field of nonzero tangent vectors if and only if n is odd

***Proof*:**
Let $V : S^n \to TS^n$ where $x \mapsto v(x)$ be the vector-field map. Thinking of $v(x)$ as a vector at the origin and of x as a vector, x and $v(x)$ are orthogonal in $\mathbb{R}^{n+1}$.

Now, if $v(x) \neq 0$ for all x, then

$$\frac{v(x)}{|v(x)|}$$

is well-defined. Replacing $v(x)$ by this normalization, the vectors

$$\cos(t)x + \sin(t)v(x)$$

are in the unit circle in the plane spanned by x and $v(x)$. Taking t from 0 to π, we get a homotopy:

$$f_t(x) = \cos(t)x + \sin(t)v(x)$$

from id_{S^n} to the antipodal map $-\mathrm{id}$. But then, by proposition 3.5.2(3) and (6):

$$(-1)^{n+1} = \deg(-\mathrm{id}) = \deg(\mathrm{id}) = 1$$

implying n is odd. Conversely if n is odd so that $n = 2k-1$, we can take

$$v(\vec{x}) = (-x_2, x_1, ..., -x_{2k}, x_{2k-1})$$

Then $v(x)$ is orthogonal to x, so v is a tangent vector field on S^n and $|v(x)| = 1$ for all $x \in S^n$.

Proposition 3.5.4: Free Action on Sphere

$\mathbb{Z}/2\mathbb{Z}$ is the only nontrivial group that can act freely on S^n if n is even

***Proof*:**
Let $G \curvearrowright S^n$ so that there is a homomorphism $\rho : G \to \mathrm{Aut}_{\mathsf{Top}}(S^n)$. Since homeomorphisms have degree ± 1, an action by a group G on S^n induces a "degree" function $d : G \to \{\pm 1\}$, which is a homomorphism since $\deg fg = \deg f \deg g$. If the action is free, d sends nontrivial elements of G to $(-1)^{n+1}$ by proposition 3.5.2(7). But then, when n is even, d has trivial kernel, hence G embeds in $\mathbb{Z}/2\mathbb{Z}$. Conversely, $\mathbb{Z}/2\mathbb{Z}$ acts freely on S^n by the antipodal map.

The degree of a map can be computed locally in the most common case, that is when for some $y \in S^n$, $f^{-1}(y)$ is finite. Say $x_1, x_2, ..., x_m$ are the points of $f^{-1}(y)$, with disjoint neighbourhoods $U_1, U_2, ..., U_m$ mapped by f into a neighbourhood V of y. Then $f(U_i - x_i) \subseteq V - y$ for each i, and we have:

$$\begin{array}{ccccc} & & H_n(U_i, U_i - x_i) & \xrightarrow{f_*} & H_n(V, V-y) \\ & \overset{\cong}{\swarrow} & \downarrow{\scriptstyle k_i} & & \downarrow{\scriptstyle \cong} \\ H_n(S^n, S^n - x_i) & \xleftarrow[\pi_i]{} & H_n(S^n, S^n - f^{-1}(y)) & \xrightarrow{f_*} & H_n(S^n, S^n - y) \\ & \underset{\cong}{\nwarrow} & \uparrow{\scriptstyle j} & & \uparrow{\scriptstyle \cong} \\ & & H_n(S^n) & \xrightarrow{f_*} & H_n(S^n) \end{array} \tag{3.11}$$

where k_i and π_i are induced by inclusion, and all triangles and squares commute. The two isomorphisms in the upper half are given by excision, and the lower two by the long exact sequences of pairs. These four isomorphisms identify the top two groups in the diagram with $H_n(S^n) \cong \mathbb{Z}$, and the top homomorphism f_* becomes multiplication by an integer:

Definition 3.5.5: Local Degree

Given the above setup, the *local degree* at x_i is the degree of the map f_* and is written as:

$$\deg f|_{x_i}$$

For example, if f is a homeomorphism, then $f^{-1}(y) = x$ is a single point, and so $\deg f|_{x_i} = \deg f = \pm 1$. More generally, if f maps each U_i homeomorphically onto V (if f is a local homeomorphism), then f induces an isomorphism $H_n(U_i, U_i - x_i) \to H_n(V, V - y)$ of infinite cyclic groups, so

$$\deg f|_{x_i} = \pm 1 \qquad 1 \leq i \leq m$$

The local degrees determine the degree:

Proposition 3.5.6: Computing Degree

Given the setup as above, we have:

$$\deg f = \sum_i (\deg f|_{x_i})$$

***Proof*:**
In the diagram (3.11), the term $H_n(S^n, S^n - f^{-1}(y))$ is the direct sum of the groups $H_n(U_i, U_i - x_i) \cong \mathbb{Z}$, by excision of $S^n - \bigcup_i U_i$ followed by the splitting of the relative chains of the disjoint open sets U_i as in Step 2 of the proof of corollary 3.4.19, with k_i the inclusion of the ith summand. The π_i is the projection onto the ith summand since the upper triangle commutes and

$$\pi_i k_j = 0 \qquad j \neq i$$

since $\pi_i k_j$ factors through $H_n(U_j, U_j) = 0$, as $U_j \subseteq S^n - x_i$. Identifying the outer group in the diagram with $\mathbb{Z}$, commutativity of the lower triangles gives us that

$$\pi_i j(1) = 1 \text{ for each } i \implies j(1) = (1, ..., 1) = \sum_i k_i(1)$$

Commutativity of the upper square gives that the middle f_* takes $k_i(1)$ to $\deg f|_{x_i}$, thus

$$\sum_i k_i(1) = j(1) \mapsto \sum_i \deg f|_{x_i}$$

Commutativity of the lower square finally gives us:

$$\deg f = \sum_i \deg f|_{x_i}$$

as we sought to show.

Proposition 3.5.7: Suspension And Degree

Let $f : S^n \to S^n$ be a continuous map, $Sf : S^{n+1} \to S^{n+1}$ its suspension. Then:

$$\deg Sf = \deg f$$

***Proof*:**
The statement follows once the suspension is shown to induce, naturally in X, an isomorphism between the reduced homology of X and that of SX shifted up one degree. Applying this to S^n, where $SS^n = S^{n+1}$, converts the equality of degrees into the equality of two integers read off the same isomorphism.

Step 1: The cone decomposition of the suspension. Recall from definition 1.2.13 that SX is the quotient of $X \times I$ collapsing $X \times \{0\}$ to a point p_- and $X \times \{1\}$ to a point p_+. Write $q\colon X \times I \to SX$ for the quotient map and set

$$C_+ = q\big(X \times (0, 1]\big), \qquad C_- = q\big(X \times [0, 1)\big)$$

so that $SX = C_+ \cup C_-$. Each piece is a cone on X: the formula $h_+\big(q(x, s), t\big) = q\big(x, t + (1-t)s\big)$ is well-defined on C_+ (it fixes the cone point $p_+ = q(X \times \{1\})$ and is continuous because it descends from a continuous map on $X \times (0, 1] \times I$) and gives a deformation retraction of C_+ onto p_+, so C_+ is contractible. Symmetrically C_- is contractible. The interiors $q(X \times (0, 1])$ and $q(X \times [0, 1))$ already cover SX, since their union is $q(X \times [0, 1]) = SX$ and each is open (its preimage under q is the saturated open set $X \times (0, 1]$, respectively $X \times [0, 1)$). Thus (C_+, C_-)

satisfies the hypothesis of theorem 3.4.22: SX is the union of the interiors of C_+ and C_-.

Step 2: The suspension isomorphism. The set $X \times (0,1)$ is open and saturated for q, so q restricts to a quotient map $X \times (0,1) \to C_+ \cap C_- = q(X \times (0,1))$ [16, Subspaces And Quotient Maps], which is injective and hence a homeomorphism. So $p_X \colon C_+ \cap C_- \to X$, $q(x,s) \mapsto x$, is continuous. With $e_X(x) = q(x, 1/2)$ we have $p_X \circ e_X = \mathrm{id}_X$, and $e_X \circ p_X$ is homotopic to the identity through $\big(q(x,s),t\big) \mapsto q\big(x,(1-t)s + t/2\big)$, so p_X is a homotopy equivalence. The reduced Mayer-Vietoris sequence for the decomposition $SX = C_+ \cup C_-$ reads, in each degree,

$$\tilde{H}_{n+1}(C_+) \oplus \tilde{H}_{n+1}(C_-) \to \tilde{H}_{n+1}(SX) \xrightarrow{\partial} \tilde{H}_n(C_+ \cap C_-) \to \tilde{H}_n(C_+) \oplus \tilde{H}_n(C_-)$$

Since C_+ and C_- are contractible, $\tilde{H}_k(C_\pm) = 0$ for all k, so the two outer terms vanish and the connecting homomorphism ∂ is an isomorphism. Composing with the isomorphism $(p_X)_* \colon \tilde{H}_n(C_+ \cap C_-) \to \tilde{H}_n(X)$ (corollary 3.3.7), define

$$s_X = (p_X)_* \circ \partial \colon \tilde{H}_{n+1}(SX) \xrightarrow{\cong} \tilde{H}_n(X)$$

the *suspension isomorphism*, an isomorphism in every degree n.

Step 3: Naturality. A continuous map $f \colon X \to Y$ suspends to $Sf \colon SX \to SY$, the map induced by $f \times \mathrm{id}_I$ on the quotients. It carries C_+^X into C_+^Y and C_-^X into C_-^Y, hence $C_+^X \cap C_-^X$ into $C_+^Y \cap C_-^Y$, and on the intersections $p_Y \circ Sf = f \circ p_X$, both sending $q(x,s)$ to $f(x)$. By corollary 3.4.23 for the reduced sequences, applied to Sf and the open covers of Step 1, $\partial_Y \circ (Sf)_* = (Sf|)_* \circ \partial_X$, where $Sf|$ is the restriction to $C_+^X \cap C_-^X$. Hence $f_* \circ s_X = f_*(p_X)_*\partial_X = (p_Y)_*(Sf|)_*\partial_X = (p_Y)_*\partial_Y(Sf)_* = s_Y \circ (Sf)_*$, that is, the square

$$\begin{array}{ccc} \tilde{H}_{n+1}(SX) & \xrightarrow{s_X} & \tilde{H}_n(X) \\ {\scriptstyle (Sf)_*}\downarrow & & \downarrow{\scriptstyle f_*} \\ \tilde{H}_{n+1}(SY) & \xrightarrow[s_Y]{} & \tilde{H}_n(Y) \end{array}$$

commutes.

Step 4: Reading off the degree. Now take $X = Y = S^n$ and the given map $f \colon S^n \to S^n$, so that $Sf \colon S^{n+1} \to S^{n+1}$ since $SS^n = S^{n+1}$. By corollary 3.4.15 we have $\tilde{H}_n(S^n) \cong \mathbb{Z}$ and $\tilde{H}_{n+1}(S^{n+1}) \cong \mathbb{Z}$, and for $n \geq 1$ the reduced and unreduced groups agree, so the degree of definition 3.5.1 is multiplication by the same integer whether computed on H_n or on $\tilde{H}_n$: $f_* = \deg f$ on $\tilde{H}_n(S^n) \cong \mathbb{Z}$ and $(Sf)_* = \deg Sf$ on $\tilde{H}_{n+1}(S^{n+1}) \cong \mathbb{Z}$. Specializing the commuting square of Step 3 to $X = Y = S^n$ gives

$$\deg f \cdot s_{S^n} = f_* \circ s_{S^n} = s_{S^n} \circ (Sf)_* = \deg Sf \cdot s_{S^n}$$

as homomorphisms $\tilde{H}_{n+1}(S^{n+1}) \to \tilde{H}_n(S^n)$ between groups isomorphic to $\mathbb{Z}$, where s_{S^n} is the suspension isomorphism. Since s_{S^n} is an isomorphism between infinite cyclic groups, it is injective and nonzero, so the two integers acting through it must agree, namely $\deg Sf = \deg f$.

For $n = 0$, the map $f \colon S^0 \to S^0$ acts on $\tilde{H}_0(S^0) \cong \mathbb{Z}$ and its degree is again the integer by which f_* multiplies. The suspension isomorphism $s_{S^0} \colon \tilde{H}_1(S^1) \to \tilde{H}_0(S^0)$ and the same square give $\deg Sf = \deg f$ verbatim, so the identity holds in every dimension, completing the proof.

Example 3.5.8: Maps of Every Degree

1. For a map $f : S^n \to S^n$, let $q : S^n \to \bigvee_k S^n$ be the quotient map given by collapsing the complement of k disjoint open balls B_i in S^n to a point, and let $\pi : \bigvee_k S^n \to S^n$ identify all the summands to a single sphere. Then take $f = \pi \circ q$. Then the local degree of f at x_i is ± 1 since f is a homeomorphism near x_i. By precomposing π with reflections of the summands of $\bigvee_k S^n$ if necessary, we can make each local degree either 1 or -1, giving us a map with degree $\pm k$ (proposition 3.5.6).

2. Let $n = 1$, so that $S^1 \subseteq \mathbb{C}$ is the unit circle. The map $f(z) = z^k$ has degree k for every $k \in \mathbb{Z}$. For $k = 0$ it is constant, of degree 0. For $k > 0$ every point has exactly k preimages, and near each of them f is an orientation-preserving local homeomorphism, since it wraps a small arc once counterclockwise around its image. Each local degree is 1, and the degrees add to k. For $k < 0$, f is the composite of $z \mapsto z^{|k|}$, of degree $|k|$, with $z \mapsto z^{-1} = \bar{z}$, a reflection of degree -1, so its degree is $-|k| = k$. Since suspension preserves degrees (proposition 3.5.7), another way of getting a degree k map from S^n to S^n is by taking repeated suspensions of the map $z \mapsto z^k$.

Note that for a map $f : S^n \to S^n$, the suspension Sf maps only 1 point to each of the poles of S^{n+1}, hence the local degree of Sf at each pole equals the global degree of Sf (proposition 3.5.6). Hence the local degree of a map can be any integer if $n \geq 2$, as can the global degree.

3.5.2 Cellular Homology: Equivalent to Singular

Cellular homology computes the homology groups of a CW complex, using the degree calculations above. It rests on the following lemma:

Lemma 3.5.9: Cellular Homology Lemma

Let X be a CW complex. Then:

1. $H_k(X^n, X^{n-1}) = 0$ for $k \neq n$, and $H_n(X^n, X^{n-1})$ is free abelian with basis the classes $(\Phi_\alpha)_*[D^n]$, one for each n-cell e^n_α (lemma 3.4.31). In particular, if there are m cells of dimension n,
$$H_n(X^n, X^{n-1}) = \mathbb{Z}^m$$

2. $H_k(X^n) = 0$ for all $k > n$. In particular, if X is finite-dimensional, then $H_k(X) = 0$ for $k > \dim X$

3. The map $H_k(X^n) \to H_k(X)$ induced by the inclusion $X^n \hookrightarrow X$ is an isomorphism for $k < n$, and surjects for $k = n$
$$\begin{aligned} H_k(X^n) &\cong H_k(X) && k < n \\ H_n(X^n) &\twoheadrightarrow H_n(X) && k = n \end{aligned}$$
Note that X can be finite or infinite dimensional.

***Proof*:**

1. This is lemma 3.4.31.

2. Apply the long exact sequence of the pair (X^n, X^{n-1}) from the algebraic snake lemma (theorem 3.4.3),

$$\cdots \to H_{k+1}(X^n, X^{n-1}) \to H_k(X^{n-1}) \to H_k(X^n) \to H_k(X^n, X^{n-1}) \to \cdots$$

   By part (1) the relative term $H_k(X^n, X^{n-1})$ vanishes unless $k = n$. Fix k and suppose $n \neq k$ and $n \neq k+1$: then both $H_k(X^n, X^{n-1})$ and $H_{k+1}(X^n, X^{n-1})$ vanish, so exactness forces the inclusion-induced map to be an isomorphism,

$$H_k(X^{n-1}) \xrightarrow{\cong} H_k(X^n) \qquad n \notin \{k, k+1\}$$

   Reading these isomorphisms along the skeletal filtration, the only places where the map $H_k(X^{n-1}) \to H_k(X^n)$ can fail to be an isomorphism are $n = k$ and $n = k+1$. Thus for k fixed the sequence of inclusion-induced maps

$$H_k(X^0) \to H_k(X^1) \to \cdots \to H_k(X^{k-1}) \to H_k(X^k) \to H_k(X^{k+1}) \to \cdots$$

   consists of isomorphisms except possibly at $H_k(X^{k-1}) \to H_k(X^k)$ (where the relative term $H_k(X^k, X^{k-1})$ sits to the right and may obstruct surjectivity) and at $H_k(X^k) \to H_k(X^{k+1})$ (where $H_{k+1}(X^{k+1}, X^k)$ sits to the left and may obstruct injectivity). In particular,

$$H_k(X^{k-1}) \cong H_k(X^{k-2}) \cong \cdots \cong H_k(X^0)$$

   Now assume $k > 0$. The space X^0 is a discrete set of 0-cells, so $H_k(X^0) = 0$ for $k > 0$, whence $H_k(X^{k-1}) = 0$. This is the case $n = k-1$ of part (2). Fix n with $k > n$, equivalently $n \leq k-1$, and run the inclusion-induced maps from X^n up to X^{k-1},

$$H_k(X^n) \to H_k(X^{n+1}) \to \cdots \to H_k(X^{k-1})$$

   Each map here is $H_k(X^{m-1}) \to H_k(X^m)$ for some m with $n+1 \leq m \leq k-1$, so $m < k$ and in particular $m \notin \{k, k+1\}$. By the display $H_k(X^{m-1}) \cong H_k(X^m)$ valid for $m \notin \{k, k+1\}$, every map in the chain is an isomorphism. The terminal group $H_k(X^{k-1})$ has just been shown to vanish, so every group in the chain is zero, and in particular $H_k(X^n) = 0$. Hence $H_k(X^n) = 0$ for all $k > n$. If X is finite-dimensional with $\dim X = d$, then $X = X^d$, and taking $n = d$ gives $H_k(X) = H_k(X^d) = 0$ for all $k > d$.

3. The inclusion $X^n \hookrightarrow X$ factors through every higher skeleton, and we treat the finite- and infinite-dimensional cases in turn.

   Finite-dimensional X. Suppose $\dim X = d < \infty$, so $X = X^d$. For $k < n$ the isomorphisms of part (2), applied along the filtration from X^n to X^d, give

$$H_k(X^n) \cong H_k(X^{n+1}) \cong \cdots \cong H_k(X^d) = H_k(X)$$

   since each map $H_k(X^m) \to H_k(X^{m+1})$ with $m \geq n > k$ avoids both exceptional positions (the obstructing relative terms $H_k(X^{m+1}, X^m)$ and $H_{k+1}(X^{m+1}, X^m)$ vanish for $m \geq n > k$ by part (1)). For $k = n$ the map $H_n(X^n) \to H_n(X^{n+1})$ is the one possibly failing injectivity but is surjective, because the term to its right, $H_n(X^{n+1}, X^n)$, vanishes

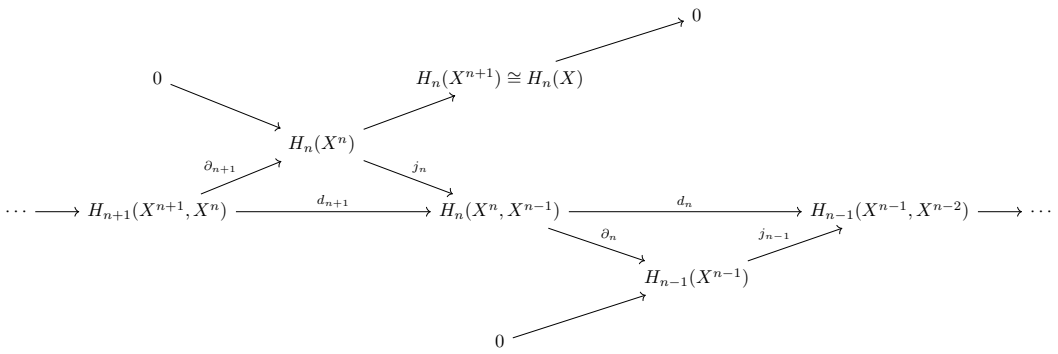
by part (1) ($n \neq n+1$). Composing the surjection $H_n(X^n) \twoheadrightarrow H_n(X^{n+1})$ with the isomorphisms $H_n(X^{n+1}) \cong \cdots \cong H_n(X^d) = H_n(X)$ for the higher skeleta (each map there has $k = n < m$, hence is an isomorphism) shows $H_n(X^n) \twoheadrightarrow H_n(X)$.

Infinite-dimensional X. When X is infinite-dimensional the chain $H_k(X^n) \to H_k(X^{n+1}) \to \cdots$ no longer terminates, and the comparison with $H_k(X)$ rests on the fact that singular homology is detected on finite skeleta. A singular simplex $\sigma\colon \Delta^j \to X$ has compact image, and a compact subset of a CW complex meets only finitely many open cells, hence lies in some skeleton X^m (lemma 1.2.4). Consequently any singular chain, being a finite sum of simplices, is supported in a single skeleton. Therefore every cycle representing a class in $H_k(X)$ lies in some $H_k(X^m)$, and every chain witnessing that such a cycle bounds lies in some $H_k(X^{m'})$. This is exactly the statement that the natural map from the colimit over the skeletal filtration is an isomorphism,

$$\varinjlim_{m} H_k(X^m) \xrightarrow{\cong} H_k(X)$$

surjective because each class is supported in some skeleton and injective because each bounding relation is supported in some skeleton (this is the same compact-support argument used in Step 4 of the proof of theorem 3.4.32, where the colimit-over-skeleta isomorphism for singular homology under the weak topology is established. The purely algebraic input, that a filtered colimit of abelian groups is computed termwise and preserves exactness, is [13]). The colimit is computed along the maps $H_k(X^m) \to H_k(X^{m+1})$. By part (2), for $m \geq n > k$ in the case $k < n$ every such map is an isomorphism, so the colimit stabilizes at $H_k(X^n)$ and the composite $H_k(X^n) \to \varinjlim_m H_k(X^m) \cong H_k(X)$ is an isomorphism. For $k = n$ the first map $H_n(X^n) \to H_n(X^{n+1})$ is surjective and the subsequent maps $H_n(X^m) \to H_n(X^{m+1})$ for $m \geq n+1$ are isomorphisms (each has $k = n < m$), so the colimit is a quotient of $H_n(X^n)$ and the composite $H_n(X^n) \to H_n(X)$ is surjective. Thus in both the finite- and infinite-dimensional cases the inclusion induces an isomorphism for $k < n$ and a surjection for $k = n$, completing the proof.

By using the above lemma, we get the following commutative diagram:



where $d_{n+1} = j_n \partial_{n+1}$ and $d_n = j_{n-1}\partial_n$. These are the "relative boundary maps" of ∂_{n+1} and ∂_n. Notice that $d_n d_{n+1} = j_{n-1}\partial_n j_n \partial_{n+1} = 0$, since $\partial_n j_n = 0$ by exactness of the long exact sequence of the pair (X^n, X^{n-1}). We thus give a name to the horizontal row:

Definition 3.5.10: Cellular Chain Complex

In the above diagram, the chain complex is called the *cellular chain complex*.

Its homology groups are denoted $H_n^{CW}(X)$, the *cellular homology groups* of X.

Theorem 3.5.11: Cellular Homology

Let X be a CW complex. Then for every n

$$H_n^{CW}(X) \cong H_n(X)$$

The isomorphism is natural for cellular maps: if $f\colon X \to Y$ is a map of CW complexes with $f(X^k) \subseteq Y^k$ for every k, the maps $f_*\colon H_n(X^n, X^{n-1}) \to H_n(Y^n, Y^{n-1})$ form a chain map between the cellular chain complexes, and the isomorphisms carry the homomorphism it induces on cellular homology to $f_*\colon H_n(X) \to H_n(Y)$.

***Proof*:**

The cellular chain complex (definition 3.5.10) has n-th group $H_n(X^n, X^{n-1})$ and boundary maps $d_n = j_{n-1}\partial_n$, where $\partial_n\colon H_n(X^n, X^{n-1}) \to H_{n-1}(X^{n-1})$ is the connecting map of the pair (X^n, X^{n-1}) and $j_{n-1}\colon H_{n-1}(X^{n-1}) \to H_{n-1}(X^{n-1}, X^{n-2})$ is the map in the long exact sequence of the pair (X^{n-1}, X^{n-2}). By definition

$$H_n^{CW}(X) = \frac{\ker d_n}{\mathrm{im} d_{n+1}}$$

so the claim amounts to identifying this subquotient of $H_n(X^n, X^{n-1})$ with $H_n(X)$. The argument reads both $\ker d_n$ and $\mathrm{im} d_{n+1}$ off the diagram preceding definition 3.5.10, whose commutativity and exactness along its diagonals are exactly the content of lemma 3.5.9 and the long exact sequences of the pairs (X^n, X^{n-1}).

Step 1: The maps j_n are injective and $\mathrm{im} d_{n+1} = \mathrm{im}(j_n\partial_{n+1})$. Fix n and consider the long exact sequence of the pair (X^n, X^{n-1}) (theorem 3.4.3 applied to the short exact sequence of relative chains, definition 3.4.2):

$$H_n(X^{n-1}) \to H_n(X^n) \xrightarrow{j_n} H_n(X^n, X^{n-1}) \xrightarrow{\partial_n} H_{n-1}(X^{n-1})$$

By lemma 3.5.9(2) the group $H_n(X^{n-1})$ vanishes, since $n > n-1$. Exactness at $H_n(X^n)$ then forces $\ker j_n = \mathrm{im}\big(H_n(X^{n-1}) \to H_n(X^n)\big) = 0$, so

$$j_n\colon H_n(X^n) \hookrightarrow H_n(X^n, X^{n-1})$$

is injective. The same applies one degree up: j_{n+1} is injective. Now $d_{n+1} = j_n\partial_{n+1}$, so $\mathrm{im} d_{n+1} = j_n\big(\mathrm{im}\partial_{n+1}\big)$ inside $H_n(X^n, X^{n-1})$, where $\partial_{n+1}\colon H_{n+1}(X^{n+1}, X^n) \to H_n(X^n)$ is the connecting map of (X^{n+1}, X^n).

Step 2: Computing $\mathrm{im}d_{n+1}$. Exactness of the long exact sequence of (X^{n+1}, X^n) at $H_n(X^n)$,

$$H_{n+1}(X^{n+1}, X^n) \xrightarrow{\partial_{n+1}} H_n(X^n) \to H_n(X^{n+1})$$

identifies $\mathrm{im}\partial_{n+1}$ with the kernel of the inclusion-induced map $H_n(X^n) \to H_n(X^{n+1})$. Applying the injective j_n of Step 1 to this equality of subgroups of $H_n(X^n)$ gives

$$\mathrm{im}d_{n+1} = j_n\big(\mathrm{im}\partial_{n+1}\big) = j_n\Big(\ker\big(H_n(X^n) \to H_n(X^{n+1})\big)\Big)$$

By lemma 3.5.9(3) the inclusion induces an isomorphism $H_n(X^{n+1}) \cong H_n(X)$, and the surjection $H_n(X^n) \twoheadrightarrow H_n(X)$ of part (3) factors through $H_n(X^{n+1})$, so the kernel of $H_n(X^n) \to H_n(X^{n+1})$ coincides with the kernel of the surjection $H_n(X^n) \to H_n(X)$. Writing $q\colon H_n(X^n) \twoheadrightarrow H_n(X)$ for that surjection, we have recorded

$$\mathrm{im}d_{n+1} = j_n(\ker q)$$

Step 3: Computing $\ker d_n$. Here $d_n = j_{n-1}\partial_n$ with $\partial_n\colon H_n(X^n, X^{n-1}) \to H_{n-1}(X^{n-1})$ and $j_{n-1}\colon H_{n-1}(X^{n-1}) \to H_{n-1}(X^{n-1}, X^{n-2})$. By Step 1 (one degree lower) the map j_{n-1} is injective, so $d_n(x) = j_{n-1}(\partial_n x) = 0$ if and only if $\partial_n x = 0$. Therefore

$$\ker d_n = \ker \partial_n$$

Exactness of the long exact sequence of (X^n, X^{n-1}) at $H_n(X^n, X^{n-1})$,

$$H_n(X^n) \xrightarrow{j_n} H_n(X^n, X^{n-1}) \xrightarrow{\partial_n} H_{n-1}(X^{n-1})$$

gives $\ker \partial_n = \mathrm{im}j_n$. Combining,

$$\ker d_n = \mathrm{im}j_n = j_n\big(H_n(X^n)\big)$$

where the last equality uses that j_n is injective, so its image is a faithful copy of $H_n(X^n)$.

Step 4: Identifying the subquotient. By Steps 2 and 3, both $\ker d_n$ and $\mathrm{im}d_{n+1}$ lie inside $H_n(X^n, X^{n-1})$ as images under the injection j_n of subgroups of $H_n(X^n)$, namely of $H_n(X^n)$ itself and of $\ker q$ respectively. An injective homomorphism induces an isomorphism onto its image and carries the quotient by a subgroup to the quotient of the image by the image of that subgroup, so j_n descends to an isomorphism

$$\frac{\ker d_n}{\mathrm{im}d_{n+1}} = \frac{j_n\big(H_n(X^n)\big)}{j_n(\ker q)} \cong \frac{H_n(X^n)}{\ker q}$$

The map $q\colon H_n(X^n) \to H_n(X)$ is surjective by lemma 3.5.9(3), so by the first isomorphism theorem ([10, First Isomorphism Theorem]) it induces an isomorphism $H_n(X^n)/\ker q \cong H_n(X)$. Composing,

$$H_n^{CW}(X) = \frac{\ker d_n}{\mathrm{im}d_{n+1}} \cong \frac{H_n(X^n)}{\ker q} \cong H_n(X)$$

The composite isomorphism is $\Theta_X : j_n(z) + \mathrm{im} d_{n+1} \mapsto q(z)$.

Step 5: Naturality. Let $f : X \to Y$ be cellular. Then f is a map of pairs $(X^n, X^{n-1}) \to (Y^n, Y^{n-1})$, and $f_\sharp$ gives a morphism from the short exact sequence $0 \to C_\bullet(X^{n-1}) \to C_\bullet(X^n) \to C_\bullet(X^n, X^{n-1}) \to 0$ to the corresponding sequence for Y, so proposition 3.4.4 gives $f_* \partial_n = \partial_n f_*$. The maps j_n and q are induced by inclusions of pairs, and f commutes with these inclusions, so the chain-level identities of proposition 3.3.3, which pass to relative chains, give $f_* j_n = j_n f_*$ and $f_* q = q f_*$. Hence $f_* d_n = f_* j_{n-1} \partial_n = j_{n-1} \partial_n f_* = d_n f_*$, so the maps f_* form a chain map of cellular chain complexes, and

$$\Theta_Y\big(f_*(j_n(z)) + \mathrm{im} d_{n+1}\big) = \Theta_Y\big(j_n(f_* z) + \mathrm{im} d_{n+1}\big) = q(f_* z) = f_*(q(z)) = f_* \Theta_X\big(j_n(z) + \mathrm{im} d_{n+1}\big)$$

This gives the asserted natural isomorphism $H_n^{CW}(X) \cong H_n(X)$, as we sought to show.

Corollary 3.5.12: Consequence Of CW Homology

1. $H_n(X) = 0$ if X is a CW complex with no n-cells
2. If X is a CW complex with k n-cells, then $H_n(X)$ is generated by at most k elements.
3. If X is a CW complex having no two of its cells in adjacent dimensions, then $H_n(X)$ is a free abelian group with basis in one-to-one correspondence with the n-cells of X.

***Proof*:**
In each part, $H_n(X) \cong H_n^{CW}(X)$ by theorem 3.5.11, and $H_n(X^n, X^{n-1})$ is free abelian on the n-cells by lemma 3.5.9(1).

1. If X has no n-cells, then $H_n(X^n, X^{n-1}) = 0$, so $H_n^{CW}(X) = 0$.
2. Since $H_n(X^n, X^{n-1})$ is free abelian on k generators, the subgroup $\ker d_n$ is generated by at most k elements (a subgroup of $\mathbb{Z}^k$ is free of rank at most k), hence similarly for the quotient $\ker d_n / \mathrm{im} d_{n+1}$.
3. Each cellular boundary map d_n has source or target 0, since no two cells lie in adjacent dimensions. So all d_n are zero, and $H_n^{CW}(X) = H_n(X^n, X^{n-1})$, giving the result.

Example 3.5.13: CW Homology
Take $X = \mathbb{C}\mathrm{P}^n$, which has a CW structure with one cell in each even dimension $0, 2, \ldots, 2n$ (example 1.2.5). By corollary 3.5.12(3):

$$H_k(\mathbb{C}\mathrm{P}^n) = \begin{cases} \mathbb{Z} & k \in \{0, 2, 4, ..., 2n\} \\ 0 & \text{otherwise} \end{cases}$$

In general, the cellular boundary maps are computed from degrees of maps of spheres:

Proposition 3.5.14: Cellular Boundary Formula

Let X be a CW complex, and consider the cellular homology with boundary map $\partial_n : H_n(X^n, X^{n-1}) \to H_{n-1}(X^{n-1}, X^{n-2})$. For a cell e^n_α with attaching map:

$$\varphi_\alpha : \partial e^n_\alpha \to X^{n-1}$$

and

$$\partial_{\alpha\beta} : S^{n-1} \xrightarrow{\cong} \partial e^n_\alpha \xrightarrow{\varphi_\alpha} X^{n-1} \xrightarrow{\pi} X^{n-1}/(X^{n-1} \setminus e^{n-1}_\beta) \xrightarrow{\cong} S^{n-1}$$

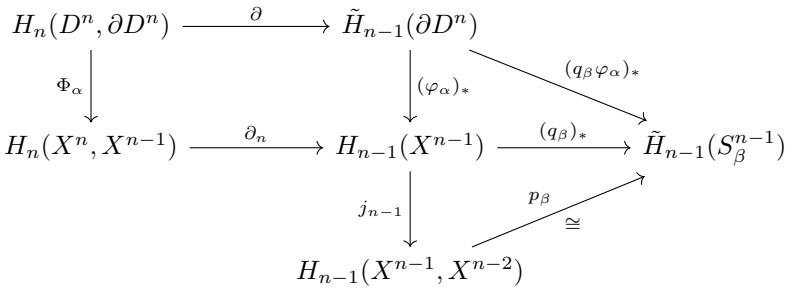
representing the attaching map φ_α with respect to an $(n-1)$-cell e^{n-1}_β, then ∂_n is determined by how it acts on each e_α and

$$\partial_n(e^n_\alpha) = \sum_\beta \deg(\partial_{\alpha\beta}) e^{n-1}_\beta$$

***Proof*:**

Since ∂_n is a homomorphism out of the free abelian group $H_n(X^n, X^{n-1})$, it is determined by its values on the basis elements e^n_α given by lemma 3.5.9. Fix one n-cell e^n_α with characteristic map $\Phi_\alpha \colon (D^n, \partial D^n) \to (X^n, X^{n-1})$ (definition 1.2.2), whose restriction to $\partial D^n = S^{n-1}$ is the attaching map φ_α. The cellular boundary is $d_n = j_{n-1}\partial_n$, where $\partial_n \colon H_n(X^n, X^{n-1}) \to H_{n-1}(X^{n-1})$ is the connecting homomorphism of the pair (X^n, X^{n-1}) and $j_{n-1} \colon H_{n-1}(X^{n-1}) \to H_{n-1}(X^{n-1}, X^{n-2})$ is induced by the inclusion of pairs. Computing $d_n(e^n_\alpha)$ means computing the coefficient of each basis element e^{n-1}_β of $H_{n-1}(X^{n-1}, X^{n-2})$, and the claim is that this coefficient is $\deg(\partial_{\alpha\beta})$. The argument assembles one large commutative diagram from the naturality of the connecting homomorphism and an excision that isolates the cell e^{n-1}_β. The identification of the relevant map of spheres with $\partial_{\alpha\beta}$ then converts the coefficient into a degree, which the local-degree calculus evaluates.

Step 1: The master diagram. For each $(n-1)$-cell e^{n-1}_β let $q_\beta \colon X^{n-1} \to X^{n-1}/(X^{n-1} \setminus e^{n-1}_\beta)$ be the quotient collapsing everything outside the open cell to a point. The target is a sphere S^{n-1}_β, since $X^{n-1}/(X^{n-1} \setminus e^{n-1}_\beta)$ is, through the characteristic map of e^{n-1}_β, the quotient $D^{n-1}/\partial D^{n-1}$. Consider the diagram



The left square commutes by naturality of the connecting homomorphism (proposition 3.4.4) applied to the map of pairs $\Phi_\alpha \colon (D^n, \partial D^n) \to (X^n, X^{n-1})$: the connecting map is natural for maps of pairs, and Φ_α restricts on ∂D^n to φ_α. The top triangle commutes by functoriality of the induced map, since $q_\beta\varphi_\alpha = q_\beta \circ \varphi_\alpha$. The bottom triangle is the definition of p_β: it is the

composite

$$H_{n-1}(X^{n-1}, X^{n-2}) \cong \bigoplus_{\gamma} \tilde{H}_{n-1}(S^{n-1}_{\gamma}) \xrightarrow{\mathrm{pr}_\beta} \tilde{H}_{n-1}(S^{n-1}_{\beta})$$

of the basis identification of lemma 3.5.9 with the projection onto the summand indexed by β. The equality $(q_\beta)_* = p_\beta \circ j_{n-1}$ holds because q_β factors as the quotient map $X^{n-1} \to X^{n-1}/X^{n-2} = \bigvee_\gamma S^{n-1}_\gamma$ followed by the map $\bigvee_\gamma S^{n-1}_\gamma \to S^{n-1}_\beta$ collapsing the spheres with $\gamma \neq \beta$. By naturality of the quotient maps of pairs, the first induces j_{n-1} followed by the isomorphism $H_{n-1}(X^{n-1}, X^{n-2}) \cong \tilde{H}_{n-1}(\bigvee_\gamma S^{n-1}_\gamma)$ of corollary 3.4.12. The second is the identity on S^{n-1}_β and constant on the other summands, so under corollary 3.4.19 it induces the projection onto the β summand.

Step 2: Reading off a single coefficient. Write $[D^n]$ for the generator of $H_n(D^n, \partial D^n) \cong \mathbb{Z}$ (corollary 3.4.15) corresponding under Φ_α to the basis element e^n_α, and $[\partial D^n] = \partial[D^n]$ for the resulting generator of $\tilde{H}_{n-1}(\partial D^n) \cong \mathbb{Z}$. The coefficient of e^{n-1}_β in $d_n(e^n_\alpha) = j_{n-1}\partial_n(e^n_\alpha)$ is, by definition of the basis, the integer $m_{\alpha\beta}$ with

$$p_\beta\big(j_{n-1}\partial_n(e^n_\alpha)\big) = m_{\alpha\beta}\,[S^{n-1}_\beta]$$

where $[S^{n-1}_\beta]$ is the chosen generator of $\tilde{H}_{n-1}(S^{n-1}_\beta) \cong \mathbb{Z}$. Chasing $[D^n]$ through the master diagram of Step 1 in two ways gives this integer. Down-then-right-then-project sends

$$[D^n] \longmapsto e^n_\alpha \longmapsto \partial_n(e^n_\alpha) \longmapsto p_\beta\big(j_{n-1}\partial_n(e^n_\alpha)\big) = m_{\alpha\beta}\,[S^{n-1}_\beta]$$

using $p_\beta j_{n-1} = (q_\beta)_*$ from the bottom triangle and the commuting left square. Right-then-diagonal sends

$$[D^n] \longmapsto [\partial D^n] \longmapsto (q_\beta\varphi_\alpha)_*[\partial D^n]$$

using the top triangle. Commutativity of the whole diagram forces these two outputs to agree:

$$m_{\alpha\beta}\,[S^{n-1}_\beta] = (q_\beta\varphi_\alpha)_*[\partial D^n]$$

Thus $m_{\alpha\beta}$ is exactly the integer by which the map $q_\beta\varphi_\alpha \colon \partial D^n \to S^{n-1}_\beta$ multiplies on $\tilde{H}_{n-1} \cong \mathbb{Z}$.

Step 3: Identifying the map with $\partial_{\alpha\beta}$ and its degree. The composite $q_\beta\varphi_\alpha$ is, after the canonical identifications $\partial D^n \cong S^{n-1}$ and the target $X^{n-1}/(X^{n-1} \setminus e^{n-1}_\beta) \cong S^{n-1}$, precisely the map

$$\partial_{\alpha\beta} \colon S^{n-1} \xrightarrow{\cong} \partial e^n_\alpha \xrightarrow{\varphi_\alpha} X^{n-1} \xrightarrow{\pi} X^{n-1}/(X^{n-1} \setminus e^{n-1}_\beta) \xrightarrow{\cong} S^{n-1}$$

of the statement, since q_β is the quotient map π followed by the homeomorphism to S^{n-1}. A self-map of S^{n-1} acts on $\tilde{H}_{n-1}(S^{n-1}) \cong \mathbb{Z}$ as multiplication by its degree (definition 3.5.1), so the integer extracted in Step 2 is

$$m_{\alpha\beta} = \deg(\partial_{\alpha\beta})$$

once the generators $[\partial D^n]$ and $[S^{n-1}_\beta]$ are taken to correspond under the chosen identifications. This is the coefficient we sought.

Step 4: Finiteness of the sum and assembly. For fixed α the integers $\deg(\partial_{\alpha\beta})$ vanish for all but finitely many β. The image $\varphi_\alpha(S^{n-1})$ is compact, hence meets only finitely many open

$(n-1)$-cells e_β^{n-1} by lemma 1.2.4. For any β whose open cell is disjoint from this image, the map $q_\beta \varphi_\alpha$ is constant, so $\partial_{\alpha\beta}$ factors through a point, whose reduced homology vanishes, and $\deg(\partial_{\alpha\beta}) = 0$. Therefore the sum below is finite. Reassembling the coordinates computed in Step 3,

$$d_n(e_\alpha^n) = \sum_\beta m_{\alpha\beta}\, e_\beta^{n-1} = \sum_\beta \deg(\partial_{\alpha\beta})\, e_\beta^{n-1}$$

which is the cellular boundary formula. (When several distinct points of S^{n-1} map into the same cell e_β^{n-1}, the degree $\deg(\partial_{\alpha\beta})$ is computed as the sum of the local degrees of $\partial_{\alpha\beta}$ over the preimages of a regular value, by proposition 3.5.6 and definition 3.5.5. The formula already records the total degree, so no separate bookkeeping is needed.) Writing ∂_n for the cellular boundary d_n as in the statement, this establishes

$$\partial_n(e_\alpha^n) = \sum_\beta \deg(\partial_{\alpha\beta})\, e_\beta^{n-1}$$

for every n-cell e_α^n, completing the proof.

By proposition 3.5.14, the homology of a CW complex is determined by the degrees of its attaching maps:

Example 3.5.15: Computing Cellular Boundary Map

1. Let $X = M_g$ be a closed orientable surface of genus g with the usual CW structure of one 0-cell, $2g$ 1-cells, and one 2-cell attached by the product of the commutators $[a_1, b_1], ..., [a_g, b_g]$. Then the associated cellular chain complex is:

   $$0 \to \mathbb{Z} \xrightarrow{d_2} \mathbb{Z}^{2g} \xrightarrow{d_1} \mathbb{Z} \to 0$$

   Then d_1 must be zero since there is only one 0-cell. d_2 is also zero since each a_i or b_i appears with its inverse in $[a_1, b_1], ..., [a_g, b_g]$, hence each map $\partial_{\alpha\beta}$ has local degrees 1 and -1 at the two preimages of a point of the corresponding 1-cell, and degree 0 (proposition 3.5.6). Since d_1, d_2 are both zero, the homology groups of M_g are the same as the cellular chain groups, hence:

   $$H_n(M_g) = \begin{cases} \mathbb{Z} & n = 0 \\ \mathbb{Z}^{2g} & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & n \geq 3 \end{cases}$$

2. Let $X = N_g$ be the closed nonorientable surface of genus g, which has one 0-cell, g 1-cells, and one 2-cell attached by the word $a_1^2 \cdots a_g^2$. We again have $d_1 = 0$ and $d_2 : \mathbb{Z} \to \mathbb{Z}^g$ is given by $d_2(1) = (2, ..., 2)$ since each a_i appears in the attaching word of the 2-cell with total exponent 2, which means $\partial_{\alpha\beta}$ has local degree 1 at each of the two preimages of a point of the 1-cell a_i, and degree 2 (proposition 3.5.6). Since $d_2(1) = (2, ..., 2)$, we get that d_2 is injective, hence $H_2(N_g) = 0$. If we change the basis for $\mathbb{Z}^g$ by replacing the last standard basis element $(0, ..., 0, 1)$ by $(1, ..., 1)$ we see that

   $$H_1(N_g) \cong \mathbb{Z}^{g-1} \oplus \mathbb{Z}_2$$

Hence:

$$H_n(N_g) = \begin{cases} \mathbb{Z} & n = 0 \\ \mathbb{Z}^{g-1} \oplus \mathbb{Z}_2 & n = 1 \\ 0 & n \geq 2 \end{cases}$$

Notice that we have detected orientability of a closed connected n-manifold by looking at $H_n(M)$, which was $\mathbb{Z}$ when M is orientable and 0 otherwise. This pattern generalizes [21, Theorem 3.26].

3. (Moore space) Let G be an abelian group and take $n \in \mathbb{N}$. Then we can construct a CW complex such that $H_n(X) \cong G$ and $\tilde{H}_i(X) = 0$ for all $i \neq n$. The space constructed is called the *Moore space*, and is often denoted $M(G, n)$.

   If G is finitely generated, then $G = F \oplus T$ for some torsion part T and free part F. The simplest case is $G = \mathbb{Z}$. Then for any n we have $X = S^n$. If $G = \mathbb{Z}/m\mathbb{Z}$, then for any n take S^n and attach e^{n+1} via $\varphi : S^n \to S^n$ of degree m (for example, an iterated suspension of $z \mapsto z^m$, by proposition 3.5.7). For the general case (not necessarily finitely generated), take a surjection $F \to G$ from a free abelian group F. The kernel of this map is free since subgroups of free abelian groups are free. Then if $\{x_\alpha\}$ is a basis for F, $\{y_\beta\}$ is a basis for the kernel K, and $y_\beta = \sum_\alpha d_{\beta\alpha} x_\alpha$, we define

   $$X^n = \bigvee_\alpha S^n_\alpha \qquad \Longrightarrow \qquad H_n(X^n) \cong F$$

   To construct X from X^n, attach cells e^{n+1}_β via maps $f_\beta : S^n \to X^n$ whose composite with the projection onto the summand S^n_α has degree $d_{\beta\alpha}$. By proposition 3.5.14 the cellular boundary map d_{n+1} is the inclusion $K \hookrightarrow F$, hence X has the desired homology.

   Now, by taking the wedge sum of Moore spaces for varying n, we can get a connected CW complex with arbitrary abelian groups G_n at every dimension.

4. Let $X = \mathbb{R}\mathrm{P}^n$. Then we saw that $\mathbb{R}\mathrm{P}^n$ has a CW structure with one cell e^k for each $k \leq n$, and the attaching map of e^k is the 2-sheeted covering $\varphi\colon S^{k-1} \to \mathbb{R}\mathrm{P}^{k-1}$. To compute d_k, we compute the degree of the composite

   $$S^{k-1} \xrightarrow{\varphi} \mathbb{R}\mathrm{P}^{k-1} \xrightarrow{q} \mathbb{R}\mathrm{P}^{k-1}/\mathbb{R}\mathrm{P}^{k-2} \cong S^{k-1}$$

   The map $q \circ \varphi$ restricts to a homeomorphism from each of the two components of $S^{k-1} - S^{k-2}$ onto $\mathbb{R}\mathrm{P}^{k-1} - \mathbb{R}\mathrm{P}^{k-2}$, and these two homeomorphisms differ by precomposition with the antipodal map of S^{k-1}, which has degree $(-1)^k$. So the local degrees at the two preimages of a point are 1 and $(-1)^k$, for a suitable identification of the target with S^{k-1}, and proposition 3.5.6 gives

   $$\deg q \circ \varphi = \deg \mathrm{id} + \deg(-\mathrm{id}) = 1 + (-1)^k$$

   Hence d_k is multiplication by 2 for even k and is 0 for odd k. Thus, we get:

   $$0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \cdots \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0 \qquad n \text{ is even}$$

   $$0 \to \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \cdots \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0 \qquad n \text{ is odd}$$

Thus, if n is *odd*:

$$H_k(\mathbb{R}\mathrm{P}^n) = \begin{cases} \mathbb{Z} & k = 0,\ k = n \\ \mathbb{Z}/2\mathbb{Z} & k \text{ odd},\ 0 < k < n \\ 0 & \text{otherwise} \end{cases}$$

and if n is *even*:

$$H_k(\mathbb{R}\mathrm{P}^n) = \begin{cases} \mathbb{Z} & k = 0 \\ \mathbb{Z}/2\mathbb{Z} & k \text{ odd},\ 0 < k < n \\ 0 & \text{otherwise} \end{cases}$$

As we noted under definition 2.1.18, having a trivial fundamental group does not imply the space is contractible. Likewise, a space whose reduced homology groups all vanish need not be contractible. Such spaces have a name:

Definition 3.5.16: Acyclic Space

Let X be a space. Then if $\tilde{H}_i(X) = 0$ for all i, we call such a space *acyclic*.

Proposition 3.5.17: Vanishing Reduced H_0 and Path-Connectedness

A nonempty space Y with $\tilde{H}_0(Y) = 0$ is path-connected, and then the augmentation induces an isomorphism $H_0(Y) \cong \mathbb{Z}$. In particular an acyclic space is nonempty and path-connected.

***Proof*:**
By proposition 3.2.3 and proposition 3.2.4, $H_0(Y)$ is free abelian on the classes of one point from each path component, each of augmentation 1. Two points y_1, y_2 in different path components would give the class $[y_1] - [y_2]$, which is nonzero and has augmentation 0, so it would be a nonzero element of $\tilde{H}_0(Y)$. Hence Y has a single path component, and by proposition 3.2.4 the augmentation is an isomorphism $H_0(Y) \cong \mathbb{Z}$. An acyclic space has $\tilde{H}_{-1} = 0$, which forces it to be nonempty (the augmented complex of the empty space has $\tilde{H}_{-1} = \mathbb{Z}$), and $\tilde{H}_0 = 0$, as we sought to show.

Theorem 3.5.18: Excised Sphere Is Acyclic

Let $n, k \geq 0$ and let $B \subseteq S^n$ be a subspace homeomorphic to the disk D^k. Then $S^n - B$ is acyclic.

***Proof*:**
Step 1: Reductions. Write $\|x\|_2$ and $\|x\|_\infty$ for the Euclidean and the maximum norm on $\mathbb{R}^k$, and set $\varphi(x) = (\|x\|_2/\|x\|_\infty)\, x$ and $\psi(y) = (\|y\|_\infty/\|y\|_2)\, y$ for nonzero x, y, with $\varphi(0) = \psi(0) = 0$. Homogeneity of the norms gives $\|\varphi(x)\|_\infty = \|x\|_2$ and $\|\varphi(x)\|_2 = \|x\|_2^2/\|x\|_\infty$, hence $\psi(\varphi(x)) = (\|x\|_\infty/\|x\|_2)\, \varphi(x) = x$, and in the same way $\|\psi(y)\|_2 = \|y\|_\infty$ and $\|\psi(y)\|_\infty = \|y\|_\infty^2/\|y\|_2$ give $\varphi(\psi(y)) = y$. The identities $\|\varphi(x)\|_\infty = \|x\|_2$ and $\|\psi(y)\|_2 = \|y\|_\infty$ also show that φ and ψ are continuous at 0, that φ maps D^k into the cube $[-1,1]^k$ and that ψ maps $[-1,1]^k$ into D^k. So D^k is homeomorphic to $[-1,1]^k$, hence to I^k, and $B = h(I^k)$ for an embedding $h\colon I^k \to S^n$.

Compact subsets of the Hausdorff space S^n are closed, so the complement in S^n of the image under h of a compact subset of I^k is open. The complement $S^n - B$ is nonempty: B is homotopy equivalent to a point, so $\tilde{H}_n(B) = 0$ by corollary 3.3.9, while $\tilde{H}_n(S^n) \cong \mathbb{Z}$ by corollary 3.4.15, so $B = S^n$ would contradict corollary 3.3.7, a homeomorphism being a homotopy equivalence. For a nonempty space the augmentation is surjective, so its reduced homology vanishes in every negative degree, and only the degrees $i \geq 0$ need attention.

Step 2: The case $k = 0$. Here B is a single point a, and $S^n - \{a\}$ is homeomorphic to $a^\perp \cong \mathbb{R}^n$ by lemma 3.4.26. The straight-line homotopy $(w, t) \mapsto (1 - t)w$ contracts $\mathbb{R}^n$ to a point, so $S^n - \{a\}$ is acyclic by corollary 3.3.9.

Step 3: Halving the last coordinate. Let $k \geq 1$ and assume the theorem for disks of dimension $k - 1$. For a closed interval $J \subseteq I$, possibly a single point, set $X_J = S^n - h(I^{k-1} \times J)$, an open subset of S^n containing $X_I = S^n - B$. The restriction of h to $I^{k-1} \times \{t\}$ embeds a cube of dimension $k - 1$, so by the inductive hypothesis and Step 1 the space $X_{\{t\}}$ is acyclic for every $t \in I$. Let $J = [u, v]$ with $u < v$, let $c = (u + v)/2$, and let $J' = [u, c]$ and $J'' = [c, v]$. Since h is injective,

$$X_{J'} \cap X_{J''} = S^n - h(I^{k-1} \times J) = X_J \qquad X_{J'} \cup X_{J''} = S^n - h(I^{k-1} \times \{c\}) = X_{\{c\}}$$

The sets $X_{J'}$ and $X_{J''}$ are open, so theorem 3.4.22 applies to them as subsets of their union $X_{\{c\}}$, giving for each $i \geq 0$ the exact segment

$$H_{i+1}(X_{\{c\}}) \xrightarrow{\partial} H_i(X_J) \xrightarrow{\Phi} H_i(X_{J'}) \oplus H_i(X_{J''})$$

whose first term is $\tilde{H}_{i+1}(X_{\{c\}}) = 0$. Hence Φ is injective. Its components are, up to sign, the maps induced by the inclusions $X_J \hookrightarrow X_{J'}$ and $X_J \hookrightarrow X_{J''}$, and these carry the reduced group $\tilde{H}_i$, which is H_i for $i \geq 1$ and the kernel of the augmentation for $i = 0$, into the reduced groups. So a nonzero class in $\tilde{H}_i(X_J)$ has nonzero image in $\tilde{H}_i(X_{J'})$ or in $\tilde{H}_i(X_{J''})$.

Step 4: Shrinking to a slice. Suppose $\alpha \in \tilde{H}_i(X_I)$ is nonzero for some $i \geq 0$. Applying Step 3 repeatedly produces closed intervals $I = J_0 \supseteq J_1 \supseteq J_2 \supseteq \cdots$, each a half of the previous one, such that the image α_m of α under the inclusion $X_I \hookrightarrow X_{J_m}$ is nonzero for every m. The lengths tend to 0, so $\bigcap_m J_m = \{p\}$ for a point $p \in I$, and injectivity of h gives

$$X_{\{p\}} = S^n - \bigcap_m h(I^{k-1} \times J_m) = \bigcup_m X_{J_m}$$

Represent α by a cycle z of the augmented chain complex of X_I. By Step 3 the space $X_{\{p\}}$ is acyclic, so $z = \partial w$ for a chain w of the augmented complex of $X_{\{p\}}$. The chain w is a finite combination of singular simplices, the union of whose images is a compact subset of $X_{\{p\}}$, and the increasing open sets X_{J_m} cover it, so a single X_{J_m} contains it. Then $z = \partial w$ holds in the augmented chain complex of X_{J_m}, so $\alpha_m = 0$, a contradiction. Hence $\tilde{H}_i(S^n - B) = 0$ for every i, and $S^n - B$ is acyclic, as we sought to show.

Acyclic spaces need not be contractible. Attaching two 2-cells to $S^1 \vee S^1$ along the words a^5b^{-3} and $b^3(ab)^{-2}$ gives a space whose cellular boundary $d_2 \colon \mathbb{Z}^2 \to \mathbb{Z}^2$ has determinant -1, so all its reduced homology vanishes, while its fundamental group has order 120 [21, Example 2.38, p. 142].

Theorem 3.5.19: More Homology calculations

Let $0 \leq k < n$ and let $h : S^k \to S^n$ be an embedding. Then

$$\tilde{H}_i(S^n - h(S^k)) \cong \begin{cases} \mathbb{Z} & i = n-k-1 \\ 0 & \text{otherwise} \end{cases}$$

***Proof*:**

Induct on k. The space $S^n - h(S^k)$ is nonempty, since $h(S^k) = S^n$ would make S^k homeomorphic to S^n, contradicting corollary 3.3.7, as $\tilde{H}_n(S^k) = 0$ and $\tilde{H}_n(S^n) \cong \mathbb{Z}$ for $k \neq n$ (corollary 3.4.15). So its reduced homology vanishes in negative degrees, as does the right-hand side of the statement, $n-k-1$ being nonnegative.

Step 1: The case $k = 0$. Here $h(S^0) = \{a, b\}$ with $a \neq b$. The stereographic projection s_a of lemma 3.4.26 restricts to a homeomorphism from $S^n - \{a, b\}$ onto $a^\perp - \{w_0\} \cong \mathbb{R}^n - \{w_0\}$, where $w_0 = s_a(b)$. The homotopy

$$r_t(w) = w_0 + \left(1 - t + \frac{t}{|w - w_0|}\right)(w - w_0)$$

is a deformation retraction of $\mathbb{R}^n - \{w_0\}$ onto the sphere of radius 1 about w_0, a copy of S^{n-1}: the coefficient of $w - w_0$ is positive, r_0 is the identity, r_1 lands in that sphere, and every r_t fixes it. Hence $S^n - h(S^0)$ is homotopy equivalent to S^{n-1}, and by corollary 3.3.7 and corollary 3.4.15 its reduced homology is $\mathbb{Z}$ in degree $n-1$ and 0 otherwise.

Step 2: The inductive step. Let $1 \leq k < n$ and assume the statement for embeddings of S^{k-1}. Write $S^k = D^k_+ \cup D^k_-$ for the closed hemispheres $x_{k+1} \geq 0$ and $x_{k+1} \leq 0$, which meet in the equator S^{k-1}. Each hemisphere is homeomorphic to D^k by forgetting the last coordinate, with inverse $x \mapsto (x, \pm\sqrt{1 - |x|^2})$. Set $A = S^n - h(D^k_+)$ and $B = S^n - h(D^k_-)$, open subsets of S^n since compact subsets of S^n are closed. Since h is injective,

$$A \cap B = S^n - h(S^k) \qquad A \cup B = S^n - h(S^{k-1})$$

By theorem 3.5.18 the spaces A and B are acyclic, so $H_j(A) = H_j(B) = 0$ for $j \geq 1$, and by proposition 3.5.17 each of them is path-connected, with the augmentation an isomorphism $H_0(A) \cong \mathbb{Z}$, and likewise for B. For $i \geq 1$ the Mayer-Vietoris sequence of theorem 3.4.22 for the open sets A and B contains

$$H_{i+1}(A) \oplus H_{i+1}(B) \to H_{i+1}(A \cup B) \xrightarrow{\partial} H_i(A \cap B) \to H_i(A) \oplus H_i(B)$$

with both outer terms zero, so ∂ is an isomorphism $\tilde{H}_{i+1}(A \cup B) \to \tilde{H}_i(A \cap B)$. For $i = 0$ it contains

$$H_1(A) \oplus H_1(B) \to H_1(A \cup B) \xrightarrow{\partial} H_0(A \cap B) \xrightarrow{\Phi} H_0(A) \oplus H_0(B)$$

with first term zero, so ∂ maps $H_1(A \cup B)$ isomorphically onto $\ker \Phi$. The components of Φ are, up to sign, induced by the inclusions of $A \cap B$ into A and into B, and these commute with the augmentations, which are injective on $H_0(A)$ and on $H_0(B)$. So a class lies in $\ker \Phi$ exactly

when its augmentation vanishes, that is $\ker \Phi = \tilde{H}_0(A \cap B)$. Hence for every $i \geq 0$

$$\tilde{H}_i\big(S^n - h(S^k)\big) \cong \tilde{H}_{i+1}\big(S^n - h(S^{k-1})\big)$$

The inductive hypothesis, applied to the embedding $h|_{S^{k-1}}$ with $k-1 < n$, says that the right-hand side is $\mathbb{Z}$ when $i+1 = n-(k-1)-1$, that is $i = n-k-1$, and 0 otherwise, as we sought to show.

Theorem 3.5.20: Generalized Jordan Curve Theorem

Let $n \geq 1$ and let $C \subseteq S^n$ be a subspace homeomorphic to the sphere S^{n-1}. Then $S^n - C$ has exactly two path components U and V. Both are open in S^n and acyclic, they are also the two connected components of $S^n - C$, and C is the boundary of each of them: $\overline{U} - U = \overline{V} - V = C$.

***Proof*:**

Step 1: Two acyclic path components. Let $g\colon S^{n-1} \to C$ be a homeomorphism, an embedding $S^{n-1} \to S^n$. By theorem 3.5.19 with $k = n-1$, the reduced homology of $S^n - C$ is $\mathbb{Z}$ in degree 0 and 0 in every other degree. In particular $S^n - C$ is nonempty, so the augmentation $H_0(S^n - C) \to \mathbb{Z}$ is onto, with kernel $\tilde{H}_0(S^n - C)$ and with the section $1 \mapsto [x_0]$ for a point x_0. By lemma 3.4.34, $H_0(S^n - C) \cong \tilde{H}_0(S^n - C) \oplus \mathbb{Z} \cong \mathbb{Z}^2$. On the other hand $H_0(S^n - C)$ is free abelian with a basis indexed by the set P of path components (proposition 3.2.3, proposition 3.2.4), and tensoring with $\mathbb{Q}$ ([10, Arbitrary Direct Sums Commute with Tensor Products]) gives $|P| = \dim_{\mathbb{Q}}\big(H_0(S^n - C) \otimes_{\mathbb{Z}} \mathbb{Q}\big) = 2$. So $S^n - C$ has exactly two path components U and V. For $i \geq 1$, proposition 3.2.3 gives $0 = H_i(S^n - C) \cong H_i(U) \oplus H_i(V)$, and U and V are nonempty and path-connected, so their reduced H_0 vanishes by proposition 3.2.4, and both are acyclic.

Step 2: Openness and connected components. Every point $y \in S^n$ lies in the open set $S^n - \{-y\}$, which is homeomorphic to $\mathbb{R}^n$ by lemma 3.4.26, and the images of the open balls about the image of y form a basis of path-connected open neighbourhoods of y in S^n. The set C is compact, hence closed in S^n. Let Q be one of U and V and let $y \in Q$. Some path-connected open neighbourhood of y lies in the open set $S^n - C$, and it lies in the path component Q, so Q is open in S^n. Thus $S^n - C$ is the union of the disjoint nonempty open sets U and V, each connected, being path-connected. A connected subset of $S^n - C$ meeting both would be disconnected by U and V, so U and V are the connected components of $S^n - C$.

Step 3: C is the boundary of U and of V. Let Q be one of U and V and Q' the other. The set Q' is open and disjoint from Q, so $\overline{Q} \cap Q' = \emptyset$ and $\overline{Q} - Q \subseteq C$. Conversely let $x \in C$ and let N be an open neighbourhood of x in S^n. Choose a subspace $D \subseteq C$ homeomorphic to D^{n-1} with $x \notin D$ and $C - D \subseteq N$, as follows. If $n = 1$, then $C = \{x, x'\}$ and $D = \{x'\}$. If $n \geq 2$, let $y = g^{-1}(x)$ and choose $\varepsilon \in (0,1)$ with $\big\{z \in S^{n-1} \;\big|\; z \cdot y > 1 - \varepsilon\big\} \subseteq g^{-1}(N \cap C)$, which is possible because $|z - y|^2 = 2 - 2\,z \cdot y$. Set $D' = \big\{z \in S^{n-1} \;\big|\; z \cdot y \leq 1 - \varepsilon\big\}$ and $D = g(D')$. The stereographic projection s_y of lemma 3.4.26 carries D' homeomorphically onto the closed ball $\big\{w \in y^\perp \;\big|\; |w|^2 \leq (2-\varepsilon)/\varepsilon\big\}$, because $|s_y(z)|^2 = (1 + z \cdot y)/(1 - z \cdot y)$ is a strictly increasing function of $z \cdot y$, and that ball is homeomorphic to D^{n-1} by scaling. By theorem 3.5.18 and proposition 3.5.17, $S^n - D$ is path-connected, and it contains U and V. Choose $p \in Q$, $q \in Q'$ and a path $\gamma\colon I \to S^n - D$ from p to q. Since Q and Q' are different path components of $S^n - C$, the

closed set $T = \gamma^{-1}(C)$ is nonempty, and $t_0 = \min T$ is positive. The path-connected set $\gamma([0, t_0))$ lies in $S^n - C$ and contains p, so it lies in Q, and therefore $\gamma(t_0) \in \overline{Q}$. Also $\gamma(t_0) \in C - D \subseteq N$. So the open set N meets $\overline{Q}$, and hence meets Q. As N was arbitrary, $x \in \overline{Q}$, and $x \notin Q$. Hence $\overline{Q} - Q = C$, as we sought to show.

Theorem 3.5.21: Invariance Of Domain

Let $U \subseteq \mathbb{R}^n$ be open, and $f : U \to \mathbb{R}^n$ an injective continuous map. Then $f(U)$ is open in $\mathbb{R}^n$, and f is an embedding onto its image.

***Proof*:**
For $n = 0$ there is nothing to prove, so let $n \geq 1$. Fix a point $N \in S^n$ and identify $\mathbb{R}^n$ with the open subset $S^n - \{N\}$ of S^n through the stereographic projection s_N of lemma 3.4.26. A subset of $\mathbb{R}^n$ is then open in $\mathbb{R}^n$ exactly when it is open in S^n. Let $x \in U$, choose $\varepsilon > 0$ such that the closed ball $\overline{B}$ of radius ε about x lies in U, and let B be the open ball and $S = \overline{B} - B$ the sphere of the same radius. Since $\overline{B}$ is compact and S^n is Hausdorff, the continuous injection $f|_{\overline{B}}$ is a homeomorphism onto its image ([16, making continuous function homeomorphisms]), so $f(S) \subseteq S^n$ is homeomorphic to S^{n-1} and $f(\overline{B})$ to D^n. By theorem 3.5.20, $S^n - f(S)$ has exactly two path components W_1 and W_2, both open in S^n. By injectivity of f, $S^n - f(S)$ is the disjoint union of $f(B)$ and $S^n - f(\overline{B})$. The set $f(B)$ is nonempty and path-connected, as the image of a ball. The set $S^n - f(\overline{B})$ contains N, and it is path-connected because $\tilde{H}_0(S^n - f(\overline{B})) = 0$ by theorem 3.5.18 with $k = n$, and a nonempty space with vanishing $\tilde{H}_0$ is path-connected (proposition 3.5.17). Each of the two sets lies in a single path component of $S^n - f(S)$, and not both in the same one, since their union also contains the other component. Say $f(B) \subseteq W_1$ and $S^n - f(\overline{B}) \subseteq W_2$. The two sets cover $W_1 \cup W_2$, so $f(B) = W_1$, which is open in S^n, hence in $\mathbb{R}^n$. Since $f(x) \in f(B) \subseteq f(U)$, the set $f(U)$ is open. Applying this to the restriction of f to an arbitrary open subset of U shows that f carries open subsets of U to open subsets of $\mathbb{R}^n$, so the inverse $f(U) \to U$ is continuous and f is an embedding onto its image, as we sought to show.

3.5.3 Euler-Characteristic

Recall Euler's formula for convex polyhedra in $\mathbb{R}^3$:

$$V - E + F = 2$$

We can define a similar constant for any finite CW complex X, namely its *Euler Characteristic*, defined to be

$$\chi(X) = \sum_n (-1)^n c_n$$

where c_n is the number of n-cells of X. This quantity has a purely homological description, so it is independent of the CW structure:

Theorem 3.5.22: Euler's Characteristic

Let X be a finite CW complex. Then

$$\chi(X) = \sum_n (-1)^n \operatorname{rank} H_n(X)$$

***Proof*:**
Rank is additive along a short exact sequence of finitely generated abelian groups, and the theorem is that fact applied to the cellular chain complex. Here $\operatorname{rank} G$ is the number of $\mathbb{Z}$ summands of G, the quantity called the Betti number in definition 3.1.15.

Step 1: Rank is a dimension. Let G be a finitely generated abelian group. Applying [10, Invariant Factor Form of Modules over PID's] with $R = \mathbb{Z}$ gives

$$G \cong \mathbb{Z}^r \oplus \mathbb{Z}/(a_1) \oplus \cdots \oplus \mathbb{Z}/(a_m)$$

with every a_i a nonzero non-unit, so the summands after the free part form a finite group T. Tensoring over $\mathbb{Z}$ with $\mathbb{Q}$ commutes with finite direct sums, and it kills T: an element $t \in T$ of order s satisfies

$$t \otimes q = t \otimes s\tfrac{q}{s} = st \otimes \tfrac{q}{s} = 0$$

for every $q \in \mathbb{Q}$. Since $\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q} = \mathbb{Q}$, this leaves

$$\operatorname{rank} G = r = \dim_{\mathbb{Q}}(G \otimes_{\mathbb{Z}} \mathbb{Q})$$

The right-hand side does not refer to any decomposition, so r is independent of the one chosen.

Step 2: Additivity. Let $0 \to A \to B \to C \to 0$ be a short exact sequence of finitely generated abelian groups. The group $\mathbb{Q}$ is torsion-free over the principal ideal domain $\mathbb{Z}$, hence flat ([10, Flat Equals Torsion-Free Over a PID]), so

$$0 \to A \otimes_{\mathbb{Z}} \mathbb{Q} \to B \otimes_{\mathbb{Z}} \mathbb{Q} \to C \otimes_{\mathbb{Z}} \mathbb{Q} \to 0$$

is an exact sequence of $\mathbb{Q}$-vector spaces. Dimension is additive along such a sequence, so Step 1 gives

$$\operatorname{rank} B = \operatorname{rank} A + \operatorname{rank} C$$

Step 3: The alternating sum of a chain complex. Let

$$0 \to C_k \xrightarrow{d_k} C_{k-1} \to \cdots \to C_1 \xrightarrow{d_1} C_0 \to 0$$

be a chain complex of finitely generated abelian groups, with cycles $Z_n = \ker d_n$, boundaries $B_n = \operatorname{im} d_{n+1}$ and homology $H_n = Z_n/B_n$. Each Z_n and each B_n is a subgroup of the finitely

generated group C_n, hence finitely generated, because $\mathbb{Z}$ is a principal ideal domain and a submodule of a finitely generated module over such a ring is finitely generated. The map d_n has kernel Z_n and image B_{n-1}, and H_n is by definition the quotient of Z_n by B_n, so there are two short exact sequences

$$0 \to Z_n \to C_n \xrightarrow{d_n} B_{n-1} \to 0, \qquad 0 \to B_n \to Z_n \to H_n \to 0$$

Step 2 applied to each gives $\operatorname{rank} C_n = \operatorname{rank} Z_n + \operatorname{rank} B_{n-1}$ and $\operatorname{rank} Z_n = \operatorname{rank} B_n + \operatorname{rank} H_n$. Substituting the second into the first,

$$\operatorname{rank} C_n = \operatorname{rank} B_n + \operatorname{rank} H_n + \operatorname{rank} B_{n-1}$$

Multiply by $(-1)^n$ and sum over n, a finite sum since $C_n = 0$ outside $0 \le n \le k$. Reindexing the last term by $m = n - 1$,

$$\sum_n (-1)^n \operatorname{rank} B_{n-1} = \sum_m (-1)^{m+1} \operatorname{rank} B_m = -\sum_m (-1)^m \operatorname{rank} B_m$$

so the two boundary contributions cancel and what remains is

$$\sum_n (-1)^n \operatorname{rank} C_n = \sum_n (-1)^n \operatorname{rank} H_n$$

Step 4: Applying this to X. Take for $C_\bullet$ the cellular chain complex of definition 3.5.10, whose n-th group is $H_n(X^n, X^{n-1})$. By lemma 3.5.9(1) that group is free abelian with basis in bijection with the n-cells of X, so $\operatorname{rank} C_n = c_n$, and it is finitely generated with only finitely many n contributing because X is finite. By theorem 3.5.11 the homology of this complex is $H_n(X)$. Step 3 therefore reads

$$\chi(X) = \sum_n (-1)^n c_n = \sum_n (-1)^n \operatorname{rank} H_n(X)$$

as we sought to show. The right-hand side makes no reference to the CW structure, so $\chi(X)$ is independent of the structure chosen.

Exercise 3.5.1

1. Let X and Y be finite CW complexes, and give $X \times Y$ the cell structure of definition 1.2.8, whose cells are the products $e^i_\alpha \times e^j_\beta$. Show that $\chi(X \times Y) = \chi(X)\,\chi(Y)$.

3.5.4 Simplicial Approximation

In a Δ-complex two simplices may meet along several faces, and a simplex may be glued to itself. In a simplicial complex two simplices meet in a single common face or not at all. Between finite simplicial complexes, every continuous map is homotopic to a map that is affine on each simplex of a fine subdivision of the domain, the simplicial approximation theorem, and this is what the Lefschetz fixed point theorem of the next subsection needs. Only finite complexes in a Euclidean space occur.

Definition 3.5.23: Finite Simplicial Complex

A *finite simplicial complex* is a finite set K of geometric simplices in some $\mathbb{R}^N$ such that every face of a simplex of K belongs to K, and the intersection of two simplices of K is either empty or a common face of both. Its *polyhedron* is the union $|K| \subseteq \mathbb{R}^N$ of its simplices, with the subspace topology. The 0-simplices are the *vertices* of K, K^n is the subcomplex of simplices of dimension at most n, and the *dimension* of K is the largest dimension of a simplex of K. The *open simplex* $\text{int}\sigma$ of a simplex σ is the set of points of σ with all barycentric coordinates positive.

Proposition 3.5.24: Simplicial Complexes Are CW Complexes

Let K be a finite simplicial complex.

1. Every point $x \in |K|$ lies in $\text{int}\sigma$ for exactly one $\sigma \in K$, the *carrier* of x. For a vertex v of K, the barycentric coordinate $t_v(x)$ of x with respect to a simplex containing x, taken to be 0 if v is not a vertex of that simplex, does not depend on the simplex, and $t_v \colon |K| \to [0,1]$ is continuous.
2. For every choice of homeomorphisms $\Psi_\sigma \colon (D^n, S^{n-1}) \to (\sigma, \partial\sigma)$, one for each simplex σ of each dimension n (lemma 3.1.5), $|K|$ is a finite CW complex whose n-skeleton is $|K^n|$, with one n-cell $\text{int}\sigma$ for each n-simplex σ and characteristic map Ψ_σ.

***Proof*:**

1. A point x of a simplex σ lies in the open simplex of the face of σ spanned by the vertices at which its barycentric coordinates are positive. If $x \in \text{int}\tau \cap \text{int}\tau'$, then $\tau \cap \tau'$ is a common face of τ and τ' containing x. A proper face of τ misses $\text{int}\tau$, since barycentric coordinates in τ are unique and some of them vanish on a proper face. Hence $\tau \cap \tau' = \tau$, likewise $\tau \cap \tau' = \tau'$, and $\tau = \tau'$. If x lies in simplices σ and σ', it lies in the common face $\sigma \cap \sigma'$, and its coordinates in that face, extended by zero, are coordinates of x in σ and in σ', so by uniqueness they are its coordinates in both. Each t_v is affine on each simplex, hence continuous there, and the simplices form a finite closed cover of $|K|$, so t_v is continuous by [16, Pasting Lemma], applied by induction on the number of simplices.
2. The 0-skeleton $|K^0|$ is a finite subset of $\mathbb{R}^N$, hence discrete. Let $n \geq 1$. The inclusion of $|K^{n-1}|$ and the maps Ψ_σ for the n-simplices σ define a continuous map from $|K^{n-1}| \sqcup_\sigma D^n_\sigma$ onto $|K^n|$. By (1) the open n-simplices are disjoint from each other and from $|K^{n-1}|$, and Ψ_σ maps the open disk bijectively onto $\text{int}\sigma$, so this map identifies exactly each $x \in S^{n-1} \subseteq D^n_\sigma$ with $\Psi_\sigma(x) \in \partial\sigma \subseteq |K^{n-1}|$. The induced map from the quotient space of definition 1.2.1 to $|K^n|$ is therefore a continuous bijection from a compact space to a Hausdorff space, hence a homeomorphism ([16, making continuous function homeomorphisms]). So $|K^n|$ is obtained from $|K^{n-1}|$ by attaching the n-cells along the maps $\Psi_\sigma|_{S^{n-1}}$, and the construction stops at the dimension of K.

This completes the proof.

Definition 3.5.25: Barycentric Subdivision

Let K be a finite simplicial complex and write $b(\sigma)$ for the barycentre of $\sigma \in K$. The *barycentric subdivision* $\operatorname{sd} K$ is the set of the simplices $[b(\sigma_0), \dots, b(\sigma_k)]$ for all chains $\sigma_0 \subsetneq \sigma_1 \subsetneq \cdots \subsetneq \sigma_k$ of simplices of K. Its iterates are $\operatorname{sd}^0 K = K$ and $\operatorname{sd}^r K = \operatorname{sd}(\operatorname{sd}^{r-1} K)$.

Proposition 3.5.26: Properties of Barycentric Subdivision

Let K be a finite simplicial complex of dimension n, and let $\mu(K)$ be the largest diameter of a simplex of K.

1. $\operatorname{sd} K$ is a finite simplicial complex with $|\operatorname{sd} K| = |K|$.
2. Each simplex $[b(\sigma_0), \dots, b(\sigma_k)]$ of $\operatorname{sd} K$ lies in σ_k, and each simplex of K is the union of the simplices of $\operatorname{sd} K$ that it contains, all of dimension at most its own.
3. $\mu(\operatorname{sd} K) \leq \frac{n}{n+1}\mu(K)$.

Consequently, for every $r \geq 0$, $\operatorname{sd}^r K$ is a finite simplicial complex with $|\operatorname{sd}^r K| = |K|$ and $\mu(\operatorname{sd}^r K) \leq \left(\frac{n}{n+1}\right)^r \mu(K)$, each simplex of $\operatorname{sd}^r K$ lies in a simplex of K, and each simplex of K is a union of simplices of $\operatorname{sd}^r K$ of no larger dimension.

***Proof*:**
Write t_v for the barycentric coordinates of proposition 3.5.24(1) and $|\sigma|$ for the number of vertices of σ, so that $t_v(b(\sigma))$ is $1/|\sigma|$ for the vertices v of σ and 0 for the other vertices.

1. Let $\sigma_0 \subsetneq \cdots \subsetneq \sigma_k$ be a chain and choose a vertex v_j of σ_j not in σ_{j-1}, with $\sigma_{-1} = \emptyset$. Suppose $\sum_j \lambda_j b(\sigma_j) = 0$ with $\sum_j \lambda_j = 0$. Each t_v is affine on σ_k, so $\sum_j \lambda_j t_v(b(\sigma_j)) = 0$ for every vertex v of σ_k. For $v = v_k$ only the term $j = k$ is nonzero, which gives $\lambda_k = 0$, and descending in the same way through $v_{k-1}, \dots, v_0$ gives $\lambda_j = 0$ for every j. So the barycentres of a chain span a geometric simplex, lying in σ_k by convexity, whose faces are the simplices of the subchains.

   Each point lies in exactly one of these open simplices. Let $x \in |K|$ have carrier σ, let $s_1 > \cdots > s_m > 0$ be the distinct values of $t_v(x)$ at the vertices of σ, put $s_{m+1} = 0$, and let σ_j be the face of σ spanned by the vertices with $t_v(x) \geq s_{j+1}$, for $0 \leq j \leq m-1$. With $c_j = |\sigma_j|(s_{j+1} - s_{j+2}) > 0$, a vertex v with $t_v(x) = s_{i+1}$ satisfies $\sum_{j \geq i} c_j/|\sigma_j| = s_{i+1}$, so $x = \sum_j c_j b(\sigma_j)$ with $\sum_j c_j = \sum_v t_v(x) = 1$, and x lies in the open simplex of $[b(\sigma_0), \dots, b(\sigma_{m-1})]$. Conversely, if $x = \sum_j c_j b(\sigma_j)$ with all $c_j > 0$ and $\sum_j c_j = 1$ for a chain $\sigma_0 \subsetneq \cdots \subsetneq \sigma_k$, then $t_v(x) = \sum_{j \geq i} c_j/|\sigma_j|$ for the vertices v of σ_i not in σ_{i-1}, a value strictly decreasing in i, and $t_v(x) = 0$ off σ_k. So the members of the chain are the sets of vertices at which $t_v(x)$ is at least one of its values, and the c_j are determined by these values.

   A simplex of $\operatorname{sd} K$ is the disjoint union of the open simplices of its faces. So a point in two simplices τ, τ' of $\operatorname{sd} K$ lies in the open simplex of a common face, and $\tau \cap \tau'$ is the union of the open simplices of the common faces. These belong to the subchains of the chain of members common to the two chains, so $\tau \cap \tau'$ is empty or the simplex of that chain, a

common face. Hence $\operatorname{sd} K$ is a finite simplicial complex, with $|\operatorname{sd} K| = |K|$.

2. The first assertion was shown in (1). For $\sigma \in K$ and $x \in \sigma$, the carrier of x is a face of σ, so the chain constructed in (1) consists of faces of σ, and the simplex of $\operatorname{sd} K$ whose open simplex contains x lies in σ. Its dimension is at most $\dim \sigma$, a chain of faces of σ having at most $\dim \sigma + 1$ members.

3. The simplex $[b(\sigma_0), \ldots, b(\sigma_k)]$ is the convex hull of the barycentres of a chain of faces of σ_k, so by lemma 3.4.10 its diameter is at most $\frac{m}{m+1} \operatorname{diam} \sigma_k \leq \frac{n}{n+1}\mu(K)$, where $m = \dim \sigma_k \leq n$.

The final assertions follow by induction on r, since $\operatorname{sd}^r K$ has dimension at most n by (2), completing the proof.

Definition 3.5.27: Stars

Let K be a finite simplicial complex. For a vertex v of K, the *open star* $\operatorname{st} v$ is the union of the open simplices $\operatorname{int}\tau$ over the simplices $\tau \in K$ having v as a vertex. For a simplex $\rho \in K$, the *closed star* $\operatorname{St} \rho$ is the union of the simplices of K having ρ as a face.

Lemma 3.5.28: Open Stars

Let K be a finite simplicial complex.

1. Each open star $\operatorname{st} v$ is open in $|K|$, and every point of $|K|$ lies in some open star.

2. For vertices $v_0, \ldots, v_k$ of K, the intersection $\operatorname{st} v_0 \cap \cdots \cap \operatorname{st} v_k$ is nonempty only if $v_0, \ldots, v_k$ are vertices of a common simplex of K, and it is the union of the open simplices $\operatorname{int}\tau$ over the simplices τ having all of $v_0, \ldots, v_k$ as vertices.

3. If every simplex of K has diameter less than δ, then every closed star has diameter less than 2δ, and $\operatorname{st} v \subseteq \operatorname{St} v$ for every vertex v.

***Proof*:**
By proposition 3.5.24(1), a point x lies in $\operatorname{st} v$ exactly when v is a vertex of the carrier of x.

1. A point outside $\operatorname{st} v$ lies in its carrier, a simplex without the vertex v. Conversely, a point of a simplex τ without the vertex v has as carrier a face of τ, so it is not in $\operatorname{st} v$. Hence $|K| - \operatorname{st} v$ is the union of the finitely many simplices not having v as a vertex, a closed set. Every point lies in the open star of each vertex of its carrier.

2. A point lies in every $\operatorname{st} v_i$ exactly when every v_i is a vertex of its carrier, and then the v_i are vertices of a common simplex.

3. Points $p \in \tau_1$ and $q \in \tau_2$ of simplices having ρ as a face satisfy $|p-q| \leq |p-c| + |c-q| < 2\delta$ for any $c \in \rho$. The inclusion holds since $\operatorname{int}\tau \subseteq \tau$.

This completes the proof.

Definition 3.5.29: Simplicial Map

Let K and L be finite simplicial complexes. A *simplicial map* $g\colon K \to L$ is a map from the vertices of K to the vertices of L such that the images of the vertices of each simplex of K are vertices of a single simplex of L, together with its extension $g\colon |K| \to |L|$ given by $g(x) = \sum_v t_v(x)\, g(v)$. For $\sigma \in K$ the image $g(\sigma)$ is the simplex of L spanned by the images of the vertices of σ, of dimension at most $\dim \sigma$.

The extension is continuous, each t_v being continuous by proposition 3.5.24(1), and it takes values in $|L|$: for x in a simplex σ, the point $g(x)$ is a convex combination of the images of the vertices of σ, which lie in a single simplex of L.

Theorem 3.5.30: Simplicial Approximation

Let K and L be finite simplicial complexes and $h\colon |K| \to |L|$ continuous. Then for every $\delta > 0$ there are $r \geq 0$ and a simplicial map $g\colon \mathrm{sd}^r K \to L$ such that every simplex of $\mathrm{sd}^r K$ has diameter less than δ and

1. for every simplex σ of $\mathrm{sd}^r K$ and every $x \in \mathrm{int}\sigma$, the point $h(x)$ lies in a simplex of L having $g(\sigma)$ as a face, and in particular $h(x)$ and $g(x)$ lie in a common simplex of L
2. the straight-line homotopy $(x,t) \mapsto (1-t)h(x) + tg(x)$ is a homotopy from h to g through maps $|K| \to |L|$

***Proof*:**

1. The open sets $h^{-1}(\mathrm{st}\, w)$, for the vertices w of L, cover $|K|$ by lemma 3.5.28(1), and $|K|$ is a compact metric space, so [16, Lebesgue Number Lemma] gives $\varepsilon > 0$ such that every subset of $|K|$ of diameter less than ε lies in one of them. By proposition 3.5.26 choose r such that every simplex of $K' = \mathrm{sd}^r K$ has diameter less than $\min(\delta, \varepsilon/2)$. For a vertex v of K' the closed star $\mathrm{St}\, v$ then has diameter less than ε by lemma 3.5.28(3), so $h(\mathrm{St}\, v) \subseteq \mathrm{st}\, w$ for some vertex w of L, and we choose such a vertex as $g(v)$. Since $\mathrm{st}\, v \subseteq \mathrm{St}\, v$, this gives $h(\mathrm{st}\, v) \subseteq \mathrm{st}\, g(v)$. Let $\sigma \in K'$ have vertices $v_0, \ldots, v_k$, and let $x \in \mathrm{int}\sigma$. Then $x \in \mathrm{st}\, v_i$ for every i, so $h(x) \in \mathrm{st}\, g(v_0) \cap \cdots \cap \mathrm{st}\, g(v_k)$. By lemma 3.5.28(2) the $g(v_i)$ are vertices of a common simplex of L, so g is a simplicial map, and $h(x)$ lies in $\mathrm{int}\rho$ for a simplex ρ having every $g(v_i)$ as a vertex, that is, having $g(\sigma)$ as a face. Since $g(x) \in g(\sigma) \subseteq \rho$, the points $h(x)$ and $g(x)$ lie in ρ, and every point of $|K|$ lies in some $\mathrm{int}\sigma$ by proposition 3.5.24(1).
2. With ρ as in (1) for the carrier of x, convexity gives $(1-t)h(x) + tg(x) \in \rho \subseteq |L|$. The formula is continuous as a map $|K| \times I \to \mathbb{R}^M$, where $|L| \subseteq \mathbb{R}^M$ carries the subspace topology, so it is a homotopy from h to g in $|L|$.

This completes the proof.

3.5.5 Lefschetz Fixed Point Theorem

The trace of an endomorphism of a free abelian group of finite rank is the trace of its matrix in a basis, and it does not depend on the basis, since a change of basis conjugates the matrix. An endomorphism φ of a finitely generated abelian group A carries the torsion subgroup T into itself, elements of finite order going to elements of finite order, and A/T is free of finite rank by [10, Invariant Factor Form of Modules over PID's]. Write $\mathrm{tr}(\varphi, A/T)$ for the trace of the endomorphism of A/T induced by φ.

Theorem 3.5.31: Hopf Trace Theorem

Let $C_\bullet$ be a chain complex of free abelian groups of finite rank with $C_n = 0$ for $n < 0$ and for $n > N$, such as the cellular chain complex of a finite CW complex of dimension N (lemma 3.5.9(1)), and let $\varphi\colon C_\bullet \to C_\bullet$ be a chain map with components $\varphi_n\colon C_n \to C_n$. Write H_n for the homology of $C_\bullet$, T_n for its torsion subgroup, and φ_* for the endomorphism of H_n induced by φ. Then

$$\sum_{n=0}^{N}(-1)^n \mathrm{tr}\,(\varphi_n, C_n) = \sum_{n=0}^{N}(-1)^n \mathrm{tr}\,(\varphi_*, H_n/T_n)$$

***Proof*:**
Write $Z_n = \ker \partial_n$ and $B_n = \mathrm{im}\partial_{n+1}$, so that $H_n = Z_n/B_n$. As a subgroup of the free group C_n of finite rank, Z_n is free of finite rank ([10, Submodule Of Free Module Over PID]), so H_n is finitely generated and the traces in the statement are defined.

Step 1: Traces over $\mathbb{Z}$ and over $\mathbb{Q}$. Let F be free abelian with basis $e_1, \dots, e_r$ and let $g\colon F \to F$ have matrix M in this basis. Then $F \otimes_{\mathbb{Z}} \mathbb{Q}$ is a $\mathbb{Q}$-vector space with basis $e_1 \otimes 1, \dots, e_r \otimes 1$, by [10, Tensor Products of Direct Sums] and $\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q} = \mathbb{Q}$, and $g \otimes 1$ has the same matrix M in it. So $\mathrm{tr}(g) = \mathrm{tr}(g \otimes 1)$.

Step 2: Additivity of trace. Let $0 \to U \xrightarrow{a} V \xrightarrow{b} W \to 0$ be a short exact sequence of finite-dimensional $\mathbb{Q}$-vector spaces with endomorphisms f_U, f_V, f_W satisfying $f_V a = a f_U$ and $b f_V = f_W b$. Choose a basis $u_1, \dots, u_p$ of U, a basis $w_1, \dots, w_q$ of W, and vectors $v_j \in V$ with $b(v_j) = w_j$. These give a basis $a(u_1), \dots, a(u_p), v_1, \dots, v_q$ of V: for $v \in V$, writing $b(v) = \sum_j c_j w_j$ puts $v - \sum_j c_j v_j$ in $\ker b = \mathrm{im} a$, and a vanishing combination $\sum_i \lambda_i a(u_i) + \sum_j c_j v_j = 0$ has all $c_j = 0$ after applying b and then all $\lambda_i = 0$ by injectivity of a. If $f_U(u_i) = \sum_{i'} m_{i'i} u_{i'}$ and $f_W(w_j) = \sum_{j'} n_{j'j} w_{j'}$, then $f_V(a(u_i)) = a(f_U u_i) = \sum_{i'} m_{i'i} a(u_{i'})$, and $b\big(f_V v_j - \sum_{j'} n_{j'j} v_{j'}\big) = f_W(w_j) - \sum_{j'} n_{j'j} w_{j'} = 0$, so $f_V v_j$ is $\sum_{j'} n_{j'j} v_{j'}$ plus an element of $\mathrm{im} a$. In this basis the matrix of f_V is block upper triangular with diagonal blocks $(m_{i'i})$ and $(n_{j'j})$, so $\mathrm{tr} f_V = \mathrm{tr} f_U + \mathrm{tr} f_W$.

Step 3: Tensoring with $\mathbb{Q}$. The group $\mathbb{Q}$ is torsion-free, hence flat over the principal ideal domain $\mathbb{Z}$ ([10, Flat Equals Torsion-Free Over a PID]), so $- \otimes_{\mathbb{Z}} \mathbb{Q}$ carries short exact sequences of abelian groups to short exact sequences of $\mathbb{Q}$-vector spaces. Since φ is a chain map, $\varphi_n(Z_n) \subseteq Z_n$ and $\varphi_n(B_n) \subseteq B_n$, so φ induces endomorphisms of Z_n, B_n and H_n, and of T_n and H_n/T_n, a homomorphism carrying elements of finite order to elements of finite order. All maps in the

short exact sequences

$$0 \to Z_n \to C_n \xrightarrow{\partial_n} B_{n-1} \to 0 \qquad 0 \to B_n \to Z_n \to H_n \to 0 \qquad 0 \to T_n \to H_n \to H_n/T_n \to 0$$

commute with these endomorphisms, using $\partial_n \varphi_n = \varphi_{n-1}\partial_n$ for the first. After tensoring with $\mathbb{Q}$ they are short exact sequences of finite-dimensional vector spaces, each term being obtained from the finite-dimensional spaces $C_n \otimes_{\mathbb{Z}} \mathbb{Q}$ and $C_{n-1} \otimes_{\mathbb{Z}} \mathbb{Q}$ by passing to subspaces and quotients, and Step 2 applies to each. Moreover $T_n \otimes_{\mathbb{Z}} \mathbb{Q} = 0$, since $t \otimes q = st \otimes (q/s) = 0$ for $t \in T_n$ of order s. Writing $\mathrm{tr}_{\mathbb{Q}}(Y)$ for the trace of the endomorphism of $Y \otimes_{\mathbb{Z}} \mathbb{Q}$ induced by φ, Step 2 gives

$$\mathrm{tr}_{\mathbb{Q}}(C_n) = \mathrm{tr}_{\mathbb{Q}}(Z_n) + \mathrm{tr}_{\mathbb{Q}}(B_{n-1}) \qquad \mathrm{tr}_{\mathbb{Q}}(Z_n) = \mathrm{tr}_{\mathbb{Q}}(B_n) + \mathrm{tr}_{\mathbb{Q}}(H_n) \qquad \mathrm{tr}_{\mathbb{Q}}(H_n) = \mathrm{tr}_{\mathbb{Q}}(H_n/T_n)$$

Step 4: Summation. Substituting, $\mathrm{tr}_{\mathbb{Q}}(C_n) = \mathrm{tr}_{\mathbb{Q}}(B_n) + \mathrm{tr}_{\mathbb{Q}}(H_n/T_n) + \mathrm{tr}_{\mathbb{Q}}(B_{n-1})$. Multiply by $(-1)^n$ and sum over $0 \le n \le N$. Reindexing by $m = n - 1$ turns $\sum_{n=0}^{N}(-1)^n \mathrm{tr}_{\mathbb{Q}}(B_{n-1})$ into $-\sum_{m=-1}^{N-1}(-1)^m \mathrm{tr}_{\mathbb{Q}}(B_m)$, so the boundary terms contribute

$$\sum_{n=0}^{N}(-1)^n \mathrm{tr}_{\mathbb{Q}}(B_n) - \sum_{m=-1}^{N-1}(-1)^m \mathrm{tr}_{\mathbb{Q}}(B_m) = (-1)^N \mathrm{tr}_{\mathbb{Q}}(B_N) + \mathrm{tr}_{\mathbb{Q}}(B_{-1})$$

and both remaining terms vanish because $B_N = 0$ and $B_{-1} = 0$, as $C_{N+1} = 0$ and $C_{-1} = 0$. This leaves

$$\sum_{n=0}^{N}(-1)^n \mathrm{tr}_{\mathbb{Q}}(C_n) = \sum_{n=0}^{N}(-1)^n \mathrm{tr}_{\mathbb{Q}}(H_n/T_n)$$

The group H_n/T_n is finitely generated and torsion-free, hence free by [10, Invariant Factor Form of Modules over PID's] with $R = \mathbb{Z}$, so Step 1 applies to it and to C_n, and identifies the rational traces with the traces of the statement, as we sought to show.

For the identity chain map of the cellular chain complex of a finite CW complex X, the left side of the Hopf trace formula is $\sum_n (-1)^n c_n$, where c_n is the number of n-cells, the n-th chain group being free on the n-cells (lemma 3.5.9(1)), and the right side is $\sum_n (-1)^n \operatorname{rank} H_n(X)$ by theorem 3.5.11, the identity of a free group of finite rank having trace equal to the rank. This recovers theorem 3.5.22. For example, a triangle divided into two triangles by the segment from one vertex to the midpoint of the opposite side is a finite simplicial complex with 4 vertices, 5 edges and 2 triangles, hence by proposition 3.5.24 a finite CW complex, and $\chi = 4 - 5 + 2 = 1$, in agreement with the homology of this contractible space (corollary 3.3.9).

Let K be a finite simplicial complex. By proposition 3.5.24, $|K|$ is a finite CW complex, so by lemma 3.5.9(1) and theorem 3.5.11 each singular homology group $H_i(|K|)$ is a subquotient of a free abelian group of finite rank, hence finitely generated, and it vanishes for $i > \dim K$.

Definition 3.5.32: Lefschetz Number

Let K be a finite simplicial complex and $h\colon |K| \to |K|$ a continuous map. The *Lefschetz number* of h is

$$\Lambda(h) = \sum_{i=0}^{\dim K} (-1)^i \operatorname{tr}\big(h_*, H_i(|K|)/T_i\big)$$

where h_* is the map induced on singular homology and T_i is the torsion subgroup of $H_i(|K|)$.

Theorem 3.5.33: Lefschetz Fixed Point Theorem

Let K be a finite simplicial complex and $h\colon |K| \to |K|$ continuous. If $\Lambda(h) \neq 0$, then h has a fixed point.

***Proof*:**
Suppose h has no fixed point. The continuous function $x \mapsto |h(x) - x|$ is positive on the compact space $|K|$, so it has a positive minimum ε.

Step 1: A simplicial approximation that moves every simplex off itself. By proposition 3.5.26 choose s such that every simplex of $L = \mathrm{sd}^s K$ has diameter less than $\varepsilon/4$, so that every closed star of L has diameter less than $\varepsilon/2$ by lemma 3.5.28(3). Apply theorem 3.5.30 to $h\colon |L| = |K| \to |L|$. It gives r and a simplicial map $g\colon K' \to L$, where $K' = \mathrm{sd}^r L$, homotopic to h and satisfying theorem 3.5.30(1). Let $\sigma \in K'$, suppose some point y lies in $g(\sigma) \cap \sigma$, and choose $x \in \mathrm{int}\sigma$. Then $h(x)$ lies in a simplex of L having $g(\sigma)$ as a face, so $h(x)$ and y both lie in the closed star $\mathrm{St}\, g(\sigma)$, and $|h(x) - y| < \varepsilon/2$. Moreover σ lies in a simplex of L by proposition 3.5.26, so $|x - y| < \varepsilon/4$. Hence $|h(x) - x| < \varepsilon$, contradicting the choice of ε. So $g(\sigma) \cap \sigma = \emptyset$ for every simplex σ of K'.

Step 2: Reduction to a trace on cellular chains. Give $|K| = |K'|$ the CW structure of proposition 3.5.24 for K', with characteristic maps $\Psi_\tau\colon (D^n, S^{n-1}) \to (\tau, \partial\tau)$ for the n-simplices τ of K'. The simplicial map g carries each simplex onto a simplex of L of no larger dimension, and each simplex of L of dimension at most n is a union of simplices of K' of dimension at most n (proposition 3.5.26), so $g(|K'^n|) \subseteq |L^n| \subseteq |K'^n|$ for every n, and g is a cellular map of $|K'|$ to itself. By theorem 3.5.11, the maps g_* on the cellular chain groups $C_n = H_n(|K'^n|, |K'^{n-1}|)$ form a chain map, and the isomorphisms between the homology of $C_\bullet$ and $H_n(|K|)$ carry the endomorphisms induced by this chain map to the maps g_* on singular homology. An isomorphism carries torsion onto torsion, so it induces an isomorphism of the free quotients conjugating the two endomorphisms, which therefore have the same trace. The groups C_n are free of finite rank by lemma 3.5.9(1) and vanish for $n < 0$ and $n > \dim K$, so theorem 3.5.31 gives

$$\Lambda(g) = \sum_n (-1)^n \operatorname{tr}\big(g_*, H_n(|K|)/T_n\big) = \sum_n (-1)^n \operatorname{tr}(g_*, C_n)$$

Step 3: The traces on chains vanish. By lemma 3.5.9(1), C_n is free with basis the classes $e_\tau = (\Psi_\tau)_*[D^n]$ over the n-simplices τ of K'. Fix an n-simplex σ and let K'_σ be the set of simplices of K' of dimension at most n other than σ, a finite simplicial complex with $(n-1)$-skeleton K'^{n-1} and polyhedron $|K'_\sigma| = |K'^n| - \mathrm{int}\sigma$. By proposition 3.5.24 and lemma 3.5.9(1), applied to K'_σ with the same maps Ψ_τ, the group $H_n(|K'_\sigma|, |K'^{n-1}|)$ is free with basis the classes

$(\Psi_\tau)_*[D^n]$ for $\tau \neq \sigma$, and the inclusion of pairs carries them to the e_τ with $\tau \neq \sigma$. By Steps 1 and 2, $g(\sigma)$ lies in $|L^n|$ and misses $\mathrm{int}\sigma$, so $g(\sigma) \subseteq |K'_\sigma|$, and $g(\partial\sigma) \subseteq |K'^{n-1}|$. So $g \circ \Psi_\sigma$ is a map of pairs $(D^n, S^{n-1}) \to (|K'_\sigma|, |K'^{n-1}|)$ followed by the inclusion of pairs, and $g_*(e_\sigma) = (g\circ\Psi_\sigma)_*[D^n]$ lies in the span of the e_τ with $\tau \neq \sigma$. Its coefficient on e_σ is zero, so the matrix of g_* on C_n in the basis (e_τ) has zero diagonal, and $\mathrm{tr}(g_*, C_n) = 0$ for every n.

Step 4: Conclusion. By Steps 2 and 3, $\Lambda(g) = 0$. Since $h \simeq g$, theorem 3.3.6 gives $h_* = g_*$ on singular homology, so $\Lambda(h) = \Lambda(g) = 0$. Hence $\Lambda(h) \neq 0$ forces a fixed point, as we sought to show.

Corollary 3.5.34: Fixed Points of Maps of Acyclic Complexes

Let K be a finite simplicial complex such that $|K|$ is acyclic. Then every continuous map $h\colon |K| \to |K|$ has a fixed point.

***Proof*:**

Since $|K|$ is acyclic, $H_i(|K|) = \tilde{H}_i(|K|) = 0$ for $i \geq 1$, and by proposition 3.5.17 $|K|$ is nonempty and path-connected with the augmentation an isomorphism $H_0(|K|) \cong \mathbb{Z}$, so $H_0(|K|)$ is generated by the class of any point. For a point x, the map h_* sends the class of the 0-simplex x to the class of the 0-simplex $h(x)$, and a path from x to $h(x)$, read as a singular 1-simplex τ, has $\partial\tau = h(x) - x$, so the two classes agree. Hence h_* is the identity of $H_0(|K|) \cong \mathbb{Z}$, of trace 1, the groups $H_i(|K|)/T_i$ vanish for $i \geq 1$, and $\Lambda(h) = 1 \neq 0$. Theorem 3.5.33 then gives a fixed point, as we sought to show.

Example 3.5.35: Lefschetz Numbers of Spheres and of the Projective Plane

The theorem applies to every space X with a homeomorphism $\varphi\colon X \to |K|$ for a finite simplicial complex K, with $\Lambda(h)$ defined for $h\colon X \to X$ by the same formula, because φ matches the fixed points of h with those of $\varphi h \varphi^{-1}$, and $(\varphi h \varphi^{-1})_* = \varphi_* h_* \varphi_*^{-1}$ has the same traces as h_*.

1. Let $n \geq 1$. The proper faces of Δ^{n+1} form a finite simplicial complex with polyhedron $\partial\Delta^{n+1}$, which is homeomorphic to S^n by lemma 3.1.5. For a map $f\colon S^n \to S^n$, the induced map on $H_0(S^n) \cong \mathbb{Z}$ is the identity, by the argument in the proof of corollary 3.5.34, on $H_n(S^n) \cong \mathbb{Z}$ it is multiplication by $\deg f$, and the other homology groups vanish (corollary 3.4.15). Hence $\Lambda(f) = 1 + (-1)^n \deg f$, and the theorem says that f has a fixed point unless $\deg f = (-1)^{n+1}$, which is proposition 3.5.2(7) again. The antipodal map has degree $(-1)^{n+1}$ by proposition 3.5.2(6), so its Lefschetz number is 0, as it must be for a map without fixed points.

2. Torsion does not enter Λ. If $|K|$ is path-connected and every $H_i(|K|)$ with $i \geq 1$ is finite, then $\Lambda(h) = 1$ for every map h, by the computation in the proof of corollary 3.5.34, so every map $|K| \to |K|$ has a fixed point. The real projective plane is homeomorphic to $|K|$ for the simplicial complex K whose six vertices $0, \ldots, 5$ are the standard basis vectors of $\mathbb{R}^6$ and whose triangles are 013, 032, 143, 345, 235, 251, 124, 042, 054 and 015, with all their faces. This complex is the image of the icosahedral triangulation of S^2 under the antipodal quotient [4, Chapter IV, Section 21, Figure IV-16, p. 248, with the ten triangles listed on p. 249]. Its homology is $\mathbb{Z}$, $\mathbb{Z}/2$ and 0 in degrees 0, 1 and 2 (example 3.5.15), so every map $\mathbb{R}\mathrm{P}^2 \to \mathbb{R}\mathrm{P}^2$ has a fixed point.

3.5.6 Homology with Coefficients

Definition 3.5.36: Chains with Coefficients

Let G be an abelian group and X a space. The group $C_n(X;G)$ of *singular n-chains with coefficients in G* consists of the finite sums $\sum_i g_i\sigma_i$ with $g_i \in G$ and each $\sigma_i\colon \Delta^n \to X$ a singular n-simplex, that is, the direct sum of copies of G indexed by the singular n-simplices of X. Its boundary map is

$$\partial\Big(\sum_i g_i\sigma_i\Big) = \sum_i \sum_{j=0}^{n} (-1)^j g_i\, \sigma_i|_{[v_0,\ldots,\hat{v}_j,\ldots,v_n]}$$

For $A \subseteq X$ the subgroup $C_n(A;G)$ consists of the chains whose simplices have image in A, and $C_n(X,A;G) = C_n(X;G)/C_n(A;G)$. The homology groups $H_n(X;G)$ and $H_n(X,A;G)$ of these complexes are the *homology groups with coefficients in G*. The reduced groups $\tilde{H}_n(X;G)$ are those of the complex augmented by $\epsilon\colon C_0(X;G) \to G$, $\epsilon(\sum_i g_i\sigma_i) = \sum_i g_i$.

The computation of $\partial\partial = 0$ for integer chains applies with the coefficients g_i carried along, so these are chain complexes, and for $G = \mathbb{Z}$ they are the singular chain complexes of definition 3.2.1. A map $f\colon X \to Y$ induces the chain map $f_\sharp(\sum_i g_i\sigma_i) = \sum_i g_i\,(f\circ\sigma_i)$, carrying $C_\bullet(A;G)$ into $C_\bullet(B;G)$ when $f(A) \subseteq B$, and hence homomorphisms f_* on these groups. The results of this chapter on singular homology, among them homotopy invariance (theorem 3.3.6), the long exact sequence of a pair (theorem 3.4.3), excision (theorem 3.4.9) and the Mayer-Vietoris sequences (theorem 3.4.22, proposition 3.4.24), hold with coefficients in G, by the same proofs with the coefficients carried along [21, Section 2.2, p. 153]. When X is a point there is a single singular simplex σ_n in each degree, and $\partial\sigma_n = \sum_{j=0}^{n}(-1)^j\sigma_{n-1}$ is σ_{n-1} for even $n \geq 2$ and 0 for odd n. The complex is $\cdots \to G \xrightarrow{\mathrm{id}} G \xrightarrow{0} G \xrightarrow{\mathrm{id}} G \xrightarrow{0} G \to 0$, so

$$H_n(\mathrm{pt};G) \cong \begin{cases} G & n = 0 \\ 0 & n \geq 1 \end{cases}$$

Lemma 3.5.37: Degree For Abelian Groups

Let $k \geq 1$, let G be an abelian group and let $f : S^k \to S^k$ be of degree m. Then $f_* : H_k(S^k;G) \to H_k(S^k;G)$ is multiplication by m.

***Proof*:**
Since $C_n(X)$ is free on the singular n-simplices, sending $\sum_i g_i\sigma_i$ to $\sum_i \sigma_i \otimes g_i$ identifies $C_n(X;G)$ with $C_n(X)\otimes_{\mathbb{Z}} G$ ([10, Arbitrary Direct Sums Commute with Tensor Products], with $\mathbb{Z}\otimes_{\mathbb{Z}} G = G$). Under this identification the boundary of definition 3.5.36 is $\partial\otimes 1$ and the map induced by f is $f_\sharp \otimes 1$. Write $C_n = C_n(S^k)$, with cycles Z_n and boundaries B_n, and fix a cycle $z \in Z_k$ whose class generates $H_k(S^k) \cong \mathbb{Z}$ (corollary 3.4.15, reduced and unreduced groups agreeing in degree $k \geq 1$).

Step 1: Splittings. Subgroups of free abelian groups are free ([10, Submodule Of Free Module Over PID, General Case]), so B_{n-1} is free, and choosing a preimage of each of its basis elements

([10, Universal Property of Free Modules]) gives a section of the surjection $\partial\colon C_n \to B_{n-1}$. By lemma 3.4.34 there is a subgroup $D_n \subseteq C_n$, the image of that section, with $C_n = Z_n \oplus D_n$ and with ∂ restricting to an isomorphism $D_n \to B_{n-1}$. Moreover $H_{k-1}(S^k)$ is free, being $\mathbb{Z}$ for $k = 1$ and 0 for $k \geq 2$, so the surjection $Z_{k-1} \to H_{k-1}(S^k)$ has a section by the same universal property, and lemma 3.4.34 gives $Z_{k-1} = B_{k-1} \oplus E$ for a subgroup E. Together with $C_{k-1} = Z_{k-1} \oplus D_{k-1}$, this makes B_{k-1} a direct summand of C_{k-1}.

Step 2: Every class is represented by some $z \otimes g$. Tensor products commute with finite direct sums ([10, Tensor Products of Direct Sums]), so $C_k \otimes G = (Z_k \otimes G) \oplus (D_k \otimes G)$, and the inclusion $B_{k-1} \otimes G \to C_{k-1} \otimes G$ of a direct summand is injective. Let $x \in C_k \otimes G$ be a cycle and write $x = x_1 + x_2$ with $x_1 \in Z_k \otimes G$ and $x_2 \in D_k \otimes G$. Since ∂ vanishes on Z_k, $0 = (\partial \otimes 1)(x) = (\partial \otimes 1)(x_2)$, and $\partial \otimes 1$ maps $D_k \otimes G$ isomorphically onto $B_{k-1} \otimes G$, which injects into $C_{k-1} \otimes G$. Hence $x_2 = 0$, and $x = x_1 = \sum_j z_j \otimes g_j$ for finitely many $z_j \in Z_k$ and $g_j \in G$. Since $[z]$ generates $H_k(S^k)$, each $z_j = n_j z + \partial w_j$ with $n_j \in \mathbb{Z}$ and $w_j \in C_{k+1}$, so

$$x = z \otimes g + (\partial \otimes 1)\Big(\sum_j w_j \otimes g_j\Big) \qquad g = \sum_j n_j g_j$$

and $[x] = [z \otimes g]$ in $H_k(S^k; G)$.

Step 3: The action of f_*. By definition 3.5.1, $f_*[z] = m[z]$, so $f_\sharp z = mz + \partial w$ for some $w \in C_{k+1}$. For $g \in G$,

$$(f_\sharp \otimes 1)(z \otimes g) = z \otimes mg + (\partial \otimes 1)(w \otimes g)$$

so $f_*[z \otimes g] = m[z \otimes g]$. By Step 2 every class in $H_k(S^k; G)$ has the form $[z \otimes g]$, hence f_* is multiplication by m, as we sought to show.

Proposition 3.5.38: Cellular Homology with Coefficients

Let X be a CW complex and G an abelian group. Then $H_n(X; G)$ is the homology of the cellular chain complex whose n-th group is $H_n(X^n, X^{n-1}; G)$, a direct sum of copies of G, one for each n-cell e^n_α, with boundary

$$d_n\Big(\sum_\alpha g_\alpha e^n_\alpha\Big) = \sum_{\alpha,\beta} \deg(\partial_{\alpha\beta})\, g_\alpha e^{n-1}_\beta$$

for the degrees $\deg(\partial_{\alpha\beta})$ of proposition 3.5.14.

***Proof Key ideas*:**
The proofs of lemma 3.5.9, theorem 3.5.11 and proposition 3.5.14 use homotopy invariance, the long exact sequences of pairs, excision and the homology of spheres, which hold with coefficients in G as recorded after definition 3.5.36. They go through with the coefficients carried along [21, Section 2.2, the discussion before Lemma 2.49]. For $n \geq 2$ the new input is lemma 3.5.37, by which a map of spheres of degree m acts on $H_{n-1}(S^{n-1}; G)$ as multiplication by m. For $n = 1$, where that lemma does not apply, $d_1(g\, e^1_\alpha)$ is g times the difference of the two endpoint 0-cells of e^1_α, which is the displayed formula with the degrees read on $\tilde{H}_0(S^0)$ as in proposition 3.5.14.

Example 3.5.39: Homology With Coefficients

Take $X = \mathbb{R}\mathrm{P}^n$ with its CW structure of one cell e^k in each dimension $0 \leq k \leq n$, and $G = \mathbb{Z}/2$. By example 3.5.15, the integral cellular boundary d_k is multiplication by the degree $1 + (-1)^k$, which is 2 for even k and 0 for odd k. By proposition 3.5.38, with coefficients in $\mathbb{Z}/2$ each cellular chain group is $\mathbb{Z}/2$, and d_k is multiplication by $1 + (-1)^k \equiv 0 \pmod 2$, so every boundary map vanishes and

$$H_k(\mathbb{R}\mathrm{P}^n; \mathbb{Z}/2) \cong \begin{cases} \mathbb{Z}/2 & 0 \leq k \leq n \\ 0 & \text{otherwise} \end{cases}$$

With integer coefficients and $n \geq 2$, $H_2(\mathbb{R}\mathrm{P}^n) = 0$ (example 3.5.15), so the two coefficient groups give different answers.

3.6 Summary

The singular, simplicial, and cellular homologies of this chapter are three models of one theory, determined by the *Eilenberg-Steenrod axioms*: a homology theory is a sequence of functors $H_n(-)$ on pairs of spaces with natural boundary maps, satisfying homotopy invariance, exactness, excision, additivity, and the *dimension axiom* fixing the homology of a point. Any two theories satisfying these axioms agree on all CW complexes, which is what turns the coincidence of the models into a theorem. Relaxing the dimension axiom enlarges the notion to a *generalized* (or extraordinary) homology theory, with K-theory and cobordism as the prototypes. These axioms, the uniqueness theorem characterizing ordinary theories, and the generalized theories obtained by dropping the dimension axiom are developed in full in section 8.2.

4

Cohomology

We shall now develop a dual theory to homological algebra focusing on the functionals on the simplices for the favourable algebraic advantages. A homology was defined given a complex $C_\bullet$. We always have access to the contravariant $\mathrm{Hom}(C_\bullet, G)$ functor that we may wish to apply to study some properties of $C_\bullet$. Often the dual category does not bring us much direct new information.

The main thing that cohomology gives us is a new product called the *cup product*. This product is technically present in homology, but is not natural and is computationally much harder to find: the natural function $X \to X \times X$ given by the diagonal map $x \mapsto (x, x)$ exists, but for a general topological space X, there is no natural map $X \times X \to X$. This shall introduce a new algebraic structure, a graded-commutative R-algebra structure on $H^\bullet(X, R)$ (given R is a commutative ring) which shall give us more insights on our spaces.

4.1 Cohomology of Spaces

To intuitively understand how to think of $\mathrm{Hom}(C^\bullet, G)$, let X be a 1-dimensional Δ-complex, i.e. a graph. Given a fixed abelian group G (or really any abelian object), the set $\mathrm{Hom}_{\mathsf{Set}}(X^0, G)$ of functions on the vertex set X^0 forms an abelian group, which we denote $\Delta^0(X; G)$ (if $G = \mathbb{Z}$ and X has finitely many vertices, this group is [not necessarily naturally] isomorphic to $\Delta_0(X)$). We may do the same for edges defining $\Delta^1(X; G)$ and more generally $\Delta^n(X; G)$. Given these function groups, we shall define a function δ between them.

Starting by understanding $\Delta^i(X; G)$ where X is a simplicial complex[1]. Then $\varphi \in \Delta^0(X; G)$ is a function from the vertices of X to G, and $\psi \in \Delta^1(X; G)$ is a function from the edges of X to G, and similarly for higher dimensions. Consider the map:

[1] just like for homology, the singular and simplicial will be shown to have the same homology, and hence we use simplicial complexes for simpler visualizations

$$\delta : \Delta^0(X;G) \to \Delta^1(X;G) \tag{4.1}$$

where φ maps to $\delta(\varphi)$ where if $e = [v_0, v_1]$ is and:

$$\delta(\varphi)(e) = \varphi(v_1) - \varphi(v_0)$$

Note how the orientability was also important to have a natural choice of which vertex to take the difference of. If we take (4.1) and embedded into a chain complex with zero's before and after it:

$$\cdots \to 0 \to \Delta^0(X;G) \xrightarrow{\delta} \Delta^1(X;G) \to 0 \to \cdots$$

Then

$$H^0(X;G) = \ker \delta \qquad \text{and} \qquad H^1(X;G) = \Delta^1(X;G)/\text{im}\delta$$

Now, $H^0(X;G)$ describes maps that have the same value on the two vertices of an edge, that is

$$\delta(\varphi) = 0 \qquad \text{iff} \qquad \varphi(v_i) = \varphi(v_j),\ e = [v_i, v_j]$$

that is φ is *constant* on each component of X (so φ is "locally constant" with respect to edges). Hence, $H^0(X;G)$ is all functions from the sets of component of X to G (similarly to how $H_0(X)$ counts the number of path components). Notice how this is similar to how $\mathrm{d}\, f = 0$ if f is constant.

Next, take $\overline{\psi} \in H^1(X;G)$ where $\psi : X \to G$ is a function where $\psi(e) \in G$ for each edge. Consider the case where $\overline{\psi} = 0$ so that $\psi \in \text{im}\delta$. If $\psi = \delta(\varphi)$ for some $\varphi \in \Delta^0(X;G)$, then ψ is purely determined by it's values at it's "extrema", in this case the value of ψ is determine by the change of value between each each corresponding to a function φ. For future reference, notice how this is similar to finding a function F given a change-in-value function f (usually the derivative). This connection is made explicit by *de Rham cohomology* (see [32]). This parallel goes even further. If there exists a φ such that $\delta\varphi = \psi$, then it is unique up to adding an element of the kernel of δ.

When X is a tree, this equation is always solvable. In this case we have $H^1(X;G) = 0$. If X is not a tree (hence it has cycles), then the equation may fail (ex. consider the each edge in a closed loop having difference 1). To measure to what degree this fails, we first pick a maximal tree. Then a ψ can be determined by a φ on this maximal tree, however the value of ψ on each edge not in the maximal tree must equal the different of already-determined values of φ at two ends of edges, which may not work since ψ can have, in general, arbitrary values on edges. Thus, $H^1(X;G)$ can be thought of as the direct product of G's, one for each edge of X not in the chosen maximal tree. This is similar to $H_1(X;G)$ which consists of a direct sum of copies of G for each edge X not in the maximal tree as seen in example 2.2.5 (abelianized).

Now, moving up a dimension, let $\Delta^2(X;G)$ be all functions from faces to G, and define

$$\delta : \Delta^1(X;G) \to \Delta^2(X;G)$$

$$\delta\psi([v_0, v_1, v_2]) = \psi([v_0, v_1]) + \psi([v_1, v_2]) - \psi([v_0, v_2])$$

i.e. a signed sum of the values of ψ on the edges. Now, combining to form a chain:

$$\cdots \to 0 \to \Delta^0(X;G) \xrightarrow{\delta} \Delta^1(X;G) \xrightarrow{\delta} \Delta^2(X;G) \to 0 \to \cdots \tag{4.2}$$

Notice that

$$\delta\delta(\varphi) = (\varphi(v_1) - \varphi(v_0)) + (\varphi(v_2) - \varphi(v_1)) - (\varphi(v_2) - \varphi(v_0)) = 0$$

Showing that $\text{im}\delta_1 \subseteq \ker \delta_2$. Again making a chain complex by adding 0 before and after in (4.2) we may define $H^2(X;G)$. There are many ways of thinking of elements of $H^2(X;G)$, each giving a useful insight.

First, notice that is $\delta\psi = 0$, then:

$$\psi([v_0, v_2]) = \psi([v_0, v_1]) + \psi([v_1, v_2])$$

showing that the cohomology groups measures the failure of ψ being additive. Another more "geometric" interpretation is that it measure the degree of failure of finding a $\varphi \in \Delta^0(X;G)$ such that $\psi = \delta\varphi$, since if $\psi = \delta\varphi$, then $\delta\psi = 0$ (since $\delta\delta\varphi = 0$). In particular, since the value of the equation $\psi = \delta\varphi$ only depends on values defined on elements, we may think of it as measuring the local obstruction. But since $H^1(X;G)$ is zero if and only if $\psi = \delta\varphi$, we can think of $H^1(X;G)$ as measuring global obstruction. Think of it as trying to find when a vector fields is the gradient of a differentiable function.

Since $\Delta_n(X)$ is a free abelian group with basis the n-simplices of X, and a homomorphism out of a free abelian group is determined by its values on a basis, which may be prescribed arbitrarily, the functions on the n-simplices are the homomorphisms out of $\Delta_n(X)$:

$$\Delta^n(X;G) = \text{Hom}(\Delta_n(X), G)$$

The same construction applied to the singular chain groups is given a name:

Definition 4.1.1: Dual Group

Let X be a topological space and $C_\bullet(X)$ the singular chain groups. Then:

$$C^\bullet(X;G) := \text{Hom}(C_\bullet(X), G)$$

is the *singular cochain groups* with coefficients in G.

The boundary homomorphism ∂_n generalizes to:

$$\delta\varphi([v_0, ..., v_{n+1}]) = \sum_i (-1)^i \varphi([v_0, ..., \hat{v}_i, ..., v_{n+1}]) = \varphi(\partial[v_0, ..., v_{n+1}])$$

Hence:

$$\delta(\varphi) = \varphi \circ \partial$$

meaning δ is a dual map, since the $\text{Hom}(-, G)$ functor maps $C_n \mapsto \text{Hom}(C_n, G)$ and $\partial \mapsto (\varphi \mapsto \varphi \circ \partial)$. Now, to form a complex, let X be a topological. We may then form a chain:

$$\cdots \to C_n \xrightarrow{\partial_n} C_{n-1} \to \cdots$$

whether singular, simplicial, or cellular. If we take the homology, we get $H_n(X)$. On the other hand, if we apply the [contravariant] hom functor:

$$\cdots \leftarrow \text{Hom}(C_n, G) \xleftarrow{\delta_n} \text{Hom}(C_{n-1}, G) \leftarrow \cdots$$

In particular, each $\varphi \in C^n(X;G)$ assigned to $\sigma : \Delta^n \to X$ a value $\varphi(\sigma) \in G$. Then $\delta = \partial^* : C^n(X;G) \to C^{n+1}(X;G)$ is given by composition:

$$\delta(\varphi) = \partial^*(\varphi) = \varphi \circ \partial$$

In particular:

$$\delta(\varphi)(\sigma) = \sum_i (-1)^i \varphi(\sigma|_{v_0,\dots,\hat{v_i},\dots,v_{n+1}})$$

hence $\delta^2 = 0$, hence the cohomology groups are well-defined.

Definition 4.1.2: Cocycles And Coboundaries

Let δ be defined as above, then element of $\ker\delta$ are called *cocycles* and element of $\text{im}\delta$ are called *coboundaries*. A cochain φ is a cocycle if

$$\delta(\varphi) = \varphi \circ \partial = 0$$

i.e. $\varphi \in \text{Hom}(C_n(X), G)$ vanishes on the boundaries $\partial C_{n+1}(X)$.

We give the image of $\text{Hom}(-, G)$ a name:

Definition 4.1.3: Cochain Complex

Let X be a topological and $C_\bullet$ a complex. Then $C^*_\bullet = \text{Hom}(C_\bullet, G)$ is a *cochain complex* with respect to G.

From this we get the comology group:

Definition 4.1.4: Cohomology Group

Let $C_\bullet$ be a chain complex. Then for some abelian group G, we may form a chain and dualize with $\text{Hom}(-, G)$ to form the cochain $C^\bullet$. Then:

$$H^n(C; G) = \frac{\ker(\delta)}{\text{im}\delta}$$

is the nth *cohomology group* of X.

We will show that the cohomology group is algebraically determined by the homology group and G. A first hope may be that $\text{Hom}(-, G)$ commutes with taking the quotient:

$$H^n(C; G) \cong \text{Hom}(H_n(C), G)$$

However, this need not be the case:

Example 4.1.5: Hom Functor Not Commuting With Quotienting

Take the chain:

$$0 \to \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0$$

where the 2 map sends $x \mapsto 2x$. Then dualizing with $\text{Hom}_\mathbb{Z}(-, \mathbb{Z})$ we get the (isomorphic) cochain:

$$0 \leftarrow \mathbb{Z} \xleftarrow{0} \mathbb{Z} \xleftarrow{2} \mathbb{Z} \xleftarrow{0} \mathbb{Z} \leftarrow 0$$

Then $H_2(C) = 0$, but $H^2(C) = \mathbb{Z}/2\mathbb{Z}$, while $\text{Hom}(H_2(C), \mathbb{Z}) = 0$.

Though this is not the case, this is not a bad first guess, and there is at minimum a homomorphism $h : H^n(C;G) \to \mathrm{Hom}(H_n(C), G)$. In fact, we can be more precise. First, we require a lemma:

Lemma 4.1.6: Free Resolution Homomorphism

Let F, F' be free resolutions of abelian groups H, H'. Then every homomorphism $\alpha : H \to H'$ can be extended to a chain map from F to F':

$$\begin{array}{ccccccccccc} \cdots & \longrightarrow & F_2 & \longrightarrow & F_1 & \longrightarrow & F_0 & \longrightarrow & H & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \alpha_2} & & \downarrow{\scriptstyle \alpha_1} & & \downarrow{\scriptstyle \alpha_0} & & \downarrow{\scriptstyle \alpha} & & \\ \cdots & \longrightarrow & F_2' & \longrightarrow & F_1' & \longrightarrow & F_0' & \longrightarrow & H' & \longrightarrow & 0 \end{array}$$

***Proof*:**
Write $f_i \colon F_i \to F_{i-1}$ for the maps of the resolution F, with the convention $F_{-1} = H$, so that $f_0 \colon F_0 \to H$ is the augmentation and the sequence $\cdots \to F_1 \xrightarrow{f_1} F_0 \xrightarrow{f_0} H \to 0$ is exact. Write f_i' and $F_{-1}' = H'$ for the same data on the side of F', and set $\alpha_{-1} = \alpha$. Exactness of F' says that f_0' is surjective and that $\mathrm{im} f_i' = \ker f_{i-1}'$ for every $i \geq 1$. The maps α_i are built by induction on i, and at each stage the freeness of F_i is what makes the choice possible: a homomorphism out of a free abelian group may be prescribed arbitrarily on a basis and extends uniquely ([10, Universal Property of Free Modules]).

Suppose $\alpha_{-1}, \ldots, \alpha_{i-1}$ have been constructed with $f_j' \alpha_j = \alpha_{j-1} f_j$ for all $0 \leq j \leq i-1$, which for $i = 0$ is the empty hypothesis beyond $\alpha_{-1} = \alpha$. Let x be a basis element of F_i. The element $\alpha_{i-1} f_i(x)$ of F_{i-1}' lies in $\mathrm{im} f_i'$. For $i = 0$ this is the surjectivity of f_0', and for $i \geq 1$ it holds because

$$f_{i-1}' \alpha_{i-1} f_i = \alpha_{i-2} f_{i-1} f_i = 0$$

using the inductive relation in the first equality and $f_{i-1} f_i = 0$ in the second, so that $\alpha_{i-1} f_i(x) \in \ker f_{i-1}' = \mathrm{im} f_i'$. Choose $x' \in F_i'$ with $f_i'(x') = \alpha_{i-1} f_i(x)$ and declare $\alpha_i(x) = x'$. Extending over a basis defines $\alpha_i \colon F_i \to F_i'$, and the relation $f_i' \alpha_i = \alpha_{i-1} f_i$ holds on a basis of F_i, hence on all of F_i, since both sides are homomorphisms.

The resulting family $(\alpha_i)_{i \geq 0}$ satisfies $f_i' \alpha_i = \alpha_{i-1} f_i$ in every degree, which is exactly the commutativity of each square of the diagram, the rightmost square being $f_0' \alpha_0 = \alpha f_0$. So (α_i) is a chain map from F to F' extending α, as we sought to show.

Recall every abelian group has a free resolution of *length one*, that is one with two terms $0 \to F_1 \to F_0 \to A \to 0$: take F_0 free on a generating set and $F_1 = \ker(F_0 \to A)$, which is free because $\mathbb{Z}$ is a PID. (The ranks are unconstrained. The resolution $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$ has both of rank one.)

Theorem 4.1.7: Universal Coefficient Theorem For Homology

Let $C_\bullet$ be a chain complex of free abelian groups and G an abelian group. For each n there is a short exact sequence

$$0 \to H_n(C) \otimes G \to H_n(C;G) \to \operatorname{Tor}(H_{n-1}(C), G) \to 0$$

where $H_n(C;G)$ is the homology of $C_\bullet \otimes G$. The sequence is natural in $C_\bullet$ and in G. It splits, but the splitting is not natural.

***Proof*:**
Write $Z_n = \ker(\partial\colon C_n \to C_{n-1})$ and $B_n = \operatorname{im}(\partial\colon C_{n+1} \to C_n)$, both subgroups of the free abelian group C_n, hence themselves free, a submodule of a free module over the PID $\mathbb{Z}$ being free. No finite-generation hypothesis is needed, which matters because the chain complexes this is applied to are usually of infinite rank. The strategy is the one used for cohomology below: replace $C_\bullet$ by the cycles and boundaries, tensor there, and only then take homology.

Step 1: a free resolution of the homology. For each n the sequence

$$0 \to B_n \to Z_n \to H_n(C) \to 0$$

is exact by definition of $H_n(C) = Z_n/B_n$, and B_n, Z_n are free. So it is a free resolution of $H_n(C)$ of length one, which is what computes Tor: tensoring with G gives

$$0 \to \operatorname{Tor}(H_n(C), G) \to B_n \otimes G \to Z_n \otimes G \to H_n(C) \otimes G \to 0$$

the left-hand term being Tor precisely because the resolution has length one. That the answer does not depend on the chosen resolution is [10, The Derived Functors Are Well Defined](1). Lemma 4.1.6 gives only the *existence* of a comparison chain map, which is one half of what independence needs, the other being that any two such lifts are chain homotopic.

Step 2: the complex of cycles and boundaries. Regard $Z_\bullet$ and $B_\bullet$ as chain complexes with zero differential. Since C_n is free and $B_{n-1} \subseteq C_{n-1}$ is free, the sequence

$$0 \to Z_n \to C_n \xrightarrow{\partial} B_{n-1} \to 0$$

splits, so tensoring with G keeps it exact:

$$0 \to Z_n \otimes G \to C_n \otimes G \to B_{n-1} \otimes G \to 0$$

This is a short exact sequence of chain complexes, the outer two having zero differential.

Step 3: the long exact sequence. Its homology long exact sequence reads

$$B_n \otimes G \xrightarrow{i_n \otimes 1} Z_n \otimes G \to H_n(C;G) \to B_{n-1} \otimes G \xrightarrow{i_{n-1} \otimes 1} Z_{n-1} \otimes G$$

writing $i_k\colon B_k \rightarrowtail Z_k$ for the inclusions. The connecting map is $i_{n-1} \otimes 1$: an element of $B_{n-1} \otimes G$ lifts to $c \otimes g$ with $\partial c = b$, and applying the differential of $C_\bullet \otimes G$ returns $b \otimes g$ viewed inside $Z_{n-1} \otimes G$. Breaking the sequence up, $H_n(C;G)$ is an extension of $\ker(i_{n-1} \otimes 1)$ by $\operatorname{coker}(i_n \otimes 1)$. By Step 1 those are exactly $\operatorname{Tor}(H_{n-1}(C), G)$ and $H_n(C) \otimes G$, which gives the displayed sequence. Naturality in $C_\bullet$ and G holds for the middle and left terms by naturality

of the long exact sequence (proposition 3.4.4) and of $\otimes$. For the right-hand term it needs one thing more: the identification $\ker(i_{n-1}\otimes 1)\cong \operatorname{Tor}(H_{n-1}(C),G)$ was made *through* the resolution $0\to B_{n-1}\to Z_{n-1}\to H_{n-1}(C)\to 0$. A chain map $C_\bullet\to C'_\bullet$ carries boundaries to boundaries and cycles to cycles, so it restricts to a comparison chain map between these resolutions lifting the induced map on H_{n-1}. The canonical map on Tor is computed from any such lift: any two lifts are chain homotopic ([13, Projective Resolution As Initial Objects]), the homotopy persists after tensoring with G, and chain homotopic maps induce the same map on homology (proposition 3.3.5). So the map the chain map induces on $\ker(i_{n-1}\otimes 1)$ is the canonical one, and the sequence is natural in the right-hand term as well.

Step 4: splitting. The sequence $0\to Z_n\to C_n\to B_{n-1}\to 0$ of Step 2 splits, so choose a retraction $p_n\colon C_n\to Z_n$ with $p_n|_{Z_n}=\mathrm{id}$. It is tempting to say this retraction splits the displayed sequence directly, but it does not: p is *not* a chain map $C_\bullet\to Z_\bullet$ when $Z_\bullet$ carries the zero differential, since for $c\in C_n$ the element ∂c lies in $B_{n-1}\subseteq Z_{n-1}$, where p_{n-1} is the identity, so $p_{n-1}(\partial c)=\partial c\neq 0$.

Composing with the projection fixes it. Put $\varphi_n:=\pi_n\circ p_n\colon C_n\to H_n(C)$, where $\pi_n\colon Z_n\twoheadrightarrow Z_n/B_n=H_n(C)$. Now φ *is* a chain map into $H_\bullet(C)$ with zero differential, because $\varphi_{n-1}(\partial c)=\pi_{n-1}(\partial c)=0$, the class of a boundary being zero. Tensoring with G and passing to homology gives $(\varphi\otimes 1)_*\colon H_n(C;G)\to H_n(C)\otimes G$, and for a cycle $z\in Z_n$ one has $\varphi_n(z)=[z]$, so this map restricts to the identity on the image of $H_n(C)\otimes G\hookrightarrow H_n(C;G)$, that inclusion being $[z]\otimes g\mapsto[z\otimes g]$ under the Step 1 identification $\operatorname{coker}(i_n\otimes 1)\cong H_n(C)\otimes G$. It is therefore a retraction, and the sequence splits.

The splitting is not natural. Non-canonicity of the choice does not by itself show this, so here is a counterexample. Take $G=\mathbb{Z}/2$, let $C_\bullet$ be $\mathbb{Z}\xrightarrow{2}\mathbb{Z}$ in degrees $1,0$ and $C'_\bullet$ be $\mathbb{Z}\to 0$ in degrees $1,0$, and let φ be the identity in degree 1 and zero in degree 0. Then $H_1(C)\otimes G=0$, so any splitting s_C of $H_1(C;G)\to\operatorname{Tor}(H_0(C),G)$ is an isomorphism $\mathbb{Z}/2\to\mathbb{Z}/2$. By contrast $\operatorname{Tor}(H_0(C'),G)=0$ forces $s_{C'}=0$. But $\varphi_*\colon H_1(C;G)\to H_1(C';G)$ is an isomorphism $\mathbb{Z}/2\to\mathbb{Z}/2$, so naturality would require $\varphi_* s_C=s_{C'}\operatorname{Tor}(\varphi_*)=0$, which is false. As we sought to show.

The two coefficient theorems answer the same question in opposite variances: Tor measures the failure of $\otimes G$ to commute with homology, and Ext the failure of $\operatorname{Hom}(-,G)$ to do so. The cohomology statement is the one used most often below, and is next.

Theorem 4.1.8: Universal Coefficient Theorem For Cohomology

Let $C_\bullet$ be a chain complex of free abelian groups with homology $H_\bullet(C)$, and G an abelian group. For each n the cohomology $H^n(C;G)$ of the cochain complex $\mathrm{Hom}(C_\bullet, G)$ sits in a short exact sequence

$$0 \to \mathrm{Ext}(H_{n-1}(C), G) \to H^n(C;G) \xrightarrow{h} \mathrm{Hom}(H_n(C), G) \to 0$$

where $h[\varphi]$ is the homomorphism $[z] \mapsto \varphi(z)$. The sequence is natural in $C_\bullet$: a chain map $\alpha\colon C_\bullet \to C'_\bullet$ between chain complexes of free abelian groups induces a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{Ext}(H_{n-1}(C), G) & \longrightarrow & H^n(C;G) & \longrightarrow & \mathrm{Hom}(H_n(C), G) & \longrightarrow & 0 \\ & & {\scriptstyle (\alpha_*)^*}\uparrow & & {\scriptstyle \alpha^*}\uparrow & & {\scriptstyle (\alpha_*)^*}\uparrow & & \\ 0 & \longrightarrow & \mathrm{Ext}(H_{n-1}(C'), G) & \longrightarrow & H^n(C';G) & \longrightarrow & \mathrm{Hom}(H_n(C'), G) & \longrightarrow & 0 \end{array}$$

The sequence splits, but the splitting is not natural.

***Proof*:**

Throughout write $Z_n = \ker(\partial\colon C_n \to C_{n-1})$ for the cycles and $B_n = \mathrm{im}(\partial\colon C_{n+1} \to C_n)$ for the boundaries, both subgroups of the free abelian group C_n. The strategy is to first replace the chain complex $C_\bullet$ by the much simpler complexes of cycles and boundaries, dualize there, and only then reassemble. The single algebraic fact that makes this possible is that a subgroup of a free abelian group is free (a submodule of a free $\mathbb{Z}$-module is free, since $\mathbb{Z}$ is a PID), so that Z_n and B_n are themselves free.

Step 1: Splitting the cycles and boundaries. For each n the boundary map $\partial\colon C_n \to C_{n-1}$ has image B_{n-1} and kernel Z_n, giving the short exact sequence

$$0 \to Z_n \to C_n \xrightarrow{\partial} B_{n-1} \to 0 \tag{4.3}$$

Its right-hand term B_{n-1} is a subgroup of the free abelian group C_{n-1}, hence free, so (4.3) splits: there is a retraction $p\colon C_n \to Z_n$ with $p|_{Z_n} = \mathrm{id}_{Z_n}$, equivalently a direct sum decomposition $C_n \cong Z_n \oplus B_{n-1}$. This splitting is the only non-canonical choice in the proof, and it is the source of the (non-natural) splitting of the final sequence. Regard $Z_\bullet$ and $B_\bullet$ as chain complexes with *zero* differential. The inclusions $Z_n \hookrightarrow C_n$ and the boundary maps $C_n \to B_{n-1}$ assemble into a short exact sequence of chain complexes

$$0 \to Z_\bullet \to C_\bullet \xrightarrow{\partial} B_{\bullet-1} \to 0 \tag{4.4}$$

(in which the degree of the boundary complex is shifted down by one, matching $\partial\colon C_n \to B_{n-1}$). The point of (4.4) is that $Z_\bullet$ and $B_\bullet$ carry the zero differential, so their cochain complexes are transparent.

Step 2: Dualizing. Apply the contravariant functor $\mathrm{Hom}(-, G)$ to (4.4). Because each sequence (4.3) splits, $\mathrm{Hom}(-, G)$ carries it to a (split) short exact sequence of abelian groups in every degree, so dualizing the complexes is exact:

$$0 \to \mathrm{Hom}(B_{\bullet-1}, G) \xrightarrow{\partial^*} \mathrm{Hom}(C_\bullet, G) \to \mathrm{Hom}(Z_\bullet, G) \to 0 \tag{4.5}$$

is a short exact sequence of cochain complexes. Here $\mathrm{Hom}(Z_\bullet, G)$ and $\mathrm{Hom}(B_\bullet, G)$ carry the zero differential, since $Z_\bullet$ and $B_\bullet$ do. The cochain complex $\mathrm{Hom}(C_\bullet, G)$ is the one whose cohomology is $H^n(C; G)$, and (4.5) expresses it as an extension of two complexes with vanishing differential, whose cohomology is computed without any further work: $H^n(\mathrm{Hom}(Z_\bullet, G)) = \mathrm{Hom}(Z_n, G)$ and $H^n(\mathrm{Hom}(B_\bullet, G)) = \mathrm{Hom}(B_n, G)$.

Step 3: The long exact sequence and its connecting map. The short exact sequence of cochain complexes (4.5) produces a long exact sequence in cohomology. Using the identification of the outer cohomology groups from Step 2, and accounting for the degree shift by one in the boundary complex, the long exact sequence reads

$$\begin{aligned} \cdots \to \mathrm{Hom}(Z_{n-1}, G) \xrightarrow{i_{n-1}^*} \mathrm{Hom}(B_{n-1}, G) \to H^n(C; G) \\ \to \mathrm{Hom}(Z_n, G) \xrightarrow{i_n^*} \mathrm{Hom}(B_n, G) \to \cdots \end{aligned} \tag{4.6}$$

The connecting homomorphism $i_n^* \colon \mathrm{Hom}(Z_n, G) \to \mathrm{Hom}(B_n, G)$ is, by the standard identification of the connecting map of (4.5), the map induced by the inclusion $i \colon B_n \hookrightarrow Z_n$ of boundaries into cycles, that is $i_n^* = \mathrm{Hom}(i, G)$ (restriction of a homomorphism on Z_n to the subgroup B_n). This is the one computational point of the proof and is verified by chasing a representative cochain through (4.5): a cocycle of $\mathrm{Hom}(C_\bullet, G)$ whose image in $\mathrm{Hom}(Z_n, G)$ is a prescribed homomorphism on cycles has connecting image equal to the restriction of that homomorphism along $B_n \hookrightarrow Z_n$.

Step 4: Extracting the short exact sequence. The long exact sequence (4.6) breaks into short exact sequences in the usual way, by splitting off the cokernel of the incoming map and the kernel of the outgoing map:

$$0 \to \mathrm{coker}(i_{n-1}^*) \to H^n(C; G) \xrightarrow{h} \ker(i_n^*) \to 0 \tag{4.7}$$

where $i_{n-1}^* \colon \mathrm{Hom}(Z_{n-1}, G) \to \mathrm{Hom}(B_{n-1}, G)$ and $i_n^* \colon \mathrm{Hom}(Z_n, G) \to \mathrm{Hom}(B_n, G)$ are the dual restriction maps of Step 3. It remains to identify the two outer terms with $\mathrm{Ext}(H_{n-1}(C), G)$ and $\mathrm{Hom}(H_n(C), G)$ respectively, and to check that the right-hand map of (4.7) is the map h named in the statement.

Step 5: The free resolution of H_n and the identification of the outer terms. Fix n and consider the exact sequence

$$0 \to B_n \xrightarrow{i} Z_n \to H_n(C) \to 0 \tag{4.8}$$

It is exact by the very definition $H_n(C) = Z_n / B_n$, and both B_n and Z_n are free by the freeness fact of Step 1. Hence (4.8) is a free resolution of $H_n(C)$ of length one (every abelian group has such a resolution, since $\mathbb{Z}$ is a PID, as recalled before lemma 4.1.6). The functor $\mathrm{Ext}(H_n(C), G)$ is computed by dropping the term $H_n(C)$, applying $\mathrm{Hom}(-, G)$, and taking cohomology of the resulting two-term complex $0 \to \mathrm{Hom}(Z_n, G) \xrightarrow{i_n^*} \mathrm{Hom}(B_n, G) \to 0$ (this is the definition of Ext as a derived functor of Hom, computed from any free resolution, and is independent of the resolution by lemma 4.1.6 and [13, Unique Projection Resolution]). By left exactness of Hom this two-term complex has

$$\ker(i_n^*) = \mathrm{Hom}(H_n(C), G), \qquad \mathrm{coker}(i_n^*) = \mathrm{Ext}(H_n(C), G) \tag{4.9}$$

The first equality holds because $\mathrm{Hom}(-, G)$ is left exact: applying it to the right-exact sequence $B_n \xrightarrow{i} Z_n \to H_n(C) \to 0$ gives the left-exact sequence $0 \to \mathrm{Hom}(H_n(C), G) \to \mathrm{Hom}(Z_n, G) \xrightarrow{i_n^*}$

$\mathrm{Hom}(B_n, G)$, so a homomorphism $Z_n \to G$ lies in $\ker(i_n^*)$ (vanishes on B_n) precisely when it factors through the quotient $Z_n/B_n = H_n(C)$. The second equality is the definition of $\mathrm{Ext}(H_n(C), G)$ as the first cohomology of the dualized resolution.

Step 6: Assembling the sequence. Substituting (4.9) into (4.7) (using the index $n-1$ for the cokernel term and n for the kernel term) yields

$$0 \to \mathrm{Ext}(H_{n-1}(C), G) \to H^n(C; G) \xrightarrow{h} \mathrm{Hom}(H_n(C), G) \to 0$$

which is the asserted sequence. The cokernel of i_{n-1}^* is $\mathrm{Ext}(H_{n-1}(C), G)$ by (4.9) with n replaced by $n-1$, and the kernel of i_n^* is $\mathrm{Hom}(H_n(C), G)$, sitting in degree n. The map h is the composite of the map $H^n(C; G) \to \ker(i_n^*)$ from (4.7) (induced by the surjection $\mathrm{Hom}(C_\bullet, G) \to \mathrm{Hom}(Z_\bullet, G)$ of (4.5), that is, restriction of a cochain to cycles) with the identification $\ker(i_n^*) = \mathrm{Hom}(H_n(C), G)$, and it is exactly the evaluation map of the statement: a cocycle $\varphi\colon C_n \to G$ (with $\varphi \circ \partial = 0$, so φ vanishes on B_n) restricts to $\varphi|_{Z_n}$, which factors through $Z_n/B_n = H_n(C)$ to give $\overline{\varphi}\colon H_n(C) \to G$, and $h([\varphi]) = \overline{\varphi}$. This is well-defined on cohomology classes: a coboundary $\varphi = \psi \circ \partial$ vanishes on Z_n, so its associated $\overline{\varphi}$ is zero. Thus h is the homomorphism $[\varphi] \mapsto \overline{\varphi|_{Z_n}}$, as named in the statement.

Step 7: The splitting. The sequence (4.7) splits because the retraction $p\colon C_n \to Z_n$ of (4.3) chosen in Step 1 furnishes a section of h directly. Given a homomorphism $\bar{\psi} \in \mathrm{Hom}(H_n(C), G)$, let $\psi_0\colon Z_n \to G$ be its composite with the quotient $Z_n \twoheadrightarrow Z_n/B_n = H_n(C)$, so that ψ_0 vanishes on B_n and descends to $\bar{\psi}$. Set $\varphi = \psi_0 \circ p\colon C_n \to G$. This φ is a cocycle: for $c \in C_{n+1}$ the element ∂c lies in Z_n, so the retraction property $p|_{Z_n} = \mathrm{id}_{Z_n}$ gives $\varphi(\partial c) = \psi_0(p(\partial c)) = \psi_0(\partial c)$, and $\partial c \in B_n = \ker(Z_n \to H_n(C))$ forces $\psi_0(\partial c) = 0$. Hence $\varphi \circ \partial = 0$. Moreover $\overline{\varphi|_{Z_n}} = \psi_0 \circ p|_{Z_n} = \psi_0$, which descends to $\bar{\psi}$, so by the description of h in Step 6, $h([\varphi]) = \overline{\varphi|_{Z_n}} = \bar{\psi}$. Thus

$$k\colon \mathrm{Hom}(H_n(C), G) \to H^n(C; G), \qquad \bar{\psi} \mapsto [\psi_0 \circ p]$$

is a homomorphism with $h \circ k = \mathrm{id}$, a section of h. A section of the surjection in (4.7) splits the sequence, so

$$H^n(C; G) \cong \mathrm{Ext}(H_{n-1}(C), G) \oplus \mathrm{Hom}(H_n(C), G)$$

as abelian groups. The section k depends on the chosen retraction p, equivalently on the chosen splitting of (4.3), and no choice of sections is natural. Take $G = \mathbb{Z}/2$, let $C'_\bullet$ be $\mathbb{Z} \xrightarrow{2} \mathbb{Z}$ in degrees $1, 0$ and $C_\bullet$ be $\mathbb{Z}$ in degree 1 and 0 elsewhere, and let $\alpha\colon C'_\bullet \to C_\bullet$ be the identity in degree 1 and zero in degree 0, a chain map. In degree $n = 1$, $H_1(C) = \mathbb{Z}$ and $H_0(C) = 0$, so h is an isomorphism for C and any section k_C is its inverse, an isomorphism onto $H^1(C; \mathbb{Z}/2) \cong \mathbb{Z}/2$. Since $H_1(C') = 0$, any section $k_{C'}$ is zero. Every 1-cochain of either complex is a cocycle, and none but 0 is a coboundary, the coboundary into degree 1 of $C'_\bullet$ being the dual of multiplication by 2, which is zero modulo 2. So $\alpha^* = \mathrm{Hom}(\alpha_1, \mathbb{Z}/2)$ is an isomorphism $H^1(C; \mathbb{Z}/2) \to H^1(C'; \mathbb{Z}/2)$, and naturality would require $\alpha^* k_C = k_{C'}(\alpha_*)^* = 0$, which is false. The short exact sequence itself is natural in $C_\bullet$. Its middle and right terms are natural by the naturality of the long exact sequence (4.6), that is proposition 3.4.4 read for cochain complexes after reindexing. For the left term, a chain map carries cycles to cycles and boundaries to boundaries, so it restricts to a map between the resolutions (4.8) lifting α_*, and the map it induces on Ext is the canonical one by the argument for Tor in Step 3 of the proof of theorem 4.1.7, completing the proof.

To comput Ext, we have a couple of nice shorhands:

Proposition 4.1.9: Computing Ext Groups

Let H, H' and G be abelian groups. Then

1. $\mathrm{Ext}(H \oplus H', G) \cong \mathrm{Ext}(H, G) \oplus \mathrm{Ext}(H', G)$
2. $\mathrm{Ext}(H, G) = 0$ if H is free
3. $\mathrm{Ext}(\mathbb{Z}_n, G) \cong G/nG$

In particular, if $C_\bullet$ is a chain complex of free abelian groups whose homology groups H_n, H_{n-1} are finitely generated, with torsion subgroups $T_n \subseteq H_n$ and $T_{n-1} \subseteq H_{n-1}$, then:

$$H^n(C; \mathbb{Z}) \cong (H_n/T_n) \oplus T_{n-1}$$

***Proof*:**
Throughout, $\mathrm{Ext}(H, G)$ is the first cohomology of the complex obtained from a free resolution of H by deleting the term H and applying $\mathrm{Hom}(-, G)$, and the answer is independent of the resolution, as recalled in Step 5 of the proof of theorem 4.1.8. Each part is proved by exhibiting a convenient resolution.

Part (1). Let $F \to H$ and $F' \to H'$ be free resolutions. The degreewise direct sum $F \oplus F'$ is again a complex of free abelian groups, and a direct sum of exact sequences is exact, so $F \oplus F' \to H \oplus H'$ is a free resolution. Applying $\mathrm{Hom}(-, G)$ to a finite direct sum gives

$$\mathrm{Hom}(F_i \oplus F_i', G) \cong \mathrm{Hom}(F_i, G) \oplus \mathrm{Hom}(F_i', G)$$

compatibly with the differentials, since a homomorphism out of a direct sum is the pair of its restrictions. The dualized complex is therefore the direct sum of the two dualized complexes, and kernels and images are formed summandwise, so its first cohomology is the direct sum of theirs. That is the asserted isomorphism.

Part (2). If H is free, take the resolution with $F_0 = H$, with $F_0 \to H$ the identity, and $F_i = 0$ for $i \geq 1$. It is exact and free. Deleting H and dualizing gives a complex whose degree-one term is $\mathrm{Hom}(0, G) = 0$, so its first cohomology vanishes and $\mathrm{Ext}(H, G) = 0$.

Part (3). Here $\mathbb{Z}_n$ is the cyclic group of order n, so $n \geq 1$ and multiplication by n on $\mathbb{Z}$ is injective with cokernel $\mathbb{Z}_n$. Hence

$$0 \to \mathbb{Z} \xrightarrow{n} \mathbb{Z} \to \mathbb{Z}_n \to 0$$

is a free resolution of $\mathbb{Z}_n$, with $F_1 = F_0 = \mathbb{Z}$ and $F_i = 0$ for $i \geq 2$. Deleting $\mathbb{Z}_n$ and applying $\mathrm{Hom}(-, G)$ gives the two-term complex $\mathrm{Hom}(\mathbb{Z}, G) \to \mathrm{Hom}(\mathbb{Z}, G)$. Evaluation at 1 identifies $\mathrm{Hom}(\mathbb{Z}, G)$ with G, and under that identification the dual of multiplication by n sends φ to

$\varphi \circ (n \cdot -)$, that is $g \mapsto ng$. The first cohomology of $0 \to G \xrightarrow{n} G \to 0$, concentrated in degrees 0 and 1, is the cokernel of that map, namely G/nG.

The consequence for $\mathbb{Z}$ coefficients. Let H be finitely generated with torsion subgroup T. By [10, Invariant Factor Form of Modules over PID's] applied with $R = \mathbb{Z}$,

$$H \cong \mathbb{Z}^r \oplus \mathbb{Z}/(a_1) \oplus \cdots \oplus \mathbb{Z}/(a_m)$$

with $T \cong \mathbb{Z}/(a_1) \oplus \cdots \oplus \mathbb{Z}/(a_m)$ and $H/T \cong \mathbb{Z}^r$. Parts (1), (2) and (3) with $G = \mathbb{Z}$ now give

$$\operatorname{Ext}(H, \mathbb{Z}) \cong \operatorname{Ext}(\mathbb{Z}^r, \mathbb{Z}) \oplus \bigoplus_{i=1}^{m} \operatorname{Ext}(\mathbb{Z}/(a_i), \mathbb{Z}) \cong 0 \oplus \bigoplus_{i=1}^{m} \mathbb{Z}/(a_i) \cong T$$

The same decomposition computes $\operatorname{Hom}(H, \mathbb{Z})$. A homomorphism from a finite group to $\mathbb{Z}$ is zero, every element of the source having finite order while $\mathbb{Z}$ is torsion-free, so

$$\operatorname{Hom}(H, \mathbb{Z}) \cong \operatorname{Hom}(\mathbb{Z}^r, \mathbb{Z}) \cong \mathbb{Z}^r \cong H/T$$

Feeding these two computations into the universal coefficient sequence of theorem 4.1.8 with $G = \mathbb{Z}$ turns

$$0 \to \operatorname{Ext}(H_{n-1}, \mathbb{Z}) \to H^n(C; \mathbb{Z}) \to \operatorname{Hom}(H_n, \mathbb{Z}) \to 0$$

into $0 \to T_{n-1} \to H^n(C; \mathbb{Z}) \to H_n/T_n \to 0$. Its right-hand term $\operatorname{Hom}(H_n, \mathbb{Z}) \cong \mathbb{Z}^{r_n}$ is free abelian, so choosing a preimage of each basis element and extending ([10, Universal Property of Free Modules]) produces a section of the surjection. By lemma 3.4.34 the sequence splits, giving

$$H^n(C; \mathbb{Z}) \cong (H_n/T_n) \oplus T_{n-1}$$

as we sought to show.

Corollary 4.1.10: Homology Isomorphism, then Cohomology Isomorphism

Let $\alpha \colon C_\bullet \to C'_\bullet$ be a chain map between chain complexes of free abelian groups. If $\alpha_* \colon H_n(C) \to H_n(C')$ is an isomorphism for every n, then $\alpha^* \colon H^n(C'; G) \to H^n(C; G)$ is an isomorphism for every n and every abelian group G.

***Proof*:**
Step 1: The mapping cone is acyclic. Let $M_n = C_{n-1} \oplus C'_n$ with $d(c, c') = (-\partial c,\ \partial c' + \alpha(c))$. Then

$$d^2(c, c') = \big(\partial\partial c,\ \partial\partial c' + \partial\alpha(c) - \alpha(\partial c)\big) = 0$$

because α is a chain map, so $M_\bullet$ is a chain complex of free abelian groups. Let $\Sigma C_\bullet$ be the complex with $(\Sigma C)_n = C_{n-1}$ and differential $-\partial$, whose homology in degree n is $H_{n-1}(C)$. The maps $\iota(c') = (0, c')$ and $\pi(c, c') = c$ are chain maps and form a short exact sequence of chain complexes

$$0 \to C'_\bullet \xrightarrow{\iota} M_\bullet \xrightarrow{\pi} \Sigma C_\bullet \to 0$$

The connecting homomorphism $H_{n-1}(C) \to H_{n-1}(C')$ of its long exact sequence (theorem 3.4.3)

is α_*: a cycle $c \in C_{n-1}$ lifts to $(c,0) \in M_n$, and $d(c,0) = (-\partial c, \alpha(c)) = \iota(\alpha(c))$. In the exact segment

$$H_n(C) \xrightarrow{\alpha_*} H_n(C') \xrightarrow{\iota_*} H_n(M) \xrightarrow{\pi_*} H_{n-1}(C) \xrightarrow{\alpha_*} H_{n-1}(C')$$

the first α_* is onto, so $\iota_* = 0$ and π_* is injective, while the second α_* is injective, so $\pi_* = 0$. Hence $H_n(M) = 0$ for every n.

Step 2: The dual of an acyclic complex of free abelian groups is acyclic. Write Z_n and B_n for the cycles and boundaries of $M_\bullet$, so that $Z_n = B_n$ by Step 1. The group B_{n-1} is a subgroup of the free group M_{n-1}, hence free ([10, Submodule Of Free Module Over PID, General Case]), and choosing a preimage of each of its basis elements ([10, Universal Property of Free Modules]) gives a section of the surjection $d\colon M_n \to B_{n-1}$. By lemma 3.4.34, $M_n = Z_n \oplus D_n$ with D_n the image of the section, and d restricts to an isomorphism $D_n \to B_{n-1} = Z_{n-1}$. Let $\varphi \in \mathrm{Hom}(M_n, G)$ be a cocycle, $\varphi \circ d = 0$. Then φ vanishes on $B_n = Z_n$. Define $\psi \in \mathrm{Hom}(M_{n-1}, G)$ to be $\varphi \circ (d|_{D_n})^{-1}$ on Z_{n-1} and 0 on D_{n-1}. For $u \in Z_n$ and $v \in D_n$,

$$(\psi \circ d)(u+v) = \psi(dv) = \varphi(v) = \varphi(u+v)$$

so $\varphi = \psi \circ d$ is a coboundary. Hence the cohomology of $\mathrm{Hom}(M_\bullet, G)$ vanishes in every degree.

Step 3: Dualizing the cone sequence. In each degree the sequence of Step 1 is $0 \to C'_n \to C_{n-1} \oplus C'_n \to C_{n-1} \to 0$, the inclusion of and the projection onto direct summands. A homomorphism out of a direct sum is the pair of its restrictions, so applying $\mathrm{Hom}(-, G)$ gives a short exact sequence of cochain complexes

$$0 \to \mathrm{Hom}(\Sigma C_\bullet, G) \xrightarrow{\pi^*} \mathrm{Hom}(M_\bullet, G) \xrightarrow{\iota^*} \mathrm{Hom}(C'_\bullet, G) \to 0$$

the dual maps being cochain maps since, for instance, $(\theta \circ \pi) \circ d = (\theta \circ (-\partial)) \circ \pi$. A cochain complex is a chain complex after reindexing, so theorem 3.4.3 applies. The cohomology of $\mathrm{Hom}(\Sigma C_\bullet, G)$ in degree $n+1$ is $H^n(C;G)$, the sign of the differential changing no kernel and no image, and by Step 2 the terms belonging to $M_\bullet$ vanish, so the connecting homomorphisms $H^n(C';G) \to H^n(C;G)$ are isomorphisms. To identify them, let $\varphi' \in \mathrm{Hom}(C'_n, G)$ be a cocycle and extend it to $\tilde{\varphi} \in \mathrm{Hom}(M_n, G)$ by $\tilde{\varphi}(c,c') = \varphi'(c')$. For $(c,c') \in M_{n+1} = C_n \oplus C'_{n+1}$,

$$(\tilde{\varphi} \circ d)(c,c') = \varphi'\big(\partial c' + \alpha(c)\big) = \varphi'\big(\alpha(c)\big)$$

since $\varphi' \circ \partial = 0$. So $\tilde{\varphi} \circ d = \pi^*(\varphi' \circ \alpha)$, and the connecting homomorphism sends $[\varphi']$ to $[\varphi' \circ \alpha] = \alpha^*[\varphi']$. Hence α^* is an isomorphism in every degree, completing the proof.

Note that instead of using abelian groups (i.e. $\mathbb{Z}$-modules), we could use general R-modules. If R is a PID, then then the universal coefficient theorem is exactly the same, while if R is not then we may have longer free resolutions. If $R = F$ is a field, then any F-module is always free, hence we always have length 0 free resolutions.

Some remarks are in order: when $n = 0$, then Ext $= 0$, and so we simply get

$$H^0(X;G) \cong \mathrm{Hom}(H_0(X), G)$$

Explicitly, we can see this since singular 0-simplices are just points, and a cochain in $C^0(X;G)$ is an arbitrary function $\varphi : X \to G$, not necessarily continuous. For this to be a cocycle, we have that

each 1-simplex $\sigma[v_0, v_1] \to X$ w have

$$\delta(\varphi)(\sigma) = \varphi(\partial\sigma) = \varphi(\sigma(v_1)) - \varphi(\sigma(v_1)) = 0$$

Hence, it is equivalent to saying that φ is constant on path components of X. Thus, $H^0(X; G)$ is all the function from path-components of X to G, which is the same as $\mathrm{Hom}(H_0(X), G)$.

Similarly, since $H_0(X)$ is free, $\mathrm{Ext}(H_0(X), G)0$, so

$$H^1(X; G) \cong \mathrm{Hom}(H_1(X), G)$$

Finally, when $G = F$ is a field, then Ext_F vanishes, and so

$$H^n(X; F) \cong \mathrm{Hom}_F(H_n(X; F), F)$$

Over a field the theorem becomes a duality between homology and cohomology.

Corollary 4.1.11: Cohomology with Field Coefficients is the Dual of Homology

Let F be a field and X a topological space. For every n the Kronecker evaluation pairing induces a natural isomorphism of F-vector spaces

$$H^n(X; F) \xrightarrow{\cong} \mathrm{Hom}_F\big(H_n(X; F), F\big)$$

so cohomology with coefficients in F is the F-linear dual of homology with coefficients in F. When $H_n(X; F)$ is finite-dimensional the two spaces have the same dimension,

$$\dim_F H^n(X; F) = \dim_F H_n(X; F)$$

so the n-th Betti number is computed equally by homology or cohomology over F.

***Proof*:**
Over the field F the singular chain groups $C_\bullet(X; F)$ are F-vector spaces and the cochain complex is their F-dual, $C^\bullet(X; F) = \mathrm{Hom}_F(C_\bullet(X; F), F)$. Dualizing a linear map is exact on vector spaces, since every short exact sequence of F-vector spaces splits, so $\mathrm{Hom}_F(-, F)$ commutes with passage to homology and $H^n\big(\mathrm{Hom}_F(C_\bullet(X; F), F)\big) \cong \mathrm{Hom}_F\big(H_n(X; F), F\big)$. Equivalently, in the universal coefficient theorem (theorem 4.1.8) the term $\mathrm{Ext}_F(H_{n-1}(X; F), F)$ vanishes because every F-module is free, leaving the evaluation map $H^n(X; F) \to \mathrm{Hom}_F(H_n(X; F), F)$ an isomorphism whose naturality is that of the coefficient sequence displayed above. A linear map and its dual have the same rank, so the two dimensions agree whenever they are finite, completing the proof.

Theorem 4.1.12: Simplicial and Singular Cohomology

Let X be a Δ-complex, $A \subseteq X$ a subcomplex and G an abelian group. Write $H^n_\Delta(X, A; G)$ for the cohomology of $\mathrm{Hom}(\Delta_\bullet(X, A), G)$, where $\Delta_\bullet(X, A) = \Delta_\bullet(X)/\Delta_\bullet(A)$, and $H^n(X, A; G)$ for the cohomology of $\mathrm{Hom}(C_\bullet(X, A), G)$, both in the sense of definition 4.1.4. Then the canonical chain map $\theta\colon \Delta_\bullet(X, A) \to C_\bullet(X, A)$, which sends each simplex to its characteristic map, induces isomorphisms

$$\theta^*\colon H^n(X, A; G) \to H^n_\Delta(X, A; G)$$

for every n. For $A = \emptyset$ these are the absolute groups $H^n(X; G)$ and $H^n_\Delta(X; G)$.

***Proof*:**
Both chain complexes consist of free abelian groups: $\Delta_n(X, A)$ is free on the n-simplices of X not in A, and $C_n(X, A)$ is free on the singular n-simplices of X whose image is not contained in A, each being the quotient of a free group by the subgroup spanned by part of its basis. By theorem 3.4.32 the chain map θ induces isomorphisms $H^\Delta_n(X, A) \to H_n(X, A)$ for every $n \geq 0$, and both complexes vanish in negative degrees. So corollary 4.1.10 makes θ^* an isomorphism for every n and every G, as we sought to show.

Definition 4.1.13: Reduced Cohomology Groups

Let X be a nonempty space and G an abelian group, and let $\tilde{C}_\bullet(X)$ be the augmented chain complex of definition 3.2.6, with $\tilde{C}_{-1}(X) = \mathbb{Z}$ and boundary $\epsilon\colon C_0(X) \to \mathbb{Z}$ out of degree 0. The *reduced cohomology groups* of X with coefficients in G are

$$\tilde{H}^n(X; G) := H^n\big(\tilde{C}_\bullet(X); G\big)$$

the cohomology of the cochain complex $\mathrm{Hom}(\tilde{C}_\bullet(X), G)$ in the sense of definition 4.1.4.

Proposition 4.1.14: Reduced and Unreduced Cohomology

Let X be a nonempty space and G an abelian group.

1. $\tilde{H}^n(X; G) = H^n(X; G)$ for $n \geq 1$, and $\tilde{H}^n(X; G) = 0$ for $n < 0$.
2. Sending $g \in G$ to the class of the constant 0-cochain with value g defines an injective homomorphism $\epsilon^*\colon G \to H^0(X; G)$ whose cokernel is $\tilde{H}^0(X; G)$, and $H^0(X; G) \cong \tilde{H}^0(X; G) \oplus G$.
3. If $\tilde{H}_n(X) = 0$ for every n, then $\tilde{H}^n(X; G) = 0$ for every n.

***Proof*:**
Evaluation at 1 identifies $\mathrm{Hom}(\tilde{C}_{-1}(X), G) = \mathrm{Hom}(\mathbb{Z}, G)$ with G, and under this identification the coboundary out of degree -1 is the dual ϵ^* of the augmentation, sending g to the cochain $g\epsilon$, which takes the value g on every singular 0-simplex. In degrees $n \geq 0$ the augmented cochain

complex has the groups and the coboundaries of $C^\bullet(X;G)$.

1. For $n \geq 1$ the groups in degrees $n-1$, n and $n+1$ and the coboundaries between them are those of $C^\bullet(X;G)$, so $\tilde{H}^n(X;G) = H^n(X;G)$. There are no cochains in degrees below -1. In degree -1, $\tilde{H}^{-1}(X;G) = \ker \epsilon^*$, which is zero: X has a point x_0, and $(g\epsilon)(x_0) = g$ for x_0 read as a singular 0-simplex.
2. Since $\epsilon \circ \partial_1 = 0$, each $g\epsilon$ is a 0-cocycle, and $\tilde{H}^0(X;G) = \ker \delta^0 / \mathrm{im}\epsilon^* = H^0(X;G)/\epsilon^*(G)$. Evaluation at the 0-simplex x_0 defines a homomorphism $r\colon H^0(X;G) = \ker \delta^0 \to G$ with $r \circ \epsilon^* = \mathrm{id}_G$. Hence ϵ^* is injective, and lemma 3.4.34, applied to $0 \to G \xrightarrow{\epsilon^*} H^0(X;G) \to \tilde{H}^0(X;G) \to 0$ with the retraction r, gives $H^0(X;G) \cong \tilde{H}^0(X;G) \oplus G$.
3. The augmented complex consists of free abelian groups. If its homology vanishes, the zero chain map from the zero complex into it is an isomorphism on homology, so by corollary 4.1.10 its cohomology is that of the zero complex, which is 0.

This completes the proof.

Applying $\mathrm{Hom}(-,G)$ to the relative chain complex of definition 3.4.1 produces cohomology groups for a pair, and because that complex is free the dualization loses no exactness, so the pair still has a long exact sequence.

Definition 4.1.15: Relative Cohomology Groups

Let X be a topological space, $A \subseteq X$ a subspace and G an abelian group. Write $C^\bullet(X,A;G) = \mathrm{Hom}(C_\bullet(X,A),G)$ for the dual of the relative chain complex of definition 3.4.1. The *relative cohomology groups* of the pair (X,A) with coefficients in G are

$$H^n(X,A;G) := H^n\big(C_\bullet(X,A);G\big)$$

in the sense of definition 4.1.4. Since $C_n(X,A) = C_n(X)/C_n(A)$, a relative n-cochain is a homomorphism $C_n(X) \to G$ vanishing on $C_n(A)$. For $A = \emptyset$ the relative chain complex is $C_\bullet(X)$ and the relative groups are the absolute groups $H^n(X;G)$.

Proposition 4.1.16: Cochains and Cohomology Are Contravariant Functors

Let G be an abelian group and $f\colon (X,A) \to (Y,B)$ a map of pairs. Then $f^\sharp = \mathrm{Hom}(f_\sharp, G)$, $f^\sharp(\varphi) = \varphi \circ f_\sharp$, is a cochain map $C^\bullet(Y,B;G) \to C^\bullet(X,A;G)$, and it induces homomorphisms $f^*\colon H^n(Y,B;G) \to H^n(X,A;G)$. These satisfy $(g \circ f)^\sharp = f^\sharp \circ g^\sharp$, $(g \circ f)^* = f^* \circ g^*$ and $\mathrm{id}^* = \mathrm{id}$, so $C^\bullet(-;G)$ and each $H^n(-;G)$ are contravariant functors on pairs of spaces and maps of pairs, and on spaces when $A = B = \emptyset$.

***Proof*:**
By theorem 3.4.5(4), $f_\sharp\colon C_\bullet(X,A) \to C_\bullet(Y,B)$ is a chain map with $(g \circ f)_\sharp = g_\sharp \circ f_\sharp$ and $\mathrm{id}_\sharp = \mathrm{id}$. Since $\mathrm{Hom}(-,G)$ is a contravariant functor (example 1.1.4), $(g \circ f)^\sharp = f^\sharp \circ g^\sharp$, $\mathrm{id}^\sharp = \mathrm{id}$, and $f^\sharp\delta(\varphi) = \varphi \circ \partial \circ f_\sharp = \varphi \circ f_\sharp \circ \partial = \delta f^\sharp(\varphi)$, so $f^\sharp$ is a cochain map. A cochain complex is a chain complex after reindexing, so proposition 3.1.13 gives the induced maps f^* and the identities for

them, completing the proof.

Theorem 4.1.17: The Long Exact Sequence of a Pair in Cohomology

Let X be a topological space, $A \subseteq X$ a subspace and G an abelian group. Then the sequence

$$\cdots \to H^n(X, A; G) \to H^n(X; G) \to H^n(A; G) \xrightarrow{\delta} H^{n+1}(X, A; G) \to \dots$$

is exact, the map $H^n(X, A; G) \to H^n(X; G)$ being the one induced by the quotient chain map $C_\bullet(X) \to C_\bullet(X, A)$ and the map $H^n(X; G) \to H^n(A; G)$ the restriction induced by the inclusion $A \hookrightarrow X$. The connecting map δ carries the class of a cocycle $\varphi \in \mathrm{Hom}(C_n(A), G)$ to the class of $\tilde{\varphi} \circ \partial$, where $\tilde{\varphi} \in \mathrm{Hom}(C_n(X), G)$ is any extension of φ. The sequence is natural in the pair: a map of pairs $f\colon (X, A) \to (Y, B)$ induces homomorphisms f^* from each group of the sequence of (Y, B) to the corresponding group of the sequence of (X, A), and these commute with all maps of the two sequences, δ included.

***Proof*:**
The relative chain complex $C_\bullet(X, A) = C_\bullet(X)/C_\bullet(A)$ of definition 3.4.1 is free abelian: $C_n(X)$ is free on the singular n-simplices of X and $C_n(A)$ is the subgroup spanned by those whose image lies in A, a subset of that basis, so the quotient is free on the remaining basis elements. Freeness of $C_n(X, A)$ lets one choose a preimage in $C_n(X)$ for each of its basis elements, which produces a section of $C_n(X) \twoheadrightarrow C_n(X, A)$ and hence, by lemma 3.4.34, a decomposition $C_n(X) \cong C_n(A) \oplus C_n(X, A)$. A homomorphism out of a direct sum is the pair of its restrictions, so applying $\mathrm{Hom}(-, G)$ returns a short exact sequence in each degree and therefore a short exact sequence of cochain complexes

$$0 \to C^\bullet(X, A; G) \to C^\bullet(X; G) \to C^\bullet(A; G) \to 0$$

A cochain complex is a chain complex after reindexing, so the algebraic snake lemma (theorem 3.4.3) applies and yields the displayed long exact sequence. The connecting map is computed by the chase of theorem 3.4.3: a cocycle $\varphi \in \mathrm{Hom}(C_n(A), G)$ is extended to $\tilde{\varphi} \in \mathrm{Hom}(C_n(X), G)$, and $\delta[\varphi]$ is the class of $\tilde{\varphi} \circ \partial$, which vanishes on $C_{n+1}(A)$ and so lives in $\mathrm{Hom}(C_{n+1}(X, A), G)$.

For naturality, a map of pairs $f\colon (X, A) \to (Y, B)$ has $f_\sharp(C_\bullet(A)) \subseteq C_\bullet(B)$, so $f_\sharp$ gives a commutative diagram of short exact sequences of chain complexes from $0 \to C_\bullet(A) \to C_\bullet(X) \to C_\bullet(X, A) \to 0$ to $0 \to C_\bullet(B) \to C_\bullet(Y) \to C_\bullet(Y, B) \to 0$, the left square commuting since $f(A) \subseteq B$ and the right square since the relative $f_\sharp$ is induced on the quotients. Applying $\mathrm{Hom}(-, G)$ reverses every arrow and keeps the squares commutative, since $\mathrm{Hom}(u \circ v, G) = \mathrm{Hom}(v, G) \circ \mathrm{Hom}(u, G)$. This gives a commutative diagram of short exact sequences of cochain complexes from the sequence of (Y, B) to the sequence of (X, A), whose vertical maps $f^\sharp\colon \varphi \mapsto \varphi \circ f_\sharp$ are cochain maps because $f_\sharp$ commutes with ∂. Proposition 3.4.4, applied to this diagram after reindexing cochain complexes as chain complexes, gives the commuting ladder between the two long exact sequences, including the squares that contain δ, as we sought to show.

Proposition 4.1.18: Reduced Cohomology as Relative Cohomology

Let X be a nonempty space and G an abelian group.

1. For every point $x_0 \in X$, the chain map $\rho\colon \tilde{C}_\bullet(X) \to C_\bullet(X, x_0)$ that is the quotient map in degrees $n \geq 0$ and zero in degree -1 induces isomorphisms $\rho^*\colon H^n(X, x_0; G) \to \tilde{H}^n(X; G)$ for every n.
2. For every nonempty subspace $A \subseteq X$ there is a long exact sequence
$$\cdots \to H^n(X, A; G) \to \tilde{H}^n(X; G) \to \tilde{H}^n(A; G) \xrightarrow{\delta} H^{n+1}(X, A; G) \to \cdots$$
whose maps are induced, as in theorem 4.1.17, by the quotient chain map, by the inclusion $A \hookrightarrow X$, and by the connecting homomorphism.
3. If (X, A) is a good pair (definition 3.4.13), the quotient map $q\colon (X, A) \to (X/A, A/A)$ induces isomorphisms $q^*\colon H^n(X/A, A/A; G) \to H^n(X, A; G)$, so that $H^n(X, A; G) \cong \tilde{H}^n(X/A; G)$ for every n by (1).

***Proof*:**

The augmented complex $\tilde{C}_\bullet(X)$ and every relative chain complex consist of free abelian groups, the latter as shown in the proof of theorem 4.1.17.

1. The augmented complex $\tilde{C}_\bullet(x_0)$ of the one-point space is a subcomplex of $\tilde{C}_\bullet(X)$, equal to it in degree -1, and the quotient is $C_n(X)/C_n(x_0) = C_n(X, x_0)$ in degrees $n \geq 0$ and zero in degree -1. So
$$0 \to \tilde{C}_\bullet(x_0) \to \tilde{C}_\bullet(X) \xrightarrow{\rho} C_\bullet(X, x_0) \to 0$$
is a short exact sequence of chain complexes. The reduced homology of a point vanishes in every degree (corollary 3.3.9), so its long exact sequence (theorem 3.4.3) makes $\rho_*\colon \tilde{H}_n(X) \to H_n(X, x_0)$ an isomorphism for every n, and corollary 4.1.10 makes ρ^* an isomorphism for every n.
2. The quotient $\tilde{C}_\bullet(X)/\tilde{C}_\bullet(A)$ is $C_\bullet(X, A)$ in degrees $n \geq 0$ and zero in degree -1, so $0 \to \tilde{C}_\bullet(A) \to \tilde{C}_\bullet(X) \to C_\bullet(X, A) \to 0$ is a short exact sequence of chain complexes. Its quotient term is free in each degree, so choosing a preimage of each basis element gives a section of the quotient map, and lemma 3.4.34 decomposes each $\tilde{C}_n(X)$ as $\tilde{C}_n(A) \oplus C_n(X, A)$ compatibly with the two maps. As in the proof of theorem 4.1.17, applying $\mathrm{Hom}(-, G)$ then gives a short exact sequence of cochain complexes
$$0 \to C^\bullet(X, A; G) \to \mathrm{Hom}\big(\tilde{C}_\bullet(X), G\big) \to \mathrm{Hom}\big(\tilde{C}_\bullet(A), G\big) \to 0$$
and theorem 3.4.3, applied after reindexing, gives the displayed exact sequence, A being nonempty so that $\tilde{H}^n(A; G)$ is defined.
3. The pair $(X/A, A/A)$ consists of a space and a point. By corollary 3.4.12, which applies because (X, A) is a good pair, the chain map $q_\sharp\colon C_\bullet(X, A) \to C_\bullet(X/A, A/A)$ induces isomorphisms on homology in every degree. Both complexes are free, so corollary 4.1.10 makes q^* an isomorphism. Part (1), applied to X/A with $x_0 = A/A$, identifies $H^n(X/A, A/A; G)$ with $\tilde{H}^n(X/A; G)$.

This completes the proof.

Theorem 4.1.19: Excision in Cohomology

Let G be an abelian group and let $Z \subseteq A \subseteq X$ with $\overline{Z} \subseteq \text{int}A$. Then the inclusion $\iota\colon (X-Z, A-Z) \hookrightarrow (X, A)$ induces isomorphisms

$$\iota^*\colon H^n(X, A; G) \to H^n(X-Z, A-Z; G) \qquad \forall n$$

Equivalently, for subspaces $A, B \subseteq X$ with $\text{int}A \cup \text{int}B = X$, the inclusion $(B, A\cap B) \hookrightarrow (X, A)$ induces isomorphisms $H^n(X, A; G) \to H^n(B, A\cap B; G)$ for every n.

***Proof*:**
The relative chain complexes $C_\bullet(X-Z, A-Z)$ and $C_\bullet(X, A)$ consist of free abelian groups (as in the proof of theorem 4.1.17), and by theorem 3.4.9 the chain map $\iota_\sharp$ between them induces isomorphisms on homology in every degree $n \geq 0$, both complexes vanishing in negative degrees. By corollary 4.1.10 the dual map ι^* is an isomorphism on cohomology with coefficients in G in every degree. For the second form, the chain map $C_\bullet(B, A\cap B) \to C_\bullet(X, A)$ induced by the inclusion is again a map of free complexes, and it is an isomorphism on homology by the second form of theorem 3.4.9, so corollary 4.1.10 applies to it as well, as we sought to show.

The skeletal filtration of a CW complex computes cohomology as well as homology, and the cellular cochain complex it produces is the $\text{Hom}(-, G)$ dual of the cellular chain complex.

Proposition 4.1.20: Cellular Cohomology

Let X be a CW complex and G an abelian group, and write $d^n\colon H^n(X^n, X^{n-1}; G) \to H^{n+1}(X^{n+1}, X^n; G)$ for the cellular coboundary maps constructed in the proof below. Then

$$H^n(X; G) \cong \ker d^n / \text{im} d^{n-1}$$

Furthermore

$$\left(H^\bullet(X^\bullet, X^{\bullet-1}; G),\ d^\bullet\right) \cong \left(\text{Hom}(H_\bullet(X^\bullet, X^{\bullet-1}), G),\ \text{Hom}(d_\bullet, G)\right)$$

i.e. the cellular cochain complex is isomorphic to the dual of the cellular chain complex after applying the $\text{Hom}(-, G)$ functor.

***Proof*:**
Throughout, the relative groups $H^n(X, A; G)$ are those of definition 4.1.15, and every long exact sequence of a pair invoked below is the one of theorem 4.1.17.

Step 1: The relative groups of a skeletal pair. The complex $C_\bullet(X^n, X^{n-1})$ is free abelian, so theorem 4.1.8 gives, for each k, the exact sequence

$$0 \to \text{Ext}(H_{k-1}(X^n, X^{n-1}), G) \to H^k(X^n, X^{n-1}; G) \xrightarrow{h} \text{Hom}(H_k(X^n, X^{n-1}), G) \to 0$$

By lemma 3.5.9(1) the group $H_j(X^n, X^{n-1})$ vanishes for $j \neq n$ and is free abelian for $j = n$. Take $k \neq n$. Then $\mathrm{Hom}(H_k(X^n, X^{n-1}), G) = 0$, and the Ext term vanishes too: if $k-1 \neq n$ its argument is the zero group, and if $k-1 = n$ its argument is free, so proposition 4.1.9(2) applies. Hence

$$H^k(X^n, X^{n-1}; G) = 0 \qquad (k \neq n)$$

For $k = n$ the Ext term has argument $H_{n-1}(X^n, X^{n-1}) = 0$, so

$$h\colon H^n(X^n, X^{n-1}; G) \xrightarrow{\cong} \mathrm{Hom}(H_n(X^n, X^{n-1}), G)$$

is an isomorphism.

Step 2: Vanishing above the dimension of the skeleton. The long exact sequence of (X^n, X^{n-1}) contains

$$H^k(X^n, X^{n-1}; G) \to H^k(X^n; G) \to H^k(X^{n-1}; G) \to H^{k+1}(X^n, X^{n-1}; G)$$

whose outer terms vanish by Step 1 whenever $k \neq n$ and $k+1 \neq n$, that is whenever $k \notin \{n-1, n\}$. For such k the restriction $H^k(X^n; G) \to H^k(X^{n-1}; G)$ is an isomorphism. Now induct on n to prove $H^k(X^n; G) = 0$ for $k > n$. For $n = 0$ the skeleton X^0 is a discrete set of points, so $H_k(X^0) = 0$ for $k > 0$ and $H_0(X^0)$ is free abelian, and the universal coefficient sequence makes $H^k(X^0; G) = 0$ for $k > 0$, the Ext term vanishing by proposition 4.1.9(2) when $k = 1$ and because the Ext argument group vanishes when $k > 1$. For the inductive step let $k > n$. Then $k \notin \{n-1, n\}$, so $H^k(X^n; G) \cong H^k(X^{n-1}; G)$, which is zero by the inductive hypothesis since $k > n > n-1$.

Step 3: Passing from X to the $(n+1)$-skeleton. The long exact sequence in homology of the pair (X, X^{n+1}) (definition 3.4.2 and theorem 3.4.3) contains

$$H_k(X^{n+1}) \to H_k(X) \to H_k(X, X^{n+1}) \to H_{k-1}(X^{n+1}) \to H_{k-1}(X)$$

For $k \leq n+1$ the left map is surjective, being an isomorphism when $k < n+1$ and a surjection when $k = n+1$ by lemma 3.5.9(3), while the right map is injective, being an isomorphism because $k - 1 < n + 1$. Exactness forces $H_k(X, X^{n+1}) = 0$ for $k \leq n+1$. The complex $C_\bullet(X, X^{n+1})$ is free abelian, so theorem 4.1.8 gives $H^k(X, X^{n+1}; G) = 0$ for $k \leq n+1$, both outer terms of its sequence having zero argument. The cohomology long exact sequence of (X, X^{n+1}) then contains

$$H^n(X, X^{n+1}; G) \to H^n(X; G) \to H^n(X^{n+1}; G) \to H^{n+1}(X, X^{n+1}; G)$$

with both outer terms zero, so restriction is an isomorphism $H^n(X; G) \cong H^n(X^{n+1}; G)$.

Step 4: The cellular cochain complex. Write $j^n\colon H^n(X^n, X^{n-1}; G) \to H^n(X^n; G)$ for the map in the long exact sequence of (X^n, X^{n-1}) and $\delta^n\colon H^n(X^n; G) \to H^{n+1}(X^{n+1}, X^n; G)$ for the connecting map of (X^{n+1}, X^n), and set

$$d^n = \delta^n \circ j^n$$

These are the maps named in the statement. They compose to zero: $d^n d^{n-1} = \delta^n j^n \delta^{n-1} j^{n-1}$, and δ^{n-1} followed by j^n are consecutive maps of the long exact sequence of (X^n, X^{n-1}), so $j^n \delta^{n-1} = 0$ and the whole composite vanishes.

Step 5: Computing the cohomology of that complex. The long exact sequence of (X^{n+1}, X^n) contains

$$H^n(X^{n+1}, X^n; G) \to H^n(X^{n+1}; G) \xrightarrow{r} H^n(X^n; G) \xrightarrow{\delta^n} H^{n+1}(X^{n+1}, X^n; G)$$

whose first term vanishes by Step 1, since $n \neq n+1$. So r is injective with image $\ker \delta^n$, giving $H^n(X^{n+1}; G) \cong \ker \delta^n$. The long exact sequence of (X^n, X^{n-1}) contains

$$H^{n-1}(X^{n-1}; G) \xrightarrow{\delta^{n-1}} H^n(X^n, X^{n-1}; G) \xrightarrow{j^n} H^n(X^n; G) \to H^n(X^{n-1}; G)$$

whose last term vanishes by Step 2, since $n > n-1$. So j^n is surjective with kernel $\mathrm{im}\delta^{n-1}$. Since $\ker d^n = (j^n)^{-1}(\ker \delta^n)$ and j^n is surjective, j^n carries $\ker d^n$ onto $\ker \delta^n$ with kernel $\ker j^n = \mathrm{im}\delta^{n-1}$, which is contained in $\ker d^n$. Hence

$$\frac{\ker d^n}{\mathrm{im}\delta^{n-1}} \cong \ker \delta^n$$

Finally $\mathrm{im}\delta^{n-1} = \mathrm{im}d^{n-1}$, because $d^{n-1} = \delta^{n-1} j^{n-1}$ and j^{n-1} is surjective by the same argument one degree down. Chaining this with Step 3,

$$H^n(X; G) \cong H^n(X^{n+1}; G) \cong \ker \delta^n \cong \frac{\ker d^n}{\mathrm{im}d^{n-1}}$$

Step 6: The isomorphism with the dual cellular chain complex. Step 1 makes each h an isomorphism, so what is left is that h commutes with the differentials. Recall from Step 6 of the proof of theorem 4.1.8 the description of h: for a complex $D_\bullet$ of free abelian groups and a cocycle $\varphi \in \mathrm{Hom}(D_n, G)$, the homomorphism $h([\varphi])$ sends a homology class $[z]$ to $\varphi(z)$. Two consequences follow at once.

First, h is natural. If $u_\bullet \colon D_\bullet \to E_\bullet$ is a chain map of free complexes and φ is a cocycle on $E_\bullet$, then $u^*[\varphi] = [\varphi \circ u]$ and

$$h(u^*[\varphi])([z]) = \varphi(u(z)) = h([\varphi])(u_*[z])$$

so $h \circ u^* = \mathrm{Hom}(u_*, G) \circ h$. Applying this to the quotient chain map $C_\bullet(X^n) \to C_\bullet(X^n, X^{n-1})$, whose induced map on homology is the map j_n of definition 3.5.10 and whose dual induces j^n, gives

$$h(j^n \alpha) = h(\alpha) \circ j_n$$

Second, h turns δ^n into the dual of ∂_{n+1}. Let $\beta \in H^n(X^n; G)$ be represented by a cocycle $\varphi \in \mathrm{Hom}(C_n(X^n), G)$. By theorem 4.1.17, $\delta^n \beta$ is represented by $\tilde{\varphi} \circ \partial$ for any extension $\tilde{\varphi}$ of φ to $C_n(X^{n+1})$. A class in $H_{n+1}(X^{n+1}, X^n)$ is represented by a chain $c \in C_{n+1}(X^{n+1})$ with

$\partial c \in C_n(X^n)$, and $\partial_{n+1}[c] = [\partial c]$ by the same chase in homology. Since $\tilde{\varphi}$ agrees with φ on $C_n(X^n)$,

$$h(\delta^n\beta)([c]) = (\tilde{\varphi} \circ \partial)(c) = \varphi(\partial c) = h(\beta)([\partial c]) = h(\beta)(\partial_{n+1}[c])$$

so $h(\delta^n\beta) = h(\beta) \circ \partial_{n+1}$.

Combining the two, and using $d_{n+1} = j_n\partial_{n+1}$ from definition 3.5.10,

$$h(d^n\alpha) = h(\delta^n j^n\alpha) = h(j^n\alpha) \circ \partial_{n+1} = h(\alpha) \circ j_n\partial_{n+1} = h(\alpha) \circ d_{n+1}$$

which says exactly that h is a map of cochain complexes from $(H^\bullet(X^\bullet, X^{\bullet-1}; G), d^\bullet)$ to the $\mathrm{Hom}(-, G)$ dual of the cellular chain complex. Being an isomorphism in each degree, it is an isomorphism of cochain complexes, as we sought to show.

Example 4.1.21: Cohomology Groups

The Δ-complexes of example 3.1.14 are used again.

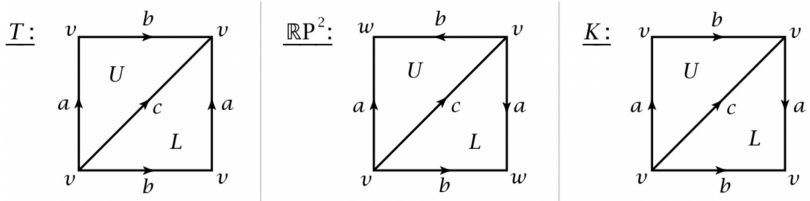


Figure 4.1: Reference Figure for Torus, Projective space, and Klein Bottle

A computation may use either the Δ-complex or the cellular complex: the Δ-complex computation is singular cohomology by theorem 4.1.12, and the cellular one by proposition 4.1.20.

1. Let $X = T = S^1 \times S^1$ with the Δ-complex of example 3.1.14: one vertex v, three edges a, b, c and two 2-simplices U, L, with $\partial a = \partial b = \partial c = v - v = 0$ and $\partial U = \partial L = a + b - c$. Write $v^*, a^*, b^*, c^*, U^*, L^*$ for the dual bases, x^* taking the value 1 on x and 0 on the other basis elements. Dualizing the chain complex $0 \to \Delta_2(T) \cong \mathbb{Z}^2 \to \Delta_1(T) \cong \mathbb{Z}^3 \to \Delta_0(T) \cong \mathbb{Z} \to 0$ gives

   $$0 \to \mathbb{Z} \xrightarrow{\delta^0} \mathbb{Z}^3 \xrightarrow{\delta^1} \mathbb{Z}^2 \to 0$$

   with $\delta^0 = 0$, since $(\delta^0 v^*)(e) = v^*(\partial e) = 0$ for every edge e, and with $\delta^1\varphi = \varphi \circ \partial_2$, so that

   $$\delta^1 a^* = U^* + L^* \qquad \delta^1 b^* = U^* + L^* \qquad \delta^1 c^* = -U^* - L^*$$

   Hence $H^0(T; \mathbb{Z}) = \mathbb{Z}$. The kernel of δ^1 consists of the cochains $xa^* + yb^* + zc^*$ with $x + y - z = 0$, free with basis $a^* + c^*$, $b^* + c^*$, so $H^1(T; \mathbb{Z}) \cong \mathbb{Z}^2$. The image of δ^1 is generated by $U^* + L^*$, which together with L^* forms a basis of $\mathbb{Z}^2$, so $H^2(T; \mathbb{Z}) \cong \mathbb{Z}$, and

   $$H^n(T; \mathbb{Z}) = \begin{cases} \mathbb{Z} & n = 0 \\ \mathbb{Z}^2 & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & n \geq 3 \end{cases}$$

With $G = \mathbb{Z}/2$ the same formulas hold modulo 2: $\delta^0 = 0$, and δ^1 sends each of a^*, b^*, c^* to $U^* + L^*$. Its kernel has dimension 2 and its image dimension 1, so

$$H^n(T; \mathbb{Z}/2) = \begin{cases} \mathbb{Z}/2 & n = 0 \\ (\mathbb{Z}/2)^2 & n = 1 \\ \mathbb{Z}/2 & n = 2 \\ 0 & n \geq 3 \end{cases}$$

2. Let $X = K$ be the Klein bottle, with the cell structure of example 3.5.15 for $N_2 = K$: one 0-cell, two 1-cells a_1, a_2 and one 2-cell e attached by the word $a_1^2 a_2^2$, so that $d_1 = 0$ and $d_2(e) = 2a_1 + 2a_2$, and

$$H_n(K) = \begin{cases} \mathbb{Z} & n = 0 \\ \mathbb{Z} \oplus \mathbb{Z}/2 & n = 1 \\ 0 & n \geq 2 \end{cases}$$

By proposition 4.1.20 the cellular cochain complex is the dual of the cellular chain complex,

$$0 \to \mathbb{Z} \xrightarrow{\delta^0} \mathbb{Z}^2 \xrightarrow{\delta^1} \mathbb{Z} \to 0$$

with $\delta^0 = 0$ and $\delta^1 a_1^* = \delta^1 a_2^* = 2e^*$. The kernel of δ^1 is generated by $a_1^* - a_2^*$ and its image is $2\mathbb{Z}$, so

$$H^n(K; \mathbb{Z}) = \begin{cases} \mathbb{Z} & n = 0 \\ \mathbb{Z} & n = 1 \\ \mathbb{Z}/2 & n = 2 \\ 0 & n \geq 3 \end{cases}$$

3. For $n \geq 1$ and the reduced cohomology of definition 4.1.13, the cell structure of S^n with one 0-cell and one n-cell and the cellular computation of proposition 4.1.20, and, in degree 0, proposition 4.1.14(2), a 0-cocycle taking equal values at the ends of every path and hence being constant on the path-connected S^n, so that ϵ^* is onto, give

$$\tilde{H}^k(S^n; G) = \begin{cases} G & n = k \\ 0 & \text{otherwise} \end{cases}$$

One translation deserves a full treatment, since later chapters lean on it: the Mayer-Vietoris sequence of theorem 3.4.22 has a cohomology counterpart, with the arrows reversed and the connecting homomorphism raising degree. The proof dualizes the chain-level machinery already assembled for the homology version, and the dualization rests on two small algebraic facts, recorded first.

Lemma 4.1.22: Dualizing Split Exactness and Chain Homotopies

Let G be an abelian group.

1. If $0 \to A \xrightarrow{i} B \xrightarrow{j} C \to 0$ is a split short exact sequence of abelian groups, then
$$0 \to \mathrm{Hom}(C, G) \xrightarrow{j^*} \mathrm{Hom}(B, G) \xrightarrow{i^*} \mathrm{Hom}(A, G) \to 0$$
is a short exact sequence, and it splits.
2. If $f, g : C_\bullet \to D_\bullet$ are chain homotopic chain maps, then the dual maps $f^*, g^* : \mathrm{Hom}(D_\bullet, G) \to \mathrm{Hom}(C_\bullet, G)$ are cochain homotopic cochain maps. In particular, $\mathrm{Hom}(-, G)$ carries a chain homotopy equivalence to a cochain homotopy equivalence, and hence induces isomorphisms on cohomology.

***Proof*:**
For (1), exactness needs the splitting only at the last step. The map j^* is injective: if $\theta \circ j = 0$ then $\theta = 0$, since j is surjective. For exactness at $\mathrm{Hom}(B, G)$, the composite $i^* j^* = (ji)^* = 0$ gives $\mathrm{im} j^* \subseteq \ker i^*$. Conversely, if $\eta : B \to G$ vanishes on $\mathrm{im} i = \ker j$, then $\eta(b)$ depends only on $j(b)$, so η factors as $\theta \circ j$ for a homomorphism $\theta : C \to G$, that is $\eta \in \mathrm{im} j^*$. For surjectivity of i^*, choose a retraction $r : B \to A$ with $ri = \mathrm{id}_A$, which the splitting provides (lemma 3.4.34 converts a section of the quotient map into such a retraction). Then for any $h : A \to G$ the homomorphism $h \circ r$ restricts along i to h, so $i^*(h \circ r) = h$. The same r shows that the dual sequence splits: $r^* : \mathrm{Hom}(A, G) \to \mathrm{Hom}(B, G)$ satisfies $i^* r^* = (ri)^* = \mathrm{id}$, a section of i^*.

For (2), a dual map is a cochain map: for $\theta \in \mathrm{Hom}(D_n, G)$,
$$\delta(f^*\theta) = \theta \circ f_n \circ \partial_{n+1} = \theta \circ \partial_{n+1} \circ f_{n+1} = f^*(\delta\theta)$$
using that f is a chain map in the middle equality. Now let P be a chain homotopy from f to g (definition 3.3.4), a collection of homomorphisms $P_n : C_n \to D_{n+1}$ with
$$g_n - f_n = \partial_{n+1} P_n + P_{n-1} \partial_n$$
Define $Q : \mathrm{Hom}(D_n, G) \to \mathrm{Hom}(C_{n-1}, G)$ by $Q\theta = \theta \circ P_{n-1}$. Then for $\theta \in \mathrm{Hom}(D_n, G)$,
$$g^*\theta - f^*\theta = \theta \circ (g_n - f_n) = \theta \circ \partial_{n+1} \circ P_n + \theta \circ P_{n-1} \circ \partial_n = Q(\delta\theta) + \delta(Q\theta)$$
since $\theta \circ \partial_{n+1} = \delta\theta$ and $(Q\theta) \circ \partial_n = \delta(Q\theta)$. So Q is a cochain homotopy from f^* to g^*. Finally, if ρ is a homotopy inverse of a chain homotopy equivalence ι, so that $\rho\iota$ and $\iota\rho$ are chain homotopic to identities, then dualizing gives that $\iota^*\rho^* = (\rho\iota)^*$ and $\rho^*\iota^* = (\iota\rho)^*$ are cochain homotopic to identities, so ι^* is a cochain homotopy equivalence. A cochain complex is a chain complex after reindexing, so proposition 3.3.5 applies: ι^* and ρ^* induce mutually inverse maps on cohomology, so the maps ι^* induces are isomorphisms, completing the proof.

Proposition 4.1.23: Homotopy Invariance of Cohomology

Let $f, g\colon (X, A) \to (Y, B)$ be maps of pairs that are homotopic through maps of pairs, and let G be an abelian group. Then $f^* = g^*\colon H^n(Y, B; G) \to H^n(X, A; G)$ for every n.

***Proof*:**
By proposition 3.4.8 the chain maps $f_\sharp, g_\sharp\colon C_\bullet(X, A) \to C_\bullet(Y, B)$ are chain homotopic. By lemma 4.1.22(2) the dual maps $f^\sharp, g^\sharp\colon C^\bullet(Y, B; G) \to C^\bullet(X, A; G)$ are cochain homotopic. A cochain complex is a chain complex after reindexing, so by proposition 3.3.5 they induce the same homomorphisms $f^* = g^*$ on cohomology, as we sought to show.

Theorem 4.1.24: Mayer-Vietoris Sequence in Cohomology

Let $A, B \subseteq X$ with $X = \mathrm{int}A \cup \mathrm{int}B$, and let G be an abelian group. Then there is a long exact sequence

$$0 \to H^0(X;G) \to \cdots \to H^n(X;G) \xrightarrow{\psi^*} H^n(A;G) \oplus H^n(B;G) \xrightarrow{\varphi^*} H^n(A\cap B;G) \xrightarrow{\delta} H^{n+1}(X;G) \to \cdots$$

in which ψ^* is the pair of restrictions induced by the inclusions $A \hookrightarrow X$ and $B \hookrightarrow X$, and φ^* is the difference of the restrictions induced by the inclusions $A \cap B \hookrightarrow A$ and $A \cap B \hookrightarrow B$.

***Proof*:**
The proof dualizes the two chain-level ingredients of theorem 3.4.22, the short exact sequence of chain complexes built there and the small-chain comparison of lemma 3.4.11, with lemma 4.1.22 giving exactly what remains after dualization.

Stage 1: the chain-level sequence and its degreewise splitting. Write $C_n(A + B) = C_n(A) + C_n(B) \subseteq C_n(X)$ for the subcomplex of chains each of whose simplices has image in A or in B, as in Stage 1 of the proof of theorem 3.4.22. Stage 2 of that proof established the short exact sequence of chain complexes

$$0 \to C_\bullet(A \cap B) \xrightarrow{\varphi} C_\bullet(A) \oplus C_\bullet(B) \xrightarrow{\psi} C_\bullet(A + B) \to 0 \tag{4.10}$$

with $\varphi(x) = (x, -x)$ and $\psi(a, b) = a + b$. In each degree this is a sequence of free abelian groups: $C_n(A \cap B)$ is free on the simplices with image in $A \cap B$, the direct sum $C_n(A) \oplus C_n(B)$ is free on the two embedded copies of the simplices of A and of B, and $C_n(A + B)$ is free on the small simplices themselves, these being a subset of the basis of $C_n(X)$. A short exact sequence of abelian groups whose quotient is free splits: choose a preimage under ψ of each basis element of $C_n(A + B)$ and extend freely to a section of ψ. So (4.10) splits degreewise.

Stage 2: dualizing to a short exact sequence of cochain complexes. Apply $\mathrm{Hom}(-, G)$ to (4.10). By lemma 4.1.22(1) each degree yields a short exact sequence of abelian groups, and the dual maps are cochain maps by lemma 4.1.22(2), so the result is a short exact sequence of cochain complexes

$$0 \to C^\bullet_{A+B}(X; G) \xrightarrow{\psi^*} C^\bullet(A; G) \oplus C^\bullet(B; G) \xrightarrow{\varphi^*} C^\bullet(A \cap B; G) \to 0 \tag{4.11}$$

where $C^n_{A+B}(X; G) := \mathrm{Hom}(C_n(A+B), G)$ and the middle term uses the canonical identification $\mathrm{Hom}(M \oplus N, G) \cong \mathrm{Hom}(M, G) \oplus \mathrm{Hom}(N, G)$, a homomorphism on a direct sum being the pair

of its restrictions to the summands, an identification compatible with the coboundaries since the boundary of the middle term of (4.10) is $\partial \oplus \partial$. Under this identification the dual maps are explicit: $\psi^*\theta = (\theta|_{C_\bullet(A)}, \theta|_{C_\bullet(B)})$ restricts a small cochain to the chains lying in A and to those lying in B, and $\varphi^*(\alpha, \beta) = \alpha|_{C_\bullet(A\cap B)} - \beta|_{C_\bullet(A\cap B)}$, the sign coming from $\varphi(x) = (x, -x)$.

Stage 3: small cochains compute $H^n(X;G)$. By lemma 3.4.11 applied to the collection $U = \{A, B\}$, whose interiors cover X by hypothesis, the inclusion $\iota : C_\bullet(A+B) \hookrightarrow C_\bullet(X)$ is a chain homotopy equivalence. By lemma 4.1.22(2) its dual

$$\iota^* : C^\bullet(X;G) \to C^\bullet_{A+B}(X;G)$$

is a cochain homotopy equivalence and induces isomorphisms $\iota^* : H^n(X;G) \xrightarrow{\sim} H^n_{A+B}(X;G)$ for every n, writing $H^n_{A+B}(X;G)$ for the cohomology of $C^\bullet_{A+B}(X;G)$. As in the homology proof, this is the single point where the covering hypothesis enters.

Stage 4: assembly. A cochain complex is a chain complex after reindexing, so the algebraic snake lemma (theorem 3.4.3) applies to (4.11), with the connecting homomorphism now raising degree by one. The resulting long exact sequence reads

$$\cdots \to H^n_{A+B}(X;G) \xrightarrow{\psi^*} H^n(A;G) \oplus H^n(B;G) \xrightarrow{\varphi^*} H^n(A\cap B;G) \xrightarrow{\delta} H^{n+1}_{A+B}(X;G) \to \cdots$$

the middle term being identified by the fact that cohomology commutes with direct sums of cochain complexes, kernels and images being computed coordinatewise. The sequence begins with $0 \to H^0_{A+B}(X;G)$, since there are no cochains in negative degrees. Now substitute $H^n(X;G)$ for $H^n_{A+B}(X;G)$ throughout, using the isomorphisms ι^* of Stage 3. Replacing groups along isomorphisms, with the adjacent maps composed with the isomorphisms and their inverses, preserves exactness, so it remains only to identify the substituted maps. The composite $\psi^*\iota^*$ sends a cochain $\theta \in C^n(X;G)$ to the pair of its restrictions to chains in A and to chains in B, which is exactly the pair of maps induced by the inclusions $A \hookrightarrow X$ and $B \hookrightarrow X$, and the new connecting map is the composite $(\iota^*)^{-1}\delta$. The result is the long exact sequence of the statement, completing the proof.

Corollary 4.1.25: Naturality of the Mayer-Vietoris Sequence in Cohomology

The sequence of theorem 4.1.24 is natural in the triple (X, A, B): if $f : X \to Y$ is continuous with $f(A) \subseteq A'$ and $f(B) \subseteq B'$, where $X = \mathrm{int}A \cup \mathrm{int}B$ and $Y = \mathrm{int}A' \cup \mathrm{int}B'$, then the diagram

$$\begin{array}{ccccccccc}
\cdots \to & H^n(Y;G) & \to & H^n(A';G) \oplus H^n(B';G) & \to & H^n(A'\cap B';G) & \xrightarrow{\delta} & H^{n+1}(Y;G) & \to \cdots \\
& \downarrow{\scriptstyle f^*} & & \downarrow & & \downarrow & & \downarrow{\scriptstyle f^*} & \\
\cdots \to & H^n(X;G) & \to & H^n(A;G) \oplus H^n(B;G) & \to & H^n(A\cap B;G) & \xrightarrow{\delta} & H^{n+1}(X;G) & \to \cdots
\end{array}$$

commutes, the unlabelled vertical maps being induced by the restrictions of f.

***Proof*:**
The chain map $f_\sharp : C_\bullet(X) \to C_\bullet(Y)$ carries $C_\bullet(A)$ into $C_\bullet(A')$, $C_\bullet(B)$ into $C_\bullet(B')$, $C_\bullet(A\cap B)$ into $C_\bullet(A'\cap B')$, and hence $C_\bullet(A+B)$ into $C_\bullet(A'+B')$. The resulting vertical maps form

a commutative diagram from the sequence (4.10) of (X, A, B) to that of (Y, A', B'), both squares commuting at the level of chains: for instance $\psi(f_\sharp a, f_\sharp b) = f_\sharp a + f_\sharp b = f_\sharp \psi(a, b)$. Applying $\mathrm{Hom}(-, G)$ reverses all arrows and preserves commutativity, by functoriality, giving a commutative diagram from the dualized sequence (4.11) of (Y, A', B') to that of (X, A, B). Proposition 3.4.4, applied after the reindexing of Stage 4 above, yields a commuting ladder between the two long exact sequences in the $H^\bullet_{A+B}$ form. It remains to transport the ladder along the isomorphisms ι^* of Stage 3, and for that it suffices that the square

$$\begin{array}{ccc} H^n(Y;G) & \xrightarrow{\iota_Y^*} & H^n_{A'+B'}(Y;G) \\ \downarrow{\scriptstyle f^*} & & \downarrow \\ H^n(X;G) & \xrightarrow{\iota_X^*} & H^n_{A+B}(X;G) \end{array}$$

commutes. At the chain level $f_\sharp \iota_X = \iota_Y f_\sharp$, both composites being $f_\sharp$ applied to small chains, so the dual square commutes already at the level of cochains. Every square of the transported ladder is then a composite of commuting squares, including those containing the connecting maps $(\iota^*)^{-1}\delta$, completing the proof.

This is the sequence that the de Rham comparison in [32, The de Rham Map is a Morphism of Mayer-Vietoris Sequences] runs against the Mayer-Vietoris sequence of differential forms.

Proposition 4.1.26: Relative Mayer-Vietoris Sequence in Cohomology

Let $X = A \cup B$, $C \subseteq A$, $D \subseteq B$ and $Y = C \cup D$ be as in proposition 3.4.24, with $X = \mathrm{int}A \cup \mathrm{int}B$ and with Y the union of the interiors of C and D in Y, and let G be an abelian group. Write $C^n_{A+B}(X, Y; G) = \mathrm{Hom}(C_n(A+B, C+D), G)$, the group of homomorphisms $C_n(A+B) \to G$ vanishing on $C_n(C+D)$.

1. There is a short exact sequence of cochain complexes

   $$0 \to C^\bullet_{A+B}(X, Y; G) \xrightarrow{\psi^*} C^\bullet(A, C; G) \oplus C^\bullet(B, D; G) \xrightarrow{\varphi^*} C^\bullet(A \cap B, C \cap D; G) \to 0$$

   with $\psi^*\theta = (\theta|_{C_\bullet(A)}, \theta|_{C_\bullet(B)})$ and $\varphi^*(\alpha, \beta) = \alpha|_{C_\bullet(A\cap B)} - \beta|_{C_\bullet(A\cap B)}$.

2. The restriction $\iota^* \colon C^\bullet(X, Y; G) \to C^\bullet_{A+B}(X, Y; G)$, $\theta \mapsto \theta|_{C_\bullet(A+B)}$, induces isomorphisms on cohomology.

3. There is a long exact sequence

   $$\cdots \to H^n(X,Y;G) \to H^n(A,C;G) \oplus H^n(B,D;G) \to H^n(A\cap B, C\cap D;G) \xrightarrow{\delta} H^{n+1}(X,Y;G) \to \cdots$$

   whose first map is the pair of restrictions and whose second map is the difference of the restrictions, with $\delta = (\iota^*)^{-1}\delta'$ for the connecting map δ' of the sequence in (1).

***Proof*:**

1. In each degree the sequence of proposition 3.4.24(1) is a short exact sequence of free abelian groups. It splits, since a choice of preimage of each basis element of the third term defines a section of ψ, to which lemma 3.4.34 applies. By lemma 4.1.22(1), $\mathrm{Hom}(-, G)$

turns it into a short exact sequence in each degree, and the dual maps are cochain maps by lemma 4.1.22(2). Identify $\mathrm{Hom}(M \oplus N, G)$ with $\mathrm{Hom}(M,G) \oplus \mathrm{Hom}(N,G)$ and $\mathrm{Hom}(C_n(A,C),G)$ with $C^n(A,C;G)$ (definition 4.1.15). From $\psi(a,b) = a+b$ the dual map sends θ to $(a,b) \mapsto \theta(a) + \theta(b)$, the pair of restrictions, and from $\varphi(x) = (x,-x)$ the dual sends (α,β) to $x \mapsto \alpha(x) - \beta(x)$.

2. The complex $C_\bullet(X,Y)$ is free on the simplices of X not lying in Y, and $C_\bullet(A+B, C+D)$ is free by proposition 3.4.24(1). The chain map ι of proposition 3.4.24(2) induces isomorphisms on homology, so its dual ι^* induces isomorphisms on cohomology by corollary 4.1.10. Reading a cochain on $C_\bullet(X,Y)$ as a cochain on X vanishing on chains in Y, the dual of ι is restriction to $C_\bullet(A+B)$.

3. A cochain complex is a chain complex after reindexing, so theorem 3.4.3 applies to the sequence of (1), with connecting map δ' raising degree by one. Its middle terms are $H^n(A,C;G) \oplus H^n(B,D;G)$, cohomology commuting with direct sums of cochain complexes. Substituting $H^n(X,Y;G)$ for the cohomology of $C^\bullet_{A+B}(X,Y;G)$ along the isomorphisms of (2) preserves exactness. The composite $\psi^*\iota^*$ sends a cocycle on X vanishing on Y to the pair of its restrictions to A and to B, which induces the pair of restrictions on cohomology.

This completes the proof.

4.2 Cup Product

Let $G = R$ now be a ring. We shall show that we can put a graded ring structure on $C^\bullet(X;R)$. If you have done DeRham cohomology, the product we shall define can be thought of as the generalization of the wedge product and shall be behave very similarly:

Definition 4.2.1: Cup Product [on chains]

Let $f \in C^k(X;R)$ and $g \in C^\ell(X;R)$. Then the *cup product*, denoted $f \smile g \in C^{k+\ell}(X;R)$ is the cochain that evaluates on a singular simplex $\sigma : \Delta^{k+\ell} \to X$ is given by:

$$(f \smile g)(\sigma) = f(\sigma|_{[v_0,\dots,v_k]}) g(\sigma|_{[v_k,\dots,v_{k+\ell}]}) \in R$$

To see that the cup product works on cohomology classes, we require the following

Lemma 4.2.2: Derivation Property Of Cup Product

$$\delta(f \smile g) = (\delta f) \smile g + (-1)^k f \smile (\delta g)$$

***Proof*:**
Let $\sigma : \Delta^{k+\ell+1} \to X$. Then:

$$(\delta f) \smile g(\sigma) = \sum_{i}^{k+1} (-1)^i f(\sigma|_{v_0,\ldots,\hat{v}_i,\ldots,v_{k+1}}) g(\sigma|_{v_{k+1},\ldots,v_{k+\ell+1}})$$

$$(-1)^k (f \smile \delta g)(\sigma) = \sum_{i=k}^{k+\ell+1} (-1)^i f(\sigma|_{v_0,\ldots,\hat{v}_i,\ldots,v_k}) g(\sigma|_{v_k,\ldots,v_{k+\ell+1}})$$

Adding these together, the last term of the first some cancels with the first term of the second sum, the remaining terms are then exactly

$$\delta(f \smile g)(\sigma) = (f \smile g)(\partial\sigma)$$

since

$$\partial\sigma = \sum_{i}^{k+\ell+1} (-1)^i \sigma|_{[v_0,\ldots,\hat{v}_i,\ldots,v_{k+\ell+1}]}$$

completing the proof.

From this, we see that the cup product of two cocycles is a cocycles, and the product of a cocycles and a coboundary (or a coboundary and a cocycle) is a coboundary since

$$f \smile (\delta g) = \pm\delta(f \smile g)$$

if $\delta f = 0$ and

$$(\delta f) \smile g = \delta(f \smile g)$$

if $\delta g = 0$. Thus, the cup-product is well-defined on cohomology groups:

$$H^k(X; R) \times H^\ell(X; R) \xrightarrow{\smile} H^{k+\ell}(X; R)$$

Since on the level of chains the cup product is associative and distributive, so too it is on cohomologies. Furthermore, since R has an identity, so too does the cup product with the class $1 \in H^0(X; R)$ which is the 0-cocycle taking the value 1 and each singular 0-simplex.

Proposition 4.2.3: Relative Cup Product

Let $A \subseteq X$ and let R be a ring. Regard $C^k(X, A; R)$ as the group of cochains in $C^k(X; R)$ vanishing on $C_k(A)$, as in definition 4.1.15.

1. If $\varphi \in C^k(X, A; R)$ and $\psi \in C^\ell(X; R)$, then $\varphi \smile \psi \in C^{k+\ell}(X, A; R)$ and $\psi \smile \varphi \in C^{k+\ell}(X, A; R)$.
2. The coboundary of $C^\bullet(X, A; R)$ is the restriction of the coboundary of $C^\bullet(X; R)$.
3. The cup product of cochains induces a well-defined bilinear product
$$H^k(X, A; R) \times H^\ell(X, A; R) \xrightarrow{\smile} H^{k+\ell}(X, A; R)$$
sending $([\varphi], [\psi])$ to $[\varphi \smile \psi]$, the *relative cup product*. For $A = \emptyset$ it is the cup product on $H^\bullet(X; R)$.
4. The same formula induces well-defined bilinear products
$$H^k(X; R) \times H^\ell(X, A; R) \xrightarrow{\smile} H^{k+\ell}(X, A; R)$$
$$H^k(X, A; R) \times H^\ell(X; R) \xrightarrow{\smile} H^{k+\ell}(X, A; R)$$
5. Let $A, B \subseteq X$ be open. Write $C^\bullet(X, A + B; R)$ for the cochains vanishing on $C_\bullet(A) + C_\bullet(B)$. The inclusion $C^\bullet(X, A \cup B; R) \subseteq C^\bullet(X, A + B; R)$ induces isomorphisms on cohomology, and the same formula induces a well-defined bilinear product
$$H^k(X, A; R) \times H^\ell(X, B; R) \xrightarrow{\smile} H^{k+\ell}(X, A \cup B; R)$$
sending $([\varphi], [\psi])$ to the class corresponding under that isomorphism to the class of $\varphi \smile \psi$.

***Proof*:**

1. Let $\sigma\colon \Delta^{k+\ell} \to X$ have image in A. Its front face $\sigma|_{[v_0,\dots,v_k]}$ also has image in A, so φ vanishes on it, and $(\varphi \smile \psi)(\sigma) = \varphi(\sigma|_{[v_0,\dots,v_k]})\,\psi(\sigma|_{[v_k,\dots,v_{k+\ell}]}) = 0$ by definition 4.2.1. Hence $\varphi \smile \psi$ vanishes on $C_{k+\ell}(A)$. Likewise $(\psi \smile \varphi)(\sigma) = \psi(\sigma|_{[v_0,\dots,v_\ell]})\,\varphi(\sigma|_{[v_\ell,\dots,v_{k+\ell}]}) = 0$, the back face having image in A.
2. The boundary of $C_\bullet(X, A) = C_\bullet(X)/C_\bullet(A)$ is induced by ∂, so a relative cochain φ, read as a homomorphism $C_k(X) \to R$ vanishing on $C_k(A)$, has relative coboundary $\varphi \circ \partial$, which is its coboundary $\delta\varphi$ in $C^\bullet(X; R)$.
3. By (2) the relative cocycles and coboundaries are the cocycles of $C^\bullet(X; R)$ lying in the relative subgroups and the coboundaries of relative cochains. Let $\varphi \in C^k(X, A; R)$ and $\psi \in C^\ell(X, A; R)$ be cocycles. By lemma 4.2.2, $\delta(\varphi \smile \psi) = 0$, and $\varphi \smile \psi$ is relative by (1), so it is a relative cocycle. If $\eta \in C^{k-1}(X, A; R)$, the same lemma gives $(\delta\eta) \smile \psi = \delta(\eta \smile \psi)$ because $\delta\psi = 0$, and $\eta \smile \psi$ is relative by (1). If $\eta \in C^{\ell-1}(X, A; R)$, it gives $\varphi \smile (\delta\eta) = (-1)^k \delta(\varphi \smile \eta)$ because $\delta\varphi = 0$, and $\varphi \smile \eta$ is relative by (1). By bilinearity of the cochain product, changing either cocycle by a relative coboundary

therefore changes $\varphi \smile \psi$ by a relative coboundary, so the product of classes is well defined, and it is bilinear. For $A = \emptyset$ every cochain is relative, and the product is the one of the absolute case.

4. Let $\psi \in C^k(X;R)$ and $\varphi \in C^\ell(X,A;R)$ be cocycles. Then $\psi \smile \varphi$ is a cocycle by lemma 4.2.2 and relative by (1). For $\eta \in C^{k-1}(X;R)$ the same lemma gives $(\delta\eta) \smile \varphi = \delta(\eta \smile \varphi)$, and for $\eta \in C^{\ell-1}(X,A;R)$ it gives $\psi \smile (\delta\eta) = (-1)^k \delta(\psi \smile \eta)$, where $\eta \smile \varphi$ and $\psi \smile \eta$ are relative by (1). So changing ψ by an absolute coboundary, or φ by a relative one, changes $\psi \smile \varphi$ by a relative coboundary, and the first product is well defined and bilinear. For the second, let $\varphi \in C^k(X,A;R)$ and $\psi \in C^\ell(X;R)$ be cocycles. Then $\varphi \smile \psi$ is a relative cocycle by (1) and lemma 4.2.2, $(\delta\eta) \smile \psi = \delta(\eta \smile \psi)$ for $\eta \in C^{k-1}(X,A;R)$, and $\varphi \smile (\delta\eta) = (-1)^k \delta(\varphi \smile \eta)$ for $\eta \in C^{\ell-1}(X;R)$, with $\eta \smile \psi$ and $\varphi \smile \eta$ relative by (1), so it is also well defined and bilinear.

5. The group $C_n(A) + C_n(B)$ is the subgroup $C_n(A+B)$ of theorem 3.4.22, spanned by the simplices with image in A or in B, and $C^\bullet(X, A+B;R)$ is the dual of $C_\bullet(X)/C_\bullet(A+B)$. The inclusion $C_\bullet(A+B) \to C_\bullet(A \cup B)$ induces isomorphisms on homology by lemma 3.4.11, applied to the space $A \cup B$ and the collection $\{A, B\}$ of its open subsets. By proposition 3.4.4 the long exact sequences of $0 \to C_\bullet(A+B) \to C_\bullet(X) \to C_\bullet(X)/C_\bullet(A+B) \to 0$ and of the pair $(X, A \cup B)$ form a commuting ladder, so the five lemma (lemma 3.4.6) makes the quotient map $C_\bullet(X)/C_\bullet(A+B) \to C_\bullet(X, A \cup B)$ a homology isomorphism. Both complexes are free, on the simplices of X lying in neither A nor B, respectively not in $A \cup B$, so by corollary 4.1.10 the dual map, which is the inclusion $C^\bullet(X, A \cup B;R) \subseteq C^\bullet(X, A+B;R)$, induces isomorphisms on cohomology. For $\varphi \in C^k(X,A;R)$ and $\psi \in C^\ell(X,B;R)$, the cochain $\varphi \smile \psi$ vanishes on the simplices in A, whose front faces lie in A, and on those in B, whose back faces lie in B, so it lies in $C^{k+\ell}(X, A+B;R)$. For cocycles it is a cocycle by lemma 4.2.2, and that lemma gives $(\delta\eta) \smile \psi = \delta(\eta \smile \psi)$ for $\eta \in C^{k-1}(X,A;R)$ and $\varphi \smile (\delta\eta) = (-1)^k \delta(\varphi \smile \eta)$ for $\eta \in C^{\ell-1}(X,B;R)$, where $\eta \smile \psi$ and $\varphi \smile \eta$ lie in $C^\bullet(X, A+B;R)$ by the same face argument. So the class of $\varphi \smile \psi$ in the cohomology of $C^\bullet(X, A+B;R)$ depends only on $[\varphi]$ and $[\psi]$, bilinearly.

This completes the proof.

Proposition 4.2.4: Simplicial and Singular Cup Products

Let X be a Δ-complex and R a ring. For simplicial cochains $f \in \mathrm{Hom}(\Delta_k(X), R)$ and $g \in \mathrm{Hom}(\Delta_\ell(X), R)$, define $f \smile g$ on each $(k+\ell)$-simplex σ of X by the formula of definition 4.2.1, $\sigma|_{[v_0,\dots,v_k]}$ and $\sigma|_{[v_k,\dots,v_{k+\ell}]}$ being its front and back faces, again simplices of X. This product induces a well-defined product on the simplicial cohomology $H^\bullet_\Delta(X;R)$, and the isomorphism $\theta^* \colon H^\bullet(X;R) \to H^\bullet_\Delta(X;R)$ of theorem 4.1.12 preserves cup products.

***Proof*:**
For a simplex e of X write σ_e for its characteristic map, so that $\theta(e) = \sigma_e$ and the dual cochain map is $\theta^\sharp(\varphi)(e) = \varphi(\sigma_e)$. By definition 3.1.4(2), applied repeatedly, the restrictions of σ_e to its front and back faces are the characteristic maps of the front and back faces of e. Hence for $\varphi \in C^k(X;R)$ and $\psi \in C^\ell(X;R)$,

$$\theta^\sharp(\varphi \smile \psi)(e) = \varphi(\sigma_e|_{[v_0,\dots,v_k]})\, \psi(\sigma_e|_{[v_k,\dots,v_{k+\ell}]}) = (\theta^\sharp\varphi \smile \theta^\sharp\psi)(e)$$

that is $\theta^\sharp(\varphi \smile \psi) = \theta^\sharp\varphi \smile \theta^\sharp\psi$. Distinct simplices have distinct characteristic maps, their open simplices being disjoint (definition 3.1.4(1)), so θ carries the basis of $\Delta_n(X)$ injectively into the basis of $C_n(X)$, and every simplicial cochain f is $\theta^\sharp\varphi$ for the singular cochain φ with $\varphi(\sigma_e) = f(e)$ that vanishes on the remaining singular simplices. For simplicial cochains $f = \theta^\sharp\varphi$ and $g = \theta^\sharp\psi$, the identity above, lemma 4.2.2 and the fact that $\theta^\sharp$ is a cochain map give

$$\delta(f \smile g) = \theta^\sharp\delta(\varphi \smile \psi) = \theta^\sharp\big((\delta\varphi) \smile \psi + (-1)^k\varphi \smile (\delta\psi)\big) = (\delta f) \smile g + (-1)^k f \smile (\delta g)$$

So the product of two simplicial cocycles is a cocycle, $(\delta\eta) \smile g = \delta(\eta \smile g)$ for a cocycle g, and $f \smile (\delta\eta) = (-1)^k\delta(f \smile \eta)$ for a cocycle f of degree k, and the product passes to $H^\bullet_\Delta(X;R)$. The cochain identity $\theta^\sharp(\varphi \smile \psi) = \theta^\sharp\varphi \smile \theta^\sharp\psi$ for cocycles φ and ψ then gives $\theta^*(\alpha \smile \beta) = \theta^*\alpha \smile \theta^*\beta$, as we sought to show.

Proposition 4.2.5: Cup Product Commutes With Continuous Map

Let $f : X \to Y$ be a continuous map, and let $f^* : H^n(Y;R) \to H^n(X;R)$ the induced cohomology maps. Then

$$f^*(\alpha \smile \beta) = f^*(\alpha) \smile f^*(\beta)$$

for all $\alpha, \beta \in H^\bullet(Y;R)$. If moreover f is a map of pairs $(X,A) \to (Y,B)$, the same identity holds for $\alpha, \beta \in H^\bullet(Y,B;R)$, with $f^*\colon H^n(Y,B;R) \to H^n(X,A;R)$ and the relative cup product of proposition 4.2.3, and likewise when one of α, β lies in $H^\bullet(Y;R)$ and the other in $H^\bullet(Y,B;R)$, with the products of proposition 4.2.3(4). For open $B_1, B_2 \subseteq Y$ and open $A_1, A_2 \subseteq X$ with $f(A_i) \subseteq B_i$, the identity also holds for $\alpha \in H^\bullet(Y,B_1;R)$ and $\beta \in H^\bullet(Y,B_2;R)$, with the product of proposition 4.2.3(5).

***Proof*:**

Let $f_\sharp\colon C_\bullet(X) \to C_\bullet(Y)$ be the chain map of definition 3.3.1 and $f^\sharp\colon C^\bullet(Y;R) \to C^\bullet(X;R)$ its dual, $f^\sharp(\varphi) = \varphi \circ f_\sharp$, which is the cochain map inducing f^*. Everything follows from an identity of cochains,

$$f^\sharp(\varphi) \smile f^\sharp(\psi) = f^\sharp(\varphi \smile \psi)$$

for $\varphi \in C^k(Y;R)$ and $\psi \in C^\ell(Y;R)$. To see it, evaluate both sides on a singular simplex $\sigma\colon \Delta^{k+\ell} \to X$. Restricting to a face commutes with composing with f, in the sense that $f \circ (\sigma|_{[v_0,\dots,v_k]}) = (f \circ \sigma)|_{[v_0,\dots,v_k]}$, both being $f\sigma$ precomposed with the same affine inclusion of Δ^k. Hence, by definition 4.2.1,

$$\begin{aligned}\big(f^\sharp(\varphi) \smile f^\sharp(\psi)\big)(\sigma) &= f^\sharp(\varphi)\big(\sigma|_{[v_0,\dots,v_k]}\big)\, f^\sharp(\psi)\big(\sigma|_{[v_k,\dots,v_{k+\ell}]}\big)\\ &= \varphi\big((f\sigma)|_{[v_0,\dots,v_k]}\big)\,\psi\big((f\sigma)|_{[v_k,\dots,v_{k+\ell}]}\big)\\ &= (\varphi \smile \psi)(f\sigma) = f^\sharp(\varphi \smile \psi)(\sigma)\end{aligned}$$

Now let α, β be represented by cocycles φ, ψ. Then $f^\sharp(\varphi)$ and $f^\sharp(\psi)$ are cocycles representing $f^*(\alpha)$ and $f^*(\beta)$, and the identity says that their cup product is $f^\sharp(\varphi \smile \psi)$, a cocycle representing $f^*(\alpha \smile \beta)$. Passing to classes gives $f^*(\alpha \smile \beta) = f^*(\alpha) \smile f^*(\beta)$.

The relative case needs no new computation. A map of pairs $f\colon (X,A) \to (Y,B)$ has $f_\sharp(C_\bullet(A)) \subseteq C_\bullet(B)$, so $f^\sharp$ carries a cochain vanishing on chains in B to one vanishing on chains in A. The displayed identity is an identity of cochains, so it restricts to these subgroups, whether both cochains are relative or only one of them is, and passing to classes with the products of

proposition 4.2.3 gives the relative identities. For the product of proposition 4.2.3(5), $f^\sharp$ also carries cochains vanishing on $C_\bullet(B_1) + C_\bullet(B_2)$ to cochains vanishing on $C_\bullet(A_1) + C_\bullet(A_2)$, compatibly with the inclusions of $C^\bullet(Y, B_1 \cup B_2; R)$ and $C^\bullet(X, A_1 \cup A_2; R)$, so the same identity of cochains gives the identity of classes, as we sought to show.

Theorem 4.2.6: Graded-Commutative Structure

Let $A \subseteq X$, let R be a commutative ring, and let $H^\bullet(X, A; R)$ be the relative cohomology of definition 4.1.15 with the relative cup product of proposition 4.2.3. Then

$$\alpha \smile \beta = (-1)^{k\ell} \beta \smile \alpha$$

for all $\alpha \in H^k(X, A; R)$ and $\beta \in H^\ell(X, A; R)$.

In particular, taking $\beta = \alpha$ with k odd, this means that $2(\alpha \smile \alpha) = 0$, which can more concisely be written as $2\alpha^2 = 0$. So if $H^{2k}(X, A; R)$ has no element of order two, then $\alpha^2 = 0$ for every α of odd degree k.

***Proof*:**
Take $A = \emptyset$ first. The cup products $\varphi \smile \psi$ and $\psi \smile \varphi$ of two cochains differ only by a permutation of the vertices of $\Delta^{k+\ell}$, and the permutation that makes the bookkeeping work is the one reversing the order of all vertices, because it reverses the order of the vertices of every face as well.

Step 1: The reversal operator. For a singular n-simplex $\sigma\colon \Delta^n = [v_0, \dots, v_n] \to X$ let $\overline{\sigma} = \sigma|_{[v_n, \dots, v_0]}$, the simplex obtained by precomposing σ with the affine homeomorphism of Δ^n that reverses the vertex order. Reversing $n+1$ vertices is a product of $n+(n-1)+\cdots+1 = n(n+1)/2$ transpositions of adjacent vertices, each of which reverses orientation, so the sign

$$\varepsilon_n = (-1)^{n(n+1)/2}$$

is the one to carry along. Define $\rho\colon C_n(X) \to C_n(X)$ on basis elements by $\rho(\sigma) = \varepsilon_n \overline{\sigma}$ and extend linearly. Two identities about these signs are immediate from the exponent. First

$$\varepsilon_n = (-1)^n \varepsilon_{n-1}$$

since $n(n+1)/2 - (n-1)n/2 = n$. Second

$$\varepsilon_{k+\ell} = (-1)^{k\ell} \varepsilon_k \varepsilon_\ell$$

since $(k+\ell)(k+\ell+1)/2 - k(k+1)/2 - \ell(\ell+1)/2 = k\ell$.

Step 2: ρ is a chain map. The i-th vertex of the reversed list $[v_n, \dots, v_0]$ is v_{n-i}, so

$$\partial\rho(\sigma) = \varepsilon_n \sum_i (-1)^i \sigma|_{[v_n, \dots, \widehat{v_{n-i}}, \dots, v_0]} = \varepsilon_n (-1)^n \sum_j (-1)^j \sigma|_{[v_n, \dots, \hat{v}_j, \dots, v_0]}$$

after substituting $j = n - i$ and using $(-1)^{n-j} = (-1)^n(-1)^j$. On the other side, ρ applied to the j-th face of σ, an $(n-1)$-simplex, reverses its vertex list and contributes the sign ε_{n-1}:

$$\rho\partial(\sigma) = \varepsilon_{n-1}\sum_j (-1)^j \sigma|_{[v_n,\dots,\hat{v}_j,\dots,v_0]}$$

The two agree because $\varepsilon_n(-1)^n = \varepsilon_{n-1}$, which is the first identity of Step 1.

Step 3: The dual identity. Write ρ^* for $\mathrm{Hom}(\rho, R)$, so $(\rho^*\varphi)(\tau) = \varphi(\rho\tau)$. For $\varphi \in C^k(X;R)$, $\psi \in C^\ell(X;R)$ and $\sigma\colon \Delta^{k+\ell} \to X$,

$$\big(\rho^*\varphi \smile \rho^*\psi\big)(\sigma) = \varepsilon_k\varepsilon_\ell\, \varphi\big(\sigma|_{[v_k,\dots,v_0]}\big)\, \psi\big(\sigma|_{[v_{k+\ell},\dots,v_k]}\big)$$

by definition 4.2.1 and the definition of ρ. On the other hand $\rho\sigma = \varepsilon_{k+\ell}\sigma|_{[v_{k+\ell},\dots,v_0]}$, and the front ℓ-face of the reversed list $[v_{k+\ell},\dots,v_0]$ is $[v_{k+\ell},\dots,v_k]$ while its back k-face is $[v_k,\dots,v_0]$, so

$$\big(\rho^*(\psi \smile \varphi)\big)(\sigma) = \varepsilon_{k+\ell}\, \psi\big(\sigma|_{[v_{k+\ell},\dots,v_k]}\big)\, \varphi\big(\sigma|_{[v_k,\dots,v_0]}\big)$$

Since R is commutative the two products of ring elements coincide, so the two displays differ only by the sign $\varepsilon_k\varepsilon_\ell\varepsilon_{k+\ell}$, the signs being their own inverses. The second identity of Step 1 evaluates that sign as $(-1)^{k\ell}$, so

$$\rho^*\varphi \smile \rho^*\psi = (-1)^{k\ell}\rho^*(\psi \smile \varphi)$$

Step 4: ρ is chain homotopic to the identity. Use the subdivision of $\Delta^n \times I$ into $(n+1)$-simplices with vertices $v_0,\dots,v_n$ in $\Delta^n \times \{0\}$ and $w_0,\dots,w_n$ in $\Delta^n \times \{1\}$, the one already used to build the prism operator for homotopy invariance, and let $\pi\colon \Delta^n \times I \to \Delta^n$ be the projection. Define $P\colon C_n(X) \to C_{n+1}(X)$ by

$$P(\sigma) = \sum_i (-1)^i \varepsilon_{n-i}\, (\sigma\pi)|_{[v_0,\dots,v_i,w_n,\dots,w_i]}$$

The w vertices are listed in reverse order, which is what the signs ε_{n-i} compensate for. Suppressing $(\sigma\pi)$ from the notation, and noting that w_j occupies position $i+1+(n-j)$ in the list $[v_0,\dots,v_i,w_n,\dots,w_i]$,

$$\begin{aligned}\partial P = &\sum_{j\le i}(-1)^i(-1)^j\varepsilon_{n-i}[v_0,\dots,\hat{v}_j,\dots,v_i,w_n,\dots,w_i]\\ &+\sum_{j\ge i}(-1)^i(-1)^{i+1+n-j}\varepsilon_{n-i}[v_0,\dots,v_i,w_n,\dots,\hat{w}_j,\dots,w_i]\end{aligned}$$

The terms with $j = i$ are

$$\begin{aligned}&\varepsilon_n[w_n,\dots,w_0] + \sum_{i>0}\varepsilon_{n-i}[v_0,\dots,v_{i-1},w_n,\dots,w_i]\\ &\quad+\sum_{i<n}(-1)^{n+i+1}\varepsilon_{n-i}[v_0,\dots,v_i,w_n,\dots,w_{i+1}] - [v_0,\dots,v_n]\end{aligned}$$

the first and last coming from $i = 0$ in the first sum and $i = n$ in the second. The two summations cancel: replacing i by $i - 1$ in the second produces the sign $(-1)^{n+i}\varepsilon_{n-i+1}$, which equals $-\varepsilon_{n-i}$ by the first identity of Step 1, and the simplices then match term for term over the same range. What is left is $\varepsilon_n[w_n, \dots, w_0] - [v_0, \dots, v_n]$. Since π carries w_j to v_j, the first of these is $\varepsilon_n \sigma|_{[v_n,\dots,v_0]} = \rho(\sigma)$ and the second is σ.

It remains to see that the terms of ∂P with $j \neq i$ give $-P\partial$. Computing $P\partial$ from the definitions, and indexing by the vertex v_i and not by its position in the face $[v_0, \dots, \hat{v}_j, \dots, v_n]$,

$$\begin{aligned} P\partial = &\sum_{i<j} (-1)^i (-1)^j \varepsilon_{n-i-1}[v_0, \dots, v_i, w_n, \dots, \hat{w}_j, \dots, w_i] \\ &+ \sum_{i>j} (-1)^{i-1} (-1)^j \varepsilon_{n-i}[v_0, \dots, \hat{v}_j, \dots, v_i, w_n, \dots, w_i] \end{aligned}$$

the shift from ε_{n-i} to ε_{n-i-1} in the first sum being the drop in dimension, and the shift from $(-1)^i$ to $(-1)^{i-1}$ in the second reflecting that v_i sits at position $i - 1$ once v_j with $j < i$ has been deleted. The $j < i$ terms of ∂P already match the $i > j$ terms of $-P\partial$ on the nose. For the others, the required identity of signs is

$$(-1)^{n+1-j}\varepsilon_{n-i} = -(-1)^{i+j}\varepsilon_{n-i-1}$$

which holds because $\varepsilon_{n-i} = (-1)^{n-i}\varepsilon_{n-i-1}$ turns the left side into $(-1)^{2n+1-i-j}\varepsilon_{n-i-1}$, and $(-1)^{2n+1-i-j} = -(-1)^{i+j}$. So $\partial P + P\partial = \rho - \mathrm{id}$, and P is a chain homotopy from the identity to ρ in the sense of definition 3.3.4.

Step 5: Conclusion. By lemma 4.1.22(2) the dual maps ρ^* and id are cochain homotopic, so by proposition 3.3.5, read for cochain complexes after reindexing, ρ^* induces the identity on $H^\bullet(X; R)$. Let $\alpha \in H^k(X; R)$ and $\beta \in H^\ell(X; R)$ be represented by cocycles φ and ψ. Step 3 gives the cochain identity $\rho^*\varphi \smile \rho^*\psi = (-1)^{k\ell}\rho^*(\psi \smile \varphi)$, and passing to cohomology classes, where every ρ^* acts as the identity, leaves

$$\alpha \smile \beta = (-1)^{k\ell}\beta \smile \alpha$$

For $A \neq \emptyset$ nothing changes. Both ρ and P send a chain in A to a chain in A, every simplex they produce having image inside the image of σ, so the dual homomorphisms ρ^* and P^* carry cochains vanishing on $C_\bullet(A)$ to cochains vanishing on $C_\bullet(A)$. They therefore act on relative cochains, and Steps 3 to 5 run verbatim there, as we sought to show.

Definition 4.2.7: Cohomology Ring and Algebra

Let X be a topological space. Then $H^\bullet(X; R) = \bigoplus_k H^k(X; R)$ is a ring with the $+$ and $\smile$ operations, and is called the *cohomology ring* with respect to R and X. Multiplying by the scalars R, we get the *cohomology algebra* with respect to R and X.

Naturally, these rings have a graded structure. If A is a graded ring and $a \in A$, then we usually write $|a| = k$ to say that $a \in A_k$, and call it the *dimension* or *degree* of a. By proposition 4.2.5, for a

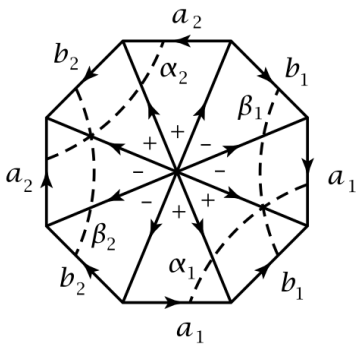
continuous map $f\colon X \to Y$,

$$f^* : H^\bullet(Y;R) \to H^\bullet(X;R)$$

is a ring homomorphism.

Example 4.2.8: Cup Product and Cohomology Ring

1. Let M^g be a closed orientable surface of genus $g \geq 1$ with the usual Δ-complex structure. To enforce orientability and for reference, here is the Δ-complex when $g = 2$:



Naturally, grading of interest is:

$$H^1(M^g) \times H^1(M^g) \to H^2(M)$$

Let $\mathbb{Z}$ be the coefficients. By the universal coefficient theorem, $H^1(M) \cong \mathrm{Hom}(H_1(M), \mathbb{Z})$. Given any basis for $H_1(M)$, let $\alpha_1, \alpha_2, ..., \alpha_g, \beta_1, \beta_2, ..., \beta_g$ be the basis for $H^1(M)$. For each α_i or β_j, we can choose a representative cocycle, namely a $\varphi_i \in H^1(M)$ such that $\delta\varphi_i = 0$. Let α_i be the loop in M meeting a_i in one point and disjoint form all other basis elements a_j and b_j. Then define φ_i to have 1 on edges meeting the arc α_i and the value 0 on all other edges. Then φ_i counts the number of intersections for each edges with the arc α_i. We may similarly find representatives ψ_i for each arc β_i. We thus have a list of representatives $\varphi_1, \varphi_2, ..., \varphi_g$ and $\psi_1, \psi_2, ..., \psi_g$.

Now, applying we see that $\varphi_1 \smile \psi_1(\sigma)$ is zero when σ does not take have the outer edge b_1, as seen in the lower-right of the above figure. Now, if c is the 2-chain formed by the sum of all the 2-simplices with the signs indicated in the center of the figure, we see that

$$\varphi_1 \smile \psi_1(c) = 1$$

Since M is 2-dimensional, $\partial c = 0$ (since c is not a boundary), $0 \neq c \in H_2(M)$. Since $(\varphi_1 \smile \psi_1)(c)$ is a generators of $\mathbb{Z}$, c represents a generator of $H_2(M) \cong \mathbb{Z}$, hence $\varphi_1 \smile \psi_1$ sends (c) to the dual $\gamma \in H^2(M) \cong \mathrm{Hom}(H_2(M), \mathbb{Z}) \cong \mathbb{Z}$ which generates it. Hence:

$$\alpha_1 \smile \alpha_2 = \gamma$$

Doing a similar computation, we get that :

$$-(\beta_i \smile \alpha_j) = \alpha_i \smile \beta_j = \begin{cases} \gamma & i = j \\ 0 & i \neq j \end{cases}, \qquad \alpha_i\alpha_j = 0, \qquad \beta_i\beta_j = 0$$

Which gives us all the relations we need to determine the cup product since it it distributive. This cup product is *not* commutative since $\alpha_i \smile \beta_i = -(\beta_i \smile \alpha_i)$.

The cohomology ring we have described is called the *exterior algebra*, if $\Lambda_{\mathbb{Z}}[\alpha_1, \alpha_2, ..., \alpha_n]$ is the exterior algebra on n generator respecting the relation

$$\alpha_i \alpha_j = -\alpha_j \alpha_i, \ i \neq j \qquad \alpha_i^2 = 0$$

Then

$$H^\bullet(T; \mathbb{Z}) \cong \Lambda_{\mathbb{Z}}[\alpha, \beta] \qquad |\alpha| = |\beta| = 1$$

More generally, we can see that

$$H^\bullet(T^n; R) \cong \Lambda_R[\alpha_1, \alpha_2, ..., \alpha_n]$$

The same is true for any product of odd-dimensional spheres where $\alpha_i|$ is the dimension of the ith sphere.

2. For the closed nonorientable surface N_g with $\mathbb{Z}_2$ coefficients, the classes $\alpha_i \in H^1(N_g; \mathbb{Z}_2)$ dual to the edges of a Δ-complex structure satisfy $\alpha_i \smile \alpha_i \neq 0$ and $\alpha_i \smile \alpha_j = 0$ for $i \neq j$ [21, Example 3.8, p. 208]. For $g = 1$, $N_1 = \mathbb{R}\mathrm{P}^2$, the cup square of the generator α of $H^1(\mathbb{R}\mathrm{P}^2; \mathbb{Z}_2)$ generates $H^2(\mathbb{R}\mathrm{P}^2; \mathbb{Z}_2)$, and $H^3(\mathbb{R}\mathrm{P}^2; \mathbb{Z}_2) = 0$, so

$$H^\bullet(\mathbb{R}\mathrm{P}^2; \mathbb{Z}_2) \cong \mathbb{Z}_2[\alpha]/(\alpha^3)$$

3. Let X be a 2-dimensional CW complex by attaching the boundary of the 2-cell S^1 by the map $z \mapsto z^m$. We see using cellular cohomology (or cellular homology and the universal coefficient theorem) that

$$H^n(X; \mathbb{Z}) = \begin{cases} \mathbb{Z} & n = 0 \\ 0 & n = 1 \\ \mathbb{Z}_m & n = 2 \\ 0 & n \geq 3 \end{cases}$$

which might suggest the cup product must be trivial. However, if we take $\mathbb{Z}_m$ coefficients, we get:

$$H^n(X; \mathbb{Z}_m) = \begin{cases} \mathbb{Z}_m & n \in \{0, 1, 2\} \\ 0 & n \geq 3 \end{cases}$$

suggesting in this case that the cup product may be nontrivial. For generators α of $H^1(X; \mathbb{Z}_m)$ and β of $H^2(X; \mathbb{Z}_m)$, a Δ-complex computation gives $\alpha \smile \alpha = 0$ for odd m and $\alpha \smile \alpha = k\beta$ for $m = 2k$ [21, Example 3.9, pp. 208 and 209].

Proposition 4.2.3 gives the cup product of two classes relative to the same subspace, of an absolute class with a relative one, and, for open A and B, of classes relative to A and to B, with values in $H^{k+\ell}(X, A \cup B; R)$, the construction of [21, §3.2].

Naturally, it may be asked what graded commutative R-algebras occurs as cup product algebras on $H^\bullet(X; R)$ for some space X. This is called the *Realization problem*, and is in general quite difficult. If $R = \mathbb{Q}$, the problem is almost solved, if $R = \mathbb{Z}/p\mathbb{Z}$, there is some progress, and for $R = \mathbb{Z}$, very little is known.

An interesting application of cohomology rings is showing a space cannot be split into a wedge-sum up to homotopy equivalent. For example:

Example 4.2.9: Cohomology Ring And Products
There is a natural isomorphism

$$H^\bullet(\coprod_\alpha X_\alpha; R) \cong \prod_\alpha H^\bullet(X_\alpha; R)$$

and:

$$\tilde{H}^\bullet\left(\bigvee_\alpha X_\alpha; R\right) \cong \prod_\alpha \tilde{H}^\bullet(X_\alpha; R)$$

where we take the reduced cohomology in the 2nd isomorphism to be relative to a basepoint (proposition 4.1.18(1)) and take the relative cup product of proposition 4.2.3[a]

Let's now show that $\mathbb{C}\mathrm{P}^2$ is not homotopy equivalent to $S^2 \vee S^4$. In particular, with $\mathbb{Z}$ coefficients, we have:

$$H^\bullet(S^2; \mathbb{Z}) \cong H^\bullet(S^4; \mathbb{Z}) \cong \mathbb{Z}[\alpha]/(\alpha^2)$$

while

$$H^\bullet(\mathbb{C}\mathrm{P}^2) \cong \mathbb{Z}[\alpha]/(\alpha^3)$$

But certainly:

$$H^\bullet(S^2 \vee S^4; \mathbb{Z}) \cong \left(\mathbb{Z}[\alpha]/(\alpha)^2 \oplus \mathbb{Z}[\alpha]/(\alpha)^2\right) \ncong \mathbb{Z}[\alpha]/(\alpha)^3 \cong H^\bullet(\mathbb{C}\mathrm{P}^2; \mathbb{Z})$$

Giving us the result. The same computation shows that the Hopf map $\eta\colon S^3 \to S^2$, the attaching map of the 4-cell of $\mathbb{C}\mathrm{P}^2 = S^2 \cup_\eta D^4$, is not nullhomotopic. If η were homotopic to a constant map, theorem 1.3.7 would give $\mathbb{C}\mathrm{P}^2 \simeq S^2 \cup_{\text{const}} D^4 = S^2 \vee S^4$, whose cup products differ from those of $\mathbb{C}\mathrm{P}^2$ as shown above.

Two spaces may have the same cohomology ring over one coefficient ring and not over another. For example, consider if $\mathbb{R}\mathrm{P}^5$ is homotopic to $\mathbb{R}\mathrm{P}^4 \vee S^5$. Notice that these have the same cohomology groups (exercise), and hence cannot be distinguished that way. They even have the same cohomology ring over $\mathbb{Z}$. However, they do not have isomorphic cohomology rings over $\mathbb{Z}/2\mathbb{Z}$.

[a]The basepoints are also assumed to be deformation retracts of some neighborhoods, as the isomorphism requires

4.3 The Künneth Formula

The homology of a disjoint union splits as a direct sum, $H_n(\coprod_\alpha X_\alpha) \cong \bigoplus_\alpha H_n(X_\alpha)$, and this was settled in chapter 3 by a direct argument on chains. The question deferred there was the homology of a *product* $X \times Y$, and the answer is genuinely more delicate. A singular simplex in $X \times Y$ is a map $\Delta^n \to X \times Y$, equivalently a pair of maps into X and into Y sharing a domain. Such a simplex carries no canonical decomposition into a simplex of X times a simplex of Y, because a simplex is not a product of simplices. The resolution proceeds in two stages, one topological and one algebraic. The topological input is the Eilenberg-Zilber theorem, which compares the singular chains of $X \times Y$ with the algebraic tensor product of the singular chains of X and of Y. The two chain complexes are chain homotopy equivalent even though they are not equal. The algebraic input is the Künneth

theorem for chain complexes over a principal ideal domain, which computes the homology of a tensor product of complexes from the homologies of the factors, with a correction term measured by Tor. Combining the two produces the Künneth formula for spaces, the statement that $H_*(X \times Y)$ is built from $H_*(X)$ and $H_*(Y)$ by a tensor product and a single Tor correction.

Throughout this section R is a principal ideal domain (a PID), and homology is taken with coefficients in R. The abelian-group case is $R = \mathbb{Z}$. All tensor products and Tor groups are over R unless decorated otherwise. The graded modules $H_*(X;R) = \bigoplus_n H_n(X;R)$ and $H_*(Y;R)$ are the homologies of the singular chain complexes $C_*(X;R)$ and $C_*(Y;R)$, which are complexes of free R-modules: $C_n(X;R)$ is free on the singular n-simplices of X.

The first object to construct is the map that turns a homology class on X and one on Y into a class on the product. On chains it is induced by the algebraic tensor product of complexes, transported to the product space by Eilenberg-Zilber.

Theorem 4.3.1: Eilenberg-Zilber Theorem

For topological spaces X and Y there are natural chain maps

$$\mathrm{EZ}\colon C_*(X;R) \otimes_R C_*(Y;R) \longrightarrow C_*(X \times Y;R),$$
$$\mathrm{AW}\colon C_*(X \times Y;R) \longrightarrow C_*(X;R) \otimes_R C_*(Y;R)$$

that are mutually inverse up to natural chain homotopy. Consequently $C_*(X \times Y;R)$ and $C_*(X;R) \otimes_R C_*(Y;R)$ are naturally chain homotopy equivalent, and EZ, AW induce mutually inverse isomorphisms on homology. In degree zero EZ is the identification of a pair of points with the corresponding point of $X \times Y$, and EZ is associative and commutative up to the evident chain homotopies.

Here $C_*(X;R) \otimes_R C_*(Y;R)$ denotes the tensor product of chain complexes: its degree-n term is $\bigoplus_{i+j=n} C_i(X;R) \otimes_R C_j(Y;R)$, with differential $\partial(a \otimes b) = \partial a \otimes b + (-1)^{|a|} a \otimes \partial b$. The map EZ admits an explicit formula, the *shuffle product*, sending a tensor of a singular i-simplex with a singular j-simplex to a signed sum over the $\binom{i+j}{i}$ ways of interleaving their vertices into a singular $(i+j)$-simplex of the product. The map AW is the *Alexander-Whitney map*, sending a singular n-simplex $\sigma = (\sigma_X, \sigma_Y)$ of $X \times Y$ to $\sum_{i+j=n} (\sigma_X|_{[v_0,\dots,v_i]}) \otimes (\sigma_Y|_{[v_i,\dots,v_n]})$, the front-face/back-face splitting already used to define the cup product (definition 4.2.1). That these are inverse up to homotopy is the substance of the theorem.

***Proof Key ideas*:**
The proof is by the method of acyclic models and is purely formal once the two explicit maps are written down. The verification that the shuffle and Alexander-Whitney maps are chain maps inverse up to a natural chain homotopy is a bookkeeping computation with no conceptual obstruction, but it is long. The acyclic-models mechanism runs as follows. Both functors $X, Y \mapsto C_*(X \times Y;R)$ and $X, Y \mapsto C_*(X;R) \otimes_R C_*(Y;R)$ are free on the models (Δ^i, Δ^j) (the representing pairs of simplices) and acyclic on those models (the chains of $\Delta^i \times \Delta^j$ and the tensor product of the chains of two simplices are both resolutions of R in degree zero, since simplices are contractible). A natural transformation agreeing with a prescribed map in degree zero exists and is unique up to natural chain homotopy whenever the source is free on models and the target is acyclic on those models. Applying this in both directions and to the two composites produces EZ, AW and the homotopies $\mathrm{EZ} \circ \mathrm{AW} \simeq \mathrm{id}$, $\mathrm{AW} \circ \mathrm{EZ} \simeq \mathrm{id}$. The full acyclic-models theorem

and the explicit homotopies are carried out in [21, §3.B] and in [13]. The argument uses only contractibility of simplices and the freeness of singular chains, both established in chapter 3. The remaining assertions (degree-zero behaviour, associativity, commutativity up to homotopy) follow from the same uniqueness clause applied to the relevant pairs of natural maps.

The content of Eilenberg-Zilber is that, on the level of homology, the product space behaves exactly as the algebraic tensor product of complexes does, despite the absence of a simplexwise product decomposition. This converts the topological problem into an algebraic one: to compute $H_*(X\times Y;R)$, compute the homology of the tensor product complex $C_*(X;R)\otimes_R C_*(Y;R)$. The map EZ is the geometric source of the cross product, and the map AW is the source of the cup product. The two products are dual aspects of the single comparison.

With the comparison in hand, the cross product is the composite of the obvious map from a tensor of homology classes into the homology of the tensor complex with the Eilenberg-Zilber isomorphism.

Definition 4.3.2: Homology Cross Product

Let X and Y be spaces and R a PID. For $\alpha\in H_i(X;R)$ and $\beta\in H_j(Y;R)$, choose cycles $a\in Z_i(X;R)$, $b\in Z_j(Y;R)$ representing them. Then $a\otimes b$ is a cycle in $\big(C_*(X;R)\otimes_R C_*(Y;R)\big)_{i+j}$, and its image under the homology isomorphism induced by EZ of theorem 4.3.1 is the *homology cross product*

$$\alpha\times\beta := \mathrm{EZ}_*[a\otimes b]\in H_{i+j}(X\times Y;R)$$

The cross product is R-bilinear, natural in X and Y, and assembles into a homomorphism of graded R-modules

$$\times\colon \bigoplus_{i+j=n} H_i(X;R)\otimes_R H_j(Y;R)\longrightarrow H_n(X\times Y;R)$$

The assignment $[a]\otimes[b]\mapsto[a\otimes b]$ is well defined on homology because the differential of the tensor complex satisfies the graded Leibniz rule: if a,b are cycles then so is $a\otimes b$, and altering a by a boundary ∂c changes $a\otimes b$ by $\partial(c\otimes b)\mp c\otimes\partial b=\partial(c\otimes b)$, a boundary (using $\partial b=0$). The Künneth theorem determines how far the cross product is from being an isomorphism. Naturality means that for maps $f\colon X\to X'$ and $g\colon Y\to Y'$ one has $(f\times g)_*(\alpha\times\beta)=f_*\alpha\times g_*\beta$, which follows from naturality of EZ.

Example 4.3.3: Cross Product of Sphere Classes

Take $X=S^1$ and $Y=S^1$, each with its standard generator. Write $\mu\in H_1(S^1;\mathbb{Z})\cong\mathbb{Z}$ for the fundamental class of the circle, represented by a 1-cycle wrapping once around. The cross product $\mu\times\mu\in H_2(S^1\times S^1;\mathbb{Z})$ is represented by $\mathrm{EZ}(a\otimes b)$ where a,b are wrapping 1-cycles. Concretely, give S^1 the Δ-complex with one vertex v and one edge, and the torus $T^2=S^1\times S^1$ the standard square model of example 3.1.14, with two triangular 2-simplices U,L. The shuffle product of the two edge cycles is, up to sign, the cycle $U-L$, which is exactly the generator of $H_2(T^2;\mathbb{Z})\cong\mathbb{Z}$ computed there (the only 2-cycle, since $\partial(pU+qL)=(p+q)(a+b-c)$ vanishes iff $p=-q$). Thus $\mu\times\mu$ generates $H_2(T^2;\mathbb{Z})$: the product of the two circle classes is the top class of the torus. In degree zero, with $p\in H_0(S^1;\mathbb{Z})$ the class of a point, $p\times\mu$ and $\mu\times p$ are the two coordinate circle classes generating $H_1(T^2;\mathbb{Z})\cong\mathbb{Z}^2$, and $p\times p$ is the point class generating H_0.

The example exhibits the cross product as an isomorphism for the torus: every homology class of T^2

is a cross product of classes from the two circle factors, and there is no correction term. The general theorem identifies exactly when this happens and what the correction is. The mechanism is entirely algebraic, so the statement and proof are first given for chain complexes.

Theorem 4.3.4: Algebraic Künneth Theorem

Let R be a PID, let C_* be a chain complex of free R-modules, and let D_* be any chain complex of R-modules. Then for each n there is a short exact sequence of R-modules

$$0 \longrightarrow \bigoplus_{i+j=n} H_i(C) \otimes_R H_j(D) \xrightarrow{\times} H_n(C \otimes_R D) \longrightarrow \bigoplus_{i+j=n-1} \operatorname{Tor}_R\big(H_i(C), H_j(D)\big) \longrightarrow 0$$

natural in C and D. The sequence splits, but the splitting is not natural. If in addition R is a field, the Tor term vanishes and the cross product is an isomorphism $H_n(C \otimes_R D) \cong \bigoplus_{i+j=n} H_i(C) \otimes_R H_j(D)$.

The map labelled $\times$ is the same construction as in definition 4.3.2: a cycle of C tensored with a cycle of D is a cycle of $C \otimes_R D$, and $[a] \otimes [b] \mapsto [a \otimes b]$. The theorem says this map is injective with cokernel measured by a single Tor group, shifted down by one in degree. Over a field every module is flat and Tor vanishes, recovering the clean statement that homology commutes with tensor product. The Tor term is the precise obstruction to that over a general PID, and it is supported on torsion, since $\operatorname{Tor}_R(A, B) = 0$ whenever either argument is free (equivalently, torsion-free over a PID).

***Proof*:**

Step 1: Splitting the cycles and boundaries. For each n write $Z_n = \ker(\partial\colon C_n \to C_{n-1})$ and $B_n = \operatorname{im}(\partial\colon C_{n+1} \to C_n)$, both submodules of the free module C_n. Over a PID a submodule of a free module is free, so Z_n and B_n are free, hence projective. The short exact sequence

$$0 \longrightarrow Z_n \longrightarrow C_n \xrightarrow{\partial} B_{n-1} \longrightarrow 0 \tag{4.12}$$

has free right-hand term B_{n-1}, so it splits: $C_n \cong Z_n \oplus B_{n-1}$ as R-modules. Regard Z_* and B_* as chain complexes with *zero* differential. The inclusion $Z_* \hookrightarrow C_*$ and the boundary map $C_* \to B_{*-1}$ (lowering degree by one onto the boundaries) then assemble into a short exact sequence of chain complexes

$$0 \longrightarrow Z_* \longrightarrow C_* \xrightarrow{\partial} B_{*-1} \longrightarrow 0 \tag{4.13}$$

in which Z_* and B_* carry the zero differential. This is the key step of the proof: it presents C_* as an extension of complexes whose homology is transparent.

Step 2: Tensoring with D. Each $C_n \cong Z_n \oplus B_{n-1}$ is free, hence flat, so tensoring (4.13) over R with the complex D_* preserves exactness in every degree: because B_{*-1} is a complex of free (hence flat) modules, the sequence

$$0 \longrightarrow Z_* \otimes_R D_* \longrightarrow C_* \otimes_R D_* \xrightarrow{\partial \otimes 1} (B_{*-1}) \otimes_R D_* \longrightarrow 0 \tag{4.14}$$

is a short exact sequence of chain complexes. Since Z_* and B_* have zero differential, the differential on $Z_* \otimes_R D_*$ is $\pm 1 \otimes \partial_D$, so this complex is just D_* with its degree shifted and grouped: in each total degree n, $(Z_* \otimes_R D_*)_n = \bigoplus_i Z_i \otimes_R D_{n-i}$, and because Z_i is free,

$$H_n(Z_* \otimes_R D_*) = \bigoplus_i Z_i \otimes_R H_{n-i}(D), \qquad H_n(B_* \otimes_R D_*) = \bigoplus_i B_i \otimes_R H_{n-i}(D) \tag{4.15}$$

the freeness of Z_i and B_i being what lets the functor $Z_i \otimes_R (-)$ commute with taking homology of D_*.

Step 3: The long exact sequence and its connecting map. The short exact sequence of complexes (4.14) yields a long exact homology sequence. Using (4.15) to identify the outer terms (and accounting for the degree shift by one in the B-complex), it reads

$$\begin{aligned}\cdots \longrightarrow \bigoplus_i B_i \otimes_R H_{n-i}(D) &\xrightarrow{\iota} \bigoplus_i Z_i \otimes_R H_{n-i}(D)\\ &\longrightarrow H_n(C \otimes_R D)\\ &\longrightarrow \bigoplus_i B_i \otimes_R H_{n-1-i}(D) \xrightarrow{\iota} \cdots\end{aligned}$$

The connecting homomorphism ι is, degreewise, the map $B_i \otimes_R H_{n-i}(D) \to Z_i \otimes_R H_{n-i}(D)$ induced by the inclusion $j\colon B_i \hookrightarrow Z_i$ of boundaries into cycles tensored with the identity, that is $\iota = (j \otimes 1)$. This identification of the connecting map is the one computational point and is checked by chasing a representative through (4.14).

Step 4: Extracting the Künneth sequence. The long exact sequence breaks into short exact sequences

$$0 \longrightarrow \operatorname{coker}(\iota_n) \longrightarrow H_n(C \otimes_R D) \longrightarrow \ker(\iota_{n-1}) \longrightarrow 0 \tag{4.16}$$

where $\iota_n\colon \bigoplus_i B_i \otimes_R H_{n-i}(D) \to \bigoplus_i Z_i \otimes_R H_{n-i}(D)$. It remains to identify $\operatorname{coker}(\iota_n)$ and $\ker(\iota_{n-1})$. Fix i and consider the free resolution

$$0 \longrightarrow B_i \xrightarrow{j} Z_i \longrightarrow H_i(C) \longrightarrow 0 \tag{4.17}$$

of $H_i(C)$: it is exact by definition of $H_i(C) = Z_i/B_i$, and both B_i, Z_i are free, so this is a length-one free resolution of $H_i(C)$ over the PID R. Tensoring (4.17) with the module $H_{n-i}(D)$ and taking homology is the definition of Tor_R: the complex $0 \to B_i \otimes_R H_{n-i}(D) \xrightarrow{j\otimes 1} Z_i \otimes_R H_{n-i}(D) \to 0$ has

$$\begin{aligned}\operatorname{coker}(j \otimes 1) &= Z_i \otimes_R H_{n-i}(D) \big/ \operatorname{im}(j \otimes 1) = H_i(C) \otimes_R H_{n-i}(D),\\ \ker(j \otimes 1) &= \operatorname{Tor}_R\big(H_i(C), H_{n-i}(D)\big)\end{aligned}$$

the first because $\otimes$ is right exact, the second because the kernel of the tensored inclusion is exactly the first derived functor computed from this resolution. Summing over i gives

$$\operatorname{coker}(\iota_n) = \bigoplus_{i+j=n} H_i(C) \otimes_R H_j(D), \qquad \ker(\iota_n) = \bigoplus_{i+j=n} \operatorname{Tor}_R\big(H_i(C), H_j(D)\big)$$

Substituting into (4.16) produces the asserted short exact sequence, with the cokernel term sitting in degree n and the Tor term, coming from $\ker(\iota_{n-1})$, sitting in degree $n-1$. That the left map is the cross product $\times$ follows from tracing the construction: the image of $[a] \otimes [b]$ under $\operatorname{coker}(\iota_n) \hookrightarrow H_n(C \otimes_R D)$ is $[a \otimes b]$.

Step 5: The splitting. For the (non-natural) splitting, return to the split short exact sequence $0 \to Z_n \to C_n \xrightarrow{\partial} B_{n-1} \to 0$ of (4.12). Its right-hand term B_{n-1} is free, so the sequence splits, and a choice of splitting is the same datum as a retraction $r_n\colon C_n \twoheadrightarrow Z_n$ of the inclusion $Z_n \hookrightarrow C_n$ (that is, $r_n|_{Z_n} = \mathrm{id}_{Z_n}$). Such a retraction exists for the same reason the sequence split in Step 1, and the choice is non-canonical. Compose r_n with the projection $\pi_n\colon Z_n \twoheadrightarrow Z_n/B_n = H_n(C)$ to

obtain $\varphi_n := \pi_n \circ r_n \colon C_n \to H_n(C)$. The maps φ_* assemble into a chain map $\varphi \colon C_* \to H_*(C)$, where $H_*(C)$ carries the zero differential: for $c \in C_n$ the element ∂c already lies in Z_{n-1}, so $r_{n-1}(\partial c) = \partial c$ (the retraction restricts to the identity on Z_{n-1}), whence $\varphi_{n-1}(\partial c) = \pi_{n-1}(\partial c) = 0$ since $\partial c \in B_{n-1} = \ker \pi_{n-1}$. On homology φ induces the identity $H_n(C) \to H_n(C)$, because for a cycle $z \in Z_n$ one has $r_n(z) = z$ and hence $\varphi_n(z) = \pi_n(z) = [z]$. Choosing such a φ for C and the analogous chain map $\psi \colon D_* \to H_*(D)$ for D, the composite

$$C_* \otimes_R D_* \xrightarrow{\varphi \otimes \psi} H_*(C) \otimes_R H_*(D)$$

is a chain map to a complex with zero differential, and on degree-n homology its restriction to the image of the cross product is the identity of $\bigoplus_{i+j=n} H_i(C) \otimes_R H_j(D)$. Thus $(\varphi \otimes \psi)_*$ is a retraction of the injection $\bigoplus_{i+j=n} H_i(C) \otimes_R H_j(D) \hookrightarrow H_n(C \otimes_R D)$, which splits the short exact sequence (4.16). The splitting depends on the choice of retraction r_*, equivalently on the choice of splitting of (4.12). That the choice is non-canonical does not by itself show no natural choice exists, so the non-naturality needs a counterexample rather than an appeal to the construction. One is already available: taking $D_\bullet$ to be G concentrated in degree 0 turns this theorem into theorem 4.1.7, whose proof exhibits a chain map for which no natural splitting can exist.

Step 6: The field case. If R is a field, every R-module is free, hence flat, so $\operatorname{Tor}_R(-,-) = 0$ identically. The Tor term in the short exact sequence vanishes, the sequence degenerates to the isomorphism $H_n(C \otimes_R D) \cong \bigoplus_{i+j=n} H_i(C) \otimes_R H_j(D)$, and the cross product is that isomorphism, as we sought to show.

The proof isolates exactly why the answer has two pieces. The tensor product of the homologies accounts for cycles that are products of cycles. The Tor term accounts for the failure of the inclusion of boundaries into cycles to remain injective after tensoring, which is a torsion phenomenon and disappears over a field. The degree shift in the Tor term, with Tor of degree-i and degree-j classes contributing in total degree $i + j + 1$, is forced by the connecting homomorphism of the long exact sequence and is the source of the most common errors in computation: a class of $H_i(X)$ with $H_j(Y)$ produces a tensor contribution in degree $i + j$ and a Tor contribution in degree $i + j + 1$.

Feeding the Eilenberg-Zilber equivalence into the algebraic theorem gives the topological Künneth formula at once. This is the central result of the section.

Theorem 4.3.5: Künneth Formula for Homology

Let X and Y be topological spaces and R a PID. Then for each n the homology cross product fits into a natural short exact sequence

$$\begin{aligned} 0 &\longrightarrow \bigoplus_{i+j=n} H_i(X;R) \otimes_R H_j(Y;R) \xrightarrow{\times} H_n(X \times Y; R) \\ &\longrightarrow \bigoplus_{i+j=n-1} \operatorname{Tor}_R\big(H_i(X;R), H_j(Y;R)\big) \\ &\longrightarrow 0 \end{aligned}$$

The sequence splits, though not naturally. If R is a field, the Tor term vanishes and the cross product is an isomorphism of graded R-modules

$$H_*(X \times Y; R) \cong H_*(X;R) \otimes_R H_*(Y;R)$$

***Proof*:**
Apply theorem 4.3.4 with $C_* = C_*(X;R)$ and $D_* = C_*(Y;R)$. The complex $C_*(X;R)$ is free in each degree, since $C_n(X;R)$ is free on the singular n-simplices of X, so the hypothesis of theorem 4.3.4 is met. The second factor $D_* = C_*(Y;R)$ may be an arbitrary complex of R-modules, which is all the theorem requires of it. This produces, for each n, the natural short exact sequence

$$\begin{aligned} 0 &\longrightarrow \bigoplus_{i+j=n} H_i(X;R) \otimes_R H_j(Y;R) \\ &\longrightarrow H_n\big(C_*(X;R) \otimes_R C_*(Y;R)\big) \\ &\longrightarrow \bigoplus_{i+j=n-1} \operatorname{Tor}_R\big(H_i(X;R), H_j(Y;R)\big) \\ &\longrightarrow 0 \end{aligned}$$

using $H_i(C_*(X;R)) = H_i(X;R)$ and likewise for Y. By the Eilenberg-Zilber theorem (theorem 4.3.1), EZ induces a natural isomorphism $H_n\big(C_*(X;R) \otimes_R C_*(Y;R)\big) \cong H_n(X \times Y; R)$. Transporting the middle term along this isomorphism replaces $H_n\big(C_*(X;R) \otimes_R C_*(Y;R)\big)$ by $H_n(X \times Y; R)$, and the composite of the left map with EZ_* is by construction the homology cross product of definition 4.3.2. The splitting and its non-naturality, and the field case, are inherited verbatim from theorem 4.3.4, completing the proof.

The formula reduces the homology of any product to the homologies of its factors, mechanically. The only subtlety is the Tor term, which is nonzero precisely when both factors carry torsion in complementary degrees. For spaces with free homology, such as spheres, tori, and orientable surfaces, the cross product is an isomorphism on the nose and $H_*(X \times Y) \cong H_*(X) \otimes H_*(Y)$. The non-naturality of the splitting is a genuine feature: the extension can be nonsplit as a sequence *varying* with X and Y, even though each individual extension splits because the relevant groups are over a PID. The two worked computations below display both behaviours.

Example 4.3.6: Künneth Computations

Throughout, recall from chapter 3 that $\tilde{H}_*(S^n;\mathbb{Z})$ is $\mathbb{Z}$ in degree n and zero otherwise, so the unreduced homology of S^n ($n \geq 1$) is $\mathbb{Z}$ in degrees 0 and n and zero elsewhere.

1. *The torus* $T^2 = S^1 \times S^1$. The homology of S^1 is $H_0 = H_1 = \mathbb{Z}$, free in every degree, so the Tor term in theorem 4.3.5 vanishes and the cross product is an isomorphism $H_*(T^2;\mathbb{Z}) \cong H_*(S^1;\mathbb{Z}) \otimes_{\mathbb{Z}} H_*(S^1;\mathbb{Z})$. Writing $1 \in H_0(S^1)$ and $\mu \in H_1(S^1)$ for the generators, the tensor product is free on $1\otimes 1$, $\mu \otimes 1$, $1 \otimes \mu$, $\mu \otimes \mu$ in degrees $0, 1, 1, 2$, so

$$H_n(T^2;\mathbb{Z}) \cong \begin{cases} \mathbb{Z} & n = 0 \\ \mathbb{Z}^2 & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & n \geq 3 \end{cases}$$

   The two degree-one generators are the coordinate circles $\mu \times 1$ and $1 \times \mu$, and the degree-two generator is $\mu \times \mu$, the top class computed in example 4.3.3. This matches the direct Δ-complex computation of chapter 3.

2. *Products of spheres* $S^m \times S^n$. Both factors have free homology ($\mathbb{Z}$ in degrees 0 and m, respectively 0 and n), so again Tor vanishes and $H_*(S^m \times S^n;\mathbb{Z}) \cong H_*(S^m;\mathbb{Z}) \otimes_{\mathbb{Z}} H_*(S^n;\mathbb{Z})$. The tensor product is free on $1\otimes 1$, $a \otimes 1$, $1 \otimes b$, $a \otimes b$ in degrees $0, m, n, m+n$, where a, b generate $H_m(S^m)$, $H_n(S^n)$. When $m \neq n$,

$$H_k(S^m \times S^n;\mathbb{Z}) \cong \begin{cases} \mathbb{Z} & k \in \{0, m, n, m+n\} \\ 0 & \text{otherwise} \end{cases}$$

   and when $m = n$ the two middle generators coincide in degree, giving $H_m(S^m \times S^m;\mathbb{Z}) \cong \mathbb{Z}^2$ with the other nonzero groups $\mathbb{Z}$ in degrees 0 and $2m$. For instance $H_*(S^2 \times S^2;\mathbb{Z})$ is $\mathbb{Z}, 0, \mathbb{Z}^2, 0, \mathbb{Z}$ in degrees 0 through 4.

3. *A nonzero* Tor *term:* $\mathbb{RP}^2 \times \mathbb{RP}^2$. The integral homology of $\mathbb{RP}^2$ is $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}/2$, $H_2 = 0$. The cross product accounts for the tensor contributions, and the Tor term contributes in one higher degree. In degree 2,

$$H_2(\mathbb{RP}^2 \times \mathbb{RP}^2;\mathbb{Z}) \;\cong\; \underbrace{(H_1 \otimes H_1)}_{\mathbb{Z}/2 \otimes \mathbb{Z}/2 = \mathbb{Z}/2} \;\oplus\; \underbrace{\bigoplus_{i+j=1} \operatorname{Tor}_{\mathbb{Z}}(H_i, H_j)}_{0}$$

   is $\mathbb{Z}/2$, the Tor summands in degree 2 vanishing because $H_0(\mathbb{RP}^2) = \mathbb{Z}$ is free. The Tor correction from the two degree-one classes instead appears in degree 3:

$$H_3(\mathbb{RP}^2 \times \mathbb{RP}^2;\mathbb{Z}) \;\cong\; \operatorname{Tor}_{\mathbb{Z}}\big(H_1(\mathbb{RP}^2), H_1(\mathbb{RP}^2)\big) = \operatorname{Tor}_{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{Z}/2) = \mathbb{Z}/2$$

   so $\mathbb{RP}^2 \times \mathbb{RP}^2$ has homology in degree 3 even though each factor is two-dimensional. This is the Tor term made visible: a product of two-dimensional complexes acquires three-dimensional homology purely from the interaction of torsion classes.

The third computation shows that the Tor term is not a formal nuisance but produces homology

classes that are present in no factor and in no tensor product, arising solely from torsion crossing torsion. It also displays the degree bookkeeping concretely: two torsion classes in degree 1 produce a Tor class in degree $1+1+1=3$, the extra unit being the degree shift built into the connecting map.

Cohomology carries a dual cross product, and over a field or under a finiteness hypothesis it satisfies the analogous formula. The cohomology cross product is built from the cup product (definition 4.2.1) and the two projections of the product space. That construction is what makes the cohomology Künneth theorem a statement about *rings*, not just modules.

Definition 4.3.7: Cohomology Cross Product

Let X, Y be spaces with projections $p\colon X\times Y\to X$ and $q\colon X\times Y\to Y$, and let R be a commutative ring. For $\alpha\in H^k(X;R)$ and $\beta\in H^\ell(Y;R)$, the *cohomology cross product* is

$$\alpha\times\beta := p^*\alpha\smile q^*\beta\in H^{k+\ell}(X\times Y;R)$$

where $\smile$ is the cup product of definition 4.2.1. This is R-bilinear and natural, and assembles into a homomorphism of graded R-modules

$$\times\colon \bigoplus_{k+\ell=n} H^k(X;R)\otimes_R H^\ell(Y;R)\longrightarrow H^n(X\times Y;R)$$

which is moreover a homomorphism of graded rings $H^\bullet(X;R)\otimes_R H^\bullet(Y;R)\to H^\bullet(X\times Y;R)$, where the source carries the graded tensor-product ring structure $(\alpha\otimes\beta)(\alpha'\otimes\beta')=(-1)^{|\beta||\alpha'|}(\alpha\smile\alpha')\otimes(\beta\smile\beta')$.

That the cross product is a ring map records the interaction of cup and cross: cupping in the product corresponds to cupping in each factor, with a Koszul sign from moving a degree-$|\beta|$ class past a degree-$|\alpha'|$ class. The sign is exactly the one forced by graded commutativity of the cup product (theorem 4.2.6). That sign is what allows the source to be a graded-commutative ring isomorphic, when the cross product is an isomorphism, to the cohomology ring of the product. The verification that $\times$ is a ring map is the identity $p^*(\alpha\smile\alpha')\smile q^*(\beta\smile\beta')=\pm(p^*\alpha\smile q^*\beta)\smile(p^*\alpha'\smile q^*\beta')$, which follows from p^*, q^* being ring homomorphisms (proposition 4.2.5) and the graded commutativity used to interleave the four factors.

Example 4.3.8: The Cohomology Cross Product on the Torus

Take $X=Y=S^1$ with $R=\mathbb{Z}$. Then $H^\bullet(S^1;\mathbb{Z})=\mathbb{Z}[\alpha]/(\alpha^2)$ with $|\alpha|=1$ (an exterior algebra on one degree-one generator). The graded tensor product ring is $\Lambda_{\mathbb{Z}}[\alpha]\otimes_{\mathbb{Z}}\Lambda_{\mathbb{Z}}[\alpha']\cong\Lambda_{\mathbb{Z}}[\alpha,\alpha']$, the exterior algebra on two degree-one generators, with $\alpha\alpha'=-\alpha'\alpha$ from the Koszul sign. Setting $a=\alpha\times 1=p^*\alpha$ and $b=1\times\alpha=q^*\alpha$ in $H^\bullet(T^2;\mathbb{Z})$, the cross product gives $a\smile b=\alpha\times\alpha$ generating $H^2(T^2;\mathbb{Z})\cong\mathbb{Z}$, while $a^2=p^*(\alpha^2)=0$ and $b^2=q^*(\alpha^2)=0$. This reproduces, from the Künneth viewpoint, the exterior-algebra structure $H^\bullet(T^2;\mathbb{Z})\cong\Lambda_{\mathbb{Z}}[a,b]$ found by the direct cup-product computation of example 4.2.8: the generator of H^2 is the cross product of the two circle generators, and the relations $a^2=b^2=0$, $ab=-ba$ are the tensor-product relations.

The cohomology Künneth theorem requires a finiteness hypothesis absent from the homology statement, because the cohomology of a product is built from the homology of the product, and dualizing a direct sum produces a direct product unless the sum is finite.

Theorem 4.3.9: Künneth Formula for Cohomology

Let R be a PID and let X, Y be spaces such that $H_k(X;R)$ is a finitely generated R-module for every k (one says X is of *finite type* over R). Then for each n the cohomology cross product fits into a natural short exact sequence

$$\begin{aligned} 0 &\longrightarrow \bigoplus_{k+\ell=n} H^k(X;R) \otimes_R H^\ell(Y;R) \xrightarrow{\times} H^n(X\times Y;R) \\ &\longrightarrow \bigoplus_{k+\ell=n+1} \operatorname{Tor}_R\big(H^k(X;R), H^\ell(Y;R)\big) \\ &\longrightarrow 0 \end{aligned}$$

which splits, not naturally. If R is a field, the Tor term vanishes and the cross product is an isomorphism of graded R-algebras

$$H^\bullet(X\times Y;R) \cong H^\bullet(X;R) \otimes_R H^\bullet(Y;R)$$

the tensor product carrying its graded-commutative ring structure as in definition 4.3.7.

***Proof*:**

Step 1: Reduction to a tensor product of cochain complexes. The cohomology of $X\times Y$ is the cohomology of $\operatorname{Hom}_R\big(C_*(X\times Y;R),R\big)$. By the Eilenberg-Zilber theorem (theorem 4.3.1), $C_*(X\times Y;R) \simeq C_*(X;R)\otimes_R C_*(Y;R)$ by a natural chain homotopy equivalence, and applying the homotopy-invariant functor $\operatorname{Hom}_R(-,R)$ gives a cochain homotopy equivalence $C^*(X\times Y;R) \simeq \operatorname{Hom}_R\big(C_*(X;R)\otimes_R C_*(Y;R),R\big)$. The finite-type hypothesis on X enters here: when $C_*(X;R)$ has finitely generated homology, the cochain complex $C^*(X;R) = \operatorname{Hom}_R(C_*(X;R),R)$ is, up to chain homotopy equivalence, a complex of finitely generated free R-modules, and there is a natural cochain map

$$\begin{aligned} C^*(X;R)\otimes_R C^*(Y;R) &\longrightarrow \operatorname{Hom}_R\big(C_*(X;R)\otimes_R C_*(Y;R),R\big), \\ (f\otimes g)(a\otimes b) &= (-1)^{|g||a|} f(a)\,g(b) \end{aligned}$$

which is a chain homotopy equivalence precisely because $H_*(X;R)$ is finitely generated in each degree (so that the natural map $\operatorname{Hom}_R(M,R)\otimes_R \operatorname{Hom}_R(N,R) \to \operatorname{Hom}_R(M\otimes_R N,R)$ is an isomorphism for the relevant finitely generated free M). Thus $H^n(X\times Y;R) \cong H^n\big(C^*(X;R)\otimes_R C^*(Y;R)\big)$.

Step 2: The algebraic Künneth theorem applied to cochains. The cochain complex $C^*(X;R)$ may be replaced, up to chain homotopy equivalence, by a complex of free R-modules with cohomology $H^*(X;R)$ (using the finite-type hypothesis to make these free of finite rank), and $C^*(Y;R)$ is an arbitrary complex of R-modules. Re-indexing cohomological degree as homological degree, theorem 4.3.4 applied to these two complexes yields the natural short exact sequence

$$\begin{aligned} 0 &\longrightarrow \bigoplus_{k+\ell=n} H^k(X;R) \otimes_R H^\ell(Y;R) \xrightarrow{\times} H^n\big(C^*(X;R)\otimes_R C^*(Y;R)\big) \\ &\longrightarrow \bigoplus_{k+\ell=n+1} \operatorname{Tor}_R\big(H^k(X;R), H^\ell(Y;R)\big) \\ &\longrightarrow 0 \end{aligned}$$

the Tor term landing in cohomological degree $n+1$ (one *higher*, since the connecting map of the algebraic theorem raises cohomological degree by one). Compose the middle term with the isomorphism of Step 1, computing the inverse of EZ^* as AW^*, which theorem 4.3.1 and lemma 4.1.22(2) allow. For cocycles $a \in C^k(X;R)$ and $b \in C^\ell(Y;R)$ and a singular $(k+\ell)$-simplex σ of $X \times Y$, only one term of $\mathrm{AW}(\sigma)$ pairs nontrivially with $a \otimes b$, and

$$(a \otimes b)\big(\mathrm{AW}(\sigma)\big) = (-1)^{k\ell}\, a\big((p \circ \sigma)|_{[v_0,\dots,v_k]}\big)\, b\big((q \circ \sigma)|_{[v_k,\dots,v_{k+\ell}]}\big) = (-1)^{k\ell}\big(p^\sharp a \smile q^\sharp b\big)(\sigma)$$

the sign coming from the interchange map of Step 1. So the left-hand map sends $\alpha \otimes \beta$ to $(-1)^{k\ell}\, p^*\alpha \smile q^*\beta$. Precomposing it with the automorphism $\alpha \otimes \beta \mapsto (-1)^{k\ell}\alpha \otimes \beta$ of the left-hand term, which preserves exactness, turns it into the cohomology cross product of definition 4.3.7.

Step 3: Ring structure and the field case. When R is a field, Tor vanishes and the cross product is an isomorphism of graded R-modules. That it is a ring isomorphism is definition 4.3.7, since the cross product is always a ring map and here it is bijective. The splitting and its non-naturality are inherited from theorem 4.3.4, completing the proof.

The cohomology Künneth theorem is the statement that the cohomology *ring* of a product is the graded tensor product of the rings, when one factor has finitely generated cohomology and (for the clean isomorphism) coefficients lie in a field. Over $\mathbb{Z}$ with finite type and free cohomology, the same isomorphism holds and is the standard route to computing cup-product structures on products: the relation $H^\bullet(T^n;\mathbb{Z}) \cong \Lambda_{\mathbb{Z}}[\alpha_1,\dots,\alpha_n]$ recorded in example 4.2.8 is the n-fold iterate of example 4.3.8, and the structure of $H^\bullet$ of any finite product of spheres follows the same way. The finiteness hypothesis cannot be dropped: for an infinite wedge or a space with infinitely generated homology in some degree, the natural map $H^\bullet(X) \otimes H^\bullet(Y) \to H^\bullet(X \times Y)$ can fail to be surjective, as the cohomology of the product involves a product where the tensor product involves a sum.

The finite-type hypothesis appears only on cohomology because cochains dualize chains: a direct sum of homology groups dualizes to a direct *product* of cohomology groups, and only finite generation forces sum and product to agree. Homology has no such constraint, which is why theorem 4.3.5 holds for all spaces while theorem 4.3.9 requires finite type. With these two theorems the homology and cohomology of any product, and the cup-product ring of a finite-type product, are computable from the factors.

Exercise 4.3.1

1. Compute $H_*(S^2 \times S^4;\mathbb{Z})$ and $H_*(S^3 \times S^3;\mathbb{Z})$ in every degree using theorem 4.3.5, and identify a generator of each nonzero group as a cross product of sphere classes.
2. Let $X = S^1 \times \mathbb{R}\mathrm{P}^2$. Compute $H_n(X;\mathbb{Z})$ for all n, being careful with the Tor term. In which degree, if any, does a Tor contribution appear, and why does it (or does it not) vanish?
3. Using the cohomology Künneth theorem over the field $\mathbb{Z}/2$, compute $H^\bullet(\mathbb{R}\mathrm{P}^\infty \times \mathbb{R}\mathrm{P}^\infty;\mathbb{Z}/2)$ as a graded ring, given that $H^\bullet(\mathbb{R}\mathrm{P}^\infty;\mathbb{Z}/2) \cong \mathbb{Z}/2[x]$ with $|x| = 1$. (Recall $\mathbb{R}\mathrm{P}^\infty$ has finite type over $\mathbb{Z}/2$.)
4. Show directly from theorem 4.3.5 that if X and Y both have free R-homology in every degree, then the cross product $H_*(X;R) \otimes_R H_*(Y;R) \to H_*(X \times Y;R)$ is an isomorphism. Where is freeness used?
5. The Künneth splitting is not natural. Exhibit the issue concretely: explain why a single choice of splitting of the short exact sequence for $H_*(X \times Y)$ cannot be made compatible

with all maps $X \to X'$ at once, by reference to the dependence of the splitting on the chosen retraction $r_n \colon C_n \twoheadrightarrow Z_n$ (equivalently the splitting of (4.12)) in the proof of theorem 4.3.4.

6. Compute the cohomology ring $H^\bullet(S^2 \times S^2; \mathbb{Z})$ using theorem 4.3.9, writing the generators and all relations. Contrast its ring with that of $\mathbb{C}\mathrm{P}^2$ from example 4.2.9, and conclude that $S^2 \times S^2$ and $\mathbb{C}\mathrm{P}^2$ are not homotopy equivalent.

4.4 Orientations and the Fundamental Class

The symmetry between homology and cohomology that Poincaré duality records is not a property of an arbitrary space. It is a property of *manifolds*, and it depends on an extra piece of geometric data that a bare topological space does not carry: an *orientation*. Before duality can be stated, this data must be defined, its existence analyzed, and the single homology class it produces (the *fundamental class*) constructed. That is the work of this section. The strategy is local-to-global throughout. The local homology groups of definition 3.4.21 already see the manifold one point at a time, and the first result computes them: at every point of an n-manifold the local homology with coefficients in a ring R is a free R-module of rank one. A choice of generator at a point is a local orientation. The problem is to choose generators at all points at once, compatibly, and then to package the compatible choice into a global homology class. The compatibility is governed by a covering space, the obstruction to a global choice is a $\mathbb{Z}/2$ phenomenon, and the packaging is carried out by a compactness argument that glues the local data over the whole manifold.

Throughout, an *n-manifold* is a Hausdorff space M in which every point has an open neighbourhood homeomorphic to $\mathbb{R}^n$, and *closed* means compact without boundary. The coefficient object is a commutative ring R with unit. The two cases the reader should keep in mind are $R = \mathbb{Z}$ and $R = \mathbb{Z}/2$. All homology is singular homology with coefficients in R, as in chapter 3.

Theorem 4.4.1: Local Homology of a Manifold

Let M be an n-manifold and $x \in M$. Then the local homology group (definition 3.4.21) with coefficients in a ring R is

$$H_i\big(M, M \setminus \{x\}; R\big) \cong \begin{cases} R & i = n \\ 0 & i \neq n \end{cases}$$

More generally, if $U \subseteq M$ is open and homeomorphic to $\mathbb{R}^n$ and $x \in U$, then the inclusion induces an isomorphism $H_i(U, U \setminus \{x\}; R) \cong H_i(M, M \setminus \{x\}; R)$ for all i.

***Proof*:**
Step 1: Reduction to a Euclidean neighbourhood. Choose an open set $U \ni x$ with $U \cong \mathbb{R}^n$. Set $Z = M \setminus U$ and $A = M \setminus \{x\}$. Then $Z \subseteq A$, and $\overline{Z} \subseteq Z \subseteq A = \mathrm{int} A$ since A is open, so the closure of Z lies in the interior of A. The excision theorem (theorem 3.4.9) applied to $Z \subseteq A \subseteq M$ gives

$$H_i\big(M \setminus Z, A \setminus Z; R\big) \cong H_i\big(M, A; R\big)$$

Now $M \setminus Z = U$ and $A \setminus Z = (M \setminus \{x\}) \cap U = U \setminus \{x\}$, so

$$H_i\big(U, U \setminus \{x\}; R\big) \cong H_i\big(M, M \setminus \{x\}; R\big)$$

This is the second assertion, and it reduces the computation to the model pair $(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$ after transporting x to 0 by the homeomorphism $U \cong \mathbb{R}^n$.

Step 2: The local homology of $\mathbb{R}^n$ at the origin. Consider the long exact sequence of the pair $(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$ with coefficients in R (definition 3.4.2):

$$\cdots \to H_i(\mathbb{R}^n; R) \to H_i\big(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R\big) \xrightarrow{\partial} H_{i-1}\big(\mathbb{R}^n \setminus \{0\}; R\big) \to H_{i-1}(\mathbb{R}^n; R) \to \cdots$$

The space $\mathbb{R}^n$ is contractible, so $H_i(\mathbb{R}^n; R) = 0$ for $i > 0$ and $H_0(\mathbb{R}^n; R) = R$. For $i \geq 2$ the two outer terms vanish, so the connecting map ∂ is an isomorphism

$$H_i\big(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R\big) \cong H_{i-1}\big(\mathbb{R}^n \setminus \{0\}; R\big) \cong \widetilde{H}_{i-1}\big(\mathbb{R}^n \setminus \{0\}; R\big)$$

the reduced and unreduced groups agreeing in positive degree. The radial deformation retraction $\mathbb{R}^n \setminus \{0\} \simeq S^{n-1}$ identifies $\widetilde{H}_{i-1}(\mathbb{R}^n \setminus \{0\}; R) \cong \widetilde{H}_{i-1}(S^{n-1}; R)$, which is R for $i - 1 = n - 1$ and 0 otherwise. Hence $H_i(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R) \cong R$ for $i = n$ and 0 for $i \geq 2$, $i \neq n$.

Step 3: The low-degree end. It remains to treat $i = 0$ and $i = 1$, and the case $n = 1$ where $S^{n-1} = S^0$ is disconnected. For $i = 0$ the relative group $H_0(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R)$ is the cokernel of $H_0(\mathbb{R}^n \setminus \{0\}; R) \to H_0(\mathbb{R}^n; R)$. For $n \geq 1$ both $\mathbb{R}^n \setminus \{0\}$ and $\mathbb{R}^n$ are nonempty and the induced map on H_0 is surjective (an inclusion of nonempty spaces into a path-connected one is onto H_0), so the relative H_0 vanishes. For $i = 1$ the relevant segment is

$$H_1(\mathbb{R}^n; R) \to H_1\big(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R\big) \xrightarrow{\partial} \widetilde{H}_0\big(\mathbb{R}^n \setminus \{0\}; R\big) \to \widetilde{H}_0(\mathbb{R}^n; R) = 0$$

with $H_1(\mathbb{R}^n; R) = 0$, so $H_1(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R) \cong \widetilde{H}_0(\mathbb{R}^n \setminus \{0\}; R) \cong \widetilde{H}_0(S^{n-1}; R)$. Now $\widetilde{H}_0(S^{n-1}; R)$ is R when $n = 1$ (where S^0 is two points) and 0 when $n \geq 2$ (where S^{n-1} is path-connected), so the degree-one group $H_1(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R)$ is R for $n = 1$ and 0 for $n \geq 2$, agreeing in both cases with "R in degree n, zero otherwise." Combining the three steps, $H_i(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R)$ is R in degree $i = n$ and 0 in every other degree, for all $n \geq 1$, as we sought to show.

The computation says that a manifold looks, homologically and one point at a time, exactly like Euclidean space: the only nonzero local homology sits in the top degree n, where it is a free R-module on a single generator. The degree n is the dimension, recovered intrinsically. This is the homological content of invariance of dimension (theorem 3.4.20), now with arbitrary coefficients. The single rank-one group in degree n is what makes orientation a binary choice: a free R-module of rank one has exactly the generators that R permits, and choosing one at each point is the data to be organized.

Example 4.4.2: Local Orientations of the Plane

Take $M = \mathbb{R}^2$, $x = 0$, and $R = \mathbb{Z}$. By theorem 4.4.1, $H_2(\mathbb{R}^2, \mathbb{R}^2 \setminus \{0\}; \mathbb{Z}) \cong \mathbb{Z}$. A generator is produced concretely as follows. Let $\Delta \subseteq \mathbb{R}^2$ be a small triangle with 0 in its interior, regarded as a singular 2-simplex $\sigma\colon \Delta^2 \to \mathbb{R}^2$ via an affine homeomorphism onto Δ. Its boundary $\partial\sigma$ is a 1-cycle encircling 0, a loop in $\mathbb{R}^2 \setminus \{0\}$ representing a generator of $H_1(\mathbb{R}^2 \setminus \{0\}; \mathbb{Z}) \cong \mathbb{Z}$. Under the connecting isomorphism $\partial\colon H_2(\mathbb{R}^2, \mathbb{R}^2 \setminus \{0\}; \mathbb{Z}) \xrightarrow{\cong} \widetilde{H}_1(\mathbb{R}^2 \setminus \{0\}; \mathbb{Z})$ of Step 2 above, the relative class $[\sigma]$ maps to $[\partial\sigma]$, so $[\sigma]$ is a generator. Reversing the vertex order of σ (a transposition, of sign -1) replaces $[\sigma]$ by $-[\sigma]$, the other generator. The two generators $\pm[\sigma]$ are the two local orientations at 0: the triangle traversed counterclockwise and clockwise. With $R = \mathbb{Z}/2$ instead, $H_2(\mathbb{R}^2, \mathbb{R}^2 \setminus \{0\}; \mathbb{Z}/2) \cong \mathbb{Z}/2$ has a unique nonzero element, $[\sigma] = -[\sigma]$, so there is exactly one local $\mathbb{Z}/2$-orientation: over $\mathbb{Z}/2$ clockwise and counterclockwise are the same.

The example isolates the binary choice at a single point. An orientation of the whole manifold must make this choice everywhere, and the choices at nearby points must agree in the only sense that makes sense: through a chart that identifies the local homology groups of neighbouring points. The device that records this agreement is a covering space whose fibre over x is the local homology group at x.

Definition 4.4.3: Orientation Covering

Let M be an n-manifold and R a ring. The *R-orientation covering* of M is the set

$$\widetilde{M}_R = \coprod_{x \in M} H_n\big(M, M \setminus \{x\}; R\big)$$

with projection $p\colon \widetilde{M}_R \to M$ sending an element of $H_n(M, M \setminus \{x\}; R)$ to x. Its topology is generated by the following basic open sets. For each open $U \subseteq M$ homeomorphic to $\mathbb{R}^n$, each ball $B \subseteq U$ with $\overline{B}$ compact and contained in U, and each $\alpha_B \in H_n(M, M \setminus B; R)$, let

$$\langle \alpha_B \rangle = \big\{\, \rho_x(\alpha_B) \;:\; x \in B \,\big\}$$

where $\rho_x\colon H_n(M, M \setminus B; R) \to H_n(M, M \setminus \{x\}; R)$ is the map induced by the inclusion $M \setminus B \hookrightarrow M \setminus \{x\}$.

The map $p\colon \widetilde{M}_R \to M$ is a covering space whose fibre over x is the R-module $H_n(M, M \setminus \{x\}; R) \cong R$. The reason the sets $\langle \alpha_B \rangle$ form a basis is that over a Euclidean ball B the restriction maps ρ_x are all isomorphisms with a common source $H_n(M, M \setminus B; R) \cong R$ (this is the $A = \overline{B}$ case of the gluing lemma below), so each $\langle \alpha_B \rangle$ is a sheet over B and these sheets partition $p^{-1}(B)$ into $|R|$ copies of B. A global orientation is exactly a continuous choice of one element from each fibre, made into a generator: a section of p landing in generators. This reformulates the everywhere-compatible local choice as a single continuous map.

Definition 4.4.4: Orientation

Let M be an n-manifold and R a ring. A *local R-orientation of M at x* is a choice of generator μ_x of the free rank-one R-module $H_n(M, M \setminus \{x\}; R)$ (theorem 4.4.1). An *R-orientation of M* is a function $x \mapsto \mu_x$ assigning to each $x \in M$ a local R-orientation, subject to the local consistency condition: each $x \in M$ has an open neighbourhood $U \cong \mathbb{R}^n$ and a ball $B \ni x$ with $\overline{B} \subseteq U$ compact, together with a class $\mu_B \in H_n(M, M \setminus B; R)$, such that $\rho_y(\mu_B) = \mu_y$ for every $y \in B$. The manifold M is *R-orientable* if an R-orientation exists. An R-orientation is the same datum as a section $x \mapsto \mu_x$ of the orientation covering $\widetilde{M}_R$ (definition 4.4.3) taking values in generators of the fibres. For $R = \mathbb{Z}$ one speaks of an *orientation* and of M being *orientable*.

The consistency condition is the precise meaning of "the choice varies continuously." Over a single chart $U \cong \mathbb{R}^n$ all the local homology groups are canonically identified through a class μ_B defined on a ball, and consistency requires that the chosen generators be the restrictions of one such μ_B. Equivalently, the function $x \mapsto \mu_x$ is a continuous section of $\widetilde{M}_R$, since the basic open set $\langle \mu_B \rangle$ is exactly a neighbourhood through which the section is constant. For connected M a section taking generator values is determined by its value at a single point, because $\widetilde{M}_R$ restricted to generators is a covering of M with fibre the unit group worth of generators and a section of a covering over

a connected base is rigid once a sheet is chosen. The case $R = \mathbb{Z}$ is the classical one: each fibre $H_n(M, M\setminus\{x\}; \mathbb{Z}) \cong \mathbb{Z}$ has exactly two generators $\pm\mu_x$, so an orientation is a consistent choice of sign, and the obstruction to making it is whether the two-sheeted covering of generators is connected (no orientation) or split (orientation exists). This corrects the earlier provisional definition: orientability is not a statement about the torsion of H_{n-1}, but about the existence of a consistent generator of the top local homology at every point.

Example 4.4.5: The Möbius Band is Non-Orientable

Let M be the open Möbius band, the quotient of $\mathbb{R} \times (-1, 1)$ by $(s, t) \sim (s + 1, -t)$. It is a 2-manifold. Fix the point $x_0 = [(0, 0)]$ on the core circle and a local orientation μ_{x_0}, realized as in example 4.4.2 by a small counterclockwise triangle in a chart around x_0. Transport μ_{x_0} along the core circle $t = 0$, parametrized by $s \in [0, 1]$: at each stage extend the chosen generator continuously into the next overlapping chart using a class μ_B as in definition 4.4.4. After traversing the full circle, the parameter returns from $s = 0^+$ to $s = 1^-$, but the gluing $(s, t) \sim (s + 1, -t)$ reflects the second coordinate, sending a counterclockwise triangle to a clockwise one. The transported generator that returns to x_0 is therefore $-\mu_{x_0}$, not μ_{x_0}. A $\mathbb{Z}$-orientation would require a globally consistent choice, contradicting the sign reversal along the loop, so the Möbius band admits no $\mathbb{Z}$-orientation: it is non-orientable. The same loop traversed with $R = \mathbb{Z}/2$ returns μ_{x_0} to itself, since $-\mu_{x_0} = \mu_{x_0}$ in $\mathbb{Z}/2$. The Möbius band is $\mathbb{Z}/2$-orientable, as every manifold is.

The example exhibits the orientation covering directly: the two-sheeted covering of generators of $\widetilde{M}_{\mathbb{Z}}$ over the Möbius band is the connected double cover (the annulus mapping onto the band), and the failure of orientability is the failure of that double cover to split into two sheets. For an orientable manifold the double cover is two disjoint copies of M, and choosing one is choosing an orientation. With $R = \mathbb{Z}/2$ there is only one generator per fibre, so the covering of generators is a copy of M itself, always split, which is why $\mathbb{Z}/2$-orientations always exist. This last point is what makes mod-2 Poincaré duality unconditional.

The local data is now organized. The remaining task is global: to convert an orientation, a compatible family of local generators, into a single homology class on M living in $H_n(M; R)$. The mechanism is a gluing lemma that builds a class over any compact set out of its pointwise restrictions, controlling at the same time the higher-degree groups. This is the technical heart of the subject.

Lemma 4.4.6: Compact Gluing of Local Homology

Let M be an n-manifold, R a ring, and $A \subseteq M$ compact. Then:

1. $H_i(M, M \setminus A; R) = 0$ for all $i > n$.
2. A class $\alpha \in H_n(M, M \setminus A; R)$ is zero if and only if its restriction $\rho_x(\alpha) \in H_n(M, M \setminus \{x\}; R)$ is zero for every $x \in A$.
3. Given any section $x \mapsto \mu_x$ of the orientation covering over A (a function with $\rho_y(\mu_B) = \mu_y$ locally, as in definition 4.4.4), there is a unique class $\mu_A \in H_n(M, M \setminus A; R)$ with $\rho_x(\mu_A) = \mu_x$ for all $x \in A$.

***Proof*:**
The argument is an induction on the number of pieces in a finite cover of A by Euclidean balls, with the relative Mayer-Vietoris sequence (proposition 3.4.24) stitching the pieces together. Write $\alpha_x := \rho_x(\alpha)$ for the restriction of a relative class to the local homology at x.

Step 1: Two reductions. First, the three statements are stable under the Mayer-Vietoris induction in the following sense. Suppose $A = A_1 \cup A_2$ with $A_1, A_2, A_1 \cap A_2$ all compact and the lemma known for each of A_1, A_2, $A_1 \cap A_2$. The relative Mayer-Vietoris sequence of proposition 3.4.24, taken with both covering sets equal to M and with the open sets $M \setminus A_1$, $M \setminus A_2$ (whose union is $M \setminus (A_1 \cap A_2)$ and whose intersection is $M \setminus A$) in the roles of C and D, reads

$$\cdots \to H_i(M, M\setminus A; R) \to H_i(M, M\setminus A_1; R) \oplus H_i(M, M\setminus A_2; R) \to H_i(M, M\setminus(A_1\cap A_2); R) \to \cdots$$

For $i > n$ the two middle and the right terms vanish by inductive hypothesis (1), so the left term vanishes, proving (1) for A. For (2) and (3), the segment

$$0 \to H_n(M, M \setminus A; R) \xrightarrow{\Phi} H_n(M, M \setminus A_1; R) \oplus H_n(M, M \setminus A_2; R) \xrightarrow{\Psi} H_n(M, M \setminus (A_1 \cap A_2); R)$$

is exact on the left because the preceding term $H_{n+1}(M, M \setminus (A_1 \cap A_2); R)$ vanishes by (1). Here $\Phi(\alpha) = (\rho^{A_1}\alpha, \rho^{A_2}\alpha)$ and $\Psi(\beta_1, \beta_2) = \rho^{A_1\cap A_2}\beta_1 - \rho^{A_1\cap A_2}\beta_2$, the maps induced by the inclusions of complements, obtained from those of proposition 3.4.24(3) by changing the sign of the second summand, which preserves exactness. Injectivity of Φ gives statement (2): a class restricting to zero at every point of A restricts to zero on each A_j (by inductive (2) for A_j), hence is zero in each summand, hence zero by injectivity. For statement (3), the given section over A restricts to sections over A_1 and A_2, producing μ_{A_1} and μ_{A_2} by inductive (3). These agree over $A_1 \cap A_2$ by uniqueness of $\mu_{A_1\cap A_2}$, so $\Psi(\mu_{A_1}, \mu_{A_2}) = 0$ and exactness yields a class μ_A with $\Phi(\mu_A) = (\mu_{A_1}, \mu_{A_2})$, restricting correctly at every point, and unique by (2). Thus the lemma propagates across unions.

Second, by compactness A is covered by finitely many Euclidean balls, and the union reduction lets one induct on their number once the base case (a single compact convex set in a chart) is settled.

Step 2: The base case, A a compact convex subset of a chart. Let $A \subseteq U \cong \mathbb{R}^n$ be compact and convex. By excision (the Euclidean reduction of theorem 4.4.1 applied to the pair, replacing the point by A), $H_i(M, M\setminus A; R) \cong H_i(\mathbb{R}^n, \mathbb{R}^n \setminus A; R)$. Since A is convex, $\mathbb{R}^n \setminus A$ deformation retracts onto a sphere S^{n-1} at large radius, so the pair $(\mathbb{R}^n, \mathbb{R}^n \setminus A)$ has the same relative homology as $(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$: namely R in degree n and 0 elsewhere, by the computation in the proof of theorem 4.4.1. This gives (1), with vanishing above degree n, and shows $H_n(M, M \setminus A; R) \cong R$. For (2) and (3), the restriction map $\rho_x \colon H_n(M, M \setminus A; R) \to H_n(M, M \setminus \{x\}; R)$ is an isomorphism for each $x \in A$, because both groups are computed by the same retraction of $\mathbb{R}^n \setminus A \simeq \mathbb{R}^n \setminus \{x\} \simeq S^{n-1}$ and ρ_x is the induced identity. Hence a class is zero iff its (single, up to the isomorphism) restriction is zero, giving (2). A section over A, being the consistent family of images of one generator $\mu_A \in H_n(M, M \setminus A; R) \cong R$, is realized by that unique μ_A, giving (3).

Step 3: The general compact set. Cover A by finitely many balls $B_1, \ldots, B_k$ contained in charts, with each $\overline{B_j}$ compact. Replacing each B_j by a compact convex set $A_j = A \cap \overline{B_j}$ inside its chart (intersections of A with closed balls, which are compact, and which one further subdivides into convex pieces inside a single chart if necessary), write $A = A_1 \cup \cdots \cup A_k$ with each A_j a finite union of compact convex chart-pieces. The base case settles each A_j and each of their pairwise (and higher) intersections, which are again compact convex chart-pieces. Inducting on k via

the union reduction of Step 1 establishes all three statements for A. The hardest bookkeeping is the verification that the iterated Mayer-Vietoris assembly produces a genuinely well-defined section-to-class correspondence independent of the chosen cover. This is carried out in full in [21, Lemma 3.27], whose argument is exactly the induction outlined here. This completes the proof.

The lemma is what the whole theory rests on. Part (1) says a manifold has no homology above its dimension, even relative to the complement of a compact set. Part (2) says top-degree relative classes are detected pointwise, so the local homology groups separate them. Part (3) says a compatible family of local generators over a compact set always assembles into a single class. Applying part (3) to $A = M$ when M itself is compact converts an orientation into a homology class, which is exactly the fundamental class. The next theorem carries this out and, in the process, reads off the entire top homology of a closed connected manifold.

Theorem 4.4.7: Top Homology of a Closed Manifold

Let M be a closed connected n-manifold and R a ring. For each $x \in M$ the restriction map

$$\rho_x \colon H_n(M;R) \longrightarrow H_n\big(M, M \setminus \{x\}; R\big) \cong R$$

is injective, and:

1. if M is R-orientable, ρ_x is an isomorphism, so $H_n(M;R) \cong R$;
2. if M is not R-orientable, the image of ρ_x is $\{\, r \in R : 2r = 0 \,\}$, the 2-torsion of R.

Moreover $H_i(M;R) = 0$ for all $i > n$. In particular $H_n(M;\mathbb{Z}) \cong \mathbb{Z}$ when M is orientable and $H_n(M;\mathbb{Z}) = 0$ when M is not, while $H_n(M;\mathbb{Z}/2) \cong \mathbb{Z}/2$ always.

***Proof*:**
Apply lemma 4.4.6 with $A = M$, which is compact since M is closed. Here $M \setminus A = \varnothing$, so $H_i(M, M \setminus A; R) = H_i(M, \varnothing; R) = H_i(M;R)$, and the restriction ρ_x of the lemma is the map in the statement.

Step 1: Vanishing above the dimension. Part (1) of lemma 4.4.6 gives $H_i(M;R) = H_i(M, M \setminus M; R) = 0$ for $i > n$, the last assertion.

Step 2: Injectivity. Let $\alpha \in H_n(M;R)$ with $\rho_x(\alpha) = 0$ for the given x. Because M is connected, the function $y \mapsto \rho_y(\alpha)$ is a section of the orientation covering $\widetilde{M}_R$ over the connected base M. Sections are locally constant (the consistency neighbourhoods $\langle \mu_B \rangle$ of definition 4.4.3 are exactly the sheets through which $y \mapsto \rho_y(\alpha)$ is constant), hence constant on the connected M. If $\rho_x(\alpha) = 0$ then $\rho_y(\alpha) = 0$ for all $y \in M$, and part (2) of lemma 4.4.6 forces $\alpha = 0$. Thus ρ_x is injective for every x.

Step 3: The image. Fix x and identify $H_n(M, M \setminus \{x\}; R) \cong R$. By Step 2 and the constancy of $y \mapsto \rho_y(\alpha)$, the image $\rho_x(H_n(M;R)) \subseteq R$ consists exactly of those $r \in R$ that extend to a constant section $y \mapsto r$ (under the local identifications $H_n(M, M \setminus \{y\}; R) \cong R$) which lifts to a global class. By part (3) of lemma 4.4.6 applied to $A = M$, a section over M lifts to a class precisely when it *is* a section, that is, when $y \mapsto r$ is consistent across overlaps. Two charts overlap with their local identifications differing by the sign of the Jacobian: an orientation-preserving overlap identifies r with r, an orientation-reversing overlap identifies r with $-r$. Consistency of the constant section $y \mapsto r$ therefore requires $r = -r$, that is $2r = 0$, exactly along every

reversing overlap.

1. If M is R-orientable, all overlaps can be taken orientation-preserving (the orientation μ trivializes the covering), so the constraint $2r = 0$ is vacuous and every $r \in R$ extends to a global class: ρ_x is onto R, hence an isomorphism by Step 2, and $H_n(M; R) \cong R$.

2. If M is not R-orientable, the generator-covering of $\widetilde{M}_R$ is nontrivial, so there is a loop along which the local identification reverses sign. A constant section $y \mapsto r$ is consistent around this loop only if $r = -r$. Hence the image is contained in $\{r : 2r = 0\}$. Conversely every such r does extend, since the sign ambiguity $r = -r$ is then immaterial and the section is globally consistent. So the image is exactly the 2-torsion of R.

Step 4: Specializations. For $R = \mathbb{Z}$ the 2-torsion is $\{0\}$, so ρ_x has image $\mathbb{Z}$ (orientable, giving $H_n(M; \mathbb{Z}) \cong \mathbb{Z}$) or image 0 (non-orientable, giving $H_n(M; \mathbb{Z}) = 0$, since ρ_x is injective with zero image). For $R = \mathbb{Z}/2$ every element satisfies $2r = 0$, so M is automatically $\mathbb{Z}/2$-orientable and $H_n(M; \mathbb{Z}/2) \cong \mathbb{Z}/2$ in all cases, as we sought to show.

The theorem reads the orientability of M directly off its top homology: a closed connected n-manifold is orientable exactly when $H_n(M; \mathbb{Z}) \cong \mathbb{Z}$, and non-orientable exactly when $H_n(M; \mathbb{Z}) = 0$. The injectivity of ρ_x is the structural fact underneath: top homology of a closed connected manifold injects into a single copy of R, so it is at most rank one, and orientability decides whether the rank is one or zero. The mod-2 statement explains why $\mathbb{Z}/2$-coefficients are the universal setting for duality on possibly-non-orientable manifolds: $H_n(M; \mathbb{Z}/2) \cong \mathbb{Z}/2$ gives a fundamental class no matter what, though it forgets signs. With the top homology pinned down, the distinguished generator can be named.

Definition 4.4.8: [Manifold] Fundamental Class

Let M be a closed connected R-orientable n-manifold equipped with an R-orientation $x \mapsto \mu_x$ (definition 4.4.4). The *fundamental class* of M with respect to this orientation is the unique class

$$[M] \in H_n(M; R)$$

whose restriction $\rho_x([M]) = \mu_x$ for every $x \in M$. When $R = \mathbb{Z}$ and an orientation is fixed, $[M]$ is *the* fundamental class. For $R = \mathbb{Z}/2$ the (unique nonzero) class $[M] \in H_n(M; \mathbb{Z}/2) \cong \mathbb{Z}/2$ is the mod-2 fundamental class, requiring no choice of orientation.

Theorem 4.4.9: Existence and Uniqueness of the Fundamental Class

With M as in definition 4.4.8, the fundamental class $[M]$ exists and is unique. It is a generator of $H_n(M; R) \cong R$, and reversing the orientation $\mu_x \mapsto -\mu_x$ replaces $[M]$ by $-[M]$.

***Proof*:**
The orientation $x \mapsto \mu_x$ is, by definition 4.4.4, a section of the orientation covering over M taking generator values. Apply part (3) of lemma 4.4.6 with $A = M$ (compact, since M is closed): there is a unique class $\mu_M = [M] \in H_n(M, M \setminus M; R) = H_n(M; R)$ with $\rho_x([M]) = \mu_x$ for all x. Uniqueness is part of the lemma and also follows from the injectivity of ρ_x in theorem 4.4.7: two classes with the same restriction at one point are equal. By that theorem $\rho_x \colon H_n(M; R) \to R$

is an isomorphism (the R-orientable case), and $\rho_x([M]) = \mu_x$ is a generator of R, so $[M]$ is a generator of $H_n(M; R)$. Replacing the orientation by its negative replaces each μ_x by $-\mu_x$. By uniqueness the new fundamental class restricts to $-\mu_x$ everywhere, and $-[M]$ does exactly this, so the reversed orientation has fundamental class $-[M]$, completing the proof.

The fundamental class is the algebraic embodiment of the manifold as a whole: a single homology cycle that simultaneously orients every point and generates the top homology. Its existence is what allows Poincaré duality to be phrased as cap product with one class, $D(\alpha) = [M] \frown \alpha$ in the notation of the next section. The sign ambiguity $[M] \leftrightarrow -[M]$ is precisely the two orientations. Over $\mathbb{Z}/2$, where $[M] = -[M]$, the ambiguity disappears and the fundamental class is canonical. The next example computes $[M]$ in the cases that recur throughout duality theory.

Example 4.4.10: Fundamental Classes of Spheres and Projective Planes

1. *The sphere S^n.* The n-sphere is a closed connected orientable n-manifold. Its homology is $H_n(S^n; \mathbb{Z}) \cong \mathbb{Z}$ and $H_i(S^n; \mathbb{Z}) = 0$ for $0 < i < n$ (chapter 3), consistent with theorem 4.4.7. A generator of $H_n(S^n; \mathbb{Z}) \cong \mathbb{Z}$ is a fundamental class $[S^n]$. The two choices $\pm[S^n]$ are the two orientations, and the standard one is given by the degree-one generator under which the identity map $\mathrm{id}\colon S^n \to S^n$ has degree $+1$. Concretely, writing $S^n = \partial\Delta^{n+1}$ as the boundary of a simplex, the alternating sum of the n-faces with the boundary orientation is a cycle representing $[S^n]$. Restriction $\rho_x([S^n])$ at any x is the local orientation μ_x, recovering the local generator of example 4.4.2 in the $n = 2$ case.

2. *The projective plane $\mathbb{R}\mathrm{P}^2$.* The real projective plane is a closed connected non-orientable 2-manifold. Its integral homology is $H_0(\mathbb{R}\mathrm{P}^2; \mathbb{Z}) = \mathbb{Z}$, $H_1(\mathbb{R}\mathrm{P}^2; \mathbb{Z}) = \mathbb{Z}/2$, and $H_2(\mathbb{R}\mathrm{P}^2; \mathbb{Z}) = 0$ (chapter 3). The vanishing $H_2(\mathbb{R}\mathrm{P}^2; \mathbb{Z}) = 0$ is exactly the non-orientability predicted by theorem 4.4.7(2): there is no integral fundamental class. Non-orientability can be seen geometrically as in example 4.4.5, since $\mathbb{R}\mathrm{P}^2$ contains a Möbius band (the complement of an open disk), and transporting a local orientation around its core circle reverses the sign. With $\mathbb{Z}/2$ coefficients, however, $H_2(\mathbb{R}\mathrm{P}^2; \mathbb{Z}/2) \cong \mathbb{Z}/2$, and its nonzero element is the mod-2 fundamental class $[\mathbb{R}\mathrm{P}^2]_2$, which exists with no choice required. This is the class on which mod-2 Poincaré duality for $\mathbb{R}\mathrm{P}^2$ rests.

3. *The torus $T^2 = S^1 \times S^1$.* The torus is closed, connected, and orientable, with $H_2(T^2; \mathbb{Z}) \cong \mathbb{Z}$ (chapter 3). Its fundamental class $[T^2]$ is the generator corresponding to the product orientation. Under the cellular decomposition with one 2-cell, $[T^2]$ is the class of that oriented 2-cell, with $\rho_x([T^2]) = \mu_x$ the product of the two circle orientations at each point. Reversing either circle's orientation reverses $[T^2]$.

These three computations are the templates for everything that follows. Spheres and tori carry honest integral fundamental classes and obey integral Poincaré duality. The projective plane $\mathbb{R}\mathrm{P}^2$ has only a mod-2 fundamental class and obeys only the mod-2 duality. The dichotomy is exactly the orientability dichotomy of theorem 4.4.7, and the fundamental class is the bridge from this section's local analysis to the global duality isomorphism developed next.

Exercise 4.4.1

1. Let M be an n-manifold and $x \in M$. Using theorem 4.4.1, compute $H_i(M, M \setminus \{x\}; \mathbb{Z}/2)$ in every degree i, and explain why the result is independent of whether M is orientable.

Then state the analogous computation for $H_i(M, M \setminus \{x\}; \mathbb{Q})$ and verify that the rank-one conclusion is unchanged.

2. Prove that a connected n-manifold M is orientable if and only if its orientation covering $\widetilde{M}_{\mathbb{Z}}$, restricted to the two generators $\pm\mu_x$ in each fibre, is a *disconnected* (equivalently, trivial two-sheeted) covering of M. Deduce that every simply connected manifold is orientable.

3. Let M be a closed connected n-manifold that is *not* $\mathbb{Z}$-orientable. Using theorem 4.4.7, determine $H_n(M;\mathbb{Z})$, $H_n(M;\mathbb{Z}/2)$, and $H_n(M;\mathbb{Z}/3)$. Explain in each case whether a fundamental class exists, and reconcile your answer with the Möbius-band sign-reversal mechanism of example 4.4.5.

4. Let M and N be closed connected oriented n- and m-manifolds with fundamental classes $[M]$ and $[N]$. Assuming the homology cross product $H_n(M) \otimes H_m(N) \to H_{n+m}(M \times N)$ of chapter 3, argue that $M \times N$ is a closed connected orientable $(n+m)$-manifold and identify its fundamental class $[M \times N]$ in terms of $[M]$ and $[N]$. Use this to write down $[T^n]$ for the n-torus $T^n = (S^1)^n$.

5. The orientability of a closed connected n-manifold can fail only through a sign. Prove that for any closed connected n-manifold M, the group $H_n(M;\mathbb{Z})$ is either $\mathbb{Z}$ or 0, and that $H_n(M;\mathbb{Z}) \cong \mathbb{Z}$ if and only if M is orientable. (Use the injectivity in theorem 4.4.7 and the structure of the 2-torsion of $\mathbb{Z}$.)

6. Let M be a closed connected orientable n-manifold with fundamental class $[M]$ and let $f\colon M \to M$ be a continuous map. The *degree* of f is the integer $\deg f$ with $f_*[M] = (\deg f)[M]$ in $H_n(M;\mathbb{Z}) \cong \mathbb{Z}$. Show this is well defined, that $\deg(\mathrm{id}) = 1$, that $\deg(g \circ f) = \deg g \cdot \deg f$, and that a map of nonzero degree is surjective. (For surjectivity, suppose f misses a point y and factor f_* through $H_n(M \setminus \{y\};\mathbb{Z})$.)

4.5 The Cap Product and Poincaré Duality

The cup product of definition 4.2.1 multiplies two cochains to produce a cochain, turning $H^\bullet(X;R)$ into the graded ring of definition 4.2.7. There is a companion operation that lets cohomology act on homology: a cochain and a chain combine to produce a chain of lower degree. This is the *cap product*, the operation through which Poincaré duality relates the homology and cohomology of a manifold. On a closed oriented n-manifold M there is a distinguished homology class, the fundamental class $[M] \in H_n(M;R)$ of definition 4.4.8, and capping with $[M]$ converts a degree-k cohomology class into a degree-$(n-k)$ homology class. The content of Poincaré duality is that this single operation $\alpha \mapsto [M] \frown \alpha$ is an isomorphism in every degree. The dimensions k and $n-k$ pair off, the Betti numbers of M are symmetric about the middle dimension, and the cup product becomes a perfect pairing. This section constructs the cap product at the chain level, proves it descends to (co)homology, establishes its formal properties, and proves the duality theorem in full for the compact case, deferring only the direct-limit bookkeeping of the non-compact case to a named source.

The starting point is a chain-level formula. A singular n-simplex has $n+1$ ordered vertices, and a cochain of degree k can be fed the front k-face $[v_0, \dots, v_k]$ while the back $(n-k)$-face $[v_k, \dots, v_n]$ is retained as a chain. The first records a scalar in R, the second a simplex of dimension $n-k$, and their product is the cap.

Definition 4.5.1: Cap Product [on chains]

Let X be a space and R a commutative ring. For a singular n-simplex $\sigma\colon \Delta^n \to X$ and a cochain $\varphi \in C^k(X;R)$ with $k \le n$, the *cap product* $\sigma \frown \varphi \in C_{n-k}(X;R)$ is the chain

$$\sigma \frown \varphi = \varphi\big(\sigma|_{[v_0,\dots,v_k]}\big)\ \sigma|_{[v_k,\dots,v_n]}$$

where $\sigma|_{[v_0,\dots,v_k]}$ is the restriction of σ to the front k-face and $\sigma|_{[v_k,\dots,v_n]}$ the restriction to the back $(n-k)$-face. Extending R-bilinearly over chains and cochains gives an R-bilinear map

$$\frown\colon C_n(X;R) \otimes_R C^k(X;R) \longrightarrow C_{n-k}(X;R)$$

When $k > n$ the cap product $\sigma \frown \varphi$ is declared to be 0.

The cap product runs against the grain of degree: it lowers homological dimension by the cohomological dimension of the cochain. The cochain φ "eats" the front face of the simplex and leaves the back face behind, scaled by the value it read off. The asymmetry of front and back is the source of the sign that will appear in the boundary formula, and it mirrors the splitting $[v_0,\dots,v_k] \cup [v_k,\dots,v_n]$ that already governed the cup product of definition 4.2.1: the two operations are built from the same combinatorial subdivision of a simplex into a front and a back face sharing the vertex v_k. The convention places the chain on the left and the cochain on the right, so that the preserved expression $[M] \frown \alpha$ of the duality theorem is the cap of the fundamental cycle with a cohomology class.

Example 4.5.2: The Cap Product on a Single Simplex

Take $X = \Delta^2$, the standard 2-simplex with vertices v_0, v_1, v_2, and let $\sigma = \mathrm{id}\colon \Delta^2 \to \Delta^2$ be the identity, a singular 2-simplex. Work with coefficients in $R = \mathbb{Z}$.

Degree 0. For $\varphi \in C^0(\Delta^2;\mathbb{Z})$ the front 0-face is the vertex $[v_0]$, and

$$\sigma \frown \varphi = \varphi([v_0])\ \sigma|_{[v_0,v_1,v_2]} = \varphi(v_0)\,\sigma$$

so capping with a 0-cochain rescales the simplex by the value of φ at the initial vertex. The unit cochain $1 \in C^0$, taking the value 1 on every vertex, gives $\sigma \frown 1 = \sigma$.

Degree 1. For $\varphi \in C^1(\Delta^2;\mathbb{Z})$ the front 1-face is the edge $[v_0, v_1]$ and the back face is the edge $[v_1, v_2]$, so

$$\sigma \frown \varphi = \varphi([v_0, v_1])\ [v_1, v_2]$$

a 1-chain. If φ is the cochain dual to the edge $[v_0, v_1]$ (value 1 there, 0 on the other edges), then $\sigma \frown \varphi = [v_1, v_2]$.

Degree 2. For $\varphi \in C^2(\Delta^2;\mathbb{Z})$ the front 2-face is all of $[v_0, v_1, v_2]$ and the back face is the vertex $[v_2]$, so

$$\sigma \frown \varphi = \varphi([v_0, v_1, v_2])\ [v_2] \in C_0(\Delta^2;\mathbb{Z})$$

If φ is dual to the 2-simplex itself, this is the 0-chain $[v_2]$. In each case the cap lowered the dimension from 2 to $2 - k$, as the definition requires.

For the cap product to descend to homology and cohomology, it must interact with the boundary ∂ and coboundary δ in a controlled way. The cup product obeyed the Leibniz rule of lemma 4.2.2. The cap product obeys a dual identity that is again a Leibniz rule, now mixing ∂ on the chain side and δ on the cochain side.

Lemma 4.5.3: Boundary Formula for the Cap Product

For σ a singular n-simplex and $\varphi \in C^k(X;R)$ with $k \le n$,

$$\partial(\sigma \frown \varphi) = (-1)^k(\partial\sigma \frown \varphi - \sigma \frown \delta\varphi)$$

***Proof*:**
Write $\sigma_{[a,\dots,b]}$ for $\sigma|_{[v_a,\dots,v_b]}$. By definition $\sigma \frown \varphi = \varphi(\sigma_{[0,\dots,k]})\,\sigma_{[k,\dots,n]}$, so its boundary is

$$\partial(\sigma \frown \varphi) = \varphi(\sigma_{[0,\dots,k]}) \sum_{i=k}^{n} (-1)^{i-k}\, \sigma_{[k,\dots,\hat{v}_i,\dots,n]}$$

the sign $(-1)^{i-k}$ arising because v_i is the $(i-k)$-th vertex of the back face $[v_k,\dots,v_n]$. Pulling the scalar $(-1)^{-k} = (-1)^k$ out front,

$$\partial(\sigma \frown \varphi) = (-1)^k \varphi(\sigma_{[0,\dots,k]}) \sum_{i=k}^{n} (-1)^{i}\, \sigma_{[k,\dots,\hat{v}_i,\dots,n]} \tag{4.18}$$

Now expand the right-hand side. First,

$$\partial\sigma \frown \varphi = \sum_{i=0}^{n} (-1)^i\, (\sigma_{[0,\dots,\hat{v}_i,\dots,n]} \frown \varphi)$$

Capping $\sigma_{[0,\dots,\hat{v}_i,\dots,n]}$, an $(n-1)$-simplex, with the k-cochain φ reads off its front k-face. For $i \le k$ the deleted vertex lies among the front $k+1$ vertices, so the front k-face of $\sigma_{[0,\dots,\hat{v}_i,\dots,n]}$ is $\sigma_{[0,\dots,\hat{v}_i,\dots,k+1]}$ and the back face is $\sigma_{[k+1,\dots,n]}$. For $i \ge k+1$ the front k-face is the unchanged $\sigma_{[0,\dots,k]}$ and the deletion happens in the back face, giving back face $\sigma_{[k,\dots,\hat{v}_i,\dots,n]}$. Splitting the sum at $i = k$,

$$\partial\sigma \frown \varphi = \sum_{i=0}^{k} (-1)^i \varphi(\sigma_{[0,\dots,\hat{v}_i,\dots,k+1]})\, \sigma_{[k+1,\dots,n]} + \sum_{i=k+1}^{n} (-1)^i \varphi(\sigma_{[0,\dots,k]})\, \sigma_{[k,\dots,\hat{v}_i,\dots,n]} \tag{4.19}$$

the index $i = k$ being placed with the first block by convention. Second, the coboundary $\delta\varphi \in C^{k+1}$ evaluated on the front $(k+1)$-face gives

$$\sigma \frown \delta\varphi = (\delta\varphi)(\sigma_{[0,\dots,k+1]})\, \sigma_{[k+1,\dots,n]} = \left(\sum_{i=0}^{k+1} (-1)^i \varphi(\sigma_{[0,\dots,\hat{v}_i,\dots,k+1]}) \right) \sigma_{[k+1,\dots,n]}$$

using $(\delta\varphi)(\tau) = \varphi(\partial\tau)$ on the simplex $\tau = \sigma_{[0,\dots,k+1]}$. Therefore

$$\sigma \frown \delta\varphi = \sum_{i=0}^{k+1} (-1)^i \varphi(\sigma_{[0,\dots,\hat{v}_i,\dots,k+1]})\, \sigma_{[k+1,\dots,n]} \tag{4.20}$$

Now form $\partial\sigma \frown \varphi - \sigma \frown \delta\varphi$ using (4.19) and (4.20). The first block of (4.19) runs i from 0 to k. The sum (4.20) runs i from 0 to $k+1$. Their difference cancels the overlapping terms $i = 0,\dots,k$ exactly, leaving from (4.20) only the single term $i = k+1$,

$$-(-1)^{k+1} \varphi(\sigma_{[0,\dots,k]})\, \sigma_{[k+1,\dots,n]} = (-1)^k \varphi(\sigma_{[0,\dots,k]})\, \sigma_{[k+1,\dots,n]}$$

(noting $\sigma_{[0,\dots,\widehat{v_{k+1}},\dots,k+1]} = \sigma_{[0,\dots,k]}$). Combined with the surviving second block of (4.19), the difference equals

$$\varphi(\sigma_{[0,\dots,k]}) \left(\sum_{i=k+1}^{n} (-1)^i \, \sigma_{[k,\dots,\hat{v}_i,\dots,n]} + (-1)^k \, \sigma_{[k+1,\dots,n]} \right)$$

The trailing term is precisely the $i = k$ term of the sum $\sum_{i=k}^{n}(-1)^i\sigma_{[k,\dots,\hat{v}_i,\dots,n]}$ (deleting v_k from the back face leaves $\sigma_{[k+1,\dots,n]}$ with sign $(-1)^k$), so the parenthesized expression is exactly $\sum_{i=k}^{n}(-1)^i\sigma_{[k,\dots,\hat{v}_i,\dots,n]}$. Multiplying by $(-1)^k$ recovers the right-hand side of (4.18), that is

$$(-1)^k(\partial\sigma \frown \varphi - \sigma \frown \delta\varphi) = \partial(\sigma \frown \varphi)$$

as we sought to show.

The boundary formula is the single computation on which everything rests. Read it as a derivation property: the boundary of a cap distributes over the two factors, with ∂ acting on the chain and δ on the cochain, up to the global sign $(-1)^k$ and the minus sign between the two terms. From it the descent to (co)homology is immediate.

Proposition 4.5.4: The Cap Product Descends to (Co)homology

The chain-level cap product of definition 4.5.1 induces an R-bilinear map

$$\frown\colon H_n(X;R) \otimes_R H^k(X;R) \longrightarrow H_{n-k}(X;R)$$

***Proof*:**
Let $z \in C_n(X;R)$ be a cycle ($\partial z = 0$) and $\varphi \in C^k(X;R)$ a cocycle ($\delta\varphi = 0$). The boundary formula of lemma 4.5.3, extended R-bilinearly from simplices to the cycle z, gives

$$\partial(z \frown \varphi) = (-1)^k(\partial z \frown \varphi - z \frown \delta\varphi) = (-1)^k(0 - 0) = 0$$

so $z \frown \varphi$ is a cycle and represents a class in $H_{n-k}(X;R)$. It remains to check that this class depends only on the homology class of z and the cohomology class of φ. Replacing z by $z + \partial w$ for $w \in C_{n+1}(X;R)$, and using $\delta\varphi = 0$,

$$(z + \partial w) \frown \varphi = z \frown \varphi + \partial w \frown \varphi = z \frown \varphi + (-1)^k\partial(w \frown \varphi)$$

where the last equality is again lemma 4.5.3 applied to w (the term $w \frown \delta\varphi$ vanishes). Thus $(z + \partial w) \frown \varphi$ and $z \frown \varphi$ differ by a boundary, hence represent the same class. Replacing φ by $\varphi + \delta\psi$ for $\psi \in C^{k-1}(X;R)$, and using $\partial z = 0$,

$$\begin{aligned} z \frown (\varphi + \delta\psi) &= z \frown \varphi + z \frown \delta\psi \\ &= z \frown \varphi + (-1)^{k-1}(\partial z \frown \psi - \partial(z \frown \psi)) \\ &= z \frown \varphi - (-1)^{k-1}\partial(z \frown \psi) \end{aligned}$$

again differing by a boundary. Hence the induced map on classes is well defined, and it is R-bilinear because the chain-level operation is, completing the proof.

With the cap product established on homology and cohomology, its formal properties follow. Three are needed for duality: a module structure making $H_\bullet(X;R)$ a module over the cohomology ring $H^\bullet(X;R)$, an adjunction tying the cap to the cup product and the Kronecker evaluation pairing, and a naturality (projection) formula governing the behaviour under continuous maps. The Kronecker pairing referenced below is the evaluation $\langle \varphi, z\rangle = \varphi(z) \in R$ of a cochain on a chain, descending to $\langle \cdot,\cdot\rangle \colon H^k(X;R)\otimes H_k(X;R)\to R$ exactly as in the universal coefficient theorem theorem 4.1.8.

Proposition 4.5.5: Properties of the Cap Product

Let X, Y be spaces and R a commutative ring. The cap product of proposition 4.5.4 satisfies the following.

1. *Module structure.* For $\varphi \in H^k(X;R)$, $\psi \in H^\ell(X;R)$ and $z \in H_n(X;R)$,
$$z \frown (\varphi \smile \psi) = (z \frown \varphi) \frown \psi$$
and $z \frown 1 = z$ for the unit $1 \in H^0(X;R)$. Thus $H_\bullet(X;R)$ is a graded right module over the cohomology ring $H^\bullet(X;R)$ of definition 4.2.7.

2. *Adjunction with the Kronecker pairing.* For $\varphi \in H^k(X;R)$, $\psi \in H^\ell(X;R)$ and $z \in H_{k+\ell}(X;R)$,
$$\langle \psi, z \frown \varphi\rangle = \langle \varphi \smile \psi, z\rangle$$
where $\langle \cdot,\cdot\rangle$ is the Kronecker evaluation pairing.

3. *Naturality (projection formula).* For a continuous $f\colon X\to Y$, $z \in H_n(X;R)$ and $\psi \in H^k(Y;R)$,
$$f_*(z \frown f^*(\psi)) = f_*(z) \frown \psi$$

***Proof*:**
Each identity is established at the chain level on a single singular simplex $\sigma\colon \Delta^n \to X$ and then extended R-bilinearly. Passage to classes is guaranteed by proposition 4.5.4.

1. *Module structure.* Let $\varphi \in C^k$, $\psi \in C^\ell$ and σ an n-simplex with $n = k+\ell+m$. Computing the right side,
$$\sigma \frown \varphi = \varphi(\sigma_{[0,\dots,k]})\,\sigma_{[k,\dots,n]}$$
and capping this $(\ell+m)$-chain with ψ reads off the front ℓ-face of $\sigma_{[k,\dots,n]}$, namely $\sigma_{[k,\dots,k+\ell]}$, leaving $\sigma_{[k+\ell,\dots,n]}$,
$$(\sigma \frown \varphi) \frown \psi = \varphi(\sigma_{[0,\dots,k]})\,\psi(\sigma_{[k,\dots,k+\ell]})\,\sigma_{[k+\ell,\dots,n]}$$
Computing the left side, the cup product of definition 4.2.1 is
$$(\varphi \smile \psi)(\tau) = \varphi(\tau_{[0,\dots,k]})\psi(\tau_{[k,\dots,k+\ell]})$$
so capping σ with $\varphi \smile \psi$ reads the front $(k+\ell)$-face,
$$\sigma \frown (\varphi \smile \psi) = (\varphi \smile \psi)(\sigma_{[0,\dots,k+\ell]})\,\sigma_{[k+\ell,\dots,n]} = \varphi(\sigma_{[0,\dots,k]})\,\psi(\sigma_{[k,\dots,k+\ell]})\,\sigma_{[k+\ell,\dots,n]}$$
The two expressions agree. For the unit, $1 \in C^0$ takes the value 1 on every vertex, so $\sigma \frown 1 = 1(\sigma_{[0]})\,\sigma_{[0,\dots,n]} = \sigma$. Associativity and the unit law make $H_\bullet(X;R)$ a graded right module over $H^\bullet(X;R)$, the degree of $z \frown \varphi$ being $n-k$ when $|z| = n$, $|\varphi| = k$.

2. *Adjunction.* Let $\varphi \in C^k$, $\psi \in C^\ell$ and σ a $(k+\ell)$-simplex. The cap $\sigma \frown \varphi = \varphi(\sigma_{[0,\dots,k]})\,\sigma_{[k,\dots,k+\ell]}$ is an ℓ-chain, so evaluating ψ on it,

$$\langle \psi, \sigma \frown \varphi \rangle = \psi(\sigma \frown \varphi) = \varphi(\sigma_{[0,\dots,k]})\,\psi(\sigma_{[k,\dots,k+\ell]})$$

On the other side, the cup product evaluated on σ is

$$\langle \varphi \smile \psi, \sigma \rangle = (\varphi \smile \psi)(\sigma) = \varphi(\sigma_{[0,\dots,k]})\,\psi(\sigma_{[k,\dots,k+\ell]})$$

which is the same scalar in R. Extending bilinearly over chains gives $\langle \psi, z \frown \varphi \rangle = \langle \varphi \smile \psi, z \rangle$ for all $z \in C_{k+\ell}$, and the identity passes to (co)homology because both sides are defined there by proposition 4.5.4 and the descent of the Kronecker pairing.

3. *Naturality.* Let $f\colon X \to Y$ be continuous with chain map $f_\sharp\colon C_\bullet(X) \to C_\bullet(Y)$ and cochain map $f^\sharp = \operatorname{Hom}(f_\sharp, R)$, so $f^\sharp(\psi) = \psi \circ f_\sharp$ on cochains. For an n-simplex σ in X and $\psi \in C^k(Y;R)$, write $f \circ \sigma$ for the composite simplex in Y. Front and back faces are preserved by composition, $(f\circ\sigma)_{[a,\dots,b]} = f \circ (\sigma_{[a,\dots,b]})$, so

$$f_\sharp(\sigma \frown f^\sharp\psi) = f_\sharp\big((f^\sharp\psi)(\sigma_{[0,\dots,k]})\,\sigma_{[k,\dots,n]}\big) = \psi\big(f\circ\sigma_{[0,\dots,k]}\big)\ (f\circ\sigma_{[k,\dots,n]})$$

using $(f^\sharp\psi)(\tau) = \psi(f\circ\tau)$ and that $f_\sharp$ is R-linear, sending the simplex $\sigma_{[k,\dots,n]}$ to $f\circ\sigma_{[k,\dots,n]}$. On the other hand,

$$f_\sharp(\sigma) \frown \psi = (f\circ\sigma) \frown \psi = \psi\big((f\circ\sigma)_{[0,\dots,k]}\big)\ (f\circ\sigma)_{[k,\dots,n]} = \psi\big(f\circ\sigma_{[0,\dots,k]}\big)\ (f\circ\sigma_{[k,\dots,n]})$$

The two right-hand sides are identical, so $f_\sharp(\sigma \frown f^\sharp\psi) = f_\sharp(\sigma) \frown \psi$ on chains. Passing to classes gives $f_*(z \frown f^*\psi) = f_*(z) \frown \psi$, completing the proof.

The three properties say, in turn, that capping is an action, that it is adjoint to cupping, and that it commutes with maps in the only way the variances allow. The module structure of part (1) makes the entire homology of X into a module over its cohomology ring, so that Poincaré duality can be read as an isomorphism of $H^\bullet(M;R)$-modules. The adjunction of part (2) is the bridge between the two products: it says the cap product is the operation $H^\bullet \otimes H_\bullet \to H_\bullet$ uniquely determined by the requirement that pairing the result against any further cohomology class reproduces a cup product. The projection formula of part (3) has the cohomology class pulled back along f on the left and pushed nowhere on the right, which is forced because cohomology is contravariant and homology covariant. That formula is what lets duality be checked on the pieces of an open cover.

The adjunction has an immediate consequence worth isolating, because it is what converts the duality isomorphism into a statement about pairings. Capping with a fixed class $z \in H_n(X;R)$ and then pairing is the same as cupping and pairing with z.

Corollary 4.5.6: Duality Map and the Cup Pairing

Let $z \in H_n(X;R)$ and let $D_z \colon H^k(X;R) \to H_{n-k}(X;R)$ be $D_z(\varphi) = z \frown \varphi$. For all $\varphi \in H^k(X;R)$ and $\psi \in H^{n-k}(X;R)$,

$$\langle \psi, D_z(\varphi) \rangle = \langle \varphi \smile \psi, z \rangle$$

so the composite $H^k \otimes H^{n-k} \xrightarrow{\smile} H^n \xrightarrow{\langle \cdot, z \rangle} R$ equals the pairing of ψ against $D_z(\varphi)$ via the Kronecker pairing.

***Proof*:**
This is proposition 4.5.5(2) with $\ell = n - k$ and the substitution $D_z(\varphi) = z \frown \varphi$, read directly: $\langle \psi, z \frown \varphi \rangle = \langle \varphi \smile \psi, z \rangle$, as required.

The reading of corollary 4.5.6 is that the duality map D_z and the cup-product pairing $(\varphi, \psi) \mapsto \langle \varphi \smile \psi, z \rangle$ carry the same information: D_z is an isomorphism if and only if, over a field, the cup pairing is nondegenerate. This is the form of duality that controls the cohomology rings of manifolds, and it is recorded among the consequences below.

Before proving the theorem, the geometric input is recalled. A closed manifold is a compact manifold without boundary. For such an M of dimension n, the local homology groups of definition 3.4.21 satisfy $H_i(M, M - \{x\}; R) \cong R$ for $i = n$ and 0 otherwise, since M is locally Euclidean and $H_i(\mathbb{R}^n, \mathbb{R}^n - \{0\}; R) \cong H_{i-1}(S^{n-1}; R)$ by the long exact sequence of the pair. An R-orientation is a choice, continuous in x, of generator of each $H_n(M, M - \{x\}; R) \cong R$. This is the orientation of definition 4.4.4. For M closed and R-orientable there is a unique class $[M] \in H_n(M;R)$, the fundamental class of definition 4.4.8, restricting to the chosen generator in every $H_n(M, M - \{x\}; R)$. The duality theorem asserts that capping with $[M]$ is an isomorphism.

Theorem 4.5.7: Poincaré Duality

Let M be a closed R-orientable n-manifold with fundamental class $[M] \in H_n(M;R)$ of definition 4.4.8. Then the map $D \colon H^k(M;R) \to H_{n-k}(M;R)$ given by $D(\alpha) = [M] \frown \alpha$ is an isomorphism for all k, that is

$$H^k(M;R) \cong H_{n-k}(M;R)$$

The strategy is local-to-global. Duality holds trivially for the open ball $\mathbb{R}^n$ (in the appropriate compactly-supported formulation), the Mayer-Vietoris sequences of theorem 3.4.22 for homology and for cohomology with compact supports glue two pieces on which duality holds into their union, and an induction over a finite cover of M by Euclidean opens assembles the global statement. The one ingredient that must be widened beyond ordinary cohomology is the cohomology theory: gluing open sets forces the use of cohomology with compact supports, for which the duality map is contravariant in inclusions of opens and admits its own Mayer-Vietoris sequence.

Definition 4.5.8: Cohomology with Compact Supports

Let X be a space and R a ring. A cochain $\varphi \in C^k(X;R)$ has *compact support* if there is a compact $K \subseteq X$ with φ vanishing on every simplex in $X - K$, equivalently $\varphi \in C^k(X, X-K; R) \subseteq C^k(X;R)$. The compactly supported cochains form a subcomplex, and its cohomology is the *cohomology with compact supports*

$$H_c^k(X;R) = \varinjlim_K H^k(X, X-K; R)$$

the colimit over compact $K \subseteq X$ ordered by inclusion, with the maps induced by $H^k(X, X-K) \to H^k(X, X-K')$ for $K \subseteq K'$.

For compact X one may take $K = X$, so $H_c^k(X;R) = H^k(X;R)$ and the compactly supported theory agrees with ordinary cohomology. The point of the definition is its behaviour for open subsets and inclusions of opens. If $U \subseteq V$ is an inclusion of open subsets of M, a compactly supported cochain on U extends by zero to one on V, giving a covariant map $H_c^k(U;R) \to H_c^k(V;R)$, the opposite variance to ordinary cohomology. This is exactly the variance of homology, which is why H_c^k pairs with homology in the duality map and why the inductive gluing works.

Example 4.5.9: Compactly Supported Cohomology of $\mathbb{R}^n$

Take $X = \mathbb{R}^n$ with coefficients R, and identify $\mathbb{R}^n$ with $S^n - \{N\}$ by the stereographic projection s_N of lemma 3.4.26. Every compact $K \subseteq \mathbb{R}^n$ lies in a closed ball B about the origin, so these balls are cofinal among the compact subsets and the colimit defining H_c^k is computed over them. For such a ball the formula for $|s_N(z)|$ shows that $S^n - B$ is a cap $\{z \in S^n \mid z \cdot N > c\}$ with $-1 < c < 1$, which s_{-N} carries onto an open ball, so $S^n - B$ is contractible and its reduced cohomology vanishes (corollary 3.3.9, proposition 4.1.14(3)). Excision in cohomology (theorem 4.1.19) for the closed set $Z = \{N\}$, which lies in the open set $S^n - B$, and the reduced long exact sequence of the pair $(S^n, S^n - B)$ (proposition 4.1.18(2)) then give isomorphisms

$$H^k(\mathbb{R}^n, \mathbb{R}^n - B; R) \xleftarrow{\cong} H^k(S^n, S^n - B; R) \xrightarrow{\cong} \widetilde{H}^k(S^n; R)$$

both induced by maps of pairs or of chain complexes that commute with the restriction from a ball to a larger ball. So the restriction maps between balls are isomorphisms, the colimit is attained at any single ball B, and

$$H_c^k(\mathbb{R}^n;R) \cong H^k(\mathbb{R}^n, \mathbb{R}^n - B; R) \cong \widetilde{H}^k(S^n;R) = \begin{cases} R & k = n \\ 0 & k \neq n \end{cases}$$

the last value by example 4.1.21(3). Contrast this with ordinary cohomology $H^k(\mathbb{R}^n;R)$, which is R for $k = 0$ and 0 otherwise: on this non-compact space the compactly supported theory concentrates the class in the top degree n, and the ordinary one in degree 0. The single nonzero group $H_c^n(\mathbb{R}^n;R) = R$ is generated by a compactly supported n-cocycle, the orientation class of the ball, and it is this class that the Euclidean case of duality pairs against the local fundamental class.

For an open subset U of an oriented M the fundamental class restricts to a compatible family of

local classes, defining a duality map

$$D_U \colon H_c^k(U;R) \longrightarrow H_{n-k}(U;R)$$

by capping with these local fundamental classes. For $U = M$ compact this is the map D of the theorem. The proof of the theorem is the assertion that D_U is an isomorphism for every open U that is a finite union of Euclidean opens, in particular for M itself.

Proof of theorem 4.5.7:

The argument establishes that $D_U \colon H_c^k(U;R) \to H_{n-k}(U;R)$ is an isomorphism for all open $U \subseteq M$ that admit a finite cover by open sets homeomorphic to $\mathbb{R}^n$. Taking $U = M$, compact, gives the theorem with $H_c^k(M) = H^k(M)$ and $D_M = D$.

Step 1: The Euclidean case. Let $U \cong \mathbb{R}^n$, an open ball. Then $H_{n-k}(U;R) = H_{n-k}(\mathbb{R}^n;R)$ is R for $k = n$ and 0 otherwise, since $\mathbb{R}^n$ is contractible. For compactly supported cohomology, example 4.5.9 gives

$$H_c^k(\mathbb{R}^n;R) \cong H^k(\mathbb{R}^n, \mathbb{R}^n - B;R) \cong \widetilde{H}^k(S^n;R)$$

for every closed ball B about the origin, the first isomorphism because the colimit is attained at B and the second by excision in cohomology (theorem 4.1.19) and the reduced long exact sequence of the pair $(S^n, S^n - B)$ (proposition 4.1.18(2)), and this group is R for $k = n$ and 0 otherwise. Thus both $H_c^k(\mathbb{R}^n;R)$ and $H_{n-k}(\mathbb{R}^n;R)$ vanish except at $k = n$, where both equal R. At $k = n$ the map D_U caps the generator of $H_c^n(\mathbb{R}^n;R)$ against the local fundamental class. Under the identifications above this is the evaluation of a generator of $H^n(\mathbb{R}^n, \mathbb{R}^n - B;R) \cong R$ against the generator of $H_n(\mathbb{R}^n, \mathbb{R}^n - B;R) \cong R$ defining the orientation, which is the Kronecker pairing $R \times R \to R$ sending $(1,1) \mapsto 1$ by the choice of orientation. Hence D_U is an isomorphism (an isomorphism $R \to R$ at $k = n$, and $0 \to 0$ elsewhere).

Step 2: The Mayer-Vietoris gluing. Suppose $U = A \cup B$ with A, B open and the theorem known for A, B, and $A \cap B$. There is a Mayer-Vietoris sequence for homology, theorem 3.4.22,

$$\cdots \to H_{n-k}(A \cap B) \to H_{n-k}(A) \oplus H_{n-k}(B) \to H_{n-k}(U) \to H_{n-k-1}(A \cap B) \to \cdots$$

and a Mayer-Vietoris sequence for cohomology with compact supports, with the variance reversed so that the maps run in the same direction as in homology,

$$\cdots \to H_c^k(A \cap B) \to H_c^k(A) \oplus H_c^k(B) \to H_c^k(U) \to H_c^{k+1}(A \cap B) \to \cdots$$

This compactly supported sequence arises from the short exact sequence of compactly supported cochain complexes

$$0 \to C_c^\bullet(A \cap B) \to C_c^\bullet(A) \oplus C_c^\bullet(B) \to C_c^\bullet(U) \to 0$$

the first map the difference of extensions by zero, the second the sum, exactness following because a compactly supported cochain on U supported in the compact $K \subseteq A \cup B$ splits, by a partition of the support into the parts inside A and inside B, into a sum of compactly supported cochains on A and on B agreeing on $A \cap B$. The duality maps $D_A, D_B, D_{A\cap B}, D_U$ commute with these two sequences: the squares relating them commute by the naturality of the cap product, proposition 4.5.5(3), applied to the inclusions $A \cap B \hookrightarrow A, B \hookrightarrow U$, together with the compatibility of the local fundamental classes under restriction. (The squares involving the connecting homomorphisms commute up to sign. Replacing D by $\pm D$ on alternate groups, a change of sign that does not affect being an isomorphism, makes the ladder strictly commute,

as detailed in [21, Lemma 3.36].) The result is a commuting ladder between the two long exact sequences with $D_A \oplus D_B$ and $D_{A\cap B}$ isomorphisms by hypothesis. By the five lemma (lemma 3.4.6), D_U is an isomorphism.

Step 3: Induction on a convex cover. The induction must be set up so that the intersections that arise stay inside the class being inducted on. The class of arbitrary Euclidean opens is not closed under intersection: an open subset of $\mathbb{R}^n$ need not be Euclidean, nor even a finite union of Euclidean opens, since in general it is a countably infinite union of convex opens (for instance a disjoint union of infinitely many balls). Convex opens behave better, because a finite intersection of convex sets is convex. The induction is therefore run over finite unions of convex opens within a single chart, and the passage to arbitrary opens and to M is made afterward by a colimit.

Fix a chart $\mathbb{R}^n \cong V \subseteq M$ and consider opens of the form $W = C_1 \cup \cdots \cup C_m$ with each $C_i \subseteq V$ a convex open subset of the chart. The claim is that D_W is an isomorphism for every such W, by induction on m. For $m = 1$, $W = C_1$ is convex, hence homeomorphic to $\mathbb{R}^n$, and Step 1 applies. For the inductive step set $A = C_1 \cup \cdots \cup C_{m-1}$ and $B = C_m$. Then A is a union of $m-1$ convex opens, so D_A is an isomorphism by induction, and D_B is an isomorphism by Step 1. The intersection

$$A \cap B = (C_1 \cap C_m) \cup \cdots \cup (C_{m-1} \cap C_m)$$

is a union of the $m-1$ opens $C_i \cap C_m$, each of which is convex (a finite intersection of convex sets), hence a union of $m-1$ convex opens. By the inductive hypothesis $D_{A\cap B}$ is an isomorphism. Step 2 then gives that D_W is an isomorphism, completing the induction over finite unions of convex opens.

The extension to an arbitrary open $U \subseteq M$ that admits a finite cover by Euclidean opens is by colimit. Each Euclidean chart, and hence U, is an increasing union $U = \bigcup_j U_j$ where each U_j is a finite union of convex opens drawn from finitely many charts (take U_j to be the convex pieces of a fixed locally finite convex refinement of the cover that meet a compact exhaustion $K_1 \subseteq K_2 \subseteq \cdots$ of U). The case where a U_j spans several charts is handled by one more pass of the Mayer-Vietoris induction of Step 2 across the charts. Duality for U follows from duality for the U_j by passing to the colimit: both $H^k_c(-;R)$ and $H_{n-k}(-;R)$ commute with such increasing unions, H^k_c by its defining colimit over compact supports and H_{n-k} because every singular chain has compact image, and the duality maps D_{U_j} assemble into D_U. This colimit-commutes-with-(co)homology bookkeeping is the direct-limit argument carried out in [21, Theorem 3.35 and Lemma 3.36], the same argument deferred for the non-compact assembly of Step 4 below. Taking $U = M$, which is compact and so covered by finitely many Euclidean charts, gives that

$$D\colon H^k(M;R) = H^k_c(M;R) \xrightarrow{\cong} H_{n-k}(M;R)$$

is an isomorphism for all k.

Step 4: The non-compact case. When M is non-compact (and R-orientable, not assumed compact), the same local argument applies, but the global statement reads $D\colon H^k_c(M;R) \to H_{n-k}(M;R)$, and assembling it requires the same passage to the colimit over compact subsets that appeared in Step 3, now over an exhaustion of the whole manifold rather than of a single open. The conceptual content is entirely the local case of Step 1 and the Mayer-Vietoris gluing of Step 2, both proved here in full. The only deferral, for the compact M of the theorem statement as well as for the non-compact case, is the direct-limit bookkeeping (commuting the colimit past homology and checking that the duality maps assemble) of [21, Theorem 3.35 and Lemma 3.36], which is routine but lengthy. This completes the proof.

The proof is the local-to-global pattern in its purest form. Duality is a statement about a single ball, where it is the self-pairing of R recording the orientation. Mayer-Vietoris glues balls, and induction exhausts the manifold. The only place where ordinary cohomology had to be replaced by the compactly supported version is the gluing, where the extension-by-zero of a compactly supported cochain provides the covariant maps that match the variance of homology. For compact M the two cohomologies coincide and the theorem is the clean isomorphism $H^k(M;R) \cong H_{n-k}(M;R)$.

The consequences of duality are statements about the shape of the (co)homology of a closed manifold: a symmetry of Betti numbers, a nondegenerate pairing on cohomology, and a vanishing of the Euler characteristic in odd dimensions.

Corollary 4.5.10: Consequences of Poincaré Duality

Let M be a closed R-orientable n-manifold.

1. *Symmetry of Betti numbers.* If $R = \mathbb{F}$ is a field, then $\dim_{\mathbb{F}} H^k(M;\mathbb{F}) = \dim_{\mathbb{F}} H^{n-k}(M;\mathbb{F})$ for all k. In particular, over $\mathbb{Q}$ the Betti numbers satisfy $b_k(M) = b_{n-k}(M)$.
2. *Nondegeneracy of the cup pairing.* If $R = \mathbb{F}$ is a field, the bilinear pairing
$$H^k(M;\mathbb{F}) \otimes_{\mathbb{F}} H^{n-k}(M;\mathbb{F}) \xrightarrow{\smile} H^n(M;\mathbb{F}) \xrightarrow{\langle\,\cdot\,,[M]\rangle} \mathbb{F}$$
is nondegenerate.
3. *Vanishing Euler characteristic in odd dimensions.* If n is odd, the Euler characteristic $\chi(M) = 0$.

***Proof*:**

1. *Symmetry of Betti numbers.* Over a field $\mathbb{F}$ the universal coefficient theorem theorem 4.1.8 has vanishing Ext term, so $H^j(M;\mathbb{F}) \cong \mathrm{Hom}_{\mathbb{F}}(H_j(M;\mathbb{F}),\mathbb{F})$ and $\dim_{\mathbb{F}} H^j(M;\mathbb{F}) = \dim_{\mathbb{F}} H_j(M;\mathbb{F})$ for every j. Poincaré duality theorem 4.5.7 gives $H^k(M;\mathbb{F}) \cong H_{n-k}(M;\mathbb{F})$, so
$$\dim_{\mathbb{F}} H^k(M;\mathbb{F}) = \dim_{\mathbb{F}} H_{n-k}(M;\mathbb{F}) = \dim_{\mathbb{F}} H^{n-k}(M;\mathbb{F})$$
For $\mathbb{F} = \mathbb{Q}$ these dimensions are the Betti numbers $b_k(M)$, giving $b_k(M) = b_{n-k}(M)$.
2. *Nondegeneracy.* By corollary 4.5.6 with $z = [M]$, the pairing in question equals $(\varphi,\psi) \mapsto \langle \psi, D(\varphi)\rangle$, where $D(\varphi) = [M] \frown \varphi$. Fix $\varphi \neq 0$ in $H^k(M;\mathbb{F})$. By theorem 4.5.7, $D(\varphi) \in H_{n-k}(M;\mathbb{F})$ is nonzero. Over a field the Kronecker pairing $\langle\,\cdot\,,\cdot\,\rangle\colon H^{n-k}(M;\mathbb{F}) \otimes H_{n-k}(M;\mathbb{F}) \to \mathbb{F}$ is the evaluation of $\mathrm{Hom}_{\mathbb{F}}(H_{n-k},\mathbb{F})$ on H_{n-k}, which is nondegenerate, so there exists $\psi \in H^{n-k}(M;\mathbb{F})$ with $\langle \psi, D(\varphi)\rangle \neq 0$. Hence no nonzero φ pairs trivially with all ψ. Symmetrically, since the pairing $\langle\,\cdot\,,[M]\rangle \circ \smile$ has the cup product graded-commutative up to sign (theorem 4.2.6), no nonzero ψ pairs trivially with all φ. The pairing is therefore nondegenerate on both sides.
3. *Euler characteristic.* Compute $\chi(M)$ over $\mathbb{Q}$, where $\chi(M) = \sum_{k=0}^{n} (-1)^k b_k(M)$ with $b_k = \dim_{\mathbb{Q}} H_k(M;\mathbb{Q})$ (the Euler characteristic is independent of the field, and over $\mathbb{Q}$ it is the alternating sum of Betti numbers). If M is $\mathbb{Q}$-orientable, part (1) gives $b_k = b_{n-k}$. With n odd, the substitution $k \mapsto n-k$ sends $(-1)^k$ to $(-1)^{n-k} = -(-1)^k$, so pairing the

terms k and $n-k$,

$$\chi(M) = \sum_{k=0}^{n}(-1)^k b_k = \sum_{k=0}^{n}(-1)^{n-k} b_{n-k} = -\sum_{k=0}^{n}(-1)^k b_k = -\chi(M)$$

whence $2\chi(M) = 0$ and $\chi(M) = 0$. For a non-orientable closed M of odd dimension the same conclusion holds by applying the orientable case to the orientable double cover $\widetilde{M}$, for which $\chi(\widetilde{M}) = 2\chi(M)$ and $\chi(\widetilde{M}) = 0$ force $\chi(M) = 0$. This completes the proof.

The symmetry of part (1) is the original form in which Poincaré stated his duality, before the cohomological machinery existed: the k-th and $(n-k)$-th Betti numbers of a closed orientable manifold coincide. Part (2) is the form that constrains cohomology rings. Over a field the cup product makes $H^\bullet(M;\mathbb{F})$ a graded-commutative ring carrying a nondegenerate pairing between complementary degrees, which forces, for instance, that a nonzero class in H^k has a partner in H^{n-k} multiplying with it to the generator of H^n. Part (3) is the cleanest numerical consequence: an odd-dimensional closed manifold has Euler characteristic zero, recovered here from the Betti-number symmetry alone, and extended to the non-orientable case through the orientable double cover.

The theorem and its corollary are best absorbed through computation. The following example verifies duality directly for the spheres and reads off the cohomology pairings of the torus and of complex projective space from the duality isomorphism and the rings of definition 4.2.7.

Example 4.5.11: Poincaré Duality for Spheres, Tori, and Projective Spaces

1. *The sphere S^n.* The sphere is a closed orientable n-manifold with $H_k(S^n;\mathbb{Z}) = \mathbb{Z}$ for $k \in \{0, n\}$ and 0 otherwise. By the universal coefficient theorem the cohomology is the same. Duality predicts $H^k(S^n) \cong H_{n-k}(S^n)$. Checking degrees: $H^0(S^n) = \mathbb{Z} \cong H_n(S^n) = \mathbb{Z}$ and $H^n(S^n) = \mathbb{Z} \cong H_0(S^n) = \mathbb{Z}$, while all intermediate groups vanish on both sides. The duality map sends the generator $1 \in H^0(S^n)$ to $[S^n] \frown 1 = [S^n] \in H_n(S^n)$ by the unit law of proposition 4.5.5(1), the fundamental class itself, and sends the generator $\alpha \in H^n(S^n)$ to $[S^n] \frown \alpha \in H_0(S^n)$, the class whose Kronecker pairing computes $\langle \alpha, [S^n] \rangle$, a generator of $H_0(S^n) = \mathbb{Z}$. Both maps are isomorphisms, confirming the theorem.

2. *The torus T^2.* The torus is a closed orientable surface with $H^\bullet(T^2;\mathbb{Z}) \cong \Lambda_{\mathbb{Z}}[\alpha, \beta]$, the exterior algebra on generators $\alpha, \beta \in H^1(T^2;\mathbb{Z})$ of example 4.2.8, with $\alpha \smile \beta = \gamma$ the generator of $H^2(T^2;\mathbb{Z}) \cong \mathbb{Z}$ and $\alpha^2 = \beta^2 = 0$. Here $n = 2$ and the significant pairing is in the middle degree $k = n - k = 1$,

$$H^1(T^2;\mathbb{Z}) \otimes H^1(T^2;\mathbb{Z}) \xrightarrow{\smile} H^2(T^2;\mathbb{Z}) \xrightarrow{\langle\,\cdot\,,[T^2]\rangle} \mathbb{Z}$$

On the basis $\{\alpha, \beta\}$ this pairing has matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, since $\alpha \smile \alpha = 0$, $\beta \smile \beta = 0$, $\alpha \smile \beta = \gamma$ and $\beta \smile \alpha = -\gamma$ by graded-commutativity theorem 4.2.6, and $\langle \gamma, [T^2] \rangle = 1$. This matrix is invertible over $\mathbb{Z}$ (determinant 1), so the pairing is nondegenerate, exactly as corollary 4.5.10(2) requires. The skew-symmetry is forced by the odd middle degree.

3. *Complex projective space $\mathbb{C}\mathrm{P}^n$.* The space $\mathbb{C}\mathrm{P}^n$ is a closed orientable $2n$-manifold with cohomology ring $H^\bullet(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}[u]/(u^{n+1})$, where $|u| = 2$, so $H^{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) = \mathbb{Z}\langle u^k \rangle$ for $0 \le k \le n$ and the odd groups vanish. (The additive groups are the CW computation of example 3.5.13, and the ring structure is derived in theorem 4.5.12 below.) Duality pairs

degree $2k$ with degree $2n-2k$,

$$H^{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \otimes H^{2n-2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \xrightarrow{\smile} H^{2n}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \xrightarrow{\langle\,\cdot\,,[\mathbb{C}\mathrm{P}^n]\rangle} \mathbb{Z}$$

Here $u^k \smile u^{n-k} = u^n$, the generator of $H^{2n}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}$, and $\langle u^n, [\mathbb{C}\mathrm{P}^n]\rangle = 1$, so each complementary pair (u^k, u^{n-k}) multiplies to the top generator and pairs to 1. The pairing matrix in the bases $\{u^k\}$ and $\{u^{n-k}\}$ is therefore the 1×1 identity in each complementary pair of degrees, nondegenerate, and the duality map $D(u^k) = [\mathbb{C}\mathrm{P}^n] \frown u^k$ generates $H_{2n-2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}$. The ring structure $\mathbb{Z}[u]/(u^{n+1})$ is precisely the structure that the nondegenerate duality pairing forces: u^k must have a multiplicative partner reaching the top class, and u^{n-k} is that partner.

The three computations show duality in its two registers. For the sphere it is the bare statement that $H^k \cong H_{n-k}$, with the fundamental class as the image of the unit. For the torus and for $\mathbb{C}\mathrm{P}^n$ it is the nondegeneracy of the cup pairing, which reads the multiplicative structure of the cohomology ring off the requirement that complementary degrees pair perfectly: in the torus the middle pairing is the symplectic form $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, in $\mathbb{C}\mathrm{P}^n$ it is the statement that u^k and u^{n-k} multiply to the generator. The cohomology ring and the duality pairing determine one another.

The last computation took the ring $\mathbb{Z}[u]/(u^{n+1})$ as given. Poincaré duality in fact forces it, and the same cell structure identifies the additive generators with the projective subspaces they bound.

Theorem 4.5.12: The Cohomology Ring of Complex Projective Space

For $n \geq 0$ the integral cohomology ring of complex projective space is the truncated polynomial ring

$$H^\bullet(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}[u]/(u^{n+1}), \qquad |u| = 2$$

on a generator $u \in H^2(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$, so that u^k generates $H^{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}$ for $0 \leq k \leq n$ and every other group vanishes. Dually, $H_{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}$ is generated by the class $[\mathbb{C}\mathrm{P}^k]$ carried by a linearly embedded projective subspace $\mathbb{C}\mathrm{P}^k \subseteq \mathbb{C}\mathrm{P}^n$, and u^{n-k} is Poincaré dual to $[\mathbb{C}\mathrm{P}^k]$, in the sense that $[\mathbb{C}\mathrm{P}^n] \frown u^{n-k}$ generates $H_{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$.

***Proof*:**

Additive structure and the geometric generators. The standard CW structure of example 3.5.13 gives $\mathbb{C}\mathrm{P}^n$ one cell in each dimension $0, 2, \ldots, 2n$ and none in odd dimensions, so $H_{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}$, the odd groups vanish, and by the universal coefficient theorem (theorem 4.1.8) the cohomology groups match. In that structure the $2k$-skeleton is the linearly embedded $\mathbb{C}\mathrm{P}^k = \{[z_0 : \cdots : z_k : 0 : \cdots : 0]\}$, whose single top cell is the unique $2k$-cell of $\mathbb{C}\mathrm{P}^n$. Hence the inclusion $\iota\colon \mathbb{C}\mathrm{P}^k \hookrightarrow \mathbb{C}\mathrm{P}^n$ carries the fundamental class of the closed orientable $2k$-manifold $\mathbb{C}\mathrm{P}^k$ to a generator $[\mathbb{C}\mathrm{P}^k]$ of $H_{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$.

Ring structure, by induction on n. For $n = 0$ the ring is $\mathbb{Z}$, and for $n = 1$ the space $\mathbb{C}\mathrm{P}^1 = S^2$ has ring $\mathbb{Z}[u]/(u^2)$ with $|u| = 2$. Assume the claim for $n-1$. The pair $(\mathbb{C}\mathrm{P}^n, \mathbb{C}\mathrm{P}^{n-1})$ has a single relative cell, in dimension $2n$, so it is a good pair (lemma 3.4.17) whose quotient $\mathbb{C}\mathrm{P}^n/\mathbb{C}\mathrm{P}^{n-1} = D^{2n}/\partial D^{2n}$ is S^{2n}, and $H^i(\mathbb{C}\mathrm{P}^n, \mathbb{C}\mathrm{P}^{n-1};\mathbb{Z}) \cong \tilde{H}^i(S^{2n};\mathbb{Z})$ (proposition 4.1.18(3)) vanishes for $i \neq 2n$ by example 4.1.21(3), and the cohomology long exact sequence of the pair (theorem 4.1.17) makes restriction $H^i(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \to H^i(\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z})$ an isomorphism for $i \leq 2n-2$. Let u generate $H^2(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$. It restricts to a generator $\bar{u}$ of $H^2(\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z})$. By the inductive

hypothesis $\bar{u}^k$ generates $H^{2k}(\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z})$ for $0 \le k \le n-1$, and since restriction is a ring homomorphism and an isomorphism in these degrees, u^k generates $H^{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$ for $0 \le k \le n-1$. Only $H^{2n}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}$ remains. Since $\mathbb{C}\mathrm{P}^n$ is a closed orientable $2n$-manifold and $H_*(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$ is torsion-free (so the Kronecker pairing $H^{2n}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \otimes H_{2n}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \to \mathbb{Z}$ is perfect), Poincaré duality (theorem 4.5.7), through the cup-cap adjunction $\langle \alpha \smile \beta, [\mathbb{C}\mathrm{P}^n]\rangle = \langle \alpha, \beta \frown [\mathbb{C}\mathrm{P}^n]\rangle$, makes the cup-product pairing

$$H^2(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \otimes H^{2n-2}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \xrightarrow{\smile} H^{2n}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}$$

nonsingular. As u generates the first factor and u^{n-1} the second, the product $u \smile u^{n-1} = u^n$ generates $H^{2n}(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$. The odd groups vanish and $H^i(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) = 0$ for $i > 2n$, so the ring is $\mathbb{Z}[u]/(u^{n+1})$.

The duality identification. Capping with the fundamental class is the duality isomorphism $D\colon H^{2n-2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \xrightarrow{\cong} H_{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$ of theorem 4.5.7, so $D(u^{n-k}) = [\mathbb{C}\mathrm{P}^n] \frown u^{n-k}$ generates $H_{2k}(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$, exhibiting u^{n-k} as the Poincaré dual of $[\mathbb{C}\mathrm{P}^k]$, completing the proof.

4.5.1 Manifolds with Boundary

The proof just given establishes more than the theorem states. What the induction verifies is that $D_U\colon H^k_c(U;R) \to H_{n-k}(U;R)$ is an isomorphism for every open $U \subseteq M$, and compactness of M entered only at the last step, to identify $H^k_c(M;R)$ with $H^k(M;R)$. Recording the general statement needs no extra work and is what the boundary case will run on.

Corollary 4.5.13: Duality for Non-Compact Manifolds

Let M be an R-orientable n-manifold, not necessarily compact. Then

$$D_M\colon H^k_c(M;R) \longrightarrow H_{n-k}(M;R)$$

is an isomorphism for every k.

***Proof*:**
This is the conclusion of the induction in the proof of theorem 4.5.7 applied to $U = M$, which nowhere used compactness of M: the base case is example 4.5.9, the inductive step is the Mayer-Vietoris comparison, and the direct-limit step covers an arbitrary open by Euclidean pieces.

A manifold with boundary is not a manifold, so duality cannot apply to it directly. Its interior is one, and the boundary is glued on in a way that changes neither the homotopy type nor the compactly supported cohomology in a manner one cannot control. The control is given by a collar.

Definition 4.5.14: Collared Boundary

Let M be a topological n-manifold with boundary ∂M. A *collar* is an embedding $c\colon \partial M \times [0,1) \to M$ onto a neighbourhood of ∂M with $c(x,0) = x$ for every $x \in \partial M$. The boundary is *collared* when such a c exists.

Every smooth compact manifold with boundary has a collar, obtained by flowing along an inward-pointing vector field. That construction is differential-topological rather than algebraic and is not carried out here. In this section the collar is a hypothesis.

Duality needs a class to cap with, and definition 4.4.8 produces one only for closed manifolds. The boundary case has its own, living in the relative group.

Proposition 4.5.15: Relative Fundamental Class

Let M be a compact R-orientable n-manifold with collared boundary, with an R-orientation $x \mapsto \mu_x$ on its interior. There is a unique class

$$[M, \partial M] \in H_n(M, \partial M; R)$$

whose image under $H_n(M, \partial M; R) \to H_n(M, M - x; R) \cong R$ is μ_x for every interior point x.

***Proof*:**

Existence. Let c be a collar and let $DM = M \cup_{\partial M} M$ be the double, two copies of M glued along the boundary. The two collars fit together into an embedding $\partial M \times (-1, 1) \to DM$, so DM is a closed n-manifold, and it is R-orientable: orient one copy by μ and the other by its mirror image, the two agreeing on the overlap collar. By theorem 4.4.9 it carries a fundamental class $[DM]$. Excising the interior of the second copy from the pair $(DM, \text{second copy})$ gives

$$H_n(DM, \text{second copy}; R) \cong H_n(M, \partial M; R)$$

and the image of $[DM]$ under $H_n(DM; R) \to H_n(DM, \text{second copy}; R) \cong H_n(M, \partial M; R)$ restricts at each interior $x \in M$ to μ_x, because restriction to $H_n(DM, DM - x; R)$ factors through the same map.

Uniqueness. A difference of two such classes restricts to 0 at every interior point. The local-to-global argument of theorem 4.4.9 shows that a class in $H_n(M, \partial M; R)$ is determined by its restrictions to $H_n(M, M - x; R)$ over interior x: that argument is a statement about the compactly supported sections of the orientation covering over the interior, and the collar makes ∂M contribute nothing to it. Hence the difference is 0.

Theorem 4.5.16: Poincaré-Lefschetz Duality

Let M be a compact R-orientable n-manifold with collared boundary ∂M, with relative fundamental class $[M, \partial M] \in H_n(M, \partial M; R)$. Then capping with $[M, \partial M]$ gives isomorphisms

$$H^k(M; R) \xrightarrow{\cong} H_{n-k}(M, \partial M; R), \qquad H^k(M, \partial M; R) \xrightarrow{\cong} H_{n-k}(M; R)$$

for every k.

***Proof*:**

Write $\mathring{M} = M - \partial M$ for the interior, an R-orientable n-manifold without boundary, and let $c\colon \partial M \times [0, 1) \to M$ be a collar.

Step 1: the collar retracts M onto its interior. Pushing in along the collar, $r_t(c(x, s)) =$

$c(x, s + t(1 - s)/2)$ on the image of c and $r_t = \mathrm{id}$ outside it, defines a homotopy from id_M to a map into $\mathring{M}$. The homotopy is continuous because the two formulas agree near $s = 1$. The inclusion $\mathring{M} \hookrightarrow M$ is therefore a homotopy equivalence, so $H_{n-k}(\mathring{M}; R) \cong H_{n-k}(M; R)$.

Step 2: the collar computes compactly supported cohomology of the interior. The compact subsets $K_\epsilon = M - c(\partial M \times [0, \epsilon))$ are cofinal among compact subsets of $\mathring{M}$, so by definition 4.5.8

$$H^k_c(\mathring{M}; R) = \varinjlim_{\epsilon} H^k(\mathring{M}, \mathring{M} - K_\epsilon; R)$$

Each pair $(\mathring{M}, \mathring{M} - K_\epsilon)$ includes into $(M, \partial M)$, and both inclusions induce isomorphisms: on the first factor by Step 1, and on the second because $\mathring{M} - K_\epsilon = c(\partial M \times (0, \epsilon))$ deformation retracts to a slice of the collar, as does the open collar neighbourhood of ∂M in M. Excising the closed complement of the collar and passing to the colimit gives $H^k_c(\mathring{M}; R) \cong H^k(M, \partial M; R)$.

Step 3: assemble. Apply corollary 4.5.13 to $\mathring{M}$ and substitute Steps 1 and 2:

$$H^k(M, \partial M; R) \cong H^k_c(\mathring{M}; R) \xrightarrow[\cong]{D_{\mathring{M}}} H_{n-k}(\mathring{M}; R) \cong H_{n-k}(M; R)$$

Under these identifications $D_{\mathring{M}}$ is cap product with $[M, \partial M]$, since the local fundamental classes assembling $[M, \partial M]$ restrict to those of $\mathring{M}$ at every interior point. The other isomorphism is the same argument with the roles of the absolute and relative theories exchanged, using that $H^k(M; R) \cong H^k(\mathring{M}; R)$ by Step 1 and that the long exact sequences of the pair on the two sides match under cap product by proposition 4.5.5.

The corollary below is the form in which the theorem is usually used, and the one that matters for intersection theory on an even-dimensional manifold with boundary: the middle-dimensional pairing is not merely nondegenerate over a field but *unimodular* over $\mathbb{Z}$ once torsion is discarded.

Corollary 4.5.17: Integral Unimodularity of the Intersection Pairing

Let M be a compact oriented $2m$-manifold with collared boundary. Then the intersection pairing

$$H_m(M; \mathbb{Z})/\mathrm{tors} \times H_m(M, \partial M; \mathbb{Z})/\mathrm{tors} \longrightarrow \mathbb{Z}$$

is unimodular: it identifies each factor with the full $\mathbb{Z}$-dual of the other. Equivalently, in bases of the two free quotients its matrix has determinant ± 1.

***Proof*:**
The Kronecker pairing $H^m(M, \partial M; \mathbb{Z}) \times H_m(M, \partial M; \mathbb{Z}) \to \mathbb{Z}$ induces, by the universal coefficient theorem theorem 4.1.8, an isomorphism

$$H^m(M, \partial M; \mathbb{Z})/\mathrm{tors} \xrightarrow{\cong} \mathrm{Hom}_{\mathbb{Z}}\big(H_m(M, \partial M; \mathbb{Z}), \mathbb{Z}\big)$$

the Ext term being torsion and so dying in the free quotient. Composing its inverse with the duality isomorphism $H^m(M, \partial M; \mathbb{Z}) \cong H_m(M; \mathbb{Z})$ of theorem 4.5.16 identifies $H_m(M; \mathbb{Z})/\mathrm{tors}$ with $\mathrm{Hom}_{\mathbb{Z}}(H_m(M, \partial M; \mathbb{Z}), \mathbb{Z})$, and the pairing implementing that identification is cap-then-evaluate, which is the intersection pairing. A perfect pairing of finitely generated free $\mathbb{Z}$-modules has unimodular matrix, since the induced map to the dual is an isomorphism of free modules of the same rank.

Exercise 4.5.1

1. Using the boundary formula of lemma 4.5.3, verify directly that if $z \in C_n(X;R)$ is a cycle and $\varphi \in C^k(X;R)$ is a cocycle, then for any $\psi \in C^{k-1}(X;R)$ the chains $z \frown \varphi$ and $z \frown (\varphi + \delta\psi)$ are homologous. (This is the cochain-independence half of proposition 4.5.4. Reproduce it in detail.)

2. Let $M = S^1 \times S^1$ be the torus with its standard cohomology ring $\Lambda_{\mathbb{Z}}[\alpha,\beta]$ from example 4.2.8. Compute the duality images $D(\alpha) = [M] \frown \alpha$ and $D(\beta) = [M] \frown \beta$ as elements of $H_1(M;\mathbb{Z}) \cong \mathbb{Z}^2$, and verify that $D\colon H^1(M;\mathbb{Z}) \to H_1(M;\mathbb{Z})$ is an isomorphism by exhibiting the matrix of D in suitable bases.

3. Let M be a closed orientable n-manifold with n even, say $n = 2m$. Show that the middle cup pairing $H^m(M;\mathbb{Q}) \otimes H^m(M;\mathbb{Q}) \to \mathbb{Q}$ is symmetric when m is even and skew-symmetric when m is odd, and deduce that when m is odd, $\dim_{\mathbb{Q}} H^m(M;\mathbb{Q})$ is even.

4. Prove that no closed orientable n-manifold M with n odd can have $\chi(M) \neq 0$, and compute $\chi(S^2 \times S^1)$ from corollary 4.5.10(3), confirming it against a cell-count. (Contrast: $\mathbb{R}\mathrm{P}^2$ is a closed surface with $\chi(\mathbb{R}\mathrm{P}^2) = 1 \neq 0$, consistent with its even dimension. Explain why this nonzero Euler characteristic is not in conflict with the result just proved.)

5. Using the projection formula proposition 4.5.5(3), show that if $f\colon M \to N$ is a map of closed oriented n-manifolds with $f_*[M] = d\,[N]$ for an integer d (the degree of f), then for every $\psi \in H^k(N;\mathbb{Z})$ one has $f_*([M] \frown f^*\psi) = d\,([N] \frown \psi)$ in $H_{n-k}(N;\mathbb{Z})$.

6. Let $\mathbb{F}$ be a field and M a closed $\mathbb{F}$-orientable n-manifold. Show that if $H^k(M;\mathbb{F}) = 0$ for some $0 < k < n$, then $H^{n-k}(M;\mathbb{F}) = 0$ as well, and give an example with $M = S^2$ where this forces a vanishing group from a nonvanishing one in the complementary degree.

4.6 Homology and Cohomology with Local Coefficients

The (co)homology theories built so far attach to a space X a single fixed abelian group of coefficients, the same copy of G over every point. The orientation theory of this chapter already pressed against that rigidity. The local orientations of an n-manifold are the groups $H_n(M, M \setminus \{x\}) \cong \mathbb{Z}$ of theorem 4.4.1, one over each point, and transporting a generator around an orientation-reversing loop returns it negated (definition 4.4.4). The Möbius band (example 4.4.5) is the first space where no global choice is consistent. For a closed non-orientable manifold the top homology $H_n(M;\mathbb{Z})$ vanishes (theorem 4.4.7), so there is no fundamental class and the Poincaré duality of section 4.5 has nothing to pair against. The coefficients were required to be constant, while the geometry asks for a group that the fundamental group is allowed to permute. This section builds the resulting twisted (co)homology, and with coefficients in the orientation system a closed non-orientable manifold regains a fundamental class (exercise 4.6.1 (5)).

The organizing idea is that the coefficients should be a representation of $\pi_1(X)$. Such a representation is a *local coefficient system*, and the construction that turns it into homology and cohomology runs entirely through one object already in hand: the universal cover $\widetilde{X}$, on which $\pi_1(X)$ acts by deck transformations (definition 2.3.8). The cellular chains of $\widetilde{X}$ form a complex of free modules over the group ring $\mathbb{Z}[\pi_1(X)]$, and tensoring or homming this complex against the chosen representation produces homology and cohomology with local coefficients. When the representation is trivial, both

reduce to the ordinary theories of chapter 3 and chapter 4. The orientation system is the motivating case, but the same construction is exactly what two later developments will call for: the Serre spectral sequence of a fibration whose fundamental group acts nontrivially on the fiber, where the E^2 page is the homology of the base with coefficients in the local system $H_q(F)$ (chapter 6), and obstruction theory for a non-simple fiber, where the obstruction to extending a map lives in a twisted group $H^{n+1}(X;\pi_n(F))$ (chapter 5).

Definition 4.6.1: Local Coefficient System

Let X be a path-connected space with basepoint x_0 and fundamental group $\pi = \pi_1(X, x_0)$. A *local coefficient system* (or *local system*) of abelian groups on X is a functor

$$\mathcal{M}\colon \Pi_1(X) \to \mathsf{Ab}$$

from the fundamental groupoid $\Pi_1(X)$ (the category whose objects are the points of X and whose morphisms $x \to y$ are homotopy classes rel endpoints of paths from x to y) to the category of abelian groups. Equivalently, since X is path-connected the groupoid $\Pi_1(X)$ is equivalent to the one-object category π, and a local system is the same datum as a *left $\mathbb{Z}[\pi]$-module* $M = \mathcal{M}(x_0)$: an abelian group M together with an action of π by automorphisms, the *monodromy action*, extended $\mathbb{Z}$-linearly to the group ring $\mathbb{Z}[\pi]$. The system is *trivial* (or *constant*) if the monodromy action is trivial, so that every $[\gamma] \in \pi$ acts as id_M.

The two descriptions are equivalent. The functorial description assigns an abelian group $\mathcal{M}(x)$ to each point and an isomorphism $\mathcal{M}(x) \to \mathcal{M}(y)$ to each homotopy class of paths, compatibly with concatenation. This is the form in which local systems arise geometrically, as a family of groups varying continuously over X. The module description collapses the family to its value $M = \mathcal{M}(x_0)$ at the basepoint, where the loops at x_0, which are the morphisms $x_0 \to x_0$ in $\Pi_1(X)$, act as automorphisms and make M a representation of π. Path-connectedness is what makes the two equivalent: any two points are joined by a path, so every $\mathcal{M}(x)$ is isomorphic to M, and the only invariant left is the action of the loops at the basepoint. The equivalence of categories $\Pi_1(X) \simeq \pi$ underlying this is the groupoid form of the statement that π_1 is independent of basepoint up to the conjugation ambiguity already recorded in chapter 2. Passing to the value at x_0 rigidifies the choice.

There is a third description, and it is the one under which local systems arise in geometry. A family of groups varying over X is, before anything else, a sheaf. The functor on $\Pi_1(X)$ appears only after paths are chosen, and the π-module only after a basepoint is. A variation of Hodge structure, a family of étale sheaves and a D-module are each given as a sheaf and used as a representation of π, and theorem 4.6.3 is the passage between the two.

Definition 4.6.2: Locally Constant Sheaf

Let X be a space and R a ring. A sheaf $\mathcal{L}$ of R-modules on X is *locally constant*, or a *local system of R-modules*, if every point has an open neighbourhood V with $\mathcal{L}|_V \cong \underline{M}_V$ for some R-module M, where $\underline{M}_V$ is the constant sheaf on V with value M. When each such M may be taken to be a finite-dimensional vector space over a field, $\mathcal{L}$ has finite rank.

Theorem 4.6.3: Locally Constant Sheaves Are Local Systems

Let X be path connected and locally simply connected with basepoint x_0. Sending a locally constant sheaf $\mathcal{L}$ to the assignment $x \mapsto \mathcal{L}_x$, together with transport of stalks along paths, is an equivalence between the category of locally constant sheaves of R-modules on X and the category of functors $\Pi_1(X) \to R\text{-}\mathsf{Mod}$. Composing with definition 4.6.1 gives an equivalence with the category of $R[\pi]$-modules.

***Proof*:**

Transport. Let $\gamma\colon [0,1] \to X$ be a path. Cover $\gamma([0,1])$ by opens on which $\mathcal{L}$ is constant and choose $0 = t_0 < \cdots < t_N = 1$ with $\gamma([t_{i-1}, t_i]) \subseteq V_i$ for such an open V_i. This is possible by compactness of $[0,1]$ and the Lebesgue number lemma. On V_i the isomorphism $\mathcal{L}|_{V_i} \cong \underline{M_i}$ identifies every stalk over V_i canonically with M_i, so composing along the chain gives an isomorphism $\tau_\gamma\colon \mathcal{L}_{\gamma(0)} \to \mathcal{L}_{\gamma(1)}$. Two chains admit a common refinement, on which the two composites visibly agree, so τ_γ is independent of the choices.

Homotopy invariance. Let $H\colon [0,1]^2 \to X$ be a homotopy rel endpoints. Subdivide the square into finitely many closed subsquares each mapping into a single open on which $\mathcal{L}$ is constant, again by compactness. Transport around the boundary of one such subsquare is the identity, since all four stalk identifications factor through the single module attached to that open. Composing subsquares across the subdivision shows transport along the two horizontal edges agrees, so τ_γ depends only on the homotopy class. Functoriality under concatenation holds by construction, so $x \mapsto \mathcal{L}_x$ is a functor on $\Pi_1(X)$.

Fully faithful. A morphism $\mathcal{L} \to \mathcal{L}'$ induces stalk maps commuting with transport. Conversely a compatible family of stalk maps is, over each open on which both sheaves are constant, a single module map, and these agree on overlaps by compatibility. A morphism of sheaves may be glued from local data, so the family comes from a unique morphism.

Essentially surjective. Given a functor $\mathcal{M}$ on $\Pi_1(X)$, cover X by simply connected opens V_i, available because X is locally simply connected. On V_i any two paths between the same points are homotopic, so the transport isomorphisms of $\mathcal{M}$ give a canonical identification of all its values, and $\underline{\mathcal{M}(x_i)}_{V_i}$ is a well-defined constant sheaf for a chosen $x_i \in V_i$. On $V_i \cap V_j$ transport gives isomorphisms between these constant sheaves, locally constant hence constant on each component, and the cocycle condition holds by functoriality of $\mathcal{M}$. Gluing produces a locally constant sheaf whose associated functor is $\mathcal{M}$.

The equivalence is the reason the words "local system" are used for all three objects without comment. It also explains a phenomenon that looks surprising from the sheaf side: a locally constant sheaf on a simply connected space is constant, since the corresponding representation of the trivial group is trivial.

Example 4.6.4: The Orientation System of a Manifold

Let M be a connected n-manifold and R a commutative ring. theorem 4.4.1 computes the local homology $H_n(M, M \setminus \{x\}; R) \cong R$ at each point, a free R-module of rank one. As x varies these groups assemble into a local system on M: the orientation system

$$\mathcal{O}_R\colon \Pi_1(M) \to \mathsf{Ab}, \qquad x \mapsto H_n(M, M \setminus \{x\}; R)$$

A homotopy class of paths from x to y transports a local orientation μ_x to μ_y by the continuation

that already governs definition 4.4.4, and this transport is the structure map $\mathcal{O}_R(x) \to \mathcal{O}_R(y)$. As a $\mathbb{Z}[\pi_1(M)]$-module the system is R with $\pi_1(M)$ acting through the sign of the determinant of the transition along a loop: a loop preserving the local orientation acts as $+1$, a loop reversing it acts as -1. Concretely, the action factors through a homomorphism

$$w_1 \colon \pi_1(M) \to \{\pm 1\}$$

acting on R through $\pm \mathrm{id}_R \in \mathrm{Aut}(R)$, the first Stiefel-Whitney class read as a representation. The system $\mathcal{O}_R$ is trivial precisely when M is R-orientable: a global orientation in the sense of definition 4.4.4 is exactly a $\pi_1(M)$-invariant generator, which exists if and only if every loop acts as $+1$. For the open Möbius band of example 4.4.5 the core circle reverses orientation, so the generator of $\pi_1 \cong \mathbb{Z}$ acts as -1 on $\mathbb{Z}$ and the system is nontrivial. This is the nonorientability of the band expressed as a nontrivial monodromy representation. With $R = \mathbb{Z}/2$ the only automorphism of $\mathbb{Z}/2$ is the identity, so $\mathcal{O}_{\mathbb{Z}/2}$ is always trivial, recovering the fact that every manifold is $\mathbb{Z}/2$-orientable.

With the coefficients defined, the chain-level construction can be assembled. The single geometric input is the deck action on the universal cover, which makes the cellular chains into free $\mathbb{Z}[\pi]$-modules.

Proposition 4.6.5: The Cellular Chains of the Universal Cover are Free over $\mathbb{Z}[\pi]$

Let X be a connected CW complex with universal cover $q \colon \widetilde{X} \to X$ and $\pi = \pi_1(X, x_0)$, identified with the deck group $\mathrm{Deck}\,(q)$ via proposition 2.3.11. Give $\widetilde{X}$ the CW structure lifting that of X, in which the cells of $\widetilde{X}$ are the connected components of the preimages $q^{-1}(e)$ of the open cells e of X. Then the deck action of π permutes the cells of $\widetilde{X}$ freely, and for each n the cellular chain group $C_n^{CW}(\widetilde{X})$ is a free left $\mathbb{Z}[\pi]$-module with one basis element for each n-cell of X. The cellular boundary maps are $\mathbb{Z}[\pi]$-linear, so $C_*^{CW}(\widetilde{X})$ is a complex of free $\mathbb{Z}[\pi]$-modules.

***Proof*:**

Step 1: The cells of $\widetilde{X}$ over a cell of X form a free π-set. A cell e of X has characteristic map $\Phi \colon D^n \to X$ whose restriction to the open disk is a homeomorphism onto e. Because D^n is simply connected, the lifting criterion of chapter 2 lifts Φ to $\widetilde{X}$, and the lifts are exactly the maps $\sigma \cdot \widetilde{\Phi}$ for $\sigma \in \pi$ and one fixed lift $\widetilde{\Phi}$, since two lifts of the same map differ by a deck transformation. Distinct σ give distinct lifts, as a nontrivial deck transformation of the universal cover fixes no point (definition 2.3.8). Hence the open cells of $\widetilde{X}$ lying over e are in bijection with π, and π acts on them by left translation, freely and transitively.

Step 2: Freeness of the chain group. Choosing one lifted cell $\widetilde{e}$ over each cell e of X selects a basis. By Step 1 every n-cell of $\widetilde{X}$ is $\sigma \cdot \widetilde{e}$ for a unique $\sigma \in \pi$ and a unique cell e of X, so the cellular chain group $C_n^{CW}(\widetilde{X})$, free abelian on the n-cells of $\widetilde{X}$, is free abelian on the symbols $\{\sigma \cdot \widetilde{e}\}$. Collecting the translates of each fixed $\widetilde{e}$, this is precisely the free $\mathbb{Z}[\pi]$-module on the chosen lifts $\{\widetilde{e}\}$, one generator per n-cell of X. The deck action on chains is the left regular action permuting the σ, which is the $\mathbb{Z}[\pi]$-module structure.

Step 3: Equivariance of the boundary. The cellular boundary of definition 3.5.10 is induced by the attaching maps and the quotient maps to spheres, all of which are deck-equivariant, since the CW structure on $\widetilde{X}$ was lifted from X and the deck group acts cellularly. Concretely, $\partial(\sigma \cdot \widetilde{e}) = \sigma \cdot \partial(\widetilde{e})$ because applying a deck transformation commutes with restricting to a

subcomplex and collapsing, so ∂ is $\mathbb{Z}[\pi]$-linear. Thus $C_*^{CW}(\widetilde{X})$ is a chain complex of free $\mathbb{Z}[\pi]$-modules, completing the proof.

The proposition is the structural fact that powers the whole section. The universal cover is the maximal symmetric cover, and its symmetry is exactly the action of π that the cellular chains record. Choosing a lift of each cell of X trivializes the chains into a free $\mathbb{Z}[\pi]$-module, with rank in each degree equal to the number of cells of X in that degree, no more and no less than the ordinary cellular chains, but now remembering the deck symmetry as a module structure. A local system is a $\mathbb{Z}[\pi]$-module. Pairing the two through $\otimes$ and Hom over $\mathbb{Z}[\pi]$ is what produces the twisted theories.

Definition 4.6.6: Homology and Cohomology with Local Coefficients

Let X be a connected CW complex with universal cover $\widetilde{X}$ and $\pi = \pi_1(X, x_0)$, and let $\mathcal{M}$ be a local coefficient system, regarded as a $\mathbb{Z}[\pi]$-module M as in definition 4.6.1. The *homology of X with local coefficients in $\mathcal{M}$* is the homology of the chain complex obtained by tensoring the free $\mathbb{Z}[\pi]$-complex $C_*^{CW}(\widetilde{X})$ of proposition 4.6.5 with M over $\mathbb{Z}[\pi]$,

$$H_*(X;\mathcal{M}) \;=\; H_*\big(C_*^{CW}(\widetilde{X}) \otimes_{\mathbb{Z}[\pi]} M\big)$$

where π acts on $C_*^{CW}(\widetilde{X})$ on the right through the deck action (using $\sigma \cdot c = c \cdot \sigma^{-1}$ to convert the left deck action to a right action) and on M on the left through the monodromy. The *cohomology of X with local coefficients in $\mathcal{M}$* is the cohomology of the cochain complex of $\mathbb{Z}[\pi]$-module homomorphisms,

$$H^*(X;\mathcal{M}) \;=\; H^*\big(\mathrm{Hom}_{\mathbb{Z}[\pi]}(C_*^{CW}(\widetilde{X}), M)\big)$$

with coboundary dual to the cellular boundary. When $\mathcal{M}$ is the trivial system on an abelian group M, these are written $H_*(X;M)$ and $H^*(X;M)$ in the ordinary notation.

The construction is the ordinary cellular recipe with the deck symmetry inserted as a tensor and a Hom. Ordinary homology with coefficients in M tensors the cellular chains $C_*^{CW}(X)$ with M over $\mathbb{Z}$. The local version tensors the *lifted* chains $C_*^{CW}(\widetilde{X})$ with M over $\mathbb{Z}[\pi]$, so that the deck action and the monodromy action are identified in the tensor product. The effect is to twist the boundary maps by the monodromy: where the ordinary boundary multiplies a coefficient by an integer incidence number, the local boundary multiplies it by an element of $\mathbb{Z}[\pi]$, whose action on M depends on how π_1 moves the coefficients. Cohomology is the dual story, with $\mathrm{Hom}_{\mathbb{Z}[\pi]}$ in place of $\otimes_{\mathbb{Z}[\pi]}$. The definition uses a CW structure and the universal cover, both of which are choices. That the resulting groups are independent of these choices, and agree with the singular-chain definition of local (co)homology, is the cover-independence statement of box 4.6.11, which names the one step left unverified and its source.

The first property to check is that the twisted theory specializes to the ordinary one when the twist is switched off. This is what makes the local theory a genuine extension rather than a parallel construction.

Proposition 4.6.7: Reduction to Ordinary (Co)homology for a Trivial System

Let X be a connected CW complex and M an abelian group, regarded as a trivial local system on which $\pi = \pi_1(X)$ acts by the identity. Then there are natural isomorphisms

$$H_*(X;\mathcal{M}) \cong H_*(X;M), \qquad H^*(X;\mathcal{M}) \cong H^*(X;M)$$

with the ordinary cellular homology and cohomology of section 3.5 and chapter 4.

***Proof*:**
Write $\varepsilon\colon \mathbb{Z}[\pi] \to \mathbb{Z}$ for the augmentation, the ring map sending every group element to 1. The quotient of $\widetilde{X}$ by the deck action is X, and on cellular chains this quotient is exactly the base change along ε: the chosen $\mathbb{Z}[\pi]$-basis $\{\widetilde{e}\}$ of $C_*^{CW}(\widetilde{X})$ from proposition 4.6.5 maps to the cellular basis $\{e\}$ of $C_*^{CW}(X)$, and the deck-equivariant boundary descends to the cellular boundary of X under ε. Hence there is a natural isomorphism of chain complexes

$$C_*^{CW}(\widetilde{X}) \otimes_{\mathbb{Z}[\pi]} \mathbb{Z} \;\cong\; C_*^{CW}(X)$$

When π acts trivially on M, the $\mathbb{Z}[\pi]$-module M is the base change $\mathbb{Z} \otimes_{\mathbb{Z}} M$ with π acting through ε, so

$$C_*^{CW}(\widetilde{X}) \otimes_{\mathbb{Z}[\pi]} M \;\cong\; \big(C_*^{CW}(\widetilde{X}) \otimes_{\mathbb{Z}[\pi]} \mathbb{Z}\big) \otimes_{\mathbb{Z}} M \;\cong\; C_*^{CW}(X) \otimes_{\mathbb{Z}} M$$

the right-hand complex being the one whose homology is ordinary $H_*(X;M)$. Taking homology gives the first isomorphism. For cohomology, a $\mathbb{Z}[\pi]$-linear map $C_*^{CW}(\widetilde{X}) \to M$ into a module with trivial action is the same as a $\mathbb{Z}$-linear map $C_*^{CW}(\widetilde{X}) \otimes_{\mathbb{Z}[\pi]} \mathbb{Z} \to M$, that is a $\mathbb{Z}$-linear map $C_*^{CW}(X) \to M$, by the adjunction between base change and restriction along ε. This identifies the cochain complex $\mathrm{Hom}_{\mathbb{Z}[\pi]}(C_*^{CW}(\widetilde{X}), M)$ with the ordinary cochain complex $\mathrm{Hom}_{\mathbb{Z}}(C_*^{CW}(X), M)$, and taking cohomology gives the second isomorphism. Both identifications are natural in M and in X, as the augmentation and the basis correspondence are, completing the proof.

The reduction confirms that the local theory deserves the same notation: a trivial system reproduces the ordinary groups, so writing $H_*(X;M)$ for the trivial system creates no ambiguity. The mechanism is the augmentation $\mathbb{Z}[\pi] \to \mathbb{Z}$, which collapses the deck symmetry and returns the chains of X from the chains of $\widetilde{X}$. Every twisting phenomenon to come is a deviation from this collapse, controlled by how far the monodromy action is from trivial.

Functoriality is the second property. A local system pulls back along a map, and the twisted groups are functorial for the pullback.

Proposition 4.6.8: Functoriality of Local Coefficients

Let $f\colon X \to Y$ be a cellular map of connected CW complexes inducing $f_*\colon \pi_1(X) \to \pi_1(Y)$, and let $\mathcal{N}$ be a local system on Y given by a $\mathbb{Z}[\pi_1(Y)]$-module N. The *pullback system* $f^*\mathcal{N}$ is the $\mathbb{Z}[\pi_1(X)]$-module N with $\pi_1(X)$ acting through f_*, that is $\sigma \cdot n = f_*(\sigma)\cdot n$. Then f induces natural homomorphisms

$$f_*\colon H_*(X; f^*\mathcal{N}) \to H_*(Y;\mathcal{N}), \qquad f^*\colon H^*(Y;\mathcal{N}) \to H^*(X; f^*\mathcal{N})$$

compatible with composition: $(g\circ f)_* = g_*\circ f_*$ and $(g\circ f)^* = f^*\circ g^*$, and reducing to the ordinary induced maps when $\mathcal{N}$ is trivial.

***Proof*:**

The map f lifts to a f_*-equivariant map of universal covers $\widetilde{f}\colon \widetilde{X} \to \widetilde{Y}$, unique once basepoint lifts are fixed, by the lifting criterion of chapter 2. Equivariance means $\widetilde{f}(\sigma\cdot\widetilde{x}) = f_*(\sigma)\cdot\widetilde{f}(\widetilde{x})$ for $\sigma \in \pi_1(X)$. Since f is cellular, $\widetilde{f}$ is cellular, and it induces a chain map

$$\widetilde{f}_\sharp\colon C_*^{CW}(\widetilde{X}) \to C_*^{CW}(\widetilde{Y})$$

which is f_*-equivariant: $\widetilde{f}_\sharp(\sigma\cdot c) = f_*(\sigma)\cdot\widetilde{f}_\sharp(c)$. Regard $C_*^{CW}(\widetilde{Y})$ as a $\mathbb{Z}[\pi_1(X)]$-complex by restriction along f_*. Then $\widetilde{f}_\sharp$ is $\mathbb{Z}[\pi_1(X)]$-linear, and tensoring with $N = f^*\mathcal{N}$ over $\mathbb{Z}[\pi_1(X)]$ gives a chain map $C_*^{CW}(\widetilde{X})\otimes_{\mathbb{Z}[\pi_1(X)]} N \to C_*^{CW}(\widetilde{Y})\otimes_{\mathbb{Z}[\pi_1(X)]} N$. The canonical surjection $C_*^{CW}(\widetilde{Y})\otimes_{\mathbb{Z}[\pi_1(X)]} N \to C_*^{CW}(\widetilde{Y})\otimes_{\mathbb{Z}[\pi_1(Y)]} N$, $c\otimes n \mapsto c\otimes n$, is well defined because each defining relation $c\cdot\sigma\otimes n = c\otimes\sigma\cdot n$ of the source, for $\sigma\in\pi_1(X)$, reads $c\cdot f_*(\sigma)\otimes n = c\otimes f_*(\sigma)\cdot n$ and holds in the target, and it commutes with the boundary maps $\partial\otimes 1$. Their composite is a chain map

$$C_*^{CW}(\widetilde{X})\otimes_{\mathbb{Z}[\pi_1(X)]} N \to C_*^{CW}(\widetilde{Y})\otimes_{\mathbb{Z}[\pi_1(Y)]} N$$

whose induced map on homology is f_*. Dually, precomposition with $\widetilde{f}_\sharp$ sends a $\mathbb{Z}[\pi_1(Y)]$-linear map $C_*^{CW}(\widetilde{Y}) \to N$ to the $\mathbb{Z}[\pi_1(X)]$-linear map $C_*^{CW}(\widetilde{X}) \to N$ obtained by restriction along f_*, giving the cochain map and the induced f^*. Composition compatibility follows because lifts compose, $\widetilde{g\circ f} = \widetilde{g}\circ\widetilde{f}$ for compatible basepoint choices, so the chain maps compose. When $\mathcal{N}$ is trivial the equivariance is vacuous and the construction is the ordinary induced map under proposition 4.6.7, completing the proof.

Functoriality makes the twisted theory a genuine invariant: a map of spaces with a local system on the target induces maps on the twisted groups in the expected variance, homology covariantly and cohomology contravariantly, and the pullback of coefficients tracks how the fundamental groups are related. The one subtlety absent from the untwisted case is that the coefficient system on the source is forced to be the pullback $f^*\mathcal{N}$. One cannot compare twisted groups for unrelated systems, just as one cannot compare ordinary groups for unrelated coefficient groups without a map between them. With the formal properties in place, the smallest nontrivial example can be computed in full.

Example 4.6.9: Twisted Homology of the Circle

Take $X = S^1$, with $\pi = \pi_1(S^1) \cong \mathbb{Z}$ generated by the loop t, and universal cover $q\colon \mathbb{R} \to S^1$ with deck group $\mathbb{Z}$ acting by integer translations (example 2.3.9). Give S^1 the CW structure with one 0-cell and one 1-cell. The lifted structure on $\mathbb{R}$ has 0-cells at the integers and 1-cells between

consecutive integers. By proposition 4.6.5 the cellular chain complex of $\mathbb{R}$ over $\mathbb{Z}[\pi] = \mathbb{Z}[t, t^{-1}]$ is free of rank one in each of degrees 0 and 1,

$$C_*^{CW}(\mathbb{R}): \quad 0 \to \mathbb{Z}[t, t^{-1}] \xrightarrow{\cdot(t-1)} \mathbb{Z}[t, t^{-1}] \to 0$$

with the boundary of the chosen 1-cell $\widetilde{e}_1$ equal to $t \cdot \widetilde{e}_0 - \widetilde{e}_0 = (t-1)\widetilde{e}_0$, the difference of its two endpoints, which are the integer 0-cell and its translate by the generator. This is the standard free resolution of $\mathbb{Z}$ over the group ring of $\mathbb{Z}$.

The twisted system. Let $M = \mathbb{Z}$ with $\pi = \mathbb{Z}$ acting through the generator t by multiplication by -1, so that $t \cdot m = -m$. This is the sign representation of $\mathbb{Z}$, the orientation system of example 4.6.4 for the Möbius band realized over its core circle. Tensoring the resolution with M over $\mathbb{Z}[\pi]$ replaces each free rank-one module by $M = \mathbb{Z}$ and the boundary $\cdot(t-1)$ by the action of $t-1$ on M, which is $t \cdot m - m = -m - m = -2m$. Thus

$$C_*^{CW}(\mathbb{R}) \otimes_{\mathbb{Z}[\pi]} \mathbb{Z}: \quad 0 \to \mathbb{Z} \xrightarrow{\cdot(-2)} \mathbb{Z} \to 0$$

with the $\mathbb{Z}$ in degrees 1 and 0. The homology with local coefficients is

$$H_0(S^1; \mathcal{M}) = \mathbb{Z}/2\mathbb{Z}, \qquad H_1(S^1; \mathcal{M}) = \ker(\cdot(-2)) = 0$$

since multiplication by -2 on $\mathbb{Z}$ has cokernel $\mathbb{Z}/2\mathbb{Z}$ and trivial kernel. For cohomology, the cochain complex $\mathrm{Hom}_{\mathbb{Z}[\pi]}(C_*^{CW}(\mathbb{R}), \mathbb{Z})$ is again $\mathbb{Z}$ in degrees 0 and 1, with coboundary the dual of $\cdot(t-1)$, which is once more multiplication by -2, so

$$H^0(S^1; \mathcal{M}) = \ker(\cdot(-2)) = 0, \qquad H^1(S^1; \mathcal{M}) = \mathbb{Z}/2\mathbb{Z}$$

the degrees of the nonzero group swapped relative to homology, as the universal coefficient pattern of theorem 4.1.8 predicts.

Contrast with the untwisted circle. For the trivial system $M = \mathbb{Z}$ with t acting as $+1$, the boundary $\cdot(t-1)$ acts on M as $1 - 1 = 0$, so the complex is $0 \to \mathbb{Z} \xrightarrow{0} \mathbb{Z} \to 0$ and

$$H_0(S^1; \mathbb{Z}) = \mathbb{Z}, \qquad H_1(S^1; \mathbb{Z}) = \mathbb{Z}$$

recovering the ordinary homology of the circle from chapter 3, in agreement with proposition 4.6.7. The twist has collapsed the free $H_0 = \mathbb{Z}$ to the torsion group $\mathbb{Z}/2\mathbb{Z}$ and killed H_1 entirely: the generator t of π acts on the twisted coefficients by -1, so an invariant 0-cycle must satisfy $m = -m$, forcing $2m = 0$ on classes and $m = 0$ on cycles. The single sign flip is the whole difference between the cylinder ($H_* = H_*(S^1; \mathbb{Z})$ of the trivial bundle) and the Möbius band, whose twisted H_0 records that the band has no consistent transverse orientation.

The example exhibits the mechanism in its purest form. The computation uses a free resolution of $\mathbb{Z}$ over $\mathbb{Z}[\pi]$, here the length-one resolution coming from the contractible cover $\mathbb{R}$. Tensoring with the coefficient module evaluates the boundary on the monodromy, and a single nontrivial sign converts the homology of the circle into the group cohomology of $\mathbb{Z}$ with sign coefficients. The computation is identical in form to a group-cohomology computation, and for an aspherical space $X = K(\pi, 1)$ the twisted (co)homology of definition 4.6.6 is exactly the group (co)homology of π with coefficients in M, since the universal cover is contractible and $C_*^{CW}(\widetilde{X})$ is a free resolution of $\mathbb{Z}$ over $\mathbb{Z}[\pi]$. Proposition 5.5.8 proves this once group (co)homology is defined, and the circle is the case

$\pi = \mathbb{Z}$. For a general X the cover need not be contractible, so $C_*^{CW}(\widetilde{X})$ is no longer a resolution of $\mathbb{Z}$. What carries over is the same algebraic pattern with the resolution replaced by the free $\mathbb{Z}[\pi]$-complex of the cover tensored against the representation, the hyper-derived reading recorded in box 4.6.11.

Two structural remarks place the construction in the wider theory and close the loop with the deferrals that opened the section.

4.6.10 Remark: Local Coefficients as Twisted Coefficients in the Two Deferred Theorems The twisted Serre spectral sequence of box 6.2.12 has E^2 page $H_p(B;\mathcal{H}_q(F))$, the homology of the base B with coefficients in the fiber-homology system $\mathcal{H}_q(F)$ of example 6.2.11. definition 4.6.6 is exactly the definition of this group: $H_p(B;\mathcal{H}_q(F)) = H_p\big(C_*^{CW}(\widetilde{B})\otimes_{\mathbb{Z}[\pi_1(B)]}H_q(F)\big)$, the cellular chains of the universal cover of B tensored against the monodromy representation on $H_q(F)$. When $\pi_1(B)$ acts trivially, proposition 4.6.7 returns the ordinary E^2 page $H_p(B;H_q(F))$ of theorem 6.2.7. Likewise the obstruction theory of theorem 5.6.7 and proposition 5.6.9, applied to a fibration with non-simple fiber F, has its obstruction cochain valued in the local system $\pi_n(F)$ on X, and its obstruction class lives in $H^{n+1}(X;\pi_n(F))$ in the sense of definition 4.6.6. The simplicity hypothesis of definition 5.1.13 is precisely what trivializes this system and returns the untwisted obstruction group used throughout chapter 5. Both deferrals are now coefficient settings of one construction.

4.6.11 Remark: The Derived Interpretation and Cover-Independence The complex $C_*^{CW}(\widetilde{X})$ is a complex of free, hence projective, $\mathbb{Z}[\pi]$-modules. Its $\mathbb{Z}[\pi]$-module structure makes precise the sense in which the twisted theory is a derived functor, but the form of that statement depends on whether X is aspherical.
When X is aspherical, that is $X = K(\pi, 1)$, the universal cover $\widetilde{X}$ is contractible, so $C_*^{CW}(\widetilde{X}) \to \mathbb{Z}$ is a free resolution of the trivial $\mathbb{Z}[\pi]$-module $\mathbb{Z}$. Tensoring or homming this resolution against M computes the classical derived functors of a *module*, and the displayed identity has $\mathbb{Z}$ (not the complex) in the resolved slot, as proposition 5.5.8 proves,

$$H_*(X;\mathcal{M}) = \operatorname{Tor}_*^{\mathbb{Z}[\pi]}(\mathbb{Z}, M) = H_*(\pi; M), \qquad H^*(X;\mathcal{M}) = \operatorname{Ext}^*_{\mathbb{Z}[\pi]}(\mathbb{Z}, M) = H^*(\pi; M)$$

the group homology and cohomology of π with coefficients in M of definition 5.5.7. The circle of example 4.6.9 is the case $\pi = \mathbb{Z}$, where the length-one resolution is the one displayed there.
For a general X the complex $C_*^{CW}(\widetilde{X})$ is *not* a resolution of $\mathbb{Z}$: its homology is $H_*(\widetilde{X})$, which is nonzero in positive degrees as soon as $\widetilde{X}$ is not contractible, so $\mathbb{Z}$ does not sit in the resolved slot and the classical $\operatorname{Tor}_*^{\mathbb{Z}[\pi]}(\mathbb{Z}, M)$ no longer computes the twisted theory. What survives is the hyper-derived functor of the *complex*: writing $\otimes^{\mathbf{L}}_{\mathbb{Z}[\pi]}$ and $\mathbf{R}\mathrm{Hom}_{\mathbb{Z}[\pi]}$ for the derived tensor product and derived Hom of *Homological Algebra* [13, Derived Tensor Product, *R*Hom], the definitions of definition 4.6.6 read

$$H_*(X;\mathcal{M}) = H_*\big(C_*^{CW}(\widetilde{X}) \otimes^{\mathbf{L}}_{\mathbb{Z}[\pi]} M\big), \qquad H^*(X;\mathcal{M}) = H^*\big(\mathbf{R}\mathrm{Hom}_{\mathbb{Z}[\pi]}(C_*^{CW}(\widetilde{X}), M)\big)$$

the underived $\otimes_{\mathbb{Z}[\pi]}$ and $\mathrm{Hom}_{\mathbb{Z}[\pi]}$ of the definition agreeing with their derived counterparts here precisely because $C_*^{CW}(\widetilde{X})$ is already a complex of *free* modules in every degree, so no further flatness or projectivity hypothesis on M is needed. These hyper-Tor and hyper-Ext groups reduce to the classical $\operatorname{Tor}_*^{\mathbb{Z}[\pi]}(\mathbb{Z}, M)$ and $\operatorname{Ext}^*_{\mathbb{Z}[\pi]}(\mathbb{Z}, M)$ exactly when $C_*^{CW}(\widetilde{X})$ is quasi-isomorphic to $\mathbb{Z}$, that is in the aspherical case above.
That $H_*(X;\mathcal{M})$ and $H^*(X;\mathcal{M})$ depend only on X and $\mathcal{M}$, and not on the CW structure or the chosen lifts of cells, is used in no proof here. It would follow from a $\mathbb{Z}[\pi]$-linear quasi-isomorphism between the cellular chain complexes of $\widetilde{X}$ for two cell structures, for instance through the singular chain complex $C_*(\widetilde{X})$, on which π also acts freely. A quasi-isomorphism between complexes of projective $\mathbb{Z}[\pi]$-modules concentrated in nonnegative degrees is a chain homotopy equivalence [13, Projective, Quasi-isomorphism Becomes Homotopy], and a chain homotopy equivalence survives $- \otimes_{\mathbb{Z}[\pi]} M$ and $\mathrm{Hom}_{\mathbb{Z}[\pi]}(-, M)$. The equivariant comparison of cellular with singular chains is the missing step: Hatcher states that cellular and singular homology with local coefficients agree and leaves the verification to the reader [21, §3.H, p. 333], and for cohomology says only that the basic properties extend to local coefficients [21, §3.H, p. 334].

Exercise 4.6.1

1. Let $X = S^1$ and $\mathcal{M}$ the local system on which the generator $t \in \pi_1(S^1) \cong \mathbb{Z}$ acts on $M = \mathbb{Z}$ by multiplication by a fixed integer k. Compute $H_0(S^1;\mathcal{M})$ and $H_1(S^1;\mathcal{M})$ as a function of k, and identify the values $k = 1$ and $k = -1$ with chapter 3 and example 4.6.9.

2. Show that for any connected CW complex X and any local system $\mathcal{M}$ with module M, the degree-zero homology is the group of coinvariants $H_0(X;\mathcal{M}) \cong M_\pi = M/\langle \sigma m - m : \sigma \in \pi, m \in M\rangle$, and the degree-zero cohomology is the group of invariants $H^0(X;\mathcal{M}) \cong M^\pi =$

$\{m \in M : \sigma m = m \text{ for all } \sigma \in \pi\}$.

3. Let $\mathcal{M}$ be a local system on X and let $q\colon \widetilde{X} \to X$ be the universal cover, with the trivial system $q^*\mathcal{M}$ on $\widetilde{X}$ obtained by pulling back along q (so $\pi_1(\widetilde{X}) = 0$ acts trivially). Identify $H_*(\widetilde{X}; q^*\mathcal{M})$ with the ordinary homology $H_*(\widetilde{X}; M)$, and explain why the local coefficients on X become ordinary coefficients upstairs.

4. Prove that if $\pi = \pi_1(X)$ is finite of order n and M is a $\mathbb{Z}[\pi]$-module that is a $\mathbb{Q}$-vector space (so n is invertible), then $H_k(X; \mathcal{M}) \cong H_k(\widetilde{X}; M)_\pi$ and the twisted homology agrees with the π-coinvariants of the ordinary homology of the cover. (Hint: averaging over π splits the relevant sequences. Use that $\mathbb{Q}[\pi]$ is semisimple.)

5. Let M be a closed connected non-orientable n-manifold with a CW structure, and let $\mathcal{M}$ be the orientation system $\mathcal{O}_\mathbb{Z}$ of example 4.6.4. Show that $H_n(M; \mathcal{M}) \cong \mathbb{Z}$ even though the ordinary top homology $H_n(M; \mathbb{Z}) = 0$, so the twisted top homology recovers a fundamental class with coefficients in the orientation system. (Hint: compare the chains of M, of the connected double cover $\widehat{M}$ corresponding to $\ker w_1$, and of M with coefficients in $\mathcal{O}_\mathbb{Z}$ in one short exact sequence.)

Part II

Homotopy Theory and Bundles

The invariants of Part I are computed from cell structures, and they can miss a map entirely. Every map $S^3 \to S^2$ induces zero on reduced homology and cohomology, since the two spheres have no nonzero reduced groups in a common degree. Yet the Hopf map, which sends a unit vector of $\mathbb{C}^2$ to the complex line it spans, is not homotopic to a constant map. Its fibers are circles, any two of which link once in S^3, and Hopf used that linking in 1931 to prove the map essential. The higher homotopy groups that Hurewicz introduced in 1935 detect such maps: $\pi_n(X)$ is the group of homotopy classes of based maps $S^n \to X$, and the Hopf map generates $\pi_3(S^2) \cong \mathbb{Z}$.

Homotopy classes of maps $X \to Y$, for three kinds of space, are the subject of chapter 5 and chapter 7, with the spectral sequence of chapter 6 between them as the means of computation. Chapter 5 takes X to be a sphere, where the classes form the homotopy groups of Y, and studies them through fibrations, the maps whose long exact sequences relate the homotopy groups of a total space, a base and a fiber. It also takes Y to be an Eilenberg-MacLane space $K(G, n)$, where the based classes are the cohomology groups, $\widetilde{H}^n(X; G) \cong [X, K(G, n)]_*$ for a CW complex X (theorem 5.5.12). Chapter 6 computes the cohomology of the total space of a fibration from that of its base and fiber, with the spectral sequence that Serre adapted in 1951 from Leray's. Chapter 7 takes Y to be a classifying space, whose cohomology classes are the characteristic classes of bundles. For the orthogonal and unitary groups these are the Stiefel-Whitney, Chern and Pontryagin classes of vector bundles, introduced between 1935 and 1947.

5

Homotopy Theory

The fundamental group of chapter 2 measures failure of contractions of loops. Generalising, mapping the n-sphere into a space and remembering the base produces the higher homotopy groups $\pi_n(X, x_0)$, the central objects of this chapter. They are notoriously hard to compute, even for the spheres themselves: as of writing this book the answer is still unknown in general, and the search for computable results motivated the development of a new field known as *stable homotopy theory*.[1]

This chapter builds the machinery of higher homotopy theory. It starts by defining π_n and the induced long exact sequence of a pair, the tool that, together with fibrations in the next section, converts geometric decompositions into algebra. It then develops the comparison between homotopy and homology through the *Hurewicz theorem* which says that the first nonvanishing homotopy group agrees with the first nonvanishing homology group. The chapter culminates in the Eilenberg-MacLane spaces $K(\pi, n)$, spaces constructed to have a *single* nonzero homotopy group, and using these spaces to show [ordinary] cohomology is a *representable homotopy functor*: cohomology classes on X are but homotopy classes of maps from X into a $K(\pi, n)$ (section 5.5). That identification bridges the geometry of spaces to the algebra of cohomology, and it sets the stage for the theory of spectra in chapter 8. For a discrete group G the space $K(G, 1)$ is the classifying space BG, and its cohomology with coefficients in a $\mathbb{Z}[G]$-module is the group cohomology of G.

5.1 Higher Homotopy Groups

Throughout, $I = [0, 1]$ denotes the unit interval, I^n the n-cube, and ∂I^n its boundary, the set of points with at least one coordinate equal to 0 or 1. A based space is a pair (X, x_0) with $x_0 \in X$ a chosen basepoint. A based map $f\colon (X, x_0) \to (Y, y_0)$ is a continuous map with $f(x_0) = y_0$, and a based homotopy is a homotopy through such maps.

[1] Stable homotopy theory had other motivations as well: stability is a natural phenomenon of high-degree homotopy groups, as section 5.7 shows.

Definition 5.1.1: Higher Homotopy Groups

Let (X, x_0) be a based space and $n \geq 1$. The *n-th homotopy group* $\pi_n(X, x_0)$ is the set of homotopy classes of maps

$$f\colon (I^n, \partial I^n) \to (X, x_0)$$

that is, continuous maps $f\colon I^n \to X$ sending the entire boundary ∂I^n to x_0, where two such maps are identified when they are homotopic through maps carrying ∂I^n to x_0 at every stage. The group operation is

$$(f+g)(t_1, \dots, t_n) = \begin{cases} f(2t_1, t_2, \dots, t_n) & t_1 \in [0, \frac{1}{2}] \\ g(2t_1 - 1, t_2, \dots, t_n) & t_1 \in [\frac{1}{2}, 1] \end{cases}$$

concatenating in the first coordinate.

Collapsing ∂I^n to a point turns the cube I^n into the quotient $I^n/\partial I^n$, which is homeomorphic to the sphere S^n with the image of ∂I^n as basepoint. A map $(I^n, \partial I^n) \to (X, x_0)$ is therefore the same as a based map $(S^n, *) \to (X, x_0)$, and $\pi_n(X, x_0)$ is equally the set of based-homotopy classes of maps $S^n \to X$. The cube description makes the group law visible (concatenation glues two cubes along a face), while the sphere description makes the geometry visible (an element of π_n is a way of wrapping a sphere through the space). For $n = 1$ this is the fundamental group of definition 2.1.12: a map $(I, \partial I) \to (X, x_0)$ is a loop at x_0, and the operation is concatenation of loops.

The sum $f + g$ is continuous by the pasting lemma ([16, Pasting Lemma]), and $f + g$ that carries ∂I^n to x_0 is immediate, since each branch does.

Example 5.1.2: The Homotopy Groups of Euclidean Space

Take $X = \mathbb{R}^m$ with basepoint the origin 0. Any map $f\colon (I^n, \partial I^n) \to (\mathbb{R}^m, 0)$ is null-homotopic through the straight-line homotopy

$$F(t_1, \dots, t_n, s) = (1 - s)\, f(t_1, \dots, t_n)$$

At $s = 0$ this is f, at $s = 1$ it is the constant map 0, and at every stage the boundary ∂I^n is sent to $(1 - s) \cdot 0 = 0$, so the homotopy is through based maps. Every element of $\pi_n(\mathbb{R}^m, 0)$ therefore equals the class of the constant map, and

$$\pi_n(\mathbb{R}^m, 0) = 0 \qquad \text{for all } n \geq 1$$

The same argument works verbatim for any convex subset of $\mathbb{R}^m$, and more generally for any contractible space: if H contracts X to x_0, composing f with H produces a based null-homotopy. Said geometrically, Euclidean space has no holes in any dimension.

Proposition 5.1.3: π_n is a Group

For every based space (X, x_0) and every $n \geq 1$, the operation $+$ of definition 5.1.1 is well defined on homotopy classes and makes $\pi_n(X, x_0)$ a group. The identity is the class 0 of the constant map c_{x_0}, and the inverse of $[f]$ is the class of $\bar{f}(t_1, t_2, \dots, t_n) = f(1 - t_1, t_2, \dots, t_n)$.

***Proof*:**

1. *Well-definedness.* Suppose $f \simeq f'$ and $g \simeq g'$ through based homotopies $F, G\colon I^n \times I \to X$, each carrying $\partial I^n \times I$ to x_0. Define $H\colon I^n \times I \to X$ by concatenating in the first coordinate at each homotopy time s,

$$H(t_1, \dots, t_n, s) = \begin{cases} F(2t_1, t_2, \dots, t_n, s) & t_1 \in [0, \frac{1}{2}] \\ G(2t_1 - 1, t_2, \dots, t_n, s) & t_1 \in [\frac{1}{2}, 1] \end{cases}$$

   The two branches agree on $t_1 = \frac{1}{2}$ (both equal x_0 there), so H is continuous by the pasting lemma, it carries $\partial I^n \times I$ to x_0, and $H(\,\cdot\,, 0) = f + g$ while $H(\,\cdot\,, 1) = f' + g'$. Hence $f + g \simeq f' + g'$ and $[f] + [g] := [f + g]$ is well defined.

2. *Associativity.* For $[f], [g], [h]$ the two bracketings $(f + g) + h$ and $f + (g + h)$ differ only by a reparametrization of the first coordinate. Explicitly, $(f + g) + h$ runs f on $[0, \frac{1}{4}]$, g on $[\frac{1}{4}, \frac{1}{2}]$, h on $[\frac{1}{2}, 1]$, while $f + (g + h)$ runs f on $[0, \frac{1}{2}]$, g on $[\frac{1}{2}, \frac{3}{4}]$, h on $[\frac{3}{4}, 1]$. Let $\varphi\colon I \to I$ be the piecewise-linear homeomorphism with $\varphi(\frac{1}{2}) = \frac{1}{4}$ and $\varphi(\frac{3}{4}) = \frac{1}{2}$, fixing 0 and 1. Then $\big((f + g) + h\big) \circ (\varphi \times \mathrm{id}_{I^{n-1}})$ equals $f + (g + h)$, since φ carries the breakpoints $\frac{1}{2}, \frac{3}{4}$ of the right bracketing to the breakpoints $\frac{1}{4}, \frac{1}{2}$ of the left and is affine between them. Writing $\varphi_s = (1 - s)\mathrm{id} + s\,\varphi$ (a homeomorphism of I fixing 0 and 1 for each s), the assignment $s \mapsto \big((f + g) + h\big) \circ (\varphi_s \times \mathrm{id})$ is a based homotopy from $(f + g) + h$ (at $s = 0$) to $f + (g + h)$ (at $s = 1$), the boundary mapping to x_0 throughout because $\varphi_s \times \mathrm{id}$ preserves ∂I^n.

3. *Identity* Let c_{x_0} be the constant map. Then $f + c_{x_0}$ runs f at double speed on $[0, \frac{1}{2}]$ and is constant at x_0 on $[\frac{1}{2}, 1]$. The homotopy

$$J(t_1, \dots, t_n, s) = \begin{cases} f\Big(\frac{2t_1}{1+s}, t_2, \dots, t_n\Big) & t_1 \in [0, \frac{1+s}{2}] \\ x_0 & t_1 \in [\frac{1+s}{2}, 1] \end{cases}$$

   deforms $f + c_{x_0}$ (at $s = 0$) to f (at $s = 1$) through based maps, the reparametrization $\frac{2t_1}{1+s}$ stretching the active interval $[0, \frac{1+s}{2}]$ back onto $[0, 1]$. The boundary stays at x_0 throughout. The mirror homotopy gives $c_{x_0} + f \simeq f$, so $0 = [c_{x_0}]$ is a two-sided identity.

4. *Inverses* With $\bar{f}(t_1, \dots, t_n) = f(1 - t_1, t_2, \dots, t_n)$, consider $f + \bar{f}$, which runs f forward in the first coordinate and then f backward. The contraction pulls the turning point in from $t_1 = 1$ to $t_1 = 0$ as s increases, holding the cube constant on a growing middle band. Define

$$K(t_1, \dots, t_n, s) = \begin{cases} f(2t_1, t_2, \dots, t_n) & t_1 \in [0, \frac{1-s}{2}], \\ f(1 - s, t_2, \dots, t_n) & t_1 \in [\frac{1-s}{2}, \frac{1+s}{2}], \\ f(2 - 2t_1, t_2, \dots, t_n) & t_1 \in [\frac{1+s}{2}, 1] \end{cases}$$

   At $s = 0$ the middle interval degenerates and $K(\,\cdot\,, 0) = f + \bar{f}$. At $s = 1$ the outer intervals degenerate and $K(\,\cdot\,, 1)$ is the constant map at $f(0, t_2, \dots, t_n) = x_0$ (since $(0, t_2, \dots, t_n) \in \partial I^n$). On the overlaps the branches agree (at $t_1 = \frac{1-s}{2}$ both give $f(1-s, \dots)$, similarly on the right), so K is continuous, and it carries $\partial I^n \times I$ to x_0 because at $t_1 \in \{0, 1\}$ the first and third branches give f of a boundary point. Hence $f + \bar{f} \simeq c_{x_0}$. Applying this to $\bar{f}$ in place of f (and noting $\bar{\bar{f}} = f$) gives $\bar{f} + f \simeq c_{x_0}$ as well, so $[\bar{f}] = -[f]$, completing the proof.

The proof is the one-variable theory of chapter 2 carried out in the first coordinate, with the remaining coordinates carried along unchanged. The reparametrizations of associativity and the identity are exactly the loop reparametrizations from π_1. The inverse is the path run backwards in the first coordinate. Nothing in this argument used $n \geq 2$, which is why π_n is a group for every $n \geq 1$ and reduces to the fundamental group when $n = 1$.

The key way $\pi_n(X)$ differs for $n \geq 2$ is that these groups are *abelian*, requiring the second coordinate to prove. For $n \geq 2$ one can concatenate in any of them. Concatenation in the second coordinate:

$$(f +_2 g)(t_1, t_2, t_3, \dots, t_n) = \begin{cases} f(t_1, 2t_2, t_3, \dots, t_n) & t_2 \in [0, \frac{1}{2}], \\ g(t_1, 2t_2 - 1, t_3, \dots, t_n) & t_2 \in [\frac{1}{2}, 1] \end{cases}$$

By the same proof as above applied to the second slot, another group operation on the same set $\pi_n(X, x_0)$, with the same identity 0. The two operations $+$ and $+_2$ are not a priori related, but they satisfy a single compatibility, and that compatibility forces both to coincide and to be commutative:

Theorem 5.1.4: Higher Homotopy Groups are Abelian

For every based space (X, x_0) and every $n \geq 2$, the group $\pi_n(X, x_0)$ is abelian.

This proof is sometimes called the *Eckmann-Hilton argument.*

***Proof*:**
Write $+_1 := +$ for concatenation in the first coordinate and $+_2$ for concatenation in the second. By proposition 5.1.3 applied to the respective slot both are group operations on $\pi_n(X, x_0)$ with the same identity element $0 = [c_{x_0}]$.

1. *The interchange law.* Fix four classes represented by maps $a, b, c, d \colon (I^n, \partial I^n) \to (X, x_0)$. Consider the map $w \colon I^n \to X$ that subdivides the unit square in the (t_1, t_2)-coordinates into four quadrants and places a in the lower-left, b in the lower-right, c in the upper-left, d in the upper-right, each precomposed with the linear rescaling of its quadrant back to I^n in the first two coordinates and the identity in the remaining coordinates. Concretely, on the lower-left quadrant $[0, \frac{1}{2}] \times [0, \frac{1}{2}]$ the map is $(t_1, t_2, \dots) \mapsto a(2t_1, 2t_2, t_3, \dots, t_n)$, and similarly for the others. The boundary between quadrants is sent to x_0 (each map sends its quadrant boundary to x_0), so w is continuous and represents an element of π_n.

   Reading w as "first combine left-right, then top-bottom" computes it as $(a +_1 b) +_2 (c +_1 d)$: the lower half is $a +_1 b$, the upper half is $c +_1 d$, and the two halves are stacked in the t_2-direction. Reading w as "first combine top-bottom, then left-right" computes it as $(a +_2 c) +_1 (b +_2 d)$: the left half is $a +_2 c$, the right half is $b +_2 d$, stacked in the t_1-direction. The two readings are the same map w, so

   $$(a +_1 b) +_2 (c +_1 d) = (a +_2 c) +_1 (b +_2 d)$$

   holds in $\pi_n(X, x_0)$.

2. *The two operations agree and are commutative.* Apply the interchange law with $b = c = c_{x_0}$, so that their classes are the common identity 0. Using that 0 is the identity for both $+_1$ and $+_2$, the left side becomes $(a +_1 0) +_2 (0 +_1 d) = a +_2 d$, and the right side becomes $(a +_2 0) +_1 (0 +_2 d) = a +_1 d$. Hence

   $$a +_2 d = a +_1 d \qquad \text{for all } [a], [d]$$

so the two operations coincide. Write $+$ for the common operation. Now apply the interchange law again, this time with $a = d = c_{x_0}$. The left side becomes $(0 +_1 b) +_2 (c +_1 0) = b +_2 c = b + c$, and the right side becomes $(0 +_2 c) +_1 (b +_2 0) = c +_1 b = c + b$. Hence

$$b + c = c + b$$

for all classes $[b], [c]$, so $\pi_n(X, x_0)$ is abelian, as we sought to show.

Notice the proof is purely algebraic: any set carrying two unital binary operations that satisfy the interchange law has both operations equal, associative, and commutative. This is the structural reason higher homotopy groups are abelian while π_1 need not be: a one-dimensional cube has a single direction, so there is no second operation to interchange with. The same argument explains why the fundamental group of a topological group is abelian (multiplication of loops provides a second operation), and it recurs throughout topology wherever two compatible compositions meet.

Example 5.1.5: π_2 of the 2-Sphere is Nontrivial

The identity map $\mathrm{id}\colon S^2 \to S^2$, viewed as a based map $(S^2, *) \to (S^2, *)$, represents a class in $\pi_2(S^2, *)$. This class is nonzero: were id based null-homotopic, S^2 would be contractible, contradicting $H_2(S^2) \cong \mathbb{Z} \neq 0$ from chapter 3, since a contractible space has vanishing reduced homology. In fact $\pi_2(S^2, *) \cong \mathbb{Z}$, generated by [id], with an element's integer being its degree as a self-map of S^2 in the sense of section 3.5. The group is abelian, infinite cyclic, and the addition is concatenation of cube maps, the integer recording how many times (with sign) the sphere is wrapped. The full computation that $\pi_2(S^2) \cong \mathbb{Z}$ follows from the Hurewicz theorem, theorem 5.4.9.

Homotopy groups would be of little use as invariants if continuous maps did not act on them: a based map $\varphi\colon (X, x_0) \to (Y, y_0)$ sends an n-cube $f\colon (I^n, \partial I^n) \to (X, x_0)$ to the composite $\varphi \circ f\colon (I^n, \partial I^n) \to (Y, y_0)$, and this respects homotopy and the group law. The resulting assignment is functorial, turning a homotopy equivalence into an isomorphism of homotopy groups, making π_n a topological invariance:

Proposition 5.1.6: Functoriality of π_n

Let $n \geq 1$.

1. A based map $\varphi\colon (X, x_0) \to (Y, y_0)$ induces a group homomorphism
$$\varphi_*\colon \pi_n(X, x_0) \to \pi_n(Y, y_0), \qquad \varphi_*[f] = [\varphi \circ f]$$
2. Functoriality holds: $(\mathrm{id}_X)_* = \mathrm{id}$ and $(\psi \circ \varphi)_* = \psi_* \circ \varphi_*$ for composable based maps.
3. If $\varphi \simeq \varphi'$ through based maps, then $\varphi_* = \varphi'_*$. Consequently a based homotopy equivalence induces an isomorphism on every π_n, and π_n is an invariant of based homotopy type.

***Proof*:**

1. If $f \simeq f'$ through a based homotopy F, then $\varphi \circ F$ is a based homotopy $\varphi \circ f \simeq \varphi \circ f'$, so φ_* is well defined on classes. For the group law, $\varphi \circ (f + g) = (\varphi \circ f) + (\varphi \circ g)$ as

maps, because φ acts pointwise and concatenation in the first coordinate commutes with postcomposition: on $t_1 \in [0, \frac{1}{2}]$ both sides equal $\varphi(f(2t_1, \dots))$, and on $t_1 \in [\frac{1}{2}, 1]$ both equal $\varphi(g(2t_1 - 1, \dots))$. Hence $\varphi_*([f] + [g]) = \varphi_*[f] + \varphi_*[g]$.

2. For $\varphi = \mathrm{id}_X$, $\varphi \circ f = f$, so $(\mathrm{id}_X)_* = \mathrm{id}$. For composites, $(\psi \circ \varphi) \circ f = \psi \circ (\varphi \circ f)$, so $(\psi \circ \varphi)_*[f] = [\psi \circ \varphi \circ f] = \psi_*[\varphi \circ f] = (\psi_* \circ \varphi_*)[f]$.

3. Suppose $G\colon X \times I \to Y$ is a based homotopy from φ to φ', so $G(x_0, s) = y_0$ for all s. For a representative $f\colon (I^n, \partial I^n) \to (X, x_0)$, the map

$$(t, s) \mapsto G(f(t), s)\colon I^n \times I \to Y$$

is a homotopy from $\varphi \circ f$ to $\varphi' \circ f$, and it is based: when $t \in \partial I^n$ one has $f(t) = x_0$, hence $G(f(t), s) = G(x_0, s) = y_0$ for all s. Therefore $[\varphi \circ f] = [\varphi' \circ f]$, that is $\varphi_* = \varphi'_*$. If φ is a based homotopy equivalence with based homotopy inverse ψ, then $\psi_*\varphi_* = (\psi\varphi)_* = (\mathrm{id}_X)_* = \mathrm{id}$ and likewise $\varphi_*\psi_* = \mathrm{id}$ by parts (2) and (3), so φ_* is an isomorphism, completing the proof.

Thus, to compute $\pi_n(X)$ one may replace X by any homotopy-equivalent space. It also delivers the standard nonexistence results in the same form as for π_1 (ex. if S^n retracted onto a contractible subspace the retraction would split a nonzero homotopy group through zero, an impossibility).

Example 5.1.7: Homotopy Groups of a Product

A projection-based computation shows that π_n converts products into products. Let (X, x_0) and (Y, y_0) be based spaces, give $X \times Y$ the basepoint (x_0, y_0), and let $p\colon X \times Y \to X$ and $q\colon X \times Y \to Y$ be the projections. The claim is

$$\pi_n(X \times Y, (x_0, y_0)) \;\cong\; \pi_n(X, x_0) \times \pi_n(Y, y_0)$$

for every $n \geq 1$. Define $\Phi = (p_*, q_*)$, sending $[h]$ to $([p \circ h], [q \circ h])$. This is a homomorphism by proposition 5.1.6. A continuous map into a product is exactly a pair of continuous maps into the factors, so $h \mapsto (p \circ h, q \circ h)$ is a bijection from maps $I^n \to X \times Y$ to pairs of maps $(I^n \to X,\, I^n \to Y)$, and it carries $\partial I^n \to (x_0, y_0)$ to the pair of conditions $\partial I^n \to x_0$ and $\partial I^n \to y_0$. The same bijection applied to homotopies (a homotopy into $X \times Y$ is a pair of homotopies) shows Φ is injective: if $p \circ h \simeq p \circ h'$ and $q \circ h \simeq q \circ h'$ through based homotopies F, G, then $(F, G)\colon I^n \times I \to X \times Y$ is a based homotopy $h \simeq h'$. For surjectivity, given $[f] \in \pi_n(X)$ and $[g] \in \pi_n(Y)$, the map $h = (f, g)\colon I^n \to X \times Y$, $h(t) = (f(t), g(t))$, sends ∂I^n to (x_0, y_0) and satisfies $p \circ h = f$, $q \circ h = g$, so $\Phi[h] = ([f], [g])$. Thus Φ is an isomorphism. For instance, the n-torus $T^n = (S^1)^n$ has $\pi_1(T^n) \cong \mathbb{Z}^n$ and $\pi_k(T^n) = 0$ for $k \geq 2$, the latter because $\pi_k(S^1) = 0$ for $k \geq 2$ (the universal cover $\mathbb{R} \to S^1$ identifies $\pi_k(S^1)$ with $\pi_k(\mathbb{R}) = 0$ once covering-space theory from chapter 2 is in hand).

The product formula is the higher-dimensional version that π_1 of a product is the product of the fundamental groups, and its proof is the universal property of the product respecting homotopy (i.e. working in the homotopy category). It is the first useful computation tool: combined with the homotopy groups of spheres it determines π_n for tori, products of spheres, and any finite product whose factors are understood.

So far every group has been computed from contractibility or from products. Already the homotopy groups of spheres, even the lowest degrees, are not elementary. Two facts will be proved in section 5.4

as consequences of the Hurewicz theorem and are recorded here so that they can be used for examples. First, $\pi_k(S^n) = 0$ for $k < n$: a sphere of dimension n has no nontrivial maps from lower-dimensional spheres up to homotopy, reflecting that S^n is $(n-1)$-connected. Second, $\pi_n(S^n) \cong \mathbb{Z}$, generated by the identity, with the isomorphism given by the degree. The off-diagonal groups $\pi_k(S^n)$ with $k > n$ are mostly nonzero, often finite, and their determination is a major open problem (as of writing this book). The first such, $\pi_3(S^2) \cong \mathbb{Z}$ generated by the Hopf map, appears in section 5.2.

The absolute groups $\pi_n(X, x_0)$ measure a single space. To compare a space with a subspace, and to set up the inductive computations that follow, one needs a relative version similar to what was worked out from theorem 3.4.9. The relative homotopy group $\pi_n(X, A, x_0)$ measures maps of a cube that send most of the boundary to the basepoint but are allowed to land anywhere in A along one distinguished face. The precise device is to single out, inside ∂I^n, the face I^{n-1} where the last coordinate vanishes, and to require that the rest of the boundary go to x_0 while I^{n-1} maps into A.

Write $I^n = I^{n-1} \times I$ with last coordinate t_n. Let

$$I^{n-1} = \{t \in \partial I^n : t_n = 0\}$$

be the bottom face, and let $J^{n-1} = \overline{\partial I^n \setminus I^{n-1}}$ be the closure of the rest of the boundary, the union of the top face $t_n = 1$ with the $2(n-1)$ side faces where some coordinate t_i $(i < n)$ is 0 or 1. Then $\partial I^n = I^{n-1} \cup J^{n-1}$ and $I^{n-1} \cap J^{n-1} = \partial I^{n-1}$.

Definition 5.1.8: Relative Homotopy Groups

Let $A \subseteq X$ be a subspace containing the basepoint x_0, and let $n \geq 1$. The *n-th relative homotopy group* $\pi_n(X, A, x_0)$ is the set of homotopy classes of maps

$$f\colon (I^n,\, \partial I^n,\, J^{n-1}) \to (X,\, A,\, x_0)$$

that is, maps $f\colon I^n \to X$ with $f(I^{n-1}) \subseteq A$ and $f(J^{n-1}) = x_0$, where the homotopies are through maps of the same form. For $n \geq 2$ the operation is concatenation in the first coordinate, exactly as in definition 5.1.1, and $\pi_n(X, A, x_0)$ is a group, abelian for $n \geq 3$. For $n = 1$ the set $\pi_1(X, A, x_0)$ is pointed but not in general a group.

Unwinding the conditions: f maps the bottom face I^{n-1} into the subspace A, sends the entire rest of the boundary to the basepoint, and is otherwise unconstrained inside X. When $A = \{x_0\}$ the condition $f(I^{n-1}) \subseteq A$ forces all of ∂I^n to x_0, and the relative group collapses to the absolute group $\pi_n(X, x_0)$. The relative group thus interpolates between X and the pair (X, A), recording maps of a disk into X whose boundary sphere lands in A. The operation concatenates in the first coordinate t_1, which is one of the directions *not* singled out as t_n. This is why concatenation is available for $n \geq 2$ (two free directions: t_1 exists and is distinct from t_n) but $\pi_1(X, A, x_0)$ carries no natural group structure, since for $n = 1$ the first and last coordinates coincide. The shift in the commutativity threshold, abelian for $n \geq 3$ rather than $n \geq 2$, comes from the same count: one coordinate is reserved for the A-face, leaving two free directions only when $n \geq 3$, so the Eckmann-Hilton argument applies one dimension later.

The geometric content is cleaner in the sphere-and-disk picture. Collapsing J^{n-1} to a point turns (I^n, I^{n-1}) into (D^n, S^{n-1}) with a basepoint $* \in S^{n-1}$, so an element of $\pi_n(X, A, x_0)$ is a based-homotopy class of maps $(D^n, S^{n-1}, *) \to (X, A, x_0)$: a disk in X whose boundary circle (more precisely boundary sphere S^{n-1}) is constrained to lie in A, the basepoint of that sphere pinned at x_0.

Example 5.1.9: Relative Homotopy of a Pair with $A = X$

When $A = X$ the relative group is trivial in every degree: $\pi_n(X, X, x_0) = 0$ for all $n \geq 1$. Given a relative cube $f\colon (I^n, \partial I^n, J^{n-1}) \to (X, X, x_0)$, set

$$F(t, s) = f\big(r(t, s)\big)$$

Then $F(\,\cdot\,, 0) = f$, and at every stage F is a relative cube: on J^{n-1}, r fixes the point, so $F(J^{n-1} \times \{s\}) = f(J^{n-1}) = x_0$. On the bottom face I^{n-1} there is no constraint to violate, since $A = X$. At $s = 1$, $r(I^n, 1) \subseteq J^{n-1}$ forces $F(\,\cdot\,, 1) = f|_{J^{n-1}} \circ r(\,\cdot\,, 1) = x_0$ everywhere, the constant map. Hence $[f] = 0$, and since f was arbitrary, $\pi_n(X, X, x_0) = 0$.

The relative group comes equipped with a boundary homomorphism that restricts a relative cube to its A-face. Given $f\colon (I^n, \partial I^n, J^{n-1}) \to (X, A, x_0)$, its restriction to the bottom face I^{n-1} is a map $I^{n-1} \to A$ sending $\partial I^{n-1} = I^{n-1} \cap J^{n-1}$ to x_0, hence a representative of an element of $\pi_{n-1}(A, x_0)$. This restriction is the connecting map of the pair.

Definition 5.1.10: Boundary Map of a Pair

The *boundary homomorphism*

$$\partial\colon \pi_n(X, A, x_0) \longrightarrow \pi_{n-1}(A, x_0)$$

is defined by $\partial[f] = [f|_{I^{n-1}}]$, the homotopy class of the restriction of f to the bottom face I^{n-1}, viewed as a map $(I^{n-1}, \partial I^{n-1}) \to (A, x_0)$. For $n \geq 2$ it is a group homomorphism.

That ∂ is well defined and a homomorphism is a matter of computation: a homotopy of relative cubes restricts to a homotopy of their bottom faces, and concatenation in the first coordinate of I^n restricts to concatenation in the first coordinate of I^{n-1} (the first coordinate is never the distinguished last one), so $\partial(f + g) = \partial f + \partial g$. The boundary map, the inclusion $A \hookrightarrow X$, and the map $\pi_n(X) \to \pi_n(X, A)$ that views an absolute cube as a relative one assemble into the long exact sequence, the key computational instrument of homotopy theory:

Proposition 5.1.11: Long Exact Sequence of a Pair

Let $A \subseteq X$ with basepoint $x_0 \in A$, let $i\colon A \hookrightarrow X$ be the inclusion and $j\colon (X, x_0, x_0) \hookrightarrow (X, A, x_0)$ the map sending an absolute cube to the same cube regarded as relative. Then the sequence

$$\cdots \to \pi_n(A, x_0) \xrightarrow{i_*} \pi_n(X, x_0) \xrightarrow{j_*} \pi_n(X, A, x_0) \xrightarrow{\partial} \pi_{n-1}(A, x_0) \xrightarrow{i_*} \cdots$$

is exact, terminating in $\cdots \to \pi_1(X, A, x_0) \xrightarrow{\partial} \pi_0(A, x_0) \xrightarrow{i_*} \pi_0(X, x_0)$, where the last three terms are pointed sets and exactness means equality of image and preimage of the basepoint.

***Proof*:**

Exactness must be verified at each of the three types of term. Throughout, the identity element (or basepoint) of each group is the class 0 of the constant map, and a class is 0 exactly when its representative is null-homotopic through maps of the prescribed type. The composite of any two consecutive maps is checked to be zero first. The harder containment, that a class killed by the

next map lies in the image of the previous one, is then established by an explicit homotopy.

Step 1: Exactness at $\pi_n(X, x_0)$. First $j_* i_* = 0$. If $[g] \in \pi_n(A, x_0)$, then $i_*[g]$ is g regarded as a map into X, and $j_* i_*[g]$ is that same map viewed as a relative cube $(I^n, \partial I^n, J^{n-1}) \to (X, A, x_0)$, which is legitimate because g has image in A (so the bottom face lands in A) and $g(J^{n-1}) = x_0$. This relative cube is null: with $r\colon I^n \times I \to I^n$ the deformation retraction of I^n onto J^{n-1} from example 5.1.9, the homotopy $(t, s) \mapsto g(r(t, s))$ is through relative cubes, since g maps the whole cube into A (so the bottom-face constraint is met at every stage) and r fixes J^{n-1} pointwise (so $g \circ r$ equals x_0 on J^{n-1} throughout). At $s = 1$ it is constant at x_0. Hence $j_* i_*[g] = 0$, and $\mathrm{im} i_* \subseteq \ker j_*$.

Conversely suppose $[f] \in \pi_n(X, x_0)$ has $j_*[f] = 0$. The kernel of j_* is characterized as follows: $j_*[f] = 0$ if and only if f is homotopic, rel ∂I^n, to a map whose image lies in A. Granting this characterization, such a homotopy keeps ∂I^n at x_0 (since $f(\partial I^n) = x_0$), so the end map f' has $f'(\partial I^n) = x_0$ and image in A, hence represents a class $[g] \in \pi_n(A, x_0)$ with $i_*[g] = [f' \text{ in } X] = [f]$, giving $[f] \in \mathrm{im} i_*$ and $\ker j_* \subseteq \mathrm{im} i_*$.

The characterization itself is proved from the relative null-homotopy. In one direction, if f is homotopic rel ∂I^n to a map f' into A, then $j_*[f] = j_*[f']$, and f' is a relative cube with image in A, hence null in $\pi_n(X, A, x_0)$ by the deformation-retraction argument of the forward direction (apply $g \mapsto g \circ r$ with $g = f'$). Hence $j_*[f] = 0$. In the other direction, let $F\colon I^n \times I \to X$ be a relative null-homotopy from f to the constant map, with $F(I^{n-1} \times \{s\}) \subseteq A$ and $F(J^{n-1} \times \{s\}) = x_0$ for all s. On the boundary of the solid $I^n \times I$, the map F takes values in A on the face $I^{n-1} \times I$, equals x_0 on the top $I^n \times \{1\}$ and on the sides $J^{n-1} \times I$, and equals f on the bottom $I^n \times \{0\}$. The union C of all boundary faces of $I^n \times I$ other than the open bottom is a closed n-disk and a deformation retract of the solid $I^n \times I$, by a retraction ρ_u that pushes radially away from the center of the bottom face and fixes C pointwise. Then $(t, u) \mapsto F(\rho_u(t, 0))$ is a homotopy from f (at $u = 0$, where ρ_0 is the identity on the bottom) to $F|_C$ reparametrized over I^n (at $u = 1$), and it is rel ∂I^n, because $\partial I^n \times \{0\} \subseteq J^{n-1} \times \{0\} \subseteq C$ is fixed by every ρ_u and there $F = x_0$. The end map has image in $F(C) \subseteq A \cup \{x_0\} = A$, since C consists of the A-valued face together with x_0-valued faces. This deforms f rel ∂I^n into A, completing the characterization and the proof that $\ker j_* \subseteq \mathrm{im} i_*$.

Step 2: Exactness at $\pi_n(X, A, x_0)$. First $\partial j_* = 0$. For $[f] \in \pi_n(X, x_0)$, the relative cube $j_* f$ has the same underlying map f, whose restriction to the bottom face I^{n-1} is constant at x_0 (because $f(\partial I^n) = x_0$ and $I^{n-1} \subseteq \partial I^n$). Thus $\partial j_*[f] = [c_{x_0}] = 0$, so $\mathrm{im} j_* \subseteq \ker \partial$.

Conversely, let $[f] \in \pi_n(X, A, x_0)$ have $\partial[f] = 0$, so the restriction $g = f|_{I^{n-1}}\colon (I^{n-1}, \partial I^{n-1}) \to (A, x_0)$ is null-homotopic in A: there is $G\colon I^{n-1} \times I \to A$ with $G(\,\cdot\,, 0) = g$, $G(\,\cdot\,, 1) = x_0$, and $G(\partial I^{n-1} \times I) = x_0$. Use G to homotope f rel J^{n-1} so that its bottom face becomes constant: extend G to a homotopy of f supported in a collar $I^{n-1} \times [0, \epsilon]$ of the bottom face, replacing the values of f on the collar by a copy of G that slides the A-face down to x_0. The result f' has $f'(I^{n-1}) = x_0$, hence $f'(\partial I^n) = x_0$, so f' is an absolute cube and represents a class in $\pi_n(X, x_0)$ with $j_*[f'] = [f' \text{ relative}] = [f]$ (the homotopy was through relative cubes). Thus $[f] \in \mathrm{im} j_*$, giving $\ker \partial \subseteq \mathrm{im} j_*$.

Step 3: Exactness at $\pi_{n-1}(A, x_0)$. First $i_* \partial = 0$. For $[f] \in \pi_n(X, A, x_0)$, the class $\partial[f] = [f|_{I^{n-1}}]$ becomes, under i_*, the class of $f|_{I^{n-1}}$ regarded as a map into X. But f itself is a homotopy, rel basepoint conditions, contracting $f|_{I^{n-1}}$ inside X: slide the bottom face up through the cube I^n. Precisely, $H(t_1, \ldots, t_{n-1}, s) = f(t_1, \ldots, t_{n-1}, s)$ defines a homotopy in X from $f|_{I^{n-1}}$ (at $s = 0$) to $f|_{t_n = 1}$, and the latter lies in J^{n-1}, hence is constant at x_0. This H keeps ∂I^{n-1} at x_0 for every s (those points lie in J^{n-1} at every height $t_n = s$). Thus $i_* \partial[f] = [f|_{I^{n-1}} \text{ in } X] = 0$,

giving $\mathrm{im}\partial \subseteq \ker i_*$.

Conversely suppose $[g] \in \pi_{n-1}(A, x_0)$ has $i_*[g] = 0$ in $\pi_{n-1}(X, x_0)$, witnessed by a null-homotopy $H\colon I^{n-1} \times I \to X$ with $H(\cdot, 0) = g$, $H(\cdot, 1) = x_0$, and $H(\partial I^{n-1} \times I) = x_0$. View H as a map on I^n by setting $t_n =$ the homotopy parameter: define $f\colon I^n \to X$ by $f(t_1, \ldots, t_{n-1}, t_n) = H(t_1, \ldots, t_{n-1}, t_n)$. Then $f(I^{n-1}) = H(\cdot, 0) = g(I^{n-1}) \subseteq A$, f on the top face $t_n = 1$ is x_0, and on the side faces ($t_i \in \{0, 1\}$, $i < n$) is $H(\partial I^{n-1} \times I) = x_0$. Together these say $f(J^{n-1}) = x_0$. Hence f is a legitimate relative cube, $[f] \in \pi_n(X, A, x_0)$, and $\partial[f] = [f|_{I^{n-1}}] = [g]$. Thus $[g] \in \mathrm{im}\partial$, giving $\ker i_* \subseteq \mathrm{im}\partial$.

The three steps establish exactness at all three term types and at every degree, and the low-degree tail involving π_0 is the same statement read as an equality of based sets, completing the proof.

Thus, whenever a space X contains a subspace A whose homotopy is understood, the sequence relates $\pi_n(X)$, $\pi_n(A)$, and the relative groups $\pi_n(X, A)$ in a single exact chain, and exactness lets one solve for an unknown group from its neighbors. Three uses recur:

1. When the relative groups vanish in a range, i_* is an isomorphism there, so A and X have the same homotopy in that range.
2. When A is contractible, $\pi_n(A) = 0$ and the sequence forces $\pi_n(X) \cong \pi_n(X, A)$.
3. when the pair (X, A) is highly connected, the sequence measures exactly how far the inclusion $A \hookrightarrow X$ is from inducing isomorphisms. This same sequence reappears in section 5.2 as the long exact sequence of a fibration, where A is replaced by a fiber, and it is the core of the Hurewicz and Whitehead theorems later in the chapter.

One dependence on the basepoint remains to be addressed. In chapter 2 a path γ from x_0 to x_1 produced an isomorphism $\pi_1(X, x_0) \to \pi_1(X, x_1)$ by conjugating loops with γ. The same construction acts on higher homotopy groups, and when γ is a loop (so $x_1 = x_0$) it yields an action of $\pi_1(X, x_0)$ on $\pi_n(X, x_0)$. Given a loop γ at x_0 and a cube $f\colon (I^n, \partial I^n) \to (X, x_0)$, shrink the cube toward its center and use γ to carry the now-free boundary along the loop: there is a map $\gamma \cdot f\colon (I^n, \partial I^n) \to (X, x_0)$ that on a small central subcube agrees with f and on the collar between that subcube and ∂I^n traverses γ radially outward. The construction is the higher-dimensional analogue of basepoint change: classes involved.

Proposition 5.1.12: The Action of π_1 on π_n

Let (X, x_0) be a based space and $n \geq 1$. The assignment $([\gamma], [f]) \mapsto [\gamma \cdot f]$ above is well defined on homotopy classes and defines a left action of the group $\pi_1(X, x_0)$ on $\pi_n(X, x_0)$ by group automorphisms, that is, $[\gamma] \cdot ([\gamma'] \cdot [f]) = ([\gamma][\gamma']) \cdot [f]$, the constant loop acts trivially, and each $[\gamma]$ acts by a group automorphism of $\pi_n(X, x_0)$. For $n \geq 2$ the group $\pi_n(X, x_0)$ is abelian, so this action makes it a left module over the group ring $\mathbb{Z}[\pi_1(X, x_0)]$. For $n = 1$ the action is conjugation: $[\gamma] \cdot [f] = [\gamma][f][\gamma]^{-1}$.

We shall often abbreviate and just say $\pi_n(X, x_0)$ is a module over $\pi_1(X, x_0)$ instead of $\mathbb{Z}[\pi_1(X, x_0)]$.

***Proof*:**
For well-definedness, a homotopy $\gamma \simeq \gamma'$ relative to endpoints together with a homotopy $f \simeq f'$ relative to ∂I^n combine, applied collarwise, into a homotopy $\gamma \cdot f \simeq \gamma' \cdot f'$ relative to ∂I^n. The construction used only homotopy classes of the radial path and of the central cube. The module

identities are basepoint-change identities transported from chapter 2: traversing γ then γ' along the collar is, up to reparametrizing the collar, the same as traversing the concatenation $\gamma * \gamma'$, giving $[\gamma] \cdot ([\gamma'] \cdot [f]) = [\gamma * \gamma'] \cdot [f]$; the constant loop leaves the collar at x_0 and returns f; and $[\gamma] \cdot ([f] + [g]) = [\gamma] \cdot [f] + [\gamma] \cdot [g]$ because the collar construction is performed independently on the two halves of $f + g$ in the first coordinate, with the same γ on both collars. Hence each $[\gamma]$ acts as an automorphism. For $n = 1$, the collar construction runs γ on the collar at each endpoint of f, producing $\gamma * f * \bar{\gamma}$, whose class is $[\gamma][f][\gamma]^{-1}$, the conjugation formula, completing the proof.

The action records how the higher homotopy of a space is twisted by its fundamental group. When the action is nontrivial, $\pi_n(X, x_0)$ is a module over the (possibly nonabelian) group ring $\mathbb{Z}[\pi_1(X, x_0)]$, not just an abelian group, and many statements (the Hurewicz theorem among them) require the action to be trivial to take their cleanest form. Spaces where this complication is absent deserve a name:

Definition 5.1.13: n-Simple and Simple Spaces

A path-connected space X is *n-simple* if the action of $\pi_1(X, x_0)$ on $\pi_n(X, x_0)$ from proposition 5.1.12 is trivial, that is, $[\gamma] \cdot [f] = [f]$ for every loop $[\gamma] \in \pi_1(X, x_0)$ and every $[f] \in \pi_n(X, x_0)$. The space is *simple* (or *abelian*) if it is n-simple for all $n \geq 1$. A space is 1-simple exactly when $\pi_1(X, x_0)$ is abelian, since for $n = 1$ the action is conjugation.

Simplicity is automatic in the most common situations. Any simply connected space is simple, since π_1 is trivial and acts trivially on everything. Any topological group, and more generally any H-space, is simple: the multiplication makes basepoint change inessential, as the abelian conclusion for π_1 already signaled. For a simple space the homotopy groups $\pi_n(X, x_0)$ are independent of the basepoint not just up to isomorphism but canonically, and the comparison maps between homotopy and homology in section 5.4 will hold without correction terms. When the action is nontrivial, those corrections become the source of the local-coefficient systems studied alongside the spectral sequences of chapter 6.

Example 5.1.14: The Action for the Wedge of Circles

Let $X = S^1 \vee S^1$. By section 2.2, $\pi_1(X, x_0) \cong \mathbb{Z} * \mathbb{Z} = \langle a, b \rangle$, the free group on the two circle classes. The action of π_1 on π_1 is conjugation, so $a \cdot b = aba^{-1} \neq b$ in the free group (the two are distinct reduced words). Hence X is not 1-simple. By contrast, the circle S^1 itself is simple: $\pi_1(S^1) \cong \mathbb{Z}$ is abelian, so the conjugation action is trivial, and $\pi_n(S^1) = 0$ for $n \geq 2$ leaves nothing to act on. The torus $S^1 \times S^1$ is likewise simple, being a topological group. Its fundamental group $\mathbb{Z}^2$ is abelian and acts trivially on every π_n.

The action also measures the difference between the based and the free homotopy classes of a map out of a sphere. A based map $S^n \to X$ represents a class in $\pi_n(X, x_0)$, while forgetting the basepoint remembers only its *free* homotopy class in $[S^n, X]$, the set of homotopy classes of unbased maps. The two are related exactly by the action of proposition 5.1.12.

Proposition 5.1.15: Free and Based Homotopy Classes of Maps from a Sphere

Let X be path-connected and $n \geq 1$. The map that forgets the basepoint,

$$\pi_n(X, x_0) = [S^n, X]_* \longrightarrow [S^n, X]$$

is surjective, and two based classes have the same image if and only if they lie in one orbit of the action of $\pi_1(X, x_0)$ on $\pi_n(X, x_0)$. It therefore descends to a bijection

$$[S^n, X] \;\cong\; \pi_n(X, x_0)/\pi_1(X, x_0)$$

from the free homotopy classes onto the set of orbits of the action. In particular, if X is n-simple (definition 5.1.13), and so if X is simply connected, the forgetful map is itself a bijection $[S^n, X] \cong \pi_n(X, x_0)$.

***Proof*:**

Surjectivity. Let $f\colon S^n \to X$ be any map and $s_0 \in S^n$ the basepoint. Since X is path-connected, choose a path from $f(s_0)$ to x_0. The inclusion $\{s_0\} \hookrightarrow S^n$ is a cofibration, so this path extends to a homotopy of f ending at a map that sends s_0 to x_0. That end map is based and freely homotopic to f, so every free class has a based representative.

Fibers are orbits. If $[g] = [\gamma] \cdot [f]$ for a loop γ at x_0, the map $\gamma \cdot f$ built in proposition 5.1.12 carries the basepoint of f around γ, and that construction is a homotopy of unbased maps from f to a representative of $[g]$, so f and g are freely homotopic. Conversely, let $H\colon S^n \times I \to X$ be a free homotopy between based maps f and g. Its trace $\gamma(t) = H(s_0, t)$ is a loop at x_0, and restricting H to a collar of the basepoint exhibits g as the map obtained from f by carrying the basepoint along γ, that is $[g] = [\gamma] \cdot [f]$. Two based classes therefore agree freely exactly when one carries to the other under the action.

The bijection onto the orbit set follows. When X is n-simple the action is trivial, so each orbit is a single class and the forgetful map is a bijection. A simply connected space is n-simple because its fundamental group is trivial, completing the proof.

The case $n = 1$ names a classical set. The action of π_1 on itself is conjugation, so its orbits are the conjugacy classes.

Corollary 5.1.16: Free Homotopy Classes of Loops are Conjugacy Classes

Let X be path-connected. The forgetful map induces a bijection

$$[S^1, X] \;\cong\; \{\text{conjugacy classes of } \pi_1(X, x_0)\}$$

between the free homotopy classes of loops in X and the conjugacy classes of the fundamental group. Two based loops are freely homotopic if and only if their classes are conjugate in $\pi_1(X, x_0)$.

***Proof*:**
Apply proposition 5.1.15 with $n = 1$: the free homotopy classes are the orbits of the action of $\pi_1(X, x_0)$ on $\pi_1(X, x_0)$. By proposition 5.1.12 that action is conjugation, $[\gamma] \cdot [f] = [\gamma][f][\gamma]^{-1}$, whose orbits are the conjugacy classes, completing the proof.

Exercise 5.1.1

1. Let X be a path-connected topological group with identity e, with multiplication providing a second composition on loops. Using the Eckmann-Hilton argument, prove that $\pi_1(X, e)$ is abelian. Identify precisely which two operations play the roles of $+_1$ and $+_2$ and verify the interchange law.

2. Compute $\pi_n(S^1 \times S^1, *)$ for all $n \geq 1$, citing the product formula and the homotopy groups of S^1. Then compute $\pi_n(S^1 \times S^2, *)$ for $n = 1, 2$, assuming the stated values $\pi_k(S^1) = 0$ for $k \geq 2$, $\pi_1(S^2) = 0$, and $\pi_2(S^2) \cong \mathbb{Z}$.

3. Let $A \subseteq X$ with A a deformation retract of X (so the inclusion is a homotopy equivalence). Use the long exact sequence of the pair (X, A) to prove that $\pi_n(X, A, x_0) = 0$ for all $n \geq 1$. Conversely, if $\pi_n(X, A, x_0) = 0$ for all n and all basepoints, what does the sequence tell you about $i_* \colon \pi_n(A) \to \pi_n(X)$?

4. Prove that a map $f \colon (I^n, \partial I^n) \to (X, x_0)$ represents 0 in $\pi_n(X, x_0)$ if and only if it extends to a continuous map $D^{n+1} \to X$ that is constant at x_0 on a hemisphere, equivalently if and only if the corresponding map $S^n \to X$ extends over the disk D^{n+1}. (Use the sphere description $I^n/\partial I^n \cong S^n$.)

5. Show that for $n \geq 2$ the operation $+_2$ (concatenation in the second coordinate) of theorem 5.1.4 defines the same group structure on $\pi_n(X, x_0)$ as $+_1$, without invoking commutativity. That is, prove $[f] +_1 [g] = [f] +_2 [g]$ directly using the Eckmann-Hilton interchange law with two of the four entries set to the identity.

6. Recall the notion of an n-simple space from definition 5.1.13, where $\pi_1(X, x_0)$ acts trivially on $\pi_n(X, x_0)$. Give an example of a space that is *not* 1-simple, that is, where π_1 acts nontrivially on itself by conjugation, and explain why the condition "π_1 acts trivially on π_1" is the same as "π_1 is abelian."

5.2 Fibrations and the Long Exact Sequence

The higher homotopy groups of section 5.1 are easy to define and notoriously difficult to compute by hand: even $\pi_3(S^2)$ resists every elementary attack, namely since a map $S^3 \to S^2$ cannot be analyzed simplex-by-simplex the way a 1-dimensional loop can. To break this impasse, we shall break down computations into a *long exact sequence of a fibration*. A *fibration* $F \to E \to B$ is a map $p \colon E \to B$ flexible enough that homotopies in the base B can always be lifted to homotopies in the total space E. The fiber $F = p^{-1}(b_0)$ then sits inside E in a controlled way, and the relative homotopy groups $\pi_n(E, F)$ of section 5.1 collapse onto $\pi_n(B)$. Splicing this collapse into the relative long exact sequence produces a single exact sequence relating $\pi_*(F)$, $\pi_*(E)$, and $\pi_*(B)$. When two of the three spaces are understood, the third is forced. The first non-trivial computation of a nonzero higher homotopy group, $\pi_3(S^2) \cong \mathbb{Z}$, comes from this sequence.

The whole apparatus rests on the ability to lift homotopies. The path-lifting and homotopy-lifting lemmas for covering spaces (lemma 2.1.30, lemma 2.1.31) are the prototype, and the definition below abstracts exactly the feature of covering maps that those lemmas used:

Definition 5.2.1: Homotopy Lifting Property

Let $p\colon E \to B$ be a continuous map and let X be a topological space. The map p has the *homotopy lifting property* with respect to X if, for every homotopy $G\colon X \times I \to B$ and every map $\tilde{g}_0\colon X \to E$ lifting the initial stage (that is, $p \circ \tilde{g}_0 = G(\,\cdot\,, 0)$), there exists a homotopy $\tilde{G}\colon X \times I \to E$ with

$$p \circ \tilde{G} = G \qquad \text{and} \qquad \tilde{G}(\,\cdot\,, 0) = \tilde{g}_0$$

Equivalently, every commutative square of solid arrows admits a diagonal filler $\tilde{G}$ making both triangles commute:

$$\begin{array}{ccc} X \times \{0\} & \xrightarrow{\tilde{g}_0} & E \\ \downarrow & \overset{\tilde{G}}{\nearrow} & \downarrow{\scriptstyle p} \\ X \times I & \xrightarrow[G]{} & B \end{array}$$

The square encodes the only thing being asked: a homotopy downstairs in B, together with a chosen lift of its starting position, can be promoted to a homotopy upstairs in E that projects back down to the original. The lifted homotopy is not required to be unique (covering spaces are the special case where it is). What matters is existence, because existence is what lets one transport homotopical information from B to E and back. The class of test spaces X for which one requires this property is what distinguishes the two notions of fibration introduced next:

Definition 5.2.2: Hurewicz and Serre Fibrations

A continuous map $p\colon E \to B$ is a

1. *Hurewicz fibration* if it has the homotopy lifting property (definition 5.2.1) with respect to *every* topological space X.
2. *Serre fibration* (or *weak fibration*) if it has the homotopy lifting property with respect to every disk D^n, $n \geq 0$.

For a fibration $p\colon E \to B$ with B path-connected, the *fiber* over a basepoint $b_0 \in B$ is the subspace $F = p^{-1}(b_0) \subseteq E$. The data (E, p, B, F) is represented bythe notation

$$F \hookrightarrow E \xrightarrow{p} B$$

with F included into E and E projected onto B.

The fiber F depends on the choice of b_0 only up to homotopy equivalence when B is path-connected (a path from b_0 to b_1 lifts, via the homotopy lifting property, to a homotopy equivalence $p^{-1}(b_0) \simeq p^{-1}(b_1)$), so one speaks of *the* fiber.

Every Hurewicz fibration is a Serre fibration, since disks are among all spaces, but the converse fails: an example (a generalized covering map of the Hawaiian earring, which lifts homotopies parametrized

by disks but not those parametrized by an arbitrary space) is constructed in [3].[2]

The gap between Serre and Hurewicz is harmless for the homotopy theory of this book, because the homotopy groups are built from maps out of spheres and disks and the Serre condition lifts exactly those. Both notions nonetheless survive in the literature because Serre fibrations are far easier to recognize (one tests only against disks, which are compact and contractible) while Hurewicz fibrations have cleaner formal properties.

Example 5.2.3: The Real Line over the Circle, and Covering maps

1. The exponential covering map $p\colon \mathbb{R} \to S^1$, $p(t) = (\cos 2\pi t, \sin 2\pi t)$, is a Hurewicz fibration with fiber $F = p^{-1}(1,0) = \mathbb{Z} \subseteq \mathbb{R}$, a discrete set. To see the homotopy lifting property against an arbitrary space X, take a homotopy $G\colon X \times I \to S^1$ and an initial lift $\tilde{g}_0\colon X \to \mathbb{R}$. For each fixed $x \in X$ the path $t \mapsto G(x,t)$ is a path in S^1 starting at $p(\tilde{g}_0(x))$, so by unique path lifting (lemma 2.1.30) it has a unique lift $t \mapsto \tilde{G}(x,t)$ in $\mathbb{R}$ with $\tilde{G}(x,0) = \tilde{g}_0(x)$. Setting $\tilde{G}(x,t)$ to be this lift defines a function $X \times I \to \mathbb{R}$ with $p \circ \tilde{G} = G$ and the correct initial stage. Continuity in (x,t) jointly follows from the same evenly-covered-neighborhood argument used in lemma 2.1.31: near any point the lift is $\tilde{G} = (p|_V)^{-1} \circ G$ for the sheet V containing the current value, and $(p|_V)^{-1}$ is a homeomorphism. Thus the square fills, and p is a Hurewicz fibration. The fiber $\mathbb{Z}$ is discrete because the covering is evenly covered with countably many sheets.

2. Every covering map is a Hurewicz fibration by the identical argument: unique path lifting promotes any base homotopy to a total-space homotopy. Covering spaces are precisely the fibrations whose fibers are discrete. In general, the fibers may be large and connected.

The first substantial theorem of the section certifies a large, geometrically natural class of Serre fibrations: the fiber bundles. Recall that a *fiber bundle* with fiber F is a map $p : E \to B$ that is *locally trivial*, meaning B is covered by open sets U over each of which p looks like a product, i.e. there is a homeomorphism $\varphi_U\colon p^{-1}(U) \to U \times F$ with $\mathrm{pr}_U \circ \varphi_U = p$, where $\mathrm{pr}_U\colon U \times F \to U$ is the projection. The smooth refinement, with a structure group acting on the fiber, is [32, Fiber Bundle]. Projections $U \times F \to U$ trivially lift homotopies (lift in the U-coordinate using G, hold the F-coordinate fixed at its initial value), so a bundle lifts homotopies locally over the base. The following theorem shows these local lifts can be patched into a single global lift when the base of the homotopy is a cube:

Theorem 5.2.4: Fiber Bundles are Serre Fibrations

Every fiber bundle $p\colon E \to B$ is a Serre fibration. That is, p has the homotopy lifting property with respect to every cube I^n (equivalently every disk D^n).

[2] The gap closes under mild hypotheses: by Dold's local-global theorem [18], a map that is a Hurewicz fibration locally over a *numerable* open cover of the base (one admitting a subordinate partition of unity, e.g. any open cover of a paracompact Hausdorff base, by [16, Existence of Subordinate Partitions of Unity]) is a Hurewicz fibration globally. In particular a fiber bundle over a paracompact Hausdorff base is a Hurewicz fibration, sharpening the remark after theorem 5.2.4. Moreover, in the convenient category of compactly generated spaces, a Serre fibration whose base *and* total space are both CW complexes is automatically a Hurewicz fibration [35, 5]

***Proof*:**
Fix $n \geq 0$ and write $X = I^n$. Let $G\colon X \times I \to B$ be a homotopy and $\tilde{g}_0\colon X \to E$ a lift of $G(\,\cdot\,, 0)$. The cube $X \times I = I^{n+1}$ is compact, and $\{p^{-1}(U)\}$ for trivializing opens $U \subseteq B$ pulls back under G to an open cover $\{G^{-1}(U)\}$ of $X \times I$.

Step 1: A Lebesgue grid adapted to the cover. By the Lebesgue number lemma applied to the compact metric space $X \times I$, there is $m \in \mathbb{N}$ so large that every subcube of the subdivision of $X \times I = I^{n+1}$ into m^{n+1} equal closed subcubes of side $1/m$ is mapped by G into a single trivializing open set U. Subdivide the last (homotopy) coordinate at the points $0 = t_0 < t_1 < \cdots < t_m = 1$ with $t_j = j/m$, and subdivide $X = I^n$ into subcubes $C_1, \ldots, C_{m^n}$ of side $1/m$. For each pair $(C_i, [t_{j-1}, t_j])$ the box $C_i \times [t_{j-1}, t_j]$ lands in a single trivializing $U_{i,j}$ under G.

Step 2: Lifting one box at a time, in lexicographic order. We construct the lift $\tilde{G}$ on $X \times I$ by lifting it over the boxes one homotopy-layer at a time, $j = 1, 2, \ldots, m$, and within each layer cube by cube. Suppose inductively that $\tilde{G}$ has been defined and is continuous on

$$A \;=\; (X \times [0, t_{j-1}]) \;\cup\; \Big(\bigcup\nolimits_{k<i} C_k \times [t_{j-1}, t_j]\Big)$$

lifting $G|_A$ and extending $\tilde{g}_0$, and that we wish to extend it across the next box $B_0 = C_i \times [t_{j-1}, t_j]$. The lift is already defined on the part of B_0 that meets A, namely

$$L \;=\; \big(C_i \times \{t_{j-1}\}\big) \;\cup\; \Big((\partial C_i \cap \bigcup\nolimits_{k<i} C_k) \times [t_{j-1}, t_j]\Big)$$

a closed subset of the cube B_0. Concretely L is the bottom face of B_0 together with some of its vertical side faces, which is a retract of B_0: there is a retraction $r\colon B_0 \to L$ (the cube retracts onto its bottom face union any union of vertical faces).

Step 3: Trivialize and lift over the box. Over B_0 the bundle is trivial: choose a trivialization $\varphi\colon p^{-1}(U_{i,j}) \xrightarrow{\sim} U_{i,j} \times F$ with $\mathrm{pr}_U \circ \varphi = p$. Write the already-defined lift on L in trivialized coordinates,

$$\varphi \circ \tilde{G}|_L \;=\; \big(G|_L,\ \sigma\big)\colon L \to U_{i,j} \times F$$

where the first component is forced to be $G|_L$ (because $\tilde{G}$ lifts G) and $\sigma\colon L \to F$ is the fiber component. Extend σ to all of B_0 by precomposing with the retraction, $\bar{\sigma} = \sigma \circ r\colon B_0 \to F$, and define

$$\tilde{G}|_{B_0} \;=\; \varphi^{-1} \circ \big(G|_{B_0},\ \bar{\sigma}\big)$$

This is continuous (a composite of continuous maps), satisfies $p \circ \tilde{G}|_{B_0} = \mathrm{pr}_U \circ \varphi \circ \tilde{G}|_{B_0} = G|_{B_0}$, and agrees with the old lift on L because $r|_L = \mathrm{id}_L$ forces $\bar{\sigma}|_L = \sigma$ and the first coordinate already matched. By the pasting lemma the extended map is continuous on $A \cup B_0$, since A and B_0 are closed and the two definitions agree on the closed overlap L.

Step 4: Conclusion. Iterating over the m^n cubes of layer j and then over the m layers exhausts all of $X \times I$, producing a continuous $\tilde{G}\colon X \times I \to E$ with $p \circ \tilde{G} = G$ and $\tilde{G}(\,\cdot\,, 0) = \tilde{g}_0$ (the latter because the very first layer started from $\tilde{g}_0$ on $X \times \{0\}$ and every extension preserved already-defined values). Thus the lifting square fills for $X = I^n$, and p is a Serre fibration, as we sought to show.

The proof is the prototype for lifting argument in homotopy theory: compactness chops the problem into pieces small enough to see local triviality, and a retraction extends a partial lift across each piece. The only place local triviality is used is Step 3, and the only place compactness is used is the Lebesgue grid of Step 1. With more care (a partition of unity in place of the Lebesgue grid) the

same conclusion holds for Hurewicz fibrations:

Theorem 5.2.5: Fiber Bundles over Paracompact are Hurewicz Fibrations

Every fiber bundle $p\colon E \to B$ over a paracompact Hausdorff base is a Hurewicz fibration.

***Proof Key ideas*:**
The proof rests on Dold's local-to-global theorem for numerable covers, cited from [18] and [25, Chapter 7, Section 4]: if B has an open cover $\{V\}$ that is locally finite and consists of the sets $V = \lambda_V^{-1}((0,1])$ for continuous functions $\lambda_V\colon B \to [0,1]$, and p restricted to $p^{-1}(V) \to V$ is a Hurewicz fibration for every V, then p is a Hurewicz fibration. Over a paracompact Hausdorff base a partition of unity subordinate to the cover by trivializing open sets exists ([16, Existence of Subordinate Partitions of Unity]), and the sets where its functions are positive form such a cover by trivializing sets. Over a trivializing set V the map p is the projection $V \times F \to V$, which lifts every homotopy by lifting in the V-coordinate and holding the F-coordinate fixed, so the theorem applies. Both sources prove the local-to-global theorem by patching path-lifting functions over the cover with the functions λ_V.

Serre fibrations are also stable under pullback, and the fiber is unchanged. The construction of the Serre spectral sequence in section 6.2 pulls a fibration back along every singular simplex of its base.

Proposition 5.2.6: Pullbacks of Serre Fibrations

Let $p\colon E \to B$ be a Serre fibration and $f\colon B' \to B$ a continuous map. Form the pullback

$$f^*E = \{(b', e) \in B' \times E \mid f(b') = p(e)\}$$

with projection $q\colon f^*E \to B'$, $(b', e) \mapsto b'$. Then q is a Serre fibration, and for every $b'_0 \in B'$ the map $(b'_0, e) \mapsto e$ is a homeomorphism from $q^{-1}(b'_0)$ onto the fiber $p^{-1}(f(b'_0))$ of p.

***Proof*:**
Let $G\colon D^n \times I \to B'$ be a homotopy and $\tilde{g}_0\colon D^n \to f^*E$ a lift of $G(\,\cdot\,, 0)$ under q. Write $\tilde{g}_0(x) = (G(x,0), \eta(x))$ for a map $\eta\colon D^n \to E$ with $p \circ \eta = f \circ G(\,\cdot\,, 0)$, which is forced by membership in f^*E together with $q \circ \tilde{g}_0 = G(\,\cdot\,, 0)$. The composite $f \circ G\colon D^n \times I \to B$ is a homotopy in B whose initial stage is lifted by η, so, p being a Serre fibration, it lifts to $H\colon D^n \times I \to E$ with $H(\,\cdot\,, 0) = \eta$ and $p \circ H = f \circ G$. Define $\tilde{G}(x,t) = (G(x,t), H(x,t))$. It lands in f^*E because $p(H(x,t)) = f(G(x,t))$, it is continuous, it satisfies $q \circ \tilde{G} = G$, and its initial stage is $(G(\,\cdot\,, 0), \eta) = \tilde{g}_0$. So q is a Serre fibration.

For the fiber, $q^{-1}(b'_0) = \{(b'_0, e) \mid p(e) = f(b'_0)\}$, and $(b'_0, e) \mapsto e$ is a continuous bijection onto $p^{-1}(f(b'_0))$ with continuous inverse $e \mapsto (b'_0, e)$, hence a homeomorphism, completing the proof.

For a.e. homotopy-group computations of this book, the Serre version above is sufficient since those groups are detected by spheres and disks. Before extracting the long exact sequence, one example of a fiber bundle that is not a covering space deserves mention, as its long exact sequence will give $\pi_3(S^2)$.

Example 5.2.7: The Hopf Fibration

Regard $S^3 = \{(z_0, z_1) \in \mathbb{C}^2 : |z_0|^2 + |z_1|^2 = 1\}$ and $S^2 = \mathbb{CP}^1$, the complex projective line, which is the set of complex lines through the origin in $\mathbb{C}^2$. The *Hopf map* is

$$p\colon S^3 \longrightarrow \mathbb{CP}^1 = S^2, \qquad p(z_0, z_1) = [z_0 : z_1]$$

sending a unit vector to the complex line it spans. The preimage of a point $[z_0 : z_1]$ is the set of unit vectors on that line, namely $\{(\lambda z_0, \lambda z_1) : \lambda \in \mathbb{C}, |\lambda| = 1\}$, a circle. Thus the fiber is $F = S^1$. The map is a fiber bundle: over the standard chart $U_0 = \{[z_0 : z_1] : z_0 \neq 0\}$ of $\mathbb{CP}^1$, the trivialization sends $(z_0, z_1) \in p^{-1}(U_0)$ to the pair (the affine coordinate $z_1/z_0 \in \mathbb{C} \cong U_0$, the phase $z_0/|z_0| \in S^1$), with inverse rebuilding the unit vector. The same works over $U_1 = \{z_1 \neq 0\}$. These two charts cover $\mathbb{CP}^1$, so p is locally trivial with fiber S^1. By theorem 5.2.4 it is a Serre fibration, written

$$S^1 \hookrightarrow S^3 \xrightarrow{p} S^2$$

Identifying $\mathbb{CP}^1$ with S^2 uses stereographic coordinates: U_0 and U_1 are the two hemispheres' coordinate charts, glued along the equator by $w \mapsto 1/w$, which is exactly the gluing of S^2 from two disks.

The Hopf fibration is the first map $S^3 \to S^2$ that fails to be nullhomotopic, and its existence is the geometric content of $\pi_3(S^2) \neq 0$. The long exact sequence will show $\pi_3(S^2) \cong \mathbb{Z}$, generated by the Hopf class.

A fiber bundle whose fibers are vector spaces, with local trivializations that are linear on each fiber, is a vector bundle.

Definition 5.2.8: Vector Bundle

Let $\mathbb{K}$ be $\mathbb{R}$ or $\mathbb{C}$. A $\mathbb{K}$-*vector bundle* over a space B, called *real* when $\mathbb{K} = \mathbb{R}$ and *complex* when $\mathbb{K} = \mathbb{C}$, is a map $p\colon E \to B$ together with the structure of a finite-dimensional $\mathbb{K}$-vector space on each fiber $E_b = p^{-1}(b)$, such that p is *locally trivial*: every point of B has an open neighbourhood U with a homeomorphism

$$\varphi_U : p^{-1}(U) \longrightarrow U \times \mathbb{K}^n, \qquad \mathrm{pr}_U \circ \varphi_U = p$$

for some $n \geq 0$, restricting to a $\mathbb{K}$-linear isomorphism $E_b \to \{b\} \times \mathbb{K}^n$ for every $b \in U$. Such a φ_U is a *local trivialization*.

1. The dimension of E_b is constant on each such U, hence locally constant in b. If it equals n for every $b \in B$, then E has *rank* n, and a vector bundle of rank one is a *line bundle*.
2. The *trivial bundle* of rank n is the projection $B \times \mathbb{K}^n \to B$, written $\underline{\mathbb{K}}^n$, or $\underline{\mathbb{R}}^n$ and $\underline{\mathbb{C}}^n$. A vector bundle isomorphic to it is *trivial*.
3. For $\mathbb{K}$-vector bundles $p'\colon E' \to B'$ and $p\colon E \to B$, a *bundle map* covering a map $f\colon B' \to B$ is a continuous map $\tilde{f}\colon E' \to E$ with $p \circ \tilde{f} = f \circ p'$ whose restriction $E'_{b'} \to E_{f(b')}$ is $\mathbb{K}$-linear for every $b' \in B'$. A bundle map covering id_B is a *homomorphism* over B, and an *isomorphism* is a homomorphism with an inverse homomorphism.
4. A *section* of E is a map $s\colon B \to E$ with $p \circ s = \mathrm{id}_B$. The *zero section* sends b to the zero vector of E_b. It is continuous, being $b \mapsto \varphi_U^{-1}(b, 0)$ over each U.

Definition 5.2.9: Constructions on Vector Bundles

Let $p\colon E \to B$ and $q\colon F \to B$ be $\mathbb{K}$-vector bundles.

1. *Pullback.* For a map $f\colon B' \to B$, the pullback $f^*E = \{(b', e) \in B' \times E \mid f(b') = p(e)\}$ of proposition 5.2.6 has fiber $\{b'\} \times E_{f(b')}$ over b', a vector space through the second coordinate. A local trivialization φ_U of E gives the local trivialization $(b', e) \mapsto \big(b', \mathrm{pr}_2 \varphi_U(e)\big)$ of f^*E over $f^{-1}(U)$, with inverse $(b', v) \mapsto \big(b', \varphi_U^{-1}(f(b'), v)\big)$, so f^*E is a vector bundle. The projection $(b', e) \mapsto e$ is a bundle map $f^*E \to E$ covering f and an isomorphism on every fiber. For $g\colon B'' \to B'$ the homeomorphism $\big(b'', (b', e)\big) \mapsto (b'', e)$, with inverse $(b'', e) \mapsto \big(b'', (g(b''), e)\big)$, is an isomorphism $g^*(f^*E) \cong (f \circ g)^*E$.

2. *Whitney sum.* The *Whitney sum* is $E \oplus F = \{(e, v) \in E \times F \mid p(e) = q(v)\}$, with fiber $E_b \oplus F_b$ over b. Over an open set U carrying local trivializations φ_U of E and ψ_U of F, the map $(e, v) \mapsto \big(p(e), \mathrm{pr}_2\varphi_U(e), \mathrm{pr}_2\psi_U(v)\big)$ is a local trivialization of $E \oplus F$, with inverse $(b, x, y) \mapsto \big(\varphi_U^{-1}(b, x), \psi_U^{-1}(b, y)\big)$, so $E \oplus F$ is a vector bundle.

3. *Subbundle.* A *subbundle* of E is a subspace $E_0 \subseteq E$ meeting each fiber E_b in a linear subspace, such that E_0 with the restriction of p is a vector bundle with these fibers.

4. *Fiberwise linear algebra.* For local trivializations φ_U and φ_V of E with target fiber $\mathbb{K}^n$, the composite $\varphi_V \circ \varphi_U^{-1}$ on $(U \cap V) \times \mathbb{K}^n$ is $(b, v) \mapsto \big(b, g_{VU}(b)v\big)$ for a *transition function* $g_{VU}\colon U \cap V \to GL_n(\mathbb{K})$, continuous because its entries are the coordinates of $\mathrm{pr}_2\, \varphi_V\varphi_U^{-1}(b, e_j)$. Let T be one of the constructions $V \mapsto V^\vee = \mathrm{Hom}_{\mathbb{K}}(V, \mathbb{K})$, $(V, W) \mapsto V \otimes_{\mathbb{K}} W$, $(V, W) \mapsto \mathrm{Hom}_{\mathbb{K}}(V, W)$, $V \mapsto \Lambda^k V$, or, for $\mathbb{K} = \mathbb{R}$, $V \mapsto V \otimes_{\mathbb{R}} \mathbb{C}$. On linear isomorphisms g, and h in the two-variable cases, each gives a linear isomorphism $T(g)$ or $T(g, h)$ (for instance $T(g, h) = g \otimes h$ for the tensor product and $x \mapsto hxg^{-1}$ for Hom) whose matrix entries are polynomials in the entries of g, h and their inverses, hence continuous in them by Cramer's rule. Apply T fiberwise, $T(E) = \bigsqcup_b T(E_b)$, and $T(E, F) = \bigsqcup_b T(E_b, F_b)$ in the two-variable cases, using open sets U that carry local trivializations φ_U of E and ψ_U of F, with transition functions g_{VU} and h_{VU}. The charts are $T(\varphi_U)\colon T(E)|_U \to U \times T(\mathbb{K}^n)$, or $T(\varphi_U, \psi_U)$ with values in $U \times T(\mathbb{K}^n, \mathbb{K}^m)$. Give $T(E)$ the topology in which a set W is open exactly when, for each chart, the image of $W \cap T(E)|_U$ is open. The charts are then homeomorphisms, since on overlaps they differ by $(b, x) \mapsto \big(b, T(g_{VU}(b))x\big)$, or by $(b, x) \mapsto \big(b, T(g_{VU}(b), h_{VU}(b))x\big)$, which are continuous with continuous inverses. This defines the *dual* $E^\vee$ (also written E^*), the *tensor product* $E \otimes F$, the bundle $\mathrm{Hom}(E, F)$, the *exterior powers* $\Lambda^k E$, with $\det E = \Lambda^n E$ for E of rank n, and the *complexification* $E_{\mathbb{C}} = E \otimes_{\mathbb{R}} \mathbb{C}$ of a real bundle.

5. *Conjugate and underlying real bundle.* For $\mathbb{K} = \mathbb{C}$, the *conjugate bundle* $\overline{E}$ is E with the scalar multiplication of each fiber replaced by $z \cdot v = \overline{z}\, v$. Its local trivializations are those of E followed by complex conjugation on $\mathbb{C}^n$. The *underlying real bundle* $E_{\mathbb{R}}$ is E with its scalars restricted to $\mathbb{R}$, a real bundle of twice the rank, trivialized by φ_U followed by the identification $\mathbb{C}^n = \mathbb{R}^{2n}$.

Lemma 5.2.10: Fiberwise Isomorphisms of Vector Bundles

Let E' and E be $\mathbb{K}$-vector bundles over B, and let $h\colon E' \to E$ be a homomorphism over B that restricts to a linear isomorphism $E'_b \to E_b$ for every $b \in B$. Then h is an isomorphism. Consequently, if $p'\colon E' \to B'$ is a $\mathbb{K}$-vector bundle and $\tilde{f}\colon E' \to E$ is a bundle map covering $f\colon B' \to B$ that is an isomorphism on every fiber, then $e' \mapsto \big(p'(e'), \tilde{f}(e')\big)$ is an isomorphism $E' \to f^*E$ over B'.

***Proof*:**

The fiberwise inverses assemble to a map $h^{-1}\colon E \to E'$ over B, linear on every fiber, and only its continuity needs proof. Continuity is local over B, so fix $b_0 \in B$ and an open neighbourhood U of b_0 carrying local trivializations φ'_U of E' and φ_U of E. Both have target $U \times \mathbb{K}^n$ for the same n, since h is an isomorphism on each fiber. The composite $\varphi_U \circ h \circ (\varphi'_U)^{-1}$ covers id_U and is linear on fibers, so it is $(b, v) \mapsto \big(b, A(b)v\big)$ for a matrix $A(b) \in GL_n(\mathbb{K})$. The (i, j) entry of $A(b)$ is the i-th coordinate of $\mathrm{pr}_2\, \varphi_U h(\varphi'_U)^{-1}(b, e_j)$, a continuous function of b. By Cramer's rule the entries of $A(b)^{-1}$ are polynomials in the entries of $A(b)$ divided by $\det A(b) \neq 0$, so they are continuous in b as well. Over U the map h^{-1} is the composite of φ_U, the map $(b, w) \mapsto \big(b, A(b)^{-1}w\big)$ and $(\varphi'_U)^{-1}$, each continuous. Hence h^{-1} is continuous.

For the second statement, the map $g(e') = \big(p'(e'), \tilde{f}(e')\big)$ lands in f^*E because $p\,\tilde{f}(e') = f\,p'(e')$, it is continuous, and it covers $\mathrm{id}_{B'}$. On the fiber over b' it is $\tilde{f}$ followed by the identification of $E_{f(b')}$ with the fiber $\{b'\} \times E_{f(b')}$ of f^*E, a linear isomorphism. The first statement applies to g, as we sought to show.

The remaining section build up to the long exact sequence. Recall from section 5.1 the relative homotopy group $\pi_n(E, F, e_0)$, defined for the pair (E, F) with basepoint $e_0 \in F$ as homotopy classes of maps $(D^n, S^{n-1}, s_0) \to (E, F, e_0)$, where the boundary sphere lands in F and the basepoint $s_0 \in S^{n-1}$ lands at e_0. The projection p collapses F to the point b_0, so p carries such a map to a map $(D^n, S^{n-1}, s_0) \to (B, b_0, b_0)$, that is, a map of D^n sending all of ∂D^n to b_0, which is the same as a map $S^n = D^n/\partial D^n \to B$ based at b_0. This defines a homomorphism $p_*\colon \pi_n(E, F, e_0) \to \pi_n(B, b_0)$, and the next proposition says it is an isomorphism:

Proposition 5.2.11: The Lifting Isomorphism

Let $p\colon E \to B$ be a Serre fibration with fiber $F = p^{-1}(b_0)$ and basepoint $e_0 \in F$. For every $n \geq 1$ the map induced by p,

$$p_*\colon \pi_n(E, F, e_0) \xrightarrow{\cong} \pi_n(B, b_0)$$

is an isomorphism (a bijection of pointed sets when $n = 1$, of groups when $n \geq 2$).

***Proof*:**

Throughout, model $\pi_n(B, b_0)$ by maps $(I^n, \partial I^n) \to (B, b_0)$ and $\pi_n(E, F, e_0)$ by maps $(I^n, I^{n-1} \times \{0\}, J^{n-1}) \to (E, F, e_0)$, where I^n is the n-cube, $I^{n-1} \times \{0\}$ is its bottom face, and $J^{n-1} \subseteq \partial I^n$ is the union of all the *other* faces (the top face together with all side faces). In this model a relative class is a map of the cube whose bottom face lands in F and whose remaining boundary faces J^{n-1} go to the basepoint e_0. Applying p sends the bottom face into F to the point

b_0 as well, hence sends all of ∂I^n to b_0, recovering an element of $\pi_n(B, b_0)$ and defining the homomorphism p_* to be analyzed. The pair (I^n, J^{n-1}) is homeomorphic to $(I^n, I^{n-1} \times \{0\})$ as a pair of a cube with a chosen face, because a self-homeomorphism of the cube carries the union of all-but-the-bottom faces onto a single face. In particular J^{n-1} is a retract of I^n, with retraction $r\colon I^n \to J^{n-1}$ used below.

Step 1: Surjectivity. Let $f\colon (I^n, \partial I^n) \to (B, b_0)$ represent a class in $\pi_n(B, b_0)$. View f as a homotopy parametrized by the bottom-face direction: write $I^n = I^{n-1} \times I$ with bottom face $I^{n-1} \times \{0\}$ and set $G = f\colon I^{n-1} \times I \to B$. The initial stage $G(\,\cdot\,, 0)\colon I^{n-1} \to B$ is the constant map at b_0 (the bottom face lies in ∂I^n), which lifts to the constant map $\tilde{g}_0 \equiv e_0\colon I^{n-1} \to E$. Since p is a Serre fibration and I^{n-1} is a cube, the homotopy lifting property (definition 5.2.1) yields a lift $\tilde{f}\colon I^{n-1} \times I \to E$ with $p \circ \tilde{f} = f$ and $\tilde{f}(\,\cdot\,, 0) = e_0$. The bottom face goes to $e_0 \in F$ by construction, and the remaining faces J^{n-1} satisfy $p \circ \tilde{f} = f \equiv b_0$ there, so $\tilde{f}(J^{n-1}) \subseteq p^{-1}(b_0) = F$. Thus $\tilde{f}$ is a map $(I^n, I^{n-1} \times \{0\}, J^{n-1}) \to (E, F, F)$ sending the bottom face to e_0 and J^{n-1} into F. Such a map is homotopic, through maps of the same triple, to one with $J^{n-1} \to e_0$: since J^{n-1} is contractible and contains the corner $J^{n-1} \cap (I^{n-1} \times \{0\})$ that already maps to e_0, the restriction $\tilde{f}|_{J^{n-1}}\colon J^{n-1} \to F$ is nullhomotopic in F to the constant map e_0 (its image lies in the path-component of e_0). Extending this nullhomotopy across the cube via the retraction r deforms $\tilde{f}$, rel the bottom face, to a map sending J^{n-1} to e_0, the whole deformation taking place over ∂I^n and hence inside $F = p^{-1}(b_0)$. The homotoped map represents a class in $\pi_n(E, F, e_0)$, and $p_*[\tilde{f}] = [p \circ \tilde{f}] = [f]$ because the deformation occurred in $F = p^{-1}(b_0)$ and so does not change the projection b_0. Hence p_* is surjective.

Step 2: Injectivity. Suppose $g_0, g_1\colon (I^n, \partial I^n, J^{n-1}) \to (E, F, e_0)$ are relative classes with $p_*[g_0] = p_*[g_1]$ in $\pi_n(B, b_0)$. Then $p \circ g_0$ and $p \circ g_1$ are homotopic rel ∂I^n through maps into B. Let $H\colon I^n \times I \to B$ be such a homotopy, with $H(\,\cdot\,, 0) = p \circ g_0$, $H(\,\cdot\,, 1) = p \circ g_1$, and H constant at b_0 on $\partial I^n \times I$. We build a lift $\tilde{H}\colon I^n \times I \to E$ with prescribed behavior on the part of the boundary where the answer is already forced, then read off a relative homotopy from g_0 to g_1.

Consider the subspace

$$A \;=\; \big(I^n \times \{0\}\big) \cup \big(I^n \times \{1\}\big) \cup \big(J^{n-1} \times I\big) \;\subseteq\; I^n \times I$$

and define $h\colon A \to E$ by $h = g_0$ on $I^n \times \{0\}$, $h = g_1$ on $I^n \times \{1\}$, and $h \equiv e_0$ on $J^{n-1} \times I$. These agree on overlaps (on $J^{n-1} \times \{0\}$ and $J^{n-1} \times \{1\}$ both prescriptions give e_0, since g_0, g_1 send J^{n-1} to e_0), so h is a well-defined continuous map lifting $H|_A$. The pair $(I^n \times I, A)$ is homeomorphic to $(I^n \times I,\ I^n \times \{0\})$, because A is the union of the bottom, top, and the J-faces of the prism $I^n \times I$, and that union is a face-of-a-cube configuration carried to a single face by a self-homeomorphism of I^{n+1}. Under such a homeomorphism the lifting problem "extend h on A to a lift of H" becomes a standard homotopy lifting problem against the cube I^n, solved by the Serre property. The resulting lift $\tilde{H}\colon I^n \times I \to E$ satisfies $p \circ \tilde{H} = H$, restricts to g_0 and g_1 on the two ends, and is e_0 on $J^{n-1} \times I$. Finally, on the bottom face $I^{n-1} \times \{0\} \subseteq \partial I^n$ across all time, $p \circ \tilde{H} = H$ is constant at b_0, so $\tilde{H}$ maps $(\partial I^n) \times I$ into F. Therefore $\tilde{H}$ is a homotopy of relative classes from g_0 to g_1, giving $[g_0] = [g_1]$ in $\pi_n(E, F, e_0)$.

Both directions established, p_* is a bijection, and it is a homomorphism for $n \geq 2$ because p respects the concatenation operation on the first coordinate that defines the group structure (section 5.1), completing the proof.

The proposition says that, as far as homotopy groups can see, projecting to the base is the same as taking the pair (E, F) and forgetting the fiber. The relative group $\pi_n(E, F)$ measures maps of

disks into E whose boundary lies in F. Lifting says every map of a sphere into B is the projection of exactly one such disk, up to homotopy. This is the precise sense in which a fibration "unwinds" the base into the total space relative to the fiber. With this identification in hand, the relative long exact sequence of (E, F) from section 5.1 becomes a sequence in $\pi_*(F)$, $\pi_*(E)$, $\pi_*(B)$:

Theorem 5.2.12: Long Exact Sequence of a Fibration

Let $p\colon E \to B$ be a Serre fibration with fiber $F = p^{-1}(b_0)$ and basepoint $e_0 \in F$. Let $i\colon F \hookrightarrow E$ be the inclusion. Then there is a long exact sequence of homotopy groups

$$\cdots \to \pi_n(F, e_0) \xrightarrow{i_*} \pi_n(E, e_0) \xrightarrow{p_*} \pi_n(B, b_0) \xrightarrow{\partial} \pi_{n-1}(F, e_0) \xrightarrow{i_*} \cdots$$

terminating in $\cdots \to \pi_1(B, b_0) \xrightarrow{\partial} \pi_0(F) \xrightarrow{i_*} \pi_0(E)$, exact as a sequence of groups for $n \geq 2$, of groups and pointed sets at $n = 1$. The map ∂ is the composite of the boundary map $\pi_n(B, b_0) \xrightarrow{(p_*)^{-1}} \pi_n(E, F, e_0) \to \pi_{n-1}(F, e_0)$ from the relative long exact sequence.

***Proof*:**
The relative long exact sequence of the pair (E, F) with basepoint e_0 from section 5.1 reads

$$\cdots \to \pi_n(F, e_0) \xrightarrow{i_*} \pi_n(E, e_0) \xrightarrow{j_*} \pi_n(E, F, e_0) \xrightarrow{\delta} \pi_{n-1}(F, e_0) \xrightarrow{i_*} \cdots$$

and is exact (proposition 5.1.11), where $j_*\colon \pi_n(E) \to \pi_n(E, F)$ is induced by the inclusion of pairs $(E, e_0) \hookrightarrow (E, F)$ and δ is the connecting map sending a relative class $(D^n, S^{n-1}, s_0) \to (E, F, e_0)$ to the restriction of its boundary $(S^{n-1}, s_0) \to (F, e_0)$.

Step 1: Substitute the lifting isomorphism. By proposition 5.2.11, $p_*\colon \pi_n(E, F, e_0) \to \pi_n(B, b_0)$ is an isomorphism for every $n \geq 1$. Replace each occurrence of $\pi_n(E, F, e_0)$ in the relative sequence by $\pi_n(B, b_0)$ via p_*. The two adjacent maps transform as follows. The composite $\pi_n(E, e_0) \xrightarrow{j_*} \pi_n(E, F, e_0) \xrightarrow{p_*} \pi_n(B, b_0)$ equals p_* itself, because for a map $a\colon (I^n, \partial I^n) \to (E, e_0)$ the class $j_*[a]$ is a viewed as a relative map (its boundary is already at $e_0 \in F$), and applying p gives $p \circ a$, which is precisely the image of $[a]$ under the homomorphism $p_*\colon \pi_n(E) \to \pi_n(B)$ induced by p. So the arrow out of $\pi_n(E)$ becomes the genuine $p_*\colon \pi_n(E, e_0) \to \pi_n(B, b_0)$ induced by the map p.

Step 2: The connecting map. Define $\partial\colon \pi_n(B, b_0) \to \pi_{n-1}(F, e_0)$ as the composite $\partial = \delta \circ (p_*)^{-1}$. Because p_* is an isomorphism, replacing the segment $\pi_n(E, F) \xrightarrow{\delta} \pi_{n-1}(F)$ by $\pi_n(B) \xrightarrow{\partial} \pi_{n-1}(F)$ preserves exactness at every node: exactness is a statement about kernels and images, and pre/post-composing a map with an isomorphism alters neither ker nor im in a way that breaks the equalities im = ker at each spot. Concretely, exactness of the original sequence at $\pi_n(E, F, e_0)$ ($\mathrm{im} j_* = \ker \delta$) transports under the isomorphism p_* to exactness of the new sequence at $\pi_n(B, b_0)$ ($\mathrm{im} p_* = \ker \partial$), since $p_*(\mathrm{im} j_*) = \mathrm{im}(p_* j_*) = \mathrm{im} p_*$ and $p_*(\ker \delta) = \ker(\delta (p_*)^{-1}) = \ker \partial$. Likewise exactness at $\pi_n(E, e_0)$ and at $\pi_{n-1}(F, e_0)$ is unchanged, the maps there being i_*, p_*, and the transported ∂.

Step 3: Assemble and read off the low end. Splicing the transported segments for all n yields the asserted sequence

$$\cdots \to \pi_n(F, e_0) \xrightarrow{i_*} \pi_n(E, e_0) \xrightarrow{p_*} \pi_n(B, b_0) \xrightarrow{\partial} \pi_{n-1}(F, e_0) \to \cdots$$

exact at each term by Step 2, of groups for $n \geq 2$ and of pointed sets through π_0 at the bottom, exactly as the relative sequence terminates. The terminal segment $\pi_1(B) \xrightarrow{\partial} \pi_0(F) \xrightarrow{i_*} \pi_0(E)$ records that two points of the fiber lie in the same path component of E if and only if they differ by the boundary action of a loop in B. This is the long exact sequence claimed, completing the proof.

Read in either direction, this sequence converts information of two of the three spaces into knowledge of the third. If the fiber F has vanishing homotopy in a range, then p_* is an isomorphism there, so E and B have the same homotopy groups in that range. If instead the total space E is highly connected, the boundary map ∂ becomes an isomorphism and reads off $\pi_n(B)$ as $\pi_{n-1}(F)$. The three running examples below exercise both readings. The concrete meaning of each map is worth keeping in view: i_* includes a sphere in the fiber into the total space; p_* projects a sphere in the total space down to the base; and ∂ takes a sphere $S^n \to B$, lifts it to a disk $D^n \to E$ as in proposition 5.2.11, and returns the boundary sphere $S^{n-1} \to F$, which is the obstruction to the lift closing up into a sphere.

The simplest application recovers, and extends to all dimensions, a fact the covering-space theory of chapter 2 could only see in dimension one:

Example 5.2.13: Higher Homotopy of the Circle

For the covering $p\colon \mathbb{R} \to S^1$ of example 5.2.3, the fiber is $F = \mathbb{Z}$, a discrete space, so $\pi_n(F) = 0$ for all $n \geq 1$ (a discrete space has no nonconstant maps from a connected S^n, $n \geq 1$, up to homotopy) and the total space $\mathbb{R}$ is contractible, so $\pi_n(\mathbb{R}) = 0$ for all $n \geq 1$. The long exact sequence (theorem 5.2.12) at level $n \geq 2$ reads

$$\underbrace{\pi_n(\mathbb{Z})}_{0} \to \underbrace{\pi_n(\mathbb{R})}_{0} \to \pi_n(S^1) \xrightarrow{\partial} \underbrace{\pi_{n-1}(\mathbb{Z})}_{0}$$

so $\pi_n(S^1)$ is squeezed between two zero groups and is therefore 0. Hence

$$\pi_n(S^1) = 0 \qquad \text{for all } n \geq 2$$

At $n = 1$ the sequence gives $0 = \pi_1(\mathbb{R}) \to \pi_1(S^1) \xrightarrow{\partial} \pi_0(\mathbb{Z}) \to \pi_0(\mathbb{R}) = 0$, and exactness with $\pi_0(\mathbb{Z}) = \mathbb{Z}$ as a set recovers the bijection $\pi_1(S^1) \cong \mathbb{Z}$ established by covering-space theory in chapter 2.

The circle is an Eilenberg-MacLane space $K(\mathbb{Z}, 1)$: it has a single nonzero homotopy group, $\pi_1 = \mathbb{Z}$, and nothing above. The contractibility of its universal cover is exactly what kills every higher group. The same mechanism applies to any space with contractible universal cover, the *aspherical* spaces, whose higher homotopy vanishes identically. Surfaces of genus at least one, and more generally any space whose universal cover is contractible, are aspherical for this reason.

The second example is not a single fibration but a construction that turns *any* pointed space into a fibration, and in doing so reveals that the higher homotopy groups are iterated π_1's of loop spaces.

Definition 5.2.14: Path Space and Loop Space

Fix a pointed space (X, x_0). The *path space* is

$$PX = \{\gamma\colon I \to X : \gamma(0) = x_0\}$$

the space of paths in X starting at x_0, topologized by the compact-open topology. The *loop space* is

$$\Omega X = \{\gamma\colon I \to X : \gamma(0) = \gamma(1) = x_0\}$$

the subspace of loops based at x_0. The endpoint map $p\colon PX \to X$, $p(\gamma) = \gamma(1)$, evaluates a path at its far end, and its fiber over x_0 is exactly ΩX.

Example 5.2.15: The Path-Loop Fibration

For a pointed space (X, x_0) the endpoint map $p\colon PX \to X$, $\gamma \mapsto \gamma(1)$, is a Hurewicz fibration with fiber ΩX, fitting into

$$\Omega X \hookrightarrow PX \xrightarrow{p} X$$

To verify the homotopy lifting property against a space Y, take a homotopy $G\colon Y \times I \to X$ and a lift $\tilde{g}_0\colon Y \to PX$ of $G(\cdot, 0)$. Write $w_y = \tilde{g}_0(y) \in PX$, a path from x_0 to $G(y, 0)$. The idea is to traverse w_y and then follow the trace $t \mapsto G(y, t)$ of the homotopy out to time s, but the two pieces must be reparametrized so that at $s = 0$ the trailing piece has length zero and the lift returns to w_y exactly. Define, for $(y, s) \in Y \times I$, the path $\tilde{G}(y, s) \in PX$ by its value at $u \in [0, 1]$,

$$\tilde{G}(y,s)(u) \;=\; \begin{cases} w_y\!\left(\dfrac{2u}{2-s}\right), & 0 \le u \le 1 - \frac{s}{2} \\ G(y,\, 2u - 2 + s)\,, & 1 - \frac{s}{2} \le u \le 1 \end{cases}$$

The two branches agree at the join $u = 1 - \frac{s}{2}$, where both give $w_y(1) = G(y, 0)$, so $\tilde{G}(y, s)$ is a continuous path. It starts at $w_y(0) = x_0$ and ends at $G(y, s)$, hence lies in PX with $p(\tilde{G}(y, s)) = G(y, s)$. At $s = 0$ the first branch covers all of $[0, 1]$ and reduces to $w_y(u)$, while the second branch has degenerate domain, so $\tilde{G}(y, 0) = w_y = \tilde{g}_0(y)$ exactly. Joint continuity of $(y, s, u) \mapsto \tilde{G}(y, s)(u)$ gives continuity of $\tilde{G}\colon Y \times I \to PX$ in the compact-open topology. So the square fills for every Y, and p is a Hurewicz fibration. The total space PX is contractible: the homotopy $\gamma_r(t) = \gamma(rt)$ for $r \in I$ slides every path back along itself to the constant path at x_0, giving a deformation retraction of PX onto $\{x_0\}$.

Feeding this into theorem 5.2.12 with $E = PX$ contractible kills the middle terms: for every $n \ge 1$,

$$\underbrace{\pi_n(PX)}_{0} \to \pi_n(X) \xrightarrow{\partial} \pi_{n-1}(\Omega X) \to \underbrace{\pi_{n-1}(PX)}_{0}$$

is exact, so ∂ is an isomorphism for $n \ge 2$, and for $n = 1$ exactness of pointed sets makes ∂ onto. For a loop f at x_0 the map $g\colon I \to PX$, $g(t)(u) = f\big(1 - u(1-t)\big)$, is continuous by the exponential law [16, The Exponential Law], satisfies $p \circ g = f$ and $g(1) = c_{x_0}$, and starts at the reversed loop $g(0) = \bar{f}$, where $\bar{f}(u) = f(1 - u)$. So g is a relative cube for $(PX, \Omega X, c_{x_0})$ with $p_*[g] = [f]$, and $\partial[f]$, which is $(p_*)^{-1}$ followed by restriction to the bottom face, is the path component of $\bar{f}$. If $\partial[f] = \partial[f']$, a path $r \mapsto \lambda_r$ in ΩX joins $\bar{f}$ to $\bar{f}'$, and $(u, r) \mapsto \lambda_r(1 - u)$, continuous by the exponential law, is a homotopy rel endpoints from f to f'. So $[f] = [f']$, and ∂ is a bijection at

$n = 1$ as well, yielding the dimension shift

$$\pi_n(X, x_0) \;\cong\; \pi_{n-1}(\Omega X, c_{x_0}) \qquad (n \geq 1)$$

Iterating, $\pi_n(X) \cong \pi_{n-1}(\Omega X) \cong \pi_{n-2}(\Omega^2 X) \cong \cdots \cong \pi_0(\Omega^n X)$.

The path-loop fibration is the source of the slogan that "homotopy groups are abelian above degree one because they are double loop spaces". The isomorphism $\pi_n(X) \cong \pi_{n-1}(\Omega X)$ trades a sphere of dimension n in X for a sphere of dimension $n-1$ in the loop space, and after two iterations one reaches $\pi_{n-2}(\Omega^2 X)$. The two independent loop coordinates of $\Omega^2 X$ commute, which is the conceptual reason behind the commutativity proved combinatorially in theorem 5.1.4. Concretely, $\pi_2(X) \cong \pi_1(\Omega X)$ identifies the second homotopy group of X with the ordinary fundamental group of its loop space.

The path-loop fibration turns the inclusion of a point into a fibration, and the same construction applies to any map $f\colon X \to Y$. Replace X by the space of points of X together with a path in Y leaving their image. Nothing about X changes up to homotopy, and the endpoint map becomes a fibration.

Proposition 5.2.16: The Mapping Path Space

Let $f\colon X \to Y$ be a continuous map, give Y^I the compact-open topology, and set

$$E_f = \{(x, \gamma) \in X \times Y^I : \gamma(0) = f(x)\}, \qquad p_f(x, \gamma) = \gamma(1)$$

Then:

1. the map $i\colon X \to E_f$, $x \mapsto (x, c_{f(x)})$, where c_y denotes the constant path at y, is a homotopy equivalence with homotopy inverse $r(x, \gamma) = x$, and $p_f \circ i = f$;
2. $p_f\colon E_f \to Y$ is a Hurewicz fibration.

For $y_0 \in Y$, the fiber $F_f = p_f^{-1}(y_0)$, consisting of the pairs (x, γ) with γ a path from $f(x)$ to y_0, is the *homotopy fiber* of f over y_0.

***Proof*:**

Item (1). The composite $r \circ i$ is the identity of X, and $p_f(i(x)) = c_{f(x)}(1) = f(x)$. For the other composite, shrink each path back to its starting point: define $H\colon E_f \times I \to E_f$ by $H((x, \gamma), t) = (x, \gamma_t)$ with $\gamma_t(u) = \gamma(tu)$. Each γ_t starts at $\gamma(0) = f(x)$, so H lands in E_f. The path γ_0 is constant at $f(x)$, so $H(\,\cdot\,, 0) = i \circ r$, and $H(\,\cdot\,, 1)$ is the identity. Evaluation $Y^I \times I \to Y$ is continuous because I is compact Hausdorff, so $(\gamma, t, u) \mapsto \gamma(tu)$ is continuous, and therefore so is its adjoint $(\gamma, t) \mapsto \gamma_t$. Hence H is a homotopy from $i \circ r$ to id_{E_f}.

Item (2). Let $G\colon Z \times I \to Y$ be a homotopy and $\tilde{g}_0\colon Z \to E_f$ a lift of $G(\,\cdot\,, 0)$, written $\tilde{g}_0(z) = (x_z, \gamma_z)$ with $\gamma_z(1) = G(z, 0)$. Keep the point x_z and lengthen the path γ_z by the trace of the homotopy, reparametrized as for the path-loop fibration (example 5.2.15):

$$\tilde{G}(z, s) = (x_z, \delta_{z,s}), \qquad \delta_{z,s}(u) = \begin{cases} \gamma_z\left(\dfrac{2u}{2 - s}\right), & 0 \leq u \leq 1 - \frac{s}{2}, \\ G(z,\, 2u - 2 + s), & 1 - \frac{s}{2} \leq u \leq 1. \end{cases}$$

> The two branches agree at $u = 1 - \frac{s}{2}$, where both equal $\gamma_z(1) = G(z, 0)$. The path $\delta_{z,s}$ starts at $\gamma_z(0) = f(x_z)$, so $\tilde{G}(z, s) \in E_f$, and it ends at $G(z, s)$, so $p_f \circ \tilde{G} = G$. At $s = 0$ the second branch has degenerate domain and $\delta_{z,0} = \gamma_z$, so $\tilde{G}(\,\cdot\,, 0) = \tilde{g}_0$. The map $z \mapsto x_z$ is continuous as a component of $\tilde{g}_0$. Each branch of $(z, s, u) \mapsto \delta_{z,s}(u)$ is continuous on its closed piece of $Z \times I \times I$, the first because $z \mapsto \gamma_z$ and evaluation are continuous, and the two branches agree where the pieces meet, so $\tilde{G}$ is continuous. Hence p_f has the homotopy lifting property with respect to every space Z.

Every map is therefore a fibration up to homotopy: f factors as the homotopy equivalence i followed by the fibration p_f. Applying theorem 5.2.12 to $F_f \to E_f \to Y$, with $\pi_n(E_f) \cong \pi_n(X)$ through i, gives a long exact sequence of homotopy groups for an arbitrary map into a path-connected space Y, and the homotopy fiber measures how far f is from a weak homotopy equivalence. When f is the inclusion of the basepoint $\{y_0\} \hookrightarrow Y$, the space E_f is the space of paths leaving y_0 and $F_f = \Omega Y$, which recovers the path-loop fibration.

Read along its last coordinate, a relative cube $f\colon (I^n, \partial I^n, J^{n-1}) \to (X, A, x_0)$ of definition 5.1.8 is a family of paths in X, parametrized by I^{n-1}, each running from a point of A to x_0. Such a path, paired with its starting point, is a point of the homotopy fiber of the inclusion $A \hookrightarrow X$.

Proposition 5.2.17: Relative Homotopy Groups as Homotopy Groups of the Homotopy Fiber

Let $A \subseteq X$ be a subspace containing the basepoint x_0, let $i\colon A \hookrightarrow X$ be the inclusion, and let

$$F = F_i = \{(a,\gamma) \in A \times X^I : \gamma(0) = a,\ \gamma(1) = x_0\}$$

be the homotopy fiber of i over x_0 (proposition 5.2.16), based at (x_0, c_{x_0}), with projection $q\colon F \to A$, $q(a,\gamma) = a$. Let $n \geq 1$ and write points of $I^n = I^{n-1} \times I$ as (s,u) with $u = t_n$, so that the bottom face of definition 5.1.8 is $I^{n-1} \times \{0\}$. For a map $f\colon (I^n, \partial I^n, J^{n-1}) \to (X, A, x_0)$ set

$$\widehat{f}\colon I^{n-1} \to F, \qquad \widehat{f}(s) = \big(f(s,0),\ u \mapsto f(s,u)\big)$$

Then:

1. The map $\widehat{f}$ is continuous and sends ∂I^{n-1} to (x_0, c_{x_0}), and $\Phi[f] = [\widehat{f}]$ defines a bijection

$$\Phi\colon \pi_n(X, A, x_0) \longrightarrow \pi_{n-1}\big(F, (x_0, c_{x_0})\big)$$

   which is an isomorphism of groups for $n \geq 2$ and a bijection of pointed sets for $n = 1$.

2. The bijection Φ is natural. A map of pairs $\varphi\colon (X, A, x_0) \to (Y, B, y_0)$ induces a continuous map $\varphi_F\colon F \to F'$, $\varphi_F(a,\gamma) = (\varphi(a), \varphi \circ \gamma)$, into the homotopy fiber F' of $B \hookrightarrow Y$ over y_0, and $\Phi \circ \varphi_* = (\varphi_F)_* \circ \Phi$, where $\varphi_*[f] = [\varphi \circ f]$.

3. The boundary map of definition 5.1.10 is $\partial = q_* \circ \Phi\colon \pi_n(X, A, x_0) \to \pi_{n-1}(A, x_0)$.

4. For the pair $(X, \{x_0\}, x_0)$ the homotopy fiber is $\{x_0\} \times \Omega X$, identified with ΩX through $\gamma \mapsto (x_0, \gamma)$, and (1) for this pair, where $\pi_n(X, \{x_0\}, x_0) = \pi_n(X, x_0)$, is a bijection $\Phi_0\colon \pi_n(X, x_0) \to \pi_{n-1}(\Omega X, c_{x_0})$ sending $[f]$ to the class of $s \mapsto f(s, \cdot\,)$. The map j_* of proposition 5.1.11 satisfies $\Phi \circ j_* = \iota_* \circ \Phi_0$, where $\iota\colon \Omega X \to F$ is the inclusion $\iota(\gamma) = (x_0, \gamma)$.

***Proof*:**

By definition 5.1.8, $\partial I^n = (I^{n-1} \times \{0\}) \cup J^{n-1}$, where $J^{n-1} = (I^{n-1} \times \{1\}) \cup (\partial I^{n-1} \times I)$ is the top face together with the side faces. So a continuous $f\colon I^n \to X$ is a relative cube exactly when $f(s,0) \in A$ and $f(s,1) = x_0$ for all $s \in I^{n-1}$, and $f(s,u) = x_0$ for all $s \in \partial I^{n-1}$ and $u \in I$. For $n = 1$ the cube I^0 is a point, ∂I^0 is empty, and $\pi_0(F, (x_0, c_{x_0}))$ is the set of path components of F, pointed by the component of (x_0, c_{x_0}). The space F carries the subspace topology from $A \times X^I$, so a map into F is continuous exactly when its components into A and into X^I are continuous.

Step 1: Maps into F are families of paths. Let Z be any space. The assignment $G \mapsto \widehat{G}$, where $\widehat{G}(z) = (G(z,0),\ u \mapsto G(z,u))$, is a bijection from the set of continuous maps $G\colon Z \times I \to X$ with $G(z,0) \in A$ and $G(z,1) = x_0$ for every $z \in Z$ onto the set of continuous maps $Z \to F$, and $\widehat{G}(z) = (x_0, c_{x_0})$ exactly when $G(z,u) = x_0$ for every $u \in I$. Since I is locally compact Hausdorff, the exponential law [16, The Exponential Law] says that G is continuous if and only if $z \mapsto G(z, \cdot\,)$ is well defined and continuous into X^I. The first component $z \mapsto G(z,0)$ is G composed with $z \mapsto (z,0)$, so it is continuous into X, and it is continuous into the subspace A because it takes values there. The path $G(z, \cdot\,)$ runs from $G(z,0)$ to x_0, so $\widehat{G}(z) \in F$, and $\widehat{G}$ is

continuous. Conversely, write a continuous $g\colon Z \to F$ as $g(z) = (a_z, \gamma_z)$. Its second component $z \mapsto \gamma_z$ is continuous, so the exponential law makes $G(z,u) = \gamma_z(u)$ continuous. This G satisfies $G(z,0) = \gamma_z(0) = a_z \in A$ and $G(z,1) = x_0$, and $\widehat{G} = g$ because $\gamma_z(0) = a_z$. The two assignments are mutually inverse, since a point (a,γ) of F has $a = \gamma(0)$. Finally $\widehat{G}(z) = (x_0, c_{x_0})$ says that $G(z,\cdot\,)$ is the constant path at x_0.

Step 2: Relative cubes and based cubes in F. Apply Step 1 with $Z = I^{n-1}$ to a continuous $f\colon I^n \to X$. The conditions $f(s,0) \in A$ and $f(s,1) = x_0$ are the hypotheses of Step 1, and the condition on the side faces says $\widehat{f}(\partial I^{n-1}) = (x_0, c_{x_0})$. So $f \mapsto \widehat{f}$ is a bijection from the relative cubes $(I^n, \partial I^n, J^{n-1}) \to (X, A, x_0)$ onto the maps $(I^{n-1}, \partial I^{n-1}) \to (F, (x_0, c_{x_0}))$, which proves the first assertion of (1).

Step 3: Homotopies. Let $H\colon I^n \times I \to X$ be a homotopy through relative cubes, with $H_r = H(\,\cdot\,,\,\cdot\,,r)$. Apply Step 1 with $Z = I^{n-1} \times I$ to $G((s,r),u) = H(s,u,r)$, which is H composed with a permutation of coordinates and so is continuous. Each H_r satisfies the conditions of Step 2, so $\widehat{G}\colon I^{n-1} \times I \to F$ is a homotopy from $\widehat{H_0}$ to $\widehat{H_1}$ sending $\partial I^{n-1} \times I$ to (x_0, c_{x_0}). Conversely, Step 1 writes a homotopy $K\colon I^{n-1} \times I \to F$ rel ∂I^{n-1} as $\widehat{G}$ for a unique G, and $H(s,u,r) = G((s,r),u)$ is then a homotopy through relative cubes with $\widehat{H_r} = K(\,\cdot\,,r)$. By the bijection of Step 2, $f \simeq f'$ through relative cubes if and only if $\widehat{f} \simeq \widehat{f}'$ rel ∂I^{n-1}. Hence Φ is well defined and injective, and Step 2 makes it surjective. The constant cube at x_0 goes to the constant map at (x_0, c_{x_0}), so Φ preserves basepoints. For $n = 1$ a homotopy of maps $I^0 \to F$ is a path in F, so the same argument gives a bijection of pointed sets onto the path components.

Step 4: Sums. Let $n \geq 2$ and $s = (t_1, \dots, t_{n-1})$, so the first coordinate t_1 is a coordinate of I^{n-1}, distinct from $u = t_n$. By definition 5.1.8 the sum of relative cubes f and g is $(f+g)(s,u) = f(2t_1, t_2, \dots, t_{n-1}, u)$ for $t_1 \leq \frac{1}{2}$ and $g(2t_1 - 1, t_2, \dots, t_{n-1}, u)$ for $t_1 \geq \frac{1}{2}$. For fixed s with $t_1 \leq \frac{1}{2}$, evaluating the definition of $\widehat{f+g}$ gives $\widehat{f+g}(s) = \widehat{f}(2t_1, t_2, \dots, t_{n-1})$, and likewise $\widehat{f+g}(s) = \widehat{g}(2t_1 - 1, t_2, \dots, t_{n-1})$ for $t_1 \geq \frac{1}{2}$. This is the sum $\widehat{f} + \widehat{g}$ of definition 5.1.1 in $\pi_{n-1}(F)$, so $\widehat{f+g} = \widehat{f} + \widehat{g}$ as maps and Φ is a homomorphism. A bijective homomorphism is an isomorphism, which completes (1).

Step 5: Naturality. Step 1 used only that the basepoint lies in the subspace, so it applies to the pair (Y, B, y_0). Take $Z = F$ and $G((a,\gamma),u) = \varphi(\gamma(u))$. This G is continuous because evaluation $X^I \times I \to X$ is continuous for the locally compact Hausdorff space I [16, Continuity of Evaluation], and it satisfies $G((a,\gamma),0) = \varphi(a) \in B$ and $G((a,\gamma),1) = \varphi(x_0) = y_0$. Its associated map is $\widehat{G} = \varphi_F$, which is therefore continuous, and $\varphi_F(x_0, c_{x_0}) = (y_0, c_{y_0})$. For a relative cube f and $s \in I^{n-1}$,

$$\widehat{\varphi \circ f}(s) = \big(\varphi(f(s,0)),\ u \mapsto \varphi(f(s,u))\big) = \varphi_F\big(\widehat{f}(s)\big)$$

so $\Phi(\varphi_*[f]) = [\varphi_F \circ \widehat{f}] = (\varphi_F)_*\Phi[f]$, proving (2).

Step 6: The boundary map. The projection q is continuous as the restriction of the projection $A \times X^I \to A$, and $q(x_0, c_{x_0}) = x_0$. For a relative cube f, $(q \circ \widehat{f})(s) = f(s,0)$, so $q \circ \widehat{f}$ is the restriction of f to the bottom face, viewed as a map $(I^{n-1}, \partial I^{n-1}) \to (A, x_0)$. Its class is $\partial[f]$ by definition 5.1.10, so $q_*\Phi[f] = \partial[f]$, proving (3).

Step 7: The map j_*. For $A = \{x_0\}$ the conditions $\gamma(0) = x_0$ and $\gamma(1) = x_0$ say that the homotopy fiber is $\{x_0\} \times \Omega X$, and $\gamma \mapsto (x_0, \gamma)$ is a homeomorphism from ΩX onto it, with inverse the restriction of the projection to X^I. A relative cube for $(X, \{x_0\}, x_0)$ is a map sending $\partial I^n = (I^{n-1} \times \{0\}) \cup J^{n-1}$ to x_0, the homotopies through such cubes are those of definition 5.1.1, and the two sums agree, so $\pi_n(X, \{x_0\}, x_0) = \pi_n(X, x_0)$. Under the identification,

$\widehat{f}(s) = (x_0, f(s,\cdot))$ becomes $f(s,\cdot)$, which gives the formula for Φ_0. The identity of X is a map of pairs $(X,\{x_0\},x_0) \to (X,A,x_0)$, it induces j_* by the definition of j in proposition 5.1.11, and its map of homotopy fibers is $(x_0,\gamma) \mapsto (x_0,\gamma)$, which is ι under the identification. Item (2) for this map of pairs gives $\Phi \circ j_* = \iota_* \circ \Phi_0$, proving (4) and completing the proof.

By (3) and (4), the long exact sequence of the pair (proposition 5.1.11) becomes, after replacing $\pi_n(X,x_0)$ by $\pi_{n-1}(\Omega X, c_{x_0})$ through Φ_0 and $\pi_n(X,A,x_0)$ by $\pi_{n-1}(F)$ through Φ, the sequence

$$\cdots \to \pi_n(A,x_0) \xrightarrow{\Phi_0 i_*} \pi_{n-1}(\Omega X, c_{x_0}) \xrightarrow{\iota_*} \pi_{n-1}(F) \xrightarrow{q_*} \pi_{n-1}(A,x_0) \to \cdots$$

which is exact because Φ and Φ_0 are bijections preserving basepoints, so images and kernels correspond. The right-hand side $\pi_{n-1}(F_h)$ of (1) makes sense for an arbitrary based map $h\colon (A,a_0) \to (X,x_0)$, and the proposition identifies it with the relative group when h is an inclusion. Hatcher states the bijection of (1) in this form [21, §4.3, after Proposition 4.64, p. 407].

Example 5.2.18: The Pairs $(X,\{x_0\})$ and (X,X)

For $A = \{x_0\}$, item (4) of proposition 5.2.17 gives an isomorphism $\pi_n(X,x_0) \cong \pi_{n-1}(\Omega X, c_{x_0})$ by an explicit formula, the class of a cube going to the class of its adjoint, with no lifting. The boundary isomorphism of example 5.2.15 instead reverses the loops. For $[f] \in \pi_n(X,x_0)$ let

$$g\colon I^n \to PX, \qquad g(s,t_n)(u) = f\big(s,\, 1 - u(1-t_n)\big)$$

continuous by the exponential law [16, The Exponential Law]. Each path $g(s,t_n)$ starts at $f(s,1) = x_0$ and ends at $f(s,t_n)$, so g lands in PX and $p \circ g = f$. On the bottom face $g(s,0)$ is the path $u \mapsto f(s,1-u)$, a loop because $f(s,0) = x_0$. On J^{n-1} the map g is the constant path c_{x_0}, because f is constant at x_0 on the top face and on the side faces. So g is a relative cube for $(PX,\Omega X, c_{x_0})$ with $p_*[g] = [f]$, and the boundary map of theorem 5.2.12, which is $(p_*)^{-1}$ followed by restriction to the bottom face, sends $[f]$ to the class of $s \mapsto \overline{f(s,\cdot)}$, where $\bar\gamma(u) = \gamma(1-u)$. Thus the boundary isomorphism of example 5.2.15 is $\nu_* \circ \Phi_0$, where the reversal $\nu\colon \Omega X \to \Omega X$, $\nu(\gamma) = \bar\gamma$, is continuous by the exponential law applied to the map $(\gamma,u) \mapsto \gamma(1-u)$, which is continuous by continuity of evaluation [16, Continuity of Evaluation]. For $n \geq 2$ the map ν_* is negation on $\pi_{n-1}(\Omega X)$. The class $\nu_*\Phi_0[f]$ is $\Phi_0[f^-]$ for the reflected cube $f^-(s,t_n) = f(s,1-t_n)$. The proof of proposition 5.1.3, run in the last coordinate, shows that $[f^-]$ is the inverse of $[f]$ for concatenation in that coordinate, and by the Eckmann-Hilton argument of theorem 5.1.4, run in the first and last coordinates, that concatenation is the sum of definition 5.1.1. So $\nu_*\Phi_0[f] = \Phi_0(-[f]) = -\Phi_0[f]$, since Φ_0 is an isomorphism of groups: it is item (1) of proposition 5.2.17 for the pair $(X,\{x_0\})$, by item (4), and Step 7 of that proof identifies the sum on $\pi_n(X,\{x_0\},x_0)$ with the sum on $\pi_n(X,x_0)$.

For $A = X$ the homotopy fiber F is the space of paths ending at x_0, each paired with its starting point. The formula

$$K\big((a,\gamma),r\big) = \big(\gamma(r),\ u \mapsto \gamma(r + (1-r)u)\big)$$

shrinks each path toward its end. It defines a continuous map $K\colon F \times I \to F$ by Step 1 of the proof of proposition 5.2.17 with $Z = F \times I$, since $((a,\gamma),r,u) \mapsto \gamma(r+(1-r)u)$ is continuous by continuity of evaluation [16, Continuity of Evaluation], takes the value $\gamma(r) \in X$ at $u = 0$, and takes the value x_0 at $u = 1$. At $r = 0$ it is the identity, since $\gamma(0) = a$, at $r = 1$ it is constant at (x_0, c_{x_0}), and it fixes (x_0,c_{x_0}) for every r. Composing a map $(I^{n-1},\partial I^{n-1}) \to (F,(x_0,c_{x_0}))$ with K gives a null-homotopy rel ∂I^{n-1}, and each point $e \in F$ is joined to the basepoint by the path $r \mapsto K(e,r)$. Hence $\pi_{n-1}(F)$ is trivial for every $n \geq 1$, and Φ recovers $\pi_n(X,X,x_0) = 0$ from example 5.1.9.

The third running example is the first nonzero higher homotopy group of a sphere, where we shall assume $\pi_k(S^m) = 0$ for $k < m$ (proven in section 5.4):

Example 5.2.19: The Homotopy of S^2 via the Hopf Fibration

Apply theorem 5.2.12 to the Hopf fibration $S^1 \hookrightarrow S^3 \xrightarrow{p} S^2$ of example 5.2.7. The relevant inputs are $\pi_n(S^1) = 0$ for $n \geq 2$ (example 5.2.13) and the connectivity of spheres, $\pi_k(S^m) = 0$ for $k < m$ (anticipating corollary 5.4.11 in section 5.4), in particular $\pi_1(S^3) = \pi_2(S^3) = 0$ since S^3 is 2-connected. The sequence around degrees 2 and 3 reads, with $F = S^1$ and $E = S^3$,

$$\pi_3(S^1) \to \pi_3(S^3) \xrightarrow{p_*} \pi_3(S^2) \xrightarrow{\partial} \pi_2(S^1) \to \pi_2(S^3) \xrightarrow{p_*} \pi_2(S^2) \xrightarrow{\partial} \pi_1(S^1) \to \pi_1(S^3)$$

Substituting the known zeros $\pi_3(S^1) = 0$, $\pi_2(S^1) = 0$, $\pi_2(S^3) = 0$, and $\pi_1(S^3) = 0$, this becomes

$$0 \to \pi_3(S^3) \xrightarrow{p_*} \pi_3(S^2) \to 0 \qquad \text{and} \qquad 0 \to \pi_2(S^2) \xrightarrow{\partial} \pi_1(S^1) \to 0$$

Exactness of the first short sequence (a map flanked by zeros is an isomorphism) gives $p_* \colon \pi_3(S^3) \xrightarrow{\cong} \pi_3(S^2)$, and $\pi_3(S^3) \cong \mathbb{Z}$ by the Hurewicz theorem (S^3 is 2-connected, so $\pi_3(S^3) \cong H_3(S^3) \cong \mathbb{Z}$, anticipating section 5.4, with generator $[\mathrm{id}_{S^3}]$). Therefore

$$\pi_3(S^2) \;\cong\; \mathbb{Z}$$

generated by the class of the Hopf map $[p]$, since $p_*[\mathrm{id}_{S^3}] = [p \circ \mathrm{id}] = [p]$. Exactness of the second short sequence gives $\partial \colon \pi_2(S^2) \xrightarrow{\cong} \pi_1(S^1) \cong \mathbb{Z}$, so

$$\pi_2(S^2) \;\cong\; \mathbb{Z}$$

generated by $[\mathrm{id}_{S^2}]$, recovering by a fibration argument the degree isomorphism $\pi_2(S^2) \cong \mathbb{Z}$.

The computation $\pi_3(S^2) \cong \mathbb{Z}$ is the first example that homotopy groups *do not necessarily* vanish once the dimension exceeds that of the space. A 3-sphere can wrap *nontrivially* around a 2-sphere, and the Hopf map is the generating wrapping. Its $\mathbb{Z}$ worth of homotopy classes are detected by the linking number of the preimages of two distinct points (each a circle in S^3), the *Hopf invariant* (if it still exists, see this animation on youtube of Hopf Fibrations). That two great circles in S^3 corresponding to distinct points of S^2 are linked exactly once is the geometric content of the generator $[p]$. This single example already shows that the homotopy groups of spheres, the $\pi_k(S^m)$, are more intricate than their homology, and computing them is a central thread of the chapters that follow.

Exercise 5.2.1

1. Let $p \colon E \to B$ and $p' \colon E' \to B'$ be Serre fibrations. Show that the product map $p \times p' \colon E \times E' \to B \times B'$, $(e, e') \mapsto (p(e), p'(e'))$, is a Serre fibration, and identify its fiber over (b_0, b_0') in terms of the fibers of p and p'.

2. Let $p \colon E \to B$ be a Serre fibration, $f \colon B'' \to B'$ and $g \colon B' \to B$ continuous. Using proposition 5.2.6, show that the two ways of pulling p back to B'' agree: $f^*(g^*E) \cong (g \circ f)^*E$ as spaces over B'', compatibly with the projections. Deduce that a composite of pullbacks of a Serre fibration is again one.

3. Suppose the Serre fibration $p \colon E \to B$ admits a *section*, a map $s \colon B \to E$ with $p \circ s = \mathrm{id}_B$

and $s(b_0) = e_0$. Using the long exact sequence (theorem 5.2.12), prove that for every $n \geq 1$ the sequence splits, giving $\pi_n(E, e_0) \cong \pi_n(F, e_0) \oplus \pi_n(B, b_0)$ for $n \geq 2$.

4. Use the Hopf fibration and example 5.2.19 to show that S^3 is *not* homotopy equivalent to $S^2 \times S^1$, even though both are closed 3-manifolds, by comparing π_2 of each.

5. Let $X = S^m$ with $m \geq 2$. Using the path-loop fibration (example 5.2.15), express $\pi_k(\Omega S^m)$ in terms of homotopy groups of spheres, and deduce the value of $\pi_{m-1}(\Omega S^m)$.

6. Let $p\colon \tilde{X} \to X$ be a covering map with $\tilde{X}$ path-connected. Using that a covering map is a Serre fibration with discrete fiber, derive from theorem 5.2.12 that $p_*\colon \pi_n(\tilde{X}) \to \pi_n(X)$ is an isomorphism for all $n \geq 2$.

5.3 Whitehead's Theorem and CW Approximation

A natural question to ask is how much of a space do homotopy groups detect. A continuous map $f\colon X \to Y$ induces homomorphisms $f_*\colon \pi_n(X, x_0) \to \pi_n(Y, f(x_0))$ for every n and every basepoint. If all of these are isomorphisms, the two spaces are indistinguishable by homotopy groups. The central result of this section, due to Whitehead, is that for the class of CW complexes this indistinguishability is a homotopy equivalence. The CW hypothesis cannot be dropped due to the Warsaw circle. The section closes with the converse construction, CW approximation, which replaces an arbitrary space by a CW complex with the same homotopy groups, so that the homotopy-theoretic study of *any* space reduces to the study of CW complexes.

Definition 5.3.1: Weak Homotopy Equivalence

A map $f\colon X \to Y$ is a *weak homotopy equivalence* if the induced map

$$f_*\colon \pi_n(X, x_0) \longrightarrow \pi_n\big(Y, f(x_0)\big)$$

is a bijection for every $n \geq 0$ and every choice of basepoint $x_0 \in X$. (For $n = 0$ the condition is that f induces a bijection on path components. For $n \geq 1$ it is that f_* is a group isomorphism.)

The quantifier over all basepoints is not redundant. Homotopy groups based at points in different path components carry independent information, and a map can be an isomorphism on the homotopy groups of one component while failing on another. The condition on π_0 forces f to match up the components of X and Y, and within each matched pair the higher conditions then compare the groups based at a point of that component. A homotopy equivalence is always a weak homotopy equivalence (lemma 5.3.2(1) below): a homotopy inverse inverts f_* up to the change of basepoint along the paths that the homotopies trace. The content of Whitehead's theorem is the converse for CW complexes. The content of the Warsaw circle example is that the converse fails in general.

A homotopy moves each point along a path, and along a path the collar construction of proposition 5.1.12 carries a map of a sphere based at one end to a map based at the other. The homotopy groups at the two ends of a path are therefore identified, compatibly with every map, so a weak homotopy equivalence stays one under homotopy and can be tested at one basepoint of a path-connected space.

Lemma 5.3.2: Weak Equivalences, Homotopies and Basepoints

Let $f\colon Y \to Z$ be a map.

1. If f is homotopic to a weak homotopy equivalence, then f is a weak homotopy equivalence. In particular a homotopy equivalence is a weak homotopy equivalence.
2. If Y and Z are path-connected and $f_*\colon \pi_k(Y, y_0) \to \pi_k(Z, f(y_0))$ is bijective for every $k \geq 1$ at one point $y_0 \in Y$, then f is a weak homotopy equivalence.

***Proof*:**
For a path δ in a space W from w to w' and $k \geq 1$, the collar construction of proposition 5.1.12, run along δ in place of a loop, turns a map $(S^k, s_0) \to (W, w')$ into one based at w. It gives a bijection $\delta_{\#}\colon \pi_k(W, w') \to \pi_k(W, w)$, inverse to the construction along the reversed path, and a map $h\colon W \to W'$ carries the collar along δ to the collar along $h \circ \delta$, so $h_* \circ \delta_{\#} = (h \circ \delta)_{\#} \circ h_*$.

Part (1). Let $F\colon Y \times I \to Z$ be a homotopy from a weak homotopy equivalence g to f, let $y \in Y$, and let $\delta(t) = F(y, t)$. For a based map $\alpha\colon (S^k, s_0) \to (Y, y)$, the homotopy $F \circ (\alpha \times \mathrm{id}_I)$ from $g \circ \alpha$ to $f \circ \alpha$ moves the basepoint along δ, so the collar argument in the proof of proposition 5.1.15 gives $[g \circ \alpha] = \delta_{\#}[f \circ \alpha]$. Hence $g_* = \delta_{\#} \circ f_*$ on $\pi_k(Y, y)$, and $f_* = \delta_{\#}^{-1} \circ g_*$ is bijective. The path δ joins $g(y)$ to $f(y)$, so f and g agree on path components. Now let g be a homotopy equivalence with homotopy inverse h. The composites $h \circ g$ and $g \circ h$ are homotopic to identity maps, hence weak homotopy equivalences by the first assertion. For $y \in Y$, bijectivity of $(h \circ g)_* = h_* \circ g_*$ on $\pi_k(Y, y)$ makes h_* onto at $g(y)$, and bijectivity of $(g \circ h)_* = g_* \circ h_*$ on $\pi_k(Z, g(y))$ makes h_* injective there. So h_* is bijective at $g(y)$, and $g_* = h_*^{-1} \circ (h \circ g)_*$ is bijective at y. On path components g and h induce inverse bijections.

Part (2). For $y \in Y$ choose a path δ from y to y_0. Then $f_* \circ \delta_{\#} = (f \circ \delta)_{\#} \circ f_*$, with f_* on the right taken at y_0, so f_* is bijective on $\pi_k(Y, y)$. Both spaces have a single path component, completing the proof.

Example 5.3.3: The Inclusion of a Deformation Retract
Let $A \subseteq X$ be a deformation retract, so there is a retraction $r\colon X \to A$ and a homotopy $H\colon X \times I \to X$ from id_X to $\iota \circ r$ fixing A pointwise, where $\iota\colon A \hookrightarrow X$ is the inclusion. Then ι is a weak homotopy equivalence, and in fact a homotopy equivalence. Concretely, take $A = S^1 \subseteq \mathbb{R}^2 \setminus \{0\} = X$, the punctured plane, with $r(v) = v/|v|$ and $H(v, t) = (1 - t)v + t\, v/|v|$, a straight-line homotopy that never crosses the origin. On homotopy groups, $r_* \circ \iota_* = (\mathrm{id}_A)_* = \mathrm{id}$ and $\iota_* \circ r_* = (\iota r)_* = (\mathrm{id}_X)_*$ because $\iota r \simeq \mathrm{id}_X$, so ι_* is an isomorphism on every π_n at every basepoint of S^1. For instance $\pi_1(S^1) \cong \mathbb{Z}$ maps isomorphically onto $\pi_1(\mathbb{R}^2 \setminus \{0\}) \cong \mathbb{Z}$, the winding number being preserved because the deformation retraction does not change how a loop wraps around the puncture. Every higher group vanishes on both sides, since S^1 is a $K(\mathbb{Z}, 1)$.

The key result behind both proceeding theorems is the compression lemma, which says that maps into a target can be pushed off the cells of a CW pair as soon as the relevant relative homotopy groups vanish. Everything in this section is an application of that lemma together with the homotopy extension property of CW pairs established in proposition 1.3.4.

Lemma 5.3.4: Compression Lemma

Let (X, A) be a CW pair and let (Y, B) be any pair with $B \neq \varnothing$. Suppose that for every n such that $X \setminus A$ has cells of dimension n, the relative homotopy group $\pi_n(Y, B, y_0)$ is trivial for all basepoints $y_0 \in B$. Then every map $f\colon (X, A) \to (Y, B)$ is homotopic rel A to a map $X \to Y$ whose image lies entirely in B.

In short, one wants to homotope a map so that it avoids a chosen subspace B of the target, and the obstruction to pushing the map off each n-cell of the domain lives in the relative homotopy group $\pi_n(Y, B)$. When those groups vanish, every obstruction disappears and the compression goes through cell by cell.

***Proof*:**
The strategy is to push f into B one skeleton at a time, using the vanishing relative homotopy groups to clear the cells of each dimension, and assembling the partial homotopies with the homotopy extension property. Throughout, f already maps A into B, and the homotopy will fix A.

Step 1: Setting up the induction on skeleta. Write X^k for the k-skeleton of X. The plan is to construct, by induction on k, a homotopy of f rel A that moves f to a map carrying $A \cup X^k$ into B. Since the homotopy will be performed rel A and the cells of A are already sent into B, only the cells of $X \setminus A$ require attention. Assume inductively that f has already been homotoped rel A so that $A \cup X^{k-1}$ maps into B. The base case $k = 0$ requires clearing the 0-cells of $X \setminus A$, which is handled by the same argument below with $n = 0$ (a relative homotopy group $\pi_0(Y, B)$ being trivial means every point of Y in the relevant component can be joined to B by a path).

Step 2: Clearing a single n-cell. Let e^n_α be an n-cell of $X \setminus A$ with characteristic map $\Phi_\alpha\colon D^n \to X$ and attaching map $\varphi_\alpha = \Phi_\alpha|_{\partial D^n}\colon S^{n-1} \to X^{n-1}$. By the inductive hypothesis f already sends X^{n-1}, hence the image of φ_α, into B. Therefore the composite

$$f \circ \Phi_\alpha\colon (D^n, \partial D^n) \longrightarrow (Y, B)$$

is a map of pairs, and as such it represents an element of the relative homotopy group $\pi_n(Y, B, y_0)$, where $y_0 = f(\Phi_\alpha(s_0))$ for a chosen basepoint $s_0 \in \partial D^n$. By hypothesis this group is trivial, so $f \circ \Phi_\alpha$ is homotopic, as a map of pairs $(D^n, \partial D^n) \to (Y, B)$, to a map with image contained in B. The triviality of a relative homotopy group means precisely that every map of the pair $(D^n, \partial D^n)$ into (Y, B) can be deformed, through maps of pairs, into one landing in B. This is the standard characterisation of the zero element established for relative homotopy groups in definition 5.1.8.

Step 3: Assembling the homotopy over all n-cells. Performing the homotopy of Step 2 simultaneously over all n-cells of $X \setminus A$ defines a homotopy

$$G\colon \big(A \cup X^{n-1}\big) \times I \;\cup\; \big(n\text{-cells of } X \setminus A\big) \times I \longrightarrow Y$$

The homotopy is constant on $A \cup X^{n-1}$ (those cells already lie in B and need not be moved) and on each new n-cell it is the cell-by-cell homotopy of Step 2, which agrees with the constant homotopy on the boundary ∂D^n because that boundary maps into X^{n-1}, already cleared. Thus the pieces match on overlaps and define a homotopy on $A \cup X^n$, starting at $f|_{A \cup X^n}$ and ending at a map sending $A \cup X^n$ into B.

Step 4: Extending over X via the homotopy extension property. The pair $(X, A \cup X^n)$ is a CW pair, hence has the homotopy extension property by proposition 1.3.4. The map $f\colon X \to Y$

together with the homotopy G just constructed on $A \cup X^n$ therefore extends to a homotopy $\widetilde{G}\colon X \times I \to Y$ of f. Replacing f by $\widetilde{G}(\,\cdot\,, 1)$ advances the induction: the new f sends $A \cup X^n$ into B, and the homotopy has been performed rel A.

Step 5: Passing to the colimit. Carrying out Steps 2-4 for $k = 0, 1, 2, \ldots$ produces a sequence of homotopies, the k-th of which is performed on the time interval $[1 - 2^{-k}, 1 - 2^{-(k+1)}]$ and is stationary on $A \cup X^{k-1}$. Concatenating them gives a single homotopy $X \times I \to Y$, rel A. Continuity at the final time $t = 1$ holds because X carries the weak topology with respect to its skeleta (definition 1.2.1): a map out of $X \times I$ is continuous if and only if its restriction to each $X^k \times I$ is, and on $X^k \times I$ the homotopy is eventually stationary, so it is continuous there. The end map of this concatenated homotopy sends all of $X = A \cup \bigcup_k X^k$ into B, completing the proof.

Example 5.3.5: Compressing a Map into a Highly Connected Subspace

Take $X = D^2$ with the standard CW structure $X = e^0 \cup e^1 \cup e^2$ (one 0-cell and one 1-cell forming $\partial D^2 = S^1$, and one 2-cell), and let $A = \varnothing$. Suppose (Y, B) is a pair with $\pi_0(Y, B) = \pi_1(Y, B) = \pi_2(Y, B) = 0$ for all basepoints in B, meaning $B \hookrightarrow Y$ is 2-connected: every component of Y meets B, and the relative groups up to degree 2 vanish. The compression lemma then says any map $f\colon D^2 \to Y$ is homotopic to a map landing in B. Concretely, let $Y = \mathbb{R}^3$ and $B = \mathbb{R}^2 \times \{0\}$. Since $\mathbb{R}^3$ deformation retracts onto $\mathbb{R}^2 \times \{0\}$, every relative group $\pi_n(\mathbb{R}^3, \mathbb{R}^2)$ vanishes, so the hypotheses hold in all dimensions. Any map $f\colon D^2 \to \mathbb{R}^3$ is compressed into the plane by the straight-line homotopy $(v, t) \mapsto (f(v)_1, f(v)_2, (1 - t)f(v)_3)$, which collapses the third coordinate to zero.

With the compression lemma in hand, Whitehead's theorem is almost immediate. The idea is to replace the map f by an inclusion (its mapping cylinder), observe that the vanishing of all relative homotopy groups of the inclusion is exactly the statement that f is a weak homotopy equivalence, and then compress the identity of the mapping cylinder onto the source. This produces a deformation retraction, which is the homotopy equivalence sought.

Theorem 5.3.6: Whitehead's Theorem

Let X and Y be CW complexes and let $f\colon X \to Y$ be a weak homotopy equivalence. Then f is a homotopy equivalence. If f is moreover the inclusion of a subcomplex, then X is a deformation retract of Y.

***Proof*:**

The proof reduces the general case to the case of an inclusion by means of the mapping cylinder, then applies the compression lemma to that inclusion.

Step 1: Reduction to an inclusion via the mapping cylinder. Recall the mapping cylinder $M_f = \big((X \times I) \sqcup Y\big)/\big((x, 1) \sim f(x)\big)$. There is a deformation retraction of M_f onto Y, so the inclusion $Y \hookrightarrow M_f$ is a homotopy equivalence, and f factors as

$$X \xhookrightarrow{\iota} M_f \xrightarrow{r} Y$$

where $\iota(x) = (x, 0)$ is the inclusion of X as the top of the cylinder and r is the retraction onto Y, with $r \circ \iota = f$. The retraction r is a homotopy equivalence, so f is a homotopy equivalence if and only if ι is. Moreover, when X and Y are CW complexes the mapping cylinder M_f is a CW complex containing X as a subcomplex (after a homotopy of f to a cellular map, which

by the cellular approximation theorem ([21, Theorem 4.8]) changes nothing up to homotopy), so (M_f, X) is a CW pair. Thus it suffices to prove the second assertion: *if $\iota\colon X \hookrightarrow Z$ is an inclusion of CW complexes that is a weak homotopy equivalence, then X is a deformation retract of Z*, with $Z = M_f$. The first assertion follows because a deformation retract inclusion is in particular a homotopy equivalence.

Step 2: The relative homotopy groups of the pair vanish. Let $\iota\colon X \hookrightarrow Z$ be an inclusion of CW complexes that is a weak homotopy equivalence. The long exact sequence of the pair (Z, X) from proposition 5.1.11 reads

$$\cdots \longrightarrow \pi_n(X, x_0) \xrightarrow{\iota_*} \pi_n(Z, x_0) \longrightarrow \pi_n(Z, X, x_0) \longrightarrow \pi_{n-1}(X, x_0) \xrightarrow{\iota_*} \pi_{n-1}(Z, x_0) \longrightarrow \cdots$$

for each basepoint $x_0 \in X$. Because ι is a weak homotopy equivalence, every map $\iota_*\colon \pi_n(X, x_0) \to \pi_n(Z, x_0)$ is an isomorphism. A short exactness argument then forces the relative groups to vanish: the map $\pi_n(Z, X) \to \pi_{n-1}(X)$ has image contained in the kernel of the injective ι_*, hence is zero, so $\pi_n(Z, X) \to \pi_{n-1}(X)$ is the zero map. The map $\pi_n(Z) \to \pi_n(Z, X)$ is surjective onto the kernel of that zero map, which is all of $\pi_n(Z, X)$, while its source $\pi_n(Z)$ is hit isomorphically from $\pi_n(X)$ by ι_*, so the composite $\pi_n(Z) \to \pi_n(Z, X)$ is zero. A surjective zero map has trivial target, so $\pi_n(Z, X, x_0) = 0$ for all n and all basepoints $x_0 \in X$.

Step 3: Compressing the identity of Z onto X. Apply the compression lemma (lemma 5.3.4) to the identity map

$$\mathrm{id}_Z\colon (Z, X) \longrightarrow (Z, X)$$

with $(X, A) = (Z, X)$ in the notation of the lemma and $(Y, B) = (Z, X)$. The pair (Z, X) is a CW pair, the subspace X is nonempty, and the relative homotopy groups $\pi_n(Z, X)$ vanish in every dimension by Step 2, so the hypotheses are met in all dimensions. The lemma produces a homotopy, rel X, from id_Z to a map $\rho\colon Z \to Z$ whose image lies in X.

Step 4: Identifying the deformation retraction. The map $\rho\colon Z \to X$ satisfies $\rho|_X = \mathrm{id}_X$, since the homotopy was performed rel X and id_Z restricts to the identity on X. Thus ρ is a retraction of Z onto X, and the homotopy from id_Z to ρ is a homotopy rel X from the identity of Z to a map into X. This is exactly a deformation retraction of Z onto X. Therefore X is a deformation retract of Z, the inclusion $X \hookrightarrow Z$ is a homotopy equivalence, and tracing through Step 1 the original map f is a homotopy equivalence, completing the proof.

Whitehead's theorem is the statement that for CW complexes the homotopy groups are a *complete* invariant of homotopy type, in the sense that a map realising isomorphisms on all of them is invertible up to homotopy. The qualification "a map realising" is essential: two CW complexes can have abstractly isomorphic homotopy groups in every degree without being homotopy equivalent, because no single map need induce those isomorphisms simultaneously. The spaces $\mathbb{R}\mathrm{P}^2 \times S^3$ and $\mathbb{R}\mathrm{P}^3 \times S^2$ are the standard example of this phenomenon, having identical homotopy groups but distinct homotopy types.

Example 5.3.7: $\mathbb{R}^n$ Is Contractible by Whitehead

Let $X = \{*\}$, a single point with its evident CW structure, and let $Y = \mathbb{R}^n$ with the CW structure of example 1.2.5 (a product of the line's cell structure). The constant inclusion $f\colon \{*\} \hookrightarrow \mathbb{R}^n$ is a weak homotopy equivalence: $\mathbb{R}^n$ is path connected so π_0 agrees, and $\mathbb{R}^n$ is convex so every map of a sphere into it is null-homotopic by a straight-line homotopy to a constant, giving $\pi_k(\mathbb{R}^n) = 0 = \pi_k(\{*\})$ for all $k \geq 1$. Both source and target are CW complexes, so Whitehead's theorem applies and f is a homotopy equivalence: $\mathbb{R}^n$ is contractible. This particular conclusion is

visible directly from the contraction $(v,t) \mapsto (1-t)v$, but the point is methodological. Whitehead's theorem reduces the assertion "$\mathbb{R}^n$ is contractible" to the purely homotopy-group computation $\pi_k(\mathbb{R}^n) = 0$, with no need to exhibit a contraction. In less transparent examples no explicit contraction is available and the homotopy-group route is the only one.

Example 5.3.8: The House With Two Rooms Is Contractible

The previous example settled the contractibility of $\mathbb{R}^n$ by computing homotopy groups, but there the conclusion already followed from the straight-line contraction. The house with two rooms of example 1.1.17, Bing's house, is the promised case where no contraction presents itself, yet the same two-line homotopy-group argument applies verbatim. Regard it as a finite, 2-dimensional CW complex B: two stacked chambers, each entered only through a vertical tube that passes through the floor of the other, with the supporting walls, partition, and tube walls glued along their shared edges into a complex of 0-cells, 1-cells, and 2-cells. The aim is to prove B contractible without ever writing down a deformation. The route is to show B is simply connected with vanishing reduced homology and to let theorem 5.4.9 and theorem 5.3.6 do the rest.

B is simply connected. The fundamental group of a connected 2-complex is read off from its 2-skeleton by the presentation-complex recipe of section 2.2: choosing a spanning tree T of the 1-skeleton $B^{(1)}$, the group $\pi_1(B^{(1)})$ is free on the 1-cells of $B^{(1)}$ not in T (collapsing T identifies $B^{(1)}$ with a wedge of circles, whose group is free by Van Kampen's theorem, theorem 2.2.4), and each 2-cell e^2_β, attached along a loop w_β in $B^{(1)}$, imposes the single relation $w_\beta = 1$ on the group. This is the same mechanism used to realize an arbitrary group as π_1 of a 2-complex in theorem 5.5.4: attaching a 2-cell along a loop w kills exactly the class $[w]$ and nothing more. The outcome is the finite presentation

$$\pi_1(B) = \left\langle\, x_e \ (e \notin T) \;\middle|\; w_\beta \ (\beta = 1, \dots, c_2) \,\right\rangle$$

in which the generators x_e are indexed by the 1-cells outside T and the relators w_β are the attaching words of the c_2 two-cells.[a] For Bing's house this presentation collapses to the trivial group: each generator threading a tunnel appears, in some relator w_β, as the attaching word of a supporting-wall 2-cell whose boundary runs once along that tunnel edge and otherwise along edges already expressible through other relators, so successive substitution eliminates the generators one by one until none remain. Verifying this collapse for the full structure is a finite but lengthy word computation over the 23 generators and 23 relators, with no conceptual content beyond the substitution just described. The cell structure itself, with $c_0 = 29$ zero-cells, $c_1 = 51$ one-cells, and $c_2 = 23$ two-cells, is enumerated in [21, Ex. 0.2]. The generator count is $c_1 - (c_0 - 1) = 51 - 28 = 23$, the number of 1-cells left once a spanning tree of the connected 1-skeleton accounts for $c_0 - 1 = 28$ of them, equivalently the first Betti number $c_1 - c_0 + 1$ of $B^{(1)}$ (the rank of the free group $\pi_1(B^{(1)})$). The relators therefore exactly match the generators in number, the balanced presentation forced on an acyclic 2-complex, where the 23 relators must kill the rank-23 free group $\pi_1(B^{(1)})$. The presentation therefore yields $\pi_1(B) = 0$.

B has trivial reduced homology. Because B is 2-dimensional, $\tilde{H}_i(B) = 0$ for $i \geq 3$ automatically. In degree one, B is path connected and now known to be simply connected, so the dimension-one Hurewicz theorem (theorem 5.4.5) gives $H_1(B) \cong \pi_1(B)^{\mathrm{ab}} = 0$. The remaining group $H_2(B)$ is settled by an Euler-characteristic count rather than by inspecting which 2-cells fit together. Since B is 2-dimensional, its cellular chain complex is $0 \to C_2(B) \to C_1(B) \to C_0(B) \to 0$, so $H_2(B) = \ker \partial_2$ is a subgroup of the free abelian group $C_2(B)$ and is therefore itself free abelian. Its rank is the only unknown. Euler's characteristic theorem (theorem 3.5.22) computes that rank

homologically,

$$\chi(B) = \sum_n (-1)^n \mathrm{rank} H_n(B) = \mathrm{rank} H_0(B) - \mathrm{rank} H_1(B) + \mathrm{rank} H_2(B) = 1 - 0 + \mathrm{rank} H_2(B)$$

using the established $H_0(B) \cong \mathbb{Z}$ from path connectivity and $H_1(B) = 0$ from the previous paragraph. On the other hand $\chi(B)$ is computed directly from the cell counts of Bing's house by its definition $\chi(B) = \sum_n (-1)^n c_n$ as the alternating sum of the numbers c_n of n-cells, and the enumeration $c_0 = 29$, $c_1 = 51$, $c_2 = 23$ recorded above ([21, Ex. 0.2]) gives $\chi(B) = 29 - 51 + 23 = 1$. Comparing the two expressions for $\chi(B)$ forces $\mathrm{rank} H_2(B) = 0$, and since $H_2(B)$ is free abelian this gives $H_2(B) = 0$. Together with $\tilde{H}_0(B) = 0$ from path connectivity, every reduced homology group vanishes, so B is acyclic in the sense of definition 3.5.16.

Hurewicz and Whitehead finish the argument. A simply connected space with vanishing reduced homology is weakly contractible, by induction on the connectivity using theorem 5.4.9. The space B is 1-connected. Assuming B is $(n-1)$-connected for some $n \geq 2$, the Hurewicz isomorphism $\pi_n(B) \xrightarrow{\cong} H_n(B) = 0$ gives $\pi_n(B) = 0$, so B is n-connected, and the induction yields $\pi_n(B) = 0$ for every n. The unique map $B \to \{*\}$ is therefore a weak homotopy equivalence: a π_0-bijection by path connectivity and an isomorphism on each π_n, both sides being trivial. Both B and $\{*\}$ are CW complexes, so Whitehead's theorem (theorem 5.3.6) upgrades this weak equivalence to a genuine homotopy equivalence, and $B \simeq \{*\}$ is contractible, with no contraction ever exhibited.

The contrast with $\mathbb{R}^n$ of example 5.3.7 is the whole point. For $\mathbb{R}^n$ the contraction $(v, t) \mapsto (1-t)v$ is written on sight and the homotopy-group computation only confirms what the eye already sees. For B no contraction is available to inspection, yet the identical argument, vanishing π_1 and vanishing reduced homology feeding Hurewicz and then Whitehead, settles the contractibility of both spaces by the same two lines.

[a] This route to $\pi_1(B) = 0$ is deliberately independent of contractibility. Hatcher establishes contractibility of B directly, by thickening B to a neighborhood homeomorphic to the ball D^3 and retracting ([21, Ch. 0]). Quoting that for $\pi_1(B) = 0$ would make the present example circular, since contractibility is exactly the conclusion being re-derived here through Hurewicz and Whitehead. The presentation above derives $\pi_1(B) = 0$ from the cell structure alone, with no contraction used.

Without the CW assumption, a weak homotopy equivalence need not be a homotopy equivalence, and the standard witness to this is the Warsaw circle. The point of the example is that a space can be *weakly contractible* (path connected with all homotopy groups trivial, so the map to a point is a weak homotopy equivalence) while being far from contractible, because homotopy groups simply fail to register certain pathologies of non-CW spaces:

Example 5.3.9: The Warsaw Circle

The *Warsaw circle* W is the subspace of $\mathbb{R}^2$ formed from the closed topologist's sine curve together with an arc joining its ends. Explicitly, let

$$S = \{(x, \sin(1/x)) \mid 0 < x \leq 1/\pi\}, \qquad V = \{0\} \times [-1, 1]$$

so that the closure $\overline{S} = S \cup V$ is the topologist's sine curve, and let C be a simple arc in $\mathbb{R}^2$ from the point $(1/\pi, 0) \in S$ to the point $(0, -1) \in V$ that meets $\overline{S}$ only at its two endpoints and otherwise lies outside the region swept by S. Set $W = \overline{S} \cup C$.

W is path connected. The bare topologist's sine curve $\overline{S}$ is not path connected: no path in $\overline{S}$ joins a point of the oscillating part S to a point of the limit segment V, because such a path would have to traverse the infinitely many oscillations of $\sin(1/x)$ as $x \to 0$, which no continuous

parametrisation can do. The arc C repairs this. A point of S is joined to a point of V by going the "other way" along C, which lies outside $\overline{S}$ and reaches V at $(0,-1)$ without ever entering the oscillating region. Thus W is path connected, and $\pi_0(W)$ is a single point.

All homotopy groups of W are trivial. The argument rests on a confinement lemma: any continuous map from a compact space into W stays clear of a tail of the oscillations. Precisely, for $a > 0$ write

$$S_a = \{(x, \sin(1/x)) \mid 0 < x < a\}$$

for the part of the oscillating graph with x-coordinate below a, an open arc of S accumulating onto the limit segment V. The claim is that for any compact, locally connected, metrizable space K and any continuous map $\beta\colon K \to W$ there exists $a > 0$ with $\beta(K) \cap S_a = \varnothing$, so β avoids S_a entirely. (Both paths $\beta\colon [0,1] \to W$ and sphere maps $\beta\colon S^k \to W$ are instances, $[0,1]$ and S^k being compact, locally connected, and metrizable.)

Proof of the confinement lemma. It suffices to treat K locally connected and metrizable, which covers the only cases used below, $K = [0,1]$ and $K = S^k$. Suppose the conclusion fails: then for each i there is a point $p_i \in K$ with $\beta(p_i) \in S_{1/i}$, so the x-coordinate of $\beta(p_i)$ tends to 0. By compactness of K pass to a subsequence with $p_i \to p_*$. By continuity $\beta(p_*) = \lim_i \beta(p_i)$, and since the x-coordinates of $\beta(p_i)$ tend to 0 while the y-coordinates lie in $[-1,1]$, the limit has x-coordinate 0, so $\beta(p_*) \in V$, say $\beta(p_*) = (0,c)$. Continuity of β at p_* together with local connectedness of K gives a connected open neighbourhood U of p_* in K with $\beta(U)$ contained in the ball B of radius $\frac{1}{2}$ about $(0,c)$ in W. Within B the set W consists of the segment of V through $(0,c)$ together with countably many disjoint vertical strands of the graph: for $c \in (-1,1)$ the curve $y = \sin(1/x)$ crosses the height $y = c$ at a discrete set of x-values $x_1 > x_2 > \cdots \to 0$, and near V the strands run almost vertically and are separated from one another and from V by gaps bounded below, so each strand is a separate connected component of $W \cap B$ and the component containing $(0,c)$ is exactly the V-segment (the cases $c = \pm 1$ are identical, the extremal strands being even more sharply separated). The image $\beta(U)$ is connected and contains $(0,c)$, so it lies in that single V-segment and meets no strand of the graph. But $\beta(p_i) \in S_{1/i}$ lies on the graph for $p_i \in U$ with i large, a contradiction. This proves the lemma.

The lemma forces every loop and every higher sphere into a contractible part of W. Given $k \geq 1$ and a map $g\colon S^k \to W$, the lemma applied to $\beta = g$ on the compact space $K = S^k$ produces a single $a > 0$ with $g(S^k) \cap S_a = \varnothing$. Thus g lands in $W_a := W \setminus S_a$. Removing the tail S_a deletes the accumulation, leaving three pieces: the segment V, the finite-length graph arc $G_a := \{(x, \sin(1/x)) \mid a \leq x \leq 1/\pi\}$ (each a homeomorphic copy of a closed interval, now disjoint from one another since V has $x = 0$ while G_a has $x \geq a > 0$), and the joining arc C, which runs from the endpoint $(1/\pi, 0)$ of G_a to the endpoint $(0,-1)$ of V. Concatenated end to end, V, then C, then G_a form a single embedded arc, a homeomorphic copy of $[0,1]$ with no oscillation barrier remaining, and an arc is contractible. A map of S^k into a contractible space is null-homotopic, so $[g] = 0$ in $\pi_k(W, w_0)$. Hence $\pi_n(W, w_0) = 0$ for all $n \geq 1$. For the complete topologist's-sine-curve and Warsaw-circle computations underlying the confinement lemma and the contractibility of W_a, see [30, §24, §27] and [21, Ch. 1]. Together with path connectivity, the unique map $W \to \{*\}$ is therefore a weak homotopy equivalence: a π_0-bijection and an isomorphism on every π_n, $n \geq 1$, since both sides are trivial.

But W is not contractible, hence not homotopy equivalent to a point. Let $\gamma\colon [0,1] \to C$ be a homeomorphism with $\gamma(0) = (0,-1)$ and $\gamma(1) = (1/\pi, 0)$, and define $q\colon W \to S^1 \subseteq \mathbb{C}$ by

$$q(x,y) = e^{\pi i(2-\pi x)} \ \text{ for } (x,y) \in \overline{S}, \qquad q(\gamma(s)) = e^{\pi i s} \ \text{ for } s \in [0,1]$$

The sets $\overline{S}$ and C are closed in W and meet in the two points $(0,-1)$ and $(1/\pi, 0)$, where both formulas give 1 and -1 respectively, so q is continuous by [16, Pasting Lemma]. Suppose

$F\colon W \times I \to W$ were a contraction, with $F_0 = \mathrm{id}_W$ and F_1 constant. Then $H = q \circ F$ is a homotopy from q to a constant map with value $e^{2\pi i\theta_0}$. The space $W \times I$ is a compact metric space, W being closed and bounded in $\mathbb{R}^2$, so H is uniformly continuous [16, Uniform Continuity Theorem], and there are $0 = t_0 < t_1 < \cdots < t_m = 1$ with $|H(w,t_{i-1}) - H(w,t_i)| < 2$ for all w and i. Two points of S^1 at distance less than 2 are not antipodal, so $H(w,t_{i-1})/H(w,t_i)$ lies in $S^1 \setminus \{-1\}$, on which the principal argument $\mathrm{Arg}\colon S^1 \setminus \{-1\} \to (-\pi,\pi)$ is continuous. The function

$$\psi(w) = \theta_0 + \frac{1}{2\pi}\sum_{i=1}^{m} \mathrm{Arg}\,\frac{H(w,t_{i-1})}{H(w,t_i)}$$

is therefore continuous $W \to \mathbb{R}$, and $e^{2\pi i\psi} = q$ because the product of the quotients telescopes to $q(w)e^{-2\pi i\theta_0}$. The set $\overline{S}$ is connected, as the closure of the connected graph S, and $\lambda(x,y) = 1 - \pi x/2$ also satisfies $e^{2\pi i\lambda} = q$ on it, so the continuous integer-valued function $\psi - \lambda$ is a constant k on $\overline{S}$. On the arc C the function $\mu(\gamma(s)) = s/2$ satisfies $e^{2\pi i\mu} = q$, so $\psi - \mu$ is a constant l on C. At $(1/\pi, 0)$ this gives $\frac{1}{2} + k = \frac{1}{2} + l$, so $k = l$, and at $(0,-1)$ it gives $1 + k = l$, a contradiction. So W is not contractible, and the map $W \to \{*\}$ is a weak homotopy equivalence that is *not* a homotopy equivalence. By Whitehead's theorem (theorem 5.3.6) W cannot have the homotopy type of a CW complex, for if it did, weak contractibility would force genuine contractibility. The CW hypothesis of Whitehead's theorem therefore cannot be dropped.

The Warsaw circle shows that homotopy groups are blind to certain non-CW pathologies, and so weak homotopy equivalence is strictly weaker than homotopy equivalence in general. The remedy to confine to spaces where they suffice. The mechanism for doing so is CW approximation: every space, however pathological, is weakly equivalent to a CW complex, and once the comparison is made the CW model can be studied with all the tools that the CW hypothesis unlocks, Whitehead's theorem foremost among them.

Theorem 5.3.10: CW Approximation

For every topological space X there exists a CW complex Z and a weak homotopy equivalence $f\colon Z \to X$. Given moreover a point $x_0 \in X$ with X path-connected, the pair may be chosen with a 0-cell $z_0 \in Z$ satisfying $f(z_0) = x_0$, since the construction below starts from a wedge of circles at a chosen basepoint and only ever extends that map. The pair (Z,f) is called a *CW approximation* to X, and it is unique up to homotopy equivalence: if $f'\colon Z' \to X$ is another CW approximation, there is a homotopy equivalence $h\colon Z \to Z'$ with $f' \circ h \simeq f$.

***Proof*:**
The construction builds Z by attaching cells in increasing dimension, alternately creating the homotopy classes needed for f_* to be surjective and then killing the classes needed for f_* to be injective. For clarity the construction is given for path-connected X with a chosen basepoint. The general case is the disjoint union of the constructions over the path components of X.

Step 1: The base of the construction (surjectivity in degree 1). Fix a basepoint $x_0 \in X$. Begin with $Z_1 = \bigvee_\alpha S^1_\alpha$, a wedge of circles, one for each generator in a chosen set of generators of $\pi_1(X,x_0)$ (if π_1 is trivial, take $Z_1 = \{z_0\}$ a point). Define $f_1\colon Z_1 \to X$ on each circle S^1_α to be a loop representing the corresponding generator of $\pi_1(X,x_0)$. By construction $(f_1)_*\colon \pi_1(Z_1) \to \pi_1(X)$ is surjective, since its image contains a generating set. Both Z_1 and X are path connected, so

the inductive invariant holds at $n = 1$:

$$(f_1)_*\colon \pi_k(Z_1) \longrightarrow \pi_k(X) \quad \text{is an isomorphism for } k < 1 \text{ and surjective for } k = 1$$

(the $k < 1$ condition being the π_0-bijection, which holds as both spaces are connected).

Step 2: The inductive step, killing the kernel in degree n (injectivity). Suppose inductively that a CW complex Z_n and a map $f_n\colon Z_n \to X$ have been constructed so that $(f_n)_*$ is an isomorphism on π_k for $k < n$ and surjective on π_n. The map $(f_n)_*$ on π_n may have a kernel. For each element β in a generating set of $\ker\big((f_n)_*\colon \pi_n(Z_n) \to \pi_n(X)\big)$, represented by a map $\gamma_\beta\colon S^n \to Z_n$, attach an $(n+1)$-cell to Z_n along γ_β, and call the result Z_n'. Attaching an $(n+1)$-cell along γ_β sets the class $[\gamma_\beta]$ to zero in $\pi_n(Z_n')$, by the relation that a sphere bounding a disk is null-homotopic, while leaving π_k unchanged for $k < n$ (cells of dimension $n+1$ do not affect homotopy below dimension n). The map f_n extends over each attached $(n+1)$-cell because $f_n \circ \gamma_\beta$ is null-homotopic in X (its class lies in $\ker(f_n)_*$, hence is trivial in $\pi_n(X)$), and a null-homotopy is precisely an extension of $f_n \circ \gamma_\beta$ over the disk D^{n+1}. The extended map $f_n'\colon Z_n' \to X$ now induces an *isomorphism* on π_n: surjectivity is inherited from f_n, and injectivity holds because the kernel has been killed. The lower-degree isomorphisms are preserved, so $(f_n')_*$ is an isomorphism on π_k for $k \leq n$.

Step 3: The inductive step, surjectivity in degree $n+1$. To make $(f_n')_*$ surjective in degree $n+1$ while preserving the isomorphisms in degrees $\leq n$, attach $(n+1)$-spheres to create the missing classes. Choose a set of generators of $\pi_{n+1}(X)$. For each such generator δ, represented by a map $S^{n+1} \to X$, form the wedge with an $(n+1)$-sphere S^{n+1}_δ and extend f_n' to send S^{n+1}_δ by a representative of δ. Wedging on spheres of dimension $n+1$ does not change π_k for $k \leq n$, so the isomorphisms in degrees $\leq n$ are preserved, and the new spheres surject onto $\pi_{n+1}(X)$. Call the result Z_{n+1} with map f_{n+1}. By construction

$$(f_{n+1})_*\colon \pi_k(Z_{n+1}) \longrightarrow \pi_k(X) \quad \text{is an isomorphism for } k \leq n \text{ and surjective for } k = n+1$$

which is the inductive invariant with n replaced by $n+1$, advancing the induction.

Step 4: Passing to the colimit. The complexes $Z_1 \subseteq Z_2 \subseteq Z_3 \subseteq \cdots$ form an increasing sequence of CW complexes, each obtained from the previous by attaching cells, and the maps f_n are compatible with the inclusions ($f_{n+1}|_{Z_n} = f_n$). Let $Z = \bigcup_n Z_n$ with the weak topology, a CW complex, and let $f\colon Z \to X$ be the map determined by the f_n (well-defined and continuous because Z has the weak topology of definition 1.2.1 and f is continuous on each Z_n). For each fixed k, the homotopy group $\pi_k(Z)$ is computed from a compact image, so any map $S^k \to Z$ and any homotopy $S^k \times I \to Z$ meet only finitely many cells (lemma 1.2.4) and land in some finite stage Z_n with $n > k$. Consequently $\pi_k(Z) = \pi_k(Z_n)$ for all $n > k$. At such a stage $(f_n)_*$ is an isomorphism on π_k, so $f_*\colon \pi_k(Z) \to \pi_k(X)$ is an isomorphism for every k. Therefore f is a weak homotopy equivalence and Z is a CW complex, establishing existence.

Step 5: Uniqueness up to homotopy equivalence. Let $f\colon Z \to X$ and $f'\colon Z' \to X$ be two CW approximations. Form the mapping cylinder $M_{f'}$ of f' and regard X as a deformation retract of it. Then $f'\colon Z' \hookrightarrow M_{f'}$ is, up to the homotopy equivalence $M_{f'} \simeq X$, the inclusion of a subcomplex. The composite $Z \to X \hookrightarrow M_{f'}$ is a map from the CW complex Z into $M_{f'}$ that is a weak homotopy equivalence (both f and the inclusion $X \hookrightarrow M_{f'}$ are). Since Z' is a CW complex and $Z' \hookrightarrow M_{f'}$ is a weak homotopy equivalence with $M_{f'} \simeq X$, both Z and Z' are CW complexes weakly equivalent over X. The map $Z \to M_{f'}$ may be homotoped into the subcomplex Z' and, being a weak homotopy equivalence between CW complexes, is a homotopy equivalence $h\colon Z \to Z'$ by Whitehead's theorem (theorem 5.3.6), with $f' \circ h \simeq f$ by construction. This gives uniqueness up to homotopy equivalence, completing the proof.

CW approximation is the structural reason that algebraic topology can restrict to CW complexes without loss of generality at the level of homotopy theory. Any space whatsoever is replaced by a CW complex with the same homotopy groups, the comparison being a single map inducing all the isomorphisms at once. Combined with Whitehead's theorem, this means the homotopy category of all spaces, localised at the weak homotopy equivalences, is equivalent to the homotopy category of CW complexes. The cellular structure of the approximation also makes the construction compatible with cellular homology of section 3.5: the homology of the CW approximation Z computes the singular homology of X, because a weak homotopy equivalence induces isomorphisms on singular homology (proposition 5.3.15 below).

The construction admits a relative refinement that is constantly used in obstruction theory and the comparison of pairs. One often needs not just a CW approximation to a space but a CW approximation to a map, or an approximation that already agrees with a given CW structure on a subspace, controlled by a connectivity parameter.

Proposition 5.3.11: Relative and n-Connected CW Approximation

Let (X, A) be a pair of spaces and let A already be a CW complex. Then there exists a CW pair (Z, A) extending the CW structure of A and a weak homotopy equivalence $f\colon Z \to X$ restricting to the inclusion $A \hookrightarrow X$ on A. Moreover, given any integer $n \geq 0$, the approximation may be taken to be *n-connected*: Z can be built so that f induces isomorphisms on π_k for $k > n$ and an injection on π_n, with Z obtained from A by attaching cells of dimension greater than n only. Such a pair (Z, A) is called an *n-connected CW model* of (X, A).

***Proof*:**
The argument is the relative form of the construction in theorem 5.3.10, started not from a wedge of spheres but from the given subcomplex A, with all cell attachments performed away from A.

Step 1: The relative construction. Run the cell-attachment procedure of theorem 5.3.10 with A as the initial stage in place of Z_1, and with the map $f|_A$ taken to be the inclusion $A \hookrightarrow X$. At each stage the new cells are attached to the part of the complex built so far without touching A, exactly as in Steps 2 and 3 of that proof: surjectivity in each degree is arranged by wedging on spheres (attached at the basepoint, away from A), and injectivity by attaching cells of one higher dimension along maps that miss the interior of A. The inductive hypothesis is now that f restricts to the inclusion on A and induces an isomorphism on π_k for k below the current dimension and a surjection at it. Passing to the colimit as in Step 4 gives a CW pair (Z, A) with A a subcomplex and a weak homotopy equivalence $f\colon Z \to X$ extending $A \hookrightarrow X$.

Step 2: The connectivity refinement. For the n-connected version, attach cells of dimension $n+1$ and higher only, never cells of dimension $\leq n$. First attach $(n+1)$-cells along maps $S^n \to A$ representing generators of the kernel of f_* on π_n (normal generators when $n = 1$), extending f over each by a null-homotopy of its image in X. This kills the kernel, so f_* becomes injective on π_n. Then make f_* bijective in each degree $k \geq n+1$ as in theorem 5.3.10, which attaches cells of dimension $n+1$ along constant maps and cells of dimension at least $n+2$. Neither changes π_n, so injectivity on π_n survives. Then (Z, A) has no relative cells of dimension $\leq n$, so the pair is n-connected: the relative homotopy groups $\pi_k(Z, A)$ vanish for $k \leq n$ because a CW pair with cells only in dimensions $> n$ has $\pi_k(Z, A) = 0$ for $k \leq n$ (given a map $(D^k, \partial D^k) \to (Z, A)$ with $k \leq n$, first homotope its restriction $\partial D^k \to A$ to a cellular map inside A and extend the

homotopy over D^k by proposition 1.3.4, after which the resulting map of pairs is homotopic rel ∂D^k to a cellular map, by the cellular approximation theorem invoked in the proof of theorem 5.3.6, and a cellular map lands in the relative k-skeleton, which is A because Z has no relative cells in dimensions $\leq n$). The induced map f_* is then an isomorphism on π_k for $k > n$ and an injection on π_n, as we sought to show.

The relative version turns CW approximation from a statement about single spaces into a tool for comparing spaces along a map, while the connectivity parameter lets one approximate only the part of the homotopy above a chosen dimension. Both features are exactly what the next sections require: the Hurewicz theorem (theorem 5.4.9) is phrased for n-connected spaces, and the construction of Eilenberg-MacLane spaces and Postnikov towers proceeds by attaching cells in prescribed dimensions to control homotopy degree by degree, which is the n-connected model applied repeatedly.

Example 5.3.12: An n-Connected Model Kills Low Homotopy

Take $X = S^2$ and $A = \{*\}$ a basepoint, and ask for a 1-connected CW model. The construction starts from the point $A = \{*\}$ and attaches cells only in dimensions ≥ 2. The first stage attaches a single 2-cell to realise the generator of $\pi_2(S^2) \cong \mathbb{Z}$ (giving S^2 itself), then higher cells in dimensions ≥ 3 to correct $\pi_3, \pi_4, \ldots$, which are nontrivial for S^2 (for instance $\pi_3(S^2) \cong \mathbb{Z}$, generated by the Hopf map). The resulting model Z is a CW complex with no 1-cells, so $(Z, \{*\})$ is 1-connected, and $f\colon Z \to S^2$ is an isomorphism on π_k for $k \geq 2$ and the trivial (hence surjective) map on $\pi_1 = 0$. The point of the example is that the connectivity parameter controls the dimensions in which cells appear: requiring a 1-connected model forbids 1-cells, which is consistent with S^2 being simply connected, and the entire homotopical content lives in dimension 2 and above, exactly where the cells of Z sit.

Whitehead's theorem compares homotopy classes through every degree at once. A map that matches homotopy groups only in a range of degrees still compares homotopy classes of maps out of complexes of bounded dimension. Maps out of a CW complex are built and compared one cell at a time, and the obstructions met on a k-cell lie in homotopy groups of degrees $k-1$ and k.

Definition 5.3.13: n-Connected Map

Let $n \geq 1$ and let $f\colon Y \to Z$ be a map of path-connected spaces. The map f is *n-connected* if for every $y \in Y$ the induced homomorphism

$$f_*\colon \pi_k(Y, y) \longrightarrow \pi_k\big(Z, f(y)\big)$$

is an isomorphism for $1 \leq k < n$ and is surjective for $k = n$.

The quantifier over basepoints follows definition 5.3.1, and a weak homotopy equivalence of path-connected spaces is n-connected for every n. When Y and Z are simply connected a single basepoint suffices. By proposition 5.1.15, for a simply connected space W and any $w \in W$, forgetting the basepoint is a bijection $\pi_k(W, w) \to [S^k, W]$ onto the free homotopy classes, and these bijections commute with f_*. So f_* is bijective, or surjective, on $\pi_k(Y, y)$ exactly when $f_*\colon [S^k, Y] \to [S^k, Z]$ is, a condition that does not mention y.

Lemma 5.3.14: Maps out of a Complex of Bounded Dimension

Let $n \geq 1$, let $f\colon Y \to Z$ be an n-connected map of simply connected spaces, and let X be a CW complex. Composition with f,

$$f_*\colon [X,Y] \longrightarrow [X,Z], \qquad [h] \longmapsto [f \circ h]$$

is surjective when $\dim X \leq n$ and bijective when $\dim X \leq n-1$.

***Proof*:**
The proof replaces f by the inclusion of Y into the mapping cylinder of f, as in the proof of theorem 5.3.6, and pushes maps into Y with the compression lemma (lemma 5.3.4).

Step 1: The mapping cylinder. Let $M = \big((Y \times I) \sqcup Z\big)/\big((y,1) \sim f(y)\big)$ be the mapping cylinder of f (example 1.1.13), write $[y,t]$ for the image of (y,t), and regard Z as a subspace of M, with inclusion $\iota\colon Z \to M$. Define $j\colon Y \to M$ by $j(y) = [y,0]$, and $r\colon M \to Z$ by $r[y,t] = f(y)$ and $r(z) = z$ for $z \in Z$. Both are continuous, being induced by continuous maps on $(Y \times I) \sqcup Z$ that respect the identification, and $r \circ j = f$ and $r \circ \iota = \mathrm{id}_Z$. The formula

$$H([y,t],s) = [y,(1-s)t+s], \qquad H(z,s) = z$$

is well defined, since at $t = 1$ it is constant at $f(y)$, and continuous, since $\big((Y\times I)\sqcup Z\big)\times I \to M \times I$ is a quotient map, I being locally compact Hausdorff [16, Products of Quotient Maps]. It is a homotopy from id_M to $\iota \circ r$, so r and ι are mutually inverse homotopy equivalences. The formula $K(y,s) = [y,s]$ defines a homotopy $K\colon Y \times I \to M$ from j to $\iota \circ f$. Finally, the image U of $Y \times [0,1)$ in M is open, its preimage $Y \times [0,1)$ being open and saturated, and the quotient map restricts to a continuous bijection $Y \times [0,1) \to U$ that is a quotient map by [16, Subspaces And Quotient Maps], hence a homeomorphism. So the projection $p\colon U \to Y$, $p[y,t] = y$, is continuous, and $p \circ j = \mathrm{id}_Y$. Write $Y' = j(Y) \subseteq U$.

Step 2: M is simply connected. The path $s \mapsto [y, t + s(1-t)]$ joins $[y,t]$ to $f(y) \in Z$, and Z is path-connected, so M is path-connected. Since r is a homotopy equivalence, theorem 2.1.45 makes $r_*\colon \pi_1(M,m) \to \pi_1(Z, r(m)) = 0$ an isomorphism for every $m \in M$.

Step 3: The inclusion j is as connected as f. The spaces Y, Z and M are simply connected, so by proposition 5.1.15, for each $k \geq 1$, forgetting basepoints identifies $\pi_k(Y,y)$, $\pi_k(Z,f(y))$ and $\pi_k(M,j(y))$ with $[S^k,Y]$, $[S^k,Z]$ and $[S^k,M]$, compatibly with the maps induced by f and j. On free homotopy classes $j_* = \iota_* \circ f_*$, because $j \simeq \iota \circ f$ through K, and $\iota_*\colon [S^k,Z] \to [S^k,M]$ is a bijection with inverse r_*, because $r \circ \iota = \mathrm{id}_Z$ and $\iota \circ r \simeq \mathrm{id}_M$. Since f is n-connected, $f_*\colon [S^k,Y] \to [S^k,Z]$ is bijective for $k < n$ and surjective for $k = n$. Hence for every $y \in Y$ the homomorphism $j_*\colon \pi_k(Y,y) \to \pi_k(M,j(y))$ is an isomorphism for $1 \leq k < n$ and surjective for $k = n$.

Step 4: The relative homotopy groups of (M,Y') vanish through degree n. Let $y' = j(y)$ and identify Y' with Y through j. For $2 \leq k \leq n$, the long exact sequence of the pair (M,Y') at y' (proposition 5.1.11) contains the segment

$$\pi_k(Y',y') \longrightarrow \pi_k(M,y') \longrightarrow \pi_k(M,Y',y') \xrightarrow{\partial} \pi_{k-1}(Y',y') \longrightarrow \pi_{k-1}(M,y')$$

whose first map is surjective and whose last map is injective by Step 3. The second map is therefore zero, ∂ is zero, and exactness at $\pi_k(M,Y',y')$ gives $\pi_k(M,Y',y') = 0$. For $k = 1$ the

set $\pi_0(Y')$ is a single point, so exactness at $\pi_1(M, Y', y')$ says that the image of $\pi_1(M, y')$, which is trivial by Step 2, is all of $\pi_1(M, Y', y')$. Finally M is path-connected and Y' is nonempty, so every point of M is joined to Y' by a path, which is the condition lemma 5.3.4 places on 0-cells. Hence lemma 5.3.4 applies to maps into the pair (M, Y') from any CW pair whose relative cells have dimension at most n.

Step 5: Surjectivity. Let $\dim X \leq n$ and let $g\colon X \to Z$. By Step 4 and lemma 5.3.4, applied to the CW pair $(X, \varnothing)$ and the map $\iota \circ g$, there is a homotopy from $\iota \circ g$ to a map g_1 with image in Y'. Put $h = p \circ g_1\colon X \to Y$, so that $j \circ h = g_1$. Then $\iota \circ g \simeq j \circ h \simeq \iota \circ f \circ h$, the second homotopy being $K \circ (h \times \mathrm{id}_I)$, and applying r gives $g = r \circ \iota \circ g \simeq r \circ \iota \circ f \circ h = f \circ h$. Hence $[g] = f_*[h]$.

Step 6: Injectivity. Let $\dim X \leq n-1$ and let $h_0, h_1\colon X \to Y$ satisfy $f \circ h_0 \simeq f \circ h_1$ through a homotopy G. Concatenating $K \circ (h_0 \times \mathrm{id}_I)$, then $\iota \circ G$, then the reverse of $K \circ (h_1 \times \mathrm{id}_I)$ gives a homotopy $F\colon X \times I \to M$ from $j \circ h_0$ to $j \circ h_1$, which carries $X \times \partial I$ into Y'. The product cell structure of definition 1.2.8 makes $X \times I$ a CW complex in the product topology, I being a finite complex (lemma 1.2.12(3)), in which $X \times \partial I$ is a subcomplex and every other cell is a product $e \times (0,1)$ of dimension $\dim e + 1 \leq n$. By Step 4 and lemma 5.3.4, the map F is homotopic rel $X \times \partial I$ to a map F_1 with image in Y'. Then $p \circ F_1$ is a homotopy from $p \circ j \circ h_0 = h_0$ to $p \circ j \circ h_1 = h_1$, so $[h_0] = [h_1]$, completing the proof.

The two dimension bounds differ by one because injectivity is the surjectivity argument run on $X \times I$ relative to its two ends, a complex one dimension higher than X. Simple connectivity is used only in Step 3, to compare the homotopy groups of M at points of Y' with those of Z through free homotopy classes, which avoids moving basepoints along the cylinder.

When f is a weak homotopy equivalence the basepoints can be moved, and the same cylinder shows that two invariants defined on all spaces cannot tell Y from Z: singular homology, and homotopy classes of maps out of a CW complex. Step 1 of the proof of lemma 5.3.14 uses nothing about f beyond continuity.

Proposition 5.3.15: Weak Homotopy Equivalences Induce Isomorphisms on Homology

Let $f\colon Y \to Z$ be a weak homotopy equivalence and G an abelian group. Then $f_*\colon H_n(Y; G) \to H_n(Z; G)$ and $f^*\colon H^n(Z; G) \to H^n(Y; G)$ are isomorphisms for every n.

***Proof*:**
Let M, j, ι, r, H, K, p and $Y' = j(Y)$ be as in Step 1 of the proof of lemma 5.3.14, so that $r \circ j = f$, $r \circ \iota = \mathrm{id}_Z$, H is a homotopy from id_M to $\iota \circ r$ fixing Z pointwise, and j is a homeomorphism onto Y' with inverse p.

Step 1: The pair (M, Y') has trivial relative homotopy groups. At a basepoint in Z the maps ι_* and r_* on π_k are inverse isomorphisms, because $r \circ \iota = \mathrm{id}_Z$ and H fixes Z. A point $m \in M$ moves under H along the path $\delta_m(s) = H(m, s)$ from m to $\iota r(m)$. For a based map $\alpha\colon (S^k, s_0) \to (M, m)$ the homotopy $H \circ (\alpha \times \mathrm{id}_I)$ from α to $\iota r \alpha$ carries the basepoint along δ_m, so by the collar argument in the proof of proposition 5.1.15 the class $[\iota r \alpha]$ is the image of $[\alpha]$ under change of basepoint along δ_m, a bijection whose inverse is the change along the reversed path. Hence $(\iota r)_*$ is bijective on $\pi_k(M, m)$, and $r_* = \iota_*^{-1}(\iota r)_*\colon \pi_k(M, m) \to \pi_k(Z, r(m))$ is an isomorphism for every $m \in M$ and $k \geq 1$. On path components r and ι are inverse bijections,

since δ_m joins m to $\iota r(m)$. As $r \circ j = f$ with f a weak homotopy equivalence, $j_* = r_*^{-1} f_*$ is bijective on path components and on every π_k at every basepoint. The exactness argument of Step 2 of the proof of theorem 5.3.6 uses only this property of the inclusion $Y' \hookrightarrow M$ and proposition 5.1.11, and it gives $\pi_k(M, Y', y') = 0$ for all $k \geq 1$ and $y' \in Y'$. Every path component of M meets Y'. So lemma 5.3.4 applies to maps of any CW pair into (M, Y').

Step 2: $H_n(M, Y'; \mathbb{Z}) = 0$. Let $z = \sum_i c_i \sigma_i$ be a relative n-cycle, with distinct singular n-simplices σ_i of M and integers c_i, so that ∂z is a chain in Y'. Take one n-simplex Δ_i for each σ_i, and identify a face of Δ_i with a face of $\Delta_{i'}$, by the order-preserving linear map, whenever σ_i and $\sigma_{i'}$ restrict to the same singular simplex on them. The quotient is a finite Δ-complex K' (definition 3.1.4), the σ_i assemble to a map $\sigma \colon K' \to M$, and distinct simplices of K' go to distinct singular simplices of M. The chain $z' = \sum_i c_i \Delta_i$ satisfies $\sigma_\sharp z' = z$, so the coefficient of an $(n-1)$-simplex τ of K' in $\partial z'$ is the coefficient of $\sigma \circ \tau$ in ∂z. Let L be the subcomplex formed by the $(n-1)$-simplices with nonzero coefficient in $\partial z'$ and their faces. Then $\sigma(L) \subseteq Y'$, and z' is a relative cycle of (K', L). As K' is a finite CW complex with subcomplex L, Step 1 and lemma 5.3.4 make σ homotopic rel L to a map σ' with image in Y'. By proposition 3.4.8, $[z] = \sigma_*[z'] = \sigma'_*[z']$, and σ'_* factors through $H_n(Y', Y') = 0$. Hence $[z] = 0$.

Step 3: Coefficients, and the conclusion. The complex $C_\bullet(M, Y')$ is free on the singular simplices not contained in Y', so the universal coefficient theorems (theorem 4.1.7, theorem 4.1.8) turn Step 2 into $H_n(M, Y'; G) = 0$ and $H^n(M, Y'; G) = 0$ for all n. The long exact sequences of the pair, from theorem 3.4.3 applied to $0 \to C_\bullet(Y'; G) \to C_\bullet(M; G) \to C_\bullet(M, Y'; G) \to 0$ and from theorem 4.1.17, then make the inclusion $Y' \hookrightarrow M$ an isomorphism on $H_*(-; G)$ and on $H^*(-; G)$, and so is j, a homeomorphism onto Y' followed by that inclusion. The map r is a homotopy equivalence, with homotopy inverse ι. Homotopic maps agree on $H_*(-; G)$, by the chain homotopy of theorem 3.3.6 tensored with G, and on $H^*(-; G)$ by proposition 4.1.23, so r_* and r^* are isomorphisms. Therefore $f_* = r_* j_*$ and $f^* = j^* r^*$ are isomorphisms, as we sought to show.

Proposition 5.3.16: Maps out of a CW Complex See Only Weak Homotopy Type

Let $f \colon Y \to Z$ be a weak homotopy equivalence and X a CW complex. Then composition with f is a bijection

$$f_* \colon [X, Y] \longrightarrow [X, Z], \qquad [h] \longmapsto [f \circ h]$$

on free homotopy classes. If moreover X, Y, Z are pointed, the basepoint x_0 of X is a 0-cell and f is based, then $f_* \colon [X, Y]_* \to [X, Z]_*$ is a bijection on based homotopy classes.

A weak homotopy equivalence of path-connected spaces is n-connected for every n, so on free classes this is lemma 5.3.14 with no bound on $\dim X$ and no simple connectivity, the latter being replaced by Step 1 of the proof of proposition 5.3.15.

***Proof*:**
Keep the notation of the proof of proposition 5.3.15. By its Step 1, lemma 5.3.4 applies to maps of every CW pair into (M, Y').

Free classes. Steps 5 and 6 of the proof of lemma 5.3.14 use the conclusion of Step 4 there, that lemma 5.3.4 applies to maps into (M, Y') from CW pairs whose relative cells have dimension at most n. Here it holds for relative cells of every dimension, so the two steps run verbatim with no bound on $\dim X$ and prove that f_* is surjective and injective on free classes.

Based classes. Let y_0 be the basepoint of Y, so $f(y_0) = z_0$ is that of Z, and let $\delta(s) = [y_0, s]$, a path in M from $j(y_0)$ to $\iota(z_0)$ along which r is constant at z_0. For surjectivity let $g\colon X \to Z$ be based. The pair $(X, \{x_0\})$ has the homotopy extension property (proposition 1.3.4), so $\iota \circ g$ is homotopic to a map g_0 with $g_0(x_0) = j(y_0)$ through a homotopy that moves x_0 along the reverse of δ. By lemma 5.3.4 for the CW pair $(X, \{x_0\})$, g_0 is homotopic rel x_0 to a map g_1 with image in Y'. Then $h = p \circ g_1$ is based with $j \circ h = g_1$, and composing the two homotopies with r gives a based homotopy from $r\iota g = g$ to $rg_1 = rjh = fh$, because r is constant along δ. For injectivity let $h_0, h_1\colon X \to Y$ be based and let G be a based homotopy from fh_0 to fh_1. Concatenating $K \circ (h_0 \times \mathrm{id}_I)$, $\iota \circ G$ and the reverse of $K \circ (h_1 \times \mathrm{id}_I)$ gives a homotopy $F\colon X \times I \to M$ from jh_0 to jh_1 whose restriction to $\{x_0\} \times I$ runs along δ, rests at $\iota(z_0)$ and returns along δ. That loop is homotopic rel endpoints to the constant loop at $j(y_0)$, and $(X \times I,\ X \times \partial I \cup \{x_0\} \times I)$ is a CW pair, so the homotopy extension property changes F rel $X \times \partial I$ into a homotopy F' that is constant at $j(y_0)$ on $\{x_0\} \times I$. Then F' carries $X \times \partial I \cup \{x_0\} \times I$ into Y', and lemma 5.3.4 deforms F' rel that subcomplex to a map F'' with image in Y'. Hence $p \circ F''$ is a based homotopy from h_0 to h_1, completing the proof.

Exercise 5.3.1

1. Let $f\colon X \to Y$ be a map of path-connected spaces inducing an isomorphism on π_n for all $n \geq 1$ but where X and Y are not assumed to be CW complexes. Show by example that f need not be a homotopy equivalence, and explain which hypothesis of Whitehead's theorem fails.

2. Let (X, A) be a CW pair such that the inclusion $A \hookrightarrow X$ is a weak homotopy equivalence. Using the compression lemma, prove directly that A is a deformation retract of X, without invoking the mapping-cylinder reduction.

3. Suppose X is a CW complex with $\pi_n(X, x_0) = 0$ for all $n \geq 0$ (a single path component, all higher groups trivial). Prove that X is contractible. Where does the argument use that X is a CW complex?

4. In the CW approximation construction, the first stage attaches 1-cells to surject onto $\pi_1(X)$ and the second attaches 2-cells to kill the kernel. Carry out these first two stages explicitly for X a space with $\pi_1(X) \cong \mathbb{Z}/3\mathbb{Z}$ and identify the resulting 2-complex Z_2 up to homotopy.

5. Let $f\colon Z \to X$ and $f'\colon Z' \to X'$ be CW approximations (theorem 5.3.10) and let $g\colon X \to X'$ be a map. Show that there is a map $\tilde{g}\colon Z \to Z'$ with $f' \circ \tilde{g} \simeq g \circ f$, unique up to homotopy, and that $g \mapsto [\tilde{g}]$ respects composition and identities.

6. Construct an n-connected CW model of the pair (D^{n+1}, S^n) for $n \geq 1$, where $S^n = \partial D^{n+1}$ carries its standard CW structure. Identify the relative cells and verify that the model is n-connected.

5.4 The Hurewicz Theorem

The higher homotopy groups $\pi_n(X)$ of definition 5.1.1 encode the homotopy classes of maps from spheres into X, and the homology groups $H_n(X)$ of chapter 3 encode cycles modulo boundaries. These two families of invariants are manufactured from completely different starting points. The

Hurewicz theorem connects them: in the lowest dimension where a space carries any homotopy, the homotopy and homology groups agree, and the agreement is realized by an explicit map. In dimension 1 this map exhibits $H_1(X)$ as the abelianization of $\pi_1(X)$. In higher dimensions, where the homotopy groups are already abelian, it is an isomorphism on the first nonvanishing level. The theorem is the principal computational tool linking the two theories.

The construction begins with a canonical generator of the homology of a sphere. By section 3.5 the reduced homology $\tilde{H}_n(S^n)$ is infinite cyclic, so it has exactly two generators. Fixing one of them once and for all is what turns a homotopy class into a homology class.

Definition 5.4.1: Fundamental Class of the Sphere

For each $n \geq 1$ fix a generator

$$\iota_n \in H_n(S^n) \cong \mathbb{Z}$$

called the *fundamental class* of S^n. Concretely, give S^n the Δ-complex structure with two n-simplices Δ^n_1 and Δ^n_2 glued along their common boundary. Then ι_n is the homology class of the cycle $\Delta^n_1 - \Delta^n_2$, oriented so that ι_n is the image of ι_{n-1} under the connecting isomorphism $H_n(D^n, S^{n-1}) \xrightarrow{\partial} \tilde{H}_{n-1}(S^{n-1})$ of the long exact sequence of the pair (D^n, S^{n-1}).

The choice of ι_n is the choice of an orientation of S^n. Compatibility under the boundary map ∂ ties the fundamental classes in consecutive dimensions together, which is precisely what makes the Hurewicz map behave well under suspension. Nothing in the theory depends on the particular orientation, only on having fixed one. Reversing the choice in every dimension replaces the Hurewicz map by its negative.

With a fixed generator in hand, every map $f\colon S^n \to X$ produces a homology class in X: push ι_n forward along f. Homotopic maps induce the same map on homology, so this depends only on the homotopy class of f. This is the Hurewicz homomorphism.

Definition 5.4.2: Hurewicz Homomorphism

Let X be a space with basepoint x_0 and let $n \geq 1$. The *Hurewicz homomorphism* is the map

$$h\colon \pi_n(X, x_0) \longrightarrow H_n(X), \qquad h([f]) = f_*(\iota_n)$$

sending the homotopy class of a basepoint-preserving map $f\colon (S^n, s_0) \to (X, x_0)$ to the image of the fundamental class ι_n of definition 5.4.1 under $f_*\colon H_n(S^n) \to H_n(X)$.

The Hurewicz map is the device that converts the spherical, homotopy-theoretic information in π_n into the linear-algebraic information in H_n. Everything in this section is an analysis of how much is lost in the conversion.

The map h is well defined because a homotopy $f \simeq g$ of based maps is in particular a free homotopy, hence $f_* = g_*$ on homology by homotopy invariance (chapter 3), so $f_*(\iota_n) = g_*(\iota_n)$. That h is a homomorphism requires an argument: the group operation on π_n is induced by the pinch map $c\colon S^n \to S^n \vee S^n$ that collapses an equator to a point, and on homology this pinch sends ι_n to (ι_n, ι_n) because the two halves of S^n map to the two wedge summands by maps of degree 1:

Proposition 5.4.3: The Hurewicz Map is a Homomorphism

For every $n \geq 1$ the map $h\colon \pi_n(X, x_0) \to H_n(X)$ of definition 5.4.2 is a group homomorphism. It is natural in X: a based map $\varphi\colon (X, x_0) \to (Y, y_0)$ gives a commuting square $\varphi_* \circ h = h \circ \varphi_*$.

***Proof*:**

Step 1: The pinch map on homology. Recall from definition 5.1.1 that the sum $[f] + [g]$ in $\pi_n(X, x_0)$ is represented by the composite

$$S^n \xrightarrow{c} S^n \vee S^n \xrightarrow{f \vee g} X$$

where c collapses the equatorial $S^{n-1} \subset S^n$ to the wedge point, sending the upper hemisphere onto the first summand and the lower hemisphere onto the second. Write $q_1, q_2\colon S^n \vee S^n \to S^n$ for the two maps that collapse one summand to the basepoint and are the identity on the other. By the wedge axiom for homology (a consequence of the good-pair long exact sequence of theorem 3.4.14, since $S^n \vee S^n$ is the quotient of $S^n \sqcup S^n$ by a basepoint pair) the map

$$(q_{1*}, q_{2*})\colon H_n(S^n \vee S^n) \xrightarrow{\cong} H_n(S^n) \oplus H_n(S^n)$$

is an isomorphism. Each composite $q_i \circ c\colon S^n \to S^n$ collapses one hemisphere to a point and is a homeomorphism on the interior of the other, hence has degree ± 1. Orienting consistently, $q_i \circ c$ has degree 1, so $(q_i \circ c)_*(\iota_n) = \iota_n$. Therefore

$$c_*(\iota_n) = (\iota_n, \iota_n) \in H_n(S^n) \oplus H_n(S^n)$$

under the identification above.

Step 2: Adding the two maps. Under the same identification the inclusions of the two summands realize $H_n(S^n) \oplus H_n(S^n)$ inside $H_n(S^n \vee S^n)$, and $(f \vee g)_*$ acts on the two copies of ι_n by f_* and g_* respectively. Hence

$$h([f] + [g]) = (f \vee g)_*\, c_*(\iota_n) = (f \vee g)_*(\iota_n, \iota_n) = f_*(\iota_n) + g_*(\iota_n) = h([f]) + h([g])$$

so h is a homomorphism.

Step 3: Naturality. For a based map $\varphi\colon (X, x_0) \to (Y, y_0)$ and $[f] \in \pi_n(X, x_0)$ one has $\varphi_*[f] = [\varphi \circ f]$ by definition of the induced map on π_n, and functoriality of homology gives

$$h(\varphi_*[f]) = (\varphi \circ f)_*(\iota_n) = \varphi_*\, f_*(\iota_n) = \varphi_*\, h([f])$$

so the square commutes, completing the proof.

Naturality is what makes h a transformation of functors $\pi_n \Rightarrow H_n$ on the category of based spaces, and it is the key result for every computation in this section: once h is known to be an isomorphism for one space, naturality transports the conclusion across any map. The remaining work is to identify the kernel and cokernel of h in the cases that matter.

Example 5.4.4: The Hurewicz Map for the Circle

Take $X = S^1$ and $n = 1$. By the computation of the fundamental group in chapter 2, $\pi_1(S^1, s_0) \cong \mathbb{Z}$ generated by the class $[\omega]$ of the loop $\omega(t) = (\cos 2\pi t, \sin 2\pi t)$, which is the identity map of S^1. Homology gives $H_1(S^1) \cong \mathbb{Z}$ generated by the fundamental class ι_1. The Hurewicz map sends

$$h([\omega]) = \omega_*(\iota_1) = \mathrm{id}_*(\iota_1) = \iota_1$$

a generator, and more generally the class $[\omega]^k$ is represented by the degree-k map $z \mapsto z^k$, whose induced map on H_1 is multiplication by k, so $h([\omega]^k) = k\,\iota_1$. Thus $h\colon \mathbb{Z} \to \mathbb{Z}$ is the identity, an isomorphism. Since $\pi_1(S^1)$ is already abelian, no abelianization is needed here, and the circle is the simplest instance of the dimension-1 Hurewicz theorem proved next.

The circle suggests the general dimension-1 statement, but a subtlety appears as soon as the fundamental group is nonabelian: $H_1(X)$ is always abelian, while $\pi_1(X)$ need not be, so the best one can hope for in dimension 1 is that h collapses exactly the commutator subgroup. This is the abelianization theorem.

Theorem 5.4.5: Hurewicz Theorem in Dimension One

Let X be a path-connected space with basepoint x_0. Then the Hurewicz homomorphism

$$h\colon \pi_1(X, x_0) \longrightarrow H_1(X)$$

is surjective, and its kernel is the commutator subgroup $[\pi_1, \pi_1]$. Consequently h induces an isomorphism

$$\pi_1(X, x_0)^{\mathrm{ab}} = \pi_1(X, x_0)/[\pi_1, \pi_1] \xrightarrow{\cong} H_1(X)$$

from the abelianization of the fundamental group onto the first homology group.

***Proof*:**

Throughout, regard a loop $\gamma\colon I \to X$ based at x_0 simultaneously as a singular 1-simplex. Since $\gamma(0) = \gamma(1) = x_0$, this 1-simplex is a cycle: $\partial\gamma = \gamma(1) - \gamma(0) = 0$ in the singular chain group $C_0(X)$. For $n = 1$ the fundamental class $\iota_1 \in H_1(S^1)$ is represented by the identity loop, so $h([\gamma])$ is exactly the homology class $[\gamma] \in H_1(X)$ of this 1-cycle. The whole proof is the translation between loop concatenation and chain addition.

Step 1: h is well defined as a homomorphism into H_1, restated for cycles. By proposition 5.4.3 the map h is a homomorphism. Made explicit, this asserts that for loops α, β based at x_0 the 1-chains are homologous,

$$\alpha \cdot \beta \;\sim\; \alpha + \beta$$

where $\alpha \cdot \beta$ is the concatenated loop. To see this directly, consider the singular 2-simplex $\sigma\colon \Delta^2 \to X$ obtained as follows. Label the vertices of Δ^2 as v_0, v_1, v_2. Let $\rho\colon \Delta^2 \to [v_0, v_2]$ be the affine map collapsing v_1 to the midpoint of $[v_0, v_2]$ and fixing v_0, v_2, and set $\sigma = (\alpha \cdot \beta) \circ \rho$ after identifying $[v_0, v_2] \cong I$ affinely. Then ρ sends $[v_0, v_1]$ onto the first half $[v_0, \mathrm{mid}]$ and $[v_1, v_2]$ onto the second half $[\mathrm{mid}, v_2]$, so the edge $[v_0, v_1]$ of σ is α, the edge $[v_1, v_2]$ is β, and the edge $[v_0, v_2]$ is $\alpha \cdot \beta$. Hence

$$\partial\sigma = \beta - (\alpha \cdot \beta) + \alpha$$

so $(\alpha \cdot \beta) - \alpha - \beta = -\partial\sigma$ is a boundary, giving $\alpha \cdot \beta \sim \alpha + \beta$ in $H_1(X)$. In particular h is a

homomorphism, and since $H_1(X)$ is abelian the commutator subgroup $[\pi_1, \pi_1]$ lies in $\ker h$.

Step 2: Two reductions. Record three elementary homology relations among loops, all proved by exhibiting an explicit 2-chain.

1. If γ is the constant loop e_{x_0}, then $\gamma \sim 0$: the constant singular 2-simplex at x_0 has boundary $e_{x_0} - e_{x_0} + e_{x_0} = e_{x_0}$, so e_{x_0} is a boundary.
2. If $\gamma \simeq \gamma'$ rel endpoints, then $\gamma \sim \gamma'$ as 1-cycles: a path homotopy is a map $\Delta^1 \times I \to X$, and subdividing the prism $\Delta^1 \times I$ into two 2-simplices expresses $\gamma - \gamma'$ as a boundary plus chains supported at the (constant) endpoints, which vanish by (a).
3. For the reverse loop $\bar{\gamma}(t) = \gamma(1-t)$ one has $\bar{\gamma} \sim -\gamma$: by (a) and Step 1, $\gamma \cdot \bar{\gamma} \sim \gamma + \bar{\gamma}$ and $\gamma \cdot \bar{\gamma} \simeq e_{x_0}$ rel endpoints, so $\gamma + \bar{\gamma} \sim 0$.

Relations (a) and (b) say h is the well-defined map of definition 5.4.2. Relation (c) records inverses.

Step 3: Surjectivity. Let $z = \sum_i n_i \sigma_i$ be a singular 1-cycle representing a class in $H_1(X)$, where each $\sigma_i \colon I \to X$ is a singular 1-simplex (a path) and $n_i \in \mathbb{Z}$. Splitting σ_i with $n_i < 0$ into $-n_i$ copies of its reverse, using relation (c), one may assume every $n_i = 1$, so $z = \sum_i \sigma_i$ is a sum of paths. The cycle condition $\partial z = 0$ reads

$$\sum_i \big(\sigma_i(1) - \sigma_i(0)\big) = 0 \in C_0(X)$$

meaning that, counted with multiplicity, each point of X appears as an endpoint exactly as often as it appears as a starting point among the σ_i. Since X is path-connected, choose for every point p occurring among the $\sigma_i(0), \sigma_i(1)$ a path λ_p from x_0 to p, with $\lambda_{x_0} = e_{x_0}$ the constant path. Replace each σ_i by the loop

$$\tilde{\sigma}_i = \lambda_{\sigma_i(0)} \cdot \sigma_i \cdot \overline{\lambda_{\sigma_i(1)}}$$

based at x_0. By Step 1, $\tilde{\sigma}_i \sim \lambda_{\sigma_i(0)} + \sigma_i - \lambda_{\sigma_i(1)}$ as 1-chains (using relation (c) of Step 2 for the reversed path). Summing over i,

$$\sum_i \tilde{\sigma}_i \;\sim\; \sum_i \sigma_i + \sum_i \big(\lambda_{\sigma_i(0)} - \lambda_{\sigma_i(1)}\big)$$

and the second sum is $-\sum_p \big(\#\{p \text{ as terminal}\} - \#\{p \text{ as initial}\}\big)\lambda_p$, which is zero by the cycle condition $\partial z = 0$. Hence $z = \sum_i \sigma_i \sim \sum_i \tilde{\sigma}_i$. The latter is a sum of loops based at x_0, so by Step 1 it equals $h\big(\big[\prod_i \tilde{\sigma}_i\big]\big)$. Thus every homology class is in the image of h, and h is surjective.

Step 4: The kernel. Let $\Phi \colon \pi_1(X, x_0)^{\mathrm{ab}} \to H_1(X)$ be the homomorphism induced by h (well defined by Step 1, since $[\pi_1, \pi_1] \subseteq \ker h$). Step 3 shows Φ is surjective. It remains to construct an inverse, which forces Φ to be injective. Define

$$\Psi \colon C_1(X) \longrightarrow \pi_1(X, x_0)^{\mathrm{ab}}$$

on a singular 1-simplex σ (a path) by $\Psi(\sigma) = \big[\lambda_{\sigma(0)} \cdot \sigma \cdot \overline{\lambda_{\sigma(1)}}\big]$, the class of the loop obtained by closing σ up through the chosen paths λ_p as in Step 3, and extend additively to $C_1(X)$, landing in the abelian group π_1^{ab}. Two facts pin down Ψ.

Step 4a: Ψ kills boundaries. For a singular 2-simplex $\tau\colon \Delta^2 \to X$ with edges $a = \tau|_{[v_0,v_1]}$, $b = \tau|_{[v_1,v_2]}$, $c = \tau|_{[v_0,v_2]}$, the boundary is $\partial\tau = b - c + a$. Closing each edge up through the λ_p and using $\Psi(a)\Psi(b) = \Psi(c)$ in π_1^{ab} (because the closed-up loops satisfy $\hat{a}\cdot\hat{b} \simeq \hat{c}$ rel endpoints, the homotopy being τ itself restricted across the 2-simplex) gives

$$\Psi(\partial\tau) = \Psi(a) + \Psi(b) - \Psi(c) = 0$$

in the additive notation of the abelian group. Hence Ψ descends to a homomorphism $\Psi\colon H_1(X) \to \pi_1(X,x_0)^{\mathrm{ab}}$.

Step 4b: Ψ inverts Φ. For a loop γ based at x_0, viewed as a 1-cycle, $\Psi([\gamma]) = [\lambda_{x_0}\cdot\gamma\cdot\overline{\lambda_{x_0}}] = [\gamma]$ since $\lambda_{x_0} = e_{x_0}$. Therefore $\Psi\circ\Phi = \mathrm{id}$ on π_1^{ab}, because $\Phi([\gamma]) = [\gamma] \in H_1(X)$ and Ψ sends it back to $[\gamma]$. A surjective homomorphism with a left inverse is an isomorphism, so Φ is an isomorphism and $\ker h = [\pi_1,\pi_1]$, completing the proof.

Example 5.4.6: First Homology of the Figure Eight and the Genus-Two Surface
Two applications of theorem 5.4.5 compute H_1 from a known fundamental group.

1. For the wedge $X = S^1 \vee S^1$, van Kampen (section 2.2) gives $\pi_1(X) \cong F_2$, the free group on two generators a, b. Its abelianization is $\mathbb{Z}^2$, so

   $$H_1(S^1 \vee S^1) \cong F_2^{\mathrm{ab}} \cong \mathbb{Z}^2$$

   recovering the homology of the figure eight without any chain computation.

2. For the closed orientable surface Σ_2 of genus 2, the fundamental group has presentation

   $$\pi_1(\Sigma_2) = \langle a_1, b_1, a_2, b_2 \,|\, [a_1,b_1][a_2,b_2] = 1\rangle$$

   Abelianizing kills all commutators, so the single relation $[a_1,b_1][a_2,b_2] = 1$ becomes trivial, and the abelianization is the free abelian group on the four generators:

   $$H_1(\Sigma_2) \cong \mathbb{Z}^4$$

   in agreement with the homology computed via the Δ-complex structure in chapter 3.

Dimension 1 is special precisely because π_1 may fail to be abelian. In every higher dimension π_n is already abelian (theorem 5.1.4), and one expects no loss of information at the first level where homotopy appears. The general Hurewicz theorem confirms this. The notion of connectivity isolates "the first level where homotopy appears."

Definition 5.4.7: n-Connected Space

A space X is *n-connected* (for $n \geq 0$) if it is nonempty and $\pi_i(X,x_0) = 0$ for all $0 \leq i \leq n$ and all basepoints x_0. Equivalently, every map $S^i \to X$ with $i \leq n$ extends to a map $D^{i+1} \to X$. A 0-connected space is a path-connected space. A 1-connected space is a simply connected space.

To say X is $(n-1)$-connected is to say the first non-trivial homotopy is in dimension n. The Hurewicz theorem asserts that the first nontrivial homology group appears in the same dimension, and that

the two agree there. Before the theorem, one preparatory fact: connectivity in homotopy forces vanishing of low-dimensional homology, which is half of the statement and is proved by a direct cellular argument.

Lemma 5.4.8: Connectivity Kills Low Homology

Let $n \geq 1$ and let X be an $(n-1)$-connected space. Then the reduced homology groups satisfy $\tilde{H}_i(X) = 0$ for all $i < n$.

***Proof*:**

Step 1: Reduce to a CW complex. By the CW approximation theorem (theorem 5.3.10) there is a CW complex Z and a weak homotopy equivalence $\varphi\colon Z \to X$, that is, a map inducing isomorphisms on all homotopy groups. A weak homotopy equivalence induces isomorphisms on all singular homology groups by proposition 5.3.15, whose proof uses the compression lemma and not the Hurewicz theorem. Since $\pi_i(X) = 0$ for $i \leq n-1$ and φ is a π_*-isomorphism, Z is also $(n-1)$-connected and has the same homology as X. It therefore suffices to prove the lemma when $X = Z$ is a CW complex.

Step 2: Replace X by a CW complex with no low-dimensional cells. A refinement of the CW approximation construction (theorem 5.3.10) produces, for any $(n-1)$-connected CW complex, a homotopy-equivalent CW complex Y with a single 0-cell and no cells in dimensions $1, 2, \ldots, n-1$. One realizes Y by the same cell-attaching procedure used to build a CW approximation, choosing at each stage generators in the lowest dimension and cancelling redundant cells. The precise statement is [21, Corollary 4.16, Proposition 4.15]. Homotopy equivalent spaces have isomorphic homology, so it suffices to treat Y.

Step 3: Cellular homology has nothing in low degrees. For the CW complex Y the cellular chain complex computes homology (section 3.5), and its degree-i group is free abelian on the i-cells of Y. Since Y has a single 0-cell and no cells in dimensions 1 through $n-1$, the cellular chain groups satisfy $C_0^{\text{cell}}(Y) = \mathbb{Z}$ and $C_i^{\text{cell}}(Y) = 0$ for $1 \leq i \leq n-1$. Hence $H_i(Y) = 0$ for $1 \leq i \leq n-1$, and $\tilde{H}_0(Y) = 0$ because Y is path-connected. Transporting back through the homotopy equivalences of Steps 1 and 2, $\tilde{H}_i(X) = 0$ for all $i < n$, as we sought to show.

The lemma gives the vanishing half of the Hurewicz theorem. The substantive half is the isomorphism at the first nonvanishing level. The proof of the full theorem proceeds by induction on connectivity, reducing the general case to the model situation of a wedge of spheres, where both sides can be computed by hand.

Theorem 5.4.9: Hurewicz Theorem

Let $n \geq 2$ and let X be an $(n-1)$-connected space with basepoint x_0. Then:

1. the reduced homology groups vanish, $\tilde{H}_i(X) = 0$ for all $i < n$; and
2. the Hurewicz homomorphism

$$h\colon \pi_n(X, x_0) \xrightarrow{\cong} H_n(X)$$

is an isomorphism.

Together with theorem 5.4.5 (the case $n = 1$, where h induces an isomorphism on abelianization), this identifies the first nonvanishing homotopy group of a simply connected space with its first nonvanishing homology group.

***Proof*:**

Claim (1) is lemma 5.4.8. The work is claim (2). By CW approximation (theorem 5.3.10) and naturality of h (proposition 5.4.3), it suffices to prove the isomorphism for a CW complex, and by Step 2 of lemma 5.4.8 one may take X to be a CW complex with a single 0-cell and no cells in dimensions $1, \dots, n-1$. The proof is induction on the connectivity, organized around the structure of such a complex.

Step 1: The case of a wedge of n-spheres. Suppose first $X = \bigvee_\alpha S^n_\alpha$ is a wedge of n-spheres, $n \geq 2$. Each summand is simply connected (as $n \geq 2$), and a wedge of simply connected spaces is simply connected, so $\pi_1(X) = 0$ and the action of π_1 on π_n is trivial. By the wedge axiom for homology (theorem 3.4.14),

$$H_n(X) = \bigoplus_\alpha H_n(S^n_\alpha) = \bigoplus_\alpha \mathbb{Z}\, \iota_n^\alpha$$

is free abelian on the fundamental classes of the summands (theorem 3.4.14 applied to the wedge). On the homotopy side, the inclusions $S^n_\alpha \hookrightarrow X$ define classes $[j_\alpha] \in \pi_n(X)$, and $\pi_n\big(\bigvee_\alpha S^n_\alpha\big)$ is free abelian on the $[j_\alpha]$ in this first nontrivial dimension. To see this for a finite wedge, the inclusion $\bigvee_\alpha S^n_\alpha \hookrightarrow \prod_\alpha S^n_\alpha$ into the product differs from a homeomorphism only by cells of dimension $\geq 2n$ (the wedge is the n-skeleton of the product CW structure), so it induces an isomorphism on π_n for $n \geq 2$. Since $\pi_n\big(\prod_\alpha S^n_\alpha\big) = \bigoplus_\alpha \pi_n(S^n_\alpha)$ is free abelian on the factor inclusions (homotopy groups commute with products), the same holds for the wedge. An arbitrary wedge reduces to the finite case because any map $S^n \to \bigvee_\alpha S^n_\alpha$ has compact image meeting only finitely many summands (lemma 1.2.4). The Hurewicz map sends

$$h([j_\alpha]) = (j_\alpha)_*(\iota_n) = \iota_n^\alpha$$

a generator of the α-summand of $H_n(X)$, so h carries a basis to a basis and is an isomorphism. The single sphere $X = S^n$ is the case of a one-element wedge.

Step 2: Pairs and the relative reduction. For the inductive step, present X by its cells: write $X = X^{(n)} \cup (\text{cells of dimension} > n)$ where, by the normalization above, the n-skeleton $X^{(n)} = \bigvee_\alpha S^n_\alpha$ is a wedge of n-spheres (a CW complex with one 0-cell and only n-cells), and X is obtained from it by attaching cells of dimension $\geq n+1$. Cells of dimension $\geq n+2$ affect neither π_n nor H_n (attaching a k-cell with $k \geq n+2$ changes only π_i and H_i for $i \geq k-1 \geq n+1$), so for computing

π_n and H_n one may assume X is obtained from $W := \bigvee_\alpha S^n_\alpha$ by attaching $(n+1)$-cells along maps $g_\beta \colon S^n \to W$.

Step 3: Both groups as the same cokernel. Attaching the $(n+1)$-cells e_β along $g_\beta \colon S^n \to W$ gives, on homotopy and on homology, parallel exact sequences. On homology, the cellular chain complex of X in the relevant range is

$$\mathbb{Z}^{\{\beta\}} = C^{\mathrm{cell}}_{n+1}(X) \xrightarrow{d_{n+1}} C^{\mathrm{cell}}_n(X) = \bigoplus_\alpha \mathbb{Z}\,\iota_n^\alpha \to 0$$

where $d_{n+1}(e_\beta) = \sum_\alpha (\deg_{\beta\alpha})\,\iota_n^\alpha$ and $\deg_{\beta\alpha}$ is the degree of the composite $S^n \xrightarrow{g_\beta} W \to S^n_\alpha$ collapsing the other summands. Hence

$$H_n(X) = \operatorname{coker}\Big(\mathbb{Z}^{\{\beta\}} \xrightarrow{d_{n+1}} \bigoplus_\alpha \mathbb{Z}\,\iota_n^\alpha\Big)$$

On homotopy, the relative homotopy long exact sequence of the pair (X, W) (proposition 5.1.11) reads

$$\pi_{n+1}(X,W) \xrightarrow{\partial} \pi_n(W) \xrightarrow{i_*} \pi_n(X) \longrightarrow \pi_n(X,W)$$

Since X is built from W by attaching $(n+1)$-cells, the relative group $\pi_n(X,W) = 0$: every map $(D^n, S^{n-1}) \to (X, W)$ compresses into W because, after cellular approximation, its image meets no cell of $X \setminus W$ in dimension $\le n$ (by the compression lemma lemma 5.3.4). Hence $i_* \colon \pi_n(W) \to \pi_n(X)$ is surjective. Its kernel is computed directly by cellular approximation, with no appeal to the Hurewicz theorem: a based map $S^n \to X$ that is nullhomotopic in X can be pushed into W and its nullhomotopy made cellular, so the kernel of i_* is exactly the subgroup of $\pi_n(W)$ generated by the attaching classes $[g_\beta]$, where $g_\beta \colon S^n \to W$ is the attaching map of the cell e_β (these are the classes that bound a disc in X, namely the cell e_β). Therefore

$$\pi_n(X) = \pi_n(W) \big/ \langle [g_\beta] \rangle = \operatorname{coker}\big(\partial \colon \pi_{n+1}(X,W) \to \pi_n(W)\big)$$

where $\partial[e_\beta] = [g_\beta] \in \pi_n(W)$. The simple connectivity of W (here $n \ge 2$) is what makes $\langle [g_\beta] \rangle$ a subgroup rather than a normal closure with π_1-action.

Step 4: Matching the two cokernels. By Step 1 the Hurewicz map $h_W \colon \pi_n(W) \to H_n(W)$ is an isomorphism carrying $[j_\alpha] \mapsto \iota_n^\alpha$. Under this isomorphism the attaching class $[g_\beta] \in \pi_n(W) = \bigoplus_\alpha \mathbb{Z}[j_\alpha]$ has coordinates $h_W([g_\beta]) = (g_\beta)_*(\iota_n) = \sum_\alpha \deg_{\beta\alpha}\, \iota_n^\alpha = d_{n+1}(e_\beta)$, because the α-coordinate of $(g_\beta)_*(\iota_n)$ is exactly the degree $\deg_{\beta\alpha}$ of the collapsed composite. Thus h_W identifies the relation subgroup $\langle [g_\beta] \rangle \subseteq \pi_n(W)$ with the image of d_{n+1} in $H_n(W)$. By naturality of h (proposition 5.4.3) applied to the inclusion $i \colon W \hookrightarrow X$, the square

$$\begin{array}{ccc} \pi_n(W) & \xrightarrow{i_*} & \pi_n(X) \\ {\scriptstyle\cong}\downarrow{\scriptstyle h_W} & & \downarrow{\scriptstyle h} \\ H_n(W) & \xrightarrow{i_*} & H_n(X) \end{array}$$

commutes, and passing to the cokernels of the two vertical-then-horizontal relation subgroups, h is the induced map

$$\pi_n(X) = \pi_n(W)/\langle [g_\beta] \rangle \xrightarrow{\bar h_W} H_n(W)/\mathrm{im} d_{n+1} = H_n(X)$$

Since h_W is an isomorphism carrying $\langle [g_\beta] \rangle$ onto $\mathrm{im} d_{n+1}$, the induced map $\bar h_W$ on quotients is an isomorphism, and it is h. Therefore $h \colon \pi_n(X) \to H_n(X)$ is an isomorphism, completing the proof.

The theorem says homology is the first nonvanishing homotopy group, made abelian and made computable. Above the connectivity threshold the two invariants can diverge: $\pi_4(S^2) \cong \mathbb{Z}/2$ while $H_4(S^2) = 0$. What the theorem secures is a foothold: knowing that X is $(n-1)$-connected, one computes $\pi_n(X)$ by the often-easier homology functor, and this single input feeds the entire machinery of obstruction theory and Postnikov towers developed later in this chapter. The proof's mechanism, comparing the homotopy and homology cokernels of the same attaching maps, is the template for the relative version, which upgrades the statement to a pair:

Theorem 5.4.10: Relative Hurewicz Theorem

Let (X, A) be a pair of path-connected spaces with A nonempty and simply connected, and let $n \geq 2$. Suppose the pair is $(n-1)$-connected, that is $\pi_i(X, A) = 0$ for $i < n$. Then the relative Hurewicz homomorphism

$$h\colon \pi_i(X, A) \longrightarrow H_i(X, A)$$

is an isomorphism for $i = n$. Moreover $H_i(X, A) = 0$ for $i < n$. Here h sends the class of a map $(D^n, S^{n-1}) \to (X, A)$ to the image of the relative fundamental class of (D^n, S^{n-1}) under the induced map on relative homology.

***Proof Sketch*:**
The argument runs parallel to theorem 5.4.9 line for line, with the absolute pair (X, x_0) replaced by (X, A) and the relative homotopy long exact sequence proposition 5.1.11 in place of the absolute one. Reproducing it in full would repeat the cokernel comparison already carried out above, so the complete details are deferred to [21, Theorem 4.37]. The proof there uses exactly the tools assembled in this chapter: CW approximation (theorem 5.3.10) reduces to a CW pair with no relative cells below dimension n; the compression lemma (lemma 5.3.4) gives $\pi_i(X, A) = 0$ and $H_i(X, A) = 0$ for $i < n$; and the identification of $\pi_n(X, A)$ and $H_n(X, A)$ as the same cokernel of attaching maps proceeds as in Steps 3 and 4 above. The hypothesis that A be simply connected is what guarantees the action of $\pi_1(A)$ on the relative homotopy groups is trivial, so that the cokernel computation is unobstructed. Without that hypothesis the conclusion must be stated in terms of the quotient by this action, as we sought to show.

The relative theorem is the form most often invoked in practice, because pairs come directly from inclusions of subcomplexes and from the mapping cylinder of an arbitrary map. It is the relative theorem that powers the Whitehead theorem (theorem 5.3.6): a weak equivalence between simply connected CW complexes induces isomorphisms on all relative homology, which by relative Hurewicz forces all relative homotopy to vanish, hence the map is a homotopy equivalence. The most immediate consequences, however, are for spheres, where the absolute theorem alone settles the homotopy in the bottom range.

Corollary 5.4.11: Connectivity and Bottom Homotopy of Spheres

For $n \geq 1$ the sphere S^n is $(n-1)$-connected. Consequently

$$\pi_k(S^n) = 0 \quad \text{for } k < n, \qquad \text{and} \qquad \pi_n(S^n) \cong \mathbb{Z}$$

where for $n \geq 1$ the isomorphism $\pi_n(S^n) \cong \mathbb{Z}$ is realized by the Hurewicz map, sending the homotopy class of a map $S^n \to S^n$ to its degree.

***Proof*:**

Step 1: S^n is $(n-1)$-connected. Equip S^n with the CW structure consisting of a single 0-cell and a single n-cell, and give S^k its standard CW structure with k-skeleton equal to all of S^k. For $k < n$, any map $f\colon S^k \to S^n$ is homotopic to a cellular map by the cellular approximation theorem ([21, Theorem 4.8]), and a cellular map sends the k-skeleton S^k of its domain into the k-skeleton of S^n. But the k-skeleton of S^n for $k < n$ is the single 0-cell, so the cellular representative is constant. Hence every map $S^k \to S^n$ with $k < n$ is nullhomotopic, giving $\pi_k(S^n) = 0$ for $k < n$, which is precisely $(n-1)$-connectivity.

Step 2: $\pi_n(S^n) \cong \mathbb{Z}$. For $n \geq 2$, Step 1 shows S^n is $(n-1)$-connected, so the Hurewicz theorem (theorem 5.4.9) applies: the Hurewicz map

$$h\colon \pi_n(S^n) \xrightarrow{\cong} H_n(S^n) \cong \mathbb{Z}$$

is an isomorphism. By definition $h([f]) = f_*(\iota_n) = (\deg f)\,\iota_n$, so under the identification $H_n(S^n) = \mathbb{Z}\iota_n$ the Hurewicz isomorphism sends a class $[f]$ to its degree $\deg f$. For $n = 1$ the same conclusion is the computation $\pi_1(S^1) \cong \mathbb{Z}$ of chapter 2, which agrees with theorem 5.4.5 since $\pi_1(S^1)$ is abelian and $H_1(S^1) \cong \mathbb{Z}$. In all cases $\pi_n(S^n) \cong \mathbb{Z}$ generated by the identity map, of degree 1, completing the proof.

The corollary is the first computation of an infinite family of homotopy groups, and it is the payoff that justifies the entire apparatus. That $\pi_n(S^n) \cong \mathbb{Z}$ with the isomorphism given by degree is the higher-dimensional analogue of the winding-number description of $\pi_1(S^1)$: a self-map of S^n is classified up to homotopy by how many times it wraps the sphere around itself, exactly as a self-map of S^1 is classified by its winding number. The vanishing $\pi_k(S^n) = 0$ for $k < n$ records that a low-dimensional sphere cannot detect a higher-dimensional hole, which is why the interesting homotopy of spheres begins on the diagonal $k = n$ and continues, far less predictably, above it. The contrast with homology, where $\tilde{H}_k(S^n)$ is $\mathbb{Z}$ for $k = n$ and 0 otherwise in *all* degrees, is the Hurewicz map's failure to be an isomorphism above the diagonal: the map $\pi_k(S^n) \to \tilde{H}_k(S^n)$ of definition 5.4.2 is an isomorphism in the first nonvanishing degree $k = n$ (theorem 5.4.9), while for $k > n$ its target is 0 and $\pi_k(S^n)$ is in general not. Ordinary homology and stable homotopy are compared as generalized homology theories in chapter 8.

Forgetting the basepoint extends the classification from based self-maps of a sphere to all of them.

Corollary 5.4.12: Free Homotopy Classes of Self-Maps of Spheres

For $n \geq 1$ the degree is a bijection

$$\deg\colon [S^n, S^n] \xrightarrow{\cong} \mathbb{Z}$$

from the free homotopy classes of maps $S^n \to S^n$ onto the integers. In particular $[S^1, S^1] \cong \mathbb{Z}$ and $[S^n, S^n] \cong \mathbb{Z}$, and two self-maps of S^n are freely homotopic if and only if they have the same degree.

***Proof*:**
Freely homotopic maps induce the same homomorphism on $H_n(S^n) \cong \mathbb{Z}$ (homology is homotopy-invariant), so they have equal degree and deg is a well-defined function on $[S^n, S^n]$. To see it is a bijection, reduce free classes to based ones by proposition 5.1.15, which gives $[S^n, S^n] \cong \pi_n(S^n)/\pi_1(S^n)$. For $n \geq 2$ the sphere S^n is simply connected by the $(n-1)$-connectivity of corollary 5.4.11, so the quotient is trivial and $[S^n, S^n] = \pi_n(S^n)$. For $n = 1$ the group $\pi_1(S^1) \cong \mathbb{Z}$ is abelian, so by corollary 5.1.16 its conjugacy classes are its elements and again $[S^1, S^1] = \pi_1(S^1)$. In both cases the forgetful map is a bijection onto $\pi_n(S^n)$, and by corollary 5.4.11 the group $\pi_n(S^n) \cong \mathbb{Z}$ with the isomorphism given by degree. Composing, $\deg\colon [S^n, S^n] \to \mathbb{Z}$ is a bijection, completing the proof.

This gives the converse direction of proposition 3.5.2(3) for a self-map of a sphere, deferred there to Hopf's degree theorem: degree is a complete invariant of the free homotopy class, so it determines a self-map of S^n up to homotopy. The route here, through the Hurewicz isomorphism of corollary 5.4.11, uses none of the general-position arguments of the classical proof. Maps between spheres of *different* dimensions are a separate and far harder matter, recorded by the homotopy groups $\pi_m(S^n)$ with $m > n$, which degree cannot see.

Exercise 5.4.1

1. Let X be the wedge $\bigvee_{i=1}^{k} S^1$ of k circles. Using theorem 5.4.5, compute $H_1(X)$, and explain why the answer is independent of the (highly nonabelian) structure of $\pi_1(X)$.
2. Let X be the real projective plane $\mathbb{R}\mathrm{P}^2$. Its fundamental group is $\mathbb{Z}/2$. Compute $H_1(\mathbb{R}\mathrm{P}^2)$ using the dimension-one Hurewicz theorem, and compare with the cellular computation of section 3.5.
3. Show that the Hurewicz map $h\colon \pi_n(X) \to H_n(X)$ is the zero map whenever X is contractible, directly from definition 5.4.2, and reconcile this with theorem 5.4.9.
4. Let $X = S^2 \times S^2$. Using theorem 5.4.9 together with the homotopy groups of a product, identify $\pi_2(X)$ and verify the Hurewicz isomorphism against the known second homology $H_2(S^2 \times S^2) \cong \mathbb{Z}^2$.
5. Suppose X is simply connected and $\tilde{H}_i(X) = 0$ for all i. Use the Hurewicz theorem inductively to prove that $\pi_i(X) = 0$ for all i, that is, X is weakly contractible.
6. Give an example of a path-connected space X for which the Hurewicz map $h\colon \pi_2(X) \to H_2(X)$ is *not* an isomorphism, and explain which hypothesis of theorem 5.4.9 fails.

5.5 Eilenberg-MacLane Spaces and the Representation of Cohomology

A sphere S^n is the cleanest geometric object whose homotopy is concentrated in one place only after the bottom dimension, since $\pi_n(S^n) \cong \mathbb{Z}$ but the higher groups $\pi_i(S^n)$ for $i > n$ are very complicated. This section constructs spaces whose homotopy is concentrated in exactly one dimension and is trivial everywhere else. These are the *Eilenberg-MacLane spaces* $K(G, n)$, and their construction proceeds by the familiar method of attaching cells, first realizing the desired group in degree n via the Moore space of section 3.5, then sterilizing every higher homotopy group by gluing on cells that kill its generators.

The payoff is a structural statement about [ordinary] cohomology. The homotopy-concentration of $K(G, n)$ forces, via the Hurewicz theorem and the universal coefficient theorem, the existence of a distinguished cohomology class ι_n, and pulling ι_n back along a map $f\colon X \to K(G, n)$ produces a cohomology class $f^*(\iota_n) \in H^n(X; G)$. The representation theorem asserts that this assignment is a bijection between homotopy classes of maps $[X, K(G, n)]$ and the cohomology group $H^n(X; G)$. Cohomology, defined in chapter 4 through cochains and coboundaries, is thereby revealed to be a homotopy functor in the strongest possible sense: it is *represented* by a single space in each degree. This is the seed from which the entire representable viewpoint on cohomology theories is developed in chapter 8.

Definition 5.5.1: Eilenberg-MacLane Space

Let G be a group (abelian if $n \geq 2$) and let $n \geq 1$. An *Eilenberg-MacLane space of type* (G, n) is a path-connected CW complex X with

$$\pi_n(X) \cong G \qquad \text{and} \qquad \pi_i(X) = 0 \quad \text{for all } i \neq n$$

Any such space is denoted $K(G, n)$.

The defining condition isolates all of the homotopy of $K(G, n)$ in the single dimension n. For $n \geq 2$ the group π_n is abelian by theorem 5.1.4, so the restriction to abelian G in that range is forced rather than chosen. For $n = 1$ the group π_1 may be nonabelian, and the construction below produces a $K(G, 1)$ for any group G, abelian or not. The notation $K(G, n)$ is mild abuse: nothing yet guarantees that two spaces satisfying the definition are the same, and indeed they need not be homeomorphic. What theorem 5.5.4 establishes is that they are homotopy equivalent, which is the only sense in which uniqueness can be expected for an object defined purely by its homotopy groups. The reader should think of $K(G, n)$ not as a particular space but as a homotopy type, the way one thinks of "a point" or "a circle" once the topology has been forgotten modulo homotopy.

Example 5.5.2: The Circle as $K(\mathbb{Z}, 1)$

The circle S^1 is a $K(\mathbb{Z}, 1)$. Its fundamental group is $\pi_1(S^1) \cong \mathbb{Z}$, as computed in chapter 2. For the higher groups, the exponential map $p\colon \mathbb{R} \to S^1$, $t \mapsto e^{2\pi i t}$, is a covering map with contractible total space $\mathbb{R}$. A covering map induces isomorphisms $\pi_i(\mathbb{R}) \xrightarrow{\cong} \pi_i(S^1)$ for all $i \geq 2$, because the homotopy lifting property lets one lift any map $S^i \to S^1$ together with a null-homotopy through $\mathbb{R}$ (this is the long exact sequence of the fibration $\mathbb{Z} \to \mathbb{R} \to S^1$ of theorem 5.2.12, whose fiber $\mathbb{Z}$ is

discrete, hence $\pi_i(\mathbb{Z}) = 0$ for $i \geq 1$). Since $\mathbb{R}$ is contractible, $\pi_i(\mathbb{R}) = 0$ for all $i \geq 1$, so

$$\pi_i(S^1) = 0 \qquad \text{for all } i \geq 2$$

Thus S^1 has $\pi_1 \cong \mathbb{Z}$ and all higher homotopy groups trivial, which is the definition of a $K(\mathbb{Z}, 1)$.

The construction of a general $K(G, n)$ begins where the homology computation of section 3.5 left off. The Moore space $M(G, n)$ built there is a CW complex with the correct homology in degree n. What it lacks is control over its homotopy. The strategy is to start from the Moore space, whose π_n is already G in the simplest cases, and then to attach cells of dimension $n + 2$ and higher so as to annihilate every homotopy group above degree n, all without disturbing what has already been achieved in degrees $\leq n + 1$. The mechanism for killing a homotopy class is the most basic move in cell attachment: a map $\varphi\colon S^m \to X$ representing a class in $\pi_m(X)$ becomes null-homotopic in $X \cup_\varphi e^{m+1}$, because the attached cell provides exactly the disk needed to contract φ.

Lemma 5.5.3: Killing Homotopy Classes by Cell Attachment

Let X be a path-connected CW complex and let $m \geq 1$. There is a CW complex $Y \supseteq X$, obtained from X by attaching cells of dimension $m + 1$, such that the inclusion $X \hookrightarrow Y$ induces an isomorphism $\pi_i(X) \xrightarrow{\cong} \pi_i(Y)$ for $i < m$ and the surjection $\pi_m(X) \twoheadrightarrow \pi_m(Y)$ has $\pi_m(Y) = 0$.

***Proof*:**

Step 1: Attaching the cells. Choose a set of generators $\{[\varphi_\alpha]\}_{\alpha \in A}$ of $\pi_m(X, x_0)$, with representatives $\varphi_\alpha\colon S^m \to X$ that may be taken to be based maps sending the basepoint of S^m to x_0. Form

$$Y \;=\; X \cup_{\{\varphi_\alpha\}} \Big(\bigsqcup_{\alpha \in A} e^{m+1}_\alpha \Big)$$

by attaching an $(m+1)$-cell e^{m+1}_α along φ_α for each α. Since the attaching maps land in X, the space Y is a CW complex containing X as a subcomplex, and X is the m-skeleton-up portion in the sense that Y differs from X only by cells of dimension $m + 1$.

Step 2: Lower homotopy is unchanged. For a CW pair (Y, X) in which Y is obtained from X by attaching cells of dimension $\geq m + 1$, the inclusion induces an isomorphism $\pi_i(X) \to \pi_i(Y)$ for $i < m$ and a surjection for $i = m$. This is the compression lemma (lemma 5.3.4) applied to the m-connected pair (Y, X). Concretely, any map $S^i \to Y$ with $i \leq m$ is homotopic, rel basepoint, to one whose image avoids the interiors of the attached cells of dimension $\geq m + 1$, hence lies in X, because a map of a complex of dimension $i \leq m$ can be pushed off cells of strictly larger dimension. Any homotopy $S^i \times I \to Y$ with $i < m$ (so that $\dim(S^i \times I) = i + 1 \leq m$) may likewise be pushed off those cells into X. Thus $\pi_i(X) \to \pi_i(Y)$ is bijective for $i < m$ and surjective for $i = m$.

Step 3: The chosen generators die. For each α, the attached cell e^{m+1}_α is a map $D^{m+1} \to Y$ restricting on $\partial D^{m+1} = S^m$ to φ_α. This map is precisely a null-homotopy of φ_α inside Y: its restrictions to the spheres of radius t give a homotopy from φ_α at $t = 1$ to the constant map at $t = 0$. Hence each generator $[\varphi_\alpha]$ maps to 0 in $\pi_m(Y)$. Since the $[\varphi_\alpha]$ generate $\pi_m(X)$ and $\pi_m(X) \to \pi_m(Y)$ is surjective by Step 2, the whole of $\pi_m(Y)$ is generated by the images of the $[\varphi_\alpha]$, all of which vanish. Therefore $\pi_m(Y) = 0$, completing the proof.

The lemma performs surgery on a single homotopy group: it removes π_m while leaving everything below it intact. The price is that attaching $(m+1)$-cells may *create* new elements in π_{m+1}, the very next group up, because each new cell is a potential generator of a higher class. This is why the construction of $K(G,n)$ is an infinite process rather than a single step. One kills π_{n+1}, which may disturb π_{n+2}. One then kills π_{n+2}, which may disturb π_{n+3}, and so on upward forever. Because each stage only adds cells of strictly increasing dimension, the homotopy groups in any fixed range below the current stage stabilize, and the infinite union has the desired groups in every degree. This is the content of the existence half of the next theorem.

Theorem 5.5.4: Existence and Uniqueness of $K(G,n)$

Let $n \geq 1$ and let G be a group, abelian if $n \geq 2$.

1. *(Existence)* There exists a CW complex of type $K(G,n)$.
2. *(Uniqueness)* Any two CW complexes of type $K(G,n)$ are homotopy equivalent. More precisely, if X and X' are both of type $K(G,n)$, then any isomorphism $\pi_n(X) \xrightarrow{\cong} \pi_n(X')$ (of groups, respecting the π_1-action when $n=1$) is induced by a homotopy equivalence $X \to X'$, unique up to homotopy.

***Proof*:**

Step 1: Existence, base case $n \geq 2$. Let G be abelian. The Moore space construction of example 3.5.15 produces a CW complex $M = M(G,n)$ with $\tilde{H}_i(M) = 0$ for $i \neq n$ and $H_n(M) \cong G$. Building M with cells in dimensions n and $n+1$ only, one obtains a space whose $(n-1)$-skeleton is a point (a single 0-cell, no cells in dimensions $1, \ldots, n-1$), so M is $(n-1)$-connected: $\pi_i(M) = 0$ for $i < n$. By the Hurewicz theorem (theorem 5.4.9) the Hurewicz map $\pi_n(M) \xrightarrow{\cong} H_n(M) \cong G$ is an isomorphism, so M has the correct homotopy in degree n. This M is the starting point.

Step 1′: Existence, base case $n = 1$. For a possibly nonabelian group G, choose a presentation $G = \langle\, s_\alpha \,|\, r_\beta \,\rangle$. Form the wedge $\bigvee_\alpha S^1_\alpha$ of circles, one for each generator s_α. By Van Kampen's theorem (section 2.2) its fundamental group is the free group on the s_α. For each relator r_β, regarded as a loop, attach a 2-cell e^2_β along it. The resulting 2-complex K^2 has $\pi_1(K^2) \cong G$, again by Van Kampen, since attaching a 2-cell along a loop w imposes the relation $w = 1$ on π_1. Set $M = K^2$. It is 0-connected with $\pi_1(M) \cong G$.

Step 2: Killing the higher groups. Starting from M (in either base case), build an increasing sequence of CW complexes

$$M = X_{n+1} \subseteq X_{n+2} \subseteq X_{n+3} \subseteq \cdots$$

as follows. Given X_m with $\pi_i(X_m) = 0$ for $n < i < m$ and $\pi_n(X_m) \cong G$ and $\pi_i(X_m) = 0$ for $i < n$, apply lemma 5.5.3 with the homotopy group $\pi_m(X_m)$ to obtain $X_{m+1} \supseteq X_m$ by attaching $(m+1)$-cells. By that lemma $\pi_m(X_{m+1}) = 0$, and the inclusion induces isomorphisms $\pi_i(X_m) \xrightarrow{\cong} \pi_i(X_{m+1})$ for all $i < m$. Since $m \geq n+1 > n$, the groups π_n and all groups below it are preserved, so $\pi_n(X_{m+1}) \cong G$ and $\pi_i(X_{m+1}) = 0$ for $i < n$ and for $n < i \leq m$. The induction starts at $m = n+1$ with $X_{n+1} = M$, whose only possibly nonzero homotopy group above n is unconstrained. The inductive hypothesis is met vacuously since there are no integers strictly between n and $n+1$.

Step 3: The union. Set

$$X = \bigcup_{m \geq n+1} X_m$$

with the weak topology, a CW complex. Fix i. Any map $S^i \to X$ has compact image, hence meets only finitely many cells (lemma 1.2.4) and lands in some X_m with $m > i$. Likewise any homotopy $S^i \times I \to X$ lands in some X_m with $m > i+1$. Therefore $\pi_i(X) = \varinjlim_m \pi_i(X_m)$, and for $m > i$ the inclusions $X_m \hookrightarrow X_{m+1} \hookrightarrow \cdots$ induce isomorphisms on π_i by Step 2. Consequently $\pi_i(X) \cong \pi_i(X_{i+1})$, which equals G when $i = n$ and equals 0 when $i \neq n$. Thus X is of type $K(G, n)$, proving existence.

Step 4: Uniqueness, reduction and constructing the map. Let X, X' be of type $K(G, n)$ and let $\theta\colon \pi_n(X) \to \pi_n(X')$ be an isomorphism. First reduce to a convenient cell structure on the source. Since X is $(n-1)$-connected, CW approximation (theorem 5.3.10) provides a CW complex Z homotopy equivalent to X with a single 0-cell and no cells in dimensions $1, \ldots, n-1$. Constructing the map out of Z and composing with the equivalence $Z \simeq X$ suffices, since homotopy equivalence is what is to be proved. Rename Z as X, so X now has trivial $(n-1)$-skeleton.

Choose basepoints $x_0 \in X$, $x_0' \in X'$ and send the $(n-1)$-skeleton $X^{n-1} = \{x_0\}$ to x_0'. For each n-cell e_α^n of X, its attaching map is null-homotopic (it lands in the point X^{n-1}), so the characteristic map gives an element $[e_\alpha^n] \in \pi_n(X)$. Define f on e_α^n by a based map $S^n \to X'$ representing $\theta([e_\alpha^n])$. This defines f on X^n inducing θ on π_n, because $\pi_n(X)$ is generated by the images of the n-cells under the Hurewicz-cellular identification.

Step 5: Uniqueness, extending over higher cells. Extend f over the cells of dimension $> n$ inductively. Suppose f is defined on X^m for some $m \geq n$. An $(m+1)$-cell e^{m+1} of X is attached along $\varphi\colon S^m \to X^m$. To extend f over e^{m+1} amounts to null-homotoping $f \circ \varphi\colon S^m \to X'$, that is, to showing $[f \circ \varphi] = 0$ in $\pi_m(X')$. Now φ is null-homotopic in X (it bounds the cell e^{m+1}), so $[\varphi] = 0$ in $\pi_m(X)$. For $m > n$, however, $\pi_m(X') = 0$ outright, so $[f \circ \varphi] = 0$ automatically and the extension exists. For $m = n$ the class $[f \circ \varphi] = f_*[\varphi] = \theta([\varphi]) = \theta(0) = 0$ vanishes because $[\varphi] = 0$ in $\pi_n(X)$ and θ is a homomorphism. In every case $f \circ \varphi$ is null-homotopic, and choosing a null-homotopy extends f over e^{m+1}. Inducting over all cells (and taking the weak-topology union) yields $f\colon X \to X'$ with $f_* = \theta$ on π_n, while on π_i for $i \neq n$ the map f_* is the zero map between zero groups.

Step 6: Uniqueness, f is an equivalence. The map f induces an isomorphism on every homotopy group: on π_n it is θ, an isomorphism by hypothesis, and on π_i for $i \neq n$ both groups vanish so f_* is trivially an isomorphism. Thus f is a weak homotopy equivalence (definition 5.3.1). Since X and X' are CW complexes, Whitehead's theorem (theorem 5.3.6) upgrades f to a genuine homotopy equivalence.

Step 7: Uniqueness of the map up to homotopy. If $f, g\colon X \to X'$ both induce θ on π_n, the same skeletal argument shows they are homotopic. They agree up to homotopy on X^n because two based maps $S^n \to X'$ inducing the same element of $\pi_n(X')$ are based-homotopic. The obstruction to homotoping them over an $(m+1)$-cell with $m > n$ lives in $\pi_m(X') = 0$. Inducting up the skeleta produces a homotopy $f \simeq g$. Hence the homotopy equivalence realizing θ is unique up to homotopy, as we sought to show.

The two halves of the theorem are complementary uses of the same skeletal philosophy. Existence builds a space by attaching cells to enforce a homotopy condition. Uniqueness builds a map by extending over cells, the obstruction at each stage living in a homotopy group of the target that

the $K(G,n)$ condition has set to zero. The vanishing of the higher homotopy of $K(G,n)$ is what makes the extension and the homotopy unobstructed: there is simply nowhere for an obstruction to live. This is the first appearance of a pattern that the section on obstruction theory will systematize, where cohomology groups with coefficients in homotopy groups measure exactly the failure of such extensions. Here those coefficient groups all vanish, so the extension always succeeds.

Example 5.5.5: $\mathbb{C}\mathrm{P}^\infty$ as $K(\mathbb{Z},2)$

The infinite complex projective space $\mathbb{C}\mathrm{P}^\infty = \bigcup_n \mathbb{C}\mathrm{P}^n$ is a $K(\mathbb{Z},2)$. Recall from chapter 1 that $\mathbb{C}\mathrm{P}^\infty$ is a CW complex with one cell in each even dimension $0, 2, 4, \ldots$ and none in odd dimensions. Its homotopy groups are computed from the fibration

$$S^1 \longrightarrow S^\infty \longrightarrow \mathbb{C}\mathrm{P}^\infty$$

where $S^\infty = \bigcup_n S^{2n-1}$ is the unit sphere in $\mathbb{C}^\infty$ and the map is the quotient $v \mapsto [v]$ identifying v with λv for $|\lambda| = 1$. This is the infinite analogue of the Hopf fibration (example 5.2.7), and it is a fiber bundle, hence a fibration by theorem 5.2.4. The total space S^∞ is contractible: any map from a compact space into S^∞ lands in some finite S^{2n-1}, and S^{2n-1} is contractible inside $S^{2n+1} \subseteq S^\infty$ by sliding to a missing point, so $\pi_i(S^\infty) = \varinjlim_n \pi_i(S^{2n-1}) = 0$ for every i. The long exact sequence of the fibration (theorem 5.2.12) reads

$$\cdots \to \pi_i(S^1) \to \pi_i(S^\infty) \to \pi_i(\mathbb{C}\mathrm{P}^\infty) \to \pi_{i-1}(S^1) \to \cdots$$

Since $\pi_i(S^\infty) = 0$ for all i, the boundary map $\pi_i(\mathbb{C}\mathrm{P}^\infty) \to \pi_{i-1}(S^1)$ is an isomorphism for every i. The fiber S^1 is a $K(\mathbb{Z},1)$ by example 5.5.2, so $\pi_{i-1}(S^1) = \mathbb{Z}$ when $i-1 = 1$ and 0 otherwise. Therefore

$$\pi_i(\mathbb{C}\mathrm{P}^\infty) \cong \pi_{i-1}(S^1) = \begin{cases} \mathbb{Z} & i = 2 \\ 0 & i \neq 2 \end{cases}$$

which exhibits $\mathbb{C}\mathrm{P}^\infty$ as a $K(\mathbb{Z},2)$.

Example 5.5.6: $\mathbb{R}\mathrm{P}^\infty$ as $K(\mathbb{Z}/2,1)$ and the Classifying Space BG

The infinite real projective space $\mathbb{R}\mathrm{P}^\infty = \bigcup_n \mathbb{R}\mathrm{P}^n$ is a $K(\mathbb{Z}/2,1)$. The double cover $S^\infty \to \mathbb{R}\mathrm{P}^\infty$ identifies antipodal points and has total space S^∞, which is contractible by the same argument as in example 5.5.5. This is the universal cover of $\mathbb{R}\mathrm{P}^\infty$: the deck group is $\mathbb{Z}/2$ acting by the antipodal map, so $\pi_1(\mathbb{R}\mathrm{P}^\infty) \cong \mathbb{Z}/2$ by the classification of covering spaces of chapter 2. The higher groups vanish because a covering map induces isomorphisms $\pi_i(S^\infty) \xrightarrow{\cong} \pi_i(\mathbb{R}\mathrm{P}^\infty)$ for $i \geq 2$, and $\pi_i(S^\infty) = 0$. Hence

$$\pi_i(\mathbb{R}\mathrm{P}^\infty) = \begin{cases} \mathbb{Z}/2 & i = 1 \\ 0 & i \neq 1 \end{cases}$$

so $\mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2,1)$. The same mechanism produces $K(G,1)$ for any group G acting freely and properly discontinuously on a contractible CW complex EG, which is locally path-connected by lemma 2.3.2, so that $EG \to EG/G$ is a covering with deck group G ([16, Covering Quotient Theorem]). It is the universal cover of $BG = EG/G$, which is path-connected and, the covering being a local homeomorphism, locally path-connected, so $\pi_1(BG) \cong G$ as groups (proposition 2.3.11), and $\pi_i(BG) \cong \pi_i(EG) = 0$ for $i \geq 2$ as for $\mathbb{R}\mathrm{P}^\infty$. So $BG = K(G,1)$ when it is a CW complex. For $G = \mathbb{Z}/2$ one may take $EG = S^\infty$, recovering $BG = \mathbb{R}\mathrm{P}^\infty$. For $G = \mathbb{Z}$ one may take $EG = \mathbb{R}$, recovering $B\mathbb{Z} = S^1$ as in example 5.5.2.

The last example builds a $K(G,1)$ as the quotient of a contractible space by a free action of G, and every $K(G,1)$ arises in this way. If X is a CW complex of type $K(G,1)$, its universal cover $\widetilde{X}$ is simply connected, the covering map induces isomorphisms $\pi_i(\widetilde{X}) \cong \pi_i(X) = 0$ for $i \geq 2$ as in example 5.5.2, and $\pi_1(X) \cong G$ acts on $\widetilde{X}$ by deck transformations (definition 2.3.8). For a group G, write BG for a CW complex of type $K(G,1)$, which exists and is unique up to homotopy equivalence by theorem 5.5.4, and EG for its universal cover. Section 7.1 shows that the same space classifies principal G-bundles, which is the origin of the name classifying space.

The cellular chains of EG carry the deck action of G. They are a free resolution of $\mathbb{Z}$ over the group ring $\mathbb{Z}[G]$, because the action permutes the cells freely (proposition 4.6.5) and EG is contractible. A projective resolution of $\mathbb{Z}$ is the input from which Ext and Tor over $\mathbb{Z}[G]$ are computed [10, Tor, Ext], so the (co)homology of BG is an invariant of the group G alone.

Definition 5.5.7: Group Homology and Cohomology

Let G be a group with integral group ring $\mathbb{Z}[G]$, and let $\mathbb{Z}$ denote the trivial $\mathbb{Z}[G]$-module, on which every element of G acts as the identity. For a left $\mathbb{Z}[G]$-module M, the *group cohomology* and the *group homology* of G with coefficients in M are

$$H^n(G;M) := \operatorname{Ext}^n_{\mathbb{Z}[G]}(\mathbb{Z},M), \qquad H_n(G;M) := \operatorname{Tor}^{\mathbb{Z}[G]}_n(\mathbb{Z},M), \qquad n \geq 0$$

where Ext is computed from a projective resolution of $\mathbb{Z}$ by left $\mathbb{Z}[G]$-modules and Tor from one by right $\mathbb{Z}[G]$-modules, a left module P being a right module through $p \cdot g = g^{-1}p$.

By [10, The Derived Functors Are Well Defined](1) these groups do not depend on the resolution, up to isomorphism. In degree zero they are the invariants and the coinvariants of M: $H^0(G;M) = \operatorname{Hom}_{\mathbb{Z}[G]}(\mathbb{Z},M)$ is the subgroup M^G of elements fixed by G, and $H_0(G;M) = \mathbb{Z} \otimes_{\mathbb{Z}[G]} M$ is the quotient M_G of M by the subgroup generated by the elements $gm - m$ [10, Degree Zero]. The bar resolution and the applications of group cohomology to algebra are developed in *Homological Algebra* [13] and *Galois Cohomology* [11, Group Cohomology and Homology]. A $K(G,1)$ computes the groups geometrically.

Proposition 5.5.8: Cohomology of BG is Group Cohomology

Let G be a group, let X be a CW complex of type $K(G,1)$ with $\pi_1(X,x_0)$ identified with G, and let M be a left $\mathbb{Z}[G]$-module, regarded as a local coefficient system on X (definition 4.6.1). Then the cellular chain complex $C^{CW}_*(\widetilde{X})$ of the universal cover, augmented by the map $\varepsilon\colon C^{CW}_0(\widetilde{X}) \to \mathbb{Z}$ that sends every 0-cell to 1, is a free resolution of the trivial $\mathbb{Z}[G]$-module $\mathbb{Z}$, and for every $n \geq 0$

$$H^n(X;M) \cong H^n(G;M), \qquad H_n(X;M) \cong H_n(G;M)$$

where the left-hand sides are the (co)homology with local coefficients of definition 4.6.6. In particular $H^n(BG;M)$ and $H_n(BG;M)$ depend only on G and M, and when G acts trivially on M they are the singular cohomology and homology of BG with coefficients in M.

***Proof*:**

Step 1: The universal cover is contractible. Give $\widetilde{X}$ the cell structure lifted from X (proposition 4.6.5). The covering map is a fiber bundle with discrete fiber, so theorem 5.2.4 and theorem 5.2.12 make it an isomorphism on π_i for $i \geq 2$, as in example 5.5.2, and these groups vanish because X is of type $K(G,1)$. The universal cover is path-connected and simply connected. So the map $\widetilde{X} \to \{*\}$ is a weak homotopy equivalence of CW complexes, hence a homotopy equivalence by theorem 5.3.6. Consequently $H_n(\widetilde{X}) = 0$ for $n > 0$ and $H_0(\widetilde{X}) \cong \mathbb{Z}$ (corollary 3.3.7, proposition 3.2.5), and the cellular homology of $\widetilde{X}$ is the same by theorem 3.5.11.

Step 2: A free resolution. By proposition 4.6.5 each $C_n^{CW}(\widetilde{X})$ is a free left $\mathbb{Z}[G]$-module and the boundary maps are $\mathbb{Z}[G]$-linear. The deck transformations permute the 0-cells, so ε is $\mathbb{Z}[G]$-linear for the trivial action on $\mathbb{Z}$, and it is onto because $\widetilde{X}$ has a 0-cell. The group $C_0^{CW}(\widetilde{X})$ is $H_0(\widetilde{X}^0)$, on which ε is induced by the constant map $\widetilde{X}^0 \to \{*\}$, and the cellular boundary d_1 is the connecting map $H_1(\widetilde{X}^1, \widetilde{X}^0) \to H_0(\widetilde{X}^0)$ of the pair (definition 3.5.10). So $\varepsilon \circ d_1$ factors through the composite $H_1(\widetilde{X}^1, \widetilde{X}^0) \to H_0(\widetilde{X}^0) \to H_0(\widetilde{X}^1)$, which vanishes by exactness of the long exact sequence of the pair (theorem 3.4.5). Hence ε induces a surjection from $C_0^{CW}(\widetilde{X})/\mathrm{im} d_1 = H_0^{CW}(\widetilde{X}) \cong \mathbb{Z}$ onto $\mathbb{Z}$. A surjective homomorphism $\mathbb{Z} \to \mathbb{Z}$ is injective, so $\ker \varepsilon = \mathrm{im} d_1$. Exactness at $C_n^{CW}(\widetilde{X})$ for $n \geq 1$ is the vanishing of $H_n^{CW}(\widetilde{X})$ from Step 1. Thus the augmented complex is a free resolution of $\mathbb{Z}$ [10, Free Resolution].

Step 3: The isomorphisms. Free modules are projective [10, Free Modules Are Projective], so Step 2 gives a projective resolution of $\mathbb{Z}$ by left $\mathbb{Z}[G]$-modules. By definition 4.6.6, $H^n(X; M)$ is the cohomology of $\mathrm{Hom}_{\mathbb{Z}[G]}(C_*^{CW}(\widetilde{X}), M)$, which is $\mathrm{Ext}^n_{\mathbb{Z}[G]}(\mathbb{Z}, M)$ computed from this resolution [10, Ext], that is $H^n(G; M)$. Through $c \cdot g = g^{-1}c$ the same complex is a complex of right $\mathbb{Z}[G]$-modules, free on the same bases because $g \mapsto g^{-1}$ is a bijection of G, and so a projective resolution of the trivial right module $\mathbb{Z}$. By definition 4.6.6 again, $H_n(X; M)$ is the homology of $C_*^{CW}(\widetilde{X}) \otimes_{\mathbb{Z}[G]} M$, which is $\mathrm{Tor}_n^{\mathbb{Z}[G]}(\mathbb{Z}, M) = H_n(G; M)$ [10, Tor].

Step 4: Independence and trivial coefficients. The groups $H^n(G; M)$ and $H_n(G; M)$ do not depend on the resolution [10, The Derived Functors Are Well Defined](1), so Step 3 identifies the local-coefficient groups of any two CW complexes of type $K(G,1)$, with any cell structures. When G acts trivially on M, proposition 4.6.7 identifies the local-coefficient groups with the cellular cohomology and homology of X with coefficients in M, and these are the singular groups by proposition 4.1.20 and proposition 3.5.38, completing the proof.

Every $K(G,1)$ computes the (co)homology of G, so any convenient model serves: S^1 for $\mathbb{Z}$, $\mathbb{RP}^\infty$ for $\mathbb{Z}/2$, a wedge of circles for a free group (exercise 5.5.1 (7)). In the other direction, the (co)homology of a $K(G,1)$ with any local coefficients is determined by its fundamental group and the coefficient module. For $G = \mathbb{Z}$ and $X = S^1$ the resolution of Step 2 is the two-term complex of example 4.6.9, and the computation there with the sign module gives $H^0(\mathbb{Z}; \mathbb{Z}_{\mathrm{sgn}}) = 0$ and $H^1(\mathbb{Z}; \mathbb{Z}_{\mathrm{sgn}}) = \mathbb{Z}/2$.

Example 5.5.9: Cohomology of $B(\mathbb{Z}/2)$ and $B\mathbb{Z}$

1. For $G = \mathbb{Z}/2$ take $BG = \mathbb{RP}^\infty$ (example 5.5.6), with one cell in each dimension. By proposition 4.1.20 its cohomology with $\mathbb{Z}/2$ coefficients is computed by the dual of the cellular chain complex, whose k-th boundary map is multiplication by $1 + (-1)^k$ (example 3.5.15). Every coboundary vanishes modulo 2, so $H^k(\mathbb{RP}^\infty; \mathbb{Z}/2) \cong \mathbb{Z}/2$ for every $k \geq 0$, and the same computation gives $H^k(\mathbb{RP}^n; \mathbb{Z}/2) \cong \mathbb{Z}/2$ for $0 \leq k \leq n$ and 0 for $k > n$. By proposition 5.5.8

the first groups are $H^k(\mathbb{Z}/2;\mathbb{Z}/2)$.

The cup product makes these polynomial rings,

$$H^*(\mathbb{RP}^\infty;\mathbb{Z}/2) \cong \mathbb{Z}/2[w], \qquad H^*(\mathbb{RP}^n;\mathbb{Z}/2) \cong \mathbb{Z}/2[w]/(w^{n+1}), \qquad |w| = 1$$

with w the nonzero class in degree one. For $1 \leq m \leq n \leq \infty$ the inclusion of the m-skeleton $\mathbb{RP}^m \hookrightarrow \mathbb{RP}^n$ is onto on H_1 (lemma 3.5.9(3)), and $H^1(-;\mathbb{Z}/2) \cong \mathrm{Hom}(H_1(-),\mathbb{Z}/2)$ naturally, the Ext term vanishing because H_0 is free (theorem 4.1.8). So restriction is injective on $H^1(-;\mathbb{Z}/2)$ and carries w to w. The space $\mathbb{RP}^n$ is a closed n-manifold, covered two-to-one by S^n, and it is $\mathbb{Z}/2$-orientable, as every manifold is (example 4.6.4). Suppose $w^{n-1} \neq 0$ in $H^{n-1}(\mathbb{RP}^{n-1};\mathbb{Z}/2)$, as holds for $n = 2$, where $\mathbb{RP}^1 = S^1$. Restriction is a ring homomorphism, so $w^{n-1} \neq 0$ in $H^{n-1}(\mathbb{RP}^n;\mathbb{Z}/2)$. The cup pairing $H^1 \otimes H^{n-1} \to H^n$ of $\mathbb{RP}^n$ with $\mathbb{Z}/2$ coefficients is nondegenerate (corollary 4.5.10(2)) and both factors are $\mathbb{Z}/2$, so $w^n = w \smile w^{n-1} \neq 0$. By induction $w^n \neq 0$ in $H^n(\mathbb{RP}^n;\mathbb{Z}/2)$ for every n, hence $w^k \neq 0$ for $k \leq n$, which is the truncated polynomial ring. Each $w^k \in H^k(\mathbb{RP}^\infty;\mathbb{Z}/2)$ restricts to $w^k \neq 0$ on $\mathbb{RP}^k$, which gives the polynomial ring. The class w is the first Stiefel-Whitney class of the tautological line bundle (example 7.2.14).

2. For $G = \mathbb{Z}$ the classifying space is $B\mathbb{Z} = S^1$ (example 5.5.2). Its integral cohomology is $H^0(S^1;\mathbb{Z}) = H^1(S^1;\mathbb{Z}) = \mathbb{Z}$ and $H^i(S^1;\mathbb{Z}) = 0$ for $i \geq 2$, matching the group cohomology of $\mathbb{Z}$: a free resolution of $\mathbb{Z}$ over $\mathbb{Z}[\mathbb{Z}] = \mathbb{Z}[t,t^{-1}]$ is the two-term complex $0 \to \mathbb{Z}[t,t^{-1}] \xrightarrow{t-1} \mathbb{Z}[t,t^{-1}] \to \mathbb{Z} \to 0$, so $H^*(\mathbb{Z};\mathbb{Z})$ is concentrated in degrees 0 and 1, each $\cong \mathbb{Z}$. The vanishing above degree one reflects that S^1 is one-dimensional, and in the language of exercise 5.5.1 (7) the group $\mathbb{Z}$ has cohomological dimension one.

For abelian G the spaces $K(G,n)$ carry a distinguished cohomology class. The vanishing of homotopy below degree n forces, through Hurewicz and the universal coefficient theorem, a canonical identification of $H^n(K(G,n);G)$ with $\mathrm{Hom}(G,G)$, and the identity homomorphism singles out a distinguished class. Everything in the representation theorem flows from this class, so it deserves a formal name.

Definition 5.5.10: [Eilenberg-MacLane] Fundamental Class

Let G be abelian and $n \geq 1$, and let $K = K(G,n)$. The Hurewicz map gives an isomorphism $h\colon \pi_n(K) \xrightarrow{\cong} H_n(K)$, whence $H_n(K) \cong G$. For $n \geq 2$ the space K is $(n-1)$-connected, so this is the Hurewicz theorem (theorem 5.4.9), which moreover gives $H_i(K) = 0$ for $0 < i < n$. For $n = 1$ the group $\pi_1(K) = G$ is abelian, so by the dimension-one Hurewicz theorem (theorem 5.4.5) the Hurewicz map induces $H_1(K) \cong \pi_1(K)^{\mathrm{ab}} = G$. The universal coefficient theorem (theorem 4.1.8) then gives

$$H^n(K;G) \xrightarrow{\cong} \mathrm{Hom}(H_n(K),G) \cong \mathrm{Hom}(G,G)$$

the Ext term $\mathrm{Ext}(H_{n-1}(K),G)$ vanishing because $H_{n-1}(K) = 0$ when $n \geq 2$, while for $n = 1$ it is $\mathrm{Ext}(H_0(K),G) = \mathrm{Ext}(\mathbb{Z},G) = 0$ since $H_0(K) \cong \mathbb{Z}$ is free. The *fundamental class* $\iota_n \in H^n(K(G,n);G)$ is the class corresponding under this isomorphism to the identity homomorphism $\mathrm{id}_G \in \mathrm{Hom}(G,G)$.

The fundamental class is the universal cohomology class of degree n with coefficients in G: it is the cohomology class corresponding to the abstract isomorphism $\pi_n(K) \cong G$. Tracing the identifications,

ι_n is the unique class in $H^n(K;G)$ whose evaluation on the Hurewicz image of any $\alpha \in \pi_n(K)$ returns $\alpha \in G$ itself. The choice of id_G is not arbitrary decoration. It is the only homomorphism that makes the class *tautological*, so that pulling it back records a map exactly. The next theorem shows this tautological character propagates: a class on any space X is the pullback of ι_n along the map that the class itself determines.

Example 5.5.11: The Generator of $H^2(\mathbb{CP}^\infty;\mathbb{Z})$

For $K(\mathbb{Z},2) = \mathbb{CP}^\infty$ the fundamental class $\iota_2 \in H^2(\mathbb{CP}^\infty;\mathbb{Z})$ is the standard generator of the cohomology ring. The cellular chain complex of $\mathbb{CP}^\infty$ has one $\mathbb{Z}$ in each even degree and zero in odd degrees, with all boundary maps zero, so $H_i(\mathbb{CP}^\infty) \cong \mathbb{Z}$ for i even and 0 for i odd. In particular $H_2(\mathbb{CP}^\infty) \cong \mathbb{Z}$ and $H_1(\mathbb{CP}^\infty) = 0$. The universal coefficient theorem gives $H^2(\mathbb{CP}^\infty;\mathbb{Z}) \cong \mathrm{Hom}(H_2,\mathbb{Z}) \cong \mathbb{Z}$, and ι_2 corresponds to $\mathrm{id}_\mathbb{Z}$, hence to the homomorphism $H_2(\mathbb{CP}^\infty) \to \mathbb{Z}$ sending a chosen generator to 1. Thus ι_2 is a generator of $H^2(\mathbb{CP}^\infty;\mathbb{Z}) \cong \mathbb{Z}$. The cohomology ring is the polynomial ring $H^\bullet(\mathbb{CP}^\infty;\mathbb{Z}) \cong \mathbb{Z}[\iota_2]$, with ι_2^k generating H^{2k}, matching the cup-product computation of chapter 4: the fundamental class is the polynomial generator.

The representation theorem is the structural endpoint of the section. It says that the assignment $f \mapsto f^*(\iota_n)$, taking a based map into $K(G,n)$ to the pullback of the fundamental class, sets up a bijection between homotopy classes of maps and degree-n cohomology classes. Naturality means the bijection is compatible with maps of the source X, so that $H^n(-;G)$ and $[-,K(G,n)]$ are isomorphic functors. The hypotheses must be stated with care: X is required to have the homotopy type of a CW complex, the homotopy classes are based and the cohomology is reduced, and these based notions agree with the unbased ones because $K(G,n)$ is a simple space (its π_1 acts trivially on all π_n, vacuously so for $n \geq 2$, and abelian π_1 acting on itself by conjugation is trivial for $n = 1$ when G is abelian).

Theorem 5.5.12: Cohomology is Represented by $K(G,n)$

Let G be abelian, $n \geq 1$, and let X be a CW complex (or a space of CW homotopy type) with basepoint. Let $\iota_n \in H^n(K(G,n);G)$ be the fundamental class. The map (where the $*$ represents based homotopies)

$$T\colon [X,K(G,n)]_* \longrightarrow \tilde{H}^n(X;G), \qquad [f] \longmapsto f^*(\iota_n)$$

sending a based homotopy class of maps to the pullback of ι_n is a bijection, natural in X: for any based map $g\colon X' \to X$ the square

$$\begin{array}{ccc} [X,K(G,n)]_* & \xrightarrow{T} & \tilde{H}^n(X;G) \\ {\scriptstyle g^*}\downarrow & & \downarrow{\scriptstyle g^*} \\ [X',K(G,n)]_* & \xrightarrow{T} & \tilde{H}^n(X';G) \end{array}$$

commutes. Here $\tilde{H}^n(X;G)$ is the reduced cohomology of definition 4.1.13, which equals $H^n(X;G)$ because $n \geq 1$ (proposition 4.1.14(1)). Because $K(G,n)$ is a simple space, the based set $[X,K(G,n)]_*$ coincides with the free set $[X,K(G,n)]$, so the bijection holds for free homotopy classes and unreduced cohomology when X is connected as well.

***Proof*:**
Write $K = K(G, n)$ throughout. Naturality is immediate: for $g\colon X' \to X$ and $f\colon X \to K$ one has $(f \circ g)^*(\iota_n) = g^*(f^*(\iota_n))$ by functoriality of $H^n(-; G)$, which is exactly the assertion that the square commutes. The content is that T is a bijection, proved by exhibiting surjectivity and injectivity through cell-by-cell constructions on X.

Step 1: A cohomology class as a cocycle on skeleta. Take cohomology computed cellularly (proposition 4.1.20), legitimate since X is a CW complex. A class $u \in \tilde{H}^n(X; G)$ is represented by a cellular cocycle, a homomorphism $u\colon C_n^{\text{cell}}(X) \to G$ on the free abelian group on the n-cells of X that vanishes on cellular $(n+1)$-boundaries. The strategy is to manufacture a map $f\colon X \to K$ whose pullback $f^*(\iota_n)$ realizes u, by defining f on successive skeleta so that on n-cells it records the value of u.

Step 2: Surjectivity, defining f on X^n. Build a representative K with a single 0-cell and cells only in dimensions $\geq n$ (the cellular model of theorem 5.5.4), with a chosen n-cell e_0^n whose characteristic map represents a generator of $\pi_n(K) \cong G$ corresponding to a chosen generator-system of G. Send the $(n-1)$-skeleton X^{n-1} to the basepoint of K. For each n-cell e_α^n of X, the cocycle assigns a value $u(e_\alpha^n) = g_\alpha \in G$. Define f on e_α^n to be a based map $S^n \to K$ representing the element $g_\alpha \in G \cong \pi_n(K)$ (using that the attaching map of e_α^n goes to the basepoint, so the characteristic map factors through $S^n = D^n/\partial D^n$). This defines a map $f\colon X^n \to K$ on the n-skeleton.

Step 3: Surjectivity, f extends over X^{n+1}. An $(n+1)$-cell e_β^{n+1} is attached along $\varphi_\beta\colon S^n \to X^n$. To extend f over e_β^{n+1} one needs $[f \circ \varphi_\beta] = 0$ in $\pi_n(K) \cong G$. Under the identification $\pi_n(K) \cong G$, the class $[f \circ \varphi_\beta]$ equals $\sum_\alpha (\text{degree of } \varphi_\beta \text{ on } e_\alpha^n) \cdot g_\alpha$, which is precisely $u(\partial e_\beta^{n+1})$, the cellular boundary $\partial e_\beta^{n+1} = \sum_\alpha d_{\beta\alpha} e_\alpha^n$ paired with u. Because u is a cocycle, $u(\partial e_\beta^{n+1}) = (\delta u)(e_\beta^{n+1}) = 0$. Hence $[f \circ \varphi_\beta] = 0$, the map $f \circ \varphi_\beta$ is null-homotopic, and f extends over each e_β^{n+1}, giving $f\colon X^{n+1} \to K$.

Step 4: Surjectivity, f extends over higher skeleta. For $m \geq n+1$, extending f from X^m over an $(m+1)$-cell attached along $\psi\colon S^m \to X^m$ requires $[f \circ \psi] = 0$ in $\pi_m(K)$, and $\pi_m(K) = 0$ since $K = K(G, n)$ and $m > n$. So every such extension is unobstructed, and inducting up the skeleta (with the weak-topology union) produces $f\colon X \to K$.

Step 5: Surjectivity, $f^*(\iota_n) = u$. By construction f sends X^{n-1} to the basepoint and sends e_α^n to a representative of $g_\alpha = u(e_\alpha^n)$. The fundamental class ι_n is characterized by evaluating to the identity under $H^n(K; G) \cong \text{Hom}(H_n(K), G)$, that is, ι_n assigns to the generating n-cell e_0^n of K the corresponding element of G. The pullback $f^*(\iota_n)$ is the cellular cocycle whose value on e_α^n is the value of ι_n on $f_*(e_\alpha^n) \in H_n(K^n, K^{n-1})$, which is the element of G represented by $f|_{e_\alpha^n}$, namely $g_\alpha = u(e_\alpha^n)$. Thus $f^*(\iota_n)$ and u agree as cellular cocycles on n-cells, so $f^*(\iota_n) = u$ in $\tilde{H}^n(X; G)$, and T is surjective.

Step 6: Injectivity, the relative construction. Suppose $f_0, f_1\colon X \to K$ satisfy $f_0^*(\iota_n) = f_1^*(\iota_n)$ in $\tilde{H}^n(X; G)$. To show $[f_0] = [f_1]$, apply the surjectivity construction to the CW pair $(X \times I,\ X \times \{0, 1\})$. The two maps assemble to a map $F\colon X \times \{0, 1\} \to K$ on the subcomplex $X \times \{0, 1\}$, and one seeks to extend F to a homotopy $X \times I \to K$, which is exactly a based homotopy $f_0 \simeq f_1$. The obstruction to extending F over $X \times I$ is the relative analogue of the cocycle in Step 2-3: extending over the 0- and 1-cells of I is given, and extending over an $(n+1)$-cell $e \times I$ of the product requires a relative cocycle to vanish.

Step 7: Injectivity, the obstruction is the difference of the two classes. The cells of $X \times I$ not in

$X \times \{0,1\}$ are of the form $e \times \{(0,1)\}$, the product of a cell e of X with the open 1-cell of I. An n-cell of X contributes an $(n+1)$-cell $e \times I$ to the product. Extending F over $X \times I$ skeleton by skeleton, the first obstruction lives in the cellular cochains $C^{n+1}(X \times I,\ X \times \{0,1\}; \pi_n K)$ and equals, under $\pi_n K \cong G$, the difference cocycle $f_1^*(\iota_n) - f_0^*(\iota_n)$ transported to the product. Concretely, on $e \times I$ for an n-cell e of X, the obstruction value is $u_1(e) - u_0(e)$ where $u_j = f_j^*(\iota_n)$ are the cocycles. By hypothesis u_0 and u_1 are cohomologous, $u_1 - u_0 = \delta v$ for some $v \in C^{n-1}(X; G)$. The freedom in extending F over $X \times I$ at the previous stage is exactly a choice of value in $\pi_n K \cong G$ on each n-cell of $X \times I$ of the form $e \times I$ with e an $(n-1)$-cell of X. These n-cells are indexed by the $(n-1)$-cells of X, so the choices form precisely a cochain in $C^{n-1}(X; G)$. Re-choosing the extension of F over these cells by the cochain v adjusts the obstruction cocycle by δv, replacing $u_1 - u_0$ by $u_1 - u_0 - \delta v = 0$. With the obstruction cocycle now zero, F extends over the $(n+1)$-skeleton of $X \times I$.

Step 8: Injectivity, higher cells. For cells of $X \times I$ of dimension $> n+1$, the obstruction to extending F lies in $\pi_m(K) = 0$ for the relevant $m > n$, so all further extensions are unobstructed exactly as in Step 4. Inducting up the skeleta extends F to all of $X \times I$, producing a based homotopy $f_0 \simeq f_1$. Hence $[f_0] = [f_1]$ and T is injective.

Step 9: Conclusion and the based-versus-free clause. Steps 2-5 give surjectivity and Steps 6-8 give injectivity, so T is a natural bijection. Finally, $K(G,n)$ is simple: for $n \geq 2$ the action of $\pi_1(K) = 0$ on $\pi_n(K)$ is trivial, and for $n = 1$ the abelian group $G = \pi_1(K)$ acts on itself by conjugation, which is trivial because G is abelian. For a simple space Y and connected X of CW type, the forgetful map $[X, Y]_* \to [X, Y]$ from based to free homotopy classes is a bijection. Therefore the bijection T also identifies free homotopy classes with unreduced $H^n(X; G)$ on connected X, completing the proof.

The theorem turns the functor $X \mapsto H^n(X; G)$ into a geometric object. To know all degree-n cohomology of all spaces, with coefficients in G, it suffices to know one space, $K(G,n)$, together with one class ι_n on it. Every cohomology class anywhere is a pullback of ι_n, and homotopic maps give the same class while non-homotopic maps give different classes. Cohomology operations, the natural transformations $H^n(-; G) \to H^m(-; G')$, become in this language nothing but homotopy classes of maps $K(G,n) \to K(G',m)$, recovered by the same Yoneda-style argument applied to the universal class (see [9]). The computation of such operations, the Steenrod squares chief among them, rests entirely on this representability.

5.5.13 Foreshadowing: Brown Representability The representation theorem $\tilde{H}^n(X; G) \cong [X, K(G,n)]_*$ is the prototype of a general phenomenon. The proof used only formal features of $\tilde{H}^n(-; G)$: it is a homotopy-invariant functor on CW complexes, it sends wedges to products, and it has a Mayer-Vietoris-type gluing property. Brown's representability theorem, developed in chapter 8, shows that *any* functor with these properties is represented by some space, so every reasonable cohomology theory, not just ordinary cohomology, has its own sequence of representing spaces. The Eilenberg-MacLane spaces are the first instance, and the sequence $\{K(G,n)\}_n$, linked by weak homotopy equivalences $K(G,n) \to \Omega K(G,n+1)$ (exercise 5.5.1 (1)), is the first example of a spectrum, the central object of stable homotopy theory.

Exercise 5.5.1

1. Show that for an abelian group G and $n \geq 1$ there is a weak homotopy equivalence $K(G,n) \to \Omega K(G,n+1)$, where ΩY denotes the loop space of Y. (Use the long exact sequence of the path-loop fibration $\Omega Y \to PY \to Y$ from example 5.2.15 and a CW approximation of $\Omega K(G,n+1)$.)

2. Prove that $K(G \times H, n) \simeq K(G,n) \times_c K(H,n)$ for abelian G, H and $n \geq 1$, the product carrying the cell topology of lemma 1.2.12. Deduce that $K(\mathbb{Z}^k, n) \simeq (K(\mathbb{Z},n))^k$ and identify $K(\mathbb{Z}^2, 1)$ explicitly.

3. Using theorem 5.5.12, compute the set of homotopy classes $[S^m, K(G,n)]_*$ for all $m, n \geq 1$, and check the answer against the homotopy groups of $K(G,n)$ directly.

4. Let G be abelian and $n \geq 1$. Show that a cohomology operation, that is, a natural transformation $\theta \colon \tilde{H}^n(-;G) \to \tilde{H}^m(-;G')$ of functors on CW complexes, is the same datum as a single class $\theta(\iota_n) \in \tilde{H}^m(K(G,n);G')$ (this is the Yoneda lemma in a topological setting).

5. The torus $T^2 = S^1 \times S^1$ is a $K(\mathbb{Z}^2, 1)$. Verify directly from example 5.5.6 (the BG description) that T^2 is aspherical, and identify the contractible universal cover and the deck group.

6. Show that $\mathbb{R}\mathrm{P}^\infty$ and S^1 are not homotopy equivalent, even though both are Eilenberg-MacLane spaces of degree 1, by exhibiting an algebraic invariant distinguishing them. Why does this not contradict the uniqueness in theorem 5.5.4?

7. Say that a group G has *cohomological dimension* at most d if $H^k(G;M) = 0$ for every $k > d$ and every $\mathbb{Z}[G]$-module M. Using proposition 5.5.8, prove that if some CW complex of type $K(G,1)$ has dimension at most d, then G has cohomological dimension at most d. Deduce that a nontrivial free group has cohomological dimension exactly 1. (The converse holds for $d = 1$, where it is the theorem of Stallings and Swan that a group of cohomological dimension at most one is free, with pro-p analogue *Galois Cohomology* [11, Cohomological Characterization of Free Pro-p-Groups], and for $d \geq 3$, by a theorem of Eilenberg and Ganea. For $d = 2$ it is the Eilenberg-Ganea conjecture, which is open.)

5.6 Postnikov Towers and Obstruction Theory

The homotopy groups $\pi_n(X)$ assembled in section 5.1 record an enormous amount of information about a space, but they do so all at once, with no indication of how the groups in different degrees fit together. A space with $\pi_1 = \mathbb{Z}$ and $\pi_2 = \mathbb{Z}$ could be the torus-like product $S^1 \times \mathbb{C}\mathrm{P}^\infty$, or it could be a twisted assembly in which the action of π_1 on π_2 and a cohomology class entangle the two groups:

Example 5.6.1: Two Assemblies of $\pi_1 = \pi_2 = \mathbb{Z}$
The untwisted assembly is the product

$$P \;=\; S^1 \times \mathbb{C}\mathrm{P}^\infty$$

Since $\mathbb{C}\mathrm{P}^\infty = K(\mathbb{Z},2)$ (example 5.5.5) and $S^1 = K(\mathbb{Z},1)$, the product formula for homotopy groups (example 5.1.7) gives $\pi_1(P) = \mathbb{Z}$, $\pi_2(P) = \mathbb{Z}$, and $\pi_n(P) = 0$ otherwise.

The twisted assembly is a mapping torus. Let $c \colon \mathbb{C}\mathrm{P}^\infty \to \mathbb{C}\mathrm{P}^\infty$ be complex conjugation. On each $\mathbb{C}\mathrm{P}^1$ it reverses orientation, so on $\pi_2(\mathbb{C}\mathrm{P}^\infty) = H_2(\mathbb{C}\mathrm{P}^\infty) = \mathbb{Z}$ it acts by $c_* = -1$. Glue the two

ends of $\mathbb{C}\mathrm{P}^\infty \times [0,1]$ together along c:

$$M \;=\; \mathbb{C}\mathrm{P}^\infty \times [0,1]/(x,0) \sim (c(x),1)$$

This is a fiber bundle $\mathbb{C}\mathrm{P}^\infty \hookrightarrow M \to S^1$, hence a Serre fibration (theorem 5.2.4). Its long exact sequence (theorem 5.2.12)

$$\cdots \to \pi_n(\mathbb{C}\mathrm{P}^\infty) \to \pi_n(M) \to \pi_n(S^1) \to \pi_{n-1}(\mathbb{C}\mathrm{P}^\infty) \to \cdots$$

collapses using $\pi_n(S^1) = 0$ for $n \geq 2$ and $\pi_n(\mathbb{C}\mathrm{P}^\infty) = 0$ for $n \neq 2$: it forces $\pi_1(M) = \pi_1(S^1) = \mathbb{Z}$, $\pi_2(M) = \pi_2(\mathbb{C}\mathrm{P}^\infty) = \mathbb{Z}$, and $\pi_n(M) = 0$ for $n \geq 3$. Thus P and M have *identical homotopy groups in every degree.*

What separates them is the action of π_1 on π_2 (proposition 5.1.12). For the product this action is trivial. For the mapping torus, transporting the fiber once around the base circle applies the gluing map c, so the generator of $\pi_1(M) = \mathbb{Z}$ acts on $\pi_2(M) = \mathbb{Z}$ by $c_* = -1$. A homotopy equivalence $M \simeq P$ would intertwine the two actions through isomorphisms of π_1 and π_2. But the only automorphisms of $\mathbb{Z}$ are ± 1, and conjugating multiplication-by-(-1) by any of these returns multiplication by -1. The trivial action and the sign action are distinct, hence

$$M \;\not\simeq\; P \qquad \text{even though} \qquad \pi_*(M) \cong \pi_*(P) \text{ degree by degree.}$$

Note this does not contradict Whitehead's theorem (theorem 5.3.6) since there is no map $f : M \to P$ where $f_* : \pi_*(M) \to \pi_*(P)$ induces the isomorphism.

The difference is also visible in ordinary homology: a Mayer-Vietoris computation for the mapping torus (theorem 3.4.22) gives $H_2(M) \cong \mathbb{Z}/2$, whereas the Künneth theorem (theorem 4.3.5) gives $H_2(P) \cong \mathbb{Z}$.

The list of homotopy groups cannot distinguish these. A *Postnikov tower* makes that gluing data visible. It resolves a space into a sequence of approximations, each one capturing the homotopy groups up to a fixed degree and nothing above, and it records exactly how each new layer is attached to the ones below by a single cohomology class. The Eilenberg-MacLane spaces and the representability theorem of section 5.5 are the tools that attach layers using a class in a cohomology group.

This same cohomological mechanism governs a second, older problem: given a continuous map defined on part of a space, when does it extend to the whole space? Extending one cell at a time, the obstruction to pushing a map across the cells of a single dimension is a cocycle, and the map extends precisely when its cohomology class vanishes. The k-invariants that build a Postnikov tower and the obstruction classes that extend a map both live in $H^{n+1}(-;\pi_n)$, and both answer a question about the existence of a map by producing a single algebraic invariant whose vanishing is necessary *and* sufficient. This section develops both, and shows that cohomology is the natural home of obstructions.

Throughout, X denotes a connected CW complex, and (where homotopy groups of X enter the construction of a tower) a *simple* space in the sense of definition 5.1.13: the action of π_1 on every π_n is trivial. Simplicity is what lets local homotopy-theoretic data be glued into cohomology with untwisted coefficients. Without it the same constructions go through with local (twisted) coefficients, not developed here.

The first task is to manufacture, from a space X, an approximation that has the correct homotopy groups in low degrees and is trivial above a chosen degree. The mechanism is to attach cells: to kill an element of $\pi_m(X)$, glue on an $(m+1)$-cell along a representing map, and to kill all of π_m in every

degree $m > n$ while leaving π_i for $i \leq n$ untouched, attach cells of dimension $m+1 \geq n+2$. Cells of such dimension cannot create or destroy homotopy in degrees $\leq n$, so the lower groups survive. First, we give a name to such a construction:

Definition 5.6.2: Postnikov Tower

Let X be a connected CW complex. A *Postnikov tower* for X is a sequence of spaces X_n for $n \geq 1$, together with maps

$$f_n \colon X \to X_n \qquad \text{and} \qquad p_n \colon X_n \to X_{n-1}$$

for $n \geq 2$, such that the following hold.

1. The triangle commutes up to homotopy: $p_n \circ f_n \simeq f_{n-1}$ for all $n \geq 2$.
2. The map f_n induces an isomorphism $\pi_i(X) \xrightarrow{\cong} \pi_i(X_n)$ for all $i \leq n$.
3. The space X_n has $\pi_i(X_n) = 0$ for all $i > n$.

The space X_n is the *n-th Postnikov section* of X.

The defining conditions say that X_n is a space with the homotopy groups of X truncated above degree n: it sees $\pi_1(X), \ldots, \pi_n(X)$ faithfully and is blind to everything above. The maps p_n organize these truncations into a tower

$$\cdots \to X_n \xrightarrow{p_n} X_{n-1} \xrightarrow{p_{n-1}} \cdots \to X_2 \xrightarrow{p_2} X_1$$

sitting under X by the compatible family f_n. Reading the tower upward, each stage refines its predecessor by installing one more homotopy group. Reading the maps f_n, the space X maps into every stage, compatibly up to homotopy. For the tower built in theorem 5.6.4, once it is converted into a tower of fibrations, X is weakly equivalent to its limit (the remark after that theorem). The tower is therefore a method of building X, up to weak equivalence, out of its homotopy groups one degree at a time, in exact analogy with how a CW structure builds X out of cells one dimension at a time.

Example 5.6.3: The Postnikov Tower of S^2 Begins

The sphere S^2 has $\pi_1(S^2) = 0$, $\pi_2(S^2) = \mathbb{Z}$, $\pi_3(S^2) = \mathbb{Z}$ (generated by the Hopf map, example 5.2.7), and nonzero homotopy in infinitely many higher degrees. Its Postnikov tower begins

$$X_1 = *, \qquad X_2 = K(\mathbb{Z}, 2) = \mathbb{C}\mathrm{P}^\infty, \qquad X_3 \to X_2$$

Indeed X_1 must be a space with $\pi_i = 0$ for all i, so $X_1 \simeq *$. The section X_2 must have $\pi_2 = \mathbb{Z}$ and all other homotopy groups zero, so X_2 is an Eilenberg-MacLane space $K(\mathbb{Z}, 2)$, realized concretely as $\mathbb{C}\mathrm{P}^\infty$ (example 5.5.5 identifies $S^\infty \to \mathbb{C}\mathrm{P}^\infty$ as a fibration with contractible total space, forcing $\pi_i(\mathbb{C}\mathrm{P}^\infty) = \pi_{i-1}(S^1)$, which is $\mathbb{Z}$ for $i = 2$ and 0 otherwise). The map $f_2 \colon S^2 \to \mathbb{C}\mathrm{P}^\infty$ is the inclusion of the bottom cell, an isomorphism on π_2. The next stage X_3 has $\pi_2 = \pi_3 = \mathbb{Z}$ and nothing else. It is *not* a product $K(\mathbb{Z}, 2) \times K(\mathbb{Z}, 3)$, and the class that measures its failure to be a product is the first nontrivial k-invariant of S^2, computed at the end of this section.

The existence of a Postnikov tower for an arbitrary connected CW complex is not obvious: one must produce, for each n, a space with the prescribed truncated homotopy and fit these into a tower. The

construction attaches cells to X itself.

Theorem 5.6.4: Existence of Postnikov Towers

Every connected CW complex X admits a Postnikov tower. Moreover the tower may be built so that each X_n is a CW complex containing X as a subcomplex, with $f_n\colon X \hookrightarrow X_n$ the inclusion, and so that f_n is the inclusion of a subcomplex obtained from X by attaching cells of dimension $\geq n+2$.

***Proof*:**
Fix $n \geq 1$. The strategy is to enlarge X to a CW complex $X_n \supseteq X$ by attaching cells of dimension at least $n+2$, chosen to kill all homotopy of X in degrees above n while preserving it in degrees $\leq n$.

Step 1: Killing π_{n+1}. Choose a set of generators $\{[\varphi_\alpha]\}_\alpha$ of the group $\pi_{n+1}(X)$, represented by maps $\varphi_\alpha\colon S^{n+1} \to X$. Form

$$Y^{(n+1)} = X \cup \Big(\coprod_\alpha D_\alpha^{n+2} \Big)$$

attaching one $(n+2)$-cell D_α^{n+2} along each φ_α (using $\partial D_\alpha^{n+2} = S^{n+1}$). The inclusion $X \hookrightarrow Y^{(n+1)}$ is built by attaching cells of dimension $n+2$. Two facts control its effect on homotopy. First, cellular approximation (theorem 5.3.10) shows that attaching cells of dimension $\geq n+2$ does not change π_i for $i \leq n$: any map $S^i \to Y^{(n+1)}$ with $i \leq n$, and any homotopy $S^i \times I \to Y^{(n+1)}$ with $i \leq n$ (so the homotopy has source dimension $\leq n+1$), is homotopic rel endpoints into the subcomplex X, since the cells D_α^{n+2} have dimension exceeding both i and $i+1$. Hence

$$\pi_i(X) \xrightarrow{\cong} \pi_i\big(Y^{(n+1)}\big), \qquad i \leq n$$

is an isomorphism. Second, in degree $n+1$, each generator φ_α now bounds the attached cell D_α^{n+2}, so $[\varphi_\alpha] = 0$ in $\pi_{n+1}(Y^{(n+1)})$. Since the $[\varphi_\alpha]$ generate $\pi_{n+1}(X)$ and the inclusion is surjective on π_{n+1} (every $(n+1)$-sphere lands in the $(n+1)$-skeleton, which lies in X), we conclude $\pi_{n+1}(Y^{(n+1)}) = 0$.

Step 2: Killing all higher homotopy. Repeat the procedure in each higher degree, inductively. Having built $Y^{(m)}$ with $\pi_i(Y^{(m)}) \cong \pi_i(X)$ for $i \leq n$ and $\pi_i(Y^{(m)}) = 0$ for $n < i \leq m$, choose generators of $\pi_{m+1}(Y^{(m)})$, represent them by maps $S^{m+1} \to Y^{(m)}$, and attach $(m+2)$-cells along them to form $Y^{(m+1)}$. As in Step 1, attaching cells of dimension $m+2 \geq n+2$ leaves π_i unchanged for $i \leq m$ (in particular for $i \leq n$ the groups remain $\pi_i(X)$, and for $n < i \leq m$ they remain 0) and kills π_{m+1}. Set

$$X_n := \bigcup_{m \geq n+1} Y^{(m)} = X \cup \big(\text{cells of dimension} \geq n+2\big)$$

the increasing union, given the weak topology. A map from a sphere S^i into X_n has compact image, hence meets only finitely many cells (lemma 1.2.4) and lands in some finite stage $Y^{(m)}$ with $m \geq i$. The same holds for a nullhomotopy. Therefore

$$\pi_i(X_n) = \varinjlim_m \pi_i\big(Y^{(m)}\big) = \begin{cases} \pi_i(X), & i \leq n, \\ 0, & i > n. \end{cases}$$

The inclusion $f_n\colon X \hookrightarrow X_n$ is the inclusion of a subcomplex and induces the displayed isomorphisms in degrees $\leq n$. This establishes conditions (2) and (3) of definition 5.6.2, and the supplementary claim that X_n is obtained from X by attaching cells of dimension $\geq n+2$.

Step 3: The tower maps. It remains to produce, for each $n \geq 2$, a map $p_n\colon X_n \to X_{n-1}$ with $p_n \circ f_n \simeq f_{n-1}$, establishing condition (1) of definition 5.6.2. The key observation is that X_n is built from X by attaching cells whose attaching maps become nullhomotopic in X_{n-1}, so the map $f_{n-1}\colon X \to X_{n-1}$ extends across those cells.

Make this precise. By Step 2, the section X_n is obtained from X by attaching cells, all of dimension $\geq n+2$. The space X_{n-1} satisfies conditions (2) and (3) (the construction of Steps 1-2 with index $n-1$ in place of n), so $\pi_i(X_{n-1}) = 0$ for all $i \geq n$. Extend $f_{n-1}\colon X \to X_{n-1}$ over the cells of $X_n \setminus X$ one dimension at a time, starting in dimension $n+2$. Suppose a map h extending f_{n-1} has been built over X together with all cells of $X_n \setminus X$ of dimension $< d$ (where $d \geq n+2$), and let e be a d-cell of $X_n \setminus X$, attached by $\varphi\colon S^{d-1} \to X_n^{(d-1)}$. The attaching sphere lands in the domain of h, so the composite $h \circ \varphi\colon S^{d-1} \to X_{n-1}$ represents a class in $\pi_{d-1}(X_{n-1})$, and $d-1 \geq n+1 \geq n$, so $\pi_{d-1}(X_{n-1}) = 0$ and $h \circ \varphi$ is nullhomotopic. A nullhomotopy of an attaching map is exactly the datum needed to extend a map over the cell it attaches, so h extends over e. Doing this for every d-cell and letting d increase produces a map

$$p_n\colon X_n \to X_{n-1}, \qquad p_n|_X = f_{n-1}$$

defined on all of X_n. Since p_n restricts to f_{n-1} on X and $f_n\colon X \hookrightarrow X_n$ is the inclusion, $p_n \circ f_n = f_{n-1}$ on the nose, which in particular gives $p_n \circ f_n \simeq f_{n-1}$, condition (1) of definition 5.6.2. Constructing p_n for every n assembles the maps of the tower, completing the proof.

The construction is the homotopy-theoretic analogue of forming the quotient of a group by the subgroup generated by certain elements: to kill a homotopy class one attaches a cell whose boundary is a representative, exactly as one kills a group element by imposing a relation. The price is that X_n is typically infinite-dimensional even when X is a finite complex, since killing homotopy in one degree generically creates new homotopy in the next, which must in turn be killed. The tower trades the finiteness of X for the transparency of having one homotopy group visible per stage. The map $X \to X_n$ is an isomorphism on π_i for $i \leq n$ and the target has no higher homotopy, so X_n is the universal target receiving X among n-truncated spaces. This universality (which one proves by a cell-attaching extension argument: a map from X to any n-truncated space extends over the killing cells of dimension $\geq n+2$ since their attaching spheres are nullhomotopic in the target) makes the tower functorial up to homotopy. Recovering X from the whole tower takes more care than recovering its homotopy through a fixed degree. The maps p_n built in Step 3 are not fibrations, and the strict limit of this tower plays no role below. Applying proposition 5.2.16 one stage at a time replaces the tower by a tower of fibrations $\tilde{p}_n\colon \tilde{X}_n \to \tilde{X}_{n-1}$ with $\tilde{X}_n \simeq X_n$, and X still maps compatibly into it because $p_n \circ f_n = f_{n-1}$ on the nose. For a tower of fibrations, the Milnor exact sequence

$$0 \to \varprojlim{}^{1}{}_n \pi_{i+1}(\tilde{X}_n) \to \pi_i\big(\varprojlim_n \tilde{X}_n\big) \to \varprojlim_n \pi_i(\tilde{X}_n) \to 0$$

computes the homotopy groups of the limit. It is proved for Kan fibrations of simplicial sets in [20, Prop. VI.2.15]. The singular complex carries a tower of fibrations of spaces to a tower of Kan fibrations without changing any π_i, and, being a right adjoint, it commutes with the limit. Each tower $\pi_i(\tilde{X}_n)$ is constant for $n \geq i$, so the $\varprojlim^1$ term vanishes and $X \to \varprojlim_n \tilde{X}_n$ is an isomorphism on every π_i, which is a weak homotopy equivalence (definition 5.3.1). Exercise 5.6.1 (2) needs none of this, since it reassembles X from maps into the individual stages.

With existence settled, attention turns to the structural content of a Postnikov tower, which lies in the nature of the maps $p_n\colon X_n \to X_{n-1}$. The next result identifies them: each is a fibration whose fiber is an Eilenberg-MacLane space $K(\pi_n(X), n)$, and (when X is simply connected) it is *principal*, meaning it is, up to weak equivalence over X_{n-1}, pulled back from a path-loop fibration over a $K(\pi_n(X), n+1)$. Principality is what packages the entire map p_n into a single classifying map, the k-invariant.

To set up the statement, recall from section 5.2 that a fibration $F \to E \to B$ has a long exact sequence of homotopy groups (theorem 5.2.12), and from example 5.2.15 that the path space fibration $\Omega K \to PK \to K$ over any space K has contractible total space PK, so the connecting map identifies $\pi_i(\Omega K) \cong \pi_{i+1}(K)$. Recall also from definition 5.5.1 that a $K(\pi, n)$ is a path-connected CW complex with $\pi_n = \pi$ and all other homotopy groups zero, and from theorem 5.5.4 that it is unique up to homotopy equivalence. Combined with the path-loop fibration and a CW approximation, this gives a weak homotopy equivalence $K(\pi, n) \to \Omega K(\pi, n+1)$ (exercise 5.5.1 (1)).

Proposition 5.6.5: The Tower Maps are Fibrations with Eilenberg-MacLane Fiber

Let X be a connected CW complex with Postnikov tower $\{X_n, p_n\}$ as in theorem 5.6.4. For each $n \geq 2$ the map $p_n\colon X_n \to X_{n-1}$ may be converted (up to homotopy) into a fibration whose fiber F is an Eilenberg-MacLane space $K(\pi_n(X), n)$.

***Proof*:**

Replace p_n by a fibration $\tilde{p}_n\colon \tilde{X}_n \to X_{n-1}$ with $\tilde{X}_n \simeq X_n$, using the mapping path space of proposition 5.2.16: set $\tilde{X}_n = E_{p_n} = \{(x, \gamma) : x \in X_n,\ \gamma\colon I \to X_{n-1},\ \gamma(0) = p_n(x)\}$, with $\tilde{p}_n(x, \gamma) = \gamma(1)$. By that proposition the map $X_n \to \tilde{X}_n$, $x \mapsto (x, c_{p_n(x)})$, is a homotopy equivalence, and $\tilde{p}_n$ is a Hurewicz fibration. Let $F = \tilde{p}_n^{-1}(b_0)$ be the fiber over the basepoint.

The long exact sequence of the fibration $F \to \tilde{X}_n \to X_{n-1}$ (theorem 5.2.12) reads

$$\cdots \to \pi_{i+1}(X_{n-1}) \to \pi_i(F) \to \pi_i(X_n) \to \pi_i(X_{n-1}) \to \cdots$$

Insert the known homotopy groups. For $i \leq n-1$, the map $\pi_i(X_n) \to \pi_i(X_{n-1})$ is the isomorphism $\pi_i(X) \to \pi_i(X)$ induced by p_n (both equal $\pi_i(X)$ via f_n, f_{n-1}, and $p_n \circ f_n \simeq f_{n-1}$), so the connecting maps force $\pi_i(F) = 0$ for $i < n$ and the segment around $i = n$ becomes

$$\underbrace{\pi_{n+1}(X_{n-1})}_{=0} \to \pi_n(F) \to \underbrace{\pi_n(X_n)}_{=\pi_n(X)} \to \underbrace{\pi_n(X_{n-1})}_{=0}$$

so $\pi_n(F) \cong \pi_n(X)$. For $i > n$ both $\pi_i(X_n) = 0$ and $\pi_{i+1}(X_{n-1}) = 0$, forcing $\pi_i(F) = 0$. Thus F has $\pi_n(F) = \pi_n(X)$ and all other homotopy groups zero, which is exactly a $K(\pi_n(X), n)$ by definition 5.5.1, as we sought to show.

The proposition says that passing from X_{n-1} to X_n adds precisely the homotopy group $\pi_n(X)$, packaged as the fiber $K(\pi_n(X), n)$ of a fibration. To know X_n given X_{n-1}, it therefore suffices to know how this $K(\pi_n(X), n)$ is fibered over X_{n-1}. The remaining structural input is that this fibration is principal, which converts the fibration data into a single map out of X_{n-1}.

Cohomology also solves the extension problem. Suppose Y is a simple space (so $\pi_1(Y)$ acts trivially on each $\pi_n(Y)$), X is a CW complex with skeleta $X^{(n)}$, and a map $g\colon X^{(n)} \to Y$ is given on the n-skeleton. The question is whether g extends over the $(n+1)$-skeleton, and then over all of X. The

answer is a single cohomology class.

The mechanism is local. Each $(n+1)$-cell e_α^{n+1} of X is attached by a map $\varphi_\alpha\colon S^n \to X^{(n)}$. Composing with g gives a map $g\circ\varphi_\alpha\colon S^n \to Y$, a based map whose homotopy class lies in $\pi_n(Y)$. To extend g over the cell e_α^{n+1} means to fill in a map $D_\alpha^{n+1}\to Y$ agreeing with $g\circ\varphi_\alpha$ on the boundary S^n, which is possible exactly when $g\circ\varphi_\alpha$ is nullhomotopic, that is when $[g\circ\varphi_\alpha]=0$ in $\pi_n(Y)$. Assembling these obstructions over all cells produces a cellular cochain.

Definition 5.6.6: Obstruction Cochain

Let Y be a simple space, X a CW complex, and $g\colon X^{(n)}\to Y$ a map on the n-skeleton. The *obstruction cochain* of g is the cellular $(n+1)$-cochain

$$\mathfrak{o}(g)\in C^{n+1}\big(X;\,\pi_n(Y)\big),\qquad \mathfrak{o}(g)(e_\alpha^{n+1})=[\,g\circ\varphi_\alpha\,]\in\pi_n(Y)$$

assigning to each $(n+1)$-cell e_α^{n+1}, attached by $\varphi_\alpha\colon S^n\to X^{(n)}$, the homotopy class of $g\circ\varphi_\alpha$.

The obstruction cochain records, cell by cell, the local failure of g to extend. Its vanishing on a particular cell means g extends over that cell. Its vanishing identically means g extends over the whole $(n+1)$-skeleton. The substance of the theory is that $\mathfrak{o}(g)$ is automatically a cocycle, and that its cohomology class (not the cochain itself) is the honest obstruction: the class vanishes precisely when g can be *modified on the n-skeleton*, without changing it on the $(n-1)$-skeleton, so as to extend over the $(n+1)$-skeleton.

Theorem 5.6.7: The Obstruction to Extension

Let Y be a simple space, X a CW complex, and $g\colon X^{(n)}\to Y$ a map defined on the n-skeleton. Then:

1. The obstruction cochain $\mathfrak{o}(g)\in C^{n+1}(X;\pi_n(Y))$ is a cocycle, so it determines a class $[\mathfrak{o}(g)]\in H^{n+1}(X;\pi_n(Y))$.
2. The restriction $g|_{X^{(n-1)}}$ extends to a map $X^{(n+1)}\to Y$ if and only if $[\mathfrak{o}(g)]=0$.

Consequently, if $[\mathfrak{o}(g)]=0$ then g may be altered on the n-skeleton, rel $X^{(n-1)}$, to a map that extends over the $(n+1)$-skeleton.

***Proof*:**
Write $\pi=\pi_n(Y)$.

Step 1: $\mathfrak{o}(g)$ is a cocycle. Recognize $\mathfrak{o}(g)$ as a composite of two homomorphisms, after which the cocycle property reduces to exactness of a pair sequence. Write

$$\partial\colon \pi_{n+1}\big(X^{(n+1)},X^{(n)}\big)\to\pi_n\big(X^{(n)}\big),\qquad [F]\mapsto[F|_{S^n}]$$

for the connecting boundary of the pair (proposition 5.1.11), and $g_*\colon \pi_n(X^{(n)})\to\pi_n(Y)$ for the map induced by g. The relative group $\pi_{n+1}(X^{(n+1)},X^{(n)})$ is free abelian on the classes $[\chi_\alpha]$ of the characteristic maps $\chi_\alpha\colon(D^{n+1},S^n)\to(X^{(n+1)},X^{(n)})$ of the $(n+1)$-cells, one generator per cell, with $\partial[\chi_\alpha]=[\varphi_\alpha]\in\pi_n(X^{(n)})$. That relative group is abelian since $n\geq 1$, and identifies with the cellular chain group $C_{n+1}(X)$ in degree $n+1$ (definition 5.1.8, proposition 5.1.11). Simplicity of Y makes $\pi_n(Y)$ abelian with trivial $\pi_1(Y)$-action, so g_* is a well-defined homomorphism

independent of basepoint paths. Reading off the cellular basis,

$$(g_* \circ \partial)[\chi_\alpha] = g_*[\varphi_\alpha] = [g \circ \varphi_\alpha] = \mathfrak{o}(g)(e_\alpha^{n+1})$$

so the obstruction cochain *is* the homomorphism $\mathfrak{o}(g) = g_* \circ \partial \colon C_{n+1}(X) \to \pi_n(Y)$.

Now compute the coboundary on an $(n+2)$-cell e_β^{n+2}, with characteristic map χ_β and attaching map $\Phi_\beta \colon S^{n+1} \to X^{(n+1)}$. The cellular boundary is $\partial_{n+2}[\chi_\beta] = j\,[\Phi_\beta]$, where $[\Phi_\beta] \in \pi_{n+1}(X^{(n+1)})$ is the absolute class of the attaching map and j is the map $\pi_{n+1}(X^{(n+1)}) \to \pi_{n+1}(X^{(n+1)}, X^{(n)})$ induced by inclusion (the standard identification of the cellular boundary $\partial_{n+2} e_\beta^{n+2} = \sum_\alpha d_{\beta\alpha}\, e_\alpha^{n+1}$ with the degrees $d_{\beta\alpha}$ of Φ_β on the $(n+1)$-cells). Therefore

$$\delta\mathfrak{o}(g)(e_\beta^{n+2}) = \mathfrak{o}(g)\big(\partial_{n+2}[\chi_\beta]\big) = (g_* \circ \partial \circ j)[\Phi_\beta]$$

By exactness of the long exact homotopy sequence of the pair $(X^{(n+1)}, X^{(n)})$ at the term $\pi_{n+1}(X^{(n+1)}, X^{(n)})$, the composite $\partial \circ j = 0$: an absolute class pushed into the relative group has constant boundary restriction $F|_{S^n}$. Hence $\delta\mathfrak{o}(g)(e_\beta^{n+2}) = g_*(0) = 0$ for every β, so $\delta\mathfrak{o}(g) = 0$ and $\mathfrak{o}(g)$ is a cocycle.

Step 2: Recoordinatizing g on n-cells. Fix g on $X^{(n-1)}$ and consider all maps $g' \colon X^{(n)} \to Y$ agreeing with g on $X^{(n-1)}$. Two such, g and g', restrict to maps on each n-cell that agree on its boundary S^{n-1}. Gluing the two disks along their common boundary yields a map $S^n \to Y$, whose class is an element $d(g, g')(e_\alpha^n) \in \pi_n(Y)$ for each n-cell e_α^n. This defines the *difference cochain* $d(g, g') \in C^n(X; \pi)$. Every cochain arises this way: given any $c \in C^n(X; \pi)$, one constructs g' agreeing with g on $X^{(n-1)}$ and realizing $d(g, g') = c$, by modifying g on the interior of each n-cell using a representative of $c(e_\alpha^n) \in \pi_n(Y)$ (collapse a small disk in the cell's interior to a point, map the resulting pinch sphere by the representative, and keep the old map on the complement). The relation between difference cochains and obstruction cochains is the coboundary formula

$$\mathfrak{o}(g') - \mathfrak{o}(g) = \delta\, d(g, g') \tag{5.1}$$

which is verified cell by cell: on an $(n+1)$-cell e_α^{n+1} with attaching map φ_α, the classes $[g' \circ \varphi_\alpha]$ and $[g \circ \varphi_\alpha]$ differ by the sum over the boundary n-cells of e_α^{n+1} of the difference classes $d(g, g')$, weighted by the incidence degrees, which is exactly $(\delta d(g, g'))(e_\alpha^{n+1})$. The identity (5.1) shows that as g' ranges over all modifications of g rel $X^{(n-1)}$, the obstruction cochain $\mathfrak{o}(g')$ ranges over the entire coset $\mathfrak{o}(g) + \delta C^n(X; \pi)$. Therefore the cohomology class $[\mathfrak{o}(g)] \in H^{n+1}(X; \pi)$ depends only on $g|_{X^{(n-1)}}$, and is unchanged by any modification of g on the n-skeleton rel $X^{(n-1)}$.

Step 3: Extension criterion. Suppose first that $g|_{X^{(n-1)}}$ extends to a map $h \colon X^{(n+1)} \to Y$. Then $h|_{X^{(n)}}$ is a map on the n-skeleton agreeing with g on $X^{(n-1)}$, and since h extends over every $(n+1)$-cell, $\mathfrak{o}(h)(e_\alpha^{n+1}) = [h \circ \varphi_\alpha] = 0$ for all α (the map h on the cell is a nullhomotopy of $h \circ \varphi_\alpha$). Thus $\mathfrak{o}(h) = 0$, and by (5.1) with $g' = h|_{X^{(n)}}$,

$$[\mathfrak{o}(g)] = [\mathfrak{o}(h|_{X^{(n)}})] = [0] = 0$$

in $H^{n+1}(X; \pi)$. So extendability forces $[\mathfrak{o}(g)] = 0$.

Conversely, suppose $[\mathfrak{o}(g)] = 0$, so $\mathfrak{o}(g) = \delta c$ for some $c \in C^n(X; \pi)$. By the surjectivity established in Step 2, there is a map $g' \colon X^{(n)} \to Y$ agreeing with g on $X^{(n-1)}$ with $d(g, g') = -c$. Then (5.1) gives

$$\mathfrak{o}(g') = \mathfrak{o}(g) + \delta\, d(g, g') = \delta c - \delta c = 0$$

as cochains. An obstruction cochain that vanishes identically means $[g' \circ \varphi_\alpha] = 0$ for every $(n+1)$-cell, so $g' \circ \varphi_\alpha$ is nullhomotopic and g' extends over each $(n+1)$-cell. These extensions assemble to a map $X^{(n+1)} \to Y$ extending $g' \simeq g$ rel $X^{(n-1)}$. This proves both directions of (2), and the final assertion is the construction of g' just given, completing the proof.

The theorem isolates exactly what cohomology measures in this context. The cochain $\mathfrak{o}(g)$ is a *local* record, one element of $\pi_n(Y)$ per $(n+1)$-cell, of whether the map extends over that cell. The passage to the cohomology class $[\mathfrak{o}(g)]$ quotients out the freedom to modify g on the n-skeleton: a coboundary δc can always be absorbed by recoordinatizing g, so only the class survives as a genuine obstruction. This is why cohomology, and not the cochain group, is the natural home of obstructions: the cochain depends on a choice (the map g on n-cells), while the class depends only on the geometrically meaningful data (g on the $(n-1)$-skeleton). The same coefficient group $\pi_n(Y)$ and the same degree H^{n+1} appear in the k-invariant of definition 5.6.11 below, and for the same reason: building a map across cells of dimension $n+1$ tests π_n of the target, and consistency across cells of dimension $n+2$ produces the cocycle condition, landing the obstruction in H^{n+1}. The k-invariant is the special case where the "map being extended" is the section that would split the Postnikov fibration.

Example 5.6.8: Extending a Map over a Moore Space Cell

Let $Y = K(\pi, n)$ and take the two-cell complex $X = S^n \cup_d e^{n+1}$, the mapping cone of a degree-d self-map of S^n (when $\pi = \mathbb{Z}$ this is the Moore space $M(\mathbb{Z}_d, n)$ of section 3.5). Its n-skeleton is $X^{(n)} = S^n$, with the single $(n+1)$-cell attached by the degree-d map $\varphi\colon S^n \to S^n$. A map $g\colon X^{(n)} = S^n \to K(\pi, n)$ is, by representability (theorem 5.5.12), the same datum as a class in $[S^n, K(\pi,n)] = \tilde{H}^n(S^n;\pi) = \pi$. Write $a \in \pi$ for that class, so that $g_*\colon \pi_n(S^n) = \mathbb{Z} \to \pi_n(K(\pi,n)) = \pi$ carries the generator to a.

The obstruction cochain has a single value, on the lone $(n+1)$-cell:

$$\mathfrak{o}(g)(e^{n+1}) = [g \circ \varphi] = g_*([\varphi]) = g_*(d \cdot 1) = d \cdot a \in \pi$$

using that φ has degree d, so $[\varphi] = d \in \pi_n(S^n) = \mathbb{Z}$. The fixed map g extends over X exactly when $g \circ \varphi$ is nullhomotopic, that is, when the cochain value vanishes:

$$g \text{ extends over } X \iff d \cdot a = 0 \text{ in } \pi$$

the criterion governing this two-cell complex. For $\pi = \mathbb{Z}$ and $d \geq 1$ this forces $a = 0$: only the nullhomotopic g extends. For $\pi = \mathbb{Z}_d$ every a satisfies $d\,a = 0$, so every g extends.

The cohomology class refines this to a statement about the basepoint datum $g|_{X^{(n-1)}}$. The cellular cochain complex of X in the relevant degrees is $C^n(X;\pi) = \pi \xrightarrow{\times d} C^{n+1}(X;\pi) = \pi$, so $H^{n+1}(X;\pi) = \pi/d\pi$ and the coboundaries are exactly $d\pi$. Since $\mathfrak{o}(g) = da \in d\pi$, the class $[\mathfrak{o}(g)] = 0$ for *every* a, reflecting that $g|_{X^{(n-1)}}$ is the basepoint, which always extends (by the constant map) once one is permitted to alter g on the n-cell. The contrast is exactly the content of theorem 5.6.7: the vanishing cochain $\mathfrak{o}(g) = 0$ controls extension of the *fixed* g, while the vanishing class $[\mathfrak{o}(g)] = 0$ controls extension of $g|_{X^{(n-1)}}$ after modification on the n-skeleton. The two coincide precisely when the n-skeleton offers no modification freedom, here visible in $\tilde{H}^n(X;\pi) = \ker(\times d\colon \pi \to \pi)$, which is 0 for $\pi = \mathbb{Z}$ (no freedom, so $a = 0$ is forced) and all of $\mathbb{Z}_d$ for $\pi = \mathbb{Z}_d$ (full freedom).

Proposition 5.6.9: Obstructions to Sections and Lifts

Let $p\colon E \to X$ be a fibration over a connected CW complex X with path-connected fiber F, such that $\pi_1(F)$ and $\pi_1(X)$ act trivially on every $\pi_m(F)$, and let s be a section of p over the n-skeleton $X^{(n)}$, where $n \geq 1$. Over each $(n+1)$-cell, lifting through p the nullhomotopy of its attaching map given by the cell, starting from s on the attaching sphere, ends at a map of S^n into a fiber. Its class in $\pi_n(F)$ defines a cellular cocycle $\mathfrak{o}(s) \in C^{n+1}(X;\pi_n(F))$, and $[\mathfrak{o}(s)] = 0$ in $H^{n+1}(X;\pi_n(F))$ if and only if $s|_{X^{(n-1)}}$ extends to a section over $X^{(n+1)}$. If moreover $\pi_i(F) = 0$ for $i < n$, the class $[\mathfrak{o}(s)]$ does not depend on s, and for a cellular map $f\colon Y \to X$ of connected CW complexes the class for the pullback fibration $f^*E \to Y$ is $f^*[\mathfrak{o}]$, where $[\mathfrak{o}]$ denotes this common class. A lift of a map $g\colon X \to B$ across a fibration over B is a section of the pullback fibration $g^*E \to X$, so the same criterion governs lifts.

***Proof Key ideas*:**
Over a single cell the problem is the extension problem of theorem 5.6.7 with target F, and the cocycle property and the extension criterion follow the proof of that theorem cell by cell. The triviality of the two actions identifies the groups π_n of all the fibers with the single group $\pi_n(F)$. The verification that the class is well defined and has the stated property uses the homotopy lifting property repeatedly and is long, and it is not repeated here. It is carried out for fiber bundles in [23, Section 3.3, Lemma 3.18 and Proposition 3.19], with the hypotheses of this proposition stated before Lemma 3.18. The independence of the class from s when $\pi_i(F) = 0$ for $i < n$, and its naturality under cellular maps, are shown in the same section [23, p. 102]. That argument uses the bundle structure only over single cells: to lift homotopies, to straighten a lifted nullhomotopy near the fiber (p. 100), and, in the converse half of Proposition 3.19, to change a section over an n-cell by a prescribed element of $\pi_n(F)$ (p. 101). For a fibration the relative homotopy lifting property over a cell does all three.

Without the triviality of the action of $\pi_1(X)$ the same construction produces a class in cohomology with local coefficients in the system of groups π_n of the fibers [23, p. 102]. The Postnikov k-invariant of definition 5.6.11 below is the obstruction to sectioning the tower fibration p_n, which is why the same cohomology group hosts both the tower's gluing data and the obstructions.

A map $p\colon E \to B$ is *principal* if it is, up to weak equivalence over B, the pullback of a path fibration: there are a space B', a map $k\colon B \to B'$, and a weak homotopy equivalence $e\colon E \to k^*PB'$ with $q \circ e = p$, where $k^*PB' = \{(b,\gamma) \in B \times PB' : \gamma(1) = k(b)\}$, $q(b,\gamma) = b$, and $PB' \to B'$ is the path fibration of definition 5.2.14, with fiber $\Omega B'$. When $B' = K(\pi, n+1)$, the fiber $\Omega B'$ is weakly equivalent to $K(\pi, n)$ (exercise 5.5.1 (1)), matching the fiber of the tower map. The classifying map k then carries all the information of the fibration. The Postnikov tower of a simply connected CW complex is in this situation at every stage $n \geq 2$ (proposition 5.6.10).

Proposition 5.6.10: Principality of the Postnikov Fibrations

Let X be a simply connected CW complex with a Postnikov tower $\{X_n, p_n\}$ whose stages are CW complexes and whose maps p_n are based, for instance that of theorem 5.6.4. Let $n \geq 2$, $\pi = \pi_n(X)$ and $K = K(\pi, n+1)$ with fundamental class ι, and let $k_n \in H^{n+1}(X_{n-1}; \pi)$ be the obstruction class (proposition 5.6.9) to a section of the fibration $\tilde{p}_n \colon \tilde{X}_n \to X_{n-1}$ of proposition 5.6.5, whose fiber is a $K(\pi, n)$. Then p_n is principal: there are a based map $g \colon X_{n-1} \to K$ with $g^*\iota = k_n$ and a weak homotopy equivalence

$$\Psi \colon X_n \longrightarrow g^*PK = \{(b, \gamma) \in X_{n-1} \times PK : \gamma(1) = g(b)\} \qquad \text{with} \qquad q \circ \Psi = p_n$$

where $q(b, \gamma) = b$ and $PK \to K$ is the path fibration of definition 5.2.14.

***Proof*:**

Every space below is simply connected: $\pi_1(X_m) \cong \pi_1(X) = 0$ by definition 5.6.2(2), and the other spaces are checked as they appear. For maps between simply connected spaces, being bijective on every π_i is a property of the free homotopy class, because based and free classes of maps $S^i \to W$ agree (proposition 5.1.15). Let x_0 and $b_0 = p_n(x_0)$ be the basepoints, 0-cells of X_n and X_{n-1}, and take K with a single 0-cell $*$ and no cells of dimension $1, \dots, n$ (theorem 5.5.4). The coefficients of k_n are π_n of the fiber of $\tilde{p}_n$, identified with π in proposition 5.6.5.

Step 1: The cylinder. By cellular approximation [21, Theorem 4.8] there is a cellular map p_n' and a homotopy J rel x_0 from p_n' to p_n. Let M be the mapping cylinder of p_n', a CW complex with subcomplexes $X_n = X_n \times \{0\}$ and X_{n-1}, write $[x, t]$ for the image of (x, t), and let $r \colon M \to X_{n-1}$ be the retraction of Step 1 of the proof of lemma 5.3.14, a homotopy equivalence with $r|_{X_n} = p_n'$. Since p_n is an isomorphism on π_i for $i \leq n-1$ (as $p_n \circ f_n \simeq f_{n-1}$) and $\pi_i(X_{n-1}) = 0$ for $i \geq n$, the long exact sequence of the pair (proposition 5.1.11) gives $\pi_i(M, X_n) = 0$ for $i \neq n+1$ and an isomorphism $\partial \colon \pi_{n+1}(M, X_n) \to \pi_n(X_n) = \pi$.

Step 2: A map $k \colon (M, X_n) \to (K, *)$ that is an isomorphism on π_{n+1}. The pair (M, X_n) is n-connected with X_n simply connected, so by theorem 5.4.10 the Hurewicz map $h \colon \pi_{n+1}(M, X_n) \to H_{n+1}(M, X_n)$ is an isomorphism and $H_i(M, X_n) = 0$ for $i \leq n$. The relative chain complex is free, so theorem 4.1.8 gives $u \in H^{n+1}(M, X_n; \pi)$ with $\langle u, h(\beta) \rangle = \partial\beta$ for all $\beta \in \pi_{n+1}(M, X_n)$. The collapse $(M, X_n) \to (M/X_n, *)$ is an isomorphism on relative homology (corollary 3.4.12, as X_n is a deformation retract of the image of $X_n \times [0, \frac{1}{2})$), hence on cohomology (corollary 4.1.10). Since M/X_n is a CW complex based at a 0-cell, theorem 5.5.12 gives a based map $M/X_n \to K$ whose composite $k \colon M \to K$ with the collapse has $k(X_n) = *$ and $k^*\iota = u$. The homotopy extension property of the CW pair $(M, X_n \cup \{x_0\} \times I)$ deforms k rel X_n to be constant on the segment $\{x_0\} \times I$, by contracting the path $k|_{\{x_0\} \times I}$ to its initial point $*$, and cellular approximation rel that subcomplex then makes k cellular. These homotopies of maps of pairs do not change $k^*\iota$. If β is represented by $b \colon (D^{n+1}, S^n) \to (M, X_n)$, then $k_*\beta$ is represented by $k \circ b$, so $\langle \iota, h(k_*\beta) \rangle = \langle k^*\iota, h(\beta) \rangle = \partial\beta$. As $\alpha \mapsto \langle \iota, h(\alpha) \rangle$ is the isomorphism $\pi_{n+1}(K) \to \pi$ fixed by definition 5.5.10, $k_* \colon \pi_{n+1}(M, X_n) \to \pi_{n+1}(K)$ is an isomorphism.

Step 3: X_n is weakly equivalent to the homotopy fiber of k. Let $p_k \colon E_k \to K$ be the fibration of proposition 5.2.16, with fiber F_k over $*$ consisting of the pairs (m, γ) with γ a path from $k(m)$ to $*$. The homotopy equivalence $m \mapsto (m, c_{k(m)})$ from M to E_k carries X_n into F_k, and its composite with p_k is k. On the long exact sequences of the pairs (M, X_n) and (E_k, F_k) it is therefore an isomorphism on the terms $\pi_i(M) \to \pi_i(E_k)$, and on the relative terms it is k_* under

the isomorphisms $\pi_i(E_k, F_k) \cong \pi_i(K)$ of theorem 5.2.12, an isomorphism for every i by Steps 1 and 2. The five lemma (lemma 3.4.6) makes $X_n \to F_k$ an isomorphism on π_i for $i \geq 2$. The long exact sequence of p_k, with K n-connected and $\pi_1(E_k) \cong \pi_1(M) = 0$, shows F_k path-connected and simply connected, so $X_n \to F_k$ is a weak homotopy equivalence.

Step 4: The comparison map. Let $g = k|_{X_{n-1}}$, which is based and cellular, hence equal to $*$ on $X^{(n)}_{n-1}$. Reversing paths identifies g^*PK with the part of F_k over $X_{n-1} \subseteq M$. This inclusion into F_k is a map of fibrations, both pulled back from the paths in K ending at $*$, over the homotopy equivalence $X_{n-1} \hookrightarrow M$, and the identity on the fiber over b_0, so the five lemma on the sequences of theorem 5.2.12 makes it a weak homotopy equivalence. The space g^*PK is simply connected, by the long exact sequence of q and $\pi_1(\Omega K) \cong \pi_2(K) = 0$. For $x \in X_n$ the path $\gamma_x(t) = k([x,t])$ runs from $k(x) = *$ to $g(p'_n(x))$, so $\Psi'(x) = (p'_n(x), \gamma_x)$ defines $\Psi' \colon X_n \to g^*PK$ with $q \circ \Psi' = p'_n$. The formula $(x,s) \mapsto \big([x,s],\, \tau \mapsto k([x, s(1-\tau)])\big)$ is a homotopy in F_k from the map $x \mapsto (x, c_*)$ of Step 3 to Ψ' followed by the inclusion. Hence Ψ' is a weak homotopy equivalence. Lifting J through q, a pullback of the fibration $PK \to K$, starting at Ψ' gives Ψ homotopic to Ψ' with $q \circ \Psi = p_n$, again a weak homotopy equivalence.

Step 5: Matching $g^*\iota$ with k_n. Write $\Psi(x) = (p_n(x), \lambda_x)$. The map $\tilde\Psi(x,\beta) = (\beta(1), \lambda_x \cdot (g \circ \beta))$ from $\tilde X_n$ to g^*PK lies over X_{n-1}, and its composite with the homotopy equivalence $X_n \to \tilde X_n$ of proposition 5.6.5 sends x to $p_n(x)$ and λ_x followed by a constant path, so it is homotopic to Ψ by reparametrizing. So $\tilde\Psi$ is a weak homotopy equivalence, and the five lemma on the two long exact sequences over X_{n-1} shows that on the fibers over b_0 it induces an automorphism φ of π. Here π_n of the fiber ΩK of q is identified with $\pi_{n+1}(K) = \pi$ by the connecting map of example 5.2.15. A section s of $\tilde p_n$ over $X^{(n)}_{n-1}$ gives the section $\tilde\Psi \circ s$ of q. Since $\tilde\Psi$ is a map over X_{n-1}, it carries each lifted nullhomotopy in the construction of proposition 5.6.9 to a lifted nullhomotopy for q, and the end map of S^n into a fiber to its image under $\tilde\Psi$. The base is simply connected, so the identifications of the homotopy groups of the fibers with those over b_0 do not depend on paths, and on them $\tilde\Psi$ acts by φ. So the obstruction cocycle of $\tilde\Psi \circ s$ is $\varphi_* \mathfrak{o}(s)$, and the obstruction class of q is $\varphi_* k_n$. It can also be computed directly. The section $b \mapsto (b, c_*)$ of q is defined over $X^{(n)}_{n-1}$, where $g = *$. For an $(n+1)$-cell with characteristic map Φ, the radial nullhomotopy $(v,t) \mapsto \Phi((1-t)v)$ of its attaching map lifts from that section to $(v,t) \mapsto \big(\Phi((1-t)v),\, \tau \mapsto g\Phi((1-\tau t)v)\big)$, which ends at the map $v \mapsto \big(\tau \mapsto g\Phi((1-\tau)v)\big)$ into the fiber over $\Phi(0)$. Its class is $\epsilon[g\Phi]$, where $[g\Phi] \in \pi_{n+1}(K)$ is the class of $g\Phi \colon (D^{n+1}, S^n) \to (K, *)$ and the sign $\epsilon = \pm 1$ depends only on the orientation conventions. By Step 5 of the proof of theorem 5.5.12, $[g\Phi]$ is the value of $g^*\iota$ on the cell, so the obstruction class of q is $\epsilon\, g^*\iota$ and $k_n = \mu_* g^* \iota$ with $\mu = \epsilon\varphi^{-1} \in \mathrm{Aut}(\pi)$. Let $\chi \colon K \to K$ be a based map with $\chi^*\iota = \mu_*\iota$ (theorem 5.5.12). It induces μ on $\pi_{n+1}(K)$, so it is a weak homotopy equivalence, and $(b,\gamma) \mapsto (b, \chi \circ \gamma)$ is a map $g^*PK \to (\chi \circ g)^*PK$ over X_{n-1} that is $\Omega\chi$ on the fiber over b_0, hence a weak homotopy equivalence by the five lemma. Since $(\chi \circ g)^*\iota = g^*\mu_*\iota = \mu_* g^*\iota = k_n$, replacing g by $\chi \circ g$ and Ψ by its composite with this map gives $g^*\iota = k_n$ and keeps $q \circ \Psi = p_n$. Finally $g \circ q$ is nullhomotopic on g^*PK, through $((b,\gamma), s) \mapsto \gamma(1-s)$, so

$$p_n^* k_n = (g \circ q \circ \Psi)^* \iota = 0 \in H^{n+1}(X_n; \pi) \tag{5.2}$$

completing the proof.

The map $g \colon X_{n-1} \to K(\pi, n+1)$ *is* the fibration: by representability (theorem 5.5.12) it is interchangeable with a cohomology class, and pulling back the contractible path fibration over K along g reconstructs X_n up to weak homotopy equivalence over X_{n-1}. By proposition 5.3.16, composition

with Ψ is a bijection $[W, X_n] \to [W, g^*PK]$ for every CW complex W, so for maps out of CW complexes the pullback stands in for X_n. The whole transition from X_{n-1} to X_n is thus compressed into one cohomology class, the k-invariant.

Definition 5.6.11: k-invariant

Let X be a simple connected CW complex with a Postnikov tower $\{X_n, p_n\}$ whose stages are CW complexes, and let $n \geq 2$. The *n-th k-invariant* of X is the cohomology class

$$k_n \in H^{n+1}\big(X_{n-1};\, \pi_n(X)\big)$$

obstructing a section of the fibration $\tilde{p}_n \colon \tilde{X}_n \to X_{n-1}$ of proposition 5.6.5, whose fiber is a $K(\pi_n(X), n)$: the class of proposition 5.6.9, which does not depend on the section chosen over the n-skeleton, with π_n of the fiber identified with $\pi_n(X)$ as in proposition 5.6.5. Simplicity makes $\pi_1(X_{n-1})$ act trivially on that group, as proposition 5.6.9 requires. When X is simply connected and the maps p_n are based, k_n classifies the principal map p_n (proposition 5.6.10): $k_n = g^*\iota$ for a based map $g \colon X_{n-1} \to K(\pi_n(X), n+1)$, with ι the fundamental class of $K(\pi_n(X), n+1)$, and X_n maps to g^*PK by a weak homotopy equivalence over X_{n-1}.

The k-invariants are the gluing data the homotopy groups alone cannot see. The list $\pi_1(X), \pi_2(X), \dots$ together with the classes $k_2 \in H^3(X_1; \pi_2)$, $k_3 \in H^4(X_2; \pi_3)$, $\dots$ determine X up to weak equivalence, because they determine the entire tower $X_2, X_3, \dots$ inductively, and X is weakly equivalent to the limit of the tower once it is converted into a tower of fibrations (the remark after theorem 5.6.4). A simply connected space whose every k-invariant vanishes is, stage by stage, a product of Eilenberg-MacLane spaces: $k_n = 0$ makes the classifying map nullhomotopic, so $X_n \simeq X_{n-1} \times K(\pi_n, n)$, and by induction $X_n \simeq \prod_{i \leq n} K(\pi_i, i)$ (exercise 5.6.1 (2)). Products of Eilenberg-MacLane spaces are called *generalized Eilenberg-MacLane spaces*. A nonzero k-invariant is the precise measure of how far X is from being a product of Eilenberg-MacLane spaces.

Example 5.6.12: The First k-invariant of $\mathbb{R}\mathrm{P}^\infty$

Take $X = \mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}_2, 1)$, the Eilenberg-MacLane space identified in example 5.5.6 via the contractible double cover $S^\infty \to \mathbb{R}\mathrm{P}^\infty$. Its homotopy groups are $\pi_1 = \mathbb{Z}_2$ and $\pi_i = 0$ for $i \geq 2$. The Postnikov tower is therefore

$$X_1 = \mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}_2, 1), \qquad X_n = X_1 \text{ for all } n \geq 1$$

since there is no higher homotopy to install. Every k-invariant vanishes: $k_n \in H^{n+1}(X_{n-1}; \pi_n) = H^{n+1}(X_{n-1}; 0) = 0$ because $\pi_n(X) = 0$ for $n \geq 2$. The tower is constant from stage 1 onward, reflecting that X is already an Eilenberg-MacLane space. This is the degenerate case: an Eilenberg-MacLane space is its own Postnikov tower, with all k-invariants trivial because there are no nonzero coefficient groups to receive them.

Returning to the example with which the section began, the obstruction-theoretic computation pins down the first nontrivial k-invariant of S^2, the class that distinguishes its third Postnikov section from a product.

Example 5.6.13: The First k-invariant of S^2

The space S^2 has $\pi_2 = \mathbb{Z}$, $\pi_3 = \mathbb{Z}$ (generated by the Hopf map η, example 5.2.7). Its Postnikov tower has $X_2 = K(\mathbb{Z}, 2) = \mathbb{CP}^\infty$. The first nontrivial k-invariant is

$$k_3 \in H^4\big(X_2;\, \pi_3(S^2)\big) = H^4(\mathbb{CP}^\infty;\, \mathbb{Z})$$

using $\pi_3(S^2) = \mathbb{Z}$. The cohomology ring of $\mathbb{CP}^\infty$ is the polynomial ring $\mathbb{Z}[u]$ with $u \in H^2(\mathbb{CP}^\infty; \mathbb{Z})$ a generator (computed in chapter 4 via the cup product on projective space), so $H^4(\mathbb{CP}^\infty; \mathbb{Z}) = \mathbb{Z}\langle u^2 \rangle$ is infinite cyclic, generated by u^2. The class k_3 must be a multiple $k_3 = m\,u^2$ for some integer m. To identify m, use that X_3 has $\pi_2 = \pi_3 = \mathbb{Z}$ and that the fibration $K(\mathbb{Z}, 3) \to X_3 \to \mathbb{CP}^\infty$ has its connecting behavior governed by k_3. The transgression in the Serre spectral sequence of this fibration (developed in chapter 6) sends the fundamental class of the fiber $K(\mathbb{Z}, 3)$, namely $\iota_3 \in H^3(K(\mathbb{Z}, 3); \mathbb{Z}) = \mathbb{Z}$, to $k_3 \in H^4(\mathbb{CP}^\infty; \mathbb{Z})$, and matching this against the known cohomology of S^2 (which X_3 approximates through degree 3) forces k_3 to generate $H^4(\mathbb{CP}^\infty; \mathbb{Z})$ up to sign, that is $m = \pm 1$ and

$$k_3 = \pm\, u^2 \in H^4(\mathbb{CP}^\infty; \mathbb{Z})$$

a generator. The nonvanishing of k_3 is the precise statement that $X_3 \not\simeq K(\mathbb{Z}, 2) \times K(\mathbb{Z}, 3)$: the Hopf map $\eta \in \pi_3(S^2)$ is detected by the cup-square $u \mapsto u^2$, and this nontrivial cup product is exactly the obstruction to splitting the third Postnikov section of S^2 as a product of Eilenberg-MacLane spaces.

Exercise 5.6.1

1. Let X be a connected CW complex whose only nonzero homotopy group is $\pi_n(X) = \pi$ for a single $n \geq 1$. Show that every Postnikov section X_m for $m \geq n$ is an Eilenberg-MacLane space $K(\pi, n)$, that $X_m \simeq *$ for $m < n$, and that every k-invariant of X vanishes. Conclude that $X \simeq K(\pi, n)$.

2. Let X be a simply connected CW complex with the Postnikov tower of theorem 5.6.4. Show that if every k-invariant of the Postnikov tower of X vanishes, then $X_n \simeq \prod_{i=1}^{n} K(\pi_i(X), i)$ for each n, and hence X is weakly equivalent to the product $\prod_{i \geq 1} K(\pi_i(X), i)$. (Such an X is a *generalized Eilenberg-MacLane space*.)

3. Let Y be a simple space and X a CW complex with $H^{n+1}(X; \pi_n(Y)) = 0$ for all $n \geq 1$. Show that every map $g\colon X^{(1)} \to Y$ from the 1-skeleton (sending the basepoint to the basepoint) extends to a map $X \to Y$. Deduce that if X has trivial reduced cohomology in all positive degrees with all relevant coefficients, then $[X, Y] = *$ for every such Y.

4. Using proposition 5.6.9, prove that a fibration $F \to E \xrightarrow{p} B$ over a simply connected CW complex B, with fiber $F = K(\pi, n)$ for some $n \geq 2$, admits a section if and only if a single class $\theta \in H^{n+1}(B; \pi)$ vanishes. When $B = X_{n-1}$ is a stage of the Postnikov tower of a simply connected CW complex X and p is the fibration $\tilde{p}_n$ of proposition 5.6.5, identify θ with the k-invariant k_n of definition 5.6.11.

5. Let $X = S^n$ for $n \geq 2$. Compute the first two stages X_{n-1} and X_n of the Postnikov tower of S^n, and identify the n-th section X_n as an Eilenberg-MacLane space. Determine the lowest degree in which a nonzero k-invariant can occur.

6. Show that the obstruction class $[\mathfrak{o}(g)] \in H^{n+1}(X; \pi_n(Y))$ of theorem 5.6.7 is natural:

if $h\colon X' \to X$ is a cellular map and $g\colon X^{(n)} \to Y$, then $[\mathfrak{o}(g \circ h|_{X'^{(n)}})] = h^*[\mathfrak{o}(g)]$ in $H^{n+1}(X'; \pi_n(Y))$. Explain why this naturality is what makes obstruction classes computable.

5.7 The Freudenthal Suspension Theorem

Computing the homotopy groups of spheres has been the hardest problem of this chapter. The Hurewicz theorem (theorem 5.4.9) pins down the first nonzero group, the long exact sequence of the Hopf fibration (example 5.2.19) reaches $\pi_3(S^2) \cong \mathbb{Z}$, and beyond that the groups $\pi_i(S^n)$ resist every direct method. Against this difficulty one regularity stands out. Laid out along diagonals, the groups $\pi_{n+k}(S^n)$ for a fixed k appear to stop depending on n once n is large: the value settles down as one climbs the diagonal. The Freudenthal suspension theorem is the explanation. It studies the suspension homomorphism $\sigma\colon \pi_i(X) \to \pi_{i+1}(\Sigma X)$, which on spheres is exactly the comparison $\pi_{n+k}(S^n) \to \pi_{n+1+k}(S^{n+1})$ moving one step up the diagonal, and identifies the precise range in which σ is an isomorphism. This section defines that homomorphism, proves the theorem, and draws its first consequences. The single hard input is homotopy excision (the Blakers-Massey theorem), whose long technical proof is the one book-length deferral in the argument. Everything else is established here. The reward is that the stable value along each diagonal is a genuine, computable invariant, the *stable homotopy group* $\pi_k^s = \varinjlim_n \pi_{n+k}(S^n)$. These groups are the foundation on which the stable homotopy category and the theory of spectra are later built (chapter 8). The present section is what makes them well defined.

The reduced suspension ΣX was introduced in chapter 1 as the quotient of SX collapsing the segment $\{x_0\} \times I$ through the basepoint, homotopy equivalent to the unreduced suspension for a CW complex and equal to the smash product $X \wedge S^1$. The construction needed here presents ΣX as two cones on X glued along their common base, which is the form to which homotopy excision applies.

Definition 5.7.1: Reduced Suspension as a Union of Cones

Let (X, x_0) be a pointed CW complex. The *reduced cone* on X is

$$CX = (X \times I)/\big((X \times \{1\}) \cup (\{x_0\} \times I)\big)$$

the quotient of the cylinder collapsing the top $X \times \{1\}$ to the cone point and the segment $\{x_0\} \times I$ to the basepoint, with X included as the base $X \times \{0\}$. The reduced cone is contractible, the contraction sliding each point up to the cone point. The *reduced suspension* ΣX is the union of two reduced cones glued along their common base copy of X:

$$\Sigma X = C_- X \cup_X C_+ X$$

where C_-X and C_+X are two copies of CX and the gluing identifies the base $X \times \{0\}$ of one with the base of the other. Equivalently $\Sigma X = (X \times I)/\!\sim$, collapsing $X \times \{0\}$ to one point, $X \times \{1\}$ to another, and $\{x_0\} \times I$ to the basepoint. The two cones are the images of $X \times [0, \tfrac{1}{2}]$ and $X \times [\tfrac{1}{2}, 1]$, meeting in the equatorial copy $X \times \{\tfrac{1}{2}\}$.

The decomposition $\Sigma X = C_-X \cup_X C_+X$ presents the suspension as a space covered by two contractible pieces whose intersection is X (more precisely the equatorial copy $X \times \{\tfrac{1}{2}\}$, which the gluing identifies with the base X). This is the topological analogue of writing a sphere as the union of its two closed hemispheres meeting along the equator, and it is exactly the configuration of an excisive triad to

which the homotopy excision theorem will apply. Each cone contributes its base X on the boundary and nothing in its interior up to homotopy, so the relative homotopy of the pair $(\Sigma X, C_+X)$ will turn out to record the homotopy of ΣX itself, while the relative homotopy of (C_-X, X) records the homotopy of X shifted by one.

Example 5.7.2: Spheres as Iterated Suspensions

Take $X = S^n$ with $n \geq 0$. The two cones $C_\pm S^n$ are copies of the disk D^{n+1} (the reduced cone on a sphere is a disk, the cone point being the disk's center and the base S^n its boundary), glued along their common boundary S^n. Two disks glued along their boundary sphere form S^{n+1}, so

$$\Sigma S^n = S^{n+1}$$

recovering the prototypical suspension $S(S^n) = S^{n+1}$ of chapter 1. Iterating, $\Sigma^k S^n = S^{n+k}$. The upper cone $C_+S^n = D^{n+1}$ is the closed upper hemisphere of S^{n+1}, the lower cone $C_-S^n = D^{n+1}$ the closed lower hemisphere, and their intersection the equator $S^n \subset S^{n+1}$, matching the hemisphere decomposition exactly. This is the case that drives the entire theory: the suspension homomorphism on spheres is the comparison $\pi_i(S^n) \to \pi_{i+1}(S^{n+1})$ whose stabilization is the subject of the theorem.

Suspension is a functor: a based map $f\colon X \to Y$ induces $\Sigma f\colon \Sigma X \to \Sigma Y$ by suspending in the cone coordinate, and a based homotopy of maps suspends to a based homotopy. Applying this to a representative $S^i \to X$ of a homotopy class and using $\Sigma S^i = S^{i+1}$ produces a map $S^{i+1} \to \Sigma X$, hence a homomorphism on homotopy groups. This is the central construction of the section.

Definition 5.7.3: The Suspension Homomorphism

Let (X, x_0) be a pointed space and $i \geq 1$. The *suspension homomorphism*

$$\sigma\colon \pi_i(X, x_0) \longrightarrow \pi_{i+1}(\Sigma X, *)$$

sends the class of a based map $f\colon S^i \to X$ to the class of its suspension $\Sigma f\colon S^{i+1} = \Sigma S^i \to \Sigma X$, where $*$ is the suspension basepoint. The assignment is well defined on homotopy classes because a based homotopy $f \simeq f'$ suspends to a based homotopy $\Sigma f \simeq \Sigma f'$, and it is a homomorphism because Σ carries the pinch map defining the group operation on $\pi_i(X)$ (concatenation of cubes) to the pinch map defining it on $\pi_{i+1}(\Sigma X)$, so $\Sigma(f + g)$ is based homotopic to $\Sigma f + \Sigma g$.

The suspension homomorphism is the algebraic form of the geometric operation of suspending a map of spheres. A class in $\pi_i(X)$ is a way of wrapping an i-sphere through X. Suspending wraps an $(i+1)$-sphere through ΣX by coning off the original wrapping at both poles. Whether this loses information is precisely the question the theorem answers. The composite of suspensions $\sigma\colon \pi_i(X) \to \pi_{i+1}(\Sigma X) \to \pi_{i+2}(\Sigma^2 X) \to \cdots$ is what the stable groups will be built from, and the theorem guarantees that this chain of homomorphisms is eventually constant.

Example 5.7.4: Suspension on the Bottom Class of a Sphere

Take $X = S^n$ and the generator $[\mathrm{id}] \in \pi_n(S^n) \cong \mathbb{Z}$ of corollary 5.4.11, the identity map of degree 1. Its suspension $\Sigma\mathrm{id} = \mathrm{id}\colon S^{n+1} \to S^{n+1}$ is again the identity, of degree 1, so

$$\sigma\colon \pi_n(S^n) \to \pi_{n+1}(S^{n+1}), \qquad \sigma[\mathrm{id}_{S^n}] = [\mathrm{id}_{S^{n+1}}]$$

carries a generator to a generator. More generally a degree-d map $S^n \to S^n$ suspends to a

degree-d map $S^{n+1} \to S^{n+1}$, because suspension preserves degree (proposition 3.5.7). Hence $\sigma\colon \pi_n(S^n) \to \pi_{n+1}(S^{n+1})$ is the identity homomorphism $\mathbb{Z} \to \mathbb{Z}$, an isomorphism in every degree $n \geq 1$. This is the first nontrivial instance of the theorem: on the diagonal $i = n$, suspension of $\pi_n(S^n) = \mathbb{Z}$ is always an isomorphism, consistent with the assertion that σ is an isomorphism for $i < 2n - 1$ once $n \geq 2$, and with $\pi_1(S^1) \cong \mathbb{Z} \cong \pi_2(S^2)$ at the boundary $n = 1$.

The whole weight of the theorem rests on a single comparison of relative homotopy groups across the two cones. To set it up cleanly, recall the precise statement of homotopy excision. It is the homotopy-theoretic counterpart of the excision theorem in homology (theorem 3.4.9), but with a sharp limitation: where excision in homology is an exact statement valid for any excisive pair, excision in homotopy holds only in a range of dimensions determined by the connectivities of the pieces. This failure of exactness is the entire reason the homotopy of spheres is hard, and the range it produces is the source of the "stable range" throughout the chapter.

Theorem 5.7.5: Homotopy Excision (Blakers-Massey)

Let X be a CW complex written as a union $X = A \cup B$ of subcomplexes, with $C = A \cap B$ nonempty, so that $(X; A, B)$ is a CW triad. Suppose

1. the pair (A, C) is m-connected, that is $\pi_i(A, C) = 0$ for $i \leq m$, with $m \geq 0$; and
2. the pair (B, C) is n-connected, that is $\pi_i(B, C) = 0$ for $i \leq n$, with $n \geq 0$.

Then the inclusion of pairs $(A, C) \hookrightarrow (X, B)$ induces a homomorphism

$$\pi_i(A, C) \longrightarrow \pi_i(X, B)$$

that is an isomorphism for $i < m + n$ and an epimorphism for $i = m + n$.

***Proof Key ideas*:**
The theorem is the one book-length input of this section, and its proof is deferred. The argument compresses a map of the pair (X, B) off the cells of A not in C by a multiple-general-position argument, controlling the dimensions in which the compression can fail by the connectivities m and n. The combinatorics of the failure dimensions is exactly what produces the bound $i < m + n$. A complete and self-contained account, including the sharper version giving the first obstruction as a Whitehead product, is in [21, §4.2 and §4.J]. The tools it uses (relative homotopy groups, the compression lemma lemma 5.3.4, CW approximation theorem 5.3.10) are all available from this chapter, but the full general-position argument is long and self-contained, so it is cited rather than reproduced, completing the proof.

The theorem says that excision in homotopy holds, but only below the dimension $m + n$ set by the connectivities, and only as a surjection at the borderline degree. The contrast with homology is the heart of the matter: in homology the analogous comparison $H_i(A, C) \to H_i(X, B)$ is an isomorphism in *all* degrees (this is excision, theorem 3.4.9, since $X/B = A/C$ for such a triad), whereas in homotopy the isomorphism degrades to a range. Everything specific to homotopy theory, including the entire difficulty of computing homotopy groups of spheres, flows from this degradation. The two cones on X furnish the triad to which the theorem will be applied, and the connectivity inputs come from the next two identifications.

The first identification reads the relative homotopy of a cone-pair as the homotopy of the base, shifted

by one. It is a direct consequence of the long exact sequence of the pair (proposition 5.1.11) together with the contractibility of the cone.

Proposition 5.7.6: Relative Homotopy of the Cone Pair

Let (X, x_0) be a pointed CW complex and let CX be its reduced cone, containing X as its base. Then the boundary homomorphism of the pair (CX, X) is an isomorphism

$$\partial\colon \pi_{i+1}(CX, X, x_0) \xrightarrow{\cong} \pi_i(X, x_0)$$

for every $i \geq 0$. In particular, if X is $(n-1)$-connected then the pair (CX, X) is n-connected.

***Proof*:**
The reduced cone CX is contractible, so $\pi_j(CX, x_0) = 0$ for all $j \geq 0$ (example 5.1.2, applied to any contractible space). The long exact sequence of the pair (CX, X) from proposition 5.1.11 reads

$$\cdots \to \pi_{i+1}(CX) \to \pi_{i+1}(CX, X) \xrightarrow{\partial} \pi_i(X) \to \pi_i(CX) \to \cdots$$

Both flanking absolute groups $\pi_{i+1}(CX)$ and $\pi_i(CX)$ vanish, so exactness forces the boundary map ∂ to be both surjective (its image is the kernel of $\pi_i(X) \to \pi_i(CX) = 0$, hence all of $\pi_i(X)$) and injective (its kernel is the image of $\pi_{i+1}(CX) = 0$, hence trivial). Thus ∂ is an isomorphism for every $i \geq 0$.

For the connectivity statement, suppose X is $(n-1)$-connected, so $\pi_i(X) = 0$ for $i \leq n-1$ (definition 5.4.7). The isomorphism ∂ then gives $\pi_{i+1}(CX, X) \cong \pi_i(X) = 0$ for $i \leq n-1$, that is $\pi_j(CX, X) = 0$ for $j \leq n$. By definition 5.4.7 applied to the pair, the pair (CX, X) is n-connected, as we sought to show.

The proposition is the device that converts a connectivity hypothesis on X into a connectivity hypothesis on a cone pair, exactly the input homotopy excision requires. A cone is contractible, so all of the relative homotopy of (CX, X) is the homotopy of the base X pushed up one degree by the boundary map. An $(n-1)$-connected base therefore yields an n-connected cone pair. The second identification reads the relative homotopy of the other cone pair, $(\Sigma X, C_+X)$, as the homotopy of the suspension.

Proposition 5.7.7: The Suspension Map as a Cone-Pair Inclusion

Let (X, x_0) be a pointed CW complex and write $\Sigma X = C_-X \cup_X C_+X$ as in definition 5.7.1. Then:

1. the map $j_*\colon \pi_{i+1}(\Sigma X, *) \to \pi_{i+1}(\Sigma X, C_+X, *)$ of the long exact sequence of the pair $(\Sigma X, C_+X)$ is an isomorphism for $i \geq 1$; and

2. under the cone isomorphism $\partial\colon \pi_{i+1}(C_-X, X) \xrightarrow{\cong} \pi_i(X)$ of proposition 5.7.6, the composite

$$\pi_i(X, x_0) \xrightarrow{\sigma} \pi_{i+1}(\Sigma X, *) \xrightarrow{j_*} \pi_{i+1}(\Sigma X, C_+X, *)$$

equals the inclusion-of-pairs map $\iota_*\colon \pi_{i+1}(C_-X, X) \to \pi_{i+1}(\Sigma X, C_+X)$ induced by $(C_-X, X) \hookrightarrow (\Sigma X, C_+X)$, precomposed with ∂^{-1}. That is, $j_* \circ \sigma = \iota_* \circ \partial^{-1}$.

***Proof*:**

1. *j_* is an isomorphism.* The upper cone C_+X is contractible (definition 5.7.1), so $\pi_j(C_+X) = 0$ for all j. The long exact sequence of the pair $(\Sigma X, C_+X)$ from proposition 5.1.11 reads

$$\pi_{i+1}(C_+X) \to \pi_{i+1}(\Sigma X) \xrightarrow{j_*} \pi_{i+1}(\Sigma X, C_+X) \xrightarrow{\partial} \pi_i(C_+X)$$

   The flanking terms $\pi_{i+1}(C_+X)$ and $\pi_i(C_+X)$ vanish, so j_* has trivial kernel (the image of $\pi_{i+1}(C_+X) = 0$) and is surjective (its cokernel injects into $\pi_i(C_+X) = 0$). Hence j_* is an isomorphism for $i \geq 1$.

2. *The composite is the pair inclusion.* A class $[f] \in \pi_i(X)$ has suspension $\sigma[f] = [\Sigma f] \in \pi_{i+1}(\Sigma X)$, represented by the suspended sphere $\Sigma f \colon S^{i+1} = \Sigma S^i \to \Sigma X$. Decompose $S^{i+1} = C_-S^i \cup_{S^i} C_+S^i$ into its two cones (example 5.7.2). By construction the suspended map carries C_-S^i into C_-X, carries C_+S^i into C_+X, and carries the equator S^i into the equator X by f itself. Now $j_*\sigma[f]$ is $\sigma[f]$ regarded as a class of the pair $(\Sigma X, C_+X)$: it is represented by Σf viewed as a map $(D^{i+1}, S^i) \to (\Sigma X, C_+X)$, where $D^{i+1} = C_-S^i$ is the lower hemisphere (whose boundary sphere Σf sends into C_+X, since the equator and the upper cone both land in C_+X). This is precisely the image, under ι_*, of the relative class represented by $\Sigma f|_{C_-S^i} \colon (C_-S^i, S^i) \to (C_-X, X)$. Under the cone isomorphism ∂ of proposition 5.7.6, that relative class corresponds to $\partial[\Sigma f|_{C_-S^i}] = [f] \in \pi_i(X)$, because ∂ restricts a relative cube to its X-face and the X-face of $\Sigma f|_{C_-S^i}$ is f (definition 5.1.10). Reading the equality backward, $j_*\sigma[f] = \iota_*\big(\partial^{-1}[f]\big)$, which is the claimed identity $j_* \circ \sigma = \iota_* \circ \partial^{-1}$, completing the proof.

The two propositions reduce the suspension homomorphism to the homotopy-excision comparison map. After the isomorphism j_* of part (1) and the cone isomorphism ∂ of proposition 5.7.6, the suspension $\sigma \colon \pi_i(X) \to \pi_{i+1}(\Sigma X)$ becomes the inclusion-induced map $\iota_* \colon \pi_{i+1}(C_-X, X) \to \pi_{i+1}(\Sigma X, C_+X)$, which is exactly the excision comparison of theorem 5.7.5 for the triad $(\Sigma X; C_-X, C_+X)$ in degree $i+1$ (the roles of A, B, C being C_-X, C_+X, X). The theorem then transfers the range in which excision is an isomorphism directly to the range in which suspension is one. Assembling the pieces gives the main result.

Theorem 5.7.8: Freudenthal Suspension Theorem

Let X be an $(n-1)$-connected pointed CW complex with $n \geq 1$. Then the suspension homomorphism

$$\sigma \colon \pi_i(X, x_0) \longrightarrow \pi_{i+1}(\Sigma X, *)$$

is an isomorphism for $i < 2n-1$ and an epimorphism for $i = 2n-1$.

***Proof*:**
Write $\Sigma X = C_-X \cup_X C_+X$ as in definition 5.7.1, with the equatorial intersection $C_-X \cap C_+X = X$. The proof applies homotopy excision to this triad, then translates the conclusion back to σ through the cone identifications.

Step 1: Connectivity of the two cone pairs. The space X is $(n-1)$-connected by hypothesis. By proposition 5.7.6 each cone pair (C_-X, X) and (C_+X, X) is then n-connected: the boundary

isomorphism $\pi_j(C_\pm X, X) \cong \pi_{j-1}(X)$ vanishes for $j \le n$ because $\pi_{j-1}(X) = 0$ for $j - 1 \le n - 1$. Thus in the notation of theorem 5.7.5, applied to the triad $(\Sigma X; C_- X, C_+ X)$ with $A = C_- X$, $B = C_+ X$, $C = X$, both connectivity indices equal n:

$$m = n, \qquad (A, C) = (C_- X, X) \text{ is } n\text{-connected}, \qquad (B, C) = (C_+ X, X) \text{ is } n\text{-connected}$$

Step 2: Apply homotopy excision. The triad $(\Sigma X; C_- X, C_+ X)$ satisfies the hypotheses of theorem 5.7.5 with both connectivities equal to n. The theorem gives that the inclusion of pairs $(C_- X, X) \hookrightarrow (\Sigma X, C_+ X)$ induces

$$\pi_i(C_- X, X) \longrightarrow \pi_i(\Sigma X, C_+ X)$$

an isomorphism for $i < n + n = 2n$ and an epimorphism for $i = 2n$.

Step 3: Translate to the suspension homomorphism. By proposition 5.7.7, the suspension homomorphism $\sigma\colon \pi_i(X) \to \pi_{i+1}(\Sigma X)$ is, after the isomorphism $j_*\colon \pi_{i+1}(\Sigma X) \xrightarrow{\cong} \pi_{i+1}(\Sigma X, C_+ X)$ of that proposition, identified with the excision comparison map of Step 2 in degree $i+1$. Precisely, the commuting diagram

$$\begin{array}{ccc} \pi_i(X) & \xrightarrow{\quad\sigma\quad} & \pi_{i+1}(\Sigma X) \\ {\scriptstyle\cong}\downarrow{\scriptstyle\partial^{-1}} & & {\scriptstyle j_*}\downarrow{\scriptstyle\cong} \\ \pi_{i+1}(C_- X, X) & \longrightarrow & \pi_{i+1}(\Sigma X, C_+ X) \end{array}$$

has both vertical maps isomorphisms (the left by proposition 5.7.6, the right by proposition 5.7.7 Step 1), and the bottom map is the excision comparison in degree $i + 1$. Therefore σ is an isomorphism exactly when the bottom map in degree $i + 1$ is, and an epimorphism exactly when the bottom map is.

By Step 2 the bottom map in degree $i + 1$ is an isomorphism for $i + 1 < 2n$, that is $i < 2n - 1$, and an epimorphism for $i + 1 = 2n$, that is $i = 2n - 1$. Transporting through the vertical isomorphisms, $\sigma\colon \pi_i(X) \to \pi_{i+1}(\Sigma X)$ is an isomorphism for $i < 2n - 1$ and an epimorphism for $i = 2n - 1$, as we sought to show.

The theorem is the precise measure of how much information the suspension operation preserves. An $(n - 1)$-connected space has its first possible homotopy in degree n, and the theorem says that suspension faithfully records all homotopy up to degree just below $2n - 1$, weakening to surjectivity at the borderline $i = 2n - 1$ and unconstrained above it. The doubling in the bound $i < 2n - 1$ traces back to the two cones contributing equal connectivity n to the excision range $m + n = 2n$, with the single unit subtracted by the dimension shift between $\pi_i(X)$ and $\pi_{i+1}(\Sigma X)$. Below the stable range nothing is lost. At and above it, suspension can collapse the group, and the collapse is governed by Whitehead products that the sharper form of theorem 5.7.5 computes.

5.7.9 Remark: The Stable Range For the sphere $X = S^n$, which is $(n-1)$-connected (corollary 5.4.11), the theorem reads: $\sigma\colon \pi_i(S^n) \to \pi_{i+1}(S^{n+1})$ is an isomorphism for $i < 2n-1$ and an epimorphism for $i = 2n-1$. Writing $i = n+k$ for the k-th homotopy group above the diagonal, the isomorphism condition $n+k < 2n-1$ is $k < n-1$, equivalently $n > k+1$. Thus the k-stem $\pi_{n+k}(S^n)$ stabilizes once $n > k+1$, the precise stabilization threshold. The interval $i < 2n-1$ is the *stable range* for X, and a class lying in it suspends isomorphically forever.

The first and most basic consequence is that the colimits defining the stable homotopy groups are well posed and computable: the suspension maps eventually become isomorphisms, so the colimit is attained at a finite stage.

Corollary 5.7.10: Well-Definedness and Stabilization of the Stable Stems

For each $k \geq 0$ the suspension homomorphisms

$$\pi_k(S^0) \xrightarrow{\sigma} \pi_{k+1}(S^1) \xrightarrow{\sigma} \pi_{k+2}(S^2) \xrightarrow{\sigma} \cdots \xrightarrow{\sigma} \pi_{n+k}(S^n) \xrightarrow{\sigma} \pi_{n+1+k}(S^{n+1}) \xrightarrow{\sigma} \cdots$$

are isomorphisms for all $n > k+1$. Consequently the colimit

$$\pi_k^s := \varinjlim_n \pi_{n+k}(S^n)$$

is well defined and equals $\pi_{n+k}(S^n)$ for every $n > k+1$. The colimit is attained at a finite stage, not approached in a limit. In particular $\pi_0^s \cong \mathbb{Z}$, realized by $\pi_n(S^n) \cong \mathbb{Z}$ for all $n \geq 1$.

***Proof*:**

Step 1: The maps stabilize. The sphere S^n is $(n-1)$-connected (corollary 5.4.11), so theorem 5.7.8 applies with the connectivity index n: the map $\sigma\colon \pi_i(S^n) \to \pi_{i+1}(S^{n+1})$ is an isomorphism for $i < 2n-1$. Setting $i = n+k$, the condition $n+k < 2n-1$ rearranges to $n > k+1$. Hence for every $n > k+1$ the suspension $\sigma\colon \pi_{n+k}(S^n) \to \pi_{n+1+k}(S^{n+1})$ is an isomorphism.

Step 2: The colimit is attained. The system $\{\pi_{n+k}(S^n)\}_n$ with suspension transition maps is a sequence of abelian groups in which all maps beyond the stage $n = k+2$ are isomorphisms, by Step 1. A colimit of a sequence of groups whose transition maps are eventually all isomorphisms equals the common value of the stabilized groups: explicitly, the canonical map $\pi_{n+k}(S^n) \to \varinjlim_n \pi_{n+k}(S^n)$ is an isomorphism for each $n > k+1$, since every element of the colimit is represented at some stage and any two representatives become equal at a later stage, both of which are already isomorphisms. Therefore $\pi_k^s = \varinjlim_n \pi_{n+k}(S^n) \cong \pi_{n+k}(S^n)$ for all $n > k+1$.

Step 3: The case $k = 0$. For $k = 0$ the stabilization threshold is $n > 1$, so $\pi_0^s \cong \pi_n(S^n)$ for all $n \geq 2$. By corollary 5.4.11, $\pi_n(S^n) \cong \mathbb{Z}$ generated by the degree-1 identity, and example 5.7.4 showed σ is the identity $\mathbb{Z} \to \mathbb{Z}$ on these groups, so the colimit is $\mathbb{Z}$. The stage $n = 1$ also gives $\pi_1(S^1) \cong \mathbb{Z}$ with σ an isomorphism onto $\pi_2(S^2) \cong \mathbb{Z}$, so in fact the system is constant at $\mathbb{Z}$ from the first stage. Thus $\pi_0^s \cong \mathbb{Z}$, completing the proof.

The corollary puts the stable stems on a rigorous footing. The stem π_k^s is not a delicate limiting object. It is literally the value $\pi_{n+k}(S^n)$ for any single n past the threshold $n > k+1$, so computing one such group computes the stem, and the colimit notation is bookkeeping for "the eventual value" whose eventuality Freudenthal guarantees. For $k = 0$ it gives $\pi_0^s \cong \mathbb{Z}$, the stable degree of a self-map

of a sphere. This is the value the sphere spectrum will carry in degree zero (chapter 8).

5.7.11 Historical Note: Freudenthal's 1937 Theorem The theorem was proved by Hans Freudenthal in 1937, predating both the relative homotopy groups in the form used here and the Blakers-Massey theorem (1949 to 1953) from which it is derived above. Freudenthal's original argument was a direct general-position analysis of maps of spheres, establishing the stabilization of $\pi_{n+k}(S^n)$ that makes the stable stems π_k^s well defined. Homotopy excision later subsumed the result as a special case, which is the modern route taken here. The derivation from theorem 5.7.5 isolates the geometric content (two cones, equal connectivity) and produces the range $i < 2n-1$ as the arithmetic $m+n-1 = 2n-1$ of the excision bound.

Exercise 5.7.1

1. Let X be a pointed CW complex that is 4-connected. For which i does theorem 5.7.8 guarantee that $\sigma\colon \pi_i(X) \to \pi_{i+1}(\Sigma X)$ is an isomorphism, and for which single i does it guarantee only an epimorphism? Express the stable range in terms of the connectivity.

2. Using corollary 5.7.10, determine the smallest n for which $\pi_{n+2}(S^n)$ is guaranteed to equal the stable stem π_2^s. (You are not asked to compute π_2^s, which is $\mathbb{Z}/2$ by [34, table on p. 3 and Theorem 1.1.14].) Then state, with reasons, which of $\pi_3(S^1)$, $\pi_4(S^2)$, $\pi_5(S^3)$, $\pi_6(S^4)$ are guaranteed stable.

3. Show that for $n \geq 2$ the suspension $\sigma\colon \pi_n(S^n) \to \pi_{n+1}(S^{n+1})$ is an isomorphism $\mathbb{Z} \to \mathbb{Z}$, directly from theorem 5.7.8 (identify the connectivity and check $i = n$ lies in the stable range), and reconcile this with the degree computation of example 5.7.4.

4. The space $X = S^2 \vee S^2$ is 1-connected. Compute the stable range from theorem 5.7.8, and use $\Sigma(A \vee B) = \Sigma A \vee \Sigma B$ (chapter 1) to describe ΣX. In what degree i is σ only guaranteed to be surjective?

5. Prove that a 1-connected pointed CW complex X has $\sigma\colon \pi_2(X) \to \pi_3(\Sigma X)$ an isomorphism. Deduce that $\pi_3(\Sigma X) \cong \pi_2(X)$, and check this against $X = S^2$, where $\Sigma X = S^3$.

6. Explain why the colimit $\pi_k^s = \varinjlim_n \pi_{n+k}(S^n)$ would not be guaranteed to stabilize at a finite stage if homotopy excision (theorem 5.7.5) gave isomorphisms only in degrees $i < m$ rather than $i < m+n$. Concretely, what would go wrong with the threshold $n > k+1$ of corollary 5.7.10?

6

Spectral Sequences

Spectral sequences have a reputation for opacity. The standard presentations open with a doubly-indexed family of bigraded groups and their differentials, and the reader is asked to take on faith that this apparatus computes anything at all. A spectral sequence is, before it is anything else, a bookkeeping method for computing the homology or cohomology of an object that has been built up in stages (ex. a chain complex carrying a filtration, a space assembled from a tower of subspaces, a fibration whose total space is glued from fibers over a base). It produces a sequence of *pages* $E_1, E_2, E_3, \dots$, each a bigraded group with a differential, and each page is the homology of the page before it. The first page is the crudest approximation to the answer, the associated graded of the filtration. Every subsequent page corrects it by detecting differentials of longer and longer range, until the corrections run out and the pages stabilize at a limit page E_∞ that is the associated graded of the (co)homology one set out to compute. Turning the pages is successive approximation, and the limit is the answer assembled one graded piece at a time.

Section 6.1 isolates the purely algebraic mechanism, the *exact couple* of Massey, from which every page and every differential is generated by a single operation (the passage to the derived couple). The two sections that follow construct the Serre spectral sequence, which computes the homology of the total space of a fibration from the homology of base and fiber, and turns the homotopy-theoretic machinery of chapter 5 into a computational tool capable of extracting homotopy groups of spheres. The closing sections develop the multiplicative structure that makes the cohomology spectral sequence a ring, and the Leray-Hirsch theorem. The same exact-couple construction gives the Atiyah-Hirzebruch spectral sequence of a generalized cohomology theory in section 8.6. The abelian-category generalities (spectral sequences of double complexes, of derived functors, the comparison and convergence theorems in their full generality) are developed in [13].

6.1 Exact Couples and the Spectral Sequence of a Filtration

The cleanest way to meet a spectral sequence is to watch one compute something by hand, before any of the formal apparatus is in place. The setting is a chain complex equipped with a filtration: a chain complex $C_\bullet$ together with an increasing chain of subcomplexes

$$0 = F_{-1}C_\bullet \subseteq F_0C_\bullet \subseteq F_1C_\bullet \subseteq \cdots \subseteq F_pC_\bullet \subseteq \cdots \subseteq C_\bullet$$

each $F_pC_\bullet$ closed under the boundary map ∂, with $\bigcup_p F_pC_\bullet = C_\bullet$. The homology $H_n(C_\bullet)$ is the object of interest, and the filtration is a "recipe" for building $C_\bullet$ in stages. A spectral sequence is the systematic accounting of how the homology of the stages assembles into the homology of the whole.

To see this in action, take the smallest nontrivial case, a filtration with two steps, and compute the homology directly. Namely, consider a chain complex $C_\bullet$ with a single proper subcomplex $A_\bullet = F_0C_\bullet$, so the filtration is

$$0 \subseteq A_\bullet \subseteq C_\bullet$$

with quotient $B_\bullet = C_\bullet/A_\bullet = F_1C_\bullet/F_0C_\bullet$. The associated graded complex is $\operatorname{gr} C_\bullet = A_\bullet \oplus B_\bullet$, the direct sum of the two filtration layers. A first approximation to $H_n(C_\bullet)$ is the homology of this associated graded,

$$E^1 = H_\bullet(A_\bullet) \oplus H_\bullet(B_\bullet)$$

which is what one would get if the complex genuinely split as a direct sum. It does not, in general, and the discrepancy is governed by the short exact sequence of complexes $0 \to A_\bullet \to C_\bullet \to B_\bullet \to 0$, whose long exact sequence in homology

$$\cdots \xrightarrow{\delta} H_n(A_\bullet) \xrightarrow{\iota_*} H_n(C_\bullet) \xrightarrow{q_*} H_n(B_\bullet) \xrightarrow{\delta} H_{n-1}(A_\bullet) \to \cdots$$

contains the connecting map δ. This δ is the entire correction. Reading the exact sequence, $H_n(C_\bullet)$ is built from the part of $H_n(B_\bullet)$ on which δ vanishes (the image of q_*) and the part of $H_n(A_\bullet)$ not hit by δ (the image of ι_*, isomorphic to $H_n(A_\bullet)/\mathrm{im}\delta$). Concretely, exactness gives a short exact sequence

$$0 \to \frac{H_n(A_\bullet)}{\mathrm{im}\big(\delta\colon H_{n+1}(B_\bullet) \to H_n(A_\bullet)\big)} \to H_n(C_\bullet) \to \ker\big(\delta\colon H_n(B_\bullet) \to H_{n-1}(A_\bullet)\big) \to 0$$

The connecting map δ is playing the role of a differential: forming the kernel-modulo-image of δ on the approximation E^1 produces a second, better approximation, and that second approximation is precisely the associated graded of the true homology $H_n(C_\bullet)$. This two-term computation is a spectral sequence with exactly one nontrivial differential. The page $E^1 = H_\bullet(A) \oplus H_\bullet(B)$ carries the differential δ. Its homology is the page E^2, which is the associated graded of $H_\bullet(C)$. There are no further corrections because there are only two filtration layers, so the pages stabilize at $E^2 = E^\infty$.

The general spectral sequence is this computation iterated. With more filtration layers, the analogue of δ does not capture the whole correction in one step. Instead a differential d_1 on the associated graded E^1 produces E^2. A new differential d_2 on E^2, of longer range, produces E^3, and so on, each d_r detecting the part of the assembly that crosses r filtration layers at once. The pages turn until the differentials become zero in every spot, and the limit page is the associated graded of $H_\bullet(C)$. The machine that generates this entire sequence of pages and differentials from a single piece of data is the exact couple, to which the development now turns.

Example 6.1.1: A Filtered Mapping Cone

Let $f\colon \mathbb{Z} \to \mathbb{Z}$ be multiplication by 2, regarded as a chain map between two copies of $\mathbb{Z}$ concentrated in degree 0. Its mapping cone is the complex $C_\bullet$ with $C_1 = \mathbb{Z}$, $C_0 = \mathbb{Z}$, boundary $\partial = 2$, and $C_n = 0$ otherwise, so $H_0(C_\bullet) = \mathbb{Z}/2$ and $H_1(C_\bullet) = 0$. Filter it by $A_\bullet = (C_0$ in degree 0), the subcomplex $0 \to 0 \to \mathbb{Z}$ with zero boundary, and $C_\bullet$ itself. Then $A_\bullet$ has $H_0(A) = \mathbb{Z}$, and $B_\bullet = C/A$ is $\mathbb{Z}$ in degree 1 with zero boundary, so $H_1(B) = \mathbb{Z}$. The page $E^1 = H_\bullet(A) \oplus H_\bullet(B)$ is $\mathbb{Z}$ in bidegree recording degree 0 (from A) and $\mathbb{Z}$ recording degree 1 (from B).

Placing these in the (p, q)-plane with $E^1_{p,q} = H_{p+q}(\mathrm{gr}_p C)$, the two filtration steps $p = 0$ (from A) and $p = 1$ (from B) contribute the class $H_0(A) = \mathbb{Z}$ at $(0, 0)$ and $H_1(B) = \mathbb{Z}$ at $(1, 0)$. Every other spot is zero, so the entire sequence lives in the row $q = 0$:

$$\begin{array}{lccc} & q{=}1 & 0 & 0 \\ E^1: & q{=}0 & \mathbb{Z} \xleftarrow{\times 2} & \mathbb{Z} \qquad d^1 = \delta\colon E^1_{1,0} = \mathbb{Z} \xrightarrow{\times 2} E^1_{0,0} = \mathbb{Z} \\ & & p{=}0 & p{=}1 \end{array}$$

The connecting map $\delta\colon H_1(B) = \mathbb{Z} \to H_0(A) = \mathbb{Z}$ sends the generator of $H_1(B)$, represented by the cycle $1 \in C_1$, to $\partial(1) = 2 \in C_0$, that is, $\delta = 2$. In the homological convention $d^1\colon E^1_{p,q} \to E^1_{p-1,q}$ decreases the filtration degree by one, so this connecting map is precisely the d^1 differential, the single leftward horizontal arrow drawn above. The next page is its kernel-cokernel, $E^2_{1,0} = \ker(\delta)$ and $E^2_{0,0} = \mathrm{coker}(\delta)$:

$$\begin{array}{lccc} & q{=}1 & 0 & 0 \\ E^2 = E^\infty: & q{=}0 & \mathbb{Z}/2 & 0 \\ & & p{=}0 & p{=}1 \end{array}$$

Since $d^2\colon E^2_{p,q} \to E^2_{p-2,q+1}$ has no room to be nonzero, $E^2 = E^\infty$. Concretely,

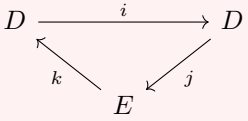
$$E^2 = \ker(\delta) \ \oplus \ \mathrm{coker}(\delta) = 0 \ \oplus \ \mathbb{Z}/2$$

The kernel records $H_1(C) = 0$ and the cokernel records $H_0(C) = \mathbb{Z}/2$, matching the homology computed directly. The single differential $\delta = 2$ is exactly the multiplication that the mapping cone was built to encode, and the spectral sequence has recovered it.

Generalizing this example, Massey observed that the long exact sequences attached to every layer of a filtration can be packaged into a single exact triangle of bigraded groups, and that this triangle regenerates itself under one formal operation, producing every page of the spectral sequence at once. The packaging is the exact couple:

Definition 6.1.2: Exact Couple

An *exact couple* is a pair of (possibly bigraded) abelian groups D and E together with three homomorphisms



such that the triangle is exact at each of its three corners: $\mathrm{im}i = \ker j$, $\mathrm{im}j = \ker k$, and $\mathrm{im}k = \ker i$. The composite

$$d := j \circ k \colon E \to E$$

is the *differential* of the couple.

The data (D, E, i, j, k) should be read as a compression of a long exact sequence. In the filtered case D collects the homologies of the subcomplexes $F_pC_\bullet$, the group E collects the homologies of the layers $F_pC_\bullet/F_{p-1}C_\bullet$, the map i is induced by the inclusions $F_{p-1} \hookrightarrow F_p$, the map j is the projection to the layer, and k is the connecting homomorphism. Exactness of the triangle is exactly the exactness of all these long exact sequences at once. The group E is the page one can see. The group D is the auxiliary data that remembers how the layers are glued, and it is what lets the page be refined. The first thing to verify is that d deserves the name differential.

Lemma 6.1.3: The Differential Squares to Zero

For any exact couple (D, E, i, j, k), the differential $d = jk$ satisfies $d \circ d = 0$, so (E, d) is a differential group and its homology $H(E, d) = \ker d/\mathrm{im}d$ is defined.

***Proof*:**
Compute the composite directly:

$$d \circ d = (jk)(jk) = j(kj)k$$

The middle composite kj is zero, because $\mathrm{im}j = \ker k$ by exactness of the couple at E, so every element in the image of j lies in the kernel of k. Hence $kj = 0$, and therefore $d^2 = j(kj)k = j \cdot 0 \cdot k = 0$, as we sought to show.

The proof uses only one of the three exactness conditions, the one at the visible corner E. The other two are spent in the construction of the next page, to which the lemma is the gateway: with $d^2 = 0$ established, the homology $H(E, d)$ is a new bigraded group, and the claim is that this new group is again the E-term of an exact couple, built from the same D by a derived process. Producing that new couple is the heart of the mechanism. The new D-term is the image $i(D) \subseteq D$, and the new maps are induced from i, j, k in a way that has to be checked for well-definedness at every step.

Definition 6.1.4: Derived Couple

Let (D, E, i, j, k) be an exact couple with differential $d = jk$. Its *derived couple* (D', E', i', j', k') is defined by

$$D' := i(D) = \mathrm{im} i, \qquad E' := H(E, d) = \frac{\ker d}{\mathrm{im} d}$$

with structure maps

- $i' := i|_{D'} \colon D' \to D'$, the restriction of i to its image;
- $j' \colon D' \to E'$ defined by $j'(ia) := [\, ja \,]$ for $a \in D$, the class of ja in $E' = \ker d / \mathrm{im} d$;
- $k' \colon E' \to D'$ defined by $k'([e]) := ke$ for $e \in \ker d$, the value of k on a cycle.

The two new groups are forced by the structure: the only way to shrink D compatibly with the differential is to pass to $\mathrm{im} i$, and the only sensible refinement of the visible page E is its homology. The subtlety is entirely in the three maps, each of which is defined by a representative and must be shown independent of the choice. The map j' takes an element $ia \in D'$, lifts it to a, and reads off the class of ja, while k' takes a homology class, lifts it to a d-cycle, and applies k. That these are well-defined and that the resulting triangle is exact is the content of the next result, on which the whole theory rests.

Theorem 6.1.5: The Derived Couple is Exact

Let (D, E, i, j, k) be an exact couple. Then the derived couple (D', E', i', j', k') of definition 6.1.4 is well-defined and is again an exact couple.

***Proof*:**
The argument proceeds in stages: first the maps land where claimed and are independent of choices, then exactness is checked at each of the three corners.

Step 1: j' is well-defined. Let $a \in D$ and consider $j'(ia) = [ja]$. Two checks are required: that ja is a d-cycle, so that its class in $E' = \ker d / \mathrm{im} d$ makes sense, and that the class is independent of the choice of a with the given image ia.

For the first, $d(ja) = jk(ja) = j(kj)(a) = 0$ since $kj = 0$ (exactness at E, used as in lemma 6.1.3). So $ja \in \ker d$.

For the second, suppose $ia = ia'$, so $i(a - a') = 0$ and $a - a' \in \ker i = \mathrm{im} k$ by exactness at D (the corner where k enters i). Write $a - a' = ke$ for some $e \in E$. Then

$$ja - ja' = j(a - a') = jke = de \in \mathrm{im} d$$

so $[ja] = [ja']$ in E'. Thus j' is well-defined on $D' = \mathrm{im} i$.

Step 2: k' is well-defined. Let $e \in \ker d$ and consider $k'([e]) = ke$. Two checks: that $ke \in D' = \mathrm{im} i$, and that the value depends only on the homology class $[e]$.

For the first, $de = 0$ means $jke = 0$, so $ke \in \ker j = \mathrm{im} i$ by exactness at D (the corner where i enters j). Hence $ke \in D'$.

For the second, if $[e] = [e']$ then $e - e' = dc = jkc$ for some $c \in E$, and

$$ke - ke' = k(e - e') = k(jkc) = (kj)(kc) = 0$$

since $kj = 0$. So $ke = ke'$, and k' is well-defined.

Step 3: i' is well-defined. The map $i' = i|_{D'}$ restricts i to $D' = \mathrm{im}i$. Its image is $i(\mathrm{im}i) = \mathrm{im}(i \circ i) \subseteq \mathrm{im}i = D'$, so i' does map D' into D'. No choices are involved.

Step 4: Exactness at D' (the corner $D' \xrightarrow{i'} D' \xrightarrow{j'} E'$). Show $\mathrm{im}i' = \ker j'$.

($\mathrm{im}i' \subseteq \ker j'$): An element of $\mathrm{im}i'$ has the form $i'(ia) = i(ia) = i^2a$ for some $a \in D$. Then $j'(i^2a) = j'\big(i(ia)\big) = [j(ia)]$. Now $j(ia) = (ji)(a) = 0$ since $\mathrm{im}i = \ker j$ by exactness at D, so $[j(ia)] = 0$ and $i^2a \in \ker j'$.

($\ker j' \subseteq \mathrm{im}i'$): Suppose $ia \in D'$ satisfies $j'(ia) = [ja] = 0$, that is, $ja \in \mathrm{im}d = \mathrm{im}(jk)$. Write $ja = jkc$ for some $c \in E$. Then $j(a - kc) = 0$, so $a - kc \in \ker j = \mathrm{im}i$ by exactness at D. Write $a - kc = ib$ for some $b \in D$. Applying i,

$$ia - ikc = i^2b$$

But $ik = 0$ since $\mathrm{im}k = \ker i$ by exactness at D, so $ikc = 0$ and $ia = i^2b = i'(ib) \in \mathrm{im}i'$. Thus $\ker j' \subseteq \mathrm{im}i'$.

Step 5: Exactness at E' (the corner $D' \xrightarrow{j'} E' \xrightarrow{k'} D'$). Show $\mathrm{im}j' = \ker k'$.

($\mathrm{im}j' \subseteq \ker k'$): An element of $\mathrm{im}j'$ is $j'(ia) = [ja]$ with $ja \in \ker d$ (Step 1). Then $k'([ja]) = k(ja) = (kj)(a) = 0$, so $[ja] \in \ker k'$.

($\ker k' \subseteq \mathrm{im}j'$): Suppose $[e] \in E'$ with $e \in \ker d$ and $k'([e]) = ke = 0$. Then $e \in \ker k = \mathrm{im}j$ by exactness at E. Write $e = ja$ for some $a \in D$. Then $[e] = [ja] = j'(ia) \in \mathrm{im}j'$.

Step 6: Exactness at D' for the corner $E' \xrightarrow{k'} D' \xrightarrow{i'} D'$. Show $\mathrm{im}k' = \ker i'$.

($\mathrm{im}k' \subseteq \ker i'$): An element of $\mathrm{im}k'$ is $k'([e]) = ke$ with $e \in \ker d$. Then $i'(ke) = i(ke) = (ik)(e) = 0$ since $\mathrm{im}k = \ker i$. So $ke \in \ker i'$.

($\ker i' \subseteq \mathrm{im}k'$): Suppose $ia \in D'$ with $i'(ia) = i(ia) = i^2a = 0$. Then $ia \in \ker i = \mathrm{im}k$, so $ia = ke$ for some $e \in E$. To use $[e]$ as an element of E', check $e \in \ker d$: $de = jke = j(ia) = (ji)(a) = 0$ since $ji = 0$ (exactness at D). Thus $e \in \ker d$ and $k'([e]) = ke = ia$, so $ia \in \mathrm{im}k'$.

All three corners are exact and all three maps are well-defined, so (D', E', i', j', k') is an exact couple, completing the proof.

Iterating the theorem turns a single exact couple into an infinite sequence of pages. Apply it once to (D, E, i, j, k) to obtain (D', E', i', j', k'). Applying it again to the derived couple gives a second derived couple, and so on indefinitely. Each iteration replaces the visible page E by its homology under the current differential, so the sequence of E-terms is exactly a sequence of pages, each the homology of the previous. The auxiliary group D shrinks at each stage to $\mathrm{im}i$, recording the progressively deeper part of the filtration that survives. Reading off the sequence of E-terms with their differentials gives the formal object the chapter is named for.

Definition 6.1.6: Spectral Sequence

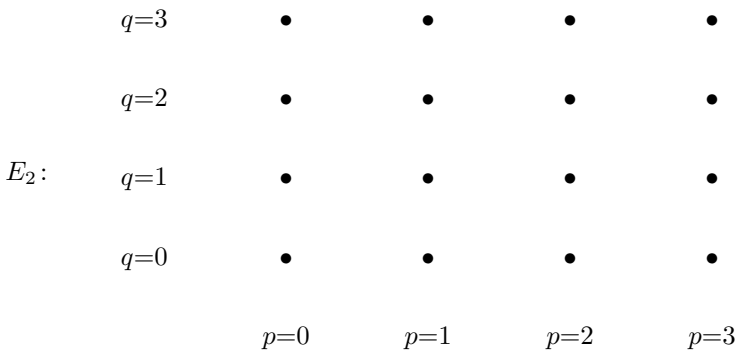
A (homological) *spectral sequence* is a sequence $\{(E_r, d_r)\}_{r \geq r_0}$ of bigraded abelian groups $E_r = \bigoplus_{p,q} E_r^{p,q}$, each equipped with a differential $d_r \colon E_r \to E_r$ satisfying $d_r \circ d_r = 0$, together with isomorphisms

$$E_{r+1} \cong H(E_r, d_r) = \frac{\ker(d_r)}{\operatorname{im}(d_r)}$$

identifying each page as the homology of the page before it. The index r is the *page number*, E_r is the *r-th page*, and d_r is the *r-th differential*. In the bigraded setting the differential has a fixed bidegree: for a homological spectral sequence $d_r \colon E_r^{p,q} \to E_r^{p-r,\, q+r-1}$, lowering the filtration index p by r and preserving the total degree $p+q$.
The spectral sequence *arising from an exact couple* (D, E, i, j, k) is the sequence whose page E_r is the E-term of the $(r-1)$-fold derived couple, with differential d_r the differential of that couple. Theorem 6.1.5 guarantees each page is the homology of its predecessor.

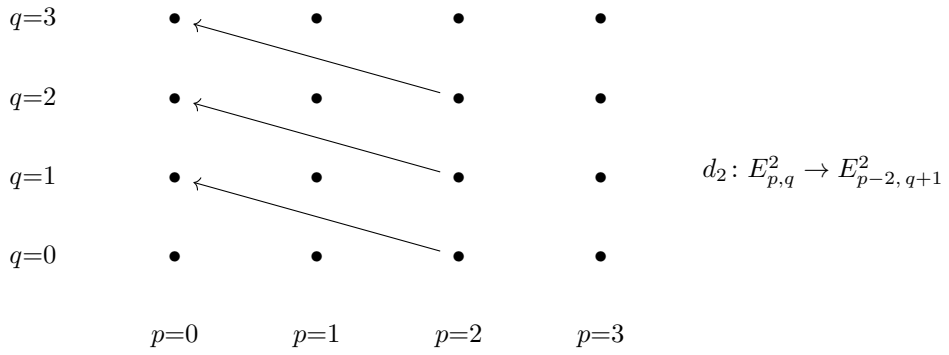
The bidegree of d_r encodes the geometry of the construction. Recall the two-step example: the connecting map δ ran from $H_n(B)$, a layer at filtration level 1, to $H_{n-1}(A)$, a layer at level 0, dropping the filtration by 1 and the homological degree by 1. After one derivation the differential reaches one filtration layer further, after two derivations two layers further, and in general d_r on the r-th page connects spots whose filtration indices differ by r. This is why higher differentials detect higher-order gluing: d_1 sees adjacent layers, d_2 sees layers two apart, and a class that survives to the r-th page only to be killed by d_r encodes an extension that spans r filtration steps. Each derivation of the couple advances the reach of the differential by one. The bookkeeping is exact and mechanical, which is the point: once the exact couple is in hand, the entire spectral sequence is determined, page by page, with no further choices.

A picture shows the loop. Strip away the particular groups and follow an abstract page through two turns. Before any differential is drawn, a page is only an array of groups, one at each lattice point (p, q). This is the input the machine receives, drawn here as the second page E_2 with a dot at each spot:

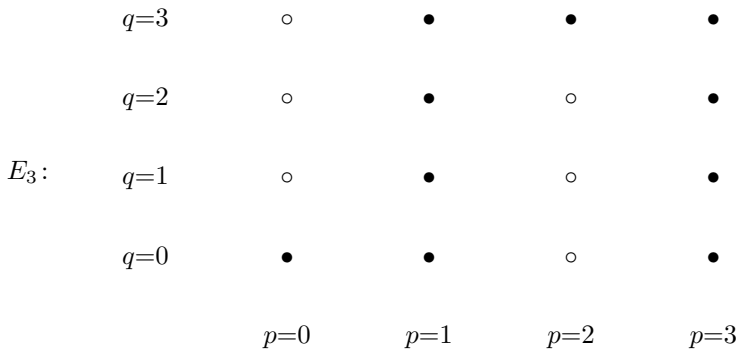


A differential is then drawn on the page. On E_2 it has bidegree $(-2, 1)$, so every arrow points two

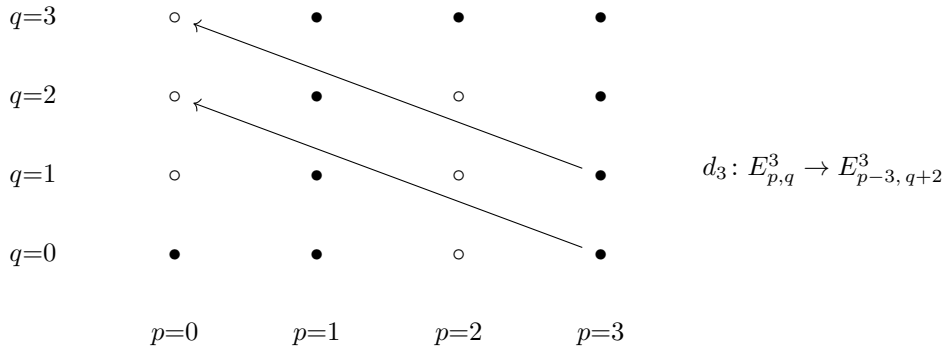
columns left and one row up, the same slope at each spot:



Turning the page passes to homology: each spot of E_3 is $\ker d_2/\mathrm{im} d_2$ computed there, so a group touched by a d_2 is cut down or killed, and the surviving array is in general smaller. The lattice of spots persists while its entries are refined. The open circles below mark the spots where a d_2 acted, each now a subquotient (possibly zero), and the solid dots are untouched:



The new behavior appears when the next differential is drawn. On E_3 it is d_3, of bidegree $(-3, 2)$: three columns left and two rows up. Its arrows are longer than those of d_2 and run at a steeper angle, reaching one filtration layer deeper into the page:



Each further page tilts the differential once more, d_r running r columns left and $r-1$ rows up, until in any fixed spot the arrows overshoot the quadrant and vanish, leaving the limit page E_∞. The

whole computation is this loop: read the page, draw its differential, pass to homology, and watch the next differential swing out to a longer, steeper reach.

Example 6.1.7: The Two-Page Spectral Sequence of a Two-Step Filtration
Return to the two-step filtration $0 = F_{-1} \subseteq F_0 = A_\bullet \subseteq F_1 = C_\bullet$ with quotient layer $B_\bullet = F_1/F_0$, and assemble its exact couple in the indexing of theorem 6.1.10. The auxiliary group collects the homologies of the filtration stages and the page collects the homologies of the layers,

$$D_{p,q} = H_{p+q}(F_pC_\bullet), \qquad E^1_{p,q} = H_{p+q}(F_pC_\bullet/F_{p-1}C_\bullet)$$

so that the only occupied filtration levels are $p = 0$, with layer $F_0/F_{-1} = A_\bullet$, and $p = 1$, with layer $F_1/F_0 = B_\bullet$. The maps are $i = \iota_*$ (induced by $F_{p-1} \hookrightarrow F_p$), $j = \pi_*$ (projection to the layer), and $k = \partial$ (the connecting homomorphism), and exactness of the triangle is the long exact sequence of $0 \to A \to C \to B \to 0$. The first page therefore reads

$$E^1_{0,q} = H_q(A_\bullet), \qquad E^1_{1,q} = H_{1+q}(B_\bullet)$$

with all other spots zero, so the entire first page occupies just two columns, $p = 0$ carrying the A-layer and $p = 1$ carrying the B-layer, with the rows indexed by the complementary degree q. The only possible differential runs from level 1 to level 0, $d_1 \colon E^1_{1,q} \to E^1_{0,q}$, that is, $H_{1+q}(B) \to H_q(A)$. Unwinding $d_1 = jk = \pi_* \circ \partial$, the connecting map $\partial = \delta$ carries $H_{1+q}(B)$ into $D_{0,q} = H_q(A)$ and then $\pi_* \colon H_q(A) = H_q(F_0) \to H_q(F_0/F_{-1}) = H_q(A)$ is the identity, so d_1 is the connecting homomorphism itself. Abbreviating it $\delta_q \colon H_{q+1}(B) \to H_q(A)$ at each height q (the map $\delta \colon H_n(B) \to H_{n-1}(A)$ with $n = q + 1$), the first page carries one such arrow per row:

$$E^1 \colon \quad \begin{array}{ccc} \vdots & \vdots & \vdots \\ q{+}1 & H_{q+1}(A) \xleftarrow{\delta_{q+1}} & H_{q+2}(B) \\ q & H_q(A) \xleftarrow{\delta_q} & H_{q+1}(B) \\ q{-}1 & H_{q-1}(A) \xleftarrow{\delta_{q-1}} & H_q(B) \\ \vdots & \vdots & \vdots \\ & p = 0 & p = 1 \end{array} \qquad d_1 = \delta_q \colon \; \underset{(1,q)}{H_{q+1}(B)} \longrightarrow \underset{(0,q)}{H_q(A)}$$

Passing to homology replaces each row by the kernel and cokernel of its δ_q: the kernel survives at

$(1, q)$ and the cokernel at $(0, q)$, so the second page has no arrows left to draw,

$$E^2 = E^\infty: \quad \begin{array}{ccc} \vdots & \vdots & \vdots \\ q{+}1 & \operatorname{coker}\delta_{q+1} & \ker\delta_{q+1} \\ q & \operatorname{coker}\delta_q & \ker\delta_q \\ q{-}1 & \operatorname{coker}\delta_{q-1} & \ker\delta_{q-1} \\ \vdots & \vdots & \vdots \\ & p=0 & p=1 \end{array} \qquad \begin{array}{l} E^2_{1,q} = \ker\delta_q \\ E^2_{0,q} = \operatorname{coker}\delta_q \\ d_2 = 0 \end{array}$$

The next differential $d_2 \colon E^2_{p,q} \to E^2_{p-2,q+1}$ would drop the filtration level by 2, from a column $p \in \{0, 1\}$ into the empty columns $p = -2$ or $p = -1$, so its source or target is always zero. Hence $d_2 = 0$, every later differential vanishes, and $E^2 = E^\infty$ is already the limit page. Reading up the anti-diagonal $p + q = n$ picks out its two nonzero spots, $E^\infty_{0,n} = \operatorname{coker}\delta_n$ at $(0, n)$ and $E^\infty_{1,n-1} = \ker\delta_{n-1}$ at $(1, n-1)$, explicitly $\operatorname{coker}(\delta \colon H_{n+1}(B) \to H_n(A))$ and $\ker(\delta \colon H_n(B) \to H_{n-1}(A))$, the two successive quotients of the filtration of $H_n(C)$, recovering the short exact sequence

$$0 \to \operatorname{coker}\big(\delta \colon H_{n+1}(B) \to H_n(A)\big) \to H_n(C) \to \ker\big(\delta \colon H_n(B) \to H_{n-1}(A)\big) \to 0$$

found by hand at the start of the section.

The two pages become concrete in the mapping cone of example 6.1.1. There $A_\bullet$ and $B_\bullet$ are copies of $\mathbb{Z}$ concentrated in degrees 0 and 1, so of all the rows above only $q = 0$ survives: the first page has $E^1_{0,0} = H_0(A) = \mathbb{Z}$ at $(0, 0)$ and $E^1_{1,0} = H_1(B) = \mathbb{Z}$ at $(1, 0)$, the differential is $\delta_0 \colon \mathbb{Z} \to \mathbb{Z}$ equal to multiplication by 2, and the second page collapses to $E^\infty_{0,0} = \operatorname{coker}\delta_0 = \mathbb{Z}/2$ and $E^\infty_{1,0} = \ker\delta_0 = 0$, the values $H_0(C) = \mathbb{Z}/2$ and $H_1(C) = 0$. The general grids are that same picture with $H_{p+q}(-)$ left unevaluated.

The example exhibits the general phenomenon that makes spectral sequences finite to compute in practice: in any fixed spot, the differentials eventually run out of room. When only finitely many filtration levels are occupied in each total degree, the differentials d_r for large r must vanish because their sources or targets are zero, and the page stabilizes. The condition that guarantees this uniformly, and that holds for the spectral sequences of the next two sections, is that the spectral sequence be concentrated in a quadrant.

Definition 6.1.8: Convergence and First-Quadrant Spectral Sequences

A spectral sequence $\{(E_r, d_r)\}$ is *first-quadrant* if $E_r^{p,q} = 0$ whenever $p < 0$ or $q < 0$, for every page r (equivalently, for $r = r_0$, since later pages are subquotients). For such a sequence, fix a spot (p, q). The differential d_r into (p, q) comes from $(p + r, q - r + 1)$ and the differential out of (p, q) goes to $(p - r, q + r - 1)$. For $r > \max(p, q + 1)$ both source and target lie outside the first quadrant, so $d_r = 0$ in and out of (p, q). Consequently the groups $E_r^{p,q}$ are equal for all sufficiently large r. This stable value is denoted $E_\infty^{p,q}$.
The spectral sequence *converges* to a graded abelian group $H_\bullet$, written

$$E_{r_0}^{p,q} \Longrightarrow H_{p+q}$$

if $H_\bullet$ carries a filtration $0 = F_{-1}H_n \subseteq F_0H_n \subseteq \cdots \subseteq F_nH_n = H_n$ (in the first-quadrant case, bounded in each degree) and there are isomorphisms

$$E_\infty^{p,q} \cong \frac{F_pH_{p+q}}{F_{p-1}H_{p+q}}$$

identifying the limit page, spot by spot, with the associated graded of $H_\bullet$.

The convergence statement is the payoff: it says the limit page E_∞ is not some abstract terminal object but the associated graded of the very homology one wanted. The spectral sequence does not hand over H_n on a plate. It hands over the successive quotients $F_pH_n/F_{p-1}H_n$ along a filtration, and recovering H_n from these requires solving the extension problems that the differentials leave behind. In favorable cases (a field of coefficients, or a filtration that splits) the extensions are trivial and $H_n \cong \bigoplus_p E_\infty^{p,n-p}$. In general the E_∞-page determines H_n only up to those extensions, which is exactly the residue of information that the factor of 2 in example 6.1.1 encoded. That a first-quadrant spectral sequence stabilizes at every spot, with no convergence hypotheses beyond the quadrant condition, is what makes it a usable computational tool, and the precise statement deserves to be recorded.

Proposition 6.1.9: Stabilization of First-Quadrant Spectral Sequences

Let $\{(E_r, d_r)\}_{r \geq 1}$ be a first-quadrant spectral sequence. Then for each fixed (p, q) there is an integer $R = R(p, q)$ such that the differentials d_r into and out of the spot (p, q) vanish for all $r \geq R$, and consequently the canonical maps

$$E_R^{p,q} = E_{R+1}^{p,q} = \cdots = E_\infty^{p,q}$$

are equalities for all $r \geq R$. Moreover, if the sequence converges to $H_\bullet$ in the sense of definition 6.1.8, then each H_n admits a finite filtration of length $n + 1$ whose successive quotients are the groups $E_\infty^{p,n-p}$ for $0 \leq p \leq n$.

***Proof*:**
Step 1: The differentials vanish for large r. Fix (p, q) with $p, q \geq 0$ (otherwise $E_r^{p,q} = 0$ for all r and there is nothing to prove). The outgoing differential is $d_r \colon E_r^{p,q} \to E_r^{p-r,\, q+r-1}$. Its target lies in the first quadrant only if $p - r \geq 0$, that is, $r \leq p$. For $r > p$ the target group is zero, so the outgoing $d_r = 0$. The incoming differential is $d_r \colon E_r^{p+r,\, q-r+1} \to E_r^{p,q}$. Its source lies in

the first quadrant only if $q - r + 1 \geq 0$, that is, $r \leq q + 1$. For $r > q + 1$ the source is zero, so the incoming $d_r = 0$. Set $R = R(p, q) := \max(p, q + 1) + 1$. For every $r \geq R$ both differentials touching (p, q) are zero.

Step 2: The page stabilizes. For $r \geq R$, the spot (p, q) has $d_r = 0$ both into and out of it, so at (p, q) one has $\ker(d_r) = E_r^{p,q}$ and $\operatorname{im}(d_r) = 0$. Therefore

$$E_{r+1}^{p,q} = \frac{\ker(d_r \colon E_r^{p,q} \to \cdot)}{\operatorname{im}(d_r \colon \cdot \to E_r^{p,q})} = \frac{E_r^{p,q}}{0} = E_r^{p,q}$$

so the groups are equal for all $r \geq R$. Their common value is by definition $E_\infty^{p,q}$, and the identifications $E_R^{p,q} = E_{R+1}^{p,q} = \cdots = E_\infty^{p,q}$ hold.

Step 3: The filtration of H_n is finite. Suppose the sequence converges to $H_\bullet$ as in definition 6.1.8, so H_n carries a filtration with $E_\infty^{p,\,n-p} \cong F_pH_n/F_{p-1}H_n$. The spot $(p, n - p)$ lies in the first quadrant only when $0 \leq p$ and $n - p \geq 0$, that is, $0 \leq p \leq n$. Outside this range $E_\infty^{p,n-p} = 0$, forcing $F_pH_n = F_{p-1}H_n$ there. Hence the filtration is constant for $p < 0$ (where it is 0) and for $p \geq n$ (where it is all of H_n), so

$$0 = F_{-1}H_n \subseteq F_0H_n \subseteq \cdots \subseteq F_nH_n = H_n$$

is a filtration of length $n + 1$ with successive quotients $F_pH_n/F_{p-1}H_n \cong E_\infty^{p,\,n-p}$ for $0 \leq p \leq n$, as claimed. This is a finite filtration recovering H_n from the limit page up to the extension problems posed by its successive quotients, completing the proof.

The proposition is the structural guarantee that underwrites every computation in the next two sections. It says that for a first-quadrant spectral sequence one never has to worry about convergence as a limiting process: at each spot the value is reached after finitely many pages, and the diagonal $p + q = n$ of the limit page contains, in its $n + 1$ nonzero spots, the associated graded pieces of H_n. A computation then reduces to drawing the first page, computing the finitely many differentials, and reading off the stable values along each diagonal. With the abstract machinery in place, the construction promised at the start can be carried out in full: every filtered chain complex produces an exact couple, hence a spectral sequence, converging to its homology.

The setup is a filtered chain complex. Let $C_\bullet$ be a chain complex of abelian groups with an increasing filtration by subcomplexes $F_pC_\bullet$, $p \in \mathbb{Z}$, with $F_pC_\bullet \subseteq F_{p+1}C_\bullet$ and $\partial(F_pC_\bullet) \subseteq F_pC_\bullet$. Assume the filtration is bounded below in each degree, meaning for each n there is an integer $s(n)$ with $F_{s(n)}C_n = 0$, and exhaustive, meaning $\bigcup_p F_pC_\bullet = C_\bullet$. To extract a spectral sequence, set

$$D_{p,q} := H_{p+q}(F_pC_\bullet), \qquad E^1_{p,q} := H_{p+q}\big(F_pC_\bullet/F_{p-1}C_\bullet\big)$$

where the second group is the homology of the p-th filtration layer, with q recording the complementary degree so that $p + q$ is the total homological degree. The short exact sequence of complexes

$$0 \to F_{p-1}C_\bullet \to F_pC_\bullet \to F_pC_\bullet/F_{p-1}C_\bullet \to 0$$

gives, for each p, a long exact sequence in homology, and these assemble into the maps of an exact couple. The inclusion $F_{p-1} \hookrightarrow F_p$ induces $i \colon D_{p-1,q+1} \to D_{p,q}$. The projection induces $j \colon D_{p,q} \to E^1_{p,q}$, and the connecting map gives $k \colon E^1_{p,q} \to D_{p-1,q}$. This is the exact couple of the filtration, and its spectral sequence is the object to be constructed.

Theorem 6.1.10: The Spectral Sequence of a Filtered Chain Complex

Let $C_\bullet$ be a chain complex with an increasing filtration $\{F_pC_\bullet\}_{p\in\mathbb{Z}}$ by subcomplexes that is exhaustive and bounded below in each degree. Then there is a spectral sequence $\{(E_r, d_r)\}_{r\geq 1}$, natural in the filtered complex, with first page

$$E^1_{p,q} = H_{p+q}\big(F_pC_\bullet/F_{p-1}C_\bullet\big)$$

the homology of the associated graded complex, and differential d_1 equal to the connecting homomorphism of the layer short exact sequences. The spectral sequence converges to the homology of $C_\bullet$,

$$E^1_{p,q} \Longrightarrow H_{p+q}(C_\bullet)$$

where $H_n(C_\bullet)$ carries the filtration $F_pH_n(C_\bullet) := \operatorname{im}\big(H_n(F_pC_\bullet) \to H_n(C_\bullet)\big)$ induced by the inclusions, and the limit page is identified with the associated graded:

$$E^{p,q}_\infty \cong \frac{F_pH_{p+q}(C_\bullet)}{F_{p-1}H_{p+q}(C_\bullet)}$$

If moreover the filtration is bounded (in each degree $F_pC_n = 0$ for p small and $F_pC_n = C_n$ for p large), the convergence is the stabilization of proposition 6.1.9: each spot reaches E_∞ after finitely many pages.

***Proof*:**
The proof builds the exact couple, reads off the first page, and then identifies the limit page with the associated graded of $H_\bullet(C)$ by tracking the iterated derived couples through the cycle-and-boundary description.

Step 1: The exact couple. For each p the short exact sequence of complexes $0 \to F_{p-1}C_\bullet \to F_pC_\bullet \to F_pC_\bullet/F_{p-1}C_\bullet \to 0$ yields a long exact sequence in homology,

$$\cdots \to H_n(F_{p-1}C) \xrightarrow{\iota_*} H_n(F_pC) \xrightarrow{\pi_*} H_n\big(\tfrac{F_pC}{F_{p-1}C}\big) \xrightarrow{\partial} H_{n-1}(F_{p-1}C) \to \cdots$$

Set $D = \bigoplus_{p,q} D_{p,q}$ with $D_{p,q} = H_{p+q}(F_pC)$ and $E = \bigoplus_{p,q} E^1_{p,q}$ with $E^1_{p,q} = H_{p+q}(F_pC/F_{p-1}C)$, and define $i = \iota_*$, $j = \pi_*$, $k = \partial$ as above. The three exactness conditions of definition 6.1.2 are precisely the exactness of these long exact sequences at the three positions $H_n(F_pC)$, $H_n(F_pC/F_{p-1}C)$, and $H_{n-1}(F_{p-1}C)$. Thus (D, E, i, j, k) is an exact couple, and by definition 6.1.6 it generates a spectral sequence with $E^1 = E$ and $d_1 = jk = \pi_*\partial$, the connecting homomorphism followed by the projection.

Step 2: First-quadrant boundedness. The hypothesis that the filtration is bounded below and exhaustive forces, for each total degree n, only finitely many p to contribute nontrivially in the bounded case, and in the bounded-below case it guarantees that for each (p, q) the incoming and outgoing differentials eventually leave the support of the spectral sequence. Concretely, $E^1_{p,q} = 0$ when $F_pC_{p+q} = F_{p-1}C_{p+q}$, which holds for p below $s(p+q)$ (boundedness below) and, in the bounded case, for p above the top of the filtration in degree $p+q$. The stabilization argument of proposition 6.1.9 applies verbatim once the support lies in a translate of the first quadrant, which the boundedness hypotheses provide. This gives a well-defined $E^{p,q}_\infty$ at each spot.

Step 3: Cycles and boundaries on the r-th page. To identify E_∞, track the iterated derived

couples explicitly. Write $Z_n^p = \ker(\partial\colon C_n \to C_{n-1})$ for the cycles and let

$$Z_r^p = \{\, x \in F_pC_n : \partial x \in F_{p-r}C_{n-1} \,\}$$

be the elements of F_pC whose boundary drops r filtration levels (the "approximate cycles" of depth r), and

$$B_r^p = F_pC_n \cap \partial\big(F_{p+r-1}C_{n+1}\big) + F_{p-1}C_n \cap Z_{r-1}^p$$

the boundaries reachable from $r-1$ levels above together with the lower-filtration approximate cycles. A direct induction on r, unwinding the definition of the derived couple, shows that the r-th page of the spectral sequence of the exact couple is

$$E_{p,q}^r \cong \frac{Z_r^p}{B_r^p} \quad \text{in total degree } n = p+q$$

with d_r induced by ∂. The base case $r=1$ is $E_{p,q}^1 = Z_1^p/B_1^p = H_{p+q}(F_pC/F_{p-1}C)$, the homology of the layer, since Z_1^p consists of chains whose boundary already lies in $F_{p-1}C$ (cycles modulo F_{p-1}) and B_1^p of the boundaries from one level up together with F_{p-1}. The inductive step is exactly the passage to the derived couple of theorem 6.1.5: forming $\ker d_r/\mathrm{im} d_r$ replaces Z_r^p/B_r^p by Z_{r+1}^p/B_{r+1}^p, because taking homology with respect to the differential induced by ∂ relaxes the boundary condition by one more filtration level (enlarging Z_r^p to Z_{r+1}^p) and enlarges the boundaries correspondingly. The full verification is the unwinding of the derived-couple definition and is carried out in the abelian-category setting in [13, Explicit Description of the Pages]. The topological content is the identification of E^1 with the layer homology, established here.

Step 4: The limit page is the associated graded. For r large (larger than the filtration length in degree n, available by boundedness, or in the limit using bounded-below exhaustiveness), the condition $\partial x \in F_{p-r}C$ becomes $\partial x = 0$, so Z_r^p stabilizes to $Z_\infty^p = F_pC_n \cap Z_n$, the genuine cycles in filtration $\le p$. Likewise B_r^p stabilizes to $B_\infty^p = F_pC_n \cap \partial C_{n+1} + F_{p-1}C_n \cap Z_n$, the genuine boundaries in filtration $\le p$ together with the lower-filtration cycles. Therefore

$$E_\infty^{p,q} = \frac{F_pC_n \cap Z_n}{F_pC_n \cap \partial C_{n+1} + F_{p-1}C_n \cap Z_n}$$

Now compare with the filtration $F_pH_n(C) = \mathrm{im}(H_n(F_pC) \to H_n(C))$. An element of $F_pH_n(C)$ is the class of a cycle $z \in F_pC_n \cap Z_n$, and two such classes coincide in $F_pH_n/F_{p-1}H_n$ exactly when they differ by a boundary from C_{n+1} or by a cycle of lower filtration. That is precisely the equivalence defining the quotient displayed above, so the natural map

$$E_\infty^{p,q} = \frac{F_pC_n \cap Z_n}{F_pC_n \cap \partial C_{n+1} + F_{p-1}C_n \cap Z_n} \xrightarrow{\;\cong\;} \frac{F_pH_n(C)}{F_{p-1}H_n(C)}$$

sending the class of a cycle z to the class of its homology class is a well-defined isomorphism: surjectivity is the definition of $F_pH_n(C)$ as the image of $H_n(F_pC)$, and injectivity is the matching of the two equivalence relations just described. This is the asserted identification $E_\infty^{p,q} \cong F_pH_{p+q}(C)/F_{p-1}H_{p+q}(C)$.

Step 5: Naturality. A filtered chain map $f\colon C_\bullet \to C'_\bullet$ (one with $f(F_pC) \subseteq F_pC'$) induces a map of short exact sequences of layers in every degree, hence a morphism of exact couples $(D, E, \dots) \to (D', E', \dots)$ commuting with i, j, k. By the functoriality of the derived-couple construction (each of i', j', k' is built from i, j, k by operations a couple morphism respects), f induces a map of spectral sequences $E_r(C) \to E_r(C')$ compatible with all differentials and with the filtration on homology, which on E^1 is the map induced by f on layer homology and on E_∞ is the associated graded of $f_*\colon H_\bullet(C) \to H_\bullet(C')$. This is the naturality claim. All assertions of the theorem are established, completing the proof.

The theorem is the prototype every later spectral sequence specializes. The Serre spectral sequence of section 6.2 arises from filtering the singular chains of the total space of a fibration by the dimension of the base simplex each one covers. The cellular filtration of a CW complex by its skeleta recovers, through this same machinery, the cellular chain complex and its homology as the degenerate spectral sequence that collapses at E^2. The spectral sequence of a double complex, central in homological algebra, is the filtered-complex spectral sequence applied to the total complex filtered by columns. In each case the input is a filtration, the E^1-page is the homology of the layers, the differentials are the higher connecting maps, and the limit is the homology one wanted, filtered by the depth at which each class first appears. The single mechanism of the exact couple and its derived couple generates them all.

Example 6.1.11: The Skeletal Filtration Recovers Cellular Homology

Let X be a CW complex and filter its singular chain complex $C_\bullet(X)$ by the skeleta, $F_pC_\bullet(X) := C_\bullet(X^{(p)})$, the singular chains supported on the p-skeleton. This is increasing, exhaustive ($\bigcup_p X^{(p)} = X$), and bounded below ($X^{(-1)} = \emptyset$). The layer is $F_pC/F_{p-1}C = C_\bullet(X^{(p)})/C_\bullet(X^{(p-1)}) = C_\bullet(X^{(p)}, X^{(p-1)})$, whose homology is the relative homology of the skeletal pair. By the computation of cellular homology (section 3.5), this relative homology is concentrated in a single degree:

$$E^1_{p,q} = H_{p+q}(X^{(p)}, X^{(p-1)}) = \begin{cases} C_p^{\mathrm{cell}}(X), & q = 0, \\ 0, & q \neq 0 \end{cases}$$

the free abelian group on the p-cells, sitting along the row $q = 0$. The differential $d_1 \colon E^1_{p,0} \to E^1_{p-1,0}$ is the connecting map $H_p(X^{(p)}, X^{(p-1)}) \to H_{p-1}(X^{(p-1)}, X^{(p-2)})$, which is exactly the cellular boundary map ∂^{cell} (section 3.5). Since the entire E^1-page lies in the single row $q = 0$, every higher differential d_r for $r \geq 2$ maps into or out of a zero group (the target of d_r on the row $q = 0$ is the row $q = r - 1 \neq 0$), so $d_r = 0$ and the sequence collapses at E^2:

$$E^2_{p,0} = \frac{\ker(\partial_p^{\mathrm{cell}})}{\mathrm{im}(\partial_{p+1}^{\mathrm{cell}})} = H_p^{\mathrm{cell}}(X) = H_p(X)$$

The spectral sequence has degenerated to a single row whose homology is cellular homology, and the convergence statement $E^2_{p,0} \cong F_pH_p(X)/F_{p-1}H_p(X) = H_p(X)$ recovers the agreement of cellular and singular homology established in chapter 3.

The skeletal example is the sanity check that the machine reproduces what is already known: when the filtration is the skeletal one, the spectral sequence collapses immediately and returns cellular homology, with the cellular boundary map appearing as the single differential d_1. The spectral sequence becomes genuinely informative only when the E^1-page spreads over more than one row, so that higher differentials have room to act. That is exactly the situation of a fibration, where the fiber contributes a second grading and the Serre spectral sequence carries differentials on every page.

Exercise 6.1.1

1. Verify in full the two identities $kj = 0$ and $ik = 0$ used throughout the derived-couple proof, citing the precise exactness condition of definition 6.1.2 each invokes. Then show, conversely, that if (D, E, i, j, k) is a triple of maps forming a triangle with $kj = 0$, $ik = 0$, and $ji = 0$, it need *not* be an exact couple, by exhibiting a triple satisfying all three composite-zero conditions whose triangle fails exactness at E.

2. Let (D, E, i, j, k) be an exact couple in which i is an isomorphism. Show that $E' = H(E, d) = 0$, so the derived couple has trivial E-term and the spectral sequence collapses to zero from the second page onward. Interpret this for the filtered-complex spectral sequence: which filtrations have i an isomorphism, and why is the homology then concentrated entirely in the limit of D?

3. Consider the chain complex $C_\bullet$ with $C_2 = \mathbb{Z}$, $C_1 = \mathbb{Z} \oplus \mathbb{Z}$, $C_0 = \mathbb{Z}$, boundaries $\partial_2(1) = (3, 0)$ and $\partial_1(a, b) = b$, and $C_n = 0$ otherwise. Filter it by $F_0 C_\bullet = (0 \to 0 \to \mathbb{Z}' \to \mathbb{Z})$ where $\mathbb{Z}'$ is the second summand of C_1, and $F_1 C_\bullet = C_\bullet$. Compute the E^1-page, the differential d_1, and the $E^2 = E^\infty$-page of the resulting spectral sequence, and check that it converges to $H_\bullet(C_\bullet)$ by computing the latter directly.

4. Let $\{(E_r, d_r)\}$ be a first-quadrant spectral sequence converging to $H_\bullet$, and suppose the E_2-page is concentrated in the two columns $p = 0$ and $p = 1$ (so $E_2^{p,q} = 0$ unless $p \in \{0, 1\}$). Show that all differentials d_r for $r \geq 2$ vanish, that $E_2 = E_\infty$, and write down the resulting filtration of H_n. Conclude that there is a short exact sequence relating H_n to the two columns, and identify the remaining extension problem.

5. Show that for a first-quadrant homological spectral sequence (differentials $d_r \colon E_r^{p,q} \to E_r^{p-r,q+r-1}$ as in definition 6.1.6) the edge spots satisfy: $E_\infty^{n,0}$ is a *subgroup* of $E_2^{n,0}$ (the bottom row emits differentials but receives none), and $E_\infty^{0,n}$ is a *quotient* of $E_2^{0,n}$ (the left column receives differentials but emits none). Identify each as an explicit subquotient in terms of the differentials touching those spots, and write down the two edge homomorphisms relating these groups to H_n. (These edge homomorphisms are developed for the Serre spectral sequence in section 6.2.)

6. A morphism of spectral sequences $f \colon E_\bullet \to E'_\bullet$ is a family of maps $f_r \colon E_r \to E'_r$ commuting with differentials and inducing f_{r+1} on homology. Prove the *mapping lemma*: if f_r is an isomorphism on some page E_r, then f_s is an isomorphism for all $s \geq r$, and (for first-quadrant sequences) f_∞ is an isomorphism on E_∞. Explain why this does *not* by itself imply that the induced map on the abutments $H_\bullet \to H'_\bullet$ is an isomorphism, and state what additional input (convergence plus a filtration-compatibility) closes that gap.

6.2 The Serre Spectral Sequence

The exact couple of section 6.1 is a machine: feed it a filtered chain complex and it returns a spectral sequence converging to the homology of the complex. The Serre spectral sequence is what comes out when the filtered complex is the singular chains of the total space of a fibration $F \to E \to B$, filtered by the dimension of the piece of the base each singular simplex covers. A singular simplex of E has filtration at most k when its projection to B factors through a simplicial operator from its simplex to Δ^k. Over a single simplex of the base a Serre fibration is homotopically a product, and that is what turns the abstract E^1 page into something recognizable: the chains of B with coefficients in the homology of the fiber. The result is a calculational device that, when two of the three homologies $H_*(F)$, $H_*(E)$, $H_*(B)$ are known, constrains the third, often forcing it completely. Dress [19, §1] built the spectral sequence from the singular simplices of the base through a double complex. The version below filters the singular chains of E itself, needs no cell structure on B, and is respected by the cup product on the nose, which is what makes the cohomology spectral sequence multiplicative in theorem 6.2.15.

The standing hypotheses are that B is path-connected and that $\pi_1(B)$ acts trivially on $H_*(F)$,

through the transport of fiber homology along loops (lemma 6.2.5). The second condition (automatic when B is simply connected) is what makes the coefficient group $H_q(F)$ a constant coefficient system over B. The version without it, recorded in box 6.2.12, replaces ordinary homology of B by homology with local coefficients. Throughout, $p\colon E \to B$ is a Serre fibration (definition 5.2.2) with fiber $F = p^{-1}(b_0)$ over the basepoint and fiber $F_b = p^{-1}(b)$ over $b \in B$, and homology is taken with coefficients in a fixed abelian group, suppressed from the notation when it is $\mathbb{Z}$.

The filtration counts base dimensions with the linear maps between standard simplices. A *simplicial operator* $\theta\colon \Delta^m \to \Delta^k$ is the affine map carrying the vertex e_i to $e_{\theta(i)}$ for a weakly increasing map $\theta\colon \{0,\dots,m\} \to \{0,\dots,k\}$, denoted by the same letter. The face inclusions $d^i\colon \Delta^{m-1} \to \Delta^m$ omitting e_i are simplicial operators, composites of simplicial operators are simplicial operators, and every simplicial operator factors uniquely as a surjective one followed by the inclusion of the face spanned by its image. A singular simplex $\tau\colon \Delta^k \to B$ is *degenerate* if $\tau = \tau' \circ \eta$ for a surjective simplicial operator $\eta\colon \Delta^k \to \Delta^{k'}$ with $k' < k$, and *nondegenerate* otherwise. Write NB_k for the set of nondegenerate singular k-simplices of B.

Lemma 6.2.1: The Nondegenerate Root of a Singular Simplex

Every singular simplex $\beta\colon \Delta^n \to B$ factors as $\beta = \tau \circ \eta$ with τ nondegenerate and $\eta\colon \Delta^n \to \Delta^k$ a surjective simplicial operator, and the pair (τ, η) is unique.

***Proof*:**
For existence choose a factorization $\beta = \tau \circ \eta$ with η a surjective simplicial operator onto Δ^k and k as small as possible (the identity of Δ^n gives one factorization). If $\tau = \tau' \circ \eta'$ with η' surjective onto a simplex of smaller dimension, then $\beta = \tau' \circ (\eta'\eta)$ contradicts minimality, so τ is nondegenerate.

For uniqueness let $\beta = \tau \circ \eta = \tau' \circ \eta'$ with $\tau\colon \Delta^k \to B$ and $\tau'\colon \Delta^{k'} \to B$ nondegenerate and η, η' surjective. A *section* of η is an injective simplicial operator $\delta\colon \Delta^k \to \Delta^n$ with $\eta\delta = \mathrm{id}$. For every vertex v of Δ^n there is a section with $\delta(\eta(v)) = v$: put $\delta(\eta(v)) = v$ and $\delta(i) = \min \eta^{-1}(i)$ for $i \neq \eta(v)$, which is increasing because η is. For a section δ, $\tau = \tau\eta\delta = \tau' \circ (\eta'\delta)$. Factor $\eta'\delta = \iota \circ \varepsilon$ with ε surjective onto Δ^j and ι injective. Then $\tau = (\tau'\iota) \circ \varepsilon$, and nondegeneracy of τ forces $j = k$, so $\varepsilon = \mathrm{id}$ and $\eta'\delta = \iota$ is injective, giving $k \leq k'$. By symmetry $k = k'$, so $\iota = \mathrm{id}$, $\tau = \tau'$, and $\eta'\delta = \mathrm{id}$ for every section δ of η. A section with $\delta(\eta(v)) = v$ then gives $\eta'(v) = \eta(v)$ for every vertex v, so $\eta' = \eta$.

Definition 6.2.2: Skeletal Filtration of the Total Space

Let $p\colon E \to B$ be a Serre fibration. For $k \geq -1$ let $F_kC_n(E)$ be the subgroup of $C_n(E)$ spanned by the singular simplices $\sigma\colon \Delta^n \to E$ whose projection $p \circ \sigma$ factors through a simplicial operator $\Delta^n \to \Delta^j$ with $j \leq k$, equivalently (lemma 6.2.1) whose nondegenerate root has dimension at most k. For an abelian group G let $F^kC^n(E;G)$ be the group of cochains vanishing on $F_{k-1}C_n(E)$. The *skeletal filtration* of E is the pair of filtrations

$$0 = F_{-1}C_*(E) \subseteq F_0C_*(E) \subseteq F_1C_*(E) \subseteq \cdots \subseteq C_*(E), \qquad C^*(E;G) = F^0C^* \supseteq F^1C^* \supseteq \cdots$$

The faces of σ have projections $p \circ \sigma \circ d^i$, which factor through the composite of the same simplicial

operator with d^i, so each $F_kC_*(E)$ is a subcomplex, and dually each $F^kC^*(E;G)$ is a subcomplex. The filtration is bounded in each degree: $F_nC_n(E) = C_n(E)$, since $p \circ \sigma$ factors through the identity of Δ^n, so $F^{n+1}C^n(E;G) = 0$. The bottom stage $F_0C_*(E)$ is spanned by the simplices with constant projection, those lying in a single fiber, so $F_0C_*(E) = \bigoplus_{b\in B} C_*(F_b)$. A map $\Psi\colon E' \to E$ of fibrations over $g\colon B' \to B$, meaning $p\Psi = gp'$, preserves the filtrations, because $p\Psi\sigma = gp'\sigma$ factors through every simplicial operator through which $p'\sigma$ factors. The filtration is by "base degree", regardless of the dimension of the chains. Two gradings are now in play, the total (homological) degree n of a chain in E and the filtration degree p recording how many dimensions of the base it covers. Their difference $q = n - p$ will turn out to be fiber degree. The spectral sequence is the bookkeeping device that disentangles these two gradings, and the whole content of the Serre spectral sequence is the computation of what the associated graded of this particular filtration is.

Example 6.2.3: The Filtration for the Path-Loop Fibration over S^2

Let $p\colon PS^2 \to S^2$ be the path fibration of example 5.2.15, with fiber the loop space $F = \Omega S^2$. A singular simplex of PS^2 lies in F_0 when its projection is constant, so $F_0C_*(PS^2) = \bigoplus_{b\in S^2} C_*\big(p^{-1}(b)\big)$ collects the chains of all the fibers. It lies in F_1 when its projection factors through a simplicial operator $\Delta^n \to \Delta^1$, so that it runs along a single path of S^2, and in F_k when its projection factors through Δ^k. The sphere has nondegenerate singular simplices in every dimension, so this filtration grows in every degree, unlike the two-step filtration a CW structure on S^2 would suggest. The first page is correspondingly large: by lemma 6.2.6 below, $E^1_{p,q}$ is a direct sum of copies of $H_q(\Omega S^2)$, one for each nondegenerate singular p-simplex of S^2. Its homology is small. The second page $E^2_{p,q} = H_p\big(S^2; H_q(\Omega S^2)\big)$ of theorem 6.2.7 vanishes outside the columns $p = 0$ and $p = 2$. With PS^2 contractible, the spectral sequence must converge to the homology of a point, and the constraint that everything cancels except in degree 0 is what will compute $H_*(\Omega S^2)$ in section 6.3.

Applying theorem 6.1.10 to the skeletal filtration produces a spectral sequence with first page $E^1_{p,q} = H_{p+q}\big(F_pC_*(E)/F_{p-1}C_*(E)\big)$, converging to $H_*(E)$ because the filtration is bounded in each degree. The work specific to fibrations is identifying the E^1 and E^2 pages. Each layer F_p/F_{p-1} splits over the nondegenerate p-simplices of B, and over a single simplex the fibration is pulled back to a fibration over Δ^p. For a singular simplex $\tau\colon \Delta^m \to B$, the pullback

$$X_\tau = \{(x, e) \in \Delta^m \times E \mid \tau(x) = p(e)\} \longrightarrow \Delta^m$$

is a Serre fibration whose fiber over a vertex e_v is a copy of $F_{\tau(e_v)}$ (proposition 5.2.6). A singular simplex of X_τ is a pair (α, σ) of maps $\alpha\colon \Delta^n \to \Delta^m$ and $\sigma\colon \Delta^n \to E$ with $\tau\alpha = p\sigma$. For a Serre fibration $g\colon X \to \Delta^m$ and a union K of faces of Δ^m, let $A_*(X|K) \subseteq C_*(X)$ be spanned by the singular simplices σ for which $g\sigma$ is a simplicial operator with image in K. Faces of such simplices are again such, so $A_*(X|K)$ is a subcomplex. Write $A_*(X) = A_*(X|\Delta^m)$, write $\dot{A}_*(X) = A_*(X|\partial\Delta^m)$ for the span of the simplices whose simplicial operator is not surjective, and write $X_v = g^{-1}(e_v)$, so that $A_*(X|e_v) = C_*(X_v)$. The relative complex $A_*(X)/\dot{A}_*(X)$ is free on the simplices over surjective simplicial operators, and its differential $\bar{\partial}$ keeps the terms of ∂ of that kind.

Lemma 6.2.4: Contraction onto a Vertex Fiber

Let $g\colon X \to \Delta^m$ be a Serre fibration, let e_v be the first or the last vertex of Δ^m, and let K be a nonempty union of faces of Δ^m each containing e_v. Then the inclusion $C_*(X_v) \to A_*(X|K)$ is a chain homotopy equivalence, with a homotopy inverse r that is the identity on $C_*(X_v)$. It induces isomorphisms on homology and cohomology with coefficients in any abelian group.

***Proof*:**
Take $e_v = e_m$, the last vertex. For $e_v = e_0$ exchange the two ends of I throughout. For a simplicial operator $\alpha\colon \Delta^n \to \Delta^m$ let $G_\alpha\colon \Delta^n \times I \to \Delta^m$ be affine on each simplex $[v_0, \dots, v_i, w_i, \dots, w_n]$ of the decomposition of the prism used in theorem 3.3.6, with $G_\alpha(v_j) = e_{\alpha(j)}$ and $G_\alpha(w_j) = e_m$. The affine pieces agree on common faces, so G_α is continuous, with $G_\alpha(x,0) = \alpha(x)$ and $G_\alpha(x,1) = e_m$. From the vertex values, $G_{\alpha d^j} = G_\alpha \circ (d^j \times \mathrm{id}_I)$, the map G_α is constant when α is constant at e_m, its image lies in the face spanned by e_m and the image of α, and $G_\alpha \circ [v_0, \dots, v_i, w_i, \dots, w_n]$ is the simplicial operator with vertex values $\alpha(0), \dots, \alpha(i), m, \dots, m$.

Choose, by induction on n, for every $\sigma \in A_n(X)$ a map $\Sigma_\sigma\colon \Delta^n \times I \to X$ with $g\Sigma_\sigma = G_{g\sigma}$, $\Sigma_\sigma(\,\cdot\,, 0) = \sigma$ and $\Sigma_{\sigma d^j} = \Sigma_\sigma \circ (d^j \times \mathrm{id}_I)$ for all j. If σ lies in X_v, let $\Sigma_\sigma(x,t) = \sigma(x)$. Otherwise the maps already chosen for the faces of σ agree on common faces and, together with σ on $\Delta^n \times \{0\}$, define a lift of $G_{g\sigma}$ on $\Delta^n \times \{0\} \cup \partial\Delta^n \times I$. This pair is homeomorphic to $(\Delta^n \times I, \Delta^n \times \{0\})$, as for the pair (I^n, J^{n-1}) in the opening paragraph of the proof of proposition 5.2.11, so the Serre property extends the lift over $\Delta^n \times I$. (Dress uses the same extension relative to the boundary in [19, Satz 2].)

Set $r(\sigma) = \Sigma_\sigma(\,\cdot\,, 1)$, a singular simplex of X_v, and $P(\sigma) = \sum_i (-1)^i \, \Sigma_\sigma \circ [v_0, \dots, v_i, w_i, \dots, w_n]$. The face relations of the Σ_σ make r a chain map, equal to the identity on $C_*(X_v)$, and the computation in Step 2 of the proof of theorem 3.3.6, which uses only those face relations, gives $\partial P + P\partial = r - \mathrm{id}$. Each simplex of $P(\sigma)$ lies over a simplicial operator. If $g\sigma$ has image in K, that image lies in one of the faces making up K (the one containing its barycentre), which contains e_v, so $P(\sigma)$ lies in $A_*(X|K)$. Hence r and P restrict to $A_*(X|K)$, and r is a homotopy inverse of the inclusion. The functors $- \otimes G$ and $\mathrm{Hom}(-, G)$ preserve chain homotopies, which gives the last sentence.

Lemma 6.2.5: Transport of Fiber Homology

Let $p\colon E \to B$ be a Serre fibration. For a path $\omega\colon \Delta^1 \to B$ let i_0 and i_1 be the inclusions of $C_*(F_{\omega(e_0)})$ and $C_*(F_{\omega(e_1)})$ into $A_*(X_\omega)$, and let r_1 be a homotopy inverse of i_1 as in lemma 6.2.4. The *transport* along ω is the chain map $T_\omega = r_1 i_0\colon C_*(F_{\omega(e_0)}) \to C_*(F_{\omega(e_1)})$. On homology with any coefficients it is the isomorphism $(i_{1*})^{-1} i_{0*}$. On cohomology it induces $T^\omega = i_0^*(i_1^*)^{-1}\colon H^*(F_{\omega(e_1)}; R) \to H^*(F_{\omega(e_0)}; R)$, an isomorphism of rings for a commutative ring R, as $A_*(X_\omega)$ contains the front and back faces of its simplices and i_0^*, i_1^* preserve cup products.

1. If $\sigma\colon \Delta^2 \to B$ has edges $\omega_{jl} = \sigma|_{[e_j, e_l]}$, then $T_{\omega_{02}} = T_{\omega_{12}} T_{\omega_{01}}$ on homology.
2. On homology T_ω depends only on the class of ω rel endpoints, is the identity for a constant path, and satisfies $T_{\omega * \omega'} = T_{\omega'} T_\omega$ for composable paths.
3. If $L\colon F_{\omega(e_0)} \times I \to E$ satisfies $pL(x,t) = \omega(t)$ and $L(x,0) = x$, then $T_\omega = (L_1)_*$, where $L_1 = L(\,\cdot\,, 1)$.
4. For a map $\Psi\colon E' \to E$ of fibrations over $g\colon B' \to B$ and a path ω in B', $\Psi_* T_\omega = T_{g\omega} \Psi_*$ on the homology of the fibers.

The same statements hold on cohomology, with the arrows reversed.

***Proof*:**
The two formulas on homology and cohomology hold because r_1 inverts i_1 up to homotopy. For (1), let $X = X_\sigma \to \Delta^2$. By lemma 6.2.4, $C_*(X_0)$ and $C_*(X_2)$ include into $A_*(X)$ by homology isomorphisms, and so do $A_*(X|[e_0, e_1])$ and $A_*(X|[e_1, e_2])$, which contain $C_*(X_0)$, respectively $C_*(X_2)$, by homology isomorphisms. Through the first edge, where e_1 is the last vertex, $C_*(X_1) \to A_*(X)$ is a homology isomorphism as well. The pullback of X along the edge $[e_j, e_l]$ is $X_{\omega_{jl}}$, with $A_*(X_{\omega_{jl}}) = A_*(X|[e_j, e_l])$, so each $T_{\omega_{jl}}$ is $i_{l*}^{-1} i_{j*}$ computed in $H_*(A_*(X))$, and $T_{\omega_{02}} = (i_{2*}^{-1} i_{1*})(i_{1*}^{-1} i_{0*})$.

For (2), a constant path at b has $X_\omega = \Delta^1 \times F_b$, and $\sigma \mapsto \mathrm{pr}_2 \circ \sigma$ is a chain map $A_*(X_\omega) \to C_*(F_b)$ left inverse to i_0 and to i_1, so $i_{0*}^{-1} = i_{1*}^{-1}$ and $T_\omega = \mathrm{id}$. A homotopy $h\colon \Delta^1 \times I \to B$ rel endpoints from ω to ω' gives the singular 2-simplices $h \circ [v_0, v_1, w_1]$ and $h \circ [v_0, w_0, w_1]$, with edges $(\omega, \mathrm{const}, \omega'')$ and $(\mathrm{const}, \omega', \omega'')$ for $\omega'' = h \circ [v_0, w_1]$. By (1), $T_{\omega''} = T_\omega$ and $T_{\omega''} = T_{\omega'}$. For composable ω, ω', a retraction of Δ^2 onto $[e_0, e_1] \cup [e_1, e_2]$ composed with $\omega \cup \omega'$ is a 2-simplex with edges ω, ω' and a path homotopic rel endpoints to $\omega * \omega'$, both being paths from end to end of the arc $[e_0, e_1] \cup [e_1, e_2]$. Then (1) gives $T_{\omega * \omega'} = T_{\omega'} T_\omega$.

For (3), the map $F_{\omega(e_0)} \times I \to X_\omega$, $(x,t) \mapsto (t, L(x,t))$, is a homotopy from i_0 to $i_1 L_1$. Its prism operator (theorem 3.3.6) carries each simplex to simplices lying over the simplicial operators with vertex values $0, \dots, 0, 1, \dots, 1$, hence into $A_*(X_\omega)$, so $i_{0*} = i_{1*}(L_1)_*$. For (4), $(x, e') \mapsto (x, \Psi(e'))$ maps X'_ω to $X_{g\omega}$ over Δ^1, carries A-chains to A-chains, and restricts to Ψ on the end fibers, so it commutes with i_{0*} and i_{1*}. Every step is a statement about chain maps and chain homotopies, so the cohomology versions follow by applying $\mathrm{Hom}(-, G)$.

Restricting (2) to loops at b_0 gives, for each q, an action of $\pi_1(B, b_0)$ on $H_q(F; G)$ and on $H^q(F; G)$,

the *monodromy*, and by (3) a loop acts through the end of a lifted homotopy whenever such a lift exists. Suppose the action on $H_*(F;G)$ is trivial. A loop γ' at any b then acts trivially too, since $T_{\omega*\gamma'*\bar{\omega}} = T_{\bar{\omega}} T_{\gamma'} T_{\omega}$ for a path ω from b_0 to b with reverse $\bar{\omega}$. Hence T_ω depends only on the endpoints of ω: for two paths ω, ω' from b to b', $T_{\omega'}^{-1} T_\omega = T_{\bar{\omega}'} T_\omega = T_{\omega*\bar{\omega}'} = \mathrm{id}$. Choosing a path ω_b from b_0 to each $b \in B$, the isomorphisms

$$\lambda_b = T_{\omega_b}^{-1} \colon H_*(F_b;G) \to H_*(F;G) \qquad \text{satisfy} \qquad \lambda_{b'} T_\omega = \lambda_b$$

for every path ω from b to b'. Likewise, when $\pi_1(B)$ acts trivially on $H^*(F;G)$, the isomorphisms $\lambda^b = T^{\omega_b} \colon H^*(F_b;G) \to H^*(F;G)$ satisfy $\lambda^b T^\omega = \lambda^{b'}$, and for a coefficient ring $G = R$ they are ring isomorphisms. These identifications are the only place the hypothesis on $\pi_1(B)$ enters.

Lemma 6.2.6: The Layers of the Skeletal Filtration

Let $p \colon E \to B$ be a Serre fibration.

1. For each k the maps $(\eta, \sigma) \mapsto \sigma$ assemble into an isomorphism of chain complexes

$$\bigoplus_{\tau \in N B_k} A_*(X_\tau)/\dot{A}_*(X_\tau) \xrightarrow{\cong} F_k C_*(E)/F_{k-1}C_*(E)$$

2. Let $g \colon X \to \Delta^k$ be a Serre fibration and let $\nu_{n,k} \colon \Delta^n \to \Delta^k$ be the simplicial operator with vertex values $0, \dots, 0, 1, 2, \dots, k$, beginning with $n-k+1$ zeros. The homomorphisms $\Phi_X \colon A_n(X)/\dot{A}_n(X) \to C_{n-k}(X_0)$ given on a simplex σ over a surjective simplicial operator by

$$\Phi_X(\sigma) = (-1)^{k(n-k)}\, \sigma|_{[e_0,\dots,e_{n-k}]} \ \text{ if } g\sigma = \nu_{n,k}, \qquad \Phi_X(\sigma) = 0 \ \text{ otherwise}$$

   satisfy $\partial \Phi_X = (-1)^k \Phi_X \bar{\partial}$, and for every abelian group G they induce isomorphisms

$$\Theta_X \colon H_n\big(A_*(X)/\dot{A}_*(X);G\big) \xrightarrow{\cong} H_{n-k}(X_0;G)$$
$$\Theta^X \colon H^n\big(A_*(X)/\dot{A}_*(X);G\big) \xrightarrow{\cong} H^{n-k}(X_0;G)$$

   the second being the inverse of the map induced by $c \mapsto c \circ \Phi_X$.

3. For $0 \le i \le k$ let $\partial_i \colon A_n(X)/\dot{A}_n(X) \to A_{n-1}(X|d^i)/\dot{A}_{n-1}(X|d^i)$ keep the terms of ∂ lying over the facet $d^i\Delta^{k-1}$ by a surjective simplicial operator, where $X|d^i$ is the restriction of X to that facet, read as a fibration over Δ^{k-1} through d^i. Then $\Phi_{X|d^i}\partial_i = (-1)^i \Phi_X$ for $1 \le i \le k$, and $\Phi_{X|d^0}\partial_0$ agrees with $T\Phi_X$ up to homotopy, T being the transport along the edge $[e_0, e_1]$. Consequently, with any coefficients,

$$\Theta_{X|d^i}\, \partial_{i*} = (-1)^i\, \Theta_X, \qquad \Theta^X \partial_i^* = (-1)^i\, \Theta^{X|d^i} \qquad (1 \le i \le k)$$

   while $\Theta_{X|d^0}\partial_{0*} = T\,\Theta_X$ and $\Theta^X\partial_0^* = T\,\Theta^{X|d^0}$, with T the transport on homology, respectively on cohomology.

***Proof*:**

Part (1). A pair (η, σ) with η surjective is a simplex of $A_*(X_\tau)$ outside $\dot{A}_*(X_\tau)$, and $\sigma \in F_k C_*(E)$ because $p\sigma = \tau\eta$. The map sends $\dot{A}_*(X_\tau)$ into $F_{k-1}C_*(E)$ and commutes with boundaries, so it is a chain map. A basis element of $F_kC_n(E)/F_{k-1}C_n(E)$ is a simplex σ whose nondegenerate root has dimension exactly k, and by lemma 6.2.1 it comes from exactly one $\tau \in NB_k$ and one surjective η. So the map is bijective on bases.

The chain identities of (2) and (3). Let σ lie over a surjective η. A face σd^j lies over ηd^j, which is surjective unless $\eta^{-1}(\eta(j)) = \{j\}$. Suppose $\eta = \nu_{n,k}$ with $n > k$. The faces with $j \leq n-k$ lie over $\nu_{n-1,k}$, the others lie in $\dot{A}_*(X)$, and $(\sigma d^j)|_{[e_0,\ldots,e_{n-k-1}]} = (\sigma|_{[e_0,\ldots,e_{n-k}]})d^j$ for $j \leq n-k$. So $\Phi_X(\bar{\partial}\sigma) = (-1)^{k(n-1-k)}\partial(\sigma|_{[e_0,\ldots,e_{n-k}]}) = (-1)^k\partial\Phi_X(\sigma)$. If $n = k$, both sides vanish. If $\eta \neq \nu_{n,k}$, then $\Phi_X(\sigma) = 0$, and a face of σ lies over $\nu_{n-1,k}$ only if η is $\nu_{n-1,k}$ with one value $c \geq 1$ repeated, at two adjacent positions $j, j+1 \geq n-k$. The faces σd^j and σd^{j+1} then have the same front face $[e_0, \ldots, e_{n-k-1}]$, so their images under Φ_X agree, and they enter $\Phi_X(\bar{\partial}\sigma)$ with opposite signs. Hence $\partial\Phi_X = (-1)^k\Phi_X\bar{\partial}$. For $1 \leq i \leq k$, a face σd^j lies over the facet $d^i\Delta^{k-1}$ by a surjective operator exactly when $\eta^{-1}(i) = \{j\}$, and in the coordinates of that facet it lies over $\nu_{n-1,k-1}$ exactly when moreover $\eta = \nu_{n,k}$. Then $j = n-k+i$ and the front face $[e_0, \ldots, e_{n-k}]$ of σd^j is that of σ, so $\Phi_{X|d^i}\partial_i(\sigma) = (-1)^{n-k+i}(-1)^{(k-1)(n-k)}\sigma|_{[e_0,\ldots,e_{n-k}]} = (-1)^i\Phi_X(\sigma)$.

Part (2). Induct on k. For $k = 0$, $A_*(X) = C_*(X_0)$, $\dot{A}_*(X) = 0$ and $\Phi_X = \mathrm{id}$. Let $k \geq 1$, and let Λ be the union of the facets $d^i\Delta^{k-1}$ with $i < k$, which are the facets containing e_k. Then $\dot{A}_*(X) = A_*(X|\Lambda) + A_*(X|d^k)$ and $A_*(X|\Lambda) \cap A_*(X|d^k) = \dot{A}_*(X|d^k)$, since a simplex over a face of $d^k\Delta^{k-1}$ lies over a face of Λ unless that face is $d^k\Delta^{k-1}$ itself. By lemma 6.2.4 with $e_v = e_k$, both $C_*(X_k) \to A_*(X|\Lambda)$ and $C_*(X_k) \to A_*(X)$ are chain homotopy equivalences, so $A_*(X)/A_*(X|\Lambda)$ is acyclic with every coefficient group. The short exact sequence

$$0 \to \dot{A}_*(X)/A_*(X|\Lambda) \to A_*(X)/A_*(X|\Lambda) \to A_*(X)/\dot{A}_*(X) \to 0$$

of free complexes splits in each degree, so it stays exact under $- \otimes G$ and $\mathrm{Hom}(-, G)$, and its connecting maps (theorem 3.4.3) are isomorphisms. On chains the connecting map sends a relative cycle z to the class of ∂z modulo $A_*(X|\Lambda)$, and under the isomorphism $\dot{A}_*(X)/A_*(X|\Lambda) \cong A_*(X|d^k)/\dot{A}_*(X|d^k)$ this class is $\partial_k z$, the terms of ∂z over other faces lying in $A_*(X|\Lambda)$. So ∂_k induces isomorphisms on homology and cohomology. By induction so does $\Phi_{X|d^k}$, hence so does $\Phi_X = (-1)^k\Phi_{X|d^k}\partial_k$.

Part (3) for $i = 0$. Call two maps f, f' of the same degree *homotopic* if $f - f' = \partial h \pm h\bar{\partial}$ for some h. Such maps induce the same maps on homology and on cohomology with any coefficients, and composing with ∂_i keeps the relation, ∂_i anticommuting with the differentials (the component of $\partial\partial z = 0$ over $d^i\Delta^{k-1}$). Let $k = 1$, let r be a homotopy inverse of $C_*(X_1) \to A_*(X)$ as in lemma 6.2.4, and let $T = ri_0$. For z in the span of the simplices over surjective operators, $\partial z = \bar{\partial}z + \partial_0 z + \partial_1 z$ with $\partial_0 z \in C_*(X_1)$ and $\partial_1 z \in C_*(X_0)$. Applying the chain map r, which fixes $\partial_0 z$, gives $\partial_0 z + T\partial_1 z = \partial r(z) - r(\bar{\partial}z)$. Here $\Phi_{X|d^0} = \mathrm{id}$ and $\Phi_X = -\partial_1$, by the case $i = k = 1$ above, so $\Phi_{X|d^0}\partial_0 - T\Phi_X = \partial r - r\bar{\partial}$. Let $k \geq 2$, and let Y be the restriction of X to the face spanned by $e_1, \ldots, e_{k-1}$, the last facet of $d^0\Delta^{k-1}$ and the facet of $d^k\Delta^{k-1}$ opposite its first vertex. No face of a simplex over a surjective operator lies over a face of codimension two, so the component of $\partial\partial z = 0$ over this face is $\partial^{(0)}_{k-1}\partial_0 z + \partial^{(k)}_0\partial_k z = 0$, the superscript naming the facet whose boundary is taken. By the identities above for $X|d^0$ and for X, and by the case

$k-1$ applied to $X|d^k$, whose edge $[e_0,e_1]$ is that of X,

$$\Phi_{X|d^0}\partial_0 = (-1)^{k-1}\Phi_Y\partial^{(0)}_{k-1}\partial_0 = (-1)^k\Phi_Y\partial^{(k)}_0\partial_k \simeq (-1)^k T\Phi_{X|d^k}\partial_k = T\Phi_X$$

The identities for Θ_X and Θ^X follow, maps that agree on chains or are homotopic inducing equal maps on homology and on cohomology.

Over each nondegenerate simplex of the base the layer carries the fiber homology shifted up by the dimension of the simplex, and the boundary between layers is the alternating sum over faces, the zeroth face transporting fiber homology from the first vertex to the second. Under the trivial-action hypothesis the identifications λ_b absorb the transports, and the first page becomes the *normalized* singular chain complex of B with coefficients in $H_*(F)$. For an abelian group M this is the complex $N_*(B;M)$ with

$$N_p(B;M) = \bigoplus_{\tau\in NB_p} M, \qquad \partial(\tau\otimes m) = \sum_i (-1)^i[\tau d^i]\otimes m$$

where $[\tau d^i]$ is τd^i if that face is nondegenerate and 0 otherwise. It is the quotient of $C_*(B;M)$ by the chains on degenerate simplices, which form a subcomplex because the boundary of a degenerate simplex consists of degenerate faces and of two equal faces with opposite signs. Dually $N^*(B;M)\subseteq C^*(B;M)$ is the subcomplex of cochains vanishing on degenerate simplices.

Theorem 6.2.7: Homology Serre Spectral Sequence

Let $p\colon E\to B$ be a Serre fibration with B path-connected and fiber F, and suppose $\pi_1(B)$ acts trivially on $H_*(F;G)$ for a fixed coefficient group G. Then there is a first-quadrant homological spectral sequence $\{E^r_{p,q},d^r\}$, with differentials d^r of bidegree $(-r,r-1)$, whose second page is

$$E^2_{p,q} \;\cong\; H_p\big(B;\,H_q(F;G)\big)$$

the homology of the base with coefficients in the fiber homology, and which converges to the homology of the total space:

$$E^2_{p,q} \;\Longrightarrow\; H_{p+q}(E;G)$$

Convergence means that the filtration of definition 6.2.2 induces on $H_n(E;G)$ a finite filtration $0=\mathcal{F}_{-1}\subseteq\mathcal{F}_0\subseteq\cdots\subseteq\mathcal{F}_n=H_n(E;G)$, with $\mathcal{F}_p=\operatorname{im}\big(H_n(F_pC_*(E);G)\to H_n(E;G)\big)$, whose successive quotients are the stable terms: $\mathcal{F}_p/\mathcal{F}_{p-1}\cong E^\infty_{p,n-p}$. The spectral sequence is natural. A map $\Psi\colon E'\to E$ of such fibrations over $g\colon B'\to B$, with g carrying the basepoint of B' to b_0, induces a morphism of spectral sequences that is Ψ_* on the abutments and on the second pages the map $H_p(B';H_q(F';G))\to H_p(B;H_q(F;G))$ induced by g and by $\Psi|_{F'}\colon F'\to F$. A homomorphism of coefficient groups likewise induces a morphism, given on the second pages by its effect on coefficients.

***Proof*:**

Step 1: The first page. Apply theorem 6.1.10 to the filtration of $C_*(E;G)=C_*(E)\otimes G$ by the subcomplexes $F_pC_*(E)\otimes G$ of definition 6.2.2. By lemma 6.2.6(1) and (2), applied to the fibrations X_τ, whose fiber over e_0 is $F_{\tau(e_0)}$,

$$E^1_{p,q} = H_{p+q}\big(F_pC_*(E)/F_{p-1}C_*(E);G\big) \cong \bigoplus_{\tau\in NB_p} H_q\big(F_{\tau(e_0)};G\big)$$

In particular $E^1_{p,q} = 0$ for $p < 0$ and for $q < 0$, so the spectral sequence is first-quadrant.

Step 2: The differential d^1. The differential $d^1 = jk$ of the exact couple is induced by the boundary followed by the projection to $F_{p-1}C_*(E)/F_{p-2}C_*(E)$. For a relative cycle z of the τ-summand, $\partial z = \sum_i \partial_i z$ with the components $\partial_i z$ of lemma 6.2.6(3), no face of a simplex over a surjective operator lying over a face of codimension two. The component $\partial_i z$ lies in the summand of τd^i when τd^i is nondegenerate and in $F_{p-2}C_*(E)$ otherwise. By lemma 6.2.6(3),

$$d^1(\tau \otimes x) = \sum_{i=0}^{p} (-1)^i \, [\tau d^i] \otimes T_i x, \qquad x \in H_q(F_{\tau(e_0)}; G)$$

where $[\tau d^i]$ is 0 when τd^i is degenerate, $T_i = \mathrm{id}$ for $i \geq 1$, and T_0 is the transport along the edge $\tau|_{[e_0,e_1]}$, from the fiber over $\tau(e_0)$ to the fiber over $\tau(e_1) = \tau d^0(e_0)$.

Step 3: Normalized chains compute homology. Apply Steps 1 and 2 to the identity fibration $B \to B$ with coefficients in an abelian group M. Its fibers are points, with identity transports, so $E^1_{p,q} = 0$ for $q \neq 0$ and $E^1_{*,0} = N_*(B; M)$ with d^1 its boundary. Every d^r with $r \geq 2$ leaves the row $q = 0$, so $E^2 = E^\infty$. In total degree p the only nonzero spot is $(p, 0)$, and $F_pC_p(B) = C_p(B)$, so $E^\infty_{p,0} = \mathcal{F}_pH_p/\mathcal{F}_{p-1}H_p = H_p(B; M)$. A cycle $z \in C_p(B; M) = F_pC_p(B; M)$ represents its homology class in $E^\infty_{p,0}$, and its image in $E^1_{p,0} = N_p(B; M)$ is the image of z under the quotient map, since Φ_X sends the identity simplex of Δ^p to the vertex e_0 when the fiber is a point. Hence the quotient map $C_*(B; M) \to N_*(B; M)$ induces isomorphisms $H_p(B; M) \cong H_p\big(N_*(B; M)\big)$.

Step 4: The second page. Under the trivial-action hypothesis, the isomorphisms λ_b that follow lemma 6.2.5 identify the summand of τ in Step 1 with $H_q(F; G)$, and $\lambda_{\tau(e_1)}T_0 = \lambda_{\tau(e_0)}$ turns the formula of Step 2 into the boundary of $N_*\big(B; H_q(F; G)\big)$. By Step 3 with $M = H_q(F; G)$,

$$E^2_{p,q} = H_p\Big(N_*\big(B; H_q(F; G)\big)\Big) \cong H_p\big(B;\, H_q(F; G)\big)$$

Step 5: Convergence and naturality. The filtration is bounded in each degree, $F_{-1}C_n = 0$ and $F_nC_n = C_n$, so theorem 6.1.10 gives the convergence with the stated filtration. A map Ψ over g preserves the filtrations (definition 6.2.2), so it induces a morphism of spectral sequences that is Ψ_* on abutments (theorem 6.1.10, Step 5). On the first page it sends the summand of $\tau' \in NB'_p$ to zero if $g\tau'$ is degenerate, the image then lying in $F_{p-1}C_*(E)$, and otherwise to the summand of $g\tau'$ through the map $X'_{\tau'} \to X_{g\tau'}$, $(x, e') \mapsto (x, \Psi(e'))$, which commutes with Φ, as Φ restricts simplices to front faces. The isomorphism $\lambda_{g(b')}$ may be computed with the path $g\omega'_{b'}$ from b_0 to $g(b')$, so lemma 6.2.5(4) gives $\lambda_{g(b')}\Psi_* = (\Psi|_{F'})_*\lambda'_{b'}$, and the morphism on the first page is the map $N_p(B'; H_q(F'; G)) \to N_p(B; H_q(F; G))$ induced by g and by $\Psi|_{F'}$. The quotient maps of Step 3 commute with g, which gives the description on the second page. A homomorphism of coefficient groups commutes with every identification above.

The theorem packages the whole construction into one statement: a first-quadrant grid whose (p, q)-entry on the second page is the base homology in degree p with fiber-homology coefficients in degree q, and which assembles, after running all the differentials, into the homology of the total space. The bottom row $q = 0$ is $H_p(B; H_0(F)) = H_p(B)$ (the fiber being path-connected), the base

homology. The left column $p = 0$ is $H_0(B; H_q(F)) = H_q(F)$, the fiber homology. The differentials d^r run leftward and upward, trading base degree for fiber degree, and the surviving E^∞ page records $H_*(E)$ only up to the extension problem of reassembling the filtration quotients. The power of the device is that the E^2 page is computable from $H_*(B)$ and $H_*(F)$ alone, so any partial knowledge of $H_*(E)$ forces relations among the differentials, and conversely.

The same construction carries over to cohomology, with the filtration directions reversed. Filtering $C^*(E)$ by the subcomplexes $F^pC^*(E)$ of cochains vanishing on $F_{p-1}C_*(E)$ produces a spectral sequence, and the cohomological half of lemma 6.2.6 identifies its second page as cohomology of the base with fiber-cohomology coefficients. The cohomological version is the one used most in practice, because cohomology carries the cup product, which makes the cohomology Serre spectral sequence into a spectral sequence of algebras (theorem 6.2.15).

Theorem 6.2.8: Cohomology Serre Spectral Sequence

Let $p\colon E \to B$ be a Serre fibration with B path-connected and fiber F, and suppose $\pi_1(B)$ acts trivially on $H^*(F;G)$ for a fixed coefficient group G. Then there is a first-quadrant cohomological spectral sequence $\{E_r^{p,q}, d_r\}$, with differentials d_r of bidegree $(r, 1-r)$, whose second page is

$$E_2^{p,q} \;\cong\; H^p\big(B;\, H^q(F;G)\big)$$

and which converges to the cohomology of the total space:

$$E_2^{p,q} \;\Longrightarrow\; H^{p+q}(E;G)$$

where convergence means $H^n(E;G)$ carries a finite decreasing filtration $H^n = \mathcal{F}^0 \supseteq \mathcal{F}^1 \supseteq \cdots \supseteq \mathcal{F}^{n+1} = 0$, with $\mathcal{F}^p = \operatorname{im}\big(H^n(F^pC^*(E;G)) \to H^n(E;G)\big)$, and $\mathcal{F}^p/\mathcal{F}^{p+1} \cong E_\infty^{p,n-p}$. The spectral sequence is natural for maps of fibrations and for homomorphisms of coefficient groups, as in theorem 6.2.7, with the arrows on base and fiber reversed.

***Proof*:**

The filtration $F^pC^*(E;G)$ of definition 6.2.2 is decreasing, by subcomplexes, with $F^0C^n = C^n$ and $F^{n+1}C^n = 0$. So [13, Convergence for a Decreasing Filtration of a Cochain Complex] gives a spectral sequence with $E_1^{p,q} = H^{p+q}\big(F^pC^*/F^{p+1}C^*\big)$ converging to $H^{p+q}(E;G)$ with the stated filtration. Restriction identifies $F^pC^n/F^{p+1}C^n$ with $\operatorname{Hom}\big(F_pC_n(E)/F_{p-1}C_n(E), G\big)$, a homomorphism on the layer extending to a cochain because $F_pC_n(E)$ is spanned by a subset of the basis of $C_n(E)$. By lemma 6.2.6(1) and (2),

$$E_1^{p,q} \cong \prod_{\tau \in NB_p} H^q\big(F_{\tau(e_0)}; G\big)$$

so the spectral sequence is first-quadrant. The differential d_1 is the connecting map of $0 \to F^{p+1}/F^{p+2} \to F^p/F^{p+2} \to F^p/F^{p+1} \to 0$, which on cochains composes with the boundary components of Step 2 of the proof of theorem 6.2.7. By lemma 6.2.6(3), $(d_1x)(\tau) = \sum_i (-1)^i T^i x(\tau d^i)$ for $\tau \in NB_{p+1}$, where $x(\tau d^i) = 0$ if τd^i is degenerate, $T^i = \mathrm{id}$ for $i \geq 1$, and T^0 is the cohomological transport along $\tau|_{[e_0,e_1]}$. The isomorphisms λ^b that follow lemma 6.2.5 then identify the row $E_1^{*,q}$ with the normalized cochain complex $N^*\big(B; H^q(F;G)\big)$. Running the same argument for the identity fibration of B with coefficients in M, exactly as in Step 3 of the proof of theorem 6.2.7, shows that the inclusion $N^*(B;M) \subseteq C^*(B;M)$ induces isomorphisms

$H^p\big(N^*(B;M)\big) \cong H^p(B;M)$, a normalized cocycle c lying in $F^pC^p(B;M)$ and representing both its class in $E_2^{p,0}$ and its class in $H^p(B;M)$. Hence $E_2^{p,q} \cong H^p\big(B;H^q(F;G)\big)$. Naturality follows as in Step 5 of that proof.

The cohomological version reverses the geometry of the differentials: d_r now raises base degree and lowers fiber degree, of bidegree $(r, 1-r)$, so the arrows on the cohomology page run rightward and downward. The filtration of $H^n(E)$ is decreasing rather than increasing, with the top quotient $\mathcal{F}^0/\mathcal{F}^1 = E_\infty^{0,n}$ a subquotient of the fiber cohomology $H^n(F)$ and the bottom piece $\mathcal{F}^n = E_\infty^{n,0}$ a subquotient of the base cohomology $H^n(B)$. Beyond this bookkeeping difference the two spectral sequences carry the same information, and which one to reach for is dictated by whether the problem at hand gives (co)homology of two of the three spaces and by whether the cup product is needed.

A spectral sequence is only useful when its edges can be read off, because the edges are where the abstract grid touches the actual homology of B, E, and F. The bottom row and left column of the E^2 page survive to E^∞ in part, and the surviving pieces map to or from the homology of the total space through the filtration. These edge maps are the source of every quick application: the Gysin and Wang sequences, the behavior of the spectral sequence for highly connected fiber or base, and the comparison with the maps $F \to E \to B$ on homology.

Theorem 6.2.9: Edge Homomorphisms of the Serre Spectral Sequence

Let $F \xrightarrow{\iota} E \xrightarrow{p} B$ be a Serre fibration as in theorem 6.2.7, with B path-connected and $\pi_1(B)$ acting trivially on $H_*(F)$. In the homology spectral sequence:

1. *(Base edge, for F path-connected.)* The bottom-row term $E_{p,0}^\infty$ is a subgroup of $E_{p,0}^2 = H_p(B;H_0(F)) = H_p(B)$, the intersection of the kernels of the differentials $d^r\colon E_{p,0}^r \to E_{p-r,r-1}^r$ leaving it, and the composite

$$H_p(E) \twoheadrightarrow \mathcal{F}_p/\mathcal{F}_{p-1} = E_{p,0}^\infty \hookrightarrow E_{p,0}^2 = H_p(B)$$

   equals the map $p_*\colon H_p(E) \to H_p(B)$ induced by the projection.

2. *(Fiber edge.)* The left-column term $E_{0,q}^\infty$ is a quotient of $E_{0,q}^2 = H_0(B;H_q(F)) = H_q(F)$, by the images of the differentials $d^r\colon E_{r,q-r+1}^r \to E_{0,q}^r$ entering it, and the composite

$$H_q(F) = E_{0,q}^2 \twoheadrightarrow E_{0,q}^\infty = \mathcal{F}_0H_q(E) \hookrightarrow H_q(E)$$

   equals the map $\iota_*\colon H_q(F) \to H_q(E)$ induced by the fiber inclusion.

Dually, under the hypotheses of theorem 6.2.8, in the cohomology spectral sequence the base edge $H^p(B) = E_2^{p,0} \twoheadrightarrow E_\infty^{p,0} \hookrightarrow H^p(E)$ is p^* for F path-connected, and the fiber edge $H^q(E) \twoheadrightarrow E_\infty^{0,q} \hookrightarrow E_2^{0,q} = H^q(F)$ is the restriction ι^*.

***Proof*:**

Step 1: The bottom row and the base edge. On the bottom row $q = 0$ the entry $E_{p,0}^2 = H_p(B;H_0(F)) = H_p(B)$ uses $H_0(F) \cong \mathbb{Z}$ for F path-connected. No differential d^r for $r \geq 2$ can enter the bottom row, since the only source available is $E_{p+r,\,1-r}^r$, which sits in negative fiber degree and so is zero. Differentials do leave the bottom row, namely $d^r\colon E_{p,0}^r \to E_{p-r,\,r-1}^r$, so each page replaces $E_{p,0}^r$ by the kernel of its outgoing d^r, a subgroup. Hence $E_{p,0}^\infty$ is a subgroup

of $E^2_{p,0} = H_p(B)$. By the convergence of theorem 6.2.7, $E^\infty_{p,0} = \mathcal{F}_p H_p(E)/\mathcal{F}_{p-1}H_p(E)$, and since $\mathcal{F}_p H_p(E) = H_p(E)$ (the top filtration step in total degree p, as $E^\infty_{p',p-p'} = 0$ for $p' > p$), the edge is the composite $H_p(E) \twoheadrightarrow E^\infty_{p,0} \hookrightarrow E^2_{p,0} = H_p(B)$.

To identify this composite with p_*, regard p as a map from the fibration $E \to B$ to the identity fibration of B, over id_B. By the naturality clause of theorem 6.2.7 it induces a morphism of spectral sequences which is p_* on the abutments and, on $E^2_{p,0}$, the map $H_p(B; H_0(F)) \to H_p(B;\mathbb{Z})$ induced by the augmentation $H_0(F) \to \mathbb{Z}$, the identity of $H_p(B)$ for F path-connected. For the identity fibration $E^\infty_{p,0} = E^2_{p,0}$ is all of $H_p(B)$, and its base edge is the identity (Step 3 of the proof of theorem 6.2.7). A morphism of spectral sequences commutes with the projections to the top filtration quotients, so the base edge of E is p_*.

Step 2: The left column and the fiber edge. On the left column $p = 0$ the entry $E^2_{0,q} = H_0(B; H_q(F)) = H_q(F)$ uses that B is path-connected: $E^1_{0,q} = \bigoplus_b H_q(F_b)$, and $E^2_{0,q}$ is its quotient by the relations $x = T_\omega x$ for paths ω, identified with $H_q(F)$ through the λ_b. No differential leaves $E^r_{0,q}$, since $d^r\colon E^r_{0,q} \to E^r_{-r,q+r-1} = 0$ lands in negative base degree. Differentials *enter* $E^r_{0,q}$ from $E^r_{r,q-r+1}$ for $r \geq 2$, so $E^\infty_{0,q}$ is a quotient of $E^2_{0,q} = H_q(F)$ by the images of those incoming differentials. By convergence, $E^\infty_{0,q} = \mathcal{F}_0 H_q(E)$, the image of $H_q\big(F_0 C_*(E)\big) = \bigoplus_b H_q(F_b)$ in $H_q(E)$, and the map $E^1_{0,q} = \bigoplus_b H_q(F_b) \to E^\infty_{0,q}$ is induced by the inclusions of the fibers. A class $x \in H_q(F) = E^2_{0,q}$ is represented by x in the summand of b_0, so its image in $H_q(E)$ is $\iota_* x$. Hence the fiber edge is ι_*.

Step 3: The cohomological edges. In cohomology the differentials d_r have bidegree $(r, 1-r)$, so on the bottom row $q = 0$ the differentials leaving $E^{p,0}_r \to E^{p+r,1-r}_r = 0$ vanish (negative fiber degree), while those entering come from $E^{p-r,r-1}_r$ and can be nonzero, making $E^{p,0}_\infty$ a *quotient* of $E^{p,0}_2 = H^p(B)$. The decreasing filtration on $H^p(E)$ has bottom step $\mathcal{F}^p H^p(E) = E^{p,0}_\infty$, and the base edge $H^p(B) = E^{p,0}_2 \twoheadrightarrow E^{p,0}_\infty = \mathcal{F}^p H^p(E) \hookrightarrow H^p(E)$ is p^* by the naturality argument of Step 1. On the left column $p = 0$, differentials entering $E^{0,q}_r$ come from $E^{-r,q+r-1}_r = 0$ and vanish, while those leaving go to $E^{r,q-r+1}_r$, so $E^{0,q}_\infty$ is a *subgroup* of $E^{0,q}_2 = H^q(F)$, cut out as the intersection of the kernels of the outgoing differentials. The map $H^q(E) \to E^{0,q}_1 = \prod_b H^q(F_b)$ restricts a class to every fiber, and $E^{0,q}_2 = H^q(F)$ reads the component at b_0, so the fiber edge $H^q(E) \twoheadrightarrow E^{0,q}_\infty \hookrightarrow E^{0,q}_2 = H^q(F)$ is the restriction ι^*. This establishes all four edge homomorphisms, completing the proof.

The edge homomorphisms are the spectral sequence's interface with the maps in the fibration. The base edge factors p_* through the bottom row, so $p_*\colon H_p(E) \to H_p(B)$ is surjective onto $E^\infty_{p,0}$, the part of $H_p(B)$ that survives all differentials. Classes of $H_p(B)$ killed by some d^r are precisely those not hit by p_*. The fiber edge factors ι_* through the left column, so the image of $H_q(F) \to H_q(E)$ is $E^\infty_{0,q}$, the part of $H_q(F)$ that survives, and classes of $H_q(F)$ killed by an incoming differential die in the total space. In cohomology the roles invert: the base edge p^* factors as the surjection $H^p(B) \twoheadrightarrow E^{p,0}_\infty$ followed by the inclusion $E^{p,0}_\infty \hookrightarrow H^p(E)$, so $\mathrm{im}(p^*) = E^{p,0}_\infty$. The fiber edge ι^* surjects $H^q(E)$ onto $E^{0,q}_\infty \subseteq H^q(F)$, the classes of the fiber that extend over the total space. The simplest consequence, used constantly, is that when the spectral sequence collapses (all differentials zero) the edges become honest injections and surjections relating the three homologies directly.

The product case is the cleanest test of the whole apparatus. When the fibration is trivial, $E = B \times F$, the total space homology is governed by the Künneth theorem, and the Serre spectral sequence must reproduce that answer. It does so by collapsing immediately, which exposes both the meaning of the E^2 page and the role of the differentials as the measure of nontriviality of the fibration.

Example 6.2.10: The Product Fibration and the Künneth Theorem

Let $E = B \times F \xrightarrow{p} B$ be the trivial fibration, with B a path-connected CW complex and F a path-connected space with $H_*(F)$ of finite type, and suppose all groups are taken with field coefficients in a field $\Bbbk$ so that Tor terms vanish. The projection p admits the section $s\colon b \mapsto (b, f_0)$ and the fiber inclusion $\iota\colon F = \{b_0\} \times F \hookrightarrow B \times F$ admits the retraction $r\colon (b, f) \mapsto f$. The E^2 page is

$$E^2_{p,q} = H_p\big(B; H_q(F;\Bbbk)\big) \cong H_p(B;\Bbbk) \otimes_\Bbbk H_q(F;\Bbbk)$$

the last isomorphism because $H_q(F;\Bbbk)$ is a $\Bbbk$-vector space and homology with vector-space coefficients is $H_p(B;\Bbbk) \otimes_\Bbbk H_q(F;\Bbbk)$ by the universal coefficient theorem over a field.

The edges collapse, forcing Künneth in a range. The projection $p\colon E \to B$ admits the section just named, and the fiber inclusion ι admits the retraction r. By theorem 6.2.9 the fiber edge $\iota_*\colon H_q(F) \to H_q(E)$ factors as $H_q(F) = E^2_{0,q} \twoheadrightarrow E^\infty_{0,q} \hookrightarrow H_q(E)$, and the retraction gives $r_*\iota_* = \mathrm{id}$, so ι_* is injective. An injective map whose first factor is the surjection $E^2_{0,q} \twoheadrightarrow E^\infty_{0,q}$ forces that surjection to be an isomorphism, hence $E^\infty_{0,q} = E^2_{0,q}$ and no differential enters the left column. Dually the section makes p_* surjective, so the inclusion $E^\infty_{p,0} \hookrightarrow E^2_{p,0} = H_p(B)$ is onto, hence $E^\infty_{p,0} = E^2_{p,0}$ and no differential leaves the bottom row.

Now restrict to total degree $n \le N$ where N is below the first interior obstruction: concretely, suppose F is $(m-1)$-connected, so the rows $1 \le q \le m-1$ of the E^2 page vanish entirely. Then in total degree $n \le m$ the only nonzero spots are the bottom row $E^2_{n,0} = H_n(B)$ and the left-column spot $E^2_{0,n} = H_n(F)$ (the latter present only for $n = 0$ or $n \ge m$), every other spot $E^2_{p,q}$ with $p+q = n$ and $0 < q < m$ being zero. All differentials in this range have a zero source or target except those on the two edges, which vanish by the previous paragraph. Hence $E^\infty_{p,q} = E^2_{p,q}$ for all (p,q) with $p+q \le m$. Over the field $\Bbbk$ the extension problem is trivial (every short exact sequence of vector spaces splits), so

$$H_n(B \times F;\Bbbk) \cong \bigoplus_{p+q=n} E^2_{p,q} = \bigoplus_{p+q=n} H_p(B;\Bbbk) \otimes_\Bbbk H_q(F;\Bbbk) \qquad (n \le m)$$

which is the Künneth theorem over a field in the stated range. The full collapse $E^\infty = E^2$ in all degrees holds as well, but its proof uses the multiplicative structure of the cohomology spectral sequence, theorem 6.2.15 below (the page is generated by the edges as an algebra, and the Leibniz rule propagates the vanishing of edge differentials to the whole page). The present sanity check needs only the edges and the connectivity range. The differentials d^r are thus the corrections that a nontrivial fibration imposes on this product answer.

When $\pi_1(B)$ acts nontrivially on $H_*(F)$, the identification of the E^2 page breaks: the transports in the formula for d^1 can no longer be absorbed into identifications λ_b, and the coefficient group is no longer constant. The repair is to use homology of B with local coefficients (section 4.6) in the homology of the fiber.

Example 6.2.11: The Fiber-Homology System of a Fibration

Let $p\colon E \to B$ be a Serre fibration (definition 5.2.2) with path-connected base B and fiber $F = p^{-1}(b_0)$. Transport along paths (lemma 6.2.5) gives isomorphisms $H_q(p^{-1}(b)) \to H_q(p^{-1}(b'))$ that depend only on the class of the path rel endpoints and compose along concatenation. On loops at b_0 this is, for each q, the monodromy action of $\pi_1(B, b_0)$ on $H_q(F)$ by automorphisms. When a loop γ lifts to a homotopy $F \times I \to E$ starting at the inclusion, as it does for a Hurewicz

fibration, γ acts through the end of the lift, a self-map of F. The transports form a local system

$$\mathcal{H}_q(F)\colon \Pi_1(B) \to \mathsf{Ab}, \qquad b \mapsto H_q(p^{-1}(b))$$

on the base, in the sense of definition 4.6.1. It is the coefficient system of the twisted Serre spectral sequence of box 6.2.12, whose E^2 page $H_p(B;\mathcal{H}_q(F))$ is homology of B with coefficients in this system. The system is trivial when $\pi_1(B)$ acts trivially on $H_*(F)$, the hypothesis under which the untwisted spectral sequence of theorem 6.2.7 holds. The smallest example where it is not is the Klein bottle K, realized as the S^1-bundle over S^1 obtained from $S^1 \times [0,1]$ by gluing the ends through the reflection $z \mapsto \bar{z}$ of the circle fiber. The reflection reverses orientation of $F = S^1$, so transport around the base circle acts on $H_1(F) = \mathbb{Z}$ by -1 and on $H_0(F) = \mathbb{Z}$ trivially. The degree-one system $\mathcal{H}_1(F)$ is the sign representation of $\pi_1(S^1) \cong \mathbb{Z}$ on $\mathbb{Z}$, nontrivial, and it is precisely the E^2 input that produces the torsion in $H_1(K;\mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ in exercise 6.2.1 (6).

6.2.12 Remark: The Local-Coefficients Serre Spectral Sequence Drop the hypothesis that $\pi_1(B)$ acts trivially on $H_*(F)$. The monodromy makes $H_q(F)$ a $\mathbb{Z}[\pi_1(B)]$-module, the local coefficient system $\mathcal{H}_q(F)$ of example 6.2.11. Steps 1 and 2 of the proof of theorem 6.2.7 go through unchanged, so the E^1 row $E^1_{*,q} = \bigoplus_\tau H_q(F_{\tau(e_0)})$, with d^1 the alternating sum over faces and the zeroth face twisted by transport, is the normalized singular chain complex of B with local coefficients in $\mathcal{H}_q(F)$. Its homology is homology with local coefficients (for a CW base, its comparison with definition 4.6.6, including the comparison of normalized with unnormalized chains with local coefficients, belongs to the cover-independence statement of section 4.6 and to the citation at the end of this remark):

$$E^2_{p,q} \cong H_p\big(B;\, \mathcal{H}_q(F)\big) \implies H_{p+q}(E)$$

Cohomologically, $E_2^{p,q} \cong H^p(B;\mathcal{H}^q(F)) \Rightarrow H^{p+q}(E)$. When B is simply connected, or more generally when the action is trivial, the local system is constant and this reduces to theorem 6.2.7. The local-coefficient homology $H_*(B;\mathcal{M})$ is the homology of the chain complex $C_*^{CW}(\widetilde{B})\otimes_{\mathbb{Z}[\pi_1]}\mathcal{M}$ built from the universal cover. This version is stated in [22, §5.1, Local Coefficients], which leaves the changes to the proof to the reader.

The spectral sequence of theorem 6.2.8 computes the cohomology groups of E. The cup product makes $H^*(E;R)$ a graded-commutative ring (definition 4.2.7), and the spectral sequence carries that structure as well: every page $E_r^{\bullet,\bullet}$ is a bigraded algebra, each differential d_r is a derivation, and the product on the second page is, up to a Koszul sign, the cup product of base and fiber transported through the identification $E_2^{p,q} \cong H^p(B;H^q(F))$. The product on $H^*(E)$ is recovered from the product on E_∞ up to the ambiguity inherent in any filtration. The computations of section 6.3 and section 6.4 use this structure through the derivation property of the differentials.

For the multiplicative structure, R is a commutative coefficient ring (taken to be $\mathbb{Z}$ or a field unless stated otherwise), and $p\colon E \to B$ is a Serre fibration whose base B is path-connected, with fiber F and $\pi_1(B)$ acting trivially on $H^*(F;R)$, so that the cohomology Serre spectral sequence of theorem 6.2.8 has second page

$$E_2^{p,q} \;\cong\; H^p\big(B;\, H^q(F;R)\big)$$

converging to $H^{p+q}(E;R)$, with the abutment carrying the decreasing filtration

$$H^n(E;R) = F^0H^n \supseteq F^1H^n \supseteq \cdots \supseteq F^nH^n \supseteq F^{n+1}H^n = 0$$

whose successive quotients are $F^pH^n/F^{p+1}H^n \cong E_\infty^{p,n-p}$. This is the filtration induced by the skeletal filtration of definition 6.2.2: F^pH^n is the image of $H^n(F^pC^*(E;R))$ in $H^n(E;R)$.

The product on the spectral sequence is tied to the cup product on $H^*(E)$ by a compatibility of the abutment filtration with products.

Definition 6.2.13: Multiplicative Filtration

Let $A = \bigoplus_n A^n$ be a graded R-algebra equipped with a decreasing filtration $A^n = F^0A^n \supseteq F^1A^n \supseteq \cdots$ by R-submodules. The filtration is *multiplicative* if the cup product respects filtration degree:

$$F^pA^m \smile F^{p'}A^{m'} \subseteq F^{p+p'}A^{m+m'}$$

for all p, p', m, m'. The associated graded object $\operatorname{gr} A = \bigoplus_{p,q} F^pA^{p+q}/F^{p+1}A^{p+q}$ is then a bigraded R-algebra, with product induced by the cup product: the class of $a \in F^pA^{p+q}$ times the class of $b \in F^{p'}A^{p'+q'}$ is the class of $a \smile b \in F^{p+p'}A^{p+q+p'+q'}$ in the quotient $F^{p+p'}/F^{p+p'+1}$.

The condition $F^p \smile F^{p'} \subseteq F^{p+p'}$ says that "deeper" classes multiply to "even deeper" ones, so the leading-order behavior of a product depends only on the leading-order behavior of the factors. This is exactly what makes the associated graded a ring: in $\operatorname{gr} A$ one multiplies leading terms and discards everything lower, and the inclusion guarantees the discarded part really is lower. The spectral-sequence product will live on $\operatorname{gr} H^*(E)$, and the gap between $\operatorname{gr} H^*(E)$ and $H^*(E)$ itself is the extension problem, both additive and multiplicative, that the spectral sequence cannot resolve on its own.

Example 6.2.14: The Two-Step Filtration on $H^*(\mathbb{C}\mathrm{P}^1)$

Take $A = H^*(\mathbb{C}\mathrm{P}^1;\mathbb{Z}) = H^*(S^2;\mathbb{Z})$, which is $\mathbb{Z}$ in degrees 0 and 2 and zero elsewhere, with $H^0 = \mathbb{Z}\langle 1\rangle$ and $H^2 = \mathbb{Z}\langle x\rangle$ where $x^2 = 0$ for dimension reasons. View $\mathbb{C}\mathrm{P}^1 = S^2$ as the base of the trivial fibration $\{*\} \to S^2 \to S^2$, whose abutment filtration on $H^2(S^2)$ is $F^0H^2 = F^1H^2 = F^2H^2 = \mathbb{Z}\langle x\rangle$ and $F^3H^2 = 0$, while on H^0 it is $F^0H^0 = \mathbb{Z}\langle 1\rangle \supseteq F^1H^0 = 0$. The product $1 \smile 1 = 1$ lands in F^0, and $1 \smile x = x$ lands in $F^2H^2 = F^{0+2}H^2$, consistent with $F^0 \smile F^2 \subseteq F^2$. There is no pair of positive-filtration classes to multiply, so the inclusion holds vacuously in the remaining cases, and the filtration is multiplicative. The associated graded recovers A itself, since each filtration jump is by a single $\mathbb{Z}$, exhibiting the degenerate case where $\operatorname{gr} A \cong A$ as rings.

The product on the spectral sequence is built so that, page by page, it is compatible with the differential and descends to the next page, terminating in a product on E_∞ that agrees with the product on $\operatorname{gr} H^*(E)$. The singular cup product respects the cochain filtration of definition 6.2.2 on the nose, because the front and back faces of a simplex split the dimension of its base simplex between them. A multiplicative filtration of a differential graded algebra carries products on all pages by [13, Multiplicative Structure on the Spectral Sequence of a Filtered DG Algebra]. The topology enters again in the product on E_2, read off from the first page of lemma 6.2.6.

Theorem 6.2.15: Multiplicativity of the Cohomology Serre Spectral Sequence

Let $p\colon E \to B$ be a Serre fibration over a path-connected base, with fiber F and trivial action of $\pi_1(B)$ on $H^*(F;R)$, with R a commutative ring. The cohomology Serre spectral sequence $\{E_r^{\bullet,\bullet}, d_r\}$ of theorem 6.2.8 admits products

$$E_r^{p,q} \otimes E_r^{p',q'} \longrightarrow E_r^{p+p',\,q+q'}, \qquad a \otimes b \longmapsto a \cdot b$$

one on each page, with the following properties.

1. *(Bigraded algebra)* On each page the product is R-bilinear and associative, with unit $1 \in E_r^{0,0}$. From the second page on it is graded-commutative for the total degree: if $r \geq 2$, $a \in E_r^{p,q}$ and $b \in E_r^{p',q'}$, then
$$a \cdot b = (-1)^{(p+q)(p'+q')}\, b \cdot a$$
2. *(Leibniz rule)* Each differential d_r is a derivation:
$$d_r(a \cdot b) = d_r(a) \cdot b + (-1)^{p+q}\, a \cdot d_r(b) \qquad \text{for } a \in E_r^{p,q}$$
and the isomorphism $E_{r+1} \cong H(E_r, d_r)$ is one of bigraded algebras.
3. *(Product on E_2)* Under the identification $E_2^{p,q} \cong H^p(B; H^q(F;R))$ of theorem 6.2.8, the product of $a \in E_2^{p,q}$ and $b \in E_2^{p',q'}$ is
$$a \cdot b = (-1)^{qp'}\, a \smile b$$
where $\smile\colon H^p(B;H^q(F;R)) \otimes H^{p'}(B;H^{q'}(F;R)) \to H^{p+p'}(B;H^{q+q'}(F;R))$ is the cup product given on cochains by $(\alpha \smile \beta)(\tau) = \alpha(\tau|_{[e_0,\dots,e_p]}) \smile \beta(\tau|_{[e_p,\dots,e_{p+p'}]})$, the values multiplied by the cup product of F. In particular the left column $E_2^{0,*} = H^*(F;R)$ carries the cup product of F. For F path-connected, the bottom row $E_2^{*,0} = H^*(B;R)$ carries the cup product of B, and the product $E_2^{p,0} \otimes E_2^{0,q} \to E_2^{p,q}$ is the coefficient map $H^p(B;R) \otimes_R H^q(F;R) \to H^p(B;H^q(F;R))$.
4. *(Convergence to the ring)* The abutment filtration on $H^*(E;R)$ is multiplicative in the sense of definition 6.2.13, and the induced product on $\operatorname{gr} H^*(E;R)$ agrees with the product on E_∞ under the isomorphism $E_\infty^{p,q} \cong F^pH^{p+q}/F^{p+1}H^{p+q}$.

Clauses (1), (2) and (4), graded commutativity aside, hold for the spectral sequence of any filtered differential graded algebra whose filtration is multiplicative and bounded in each degree, and the proof checks that the cochain filtration of definition 6.2.2 is such. Clause (3) and graded commutativity are topological, and the proof computes the product on the first page through lemma 6.2.6. That first page is the algebra of normalized cochains on B with values in $H^*(F;R)$, graded-commutative only up to coboundaries, which is why graded commutativity starts on the second page. The sign $(-1)^{qp'}$ in (3) is the Koszul sign for moving the fiber degree q of a past the base degree p' of b, the sign of the product $(\beta \otimes f)(\beta' \otimes f') = (-1)^{|f||\beta'|}\beta\beta' \otimes ff'$ in a graded tensor product.

***Proof*:**
The cohomology Serre spectral sequence is the spectral sequence of the cochain algebra $A = C^*(E;R)$, with the cup product of definition 4.2.1, filtered by the subcomplexes F^kA of definition 6.2.2.

Step 1: The filtration is multiplicative. Let $\varphi \in F^kC^m(E;R)$ and $\psi \in F^lC^{m'}(E;R)$, and let σ be a singular $(m+m')$-simplex in $F_{k+l-1}C_*(E)$, so that $p\sigma = \gamma\circ\theta$ for a surjective simplicial operator θ onto Δ^j with $j \leq k+l-1$. A weakly increasing surjection takes every value between its first and its last, so θ carries $\{0,\dots,m\}$ onto $\{0,\dots,\theta(m)\}$ and $\{m,\dots,m+m'\}$ onto $\{\theta(m),\dots,j\}$. Hence the front face $\sigma|_{[e_0,\dots,e_m]}$ lies in $F_{\theta(m)}C_*(E)$ and the back face $\sigma|_{[e_m,\dots,e_{m+m'}]}$ in $F_{j-\theta(m)}C_*(E)$. If $\theta(m) \leq k-1$, then φ vanishes on the front face. Otherwise $j-\theta(m) \leq l-1$, and ψ vanishes on the back face. Either way $(\varphi \smile \psi)(\sigma) = 0$, so $F^kA \smile F^lA \subseteq F^{k+l}A$. The algebra A is associative with unit the constant 0-cochain 1, it satisfies the Leibniz rule of lemma 4.2.2, $F^0A = A$, and $F^{n+1}A^n = 0$. So [13, Multiplicative Structure on the Spectral Sequence of a Filtered DG Algebra] applies. Every page carries an associative, unital, R-bilinear product induced by the cup product on representatives, and (2) and (4) hold.

Step 2: The product on the first page. Let $x \in E_1^{p,q}$ and $y \in E_1^{p',q'}$ be represented by $\varphi \in F^pA^{p+q}$ and $\psi \in F^{p'}A^{p'+q'}$, so that $x\cdot y$ is represented by $\varphi \smile \psi$. Fix $\tau \in NB_{p+p'}$, let $X = X_\tau \to \Delta^{p+p'}$, and let X_f and X_b be its restrictions to the front face $[e_0,\dots,e_p]$ and the back face $[e_p,\dots,e_{p+p'}]$, the pullbacks along $\tau_f = \tau|_{[e_0,\dots,e_p]}$ and $\tau_b = \tau|_{[e_p,\dots,e_{p+p'}]}$. For relative cochains α on $A_*(X_f)/\dot{A}_*(X_f)$ and β on $A_*(X_b)/\dot{A}_*(X_b)$, extend both by zero to cochains $\tilde{\alpha},\tilde{\beta}$ on $A_*(X)$, and let $\mu(\alpha,\beta)$ be $\tilde{\alpha}\smile\tilde{\beta}$ on the simplices over surjective operators. For such a simplex over η, the computation of Step 1 gives $\mu(\alpha,\beta)(\sigma) = \alpha(\text{front})\,\beta(\text{back})$ if $\eta(|\alpha|) = p$, the two faces then lying over $[e_0,\dots,e_p]$ and $[e_p,\dots,e_{p+p'}]$ by surjective operators, and $\mu(\alpha,\beta)(\sigma) = 0$ otherwise. On the same grounds the Leibniz rule for $\tilde{\alpha}\smile\tilde{\beta}$ gives $\delta\mu(\alpha,\beta) = \mu(\delta\alpha,\beta) + (-1)^{|\alpha|}\mu(\alpha,\delta\beta)$, so μ pairs cohomology classes. By Step 1 again, the τ-component of $\varphi\smile\psi$ in $E_0^{p+p',q+q'} = \prod_\tau \operatorname{Hom}\big(A_*(X_\tau)/\dot{A}_*(X_\tau),R\big)$ is $\mu(\varphi_f,\psi_b)$, where φ_f is $\sigma\mapsto\varphi(\sigma)$ on the simplices of X_f over surjective operators, zero if τ_f is degenerate, and likewise ψ_b. We claim that for $a \in H^{p+q}\big(A_*(X_f)/\dot{A}_*(X_f);R\big)$ and $b \in H^{p'+q'}\big(A_*(X_b)/\dot{A}_*(X_b);R\big)$,

$$\Theta^X\mu(a,b) = (-1)^{qp'}\,\Theta^{X_f}(a)\smile T\Theta^{X_b}(b) \qquad (*)$$

with Θ as in lemma 6.2.6(2) and $T\colon H^*(X_p;R)\to H^*(X_0;R)$ the transport along the edge $[e_0,e_p]$. Induct on p'.

Let $p'=0$, so $X = X_f$ and $X_b = X_p$. Represent b by a cocycle β on X_p and choose a cocycle Ψ on $A_*(X)$ restricting to β. One exists by lemma 6.2.4: a cocycle Ψ_0 with $\Psi_0|_{X_p} - \beta = \delta\gamma$ is corrected by the coboundary of an extension of γ. The back face in $\mu(\alpha,\beta)(\sigma)$ lies in X_p, so $\mu(\alpha,\beta) = \alpha\smile\Psi$. For a relative cocycle c of $X|d^p$, the cochain $\bar{c}\smile\Psi$, with $\bar{c}$ the extension of c by zero, extends $c\smile\Psi$ and vanishes over the faces of Δ^p missing a vertex other than e_p. So it computes the connecting map in the proof of lemma 6.2.6(2): $\partial_p^*[c\smile\Psi]$ is the class of $\delta(\bar{c}\smile\Psi) = \delta\bar{c}\smile\Psi$. On simplices over surjective operators this is $\partial_p^*(c)\smile\Psi$, because $\delta\bar{c}$ vanishes on front faces over proper faces of Δ^p, c being a relative cocycle. With $\Theta^X\partial_p^* = (-1)^p\Theta^{X|d^p}$ from lemma 6.2.6(3), induction on p gives $\Theta^X[\alpha\smile\Psi] = \Theta^X[\alpha]\smile[\Psi|_{X_0}]$. Finally $[\Psi|_{X_0}] = Tb$, the transport being computable in $A_*(X)$, which contains $A_*(X|[e_0,e_p])$ by a cohomology isomorphism (both contain $C_*(X_0)$ by isomorphisms, lemma 6.2.4).

Let $p' \geq 1$. The last facet $d^{p+p'}\Delta^{p+p'-1}$ has front face $[e_0, \dots, e_p]$ and back face the last facet of $[e_p, \dots, e_{p+p'}]$. By lemma 6.2.6(2) every b is $\partial^*_{p'}[\beta']$ for a relative cocycle β' of $X_b|d^{p'}$. Let $\kappa = \tilde{\alpha} \smile \tilde{\beta}'$, with $\tilde{\beta}'$ the extension of β' by zero to $A_*(X)$. It vanishes except on the simplices over $d^{p+p'}\Delta^{p+p'-1}$ by surjective operators, where it is $\mu(\alpha, \beta')$ for the fibration $X|d^{p+p'}$. So $\partial^*_{p+p'}[\mu(\alpha, \beta')]$ is the class of $\delta\kappa = \delta\tilde{\alpha} \smile \tilde{\beta}' + (-1)^{|\alpha|}\tilde{\alpha} \smile \delta\tilde{\beta}'$ on simplices over surjective operators. There the first term vanishes, the back face having its last vertex over $e_{p+p'}$, off the support of $\tilde{\beta}'$, and the second is $(-1)^{|\alpha|}\mu(\alpha, \partial^*_{p'}\beta')$. Hence, by lemma 6.2.6(3) twice and the case $p' - 1$,

$$\begin{aligned}\Theta^X\mu(a, b) &= (-1)^{p+q}\,\Theta^X\partial^*_{p+p'}\mu(a, [\beta']) = (-1)^{q+p'}\,\Theta^{X|d^{p+p'}}\mu(a, [\beta']) \\ &= (-1)^{q+p'+q(p'-1)}\,\Theta^{X_f}(a) \smile T\Theta^{X_b|d^{p'}}[\beta']\end{aligned}$$

and $\Theta^{X_b|d^{p'}}[\beta'] = (-1)^{p'}\Theta^{X_b}(b)$, which gives $(*)$.

Step 3: The product on E_2. The isomorphisms λ^b used in the proof of theorem 6.2.8 are ring isomorphisms with $\lambda^{\tau(e_0)}T = \lambda^{\tau(e_p)}$ for the transport along $\tau|_{[e_0,e_p]}$. By Step 2 they identify E_1 with $N^*\big(B; H^*(F; R)\big)$, the product being $(x \cdot y)(\tau) = (-1)^{qp'}x(\tau_f) \smile y(\tau_b)$, where $x(\tau_f) = 0$ for degenerate τ_f. This is $(-1)^{qp'}$ times the cup product of (3) on normalized cochains. By (2), $E_2 = H(E_1, d_1)$ as algebras, and the inclusion $N^* \subseteq C^*$ that induces $E_2^{p,q} \cong H^p(B; H^q(F; R))$ is compatible with this cup product, so (3) holds. For $p = p' = 0$ the normalized 0-cocycles are the constant functions, multiplied pointwise, which gives the left column. For F path-connected, $H^0(F; R) = R$ gives the bottom row, and $(\beta \smile f)(\tau) = \beta(\tau)f$ for a constant 0-cocycle f gives the coefficient map.

Step 4: Graded commutativity. Let $a \in E_2^{p,q}$ and $b \in E_2^{p',q'}$ correspond to α and β. The proof of theorem 4.2.6 uses the commutativity of R only to exchange the two coefficient factors, so it gives $\alpha \smile \beta = (-1)^{pp'}\beta \smile' \alpha$, where $\smile'$ multiplies the values $v \in H^{q'}(F; R)$ and $u \in H^q(F; R)$ to $u \smile v = (-1)^{qq'}v \smile u$ (theorem 4.2.6 for F). So $\alpha \smile \beta = (-1)^{pp'+qq'}\beta \smile \alpha$, and by (3)

$$a \cdot b = (-1)^{qp'+pp'+qq'}\,\beta \smile \alpha = (-1)^{qp'+pp'+qq'+q'p}\,b \cdot a = (-1)^{(p+q)(p'+q')}\,b \cdot a$$

Graded commutativity on E_2 passes to every later page and to E_∞ by [13, Multiplicative Structure on the Spectral Sequence of a Filtered DG Algebra], which completes (1).

The computations of section 6.3 and section 6.4 rest on this theorem. Its practical force is concentrated in clause (2): because d_r is a derivation, it is determined by its values on a set of algebra generators of E_r. If E_2 is generated as an algebra by a few classes, then knowing d_r on those classes determines d_r everywhere, and conversely a single nonzero differential propagates multiplicatively through all products. Clause (3) is what makes E_2 computable: it is the cohomology of the base with coefficients in the cohomology ring of the fiber, with the cup product corrected by the Koszul sign. Clause (4) is the bridge back to topology, and it carries the standard caveat of all spectral sequences: E_∞ computes only the associated graded ring, so reconstructing $H^*(E)$ requires solving extension problems, which in the multiplicative setting means determining products of classes from different filtration levels that the spectral sequence sees only separately.

6.2.16 Warning: E_∞ Gives Only the Associated Graded Ring The isomorphism $E_\infty^{p,q} \cong F^pH^{p+q}/F^{p+1}$ is one of *bigraded* algebras, and two phenomena can obstruct reading off the singly graded ring $H^*(E)$ from E_∞. First, additive extensions: two entries $\mathbb{Z}/2$ in total degree n leave open whether $H^n(E)$ is $\mathbb{Z}/2 \oplus \mathbb{Z}/2$ or $\mathbb{Z}/4$, and no such ambiguity arises when the entries are free abelian, since then every extension splits. Second, multiplicative extensions: even when the additive structure is settled, a product $a \smile b$ may equal a class of strictly higher filtration than its leading term in E_∞, so E_∞ records $a \cdot b$ only modulo higher filtration. The additive ambiguity disappears when each total degree of E_∞ holds at most one nonzero entry, which is then all of $H^n(E)$. The multiplicative one can persist even then: for a and b of filtrations p and p' the product is known only to lie in $F^{p+p'}H^{|a|+|b|}(E)$, which can be the whole group while $E_\infty^{p+p',\,|a|+|b|-p-p'}$ vanishes.

Exercise 6.2.1

1. Let $F \to E \to B$ be a Serre fibration satisfying the hypotheses of theorem 6.2.7, and suppose B is $(n-1)$-connected, so $\widetilde{H}_p(B) = 0$ for $p < n$. Show that the fiber edge homomorphism $\iota_*\colon H_q(F) \to H_q(E)$ is an isomorphism for $q < n-1$ and surjective for $q = n-1$. (This is the homology analogue of the connectivity statement for fibrations.)

2. Suppose instead the fiber F is $(m-1)$-connected, so $\widetilde{H}_q(F) = 0$ for $q < m$. Show that the base edge $p_*\colon H_p(E) \to H_p(B)$ is an isomorphism for $p < m$ and that there is an exact sequence $H_{m+1}(B) \to H_m(F) \to H_m(E) \to H_m(B) \to 0$ in low degrees. Identify the first map with a single differential of the spectral sequence.

3. Consider the Hopf fibration $S^1 \to S^3 \to S^2$ of example 5.2.7. Write out the E^2 page of the homology Serre spectral sequence, determine the single possibly-nonzero differential, and use convergence to $H_*(S^3)$ to compute that differential. Confirm the answer is consistent with $H_*(S^3)$ being that of a 3-sphere.

4. Let $F \to E \to B$ be a fibration with $B = S^n$ $(n \geq 2)$ simply connected. Using the CW structure on S^n with one 0-cell and one n-cell, show directly from theorem 6.2.7 that the E^2 page has only two nonzero columns, $p = 0$ and $p = n$, each isomorphic to $H_*(F)$. Deduce that the only possibly-nonzero differential is d^n, and write down the resulting long exact sequence relating $H_*(F)$, $H_*(E)$, and the shift $H_{*-n}(F)$.

5. In the cohomology Serre spectral sequence of theorem 6.2.8, suppose E is contractible (for instance the path fibration $\Omega B \to PB \to B$). Explain why $E_\infty^{p,q} = 0$ for $(p,q) \neq (0,0)$, and describe in words how this forces $H^*(\Omega B)$ to be determined by $H^*(B)$. (You need not compute a specific example. The point is the structural mechanism that section 6.3 exploits.)

6. The Klein bottle K is the total space of an S^1-bundle over S^1 with monodromy acting on $H_1(S^1) = \mathbb{Z}$ by -1. Using example 4.6.9 as the E^2 input, identify the row $H_p(S^1; \mathcal{H}_1(F))$ of the twisted Serre spectral sequence box 6.2.12 for this bundle, and compare $H_1(K;\mathbb{Z})$ computed this way with the known value $H_1(K;\mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$.

6.3 Computations with the Serre Spectral Sequence

The Serre spectral sequence of theorem 6.2.7 and theorem 6.2.8 is an apparatus for transferring homological information across a fibration $F \to E \to B$: knowing the homology of two of the three spaces, one reads off the third from the E_2 page and its differentials. The previous section assembled the machine. This section runs it. Three computations occupy the bulk of what follows, and each is carried out to the last differential. The first computes the homology of the loop space ΩS^n by feeding the path-loop fibration of example 5.2.15 into the cohomology spectral sequence, exploiting the contractibility of the total space PS^n to force every group of $H^*(\Omega S^n)$ into existence. The second extracts two general-purpose exact sequences, the Gysin sequence for a sphere bundle and the Wang sequence for a fibration over a sphere, directly from the shape of the E_2 page when the base or fiber has cohomology concentrated in two degrees. The third returns to the problem that opened the chapter on homotopy theory, the higher homotopy of spheres, and uses the spectral sequence together with the Hurewicz theorem (theorem 5.4.9) to compute $\pi_3(S^2) \cong \mathbb{Z}$ and the first stable stem $\pi_{n+1}(S^n) \cong \mathbb{Z}/2$ for $n \geq 3$.

The governing principle throughout is that a contractible total space, or a base or fiber whose cohomology vanishes outside a narrow band, collapses the spectral sequence so completely that the surviving differentials are forced to be isomorphisms. Every computation below is an instance of this principle: identify the page, locate the one differential that can be nonzero, and use convergence to pin it down.

The first application is the cleanest. The path-loop fibration $\Omega S^n \to PS^n \to S^n$ of example 5.2.15 has contractible total space, so the spectral sequence must converge to the cohomology of a point. The base S^n has cohomology in only two degrees, 0 and n, so the E_2 page is supported on two vertical lines, and a single family of differentials d_n connects them. Forcing the abutment to be trivial in positive degrees determines $H^*(\Omega S^n)$ completely.

Example 6.3.1: The Cohomology of ΩS^n

Fix $n \geq 2$ and consider the path-loop fibration $\Omega S^n \hookrightarrow PS^n \xrightarrow{p} S^n$ of example 5.2.15, with contractible total space PS^n. Because S^n is simply connected for $n \geq 2$, the local coefficient system is trivial, and the cohomology Serre spectral sequence (theorem 6.2.8) has

$$E_2^{p,q} = H^p\big(S^n;\, H^q(\Omega S^n;\mathbb{Z})\big) = H^p(S^n;\mathbb{Z}) \otimes H^q(\Omega S^n;\mathbb{Z})$$

converging to $H^{p+q}(PS^n;\mathbb{Z})$. Write $F = \Omega S^n$ and abbreviate $H^q = H^q(F;\mathbb{Z})$.

Step 1: The shape of the E_2 page. Since $H^p(S^n;\mathbb{Z})$ is $\mathbb{Z}$ for $p \in \{0, n\}$ and 0 otherwise, the E_2 page is concentrated in the two columns $p = 0$ and $p = n$:

$$E_2^{0,q} = H^q, \qquad E_2^{n,q} = H^q, \qquad E_2^{p,q} = 0 \text{ for } p \notin \{0, n\}$$

A differential $d_r \colon E_r^{p,q} \to E_r^{p+r,\,q-r+1}$ raises the column index by r. With only the columns 0 and n populated, the sole differential that can be nonzero is $d_n \colon E_n^{0,q} \to E_n^{n,\,q-n+1}$, joining the two columns. Every other d_r has source or target in an empty column and so vanishes. Hence $E_2 = E_n$ (no earlier differential acts) and $E_{n+1} = E_\infty$ (no later differential acts).

Step 2: Convergence forces d_n to be an isomorphism. The total space PS^n is contractible, so $H^k(PS^n;\mathbb{Z}) = 0$ for $k > 0$ and $\mathbb{Z}$ for $k = 0$. The abutment in total degree $k > 0$ is the associated graded of $H^k(PS^n)$, which is 0. Therefore $E_\infty^{p,q} = 0$ for $(p,q) \neq (0,0)$, and $E_\infty^{0,0} = \mathbb{Z}$. The only

surviving page-$(n+1)$ groups are the homology of the two-term complexes

$$E_n^{0,q} \xrightarrow{d_n} E_n^{n,\,q-n+1}$$

read across the page: $E_{n+1}^{n,\,q-n+1} = \operatorname{coker} d_n$ and $E_{n+1}^{0,q} = \ker d_n$. For these to vanish for all $(p,q) \neq (0,0)$, the map

$$d_n \colon E_n^{0,q} = H^q \longrightarrow E_n^{n,\,q-n+1} = H^{q-n+1}$$

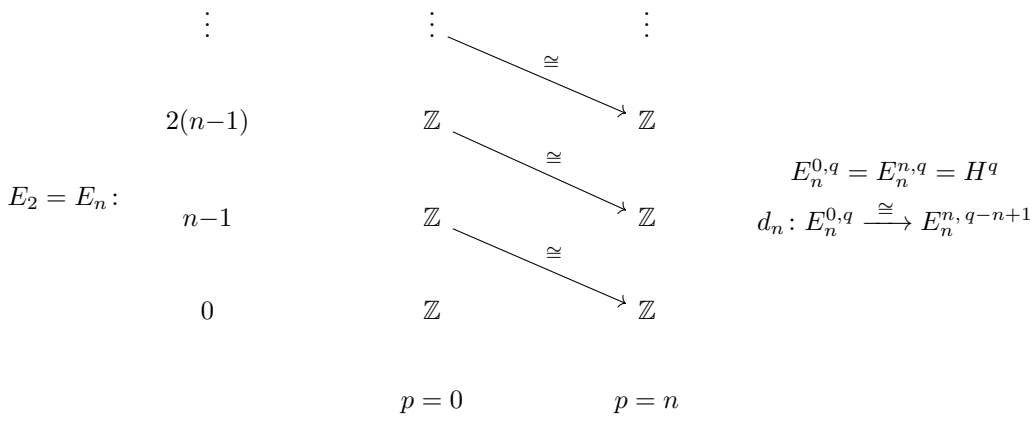
must be injective (so its kernel, sitting in column 0, dies) and surjective (so its cokernel, sitting in column n, dies) for every $q \geq 1$. Thus $d_n \colon H^q \to H^{q-n+1}$ is an isomorphism for all q with $q \geq 1$, while $E_\infty^{0,0} = \ker(d_n \colon H^0 \to H^{1-n}) = H^0 = \mathbb{Z}$ survives (the target $H^{1-n} = 0$).

Step 3: Solving the recursion. The isomorphism $d_n \colon H^q \xrightarrow{\cong} H^{q-n+1}$ for $q \geq 1$ is a recursion that shifts the degree down by $n-1$. Starting from $H^0 = \mathbb{Z}$ (the loop space is path-connected, since S^n is simply connected, so ΩS^n is connected and $H^0(\Omega S^n; \mathbb{Z}) = \mathbb{Z}$), the isomorphism with $q = n-1$ gives $H^{n-1} \cong H^0 = \mathbb{Z}$. With $q = 2(n-1)$ it gives $H^{2(n-1)} \cong H^{n-1} = \mathbb{Z}$, and inductively

$$H^q(\Omega S^n; \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & q \equiv 0 \pmod{n-1},\ q \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

For any degree q not divisible by $n-1$, the isomorphism chain reaches a degree in the open range $0 < q' < n-1$, where the target $H^{q'-n+1}$ has negative degree and is 0. Hence $H^{q'} = 0$, and tracing back, $H^q = 0$.

The recursion is a staircase. On the $E_2 = E_n$ page the columns $p = 0$ and $p = n$ each carry $H^\bullet(\Omega S^n)$, a copy of $\mathbb{Z}$ at every height divisible by $n-1$. Each transgression d_n is a down-right arrow dropping the degree by $n-1$, forced to be an isomorphism, so the two columns cancel in pairs:



Each arrow $\mathbb{Z} \xrightarrow{\cong} \mathbb{Z}$ pairs a class in column 0 with one in column n. The only class with no partner

is the generator of $E_2^{0,0}$, whose d_n lands in negative degree. It alone survives to the limit page,

$$E_\infty: \quad \begin{array}{ccc} \vdots & \vdots & \vdots \\ 2(n-1) & 0 & 0 \\ n-1 & 0 & 0 \\ 0 & \boxed{\mathbb{Z}} & 0 \\ & p=0 & p=n \end{array} \qquad \begin{array}{l} E_\infty^{0,0} = \mathbb{Z} \\ E_\infty^{p,q} = 0 \ \text{ otherwise} \end{array}$$

the associated graded of $H^*(PS^n) = \mathbb{Z}$ concentrated in degree 0, which is the contractibility the argument started from.

Step 4: Reading off homology. The groups just computed are free abelian of rank one in each degree $k(n-1)$, $k \geq 0$, so they are torsion-free in every degree. By the universal coefficient theorem (chapter 4) the integral homology agrees with the cohomology in each degree:

$$H_q(\Omega S^n; \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & q \equiv 0 \pmod{n-1},\ q \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

For $n = 2$ this reads $H_q(\Omega S^2; \mathbb{Z}) = \mathbb{Z}$ in every degree $q \geq 0$, matching the fact that ΩS^2 has the homology of $\mathbb{CP}^\infty$ shifted. For $n = 3$ it gives $\mathbb{Z}$ in every even degree, and in general the homology is concentrated in the arithmetic progression of multiples of $n-1$.

The computation is the spectral-sequence proof of a result first found by Serre, and it is the archetype of every collapse argument: a contractible total space turns the differential into an isomorphism, and an isomorphism that shifts degree by a fixed amount is solved by a recursion. The ring structure, a divided-power algebra on a generator of degree $n-1$, follows from the multiplicative structure of theorem 6.2.15 and is computed in example 6.4.5.

The next two results are not single computations but reusable exact sequences. Each arises when one of the two spaces in the fibration has cohomology concentrated in exactly two degrees: the Gysin sequence when the *fiber* is a sphere, the Wang sequence when the *base* is a sphere. In both cases the E_2 page occupies two rows or two columns, the only possibly-nonzero differential connects them, and the resulting two-term structure unwinds into a long exact sequence relating the cohomology of base, total space, and the shift induced by that differential.

Begin with the Gysin sequence. An oriented sphere bundle $S^k \to E \to B$ has fiber S^k, whose cohomology is $\mathbb{Z}$ in degrees 0 and k and zero elsewhere. The orientation hypothesis guarantees that the local coefficient system on B is trivial, so the E_2 page is the tensor product $H^*(B) \otimes H^*(S^k)$, supported on the two rows $q = 0$ and $q = k$. The single differential d_{k+1} crossing from the top row to the bottom row is, up to sign, cup product with a distinguished class, the Euler class of the bundle.

Theorem 6.3.2: The Gysin Sequence

Let $S^k \xrightarrow{i} E \xrightarrow{p} B$ be a fiber bundle with fiber the k-sphere ($k \geq 1$), over a path-connected CW complex B, oriented in the sense that the local system $H^k(S^k;\mathbb{Z})$ on B is trivial. Then there is a class $e \in H^{k+1}(B;\mathbb{Z})$, the *Euler class* of the bundle, and a long exact sequence

$$\cdots \to H^m(B) \xrightarrow{\smile e} H^{m+k+1}(B) \xrightarrow{p^*} H^{m+k+1}(E) \xrightarrow{\int_{S^k}} H^{m+1}(B) \xrightarrow{\smile e} H^{m+k+2}(B) \to \cdots$$

with $\mathbb{Z}$ coefficients, where p^* is induced by the projection, $\smile e$ is cup product with the Euler class, and $\int_{S^k}$ is the fiber-integration map lowering degree by k.

***Proof*:**

Step 1: The E_2 page lies in two rows. The bundle is a Serre fibration (theorem 5.2.4). Since B is a path-connected CW complex and the system $H^k(S^k;\mathbb{Z})$ is trivial by the orientation hypothesis, the Serre spectral sequence (theorem 6.2.8) has

$$E_2^{p,q} = H^p(B;\mathbb{Z}) \otimes H^q(S^k;\mathbb{Z})$$

The cohomology of the fiber is $H^0(S^k) = \mathbb{Z}$ and $H^k(S^k) = \mathbb{Z}$ with all other groups zero, so only the rows $q = 0$ and $q = k$ are nonzero:

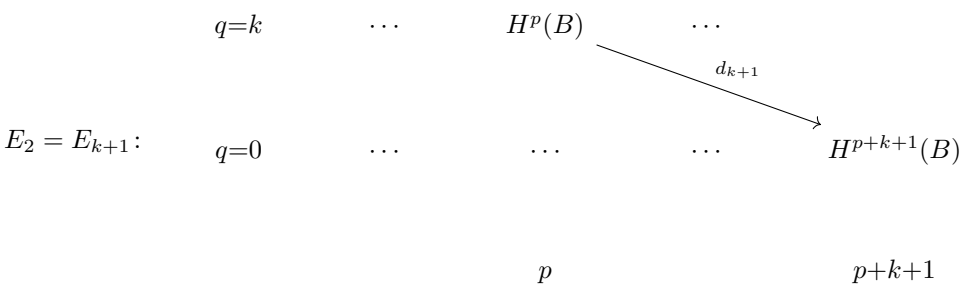
$$E_2^{p,0} = H^p(B), \qquad E_2^{p,k} = H^p(B), \qquad E_2^{p,q} = 0 \text{ for } q \notin \{0, k\}$$

Fix the generator $1 \in H^0(S^k)$ and a generator $\alpha \in H^k(S^k)$ specified by the orientation. Then $E_2^{p,0} = H^p(B) \cdot 1$ and $E_2^{p,k} = H^p(B) \cdot \alpha$.

Step 2: Only d_{k+1} is nonzero. A differential d_r shifts (p, q) to $(p + r,\, q - r + 1)$. With nonzero entries only on rows 0 and k, a differential out of row k lands in row $k - r + 1$, which is a nonzero row only when $k - r + 1 = 0$, that is $r = k + 1$. Differentials out of row 0 land in row $1 - r < 0$, always zero. Differentials *into* row 0 come from row $r - 1$, nonzero only for $r - 1 = k$, again $r = k + 1$. Hence the only possibly-nonzero differential is

$$d_{k+1} \colon E_{k+1}^{p,k} \longrightarrow E_{k+1}^{p+k+1,\,0}$$

and $E_2 = E_{k+1}$, $E_{k+2} = E_\infty$, since no earlier or later differential acts. The nonzero part of the page occupies the two rows $q = 0$ and $q = k$, joined by the single slanted differential d_{k+1} running from the top row down into the bottom row, $k + 1$ columns to the right:



Step 3: Naming the Euler class. Define the Euler class by applying d_{k+1} to the fiber generator placed over $1 \in H^0(B)$:

$$e := d_{k+1}(1 \otimes \alpha) \in E_{k+1}^{k+1,0} = H^{k+1}(B)$$

The cohomology Serre spectral sequence carries a multiplicative structure for which each differential d_r is a derivation, $d_r(xy) = d_r(x)y + (-1)^{|x|} x\, d_r(y)$, with the product on E_2 induced by the cup products of base and fiber (theorem 6.2.15, whose hypotheses hold: the base is a path-connected CW complex and the orientation makes $\pi_1(B)$ act trivially on $H^*(S^k;\mathbb{Z})$). So d_{k+1} is a derivation, and the classes in row 0 are permanent cycles (they are the image of p^* from the base, so d_{k+1} vanishes on them). For $x \in H^p(B) = E_{k+1}^{p,0}$, the Leibniz rule gives

$$d_{k+1}(x \cdot \alpha) = d_{k+1}\big((x \otimes 1) \smile (1 \otimes \alpha)\big) = (-1)^p\,(x \otimes 1) \smile d_{k+1}(1 \otimes \alpha) = (-1)^p\, x \smile e$$

using $d_{k+1}(x \otimes 1) = 0$. Therefore on row k the differential is exactly cup product with e:

$$d_{k+1} \colon H^p(B) \xrightarrow{\smile e} H^{p+k+1}(B)$$

under the identifications $E_{k+1}^{p,k} = H^p(B) \cdot \alpha \cong H^p(B)$, $(-1)^p x \cdot \alpha \leftrightarrow x$, and $E_{k+1}^{p+k+1,0} = H^{p+k+1}(B)$.

Step 4: The E_∞ page and the extension. After d_{k+1} the surviving groups are

$$E_\infty^{p,0} = \operatorname{coker}\big(\smile e \colon H^{p-k-1}(B) \to H^p(B)\big), \qquad E_\infty^{p,k} = \ker\big(\smile e \colon H^p(B) \to H^{p+k+1}(B)\big)$$

In total degree $m+k+1$ only two entries can be nonzero, namely $E_\infty^{m+k+1,0}$ (on row 0) and $E_\infty^{m+1,k}$ (on row k), so the abutment $H^{m+k+1}(E)$ sits in a short exact sequence

$$0 \to E_\infty^{m+k+1,0} \to H^{m+k+1}(E) \to E_\infty^{m+1,k} \to 0$$

the row-0 piece being the bottom of the filtration (the image of p^*) and the row-k piece the top quotient (the fiber-integration $\int_{S^k}$, evaluation against the fiber generator α). Writing these out,

$$0 \to \operatorname{coker}\big(\smile e \colon H^m(B) \to H^{m+k+1}(B)\big) \xrightarrow{p^*} H^{m+k+1}(E)$$
$$\xrightarrow{\int_{S^k}} \ker\big(\smile e \colon H^{m+1}(B) \to H^{m+k+2}(B)\big) \to 0$$

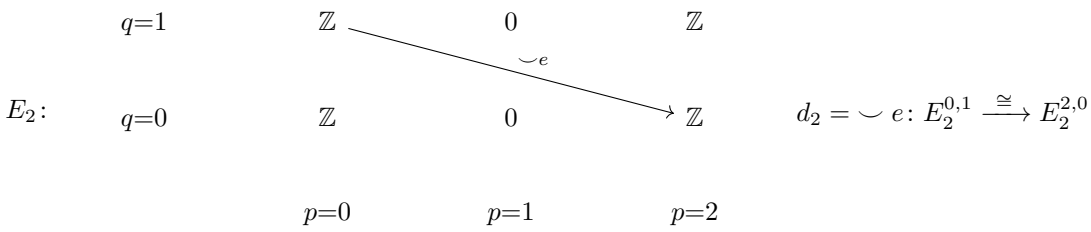
Step 5: Splicing into a long exact sequence. The short exact sequence of Step 4, for varying m, splices into the long exact sequence claimed. Concretely, the map $\int_{S^k} \colon H^{m+k+1}(E) \to H^{m+1}(B)$ followed by $\smile e \colon H^{m+1}(B) \to H^{m+2}(B)$ is zero precisely because $\int_{S^k}$ lands in $\ker(\smile e)$. Exactness there records that $\operatorname{im}(\int_{S^k}) = \ker(\smile e)$. The map $\smile e \colon H^m(B) \to H^{m+k+1}(B)$ followed by p^* is zero because p^* kills the image of $\smile e$ (that image is exactly the kernel of the projection onto the cokernel), and exactness there is $\ker(p^*) = \operatorname{im}(\smile e)$. Finally p^* followed by $\int_{S^k}$ is zero because the image of p^* is the bottom filtration piece $E_\infty^{*,0}$, which the fiber-integration annihilates, and exactness is $\ker(\int_{S^k}) = \operatorname{im}(p^*)$. The three exactness statements are the three nodes of the asserted sequence, completing the proof.

The Gysin sequence is the spectral-sequence form of the Thom isomorphism and the long exact sequence of the pair $(D(E), S(E))$ for the disk bundle of an oriented vector bundle. For the sphere bundle $S(E)$ of an oriented real vector bundle E of rank $n \geq 2$ over a path-connected CW complex of finite type, with the orientation of $S(E)$ induced by that of E (definition 7.1.25), e is the Euler

class of E studied in chapter 7: it is the restriction of the Thom class to the zero section by proposition 7.3.5(2), and that restriction is the Euler class of definition 7.2.15 by corollary 7.3.6. Its computational force is that it relates the cohomology of the total space E to that of the base B through one algebraic operation, cup product with e. When e is a non-zero-divisor the sequence breaks into short exact pieces and $H^*(E)$ is determined. When $e = 0$ the bundle is cohomologically a product. The most familiar instance is the Hopf fibration, where the base is S^2 and the Euler class is a generator of $H^2(S^2)$, giving the cohomology of S^3 in a single stroke.

Example 6.3.3: The Hopf Fibration via Gysin

Apply theorem 6.3.2 to the Hopf fibration $S^1 \hookrightarrow S^3 \to S^2$ of example 5.2.7, an oriented S^1-bundle, so $k = 1$. The base is $B = S^2$ with $H^0(S^2) = H^2(S^2) = \mathbb{Z}$ and $H^1(S^2) = 0$. The Euler class $e \in H^2(S^2) \cong \mathbb{Z}$ is a generator (the bundle is nontrivial, as $S^3 \neq S^1 \times S^2$). With $k = 1$ the E_2 page occupies the two fiber rows $q = 0$ and $q = 1$, and the single differential $d_2 = \smile e$ crosses from the top row down into the bottom row, two columns to the right. Since e generates $H^2(S^2)$ it is an isomorphism:



The Gysin sequence around the relevant degrees, with $\int = \int_{S^1}$ lowering degree by 1, reads in degree $m = 0$,

$$H^0(S^2) \xrightarrow{\smile e} H^2(S^2) \xrightarrow{p^*} H^2(S^3) \xrightarrow{\int} H^1(S^2) = 0$$

The map $\smile e \colon H^0(S^2) = \mathbb{Z} \to H^2(S^2) = \mathbb{Z}$ sends $1 \mapsto e$, a generator, hence is an isomorphism. Exactness then forces $p^* \colon H^2(S^2) \to H^2(S^3)$ to be the zero map on a surjection, so $H^2(S^3) = 0$. Likewise the segment

$$H^2(S^2) \xrightarrow{\smile e} H^4(S^2) = 0, \qquad 0 = H^3(S^2) \to H^3(S^3) \xrightarrow{\int} H^2(S^2) \xrightarrow{\smile e} H^4(S^2) = 0$$

shows $H^3(S^3) \cong H^2(S^2) = \mathbb{Z}$. Together with $H^0(S^3) = \mathbb{Z}$ from path-connectedness, this recovers $H^*(S^3; \mathbb{Z}) = \mathbb{Z}, 0, 0, \mathbb{Z}$ in degrees $0, 1, 2, 3$, exactly the cohomology of S^3.

The dual situation, where the base rather than the fiber is a sphere, produces the Wang sequence. Here the E_2 page occupies two *columns*, $p = 0$ and $p = n$, because $H^*(S^n)$ is concentrated in those two degrees. The single differential d_n now runs between columns, and the resulting long exact sequence relates the cohomology of the total space to that of the fiber, with the connecting map encoding how the fiber is twisted as one goes around the base sphere.

Theorem 6.3.4: The Wang Sequence

Let $F \xrightarrow{i} E \xrightarrow{p} S^n$ be a fibration over the n-sphere ($n \geq 2$), with path-connected fiber F, so that the simply connected base S^n gives a trivial local system. Then there is a long exact sequence with $\mathbb{Z}$ coefficients

$$\cdots \to H^m(E) \xrightarrow{i^*} H^m(F) \xrightarrow{\theta} H^{m-n+1}(F) \to H^{m+1}(E) \xrightarrow{i^*} H^{m+1}(F) \to \cdots$$

where i^* is restriction to the fiber and $\theta\colon H^m(F) \to H^{m-n+1}(F)$ is a derivation of degree $-(n-1)$ (it satisfies the Leibniz rule $\theta(xy) = \theta(x)\, y + (-1)^{(n-1)|x|} x\, \theta(y)$).

***Proof*:**
Step 1: The E_2 page lies in two columns. As S^n is simply connected for $n \geq 2$, the local system is trivial and the Serre spectral sequence (theorem 6.2.8) has

$$E_2^{p,q} = H^p(S^n; \mathbb{Z}) \otimes H^q(F; \mathbb{Z})$$

Since $H^p(S^n)$ is $\mathbb{Z}$ for $p \in \{0, n\}$ and zero otherwise, only the columns $p = 0$ and $p = n$ are nonzero:

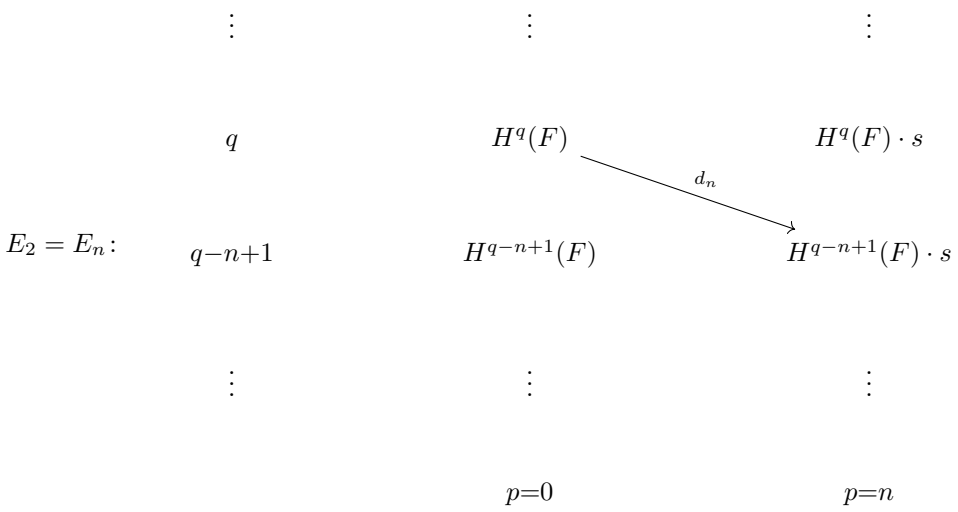
$$E_2^{0,q} = H^q(F), \qquad E_2^{n,q} = H^q(F), \qquad E_2^{p,q} = 0 \text{ for } p \notin \{0, n\}$$

Let $s \in H^n(S^n)$ be the generator, so $E_2^{n,q} = H^q(F) \cdot s$.

Step 2: Only d_n is nonzero. A differential d_r raises the column index by r. From column 0 it reaches column r, which is nonzero only for $r = n$. From column n it reaches column $n + r > n$, always zero. Differentials *into* column n come from column $n - r$, nonzero only for $r = n$. So the only possibly-nonzero differential is

$$d_n\colon E_n^{0,q} = H^q(F) \longrightarrow E_n^{n,\, q-n+1} = H^{q-n+1}(F)$$

and $E_2 = E_n$, $E_{n+1} = E_\infty$. The nonzero part of the page occupies the two columns $p = 0$ and $p = n$, joined by the single differential d_n running from the left column down into the right column, dropping the fiber degree by $n - 1$:



Step 3: The differential is the Wang derivation. Define

$$\theta := d_n \colon H^q(F) = E_n^{0,q} \longrightarrow E_n^{n,\,q-n+1} = H^{q-n+1}(F)$$

where the target identification drops the generator s. By the derivation property of the differentials in the multiplicative cohomology Serre spectral sequence (theorem 6.2.15(2)), d_n is a derivation: for $x \in H^{|x|}(F) = E_n^{0,|x|}$ and $y \in E_n^{0,|y|}$,

$$\theta(xy) = d_n(x \otimes y) = d_n(x)\, y + (-1)^{|x|}\, x\, d_n(y)$$

Tracking the sign through the identification of $E_n^{n,\,q-n+1}$ with $H^{q-n+1}(F)$, which moves the class $d_n(x)$ of column n and bidegree shift $-(n-1)$ past y, the Koszul sign becomes $(-1)^{(n-1)|x|}$, giving the stated Leibniz rule $\theta(xy) = \theta(x)y + (-1)^{(n-1)|x|} x\,\theta(y)$.

Step 4: The E_∞ page and the abutment. After d_n,

$$E_\infty^{0,q} = \ker\big(\theta \colon H^q(F) \to H^{q-n+1}(F)\big), \qquad E_\infty^{n,q} = \operatorname{coker}\big(\theta \colon H^{q+n-1}(F) \to H^q(F)\big)$$

In total degree m exactly two entries can survive, $E_\infty^{0,m}$ (column 0) and $E_\infty^{n,m-n}$ (column n), so the abutment $H^m(E)$ fits in

$$0 \to E_\infty^{n,m-n} \to H^m(E) \to E_\infty^{0,m} \to 0$$

The bottom of the filtration is the column-n piece and the top quotient is the column-0 piece $E_\infty^{0,m} = \ker(\theta) \subseteq H^m(F)$. The quotient map $H^m(E) \to E_\infty^{0,m} \hookrightarrow H^m(F)$ is the restriction i^* to the fiber (edge homomorphism along the fiber, theorem 6.2.9). Writing the short exact sequence as

$$0 \to \operatorname{coker}\big(\theta \colon H^{m-1}(F) \to H^{m-n}(F)\big) \to H^m(E) \xrightarrow{i^*} \ker\big(\theta \colon H^m(F) \to H^{m-n+1}(F)\big) \to 0$$

Step 5: Splicing. For varying m these short exact sequences splice into the long exact sequence asserted. The connecting homomorphism $H^m(F) \to H^{m-n+1}(F)$ is θ itself: an element of $H^m(F)$ lifts to $H^m(E)$ exactly when it lies in $\ker\theta$ (by the right-hand surjection above), and its obstruction to lifting is its image $\theta(x) \in H^{m-n+1}(F)$, landing in the cokernel piece $\operatorname{coker}\theta$ that forms the column-n part of $H^{m+1}(E)$. The three exactness conditions ($\operatorname{im} i^* = \ker\theta$, $\operatorname{im}\theta = \ker(\text{into } H^{m+1}(E))$, and $\ker i^* = \operatorname{im}(\text{from } H^m(E))$) are read off the short exact sequences exactly as in the Gysin case, yielding

$$\cdots \to H^m(E) \xrightarrow{i^*} H^m(F) \xrightarrow{\theta} H^{m-n+1}(F) \to H^{m+1}(E) \xrightarrow{i^*} \cdots$$

as we sought to show.

The Wang sequence is the tool of choice whenever a space fibers over a sphere, and it appears throughout the homotopy theory of Lie groups and loop spaces, where total spaces are frequently bundles over spheres. The derivation θ measures the monodromy of the fibration: it records how the cohomology of the fiber is sheared as the base sphere is traversed. When $\theta = 0$ the cohomology of E is additively the tensor product $H^*(F) \otimes H^*(S^n)$, the cohomological signature of a trivial bundle. For the path-loop fibration $\Omega S^n \to PS^n \to S^n$, whose total space is contractible, the outer terms $H^m(PS^n)$ and $H^{m+1}(PS^n)$ of the Wang sequence vanish for $m \geq 1$, so $\theta \colon H^m(\Omega S^n) \to H^{m-n+1}(\Omega S^n)$ is an isomorphism for every $m \geq 1$. This is the recursion solved in Step 3 of example 6.3.1, θ being the

differential d_n there.

The final and most substantial application returns to the homotopy groups of spheres. The fibration long exact sequence of theorem 5.2.12 computed $\pi_n(S^n) \cong \mathbb{Z}$ and, via the Hopf fibration, gave $\pi_3(S^2) \cong \mathbb{Z}$. But it stalls immediately above the diagonal because the relevant fibrations run out. The spectral sequence breaks the impasse through a different mechanism: it computes the *homology* of an Eilenberg-MacLane space $K(\mathbb{Z}, n)$, and the Hurewicz theorem (theorem 5.4.9) converts homology back into homotopy. The strategy, due to Serre, is to study the map $S^n \to K(\mathbb{Z}, n)$ inducing an isomorphism on π_n, replace it by a fibration, and analyze the fiber.

The central computation is the integral homology of $K(\mathbb{Z}, n)$ in the first range above degree n. It is built from the path-loop fibration over $K(\mathbb{Z}, n)$, whose fiber is $K(\mathbb{Z}, n-1)$, descending an induction on n that starts from the known $K(\mathbb{Z}, 1) = S^1$ and $K(\mathbb{Z}, 2) = \mathbb{CP}^\infty$.

Proposition 6.3.5: Low-Degree Homology of $K(\mathbb{Z}, n)$

For $n \geq 3$ the integral homology of $K(\mathbb{Z}, n)$ in degrees $\leq n+2$ is

$$H_i(K(\mathbb{Z}, n); \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & i = 0 \\ 0 & 0 < i < n \\ \mathbb{Z} & i = n \\ 0 & i = n+1 \\ \mathbb{Z}/2 & i = n+2 \end{cases}$$

For $n = 2$ one has instead $H_i(K(\mathbb{Z}, 2); \mathbb{Z}) = H_i(\mathbb{CP}^\infty; \mathbb{Z})$, which is $\mathbb{Z}$ in every even degree and 0 in every odd degree. In particular $H_3(K(\mathbb{Z}, 2)) = 0$ and $H_4(K(\mathbb{Z}, 2)) = \mathbb{Z}$. Thus $H_{n+1}(K(\mathbb{Z}, n); \mathbb{Z}) = 0$ for all $n \geq 2$, while $H_{n+2}(K(\mathbb{Z}, n); \mathbb{Z}) \cong \mathbb{Z}/2$ holds for $n \geq 3$ and fails at $n = 2$.

***Proof*:**

Step 1: The base case $K(\mathbb{Z}, 2) = \mathbb{CP}^\infty$. By example 5.5.5 the homology of $K(\mathbb{Z}, 2) = \mathbb{CP}^\infty$ is $\mathbb{Z}$ in every even degree and 0 in every odd degree, read off the cellular chain complex of example 5.5.11. This gives $H_3(K(\mathbb{Z}, 2)) = 0$ and $H_4(K(\mathbb{Z}, 2)) = \mathbb{Z}$, as stated, and the cohomology ring is the polynomial ring $H^*(\mathbb{CP}^\infty; \mathbb{Z}) = \mathbb{Z}[a]$ with $|a| = 2$ (example 5.5.11). The case $n \geq 3$ is the substance of the proposition.

Step 2: The path-loop fibration over $K(\mathbb{Z}, n)$. For $n \geq 3$ use the path-loop fibration of example 5.2.15 applied to $X = K(\mathbb{Z}, n)$:

$$\Omega K(\mathbb{Z}, n) \hookrightarrow PK(\mathbb{Z}, n) \xrightarrow{p} K(\mathbb{Z}, n)$$

The total space $PK(\mathbb{Z}, n)$ is contractible, so $\pi_i(\Omega K(\mathbb{Z}, n)) \cong \pi_{i+1}(K(\mathbb{Z}, n))$ is $\mathbb{Z}$ for $i + 1 = n$, that is $i = n - 1$, and 0 otherwise (example 5.2.15). By theorem 5.3.10 there is a CW complex Γ and a weak homotopy equivalence $\Gamma \to \Omega K(\mathbb{Z}, n)$. Then Γ is path-connected with the homotopy groups of $K(\mathbb{Z}, n-1)$, so $\Gamma \simeq K(\mathbb{Z}, n-1)$ by theorem 5.5.4. A weak homotopy equivalence induces isomorphisms on singular cohomology (proposition 5.3.15), so the two maps give ring isomorphisms $H^*(\Omega K(\mathbb{Z}, n); \mathbb{Z}) \cong H^*(\Gamma; \mathbb{Z}) \cong H^*(K(\mathbb{Z}, n-1); \mathbb{Z})$, and the fiber cohomology may be written as that of $K(\mathbb{Z}, n-1)$. The cohomology Serre spectral sequence (theorem 6.2.8)

of this fibration has

$$E_2^{p,q} = H^p\big(K(\mathbb{Z},n);\, H^q(K(\mathbb{Z},n-1);\mathbb{Z})\big)$$

converging to $H^{p+q}(PK(\mathbb{Z},n);\mathbb{Z})$, which is $\mathbb{Z}$ in total degree 0 and 0 in every positive total degree. Both base and fiber are simply connected (each is $(n-1)$- or $(n-2)$-connected with $n \geq 3$), so the local system is trivial and $E_2^{p,q} = H^p(B) \otimes H^q(F)$ with $B = K(\mathbb{Z},n)$, $F = K(\mathbb{Z},n-1)$.

Step 3: The transgression kills the fundamental classes. Both base and fiber have known cohomology below the band of interest: by the connectivity of $K(\mathbb{Z},m)$ and Hurewicz (theorem 5.4.9), $H^q(F) = 0$ for $0 < q < n-1$ and $H^{n-1}(F) = \mathbb{Z}$, while $H^q(B) = 0$ for $0 < q < n$ and $H^n(B) = \mathbb{Z}$. Let $a \in H^{n-1}(F) = \mathbb{Z}$ and $b \in H^n(B) = \mathbb{Z}$ be the generators. The only off-axis E_2 entry in total degree $n-1$ is $E_2^{0,n-1} = \mathbb{Z}\langle a\rangle$, and the only entry in total degree n on the base axis is $E_2^{n,0} = \mathbb{Z}\langle b\rangle$. Convergence to a point forces both to vanish on E_∞. The sole differential out of $E^{0,n-1}$ is the transgression

$$d_n \colon E_n^{0,n-1} \longrightarrow E_n^{n,0}, \qquad d_n(a) = \pm b$$

and since $E_\infty^{0,n-1} = 0$ this d_n must be injective, while $E_\infty^{n,0} = 0$ forces it onto $\mathbb{Z}\langle b\rangle$. Hence d_n is an isomorphism. Orient b so that $d_n(a) = b$.

Step 4: Degree $n+1$ gives $H^{n+1}(K(\mathbb{Z},n)) = 0$. The entries of total degree $n+1$ in the populated region are $E_2^{n+1,0} = H^{n+1}(B)$, $E_2^{0,n+1} = H^{n+1}(F)$, and $E_2^{2,n-1} = H^2(B) \otimes H^{n-1}(F) = 0$ (since $H^2(B) = 0$ as $2 < n$). All must die. Examine $E^{n+1,0}$ on the base axis: a differential into it is $d_r \colon E_r^{n+1-r,\,r-1} \to E_r^{n+1,0}$. For the source to be nonzero one needs a nonzero $H^{n+1-r}(B)$ with $0 < n+1-r$, but $H^p(B) = 0$ for $0 < p < n$ and $n+1-r < n$ for $r \geq 2$, while the fiber factor $H^{r-1}(F)$ is nonzero only at $r-1 = 0$ (giving the axis itself) or $r-1 = n-1$, i.e. $r = n$, where the source is $E_n^{1,n-1} = H^1(B) \otimes H^{n-1}(F) = 0$ because $H^1(B) = 0$. So no differential hits $E^{n+1,0}$, and none leaves it into a populated group. Therefore $E_\infty^{n+1,0} = H^{n+1}(B)$, which convergence forces to be 0. By the universal coefficient theorem (the homology of B being torsion-free in degrees $< n+1$), $H^{n+1}(B) = 0$ gives $H_{n+1}(K(\mathbb{Z},n);\mathbb{Z}) = 0$, valid for every $n \geq 2$.

Step 5: The case $n = 3$, where the $\mathbb{Z}/2$ is produced explicitly. Take $n = 3$, fiber $F = K(\mathbb{Z},2) = \mathbb{C}\mathrm{P}^\infty$ with $H^*(F) = \mathbb{Z}[a]$, $|a| = 2$, and base $B = K(\mathbb{Z},3)$ with $H^3(B) = \mathbb{Z}\langle b\rangle$, $b = d_3(a)$. The relevant E_2 entries are $E_2^{0,2} = \mathbb{Z}\langle a\rangle$, $E_2^{0,4} = \mathbb{Z}\langle a^2\rangle$, $E_2^{3,2} = \mathbb{Z}\langle ba\rangle$, $E_2^{3,0} = \mathbb{Z}\langle b\rangle$, $E_2^{6,0} = H^6(B)$. The differential d_3 is a derivation (theorem 6.2.15(2)), so

$$d_3(a^2) = 2a\, d_3(a) = 2ab = 2\,(ba) \in E_3^{3,2}, \qquad d_3(ba) = d_3(b)\, a \pm b\, d_3(a) = \pm b^2 \in E_3^{6,0} = H^6(B)$$

using $d_3(b) = 0$ (b lies on the base axis, a permanent cycle). Now $b \in H^3(B)$ has odd degree, so graded-commutativity gives $b^2 = -b^2$, whence $2b^2 = 0$: the class $b^2 \in H^6(B)$ is annihilated by 2.

Track the two-term complex on the diagonal of total degree 5, namely $E_3^{0,4} \xrightarrow{d_3} E_3^{3,2} \xrightarrow{d_3} E_3^{6,0}$, which reads

$$\mathbb{Z}\langle a^2\rangle \xrightarrow{\times 2} \mathbb{Z}\langle ba\rangle \xrightarrow{b^2} H^6(B)$$

Convergence to a point forces $E_\infty^{3,2} = 0$ (total degree $5 > 0$). The homology at the middle spot is $\ker(ba \mapsto b^2)/\mathrm{im}(\times 2)$. Since b^2 is 2-torsion, $k\,(ba) \mapsto k\,b^2 = 0$ precisely when k is even, so $\ker(ba \mapsto b^2) = 2\mathbb{Z}\langle ba\rangle$. The image $\mathrm{im}(\times 2)$ is the same subgroup $2\mathbb{Z}\langle ba\rangle$. Hence $E_4^{3,2} = 2\mathbb{Z}/2\mathbb{Z} = 0$, and no later differential is needed: the entry dies exactly because $d_3(a^2) = 2\,ba$ and $d_3(ba) = b^2$ are compatible. Consequently the class $b^2 = d_3(ba)$ is a boundary, so it becomes 0 on $E_4^{6,0}$. But $b^2 \neq 0$ on E_2, and it is exactly the 2-torsion subgroup $\mathbb{Z}/2\langle b^2\rangle \subseteq H^6(B)$ that is hit. This 2-torsion is the Ext-image of $H_5(K(\mathbb{Z},3))$ under the universal coefficient theorem: the

torsion subgroup of $H^6(B)$ is $\mathrm{Ext}(H_5(B),\mathbb{Z}) \cong (H_5(B))_{\mathrm{tors}}$, and the $\mathbb{Z}/2$ produced here gives $(H_5(B))_{\mathrm{tors}} = \mathbb{Z}/2$. The free part of $H_5(B)$ vanishes because $H^5(B) = 0$: the base-axis entry $E^{5,0} = H^5(B)$ in total degree 5 receives no differential (its only candidate source $E^{0,4} = \langle a^2 \rangle$ already died on E_4, as $\times 2$ is injective on $\mathbb{Z}$, so $E_5^{0,4} = 0$) and supports none, so $E_\infty^{5,0} = H^5(B) = 0$ by convergence, whence $\mathrm{Hom}(H_5(B),\mathbb{Z}) = 0$. Therefore $H_5(K(\mathbb{Z},3);\mathbb{Z}) \cong \mathbb{Z}/2$ is pure torsion, and $H_{n+2}(K(\mathbb{Z},3);\mathbb{Z}) = H_5(K(\mathbb{Z},3);\mathbb{Z}) \cong \mathbb{Z}/2$.

Step 6: Propagation to $n > 3$. The same fibration $K(\mathbb{Z},n-1) \to PK(\mathbb{Z},n) \to K(\mathbb{Z},n)$ propagates the $\mathbb{Z}/2$ upward: granting the inductive hypothesis $H_{(n-1)+2}(K(\mathbb{Z},n-1);\mathbb{Z}) = \mathbb{Z}/2$, the fiber $F = K(\mathbb{Z},n-1)$ carries a $\mathbb{Z}/2$ in degree $n+1$ homology, and the transgression of the fiber's torsion class lands the $\mathbb{Z}/2$ in $H_{n+2}(K(\mathbb{Z},n))$ by the identical two-term-complex mechanism of Step 5, now driven by the transgression theorem for the fundamental class and its square. The integral torsion bookkeeping that carries this through for all n, identifying the surviving differential as multiplication by 2 and ruling out cancellation by other classes in the band, is the content of Serre's computation of $H_*(K(\mathbb{Z},n);\mathbb{Z})$. It uses the transgression theorem developed in the next section in full generality. The complete argument is [22, Lemma 5.31]: for $n \geq 3$ the p-torsion of $H^i(K(\mathbb{Z},n);\mathbb{Z})$ is 0 for $i < 2p+n-1$ and $\mathbb{Z}/p$ for $i = 2p+n-1$. The homology of $K(\mathbb{Z},n)$ is finitely generated in each degree [22, Lemma 5.10], so by the universal coefficient theorem the torsion of H_{n+2} is that of H^{n+3}. The case $p = 2$ makes the 2-torsion of H^{n+3} equal to $\mathbb{Z}/2$, and an odd prime p has $2p+n-1 \geq n+5$, so it contributes no torsion in degree $n+3$. The free part of H_{n+2} vanishes by the rational computation [22, Proposition 5.21], which puts the rational cohomology in degrees $0, n, 2n, \ldots$, all different from $n+2$ when $n \geq 3$. The conclusion is $H_{n+2}(K(\mathbb{Z},n);\mathbb{Z}) \cong \mathbb{Z} \otimes \mathbb{Z}/2 \cong \mathbb{Z}/2$ for every $n \geq 3$, as we sought to show.

The proposition is the homological input to the stable computation. Its content is the appearance of 2-torsion in $H_{n+2}(K(\mathbb{Z},n))$ for $n \geq 3$, traced to the derivation rule $d_n(a^2) = 2a\,d_n(a)$ together with the graded-commutative identity $b^2 = -b^2$ for the odd-degree transgressed class. The factor of 2 is the source of the $\mathbb{Z}/2$ in the stable stem, and it is the first place in the book where torsion in a higher homotopy group is produced by an explicit algebraic mechanism. With the homology of $K(\mathbb{Z},n)$ in hand, the homotopy of spheres in the first degree above the diagonal falls out.

Example 6.3.6: $\pi_3(S^2)$ and the First Stable Stem $\pi_{n+1}(S^n)$

Two computations: first $\pi_3(S^2) \cong \mathbb{Z}$, then $\pi_{n+1}(S^n) \cong \mathbb{Z}/2$ for $n \geq 3$.

Part 1: $\pi_3(S^2) \cong \mathbb{Z}$. The Hopf fibration $S^1 \to S^3 \to S^2$ of example 5.2.7 has long exact homotopy sequence (theorem 5.2.12)

$$\pi_3(S^1) \to \pi_3(S^3) \xrightarrow{p_*} \pi_3(S^2) \xrightarrow{\partial} \pi_2(S^1)$$

Now $\pi_3(S^1) = 0$ and $\pi_2(S^1) = 0$ because $S^1 = K(\mathbb{Z},1)$ has all higher homotopy trivial (example 5.2.13), so $p_*\colon \pi_3(S^3) \to \pi_3(S^2)$ is an isomorphism. By corollary 5.4.11 $\pi_3(S^3) \cong \mathbb{Z}$, generated by the identity, hence

$$\pi_3(S^2) \cong \mathbb{Z}$$

generated by the Hopf map, recovering the result of example 5.2.19.

Part 2: The stable stem $\pi_{n+1}(S^n) \cong \mathbb{Z}/2$ for $n \geq 3$. Fix $n \geq 3$. Choose a map $f\colon S^n \to K(\mathbb{Z},n)$ representing a generator of $\pi_n(K(\mathbb{Z},n)) = \mathbb{Z}$. Since $f_*\colon \pi_n(S^n) \to \pi_n(K(\mathbb{Z},n))$ is an isomorphism (both are $\mathbb{Z}$ and f is a generator), and both spaces are $(n-1)$-connected, the map f is an isomorphism on π_i for $i \leq n$. Convert f into a fibration by proposition 5.2.16, with fiber F the

homotopy fiber of f. By the long exact homotopy sequence the fiber F is n-connected ($\pi_i(F) = 0$ for $i \leq n$) and

$$\pi_{n+1}(F) \cong \pi_{n+1}(S^n)$$

where the isomorphism is read from the long exact homotopy sequence of the fibration $F \to S^n \to K(\mathbb{Z}, n)$ in two adjacent segments. The segment $\pi_{n+1}(K(\mathbb{Z}, n)) \to \pi_n(F) \to \pi_n(S^n) \to \pi_n(K(\mathbb{Z}, n))$ reads $0 \to \pi_n(F) \to \mathbb{Z} \xrightarrow{\cong} \mathbb{Z}$, forcing $\pi_n(F) = 0$. The segment $\pi_{n+2}(K(\mathbb{Z}, n)) \to \pi_{n+1}(F) \to \pi_{n+1}(S^n) \to \pi_{n+1}(K(\mathbb{Z}, n))$ reads $0 \to \pi_{n+1}(F) \to \pi_{n+1}(S^n) \to 0$, an isomorphism, using $\pi_{n+1}(K(\mathbb{Z}, n)) = \pi_{n+2}(K(\mathbb{Z}, n)) = 0$ since $K(\mathbb{Z}, n)$ has homotopy concentrated in degree n.

Since F is n-connected, the Hurewicz theorem (theorem 5.4.9) gives

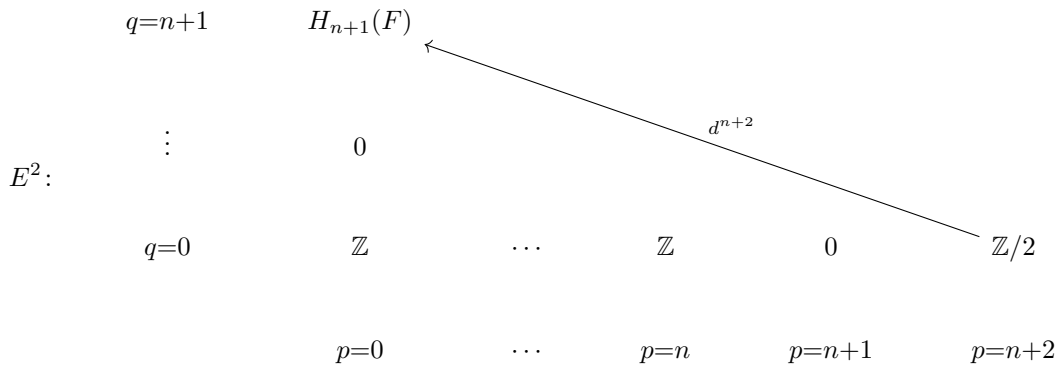
$$\pi_{n+1}(F) \cong H_{n+1}(F; \mathbb{Z})$$

so it remains to compute $H_{n+1}(F)$. Run the *homology* Serre spectral sequence (theorem 6.2.7) of the fibration $F \to S^n \xrightarrow{f} K(\mathbb{Z}, n)$, with base $B = K(\mathbb{Z}, n)$ and total space S^n, to keep all torsion in its native degree. The E^2 page is $E^2_{p,q} = H_p(K(\mathbb{Z}, n); H_q(F))$, converging to $H_{p+q}(S^n)$, which is $\mathbb{Z}$ in degrees 0 and n and 0 otherwise. The base homology, by proposition 6.3.5, is $H_0(B) = \mathbb{Z}$, $H_p(B) = 0$ for $0 < p < n$, $H_n(B) = \mathbb{Z}$, $H_{n+1}(B) = 0$, and $H_{n+2}(B) = \mathbb{Z}/2$. The fiber, being n-connected, has $H_0(F) = \mathbb{Z}$, $H_q(F) = 0$ for $0 < q \leq n$, and $H_{n+1}(F)$ the first unknown group.

The cancellation. The class in $H_n(B) = \mathbb{Z}$ survives to E^∞ and maps isomorphically to $H_n(S^n) = \mathbb{Z}$ under the edge homomorphism (theorem 6.2.9), since f induces an isomorphism on H_n in this range. Hence $E^\infty_{n,0} = \mathbb{Z}$ accounts for $H_n(S^n)$. Now read off total degree $n+1$. The only populated E^2 entries in total degree $n+1$ are $E^2_{n+1,0} = H_{n+1}(B) = 0$ and $E^2_{0,n+1} = H_{n+1}(F)$. Convergence to $H_{n+1}(S^n) = 0$ forces $E^\infty_{0,n+1} = 0$. The fiber class in $E^2_{0,n+1} = H_{n+1}(F)$ sits in the leftmost (fiber) column and receives the transgression differential

$$d^{n+2} \colon E^{n+2}_{n+2,0} = H_{n+2}(B) = \mathbb{Z}/2 \longrightarrow E^{n+2}_{0,n+1} = H_{n+1}(F)$$

the only differential touching either entry (all d^r for $r \leq n+1$ vanish on these because the intervening rows and columns are empty). On the page, the two live spots sit at opposite corners of an otherwise empty region, joined by the lone transgression running up-left from the base row to the fiber column:



Convergence forces both ends to die: $E^\infty_{0,n+1} = 0$ requires d^{n+2} surjective, and $E^\infty_{n+2,0} = 0$ (the abutment $H_{n+2}(S^n) = 0$) requires d^{n+2} injective. Hence

$$d^{n+2} \colon H_{n+2}(K(\mathbb{Z}, n); \mathbb{Z}) = \mathbb{Z}/2 \xrightarrow{\cong} H_{n+1}(F; \mathbb{Z})$$

is an isomorphism, giving $H_{n+1}(F;\mathbb{Z}) \cong \mathbb{Z}/2$ directly, with no universal-coefficient shift to track. Therefore

$$\pi_{n+1}(S^n) \cong \pi_{n+1}(F) \cong H_{n+1}(F;\mathbb{Z}) \cong \mathbb{Z}/2 \qquad (n \geq 3)$$

The group $\pi_{n+1}(S^n) \cong \mathbb{Z}/2$ is independent of n for $n \geq 3$: this is the first *stable* homotopy group of spheres, the first stable stem $\pi_1^s \cong \mathbb{Z}/2$, generated by the suspensions of the Hopf map $\eta\colon S^3 \to S^2$. The transition from the unstable $\pi_3(S^2) \cong \mathbb{Z}$ to the stable $\pi_4(S^3) \cong \pi_5(S^4) \cong \cdots \cong \mathbb{Z}/2$ is exactly the passage through one suspension, under which the integer Hopf invariant collapses to its parity.

The two parts of the example exhibit the two regimes of the homotopy of spheres. Below and on the diagonal the fibration sequence alone suffices, as in $\pi_3(S^2) \cong \mathbb{Z}$, where the contractible-fiber mechanism of the Hopf fibration reads the answer off directly. One step above the diagonal the fibration sequence is exhausted and the spectral sequence takes over: the homology of $K(\mathbb{Z},n)$, computed by collapsing the path-loop fibration, carries a $\mathbb{Z}/2$ that the Hurewicz theorem transports into $\pi_{n+1}(S^n)$. The stabilization, that the answer ceases to depend on n once $n \geq 3$, is the first visible consequence of the Freudenthal suspension theorem and the entire edifice of stable homotopy theory built on it, the subject taken up with spectra in chapter 8.

Corollary 6.3.7: The First Stable Stem is $\mathbb{Z}/2$

The first stable stem is $\pi_1^s \cong \mathbb{Z}/2$. It is realized by $\pi_{n+1}(S^n)$ for every $n \geq 3$, and the unstable group $\pi_3(S^2) \cong \mathbb{Z}$ suspends *onto* it: the suspension

$$\sigma\colon \pi_3(S^2) = \mathbb{Z} \longrightarrow \pi_4(S^3) = \mathbb{Z}/2$$

is surjective but not injective, the borderline case $i = 2n-1$ of theorem 5.7.8 with $n = 2$.

***Proof*:**

Step 1: Stabilization threshold for $k = 1$. By corollary 5.7.10 with $k = 1$, the groups $\pi_{n+1}(S^n)$ stabilize once $n > 2$, that is for all $n \geq 3$, and the common value is π_1^s. This much is Freudenthal. The value of that common group is $\mathbb{Z}/2$, generated by the iterated suspensions of the Hopf map $\eta\colon S^3 \to S^2$. This value is the Serre spectral sequence computation of example 6.3.6, which the stabilization just established permits us to read off from any single stage $n \geq 3$. Hence $\pi_1^s \cong \mathbb{Z}/2$.

Step 2: The boundary case $n = 2$. For $n = 2$ the sphere S^2 is 1-connected, so $X = S^2$ has connectivity index $n = 2$ in theorem 5.7.8, and the stable range is $i < 2 \cdot 2 - 1 = 3$ with surjectivity at $i = 3$. The group $\pi_3(S^2)$ sits exactly at $i = 3 = 2n-1$, the borderline degree, so the theorem guarantees that

$$\sigma\colon \pi_3(S^2) \to \pi_4(S^3)$$

is an epimorphism, but says nothing about injectivity. By corollary 5.4.11 and the Hopf-fibration computation of example 5.2.19, $\pi_3(S^2) \cong \mathbb{Z}$, generated by the Hopf map η. By Step 1, $\pi_4(S^3) \cong \mathbb{Z}/2$, generated by $\sigma[\eta] = [\Sigma\eta]$. A surjection $\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2$ from an infinite cyclic group onto a group of order 2 necessarily has nontrivial kernel (the kernel is $2\mathbb{Z}$), so σ is not injective. This is consistent with $\sigma[\eta]$ generating $\mathbb{Z}/2$: the integer Hopf class $[\eta]$ surjects onto its parity, and the multiples $2[\eta], 4[\eta], \ldots$ all suspend to 0.

The borderline behavior is forced: an isomorphism $\mathbb{Z} \to \mathbb{Z}/2$ is impossible, so the boundary degree $i = 2n-1$ is exactly where the theorem must weaken from isomorphism to epimorphism, and the example $\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2$ shows the weakening is genuine, completing the proof.

What Freudenthal contributes here is the structure, not the value $\mathbb{Z}/2$ itself. The group is stable because $n = 3$ exceeds the threshold $k+1 = 2$, and the unstable $\pi_3(S^2) \cong \mathbb{Z}$ falls below the threshold precisely because $n = 2$ is the borderline degree where suspension is only surjective. The value $\mathbb{Z}/2$ is given separately by the Serre spectral sequence computation of chapter 6, built from the homology of $K(\mathbb{Z}, n)$. The transition $\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2$ is the sharpest illustration of the theorem's boundary: it is the unique degree in the stem where suspension fails to be an isomorphism, and it fails in exactly the way the epimorphism clause predicts.

Example 6.3.8: The Suspension $\pi_3(S^2) \to \pi_4(S^3)$ in Detail

This is the canonical worked instance of the boundary of the stable range. Take $X = S^2$, so $n = 2$ and X is $(n-1) = 1$-connected. The relevant homotopy groups are

$$\pi_3(S^2) \cong \mathbb{Z} \quad \text{(generated by } \eta, \text{ example 5.2.19)}, \qquad \pi_4(S^3) \cong \mathbb{Z}/2 \quad \text{(example 6.3.6)}$$

The degree in question is $i = 3$. The stable range of theorem 5.7.8 for $n = 2$ is $i < 2n - 1 = 3$, so $i = 3$ is exactly the borderline $i = 2n - 1$, outside the isomorphism range and inside the epimorphism range. The theorem therefore predicts $\sigma\colon \pi_3(S^2) \to \pi_4(S^3)$ is surjective but not necessarily injective, and indeed

$$\sigma\colon \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2, \qquad [\eta] \mapsto [\Sigma\eta], \qquad \ker\sigma = 2\mathbb{Z}$$

The Hopf map suspends to a generator of $\mathbb{Z}/2$, while twice the Hopf class suspends to zero, so σ realizes the reduction $\mathbb{Z} \to \mathbb{Z}/2$ sending an integer to its parity. Suspending once more lands in the stable range: $\sigma\colon \pi_4(S^3) \to \pi_5(S^4)$ has $n = 3$ and degree $i = 4 < 2 \cdot 3 - 1 = 5$, so it is an isomorphism $\mathbb{Z}/2 \xrightarrow{\cong} \mathbb{Z}/2$, and every further suspension is an isomorphism as well. The integer information of $\pi_3(S^2)$ is destroyed at the first suspension and the stable value $\mathbb{Z}/2$ is reached immediately after, which is why $\pi_1^s = \mathbb{Z}/2$ is computed from any $\pi_{n+1}(S^n)$ with $n \geq 3$ but not from $n = 2$.

Exercise 6.3.1

1. Carry out the spectral-sequence computation of example 6.3.1 for $n = 2$ explicitly: identify every nonzero $E_2^{p,q}$ on the page for the fibration $\Omega S^2 \to PS^2 \to S^2$, draw the single family of differentials d_2, and conclude that $H_q(\Omega S^2; \mathbb{Z}) = \mathbb{Z}$ for every $q \geq 0$. Compare with the homology of $\mathbb{CP}^\infty$ and explain the relationship.

2. Use the Gysin sequence (theorem 6.3.2) for the sphere bundle $S^k \to S^{2k+1} \to \mathbb{CP}^k$ when $k = 1$ (the Hopf bundle $S^1 \to S^3 \to \mathbb{CP}^1 = S^2$) to recompute $H^*(S^3; \mathbb{Z})$, and then state what the Gysin sequence gives for the bundle $S^1 \to S^5 \to \mathbb{CP}^2$, computing $H^*(S^5; \mathbb{Z})$ from $H^*(\mathbb{CP}^2; \mathbb{Z}) = \mathbb{Z}[\alpha]/(\alpha^3)$.

3. Let $F \to E \to S^n$ be a fibration over a sphere with $n \geq 2$ and fiber F having $H^*(F; \mathbb{Z})$ free abelian. Show using the Wang sequence (theorem 6.3.4) that if the derivation θ is identically zero then $H^*(E; \mathbb{Z}) \cong H^*(F; \mathbb{Z}) \otimes H^*(S^n; \mathbb{Z})$ as graded groups. Give the example $E = F \times S^n$ to confirm $\theta = 0$ for a product.

4. In the computation of $H_{n+2}(K(\mathbb{Z}, n))$ in proposition 6.3.5, the factor of 2 arose from the derivation rule applied to a^2. Verify the rule $d_n(a^2) = 2a\, d_n(a)$ directly from the Leibniz formula $d_n(xy) = d_n(x)y + (-1)^{|x|}x\, d_n(y)$ for $x = y = a$ with $|a| = n - 1$, and explain why

the sign works out to give $+2$ rather than 0 precisely when $n-1$ is even (equivalently n odd) and what happens for n even.

5. Using example 6.3.6, identify $\pi_4(S^3)$, $\pi_5(S^4)$, and $\pi_6(S^5)$, and state which of these are stable (independent of further suspension) and which is the first stable one. Explain in one or two sentences why $\pi_3(S^2) \cong \mathbb{Z}$ does not contradict the stable value $\mathbb{Z}/2$.

6. Combine the homology computation $H_{n+1}(\Omega S^n;\mathbb{Z})$ from example 6.3.1 with the Hurewicz theorem to compute $\pi_{n-1}(\Omega S^n)$ for $n \geq 2$, and reconcile the answer with the dimension-shift isomorphism $\pi_{n-1}(\Omega S^n) \cong \pi_n(S^n)$ from example 5.2.15.

6.4 Multiplicative Structure and Transgression

The multiplicative structure of theorem 6.2.15 acts through the differentials, and one differential is distinguished. A class in the fiber column is never the target of a differential, since no column lies to its left, and of the differentials leaving it only d_{n+1} can reach the bottom row. The resulting map, the *transgression*, sends a fiber class of degree n to a base class of degree $n+1$, and the transgression theorem identifies it with the connecting behavior already present in the long exact sequences of low-dimensional fibrations. With multiplicativity, knowing the ring structure of two of $H^*(F)$, $H^*(B)$, $H^*(E)$ frequently *forces* all the differentials, because a derivation is determined by its values on algebra generators and graded-commutativity constrains those values severely. The worked computations at the end re-derive $H^*(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \cong \mathbb{Z}[x]$ and compute the ring $H^*(\Omega S^{n+1};\mathbb{Z})$ by this method.

Throughout, R and the fibration $p\colon E \to B$ with fiber F are as in theorem 6.2.15, and $H^*(E;R)$ carries the abutment filtration F^pH^n described before that theorem.

The most consequential single differential is the one carrying a fiber class to the base. A class $x \in H^n(F)$ lives, via the edge identification $E_2^{0,n} \cong H^0(B;H^n(F)) \cong H^n(F)$, in the leftmost column of the spectral sequence. No differential lands on it, since d_r would come from $E_r^{-r,\,n+r-1} = 0$, so $E_r^{0,n}$ is the subgroup of classes on which $d_2,\dots,d_{r-1}$ vanish. The differential d_r out of $E_r^{0,n}$ lands in $E_r^{r,\,n-r+1}$, which lies in the bottom row exactly when $r = n+1$. When $d_2,\dots,d_n$ vanish on x, the class survives to $E_{n+1}^{0,n}$, and $d_{n+1}(x)$ lies in $E_{n+1}^{n+1,0}$, a quotient of $E_2^{n+1,0} \cong H^{n+1}(B)$. This $d_{n+1}(x)$ is the transgression of x.

Definition 6.4.1: Transgression

Let $p\colon E \to B$ be a fibration as above with fiber F. A class $x \in H^n(F;R) \cong E_2^{0,n}$ is called *transgressive* if it survives to the page E_{n+1}, that is, if the differentials $d_2(x),\dots,d_n(x)$ all vanish (equivalently x persists to $E_{n+1}^{0,n}$). For a transgressive class the *transgression* is the differential

$$\tau = d_{n+1}\colon E_{n+1}^{0,n} \longrightarrow E_{n+1}^{n+1,0}$$

evaluated on x, giving $\tau(x) = d_{n+1}(x) \in E_{n+1}^{n+1,0}$. Since $E_{n+1}^{n+1,0}$ is a subquotient of $E_2^{n+1,0} \cong H^{n+1}(B;R)$, the transgression assigns to x a coset $\tau(x)$ in a quotient of $H^{n+1}(B;R)$. One says $\tau(x) \in H^{n+1}(B;R)$ when this coset has a canonical representative, abusing notation by the standard convention.

The transgression is the spectral-sequence form of the connecting map. A class on the fiber that

has not yet been forced to bound is examined by the longest differential that can reach the base row, and that differential measures the obstruction to extending the fiber class to the total space. In the simplest fibrations, where E is highly connected, every fiber class is transgressive and the transgression is an isomorphism onto the relevant part of $H^{*+1}(B)$, recovering base cohomology from fiber cohomology shifted by one. The bottom row $E_*^{*,0} \cong H^*(B)$ and the left column $E_*^{0,*} \cong H^*(F)$ are called the *edge* of the spectral sequence, and the transgression is the differential connecting the two edges. The qualification "transgressive" is essential: only classes that survive long enough to reach $E_{n+1}^{0,n}$ have a transgression, and the differential is well defined only up to the indeterminacy created by earlier differentials, which is why $\tau(x)$ naturally lives in a quotient.

Example 6.4.2: Transgression in the Path-Loop Fibration over S^{n+1}

Consider the path-loop fibration $\Omega S^{n+1} \to PS^{n+1} \to S^{n+1}$ of example 5.2.15, with $n \geq 1$. The base S^{n+1} has $H^*(S^{n+1};\mathbb{Z}) = \mathbb{Z}\langle 1, s\rangle$ with $|s| = n+1$, so $E_2^{p,q} = H^p(S^{n+1}; H^q(\Omega S^{n+1}))$ is concentrated in the two columns $p = 0$ and $p = n+1$. The total space PS^{n+1} is contractible, so $H^*(PS^{n+1}) = \mathbb{Z}$ in degree 0 and zero otherwise, forcing $E_\infty^{p,q} = 0$ except at $(0,0)$. Let $a \in H^n(\Omega S^{n+1};\mathbb{Z})$ be a generator of the bottom nonzero fiber cohomology (which is $\mathbb{Z}$, since ΩS^{n+1} is $(n-1)$-connected with $\pi_n(\Omega S^{n+1}) \cong \pi_{n+1}(S^{n+1}) \cong \mathbb{Z}$ by example 5.2.15). The class $a \in E_2^{0,n}$ is transgressive: the only nonzero columns are $p = 0$ and $p = n+1$, so all of $d_2, \ldots, d_n$ vanish on a for column reasons, and a survives to $E_{n+1}^{0,n}$. Because $E_\infty = 0$ off the origin, a cannot survive to E_∞, so $d_{n+1}(a)$ must be nonzero and must be an isomorphism $E_{n+1}^{0,n} \xrightarrow{\cong} E_{n+1}^{n+1,0} = H^{n+1}(S^{n+1}) = \mathbb{Z}\langle s\rangle$. Thus

$$\tau(a) = d_{n+1}(a) = s$$

up to sign: the bottom fiber generator transgresses to the base generator. This single differential, propagated multiplicatively, will compute the entire ring $H^*(\Omega S^{n+1})$ in example 6.4.5.

In the example the total space is contractible, and the transgression is the differential that cancels the fiber cohomology against the base cohomology. The transgression theorem makes the link to the connecting maps precise and explains why τ deserves to be called a transgression. The statement requires the relative cohomology of the pair (E, F) (definition 4.1.15) and the connecting homomorphism $\delta\colon H^n(F) \to H^{n+1}(E, F)$ of its long exact sequence (theorem 4.1.17), together with the projection $p^*\colon H^{n+1}(B, b_0) \to H^{n+1}(E, F)$. The transgression is the relation $\tau = (p^*)^{-1} \circ \delta$ on the classes for which this composite is defined.

Theorem 6.4.3: The Transgression Theorem

Let $p\colon E \to B$ be a Serre fibration as above, with $b_0 \in B$ a basepoint, $F = p^{-1}(b_0)$ path-connected, and $i\colon F \hookrightarrow E$ the inclusion, and let $n \geq 1$. Write $\delta\colon H^n(F;R) \to H^{n+1}(E,F;R)$ for the connecting homomorphism of the long exact cohomology sequence of the pair (E,F) (theorem 4.1.17), and $p^*\colon H^{n+1}(B,b_0;R) \to H^{n+1}(E,F;R)$ for the map induced by p. A class $x \in H^n(F;R)$ is transgressive (in the sense of definition 6.4.1) if and only if $\delta(x)$ lies in the image of p^*, and in that case the transgression $\tau(x) \in E_{n+1}^{n+1,0}$ is the image of every class $\bar{x} \in H^{n+1}(B,b_0;R) = H^{n+1}(B;R)$ with

$$p^*(\bar{x}) = \delta(x)$$

Equivalently, transgression is the "zig-zag"

$$\tau\colon H^n(F;R) \xrightarrow{\delta} H^{n+1}(E,F;R) \xleftarrow{p^*} H^{n+1}(B,b_0;R)$$

defined on the classes x with $\delta(x) \in \mathrm{im} p^*$ and valued in the quotient $E_{n+1}^{n+1,0}$ of $H^{n+1}(B;R)$ by the images of the earlier differentials $d_2,\dots,d_n$ landing in the bottom row, a quotient in which $\ker p^*$ maps to zero.

***Proof*:**

The pages are described by [13, Explicit Description of the Pages] in its cochain form: $E_r^{k,l} = Z_r^{k,l}/D_r^{k,l}$ with $Z_r^{k,l} = \{u \in F^kC^{k+l}(E;R) : \delta u \in F^{k+r}C^{k+l+1}(E;R)\}$, and the differential is $d_r[u] = [\delta u]$. The proof of that proposition identifies $E_{r+1} \cong H(E_r,d_r)$ by sending the class of $u \in Z_{r+1}^{k,l}$ to the class of $[u]$.

Step 1: Transgressive classes. No differential enters the column $k=0$, so $E_{r+1}^{0,n}$ is the subgroup $\ker d_r$ of $E_r^{0,n}$, and by the description of the pages this inclusion sends the class of $u \in Z_{r+1}^{0,n}$ to the class of the same u. Hence $E_{n+1}^{0,n} \subseteq E_1^{0,n}$ consists of the classes of the cochains $u \in C^n(E;R)$ with $\delta u \in F^{n+1}C^{n+1}(E;R)$. The class of u in $E_1^{0,n} = \prod_b H^n(F_b;R)$ is the family of the classes of its restrictions to the fibers, and $E_2^{0,n} = H^n(F;R)$ reads its component at b_0 (theorem 6.2.9). So x is transgressive if and only if $x = [u|_F]$ for a cochain u on E with $\delta u \in F^{n+1}C^{n+1}(E;R)$, and then $\tau(x) = d_{n+1}[u] = [\delta u]$.

Step 2: These coboundaries are pulled back from B. Since $F^{n+1}C^n = 0$ and $F^{n+2}C^{n+1} = 0$, the group $E_1^{n+1,0}$ is $\{v \in F^{n+1}C^{n+1}(E;R) : \delta v \in F^{n+2}C^{n+2}(E;R)\}$, with no quotient. The map p from $E \to B$ to the identity fibration of B induces a morphism of spectral sequences (theorem 6.2.8), given by $c \mapsto p^\sharp c$ on cochains. On the first pages it is $N^{n+1}(B;R) \to \prod_\tau H^0(F_{\tau(e_0)};R)$, induced by the inclusions of constants $R \to H^0(F_b;R)$, which are isomorphisms because every fiber is path-connected, its $H_0(-;\mathbb{Z})$ being isomorphic to $H_0(F;\mathbb{Z}) = \mathbb{Z}$ by transport (lemma 6.2.5). So every v as above is $p^\sharp c$ for a unique normalized cochain $c \in N^{n+1}(B;R)$, and c is a cocycle when v is, $p^\sharp$ being injective on $N^{n+2}(B;R)$ for the same reason. A normalized cochain of positive degree vanishes on the constant simplices at b_0, so c is also a cocycle of the pair (B,b_0). In $E_2^{n+1,0} = H^{n+1}(B;R)$ the class of $p^\sharp c$ is $[c]$, and $E_{n+1}^{n+1,0}$ is the quotient of $E_2^{n+1,0}$ by the images of $d_2,\dots,d_n$, the bottom row emitting no differentials.

Step 3: Transgressive implies $\delta(x) \in \operatorname{im} p^*$. Let u be as in Step 1, and by Step 2 write $\delta u = p^\sharp c$. The cocycle $u|_F$ represents x, so by the formula for δ in theorem 4.1.17 the relative cocycle δu represents $\delta(x) \in H^{n+1}(E, F; R)$. Hence $p^*[c] = [p^\sharp c] = \delta(x)$.

Step 4: The converse, and the value of τ. Let $\delta(x) = p^*(y)$ with $y \in H^{n+1}(B, b_0; R) = H^{n+1}(B; R)$, using $n \geq 1$. Normalized cochains compute $H^{n+1}(B; R)$ (proof of theorem 6.2.8), so y is represented by a normalized cocycle c, which vanishes on b_0. Represent x by a cocycle w on F and extend w to a cochain u_0 on E, so that δu_0 represents $\delta(x)$. Then $p^\sharp c - \delta u_0$ represents $p^*(y) - \delta(x) = 0$ in $H^{n+1}(E, F; R)$, so $p^\sharp c - \delta u_0 = \delta v$ for a cochain v vanishing on F. The cochain $u = u_0 + v$ has $u|_F = w$ and $\delta u = p^\sharp c$, which lies in $F^{n+1}C^{n+1}(E; R)$ because the simplices of $F_n C_{n+1}(E)$ have degenerate projections. By Step 1, x is transgressive and $\tau(x) = [\delta u] = [p^\sharp c]$, the image of y in $E_{n+1}^{n+1,0}$ by Step 2. As $\tau(x)$ depends only on x, this holds for every y with $p^*(y) = \delta(x)$, and $x = 0$ shows that $\ker p^*$ maps to zero in $E_{n+1}^{n+1,0}$, completing the proof.

The transgression theorem says that the longest differential out of the fiber column is the connecting map of the pair (E, F), read in base coordinates. In the path-loop fibration of example 6.4.2 the pair $(PX, \Omega X)$ has PX contractible, so the long exact sequence of the pair (theorem 4.1.17) gives $\delta\colon H^n(\Omega X) \xrightarrow{\cong} H^{n+1}(PX, \Omega X)$, and p^* identifies $H^{n+1}(X, x_0)$ with $H^{n+1}(PX, \Omega X)$. The transgression is then the suspension isomorphism in disguise, matching the homotopy-theoretic dimension shift $\pi_n(\Omega X) \cong \pi_{n+1}(X)$. More generally the theorem permits the computation of base cohomology from fiber cohomology whenever the total space is understood, and it is the precise statement underlying the assertion in section 5.6 that the k-invariant of a Postnikov tower is the transgression of the fundamental class of the fiber.

6.4.4 Note: Transgression and the k-Invariant In section 5.6 the first k-invariant of S^2 was identified as $\pm u^2 \in H^4(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z})$, with the assertion that the transgression in the Serre spectral sequence of the fibration $K(\mathbb{Z}, 3) \to X_3 \to \mathbb{C}\mathrm{P}^\infty$ sends the fundamental class $\iota_3 \in H^3(K(\mathbb{Z}, 3))$ to $k_3 \in H^4(\mathbb{C}\mathrm{P}^\infty)$. Theorem 6.4.3 is the theorem behind that assertion: the fiber $K(\mathbb{Z}, 3)$ has $H^3 = \mathbb{Z}\langle \iota_3 \rangle$ as its bottom class, ι_3 is transgressive because the lower columns vanish, and $\tau(\iota_3) = k_3$ is the connecting map of the pair, which is exactly the obstruction class defining the Postnikov fibration. The transgression is thus the spectral-sequence computation of the k-invariant.

The remaining work is to demonstrate the combined force of multiplicativity and transgression on concrete rings. The strategy in each case is identical: identify a transgressive generator, compute its transgression from a degeneration constraint (typically a contractible or known total space), and then let the Leibniz rule propagate the single differential across all products, with graded-commutativity giving the divided-power or polynomial relations among the propagated classes.

Example 6.4.5: Computing Cohomology Rings by Multiplicativity
Two rings are computed, the second illustrating how graded-commutativity forces divided powers.

1. *The ring $H^*(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) \cong \mathbb{Z}[x]$, $|x| = 2$, from $S^1 \to S^\infty \to \mathbb{C}\mathrm{P}^\infty$.* Use the fibration $S^1 \hookrightarrow S^\infty \to \mathbb{C}\mathrm{P}^\infty$ of example 5.5.5, with contractible total space S^∞ (so $H^*(S^\infty) = \mathbb{Z}$ in degree 0, zero above). The fiber is S^1 with $H^*(S^1; \mathbb{Z}) = \mathbb{Z}\langle 1, t \rangle$, $|t| = 1$, $t^2 = 0$. The base $\mathbb{C}\mathrm{P}^\infty$ is simply connected, so π_1 acts trivially and theorem 6.2.8 gives
$$E_2^{p,q} = H^p(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) \otimes H^q(S^1; \mathbb{Z})$$

concentrated in rows $q = 0, 1$, since $H^q(S^1) = 0$ for $q \geq 2$. Write $H^*(\mathbb{CP}^\infty)$ provisionally as an unknown graded ring A^*, which the spectral sequence will determine. The bottom row is $E_2^{*,0} = A^*$ and the top row is $E_2^{*,1} = A^* \cdot t$ (the bottom row tensored with t). The only possibly nonzero differential is $d_2 \colon E_2^{p,1} \to E_2^{p+2,0}$, since d_2 drops the fiber degree by 1 and raises base degree by 2, and all higher d_r ($r \geq 3$) vanish for bidegree reasons (only two rows). Convergence to $H^*(S^\infty) = \mathbb{Z}$ in degree 0 forces $E_3 = E_\infty = 0$ except at $(0, 0)$, so d_2 must be an isomorphism $E_2^{p,1} \xrightarrow{\cong} E_2^{p+2,0}$ for every $p \geq 0$ where the target is in positive total degree.

Now apply multiplicativity. The class $t \in E_2^{0,1}$ is transgressive with $d_2(t) = x$ for some $x \in E_2^{2,0} = A^2$. Since d_2 is an isomorphism $E_2^{0,1} \to E_2^{2,0}$ and $E_2^{0,1} = \mathbb{Z}\langle t \rangle$, the image x generates $A^2 \cong \mathbb{Z}$. The Leibniz rule (theorem 6.2.15(2)) computes d_2 on the whole top row by induction on k. Suppose $A^{2k} = \mathbb{Z}\langle x^k \rangle$ is infinite cyclic on the power x^k and $A^{2k-1} = 0$ (the base case $k = 1$ being $A^2 = \mathbb{Z}\langle x \rangle$, $A^1 = H^1(\mathbb{CP}^\infty) = 0$ since $\mathbb{CP}^\infty$ is simply connected). Then $E_2^{2k,1} = A^{2k} \cdot t = \mathbb{Z}\langle x^k t \rangle$, and graded-commutativity gives

$$d_2(x^k\, t) = d_2(x^k) \cdot t + (-1)^{2k} x^k \cdot d_2(t) = 0 + x^k \cdot x = x^{k+1}$$

because $d_2(x^k) = 0$ (the bottom row $q = 0$ consists of permanent cycles, no differential leaving it landing in a nonzero entry). Since $d_2 \colon E_2^{2k,1} \to E_2^{2k+2,0} = A^{2k+2}$ must be an isomorphism (convergence to a point in this total degree), the target A^{2k+2} is infinite cyclic generated by the image $x^{k+1} = d_2(x^k t)$, advancing the first part of the induction hypothesis. For the odd group, A^{2k+1} sits in the bottom row and can only be killed by d_2 from $E_2^{2k-1,1} = A^{2k-1} \cdot t = 0$. Receiving no differential, it must vanish for E_∞ to be zero off the origin, so $A^{2k+1} = 0$, advancing the second part. The induction gives $A^{2k} \cong \mathbb{Z}$ generated by x^k and $A^{2k+1} = 0$ for all k. Hence $A^* = \mathbb{Z}[x]$ with $|x| = 2$:

$$H^*(\mathbb{CP}^\infty; \mathbb{Z}) \cong \mathbb{Z}[x], \qquad |x| = 2$$

recovering the polynomial ring computed by cup products in chapter 4, now from the fibration alone. The ring structure is forced: multiplicativity requires $x^k = d_2(x^{k-1} t)$ generate H^{2k}, so the powers of x are exactly the generators, with no room for any relation.

2. *The ring $H^*(\Omega S^{n+1}; \mathbb{Z})$ as a divided-power algebra.* Use the path-loop fibration $\Omega S^{n+1} \to PS^{n+1} \to S^{n+1}$ of example 6.4.2, $n \geq 1$, with contractible total space. The base has $H^*(S^{n+1}) = \mathbb{Z}\langle 1, s \rangle$, $|s| = n + 1$, so $E_2^{p,q} = H^p(S^{n+1}) \otimes H^q(\Omega S^{n+1})$ is concentrated in columns $p = 0$ and $p = n + 1$. Let $a_1 \in H^n(\Omega S^{n+1})$ be the transgressive generator with $\tau(a_1) = d_{n+1}(a_1) = s$ from example 6.4.2. The spectral sequence computes the unknown fiber ring $H^*(\Omega S^{n+1})$ from the constraint $E_\infty = 0$ off the origin.

   The additive structure is example 6.3.1 for the sphere S^{n+1}: the differential $d_{n+1} \colon E_{n+1}^{0,q} \to E_{n+1}^{n+1,q-n}$ is an isomorphism for every $q \geq 1$, and $H^q(\Omega S^{n+1}; \mathbb{Z})$ is infinite cyclic in the degrees $0, n, 2n, 3n, \ldots$ and zero elsewhere. Write a_k for a generator of $H^{kn}(\Omega S^{n+1}) \cong \mathbb{Z}$ (so $a_0 = 1$, $|a_k| = kn$, and $a_k \in E_2^{0,kn}$). The entries are $E_2^{0,kn} = \mathbb{Z}\langle a_k \rangle$ and $E_2^{n+1,kn} = \mathbb{Z}\langle s\, a_k \rangle$ (column $n + 1$ is column 0 shifted by the base class s, by multiplicativity). For E_∞ to vanish off the origin, d_{n+1} must pair off $E_{n+1}^{0,(k+1)n}$ with $E_{n+1}^{n+1,kn}$ isomorphically for each $k \geq 0$, since these are the only nonzero entries and they sit in the right bidegrees: d_{n+1} sends bidegree $(0, (k+1)n)$ to $\big(n+1,\, (k+1)n - n\big) = (n+1, kn)$, which is exactly the bidegree of $s\, a_k$, confirming the pairing $a_{k+1} \leftrightarrow s\, a_k$.

   The Leibniz rule now determines the integer. Compute, using $d_{n+1}(s) = 0$ (the base class s is a permanent cycle, lying in the bottom row $q = 0$) and graded-commutativity,

$$d_{n+1}(s\, a_k) = d_{n+1}(s)\, a_k \pm s\, d_{n+1}(a_k) = \pm s \cdot s\, a_{k-1} = 0$$

since $s^2 = 0$ in $H^*(S^{n+1})$ (the base is a single sphere). So $s\,a_k$ is a d_{n+1}-cycle, consistent with its being hit from the left column. To find $d_{n+1}(a_{k+1})$, use the product structure of the fiber: write a_{k+1} in terms of lower classes. Graded-commutativity of the fiber ring constrains a_1^2: if n is odd, $a_1^2 = -a_1^2$ forces $2a_1^2 = 0$, but $H^{2n}(\Omega S^{n+1}) \cong \mathbb{Z}$ is torsion-free, so $a_1^2 = 0$ and a_1^2 *cannot* generate H^{2n}. Hence a_2 is a new generator, not a_1^2. The relation between a_1^2 and a_2 is the divided-power relation: applying d_{n+1} as a derivation to the product $a_1 \cdot a_1$,

$$d_{n+1}(a_1^2) = d_{n+1}(a_1)\,a_1 + (-1)^n a_1\,d_{n+1}(a_1) = s\,a_1 + (-1)^n a_1\,s = (1 + (-1)^n)\,s\,a_1$$

where $a_1\,s = s\,a_1$ because the total degrees satisfy $n(n+1) \equiv 0 \pmod 2$, so graded-commutativity introduces no sign. For n odd this is 0, consistent with $a_1^2 = 0$. For n even it is $2\,s\,a_1$. Meanwhile $d_{n+1}(a_2) = s\,a_1$ (the pairing isomorphism, normalized so $a_2 \mapsto s\,a_1$). Comparing, for n even $d_{n+1}(a_1^2) = 2\,s\,a_1 = 2\,d_{n+1}(a_2)$, so $d_{n+1}(a_1^2 - 2a_2) = 0$. Since the only d_{n+1}-cycle in $E_{n+1}^{0,2n}$ that survives is zero (the column-0 entry must die), $a_1^2 - 2a_2$ must itself be a boundary, but $E_{n+1}^{0,2n}$ receives no differential, so $a_1^2 - 2a_2 = 0$, giving

$$a_1^2 = 2\,a_2 \qquad (n \text{ even})$$

Iterate the computation, taking n even first so that $(-1)^{kn} = 1$ and all signs are positive. The Leibniz rule applied to $a_k \cdot a_1$ gives, using $d_{n+1}(a_k) = s\,a_{k-1}$, $d_{n+1}(a_1) = s$, and $a_k\,s = s\,a_k$,

$$d_{n+1}(a_k\,a_1) = s\,a_{k-1}\,a_1 + s\,a_k = s\,(a_{k-1}a_1) + s\,a_k$$

By the inductive recursion $a_{k-1}a_1 = k\,a_k$ this is $k\,s\,a_k + s\,a_k = (k+1)\,s\,a_k = (k+1)\,d_{n+1}(a_{k+1})$. Hence $d_{n+1}\big(a_k a_1 - (k+1)a_{k+1}\big) = 0$, and since the column-0 entry $E_{n+1}^{0,(k+1)n}$ must die (it receives no differential and cannot survive to E_∞), the cycle $a_k a_1 - (k+1)a_{k+1}$ vanishes. Therefore

$$a_1 \cdot a_k = (k+1)\,a_{k+1}, \qquad \text{equivalently} \qquad a_1^k = k!\,a_k \qquad (n \text{ even})$$

so $a_k = a_1^k/k!$ is a genuine class in integral cohomology even though a_1^k is divisible by $k!$. The ring is the *divided-power algebra* $\Gamma_{\mathbb{Z}}[a]$ on one generator $a = a_1$ of degree n, with additive basis $\{a_k = a^{[k]}\}_{k \geq 0}$ and multiplication $a^{[j]}a^{[k]} = \binom{j+k}{j} a^{[j+k]}$:

$$H^*(\Omega S^{n+1}; \mathbb{Z}) \cong \Gamma_{\mathbb{Z}}[a], \qquad |a| = n$$

For n odd the additive computation is unchanged (one $\mathbb{Z}$ in each degree kn), but graded-commutativity now forces $a_1^2 = 0$, so the odd generator a_1 is exterior and the genuine divided powers begin one degree up: the same Leibniz analysis applied to a_2 (of even degree $2n$) shows $a_2^k = k!\,a_{2k}$, so $H^*(\Omega S^{n+1}; \mathbb{Z}) \cong \Lambda_{\mathbb{Z}}[a_1] \otimes \Gamma_{\mathbb{Z}}[a_2]$ with $|a_1| = n$, $|a_2| = 2n$. In both parities this is the divided-power algebra $\Gamma_{\mathbb{Z}}[a]$ on a degree-n generator, since $\Gamma_{\mathbb{Z}}[a]$ for an odd-degree generator is by definition this exterior-times-divided-power tensor product. Over a field of characteristic zero, $\Gamma_{\mathbb{Q}}[a] \cong \mathbb{Q}[a]$ is a polynomial algebra (on a_1 for n even, on a_2 for n odd). The divided-power structure is visible only integrally, where the divisibility $a^k = k!\,a^{[k]}$ records that ΩS^{n+1} has more integral cohomology than a polynomial algebra would predict.

The two computations exhibit the two outcomes multiplicativity can force. In the first, graded-commutativity placed no constraint (the generator x has even degree, so x^k are all nonzero and

independent) and the answer is a free polynomial algebra. The spectral sequence confirms that the powers of x generate, with the differentials computing each power as the transgression of the previous one times the fiber class. In the second, graded-commutativity of an odd-degree class would force $a_1^2 = 0$, which collides with the torsion-free integral cohomology the convergence requires, and the resolution is the divided-power structure: the class a_1^k exists but is $k!$ times a primitive generator. The divided-power phenomenon is the integral fingerprint of the loop-space multiplication, invisible rationally and central to the structure of the homology of iterated loop spaces.

6.4.6 Historical Note: Serre's Thesis and the Method of Killing Homotopy Groups
The multiplicative spectral sequence and the transgression were the tools with which Jean-Pierre Serre, in his 1951 thesis, broke open the computation of homotopy groups of spheres. The path-loop fibration over S^{n+1}, computed in example 6.4.5, was Serre's lever: combined with the method of "killing homotopy groups" (the dual of the Postnikov construction of section 5.6), it gave the first proof that $\pi_i(S^n)$ is finite for $i > n$ except for $\pi_{4k-1}(S^{2k})$, and computed many stable stems. The divided-power structure of $H^*(\Omega S^{n+1})$ and its mod-p refinements, where the divided powers interact with the Steenrod operations, remain the computational heart of the subject.

Exercise 6.4.1

1. Let $E_r^{\bullet,\bullet}$ be a multiplicative spectral sequence with d_r a derivation. Suppose $a \in E_r^{p,q}$ satisfies $d_r(a) = 0$ and $b \in E_r^{p',q'}$ satisfies $b = d_r(c)$ for some c. Show directly that $a \cdot b$ is a boundary, and conclude that the product $[a] \cdot [b]$ on $E_{r+1} = H(E_r, d_r)$ is well defined.

2. Using the multiplicative Serre spectral sequence of the trivial fibration $F \to B \times F \to B$ with B a path-connected CW complex and B, F of finite type over a field k, prove the Künneth formula $H^*(B \times F; k) \cong H^*(B; k) \otimes_k H^*(F; k)$ as graded k-algebras. (Show that $E_2 = E_\infty$ and that there are no extension problems.)

3. Compute the cohomology ring $H^*(\mathbb{C}\mathrm{P}^n; \mathbb{Z})$ for finite n from the fibration $S^1 \to S^{2n+1} \to \mathbb{C}\mathrm{P}^n$, and identify it as $\mathbb{Z}[x]/(x^{n+1})$ with $|x| = 2$. Explain which differentials change relative to the $\mathbb{C}\mathrm{P}^\infty$ computation of example 6.4.5 and why the relation $x^{n+1} = 0$ appears.

4. Let n be even. Show that $H^*(\Omega S^{n+1}; \mathbb{Q}) \cong \mathbb{Q}[a]$ is a polynomial algebra on a single generator of degree n, while $H^*(\Omega S^{n+1}; \mathbb{Z})$ is the divided-power algebra $\Gamma_{\mathbb{Z}}[a]$. Exhibit explicitly the class $a^{[p]} = a^p/p!$ and verify that it is not in the image of $H^*(\Omega S^{n+1}; \mathbb{Z}) \to H^*(\Omega S^{n+1}; \mathbb{Q})$ from any integral polynomial in a.

5. In the path-loop fibration $\Omega X \to PX \to X$ of a simply connected space X, show that for $n \geq 1$ the transgression $\tau\colon H^n(\Omega X; R) \to H^{n+1}(X; R)$ defined by definition 6.4.1 coincides with the cohomology suspension isomorphism on the classes for which it is defined, using theorem 6.4.3 and the contractibility of PX.

6. Let $p\colon E \to B$ be a fibration with B simply connected, $H^*(B; \mathbb{Q})$ and $H^*(F; \mathbb{Q})$ both finite-dimensional, and suppose the rational Serre spectral sequence collapses at E_2 (all differentials vanish). Show that the Poincaré series satisfy $P_E(t) = P_B(t) \cdot P_F(t)$, where $P_X(t) = \sum_i \dim_{\mathbb{Q}} H^i(X; \mathbb{Q})\, t^i$. Give an example where collapse fails and the equality is a strict inequality $P_E(t) \leq P_B(t) P_F(t)$ coefficientwise.

6.5 The Leray-Hirsch Theorem

The Serre spectral sequence computes $H^*(E)$ from $H^*(B)$ and $H^*(F)$ only once its differentials are known, and pinning those down can be the hard part of a calculation. One recurring situation settles the outcome in advance, with no differential computed at all: when the cohomology of the fiber is free and every fiber class extends to a global class on the total space. The spectral sequence then collapses for a structural reason, and $H^*(E)$ is a free module over $H^*(B)$ on the extending classes. This is the Leray-Hirsch theorem. It is the last tool this chapter assembles and the one that opens the next: the cohomology of a projective bundle is computed by it directly, and on that computation the characteristic classes of chapter 7 are built.

The reason the hypothesis forces collapse can be given before any proof. A class defined on all of E is a permanent cycle of the spectral sequence, since it already lives in $H^*(E)$ and can be neither the source nor the target of a differential. If such global classes restrict to a basis of the fiber cohomology, multiplicativity spreads their survival across the whole spectral sequence and every differential is forced to vanish. Recall that $p^*\colon H^*(B;R) \to H^*(E;R)$ is a ring homomorphism, so $H^*(E;R)$ is a module over $H^*(B;R)$ through $\alpha \cdot \beta = p^*(\alpha) \smile \beta$ (definition 4.2.7). The theorem identifies this module completely.

Theorem 6.5.1: The Leray-Hirsch Theorem

Let $p\colon E \to B$ be a Serre fibration over a path-connected base with fiber F, and let R be a commutative ring. Suppose that

1. $H^n(F;R)$ is a free R-module of finite rank for every n, and
2. there exist classes $c_j \in H^{q_j}(E;R)$, $j = 1,\dots,k$, whose restrictions $i^*c_1,\dots,i^*c_k$ along the fiber inclusion $i\colon F \hookrightarrow E$ form an R-basis of $H^*(F;R)$.

Then $H^*(E;R)$ is a free $H^*(B;R)$-module with basis $c_1,\dots,c_k$. Equivalently, the $H^*(B;R)$-linear map

$$\Phi\colon H^*(B;R) \otimes_R H^*(F;R) \longrightarrow H^*(E;R), \qquad \alpha \otimes i^*c_j \longmapsto p^*(\alpha) \smile c_j$$

is an isomorphism.

***Proof*:**
The existence of the global classes forces the fundamental group of the base to act trivially on the fiber cohomology: each restriction i^*c_j is the image of a class defined on all of E, hence is invariant under the monodromy action of $\pi_1(B)$ on $H^*(F;R)$ (for a loop γ, the class c_j pulled back to $A_*(X_\gamma)$ restricts to i^*c_j on both end fibers, so the transport T^γ of lemma 6.2.5 fixes i^*c_j), and by hypothesis (2) the i^*c_j span $H^*(F;R)$, so the action is trivial. Theorem 6.2.8 therefore gives the cohomology Serre spectral sequence, with second page

$$E_2^{p,q} \cong H^p\big(B; H^q(F;R)\big) \cong H^p(B;R) \otimes_R H^q(F;R)$$

converging to $H^{p+q}(E;R)$. The second isomorphism holds because $H^q(F;R)$ is free of finite rank by hypothesis (1), so cohomology with these coefficients is the tensor product with $H^*(B;R)$.

Step 1: The entire fiber column survives. By theorem 6.2.9 the edge homomorphism in the

fiber direction is the restriction $i^*\colon H^q(E;R) \to H^q(F;R)$, and it factors as the projection $H^q(E) \twoheadrightarrow E_\infty^{0,q}$ onto the top filtration quotient followed by the inclusion $E_\infty^{0,q} \hookrightarrow E_2^{0,q} \cong H^q(F;R)$. In particular the image of i^* is exactly the surviving subgroup $E_\infty^{0,q} \subseteq E_2^{0,q}$. Each i^*c_j lies in this image, and by hypothesis (2) the classes i^*c_j exhaust a basis of $H^q(F;R)$ as q ranges over all degrees, so i^* is surjective and $E_\infty^{0,q} = E_2^{0,q}$ for every q. Thus no element of the fiber column $E_2^{0,\bullet}$ is killed by a differential. Since the fiber column is the leftmost, no differential maps into it either: a differential d_r into $E_r^{0,q}$ would originate in $E_r^{-r,\,q+r-1} = 0$. Hence every class in the fiber column is a permanent cycle, and d_r vanishes on $E_r^{0,\bullet}$ for all $r \geq 2$.

Step 2: All differentials vanish. The base row $E_2^{\bullet,0} \cong H^\bullet(B;R)$ also consists of permanent cycles: a differential out of $E_r^{p,0}$ lands in $E_r^{p+r,\,1-r} = 0$, so d_r vanishes on the base row. By the multiplicative structure (theorem 6.2.15(3)) the product

$$E_2^{p,0} \otimes_R E_2^{0,q} \longrightarrow E_2^{p,q}$$

is the isomorphism $H^p(B;R) \otimes_R H^q(F;R) \xrightarrow{\sim} E_2^{p,q}$, so E_2 is generated as a bigraded algebra by the base row together with the fiber column. Each differential d_r is a derivation (theorem 6.2.15(2)) that vanishes on both sets of generators by Steps 1 and 2. A derivation vanishing on algebra generators vanishes identically, so $d_2 = 0$ on all of E_2. Then $E_3 = E_2$ with the same generators, and the argument repeats verbatim: inductively $d_r = 0$ and $E_{r+1} = E_r$ for every $r \geq 2$. The spectral sequence collapses, $E_2 = E_\infty$.

Step 3: From collapse to the module isomorphism. Collapse identifies the associated graded of the abutment filtration,

$$\operatorname{gr} H^*(E;R) \cong E_\infty = E_2 = H^*(B;R) \otimes_R H^*(F;R)$$

and it remains to see that Φ realizes this identification. The map Φ is $H^*(B;R)$-linear because p^* is a ring homomorphism and the cup product is associative:

$$\Phi\big((\alpha'\alpha) \otimes i^*c_j\big) = p^*(\alpha'\alpha) \smile c_j = p^*(\alpha') \smile \big(p^*(\alpha) \smile c_j\big) = \alpha' \cdot \Phi(\alpha \otimes i^*c_j)$$

It preserves the filtration: $p^*(\alpha) \in F^{|\alpha|}H^*(E)$ and $c_j \in F^0H^*(E)$, so by multiplicativity of the abutment filtration (theorem 6.2.15(4)) the product $p^*(\alpha) \smile c_j$ lies in $F^{|\alpha|}$. On the associated graded, Φ induces the product map $E_2^{|\alpha|,0} \otimes E_2^{0,q_j} \to E_2^{|\alpha|,q_j}$ of Step 2, which is exactly the isomorphism $E_2 \cong H^*(B) \otimes H^*(F)$. Hence $\operatorname{gr}\Phi$ is an isomorphism. In each total degree the abutment filtration is finite, since the spectral sequence is first-quadrant (proposition 6.1.9), so it is bounded and exhaustive. A filtration-preserving map inducing an isomorphism on the associated graded of a finite filtration is itself an isomorphism, by induction up the filtration steps and the five lemma. Therefore Φ is an isomorphism of $H^*(B;R)$-modules, and $c_1, \dots, c_k$ are a free basis of $H^*(E;R)$ over $H^*(B;R)$, as we sought to show.

The theorem is a collapse criterion wearing the clothes of a module statement: hypothesis (2) is precisely what makes the fiber column survive, multiplicativity does the rest, and no differential is ever computed. The freeness of hypothesis (1) cannot be dropped, since torsion in the fiber cohomology introduces coefficient extensions the collapse argument does not control (exercise 6.5.1 (3)). One hypothesis deserves removal before the theorem is put to work: the base is assumed path-connected, while the characteristic-class constructions of chapter 7 run over bases with no connectivity guarantee. The assumption can be dropped wholesale, because singular cohomology sees the path components of a space separately and a fibration decomposes over the path components of its base. Both hypotheses

below are imposed at *every* fiber. Over a disconnected base no single fiber can speak for the others. The corollary is stated for a Serre fibration. Every fiber bundle, over any base whatsoever, qualifies by theorem 5.2.4.

Corollary 6.5.2: Leray-Hirsch over a Disconnected Base

Let $p\colon E \to B$ be a Serre fibration whose fibers are all nonempty, and let R be a commutative ring. Suppose that

1. for every $b \in B$ the fiber $F_b = p^{-1}(b)$ has $H^m(F_b; R)$ a free R-module of finite rank for every m, and
2. there exist classes $c_j \in H^{q_j}(E; R)$, $j = 1, \dots, k$, whose restrictions along the inclusion $F_b \hookrightarrow E$ form an R-basis of $H^*(F_b; R)$ for *every* $b \in B$.

Then $H^*(E; R)$ is a free $H^*(B; R)$-module with basis $c_1, \dots, c_k$: the $H^*(B; R)$-linear map

$$\Phi\colon \bigoplus_{j=1}^{k} H^*(B; R) \longrightarrow H^*(E; R), \qquad (\alpha_j)_j \longmapsto \sum_{j=1}^{k} p^*(\alpha_j) \smile c_j$$

is an isomorphism.

***Proof*:**

Step 1: Cohomology is the product over a saturated partition. Let Y be any space and let $\{Y_c\}_{c \in C}$ be a partition of Y into subsets each of which is a union of path components of Y. Every singular simplex $\sigma\colon \Delta^m \to Y$ has path-connected image, so it lands in a single path component, hence in a single Y_c, and the boundary of a simplex lying in Y_c again lies in Y_c. Hence the singular chain complex splits as a direct sum of subcomplexes, $C_*(Y) = \bigoplus_c C_*(Y_c)$. Applying $\mathrm{Hom}_{\mathbb{Z}}(-, R)$ converts the direct sum into a direct product of cochain complexes, $C^*(Y; R) = \prod_c C^*(Y_c; R)$, and cohomology commutes with direct products of cochain complexes, kernels and images being computed coordinatewise. The resulting isomorphism

$$H^*(Y; R) \xrightarrow{\cong} \prod_c H^*(Y_c; R)$$

is induced by the restrictions to the members of the partition. It is natural for partition-preserving maps: if $\varphi\colon Y \to Y'$ carries each member Y_c of such a partition into the member Y'_c of a like partition of Y', then restriction commutes with φ^* coordinate by coordinate. And it is a ring isomorphism, cup products being computed by restriction member by member.

Step 2: The total space partitions over the path components of the base. Let $\{B_c\}$ be the path components of B, a partition of B of the kind Step 1 asks for, and put $E_c = p^{-1}(B_c)$, a nonempty set because the fiber over any point of B_c is nonempty. A path in E projects to a path in B and so cannot leave E_c. Hence each E_c is a union of path components of E, and $\{E_c\}$ is a partition of E of the same kind. No connectivity of E_c itself is claimed, and none is needed. The restriction $p_c\colon E_c \to B_c$ is again a Serre fibration: a lifting problem for p_c is one for p whose homotopy lands in B_c, and any solution for p then lands in $p^{-1}(B_c) = E_c$.

Step 3: The connected case on each component, and reassembly. Fix a component B_c and a point $b \in B_c$. The Serre fibration $p_c\colon E_c \to B_c$ has path-connected base and fiber F_b. Hypothesis (1)

makes $H^m(F_b;R)$ free of finite rank for every m, and hypothesis (2), read at the fiber over b, makes the restrictions $c_j|_{E_c}$ a family of homogeneous classes whose further restriction to F_b is an R-basis of $H^*(F_b;R)$. Theorem 6.5.1 therefore applies to p_c and makes $H^*(E_c;R)$ a free $H^*(B_c;R)$-module with basis the $c_j|_{E_c}$. Equivalently, the map Φ_c defined like Φ with B_c, E_c, and $c_j|_{E_c}$ in place of B, E, and c_j is an isomorphism. Now assemble. The maps p and the inclusions $E_c \hookrightarrow E$, $B_c \hookrightarrow B$ all preserve the partitions of Step 2, so the naturality clause of Step 1 applies to them. Restriction commutes with the module structure, $p^*(\alpha)|_{E_c} = p_c^*(\alpha|_{B_c})$, both being induced by the square formed by p and the two inclusions, restriction being a ring map. Under the isomorphisms of Step 1 for B and for E, the map Φ is therefore identified with the composite

$$\bigoplus_{j=1}^{k} \prod_{c} H^*(B_c;R) \;\cong\; \prod_{c} \bigoplus_{j=1}^{k} H^*(B_c;R) \xrightarrow{\prod_c \Phi_c} \prod_{c} H^*(E_c;R)$$

where the first exchange of sum and product is an isomorphism because the sum is finite. A product of isomorphisms is an isomorphism, so Φ is one, and $c_1,\dots,c_k$ are a free basis of $H^*(E;R)$ over $H^*(B;R)$, completing the proof.

The quantifier in clause (2) is what makes the corollary true over a disconnected base, and it cannot be weakened to a single fiber. Two examples show why, and they fail in different ways.

First, take $B = \mathrm{pt} \sqcup \mathrm{pt}$, $E = S^1 \sqcup S^1$ the product bundle, $R = \mathbb{Z}$, and the classes $c_0 = (1,1)$, $c_1 = (u,0)$ where u generates $H^1(S^1;\mathbb{Z})$. Clause (1) holds, the fibers being circles. At the first point the restrictions are 1 and u, a basis of $H^*(S^1;\mathbb{Z})$. At the second they are 1 and 0, which are not. So clause (2) holds at one fiber and fails at the other, and the conclusion fails too: $H^*(B;\mathbb{Z})$ is concentrated in degree 0, so the $H^*(B;\mathbb{Z})$-span of c_0 and c_1 misses $(0,u)$ and they are no basis of $H^*(E;\mathbb{Z})$. Checking clause (2) at a single fiber would have admitted this.

The second example shows that clause (1) holding everywhere does not rescue matters. Take $R = \mathbb{Z}$, fiber $F = S^1$, base $B = \mathrm{pt} \sqcup S^1$, and $E = S^1 \sqcup K$ with K the Klein bottle as the circle bundle over S^1 given by the mapping torus of a reflection. Every fiber is a circle, so clause (1) holds. But $H^2(K;\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}$, while a free graded module over $H^*(B;\mathbb{Z})$ is torsion-free as an abelian group, so $H^*(E;\mathbb{Z})$ is not free and the conclusion fails. What fails in the hypotheses is clause (2) over the S^1 component: $H^1(K;\mathbb{Z}) = \mathbb{Z}$ restricts to zero on a fiber circle, that circle being 2-torsion in $H_1(K) = \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, so no choice of classes restricts to a basis there. Requiring clause (2) at every b excludes exactly this.

The corollary carries no connectivity hypothesis on base or fibers, which is the strength of the statement recorded in [23, §3.1]. The proof is not given there but in [21, Theorem 4D.1], by an induction over a CW base followed by a CW-approximation reduction. Here the spectral sequence needs no CW base, and the route through path components replaces that induction. The paradigm application, and the one the next chapter turns into a definition, is the cohomology of a projective bundle.

Example 6.5.3: The Cohomology of a Projective Bundle

Let $E \to B$ be a complex vector bundle (definition 5.2.8) of rank n over a paracompact Hausdorff base, and let $\mathbb{P}(E) \to B$ be its projectivization, the fiber bundle whose fiber over b is the space $\mathbb{P}(E_b) \cong \mathbb{C}\mathrm{P}^{n-1}$ of complex lines in E_b. The fiber cohomology $H^*(\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z}) = \mathbb{Z}[u]/(u^n)$ is free of rank n over $\mathbb{Z}$ with basis $1,u,\dots,u^{n-1}$, so hypothesis (1) of corollary 6.5.2 holds at every fiber. For hypothesis (2), the projectivization carries a tautological line bundle $\gamma \to \mathbb{P}(E)$, the subbundle

of p^*E whose fiber over a line $\ell \in \mathbb{P}(E_b)$ is ℓ itself. Set $t = -c_1(\gamma) \in H^2(\mathbb{P}(E);\mathbb{Z})$, the negative first Chern class of γ (example 7.2.3, developed in the next chapter). On each fiber $\mathbb{P}(E_b) = \mathbb{C}\mathrm{P}^{n-1}$ the bundle γ restricts to the tautological line bundle of $\mathbb{C}\mathrm{P}^{n-1}$, whose class t restricts to the generator u of $H^2(\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z})$, so $1, t, \dots, t^{n-1}$ restrict to the fiber basis $1, u, \dots, u^{n-1}$ in every fiber. Corollary 6.5.2, applied to the projective bundle, a Serre fibration with nonempty fibers by theorem 5.2.4, then gives that $H^*(\mathbb{P}(E);\mathbb{Z})$ is a free module over $H^*(B;\mathbb{Z})$ with basis $1, t, \dots, t^{n-1}$. In particular t^n is an $H^*(B)$-linear combination of the lower powers: there are unique classes $a_i \in H^{2i}(B;\mathbb{Z})$ with

$$t^n + a_1 t^{n-1} + \cdots + a_{n-1} t + a_n = 0$$

and this single relation presents the ring,

$$H^*(\mathbb{P}(E);\mathbb{Z}) \cong H^*(B;\mathbb{Z})[t] \big/ \left(t^n + a_1 t^{n-1} + \cdots + a_n\right)$$

The coefficients a_i are invariants of the bundle E. Naming them $a_i = c_i(E)$ is the projective-bundle route to the Chern classes recorded in proposition 7.4.11.

Exercise 6.5.1

1. Let B and F be spaces with $H^*(F;R)$ free of finite rank over a commutative ring R, and consider the trivial fibration $F \to B \times F \xrightarrow{p} B$, which is a Serre fibration with fiber F over every point, so that corollary 6.5.2 applies to any B. Using the projection $\mathrm{pr}_F \colon B \times F \to F$, exhibit classes on $B \times F$ satisfying the hypotheses of corollary 6.5.2, and conclude that $H^*(B \times F;R) \cong H^*(B;R) \otimes_R H^*(F;R)$ as $H^*(B;R)$-modules. Explain how this recovers the Künneth theorem in this case.

2. Show that hypothesis (2) of theorem 6.5.1 holds if and only if $H^*(F;R)$ is free of finite rank and the restriction $i^* \colon H^*(E;R) \to H^*(F;R)$ is surjective. Conclude that Leray-Hirsch applies exactly when the fiber cohomology is free and entirely restricted from the total space.

3. Locate the two places in the proof of theorem 6.5.1 where freeness of $H^*(F;R)$ is used: the identification $E_2^{p,q} \cong H^p(B;R) \otimes_R H^q(F;R)$ and the notion of an R-basis in hypothesis (2). Give a fiber F and ring R for which $H^*(F;R)$ is not free (for instance $F = \mathbb{R}\mathrm{P}^2$, $R = \mathbb{Z}$), and identify which step of the argument breaks.

7

Characteristic Classes

A characteristic class is a rule that attaches to every vector bundle a cohomology class of its base, in a way that respects pullback. The decisive observation of this chapter is that such a rule is the same thing as a single cohomology class on one universal space, the classifying space BG of section 7.1. Every principal G-bundle over a CW complex X is the pullback of one universal bundle along a classifying map $X \to BG$, unique up to homotopy by theorem 7.1.10. A characteristic class is therefore nothing more than an element of $H^*(BG)$, pulled back to X along that map. The whole theory of characteristic classes is, in this reading, the computation of the cohomology rings $H^*(BG)$ for the structure groups that arise in geometry, together with the dictionary that turns the resulting universal classes into invariants of bundles. This is the cohomological route to a subject that also admits a bundle-theoretic route (the Grassmannian and obstruction-theoretic constructions, developed in section 7.1.2 and section 7.4) and an analytic route (the Chern-Weil construction from curvature, developed in the Riemannian-geometry volume [14]). The three agree, and the present chapter develops the first.

The classifying spaces are constructed in section 7.1 from principal bundles, which generalize the covering spaces of chapter 2, and for the orthogonal and unitary groups they are Grassmannians. Their cohomology is a ring under the cup product of chapter 4, and for these groups it is a polynomial algebra. Section 7.2 computes it with the Serre spectral sequence of chapter 6, run on the universal sphere bundles relating $BU(n-1)$ to $BU(n)$ and $BO(n-1)$ to $BO(n)$, the Gysin sequence of theorem 6.3.2 adding one generator at a time. Section 7.3 constructs the Thom class, and section 7.4 identifies the classes as obstructions, the Euler class in particular as the primary obstruction to a nowhere-zero section in the sense of chapter 5. The chapter closes with the rank-two bundles on complexes of dimension at most four and with the Pontryagin classes, and chapter 8 uses the same classes to orient the Thom spectra of cobordism.

7.1 Principal Bundles and Classifying Spaces

A rank-n real vector bundle with a Euclidean inner product has over each point the set of orthonormal bases of its fiber, and the orthogonal group $O(n)$ acts on that set freely and transitively. A normal covering space has the same structure for a discrete group: the deck group acts freely and transitively on each fiber, and the base is the quotient by that action. Both are principal bundles. This section constructs, for a Hausdorff locally compact group G, a principal G-bundle $EG \to BG$ from which every principal G-bundle over a CW complex is pulled back, along a map to BG that is unique up to homotopy. For a discrete group G the base is a $K(G,1)$, the space BG of section 5.5. For the orthogonal and unitary groups it is a Grassmannian, which section 7.1.1 makes a CW complex, and section 7.1.2 classifies vector bundles over every paracompact Hausdorff base by maps into it.

Definition 7.1.1: Principal G-Bundle

Let G be a topological group. A *principal G-bundle* is a map $p\colon E \to X$ of spaces together with a continuous right action $E \times G \to E$ of G on E, written $(e,g) \mapsto e \cdot g$, satisfying:

1. the action preserves fibers and is free and transitive on each fiber: $p(e \cdot g) = p(e)$ for all g, and for each $x \in X$ the group G acts simply transitively on $p^{-1}(x)$;
2. the bundle is *locally trivial*: every point of X has an open neighbourhood U admitting a G-equivariant homeomorphism $\varphi_U\colon p^{-1}(U) \xrightarrow{\cong} U \times G$ over U, where G acts on $U \times G$ by right translation in the second factor, $(u,h)\cdot g = (u,hg)$, and "over U" means $\mathrm{pr}_U \circ \varphi_U = p$.

An *isomorphism* of principal G-bundles $p\colon E \to X$ and $p'\colon E' \to X$ over the same base is a G-equivariant homeomorphism $E \to E'$ commuting with the projections to X. The set of isomorphism classes of principal G-bundles over X is written $\mathrm{Prin}_G(X)$.

A principal G-bundle is a fiber bundle whose fiber is the group G itself, but carrying the extra datum of a free fiberwise action that turns each fiber into a *torsor*: a copy of G with no distinguished identity element. Local triviality says the bundle looks like the product $U \times G$ over small opens, exactly as for the fiber bundles of theorem 5.2.4, with the additional requirement that the trivializing homeomorphisms respect the G-action. The transition between two trivializations φ_U and φ_V over $U \cap V$ is left multiplication by a continuous function $g_{UV}\colon U \cap V \to G$, the *transition function*, because an equivariant self-map of $U \cap V \times G$ over $U \cap V$ commuting with right translation is precisely left translation by an element of G varying continuously with the base point. A principal bundle is trivial, that is, isomorphic to the product $X \times G$, if and only if it admits a global section, since a section $s\colon X \to E$ produces the equivariant trivialization $X \times G \to E$, $(x,g) \mapsto s(x)\cdot g$, and conversely the product bundle has the obvious section $x \mapsto (x,e)$. This is the first point of contact with covering-space theory: a covering map with deck group G acting freely and transitively on fibers is a principal G-bundle for the discrete group G.

Example 7.1.2: The Real Line over the Circle

Take $G = \mathbb{Z}$ with the discrete topology and the exponential covering $p\colon \mathbb{R} \to S^1$, $t \mapsto e^{2\pi i t}$, with $\mathbb{Z}$ acting on $\mathbb{R}$ by $t \cdot n = t + n$. This is a principal $\mathbb{Z}$-bundle. The action is continuous, preserves fibers (since $e^{2\pi i(t+n)} = e^{2\pi i t}$), and is free and transitive on each fiber: the fiber over $e^{2\pi i\theta}$ is the coset $\theta + \mathbb{Z}$, on which $\mathbb{Z}$ acts simply transitively by translation. For local triviality, cover S^1 by

the two arcs $U_+ = S^1 \setminus \{-1\}$ and $U_- = S^1 \setminus \{1\}$. Over U_+ the principal branch of $\frac{1}{2\pi i}\log$ gives a continuous section $s_+ \colon U_+ \to \mathbb{R}$ with values in $(-\frac{1}{2}, \frac{1}{2})$, and the equivariant homeomorphism

$$p^{-1}(U_+) \xrightarrow{\cong} U_+ \times \mathbb{Z}, \qquad t \longmapsto \big(e^{2\pi i t},\, t - s_+(e^{2\pi i t})\big)$$

trivializes the bundle over U_+, the second coordinate landing in $\mathbb{Z}$ because t and $s_+(e^{2\pi i t})$ differ by an integer. The analogous formula works over U_-. The single transition function on the overlap $U_+ \cap U_-$ (two arcs) is the locally constant map to $\mathbb{Z}$ recording the integer jump $s_+ - s_-$, which is 0 on one arc and 1 on the other. This nonconstancy is exactly the obstruction to a global section, so the bundle is nontrivial. The deck group $\mathrm{Deck}\,(p) \cong \mathbb{Z}$ of definition 2.3.8 is the structure group of the bundle, and the principal $\mathbb{Z}$-bundle and the universal cover are the same object viewed two ways.

The next task is to construct, for a fixed G, a single bundle from which every principal G-bundle is pulled back. Pulling back a bundle along a map $f \colon X \to X'$ replaces the base by X and the total space by the fiber product, and the operation is functorial and, on a paracompact Hausdorff base, homotopy-invariant (lemma 7.1.9 below). The universal object is characterized by a contractibility condition on its total space.

Definition 7.1.3: Universal Bundle and Classifying Space

Let G be a topological group. A principal G-bundle $p \colon EG \to BG$ is a *universal bundle*, and its base BG is a *classifying space* of G, if the total space EG is weakly contractible, that is, $\pi_i(EG) = 0$ for all $i \geq 0$.

The defining condition asks the total space to have no homotopy of its own, so that all of the homotopy of the bundle is concentrated in the base and the structure group. When G is discrete, a principal G-bundle is a covering map on whose fibers G acts by deck transformations. Conversely, let a discrete group G act freely on a space E so that every point has a neighbourhood U with $U \cdot g \cap U = \varnothing$ for $g \neq e$, and let $p \colon E \to E/G$ be the orbit map. Then p is open, it is injective on U, so it maps U homeomorphically onto the open set $p(U)$, and $p^{-1}(p(U))$ is the disjoint union of the open sets $U \cdot g$. The map $p(U) \times G \to p^{-1}(p(U))$ sending $(p(u), g)$ to $u \cdot g$ for $u \in U$ is therefore an equivariant homeomorphism over $p(U)$, and p is a principal G-bundle. If E is moreover weakly contractible, p is a universal bundle. If E is also a CW complex, it is contractible by Whitehead's theorem (theorem 5.3.6), and E/G is a $K(G,1)$ in the sense of definition 5.5.1 whenever it is a CW complex, by the argument of example 5.5.6. For a general topological group G the same long exact sequence of the bundle $G \to EG \to BG$, applied with $\pi_i(EG) = 0$, yields the relation

$$\pi_i(BG) \;\cong\; \pi_{i-1}(G) \qquad \text{for all } i \geq 1$$

the delooping relation recorded in corollary 7.1.11 below. Milnor's infinite-join construction gives a universal bundle for every locally compact Hausdorff group, and the classification theorem shows that BG is unique up to weak homotopy equivalence.

Proposition 7.1.4: Existence of the Universal Bundle

Let G be a topological group that is Hausdorff and locally compact, for instance a discrete group or one of the matrix groups $O(n)$ and $U(n)$. Then there exists a principal G-bundle $p\colon EG \to BG$ with EG weakly contractible. When G is discrete the spaces EG and BG may be taken to be CW complexes with EG contractible, and then BG is a $K(G,1)$.

***Proof*:**

Step 1: The infinite join and the action. For $n \geq 1$ write a point of $\Delta^{n-1} \times G^n$ as $(t;g) = (t_1,\dots,t_n;\, g_1,\dots,g_n)$, with $t_i \geq 0$ and $\sum t_i = 1$. Let EG be the quotient of $\bigsqcup_{n\geq 1} \Delta^{n-1} \times G^n$ by the relation generated by two identifications: $(t;g) \sim (t;g')$ whenever $g_i = g_i'$ for every i with $t_i > 0$, and

$$(t_1,\dots,t_n;\, g_1,\dots,g_n) \sim (t_1,\dots,t_n,0;\, g_1,\dots,g_n,h) \qquad \text{for every } h \in G$$

Write the class of $(t;g)$ as $t_1g_1 \oplus \cdots \oplus t_ng_n$, and $E_n \subseteq EG$ for the image of $\Delta^{n-1} \times G^n$, the *n-fold join*. A set is closed in EG exactly when its preimage in every $\Delta^{n-1} \times G^n$ is closed. The weights $t_i\colon EG \to [0,1]$ are well defined and continuous, and the coordinate g_i is well defined and continuous on the open set $\{t_i > 0\}$, since the preimage of that set in each $\Delta^{n-1} \times G^n$ is open and saturated, and the restriction of a quotient map to an open saturated set is a quotient map. The group G acts on the right by $(t_1g_1 \oplus \cdots \oplus t_ng_n)\cdot g = t_1(g_1g) \oplus \cdots \oplus t_n(g_ng)$, which respects both identifications. The action is continuous: it is induced by the continuous map $\big(\bigsqcup_n \Delta^{n-1} \times G^n\big) \times G \to EG$, and the product of the quotient map onto EG with id_G is a quotient map because G is locally compact Hausdorff [16, Products of Quotient Maps]. It is free, because $x \cdot g = x$ with $t_i(x) > 0$ gives $g_i(x)g = g_i(x)$, hence $g = e$.

Step 2: Each E_n is a closed subspace, and compact sets lie in some E_n. The preimage of E_n in $\Delta^{m-1} \times G^m$ is everything for $m \leq n$ and the closed set $\{t_{n+1} = \cdots = t_m = 0\}$ for $m > n$, so E_n is closed. Give E_n its quotient topology from $\Delta^{n-1} \times G^n$. A set $C \subseteq E_n$ closed in that topology has closed preimage in $\Delta^{m-1} \times G^m$: for $m > n$ it is the closed set $\{t_{n+1} = \cdots = t_m = 0\}$ intersected with the product of the preimage of C in $\Delta^{n-1} \times G^n$ and G^{m-n}, and for $m < n$ it is the preimage under the continuous padding map $(t;g) \mapsto (t,0,\dots,0;\, g,e,\dots,e)$. So the quotient topology of E_n is its subspace topology in EG. Since G is Hausdorff, the preimage of a point in each $\Delta^{m-1} \times G^m$ is closed, so points of EG are closed. If a compact set $K \subseteq EG$ lay in no E_n, it would contain points $x_k \notin E_k$ for $k \geq 1$, and every subset of $\{x_k\}$ meets each E_n, hence the image of each $\Delta^{n-1} \times G^n$, in finitely many points, so it is closed. Then $\{x_k\}$ is an infinite closed subset of K with the discrete topology, which is impossible in a compact space.

Step 3: Weak contractibility. For $n \geq 1$ the map $\Delta^{n-1} \times G^n \times I \to \Delta^n \times G^{n+1}$, $(t;g;s) \mapsto \big((1-s)t_1,\dots,(1-s)t_n,s;\, g_1,\dots,g_n,e\big)$, followed by the quotient map, respects the identifications defining E_n, so it induces a homotopy $h\colon E_n \times I \to EG$, continuous because the product of the quotient map onto E_n with id_I is a quotient map [16, Products of Quotient Maps]. At $s = 0$ it is the inclusion of E_n, and at $s = 1$ it is constant at the point $1e$ in the slot $n+1$. Let $f\colon S^k \to EG$ be a map. By Step 2 its compact image lies in some E_n and f is continuous into E_n, so $h \circ (f \times \mathrm{id}_I)$ is a homotopy from f to a constant map. The case $k = 0$ shows that EG is path-connected. For $k \geq 1$ and f based, the class of f lies in the orbit of 0 under the action of π_1, which is $\{0\}$ (proposition 5.1.15). So $\pi_k(EG) = 0$ for every k at every basepoint.

Step 4: Local triviality. Let $BG = EG/G$ with the quotient topology and p the orbit map, which is open since each g acts by a homeomorphism. For $i \geq 1$ the open set $V_i = \{t_i > 0\}$ is

saturated, since $t_i(x \cdot g) = t_i(x)$ and $g_i(x \cdot g) = g_i(x)g$. The map $V_i \to V_i$, $x \mapsto x \cdot g_i(x)^{-1}$, is continuous and constant on orbits, so it descends to a continuous section $s_i \colon p(V_i) \to EG$ of p, the restriction of p to the open saturated set V_i being a quotient map onto the open set $p(V_i)$. The map $p(V_i) \times G \to V_i$, $(b, g) \mapsto s_i(b) \cdot g$, is a continuous equivariant bijection over $p(V_i)$ with continuous inverse $x \mapsto \big(p(x), g_i(x)\big)$, because $x = s_i(p(x)) \cdot g_i(x)$. The sets $p(V_i)$ cover BG, since every point of EG has a positive weight. Hence p is a principal G-bundle with weakly contractible total space.

Step 5: The discrete case. Let G be discrete. Then $\Delta^{n-1} \times G^n$ is a disjoint union of simplices, and the identifications glue them along faces, so EG is the Δ-complex whose n-simplices are the $(n+1)$-element sets of pairs $(g, i) \in G \times \{1, 2, \dots\}$ with distinct second coordinates, a CW complex by proposition 3.1.6. For $x \in EG$ with $t_i(x) > 0$, the open set $U = \{t_i > 0,\ g_i = g_i(x)\}$ satisfies $U \cdot g \cap U = \varnothing$ for $g \neq e$. The action permutes the simplices freely and never identifies two vertices of one simplex, so $BG = EG/G$ is the Δ-complex of the orbits of simplices, again a CW complex, and by the discussion after definition 7.1.3 the map p is a covering. The weakly contractible CW complex EG is contractible by Whitehead's theorem (theorem 5.3.6), so BG is a $K(G, 1)$ by example 5.5.6, completing the proof.

7.1.5 Remark: Milnor's Topology on the Join Local compactness of G enters the proof once, to make the action $EG \times G \to EG$ continuous: the product of a quotient map with id_G need not be a quotient map otherwise. Milnor gives the infinite join instead the coarsest topology for which every weight t_i and every coordinate g_i is continuous. With that topology the join is a principal G-bundle for every topological group G, and it classifies the numerable principal G-bundles [24, Ch. 4, Theorem 11.2, Theorem 12.2 and Theorem 12.4]. The groups used in this book are discrete groups and matrix groups, all locally compact.

The join model is concrete enough to compute with but is rarely the most efficient representative. It shows that a universal bundle exists for every locally compact Hausdorff group, and it makes the weak contractibility of EG visible: a sphere in EG lies in a finite join and contracts in the next one. In practice one replaces EG by any principal G-bundle with weakly contractible total space, and corollary 7.1.11 shows that the resulting base is independent of the choice up to weak homotopy equivalence.

Example 7.1.6: Classifying Spaces of $\mathbb{Z}$, $\mathbb{Z}/2$, and $U(1)$

1. For $G = \mathbb{Z}$ take $EG = \mathbb{R}$ with the translation action $t \cdot n = t + n$, under which every interval of length less than 1 is disjoint from its translates. The space $\mathbb{R}$ is contractible and $E\mathbb{Z}/\mathbb{Z} = \mathbb{R}/\mathbb{Z} = S^1$, so by the discussion after definition 7.1.3 the bundle $\mathbb{R} \to S^1$ is universal. Thus $B\mathbb{Z} = S^1$, recovering the principal bundle of example 7.1.2 and the space $K(\mathbb{Z}, 1)$ of example 5.5.2.
2. For $G = \mathbb{Z}/2$ take $EG = S^\infty = \bigcup_n S^n$ with the antipodal action $v \cdot (-1) = -v$, under which every open set of diameter less than 2 is disjoint from its antipodal image. The infinite sphere is contractible (any compact subset lies in some S^n, which contracts inside S^{n+1} by sliding to a missing point), and $S^\infty/(\mathbb{Z}/2) = \mathbb{R}\mathrm{P}^\infty$. Thus $B(\mathbb{Z}/2) = \mathbb{R}\mathrm{P}^\infty$, so $\mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2, 1)$, matching the computation of example 5.5.6.
3. For $G = U(1) = S^1$ take $EG = S^\infty \subseteq \mathbb{C}^\infty$, the unit sphere, with $U(1)$ acting by scalar multiplication $v \cdot \lambda = \lambda v$ for $|\lambda| = 1$. The orbit space is $S^\infty/U(1) = \mathbb{C}\mathrm{P}^\infty$, and $S^\infty \to \mathbb{C}\mathrm{P}^\infty$

is the frame bundle $V_1(\mathbb{C}^\infty) \to \mathrm{Gr}_1(\mathbb{C}^\infty)$ of lemma 7.1.18(2) below, a principal $U(1)$-bundle with contractible total space. Thus $BU(1) = \mathbb{C}\mathrm{P}^\infty$. Here $U(1)$ is not discrete, so $\mathbb{C}\mathrm{P}^\infty$ is not a $K(U(1), 1)$. Instead $\pi_i(\mathbb{C}\mathrm{P}^\infty) \cong \pi_{i-1}(U(1)) = \pi_{i-1}(S^1)$, which is $\mathbb{Z}$ for $i = 2$ and 0 otherwise, exhibiting $\mathbb{C}\mathrm{P}^\infty = K(\mathbb{Z}, 2)$ as predicted by the delooping relation $\pi_i(BG) \cong \pi_{i-1}(G)$.

For the discrete groups $\mathbb{Z}$ and $\mathbb{Z}/2$ the bases are spaces $K(G, 1)$, as example 5.5.6 shows for every group acting freely and properly discontinuously on a contractible CW complex. For the circle $U(1)$ the same total space S^∞ produces $\mathbb{C}\mathrm{P}^\infty = K(\mathbb{Z}, 2)$, with the homotopy of $U(1) = S^1 = K(\mathbb{Z}, 1)$ moved up one degree. The relation $\pi_i(BG) \cong \pi_{i-1}(G)$ accounts for both: if a topological group G is a $K(A, n)$, then BG has the homotopy groups of a $K(A, n + 1)$, as in $B\mathbb{Z} = S^1 = K(\mathbb{Z}, 1)$ and $BU(1) = K(\mathbb{Z}, 2) = \mathbb{C}\mathrm{P}^\infty$.

The classification theorem states that the universal bundle deserves its name: every principal G-bundle over a CW complex X is pulled back from $EG \to BG$ along a map to BG, and two bundles are isomorphic exactly when their classifying maps are homotopic. The proof rests on two homotopy properties of pullbacks, isolated in lemma 7.1.9 below, together with the weak contractibility of EG. The homotopy invariance of pullbacks is proved with partitions of unity, which exist on a paracompact Hausdorff space [16, Existence of Subordinate Partitions of Unity]. The bases of interest are CW complexes, which are Hausdorff by theorem 1.2.3. They are also paracompact, by a skeletal induction like the one in the proof of theorem 1.2.3, now extending a partition of unity over one cell at a time.

Theorem 7.1.7: CW Complexes Are Paracompact

Let X be a CW complex and $\{U_\alpha\}$ an open cover of X. Then there is a partition of unity $\{\varphi_\beta\}$ subordinate to $\{U_\alpha\}$: continuous functions $\varphi_\beta \colon X \to [0, 1]$ with locally finite supports, each support contained in some U_α, summing to 1. Consequently X is paracompact in the sense of [16, Paracompact Space], and being Hausdorff by theorem 1.2.3, it is paracompact Hausdorff.

***Proof*:**

The partition of unity is built over the skeleta by induction, following [23, Proposition 1.20].

Step 1: The induction and its extension step. On X^0, a discrete space, choose for each point one U_α containing it and let the corresponding function be the indicator of that point. This is a partition of unity subordinate to the cover, with locally finite supports because X^0 is discrete. Suppose a partition of unity $\{\varphi_\beta\}$ on X^n subordinate to $\{U_\alpha \cap X^n\}$ has been constructed. Let $\Phi_\gamma \colon D^{n+1} \to X$ be the characteristic map of an $(n + 1)$-cell (definition 1.2.2). The composites $\{\varphi_\beta \Phi_\gamma\}$ form a partition of unity on $S^n = \partial D^{n+1}$, and since S^n is compact only finitely many of them are not identically zero.

Write points of $D^{n+1} \setminus \{0\}$ in spherical coordinates $(r, x) \in (0, 1] \times S^n$, and let $\rho_\varepsilon \colon [0, 1] \to [0, 1]$ be continuous with $\rho_\varepsilon = 0$ on $[0, 1 - \varepsilon]$ and $\rho_\varepsilon = 1$ on $[1 - \varepsilon/2, 1]$. Extend each $\varphi_\beta \Phi_\gamma$ over D^{n+1} by $(r, x) \mapsto \rho_\varepsilon(r)\, \varphi_\beta \Phi_\gamma(x)$, and by 0 at the origin. The factor ρ_ε vanishes near 0, so this is continuous. Choosing ε small enough makes these extensions subordinate to $\{\Phi_\gamma^{-1}(U_\alpha)\}$. Indeed, fix one of the finitely many β with $\varphi_\beta \Phi_\gamma \not\equiv 0$ and let $K_\beta = \mathrm{supp}(\varphi_\beta \Phi_\gamma) \subseteq S^n$, a compact set contained in the open set $W_\beta = \Phi_\gamma^{-1}(U_{\alpha(\beta)})$. The support of the extended function lies in the radial collar $\{(r, x) : r \geq 1 - \varepsilon,\ x \in K_\beta\}$, whose points are within ε of K_β in the metric of D^{n+1}. Since K_β is compact and $D^{n+1} \setminus W_\beta$ is closed and disjoint from it, their distance is positive, so the collar lies in W_β once ε is smaller than that distance. Let $\varepsilon = \varepsilon_\gamma$ be *half* the minimum of

these finitely many distances, which is strictly smaller than each of them.

Step 2: Filling the interior. The disk D^{n+1} is compact Hausdorff, hence paracompact, so the open cover $\{\Phi_\gamma^{-1}(U_\alpha)\}$ admits a subordinate partition of unity $\{\psi_{\gamma j}\}$ ([16, Existence of Subordinate Partitions of Unity]), finite after discarding the identically zero members, the disk being compact. The family

$$\{\, \rho_\varepsilon\, \varphi_\beta \Phi_\gamma \,\} \cup \{\, (1-\rho_\varepsilon)\, \psi_{\gamma j} \,\}$$

is then a partition of unity on D^{n+1} subordinate to $\{\Phi_\gamma^{-1}(U_\alpha)\}$: the first group sums to ρ_ε and the second to $1-\rho_\varepsilon$. On S^n it restricts to $\{\varphi_\beta\Phi_\gamma\}$, because $\rho_\varepsilon = 1$ there kills the second group. Since the topology of X^{n+1} is the quotient topology from $X^n \sqcup_\gamma D_\gamma^{n+1}$, a function on X^{n+1} is continuous exactly when its restriction to X^n and its composite with each Φ_γ are continuous, so these compatible extensions assemble to a partition of unity on X^{n+1} extending $\{\varphi_\beta\}$ and subordinate to $\{U_\alpha \cap X^{n+1}\}$, after adjoining the new functions $(1-\rho_{\varepsilon_\gamma})\psi_{\gamma j}$ pushed forward from each cell.

Step 3: Local finiteness, via thin collars. The neighbourhoods that witness local finiteness must be chosen inside the region where $\rho_{\varepsilon_\gamma} \equiv 1$, because on the outer annulus $1-\varepsilon_\gamma \le r < 1-\varepsilon_\gamma/2$ the factor $1-\rho_{\varepsilon_\gamma}$ is positive and the $\psi_{\gamma j}$ sum to 1, so some newly adjoined function is nonzero at every point there. Call

$$C_\gamma = \Phi_\gamma\big(\{(r,x) : r > 1-\varepsilon_\gamma/2\}\big)$$

the *thin* collar of the cell e_γ^{n+1}. On C_γ every $\psi_{\gamma j}$ contributes 0, so the only functions that can be nonzero are the extensions of the φ_β.

Now let $x \in X^n$ and let $N_n \subseteq X^n$ be a neighbourhood of x in X^n meeting the supports of only finitely many φ_β, say those indexed by a finite set S. Put

$$N_{n+1} = N_n \cup \bigcup_\gamma \Big(C_\gamma \cap \Phi_\gamma\big(\{(r,y) : y \in \Phi_\gamma^{-1}(N_n)\}\big)\Big)$$

the union of N_n with the thin collars taken only over N_n. This is a neighbourhood of x in X^{n+1}: its preimage under each Φ_γ is open, and it meets X^n in N_n. By the previous paragraph the only functions nonzero on N_{n+1} are the extensions of the φ_β with $\beta \in S$, the same finite set, because passing to a thin collar adjoins no new function. Iterating gives a nested chain $N_n \subseteq N_{n+1} \subseteq N_{n+2} \subseteq \cdots$ whose union N is open in X, since $N \cap X^m$ is open in X^m for every m and X carries the weak topology determined by its skeleta, and on which only the finitely many functions indexed by S are nonzero. A point in an open cell e_γ^{n+1} is handled by the same construction started from a neighbourhood inside that cell, where the nonzero functions are the finitely many $\psi_{\gamma j}$ together with the finitely many extended $\varphi_\beta\Phi_\gamma$.

This is what the colimit step needs, and the naive argument does not give it: local finiteness on X^n gives a neighbourhood open in X^n, and an open subset of X^n need not be open in X: a point on the boundary of an attached cell has every X-neighbourhood meeting that cell's interior. The chain above is built precisely to grow one dimension at a time without enlarging S. Continuity of each function and the fact that they sum to 1 do pass to the union pointwise, the first because X carries the weak topology and the second because every point lies in some skeleton.

Step 4: Paracompactness. Given the partition of unity, the open sets $V_\beta = \varphi_\beta^{-1}\big((0,1]\big)$ satisfy $V_\beta \subseteq \operatorname{supp}\varphi_\beta \subseteq U_{\alpha(\beta)}$, so $\{V_\beta\}$ is an open refinement of $\{U_\alpha\}$, and it is locally finite because the supports are. It covers X because the φ_β sum to 1. Hence every open cover of X has a locally finite open refinement, which is [16, Paracompact Space], and X is Hausdorff by theorem 1.2.3, completing the proof.

A CW complex is paracompact Hausdorff, so every statement of this chapter whose base is assumed paracompact Hausdorff applies to a CW complex, and the classification of vector bundles over a paracompact Hausdorff base in section 7.1.2 contains, for vector bundles, the classification over CW complexes of theorem 7.1.10. The homotopy invariance of pullbacks needs a partition of unity indexed by a countable set. On a paracompact Hausdorff space every open cover has one, each of whose members is supported in a disjoint union of open sets lying in members of the cover.

Lemma 7.1.8: Countable Covers by Disjoint Unions

Let X be a paracompact Hausdorff space and $\{U_\alpha\}_{\alpha\in A}$ an open cover of X. Then X has a countable open cover $\{V_k\}_{k\geq 1}$ and a partition of unity $\{\varphi_k\}_{k\geq 1}$ with $\operatorname{supp}\varphi_k \subseteq V_k$, such that each V_k is a disjoint union of open sets, each contained in some U_α.

***Proof*:**
Let $\{\psi_\alpha\}_{\alpha\in A}$ be a partition of unity subordinate to $\{U_\alpha\}$, given by [16, Existence of Subordinate Partitions of Unity]: continuous functions with locally finite supports, $\operatorname{supp}\psi_\alpha \subseteq U_\alpha$, summing to 1. For a finite nonempty set $S \subseteq A$ let

$$W_S = \{x \in X : \psi_\alpha(x) > 0 \text{ and } \psi_\alpha(x) > \psi_\beta(x) \text{ for all } \alpha \in S \text{ and all } \beta \notin S\}$$

Near any point only finitely many ψ_β are nonzero, so near that point W_S is cut out by finitely many strict inequalities between continuous functions, and W_S is open. For $\alpha \in S$ the function ψ_α is positive on W_S, so $W_S \subseteq \operatorname{supp}\psi_\alpha \subseteq U_\alpha$. If $S \neq S'$ have the same number of elements, choose $\alpha \in S \setminus S'$ and $\beta \in S' \setminus S$. On $W_S \cap W_{S'}$ both $\psi_\alpha > \psi_\beta$ and $\psi_\beta > \psi_\alpha$ would hold, so $W_S \cap W_{S'} = \varnothing$. For $k \geq 1$ let V_k be the union of the sets W_S over the S with exactly k elements, a disjoint union of open sets each contained in some U_α. Each x lies in $W_{S(x)}$ for $S(x) = \{\alpha : \psi_\alpha(x) > 0\}$, which is finite by local finiteness and nonempty because the ψ_α sum to 1. So $\{V_k\}_{k\geq 1}$ is a countable open cover of X, and [16, Existence of Subordinate Partitions of Unity] applied to it gives the partition of unity $\{\varphi_k\}_{k\geq 1}$, completing the proof.

Lemma 7.1.9: Homotopy Invariance of Pullback Bundles

Let G be a topological group and $p\colon E \to B$ a principal G-bundle. For a map $f\colon X \to B$ the pullback

$$f^*E \;=\; \{(x, e) \in X \times E : f(x) = p(e)\}$$

with the projection to X and the diagonal G-action $(x, e)\cdot g = (x, e\cdot g)$ is a principal G-bundle over X. If X is paracompact Hausdorff (for instance a CW complex, by theorem 7.1.7) and $f_0 \simeq f_1\colon X \to B$ are homotopic, then $f_0^*E \cong f_1^*E$ as principal G-bundles over X.

***Proof*:**
Step 1: The pullback is a principal bundle. Local triviality transports along f: if $\varphi_U : p^{-1}(U) \cong U \times G$ trivializes E over $U \subseteq B$, then over $f^{-1}(U)$ the map $(x, e) \mapsto (x, \operatorname{pr}_G\varphi_U(e))$ is an equivariant homeomorphism $(f^*E)|_{f^{-1}(U)} \cong f^{-1}(U) \times G$ over $f^{-1}(U)$, with inverse $(x, g) \mapsto \big(x, \varphi_U^{-1}(f(x), g)\big)$. The diagonal G-action is free and fiberwise transitive because the E-coordinate already is. Hence $f^*E \to X$ is a principal G-bundle.

Step 2: Reduction to a bundle over $X \times I$. Let $H\colon X \times I \to B$ be a homotopy from f_0 to f_1 and form the principal G-bundle $Q = H^*E$ over $X \times I$. Its restrictions to $X \times \{0\}$ and $X \times \{1\}$ are f_0^*E and f_1^*E. It suffices to show that for a principal G-bundle Q over $X \times I$ with X paracompact Hausdorff, the restrictions $Q|_{X\times\{0\}}$ and $Q|_{X\times\{1\}}$ are isomorphic.

Step 3: Q is trivial over $U \times I$ for the sets U of an open cover of X. Fix $x \in X$. For each $t \in I$, local triviality at (x, t) gives an open set $U_t \ni x$ of X and an interval $J_t \ni t$ open in I with Q trivial over $U_t \times J_t$. Finitely many $J_{t_1}, \dots, J_{t_r}$ cover the compact interval I. Put $U = U_{t_1} \cap \cdots \cap U_{t_r}$ and choose, by the Lebesgue number lemma, a partition $0 = s_0 < s_1 < \cdots < s_m = 1$ with each $[s_{j-1}, s_j]$ inside some J_{t_l}, so that Q is trivial over every $U \times [s_{j-1}, s_j]$. Suppose Q has equivariant trivializations h_1 over $U \times [a, c]$ and h_2 over $U \times [c, b]$. An equivariant homeomorphism of $U \times G$ over U has the form $(y, g) \mapsto (y, \lambda(y)g)$, with $\lambda(y)$ the G-coordinate of the image of (y, e), so $h_1 h_2^{-1}(y, c, g) = (y, c, \lambda(y)g)$ for a continuous $\lambda\colon U \to G$. Replacing h_2 by its composite with $(y, t, g) \mapsto (y, t, \lambda(y)g)$ makes h_1 and h_2 agree over $U \times \{c\}$. They then define an equivariant bijection from $Q|_{U\times[a,b]}$ onto $U \times [a, b] \times G$, continuous with continuous inverse by the pasting lemma [16, Pasting Lemma] applied to the closed sets lying over $U \times [a, c]$ and $U \times [c, b]$. By induction on j, Q is trivial over $U \times I$.

Step 4: Moving across the graphs of a partition of unity. Let $\{U_\alpha\}$ be an open cover of X with Q trivial over each $U_\alpha \times I$ (Step 3), and let $\{V_k\}_{k\geq 1}$ and $\{\varphi_k\}_{k\geq 1}$ be as in lemma 7.1.8. Trivializations over disjoint open sets amalgamate, so Q is trivial over each $V_k \times I$, and we fix equivariant trivializations $h_k\colon Q|_{V_k\times I} \to V_k \times I \times G$. Put $\psi_0 = 0$ and $\psi_k = \varphi_1 + \cdots + \varphi_k$, let $\Gamma_k \subseteq X \times I$ be the graph of ψ_k, and let Q_k be the restriction of Q over Γ_k, so that $Q_0 = Q|_{X\times\{0\}}$. Define $\theta_k\colon Q_k \to Q_{k-1}$ to be the identity over the open set $X \setminus \operatorname{supp}\varphi_k$, where $\psi_k = \psi_{k-1}$, and over V_k by

$$\theta_k(q) = h_k^{-1}\big(x, \psi_{k-1}(x), g\big) \qquad \text{where } h_k(q) = \big(x, \psi_k(x), g\big)$$

The two prescriptions agree over $V_k \setminus \operatorname{supp}\varphi_k$, and the parts of Q_k over the open sets $X \setminus \operatorname{supp}\varphi_k$ and V_k cover Q_k. So θ_k is continuous and G-equivariant, it covers the homeomorphism $\Gamma_k \to \Gamma_{k-1}$, $(x, \psi_k(x)) \mapsto (x, \psi_{k-1}(x))$, and the same formulas with ψ_k and ψ_{k-1} exchanged give a continuous inverse.

Step 5: The infinite composite. Each point of X has an open neighbourhood N meeting $\operatorname{supp}\varphi_k$ for only finitely many k, say only for $k \leq K$. The φ_k sum to 1, so $\psi_k = 1$ on N for $k \geq K$, the part of Γ_k over N is $N \times \{1\}$, and θ_k is the identity over N for $k > K$. Hence the composites $\theta_1 \circ \theta_2 \circ \cdots \circ \theta_k\colon Q_k \to Q_0$ agree over N for all $k \geq K$, and they define a G-equivariant map $\theta\colon Q|_{X\times\{1\}} \to Q|_{X\times\{0\}}$ covering $(x, 1) \mapsto (x, 0)$. It is continuous, being continuous over each such N, and the composites of the θ_k^{-1} assemble in the same way into a continuous inverse. Identifying $X \times \{0\}$ and $X \times \{1\}$ with X, θ is an isomorphism of principal G-bundles. Applied to $Q = H^*E$ this reads $f_1^*E \cong f_0^*E$, as we sought to show.

The lemma reduces the isomorphism problem for pulled-back bundles to a homotopy problem: a bundle depends on its classifying map only up to homotopy, so classifying all bundles reduces to computing a set of homotopy classes. The remaining content of the classification theorem is that the classifying map exists for every bundle (surjectivity onto $[X, BG]$) and is unique up to homotopy (injectivity), and both come from the weak contractibility of EG.

Theorem 7.1.10: Classification of Principal G-Bundles

Let G be a topological group with a universal bundle $p\colon EG \to BG$ (definition 7.1.3), which exists when G is locally compact Hausdorff by proposition 7.1.4, and let X be a CW complex or a paracompact Hausdorff space of CW homotopy type. Pulling back the universal bundle along a map gives a bijection

$$[X, BG] \xrightarrow{\cong} \mathrm{Prin}_G(X), \qquad [f] \longmapsto f^*EG$$

between the set of homotopy classes of maps $X \to BG$ and the set of isomorphism classes of principal G-bundles over X. The bijection is natural in X: for a map $h\colon X' \to X$ the square relating $[-, BG]$ and $\mathrm{Prin}_G(-)$ commutes, both vertical maps being induced by pulling back along h.

***Proof*:**
The base X is paracompact Hausdorff, by hypothesis or by theorem 7.1.7, so the assignment $[f] \mapsto f^*EG$ is well defined by the homotopy invariance of lemma 7.1.9, and it is natural in X because pullback is functorial: $h^*(f^*EG) = (f \circ h)^*EG$. It remains to prove the map is surjective and injective. Steps 1 and 2 treat a CW complex X, and Step 3 reduces the general case to it.

Step 1: Surjectivity, constructing a classifying map. Let $q\colon P \to X$ be a principal G-bundle. A map $f\colon X \to BG$ with $f^*EG \cong P$ is the same datum as a G-equivariant map $\tilde{f}\colon P \to EG$ covering it: given $\tilde{f}$, the induced f on orbit spaces satisfies $f^*EG \cong P$ via $e \mapsto (q(e), \tilde{f}(e))$, and conversely an isomorphism $P \cong f^*EG$ composed with the projection $f^*EG \to EG$ yields $\tilde{f}$. So it suffices to build a G-equivariant map $\tilde{f}\colon P \to EG$. This is an extension problem solved skeleton by skeleton, using that EG is weakly contractible.

Build $\tilde{f}$ over the skeleta of X pulled up to P. Over the 0-skeleton, choose for each vertex v a point e_v of the fiber P_v and a point $y_v \in EG$, and set $\tilde{f}(e_v \cdot g) = y_v \cdot g$, which is well defined and equivariant because G acts simply transitively on P_v, and continuous, $g \mapsto e_v \cdot g$ being a homeomorphism $G \to P_v$ by local triviality. Suppose $\tilde{f}$ is defined and equivariant over $P|_{X^{n-1}}$. For an n-cell e^n_α with characteristic map $\Phi_\alpha\colon D^n \to X$, the pullback Φ_α^*P is a principal bundle over the disk D^n, a finite CW complex. By lemma 7.1.9, applied to a homotopy from id_{D^n} to a constant map, it is isomorphic to the pullback along the constant map, the product $D^n \times G$. Through this trivialization, an equivariant extension of $\tilde{f}$ over the part of P lying over e^n_α is the same as a map $D^n \to EG$ extending the given map on $\partial D^n = S^{n-1}$, obtained from $\tilde{f}$ on X^{n-1}. The obstruction to extending a map $S^{n-1} \to EG$ over D^n is its class in $\pi_{n-1}(EG)$, which vanishes because EG is weakly contractible. Hence the extension exists. Performing it on every n-cell and inducting up the skeleta produces an equivariant $\tilde{f}\colon P \to EG$. It is continuous: over an open set $U \subseteq X$ with an equivariant trivialization $P|_U \cong U \times G$ it has the form $(x, g) \mapsto s(x) \cdot g$, and s is continuous because its composite with each characteristic map is, U carrying the quotient topology from the sets $\Phi_\alpha^{-1}(U)$ ([16, Subspaces And Quotient Maps]). The induced $f\colon X \to BG$ satisfies $f^*EG \cong P$. Thus the map $[X, BG] \to \mathrm{Prin}_G(X)$ is surjective.

Step 2: Injectivity. Suppose $f_0, f_1\colon X \to BG$ satisfy $f_0^*EG \cong f_1^*EG =: P$. Choose equivariant maps $\tilde{f}_0, \tilde{f}_1\colon P \to EG$ covering f_0, f_1 (their existence is Step 1 read backwards: an isomorphism $P \cong f_i^*EG$ followed by projection gives $\tilde{f}_i$). A homotopy $f_0 \simeq f_1$ is produced by an equivariant homotopy $\tilde{f}_0 \simeq \tilde{f}_1\colon P \to EG$, since such a homotopy descends to orbit spaces and pulls EG

back to the homotopy realizing $f_0 \simeq f_1$. Build the equivariant homotopy $F\colon P \times I \to EG$ by the same skeletal extension as in Step 1, now over the CW complex $X \times I$ with the bundle $P \times I \to X \times I$: F is prescribed on $P \times \{0\}$ and $P \times \{1\}$ by $\tilde{f}_0$ and $\tilde{f}_1$, and the obstruction to extending over an n-cell of $X \times I$ (rel the two ends) again lands in $\pi_{n-1}(EG) = 0$. Hence F exists, $\tilde{f}_0 \simeq \tilde{f}_1$ equivariantly, and $f_0 \simeq f_1$. Therefore the map is injective.

Combining surjectivity and injectivity, $[X, BG] \to \mathrm{Prin}_G(X)$ is a natural bijection when X is a CW complex.

Step 3: Paracompact Hausdorff bases of CW homotopy type. Let X be paracompact Hausdorff and let $h\colon X' \to X$ be a homotopy equivalence with X' a CW complex and homotopy inverse $k\colon X \to X'$. Composition with h is a bijection $[X, BG] \to [X', BG]$, with inverse given by composition with k, since $h \circ k \simeq \mathrm{id}_X$ and $k \circ h \simeq \mathrm{id}_{X'}$. Pullback along h is a bijection $\mathrm{Prin}_G(X) \to \mathrm{Prin}_G(X')$ with inverse pullback along k: for P over X, $k^*h^*P \cong (h \circ k)^*P \cong P$ by lemma 7.1.9 over the paracompact Hausdorff base X, and for Q over X', $h^*k^*Q \cong Q$ by the same lemma over the CW complex X', which is paracompact Hausdorff by theorem 7.1.7. These two bijections commute with the assignments $[f] \mapsto f^*EG$ for X and X', by the naturality above, and the assignment for X' is a bijection by Steps 1 and 2. Hence so is the assignment for X, completing the proof.

The theorem is the precise sense in which BG "classifies" principal G-bundles: the entire isomorphism theory of such bundles over a CW complex X is encoded in the homotopy-theoretic invariant $[X, BG]$. The universal bundle $EG \to BG$ is the universal example, the bundle from which all others descend, and the classifying map f records which bundle by recording how X maps into BG. The proof is an obstruction argument of the kind that recurs throughout homotopy theory: the only thing one ever needs of EG is that its homotopy groups vanish, which makes every extension over a cell unobstructed and every two extensions homotopic. Naturality means $\mathrm{Prin}_G(-)$ and $[-, BG]$ are isomorphic as functors on the homotopy category of CW complexes, so BG represents the functor "isomorphism classes of principal G-bundles" in the precise sense of representable functors.

Two structural consequences follow. The classification pins down BG: any two universal bundles for G have weakly homotopy equivalent bases, homotopy equivalent when they have the homotopy type of CW complexes. And the long exact sequence of the universal bundle identifies the homotopy groups of BG with those of G shifted up by one.

Corollary 7.1.11: Uniqueness of BG and the Delooping

Let G be a topological group.

1. If $EG \to BG$ and $E'G \to B'G$ are two universal bundles for G, there are a CW complex Z and weak homotopy equivalences $BG \leftarrow Z \to B'G$. If BG and $B'G$ have the homotopy type of CW complexes, then $BG \simeq B'G$.
2. The space BG is path-connected, and the boundary map of the long exact sequence of the universal bundle $G \to EG \to BG$ is an isomorphism $\partial\colon \pi_i(BG) \to \pi_{i-1}(G)$ for every $i \geq 2$ and a bijection of pointed sets $\partial\colon \pi_1(BG) \to \pi_0(G)$.

***Proof*:**
Part 2. The base BG is path-connected, being the image of EG, which is path-connected because $\pi_0(EG) = 0$. The bundle $p\colon EG \to BG$ is a Serre fibration, since a principal bundle is a fiber bundle and theorem 5.2.4 applies, so theorem 5.2.12 gives its long exact sequence. Its fiber $p^{-1}(b_0)$ is homeomorphic to G through $g \mapsto e_0 \cdot g$. Weak contractibility, $\pi_i(EG) = 0$ for every $i \geq 0$, makes $\partial\colon \pi_i(BG) \to \pi_{i-1}(G)$ an isomorphism for $i \geq 2$ and makes $\partial\colon \pi_1(BG) \to \pi_0(G)$ onto. Exactness of pointed sets gives only a trivial kernel at $i = 1$, and injectivity there comes from path lifting. For a loop α at b_0, the class $\partial[\alpha]$ is the path component in $p^{-1}(b_0)$ of the free end of a lift $\tilde{\alpha}$ of α whose other end is e_0. If $\partial[\alpha] = \partial[\beta]$, a path δ in $p^{-1}(b_0)$ joins the free ends of $\tilde{\alpha}$ and $\tilde{\beta}$. Then $\tilde{\alpha}$, δ and the reverse of $\tilde{\beta}$, composed in the order that makes them a loop at e_0, form a null-homotopic loop, because $\pi_1(EG) = 0$. Its image under p is α, a constant path and the reverse of β in the same order, so $[\alpha] = [\beta]$.

Part 1. Let $v\colon Z \to B'G$ be a CW approximation (theorem 5.3.10), a weak homotopy equivalence from a CW complex, and $P = v^*E'G$. By theorem 7.1.10 over the CW complex Z there is $u\colon Z \to BG$ with $u^*EG \cong P$. The projections of the pullbacks give G-equivariant maps $\tilde{u}\colon P \to EG$ and $\tilde{v}\colon P \to E'G$ covering u and v, each restricting on a fiber to a G-equivariant map $G \to G$, $g \mapsto g_0 g$, a homeomorphism. The boundary map of theorem 5.2.12 is the composite of $(p_*)^{-1}$ on $\pi_n(E, F)$ with the boundary of the pair, both natural for maps of pairs, so $\partial_{EG} \circ u_* = \partial_P$ and $\partial_{E'G} \circ v_* = \partial_P$ after these identifications of the fibers with G. By part (2), the boundary maps ∂_{EG} and $\partial_{E'G}$ are bijections, and v_* is a bijection on every π_i, so ∂_P and then $u_* = \partial_{EG}^{-1} \circ \partial_P$ are bijections on π_i for $i \geq 1$. The space Z may be taken path-connected, as $B'G$ is, so u and v are weak homotopy equivalences. If BG has the homotopy type of a CW complex Y, with homotopy equivalence $k\colon BG \to Y$, then $k \circ u$ is a weak homotopy equivalence of CW complexes, hence a homotopy equivalence by theorem 5.3.6, and so is u. The same holds for v when $B'G$ has CW homotopy type, and then $BG \simeq Z \simeq B'G$, as we sought to show.

The delooping relation is the reason B is called a delooping. The path-loop fibration of example 5.2.15 gives $\pi_{i-1}(\Omega BG) \cong \pi_i(BG)$, so the loop space of BG has the homotopy groups of G, and B undoes the loop operation Ω at the level of homotopy groups, as in example 7.1.6, where B sent $K(\mathbb{Z}, 1) = S^1$ to $K(\mathbb{Z}, 2) = \mathbb{C}\mathrm{P}^\infty$. For a discrete group G whose classifying space is a CW complex, as in proposition 7.1.4, BG is a CW complex of type $K(G, 1)$. It is therefore the space BG of section 5.5 up to homotopy equivalence (theorem 5.5.4), and its cohomology with coefficients in a $\mathbb{Z}[G]$-module is the group cohomology of G (proposition 5.5.8).

A continuous homomorphism $\varphi\colon G \to H$ of topological groups with universal bundles induces a map $B\varphi\colon BG \to BH$ when BG is a CW complex. The space $EG \times_G H$, the quotient of $EG \times H$ by $(e \cdot g, h) \sim (e, \varphi(g)h)$, is a principal H-bundle over BG, the local trivializations of EG giving ones of $EG \times_G H$, so it is classified by a map $B\varphi\colon BG \to BH$, unique up to homotopy (theorem 7.1.10). For a second homomorphism $\psi\colon H \to K$, the bundle $(EG \times_G H) \times_H K$ is $EG \times_G K$, so $B\psi \circ B\varphi \simeq B(\psi \circ \varphi)$, and B is a functor from such groups to the homotopy category.

For the matrix groups BG classifies vector bundles. A rank-n complex vector bundle with a Hermitian inner product has the principal $U(n)$-bundle of its unitary frames, and the frame bundle recovers the vector bundle by the associated-bundle construction (lemma 7.1.26). The Grassmannian $\mathrm{Gr}_n(\mathbb{C}^\infty)$ of n-planes in $\mathbb{C}^\infty$ carries a principal $U(n)$-bundle with weakly contractible total space, the infinite Stiefel manifold of unitary n-frames, so it is a model of $BU(n)$, and the real Grassmannian $\mathrm{Gr}_n(\mathbb{R}^\infty)$ is likewise a model of $BO(n)$ (lemma 7.1.18). Section 7.1.1 gives these spaces their CW structures, and section 7.1.2 classifies vector bundles over every paracompact Hausdorff base by maps into them. The colimits BU and BO over n (definition 7.1.20) classify bundles over a compact base up to the

addition of trivial summands, and $\mathbb{Z} \times BU$ represents the complex K-theory of compact spaces in section 8.5.

7.1.1 Grassmannians and Their Schubert Cells

The classifying spaces $BO(n)$ and $BU(n)$ are Grassmannians, and the chapter uses them as CW complexes: proposition 7.2.2 below needs BG to be a CW complex, the classification over a paracompact base puts an inner product on the universal bundle over $BO(n)$, the Gysin sequences that compute their cohomology need a CW base, and the Thom isomorphism needs a base with finitely many cells in each dimension. Their cells are the Schubert cells, and the construction follows [29, §6]. Throughout, $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$, $r = \dim_{\mathbb{R}} \mathbb{K}$, and $\langle x, y\rangle = \sum_j x_j \bar{y}_j$ is the standard inner product. The space $\mathbb{K}^m$ sits in $\mathbb{K}^{m+1}$ as the vectors with last coordinate zero, $\mathbb{K}^\infty = \bigcup_m \mathbb{K}^m$, and $e_1, e_2, \dots$ is the standard basis.

Definition 7.1.12: Grassmannian

Let $0 \le n \le m < \infty$. The *Stiefel manifold* $V_n(\mathbb{K}^m)$ is the space of orthonormal n-frames $(x_1, \dots, x_n)$ in $\mathbb{K}^m$, a closed and bounded subset of $(\mathbb{K}^m)^n$, hence compact. The *Grassmannian* $\mathrm{Gr}_n(\mathbb{K}^m)$ is the set of n-dimensional subspaces of $\mathbb{K}^m$ with the quotient topology of

$$q\colon V_n(\mathbb{K}^m) \longrightarrow \mathrm{Gr}_n(\mathbb{K}^m), \qquad (x_1, \dots, x_n) \longmapsto \mathrm{span}(x_1, \dots, x_n)$$

and the *oriented Grassmannian* $\widetilde{\mathrm{Gr}}_n(\mathbb{R}^m)$ is the set of oriented n-dimensional subspaces of $\mathbb{R}^m$ with the quotient topology of the map $\tilde{q}$ sending a frame to its span, oriented by the frame. The infinite Grassmannians $\mathrm{Gr}_n(\mathbb{K}^\infty) = \bigcup_m \mathrm{Gr}_n(\mathbb{K}^m)$ and $\widetilde{\mathrm{Gr}}_n(\mathbb{R}^\infty) = \bigcup_m \widetilde{\mathrm{Gr}}_n(\mathbb{R}^m)$, the Stiefel manifold $V_n(\mathbb{K}^\infty) = \bigcup_m V_n(\mathbb{K}^m)$ and $\mathbb{K}^\infty$ carry the topology in which a set is closed exactly when it meets every finite stage in a closed set. Here an orientation of a plane is a class of ordered bases under changes of basis of positive determinant. The *tautological bundle* γ_n over $\mathrm{Gr}_n(\mathbb{K}^m)$, $m \le \infty$, has total space $E(\gamma_n) = \{(V, v) : v \in V\} \subseteq \mathrm{Gr}_n(\mathbb{K}^m) \times \mathbb{K}^m$, with the subspace topology of the product, projecting to the first coordinate, so that its fiber over V is V. The *oriented tautological bundle* $\tilde{\gamma}_n$ over $\widetilde{\mathrm{Gr}}_n(\mathbb{R}^m)$ is defined in the same way, with the plane V, carrying the orientation the point records, as its fiber over an oriented plane.

Two planes are close when they have close orthonormal bases. The finite Grassmannians are compact, as images of $V_n(\mathbb{K}^m)$, and Hausdorff. The matrix $P_V = \sum_i x_i x_i^*$ of the orthogonal projection onto $V = q(x)$ is a polynomial in the entries of x and depends only on V, so $V \mapsto P_V$ is continuous, and it is injective because V is the image of P_V. A space with a continuous injection into a Hausdorff space is Hausdorff. The oriented Grassmannian is compact Hausdorff in the same way, recording with P_V the $n \times n$ minors of the matrix with columns $x_1, \dots, x_n$: two orthonormal bases of V differ by a matrix $A \in O(n)$, which multiplies these minors by $\det A$, so the minors, not all zero, determine the orientation. For $m \le m'$ the inclusion $\mathrm{Gr}_n(\mathbb{K}^m) \subseteq \mathrm{Gr}_n(\mathbb{K}^{m'})$ is induced by the inclusion of frames, so it is a continuous injection of a compact space into a Hausdorff space, hence an embedding onto a closed subspace, and likewise for the oriented Grassmannians.

Definition 7.1.13: Schubert Symbol and Schubert Cell

A *Schubert symbol* is a sequence of integers $\sigma = (\sigma_1, \dots, \sigma_n)$ with $1 \le \sigma_1 < \cdots < \sigma_n$, and its *dimension* is $d(\sigma) = r\sum_{i=1}^{n}(\sigma_i - i)$. The *Schubert symbol* of an n-plane $V \subseteq \mathbb{K}^\infty$ has σ_i the least k with $\dim(V \cap \mathbb{K}^k) \ge i$, and the *Schubert cell* $e(\sigma) \subseteq \mathrm{Gr}_n(\mathbb{K}^\infty)$ is the set of n-planes with symbol σ. Write $\bar{e}'(\sigma)$ for the set of orthonormal frames $(x_1, \dots, x_n)$ with $x_i \in \mathbb{K}^{\sigma_i}$ and the σ_i-th coordinate of x_i real and nonnegative for every i, and $e'(\sigma)$ for its subset where these coordinates are all positive.

The number $\dim(V \cap \mathbb{K}^k)$ grows from 0 to n in steps of at most one, as $V \cap \mathbb{K}^{k-1}$ is the kernel of the k-th coordinate on $V \cap \mathbb{K}^k$, and the symbol lists where it grows. So $\dim(V \cap \mathbb{K}^k)$ is the number of i with $\sigma_i \le k$, every n-plane lies in exactly one Schubert cell, and $V \subseteq \mathbb{K}^m$ exactly when $\sigma_n \le m$.

Lemma 7.1.14: Schubert Cells are Open Cells

Let σ be a Schubert symbol, $m \ge \sigma_n$ finite, and $q\colon V_n(\mathbb{K}^m) \to \mathrm{Gr}_n(\mathbb{K}^m)$ as in definition 7.1.12.

1. There is a homeomorphism $\varphi_\sigma\colon D^{d(\sigma)} \to \bar{e}'(\sigma)$ carrying the open disk onto $e'(\sigma)$.
2. The map q restricts to a homeomorphism from $e'(\sigma)$ onto $e(\sigma)$.
3. The map q carries $\bar{e}'(\sigma) \setminus e'(\sigma)$ into the union of the Schubert cells $e(\tau)$ with $d(\tau) < d(\sigma)$.

***Proof*:**

Step 1: Triangular frames. Let $(x_1, \dots, x_n)$ be an orthonormal frame with $x_i \in \mathbb{K}^{\sigma_i}$ and with nonzero σ_i-th coordinate of x_i for every i, and let V be its span. Then $V \cap \mathbb{K}^k$ is spanned by the x_i with $\sigma_i \le k$. To see this, let $v = \sum_i c_i x_i \ne 0$ lie in $\mathbb{K}^k$, and let i_0 be the largest index with $c_{i_0} \ne 0$. The x_i with $i < i_0$ lie in $\mathbb{K}^{\sigma_{i_0}-1}$, so the σ_{i_0}-th coordinate of v is c_{i_0} times that of x_{i_0}, which is nonzero. Hence $\sigma_{i_0} \le k$, and every x_i occurring in v has $\sigma_i \le k$. So $\dim(V \cap \mathbb{K}^k)$ is the number of i with $\sigma_i \le k$, and $V \in e(\sigma)$.

Step 2: q maps $e'(\sigma)$ bijectively onto $e(\sigma)$. By Step 1, $q(e'(\sigma)) \subseteq e(\sigma)$. Let $V \in e(\sigma)$ and $W_i = V \cap \mathbb{K}^{\sigma_i}$, of dimension i. As $\sigma_{i-1} \le \sigma_i - 1$, the space $V \cap \mathbb{K}^{\sigma_i - 1}$ contains W_{i-1} and has dimension $i - 1$, so it equals W_{i-1}. Let L_i be the orthogonal complement of W_{i-1} in W_i, a line. If $L_i \subseteq \mathbb{K}^{\sigma_i - 1}$ then $L_i \subseteq W_{i-1}$, which is impossible, so the σ_i-th coordinate is nonzero on $L_i \setminus 0$. The unit vectors of L_i are the multiples of one of them by the scalars of absolute value one, so exactly one of them, x_i, has positive real σ_i-th coordinate. The frame $(x_1, \dots, x_n)$ is orthonormal, lies in $e'(\sigma)$ and spans V. Conversely, if $(y_1, \dots, y_n) \in e'(\sigma)$ spans V, then Step 1 gives $\mathrm{span}(y_1, \dots, y_i) = W_i$, so y_i is a unit vector of L_i with positive real σ_i-th coordinate, and $y_i = x_i$.

Step 3: Rotations. For unit vectors $u, v \in \mathbb{K}^m$ with $c = \langle u, v\rangle$ real and $c > -1$, put

$$T(u,v)x = x - \frac{\langle x, u+v\rangle}{1+c}(u+v) + 2\langle x, u\rangle v$$

This is $\mathbb{K}$-linear in x and continuous in (u, v, x). It fixes every vector orthogonal to u and v, sends u to v, and sends v to $-u + 2cv$. If u and v are independent, the pairs (u, v) and $(v, -u + 2cv)$ have the same inner products, c being real, so $T(u, v)$ preserves inner products on $\mathrm{span}(u, v)$, which it maps to itself, and it is unitary. If they are dependent, then $v = u$,

as $c > -1$, and $T(u,u)$ is the identity. The map $T(v,u)$ sends v to u and u to $-v + 2cu$, so $T(v,u)T(u,v)$ fixes u, v and their orthogonal complement, and $T(v,u) = T(u,v)^{-1}$. If $u, v \in \mathbb{K}^k$, then $T(u,v)x - x \in \mathbb{K}^k$ for every x.

Step 4: Part (1). Induct on n for the claim that $\bar{e}'(\sigma)$ is homeomorphic to a product of closed disks of dimensions $r(\sigma_i - i)$ by a homeomorphism carrying $e'(\sigma)$ onto the product of their interiors, the case $n = 0$ being a point. Suppose the claim for $\sigma = (\sigma_1, \dots, \sigma_n)$, and let $\sigma' = (\sigma_1, \dots, \sigma_{n+1})$. For $x \in \bar{e}'(\sigma)$ put $b_i = e_{\sigma_i}$ and $T_x = T(b_n, x_n) \circ \cdots \circ T(b_1, x_1)$, defined because $\langle b_i, x_i \rangle$ is the real nonnegative σ_i-th coordinate of x_i. Then $T_x b_i = x_i$: the factors $T(b_j, x_j)$ with $j < i$ fix b_i, which is orthogonal to b_j and to $x_j \in \mathbb{K}^{\sigma_j}$, the factor $T(b_i, x_i)$ sends b_i to x_i, and the factors with $j > i$ fix x_i, which is orthogonal to x_j and to b_j. Let H be the set of unit vectors $u \in \mathbb{K}^{\sigma_{n+1}}$ orthogonal to $b_1, \dots, b_n$ with σ_{n+1}-th coordinate real and nonnegative. These u form a closed hemisphere in the unit sphere of a real vector space of dimension $r(\sigma_{n+1} - n - 1) + 1$, so orthogonal projection onto the equatorial hyperplane is a homeomorphism from H onto $D^{r(\sigma_{n+1}-n-1)}$, carrying the vectors with positive last coordinate onto the open disk. Define

$$f \colon \bar{e}'(\sigma) \times H \longrightarrow \bar{e}'(\sigma'), \qquad f(x,u) = (x_1, \dots, x_n, T_x u)$$

The vector $T_x u$ is a unit vector orthogonal to every $x_i = T_x b_i$, since T_x is unitary. The b_i and x_i lie in $\mathbb{K}^{\sigma_n}$, so $T_x u - u \in \mathbb{K}^{\sigma_n}$ by Step 3, and $T_x u$ lies in $\mathbb{K}^{\sigma_{n+1}}$ with the same σ_{n+1}-th coordinate as u. So f is well defined and continuous. For $y \in \bar{e}'(\sigma')$ write $\bar{y} = (y_1, \dots, y_n)$. The map $y \mapsto \big(\bar{y},\, T_{\bar{y}}^{-1} y_{n+1}\big)$, with $T_{\bar{y}}^{-1} = T(y_1, b_1) \circ \cdots \circ T(y_n, b_n)$, is a continuous inverse of f: the vector $T_{\bar{y}}^{-1} y_{n+1}$ is a unit vector with $\langle T_{\bar{y}}^{-1} y_{n+1}, b_i \rangle = \langle y_{n+1}, y_i \rangle = 0$, and it lies in $\mathbb{K}^{\sigma_{n+1}}$ with the same σ_{n+1}-th coordinate as y_{n+1}. As f preserves that coordinate, it carries the product of $e'(\sigma)$ with the open hemisphere onto $e'(\sigma')$, and the claim holds for σ'. Finally $D^a \times D^b$ is the unit ball of the norm $\max(|y|, |z|)$ on $\mathbb{R}^{a+b}$, and rescaling each ray from the origin carries it onto D^{a+b} and its interior onto the open disk. This proves (1).

Step 5: Parts (2) and (3). Let $y \in \bar{e}'(\sigma) \setminus e'(\sigma)$, put $V = q(y)$, and let i be an index for which the σ_i-th coordinate of y_i vanishes. Since $y_1, \dots, y_j \in \mathbb{K}^{\sigma_j}$, $\dim(V \cap \mathbb{K}^{\sigma_j}) \geq j$ for every j, and since $y_1, \dots, y_i \in \mathbb{K}^{\sigma_i - 1}$, $\dim(V \cap \mathbb{K}^{\sigma_i - 1}) \geq i$. So the symbol τ of V has $\tau_j \leq \sigma_j$ for every j and $\tau_i < \sigma_i$, whence $d(\tau) < d(\sigma)$, which is (3). For (2), $\bar{e}'(\sigma)$ is compact by (1) and $\mathrm{Gr}_n(\mathbb{K}^m)$ is Hausdorff, so q carries closed subsets of $\bar{e}'(\sigma)$ to closed sets. Let A be closed in $e'(\sigma)$ and $\bar{A}$ its closure in $\bar{e}'(\sigma)$, so that $\bar{A} \cap e'(\sigma) = A$. By (3), q carries $\bar{A} \setminus A$ outside $e(\sigma)$, so $q(A) = q(\bar{A}) \cap e(\sigma)$ is closed in $e(\sigma)$. The continuous bijection of Step 2 is therefore closed, hence a homeomorphism, completing the proof.

The cell $e(\sigma)$ is parametrized by its frames in $e'(\sigma)$, and its frontier consists of planes that meet the flag $\mathbb{K}^1 \subseteq \mathbb{K}^2 \subseteq \cdots$ earlier than σ prescribes.

Lemma 7.1.15: Increasing Unions of CW Complexes

Let $X_0 \subseteq X_1 \subseteq \cdots$ be CW complexes such that each X_k is a closed subspace of X_{k+1} and a union of open cells of X_{k+1}, the characteristic maps of X_k being characteristic maps of X_{k+1}. Give $X = \bigcup_k X_k$ the topology in which a set is closed exactly when it meets every X_k in a closed set. Then X is a CW complex with the cells and characteristic maps of the X_k, each X_k is a subcomplex of X carrying the subspace topology, and every compact subset of X lies in some X_k.

***Proof*:**
Write X^p and X_k^p for the unions of the open cells of dimension at most p. An open cell of X_j meeting X_k lies in X_k, the open cells being disjoint, so $X^p \cap X_k = X_k^p$.

Step 1: Each X_k is a closed subspace of X. Let B be closed in X_k. For $j \geq k$ the set B is closed in X_j, of which X_k is a closed subspace, and for $j < k$ the set $B \cap X_j$ is closed in X_j, a subspace of X_k. So B is closed in X, and the topology of X_k is its subspace topology. Each X^p is closed in X, meeting every X_k in the closed set X_k^p.

Step 2: The skeleta. Let $A \subseteq X^p$ meet X^{p-1} in a closed subset of X^{p-1} and have closed preimage under the characteristic map of every p-cell. By Step 1, $A \cap X_k^{p-1}$ is closed in X_k^{p-1}, and the p-cells of X_k are p-cells of X with the same characteristic maps, so $A \cap X_k$ is closed in X_k^p, which carries the quotient topology of definition 1.2.1, and hence in X_k. So A is closed in X. The characteristic maps are continuous into X by Step 1, so X^p carries the quotient topology from X^{p-1} and the p-disks. Each characteristic map is injective on the open disk, and the open p-cells are disjoint and miss X^{p-1}, so the only identifications are those along the attaching maps, and X^p is X^{p-1} with the p-cells attached. For $p = 0$ the same argument shows that every subset of X^0 is closed, so X^0 is discrete.

Step 3: Conclusion. A set meeting every X^p in a closed set meets every X_k^p in a closed set, hence meets every X_k, which carries the weak topology of its skeleta, in a closed set, and so it is closed in X. Thus X is a CW complex, and each X_k, closed and a union of cells, is a subcomplex. A compact subset of X meets only finitely many open cells (lemma 1.2.4), each lying in some X_k, so it lies in one X_k, completing the proof.

The lemma asks only for the colimit topology, which is the topology of every infinite Grassmannian and of the spaces BO and BU below.

On such a union every compact set, in particular the image of every singular simplex, lies in a finite stage. The cohomology of the union is then computed from the stages wherever the restrictions between consecutive stages are bijective, which is how the cohomology of BU and BO is obtained from that of the $BU(n)$ and $BO(n)$.

Lemma 7.1.16: Cohomology of an Increasing Union

Let (X, A) be a pair of spaces and $X_0 \subseteq X_1 \subseteq \cdots$ subspaces of X such that every compact subset of X lies in some X_m, and put $A_m = A \cap X_m$. Let G be an abelian group and j and $m_0 \geq 0$ integers, and suppose that for every $m \geq m_0$ the restriction $H^j(X_{m+1}, A_{m+1}; G) \to H^j(X_m, A_m; G)$ is bijective and the restriction $H^{j-1}(X_{m+1}, A_{m+1}; G) \to H^{j-1}(X_m, A_m; G)$ is surjective. Then the restriction $H^j(X, A; G) \to H^j(X_{m_0}, A_{m_0}; G)$ is bijective.

***Proof*:**
Write $C_m^k = C^k(X_m, A_m; G)$. As in definition 4.1.15, an element of $C^k(X, A; G)$ is a function on the singular k-simplices of X vanishing on those with image in A, and restriction to X_m keeps its values on the simplices with image in X_m. A simplex of X_m with image in A has image in A_m. A singular simplex has compact image, so it is a simplex of some X_m and of every $X_{m'}$ with $m' \geq m$. Restriction commutes with δ. A cochain $b \in C_m^k$ extends by zero to $\hat{b} \in C_{m+1}^k$, with value 0 on the simplices of X_{m+1} not lying in X_m, and $\hat{b}|_{X_m} = b$. Cochains $c_m \in C_m^k$ for $m \geq m_0$ with $c_{m+1}|_{X_m} = c_m$ define $c \in C^k(X, A; G)$ by $c(\sigma) = c_m(\sigma)$ for any $m \geq m_0$ such that

σ lies in X_m, and δc is the cochain defined in the same way by the cochains δc_m, since the faces of a simplex of X_m lie in X_m.

Step 1: Surjectivity. Let $c_{m_0} \in C^j_{m_0}$ be a cocycle, and suppose cocycles $c_{m_0}, \dots, c_m$ have been chosen with $c_{i+1}|_{X_i} = c_i$. Surjectivity in degree j gives a cocycle $c' \in C^j_{m+1}$ with $c'|_{X_m} = c_m + \delta b$ for some $b \in C^{j-1}_m$, and $c_{m+1} = c' - \delta\hat{b}$ is a cocycle with $c_{m+1}|_{X_m} = c_m$. The cocycles so chosen define a cocycle $c \in C^j(X, A; G)$ with $c|_{X_{m_0}} = c_{m_0}$, so the class of c_{m_0} lies in the image.

Step 2: Injectivity. Let $c \in C^j(X, A; G)$ be a cocycle with $c|_{X_{m_0}} = \delta b_{m_0}$ for some $b_{m_0} \in C^{j-1}_{m_0}$, and suppose $b_{m_0}, \dots, b_m$ have been chosen with $\delta b_i = c|_{X_i}$ and $b_{i+1}|_{X_i} = b_i$. The class of $c|_{X_{m+1}}$ restricts to the class of $c|_{X_m} = \delta b_m$, which is zero, so injectivity in degree j gives $b' \in C^{j-1}_{m+1}$ with $\delta b' = c|_{X_{m+1}}$. Then $z = b'|_{X_m} - b_m$ is a cocycle, and surjectivity in degree $j-1$ gives a cocycle $z' \in C^{j-1}_{m+1}$ and $e \in C^{j-2}_m$ with $z'|_{X_m} = z + \delta e$. Put $b_{m+1} = b' - z' + \delta\hat{e}$. Then $\delta b_{m+1} = \delta b' = c|_{X_{m+1}}$ and

$$b_{m+1}|_{X_m} = b'|_{X_m} - z - \delta e + \delta e = b_m$$

The cochains b_m define $b \in C^{j-1}(X, A; G)$ with $\delta b = c$, so the class of c is zero, completing the proof.

Theorem 7.1.17: The Schubert Cell Structure

Let $0 \le n \le m \le \infty$, and for each Schubert symbol σ fix a homeomorphism φ_σ as in lemma 7.1.14(1).

1. The Grassmannian $\mathrm{Gr}_n(\mathbb{K}^m)$ is a CW complex whose cells are the Schubert cells $e(\sigma)$ with $\sigma_n \le m$, with characteristic maps $q \circ \varphi_\sigma$, and $\mathrm{Gr}_n(\mathbb{K}^{m'})$ is a subcomplex for $n \le m' \le m$.
2. The Grassmannian $\mathrm{Gr}_n(\mathbb{K}^\infty)$ has finitely many cells in each dimension. In particular $BO(n) = \mathrm{Gr}_n(\mathbb{R}^\infty)$ and $BU(n) = \mathrm{Gr}_n(\mathbb{C}^\infty)$ are CW complexes of finite type, and they are paracompact Hausdorff (theorem 7.1.7).
3. The oriented Grassmannian $\widetilde{\mathrm{Gr}}_n(\mathbb{R}^m)$ is a CW complex with two cells of dimension $d(\sigma)$ lying over each Schubert cell $e(\sigma)$ of $\mathrm{Gr}_n(\mathbb{R}^m)$, so $\widetilde{\mathrm{Gr}}_n(\mathbb{R}^\infty)$ also has finitely many cells in each dimension.

***Proof*:**

Step 1: Finite m. Let X^k be the union of the cells $e(\sigma)$ with $\sigma_n \le m$ and $d(\sigma) \le k$. By lemma 7.1.14(3), $q(\bar{e}'(\sigma))$ is $e(\sigma)$ together with points of cells of smaller dimension, so X^k is the finite union of the compact sets $q(\bar{e}'(\sigma))$ with $d(\sigma) \le k$, hence compact and closed. The map from $X^{k-1} \sqcup \bigsqcup_{d(\sigma)=k} D^k$ to X^k given by the inclusion and the maps $q \circ \varphi_\sigma$ is a continuous surjection from a compact space to a Hausdorff space, hence a quotient map. It is injective on X^{k-1} and on each open disk, it carries the open disks onto the disjoint cells $e(\sigma)$, which miss X^{k-1}, and it carries their boundary spheres into X^{k-1} (lemma 7.1.14). So X^k is X^{k-1} with k-cells attached, X^0 is a finite Hausdorff space, hence discrete, and $\mathrm{Gr}_n(\mathbb{K}^m)$ is a finite CW complex in the sense of definition 1.2.1. The frames of $\bar{e}'(\sigma)$ lie in $\mathbb{K}^{\sigma_n}$, so the characteristic map $q \circ \varphi_\sigma$ does not depend on $m \ge \sigma_n$. Hence for $m' \le m$ the closed subspace $\mathrm{Gr}_n(\mathbb{K}^{m'})$, the union of the cells with $\sigma_n \le m'$, is a subcomplex.

Step 2: Infinite m. By Step 1 and the closed embeddings of finite stages shown after definition 7.1.12, lemma 7.1.15 applies to the finite Grassmannians and proves (1) for $m = \infty$. A cell of dimension d has $r(\sigma_n - n) \leq d$, and only finitely many Schubert symbols satisfy $\sigma_n \leq n + d/r$, which proves (2).

Step 3: Oriented Grassmannians. Let $m < \infty$, let $\pi\colon \widetilde{\mathrm{Gr}}_n(\mathbb{R}^m) \to \mathrm{Gr}_n(\mathbb{R}^m)$ forget the orientation, so that $\pi \circ \tilde{q} = q$, and let ϵ negate the first vector of a frame, which preserves the span and reverses the orientation. A plane $V \in e(\sigma)$ carries two orientations, that of its frame in $e'(\sigma)$ (Step 2 of the proof of lemma 7.1.14) and that of ϵ applied to this frame. So $\pi^{-1}(e(\sigma))$ is the disjoint union of $\tilde{q}(e'(\sigma))$ and $\tilde{q}(\epsilon\, e'(\sigma))$, the maps $\tilde{q} \circ \varphi_\sigma$ and $\tilde{q} \circ \epsilon \circ \varphi_\sigma$ are injective on the open disk, and by lemma 7.1.14(3) they carry the boundary sphere into π^{-1} of the cells of smaller dimension, which is the union of the cells lying over them. The space $\widetilde{\mathrm{Gr}}_n(\mathbb{R}^m)$ is compact Hausdorff, so the argument of Step 1 applies to these cells, and lemma 7.1.15 handles $m = \infty$, completing the proof.

For $n = 1$ the cells $e(\sigma_1)$ have dimensions $0, r, 2r, \ldots$, one in each, which is the cell structure of $\mathbb{R}\mathrm{P}^\infty$ and $\mathbb{C}\mathrm{P}^\infty$.

Lemma 7.1.18: The Tautological and Frame Bundles

Let $0 \leq n \leq m \leq \infty$, let G be $O(n)$ for $\mathbb{K} = \mathbb{R}$ and $U(n)$ for $\mathbb{K} = \mathbb{C}$, acting on $V_n(\mathbb{K}^m)$ from the right by $(x \cdot g)_k = \sum_i g_{ik} x_i$, and write E_M for the part of $E(\gamma_n)$ over $\mathrm{Gr}_n(\mathbb{K}^M)$.

1. For an n-plane $X \subseteq \mathbb{K}^m$ with orthogonal projection p_X onto X, the set U_X of n-planes Y with $Y \cap X^\perp = 0$ is open, and $h_X(Y, x) = (Y, y)$, where $y \in Y$ is the vector with $p_X(y) = x$, is a homeomorphism $U_X \times X \to E(\gamma_n)|_{U_X}$, linear on each fiber. In particular γ_n is a vector bundle of rank n.
2. The map $q\colon V_n(\mathbb{K}^m) \to \mathrm{Gr}_n(\mathbb{K}^m)$ is a principal G-bundle, the bundle of orthonormal frames of γ_n. The map $\pi\colon \widetilde{\mathrm{Gr}}_n(\mathbb{R}^m) \to \mathrm{Gr}_n(\mathbb{R}^m)$ forgetting the orientation is a two-sheeted covering, and $\tilde{q}\colon V_n(\mathbb{R}^m) \to \widetilde{\mathrm{Gr}}_n(\mathbb{R}^m)$ is a principal $SO(n)$-bundle.
3. Let $m = \infty$. A subset of $E(\gamma_n)$, or of $E(\gamma_n) \times I$, is open exactly when it meets every E_M, or every $E_M \times I$, in a relatively open set. Every compact subset of $V_n(\mathbb{K}^\infty)$ lies in some $V_n(\mathbb{K}^M)$.
4. The space $V_n(\mathbb{K}^\infty)$ is weakly contractible. Hence $\mathrm{Gr}_n(\mathbb{R}^\infty)$, $\mathrm{Gr}_n(\mathbb{C}^\infty)$ and $\widetilde{\mathrm{Gr}}_n(\mathbb{R}^\infty)$ are classifying spaces $BO(n)$, $BU(n)$ and $BSO(n)$ in the sense of definition 7.1.3.

***Proof*:**
Four kinds of continuous surjection are quotient maps below: the restriction of a quotient map to an open saturated subset, the product of a quotient map with the identity of a locally compact Hausdorff space [16, Products of Quotient Maps], a continuous surjection from a compact space to a Hausdorff space, and, for $m = \infty$, the map from the disjoint union of the finite stages to $\mathrm{Gr}_n(\mathbb{K}^\infty)$, $\widetilde{\mathrm{Gr}}_n(\mathbb{R}^\infty)$ or $V_n(\mathbb{K}^\infty)$, by the definition of their topologies. The maps q and $\tilde{q}$ are quotient maps for $m = \infty$ as well, since a set whose preimage is open meets every finite stage in an open set. An open subset W of one of these unions carries the topology of its intersections with the stages: a subset of W meeting every stage of W in an open set meets every stage of the union in an open set.

Step 1: Part (1). Fix an orthonormal basis $x_1, \dots, x_n$ of X, so that $p_X(v) = \sum_i \langle v, x_i \rangle x_i$, which is continuous on each $\mathbb{K}^M$ and so on $\mathbb{K}^m$. For a frame $y \in V_n(\mathbb{K}^M)$ let $A(y) = (\langle y_j, x_i \rangle)_{i,j}$, the matrix of p_X restricted to $\operatorname{span}(y)$. The kernel of that restriction is $\operatorname{span}(y) \cap X^\perp$, so $q(y) \in U_X$ exactly when $\det A(y) \neq 0$, an open condition on y. So U_X meets every $\operatorname{Gr}_n(\mathbb{K}^M)$ in an open set and is open. For $q(y) \in U_X$ and $x = \sum_i a_i x_i$, the vector of $q(y)$ projecting to x is $\sum_j c_j y_j$ with $c = A(y)^{-1}a$, continuous in (y, x) and depending on y only through $q(y)$. By the quotient maps above, h_X is continuous on each $(U_X \cap \operatorname{Gr}_n(\mathbb{K}^M)) \times X$, with values in E_M, and then on $U_X \times X$. It is a bijection onto $E(\gamma_n)|_{U_X}$, linear on each fiber, with continuous inverse $(Y, y) \mapsto (Y, p_X(y))$. The set $E(\gamma_n)|_{U_X}$ is open, the preimage of U_X under the projection, so γ_n is locally trivial. This is the argument of [29, Lemmas 5.2 and 5.4].

Step 2: Part (2). The action of G is continuous, preserves q, and is free and transitive on each fiber, two orthonormal bases of one plane differing by a unique matrix of G. Let $s_X(Y)$ be the Gram-Schmidt orthonormalization of the second coordinates of $h_X(Y, x_1), \dots, h_X(Y, x_n)$, a frame of Y depending continuously on $Y \in U_X$. The map $U_X \times G \to q^{-1}(U_X)$, $(Y, g) \mapsto s_X(Y) \cdot g$, is a G-equivariant bijection over U_X, continuous on the stages and so continuous, G being compact. Its inverse $x \mapsto \big(q(x), g(x)\big)$, $g(x)_{ik} = \langle x_k, s_X(q(x))_i \rangle$, is continuous on every stage of the open set $q^{-1}(U_X)$. So q is a principal G-bundle, and its fiber over Y is the set of orthonormal bases of Y. For $\tilde{q}$, the frames $x \in q^{-1}(U_X)$ with $\det g(x) > 0$, those orienting $q(x)$ as $s_X(q(x))$ does, form an open set saturated for $\tilde{q}$, and so do those with $\det g(x) < 0$. Their images $\widetilde{U}_X^+$ and $\widetilde{U}_X^-$ are open, disjoint and cover $\pi^{-1}(U_X)$, and π maps each homeomorphically onto U_X, with inverses $\tilde{q} \circ s_X$ and $\tilde{q} \circ \epsilon \circ s_X$, where ϵ negates the first vector of a frame. So π is a two-sheeted covering, and over $\widetilde{U}_X^\pm$ the sections $s_X \circ \pi$ and $\epsilon \circ s_X \circ \pi$ trivialize $\tilde{q}$ as before, with $SO(n)$ in place of G.

Step 3: Part (3). Let $W \subseteq E(\gamma_n)$ meet every E_M in a relatively open set. The map h_X carries $(U_X \cap \operatorname{Gr}_n(\mathbb{K}^M)) \times X$ continuously into E_M, so $h_X^{-1}(W)$ meets each of these sets in an open set and is open in $U_X \times X$. Hence $W \cap E(\gamma_n)|_{U_X}$ is open, and these sets cover W. The same argument with $h_X \times \mathrm{id}_I$ treats $E(\gamma_n) \times I$. For the second claim, a point of $V_n(\mathbb{K}^\infty)$ meets each compact Hausdorff stage in at most one point, so points are closed. If a compact set C lies in no stage, choose $y_k \in C \setminus V_n(\mathbb{K}^k)$ for each k. Then $y_k \in V_n(\mathbb{K}^M)$ forces $k < M$, so every subset of $\{y_k\}$ meets every stage in a finite, hence closed, set and is closed. This makes $\{y_k\}$ an infinite closed subset of C with the discrete topology, which is impossible in a compact space.

Step 4: Part (4). Each stage $V_n(\mathbb{K}^M)$ is a closed subspace of $V_n(\mathbb{K}^\infty)$, by the argument of Step 1 of the proof of lemma 7.1.15. Let $f\colon S^i \to V_n(\mathbb{K}^\infty)$, $i \geq 0$, be a map. By (3) it is a map into some $V_n(\mathbb{K}^M)$. For $t \in [0, 1]$ the vectors $(1-t)f_k(\xi) + te_{M+k}$, $k = 1, \dots, n$, are independent, by their components along $e_{M+1}, \dots, e_{M+n}$ when $t > 0$. Their Gram-Schmidt orthonormalization is a homotopy in $V_n(\mathbb{K}^{M+n})$ from f to the constant frame $(e_{M+1}, \dots, e_{M+n})$. The case $i = 0$ shows that $V_n(\mathbb{K}^\infty)$ is path-connected. For $i \geq 1$ and f based, the class of f lies in the orbit of 0 under the action of π_1, which is $\{0\}$ (proposition 5.1.15), so all homotopy groups vanish at every basepoint. By (2), q and $\tilde{q}$ are principal bundles with weakly contractible total space, which is definition 7.1.3, completing the proof.

By part (4) the Grassmannians are classifying spaces in the sense of definition 7.1.3, so theorem 7.1.10 classifies bundles over a CW complex by maps into them. The oriented Grassmannian $\widetilde{\operatorname{Gr}}_n(\mathbb{R}^\infty)$, with the oriented tautological bundle $\tilde{\gamma}_n$, is the model of $BSO(n)$ in this chapter.

Proposition 7.1.19: Stabilization of Grassmannians

Let $s(z_1, z_2, \dots) = (0, z_1, z_2, \dots)$ be the shift of $\mathbb{K}^\infty$, and for $n \geq 0$ let

$$i_{n+1} \colon \mathrm{Gr}_n(\mathbb{K}^\infty) \longrightarrow \mathrm{Gr}_{n+1}(\mathbb{K}^\infty), \qquad i_{n+1}(V) = \mathbb{K}e_1 \oplus s(V)$$

be the *stabilization*.

1. The map i_{n+1} is a homeomorphism onto the closed subspace of planes containing e_1, and $i_{n+1}(\mathbb{K}^n) = \mathbb{K}^{n+1}$.
2. It carries $e(\sigma)$ onto $e(1, \sigma_1 + 1, \dots, \sigma_n + 1)$, a cell of the same dimension, and $i_{n+1} \circ q \circ \varphi_\sigma$ is $q \circ \varphi$ for a homeomorphism φ as in lemma 7.1.14(1) for that cell. Every Schubert cell of $\mathrm{Gr}_{n+1}(\mathbb{K}^\infty)$ outside the image of i_{n+1} has dimension at least $r(n+1)$.
3. The map $(V, \lambda, w) \mapsto \big(V, \lambda e_1 + s(w)\big)$ is an isomorphism of vector bundles $\underline{\mathbb{K}} \oplus \gamma_n \to i_{n+1}^* \gamma_{n+1}$ over $\mathrm{Gr}_n(\mathbb{K}^\infty)$, where $\underline{\mathbb{K}}$ is the trivial line bundle.

***Proof*:**

Part (1). On frames, i_{n+1} is $(x_1, \dots, x_n) \mapsto (e_1, sx_1, \dots, sx_n)$, continuous from $V_n(\mathbb{K}^M)$ to $V_{n+1}(\mathbb{K}^{M+1})$, so i_{n+1} is continuous on every $\mathrm{Gr}_n(\mathbb{K}^M)$ and hence on $\mathrm{Gr}_n(\mathbb{K}^\infty)$. The shift is an isometry of $\mathbb{K}^\infty$ onto $e_1^\perp$, and a plane W contains e_1 exactly when $W = \mathbb{K}e_1 \oplus (W \cap e_1^\perp)$, so i_{n+1} is a bijection onto the planes containing e_1. A plane $i_{n+1}(V)$ lies in $\mathbb{K}^{M+1}$ exactly when $V \subseteq \mathbb{K}^M$. So for a closed set $C \subseteq \mathrm{Gr}_n(\mathbb{K}^\infty)$ and every M, the set $i_{n+1}(C) \cap \mathrm{Gr}_{n+1}(\mathbb{K}^{M+1}) = i_{n+1}\big(C \cap \mathrm{Gr}_n(\mathbb{K}^M)\big)$ is a continuous image of a compact set in a Hausdorff space, hence closed. Thus $i_{n+1}(C)$ is closed, and i_{n+1} is a closed embedding whose image, the case $C = \mathrm{Gr}_n(\mathbb{K}^\infty)$, is the set of planes containing e_1. Finally $i_{n+1}(\mathbb{K}^n) = \mathbb{K}e_1 \oplus \mathrm{span}(e_2, \dots, e_{n+1}) = \mathbb{K}^{n+1}$.

Part (2). For $W = i_{n+1}(V)$ and $k \geq 1$, $W \cap \mathbb{K}^k = \mathbb{K}e_1 \oplus s(V \cap \mathbb{K}^{k-1})$, so $\dim(W \cap \mathbb{K}^k) = 1 + \dim(V \cap \mathbb{K}^{k-1})$, and the symbol of W is $(1, \sigma_1 + 1, \dots, \sigma_n + 1)$, of dimension $r\sum_i\big((\sigma_i + 1) - (i+1)\big) = d(\sigma)$. A plane whose symbol begins with 1 contains e_1, so i_{n+1} maps $e(\sigma)$ onto this cell. The map $\iota(x) = (e_1, sx_1, \dots, sx_n)$ is a homeomorphism from $\bar{e}'(\sigma)$ onto $\bar{e}'(1, \sigma_1 + 1, \dots, \sigma_n + 1)$ carrying $e'(\sigma)$ onto $e'(1, \sigma_1 + 1, \dots, \sigma_n + 1)$, because a frame of the target has first vector a unit vector of $\mathbb{K}^1$ with real nonnegative coordinate, which is e_1, and remaining vectors orthogonal to e_1, hence of the form sx_i, while s preserves inner products and moves the σ_i-th coordinate to position $\sigma_i + 1$. So $i_{n+1} \circ q \circ \varphi_\sigma = q \circ \iota \circ \varphi_\sigma$, with $\varphi = \iota \circ \varphi_\sigma$ as required. A symbol τ with $\tau_1 \geq 2$ has $\tau_i \geq i + 1$ for every i, so $d(\tau) \geq r(n+1)$.

Part (3). The fiber of $i_{n+1}^* \gamma_{n+1}$ over V is $\mathbb{K}e_1 \oplus s(V)$, so the map is a linear isomorphism on every fiber. It is continuous, so it is an isomorphism of vector bundles by lemma 5.2.10, completing the proof.

Stabilization adds the line $\mathbb{K}e_1$ to every plane, and on bundles it adds a trivial summand.

Definition 7.1.20: The Spaces BO and BU

Give $BO(n) = \mathrm{Gr}_n(\mathbb{R}^\infty)$ and $BU(n) = \mathrm{Gr}_n(\mathbb{C}^\infty)$ the basepoints $\mathbb{R}^n$ and $\mathbb{C}^n$. The space BO, respectively BU, is the colimit of

$$BO(0) \xrightarrow{i_1} BO(1) \xrightarrow{i_2} BO(2) \xrightarrow{i_3} \cdots,$$
$$\text{respectively} \quad BU(0) \xrightarrow{i_1} BU(1) \xrightarrow{i_2} BU(2) \xrightarrow{i_3} \cdots$$

along the stabilizations of proposition 7.1.19: the union of the $BO(n)$, each identified with its image under the later stabilizations, in which a set is closed exactly when it meets every $BO(n)$ in a closed set, and likewise for BU. The stabilizations identify the basepoints with one another (proposition 7.1.19(1)), and they give the basepoints of BO and BU.

By proposition 7.1.19(3), passing from $BU(n)$ to $BU(n+1)$ adds a trivial line to the bundle a map classifies, so a map into BU records a bundle up to trivial summands, which is the equivalence K-theory measures.

Corollary 7.1.21: BO and BU are CW Complexes

The spaces BO and BU are CW complexes in which every $BO(n)$, respectively $BU(n)$, is a subcomplex carrying the subspace topology. Every compact subset lies in some $BO(N)$, respectively $BU(N)$, and every cell outside $BO(n)$, respectively $BU(n)$, has dimension at least $n+1$, respectively $2(n+1)$.

***Proof*:**
Choose the homeomorphisms φ_τ of theorem 7.1.17 inductively in the length of τ, taking $\varphi_\tau = \iota \circ \varphi_\sigma$ for $\tau = (1, \sigma_1 + 1, \ldots, \sigma_n + 1)$, with ι as in the proof of proposition 7.1.19(2). By proposition 7.1.19(1),(2), each $BO(n)$ is then a closed subspace of $BO(n+1)$ and a union of its open cells, with characteristic maps among those of $BO(n+1)$, so lemma 7.1.15 gives the first two claims. A cell outside $BO(n)$ first appears in some $BO(k)$ with $k \geq n+1$, outside the image of $BO(k-1)$, so its dimension is at least k by proposition 7.1.19(2). For BU the same argument gives at least $2k$, completing the proof.

In particular a map from a compact space into BU is a map into a single $BU(N)$, which is how vector bundles over finite complexes enter K-theory.

7.1.2 Vector Bundles over a Paracompact Base

Theorem 7.1.10 classifies principal G-bundles over a CW complex, or a paracompact Hausdorff space of CW homotopy type, and its proof extends a map over the skeleta one at a time, a step with no substitute for a general topological group. For *vector* bundles the classification admits an entirely different proof, one that never mentions a cell. A vector bundle over a paracompact Hausdorff base carries a map to $\mathbb{R}^\infty$ that is linear and injective on every fiber, built from a partition of unity, and such a map is the same data as a map to a Grassmannian. This subsection carries out that route, because the multiplicative theory of section 7.4 (naturality, the splitting principle, the Whitney sum formula) is applied to flag bundles, and a flag bundle carries no CW structure that this book

constructs. Without the paracompact classification the theory would be defined only on spaces it is not applied to.

Two elementary facts about vector bundles come first. Both serve to split a subbundle off, which is how the flag bundle is built one stage at a time.

Lemma 7.1.22: Inner Products on a Vector Bundle

Let $E \to X$ be a real vector bundle over a paracompact Hausdorff base. Then E admits a *Euclidean inner product*: a continuous map $E \oplus E \to \mathbb{R}$ restricting on every fiber to a positive-definite symmetric bilinear form. A complex vector bundle over such a base admits a *Hermitian inner product*, restricting on every fiber to a positive-definite conjugate-symmetric sesquilinear form.

***Proof*:**
Let $\{U_\alpha\}$ be an open cover of X by sets over which E is trivial, and for each α let $h_\alpha \colon p^{-1}(U_\alpha) \to \mathbb{R}^n$ be the fiber coordinate of a trivialization. Pulling back the standard inner product of $\mathbb{R}^n$ gives a continuous form $\langle v, w\rangle_\alpha = h_\alpha(v) \cdot h_\alpha(w)$ over U_α, symmetric and positive-definite on each fiber there. Let $\{\varphi_\alpha\}$ be a partition of unity subordinate to the cover, which exists by [16, Existence of Subordinate Partitions of Unity], and set

$$\langle v, w\rangle = \sum_\alpha \varphi_\alpha\big(p(v)\big)\, \langle v, w\rangle_\alpha$$

reading a term as 0 off $p^{-1}(\mathrm{supp}\varphi_\alpha)$. The supports are locally finite, so near any point of X the sum has finitely many nonzero terms and is continuous there. Continuity is local, so the form is continuous. On the fiber over x it is a combination of symmetric bilinear forms with nonnegative coefficients, hence symmetric and bilinear. For positive-definiteness let $v \neq 0$ lie in that fiber: each α with $\varphi_\alpha(x) > 0$ has $x \in U_\alpha$ and contributes $\varphi_\alpha(x)\,|h_\alpha(v)|^2 > 0$, no term is negative, and at least one such α exists because the φ_α sum to 1. Hence $\langle v, v\rangle > 0$. The complex case is the same construction with $h_\alpha \colon p^{-1}(U_\alpha) \to \mathbb{C}^n$ and the standard Hermitian form: the multipliers φ_α are real, so a nonnegative combination of conjugate-symmetric sesquilinear forms is again one, and the positivity argument is unchanged, as we sought to show.

Proposition 7.1.23: Subbundles Have Complements

Let $E \to X$ be a vector bundle over a paracompact Hausdorff base, real or complex, and let $E_0 \subseteq E$ be a subbundle. Then there is a subbundle $E_0^\perp \subseteq E$ with $E \cong E_0 \oplus E_0^\perp$.

***Proof*:**
Fix an inner product on E by lemma 7.1.22, Euclidean in the real case and Hermitian in the complex case, and let $E_0^\perp \subseteq E$ be the set of vectors orthogonal to E_0 within their own fiber. Each fiber $(E_0^\perp)_x$ is the orthogonal complement of $(E_0)_x$ in E_x, of dimension $n - m$ for n and m the ranks of E and E_0, so the fiberwise sum $E_0 \oplus E_0^\perp \to E$, $(v, w) \mapsto v + w$, is a linear isomorphism on every fiber. It is continuous, being a restriction of the addition of E, so once $E_0^\perp$ is known to be a subbundle this map is an isomorphism of vector bundles by lemma 5.2.10.

Local triviality of $E_0^\perp$ is a local question on X, so fix x_0 and work over a neighbourhood on

which both E and E_0 are trivial. A trivialization of E_0 gives sections $s_1, \dots, s_m$ that form a basis of $(E_0)_x$ for every nearby x. Gram-Schmidt applied to $s_1, \dots, s_m$ against the inner product yields sections $e_1, \dots, e_m$ orthonormal in every nearby fiber. The process is a composite of inner products, subtractions and square roots of positive quantities, hence continuous in x, and it is defined because the s_i are independent in each fiber. The fiberwise orthogonal projection onto the complement,

$$\pi(v) = v - \sum_{i=1}^{m} \langle v, e_i(x) \rangle \, e_i(x), \qquad v \in E_x$$

is then continuous over that neighbourhood. Extend $e_1, \dots, e_m$ to sections $e_1, \dots, e_n$ spanning each nearby fiber of E, by completing a basis at x_0 and taking the same vectors in nearby fibers under the trivialization of E, which stay independent near x_0 since independence is an open condition. Then $\pi(e_{m+1}), \dots, \pi(e_n)$ are sections of $E_0^{\perp}$ spanning each nearby fiber of it: π carries the spanning set $e_1, \dots, e_n$ of E_x onto a spanning set of $(E_0^{\perp})_x$ while killing $e_1, \dots, e_m$. As $\dim (E_0^{\perp})_x = n - m$, these $n - m$ spanning sections are a basis in every nearby fiber, and $(x, a_{m+1}, \dots, a_n) \mapsto \sum_j a_j \, \pi(e_j)(x)$ is a fiberwise-linear homeomorphism onto the part of $E_0^{\perp}$ over that neighbourhood. Hence $E_0^{\perp}$ is locally trivial, completing the proof.

The same inner product names a construction used throughout the Thom and Gysin arguments.

Definition 7.1.24: Sphere Bundle

Let $p\colon E \to B$ be a real vector bundle of rank n over a paracompact Hausdorff base, equipped with an inner product (which exists by lemma 7.1.22). The *sphere bundle* $S(E) \to B$ is the subspace of E consisting of unit vectors,

$$S(E) = \{ v \in E \mid \langle v, v \rangle = 1 \}$$

with the restricted projection. It is a fiber bundle with fiber the unit sphere $S^{n-1} \subset \mathbb{R}^n$.

Definition 7.1.25: Orientations of a Bundle and of Its Sphere Bundle

An *orientation* of a real vector space V of dimension $n \geq 1$ is a generator μ_V of $H^n(V, V \setminus 0; \mathbb{Z}) \cong \mathbb{Z}$. An *orientation* of a real vector bundle $E \to B$ of rank n is a choice of an orientation μ_b of every fiber E_b such that each point of B has a neighbourhood U, a trivialization $h\colon E|_U \to U \times \mathbb{R}^n$ and a generator μ of $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus 0; \mathbb{Z})$ with $h_b^* \mu = \mu_b$ for every $b \in U$, where h_b is the restriction of h to E_b. If $n \geq 2$ and E carries an inner product, the orientation of the sphere bundle $S(E)$ *induced* by μ is the family of classes $\alpha_b \in H^{n-1}(S(E_b); \mathbb{Z})$, where α_b is the restriction to $S(E_b)$ of the class $a_b \in H^{n-1}(E_b \setminus 0; \mathbb{Z})$ with $\delta a_b = -\mu_b$, for δ the connecting isomorphism of the pair $(E_b, E_b \setminus 0)$.

An orientation picks one of the two generators in every fiber, consistently over trivializing sets. The connecting map δ is an isomorphism because E_b is contractible, and $S(E_b)$ is a deformation retract of $E_b \setminus 0$, so α_b generates $H^{n-1}(S(E_b); \mathbb{Z})$. Over U the trivialization carries the α_b to one class of $H^{n-1}(S^{n-1}; \mathbb{Z})$, so they trivialize the local system of fiber cohomology that theorem 6.3.2 requires to be trivial. The minus sign makes the Euler class of that Gysin sequence equal to the restriction of the Thom class (proposition 7.3.5). The opposite sign would produce $-e(E)$.

The *direct-sum orientation* of $V \oplus W$ is the product $\mu_V \times \mu_W = q_V^* \mu_V \smile q_W^* \mu_W$ of proposition 4.2.3(5),

for the projections q_V, q_W and the open sets $q_V^{-1}(V \setminus 0)$ and $q_W^{-1}(W \setminus 0)$, whose union is $(V \oplus W) \setminus 0$. It is a generator by lemma 7.3.3. The *standard orientation* $\mu_{\mathbb{C}}$ of $\mathbb{C} = \mathbb{R}^2$ is the generator with $\langle \mu_{\mathbb{C}}, \tau \rangle = 1$ for the class $\tau \in H_2(\mathbb{C}, \mathbb{C} \setminus 0)$ whose boundary is the class of the counterclockwise loop $t \mapsto e^{2\pi i t}$ (example 4.4.2). The *complex orientation* of a complex vector space V of dimension n is $\varphi^*(\mu_{\mathbb{C}} \times \cdots \times \mu_{\mathbb{C}})$ for a complex-linear isomorphism $\varphi \colon V \to \mathbb{C}^n$. It does not depend on φ. For $g \in GL_n(\mathbb{C})$ the polynomial $\det\big((1-t)I + tg\big)$ in $t \in \mathbb{C}$ has finitely many roots, none at 0 or 1, so a path from 0 to 1 in $\mathbb{C}$ avoiding them joins I to g in $GL_n(\mathbb{C})$, and homotopic maps of pairs induce the same map in cohomology. For a complex vector bundle the complex orientations of the fibers form a bundle orientation, since over an open set where the bundle is trivial they are all pulled back from $\mu_{\mathbb{C}} \times \cdots \times \mu_{\mathbb{C}}$ by one trivialization. Applied to $\varphi \oplus \psi$, the definition shows that the complex orientation of $V \oplus W$ is the direct-sum orientation of the complex orientations of V and W. Finally, an orthonormal basis b of an inner product space V, read as an isometry $b \colon \mathbb{R}^n \to V$, orients V by $(b^{-1})^* \mu_n$ for a generator μ_n of $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus 0; \mathbb{Z})$ fixed once and for all. Two orthonormal bases related by $g \in O(n)$ give orientations differing by $\det g$: an orthogonal map is a product of reflections in hyperplanes, and a reflection acts on $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus 0; \mathbb{Z}) \cong \widetilde{H}^{n-1}(S^{n-1}; \mathbb{Z})$ by -1 (proposition 3.5.2). This is how the oriented tautological bundle $\tilde{\gamma}_n$ of definition 7.1.12 is oriented, through the orthonormal bases in the recorded class.

Vector bundles and principal bundles are two views of the same object, and the passage between them is what lets the classification of theorem 7.1.10 and the homotopy invariance of lemma 7.1.9, both stated for principal bundles, be applied to vector bundles.

Lemma 7.1.26: Vector Bundles and Their Frame Bundles

Let X be paracompact Hausdorff. Sending a rank-n real vector bundle $E \to X$, equipped with a Euclidean inner product, to its bundle $\mathrm{Fr}(E) \to X$ of orthonormal frames, and sending a principal $O(n)$-bundle $P \to X$ to the associated bundle $P \times_{O(n)} \mathbb{R}^n$, are mutually inverse up to isomorphism. Both constructions commute with pullback: $\mathrm{Fr}(f^*E) \cong f^*\mathrm{Fr}(E)$ and $(f^*P) \times_{O(n)} \mathbb{R}^n \cong f^*(P \times_{O(n)} \mathbb{R}^n)$. The same holds for complex bundles with $U(n)$ and a Hermitian inner product.

***Proof*:**

An inner product exists by lemma 7.1.22. The fiber of $\mathrm{Fr}(E)$ over x is the set of orthonormal bases of E_x, on which $O(n)$ acts simply transitively by change of basis, so $\mathrm{Fr}(E)$ is a principal $O(n)$-bundle. A local trivialization of E over U, made orthonormal fiberwise by Gram-Schmidt as in proposition 7.1.23, trivializes $\mathrm{Fr}(E)$ over U. Conversely $P \times_{O(n)} \mathbb{R}^n$, the quotient of $P \times \mathbb{R}^n$ by $(p \cdot g, v) \sim (p, gv)$, is locally $U \times \mathbb{R}^n$ and fiberwise linear, hence a rank-n vector bundle, and it carries the inner product descending from the standard one on $\mathbb{R}^n$, which is $O(n)$-invariant.

The two composites are the identity up to isomorphism. Starting from E, the map $\mathrm{Fr}(E) \times_{O(n)} \mathbb{R}^n \to E$ sending $[(e_1, \ldots, e_n), v]$ to $\sum_i v_i e_i$ is well defined, since replacing the frame by g-times-itself replaces v by $g^{-1}v$, and it is continuous, fiberwise linear and bijective, hence an isomorphism by lemma 5.2.10. Starting from P, a point of $\mathrm{Fr}(P \times_{O(n)} \mathbb{R}^n)$ is an orthonormal frame of a fiber, and $p \mapsto [(p, e_1)], \ldots, [(p, e_n)]$ for the standard basis identifies P with it equivariantly. Both constructions are defined fiberwise from local trivializations, and pullback replaces the fiber over x by the fiber over $f(x)$, which gives the two compatibilities. The complex case replaces orthonormal by unitary throughout, as we sought to show.

Lemma 7.1.27: Products and Frames in $\mathbb{K}^\infty$

Let $\mathbb{K}$ be $\mathbb{R}$ or $\mathbb{C}$, and give $\mathbb{K}^\infty$ and $V_n(\mathbb{K}^\infty)$ the topologies of definition 7.1.12.

1. For every $k \geq 1$ a subset of $(\mathbb{K}^\infty)^k$ is open in the product topology exactly when it meets every $(\mathbb{K}^N)^k$ in an open set. The same holds for $(\mathbb{K}^\infty)^k \times I$ and the subspaces $(\mathbb{K}^N)^k \times I$.

2. Addition $\mathbb{K}^\infty \times \mathbb{K}^\infty \to \mathbb{K}^\infty$, scalar multiplication $\mathbb{K} \times \mathbb{K}^\infty \to \mathbb{K}^\infty$ and the inner product $\mathbb{K}^\infty \times \mathbb{K}^\infty \to \mathbb{K}$ are continuous.

3. The topology of $V_n(\mathbb{K}^\infty)$ is the subspace topology of $(\mathbb{K}^\infty)^n$. Hence, if $v_1, \dots, v_n \colon Z \to \mathbb{K}^\infty$ are maps from a space Z, linearly independent at every point, then the Gram-Schmidt orthonormalization $b \colon Z \to V_n(\mathbb{K}^\infty)$ of $(v_1, \dots, v_n)$ is continuous, and so is $z \mapsto \operatorname{span}(v_1(z), \dots, v_n(z)) = q(b(z))$ into $\operatorname{Gr}_n(\mathbb{K}^\infty)$. For $\mathbb{K} = \mathbb{R}$ the plane oriented by $(v_1(z), \dots, v_n(z))$ is $\tilde{q}(b(z))$, continuous in z into $\widetilde{\operatorname{Gr}}_n(\mathbb{R}^\infty)$.

***Proof*:**

Each $\mathbb{K}^N$ is closed in $\mathbb{K}^{N+1}$, so it is a closed subspace of $\mathbb{K}^\infty$ with its own topology, by the argument of Step 1 of the proof of lemma 7.1.15. Hence $(\mathbb{K}^N)^k$ carries the subspace topology of the product $(\mathbb{K}^\infty)^k$, and a set open in the product meets it in an open set.

Part (1). Conversely let $U \subseteq (\mathbb{K}^\infty)^k$ meet every $(\mathbb{K}^N)^k$ in an open set, and let $a = (a^1, \dots, a^k) \in U$ with every a^i in $\mathbb{K}^{N_0}$. Choose open sets $A^i_{N_0} \ni a^i$ of $\mathbb{K}^{N_0}$ with compact closures and $\prod_i \overline{A^i_{N_0}} \subseteq U$. Inductively, the compact set $\prod_i \overline{A^i_N}$ lies in the open set $U \cap (\mathbb{K}^{N+1})^k$, so the tube lemma [30, §26], applied one factor at a time, and local compactness of $\mathbb{K}^{N+1}$ give open sets $A^i_{N+1} \supseteq \overline{A^i_N}$ of $\mathbb{K}^{N+1}$ with compact closures and $\prod_i \overline{A^i_{N+1}} \subseteq U$. Each union $A^i = \bigcup_N A^i_N$ is open, since $A^i \cap \mathbb{K}^M = \bigcup_{N \geq M} (A^i_N \cap \mathbb{K}^M)$, and $a \in A^1 \times \cdots \times A^k \subseteq U$ by the nesting. So U is open in the product. The map onto $(\mathbb{K}^\infty)^k$ from the disjoint union of the $(\mathbb{K}^N)^k$ is therefore a quotient map, and so is its product with the identity of the locally compact Hausdorff interval [16, Products of Quotient Maps], which is the statement for $(\mathbb{K}^\infty)^k \times I$.

Part (2). Each of the three maps is continuous on every stage $\mathbb{K}^N \times \mathbb{K}^N$, or $\mathbb{K} \times \mathbb{K}^N$, into $\mathbb{K}^N$ or $\mathbb{K}$. By (1) with $k = 2$, and for scalar multiplication by [16, Products of Quotient Maps] with the locally compact Hausdorff factor $\mathbb{K}$, each domain carries the topology of its stages, so each map is continuous.

Part (3). The set $V_n(\mathbb{K}^\infty)$ meets $(\mathbb{K}^N)^n$ in the compact set $V_n(\mathbb{K}^N)$, so it is closed in $(\mathbb{K}^\infty)^n$ by (1). A subset C of $V_n(\mathbb{K}^\infty)$ that meets every $V_n(\mathbb{K}^N)$ in a closed set meets every $(\mathbb{K}^N)^n$ in a closed set, so C is closed in $(\mathbb{K}^\infty)^n$ by (1). Conversely a set closed in the subspace topology meets each $V_n(\mathbb{K}^N)$, a subspace of $(\mathbb{K}^N)^n$, in a closed set. The two topologies therefore agree. Gram-Schmidt puts $b_1 = v_1/|v_1|$ and $b_i = u_i/|u_i|$ with $u_i = v_i - \sum_{j<i} \langle v_i, b_j \rangle b_j$, which is nonzero because the v_i are independent. By (2) and induction on i, each $b_i \colon Z \to \mathbb{K}^\infty$ is continuous, so $b = (b_1, \dots, b_n)$ is continuous into $(\mathbb{K}^\infty)^n$ with values in $V_n(\mathbb{K}^\infty)$, hence continuous into $V_n(\mathbb{K}^\infty)$. The maps q and $\tilde{q}$ are continuous on every stage, hence continuous. The frame $b(z)$ spans the span of the $v_i(z)$, and the matrix expressing it in the $v_i(z)$ is upper triangular with positive diagonal, so for $\mathbb{K} = \mathbb{R}$ it carries the same orientation, completing the proof.

The classification itself rests on one construction, a fiberwise linear injection into $\mathbb{R}^\infty$. It is what

replaces, for vector bundles over a paracompact base, the skeletal induction that proves theorem 7.1.10.

Lemma 7.1.28: Fiberwise Linear Injections into $\mathbb{R}^\infty$

Let $p\colon E \to X$ be a rank-n real vector bundle over a paracompact Hausdorff base, and let $\mathbb{R}^\infty = \bigcup_N \mathbb{R}^N$ carry the weak topology. Then there is a map $g\colon E \to \mathbb{R}^\infty$ whose restriction to every fiber is linear and injective, and any two such maps are homotopic through maps whose restriction to every fiber is linear and injective. The same statements hold for a rank-n complex vector bundle with $\mathbb{C}^\infty$ in place of $\mathbb{R}^\infty$ and complex-linear restrictions in place of linear ones.

***Proof*:**

Step 1: A countable trivializing cover with a subordinate partition of unity. Apply lemma 7.1.8 to an open cover of X by sets over which E is trivial. It gives a countable open cover $\{W_k\}_{k\geq 1}$ of X and a partition of unity $\{\varphi_k\}_{k\geq 1}$ with $\operatorname{supp}\varphi_k \subseteq W_k$, each W_k a disjoint union of open sets over each of which the rank-n bundle E is trivial. Trivializations over disjoint open sets amalgamate, so E is trivial over each W_k.

Step 2: The map. For each k fix a trivialization of E over W_k and let $h_k\colon p^{-1}(W_k) \to \mathbb{R}^n$ be its fiber coordinate, a linear isomorphism on every fiber over W_k. Define $g_k\colon E \to \mathbb{R}^n$ by $g_k(v) = \varphi_k\big(p(v)\big)\, h_k(v)$ on $p^{-1}(W_k)$ and $g_k = 0$ off $p^{-1}\big(\operatorname{supp}\varphi_k\big)$. The two prescriptions agree on the overlap, their open domains cover E because $\operatorname{supp}\varphi_k \subseteq W_k$, and so g_k is continuous. Set

$$g = (g_1, g_2, \dots)\colon E \longrightarrow \mathbb{R}^n \oplus \mathbb{R}^n \oplus \cdots = \mathbb{R}^\infty$$

Near any point of X only finitely many φ_k are nonzero, so over a suitable neighbourhood g takes values in a finite partial sum $\mathbb{R}^{nm} \subseteq \mathbb{R}^\infty$ and is continuous into it. Continuity is a local property, so g is continuous into the weak topology. On the fiber over x each coordinate g_k is the scalar $\varphi_k(x)$ times a map linear on that fiber, so g is linear there, and choosing k with $\varphi_k(x) > 0$ shows g is injective there, the block g_k already being injective on that fiber.

Step 3: Any two are homotopic through fiberwise linear injections. Let g, g' be two such maps and let $J_{\mathrm{o}}, J_{\mathrm{e}}\colon \mathbb{R}^\infty \to \mathbb{R}^\infty$ be the linear injections into the odd and the even coordinates, $J_{\mathrm{o}}(z_1, z_2, \dots) = (z_1, 0, z_2, 0, \dots)$ and $J_{\mathrm{e}}(z_1, z_2, \dots) = (0, z_1, 0, z_2, \dots)$. The straight line $L_t = (1-t)\,\mathrm{id} + t\,J_{\mathrm{o}}$ consists of injective linear maps: for $0 < t < 1$ and a nonzero z with highest nonzero coordinate z_m, if $m \geq 2$ then coordinate $2m-1 > m$ of $L_t z$ equals $t\,z_m \neq 0$, while if $m = 1$ the first coordinate of $L_t z$ equals $(1-t)z_1 + tz_1 = z_1 \neq 0$. For $t = 0, 1$ injectivity is immediate. Hence $g \simeq J_{\mathrm{o}} \circ g$ through the fiberwise linear injections $L_t \circ g$, and likewise $g' \simeq J_{\mathrm{e}} \circ g'$ through $L'_t \circ g'$ for $L'_t = (1-t)\,\mathrm{id} + t\,J_{\mathrm{e}}$. The images of J_{o} and J_{e} meet only in 0, so the straight line $(1-t)\,J_{\mathrm{o}} \circ g + t\,J_{\mathrm{e}} \circ g'$ is, for each t, linear on every fiber and injective there: a nonzero fiber vector v has $g(v) \neq 0$ and $g'(v) \neq 0$, the odd coordinates of the combination are $(1-t)$ times those of $J_{\mathrm{o}}g(v)$ and the even ones are t times those of $J_{\mathrm{e}}g'(v)$, and for every $t \in [0,1]$ at least one of the factors $1-t$, t is nonzero. Continuity of the three homotopies is not the local finiteness of Step 2, which was a property of the *constructed* map g and is unavailable for arbitrary g, g'. That continuity comes instead from the topology of $\mathbb{R}^\infty$ itself. The combination map $(y, z, t) \mapsto (1-t)\,y + t\,z$ is continuous on each $\mathbb{R}^N \times \mathbb{R}^N \times I$ into $\mathbb{R}^N$, and $\mathbb{R}^\infty \times \mathbb{R}^\infty \times I$ carries the topology in which a set is open exactly when it meets every $\mathbb{R}^N \times \mathbb{R}^N \times I$ in an open set (lemma 7.1.27(1)), so the combination map is continuous. Each of the three homotopies is the composite of this combination map with a map to $\mathbb{R}^\infty \times \mathbb{R}^\infty \times I$ having continuous coordinates, such as $(v, t) \mapsto \big(J_{\mathrm{o}}g(v),\ J_{\mathrm{e}}g'(v),\ t\big)$ for the third, hence is continuous, and their concatenation

proves the claim.

Step 4: The complex case. Replace $\mathbb{R}^n$ and $\mathbb{R}^\infty$ by $\mathbb{C}^n$ and $\mathbb{C}^\infty$ and linear by complex-linear throughout: the partitions of unity are real-valued scalars, hence complex-linear multipliers, and the interleaving maps $J_{\mathrm{o}}, J_{\mathrm{e}}$ are complex-linear. The three steps repeat verbatim, as we sought to show.

A fiberwise linear injection into $\mathbb{R}^\infty$ is the same data as a map to the infinite Grassmannian: send x to the n-plane $g(E_x)$. Since $\mathrm{Gr}_n(\mathbb{R}^\infty)$ is a model for $BO(n)$ by lemma 7.1.18(4), the lemma classifies vector bundles over a paracompact base, and this is the form in which the rest of the chapter uses it.

Theorem 7.1.29: Classification of Vector Bundles over a Paracompact Base

Let X be a paracompact Hausdorff space and $n \geq 0$. Pulling back the universal bundle $\gamma^n \to \mathrm{Gr}_n(\mathbb{R}^\infty) = BO(n)$ gives a bijection

$$[X, BO(n)] \xrightarrow{\cong} \mathrm{Vect}^n_{\mathbb{R}}(X), \qquad [f] \longmapsto f^*\gamma^n$$

from free homotopy classes of maps to isomorphism classes of rank-n real vector bundles over X, natural in X: for a map $u\colon X' \to X$ the bundle $u^*(f^*\gamma^n)$ is classified by $f \circ u$. The same holds for rank-n complex bundles with $\mathrm{Gr}_n(\mathbb{C}^\infty) = BU(n)$ and the universal complex bundle.

***Proof*:**
Well-definedness. The assignment must be shown to depend only on the homotopy class of f, that is, that $f_0 \simeq f_1$ implies $f_0^*\gamma^n \cong f_1^*\gamma^n$. Pass to principal bundles. The base $BO(n)$ is paracompact Hausdorff by theorem 7.1.17, so lemma 7.1.22 gives a Euclidean inner product on γ^n. Let $P \to BO(n)$ be its bundle of orthonormal frames, a principal $O(n)$-bundle whose associated rank-n vector bundle is γ^n again by lemma 7.1.26. Pullback commutes with both constructions by the same lemma, so $f_i^*\gamma^n$ is the vector bundle associated to f_i^*P. By lemma 7.1.9, whose hypothesis is exactly that the base be paracompact, $f_0 \simeq f_1$ gives $f_0^*P \cong f_1^*P$ as principal $O(n)$-bundles, and applying the associated-bundle construction to that isomorphism gives $f_0^*\gamma^n \cong f_1^*\gamma^n$. Hence the map on homotopy classes is defined. (In the complex case replace $O(n)$ by $U(n)$ and the Euclidean inner product by a Hermitian one.)

Surjectivity. Let $E \to X$ be a rank-n real bundle and let $g\colon E \to \mathbb{R}^\infty$ be fiberwise linear and injective, by lemma 7.1.28. Define $f\colon X \to \mathrm{Gr}_n(\mathbb{R}^\infty)$ by $f(x) = g(E_x)$. Over a neighbourhood on which E is trivial, a frame $e_1, \dots, e_n$ gives $f(x) = \mathrm{span}\big(g(e_1(x)), \dots, g(e_n(x))\big)$ with the $g(e_i(x))$ continuous in x and independent at every point, so f is continuous into the Grassmannian by lemma 7.1.27(3). The map $E \to f^*\gamma^n$ sending $v \in E_x$ to $\big(x, g(v)\big)$, legitimate because $g(v)$ lies in the plane $f(x)$, which is the fiber of γ^n over $f(x)$, is continuous and fiberwise linear and injective, hence, both fibers having dimension n, an isomorphism of vector bundles by lemma 5.2.10.

Injectivity. Suppose $f_0^*\gamma^n \cong f_1^*\gamma^n$ for maps $f_0, f_1\colon X \to \mathrm{Gr}_n(\mathbb{R}^\infty)$, and call the common bundle E. Composing each isomorphism $E \cong f_i^*\gamma^n$ with the tautological inclusion of the fiber plane into $\mathbb{R}^\infty$ gives fiberwise linear injections $g_0, g_1\colon E \to \mathbb{R}^\infty$ inducing f_0 and f_1. By the homotopy clause of lemma 7.1.28 there is a homotopy g_t from g_0 to g_1 through fiberwise linear injections. Taking images fiberwise, $(x,t) \mapsto g_t(E_x)$ is a homotopy $X \times I \to \mathrm{Gr}_n(\mathbb{R}^\infty)$ from f_0 to f_1, continuous by lemma 7.1.27(3) applied to the frames $g_t(e_i(x))$ over a trivializing neighbourhood, continuous in (x,t). Hence $[f_0] = [f_1]$.

Naturality. Pullback is functorial, $u^*(f^*\gamma^n) \cong (f \circ u)^*\gamma^n$, which is the stated compatibility. The complex case replaces $\mathbb{R}$ by $\mathbb{C}$ and linear by complex-linear throughout, on the complex clause of lemma 7.1.28, completing the proof.

Over a base that is both a CW complex and paracompact Hausdorff, theorem 7.1.29 and theorem 7.1.10 classify the same bundles and they agree, both sending a map to the pullback of the universal bundle along it. Nothing below depends on which of the two gives a classifying map.

Corollary 7.1.30: Oriented Classifying Maps over a Paracompact Base

Let X be a paracompact Hausdorff space and $E \to X$ an oriented real vector bundle of rank $n \geq 1$. There is a map $\tilde{f}\colon X \to BSO(n) = \widetilde{\mathrm{Gr}}_n(\mathbb{R}^\infty)$ with $E \cong \tilde{f}^*\tilde{\gamma}_n$ as oriented bundles, and any two such maps are homotopic.

***Proof*:**
For a linear injection $g_x\colon E_x \to \mathbb{R}^\infty$, write $g_x(\mu_x)$ for the orientation of the plane $g_x(E_x)$ whose orthonormal bases b satisfy $(b^{-1})^*\mu_n = (g_x^{-1})^*\mu_x$ (definition 7.1.25). Let $g\colon E \to \mathbb{R}^\infty$ be linear and injective on every fiber (lemma 7.1.28), and put $\tilde{f}(x) = \big(g(E_x), g_x(\mu_x)\big)$. Near a point of X choose a trivialization h of E with $h_x^*\mu = \mu_x$, the frame $e_i(x) = h_x^{-1}(\varepsilon_i)$ from the standard basis ε_i of $\mathbb{R}^n$, and the Gram-Schmidt orthonormalization $b(x)$ of $g(e_1(x)), \dots, g(e_n(x))$. The matrix $T(x) = h_x \circ g_x^{-1} \circ b(x)$ is upper triangular with positive diagonal, so the straight-line homotopy from $T(x)$ to the identity runs through invertible matrices and $T(x)^*$ is the identity on $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus 0; \mathbb{Z})$. Hence $(g_x^{-1})^*\mu_x = (b(x)^{-1})^*\mu$, and $\tilde{f}(x)$ is $\tilde{q}(b(x))$ if $\mu = \mu_n$ and $\tilde{q}(\epsilon b(x))$ if $\mu = -\mu_n$, with ϵ negating the first vector. Both are continuous in x by lemma 7.1.27(3), ϵ being continuous on $V_n(\mathbb{R}^\infty)$. The map $v \mapsto (x, g(v))$ on E_x is an isomorphism $E \to \tilde{f}^*\tilde{\gamma}_n$ carrying μ_x to the orientation of the fiber.

For uniqueness, let $E \cong \tilde{f}_i^*\tilde{\gamma}_n$ as oriented bundles for $i = 0, 1$. Composing with the inclusion of the fiber planes into $\mathbb{R}^\infty$ gives fiberwise linear injections g_i with $\tilde{f}_i(x) = \big(g_i(E_x), g_{i,x}(\mu_x)\big)$. The homotopy clause of lemma 7.1.28 joins g_0 to g_1 through fiberwise linear injections g_t, and $(x,t) \mapsto \big(g_t(E_x), g_{t,x}(\mu_x)\big)$ is continuous by the local formula above applied to the bundle $E \times I \to X \times I$. It is a homotopy from $\tilde{f}_0$ to $\tilde{f}_1$, completing the proof.

So an oriented bundle over a paracompact Hausdorff base has an oriented classifying map, unique up to homotopy, and the Euler class of definition 7.2.15 is defined over every such base.

Exercise 7.1.1

1. Let G be a topological group. Prove that a principal G-bundle $p\colon E \to X$ is trivial (isomorphic to $X \times G$ as a principal bundle) if and only if it admits a continuous section $s\colon X \to E$ with $p \circ s = \mathrm{id}_X$. Deduce that the principal $\mathbb{Z}$-bundle $\mathbb{R} \to S^1$ of example 7.1.2 is nontrivial.
2. Using the classification theorem (theorem 7.1.10) and the computation $B\mathbb{Z}/2 = \mathbb{R}\mathrm{P}^\infty$, determine the set $\mathrm{Prin}_{\mathbb{Z}/2}(S^1)$ of isomorphism classes of principal $\mathbb{Z}/2$-bundles over the circle, and identify the two bundles geometrically.
3. Let G be a path-connected topological group with BG its classifying space. Show that the principal G-bundles over the sphere S^n (for $n \geq 1$) are classified by the homotopy group

$\pi_{n-1}(G)$, and as a check recover $\mathrm{Prin}_{U(1)}(S^2) \cong \mathbb{Z}$. (The hypothesis that G is path-connected is needed: it forces BG to be simply connected, so that the free homotopy set $[S^n, BG]$ agrees with the based set $\pi_n(BG)$. Without it the count is instead the orbit set of the $\pi_0(G)$-action, as the case $G = \mathbb{Z}/2$, $n = 1$ of exercise 7.1.1 (2) shows.)

4. Let G and H be topological groups with universal bundles $EG \to BG$ and $EH \to BH$. Prove that $EG \times EH \to BG \times BH$ is a universal $(G \times H)$-bundle, and deduce from corollary 7.1.11(1) that every classifying space of $G \times H$ is weakly homotopy equivalent to $BG \times BH$, and homotopy equivalent to it when both have the homotopy type of CW complexes. Use it to identify classifying spaces of the torus $(S^1)^n$ and of $(\mathbb{Z}/2)^n$.

7.2 Characteristic Classes and the Cohomology of Classifying Spaces

The classification theorem theorem 7.1.10 states that for a topological group G the isomorphism classes of principal G-bundles over a CW complex X are in natural bijection with the homotopy classes of maps $X \to BG$, the bijection sending a classifying map $f\colon X \to BG$ to the pullback f^*EG of the universal bundle. Naturality means that for a map $g\colon X' \to X$ the classifying map of $g^*(f^*EG)$ is $f \circ g$, so the assignment of bundles to maps converts pullback of bundles into composition of maps. A characteristic class is exactly an invariant that is compatible with this structure: a rule assigning to each bundle a cohomology class of the base, natural under pullback. The point of the section is that naturality forces such a rule to be the pullback of one universal class living on BG, so the classification theorem converts the search for invariants into a cohomology computation on a single space.

Definition 7.2.1: Characteristic Class

Fix a topological group G, an abelian coefficient group A, and an integer $k \geq 0$. A *characteristic class* of degree k for principal G-bundles, with values in A, is a rule c that assigns to every principal G-bundle $P \to X$ over a CW complex X a cohomology class

$$c(P) \in H^k(X; A)$$

subject to *naturality under pullback*: for every map $g\colon X' \to X$ of CW complexes,

$$c(g^*P) = g^*\big(c(P)\big) \in H^k(X'; A)$$

where $g^*P \to X'$ is the pulled-back bundle and $g^*\colon H^k(X; A) \to H^k(X'; A)$ is the induced map on cohomology. A characteristic class for rank-n complex (respectively real) vector bundles is a characteristic class for principal $U(n)$-bundles (respectively $O(n)$-bundles), under the correspondence of lemma 7.1.26, which sends a vector bundle to its principal bundle of unitary (respectively orthonormal) frames and a principal bundle to its associated vector bundle.

The naturality condition is the whole content of the definition, and it is what makes the notion computable. It says that the value of c on a bundle is determined by the value of c on bundles into which the given one maps, and ultimately by the value of c on the one bundle through which all

others factor. The vector-bundle version is the form in which characteristic classes are usually met: the structure group of a rank-n complex vector bundle is $U(n)$ once a Hermitian metric is chosen (any two metrics give homotopic, hence isomorphic, structure-group reductions, so the choice does not affect the bundle's classification), and similarly $O(n)$ for a rank-n real bundle with a Euclidean metric. The frame bundle of orthonormal frames is then a principal bundle, and characteristic classes of the vector bundle are by definition those of its frame bundle.

The classification theorem turns definition 7.2.1 into the statement that a characteristic class is a single element of the cohomology of BG.

Proposition 7.2.2: Characteristic Classes are the Cohomology of BG

Let G be a topological group with a universal bundle $EG \to BG$ whose base BG is a CW complex, let A be an abelian group, and fix $k \geq 0$. Evaluation on the universal bundle gives a bijection

$$\left\{ \begin{array}{c} \text{degree-}k \text{ characteristic classes for} \\ G\text{-bundles with values in } A \end{array} \right\} \xrightarrow{\cong} H^k(BG; A), \qquad c \longmapsto c(EG)$$

Its inverse sends a class $u \in H^k(BG; A)$ to the characteristic class c_u defined by $c_u(P) = f_P^* u$, where $f_P \colon X \to BG$ is the classifying map of $P \to X$.

***Proof*:**

The two assignments $c \mapsto c(EG)$ and $u \mapsto c_u$ are shown to be mutually inverse.

Step 1: c_u is a characteristic class. Fix $u \in H^k(BG; A)$ and define $c_u(P) = f_P^* u$ for $P \to X$ with classifying map $f_P \colon X \to BG$. The class f_P is well defined up to homotopy by theorem 7.1.10, and homotopic maps induce equal maps on cohomology, so $c_u(P)$ depends only on the isomorphism class of P. For naturality, let $g \colon X' \to X$ and consider $g^*P \to X'$. By the naturality in theorem 7.1.10, a classifying map for g^*P is $f_P \circ g$, so

$$c_u(g^*P) = (f_P \circ g)^* u = g^*\big(f_P^* u\big) = g^*\big(c_u(P)\big)$$

using functoriality of cohomology. Thus c_u satisfies definition 7.2.1.

Step 2: The two maps are inverse. Apply $c \mapsto c(EG)$ to c_u. The universal bundle $EG \to BG$ has classifying map the identity id_{BG} (the identity classifies the bundle one started with), so

$$c_u(EG) = \mathrm{id}_{BG}^* u = u$$

Hence $u \mapsto c_u \mapsto c_u(EG) = u$ is the identity on $H^k(BG; A)$.

Conversely, start with a characteristic class c, set $u = c(EG)$, and form c_u. For any bundle $P \to X$ with classifying map f_P, the bundle P is isomorphic to $f_P^* EG$ by theorem 7.1.10. Naturality of c applied to the map $f_P \colon X \to BG$ and the bundle $EG \to BG$ gives

$$c(P) = c(f_P^* EG) = f_P^*\big(c(EG)\big) = f_P^* u = c_u(P)$$

so $c_u = c$ as functions on bundles. Thus $c \mapsto c(EG) \mapsto c_{c(EG)} = c$ is the identity on characteristic classes. The two assignments are mutually inverse, completing the proof.

The proposition is the organizing principle of the entire chapter. It collapses the problem of finding all natural invariants of G-bundles, which on its face ranges over every base space at once, into the problem of computing the single graded ring $H^*(BG;A)$. The cup product on BG even gives the characteristic classes a ring structure: the product of two characteristic classes, formed by pulling back the product $u \smile u'$, is again natural, and the bijection of proposition 7.2.2 is a ring isomorphism between the characteristic classes (with the obvious product) and $H^*(BG;A)$. The remaining work is computational: identify the rings $H^*(BU(n);\mathbb{Z})$ and $H^*(BO(n);\mathbb{Z}/2)$, name their generators, and record what those generators measure. The hypothesis that BG is a CW complex holds for the models used here: $BU(n)$ and $BO(n)$ are the infinite Grassmannians $\mathrm{Gr}_n(\mathbb{C}^\infty)$ and $\mathrm{Gr}_n(\mathbb{R}^\infty)$ by lemma 7.1.18(4), CW complexes whose cells are the Schubert cells (theorem 7.1.17).

A first instance is already in hand and worth stating in the new language, since it fixes the normalization for everything that follows.

Example 7.2.3: The First Chern Class of a Line Bundle

Take $G = U(1)$, the circle group. Its classifying space is $BU(1) = \mathbb{C}\mathrm{P}^\infty$ (example 7.1.6), and the cohomology ring was computed in example 6.4.5 by the Serre spectral sequence of the universal circle bundle $S^1 \to S^\infty \to \mathbb{C}\mathrm{P}^\infty$:

$$H^*(BU(1);\mathbb{Z}) = H^*(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \cong \mathbb{Z}[c_1], \qquad |c_1| = 2$$

a polynomial ring on one degree-two generator c_1. By proposition 7.2.2 the characteristic classes of complex line bundles are exactly the elements of this ring: the additive ones in degree 2 are the integer multiples $m\,c_1$, and the full ring of characteristic classes is $\mathbb{Z}[c_1]$. The group $H^2(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \cong \mathbb{Z}$ has two generators, and the computation does not prefer either. The *first Chern class* c_1 is fixed by a convention. Let $u \in H^2(\mathbb{C}\mathrm{P}^1;\mathbb{Z})$ be the generator dual to the fundamental class $[\mathbb{C}\mathrm{P}^1]$ whose local orientation at $[0:1]$ is the image of the counterclockwise local orientation of $\mathbb{C}$ at 0 (example 4.4.2) under the chart $w \mapsto [w:1]$. Restriction $H^2(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \to H^2(\mathbb{C}\mathrm{P}^1;\mathbb{Z})$ is an isomorphism, since $\mathbb{C}\mathrm{P}^1$ is the 3-skeleton of $\mathbb{C}\mathrm{P}^\infty$ and the two cellular cochain complexes agree through degree 3 (proposition 4.1.20). The convention is that c_1 is the generator restricting to $-u$. For a line bundle $L \to X$ with classifying map $f_L\colon X \to \mathbb{C}\mathrm{P}^\infty$, its value is $c_1(L) = f_L^* c_1 \in H^2(X;\mathbb{Z})$. So the tautological line bundle γ^1 over $\mathbb{C}\mathrm{P}^1 = S^2$, whose fiber over a line $\ell \subseteq \mathbb{C}^2$ is ℓ itself, has $c_1(\gamma^1) = -u$ and Chern number $\int_{\mathbb{C}\mathrm{P}^1} c_1(\gamma^1) = -1$, which detects γ^1 as a generator of $\mathrm{Prin}_{U(1)}(S^2) \cong \mathbb{Z}$. This sign is the one for which $c_1(L)$ is the Euler class of the underlying oriented real bundle: example 7.3.8 proves $e(\gamma^1_{\mathbb{R}}) = -u$ for the complex orientation. The dual of γ^1, the hyperplane bundle, is the Hopf bundle H of example 5.2.7, with $c_1(H) = +u$ by proposition 7.4.7. Let X be a paracompact Hausdorff space of CW homotopy type. Complex line bundles over X correspond to homotopy classes of maps $X \to \mathbb{C}\mathrm{P}^\infty$ (theorem 7.1.29 with $n = 1$). A map $X \to \mathbb{C}\mathrm{P}^\infty$ is the same thing as a based map $X_+ \to \mathbb{C}\mathrm{P}^\infty$ from X with a disjoint basepoint, and free homotopies are based homotopies of X_+, so since $\mathbb{C}\mathrm{P}^\infty = K(\mathbb{Z},2)$ (example 5.5.5) these classes correspond to $\tilde{H}^2(X_+;\mathbb{Z}) = H^2(X;\mathbb{Z})$ by $f \mapsto f^* c_1$ (theorem 5.5.12, c_1 being $\pm\iota_2$). The composite bijection sends L to $f_L^* c_1 = c_1(L)$, so c_1 is a complete invariant: two line bundles over X are isomorphic if and only if their first Chern classes agree.

The line-bundle case is the seed of the general theory, and the strategy for higher rank is to reduce to it. The cohomology of $BU(n)$ is computed below by induction on n through the universal sphere bundle, feeding the Gysin sequence of theorem 6.3.2. Once the Chern classes and the Whitney sum formula are in place, proposition 7.4.11 in section 7.4 presents the cohomology of the projectivization $\mathbb{P}(E)$ of a bundle through the Chern classes of E, by the Leray-Hirsch theorem (theorem 6.5.1). The

classes are defined over every paracompact Hausdorff base through the classification of theorem 7.1.29.

The inductive route rests on one geometric fact, the universal sphere bundle relating consecutive Grassmannians. A unit vector v in an n-plane W leaves behind the $(n-1)$-plane $W \cap v^\perp$, and the space of such pairs (W, v) is homotopy equivalent to $BU(n-1)$.

Lemma 7.2.4: The Universal Sphere Bundle for $U(n)$

Let $\mathbb{K}$ be $\mathbb{R}$ or $\mathbb{C}$, $r = \dim_{\mathbb{R}} \mathbb{K}$ and $n \geq 1$, and give $\gamma_n \to \mathrm{Gr}_n(\mathbb{K}^\infty)$ the inner product of $\mathbb{K}^\infty$.

1. The sphere bundle $\rho\colon S(\gamma_n) \to \mathrm{Gr}_n(\mathbb{K}^\infty)$, whose points are the pairs (W, v) of an n-plane W and a unit vector $v \in W$, is a fiber bundle with fiber S^{rn-1}.
2. The map $j\colon \mathrm{Gr}_{n-1}(\mathbb{K}^\infty) \to S(\gamma_n)$, $j(V) = \big(\mathbb{K}e_1 \oplus s(V), e_1\big)$, is a homotopy equivalence with $\rho \circ j = i_n$, the stabilization of proposition 7.1.19.

For $\mathbb{K} = \mathbb{C}$ this is the sphere bundle $S^{2n-1} \to S(\gamma_n) \xrightarrow{\rho} BU(n)$, whose total space is $BU(n-1)$ up to the homotopy equivalence j.

***Proof*:**

Write Gr_k for $\mathrm{Gr}_k(\mathbb{K}^\infty)$ and S_M for the part of $S(\gamma_n)$ over $\mathrm{Gr}_n(\mathbb{K}^M)$, a compact Hausdorff space. The map $V_n(\mathbb{K}^M) \to S_M$, $x \mapsto (q(x), x_1)$, is a continuous surjection from a compact space onto a Hausdorff space, hence a quotient map, and so is its product with id_I. The norm is continuous on $\mathbb{K}^\infty$, so $S(\gamma_n)$ is closed in $E(\gamma_n)$, and by lemma 7.1.18(3) a subset of $S(\gamma_n)$, or of $S(\gamma_n) \times I$, is open exactly when it meets every S_M, or every $S_M \times I$, in a relatively open set. So a map out of $S(\gamma_n)$ or $S(\gamma_n) \times I$ is continuous once it is continuous on the stages.

Part (1). Over the set U_X of lemma 7.1.18(1), the map $(Y, x) \mapsto (Y, y/|y|)$, with $(Y, y) = h_X(Y, x)$, is a bijection $U_X \times S(X) \to \rho^{-1}(U_X)$ over U_X, with inverse $(Y, y) \mapsto \big(Y, p_X(y)/|p_X(y)|\big)$. Both are continuous on the stages, hence continuous, so ρ is locally trivial with fiber the unit sphere $S(X) \cong S^{rn-1}$.

Part (2). On frames, j is $x \mapsto (e_1, sx_1, \ldots, sx_{n-1})$ followed by $y \mapsto (q(y), y_1)$, so it is continuous on each $\mathrm{Gr}_{n-1}(\mathbb{K}^M)$, hence continuous, and $\rho \circ j = i_n$ by the definition of i_n. Let $\psi\colon S(\gamma_n) \to \mathrm{Gr}_{n-1}$, $\psi(W, v) = W \cap v^\perp$. On S_M the matrix of the orthogonal projection onto $W \cap v^\perp$ is $P_W - vv^*$, continuous in (W, v), and $V \mapsto P_V$ embeds the compact space $\mathrm{Gr}_{n-1}(\mathbb{K}^M)$ into the matrices, so ψ is continuous. For $t \in [0, 1]$ the map $L_t = (1-t)\mathrm{id} + ts$ is injective on $\mathbb{K}^\infty$: if $z \neq 0$ has last nonzero coordinate z_k, the $(k+1)$-st coordinate of $L_t z$ is tz_k, and $L_0 = \mathrm{id}$. Since $\psi(j(V)) = s(V)$, the map $(V, t) \mapsto L_t(V)$ is a homotopy from the identity to $\psi \circ j$. In the other direction, $(W, v, t) \mapsto \big(L_t(W), L_t v/|L_t v|\big)$ is a homotopy from the identity of $S(\gamma_n)$ to $(W, v) \mapsto \big(s(W), s(v)\big)$. Put $u = s(v)$, a unit vector orthogonal to e_1, and $\theta = \pi t/2$. The map

$$R_t(x) = x + (\cos\theta - 1)\big(\langle x, u\rangle u + \langle x, e_1\rangle e_1\big) + \sin\theta\big(\langle x, u\rangle e_1 - \langle x, e_1\rangle u\big)$$

acts on the orthonormal pair u, e_1 as a rotation and fixes their orthogonal complement, so it is $\mathbb{K}$-linear and unitary, with $R_1(u) = e_1$. Since $s(W) = \mathbb{K}u \oplus s(W \cap v^\perp)$ with $s(W \cap v^\perp)$ orthogonal to u and e_1, the map $(W, v, t) \mapsto \big(R_t(s(W)), R_t(u)\big)$ is a homotopy from $\big(s(W), s(v)\big)$ to $\big(\mathbb{K}e_1 \oplus s(W \cap v^\perp), e_1\big) = j(\psi(W, v))$. Each of these homotopies is given on frames by Gram-Schmidt applied to the images of the frame vectors under L_t or R_t, the first vector going to the image of v, so they are continuous on the stages and hence continuous. Thus $\psi \circ j \simeq \mathrm{id}$ and $j \circ \psi \simeq \mathrm{id}$, completing the proof.

The lemma packages the geometry that drives the induction: passing from $U(n)$ to $U(n-1)$ is the same as passing from the universal rank-n bundle to its unit sphere bundle, and the cohomological cost of that passage is governed by one Euler class through the Gysin sequence. Because the fiber is the odd sphere S^{2n-1}, the Gysin sequence has its differential in degree $2n$, and the Euler class of the universal bundle lives in $H^{2n}(BU(n))$. This single class is the top Chern class c_n, and the induction shows that adjoining it to the lower Chern classes is exactly what builds $H^*(BU(n))$ from $H^*(BU(n-1))$. The base case $n=1$ is example 7.2.3.

Theorem 7.2.5: The Cohomology of $BU(n)$

For every $n \geq 0$ there is an isomorphism of graded rings

$$H^*(BU(n);\mathbb{Z}) \cong \mathbb{Z}[c_1, c_2, \ldots, c_n], \qquad |c_i| = 2i$$

a polynomial ring on n generators c_i in degree $2i$. The generators may be chosen so that the stabilization $i_n\colon BU(n-1) \to BU(n)$ of proposition 7.1.19 satisfies $i_n^* c_i = c_i$ for $i \leq n-1$ and $i_n^* c_n = 0$, and so that for $n \geq 2$ the top generator c_n is the Euler class, in the Gysin sequence of theorem 6.3.2, of the sphere bundle $\rho\colon S(\gamma_n) \to BU(n)$ of lemma 7.2.4, for the orientation of $S(\gamma_n)$ induced by the complex orientation of γ_n (definition 7.1.25). In particular the cohomology is concentrated in even degrees and is torsion-free.

***Proof*:**
The argument is induction on n, the inductive step run through the Gysin sequence of the fibration of lemma 7.2.4.

Step 1: Base cases. For $n=0$ the group $U(0)$ is trivial, $BU(0)$ is a point, and $H^*(\mathrm{pt};\mathbb{Z}) = \mathbb{Z} = \mathbb{Z}[\,]$, the empty polynomial ring. For $n=1$, example 7.2.3 gives $H^*(BU(1);\mathbb{Z}) = \mathbb{Z}[c_1]$ with $|c_1| = 2$, the case $n=1$ of the claim.

Step 2: The Gysin sequence of the universal sphere bundle. Assume $n \geq 2$ and that $H^*(BU(n-1);\mathbb{Z}) \cong \mathbb{Z}[c_1, \ldots, c_{n-1}]$ with $|c_i| = 2i$, concentrated in even degrees and torsion-free. The sphere bundle $\rho\colon S(\gamma_n) \to BU(n)$ of lemma 7.2.4 has fiber the odd sphere S^{2n-1}, with $k = 2n-1$. Its base $BU(n)$ is a CW complex with a single 0-cell (theorem 7.1.17), hence path-connected, as theorem 6.3.2 requires, and the orientation of $S(\gamma_n)$ fixed in the statement trivializes the local system of fiber cohomology (definition 7.1.25). So the Gysin sequence applies. Let $e \in H^{2n}(BU(n);\mathbb{Z})$ be its Euler class. The homotopy equivalence j of lemma 7.2.4 induces a ring isomorphism $j^*\colon H^*(S(\gamma_n)) \to H^*(BU(n-1))$ with $j^*\rho^* = i_n^*$. Through j^*, identify $H^*(S(\gamma_n))$ with $H^*(BU(n-1))$, so that ρ^* becomes i_n^*. The sequence reads

$$\cdots \to H^m(BU(n)) \xrightarrow{\smile e} H^{m+2n}(BU(n)) \xrightarrow{\rho^*} H^{m+2n}(BU(n-1))$$
$$\xrightarrow{\int} H^{m+1}(BU(n)) \xrightarrow{\smile e} H^{m+2n+1}(BU(n)) \to \cdots$$

where $\int = \int_{S^{2n-1}}$ lowers degree by $2n-1$.

Step 3: ρ^* is surjective and $\int$ vanishes. The inductive hypothesis gives $H^*(BU(n-1)) = \mathbb{Z}[c_1, \ldots, c_{n-1}]$ concentrated in even degrees. The fiber-integration $\int$ lowers degree by the odd number $2n-1$, so it carries the even-degree group $H^{m+2n}(BU(n-1))$ into the odd-degree group $H^{m+1}(BU(n))$ whenever m is even, and the even-degree group into the even-degree group only when m is odd. The strategy is to prove by induction on total degree that $H^{\mathrm{odd}}(BU(n)) = 0$

and that ρ^* is surjective. The two statements feed each other through the sequence. Suppose, as an inner induction on d, that $H^j(BU(n)) = 0$ for all odd $j < d$. Consider the segment of the Gysin sequence ending in $H^d(BU(n))$ with d odd:

$$H^{d-2n}(BU(n)) \xrightarrow{\smile e} H^d(BU(n)) \xrightarrow{\rho^*} H^d(BU(n-1))$$

Here $H^d(BU(n-1)) = 0$ because d is odd and the fiber cohomology is even, so ρ^* from $H^d(BU(n))$ is the zero map and its kernel is all of $H^d(BU(n))$. Exactness gives $H^d(BU(n)) = \mathrm{im}(\smile e)$, the image of $\smile e\colon H^{d-2n}(BU(n)) \to H^d(BU(n))$. Since e has even degree $2n$, $\smile e$ sends the odd-degree group $H^{d-2n}(BU(n))$ (note $d-2n$ is odd) into H^d, and $H^{d-2n}(BU(n)) = 0$ by the inner inductive hypothesis ($d - 2n < d$ odd). Hence $H^d(BU(n)) = 0$. By induction on d, the cohomology of $BU(n)$ vanishes in all odd degrees.

With $H^{\mathrm{odd}}(BU(n)) = 0$ established, return to the general segment

$$H^{m+2n}(BU(n)) \xrightarrow{\rho^*} H^{m+2n}(BU(n-1)) \xrightarrow{\int} H^{m+1}(BU(n))$$

When $m+2n$ is even (that is m even), the term $H^{m+1}(BU(n))$ is odd-degree, hence zero, so $\int = 0$ on $H^{m+2n}(BU(n-1))$ and exactness forces ρ^* to be surjective onto the even-degree groups. When $m + 2n$ is odd, both $H^{m+2n}(BU(n-1))$ (odd, fiber even) and the source $H^{m+2n}(BU(n))$ (odd, just shown zero) vanish, so there is nothing to check. Therefore $\rho^*\colon H^*(BU(n)) \to H^*(BU(n-1))$ is surjective in every degree, and $\int = 0$ throughout.

Step 4: $\smile e$ is injective, and the structure of the ring. With $\int = 0$, the Gysin sequence breaks into short exact sequences

$$0 \to H^m(BU(n)) \xrightarrow{\smile e} H^{m+2n}(BU(n)) \xrightarrow{\rho^*} H^{m+2n}(BU(n-1)) \to 0$$

for every m. Exactness at $H^m(BU(n))$ says $\ker(\smile e) = \mathrm{im}(\int) = 0$ (the incoming $\int$ is zero), so $\smile e$ is injective. Exactness at $H^{m+2n}(BU(n))$ says $\mathrm{im}(\smile e) = \ker(\rho^*)$, and surjectivity of ρ^* was Step 3. Thus for each m,

$$H^{m+2n}(BU(n)) \cong H^m(BU(n)) \cdot e \;\oplus\; \big(\text{a lift of } H^{m+2n}(BU(n-1))\big)$$

as groups, the first summand the image of $\smile e$ and the second a section of ρ^* (which exists because the groups are free abelian, being torsion-free of finite rank in each degree). Set $c_n := e \in H^{2n}(BU(n))$, and for $i < n$ let $c_i \in H^{2i}(BU(n))$ be any class with $\rho^* c_i = c_i$ (the lower Chern class of $BU(n-1)$). Such a lift exists by surjectivity of ρ^*. These choices give a ring homomorphism

$$\Phi\colon \mathbb{Z}[c_1, \dots, c_n] \longrightarrow H^*(BU(n); \mathbb{Z})$$

sending each polynomial generator to the chosen class. To see Φ is an isomorphism, filter $\mathbb{Z}[c_1, \dots, c_n]$ by powers of c_n: every element is uniquely $\sum_{j \geq 0} p_j\, c_n^j$ with $p_j \in \mathbb{Z}[c_1, \dots, c_{n-1}]$. The short exact sequence above says precisely that $H^*(BU(n))$ is a free module over the subring generated by $c_1, \dots, c_{n-1}$ with basis the powers $\{c_n^j\}_{j \geq 0}$: applying the short exact sequence iteratively, multiplication by $e = c_n$ is injective with cokernel ρ^*-isomorphic to $H^*(BU(n-1)) = \mathbb{Z}[c_1, \dots, c_{n-1}]$, so

$$H^*(BU(n)) \cong H^*(BU(n-1))[c_n] = \mathbb{Z}[c_1, \dots, c_{n-1}][c_n] = \mathbb{Z}[c_1, \dots, c_n]$$

by induction on the power of c_n and dimension count in each degree. This is exactly the statement that Φ is an isomorphism. The relation $\rho^* c_i = c_i$ for $i \leq n-1$ holds by construction,

and $\rho^* c_n = \rho^* e = 0$ because the Euler class restricts to zero on the total space of its own sphere bundle (the class $e = d_{2n}(1 \otimes \alpha)$ lies in the kernel of ρ^* as $\mathrm{im}(\smile e) = \ker \rho^*$ contains $e = 1 \smile e$). Through the identification of Step 2 these relations are $i_n^* c_i = c_i$ and $i_n^* c_n = 0$. The induction closes, and the cohomology is concentrated in even degrees and torsion-free, completing the proof.

The theorem is the central computation of the chapter, and its shape repays attention. The cohomology of $BU(n)$ is as simple as a graded ring can be that is not a single copy of $\mathbb{Z}$: a polynomial ring, free of relations, on one generator in each even degree up to $2n$. The generator c_n is the Euler class of the universal bundle, born as the transgression of the fiber generator in the Gysin sequence. The lower generators c_i are inherited from $BU(n-1)$ and ultimately from the line-bundle class of example 7.2.3. The absence of relations is what makes Chern classes independent invariants: no polynomial identity holds among the c_i universally, so a characteristic class of complex rank-n bundles is precisely a polynomial in the c_i with integer coefficients, and two such polynomials that differ give distinct invariants. The even-degree, torsion-free conclusion also means there is no information lost in passing to rational or complex coefficients, which is why the Chern-Weil construction of the Riemannian-geometry volume [14] over $\mathbb{R}$ recovers the same classes.

Taking the colimit over n assembles the stable theory, where the rank is allowed to grow without bound and the polynomial ring acquires infinitely many generators.

Corollary 7.2.6: The Cohomology of BU

Let BU be the colimit of the stabilizations $i_n \colon BU(n-1) \to BU(n)$ (definition 7.1.20). Then

$$H^*(BU;\mathbb{Z}) \cong \mathbb{Z}[c_1, c_2, c_3, \dots], \qquad |c_i| = 2i$$

the polynomial ring on countably many generators, one in each positive even degree, where c_i restricts to the generator c_i of theorem 7.2.5 on every $BU(n)$ with $n \geq i$.

***Proof*:**
By theorem 7.2.5, $i_n^* c_j = c_j$ for $j \leq n-1$ and $i_n^* c_n = 0$, so i_n^* is onto in every degree, with kernel the ideal generated by c_n, and it is an isomorphism in degrees below $2n$. Every compact subset of BU lies in some $BU(m)$ (corollary 7.1.21), so lemma 7.1.16, applied to $(BU, \emptyset)$ and the subspaces $BU(m)$, makes the restriction $H^k(BU;\mathbb{Z}) \to H^k(BU(m);\mathbb{Z})$ bijective whenever $2m \geq k$. Let $c_i \in H^{2i}(BU;\mathbb{Z})$ be the class restricting to c_i on $BU(i)$. It restricts to c_i on every $BU(m)$ with $m \geq i$, since the restriction from $BU(m)$ to $BU(i)$ is bijective in degree $2i$ and carries c_i to c_i. The ring homomorphism $\mathbb{Z}[c_1, c_2, \dots] \to H^*(BU;\mathbb{Z})$ so defined is bijective in each degree k: followed by the bijective restriction to $BU(m)$ with $2m \geq k$, it is the isomorphism of theorem 7.2.5 in degree k, the generators c_i with $i > m$ having degree above k. This completes the proof.

The stable ring $H^*(BU;\mathbb{Z}) = \mathbb{Z}[c_1, c_2, \dots]$ is the universal home of Chern classes: every complex vector bundle of every rank has a classifying map into BU (after stabilizing by adding trivial summands), and pulling back the universal c_i gives the Chern classes of the bundle, independent of the rank used to define them. This stability is what makes the total Chern class, introduced next, multiplicative under Whitney sum, and it is the cohomological counterpart of the fact that BU classifies the reduced K-theory $\widetilde{K}^0$ of compact connected spaces (section 8.5 and [33, Representability of K^0]).

With the ring computed, the named classes can be defined. The Chern classes are the polynomial generators, pulled back to a bundle. The total Chern class packages them into a single inhomogeneous class whose virtue is multiplicativity.

Definition 7.2.7: Chern Classes and the Total Chern Class

Let $E \to X$ be a complex vector bundle of rank n over a paracompact Hausdorff base X, with classifying map $f_E \colon X \to BU(n)$, unique up to homotopy by theorem 7.1.29. For $1 \le i \le n$ the *i-th Chern class* of E is

$$c_i(E) := f_E^* \, c_i \in H^{2i}(X;\mathbb{Z})$$

the pullback of the polynomial generator $c_i \in H^{2i}(BU(n);\mathbb{Z})$ of theorem 7.2.5, and $c_i(E) = 0$ for $i > n$ by convention. The *total Chern class* is the inhomogeneous sum

$$c(E) := 1 + c_1(E) + c_2(E) + \cdots + c_n(E) \in H^*(X;\mathbb{Z})$$

a unit in the cohomology ring (its degree-zero part is 1).

The definition is forced by proposition 7.2.2: the Chern classes are the images of the universal generators, so they are the characteristic classes from which all others (the polynomials in them) are built. The total class is a bookkeeping device whose payoff is the Whitney sum formula $c(E \oplus E') = c(E) \smile c(E')$, proved in theorem 7.4.5 by the splitting principle. The formula says that the total Chern class converts direct sum of bundles into cup product of cohomology classes, which is exactly the multiplicativity one needs to compute Chern classes of bundles built from line bundles. The convention $c_i(E) = 0$ for $i > n$ records that a rank-n bundle has no Chern classes above degree $2n$, since $H^{>2n}(BU(n))$ has no generators not already products of lower ones.

The real case runs in parallel, with two changes. Coefficients must be taken in $\mathbb{Z}/2$, because the relevant sphere bundles need not be orientable and torsion appears integrally. The generators land in every degree, not only the even ones, because the fiber sphere S^{n-1} has its top class in degree $n-1$. The structural mechanism is identical: a universal sphere bundle relates $BO(n-1)$ to $BO(n)$, and the Gysin sequence with $\mathbb{Z}/2$ coefficients adjoins one Stiefel-Whitney class at each stage.

Lemma 7.2.8: The Universal Sphere Bundles for $O(n)$ and $SO(n)$

Let $n \ge 1$. The sphere bundle $\rho \colon S(\gamma_n) \to BO(n)$ is a fiber bundle with fiber S^{n-1}, and $j \colon BO(n-1) \to S(\gamma_n)$, $j(V) = (\mathbb{R}e_1 \oplus s(V), e_1)$, is a homotopy equivalence with $\rho \circ j = i_n$. The sphere bundle $\tilde{\rho} \colon S(\tilde{\gamma}_n) \to BSO(n)$ of the oriented tautological bundle is a fiber bundle with fiber S^{n-1}, and $\tilde{\jmath}(V,o) = (\mathbb{R}e_1 \oplus s(V), e_1)$, the plane $\mathbb{R}e_1 \oplus s(V)$ oriented by e_1 followed by $s(o)$, is a homotopy equivalence $BSO(n-1) \to S(\tilde{\gamma}_n)$ with $\tilde{\rho} \circ \tilde{\jmath} = \tilde{\imath}_n$, the oriented stabilization $\tilde{\imath}_n(V,o) = (\mathbb{R}e_1 \oplus s(V),\ e_1$ followed by $s(o))$. The pullback $\tilde{\imath}_n^* \tilde{\gamma}_n$ is isomorphic to $\underline{\mathbb{R}} \oplus \tilde{\gamma}_{n-1}$.

***Proof*:**
The unoriented statement is lemma 7.2.4 for $\mathbb{K} = \mathbb{R}$. In the oriented case the same proof runs on oriented frames. Replace q by $\tilde{q}$ and ψ by $(W,o,v) \mapsto (W \cap v^\perp, o')$, where o' is the orientation for which v followed by o' is o. The maps L_t and R_t carry an oriented frame to a frame that Gram-Schmidt turns into an oriented frame of the image plane, and at the end of the last homotopy the frame $\big(e_1, R_1(sx_2), \ldots, R_1(sx_n)\big)$ carries the orientation of $\tilde{\jmath}\big(\psi(W,o,v)\big)$. The local trivializations and the topology of $S(\tilde{\gamma}_n)$ come from those of $S(\gamma_n)$ through the sheets of

the covering π of lemma 7.1.18(2). Finally $\pi \circ \tilde{\imath}_n = i_n \circ \pi$ and $\tilde{\gamma}_k = \pi^* \gamma_k$ as vector bundles, so $\tilde{\imath}_n^* \tilde{\gamma}_n = \pi^* i_n^* \gamma_n \cong \underline{\mathbb{R}} \oplus \tilde{\gamma}_{n-1}$ by proposition 7.1.19(3), completing the proof.

The real sphere bundle differs from the complex one in that its fiber S^{n-1} may be even-dimensional, and over $\mathbb{Z}$ the local coefficient system $H^{n-1}(S^{n-1};\mathbb{Z})$ need not be trivial: monodromy around a loop in $BO(n)$ can reverse the orientation of the fiber sphere, exactly when the bundle is non-orientable. This is why integral coefficients fail to give a clean polynomial answer. Over $\mathbb{Z}/2$ the obstruction evaporates, because $H^{n-1}(S^{n-1};\mathbb{Z}/2) = \mathbb{Z}/2$ has a unique nonzero element and no orientation to reverse, so the local system is automatically trivial and the Gysin sequence of theorem 6.3.2 applies with $\mathbb{Z}/2$ coefficients.

Lemma 7.2.9: The Mod-2 Euler Class of a Sphere Bundle

For a real vector bundle E of rank $m \geq 2$ over a path-connected CW complex B, let $e_2(E) \in H^m(B;\mathbb{Z}/2)$ be the Euler class of the Gysin sequence, with $\mathbb{Z}/2$ coefficients, of the sphere bundle $S(E) \to B$ of a Euclidean inner product on E.

1. The class $e_2(E)$ depends only on the isomorphism class of E, and $e_2(g^*E) = g^* e_2(E)$ for every map $g\colon B' \to B$ from a path-connected CW complex.
2. If E has a nowhere-zero section, then $e_2(E) = 0$.
3. Let $\gamma_1 \to \mathbb{R}\mathrm{P}^\infty$ be the tautological line bundle, t the generator of $H^1(\mathbb{R}\mathrm{P}^\infty;\mathbb{Z}/2)$, and $m\gamma_1$ the sum of m copies of γ_1. Then $e_2(m\gamma_1) = t^m$.

***Proof*:**

The base is paracompact Hausdorff (theorem 7.1.7), so an inner product exists (lemma 7.1.22), and the Gysin sequence applies with $\mathbb{Z}/2$ coefficients, as just noted. As in the proof of theorem 6.3.2, $e_2(E) = d_m(1 \otimes \alpha)$ for the generator α of $H^{m-1}(S^{m-1};\mathbb{Z}/2)$.

Part (1). Let $G\colon S(g^*E) \to S(E)$ be the map over g that is the identity from the fiber over b', the unit sphere of $E_{g(b')}$ for the pulled-back inner product, to the fiber over $g(b')$. By the naturality clause of theorem 6.2.8, with basepoints b'_0 and $g(b'_0)$, G^* induces a morphism of spectral sequences commuting with every differential, which is g^* on the bottom row $H^*(B;\mathbb{Z}/2)$ and carries α to the generator α of the left column, G being a homeomorphism on each fiber. Hence

$$g^* e_2(E) = G^* d_m(1 \otimes \alpha) = d_m\big(G^*(1 \otimes \alpha)\big) = d_m(1 \otimes \alpha) = e_2(g^*E)$$

The case $g = \mathrm{id}_B$ shows that a homeomorphism $S(E') \to S(E)$ over B carries $e_2(E)$ to $e_2(E')$. Two inner products on E, or an isomorphism $E' \cong E$, give such a homeomorphism by rescaling every vector to unit length, so $e_2(E)$ depends only on the isomorphism class of E.

Part (2). Normalizing a nowhere-zero section gives a section s of the projection $p\colon S(E) \to B$. Then $s^*p^* = \mathrm{id}$, so p^* is injective, and exactness $\ker p^* = \mathrm{im}(\smile e_2(E))$ at $H^m(B;\mathbb{Z}/2)$ gives $e_2(E) = 1 \smile e_2(E) = 0$.

Part (3). A point of $S(m\gamma_1)$ is a line $\ell \in \mathbb{R}\mathrm{P}^\infty$ with vectors $v_1, \ldots, v_m \in \ell$ satisfying $\sum_i |v_i|^2 = 1$, for the sum of the standard inner products. Writing $v_i = c_i x$ for a unit vector $x \in \ell$ gives a unit vector $c \in \mathbb{R}^m$, determined up to sign by the point, and $\pi(\ell; v) = \pm c$ defines a map $\pi\colon S(m\gamma_1) \to \mathbb{R}\mathrm{P}^{m-1}$. It is continuous, as x can be chosen continuously near any line, the double cover $S^\infty \to \mathbb{R}\mathrm{P}^\infty$ of example 7.1.6 having local sections. Over an open set $U \subseteq \mathbb{R}\mathrm{P}^{m-1}$

with a continuous choice $\pm c \mapsto c$, the map $(\ell; v) \mapsto \big(\pi(\ell; v), \sum_i c_i v_i\big)$ is a homeomorphism $\pi^{-1}(U) \to U \times S^\infty$, with inverse $(\pm c, x) \mapsto (\mathbb{R}x; c_1 x, \dots, c_m x)$. So π is a fiber bundle with fiber S^∞, which is contractible (example 7.1.6). By theorem 5.2.4 and theorem 5.2.12, π is a weak homotopy equivalence, so it induces isomorphisms on cohomology (proposition 5.3.15, corollary 4.1.10), and $H^m(S(m\gamma_1); \mathbb{Z}/2) \cong H^m(\mathbb{R}\mathrm{P}^{m-1}; \mathbb{Z}/2) = 0$. In the exact segment

$$H^0(\mathbb{R}\mathrm{P}^\infty; \mathbb{Z}/2) \xrightarrow{\smile e_2(m\gamma_1)} H^m(\mathbb{R}\mathrm{P}^\infty; \mathbb{Z}/2) \xrightarrow{p^*} H^m\big(S(m\gamma_1); \mathbb{Z}/2\big) = 0$$

the first map is therefore onto $H^m(\mathbb{R}\mathrm{P}^\infty; \mathbb{Z}/2)$, whose nonzero element is t^m. So $e_2(m\gamma_1) = t^m$, completing the proof.

Parts (1) and (2) say that e_2 is a natural obstruction to a nowhere-zero section, and part (3) is the one computation that shows it can be nonzero in every rank.

Theorem 7.2.10: The Cohomology of $BO(n)$

For every $n \geq 0$ there is an isomorphism of graded rings

$$H^*(BO(n); \mathbb{Z}/2) \cong \mathbb{Z}/2[w_1, w_2, \dots, w_n], \qquad |w_i| = i$$

a polynomial ring over $\mathbb{Z}/2$ on n generators w_i in degree i. The generators may be chosen so that the stabilization $i_n \colon BO(n-1) \to BO(n)$ satisfies $i_n^* w_i = w_i$ for $i \leq n-1$ and $i_n^* w_n = 0$, and so that for $n \geq 2$ the top generator w_n is the Euler class, with $\mathbb{Z}/2$ coefficients, of the Gysin sequence of the sphere bundle $\rho \colon S(\gamma_n) \to BO(n)$ of lemma 7.2.8.

***Proof*:**

The argument is induction on n via the Gysin sequence of lemma 7.2.8 with $\mathbb{Z}/2$ coefficients, parallel to theorem 7.2.5 but without the even-degree restriction. All cohomology in this proof is with $\mathbb{Z}/2$ coefficients.

Step 1: Base cases. For $n = 0$, $BO(0)$ is a point and $H^*(\mathrm{pt}; \mathbb{Z}/2) = \mathbb{Z}/2$, the empty polynomial ring. For $n = 1$, the group $O(1) = \{\pm 1\} \cong \mathbb{Z}/2$ is discrete, so $BO(1) = B\mathbb{Z}/2 = \mathbb{R}\mathrm{P}^\infty$ (example 5.5.6), and its mod-2 cohomology is

$$H^*(\mathbb{R}\mathrm{P}^\infty; \mathbb{Z}/2) \cong \mathbb{Z}/2[w_1], \qquad |w_1| = 1$$

the polynomial ring on a single degree-one class, computed in example 5.5.9. This is the case $n = 1$.

Step 2: The Gysin sequence. Assume $n \geq 2$ and $H^*(BO(n-1); \mathbb{Z}/2) \cong \mathbb{Z}/2[w_1, \dots, w_{n-1}]$ with $|w_i| = i$. The sphere bundle $\rho \colon S(\gamma_n) \to BO(n)$ of lemma 7.2.8 has fiber S^{n-1}, so $k = n-1$ in theorem 6.3.2. Its base $BO(n)$ is a CW complex with a single 0-cell (theorem 7.1.17), hence path-connected, as theorem 6.3.2 requires. With $\mathbb{Z}/2$ coefficients the orientation hypothesis holds automatically, as noted above. The homotopy equivalence j of lemma 7.2.8 induces a ring isomorphism j^* from $H^*(S(\gamma_n))$ to $H^*(BO(n-1))$ with $j^* \rho^* = i_n^*$. Through j^*, identify the two rings, so that ρ^* becomes i_n^*. The Euler class $e \in H^n(BO(n); \mathbb{Z}/2)$ and the Gysin sequence read

$$\cdots \to H^m(BO(n)) \xrightarrow{\smile e} H^{m+n}(BO(n)) \xrightarrow{\rho^*} H^{m+n}(BO(n-1))$$
$$\xrightarrow{\int} H^{m+1}(BO(n)) \xrightarrow{\smile e} H^{m+n+1}(BO(n)) \to \cdots$$

with $\int$ lowering degree by $n-1$.

Step 3: $\int = 0$ and ρ^* is surjective in every degree. Unlike the complex case, there is no parity to exploit: the Euler class e has degree n and $\int$ lowers degree by the possibly-even number $n-1$, so the vanishing of H^{odd} is unavailable and a different argument is required. The two statements "$\int = 0$ out of degree d" and "ρ^* is surjective in degree d" are equivalent at each d: the exact segment

$$H^d(BO(n)) \xrightarrow{\rho^*} H^d(BO(n-1)) \xrightarrow{\int} H^{d-n+1}(BO(n))$$

has $\mathrm{im}(\rho^*) = \ker(\int)$ by exactness, so ρ^* is onto $H^d(BO(n-1))$ if and only if $\int = 0$ there. The proof of both, by induction on d, runs through a degree count in the Gysin sequence rather than through ring generation by lower classes.

Low degrees $d < n-1$. The flanking terms of the segment

$$H^{d-n}(BO(n)) \xrightarrow{\smile e} H^d(BO(n)) \xrightarrow{\rho^*} H^d(BO(n-1)) \xrightarrow{\int} H^{d-n+1}(BO(n))$$

vanish for $d < n-1$: the source $H^{d-n}(BO(n)) = 0$ since $d-n < 0$, and the target $H^{d-n+1}(BO(n)) = 0$ since $d-n+1 < 0$. The segment collapses to $0 \to H^d(BO(n)) \xrightarrow{\rho^*} H^d(BO(n-1)) \to 0$, so ρ^* is an isomorphism, hence surjective, in every degree $d < n-1$, and $\int = 0$ there.

The degree $d = n-1$, where the content sits. Here the source $H^{d-n}(BO(n)) = H^{-1}(BO(n)) = 0$ still, but the target $H^{d-n+1}(BO(n)) = H^0(BO(n)) = \mathbb{Z}/2$ does not vanish, so flank vanishing alone gives only injectivity of ρ^*, not surjectivity. Surjectivity of ρ^* in degree $n-1$ is equivalent, by the exactness $\ker(\smile e) = \mathrm{im}(\int)$ at the node $H^0(BO(n))$, to the injectivity of $\smile e\colon H^0(BO(n)) \to H^n(BO(n))$, that is to $e \neq 0$ in $H^n(BO(n); \mathbb{Z}/2)$. The class e is $e_2(\gamma_n)$ in the notation of lemma 7.2.9, and that lemma computes its pullback to $BO(1)^n = (\mathbb{R}\mathrm{P}^\infty)^n$. This product is a CW complex (lemma 1.2.12(5)), with $H^*(BO(1)^n; \mathbb{Z}/2) = \mathbb{Z}/2[t_1, \dots, t_n]$ by the Künneth theorem over the field $\mathbb{Z}/2$, where t_i is pulled back from the generator t of the i-th factor and $H^*(\mathbb{R}\mathrm{P}^\infty; \mathbb{Z}/2) = \mathbb{Z}/2[t]$ is the base case of Step 1 (example 5.5.6). Let ℓ_i be the tautological line bundle pulled back from the i-th factor, let $f\colon BO(1)^n \to BO(n)$ classify $\ell_1 \oplus \cdots \oplus \ell_n$ (theorem 7.1.29), and put $P = f^*e$, which equals $e_2(\ell_1 \oplus \cdots \oplus \ell_n)$ by lemma 7.2.9(1) and is a homogeneous polynomial of degree n. A permutation of the factors pulls $\ell_1 \oplus \cdots \oplus \ell_n$ back to a reordering of itself, so it fixes P by lemma 7.2.9(1), and P is symmetric. The map $j\colon BO(1)^{n-1} \to BO(1)^n$, $j(x) = (x, b_0)$, pulls ℓ_n back to a trivial bundle, so $j^*(\ell_1 \oplus \cdots \oplus \ell_n)$ has a nowhere-zero section, and $j^*P = 0$ by lemma 7.2.9(1),(2). As j^* sets $t_n = 0$, the polynomial P is divisible by t_n, by symmetry by every t_i, and so by the product $t_1 \cdots t_n$ of these distinct primes of the unique factorization domain $\mathbb{Z}/2[t_1, \dots, t_n]$. Comparing degrees, $P = \lambda\, t_1 \cdots t_n$ with $\lambda \in \mathbb{Z}/2$. The diagonal $\Delta\colon \mathbb{R}\mathrm{P}^\infty \to BO(1)^n$ pulls $\ell_1 \oplus \cdots \oplus \ell_n$ back to $n\gamma_1$ and every t_i to t, so lemma 7.2.9(1),(3) give $\lambda\, t^n = \Delta^* P = e_2(n\gamma_1) = t^n \neq 0$. Hence

$$f^*e = t_1 \cdots t_n \neq 0$$

and $e \neq 0$. Therefore $\smile e$ is injective on H^0, $\int = 0$ out of degree $n-1$, and ρ^* is surjective in degree $n-1$. Choosing lifts $w_i \in H^i(BO(n))$ with $\rho^* w_i = w_i$ for $i \leq n-1$ (possible by surjectivity in degrees $\leq n-1$) realizes ρ^* as a ring map hitting every generator $w_1, \dots, w_{n-1}$ of $H^*(BO(n-1))$.

High degrees $d \geq n$. Assume inductively that $\int = 0$ and ρ^* is surjective in all degrees $< d$, so $\smile e$ is injective in all degrees $< d-n+1$ (the equivalence above, read one node lower, shows

$\smile e\colon H^j(BO(n)) \to H^{j+n}(BO(n))$ is injective whenever $\int = 0$ out of degree $j+n-1 < d$). The generators $w_1, \dots, w_{n-1}$ of $H^*(BO(n-1))$ have degrees $1, \dots, n-1$, all strictly less than d, and each lies in $\mathrm{im}(\rho^*)$ by the inductive surjectivity (their lifts w_i were fixed at the degree $n-1$ stage). Since ρ^* is a ring homomorphism and $H^d(BO(n-1))$ is spanned over $\mathbb{Z}/2$ by the degree-d monomials in $w_1, \dots, w_{n-1}$, each a product of generators of degree $< d$ already in $\mathrm{im}(\rho^*)$, the whole group $H^d(BO(n-1))$ lies in $\mathrm{im}(\rho^*)$. Thus ρ^* is surjective in degree d, and $\int = 0$ there. The induction closes: ρ^* is surjective and $\int = 0$ in every degree, and consequently, by exactness $\ker(\smile e) = \mathrm{im}(\int) = 0$ at each node, $\smile e$ is injective in every degree.

Step 4: The structure of the ring. With $\int = 0$ the Gysin sequence breaks into short exact sequences

$$0 \to H^m(BO(n)) \xrightarrow{\smile e} H^{m+n}(BO(n)) \xrightarrow{\rho^*} H^{m+n}(BO(n-1)) \to 0$$

Set $w_n := e \in H^n(BO(n))$ and keep the lifts w_i $(i < n)$ of Step 3. The short exact sequence says $H^*(BO(n))$ is a free module over the subring generated by $w_1, \dots, w_{n-1}$ on the powers $\{w_n^j\}_{j \geq 0}$, by the same module argument as Step 4 of theorem 7.2.5 ($\smile w_n$ injective with cokernel ρ^*-isomorphic to $H^*(BO(n-1))$). Therefore

$$H^*(BO(n); \mathbb{Z}/2) \cong H^*(BO(n-1); \mathbb{Z}/2)[w_n] = \mathbb{Z}/2[w_1, \dots, w_{n-1}][w_n] = \mathbb{Z}/2[w_1, \dots, w_n]$$

The relations $\rho^* w_i = w_i$ for $i \leq n-1$ hold by construction and $\rho^* w_n = \rho^* e = 0$ because $e \in \ker \rho^* = \mathrm{im}(\smile e)$. Through the identification of Step 2 these are the relations $i_n^* w_i = w_i$ and $i_n^* w_n = 0$. The induction closes, establishing the polynomial ring on generators w_i of degree i, completing the proof.

The mod-2 cohomology of $BO(n)$ has the same polynomial shape as the integral cohomology of $BU(n)$, with the single change that the generators are spread across all degrees 1 through n rather than confined to even degrees. The reason for the degree shift is the real sphere fiber S^{n-1}, whose top class sits in degree $n-1$ and produces an Euler class in degree n, against the complex fiber S^{2n-1} producing a class in degree $2n$. The reason for the mod-2 coefficients is orientation: integrally the universal real bundle is non-orientable, its sphere bundle carries a nontrivial $\mathbb{Z}$-local system, and the Gysin machinery stalls. Over $\mathbb{Z}/2$ orientation is automatic and the computation proceeds. The integral cohomology of $BO(n)$ is correspondingly more intricate, carrying 2-torsion together with the Pontryagin and Euler classes taken up in section 7.6.

Corollary 7.2.11: The Cohomology of BO

Let BO be the colimit of the stabilizations $i_n\colon BO(n-1) \to BO(n)$ (definition 7.1.20). Then

$$H^*(BO; \mathbb{Z}/2) \cong \mathbb{Z}/2[w_1, w_2, w_3, \dots], \qquad |w_i| = i$$

the polynomial ring over $\mathbb{Z}/2$ on countably many generators, one in each positive degree, where w_i restricts to the generator w_i of theorem 7.2.10 on every $BO(n)$ with $n \geq i$.

***Proof*:**
The proof of corollary 7.2.6 applies with theorem 7.2.10 in place of theorem 7.2.5. The relations $i_n^* w_j = w_j$ for $j \leq n-1$ and $i_n^* w_n = 0$ make i_n^* onto with kernel the ideal generated by w_n, an isomorphism in degrees below n, and lemma 7.1.16 makes the restriction $H^k(BO; \mathbb{Z}/2) \to$

$H^k(BO(m);\mathbb{Z}/2)$ bijective whenever $m \geq k$, completing the proof.

Corollary 7.2.12: The Mod-2 Cohomology of $BSO(n)$

Let $\pi\colon BSO(n) \to BO(n)$ forget the orientation. For every $n \geq 1$

$$H^*(BSO(n);\mathbb{Z}/2) \cong \mathbb{Z}/2[w_2,\dots,w_n], \qquad |w_i| = i$$

a polynomial ring on the pullbacks $w_i = \pi^* w_i$ of the generators of theorem 7.2.10, and $\pi^* w_1 = 0$.

***Proof*:**
All cohomology is with $\mathbb{Z}/2$ coefficients, and the proof is induction on n. The map $\tilde{q}\colon V_1(\mathbb{R}^\infty) \to BSO(1)$ is a principal bundle for the trivial group $SO(1)$ (lemma 7.1.18(2)), hence a homeomorphism, so $BSO(1)$ is the contractible sphere S^∞ (example 7.1.6), which is the case $n = 1$. Let $n \geq 2$ and assume the claim for $n-1$. The space $BSO(n)$ is a CW complex (theorem 7.1.17), path-connected as the image of $V_n(\mathbb{R}^\infty)$ (lemma 7.1.18(4)). The sphere bundle $\tilde{\rho}\colon S(\tilde{\gamma}_n) \to BSO(n)$ of lemma 7.2.8 has fiber S^{n-1}, so its Gysin sequence with $\mathbb{Z}/2$ coefficients applies, as in the proof of theorem 7.2.10. As vector bundles $\tilde{\gamma}_n = \pi^*\gamma_n$, so its Euler class is $e_2(\tilde{\gamma}_n) = \pi^* e_2(\gamma_n) = \pi^* w_n$ by lemma 7.2.9(1) and theorem 7.2.10. Through the homotopy equivalence $\tilde{\jmath}$ of lemma 7.2.8 the map $\tilde{\rho}^*$ becomes $\tilde{\imath}_n^*$. Forgetting the orientation commutes with stabilization, $\pi \circ \tilde{\imath}_n = i_n \circ \pi$, so $\tilde{\imath}_n^* \pi^* w_i = \pi^* i_n^* w_i = \pi^* w_i$ for $i \leq n-1$. By the inductive hypothesis these classes with $2 \leq i \leq n-1$ generate the ring $H^*(BSO(n-1))$, so the ring map $\tilde{\imath}_n^*$ is onto in every degree. Exactness then makes the integration map of the Gysin sequence vanish and multiplication by $\pi^* w_n$ injective, and the sequence breaks into short exact sequences

$$0 \to H^m(BSO(n)) \xrightarrow{\smile \pi^* w_n} H^{m+n}(BSO(n)) \xrightarrow{\tilde{\imath}_n^*} H^{m+n}(BSO(n-1)) \to 0$$

The module argument of Step 4 of theorem 7.2.5, with the lifts $\pi^* w_2, \dots, \pi^* w_{n-1}$ of the generators of $H^*(BSO(n-1))$, gives $H^*(BSO(n)) = H^*(BSO(n-1))[\pi^* w_n] = \mathbb{Z}/2[w_2,\dots,w_n]$. This ring is zero in degree 1, so $\pi^* w_1 = 0$, completing the proof.

The named real classes are now defined exactly as in the complex case, as the polynomial generators pulled back to a bundle.

Definition 7.2.13: Stiefel-Whitney Classes and the Total Class

Let $E \to X$ be a real vector bundle of rank n over a paracompact Hausdorff base X, with classifying map $f_E \colon X \to BO(n)$, unique up to homotopy by theorem 7.1.29. For $1 \le i \le n$ the *i-th Stiefel-Whitney class* of E is

$$w_i(E) := f_E^* \, w_i \in H^i(X; \mathbb{Z}/2)$$

the pullback of the generator $w_i \in H^i(BO(n); \mathbb{Z}/2)$ of theorem 7.2.10, with $w_i(E) = 0$ for $i > n$. The *total Stiefel-Whitney class* is

$$w(E) := 1 + w_1(E) + w_2(E) + \cdots + w_n(E) \in H^*(X; \mathbb{Z}/2)$$

a unit in the mod-2 cohomology ring.

The Stiefel-Whitney classes are the mod-2 characteristic classes of real bundles, and by proposition 7.2.2 they generate all of them: every mod-2 characteristic class of real rank-n bundles is a $\mathbb{Z}/2$-polynomial in $w_1, \dots, w_n$. The total class again satisfies a Whitney sum formula $w(E \oplus E') = w(E) \smile w(E')$ (theorem 7.4.5), which is the source of the name. The lowest class $w_1(E) \in H^1(X; \mathbb{Z}/2)$ is the obstruction to orientability: it is the image of the bundle under $BO(n) \to BO(1) = \mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2, 1)$ classifying the determinant line bundle $\det E$ (definition 5.2.9), so $w_1(E) = 0$ if and only if the structure group reduces to $SO(n)$, that is, if and only if E is orientable. This is the obstruction-theoretic reading of chapter 5: w_1 is the primary obstruction to a reduction of structure group, living in the cohomology group $H^1(X; \pi_0(O(n)/SO(n))) = H^1(X; \mathbb{Z}/2)$ predicted by obstruction theory.

Example 7.2.14: Stiefel-Whitney Class of the Tautological Bundle over $\mathbb{R}\mathrm{P}^\infty$

Take $X = \mathbb{R}\mathrm{P}^\infty = BO(1)$ and $E = \gamma_1^{\mathbb{R}}$ the universal (tautological) real line bundle, with classifying map the identity. By theorem 7.2.10 with $n = 1$,

$$H^*(BO(1); \mathbb{Z}/2) = H^*(\mathbb{R}\mathrm{P}^\infty; \mathbb{Z}/2) = \mathbb{Z}/2[w_1], \qquad |w_1| = 1$$

and $w_1(\gamma_1^{\mathbb{R}}) = \mathrm{id}^* w_1 = w_1$ is the generator of $H^1(\mathbb{R}\mathrm{P}^\infty; \mathbb{Z}/2) \cong \mathbb{Z}/2$. The total class is $w(\gamma_1^{\mathbb{R}}) = 1 + w_1$. Restricting to a finite skeleton $\mathbb{R}\mathrm{P}^k \subseteq \mathbb{R}\mathrm{P}^\infty$, the tautological line bundle $\gamma_1^{\mathbb{R}}|_{\mathbb{R}\mathrm{P}^k}$ has $w_1 =$ the generator of $H^1(\mathbb{R}\mathrm{P}^k; \mathbb{Z}/2)$, and this class is nonzero, which is the cohomological proof that the tautological bundle over $\mathbb{R}\mathrm{P}^k$ is non-orientable: $w_1 \neq 0$ obstructs orientability. The same class detects, for instance, that the Möbius band (the total space of $\gamma_1^{\mathbb{R}}|_{\mathbb{R}\mathrm{P}^1}$ over the circle $\mathbb{R}\mathrm{P}^1 = S^1$) is non-orientable, w_1 being the generator of $H^1(S^1; \mathbb{Z}/2) = \mathbb{Z}/2$.

The Euler class, which appeared in both computations as the transgressive Gysin generator, deserves a definition in its own right, in the integral oriented setting where it is most informative. For an oriented bundle the structure group is the special orthogonal group $SO(n)$, and the orientation trivializes the local coefficient system, so the integral Gysin sequence runs and produces an integral Euler class. The home of the universal Euler class is the cohomology of $BSO(n)$.

Definition 7.2.15: Euler Class of an Oriented Bundle

Let $E \to X$ be an oriented real vector bundle of rank n over a paracompact Hausdorff base, with classifying map $f_E \colon X \to BSO(n) = \widetilde{\mathrm{Gr}}_n(\mathbb{R}^\infty)$ (lemma 7.1.18(4)), so that $E \cong f_E^* \tilde{\gamma}_n$ as oriented bundles. It exists and is unique up to homotopy by corollary 7.1.30. For $n \geq 2$ the sphere bundle $S(\tilde{\gamma}_n) \to BSO(n)$ of lemma 7.2.8 lies over a CW complex (theorem 7.1.17), path-connected as the image of the path-connected space $V_n(\mathbb{R}^\infty)$ (lemma 7.1.18(4)), and the orientation induced by that of $\tilde{\gamma}_n$ (definition 7.1.25) trivializes its integral local coefficient system, so the integral Gysin sequence of theorem 6.3.2 produces a universal class

$$e \in H^n(BSO(n); \mathbb{Z}), \qquad e := d_n(1 \otimes \alpha)$$

the transgression of the fiber generator α of that orientation. For $n = 1$ the group $H^1(BSO(1); \mathbb{Z})$ is zero, $BSO(1) = \widetilde{\mathrm{Gr}}_1(\mathbb{R}^\infty)$ being the contractible sphere S^∞, and $e = 0$. The *Euler class* of E is the pullback

$$e(E) := f_E^* \, e \in H^n(X; \mathbb{Z})$$

Equivalently, $e(E)$ is the image of the Thom class $u_E \in H^n(\mathrm{Th}(E); \mathbb{Z})$ under the composite $H^n(\mathrm{Th}(E)) \to H^n(E) \xrightarrow{s_0^*} H^n(X)$, where $s_0 \colon X \to E$ is the zero section, the restriction of the Thom class to the zero section.

The Euler class is the primary obstruction to a nowhere-zero section, which is the geometric content that chapter 5 makes precise. A section of E avoiding the zero section is a lift of the classifying map through the sphere bundle $S(E) \to X$, and the primary obstruction to such a lift lives in $H^n(X; \pi_{n-1}(S^{n-1})) = H^n(X; \mathbb{Z})$, the same group as $e(E)$. The obstruction class is exactly the Euler class (theorem 7.4.17). Thus $e(E) = 0$ is necessary for a nowhere-zero section to exist, and for the tangent bundle of a closed oriented n-manifold M the evaluation $\int_M e(TM)$ is the Euler characteristic $\chi(M)$ (corollary 7.4.18). The Thom-class description connects the cohomological definition to the Thom isomorphism that organizes the cobordism theory of chapter 8: the Euler class is the self-intersection of the zero section, and its appearance as the Gysin differential is the spectral-sequence form of that intersection. The reduction mod 2 of the integral Euler class $e(E)$ is the top Stiefel-Whitney class $w_n(E)$, since both are the obstruction to a nowhere-zero section read with the respective coefficients, which links definition 7.2.15 to the $w_n = e$ of theorem 7.2.10.

Exercise 7.2.1

1. Using theorem 7.2.5, compute the Poincaré series $\sum_k \dim_{\mathbb{Q}} H^k(BU(n); \mathbb{Q})\, t^k$ of $H^*(BU(n); \mathbb{Q})$, and verify that it equals $\prod_{i=1}^n (1 - t^{2i})^{-1}$. Then do the same for $H^*(BO(n); \mathbb{Z}/2)$, computing $\sum_k \dim_{\mathbb{Z}/2} H^k(BO(n); \mathbb{Z}/2)\, t^k$ and confirming it equals $\prod_{i=1}^n (1 - t^i)^{-1}$.
2. Show that the restriction $i_n^* \colon H^*(BU(n); \mathbb{Z}) \to H^*(BU(n-1); \mathbb{Z})$ along the stabilization of theorem 7.2.5 is surjective but not injective, and identify its kernel as a principal ideal. Interpret the generator of the kernel as a characteristic class and say which bundle invariant it computes.

7.3 The Thom Isomorphism

The Euler class of definition 7.2.15 was produced by the Gysin sequence, as a universal class on $BSO(n)$ pulled back along a classifying map. That construction is efficient but it hides the geometry: it does not say what $e(E)$ measures on the bundle itself. This section gives the missing description. Attached to an oriented bundle is a single cohomology class of its total space rel the complement of the zero section, the Thom class, and cup product with it is an isomorphism raising degree by the rank. The Euler class is what that class becomes when the relative information is discarded, and the two basic properties of e that the Gysin construction does not give, multiplicativity under Whitney sum and 2-torsion in odd rank, both fall out of the Thom class in a line.

The isomorphism also computes the cohomology of the Thom spectra of section 8.5: the identification $H^*(MO;\mathbb{Z}/2) \cong H^*(BO;\mathbb{Z}/2)$ on which Thom's calculation of $\pi_*(MO)$ rests is exactly this theorem applied to the universal bundles.

Throughout, $E \to B$ is a real vector bundle of rank n over a paracompact Hausdorff base, and $E_0 \subseteq E$ denotes the complement of the zero section. Orientations are those of definition 7.1.25: generators μ_b of $H^n(E_b, (E_b)_0;\mathbb{Z}) \cong \mathbb{Z}$ that come, near each point of B, from a single generator under a trivialization.

Definition 7.3.1: Thom Class

Let $E \to B$ be an oriented real vector bundle of rank n. A *Thom class* for E is a class

$$u_E \in H^n(E, E_0;\mathbb{Z})$$

whose restriction to each fiber pair $(E_b, (E_b)_0) \cong (\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$ is the generator of $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\};\mathbb{Z}) \cong \mathbb{Z}$ determined by the orientation of E_b. For an arbitrary real bundle, a *mod-2 Thom class* $u_E \in H^n(E, E_0;\mathbb{Z}/2)$ is defined by the same condition with $\mathbb{Z}/2$ coefficients, where no orientation is needed because $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\};\mathbb{Z}/2) \cong \mathbb{Z}/2$ has a unique nonzero element.

The definition is a normalization condition, one fiber at a time. What is not obvious is that a class satisfying it exists at all, or that it is unique, and both are part of the theorem below. The local model is the following computation, which is the whole content of the trivial case.

Lemma 7.3.2: The Local Model

Let R be a commutative ring, let X be any space, let $n \geq 1$, and let u_n be a generator of the free rank-one R-module $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; R)$. Then the cohomology cross product with u_n,

$$H^k(X;R) \longrightarrow H^{k+n}\big(X \times \mathbb{R}^n,\ X \times (\mathbb{R}^n \setminus \{0\}); R\big), \qquad \alpha \longmapsto \alpha \times u_n$$

is an isomorphism for every k. Here $\alpha \times u_n = p^*(\alpha) \smile q^*(u_n)$ for the projections p and q of $X \times \mathbb{R}^n$ onto its factors, as in definition 4.3.7, the product being that of proposition 4.2.3(4).

***Proof*:**
Write $(Y, B_0) = (\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$. Chains are integral throughout, and $C^\bullet(-;R) = \mathrm{Hom}(C_\bullet(-), R)$ as in definition 4.1.15.

Step 1: The fiber pair. The space $\mathbb{R}^n$ is contractible and $\mathbb{R}^n \setminus \{0\}$ deformation retracts onto

S^{n-1}, so the long exact sequence of reduced homology of (Y, B_0), with corollary 3.3.7 and corollary 3.4.15, gives $H_i(Y, B_0) \cong \tilde{H}_{i-1}(S^{n-1})$, which is $\mathbb{Z}$ for $i = n$ and 0 otherwise. Let z be a relative cycle whose class generates $H_n(Y, B_0)$. The Ext term of theorem 4.1.8 vanishes, so evaluation on $[z]$ is an isomorphism $H^n(Y, B_0; G) \to G$ for every abelian group G, and there is a cocycle $w \in C^n(Y, B_0; \mathbb{Z})$ with $w(z) = 1$.

Step 2: The fiber chains split. The multiples of z form a subcomplex $\langle z \rangle$ concentrated in degree n. Since $w \circ \partial = 0$, the cochain w is a chain map from $C_\bullet(Y, B_0)$ to the complex $\mathbb{Z}[n]$ with $\mathbb{Z}$ in degree n and zero elsewhere. Its kernel K is a subcomplex, and $C_\bullet(Y, B_0) = \langle z \rangle \oplus K$ because $w(z) = 1$. Comparing homology, K is acyclic, and its groups are free, being subgroups of free groups ([10, Submodule Of Free Module Over PID, General Case]). An acyclic complex of free abelian groups is contractible. Its boundary groups ∂K_{m+1} are free, so a section of $\partial\colon K_{m+1} \to \partial K_{m+1}$ and lemma 3.4.34 give $K_{m+1} = \ker(\partial|_{K_{m+1}}) \oplus D_{m+1}$ with ∂ mapping D_{m+1} isomorphically onto $\partial K_{m+1} = \ker(\partial|_{K_m})$. The map h equal to the inverse of that isomorphism on $\ker(\partial|_{K_m})$ and to 0 on D_m satisfies $\partial h + h\partial = \mathrm{id}$.

Step 3: A comparison map. The Alexander-Whitney map AW of theorem 4.3.1 sends a singular m-simplex σ of $X \times Y$ to $\sum_{i+j=m} (p \circ \sigma)|_{[v_0,\dots,v_i]} \otimes (q \circ \sigma)|_{[v_i,\dots,v_m]}$, so it carries $C_\bullet(X \times B_0)$ into $C_\bullet(X) \otimes C_\bullet(B_0)$. The sequence $0 \to C_\bullet(B_0) \to C_\bullet(Y) \to C_\bullet(Y, B_0) \to 0$ splits in each degree, so it stays exact after tensoring with $C_\bullet(X)$, and AW induces a chain map $\overline{\mathrm{AW}}\colon C_\bullet(X \times Y, X \times B_0) \to C_\bullet(X) \otimes C_\bullet(Y, B_0)$. On $X \times Y$ and on $X \times B_0$ the map AW induces isomorphisms on homology by theorem 4.3.1, so by proposition 3.4.4 and the five lemma (lemma 3.4.6) so does $\overline{\mathrm{AW}}$. Let $C_\bullet(X)[n]$ be the complex with $C_{m-n}(X)$ in degree m and boundary ∂, and define $\Psi\colon C_\bullet(X) \otimes C_\bullet(Y, B_0) \to C_\bullet(X)[n]$ by $\Psi(c \otimes y) = w(y)\, c$. It is a chain map because $w \circ \partial = 0$, it maps the summand $C_\bullet(X) \otimes \langle z \rangle$ isomorphically onto $C_\bullet(X)[n]$, and it vanishes on the summand $C_\bullet(X) \otimes K$, which is contractible through $c \otimes y \mapsto (-1)^{|c|} c \otimes h(y)$ and therefore acyclic. So Ψ induces isomorphisms on homology, and so does $\Theta = \Psi \circ \overline{\mathrm{AW}}$, a chain map between complexes of free abelian groups.

Step 4: The cross product. By corollary 4.1.10 with $G = R$, the dual of Θ induces isomorphisms $\Theta^*\colon H^k(X; R) \to H^{k+n}(X \times Y, X \times B_0; R)$, the cohomology of $\mathrm{Hom}(C_\bullet(X)[n], R)$ in degree $k + n$ being $H^k(X; R)$. For $a \in C^k(X; R)$ and a singular $(k+n)$-simplex σ of $X \times Y$, only the term with $i = k$ of $\mathrm{AW}(\sigma)$ survives Ψ, and

$$(\Theta^* a)(\sigma) = a\big((p \circ \sigma)|_{[v_0,\dots,v_k]}\big)\, w\big((q \circ \sigma)|_{[v_k,\dots,v_{k+n}]}\big) = \big(p^\sharp a \smile q^\sharp w\big)(\sigma)$$

with w read as an R-valued cocycle. So $\Theta^*\alpha = \alpha \times u$, where $u \in H^n(Y, B_0; R)$ is the class of w, which evaluates to 1 on $[z]$. By Step 1, $u_n = r\, u$ for a unit r of R, so $\alpha \times u_n = r\, \Theta^*\alpha$, and $\alpha \mapsto \alpha \times u_n$ is an isomorphism, completing the proof.

No hypothesis on X is needed, because the chains of the fiber pair split off their homology, a single free group in one degree. That is what lets the induction below be run over the open sets of a trivializing cover without any further condition at each stage.

Lemma 7.3.3: The Relative Local Model

Let R, n and u_n be as in lemma 7.3.2, let $B' \subseteq B$ be open, and put $X = B \times \mathbb{R}^n$, $P = B' \times \mathbb{R}^n$ and $Q = B \times (\mathbb{R}^n \setminus \{0\})$. Then $y \mapsto y \times u_n = p^* y \smile q^* u_n$, the product of proposition 4.2.3(5), is an isomorphism $H^j(B, B'; R) \to H^{j+n}(X, P \cup Q; R)$ for every j. Consequently, if V and W are real vector spaces of dimensions $a, b \geq 1$ and μ_V, μ_W generate $H^a(V, V \setminus 0; \mathbb{Z})$ and $H^b(W, W \setminus 0; \mathbb{Z})$, then $\mu_V \times \mu_W$ generates $H^{a+b}\big(V \oplus W, (V \oplus W) \setminus 0; \mathbb{Z}\big)$.

***Proof*:**
Fix a cocycle w on $\mathbb{R}^n$ vanishing on $\mathbb{R}^n \setminus \{0\}$ and representing u_n. Let $C^*(X, P + Q)$ be the cochains vanishing on $C_*(P) + C_*(Q)$, whose cohomology is $H^*(X, P \cup Q)$ by proposition 4.2.3(5). A singular simplex lying in both P and Q lies in $P \cap Q$, so restriction to P gives a short exact sequence of cochain complexes

$$0 \to C^*(X, P + Q) \longrightarrow C^*(X, Q) \longrightarrow C^*(P, P \cap Q) \to 0$$

surjective on the right because a cochain on P vanishing on $P \cap Q$ extends by zero. The map $y \mapsto p^\sharp y \smile q^\sharp w$ carries the short exact sequence $0 \to C^*(B, B') \to C^*(B) \to C^*(B') \to 0$ into this one, raising degrees by n: a front face in B' or a back face in $\mathbb{R}^n \setminus \{0\}$ kills the product. It commutes with δ by lemma 4.2.2, as $\delta w = 0$. By proposition 3.4.4 it gives a commuting ladder between the two long exact sequences. On $H^*(B)$ and $H^*(B')$ it is the cross product with u_n into $H^{*+n}(B \times \mathbb{R}^n, Q)$ and $H^{*+n}(P, P \cap Q)$, an isomorphism by lemma 7.3.2 applied to B and to B'. The five lemma (lemma 3.4.6) makes the remaining vertical map an isomorphism, and that map is $y \mapsto y \times u_n$ by the definition of the product in proposition 4.2.3(5).

For the last claim choose linear isomorphisms $V \cong \mathbb{R}^a$ and $W \cong \mathbb{R}^b$, which reduce it to $V = \mathbb{R}^a$, $W = \mathbb{R}^b$ and $\mu_W = \pm u_b$. Apply the first claim with $B = \mathbb{R}^a$, $B' = \mathbb{R}^a \setminus 0$ and $n = b$: then $P \cup Q = (\mathbb{R}^{a+b}) \setminus 0$, and the generator μ_V goes to the generator $\mu_V \times u_b$, completing the proof.

The lemma is the Thom isomorphism of a trivial bundle over a pair, and its last clause is what makes the direct-sum and complex orientations of definition 7.1.25 orientations.

Theorem 7.3.4: The Thom Isomorphism

Let $E \to B$ be an oriented real vector bundle of rank n over a paracompact Hausdorff base, and suppose either that B admits a finite trivializing open cover (for instance B compact, or a finite CW complex) or that B is a CW complex of finite type, meaning one with finitely many cells in each dimension. The latter covers the classifying spaces $BO(n)$, $BSO(n)$ and $BU(n)$, Grassmannians with finitely many cells in each dimension (theorem 7.1.17). Then:

1. a Thom class $u_E \in H^n(E, E_0; \mathbb{Z})$ exists and is unique;
2. the map
$$\Phi\colon H^k(B;\mathbb{Z}) \longrightarrow H^{k+n}(E, E_0; \mathbb{Z}), \qquad \Phi(\alpha) = \pi^*(\alpha) \smile u_E$$
is an isomorphism for every k, where $\pi\colon E \to B$ is the projection.

The same statements hold with $\mathbb{Z}/2$ coefficients for an arbitrary real vector bundle, with no orientation hypothesis.

***Proof*:**
For a subset $W \subseteq B$ write $E_W = \pi^{-1}(W)$ and $E_{0W} = E_0 \cap E_W$, so that $E|_W \to W$ is an oriented bundle with zero-section complement E_{0W}, and let $P(W)$ be the assertion that $E|_W$ has a unique Thom class $u_W \in H^n(E_W, E_{0W}; \mathbb{Z})$ and that $\Phi_W(\alpha) = \pi^*(\alpha) \smile u_W$ defines an isomorphism $H^k(W;\mathbb{Z}) \to H^{k+n}(E_W, E_{0W}; \mathbb{Z})$ for every k, the product being that of proposition 4.2.3(4). If $W' \subseteq W$ and $P(W)$ and $P(W')$ hold, then u_W restricts to $u_{W'}$: the restriction of u_W to a fiber over a point of W' factors through $(E_{W'}, E_{0W'})$, so the restriction of u_W to $(E_{W'}, E_{0W'})$ is a Thom class, equal to $u_{W'}$ by uniqueness. In Steps 1 to 3 the base is an arbitrary space and W ranges over its open subsets, and $P(\emptyset)$ holds because all the groups involved vanish.

Step 1: $E|_W$ trivial. Let $h\colon E_W \to W \times \mathbb{R}^n$ be a trivialization, a homeomorphism over W restricting to linear isomorphisms $h_b\colon E_b \to \mathbb{R}^n$, and let u_n generate $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; \mathbb{Z})$. For $b \in W$ write $h_b^*(u_n) = \varepsilon(b)\mu_b$, where μ_b is the orientation generator of $H^n(E_b, (E_b)_0; \mathbb{Z})$ and $\varepsilon(b) = \pm 1$. The function ε is locally constant. By the convention on orientations fixed above, a point $b_1 \in W$ has an open neighbourhood V with a trivialization g over V and a generator u of $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; \mathbb{Z})$ such that $g_b^*(u) = \mu_b$ for $b \in V$, and $\tau(b) = h_b g_b^{-1}$ defines a continuous map $V \cap W \to GL_n(\mathbb{R})$, its matrix entries $b \mapsto e_i \cdot h_b g_b^{-1}(e_j)$ being continuous because $h \circ g^{-1}$ is. The invertible matrices form an open subset of the space of all matrices, so $GL_n(\mathbb{R})$ contains a ball around $\tau(b_1)$, which contains $\tau(b)$ for b near b_1. The segment from $\tau(b_1)$ to $\tau(b)$ in that ball is a homotopy through maps of pairs $(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}) \to (\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$, so $\tau(b)^* = \tau(b_1)^*$ on $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; \mathbb{Z})$ by proposition 4.1.23, and $h_b^*(u_n) = g_b^* \tau(b_1)^*(u_n)$ is the same multiple of μ_b for all such b.

The 0-cochain taking the value $\varepsilon(b)$ at each point b is a cocycle, because ε is constant along paths, and its class $e \in H^0(W;\mathbb{Z})$ satisfies $e \smile e = 1$. Let p and q be the projections of $W \times \mathbb{R}^n$, and put $u_W = h^*\big(e \times u_n\big)$, where $e \times u_n = p^*(e) \smile q^*(u_n)$ as in lemma 7.3.2. For $\beta \in H^0(W;\mathbb{Z})$ and $b \in W$, the inclusion $i_b\colon x \mapsto (b, x)$ has $p \circ i_b$ constant at b and $q \circ i_b = \mathrm{id}$, so by proposition 4.2.5 the class $\beta \times u_n$ restricts along i_b to $\beta(b)u_n$, where $\beta(b)$ is the value of the 0-cocycle β at b. Since h restricted to E_b is $i_b h_b$, the class $h^*(\beta \times u_n)$ restricts to $\beta(b)\varepsilon(b)\mu_b$ on the fiber over b. For $\beta = e$ this is μ_b, so u_W is a Thom class. By lemma 7.3.2 with $k = 0$, every class of $H^n(E_W, E_{0W}; \mathbb{Z})$ is $h^*(\beta \times u_n)$ for a unique $\beta \in H^0(W;\mathbb{Z})$, and it is a Thom class exactly when

$\beta(b) = \varepsilon(b)$ for every b, that is $\beta = e$. So u_W is unique. Finally $\pi = ph$, and h^* and p^* preserve products by proposition 4.2.5, so

$$\Phi_W(\alpha) = h^*\big(p^*(\alpha) \smile p^*(e) \smile q^*(u_n)\big) = h^*\big((\alpha \smile e) \times u_n\big)$$

The map $\alpha \mapsto \alpha \smile e$ is a bijection of $H^k(W;\mathbb{Z})$, its own inverse since $e \smile e = 1$, the map $\alpha \mapsto \alpha \times u_n$ is an isomorphism by lemma 7.3.2, and h^* is an isomorphism because h is a homeomorphism of pairs. Hence Φ_W is an isomorphism, and $P(W)$ holds.

Step 2: gluing two open sets. Let $U, V \subseteq B$ be open with $P(U)$, $P(V)$ and $P(U \cap V)$, and put $W = U \cup V$. The sets E_U and E_V are open in E_W and cover it, and $E_{0U} = E_{0W} \cap E_U$ and $E_{0V} = E_{0W} \cap E_V$ are open in E_{0W} and cover it. So proposition 4.1.26 applies to (E_W, E_{0W}) with the covering sets E_U, E_V and the subsets E_{0U}, E_{0V}, and to W with the covering sets U, V and empty subsets.

In the sequence for (E_W, E_{0W}) the group preceding $H^n(E_W, E_{0W};\mathbb{Z})$ is $H^{n-1}(E_{U\cap V}, E_{0,U\cap V};\mathbb{Z})$, which is isomorphic to $H^{-1}(U \cap V;\mathbb{Z}) = 0$ by $P(U \cap V)$. So the pair of restrictions maps $H^n(E_W, E_{0W};\mathbb{Z})$ injectively onto the kernel of the difference of restrictions. The classes u_U and u_V both restrict to $u_{U\cap V}$, so there is a unique class u_W restricting to u_U and to u_V. It is a Thom class, its restriction to a fiber over a point of U or of V factoring through u_U or u_V. Any Thom class of $E|_W$ restricts to Thom classes over U and V, which are u_U and u_V by uniqueness, so it equals u_W.

Fix a cocycle $\tilde{u}$ representing u_W, and write $C^\bullet_{U+V}(W;\mathbb{Z})$ and $C^\bullet_{E_U+E_V}(E_W, E_{0W};\mathbb{Z})$ for the left-hand terms of the two short exact sequences of proposition 4.1.26(1). For $a \in C^k_{U+V}(W;\mathbb{Z})$ and a singular $(k+n)$-simplex σ of E_U or of E_V put

$$\tilde{\Phi}_W(a)(\sigma) = a\big(\pi \circ \sigma|_{[v_0,\dots,v_k]}\big)\, \tilde{u}\big(\sigma|_{[v_k,\dots,v_{k+n}]}\big)$$

the formula of definition 4.2.1, legitimate because $\pi \circ \sigma|_{[v_0,\dots,v_k]}$ is a simplex of U or of V. It vanishes when σ lies in E_{0U} or in E_{0V}, since then the back face lies in E_0, where $\tilde{u}$ vanishes, so $\tilde{\Phi}_W(a) \in C^{k+n}_{E_U+E_V}(E_W, E_{0W};\mathbb{Z})$. The same formula defines $\tilde{\Phi}_U(a) = \pi^\sharp(a) \smile \tilde{u}|_{E_U}$ for $a \in C^k(U;\mathbb{Z})$, and likewise $\tilde{\Phi}_V$ and $\tilde{\Phi}_{U\cap V}$, where $\tilde{u}|_{E_U}$ is the restriction of $\tilde{u}$, a cocycle representing u_U. The computation proving lemma 4.2.2 involves only a simplex and its faces, all of which lie in E_U or in E_V when the simplex does, so it gives

$$\delta\tilde{\Phi}_W(a) = \tilde{\Phi}_W(\delta a) + (-1)^k \pi^\sharp(a) \smile \delta\tilde{u} = \tilde{\Phi}_W(\delta a)$$

and the same for the other three maps. The four maps commute with the maps ψ^* and φ^* of the two sequences, since all of them are restrictions of one formula to sets of simplices, and they commute with the restrictions ι^* of proposition 4.1.26(2): for $a \in C^k(W;\mathbb{Z})$ the cochain $\pi^\sharp(a) \smile \tilde{u}$, which represents $\Phi_W[a]$ when a is a cocycle, restricts on small simplices to $\tilde{\Phi}_W(\iota^* a)$.

So $(\tilde{\Phi}_W, \tilde{\Phi}_U \oplus \tilde{\Phi}_V, \tilde{\Phi}_{U\cap V})$ is a map from the short exact sequence of cochain complexes for W to the one for (E_W, E_{0W}), with degrees raised by n. By proposition 3.4.4, applied after reindexing, it induces a commuting ladder between the two long exact sequences, and transporting the ladder along the isomorphisms ι^*, with which the maps commute, gives a commuting ladder from the sequence of proposition 4.1.26(3) for W to the one for (E_W, E_{0W}), with vertical maps $\Phi_U \oplus \Phi_V$, $\Phi_{U\cap V}$ and Φ_W. In the window

$$H^{k-1}(U) \oplus H^{k-1}(V) \to H^{k-1}(U \cap V) \to H^k(W) \to H^k(U) \oplus H^k(V) \to H^k(U \cap V)$$

the four outer vertical maps are isomorphisms by $P(U)$, $P(V)$ and $P(U \cap V)$, so Φ_W is an isomorphism by the five lemma (lemma 3.4.6), and $P(W)$ holds.

Step 3: a finite trivializing cover. By induction on r, $P(W)$ holds for every open W that is the union of r open sets over each of which E is trivial. For $r = 1$ this is Step 1. If $W = W' \cup U_r$ with $W' = U_1 \cup \cdots \cup U_{r-1}$, then $W' \cap U_r$ is the union of the $r-1$ sets $U_i \cap U_r$, over each of which E is trivial, so $P(W')$ and $P(W' \cap U_r)$ hold by induction, $P(U_r)$ by Step 1, and $P(W)$ by Step 2. In particular $P(B)$ holds when B has a finite trivializing open cover, as it does when B is compact, local triviality giving a trivializing open cover and compactness a finite subcover.

Step 4: a CW base of finite type. Let B have finitely many cells in each dimension, with skeleta B_m, and write $(E_m, E_{0m}) = (E_{B_m}, E_{0B_m})$. Each B_m is a finite complex, a finite union of images of disks, hence compact, so Step 3 applied to the bundle $E|_{B_m} \to B_m$ gives $P(B_m)$, with Thom class u_m, and u_{m+1} restricts to u_m by the first paragraph. The restriction $H^i(B_{m+1}; \mathbb{Z}) \to H^i(B_m; \mathbb{Z})$ is bijective for every $i \leq m-1$: the complex $C_\bullet(B_{m+1}, B_m)$ is free, and its homology vanishes outside degree $m+1$ and is free in degree $m+1$ by lemma 3.5.9(1), so theorem 4.1.8 and proposition 4.1.9(2) give $H^i(B_{m+1}, B_m; \mathbb{Z}) = 0$ for $i \neq m+1$, and the long exact sequence of the pair (theorem 4.1.17) makes the restriction bijective whenever $i \neq m+1$ and $i+1 \neq m+1$. The square

$$\begin{array}{ccc} H^i(B_{m+1}; \mathbb{Z}) & \longrightarrow & H^i(B_m; \mathbb{Z}) \\ \downarrow{\scriptstyle \Phi_{B_{m+1}}} & & \downarrow{\scriptstyle \Phi_{B_m}} \\ H^{i+n}(E_{m+1}, E_{0,m+1}; \mathbb{Z}) & \longrightarrow & H^{i+n}(E_m, E_{0m}; \mathbb{Z}) \end{array}$$

commutes by proposition 4.2.5, as u_{m+1} restricts to u_m, and its vertical maps are isomorphisms, so the restriction $H^j(E_{m+1}, E_{0,m+1}; \mathbb{Z}) \to H^j(E_m, E_{0m}; \mathbb{Z})$ is bijective for every $j \leq m+n-1$.

A compact subset of B lies in some B_m by lemma 1.2.4, and a compact subset $K \subseteq E$ lies in E_m for any m with $\pi(K) \subseteq B_m$, the image $\pi(K)$ being compact. Fix an integer k and let $m_k = \max(k+1, 1)$. For $m \geq m_k$ the restrictions above are bijective in degrees k and $k-1$ on the base and in degrees $k+n$ and $k+n-1$ on the pairs, so lemma 7.1.16, applied to $(B, \emptyset)$ and to (E, E_0), makes the restrictions

$$H^k(B; \mathbb{Z}) \to H^k(B_{m_k}; \mathbb{Z}) \qquad H^{k+n}(E, E_0; \mathbb{Z}) \to H^{k+n}(E_{m_k}, E_{0m_k}; \mathbb{Z})$$

bijective. For $k = 0$, so $m_k = 1$, let u_E be the class restricting to u_1. For every $m \geq 1$ the restriction $H^n(E_m, E_{0m}; \mathbb{Z}) \to H^n(E_1, E_{01}; \mathbb{Z})$ is bijective, and u_E and u_m both restrict to u_1, so u_E restricts to u_m. Each fiber lies over some B_m, so u_E is a Thom class, and any Thom class of E restricts to u_1 by uniqueness over B_1, hence equals u_E. For general k, the square comparing Φ_B with $\Phi_{B_{m_k}}$ through these restrictions commutes by proposition 4.2.5, as u_E restricts to u_{m_k}, and its other three maps are bijective, so Φ_B is an isomorphism in every degree.

Step 5: the mod-2 statement. The orientation entered only through the generators μ_b and the sign function ε of Step 1. With $\mathbb{Z}/2$ coefficients the group $H^n(E_b, (E_b)_0; \mathbb{Z}/2) \cong \mathbb{Z}/2$ has a unique generator, so $\varepsilon = 1$ and no orientation is needed. Lemma 7.3.2 holds with $R = \mathbb{Z}/2$, and every other result used above holds with coefficients in $\mathbb{Z}/2$, so the same argument runs with $\mathbb{Z}/2$ in place of $\mathbb{Z}$, as we sought to show.

Proposition 7.3.5: The Gysin Sequence from the Thom Isomorphism

Let $E \to B$ be an oriented real vector bundle of rank $n \geq 1$ over a CW complex of finite type, with projection π, zero section z, Thom class u_E and sphere bundle $p\colon S(E) \to B$ for an inner product. Let $j^*\colon H^n(E, E_0; \mathbb{Z}) \to H^n(E; \mathbb{Z})$ be induced by the inclusion, and put $e_T = z^* j^*(u_E) \in H^n(B; \mathbb{Z})$.

1. Transporting the long exact sequence of the pair (E, E_0) through the Thom isomorphism Φ of theorem 7.3.4, the isomorphism π^*, and restriction to the deformation retract $S(E)$ of E_0 gives a long exact sequence

$$\cdots \to H^{m-n}(B; \mathbb{Z}) \xrightarrow{\smile e_T} H^m(B; \mathbb{Z}) \xrightarrow{p^*} H^m(S(E); \mathbb{Z}) \longrightarrow H^{m-n+1}(B; \mathbb{Z}) \to \cdots$$

2. If $n \geq 2$, B is path-connected, and $S(E)$ carries the orientation induced by that of E (definition 7.1.25), then the Euler class of the Gysin sequence of $S(E) \to B$ in theorem 6.3.2 is e_T.

3. If moreover B has one 0-cell b_0 and one n-cell, with characteristic map $\Phi_B\colon D^n \to B$, and $\tilde{s}\colon D^n \to S(E)$ satisfies $p \circ \tilde{s} = \Phi_B$, then for every generator $[D^n, S^{n-1}]$ of $H_n(D^n, S^{n-1})$

$$\left\langle e_T, (\Phi_B)_*[D^n, S^{n-1}] \right\rangle = \left\langle \alpha_{b_0}, (\tilde{s}|_{S^{n-1}})_* \partial [D^n, S^{n-1}] \right\rangle$$

where e_T is read in $H^n(B, b_0; \mathbb{Z}) \cong H^n(B; \mathbb{Z})$.

***Proof*:**

Part (1). The homotopy $v \mapsto tv$ makes z and π inverse homotopy equivalences, and $v \mapsto v/|v|$ makes $S(E)$ a deformation retract of E_0. Substituting $H^{m-n}(B) \cong H^m(E, E_0)$ (by Φ), $H^m(B) \cong H^m(E)$ (by π^*) and $H^m(E_0) \cong H^m(S(E))$ into the long exact sequence of the pair (theorem 4.1.17) gives a long exact sequence. Its map $H^{m-n}(B) \to H^m(B)$ sends x to $z^* j^*(\pi^* x \smile u_E)$. On cochains j^* only forgets that a cochain vanishes on E_0, so this is $z^*(\pi^* x \smile j^* u_E) = x \smile e_T$, using $z^* \pi^* = \mathrm{id}$. Its map $H^m(B) \to H^m(S(E))$ is π^* followed by restriction to $S(E)$, which is p^*.

Part (2). Let U be a cocycle on E vanishing on E_0 and representing u_E, so that $[U] = j^* u_E$. Normalized cochains compute $H^n(B; \mathbb{Z})$ (proof of theorem 6.2.8), so e_T is represented by a cocycle Y on B vanishing on every degenerate singular simplex. As $\pi^*[Y] = \pi^* z^*[U] = [U]$, there is $h \in C^{n-1}(E; \mathbb{Z})$ with $\delta h = \pi^* Y - U$. Put $c = h|_{S(E)}$. Since U vanishes on $S(E) \subseteq E_0$, $\delta c = p^* Y$, which vanishes on $F_{n-1} C_n(S(E))$, the simplices there having degenerate projections. In the filtration of definition 6.2.2, c lies in F^0 and δc in F^n, so c represents an element of $E_n^{0,n-1}$, and d_n carries it to the class of δc ([13, Explicit Description of the Pages]).

The element represented by c is $1 \otimes \alpha$. Its image in $E_1^{0,n-1} = \prod_{b \in B} H^{n-1}(S(E_b))$ is the family of the classes of c on the fibers. A singular simplex of E_b projects to the constant simplex at b, which is degenerate in degree $n \geq 1$, so Y vanishes on it and $\delta(h|_{E_b}) = -U|_{E_b}$. Hence $h|_{E_b \setminus 0}$ is a cocycle, and by the formula for δ in theorem 4.1.17 its class a_b satisfies $\delta a_b = [-U|_{E_b}] = -\mu_b$, the class $U|_{E_b}$ being the orientation μ_b (definition 7.3.1). By definition 7.1.25, the restriction of a_b to $S(E_b)$, which is the class of $c|_{S(E_b)}$, is α_b.

The class of δc is e_T. On the bottom row the identification $E_1^{n,0} \cong N^n(B;\mathbb{Z})$ of theorem 6.2.8 is induced by p^*, as for the base edge of theorem 6.2.9. So $\delta c = p^*Y$ corresponds to the normalized cocycle Y, whose class in $E_2^{n,0} = H^n(B)$ is $[Y] = e_T$. Only the rows $q = 0$ and $q = n-1$ are nonzero, so no d_r with $2 \le r < n$ reaches or leaves the spot $(n,0)$, and $E_n^{n,0} = E_2^{n,0}$. Hence the Euler class $d_n(1 \otimes \alpha)$ of theorem 6.3.2 is e_T.

Part (3). Keep Y, h and c from Part (2). A normalized cochain of positive degree vanishes on the constant simplices at b_0, so Y represents the class of e_T in $H^n(B, b_0)$. Let t be a relative cycle representing $[D^n, S^{n-1}]$. Then $\tilde{s}_\sharp(\partial t)$ is a cycle in $S(E_{b_0})$ representing $(\tilde{s}|_{S^{n-1}})_*\partial[D^n, S^{n-1}]$, and

$$\langle Y, (\Phi_B)_\sharp t\rangle = \langle p^*Y, \tilde{s}_\sharp t\rangle = \langle \delta c, \tilde{s}_\sharp t\rangle = \langle c, \tilde{s}_\sharp \partial t\rangle$$

which is the value of α_{b_0}, the class of $c|_{S(E_{b_0})}$, on that cycle, completing the proof.

The theorem says that the cohomology of the pair (E, E_0) is a free module of rank one over the cohomology of the base, generated by u_E. The Thom class is thus the exact analogue for a bundle of a fundamental class for a manifold, and the isomorphism is the analogue of Poincaré duality: it converts a degree shift by the rank into multiplication by a single class. The Euler class is the restriction of u_E along the zero section (corollary 7.3.6).

Corollary 7.3.6: The Euler Class is the Restriction of the Thom Class

Let $E \to B$ be an oriented real vector bundle of rank n over a CW complex of finite type, with Thom class u_E given by theorem 7.3.4. Let $j^*\colon H^n(E, E_0;\mathbb{Z}) \to H^n(E;\mathbb{Z})$ be induced by the inclusion of pairs $(E, \emptyset) \hookrightarrow (E, E_0)$ and let $z\colon B \to E$ be the zero section. Then the Euler class of definition 7.2.15 satisfies

$$e(E) = z^*\big(j^*(u_E)\big)$$

Equivalently, under the isomorphism $\pi^*\colon H^n(B;\mathbb{Z}) \xrightarrow{\cong} H^n(E;\mathbb{Z})$, the class $j^*(u_E)$ corresponds to $e(E)$.

***Proof*:**

Rank one. Here $e(E) = 0$ by definition 7.2.15. Each $E_b \setminus 0$ has two components, and the connecting map $\delta\colon H^0(E_b \setminus 0) \to H^1(E_b, E_b \setminus 0)$ kills the constant class 1, which extends over E_b, so it carries the classes equal to 1 on one component and 0 on the other to μ_b and $-\mu_b$. The unit vector $s(b)$ in the component whose class goes to μ_b depends continuously on b, by the local form of an orientation, so s is a section of E with values in E_0, homotopic to z through the sections ts. The class j^*u_E restricts to 0 on E_0 (theorem 4.1.17), so $z^*j^*u_E = s^*j^*u_E = 0$.

Rank at least two. Let $f_E\colon B \to BSO(n)$ classify E, so that $E \cong f_E^*\tilde{\gamma}_n$ as oriented bundles. The space $BSO(n)$ is a CW complex of finite type (theorem 7.1.17), path-connected as in definition 7.2.15, so by proposition 7.3.5(2) the universal class e of definition 7.2.15 is $z^*j^*u_{\tilde{\gamma}_n}$. The map of pairs $F\colon (E, E_0) \to (\tilde{\gamma}_n, (\tilde{\gamma}_n)_0)$ covering f_E is an orientation-preserving linear isomorphism on each fiber, so $F^*u_{\tilde{\gamma}_n}$ restricts to the orientation on each fiber pair and is the Thom class u_E by the uniqueness clause of theorem 7.3.4. As F commutes with the zero sections and with j,

$$e(E) = f_E^*e = f_E^*z^*j^*u_{\tilde{\gamma}_n} = z^*j^*F^*u_{\tilde{\gamma}_n} = z^*j^*u_E$$

The equivalent formulation follows because z and π are mutually inverse homotopy equivalences,

so $z^* = (\pi^*)^{-1}$ on cohomology, completing the proof.

This discharges the description asserted in definition 7.2.15, where the Euler class was said to be "equivalently" the image of the Thom class. That equivalence is now proved. Two properties of the Euler class follow at once, and neither is visible from the Gysin construction.

Proposition 7.3.7: Naturality of the Euler Class

Let $E \to X$ be an oriented real vector bundle of rank n over a CW complex of finite type, and let $f\colon X' \to X$ be a map with X' paracompact Hausdorff and of the same type, so that both Euler classes are defined. Then, giving f^*E the pulled-back orientation,

$$e(f^*E) = f^*\big(e(E)\big) \in H^n(X';\mathbb{Z})$$

***Proof*:**
The Thom class is natural: the map of pairs $(f^*E, f^*E \setminus 0) \to (E, E \setminus 0)$ covering f carries the pulled-back orientation to the given one fiberwise, so it pulls the Thom class u_E back to the Thom class u_{f^*E}, these being characterized fiberwise by definition 7.3.1. The Euler class is the restriction of the Thom class along the zero section (corollary 7.3.6), and the zero sections commute with the map of pairs, so restricting the identity $\tilde{f}^* u_E = u_{f^*E}$ along them gives $f^*e(E) = e(f^*E)$, completing the proof.

With naturality and the Thom-class description of the Euler class, the sign convention that fixed the first Chern class in example 7.2.3 can be checked against the Euler class: on a complex line bundle the two classes agree.

Example 7.3.8: The Euler Class of a Complex Line Bundle
Let $L \to X$ be a complex line bundle over a paracompact Hausdorff base, and let $L_{\mathbb{R}}$ be the underlying real rank-2 bundle (definition 5.2.9) with the complex orientation of definition 7.1.25. Its Euler class and first Chern class coincide:

$$e(L_{\mathbb{R}}) = c_1(L) \in H^2(X;\mathbb{Z})$$

A classifying map $f\colon X \to \mathbb{C}\mathrm{P}^\infty$ of L (theorem 7.1.29) gives $c_1(L) = f^*c_1$, and composed with the oriented classifying map of $\gamma_{1,\mathbb{R}}$ it classifies $L_{\mathbb{R}}$, so $e(L_{\mathbb{R}}) = f^*e(\gamma_{1,\mathbb{R}})$ by definition 7.2.15. It therefore suffices to prove $e(\gamma_{1,\mathbb{R}}) = c_1$ over $\mathbb{C}\mathrm{P}^\infty$, a CW complex of finite type. Restriction to $\mathbb{C}\mathrm{P}^1$ is injective on H^2 (theorem 4.5.12), it carries c_1 to $c_1(\gamma^1) = -u$ (example 7.2.3), and it carries $e(\gamma_{1,\mathbb{R}})$ to $e(\gamma^1_{\mathbb{R}})$ (proposition 7.3.7). It remains to show $\langle e(\gamma^1_{\mathbb{R}}), [\mathbb{C}\mathrm{P}^1]\rangle = -1$.

The 2-cell of $\mathbb{C}\mathrm{P}^1$ has the characteristic map $\Phi(w) = \big[w : \sqrt{1-|w|^2}\big]$ (example 1.2.5), with $\Phi(\partial D^2) = b_0 = [1:0]$, and $\tilde{s}(w) = \big(\Phi(w), (w, \sqrt{1-|w|^2})\big)$ is a unit vector of the line $\Phi(w)$, equal to we_1 when $|w| = 1$. Let $[D^2, S^1]$ be the generator whose boundary is the counterclockwise class of S^1. By corollary 7.3.6 and proposition 7.3.5(3),

$$\big\langle e(\gamma^1_{\mathbb{R}}), \Phi_*[D^2, S^1]\big\rangle = \big\langle \alpha_{b_0}, [\tilde{\ell}]\big\rangle, \qquad \tilde{\ell}(t) = e^{2\pi i t} e_1$$

The complex orientation of the fiber $\mathbb{C}e_1$ is $\varphi^*\mu_{\mathbb{C}}$ for $\varphi(\lambda e_1) = \lambda$, and $\tau' = (\varphi^{-1})_*\tau$ has boundary $[\tilde{\ell}]$. With a as in definition 7.1.25, so that $\delta a = -\varphi^*\mu_{\mathbb{C}}$, the formula for δ in theorem 4.1.17 gives

$$\big\langle \alpha_{b_0}, [\tilde{\ell}]\big\rangle = \langle a, \partial\tau'\rangle = \langle \delta a, \tau'\rangle = -\langle \mu_{\mathbb{C}}, \tau\rangle = -1$$

Finally $\Phi_*[D^2, S^1] = [\mathbb{C}\mathrm{P}^1]$ under $H_2(\mathbb{C}\mathrm{P}^1) \cong H_2(\mathbb{C}\mathrm{P}^1, b_0)$. On the open disk $\Phi = \kappa \circ r$ for the chart $\kappa(w) = [w : 1]$ and $r(w) = w/\sqrt{1-|w|^2}$, and on a small disk about 0 the maps $w \mapsto w\big(1 - t + t/\sqrt{1-|w|^2}\big)$ join r to the identity through maps of pairs into $(\mathbb{C}, \mathbb{C} \setminus 0)$. So $\Phi_*[D^2, S^1]$ restricts at $[0:1]$ to κ_* of the counterclockwise local orientation, which is the local orientation fixing $[\mathbb{C}\mathrm{P}^1]$ in example 7.2.3. Hence $e(\gamma^1_{\mathbb{R}}) = -u = c_1(\gamma^1)$. For $L = \gamma^1$ this says $\int_{\mathbb{C}\mathrm{P}^1} e(\gamma^1_{\mathbb{R}}) = -1$, consistent with the tautological bundle having no nowhere-zero section.

Proposition 7.3.9: Whitney Sum Formula for the Euler Class

Let $E_1, E_2 \to B$ be oriented real vector bundles over a CW complex of finite type, of ranks n_1 and n_2, and give $E_1 \oplus E_2$ the direct-sum orientation (definition 7.1.25). Then

$$e(E_1 \oplus E_2) = e(E_1) \smile e(E_2)$$

***Proof*:**
Write $E = E_1 \oplus E_2$ and let $q_i \colon E \to E_i$ be the two projections. The pair (E, E_0) receives a map from the "external" pair: a point of E lies outside the zero section exactly when at least one of its two components does, so

$$E_0 = q_1^{-1}\big((E_1)_0\big) \cup q_2^{-1}\big((E_2)_0\big)$$

The two sets on the right are open in E, the zero sections being closed, so the product of proposition 4.2.3(5) gives a class $q_1^*(u_{E_1}) \smile q_2^*(u_{E_2}) \in H^{n_1+n_2}(E, E_0; \mathbb{Z})$. By proposition 4.2.5, restricted to a fiber this is the product of the two fiber generators, which is the generator of $H^{n_1+n_2}(\mathbb{R}^{n_1+n_2}, \mathbb{R}^{n_1+n_2} \setminus \{0\}; \mathbb{Z})$ determined by the direct-sum orientation, by the definition of that orientation, a generator by lemma 7.3.3. Hence $u_E = q_1^*(u_{E_1}) \smile q_2^*(u_{E_2})$, by the uniqueness clause of theorem 7.3.4.

Now apply z^*j^*, using corollary 7.3.6. Both z^* and j^* preserve these products by proposition 4.2.5, and the zero section of E composed with q_i is the zero section of E_i followed by no further map, so

$$e(E) = z^*j^*\big(q_1^*u_{E_1} \smile q_2^*u_{E_2}\big) = z^*j^*q_1^*(u_{E_1}) \smile z^*j^*q_2^*(u_{E_2}) = e(E_1) \smile e(E_2)$$

as we sought to show.

Proposition 7.3.10: The Euler Class of an Odd-Rank Bundle is 2-Torsion

Let $E \to B$ be an oriented real vector bundle of odd rank n over a CW complex of finite type. Then $2e(E) = 0$.

***Proof*:**
Let $\sigma \colon E \to E$ be fiberwise multiplication by -1. It is an isomorphism of vector bundles over id_B, so it carries E to E and preserves the zero section, and the induced map on $H^*(B)$ is the identity because σ covers id_B. On a fiber, σ acts on $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; \mathbb{Z})$ by the degree of the antipodal map of S^{n-1}, namely $(-1)^n$, which is -1 because n is odd. So σ reverses the orientation of every fiber, and therefore $\sigma^*(u_E) = -u_E$ by the uniqueness clause of theorem 7.3.4 applied to the orientation-reversed bundle.

Applying z^*j^* and using corollary 7.3.6 together with $\sigma \circ z = z$ gives $e(E) = z^*j^*\sigma^*(u_E) = -e(E)$, hence $2e(E) = 0$, completing the proof.

Exercise 7.3.1

1. Let $E \to B$ be an oriented rank-n bundle over a CW complex and suppose E admits a nowhere-zero section s. Show directly from corollary 7.3.6 that $e(E) = 0$. (The section gives a factorization of z through E_0 up to homotopy.)
2. Verify proposition 7.3.10 is sharp: exhibit an oriented bundle of odd rank whose Euler class is nonzero. (Consider TS^2 and explain why the rank being even is what lets e be non-torsion, then find an odd-rank example over a base with 2-torsion in H^n.)
3. Let $E \to B$ be an oriented rank-n bundle over a finite CW complex with $\dim B < n$. Show that $e(E) = 0$, and identify the step at which theorem 7.3.4 is used.
4. Show that the trivial oriented bundle $\underline{\mathbb{R}}^n$ over a finite CW complex has $e(\underline{\mathbb{R}}^n) = 0$ for $n \geq 1$, and deduce from proposition 7.3.9 that $e(E \oplus \underline{\mathbb{R}}) = 0$ for every oriented bundle E. Conclude that the Euler class, unlike the Stiefel-Whitney and Pontryagin classes, is not a stable invariant.
5. Let $E_1 \to B$ and $E_2 \to B$ be oriented bundles with E_2 of odd rank. Deduce from proposition 7.3.9 and proposition 7.3.10 that $2e(E_1 \oplus E_2) = 0$, and give an example where $e(E_1)$ and $e(E_2)$ are both nonzero while their product vanishes.

7.4 Characteristic Classes as Obstructions

Section 7.2 built the characteristic classes by hand out of the cohomology of the classifying spaces: the Stiefel-Whitney classes $w_i(E) \in H^i(X; \mathbb{F}_2)$ as the pullbacks of the polynomial generators of $H^*(BO(n); \mathbb{F}_2) = \mathbb{F}_2[w_1, \ldots, w_n]$ along a classifying map $X \to BO(n)$, and the Chern classes $c_i(E) \in H^{2i}(X; \mathbb{Z})$ as the pullbacks of the generators of $H^*(BU(n); \mathbb{Z}) = \mathbb{Z}[c_1, \ldots, c_n]$. That construction is correct but opaque: it produces invariants without saying what they measure. This section gives the meaning along two complementary axes. The first axis is *formal*: a short list of axioms (naturality, the Whitney sum formula, normalization, and the identification of the top class with the Euler class) characterizes the classes uniquely and turns them into a computational calculus. The second axis is *geometric*: each characteristic class is an obstruction, in the precise sense of theorem 5.6.7, to a structure one would like the bundle to carry. The class $w_1(E)$ is the obstruction to orienting E. Once E is oriented, $w_2(E)$ is the obstruction to a spin structure. The top class, the Euler class or the top Stiefel-Whitney class, is the obstruction to a single nowhere-zero section. For the tangent bundle of a closed manifold this last obstruction is the Euler characteristic, closing the loop with Poincaré duality (section 4.5).

The starting point is the list of properties that pin the classes down. The first is naturality, which is immediate from the classifying-map definition, since a characteristic class is a pullback and pullback is functorial.

Proposition 7.4.1: Naturality of Characteristic Classes

Let $f\colon X \to Y$ be a map of paracompact Hausdorff spaces and $E \to Y$ a real (respectively complex) vector bundle. Then for the pulled-back bundle $f^*E \to X$,

$$w_i(f^*E) = f^*\big(w_i(E)\big) \in H^i(X;\mathbb{F}_2), \qquad c_i(f^*E) = f^*\big(c_i(E)\big) \in H^{2i}(X;\mathbb{Z})$$

for every i. Equivalently, the total classes satisfy $w(f^*E) = f^*w(E)$ and $c(f^*E) = f^*c(E)$.

***Proof*:**
A rank-n real bundle $E \to Y$ is classified by a map $g\colon Y \to BO(n)$ with $E \cong g^*\gamma^n$ for the universal bundle $\gamma^n \to BO(n)$, by theorem 7.1.29. The class $w_i(E)$ is by definition $g^*(w_i)$, where $w_i \in H^i(BO(n);\mathbb{F}_2)$ is the universal class of section 7.2. The pullback f^*E is classified by the composite $g \circ f\colon X \to BO(n)$, since $(g \circ f)^*\gamma^n \cong f^*(g^*\gamma^n) \cong f^*E$ by functoriality of pullback. Therefore

$$w_i(f^*E) = (g \circ f)^*(w_i) = f^*\big(g^*(w_i)\big) = f^*\big(w_i(E)\big)$$

using that pullback on cohomology is contravariantly functorial, $(g \circ f)^* = f^* \circ g^*$. The complex case is identical with $BO(n)$, γ^n, $\mathbb{F}_2$ replaced by $BU(n)$, the universal complex bundle, and $\mathbb{Z}$. Summing over i gives the total-class statement, completing the proof.

Naturality is the reason characteristic classes are computable at all: it reduces every computation to the universal case and lets one transport a known answer from one space to another along a map. The deeper axiom is multiplicativity under direct sum, the Whitney sum formula. It says that the total class converts the Whitney sum of bundles into a cup product of total classes, so that the characteristic class of a sum is determined by the classes of the summands. The proof rests on two facts: after pullback to a suitable space every bundle splits into line bundles with no loss of cohomological information, and on a sum of line bundles the formula is a direct computation over $BO(1)^n$. The first is the *splitting principle*, and it is the central technical device of the whole theory.

The principle is stated over an arbitrary paracompact Hausdorff base, which is exactly the generality section 7.1.2 establishes: the flag construction below projectivizes bundles over bases that carry no CW structure, so theorem 7.1.10 is unavailable and theorem 7.1.29 stands in its place. Every statement in this chapter whose base is paracompact requires it to be Hausdorff as well, and the hypothesis is written into each statement rather than left to a convention. This changes nothing against the chapter's sources, both of which build the Hausdorff property into the word paracompact ([23, §1.1], where paracompact means Hausdorff with subordinate partitions of unity for every open cover; [29, §5.8], where it means Hausdorff with locally finite refinements). Hausdorffness is what makes the subordinate partitions of unity of [16, Existence of Subordinate Partitions of Unity] available, and [16, Locally Trivial Maps with Compact Fiber Preserve Paracompactness] propagates both properties up the towers built here.

The splitting principle needs one fact about projectivizations: the cohomology of $\mathbb{P}(E)$ is a free module over that of the base, on the powers of one degree-one class mod 2, or degree-two class integrally, for a real or complex bundle E over a paracompact Hausdorff base with no connectivity hypothesis. The ring structure of $H^*(\mathbb{P}(E);\mathbb{Z})$ is proposition 7.4.11, which comes after the Whitney sum formula because its relation is written in the Chern classes.

Proposition 7.4.2: Projective Bundles over a Paracompact Base

Let $E \to X$ be a rank-n real vector bundle, $n \geq 1$, over a paracompact Hausdorff base. Its projectivization $\mathbb{P}(E) \xrightarrow{\pi} X$, the quotient of the complement of the zero section by fiberwise scalar multiplication, is a fiber bundle with fiber $\mathbb{R}\mathrm{P}^{n-1}$, and there is a class $x_E \in H^1(\mathbb{P}(E);\mathbb{F}_2)$ whose restriction to every fiber $\mathbb{P}(E_x) \cong \mathbb{R}\mathrm{P}^{n-1}$ is the generator of $H^1(\mathbb{R}\mathrm{P}^{n-1};\mathbb{F}_2)$ (for $n = 1$ the condition is empty). For any such class, $H^*(\mathbb{P}(E);\mathbb{F}_2)$ is a free $H^*(X;\mathbb{F}_2)$-module with basis $1, x_E, \dots, x_E^{n-1}$. In particular $\pi^*\colon H^*(X;\mathbb{F}_2) \to H^*(\mathbb{P}(E);\mathbb{F}_2)$ is a split injection. For a rank-n complex bundle the same holds with fibers $\mathbb{C}\mathrm{P}^{n-1}$, a class $t_E \in H^2(\mathbb{P}(E);\mathbb{Z})$ restricting on every fiber to a generator of $H^2(\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z})$ (for $n = 1$ the condition is empty), the module $H^*(\mathbb{P}(E);\mathbb{Z})$ free over $H^*(X;\mathbb{Z})$ with basis $1, t_E, \dots, t_E^{n-1}$, and π^* a split injection on $H^*(-;\mathbb{Z})$.

***Proof*:**

Step 1: $\mathbb{P}(E)$ is a fiber bundle. A trivialization $p^{-1}(U) \cong U \times \mathbb{R}^n$ of E is fiberwise linear, so it carries the complement of the zero section to $U \times (\mathbb{R}^n \setminus 0)$ compatibly with scalar multiplication and descends to a homeomorphism $\pi^{-1}(U) \cong U \times \mathbb{R}\mathrm{P}^{n-1}$ over U. Hence $\mathbb{P}(E) \to X$ is locally trivial with fiber $\mathbb{R}\mathrm{P}^{n-1}$.

Step 2: The class. Lemma 7.1.28 gives $g\colon E \to \mathbb{R}^\infty$ linear and injective on every fiber. It carries nonzero vectors to nonzero vectors and commutes with scalar multiplication, so it descends to a map of projectivizations

$$\mathbb{P}(g)\colon \mathbb{P}(E) \longrightarrow \mathbb{P}(\mathbb{R}^\infty) = \mathbb{R}\mathrm{P}^\infty$$

Set $x_E = \mathbb{P}(g)^*(a)$, where a is the degree-one generator of $H^*(\mathbb{R}\mathrm{P}^\infty;\mathbb{F}_2) = \mathbb{F}_2[a]$, as computed in example 5.5.9. Over a point $x \in X$ the restriction of $\mathbb{P}(g)$ to the fiber $\mathbb{P}(E_x) \cong \mathbb{R}\mathrm{P}^{n-1}$ is the projectivization $\mathbb{P}(u)$ of a linear injection $u\colon \mathbb{R}^n \to \mathbb{R}^\infty$. Applying the homotopy clause of lemma 7.1.28 to the trivial bundle $\mathbb{R}^n$ over a one-point base, u is homotopic to the standard inclusion $\mathbb{R}^n \subseteq \mathbb{R}^\infty$ through linear injections. The homotopy never vanishes on $\mathbb{R}^n \setminus 0$, and $(\mathbb{R}^n \setminus 0) \times I \to \mathbb{R}\mathrm{P}^{n-1} \times I$ is a quotient map since I is locally compact Hausdorff [16, Products of Quotient Maps], so the homotopy projectivizes, and $\mathbb{P}(u)$ is homotopic to the standard inclusion $\mathbb{R}\mathrm{P}^{n-1} \hookrightarrow \mathbb{R}\mathrm{P}^\infty$, under which a restricts to the generator of $H^1(\mathbb{R}\mathrm{P}^{n-1};\mathbb{F}_2)$ (example 5.5.9). Hence x_E restricts on every fiber to the generator. Since restriction to a fiber is a ring homomorphism and $H^*(\mathbb{R}\mathrm{P}^{n-1};\mathbb{F}_2) = \mathbb{F}_2[a]/(a^n)$ (example 5.5.9), the powers $1, x_E, \dots, x_E^{n-1}$ restrict on every fiber to the basis $1, a, \dots, a^{n-1}$. This last conclusion uses only the stated restriction property of the class, not its construction.

Step 3: Leray-Hirsch. By Step 1 the projectivization is a fiber bundle, hence a Serre fibration by theorem 5.2.4, with nonempty fibers $\mathbb{R}\mathrm{P}^{n-1}$. The group $H^m(\mathbb{R}\mathrm{P}^{n-1};\mathbb{F}_2)$ is a free $\mathbb{F}_2$-module of finite rank in every degree ($\mathbb{F}_2$ for $0 \leq m \leq n-1$ and 0 above), and the n homogeneous classes $1, x_E, \dots, x_E^{n-1}$ restrict to a basis in every fiber. Corollary 6.5.2 applies with $R = \mathbb{F}_2$ and makes $H^*(\mathbb{P}(E);\mathbb{F}_2)$ a free $H^*(X;\mathbb{F}_2)$-module with basis $1, x_E, \dots, x_E^{n-1}$. Freeness on a basis containing 1 splits π^*: the $H^*(X;\mathbb{F}_2)$-linear projection onto the coefficient of the basis element 1 is a left inverse, so π^* is a split injection.

Step 4: The complex case. The complex clause of lemma 7.1.28 gives $g\colon E \to \mathbb{C}^\infty$ complex-linear and injective on every fiber, descending as in Step 2 to $\mathbb{P}(g)\colon \mathbb{P}(E) \to \mathbb{C}\mathrm{P}^\infty$ on the complex projectivizations. Set $t_E = \mathbb{P}(g)^*(u)$ for a generator u of $H^2(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \cong \mathbb{Z}$ (example 5.5.11). The fibers are $\mathbb{C}\mathrm{P}^{n-1}$, the restriction of $\mathbb{P}(g)$ to a fiber is homotopic to the standard inclusion

$\mathbb{C}\mathrm{P}^{n-1} \hookrightarrow \mathbb{C}\mathrm{P}^{\infty}$ by the same interleaving homotopies, a generator of $H^2(\mathbb{C}\mathrm{P}^{\infty};\mathbb{Z})$ restricts to a generator of $H^2(\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z})$, and $H^*(\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z}) = \mathbb{Z}[u]/(u^n)$ (theorem 4.5.12), so the powers $1, t_E, \dots, t_E^{n-1}$ restrict to a basis of the fiber cohomology in every fiber, free of finite rank over $\mathbb{Z}$ in each degree. The complex projectivization is a fiber bundle by the argument of Step 1, hence a Serre fibration with nonempty fibers, so corollary 6.5.2 with $R = \mathbb{Z}$ gives the freeness, and the same projection argument splits π^*, completing the proof.

Theorem 7.4.3: Splitting Principle

Let $E \to X$ be a rank-n real vector bundle over a paracompact Hausdorff base. There is a space $F(E)$, the flag bundle of E, and a map $\pi\colon F(E) \to X$ such that

1. the induced map $\pi^*\colon H^*(X;\mathbb{F}_2) \to H^*(F(E);\mathbb{F}_2)$ is injective, and
2. the pulled-back bundle π^*E splits as a Whitney sum of real line bundles, $\pi^*E \cong L_1 \oplus \cdots \oplus L_n$.

The same statement holds for a rank-n complex bundle E with $\mathbb{F}_2$ coefficients replaced by $\mathbb{Z}$ coefficients and real line bundles replaced by complex line bundles.

***Proof*:**
The construction is a tower of projectivizations, and the cohomological input is proposition 7.4.2, whose base is an arbitrary paracompact Hausdorff space with no connectivity hypothesis. Form the projective bundle $\mathbb{P}(E) \xrightarrow{\pi_1} X$ whose fiber over x is the projective space of lines in E_x. Over $\mathbb{P}(E)$ the bundle π_1^*E contains a tautological line subbundle L_1 (the line each point of $\mathbb{P}(E)$ names). By proposition 7.1.23, applied over the paracompact Hausdorff base $\mathbb{P}(E)$ and resting on the inner product of lemma 7.1.22 in whichever of the two categories is in play, the subbundle L_1 has a complement: $\pi_1^*E \cong L_1 \oplus E_1$ with E_1 of rank $n-1$. By proposition 7.4.2, $H^*(\mathbb{P}(E);\mathbb{F}_2)$ is a free $H^*(X;\mathbb{F}_2)$-module on the powers $1, x_E, \dots, x_E^{n-1}$, a basis containing 1, so π_1^* is a split injection. Each stage is a fiber bundle whose fiber is a projective space, hence compact Hausdorff, so [16, Locally Trivial Maps with Compact Fiber Preserve Paracompactness] carries paracompactness and the Hausdorff property from the base of that stage to its total space. Inducting up the tower, every intermediate projectivization is again paracompact Hausdorff, so proposition 7.4.2 applies at every stage and the inner product used to split off each complement is available there. Iterating the construction on E_1, then on its rank-$(n-2)$ complement, and so on, produces after $n-1$ steps the flag bundle $F(E)$, over which π^*E has split completely into line bundles $L_1 \oplus \cdots \oplus L_n$, and the composite of split injections is a split injection, which proves clauses (1) and (2) in the real case. The complex case runs identically on the complex clause of proposition 7.4.2: complex projectivizations replace real ones, the tower splits off complex line bundles, and the split injectivity is on $H^*(-;\mathbb{Z})$, completing the proof.

The proof establishes one thing the statement does not record, and downstream arguments need it: every stage of the tower, the flag bundle included, is again paracompact Hausdorff. Without that, the characteristic classes of definition 7.2.7 are not defined on $F(E)$ and the splitting principle could not be used to prove anything about them.

Corollary 7.4.4: The Flag Bundle is Paracompact

Let $E \to X$ be a vector bundle over a paracompact Hausdorff base. Then the flag bundle $F(E)$ of theorem 7.4.3, and every intermediate projectivization in the tower that builds it, is again paracompact Hausdorff. In particular the Stiefel-Whitney and Chern classes of definition 7.2.13 and definition 7.2.7 are defined on $F(E)$.

***Proof*:**
Each stage of the tower is a fiber bundle whose fiber is a real or complex projective space, hence compact Hausdorff. By [16, Locally Trivial Maps with Compact Fiber Preserve Paracompactness], the total space of such a bundle over a paracompact Hausdorff base is again paracompact Hausdorff. Starting from X and inducting up the $n-1$ stages, every intermediate projectivization and $F(E)$ itself are paracompact Hausdorff. The last sentence is then theorem 7.1.29 applied over that base, which is what makes the definitions available, as we sought to show.

The force of the splitting principle is that to prove an identity among characteristic classes, one may pull back to $F(E)$, where the bundle is a sum of line bundles. The identity becomes a statement about line bundles, where it can be checked directly, and the injectivity of π^* carries the verified identity back down to X. The multiplicativity of the total class is the first and most important application.

Theorem 7.4.5: Whitney Sum Formula

Let $E, F \to X$ be vector bundles over a paracompact Hausdorff base, both real or both complex. Then the total Stiefel-Whitney class (in the real case) and the total Chern class (in the complex case) are multiplicative:

$$w(E \oplus F) = w(E)\, w(F) \in H^*(X; \mathbb{F}_2), \qquad c(E \oplus F) = c(E)\, c(F) \in H^*(X; \mathbb{Z})$$

where the products are cup products in the cohomology ring. Degree by degree, $w_k(E \oplus F) = \sum_{i+j=k} w_i(E) \smile w_j(F)$ and likewise for c.

***Proof*:**
The real case is written out. The complex case follows the same steps with $\mathbb{Z}$ in place of $\mathbb{F}_2$ and degrees doubled.

Step 1: The total class of a sum of line bundles. For a single real line bundle $L \to X$, classified by a map $X \to BO(1) = \mathbb{R}\mathrm{P}^\infty$, the total Stiefel-Whitney class is $w(L) = 1 + w_1(L)$, since $H^*(BO(1); \mathbb{F}_2) = H^*(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2) = \mathbb{F}_2[w_1]$ has its generator in degree one and a rank-one bundle has no higher classes (section 7.2). Consider the external sum of n line bundles, modeled universally over $BO(1)^n = (\mathbb{R}\mathrm{P}^\infty)^n$: let $a_i \in H^1(BO(1)^n; \mathbb{F}_2)$ be the first Stiefel-Whitney class of the i-th tautological line bundle ℓ_i (the pullback of the tautological line bundle along the i-th projection). The Künneth theorem (theorem 4.3.9) gives

$$H^*\big((\mathbb{R}\mathrm{P}^\infty)^n; \mathbb{F}_2\big) = \mathbb{F}_2[a_1, \dots, a_n], \qquad |a_i| = 1$$

a polynomial algebra on the n degree-one classes. The space $BO(1)^n$ is a CW complex, hence paracompact Hausdorff (theorem 7.1.7), so theorem 7.1.29 gives a map $f_n \colon BO(1)^n \to BO(n)$

classifying $\ell_1 \oplus \cdots \oplus \ell_n$. The claim is that

$$f_n^* w_k = \sigma_k(a_1, \ldots, a_n), \qquad 0 \le k \le n$$

the k-th elementary symmetric polynomial. For $k = 0$ both sides are 1. The proof is by induction on n. For $n = 1$ the bundle ℓ_1 is the tautological bundle, which the identity classifies, so $f_1 \simeq \mathrm{id}$ and $f_1^* w_1 = w_1 = a_1$. Let $n \ge 2$.

The top class. The proof of theorem 7.2.10 defines w_n as the mod-2 Euler class e of γ_n and computes $f_n^* e = a_1 \cdots a_n = \sigma_n(a_1, \ldots, a_n)$.

Symmetry. For a permutation τ of $\{1, \ldots, n\}$ let $\hat{\tau}(x_1, \ldots, x_n) = (x_{\tau(1)}, \ldots, x_{\tau(n)})$ on $BO(1)^n$. Then $\hat{\tau}^* \ell_i = \ell_{\tau(i)}$, so $\hat{\tau}^* a_i = a_{\tau(i)}$ by proposition 7.4.1, and $\hat{\tau}^*(\ell_1 \oplus \cdots \oplus \ell_n) \cong \ell_1 \oplus \cdots \oplus \ell_n$ after reordering the summands. Isomorphic bundles have homotopic classifying maps (theorem 7.1.29), so $f_n \circ \hat{\tau} \simeq f_n$ and $\hat{\tau}^* f_n^* w_k = f_n^* w_k$. Hence $f_n^* w_k$ is a symmetric polynomial in $a_1, \ldots, a_n$.

Restriction to $n-1$ factors. Fix $b_0 \in \mathbb{R}\mathrm{P}^\infty$ and let $j \colon BO(1)^{n-1} \to BO(1)^n$, $j(x) = (x, b_0)$. Then $j^* \ell_i = \ell_i$ for $i < n$, while $j^* \ell_n$ is pulled back from a point, hence trivial, so $j^* a_n = 0$ and j^* is the ring map setting $a_n = 0$. Let $i_n \colon BO(n-1) \to BO(n)$ be the stabilization, so that $i_n^* \gamma_n \cong \varepsilon^1 \oplus \gamma_{n-1}$ with ε^1 the trivial line bundle (proposition 7.1.19(3)). Thus $i_n \circ f_{n-1}$ classifies $\ell_1 \oplus \cdots \oplus \ell_{n-1} \oplus \varepsilon^1 \cong j^*(\ell_1 \oplus \cdots \oplus \ell_n)$, so $f_n \circ j \simeq i_n \circ f_{n-1}$. For $k \le n-1$, the relation $i_n^* w_k = w_k$ of theorem 7.2.10 and the inductive hypothesis give

$$j^* f_n^* w_k = f_{n-1}^* i_n^* w_k = f_{n-1}^* w_k = \sigma_k(a_1, \ldots, a_{n-1})$$

Conclusion. For $k \le n-1$ let $P = f_n^* w_k - \sigma_k(a_1, \ldots, a_n)$, a symmetric polynomial, homogeneous of degree k. Since $\sigma_k(a_1, \ldots, a_{n-1}, 0) = \sigma_k(a_1, \ldots, a_{n-1})$, the restriction step gives $P(a_1, \ldots, a_{n-1}, 0) = 0$, so a_n divides P. By symmetry every a_i divides P. The a_i are distinct primes of the unique factorization domain $\mathbb{F}_2[a_1, \ldots, a_n]$, so $a_1 \cdots a_n$ divides P. That product has degree $n > k$, so $P = 0$, which proves the claim. Summing over k,

$$w(\ell_1 \oplus \cdots \oplus \ell_n) = \sum_{k=0}^{n} \sigma_k(a_1, \ldots, a_n) = \prod_{i=1}^{n} (1 + a_i) = \prod_{i=1}^{n} w(\ell_i)$$

the second equality being the generating identity $\sum_k \sigma_k = \prod_i (1 + a_i)$ for elementary symmetric polynomials. For line bundles $L_1, \ldots, L_n$ over a paracompact Hausdorff base X, let $g_i \colon X \to BO(1)$ classify L_i and $g = (g_1, \ldots, g_n) \colon X \to BO(1)^n$. Then $g^* \ell_i \cong L_i$, so $L_1 \oplus \cdots \oplus L_n \cong g^*(\ell_1 \oplus \cdots \oplus \ell_n)$, and proposition 7.4.1 gives $w(L_1 \oplus \cdots \oplus L_n) = \prod_i (1 + w_1(L_i)) = \prod_i w(L_i)$. This is the Whitney formula for sums of line bundles.

In the complex case $H^*(BU(1)^n; \mathbb{Z}) = \mathbb{Z}[x_1, \ldots, x_n]$ with $x_i = c_1(\ell_i)$, and two inputs change. For the top class, c_n is the Euler class of the Gysin sequence of $S(\gamma_n)$ for the orientation induced by the complex orientation (theorem 7.2.5). This is $z^* j^* u_{\gamma_n}$ by proposition 7.3.5(2), hence the Euler class $e(\gamma_n)$ by corollary 7.3.6, the space $BU(n)$ being a path-connected CW complex of finite type (theorem 7.1.17). The Euler class is natural and multiplicative for direct-sum orientations (proposition 7.3.7, proposition 7.3.9, the space $BU(1)^n$ being a CW complex of finite type), the complex orientation of $\ell_1 \oplus \cdots \oplus \ell_n$ is the direct-sum orientation of the complex orientations of the ℓ_i (definition 7.1.25), and e equals c_1 on a complex line bundle (example 7.3.8), so $f_n^* c_n = x_1 \cdots x_n$. For the restriction, $i_n^* \gamma_n \cong \varepsilon^1_{\mathbb{C}} \oplus \gamma_{n-1}$ by proposition 7.1.19(3). The ring $\mathbb{Z}[x_1, \ldots, x_n]$ is again a unique factorization domain in which the x_i are distinct primes, and the rest of the argument is unchanged.

Step 2: Reduction by the splitting principle. Let E, F be arbitrary, of ranks m, n. Apply the splitting principle (theorem 7.4.3) to $E \oplus F$: there is $\pi\colon Y \to X$ with π^* injective on $\mathbb{F}_2$-cohomology and $\pi^*(E \oplus F) = \pi^*E \oplus \pi^*F$ splitting as a sum of line bundles, say $\pi^*E \cong L_1 \oplus \cdots \oplus L_m$ and $\pi^*F \cong L_{m+1} \oplus \cdots \oplus L_{m+n}$ (one may split E first, then F, building Y as a tower over X). By Step 1 applied to the full collection of line bundles,

$$w(\pi^*E \oplus \pi^*F) = \prod_{i=1}^{m+n} w(L_i) = \Big(\prod_{i=1}^{m} w(L_i)\Big)\Big(\prod_{j=m+1}^{m+n} w(L_j)\Big) = w(\pi^*E)\, w(\pi^*F)$$

where the middle factorization regroups the product and the outer equalities are Step 1 applied to π^*E, π^*F, and their sum separately. By naturality (proposition 7.4.1), $w(\pi^*E) = \pi^* w(E)$ and similarly for F and for $E \oplus F$, so

$$\pi^*\big(w(E \oplus F)\big) = \pi^*\big(w(E)\, w(F)\big)$$

using that π^* is a ring homomorphism. Since π^* is injective by theorem 7.4.3(1), the equality of the pullbacks forces $w(E \oplus F) = w(E)\, w(F)$ in $H^*(X; \mathbb{F}_2)$. The complex case runs identically over $\mathbb{Z}$, giving $c(E \oplus F) = c(E)\, c(F)$, completing the proof.

On a sum of line bundles the formula computes every Chern class from the first Chern classes of the summands.

Example 7.4.6: Chern Classes of a Sum of Line Bundles

Let $L_1, \ldots, L_n$ be complex line bundles over a paracompact Hausdorff base X with first Chern classes $x_j = c_1(L_j) \in H^2(X; \mathbb{Z})$, and let $E = L_1 \oplus \cdots \oplus L_n$. The Whitney sum formula (theorem 7.4.5) gives

$$c(E) = \prod_{j=1}^{n} c(L_j) = \prod_{j=1}^{n} (1 + x_j)$$

Expanding the product, the degree-$2i$ part is the i-th elementary symmetric polynomial in the x_j:

$$c_i(E) = \sigma_i(x_1, \ldots, x_n) = \sum_{j_1 < \cdots < j_i} x_{j_1} \cdots x_{j_i}$$

so $c_1(E) = x_1 + \cdots + x_n$, $c_2(E) = \sum_{j<k} x_j x_k$, and the top class $c_n(E) = x_1 \cdots x_n$. This is the computational meaning of the splitting principle: the universal calculation takes place in $H^*((\mathbb{CP}^\infty)^n; \mathbb{Z}) = \mathbb{Z}[x_1, \ldots, x_n]$, the cohomology of $(\mathbb{CP}^\infty)^n$, a classifying space of $U(1)^n$ (exercise 7.1.1 (4)), and the Chern classes of a sum of lines are the elementary symmetric functions of the line classes. The diagonal $U(1)^n \hookrightarrow U(n)$ induces a map $(\mathbb{CP}^\infty)^n \to BU(n)$ under which $c_i \mapsto \sigma_i(x_1, \ldots, x_n)$, so the universal Chern class $c_i \in H^*(BU(n))$ is detected by the symmetric polynomial σ_i, and exercise 7.4.1 (10) identifies $H^*(BU(n); \mathbb{Z})$ with the ring of symmetric polynomials $\mathbb{Z}[x_1, \ldots, x_n]^{S_n} = \mathbb{Z}[\sigma_1, \ldots, \sigma_n]$.

On line bundles the tensor product of definition 5.2.9 is a group law, with the trivial bundle as unit and the dual bundle as inverse, and the first Chern class converts it into the sum in H^2.

Proposition 7.4.7: The First Chern Class is Additive on Tensor Products

Let L, L' be complex line bundles over X.

1. If X is paracompact Hausdorff, then
$$c_1(L \otimes L') = c_1(L) + c_1(L')$$
and in particular $c_1(L^\vee) = -c_1(L)$.
2. If moreover X has the homotopy type of a CW complex, for instance if X is a CW complex, c_1 is an isomorphism of groups from $(\mathrm{Vect}^1_{\mathbb{C}}(X), \otimes)$ to $(H^2(X;\mathbb{Z}), +)$.

***Proof*:**

Part (1). Complex line bundles over a paracompact X are classified by homotopy classes of maps into $\mathrm{Gr}_1(\mathbb{C}^\infty) = BU(1) = \mathbb{CP}^\infty$ by theorem 7.1.29, and $c_1(L) = f_L^* x$ for a classifying map f_L, which exists by theorem 7.1.29 and defines c_1 by definition 7.2.7, where x generates $H^2(\mathbb{CP}^\infty;\mathbb{Z}) \cong \mathbb{Z}$ (example 7.2.3).

The identity is proved once on $\mathbb{CP}^\infty \times \mathbb{CP}^\infty$ and transported. Write γ_1, γ_2 for the pullbacks of the tautological line bundle along the two projections, and let $\mu\colon \mathbb{CP}^\infty \times \mathbb{CP}^\infty \to \mathbb{CP}^\infty$ be a classifying map for $\gamma_1 \otimes \gamma_2$. Such a map exists because $\mathbb{CP}^\infty \times \mathbb{CP}^\infty$ is paracompact, and it is the map Bm induced by the multiplication $m\colon U(1) \times U(1) \to U(1)$. Then $L \otimes L' \cong (f_L, f_{L'})^*(\gamma_1 \otimes \gamma_2)$, since pulling γ_1, γ_2 back along $(f_L, f_{L'})$ gives L, L' and tensor product commutes with pullback, so $L \otimes L'$ is classified by the composite

$$X \xrightarrow{(f_L, f_{L'})} \mathbb{CP}^\infty \times \mathbb{CP}^\infty \xrightarrow{\mu} \mathbb{CP}^\infty$$

and it suffices to compute $\mu^* x$. Nothing beyond theorem 7.1.29 is used of X.

Since $H^*(\mathbb{CP}^\infty;\mathbb{Z}) = \mathbb{Z}[x]$ is concentrated in even degrees and is free, the Künneth theorem gives $H^2(\mathbb{CP}^\infty \times \mathbb{CP}^\infty;\mathbb{Z}) = \mathbb{Z}x_1 \oplus \mathbb{Z}x_2$, where x_i is the pullback of x along the i-th projection, with no cross term because $H^1(\mathbb{CP}^\infty;\mathbb{Z}) = 0$. Write $\mu^* x = a\, x_1 + b\, x_2$ with $a, b \in \mathbb{Z}$. Restricting along the inclusion $\iota_1\colon \mathbb{CP}^\infty \to \mathbb{CP}^\infty \times \mathbb{CP}^\infty$, $z \mapsto (z, *)$, the composite $\mu\iota_1$ classifies the tensor product of the tautological bundle with the trivial bundle, which is the tautological bundle itself, so $\mu\iota_1$ is homotopic to the identity. Hence $x = \iota_1^* \mu^* x = a\, x$, giving $a = 1$, and symmetrically $b = 1$. Therefore $\mu^* x = x_1 + x_2$ and

$$c_1(L \otimes L') = (f_L, f_{L'})^* \mu^* x = (f_L, f_{L'})^*(x_1 + x_2) = f_L^* x + f_{L'}^* x = c_1(L) + c_1(L')$$

The bundle $L \otimes L^\vee$ is trivial, the evaluation pairing $L \otimes L^\vee \to \underline{\mathbb{C}}$ being a homomorphism that is an isomorphism on each fiber (lemma 5.2.10), and a trivial bundle has $c_1 = 0$, being pulled back from a point (proposition 7.4.1). So $c_1(L) + c_1(L^\vee) = 0$.

Part (2). For X paracompact Hausdorff of CW homotopy type, example 7.2.3 shows that c_1 is a bijection $\mathrm{Vect}^1_{\mathbb{C}}(X) \to H^2(X;\mathbb{Z})$, through the classification of line bundles and the representation of H^2 by $\mathbb{CP}^\infty = K(\mathbb{Z}, 2)$. By part (1) it is a homomorphism, hence an isomorphism of groups, completing the proof.

The Whitney sum formula is the multiplication table of the theory. Combined with naturality and

the normalization below, it determines every characteristic class of every bundle assembled from line bundles by sums and pullbacks, which by the splitting principle is effectively every bundle. The normalization axiom fixes the one free constant: it says what the classes do on the universal line bundle.

Proposition 7.4.8: Normalization on the Universal Line Bundle

Let $\gamma^1_{\mathbb{R}} \to \mathbb{R}\mathrm{P}^\infty$ be the tautological real line bundle, and let $\gamma^1_{\mathbb{C}} \to \mathbb{C}\mathrm{P}^\infty$ be the tautological complex line bundle, with dual (hyperplane) bundle $(\gamma^1_{\mathbb{C}})^* \to \mathbb{C}\mathrm{P}^\infty$ (definition 5.2.9). Then

$$w_1(\gamma^1_{\mathbb{R}}) = a \in H^1(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2), \qquad c_1\big((\gamma^1_{\mathbb{C}})^*\big) = u \in H^2(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z})$$

where a is the generator of $H^1(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2) = \mathbb{F}_2[a]$ and u is the generator of $H^2(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) = \mathbb{Z}[u]$ restricting on $\mathbb{C}\mathrm{P}^1$ to the class u dual to $[\mathbb{C}\mathrm{P}^1]$ of example 7.2.3. Equivalently $c_1(\gamma^1_{\mathbb{C}}) = -u$. In particular $w_1(\gamma^1_{\mathbb{R}}) \neq 0$ and $c_1(\gamma^1_{\mathbb{C}}) \neq 0$. The complex equality is the sign convention that fixed c_1 in example 7.2.3, the sign for which $c_1(L)$ is the Euler class of $L_{\mathbb{R}}$ (example 7.3.8).

***Proof*:**
The classifying space of $O(1) = \{\pm 1\}$ is $BO(1) = \mathbb{R}\mathrm{P}^\infty$, with universal bundle $\gamma^1_{\mathbb{R}}$, and $w_1 \in H^1(BO(1); \mathbb{F}_2)$ is by definition the generator of $H^*(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2) = \mathbb{F}_2[a]$ pulled back along the identity classifying map of $\gamma^1_{\mathbb{R}}$, which is a itself, the only nonzero element of $H^1(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2) \cong \mathbb{F}_2$. For the complex case, $\gamma^1_{\mathbb{C}}$ has the identity as classifying map, so $c_1(\gamma^1_{\mathbb{C}})$ is the generator c_1 of $H^2(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z})$, which restricts to $-u$ on $\mathbb{C}\mathrm{P}^1$ by the convention of example 7.2.3. Restriction to $\mathbb{C}\mathrm{P}^1$ is an isomorphism on H^2, so $c_1(\gamma^1_{\mathbb{C}}) = -u$. The dual of a line bundle has the opposite first Chern class (proposition 7.4.7), so $c_1\big((\gamma^1_{\mathbb{C}})^*\big) = u$, as we sought to show.

The real counterpart of proposition 7.4.7(2) replaces $\mathbb{C}\mathrm{P}^\infty = K(\mathbb{Z}, 2)$ by $\mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2, 1)$ and c_1 by w_1, which by proposition 7.4.8 is the generator of $H^1(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2)$.

Theorem 7.4.9: Real Line Bundles are Classified by the First Stiefel-Whitney Class

Let X be a CW complex. The first Stiefel-Whitney class is a bijection

$$w_1 \colon \mathrm{Vect}^1_{\mathbb{R}}(X) \xrightarrow{\cong} H^1(X; \mathbb{F}_2)$$

natural in X, from the isomorphism classes of real line bundles over X to the first cohomology with $\mathbb{F}_2$-coefficients. Under tensor product of line bundles on one side and addition on the other it is an isomorphism of abelian groups, so a real line bundle is trivial if and only if its first Stiefel-Whitney class vanishes.

***Proof*:**
A real line bundle over X is a principal $O(1)$-bundle, and $O(1) = \{\pm 1\} = \mathbb{Z}/2$, so the classification theorem (theorem 7.1.10) for the group $\mathbb{Z}/2$ gives a natural bijection $\mathrm{Vect}^1_{\mathbb{R}}(X) \cong [X, B\mathbb{Z}/2] = [X, \mathbb{R}\mathrm{P}^\infty]$ sending a line bundle to the homotopy class of its classifying map. Since $\mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2, 1)$ (example 5.5.6), representability of cohomology by Eilenberg-MacLane spaces (theorem 5.5.12) gives a natural bijection $[X, \mathbb{R}\mathrm{P}^\infty] \cong H^1(X; \mathbb{F}_2)$ sending a map to the

pullback of the generator $a \in H^1(\mathbb{R}\mathrm{P}^\infty;\mathbb{F}_2)$. That generator is the universal first Stiefel-Whitney class $w_1(\gamma^1_{\mathbb{R}})$ of section 7.2, so the composite sends a line bundle L with classifying map f_L to $f_L^*(a) = w_1(L)$, the class of definition 7.2.13. Both bijections carry the group structures to one another: the set $\mathrm{Vect}^1_{\mathbb{R}}(X)$ is an abelian group under tensor product because $\mathbb{Z}/2$ is abelian, and this operation is the one classified by the H-space multiplication on $\mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2,1)$, which represents the addition on $H^1(-;\mathbb{F}_2)$. Hence w_1 is an isomorphism of abelian groups, and a line bundle is trivial exactly when its class is the identity 0, completing the proof.

A worked instance grounds the normalization in a finite-dimensional computation.

Example 7.4.10: The Tautological Line Bundle on $\mathbb{R}\mathrm{P}^n$ and $\mathbb{C}\mathrm{P}^n$

Let $\gamma^1 \to \mathbb{R}\mathrm{P}^n$ be the tautological real line bundle, whose fiber over a line $\ell \in \mathbb{R}\mathrm{P}^n$ is ℓ itself, viewed as a one-dimensional subspace of $\mathbb{R}^{n+1}$. It is the restriction of $\gamma^1_{\mathbb{R}}$ along the inclusion $\iota\colon \mathbb{R}\mathrm{P}^n \hookrightarrow \mathbb{R}\mathrm{P}^\infty$. The cohomology ring is $H^*(\mathbb{R}\mathrm{P}^n;\mathbb{F}_2) = \mathbb{F}_2[a]/(a^{n+1})$ with $|a| = 1$ (example 5.5.9), and ι^* sends the generator of $H^1(\mathbb{R}\mathrm{P}^\infty;\mathbb{F}_2)$ to the generator a of $H^1(\mathbb{R}\mathrm{P}^n;\mathbb{F}_2)$. By naturality (proposition 7.4.1) and normalization (proposition 7.4.8),

$$w_1(\gamma^1) = \iota^*\big(w_1(\gamma^1_{\mathbb{R}})\big) = \iota^*(a) = a \neq 0 \in H^1(\mathbb{R}\mathrm{P}^n;\mathbb{F}_2)$$

so the total class is $w(\gamma^1) = 1 + a$. The nonvanishing of $w_1(\gamma^1)$ is the cohomological certificate that γ^1 is nontrivial, indeed (as theorem 7.4.13 below shows) non-orientable: a trivial bundle would have $w_1 = 0$ by naturality applied to the constant classifying map.

For the complex tautological line bundle $\gamma^1 \to \mathbb{C}\mathrm{P}^n$ (fiber over $\ell \in \mathbb{C}\mathrm{P}^n$ the complex line $\ell \subseteq \mathbb{C}^{n+1}$), the restriction $\jmath\colon \mathbb{C}\mathrm{P}^n \hookrightarrow \mathbb{C}\mathrm{P}^\infty$ sends the generator $u \in H^2(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z})$ to the generator of $H^2(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) = \mathbb{Z}\langle u\rangle$ in the ring $\mathbb{Z}[u]/(u^{n+1})$. By naturality and normalization (proposition 7.4.8), with $c_1(\gamma^1_{\mathbb{C}}) = -u$ on the universal bundle,

$$c_1(\gamma^1) = \jmath^*\big(c_1(\gamma^1_{\mathbb{C}})\big) = \jmath^*(-u) = -u$$

a generator of $H^2(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \cong \mathbb{Z}$. The total Chern class is $c(\gamma^1) = 1 - u$. The dual (hyperplane) bundle $(\gamma^1)^*$ then has $c_1\big((\gamma^1)^*\big) = +u$ (proposition 7.4.7), the positive generator dual to the fundamental class: this is the topological counterpart of the hyperplane class, and the sign convention is the one fixed in section 7.2 and example 7.2.3, where the hyperplane bundle over $\mathbb{C}\mathrm{P}^1 = S^2$ is the Hopf bundle with Euler number $+1$.

The Chern classes of a complex bundle E also present the cohomology ring of its projectivization $\mathbb{P}(E)$, the bundle of lines in the fibers. The fiber $\mathbb{C}\mathrm{P}^{n-1}$ has cohomology $\mathbb{Z}[t]/(t^n)$, proposition 7.4.2 makes $H^*(\mathbb{P}(E))$ a free module over the cohomology of the base on the powers of one class, and the Chern classes are the coefficients of the one relation among those powers.

Proposition 7.4.11: The Projective Bundle Formula

Let $E \to X$ be a complex vector bundle of rank $n \geq 1$ over a paracompact Hausdorff base X, and let $\mathbb{P}(E) \xrightarrow{\pi} X$ be its projectivization, the bundle whose fiber over x is the projective space $\mathbb{P}(E_x) \cong \mathbb{C}\mathrm{P}^{n-1}$ of lines in E_x. Let $\gamma_E \to \mathbb{P}(E)$ be the tautological line bundle (whose fiber over a line $\ell \subseteq E_x$ is ℓ itself), and set $t = -c_1(\gamma_E) = c_1(\gamma_E^*) \in H^2(\mathbb{P}(E);\mathbb{Z})$, the first Chern class of its dual. Then $H^*(\mathbb{P}(E);\mathbb{Z})$ is a free module over $H^*(X;\mathbb{Z})$ with basis $1, t, t^2, \dots, t^{n-1}$, and as a ring

$$H^*(\mathbb{P}(E);\mathbb{Z}) \cong H^*(X;\mathbb{Z})[t] \big/ \left(t^n + c_1(E)\, t^{n-1} + \cdots + c_{n-1}(E)\, t + c_n(E)\right)$$

where the $c_i(E)$ are the Chern classes of definition 7.2.7.

***Proof*:**

Step 1: The module. The space $\mathbb{P}(E)$ is paracompact Hausdorff (corollary 7.4.4, whose first stage it is), so t is defined. On a fiber $\mathbb{P}(E_x) = \mathbb{C}\mathrm{P}^{n-1}$ the bundle γ_E restricts to the tautological line bundle, whose first Chern class is minus a generator of $H^2(\mathbb{C}\mathrm{P}^{n-1};\mathbb{Z})$ (example 7.4.10), and $c_1(\gamma_E^*) = -c_1(\gamma_E)$ (proposition 7.4.7(1)). So t restricts on every fiber to a generator, and by proposition 7.4.2 the module $H^*(\mathbb{P}(E);\mathbb{Z})$ is free over $H^*(X;\mathbb{Z})$ with basis $1, t, \dots, t^{n-1}$. In particular $t^n = -\sum_{i=1}^{n} a_i(E)\, t^{n-i}$ for unique classes $a_i(E) \in H^{2i}(X;\mathbb{Z})$, and the ring is $H^*(X;\mathbb{Z})[t]$ modulo the single relation $t^n + \sum_i a_i(E)\, t^{n-i} = 0$. It remains to prove $a_i(E) = c_i(E)$.

Step 2: Naturality of the coefficients. For a map $g\colon Y \to X$ from a paracompact Hausdorff space, $\mathbb{P}(g^*E)$ is the pullback of $\mathbb{P}(E)$ and γ_{g^*E} the pullback of γ_E, so the relation pulls back to $t^n = -\sum_i g^*a_i(E)\, t^{n-i}$ in $H^*(\mathbb{P}(g^*E);\mathbb{Z})$. Uniqueness of the coefficients in the free module gives $a_i(g^*E) = g^*a_i(E)$, and $c_i(g^*E) = g^*c_i(E)$ by proposition 7.4.1.

Step 3: Sums of line bundles. Let $E = L_1 \oplus \cdots \oplus L_n$ with $x_j = c_1(L_j)$, and let $F = \pi^*E \otimes \gamma_E^* = \bigoplus_j \pi^*L_j \otimes \gamma_E^*$, a sum of line bundles with $c_1(\pi^*L_j \otimes \gamma_E^*) = \pi^*x_j + t$ (proposition 7.4.7(1)). By example 7.4.6, $c_n(F) = \prod_j (\pi^*x_j + t)$. The inclusion $\gamma_E \subseteq \pi^*E$ is a nowhere-zero section of $\mathrm{Hom}(\gamma_E, \pi^*E) \cong F$, so F contains a trivial line subbundle $\underline{\mathbb{C}}$, and $F \cong F' \oplus \underline{\mathbb{C}}$ with F' of rank $n-1$ (proposition 7.1.23). A trivial bundle is pulled back from a point, so $c(\underline{\mathbb{C}}) = 1$, and the Whitney sum formula (theorem 7.4.5) gives $c(F) = c(F')$, which vanishes in degree $2n$ because F' has rank $n-1$. Hence $\prod_j (\pi^*x_j + t) = 0$, that is $t^n + \sigma_1(x)\, t^{n-1} + \cdots + \sigma_n(x) = 0$, and $\sigma_i(x) = c_i(E)$ by example 7.4.6, so $a_i(E) = c_i(E)$ by uniqueness of the coefficients.

Step 4: The general case. By theorem 7.4.3 and corollary 7.4.4 there is a map $p\colon F(E) \to X$ from a paracompact Hausdorff space, injective on $H^*(-;\mathbb{Z})$, with p^*E a sum of line bundles. By Steps 2 and 3, $p^*a_i(E) = a_i(p^*E) = c_i(p^*E) = p^*c_i(E)$, so $a_i(E) = c_i(E)$, completing the proof.

The module half of the formula is proposition 7.4.2, the input of the splitting principle. The ring half determines the Chern classes from $\mathbb{P}(E)$ alone, and it is the definition of Chern classes that a development starting from projective bundles adopts.

The last axiom identifies the top class. For an oriented real bundle there is an integral Euler class $e(E) \in H^n(X;\mathbb{Z})$, defined via the Thom isomorphism or, equivalently, as the obstruction to a nowhere-zero section. Its mod-2 reduction is the top Stiefel-Whitney class, and for a complex bundle the top Chern class *is* the Euler class of the underlying oriented real bundle.

Theorem 7.4.12: Top Class is the Euler Class

Let $E \to X$ be a real vector bundle of rank n over a paracompact Hausdorff base.

1. If X is a path-connected CW complex and $n \geq 2$, the top Stiefel-Whitney class is the mod-2 Euler class of lemma 7.2.9, $w_n(E) = e_2(E)$, with no orientation hypothesis. If E is oriented, with Euler class $e(E) \in H^n(X;\mathbb{Z})$, then $w_n(E) = \rho_2\big(e(E)\big)$, where $\rho_2 \colon H^n(X;\mathbb{Z}) \to H^n(X;\mathbb{F}_2)$ is reduction of coefficients.
2. If E is a complex bundle of complex rank n, regarded as an oriented real bundle of real rank $2n$ through its complex orientation (definition 7.1.25), then the top Chern class equals the Euler class, $c_n(E) = e(E) \in H^{2n}(X;\mathbb{Z})$.

***Proof*:**
Both statements are proved for the universal bundles and pulled back along classifying maps, which exist over the paracompact Hausdorff base by theorem 7.1.29 and corollary 7.1.30.

1. For $n \geq 2$, theorem 7.2.10 makes $w_n \in H^n(BO(n);\mathbb{F}_2)$ the mod-2 Euler class $e_2(\gamma_n)$ of the sphere bundle of γ_n. If X is a path-connected CW complex and $f \colon X \to BO(n)$ classifies E, then lemma 7.2.9(1) gives $e_2(E) = e_2(f^*\gamma_n) = f^* e_2(\gamma_n) = f^* w_n = w_n(E)$.

   For the oriented clause let $\pi \colon BSO(n) \to BO(n)$ forget the orientation, so that $\pi^*\gamma_n = \tilde{\gamma}_n$ as vector bundles. If $\tilde{f}$ is an oriented classifying map of E, then $\pi \circ \tilde{f}$ classifies E, so $w_n(E) = \tilde{f}^*\pi^* w_n$ and $e(E) = \tilde{f}^* e$, and since ρ_2 commutes with $\tilde{f}^*$ it suffices to show $\pi^* w_n = \rho_2(e)$ in $H^n(BSO(n);\mathbb{F}_2)$. For $n = 1$ both sides vanish, $BSO(1)$ being contractible (definition 7.2.15). For $n \geq 2$, $\pi^* w_n = \pi^* e_2(\gamma_n) = e_2(\tilde{\gamma}_n)$ by lemma 7.2.9(1), the spaces $BSO(n)$ and $BO(n)$ being path-connected CW complexes. Reduction of coefficients $C^*(-;\mathbb{Z}) \to C^*(-;\mathbb{F}_2)$ preserves the filtration of theorem 6.2.8 for $S(\tilde{\gamma}_n) \to BSO(n)$, so it induces a morphism of spectral sequences commuting with every d_r (theorem 6.1.10). On the bottom row it is ρ_2, and it carries the integral fiber generator α to the nonzero class of $H^{n-1}(S^{n-1};\mathbb{F}_2)$. Hence $\rho_2(e) = \rho_2\big(d_n(1 \otimes \alpha)\big) = d_n\big(1 \otimes \rho_2(\alpha)\big) = e_2(\tilde{\gamma}_n) = \pi^* w_n$.
2. Let $f \colon X \to BU(n)$ classify E, and let $\sigma \colon BU(n) \to BSO(2n)$ classify γ_n as an oriented real bundle with its complex orientation, which exists by corollary 7.1.30 since $BU(n)$ is paracompact Hausdorff (theorem 7.1.17). The complex orientation is defined fiberwise, so it pulls back to the complex orientation, and $\sigma \circ f$ classifies E with its complex orientation. So $e(E) = f^*\sigma^* e$ and $c_n(E) = f^* c_n$, and it suffices to show $\sigma^* e = c_n$ on $BU(n)$. For $n = 1$ this is example 7.3.8. For $n \geq 2$, the class $\sigma^* e$ is the Euler class of γ_n (definition 7.2.15), which is $z^* j^* u_{\gamma_n}$ by corollary 7.3.6, $BU(n)$ being a CW complex of finite type (theorem 7.1.17). By proposition 7.3.5(2) that class is the Euler class of the Gysin sequence of $S(\gamma_n)$ for the induced orientation, which is c_n by theorem 7.2.5, completing the proof.

These four properties (naturality, Whitney sum, normalization, top class equals Euler class) do more than hold for the classes built in section 7.2. They characterize them. The uniqueness statement is that any assignment of cohomology classes to bundles satisfying naturality, the Whitney sum formula, and the normalization on the universal line bundle must coincide with the Stiefel-Whitney classes (respectively Chern classes), because the splitting principle reduces any such assignment to its values on line bundles, where normalization fixes it. The full uniqueness theorem is proved in [29, §7, §14].

The present section uses the axioms as a calculus, deriving consequences rather than re-establishing existence.

With the axioms in place as a formal calculus, the geometric meaning of these classes can be stated: each is an obstruction. The general principle is the obstruction theory of theorem 5.6.7: a structure on a bundle (an orientation, a spin structure, a nowhere-zero section, a reduction of the structure group) is a section of an associated fibration, and the primary obstruction to building such a section lives in a cohomology group $H^{k+1}(X;\pi_k(F))$ for the relevant fiber F. The characteristic classes are exactly these primary obstructions, identified through the homotopy of the fiber. The cleanest case is orientability.

Theorem 7.4.13: First Stiefel-Whitney Class is the Obstruction to Orientability

Let $E \to X$ be a real vector bundle.

1. If X is paracompact Hausdorff and E is orientable, then $w_1(E) = 0 \in H^1(X;\mathbb{F}_2)$.
2. If X is a connected CW complex and $w_1(E) = 0$, then E is orientable.
3. If E is orientable over a paracompact Hausdorff base, its orientations form a torsor under the group $C(X;\mathbb{F}_2)$ of *locally constant* functions $X \to \mathbb{F}_2$. In particular an orientable bundle over a connected base has exactly two orientations. When X is locally path-connected, for instance a CW complex, $C(X;\mathbb{F}_2) = H^0(X;\mathbb{F}_2)$. Over a general paracompact base the two differ, since H^0 sees path-components and C sees components.

In particular, over a connected CW complex E is orientable if and only if $w_1(E) = 0$.

The two clauses carry different hypotheses because they need different theorems, and the asymmetry is the one running through this whole section. Clause (1) reads a consequence off a bundle that has been trivialized, and every step of it is available wherever the classes are defined. Clause (2) must produce a trivialization from the vanishing of a cohomology class, and that is the assertion that a map into $\mathbb{RP}^\infty$ inducing zero on H^1 is nullhomotopic, which is representability (theorem 5.5.12), holding over a CW complex and not over an arbitrary paracompact space. Stating the theorem under the single hypothesis that makes both halves true would discard clause (1) over exactly the bases, flag bundles among them, where the rest of the chapter applies it.

***Proof*:**
The proof identifies orientability with the triviality of a $\mathbb{Z}/2$-bundle, identifies that $\mathbb{Z}/2$-bundle's class with $w_1(E)$, and then reads the two implications off in the two regimes.

Step 1: Orientability as a section of the orientation cover. An orientation of a rank-n bundle E is a continuous choice of orientation of each fiber E_x, that is, a choice of generator of $H_n(E_x, E_x \setminus 0;\mathbb{Z}) \cong \mathbb{Z}$ varying continuously in x. The two generators per fiber assemble into a double cover $\mathrm{Or}(E) \to X$, the *orientation cover*, whose fiber over x is the two-element set of orientations of E_x. This is the bundle associated to the principal $O(n)$-frame bundle of E along the determinant-sign homomorphism $\det\colon O(n) \to \{\pm 1\} = \mathbb{Z}/2$ (on orthogonal matrices det takes the values ± 1). An orientation of E is precisely a section of this double cover, and a double cover admits a section if and only if it is trivial, with no connectivity hypothesis on X: a section s has clopen image (it is open because s is a section of a covering and closed because its complement is the image of the other sheet, also a section), so the cover splits as the disjoint union of $s(X)$ and its complement, each mapping homeomorphically to X. Conversely a trivial

cover has the obvious section. So E is orientable if and only if its orientation cover is trivial.

Step 2: The double cover as a map to $\mathbb{R}\mathrm{P}^\infty$. A double cover of X is the unit-sphere bundle of a real line bundle, namely of the line bundle it induces along $\mathbb{Z}/2 = O(1)$ acting on $\mathbb{R}$, and this correspondence is a bijection between double covers and rank-one real bundles, and the two constructions are mutually inverse. Given a double cover $P \to X$, the associated bundle $P \times_{\mathbb{Z}/2} \mathbb{R}$ is a real line bundle whose unit-sphere bundle for the inner product induced from the standard one on $\mathbb{R}$ is P again, since $P \times_{\mathbb{Z}/2} \{\pm 1\} = P$. Given a line bundle L, fix an inner product by lemma 7.1.22. Then $S(L) \times_{\mathbb{Z}/2} \mathbb{R} \cong L$ by $(v, t) \mapsto tv$, which is well defined on the quotient and fiberwise linear and injective, hence an isomorphism by lemma 5.2.10. The choice of inner product does not matter: any two are connected by the convex path $(1-t)\langle -,-\rangle_0 + t\langle -,-\rangle_1$, positive-definite at every t, and the resulting sphere bundles are isomorphic over X by normalizing. For the orientation cover the associated line bundle is $\det E = \Lambda^n E$, since both are induced from the frame bundle along $\det\colon O(n) \to \{\pm 1\}$. Over a paracompact Hausdorff base $\mathrm{Vect}^1_{\mathbb{R}}(X) = [X, \mathbb{R}\mathrm{P}^\infty]$ by theorem 7.1.29 with $n = 1$, so the orientation cover has a classifying map $f\colon X \to \mathbb{R}\mathrm{P}^\infty = BO(1)$, and it is trivial exactly when f is nullhomotopic. Set $\omega(E) = f^*(a) \in H^1(X; \mathbb{F}_2)$ for a the generator of $H^1(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2)$. By definition $\omega(E) = w_1(\det E)$.

Step 3: Identifying $\omega(E) = w_1(E)$. It remains to see that the class $\omega(E)$ classifying the orientation cover equals the first Stiefel-Whitney class $w_1(E)$. Both are natural in E under pullback (the orientation cover of f^*E is the pullback of the orientation cover of E, and w_1 is natural by proposition 7.4.1), and both vanish on trivial bundles, so by the universal property it suffices to compare them on the universal bundle $\gamma^n \to BO(n)$. There $H^1(BO(n); \mathbb{F}_2) = \mathbb{F}_2\langle w_1 \rangle$ is one-dimensional (the degree-one part of $\mathbb{F}_2[w_1, \dots, w_n]$), so $\omega(\gamma^n)$ is either 0 or w_1. It is not 0. Let $\iota\colon BO(1) = \mathbb{R}\mathrm{P}^\infty \to BO(n)$ be induced by the inclusion $O(1) \hookrightarrow O(n)$ as the upper-left block. Then $\iota^*\gamma^n \cong \gamma^1_{\mathbb{R}} \oplus \underline{\mathbb{R}}^{n-1}$: the block inclusion splits $\mathbb{R}^n$ as the tautological line plus a fixed complement, and the restriction of a rank-n bundle along any map has rank n, so it is not $\gamma^1_{\mathbb{R}}$ itself. Taking determinants, $\det(\gamma^1_{\mathbb{R}} \oplus \underline{\mathbb{R}}^{n-1}) \cong \gamma^1_{\mathbb{R}}$, because the top exterior power of a sum with a trivial bundle is the top exterior power of the nontrivial summand. Hence by Step 2 and naturality of det, $\iota^*\omega(\gamma^n) = \omega\big(\iota^*\gamma^n\big) = w_1\big(\det \iota^*\gamma^n\big) = w_1(\gamma^1_{\mathbb{R}}) = a \neq 0$ in $H^1(\mathbb{R}\mathrm{P}^\infty; \mathbb{F}_2)$, the last step being proposition 7.4.8. A class pulling back to something nonzero is nonzero, so $\omega(\gamma^n) \neq 0$. Thus $\omega(\gamma^n) = w_1(\gamma^n)$, and by naturality $\omega(E) = w_1(E)$ for every E.

Step 4: The two implications. For clause (1), let X be paracompact Hausdorff and E orientable. By Step 1 the orientation cover is trivial, so its classifying map f of Step 2 is nullhomotopic, hence constant up to homotopy and $\omega(E) = f^*(a) = 0$. By Step 3, $w_1(E) = 0$. Nothing here identifies $[X, \mathbb{R}\mathrm{P}^\infty]$ with a cohomology group: the argument runs from a trivialization to the vanishing of a pullback, and theorem 7.1.29 gives the classifying map over any paracompact Hausdorff base.

For clause (2), let X be a connected CW complex with $w_1(E) = 0$. By Step 3, $\omega(E) = f^*(a) = 0$. Since $\mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2, 1)$ (example 5.5.6), representability (theorem 5.5.12) makes $[X, \mathbb{R}\mathrm{P}^\infty] \to H^1(X; \mathbb{F}_2)$, $[f] \mapsto f^*(a)$, a bijection, so $f^*(a) = 0$ forces f nullhomotopic. Hence the orientation cover is trivial and E is orientable by Step 1. This is the step that fails over a general paracompact base, where the same map $[X, \mathbb{R}\mathrm{P}^\infty] \to H^1(X; \mathbb{F}_2)$ need not be injective.

Step 5: The torsor clause. Suppose E is orientable, so the orientation cover $\mathrm{Or}(E) \to X$ has a section, and let $\varepsilon \in C(X; \mathbb{F}_2)$ be a locally constant function. Acting on a section s by flipping it over the locus $\{\varepsilon = 1\}$, which is clopen, gives a map that is again continuous and again a section, so $C(X; \mathbb{F}_2)$ acts on the set of orientations. The action is free: if ε fixes s then ε

vanishes, since flipping changes the sheet wherever $\varepsilon = 1$. It is transitive: given orientations s, s', the function taking the value 1 exactly where they name different sheets is locally constant, because $\mathrm{Or}(E)$ is a covering and the two sections agree or disagree locally, and it carries s to s'. Hence the orientations form a torsor under $C(X;\mathbb{F}_2)$, and when X is connected the only locally constant functions are the two constants, giving exactly two orientations. The identification $C(X;\mathbb{F}_2) = H^0(X;\mathbb{F}_2)$ needs X locally path-connected: $H^0(X;G)$ is the functions from *path*-components (section 5.3), and on a space like the topologist's sine curve, connected with two path-components, there are two locally constant functions and four elements of H^0. When X is locally path-connected its path-components are open, so a function on X is locally constant exactly when it is constant on path-components, and $C(X;\mathbb{F}_2) = H^0(X;\mathbb{F}_2)$. This completes the proof.

The theorem casts orientability as the vanishing of a single cohomology class, and it does so by exhibiting orientability as a sectioning problem. The orientation cover is the fibration, and its fiber is the discrete two-point space $\mathbb{Z}/2$. The obstruction to a section is the primary (and here only) obstruction of theorem 5.6.7, living in $H^1(X;\pi_0(\mathbb{Z}/2)) = H^1(X;\mathbb{Z}/2)$, and that obstruction is $w_1(E)$. The two-element fiber has $\pi_0 = \mathbb{Z}/2$ and no higher homotopy, so a single obstruction in H^1 controls the entire problem, which is why orientability is governed by one class rather than a tower of them. The next structure up the ladder, a spin structure, has a fiber with nontrivial π_1, and its obstruction lands one degree higher.

Proposition 7.4.14: Relating Chern and Stiefel-Whitney Classes

Let $E \to B$ be a complex n-dimensional vector bundle over a paracompact Hausdorff base, and regard it as a $2n$-dimensional real vector bundle $E_{\mathbb{R}}$. Then

$$w_{2i+1}(E_{\mathbb{R}}) = 0$$

and $w_{2i}(E_{\mathbb{R}})$ is the image of $c_i(E)$ under the coefficient homomorphism $\rho_2 \colon H^{2i}(B;\mathbb{Z}) \to H^{2i}(B;\mathbb{Z}/2\mathbb{Z})$. Equivalently, $w(E_{\mathbb{R}}) = \rho_2(c(E))$.

***Proof*:**

Step 1: Reduction to line bundles. Let $p \colon F(E) \to B$ be the complex flag bundle of theorem 7.4.3, so that p^*E splits as a sum of complex line bundles $L_1 \oplus \cdots \oplus L_n$. That theorem gives injectivity of p^* on $H^*(-;\mathbb{Z})$. Injectivity on $H^*(-;\mathbb{Z}/2\mathbb{Z})$ holds for the same reason, since $F(E) \to B$ is built as a tower of projectivizations and corollary 6.5.2 applies over any commutative coefficient ring R, in particular $R = \mathbb{Z}/2\mathbb{Z}$, making $H^*(F(E);\mathbb{Z}/2\mathbb{Z})$ a free $H^*(B;\mathbb{Z}/2\mathbb{Z})$-module on a basis containing 1. Both $E \mapsto w(E_{\mathbb{R}})$ and $E \mapsto \rho_2(c(E))$ are natural (proposition 7.4.1) and multiplicative under direct sums (theorem 7.4.5, together with the fact that ρ_2 is a ring homomorphism and that $(E \oplus E')_{\mathbb{R}} = E_{\mathbb{R}} \oplus E'_{\mathbb{R}}$). So both sides pull back to $F(E)$ as products over the line summands, and it suffices to prove the identity for a complex line bundle.

Step 2: The line bundle case. Let $L \to B$ be a complex line bundle, so $L_{\mathbb{R}}$ is a real bundle of rank 2 carrying the canonical orientation of theorem 7.4.12(2). Its first Stiefel-Whitney class vanishes by theorem 7.4.13(1), the base being paracompact Hausdorff. In the application of Step 1 the base is the flag bundle $F(E)$, which is paracompact Hausdorff by corollary 7.4.4. Its top Stiefel-Whitney class is $w_2(L_{\mathbb{R}}) = \rho_2(e(L_{\mathbb{R}}))$ by theorem 7.4.12(1), and $e(L_{\mathbb{R}}) = c_1(L)$ by theorem 7.4.12(2) applied with $n = 1$. Hence $w_2(L_{\mathbb{R}}) = \rho_2(c_1(L))$, and $w_i(L_{\mathbb{R}}) = 0$ for $i > 2$ by

definition 7.2.13 since $\mathrm{rk} L_{\mathbb{R}} = 2$. Therefore

$$w(L_{\mathbb{R}}) = 1 + \rho_2(c_1(L)) = \rho_2\big(1 + c_1(L)\big) = \rho_2(c(L))$$

Step 3: Conclusion. By Step 1 the two total classes agree after pulling back along p, and p^* is injective on $H^*(B;\mathbb{Z}/2\mathbb{Z})$, so $w(E_{\mathbb{R}}) = \rho_2(c(E))$ on B. Comparing degrees, the odd-degree components of $\rho_2(c(E))$ vanish because $c_i(E)$ sits in degree $2i$, giving $w_{2i+1}(E_{\mathbb{R}}) = 0$ and $w_{2i}(E_{\mathbb{R}}) = \rho_2(c_i(E))$, as we sought to show.

Definition 7.4.15: Spin Structure

Let $E \to X$ be an oriented real vector bundle of rank $n \geq 3$ over a paracompact Hausdorff base, with principal $SO(n)$-bundle of oriented orthonormal frames $P_{SO} \to X$. A *spin structure* on E is a principal $\mathrm{Spin}(n)$-bundle $P_{\mathrm{Spin}} \to X$ together with a map $P_{\mathrm{Spin}} \to P_{SO}$ over X restricting on each fiber to the double cover $\mathrm{Spin}(n) \to SO(n)$.

A spin structure is a globally consistent square root of the rotation group acting on the fibers. The local square roots always exist, since $\mathrm{Spin}(n) \to SO(n)$ is a covering. The obstruction to assembling them into one global choice is a mod-2 class in degree two, and the content of the next theorem is that the class is $w_2(E)$.

Theorem 7.4.16: Second Stiefel-Whitney Class is the Obstruction to a Spin Structure

Let $E \to X$ be an oriented real vector bundle of rank $n \geq 3$ over a connected CW complex (so $w_1(E) = 0$). Then E admits a spin structure if and only if $w_2(E) = 0 \in H^2(X;\mathbb{F}_2)$. When spin structures exist, they form a torsor under $H^1(X;\mathbb{F}_2)$. In particular a bundle over a connected base admits $|H^1(X;\mathbb{F}_2)|$ of them when it admits any.

***Proof*:**
A spin structure on an oriented bundle is a lift of its structure group from $SO(n)$ to the connected double cover $\mathrm{Spin}(n) \to SO(n)$, equivalently a lift of the classifying map $g\colon X \to BSO(n)$ across the fibration $B\mathrm{Spin}(n) \to BSO(n)$. The proof identifies the fiber of this fibration, locates the primary obstruction, and identifies it with w_2.

Step 1: The fibration and its fiber. For $n \geq 3$ the group $SO(n)$ has fundamental group $\pi_1(SO(n)) = \mathbb{Z}/2$, and $\mathrm{Spin}(n)$ is its universal (double) cover, a central extension

$$1 \to \mathbb{Z}/2 \to \mathrm{Spin}(n) \to SO(n) \to 1$$

Applying the classifying-space functor B to this extension yields a fibration sequence

$$B\mathbb{Z}/2 = K(\mathbb{Z}/2,1) \to B\mathrm{Spin}(n) \to BSO(n) \xrightarrow{\beta} B(B\mathbb{Z}/2) = K(\mathbb{Z}/2,2)$$

where the fiber of $B\mathrm{Spin}(n) \to BSO(n)$ is $B\mathbb{Z}/2 = K(\mathbb{Z}/2,1)$ (the delooping of corollary 7.1.11 applied to the kernel $\mathbb{Z}/2$), and the map $\beta\colon BSO(n) \to K(\mathbb{Z}/2,2)$ is the classifying map of the next stage, representing a class in $H^2(BSO(n);\mathbb{Z}/2)$. A spin structure on E is a lift of $g\colon X \to BSO(n)$ to $B\mathrm{Spin}(n)$.

Step 2: The primary obstruction. By proposition 5.6.9, the obstructions to lifting g across the fibration with fiber $F = K(\mathbb{Z}/2, 1)$ live in $H^{m+1}(X; \pi_m(F))$. Since $F = K(\mathbb{Z}/2, 1)$ has $\pi_1(F) = \mathbb{Z}/2$ and $\pi_m(F) = 0$ for $m \neq 1$, every obstruction group vanishes except in degree $m = 1$, where the single obstruction is a class

$$\theta(E) \in H^2\big(X; \pi_1(F)\big) = H^2(X; \mathbb{Z}/2)$$

This $\theta(E)$ is the complete obstruction: a lift exists over the 2-skeleton if and only if $\theta(E) = 0$, and since there are no obstructions in higher degrees ($\pi_m(F) = 0$ for $m \geq 2$), the lift then extends over all of X. Concretely, $\theta(E) = g^*\beta$ is the pullback of the universal class $\beta \in H^2(BSO(n); \mathbb{Z}/2)$ along the classifying map.

Step 3: $\theta(E) = w_2(E)$. The universal class $\beta \in H^2(BSO(n); \mathbb{Z}/2)$ is the obstruction to spinning the universal oriented bundle. By corollary 7.2.12, $H^2(BSO(n); \mathbb{F}_2) = \mathbb{F}_2\langle w_2 \rangle$, with w_2 pulled back from $BO(n)$, so β is either 0 or w_2. Suppose $\beta = 0$. Step 2, applied to the identity of $BSO(n)$, then gives a map $s\colon BSO(n) \to B\mathrm{Spin}(n)$ with $p \circ s = \mathrm{id}$, where $p\colon B\mathrm{Spin}(n) \to BSO(n)$ is the projection, so $p_* \circ s_*$ is the identity of $\pi_2(BSO(n))$. By corollary 7.1.11(2), $\pi_2(BSO(n)) \cong \pi_1(SO(n)) = \mathbb{Z}/2$, while $\pi_2(B\mathrm{Spin}(n)) \cong \pi_1(\mathrm{Spin}(n)) = 0$, the universal cover being simply connected. The identity of $\mathbb{Z}/2$ cannot factor through 0, so $\beta = w_2$. The map $\pi \circ g$, with $\pi\colon BSO(n) \to BO(n)$ forgetting the orientation, classifies E, so naturality gives $\theta(E) = g^*\beta = g^*\pi^* w_2 = w_2(E)$, and E admits a spin structure if and only if $w_2(E) = 0$, completing the proof.

Step 4: The torsor clause. Suppose E admits a spin structure $\lambda\colon P_{\mathrm{Spin}} \to P_{SO}$, and let $Q \to X$ be a principal $\mathbb{Z}/2$-bundle. Write ζ for the central element of $\mathrm{Spin}(n)$ generating the kernel of $\mathrm{Spin}(n) \to SO(n)$. Twist by Q:

$$P_{\mathrm{Spin}} \cdot Q := \big(P_{\mathrm{Spin}} \times_X Q\big)/(\mathbb{Z}/2)$$

where the generator of $\mathbb{Z}/2$ acts by ζ on the first factor and by the deck transformation on the second. The residual $\mathrm{Spin}(n)$-action on the first factor descends, since ζ is central, and makes $P_{\mathrm{Spin}} \cdot Q$ a principal $\mathrm{Spin}(n)$-bundle. The map λ descends too, because $\lambda(p\zeta) = \lambda(p)$, so $P_{\mathrm{Spin}} \cdot Q \to P_{SO}$ is again a spin structure. This defines an action of $\mathrm{Prin}_{\mathbb{Z}/2}(X)$ on the set of spin structures, associative because twisting by Q then Q' agrees with twisting by the contracted product $Q \otimes Q'$.

The action is transitive. Given two spin structures $\lambda\colon P \to P_{SO}$ and $\lambda'\colon P' \to P_{SO}$, the fiber product $P \times_{P_{SO}} P'$ carries a free action of $\mathrm{Spin}(n)$ acting diagonally, and the quotient $Q(P') := (P \times_{P_{SO}} P')/\mathrm{Spin}(n)$ is a principal $\mathbb{Z}/2$-bundle over X: over a point of P_{SO} the two fibers are $\mathbb{Z}/2$-torsors, so the fiber product is a $(\mathbb{Z}/2 \times \mathbb{Z}/2)$-torsor and the diagonal quotient is a $\mathbb{Z}/2$-torsor. Unwinding the definitions gives $P \cdot Q(P') \cong P'$ over P_{SO}.

The action is free. Because it is transitive it suffices to check one stabilizer: if $P \cdot Q \cong P$ over P_{SO} then $Q \cong Q(P \cdot Q) \cong Q(P)$, and $Q(P)$ is the trivial double cover because $p \mapsto [(p, p)]$ is a section of it. Hence the spin structures form a torsor under $\mathrm{Prin}_{\mathbb{Z}/2}(X)$, which is $H^1(X; \mathbb{F}_2)$ by theorem 7.1.10 and theorem 5.5.12, using $B\mathbb{Z}/2 = \mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2, 1)$ (example 5.5.6). It is here, not in the existence criterion, that the CW hypothesis is used a second time.

The pattern is now visible. Each successive structure on a bundle (orientation, then spin, then string, and onward) is a lift of the structure group along a tower of two-fold or higher covers, and the primary obstruction to each lift is a characteristic class one degree higher than the last. Orientation removes

w_1. With $w_1 = 0$, the spin obstruction is w_2. The framework continues, with the next obstruction (to a string structure) a class in H^4 detected by a fractional Pontryagin class. The unifying statement is that a characteristic class is the primary obstruction to a reduction of the structure group, the cohomological measurement of how far the bundle is from carrying the desired geometric structure. The most classical of these obstructions is to a single nowhere-zero section, and there the top class enters.

Theorem 7.4.17: Top Class is the Obstruction to a Nowhere-Zero Section

Let $E \to X$ be a real vector bundle of rank n over a connected CW complex.

1. If E is oriented, the Euler class $e(E) \in H^n(X;\mathbb{Z})$ is the primary obstruction to a nowhere-zero section of E, the homotopy groups π_{n-1} of the fibers of $S(E)$ being identified with $\mathbb{Z}$ through the orientation: if E has a nowhere-zero section then $e(E) = 0$, and if in addition $\dim X \leq n$ the converse holds, so that E has a nowhere-zero section if and only if $e(E) = 0$.
2. Without orientation hypotheses, a nowhere-zero section forces $w_n(E) = 0$ in $H^n(X;\mathbb{F}_2)$.

***Proof*:**

For $n = 1$ the fiber S^0 of the sphere bundle is not path-connected, and the statement is direct. An oriented line bundle has the nowhere-zero section choosing the positive unit vector of each fiber for a metric, and a line bundle with a nowhere-zero section is trivial. So in (1) the bundle E is trivial and has a nowhere-zero section, and in (2) a nowhere-zero section makes E trivial. A trivial bundle is pulled back from a point, so its characteristic classes in positive degrees vanish by naturality, which gives both assertions. Let $n \geq 2$ from now on.

A nowhere-zero section of E, normalized, is a section of the sphere bundle $p\colon S(E) \to X$ of an inner product (lemma 7.1.22, X being paracompact by theorem 7.1.7). This is a fiber bundle, hence a Serre fibration (theorem 5.2.4), with fiber S^{n-1}, path-connected since $n \geq 2$. The obstruction theory of theorem 5.6.7 in its section form (proposition 5.6.9) governs the existence of such a section.

Step 1: Locating the obstruction. The fiber S^{n-1} is $(n-2)$-connected, and $\pi_{n-1}(S^{n-1}) = \mathbb{Z}$ by corollary 5.4.11. The obstruction groups $H^{m+1}(X;\pi_m(S^{n-1}))$ with $m < n-1$ vanish, so proposition 5.6.9 builds a section s over $X^{(n-1)}$ one skeleton at a time. Let E be oriented, with the classes $\alpha_x \in H^{n-1}(S(E_x);\mathbb{Z})$ of definition 7.1.25. Fix the generator $[D^n, S^{n-1}]$ of $H_n(D^n, S^{n-1})$ through which cellular chains are read, put $[S^{n-1}] = \partial[D^n, S^{n-1}]$, and define

$$\theta_x\colon \pi_{n-1}\big(S(E_x)\big) \longrightarrow \mathbb{Z}, \qquad \theta_x[f] = -\big\langle \alpha_x, f_*[S^{n-1}]\big\rangle$$

an isomorphism by the Hurewicz theorem (theorem 5.4.9 for $n \geq 3$, theorem 5.4.5 for $n = 2$, where $\pi_1(S^1)$ is abelian), α_x being a generator. By the formula for δ in theorem 4.1.17 and $\delta a_x = -\mu_x$, the integer $\theta_x[f]$ is the value of the orientation μ_x on the class of $H_n(E_x, E_x \setminus 0)$ whose boundary is $f_*[S^{n-1}]$. Transport along a path in X preserves the orientations μ_x, hence the classes α_x and the maps θ_x, and $\pi_1(S^{n-1})$ acts trivially on $\pi_{n-1}(S^{n-1})$, the group being trivial for $n \geq 3$ and abelian for $n = 2$. So proposition 5.6.9 applies with coefficients read in $\mathbb{Z}$ through the θ_x. Its cocycle $\mathfrak{o}(s)$ takes the value $\theta_{\Phi(0)}[H_1]$ on the cellular chain $\Phi_*[D^n, S^{n-1}]$ of an n-cell with characteristic map Φ and attaching map $\varphi = \Phi|_{S^{n-1}}$. Here $H\colon S^{n-1} \times I \to S(E)$ lifts the radial nullhomotopy $(v,t) \mapsto \Phi((1-t)v)$ of φ starting at $s \circ \varphi$, and $H_1 = H(-,1)$ maps

into $S(E_{\Phi(0)})$. As $\pi_i(S^{n-1}) = 0$ for $i < n-1$, the class

$$\mathfrak{o}(E) = [\mathfrak{o}(s)] \in H^n(X;\mathbb{Z})$$

does not depend on s, and $f^*\mathfrak{o}(E) = \mathfrak{o}(f^*E)$ for a cellular map $f\colon Y \to X$ of connected CW complexes, f^*E carrying the pulled-back orientation.

Step 2: $\mathfrak{o}(E) = e(E)$ when X has finite type. Let X have finite type, let $\pi\colon E \to X$, z and E_0 be the projection, the zero section and the complement of the zero section, and let u_E be the Thom class. Let B' be the union of X^{n-1} and the centres $\Phi(0)$ of the n-cells. The centres meet every closed cell in a finite set, so they form a closed discrete subset of X, and B' is the disjoint union of X^{n-1} and these points. Hence $H^n(B';\mathbb{Z}) = 0$, using $H^n(X^{n-1};\mathbb{Z}) = 0$ (proposition 4.1.20), and by the long exact sequence of the pair, $e(E)$ is represented by a cocycle Y on X vanishing on every singular simplex of B'. By corollary 7.3.6, $e(E) = z^*j^*u_E$ for $j^*\colon H^n(E,E_0;\mathbb{Z}) \to H^n(E;\mathbb{Z})$ induced by the inclusion, so $\pi^*[Y] = j^*u_E$. As in the proof of proposition 7.3.5(2), a cocycle U on E vanishing on E_0 and representing u_E, and a cochain h with $\delta h = \pi^*Y - U$, give $c = h|_{S(E)}$ with $\delta c = p^*Y$. So c is a cocycle on $p^{-1}(B')$, and for every centre x its class on $S(E_x)$ is α_x, by the argument of that proof: Y vanishes on the constant simplices at x, so $\delta(h|_{E_x}) = -U|_{E_x}$.

Let a be the class of c on $p^{-1}(X^{n-1})$. Extend a cocycle on X^{n-1} representing $-s^*a$ by zero to a cochain w on X, and replace Y by $Y + \delta w$ and c by $c + p^*w$. This keeps $\delta c = p^*Y$, keeps Y vanishing on the simplices of B', since w restricts to a cocycle on each piece of B', and leaves c unchanged over the centres, where w vanishes on the constant simplices. The class a becomes $a - p^*s^*a$, and as $p \circ s = \mathrm{id}$ we may assume $s^*a = 0$.

Fix an n-cell with Φ, φ and H as in Step 1 and centre $x = \Phi(0)$, and let $q\colon S^{n-1} \times I \to D^n$, $q(v,t) = (1-t)v$, so that $p \circ H = \Phi \circ q$. The map q carries $(S^{n-1} \times I, S^{n-1} \times \partial I)$ into $(D^n, S^{n-1} \cup \{0\})$. Write $[S^{n-1}]_0$ and $[S^{n-1}]_1$ for the images of $[S^{n-1}]$ under the two end inclusions. These inclusions are homotopic, so $[S^{n-1}]_1 - [S^{n-1}]_0$ dies in $H_{n-1}(S^{n-1} \times I)$ and is ∂T for some $T \in H_n(S^{n-1} \times I, S^{n-1} \times \partial I)$. The map q sends $[S^{n-1}]_1$ to 0, as $n - 1 \geq 1$, and $[S^{n-1}]_0$ to $[S^{n-1}]$, so $\partial q_* T = -[S^{n-1}]$ in $H_{n-1}(S^{n-1} \cup \{0\})$. Since D^n is contractible and $n \geq 2$, the boundary map on $H_n(D^n, S^{n-1} \cup \{0\})$ is injective, so q_*T is minus the image of $[D^n, S^{n-1}]$. Represent T by a relative cycle t with $\partial t = z_0 + z_1$, where z_i is a cycle in $S^{n-1} \times \{i\}$, of class $-[S^{n-1}]_0$ for $i = 0$ and $[S^{n-1}]_1$ for $i = 1$. The cocycle Y vanishes on $B' \supseteq \Phi(S^{n-1} \cup \{0\})$, so for a relative cycle r representing $[D^n, S^{n-1}]$

$$\langle Y, \Phi_\sharp r\rangle = -\langle Y, \Phi_\sharp q_\sharp t\rangle = -\langle p^*Y, H_\sharp t\rangle = -\langle \delta c, H_\sharp t\rangle = -\langle c, H_\sharp z_1\rangle - \langle c, H_\sharp z_0\rangle$$

The chain $H_\sharp z_1$ is a cycle of $S(E_x)$ representing $(H_1)_*[S^{n-1}]$, and c represents α_x there, so $-\langle c, H_\sharp z_1\rangle = \theta_x[H_1]$. The chain $H_\sharp z_0$ is a cycle of $p^{-1}(X^{n-1})$ representing $-(s \circ \varphi)_*[S^{n-1}]$, and c represents a there, so $\langle c, H_\sharp z_0\rangle = -\langle s^*a, \varphi_*[S^{n-1}]\rangle = 0$. The cellular cocycle of Y, whose class is $e(E)$ (proposition 4.1.20), therefore agrees with $\mathfrak{o}(s)$ on every n-cell, and $e(E) = \mathfrak{o}(E)$.

Step 3: The general case. Let $f\colon X \to BSO(n)$ classify E as an oriented bundle (corollary 7.1.30). By cellular approximation, the theorem imported in the proof of theorem 5.3.6, f is homotopic to a cellular map, which pulls back an isomorphic oriented bundle (lemma 7.1.9, applied to the bundles of oriented orthonormal frames), so f may be taken cellular. The space $BSO(n)$ is a connected CW complex of finite type (theorem 7.1.17, lemma 7.1.18(4)), so Step 2 gives $\mathfrak{o}(\tilde{\gamma}_n) = e$, and Step 1 with definition 7.2.15 gives $\mathfrak{o}(E) = f^*\mathfrak{o}(\tilde{\gamma}_n) = f^*e = e(E)$. If E has a

nowhere-zero section, its restriction to $X^{(n-1)}$ extends over X, so $\mathfrak{o}(E) = 0$ by proposition 5.6.9, and $e(E) = 0$. If $\dim X \leq n$ and $e(E) = 0$, then $\mathfrak{o}(E) = 0$, and proposition 5.6.9 extends $s|_{X^{(n-2)}}$ to a section over $X^{(n)} = X$.

Step 4: Mod-2 case. Without orientation, theorem 7.4.12(1) gives $w_n(E) = e_2(E)$, since X is a connected CW complex and $n \geq 2$, and lemma 7.2.9(2) gives $e_2(E) = 0$ when E has a nowhere-zero section. Hence a nowhere-zero section forces $w_n(E) = 0$, completing the proof.

For the tangent bundle of a closed manifold, this section obstruction is the Euler characteristic, by way of the Poincaré-Hopf theorem and Poincaré duality. A nowhere-zero vector field on M is a nowhere-zero section of TM, and its obstruction $e(TM)$ evaluates against the fundamental class to give $\chi(M)$.

Corollary 7.4.18: Euler Class of the Tangent Bundle Computes the Euler Characteristic

Let M be a closed connected oriented smooth n-manifold with fundamental class $[M] \in H_n(M;\mathbb{Z})$ (definition 4.4.8). Then the Euler class of the tangent bundle evaluates to the Euler characteristic:

$$\langle e(TM), [M] \rangle = \chi(M) \in \mathbb{Z}$$

In particular M admits a nowhere-zero vector field if and only if $\chi(M) = 0$.

***Proof*:**

Step 1: The Euler number. The evaluation $\langle e(TM), [M] \rangle$ is the Euler number of the tangent bundle, and its equality with $\chi(M)$ is [29, Corollary 11.12], for a smooth compact oriented manifold and the Euler class defined as the restriction of the Thom class to the zero section. The proof there runs through the Thom class of the normal bundle of the diagonal in $M \times M$, and so uses the smooth structure of M. That class agrees with the one of definition 7.2.15: both are natural for pullback of oriented bundles, they agree on the universal oriented bundle over the CW complex of finite type $BSO(n)$ by corollary 7.3.6, and TM is pulled back from it (corollary 7.1.30). The same equality is the Poincaré-Hopf theorem, by which the index sum of a vector field with isolated zeros is $\chi(M)$, and *Differential Topology* [32, Poincaré-Hopf] develops it through the self-intersection of the diagonal.

Step 2: A CW model. The obstruction theorem theorem 7.4.17 is stated over a CW complex, so it is applied to a CW model of M. The manifold M carries a smooth function with no degenerate critical points [28, Corollary 6.7], these are isolated [28, Corollary 2.3], so the compact M has finitely many, each of index at most n, and M has the homotopy type of a finite CW complex $\bar{M}$ of dimension at most n [28, Theorem 3.5]. Let $h\colon M \to \bar{M}$ be a homotopy equivalence with homotopy inverse $\psi\colon \bar{M} \to M$. The complex $\bar{M}$ is connected, as M is, and both spaces are paracompact Hausdorff, M being compact Hausdorff and $\bar{M}$ a CW complex (theorem 7.1.7).

Step 3: Nowhere-zero sections. Let $E' = \psi^* TM$ over $\bar{M}$, with the pulled-back orientation. Since $\psi \circ h \simeq \mathrm{id}_M$, the bundles $h^* E' = (\psi \circ h)^* TM$ and TM are isomorphic, by lemma 7.1.9 applied to their frame bundles (lemma 7.1.26). A nowhere-zero section of TM pulls back to one of E', and a nowhere-zero section of E' pulls back to one of $h^* E' \cong TM$, so TM has a nowhere-zero section if and only if E' does. As $\dim \bar{M} \leq n = \mathrm{rank} E'$, theorem 7.4.17(1) says that E' has one if and only if $e(E') = 0$. If $f\colon M \to BSO(n)$ classifies TM as an oriented bundle, then $f \circ \psi$ classifies E', so $e(E') = \psi^* e(TM)$ by the uniqueness in corollary 7.1.30 and definition 7.2.15.

The map ψ^* is an isomorphism, ψ being a homotopy equivalence, so TM has a nowhere-zero section if and only if $e(TM) = 0$.

Step 4: Evaluation detects $H^n(M;\mathbb{Z})$. By theorem 4.4.7 and definition 4.4.8, $H_n(M;\mathbb{Z}) \cong \mathbb{Z}$ generated by $[M]$, and Poincaré duality (theorem 4.5.7) gives $H^n(M;\mathbb{Z}) \cong H_0(M;\mathbb{Z}) \cong \mathbb{Z}$. The homology of M is that of the finite complex $\bar{M}$, hence finitely generated (theorem 3.5.11), so $\mathrm{Ext}(H_{n-1}(M),\mathbb{Z})$ is finite, and the universal coefficient sequence (theorem 4.1.8) presents $H^n(M;\mathbb{Z}) \cong \mathbb{Z}$ as $\mathrm{Hom}(H_n(M),\mathbb{Z}) \oplus \mathrm{Ext}(H_{n-1}(M),\mathbb{Z})$. A finite direct summand of $\mathbb{Z}$ vanishes, so evaluation on $[M]$ is an isomorphism $H^n(M;\mathbb{Z}) \to \mathbb{Z}$. Hence $e(TM) = 0$ if and only if $\langle e(TM), [M]\rangle = \chi(M)$ vanishes, which with Step 3 gives the nowhere-zero vector field criterion, completing the proof.

The corollary is the bridge promised at the start of the section: the Euler class, defined cohomologically and interpreted as a section obstruction, computes the most classical numerical invariant of a closed manifold. It also explains the hairy-ball theorem as a special case. A worked computation shows the mechanism and ties the strands of this section together.

Example 7.4.19: The Total Stiefel-Whitney Class of TS^2

The tangent bundle TS^2 is a rank-2 real bundle over S^2. Its total Stiefel-Whitney class is computed by a stabilization trick: the Whitney sum $TS^2 \oplus \nu$ of the tangent bundle with the normal bundle ν of the standard embedding $S^2 \subseteq \mathbb{R}^3$ is the restriction $T\mathbb{R}^3|_{S^2}$, which is the trivial rank-3 bundle $\underline{\mathbb{R}^3}$ (the tangent bundle of Euclidean space is trivial). The normal bundle ν is the trivial line bundle $\underline{\mathbb{R}}$ (it is spanned by the outward unit normal, a nowhere-zero global section). Therefore

$$TS^2 \oplus \underline{\mathbb{R}} \cong \underline{\mathbb{R}^3}$$

Apply the Whitney sum formula (theorem 7.4.5) and naturality. A trivial bundle has total Stiefel-Whitney class 1 (it is pulled back from a point, so $w = 1$ by proposition 7.4.1), so $w(\underline{\mathbb{R}^3}) = 1$ and $w(\underline{\mathbb{R}}) = 1$. Hence

$$w(TS^2)\, w(\underline{\mathbb{R}}) = w(TS^2 \oplus \underline{\mathbb{R}}) = w(\underline{\mathbb{R}^3}) = 1$$

and since $w(\underline{\mathbb{R}}) = 1$ this forces $w(TS^2) = 1$. So all Stiefel-Whitney classes of TS^2 vanish: $w_1(TS^2) = 0$ and $w_2(TS^2) = 0$ in $H^*(S^2;\mathbb{F}_2)$.

The vanishing $w_1(TS^2) = 0$ records that S^2 is orientable (theorem 7.4.13), as expected for a sphere. The vanishing $w_2(TS^2) = 0$ does *not* mean TS^2 is trivial: the integral Euler class is nonzero. By corollary 7.4.18, $\langle e(TS^2), [S^2]\rangle = \chi(S^2) = 2$, so $e(TS^2) = 2u' \neq 0$ where u' generates $H^2(S^2;\mathbb{Z}) \cong \mathbb{Z}$. The top Stiefel-Whitney class is the mod-2 reduction $w_2(TS^2) = \rho_2(e(TS^2)) = \rho_2(2u') = 0$ (theorem 7.4.12(1)), consistent with the Whitney computation, and the reduction loses exactly the factor of 2. The conclusion is sharp: TS^2 has no nowhere-zero section, since $e(TS^2) \neq 0$, which is the hairy-ball theorem, while its mod-2 obstruction w_2 vanishes because $2 \equiv 0$. The integral Euler class is the finer invariant here, and the example shows why the oriented theory carries strictly more information than the mod-2 theory.

The companion computation is the non-orientability of the tautological line bundle, which displays the first Stiefel-Whitney class as a genuine, computable obstruction.

Example 7.4.20: Non-Orientability of the Tautological Bundle on $\mathbb{R}\mathrm{P}^n$

The tautological real line bundle $\gamma^1 \to \mathbb{R}\mathrm{P}^n$ has $w_1(\gamma^1) = a \neq 0$ in $H^1(\mathbb{R}\mathrm{P}^n; \mathbb{F}_2) = \mathbb{F}_2[a]/(a^{n+1})$ by example 7.4.10. By theorem 7.4.13, the nonvanishing of w_1 is exactly the statement that γ^1 is *non-orientable*. Geometrically the obstruction is visible already over $\mathbb{R}\mathrm{P}^1 = S^1$: the restriction $\gamma^1|_{\mathbb{R}\mathrm{P}^1}$ is the Möbius band line bundle, whose total space is a Möbius strip, the standard nonorientable line bundle over the circle (the orientation cover is the connected double cover $S^1 \to S^1$, $z \mapsto z^2$, recovered as in example 4.4.5 for the Möbius band). Pulling back along $\mathbb{R}\mathrm{P}^1 \hookrightarrow \mathbb{R}\mathrm{P}^n$ detects $w_1(\gamma^1) = a$ on the bottom cell, confirming the obstruction is nonzero.

This also computes the Stiefel-Whitney classes of $\mathbb{R}\mathrm{P}^n$ itself, that is of its tangent bundle. The standard bundle isomorphism (proved in [29, §4])

$$T\mathbb{R}\mathrm{P}^n \oplus \underline{\mathbb{R}} \cong (\gamma^1)^{\oplus(n+1)} = \gamma^1 \oplus \cdots \oplus \gamma^1 \ (n+1 \text{ copies})$$

combined with the Whitney sum formula gives

$$w(T\mathbb{R}\mathrm{P}^n) = w(T\mathbb{R}\mathrm{P}^n)\, w(\underline{\mathbb{R}}) = w\big((\gamma^1)^{\oplus(n+1)}\big) = \big(w(\gamma^1)\big)^{n+1} = (1+a)^{n+1}$$

in $\mathbb{F}_2[a]/(a^{n+1})$, since $w(\underline{\mathbb{R}}) = 1$. For $n = 2$ this is $(1+a)^3 = 1 + 3a + 3a^2 + a^3 = 1 + a + a^2$ (mod 2, with $a^3 = 0$), so $w_1(T\mathbb{R}\mathrm{P}^2) = a \neq 0$ and $w_2(T\mathbb{R}\mathrm{P}^2) = a^2 \neq 0$. The nonvanishing of $w_1(T\mathbb{R}\mathrm{P}^2)$ reconfirms that $\mathbb{R}\mathrm{P}^2$ is non-orientable, and the nonvanishing of $w_2(T\mathbb{R}\mathrm{P}^2) = a^2$, paired against the fundamental class $[\mathbb{R}\mathrm{P}^2] \in H_2(\mathbb{R}\mathrm{P}^2; \mathbb{F}_2)$, gives the Stiefel-Whitney number $w_2[\mathbb{R}\mathrm{P}^2] = 1$ that detects $\mathbb{R}\mathrm{P}^2$ as the generator of the unoriented bordism group $\mathfrak{N}_2$ (chapter 8). The single computation $w(\gamma^1) = 1 + a$ thus drives both the non-orientability statement and the bordism invariant.

The thread running through both examples is that the structure-group reductions form a hierarchy, and each rung is governed by one characteristic class. Requiring k linearly independent nowhere-zero sections, rather than one, reduces the structure group from $O(n)$ to $O(n-k)$, and the obstruction is a string of top classes of successive quotient bundles.

Proposition 7.4.21: Vanishing of Top Classes is Necessary for Independent Sections

Let $E \to X$ be a real vector bundle of rank n over a connected CW complex. If E admits k everywhere linearly independent sections (a k-frame field), then

$$w_{n-k+1}(E) = w_{n-k+2}(E) = \cdots = w_n(E) = 0 \in H^*(X; \mathbb{F}_2)$$

the top k Stiefel-Whitney classes all vanish.

***Proof*:**

A k-frame field on E is a trivial rank-k subbundle $\underline{\mathbb{R}^k} \hookrightarrow E$ spanned by the k independent sections. Choosing a metric, split E as a Whitney sum

$$E \cong \underline{\mathbb{R}^k} \oplus Q$$

where $Q = E/\underline{\mathbb{R}^k}$ is the quotient bundle, of rank $n-k$. By the Whitney sum formula (theorem 7.4.5) and $w(\underline{\mathbb{R}^k}) = 1$ (a trivial bundle is classified by a constant map, so $w = 1$ by

proposition 7.4.1),

$$w(E) = w(\underline{\mathbb{R}^k})\, w(Q) = w(Q)$$

Now Q has rank $r = n - k$, so $w_i(Q) = 0$ for every $i > n - k$. This is the rank bound on Stiefel-Whitney classes: a rank-r bundle is classified by a map $g\colon X \to BO(r)$, and $w_i(Q) = g^*(w_i)$ is the pullback of the universal class $w_i \in H^i(BO(r); \mathbb{F}_2)$. The point is that $H^*(BO(r); \mathbb{F}_2) = \mathbb{F}_2[w_1, \dots, w_r]$ has polynomial generators only in degrees $1, \dots, r$, so for $i > r$ there is no universal class w_i to pull back, and the invariant $w_i(Q)$ is by definition zero. (The cohomology ring itself is nonzero in every degree, for instance $w_r^2 \in H^{2r}(BO(r); \mathbb{F}_2)$ is nonzero. What fails to exist past degree r is the single generator w_i, not the whole group.) Therefore $w_i(E) = w_i(Q) = 0$ for all $i > n - k$, that is for $i = n - k + 1, \dots, n$, completing the proof.

The proposition is the necessary direction of the vector-field problem on a manifold: the maximal number of independent vector fields on a sphere, computed by Adams using K-theory and the Stiefel-Whitney (and higher) obstructions, is bounded below by where these classes first fail to vanish. The obstruction picture is now complete: orientation is the w_1 obstruction, spin is the w_2 obstruction, a single nowhere-zero section is the Euler/w_n obstruction, and k independent sections require the top k classes to vanish. Each characteristic class is a primary obstruction in the sense of theorem 5.6.7, measuring the failure of a reduction of the structure group, and the cohomological degree of the class is exactly the connectivity of the fiber of the associated fibration plus one.

Exercise 7.4.1

1. Using the Whitney sum formula (theorem 7.4.5) and the bundle isomorphism $T\mathbb{R}\mathrm{P}^n \oplus \underline{\mathbb{R}} \cong (\gamma^1)^{\oplus(n+1)}$, compute the total Stiefel-Whitney class $w(T\mathbb{R}\mathrm{P}^3)$ in $\mathbb{F}_2[a]/(a^4)$, and determine whether $\mathbb{R}\mathrm{P}^3$ is orientable and whether it admits a spin structure. (Recall w_1 detects orientability by theorem 7.4.13 and, once $w_1 = 0$, w_2 detects spin by theorem 7.4.16.)

2. Let $E \to X$ be a complex line bundle with $c_1(E) = u \in H^2(X; \mathbb{Z})$. Using the Whitney sum formula, compute the total Chern class $c(E \oplus \bar{E})$ of the sum of E with its conjugate $\bar{E}$ (for which $c_1(\bar{E}) = -c_1(E)$), and show that $c_2(E \oplus \bar{E}) = -u^2$. Interpret the sign in terms of the underlying real rank-4 bundle and its Euler class.

3. Prove that the tangent bundle TS^n is stably trivial for every n (that is, $TS^n \oplus \underline{\mathbb{R}}$ is trivial), and deduce that $w(TS^n) = 1$ for all n. Then explain why this does *not* contradict the fact that S^n admits no nowhere-zero vector field when n is even. (Use corollary 7.4.18.)

4. Let $E \to X$ be an oriented rank-n bundle over a connected CW complex with $\dim X < n$. Using theorem 7.4.17, prove that E admits a nowhere-zero section, regardless of any characteristic class. Conclude that every rank-n bundle over S^m with $m < n$ has a nowhere-zero section, and relate this to the stable range of the homotopy groups of $BO(n)$.

5. The top class detecting a section. Show directly from the Whitney sum formula and example 7.4.20 that the rank-2 bundle $\gamma^1 \oplus \gamma^1 \to \mathbb{R}\mathrm{P}^2$ has $w_1 = 0$ and $w_2 = a^2 \neq 0$, hence is orientable but admits no nowhere-zero section, and contrast this with the trivial bundle $\underline{\mathbb{R}^2} \to \mathbb{R}\mathrm{P}^2$, which has $w = 1$ and does have one. Explain why w_2 distinguishes these two orientable rank-2 bundles even though w_1 cannot. (Hint: compute w_1 of each first.)

6. The splitting principle in action. Let $E \to X$ be a complex rank-2 bundle that splits as $E = L_1 \oplus L_2$ with $c_1(L_i) = x_i$. Compute $c_1(E)$, $c_2(E)$ in terms of x_1, x_2, and verify that they are the elementary symmetric polynomials σ_1, σ_2. Then express the Chern character

$\mathrm{ch}(E) = e^{x_1} + e^{x_2}$ through degree 4 in terms of $c_1(E)$ and $c_2(E)$, illustrating how the splitting principle converts a symmetric-function identity into a universal formula. (You may cite the Newton identity $x_1^2 + x_2^2 = \sigma_1^2 - 2\sigma_2$.)

7. Let $\gamma \to \mathbb{C}\mathrm{P}^\infty$ be the universal line bundle and $E = \gamma \oplus \gamma \to \mathbb{C}\mathrm{P}^\infty$. Using the Whitney sum formula and example 7.2.3, compute the total Chern class $c(E)$ and each $c_i(E) \in H^*(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) = \mathbb{Z}[c_1]$ explicitly. Is E isomorphic to the trivial rank-2 bundle? Justify using a Chern class.

8. Let $T^n = (S^1)^n$ be the n-torus and consider the real rank-n bundle $E = \bigoplus_{j=1}^{n} L_j$, where $L_j \to T^n$ is the real line bundle pulled back from the j-th circle factor under the projection $T^n \to S^1 = \mathbb{R}\mathrm{P}^1 \hookrightarrow \mathbb{R}\mathrm{P}^\infty$ classifying a nontrivial line bundle. Compute the total Stiefel-Whitney class $w(E) \in H^*(T^n; \mathbb{Z}/2)$ and determine $w_n(E)$. Is E orientable?

9. Use proposition 7.4.11 to compute $H^*(\mathbb{P}(E); \mathbb{Z})$ as a ring when $E = \underline{\mathbb{C}}^3$ is the trivial rank-3 bundle over a point, and confirm the answer is $H^*(\mathbb{C}\mathrm{P}^2; \mathbb{Z}) = \mathbb{Z}[t]/(t^3)$. Then compute $H^*(\mathbb{P}(E); \mathbb{Z})$ when $E = \gamma \oplus \underline{\mathbb{C}}^2$ over $\mathbb{C}\mathrm{P}^\infty$ with γ the universal line bundle, expressing the relation in terms of $c_1 = c_1(\gamma)$.

10. Prove that $BU(1) \times \cdots \times BU(1) = (\mathbb{C}\mathrm{P}^\infty)^n$ has cohomology $\mathbb{Z}[x_1, \ldots, x_n]$ and that the map $B\iota\colon (\mathbb{C}\mathrm{P}^\infty)^n \to BU(n)$ induced by the diagonal torus $\iota\colon U(1)^n \hookrightarrow U(n)$ sends $c_i \mapsto \sigma_i(x_1, \ldots, x_n)$, the elementary symmetric polynomial. Deduce that $B\iota^*$ is injective and identifies $H^*(BU(n); \mathbb{Z})$ with the ring of symmetric polynomials $\mathbb{Z}[x_1, \ldots, x_n]^{S_n}$.

7.5 Rank-Two Bundles on Low-Dimensional Complexes

Section 7.4 read each characteristic class as an obstruction. The converse question asks how much of a bundle its characteristic classes determine. For complex line bundles over a CW complex they determine everything: the first Chern class is a bijection $\mathrm{Vect}^1_{\mathbb{C}}(X) \to H^2(X; \mathbb{Z})$ by proposition 7.4.7, because $BU(1) = \mathbb{C}\mathrm{P}^\infty$ is the Eilenberg-MacLane space $K(\mathbb{Z}, 2)$ (example 5.5.5). In rank two the first two Chern classes assemble into a map $BU(2) \to K(\mathbb{Z}, 2) \times K(\mathbb{Z}, 4)$, and the question becomes how close this map is to a homotopy equivalence. It induces isomorphisms on homotopy groups through degree four and is not injective in degree five, where $\pi_5(BU(2)) \cong \mathbb{Z}/2$. Over a CW complex of dimension at most four that range suffices: rank-two complex bundles are classified by the pair (c_1, c_2), every pair of classes occurs, and the five-sphere shows that the bound on the dimension cannot be raised.

Throughout, $BU(2) = \mathrm{Gr}_2(\mathbb{C}^\infty)$ carries the universal rank-two bundle γ_2. A CW complex X is paracompact Hausdorff by theorem 7.1.7, so theorem 7.1.29 identifies $\mathrm{Vect}^2_{\mathbb{C}}(X)$ with the set $[X, BU(2)]$ of free homotopy classes. The homotopy groups of $BU(2)$ come from the universal sphere bundle of lemma 7.2.4, the fibration

$$S^3 \longrightarrow S(\gamma_2) \xrightarrow{\rho} BU(2)$$

whose total space consists of the unit vectors in the fibers of γ_2 and is homotopy equivalent to $BU(1) = \mathbb{C}\mathrm{P}^\infty$ through the map j of that lemma.

Proposition 7.5.1: Low-Dimensional Homotopy Groups of $BU(2)$

The space $BU(2)$ is simply connected, and at every basepoint

$$\pi_2(BU(2)) \cong \mathbb{Z}, \qquad \pi_3(BU(2)) = 0, \qquad \pi_4(BU(2)) \cong \mathbb{Z}, \qquad \pi_5(BU(2)) \cong \mathbb{Z}/2$$

In the long exact sequence of the fibration ρ, the map $\rho_*\colon \pi_2(S(\gamma_2)) \to \pi_2(BU(2))$ and the boundary maps $\partial\colon \pi_4(BU(2)) \to \pi_3(S^3)$ and $\partial\colon \pi_5(BU(2)) \to \pi_4(S^3)$ are isomorphisms.

***Proof*:**
Step 1: The total space. By lemma 7.2.4 with $n = 2$, the map ρ is a fiber bundle with fiber S^3, hence a Serre fibration by theorem 5.2.4, and its total space $S(\gamma_2)$ is homotopy equivalent to $BU(1) = \mathbb{C}\mathrm{P}^\infty$, which is the Eilenberg-MacLane space $K(\mathbb{Z}, 2)$ (example 5.5.5). So $S(\gamma_2)$ is path-connected, $\pi_2(S(\gamma_2)) \cong \mathbb{Z}$, and $\pi_i(S(\gamma_2)) = 0$ for every other $i \geq 1$. The map ρ is onto, its fibers being nonempty, so $BU(2)$ is path-connected as well.

Step 2: The exact sequence. Fix $b \in BU(2)$, let $F = \rho^{-1}(b) \cong S^3$, and fix a basepoint in F. By corollary 5.4.11, $\pi_i(F) = 0$ for $i \leq 2$ and $\pi_3(F) \cong \mathbb{Z}$, and by example 6.3.6 with $n = 3$, $\pi_4(F) \cong \mathbb{Z}/2$. For each $i \geq 1$ the long exact sequence of theorem 5.2.12 contains the segment

$$\pi_i(F) \longrightarrow \pi_i(S(\gamma_2)) \xrightarrow{\rho_*} \pi_i(BU(2)) \xrightarrow{\partial} \pi_{i-1}(F) \longrightarrow \pi_{i-1}(S(\gamma_2))$$

For $i = 1$ the set $\pi_0(F)$ is a single point, so exactness at $\pi_1(BU(2))$ makes all of $\pi_1(BU(2))$ the image of $\pi_1(S(\gamma_2)) = 0$, and $BU(2)$ is simply connected. For $i = 2$ the groups $\pi_2(F)$ and $\pi_1(F)$ vanish, so ρ_* is an isomorphism and $\pi_2(BU(2)) \cong \pi_2(S(\gamma_2)) \cong \mathbb{Z}$. For $i = 3$ the group $\pi_3(BU(2))$ sits between $\pi_3(S(\gamma_2)) = 0$ and $\pi_2(F) = 0$, so it vanishes. For $i = 4$ and $i = 5$ the groups $\pi_i(S(\gamma_2))$ and $\pi_{i-1}(S(\gamma_2))$ vanish, so ∂ is an isomorphism, giving $\pi_4(BU(2)) \cong \pi_3(F) \cong \mathbb{Z}$ and $\pi_5(BU(2)) \cong \pi_4(F) \cong \mathbb{Z}/2$. The basepoint b was arbitrary, completing the proof.

Each group in the proposition comes from the total space or from the fiber of ρ. The total space, homotopy equivalent to $BU(1) = K(\mathbb{Z}, 2)$, contributes π_2 and nothing else. The boundary map carries $\pi_4(BU(2))$ onto the bottom group $\pi_3(S^3) \cong \mathbb{Z}$ of the fiber and $\pi_5(BU(2))$ onto the first stable stem $\pi_4(S^3) \cong \mathbb{Z}/2$. Through corollary 7.1.11 the same computation gives $\pi_3(U(2)) \cong \mathbb{Z}$ and $\pi_4(U(2)) \cong \mathbb{Z}/2$. The nonzero groups π_2 and π_4 match those of $K(\mathbb{Z}, 2) \times K(\mathbb{Z}, 4)$, the product of the Eilenberg-MacLane spaces in the degrees of c_1 and c_2. The group $\pi_5(BU(2))$ is sent to zero by every map into that product, whose homotopy vanishes above degree four.

Comparing $[X, BU(2)]$ with homotopy classes of maps into $K(\mathbb{Z}, 2) \times K(\mathbb{Z}, 4)$ is the comparison lemma 5.3.14 makes, for a map of simply connected spaces that matches homotopy groups in a range of degrees (definition 5.3.13).

Example 7.5.2: A Sphere Mapping to $K(\mathbb{Z}, n)$

Let $n \geq 2$ and let $\varphi\colon S^n \to K(\mathbb{Z}, n)$ be a based map representing a generator of $\pi_n(K(\mathbb{Z}, n)) \cong \mathbb{Z}$, the map of example 6.3.6, here allowed for $n \geq 2$. Both spaces are simply connected, so by the remark following definition 5.3.13 it suffices to compute at the basepoint. For $k < n$ both $\pi_k(S^n)$ (corollary 5.4.11) and $\pi_k(K(\mathbb{Z}, n))$ vanish. In degree n the class of the identity generates $\pi_n(S^n) \cong \mathbb{Z}$ and φ_* carries it to the generator $[\varphi]$, so φ_* is an isomorphism. In degree $n + 1$ the target $\pi_{n+1}(K(\mathbb{Z}, n))$ vanishes, so φ_* is surjective. Hence φ is $(n + 1)$-connected. For $n \geq 3$ it

is not $(n+2)$-connected, because $\varphi_*\colon \pi_{n+1}(S^n) \to \pi_{n+1}(K(\mathbb{Z},n)) = 0$ is not injective, its source being $\mathbb{Z}/2$ by example 6.3.6.

The spaces $BU(2)$ and $K(\mathbb{Z},2) \times K(\mathbb{Z},4)$ are both simply connected, by proposition 7.5.1 and definition 5.5.1, and that hypothesis is enough for lemma 5.3.14.

The Chern classes enter through representability. The Grassmannian $BU(2) = \mathrm{Gr}_2(\mathbb{C}^\infty)$ is a CW complex, its cells being the Schubert cells (theorem 7.1.17), and it is connected by proposition 7.5.1, so theorem 5.5.12 applies with $X = BU(2)$ and gives maps

$$\kappa_1\colon BU(2) \longrightarrow K(\mathbb{Z},2), \qquad \kappa_2\colon BU(2) \longrightarrow K(\mathbb{Z},4), \qquad \kappa_1^*\iota_2 = c_1, \qquad \kappa_2^*\iota_4 = c_2$$

each unique up to homotopy, where c_1, c_2 are the generators of theorem 7.2.5 and ι_2, ι_4 are the fundamental classes of definition 5.5.10. Write $\kappa = (\kappa_1, \kappa_2)\colon BU(2) \to K(\mathbb{Z},2) \times K(\mathbb{Z},4)$. A based map into a product is a pair of based maps, and likewise for based homotopies, so

$$\pi_k\big(K(\mathbb{Z},2) \times K(\mathbb{Z},4)\big) \cong \pi_k(K(\mathbb{Z},2)) \times \pi_k(K(\mathbb{Z},4))$$

which is $\mathbb{Z}$ for $k = 2$ and $k = 4$ and vanishes for every other $k \geq 1$ (definition 5.5.1). In particular the product is simply connected, and under this identification $\kappa_* = \big((\kappa_1)_*, (\kappa_2)_*\big)$.

Proposition 7.5.3: The First Two Chern Classes Form a 5-Connected Map

The map $\kappa = (\kappa_1, \kappa_2)\colon BU(2) \to K(\mathbb{Z},2) \times K(\mathbb{Z},4)$ representing (c_1, c_2) is 5-connected.

***Proof*:**

Fix a basepoint $b \in BU(2)$. The argument below works for every choice of b.

Step 1: Degrees one, three and five. For $k = 1$ and $k = 3$ both $\pi_k(BU(2), b)$ (proposition 7.5.1) and $\pi_k(K(\mathbb{Z},2) \times K(\mathbb{Z},4))$ vanish, so κ_* is an isomorphism. For $k = 5$ the target vanishes, so κ_* is surjective. It remains to show that $(\kappa_1)_*$ is an isomorphism on π_2 and that $(\kappa_2)_*$ is an isomorphism on π_4.

Step 2: Reduction to an evaluation. Let $m \in \{2,4\}$ and $i = m/2$, and write $[S^m] \in H_m(S^m)$ for the fundamental class of definition 5.4.1. At every basepoint of $K(\mathbb{Z},m)$ the homomorphism

$$\theta\colon \pi_m(K(\mathbb{Z},m)) \longrightarrow \mathbb{Z}, \qquad \theta(\alpha) = \langle \iota_m, h(\alpha)\rangle$$

is an isomorphism: the Hurewicz map h is an isomorphism by theorem 5.4.9, the space $K(\mathbb{Z},m)$ being $(m-1)$-connected, and evaluation on ι_m is an isomorphism $H_m(K(\mathbb{Z},m)) \to \mathbb{Z}$ by definition 5.5.10. For a based map $g\colon (S^m, s_0) \to (BU(2), b)$, the definition of h (definition 5.4.2) and the naturality of the evaluation pairing give

$$\theta\big((\kappa_i)_*[g]\big) = \big\langle \iota_m, (\kappa_i \circ g)_*[S^m]\big\rangle = \big\langle g^*\kappa_i^*\iota_m, [S^m]\big\rangle = \big\langle g^*c_i, [S^m]\big\rangle = \big\langle c_i(g^*\gamma_2), [S^m]\big\rangle$$

the last equality by definition 7.2.7, since g classifies $g^*\gamma_2$ (theorem 7.1.29). Evaluation on $[S^m]$ is an isomorphism $H^m(S^m;\mathbb{Z}) \to \mathbb{Z}$ by the universal coefficient theorem (theorem 4.1.8) and corollary 3.4.15. So $(\kappa_i)_*$ is an isomorphism on π_m if and only if $[g] \mapsto \langle g^*c_i, [S^m]\rangle$ is an isomorphism $\pi_m(BU(2), b) \to \mathbb{Z}$, and for that it suffices that g^*c_i generates $H^m(S^m;\mathbb{Z})$ for one generator $[g]$ of $\pi_m(BU(2), b) \cong \mathbb{Z}$.

Step 3: Degree two. The space $BU(2)$ is simply connected (proposition 7.5.1), so theorem 5.4.9 with $n = 2$ gives $H_1(BU(2)) = 0$ and an isomorphism $h\colon \pi_2(BU(2), b) \to H_2(BU(2))$, whence

$H_2(BU(2)) \cong \mathbb{Z}$. The Ext term $\operatorname{Ext}(H_1(BU(2)), \mathbb{Z})$ vanishes, so the universal coefficient theorem identifies $H^2(BU(2); \mathbb{Z})$ with $\operatorname{Hom}(H_2(BU(2)), \mathbb{Z})$ by evaluation. By theorem 7.2.5 the group $H^2(BU(2); \mathbb{Z})$ is infinite cyclic on c_1, so evaluation on c_1 generates $\operatorname{Hom}(H_2(BU(2)), \mathbb{Z}) \cong \operatorname{Hom}(\mathbb{Z}, \mathbb{Z})$ and is therefore an isomorphism $H_2(BU(2)) \to \mathbb{Z}$. Hence $[g] \mapsto \langle g^* c_1, [S^2] \rangle = \langle c_1, h[g] \rangle$ is an isomorphism $\pi_2(BU(2), b) \to \mathbb{Z}$, and by Step 2 so is $(\kappa_1)_*$.

Step 4: The sphere bundle over S^4. Let $g \colon (S^4, s_0) \to (BU(2), b)$ represent a generator of $\pi_4(BU(2), b) \cong \mathbb{Z}$, put $E = g^* \gamma_2$, and let

$$P = \left\{ (x, v) \in S^4 \times S(\gamma_2) \;\middle|\; g(x) = \rho(v) \right\} \xrightarrow{q} S^4$$

be the pullback of ρ along g, with q the first projection. A point of $S(\gamma_2)$ over $g(x)$ is a unit vector of $(\gamma_2)_{g(x)} = E_x$ (lemma 7.2.4), so P is the unit sphere bundle of E for the pulled-back Hermitian metric, a fiber bundle with fiber S^3 and a Serre fibration by theorem 5.2.4. The second projection $G \colon P \to S(\gamma_2)$ satisfies $\rho \circ G = g \circ q$ and maps the fiber $q^{-1}(s_0)$ homeomorphically onto $F = \rho^{-1}(b)$, in which a basepoint is fixed. The boundary map of theorem 5.2.12 is the inverse of the lifting isomorphism from the relative group of the pair (total space, fiber) to the homotopy group of the base, followed by restriction to the fiber. Both operations commute with the map of pairs $G \colon (P, q^{-1}(s_0)) \to (S(\gamma_2), F)$, so the boundary maps of q and ρ satisfy $G_* \circ \partial_q = \partial_\rho \circ g_*$ on $\pi_4(S^4, s_0)$. The identity of S^4 generates $\pi_4(S^4, s_0) \cong \mathbb{Z}$ (corollary 5.4.11), g_* carries it to the generator $[g]$, and ∂_ρ is an isomorphism by proposition 7.5.1. Hence $\partial_q \colon \pi_4(S^4) \to \pi_3(q^{-1}(s_0)) \cong \mathbb{Z}$ is an isomorphism.

Step 5: The homotopy and cohomology of P. The space P is path-connected: two points of P are joined by lifting a path between their images in S^4 through q, then moving inside the path-connected fiber over its endpoint. Take as basepoint of P the point p_0 of $q^{-1}(s_0)$ that G carries to the basepoint of F. The long exact sequence of q at p_0 (theorem 5.2.12), with $\pi_k(S^4) = 0$ for $k \leq 3$ and $\pi_k(S^3) = 0$ for $k \leq 2$, gives $\pi_1(P, p_0) = \pi_2(P, p_0) = 0$. In the segment

$$\pi_4(S^3) \longrightarrow \pi_4(P) \longrightarrow \pi_4(S^4) \xrightarrow{\partial_q} \pi_3(S^3) \longrightarrow \pi_3(P) \longrightarrow \pi_3(S^4) = 0$$

taken at p_0, surjectivity of ∂_q makes $\pi_3(S^3) \to \pi_3(P)$ zero, so $\pi_3(P, p_0) = 0$ by exactness, and injectivity of ∂_q makes $\pi_4(P) \to \pi_4(S^4)$ zero, so $\pi_4(P, p_0)$ is the image of $\pi_4(S^3) \cong \mathbb{Z}/2$ (example 6.3.6), a group of order at most two. The vanishing of π_1, π_2 and π_3 holds at every basepoint: P is simply connected by corollary 2.1.16, and then proposition 5.1.15 identifies $\pi_k(P, p)$ with $[S^k, P]$ for every $p \in P$. Thus P is 3-connected, and theorem 5.4.9 with $n = 4$ gives $H_i(P) = 0$ for $1 \leq i \leq 3$ and $H_4(P) \cong \pi_4(P, p_0)$, a finite group. By the universal coefficient theorem $H^3(P; \mathbb{Z}) = 0$ and $H^4(P; \mathbb{Z}) \cong \operatorname{Hom}(H_4(P), \mathbb{Z}) = 0$.

Step 6: Degree four. Regard E as an oriented real bundle of rank four through its complex orientation (definition 7.1.25), as in theorem 7.4.12. The sphere $S^4 = e^0 \cup e^4$ is a finite CW complex, so proposition 7.3.5(1) and corollary 7.3.6 give a long exact sequence for the sphere bundle q whose cup-product map is $\smile e(E)$, containing the exact segment

$$0 = H^3(P; \mathbb{Z}) \longrightarrow H^0(S^4; \mathbb{Z}) \xrightarrow{\smile e(E)} H^4(S^4; \mathbb{Z}) \longrightarrow H^4(P; \mathbb{Z}) = 0$$

So $1 \mapsto e(E)$ is an isomorphism $\mathbb{Z} \to H^4(S^4; \mathbb{Z})$, and $e(E)$ generates $H^4(S^4; \mathbb{Z})$. By theorem 7.4.12(2), $g^* c_2 = c_2(E) = e(E)$. By Step 2, $(\kappa_2)_*$ is an isomorphism on π_4, completing the proof.

The proof identifies the generator of $\pi_4(BU(2))$ geometrically: it classifies a rank-two bundle over S^4 whose second Chern class generates $H^4(S^4;\mathbb{Z})$, as the Gysin sequence of its sphere bundle shows. The map κ is not 6-connected, since on π_5 it is the map from $\mathbb{Z}/2$ to 0. In the terms of definition 5.6.2, κ satisfies the two conditions placed on the fourth stage $f_4\colon BU(2) \to X_4$ of a Postnikov tower: it is an isomorphism on π_i for $i \leq 4$, and its target has no homotopy above degree four. The group $\pi_5(BU(2)) \cong \mathbb{Z}/2$ is the first homotopy group of $BU(2)$ that κ does not detect.

With κ known to be 5-connected, lemma 5.3.14 applies to every CW complex of dimension at most four, and the classification of theorem 7.1.29 and the representability of theorem 5.5.12 translate its conclusion into bundles and cohomology classes.

Theorem 7.5.4: Rank-Two Bundles on Complexes of Dimension at Most Four

Let X be a CW complex of dimension at most four. Then the map

$$\mathrm{Vect}^2_{\mathbb{C}}(X) \longrightarrow H^2(X;\mathbb{Z}) \times H^4(X;\mathbb{Z}), \qquad E \longmapsto \big(c_1(E), c_2(E)\big)$$

is a bijection. That is, two rank-two complex vector bundles over X are isomorphic if and only if they have the same first and second Chern classes, and for every $a \in H^2(X;\mathbb{Z})$ and $b \in H^4(X;\mathbb{Z})$ there is a rank-two complex vector bundle E over X with $c_1(E) = a$ and $c_2(E) = b$.

***Proof*:**

Step 1: Reduction to connected X. The closure of a cell of X is the image of a disk under its characteristic map, hence path-connected, so each path component X_α of X is a union of closed cells. Each skeleton X^m carries the quotient topology from X^{m-1} and the disks attached to it (definition 1.2.1), and the preimage of X_α in each attached disk is the whole disk or empty, so induction on m shows that $X_\alpha \cap X^m$ is open and closed in X^m, hence that X_α is open and closed in X. So X is the topological disjoint union of the X_α, each a subcomplex (definition 1.2.6) and so a connected CW complex of dimension at most four. A vector bundle over X is the same as a family of bundles over the X_α, and an isomorphism is a family of isomorphisms. A singular simplex has path-connected image, so the singular chain complex of X is the direct sum of those of the X_α, and restriction gives $H^m(X;\mathbb{Z}) \cong \prod_\alpha H^m(X_\alpha;\mathbb{Z})$. By proposition 7.4.1 the Chern classes commute with these restrictions. The map of the theorem is therefore the product of the corresponding maps for the X_α, and it suffices to treat connected X.

Step 2: The chain of bijections. Let X be connected. By theorem 7.1.29, valid since X is paracompact Hausdorff (theorem 7.1.7), the assignment $[f] \mapsto f^*\gamma_2$ is a bijection $[X, BU(2)] \to \mathrm{Vect}^2_{\mathbb{C}}(X)$. Consider the composite

$$[X, BU(2)] \xrightarrow{\kappa_*} \big[X, K(\mathbb{Z},2) \times K(\mathbb{Z},4)\big] \to [X, K(\mathbb{Z},2)] \times [X, K(\mathbb{Z},4)] \to H^2(X;\mathbb{Z}) \times H^4(X;\mathbb{Z})$$

The first map is a bijection by lemma 5.3.14 with $n = 5$: the map κ is 5-connected by proposition 7.5.3, its source and target are simply connected, and $\dim X \leq 4 = n - 1$. The second map sends a map to its two components, and is a bijection because a map or a homotopy into a product is a pair of maps or homotopies. The third sends $([u_1], [u_2])$ to $(u_1^*\iota_2, u_2^*\iota_4)$ and is a bijection by theorem 5.5.12, whose statement for free homotopy classes applies because X is connected. The composite sends $[f]$ to $(f^*\kappa_1^*\iota_2, f^*\kappa_2^*\iota_4) = (f^*c_1, f^*c_2)$, which is $(c_1(E), c_2(E))$ for $E = f^*\gamma_2$ by definition 7.2.7. Composing with the inverse of $[f] \mapsto f^*\gamma_2$ shows that $E \mapsto (c_1(E), c_2(E))$ is a bijection, completing the proof.

The theorem has two halves. Injectivity says that over a complex of dimension at most four a rank-two bundle is determined by two cohomology classes. Surjectivity says that the two classes are independent: any $a \in H^2(X;\mathbb{Z})$ and $b \in H^4(X;\mathbb{Z})$ occur together as the Chern classes of one bundle. Both halves come from the agreement of $BU(2)$ with $K(\mathbb{Z},2)\times K(\mathbb{Z},4)$ on homotopy groups through degree four. The surjectivity half of lemma 5.3.14 also covers $\dim X = 5$, and the proof above, with the first map of Step 2 replaced by that surjection, shows that every pair of classes on a CW complex of dimension at most five is $(c_1(E), c_2(E))$ for some rank-two bundle E.

Injectivity fails in dimension five, and the five-sphere shows it. Since $BU(2)$ is simply connected, theorem 7.1.29 and proposition 5.1.15 give

$$\mathrm{Vect}^2_{\mathbb{C}}(S^5) \cong [S^5, BU(2)] \cong \pi_5(BU(2)) \cong \mathbb{Z}/2$$

by proposition 7.5.1, while $H^2(S^5;\mathbb{Z}) = H^4(S^5;\mathbb{Z}) = 0$. So S^5 carries a nontrivial rank-two complex bundle whose Chern classes vanish, and the dimension bound in theorem 7.5.4 cannot be raised.

Since $H^2(S^4;\mathbb{Z}) = 0$, every rank-two bundle over S^4 has $c_1 = 0$, and over S^4 the only invariant in theorem 7.5.4 is c_2.

Example 7.5.5: Rank-Two Bundles on S^4

The sphere $S^4 = e^0 \cup e^4$ is a connected CW complex of dimension four, and $H^4(S^4;\mathbb{Z}) \cong \mathbb{Z}$ is generated by the class σ with $\langle \sigma, [S^4]\rangle = 1$. By theorem 7.5.4 the assignment

$$\mathrm{Vect}^2_{\mathbb{C}}(S^4) \longrightarrow \mathbb{Z}, \qquad E \longmapsto \langle c_2(E), [S^4]\rangle$$

is a bijection. The same count comes out of homotopy theory: $\mathrm{Vect}^2_{\mathbb{C}}(S^4) \cong [S^4, BU(2)] \cong \pi_4(BU(2)) \cong \mathbb{Z}$ by proposition 5.1.15 and proposition 7.5.1, and $\pi_4(BU(2)) \cong \pi_3(U(2))$ by corollary 7.1.11.

Explicit bundles realizing every integer come from the proof of proposition 7.5.3. There, $[g] \mapsto \langle c_2(g^*\gamma_2), [S^4]\rangle$ is the isomorphism $\theta \circ (\kappa_2)_* \colon \pi_4(BU(2)) \to \mathbb{Z}$, so some based map g_1 gives a bundle $E_1 = g_1^*\gamma_2$ with $\langle c_2(E_1), [S^4]\rangle = 1$. For $d \in \mathbb{Z}$ let $f_d \colon S^4 \to S^4$ be a map of degree d (corollary 5.4.12) and put $E_d = f_d^* E_1$. By proposition 7.4.1 and definition 3.5.1,

$$\langle c_2(E_d), [S^4]\rangle = \langle f_d^* c_2(E_1), [S^4]\rangle = \langle c_2(E_1), (f_d)_*[S^4]\rangle = d\,\langle c_2(E_1), [S^4]\rangle = d$$

Every rank-two complex bundle over S^4 is therefore isomorphic to exactly one E_d.

The integer d also determines whether E_d has a nowhere-zero section and whether it splits. Regarded as an oriented real bundle of rank four over the four-dimensional complex S^4, the bundle E_d has $e(E_d) = c_2(E_d) = d\sigma$ by theorem 7.4.12(2), so by theorem 7.4.17(1) it has a nowhere-zero section if and only if $d = 0$. No E_d with $d \neq 0$ is a sum of line bundles: every complex line bundle over S^4 is trivial by proposition 7.4.7, since $H^2(S^4;\mathbb{Z}) = 0$, and a sum of trivial bundles has $c_2 = 0$.

When $H^2(X;\mathbb{Z})$ and $H^4(X;\mathbb{Z})$ are both nonzero, the Whitney sum formula and theorem 7.5.4 together determine which rank-two bundles over X are sums of line bundles. On $\mathbb{C}\mathrm{P}^2$ both groups are infinite cyclic.

Example 7.5.6: Rank-Two Bundles on $\mathbb{C}\mathrm{P}^2$

The projective plane $\mathbb{C}\mathrm{P}^2$ is a connected CW complex with one cell in each of the dimensions 0, 2 and 4, the 4-skeleton of the cell structure on $\mathbb{C}\mathrm{P}^\infty$ recalled in example 5.5.5. By theorem 4.5.12, $H^*(\mathbb{C}\mathrm{P}^2;\mathbb{Z}) = \mathbb{Z}[u]/(u^3)$, and u may be taken to be the first Chern class of the hyperplane bundle

(example 7.4.10). By theorem 7.5.4 the assignment

$$\mathrm{Vect}^2_{\mathbb{C}}(\mathbb{C}\mathrm{P}^2) \longrightarrow \mathbb{Z}^2, \qquad E \longmapsto (a,b) \quad \text{where } c_1(E) = au,\ c_2(E) = bu^2$$

is a bijection.

By proposition 7.4.7 there is for each $m \in \mathbb{Z}$ a unique line bundle L_m with $c_1(L_m) = mu$, and by example 7.4.6

$$c(L_p \oplus L_q) = (1+pu)(1+qu) = 1 + (p+q)u + pq\,u^2$$

By the display, $L_p \oplus L_q$ has invariants $(p+q, pq)$, and conversely, by the injectivity in theorem 7.5.4, a bundle with these invariants is isomorphic to $L_p \oplus L_q$. So the bundle with invariants (a,b) is isomorphic to some $L_p \oplus L_q$ if and only if $p+q=a$ and $pq=b$ have an integer solution, that is, if and only if $t^2 - at + b$ has integer roots. If p, q are such roots then $a^2 - 4b = (p-q)^2$. Conversely, if $a^2 - 4b = s^2$ with $s \geq 0$ an integer, then $s^2 \equiv a^2 \pmod 4$ forces $s \equiv a \pmod 2$, and $p = (a+s)/2$, $q = (a-s)/2$ are integer roots. So the bundle with invariants (a,b) is a sum of two line bundles if and only if $a^2 - 4b$ is the square of an integer. With $c_1 = 0$ this happens exactly when $b = -m^2$, the bundle being $L_m \oplus L_{-m}$. The bundle with $c_1 = 0$ and $c_2 = u^2$, which exists by the surjectivity in theorem 7.5.4, is not a sum of line bundles, since -4 is not a square.

The tangent bundle is another instance. Through the complex structure of $\mathbb{C}\mathrm{P}^2$, the tangent bundle $T\mathbb{C}\mathrm{P}^2$ is a complex bundle of rank two with total Chern class $(1+u)^3 = 1 + 3u + 3u^2$ [29, Theorem 14.10], where the generator written there as $-c_1$ of the tautological bundle is u by example 7.4.10. Its class $c_2(T\mathbb{C}\mathrm{P}^2) = 3u^2$ is three times a generator of $H^4(\mathbb{C}\mathrm{P}^2;\mathbb{Z})$, consistent with $\langle c_2(T\mathbb{C}\mathrm{P}^2), [\mathbb{C}\mathrm{P}^2]\rangle = \chi(\mathbb{C}\mathrm{P}^2) = 3$ from theorem 7.4.12(2) and corollary 7.4.18. The invariants are $(a,b) = (3,3)$ and $a^2 - 4b = -3$, so $T\mathbb{C}\mathrm{P}^2$ is not a sum of two complex line bundles. The split bundles with $c_1 = 3u$ are the $L_p \oplus L_{3-p}$, whose second Chern classes $p(3-p)u^2$ take the values $2u^2$, 0, $-4u^2$, $-10u^2$, and so on, never $3u^2$.

Exercise 7.5.1

1. Let $n \geq 2$, let X be a connected CW complex, and let σ_n generate $H^n(S^n;\mathbb{Z})$. Using lemma 5.3.14, the $(n+1)$-connected map $\varphi\colon S^n \to K(\mathbb{Z},n)$ of example 7.5.2 and theorem 5.5.12, show that $[f] \mapsto f^*\sigma_n$ is a bijection $[X, S^n] \to H^n(X;\mathbb{Z})$ when $\dim X \leq n$, and a surjection when $\dim X = n+1$. Check the case $X = S^n$ against corollary 5.4.12.

2. Show that the sphere bundle $\rho\colon S(\gamma_2) \to BU(2)$ of lemma 7.2.4 is 3-connected and not 4-connected. Using the description of the points of $S(\gamma_2)$ in lemma 7.2.4, show that a map $\tilde{f}\colon X \to S(\gamma_2)$ with $\rho \circ \tilde{f} = f$ is the same datum as a unit-length section of $f^*\gamma_2$. Deduce from lemma 5.3.14 that every rank-two complex bundle E over a CW complex of dimension at most three has a nowhere-zero section, and then, using proposition 7.1.23, that $E \cong L \oplus \underline{\mathbb{C}}$ for a complex line bundle L with $c_1(L) = c_1(E)$. Compare with theorem 7.4.17.

3. Let X be a CW complex of dimension at most four and E a rank-two complex bundle over X. Show that E is isomorphic to a sum $L_1 \oplus L_2$ of complex line bundles if and only if there are classes $a_1, a_2 \in H^2(X;\mathbb{Z})$ with $a_1 + a_2 = c_1(E)$ and $a_1 a_2 = c_2(E)$. Deduce that $E \cong L \oplus \underline{\mathbb{C}}$ for some line bundle L if and only if $c_2(E) = 0$.

4. Let $X = S^2 \times S^2$ with its product cell structure, and let $x, y \in H^2(X;\mathbb{Z})$ be the pullbacks of a generator of $H^2(S^2;\mathbb{Z})$ along the two projections, so that $H^*(X;\mathbb{Z}) = \mathbb{Z}[x,y]/(x^2, y^2)$ with xy generating $H^4(X;\mathbb{Z})$ (theorem 4.3.9). For $m \in \mathbb{Z}$ let E_m be the rank-two bundle with

$c_1(E_m) = 0$ and $c_2(E_m) = m\,xy$, which exists by theorem 7.5.4. Using exercise 7.5.1 (3), show that E_m is a sum of two line bundles if and only if m is even.

5. Let X be a CW complex of dimension at most four, and let E, E' be rank-two complex bundles over X with $E \oplus \underline{\mathbb{C}}^k \cong E' \oplus \underline{\mathbb{C}}^k$ for some $k \geq 0$. Show that $E \cong E'$.

6. Let $E \to S^5$ be a rank-two complex bundle that is not trivial, which exists because $\mathrm{Vect}^2_{\mathbb{C}}(S^5)$ has two elements by proposition 7.5.1 and proposition 5.1.15. For $d \in \mathbb{Z}$ let $f_d \colon S^5 \to S^5$ be a based map of degree d. Show that $f_d^* E$ is trivial if and only if d is even. (Write $E \cong g^*\gamma_2$ for a based map $g \colon S^5 \to BU(2)$ and compute $[g \circ f_d]$ in $\pi_5(BU(2))$.)

7.6 Pontryagin Classes

The Stiefel-Whitney classes are mod-2 invariants, and over $\mathbb{Z}$ or $\mathbb{Q}$ they carry no information: reduced from $H^*(BO(n); \mathbb{Z}/2)$, they cannot distinguish real bundles rationally, and in particular cannot give the characteristic numbers that separate smooth manifolds up to cobordism. Real bundles do carry integral characteristic classes, but not through the integral cohomology of $BO(n)$, which is laden with 2-torsion. The classes are obtained instead by *complexifying*: a real bundle E has a complexification $E \otimes_{\mathbb{R}} \mathbb{C}$, a complex bundle whose Chern classes are honest integral classes of E. The odd Chern classes of the complexification are 2-torsion, but the even ones survive, and they are the Pontryagin classes, the fundamental integral characteristic classes of real bundles and, as this section ends by showing, rationally the only ones. They are the invariants behind the Pontryagin numbers, the signature theorem, and the index theorem.

The construction begins with the observation that the complexification of a real bundle is isomorphic to its own conjugate, which is what forces the odd Chern classes to be torsion.

Lemma 7.6.1: Complexification is Self-Conjugate

Let $E \to X$ be a real vector bundle over a paracompact Hausdorff base, with complexification $E_{\mathbb{C}} := E \otimes_{\mathbb{R}} \mathbb{C}$ (definition 5.2.9), a complex vector bundle of the same rank. Then $E_{\mathbb{C}}$ is isomorphic to its complex conjugate, $\overline{E_{\mathbb{C}}} \cong E_{\mathbb{C}}$, and consequently its odd Chern classes are 2-torsion:

$$2\,c_{2i-1}(E_{\mathbb{C}}) = 0 \qquad \text{for all } i \geq 1$$

***Proof*:**

The conjugate $\overline{F}$ of a complex bundle F (definition 5.2.9) is the same underlying real bundle with the complex structure J replaced by $-J$, so that multiplication by z becomes multiplication by $\overline{z}$. For $E_{\mathbb{C}} = E \otimes_{\mathbb{R}} \mathbb{C}$ the map

$$\varphi \colon E_{\mathbb{C}} \to E_{\mathbb{C}}, \qquad v \otimes z \mapsto v \otimes \overline{z}$$

is a real bundle isomorphism, and it is complex-antilinear: $\varphi\big(z \cdot (v \otimes w)\big) = \varphi(v \otimes zw) = v \otimes \overline{zw} = \overline{z}\,(v \otimes \overline{w}) = \overline{z}\,\varphi(v \otimes w)$. A complex-antilinear self-isomorphism of $E_{\mathbb{C}}$ is the same datum as a complex-linear isomorphism $E_{\mathbb{C}} \xrightarrow{\sim} \overline{E_{\mathbb{C}}}$, so $E_{\mathbb{C}} \cong \overline{E_{\mathbb{C}}}$.

For the Chern classes, first take a complex line bundle L: its conjugate is the dual, $\overline{L} \cong L^*$ (a Hermitian inner product, which exists by lemma 7.1.22, gives the homomorphism $\overline{L} \to L^*$, $v \mapsto \langle -, v \rangle$, continuous because the inner product is and an isomorphism on each fiber, hence an

isomorphism by lemma 5.2.10), so $c_1(\overline{L}) = c_1(L^*) = -c_1(L)$. For a bundle F of any rank the splitting principle (theorem 7.4.3) pulls F back to a sum $L_1 \oplus \cdots \oplus L_r$ of line bundles along a map injective on cohomology. Then $\overline{F}$ pulls back to $\overline{L_1} \oplus \cdots \oplus \overline{L_r}$, and with $x_j = c_1(L_j)$ the Whitney formula gives

$$c(\overline{F}) = \prod_{j=1}^{r} \big(1 + c_1(\overline{L_j})\big) = \prod_{j=1}^{r} (1 - x_j)$$

whose degree-i part is $(-1)^i \sigma_i(x_1, \ldots, x_r) = (-1)^i c_i(F)$. Hence $c_i(\overline{F}) = (-1)^i c_i(F)$ for every complex bundle F. Applying this to $F = E_{\mathbb{C}}$ together with $\overline{E_{\mathbb{C}}} \cong E_{\mathbb{C}}$ gives $c_i(E_{\mathbb{C}}) = c_i(\overline{E_{\mathbb{C}}}) = (-1)^i c_i(E_{\mathbb{C}})$. For i odd this reads $c_i(E_{\mathbb{C}}) = -c_i(E_{\mathbb{C}})$, that is $2\, c_i(E_{\mathbb{C}}) = 0$, as we sought to show.

The lemma says the odd-degree Chern data of $E_{\mathbb{C}}$ is invisible rationally: over $\mathbb{Q}$, or after inverting 2, the odd Chern classes vanish and only the even ones $c_2(E_{\mathbb{C}}), c_4(E_{\mathbb{C}}), \ldots$ remain. These are the classes worth naming.

Definition 7.6.2: Pontryagin Classes

Let $E \to X$ be a real vector bundle of rank n. For $1 \le i \le \lfloor n/2 \rfloor$ the *i-th Pontryagin class* of E is

$$p_i(E) := (-1)^i\, c_{2i}(E \otimes_{\mathbb{R}} \mathbb{C}) \in H^{4i}(X; \mathbb{Z})$$

the sign-corrected $2i$-th Chern class of the complexification. The *total Pontryagin class* is

$$p(E) := 1 + p_1(E) + p_2(E) + \cdots + p_{\lfloor n/2 \rfloor}(E) \in H^*(X; \mathbb{Z})$$

with the convention $p_i(E) = 0$ for $2i > n$.

The sign $(-1)^i$ is a normalization: it makes the Pontryagin classes of the tangent bundles of the standard examples come out positive, and it turns the self-intersection sign in the splitting computation below into a plus. The degree jump to $4i$ reflects that a Pontryagin class packages two Chern degrees at once, and only $\lfloor n/2 \rfloor$ classes appear because $c_{2i}(E_{\mathbb{C}}) = 0$ once $2i$ exceeds the complex rank n of $E_{\mathbb{C}}$. The Whitney sum formula holds only up to 2-torsion, which is the price of the definition through the complexification.

Proposition 7.6.3: Naturality and the Whitney Sum Formula for Pontryagin Classes

Let $E, F \to X$ be real vector bundles over a paracompact Hausdorff base.

1. *(Naturality)* For every map $g\colon Y \to X$ of paracompact Hausdorff spaces, one has $p_i(g^*E) = g^* p_i(E)$.
2. *(Whitney sum, modulo 2-torsion)* The total Pontryagin class is multiplicative up to 2-torsion:
$$2\big(p(E \oplus F) - p(E) \smile p(F)\big) = 0$$
In particular $p(E \oplus F) = p(E) \smile p(F)$ in $H^*(X; \mathbb{Q})$ and in $H^*(X; \mathbb{Z}[\tfrac{1}{2}])$.

***Proof*:**
(1) Complexification commutes with pullback, $(g^*E)\otimes_{\mathbb{R}}\mathbb{C} = g^*(E\otimes_{\mathbb{R}}\mathbb{C})$, and Chern classes are natural (definition 7.2.7), so $p_i(g^*E) = (-1)^i c_{2i}\big(g^*(E_{\mathbb{C}})\big) = (-1)^i g^* c_{2i}(E_{\mathbb{C}}) = g^* p_i(E)$.

(2) Complexification is additive, $(E\oplus F)_{\mathbb{C}} = E_{\mathbb{C}}\oplus F_{\mathbb{C}}$, so the complex Whitney sum formula (theorem 7.4.5) gives $c(E_{\mathbb{C}}\oplus F_{\mathbb{C}}) = c(E_{\mathbb{C}})\smile c(F_{\mathbb{C}})$. Work modulo 2-torsion, writing $\equiv$ for congruence there. By lemma 7.6.1 the odd Chern classes of every complexification vanish modulo 2-torsion, so

$$c(E_{\mathbb{C}}) \equiv \sum_{i\geq 0} c_{2i}(E_{\mathbb{C}}) = \sum_{i\geq 0}(-1)^i p_i(E)$$

and likewise for F and for $E\oplus F$ (a product of a class with a 2-torsion class is again 2-torsion, so the congruence survives multiplication). Introduce the graded ring automorphism ψ of $H^{4\bullet}(X;\mathbb{Z})$ that multiplies H^{4i} by $(-1)^i$. It is multiplicative because $\psi(a\smile b) = (-1)^{i+j}a\smile b = \psi(a)\smile\psi(b)$ for $a\in H^{4i}$ and $b\in H^{4j}$. By construction $\psi\big(\sum_i(-1)^i p_i(E)\big) = \sum_i p_i(E) = p(E)$. Applying ψ to the congruence $c\big((E\oplus F)_{\mathbb{C}}\big)\equiv c(E_{\mathbb{C}})\smile c(F_{\mathbb{C}})$ and using that ψ is a ring homomorphism yields

$$p(E\oplus F)\equiv p(E)\smile p(F) \pmod{\text{2-torsion}}$$

which is the first assertion. Over $\mathbb{Q}$ or $\mathbb{Z}[\frac{1}{2}]$ the 2-torsion vanishes and the equality is exact, completing the proof.

Pairing a product of characteristic classes of the tangent bundle against the fundamental class of a closed manifold turns the classes into numbers.

Definition 7.6.4: Stiefel-Whitney Number

Let M be a closed smooth m-manifold with tangent bundle TM and mod-2 fundamental class $[M]\in H_m(M;\mathbb{F}_2)$ (definition 4.4.8, summed over components when M is disconnected). For a partition $I=(i_1,\dots,i_r)$ of m, the associated *Stiefel-Whitney number* is

$$w_I[M] = \big\langle w_{i_1}(TM)\smile\cdots\smile w_{i_r}(TM),\ [M]\big\rangle\in\mathbb{F}_2$$

formed from the Stiefel-Whitney classes of definition 7.2.13. Only partitions of m are considered, since the cup product must land in $H^m(M;\mathbb{F}_2)$ to be paired with $[M]$.

Definition 7.6.5: Pontryagin Number

Let M be a closed oriented smooth manifold of dimension $m=4k$ with tangent bundle TM and integral fundamental class $[M]\in H_m(M;\mathbb{Z})$. For a partition $I=(i_1,\dots,i_r)$ of k, the associated *Pontryagin number* is

$$p_I[M] = \big\langle p_{i_1}(TM)\smile\cdots\smile p_{i_r}(TM),\ [M]\big\rangle\in\mathbb{Z}$$

formed from the Pontryagin classes of definition 7.6.2. Since $p_{i_j}(TM)\in H^{4i_j}(M;\mathbb{Z})$, the product lands in $H^{4k}(M;\mathbb{Z}) = H^m(M;\mathbb{Z})$ exactly when I partitions k, so Pontryagin numbers exist only when $\dim M\equiv 0 \pmod 4$.

Stiefel-Whitney numbers are invariant under diffeomorphism, and Pontryagin numbers under orientation-preserving diffeomorphism, both being built functorially from the tangent bundle and the

fundamental class.

Definition 7.6.6: Cobordant Manifolds

Two closed smooth n-manifolds M and N are *cobordant* if $M \sqcup N$ is the boundary of a compact smooth $(n+1)$-manifold W, called a *cobordism* between them.

Lemma 7.6.7: Stiefel-Whitney Numbers Are Cobordism Invariants

Let M be a closed smooth n-manifold. If M bounds a compact smooth $(n+1)$-manifold, then every Stiefel-Whitney number $w_I[M]$ of definition 7.6.4 vanishes. Consequently cobordant closed n-manifolds have the same Stiefel-Whitney numbers.

***Proof*:**

Let W be a compact smooth $(n+1)$-manifold with $\partial W = M$, and let $j\colon M \hookrightarrow W$ be the inclusion. A collar of the boundary gives a nowhere-zero outward normal field along M, splitting off a trivial line summand:

$$j^*TW \;\cong\; TM \oplus \underline{\mathbb{R}}$$

A trivial bundle has total Stiefel-Whitney class 1 (it is pulled back from a point, so $w = 1$ by proposition 7.4.1), so the Whitney sum formula (theorem 7.4.5) and naturality give $j^*w(TW) = w(TM \oplus \underline{\mathbb{R}}) = w(TM)$. Since j^* is a ring homomorphism, every monomial satisfies $w_I(TM) = j^*\big(w_I(TW)\big)$: the classes being evaluated all restrict from W.

The fundamental class restricts too, in the dual sense. Every manifold is $\mathbb{F}_2$-orientable (definition 4.4.4 with $R = \mathbb{F}_2$, where 1 is the only generator, so the local consistency condition holds automatically), and a compact smooth manifold has a collared boundary, so W carries a relative fundamental class $[W, M] \in H_{n+1}(W, M; \mathbb{F}_2)$ by proposition 4.5.15. The connecting homomorphism $\partial\colon H_{n+1}(W, M; \mathbb{F}_2) \to H_n(M; \mathbb{F}_2)$ of the pair sends $[W, M]$ to $[M]$, and this is checked in the collar. Over $M \times [0, 1)$ the pair (W, M) is identified with $(M \times [0, 1), M \times \{0\})$, and under that identification $[W, M]$ corresponds to the cross product of $[M]$ with the generator of $H_1([0, 1), \{0\}; \mathbb{F}_2)$, whose image under ∂ is $[M]$. So $\partial[W, M]$ restricts to the nonzero element of $H_n(M, M \setminus \{x\}; \mathbb{F}_2) \cong \mathbb{F}_2$ at every $x \in M$, and the uniqueness clause of definition 4.4.8, applied on each component of M, identifies it with $[M]$, the sum of the fundamental classes of the components as in definition 7.6.4.

Now fix a monomial w_I of degree n and set $x = w_I(TW) \in H^n(W; \mathbb{F}_2)$. With field coefficients the cohomology sequence of the pair is the vector-space dual of the homology sequence, so the two connecting homomorphisms are adjoint under the Kronecker pairing, $\langle y, \partial c\rangle = \langle \delta y, c\rangle$. Applying this to $y = j^*x$ and $c = [W, M]$,

$$w_I[M] \;=\; \big\langle j^*x,\; \partial[W, M]\big\rangle \;=\; \big\langle \delta\big(j^*x\big),\; [W, M]\big\rangle$$

and the composite $H^n(W; \mathbb{F}_2) \xrightarrow{j^*} H^n(M; \mathbb{F}_2) \xrightarrow{\delta} H^{n+1}(W, M; \mathbb{F}_2)$ is zero by exactness of the cohomology sequence of (W, M). Hence $\delta(j^*x) = 0$ and $w_I[M] = 0$.

For the second assertion, a disjoint union has $T(M \sqcup N)$ restricting to TM and to TN and fundamental class $[M] + [N]$, so $w_I[M \sqcup N] = w_I[M] + w_I[N]$. If M and N are cobordant, then $M \sqcup N$ bounds, so $w_I[M] + w_I[N] = 0$ in $\mathbb{F}_2$, that is $w_I[M] = w_I[N]$, as we sought to show.

Thom's computation of the cobordism ring gives the converse for Stiefel-Whitney numbers: a closed smooth manifold whose Stiefel-Whitney numbers all vanish is a boundary (section 8.5).

A closed formula for the Pontryagin classes of a complex bundle follows from lemma 7.6.1, and it is what makes the Pontryagin numbers of a complex manifold computable from Chern classes.

Corollary 7.6.8: Pontryagin Classes of a Complex Bundle

Let $F \to X$ be a complex vector bundle of complex rank n over a paracompact Hausdorff base, regarded through its underlying real bundle $F_{\mathbb{R}}$ (definition 5.2.9) as an oriented real bundle. Then $F_{\mathbb{R}} \otimes \mathbb{C} \cong F \oplus \overline{F}$, and for each i,

$$p_i(F_{\mathbb{R}}) = (-1)^i \sum_{r+s=2i} (-1)^s \, c_r(F) \smile c_s(F) \in H^{4i}(X;\mathbb{Z})$$

Equivalently, the total classes satisfy

$$1 - p_1(F_{\mathbb{R}}) + p_2(F_{\mathbb{R}}) - \cdots = \big(1 + c_1(F) + \cdots + c_n(F)\big) \smile \big(1 - c_1(F) + c_2(F) - \cdots\big)$$

in the degrees divisible by 4. The degree-$(4i+2)$ components of the right-hand side are 2-torsion and are discarded.

***Proof*:**
Step 1: The splitting. Let $J\colon F_{\mathbb{R}} \to F_{\mathbb{R}}$ be multiplication by i in the complex structure of F, a real bundle endomorphism with $J^2 = -\mathrm{id}$. Extend J complex-linearly to $F_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$, where it still satisfies $J^2 = -\mathrm{id}$, so its minimal polynomial divides $t^2 + 1 = (t-i)(t+i)$, which has distinct roots in $\mathbb{C}$. Hence $F_{\mathbb{R}} \otimes \mathbb{C}$ splits as the direct sum of the $(+i)$- and $(-i)$-eigenbundles of J, and these are F and $\overline{F}$ respectively: on the $(+i)$-eigenbundle the extended J acts as multiplication by i, which is the original complex structure, while the $(-i)$-eigenbundle carries the conjugate one. Eigenbundle decompositions of a bundle endomorphism with locally constant eigenvalue multiplicities are subbundles, so $F_{\mathbb{R}} \otimes \mathbb{C} \cong F \oplus \overline{F}$.

Step 2: Reading off the classes. By definition 7.6.2, $p_i(F_{\mathbb{R}}) = (-1)^i c_{2i}(F_{\mathbb{R}} \otimes \mathbb{C})$. Applying the Whitney sum formula (theorem 7.4.5) to the splitting of Step 1 gives $c(F_{\mathbb{R}} \otimes \mathbb{C}) = c(F) \smile c(\overline{F})$, and $c_s(\overline{F}) = (-1)^s c_s(F)$ because conjugation reverses the sign of the first Chern class of a line bundle and the splitting principle (theorem 7.4.3) propagates this to the elementary symmetric functions. Extracting the degree-$4i$ component,

$$c_{2i}(F_{\mathbb{R}} \otimes \mathbb{C}) = \sum_{r+s=2i} c_r(F) \smile c_s(\overline{F}) = \sum_{r+s=2i} (-1)^s \, c_r(F) \smile c_s(F)$$

and multiplying by $(-1)^i$ gives the stated formula. The total-class identity is the same statement summed over i, the odd-degree-in-c components being exactly the degree-$(4i+2)$ terms discarded by definition 7.6.2, which are 2-torsion by lemma 7.6.1, completing the proof.

Example 7.6.9: Pontryagin Classes of Spheres and of $\mathbb{C}\mathrm{P}^2$

1. *The tangent bundle of a sphere.* The tangent bundle TS^n is stably trivial: $TS^n \oplus \nu \cong \underline{\mathbb{R}}^{n+1}$, where ν is the trivial normal line bundle of the standard embedding $S^n \subseteq \mathbb{R}^{n+1}$. By

proposition 7.6.3, modulo 2-torsion $p(TS^n) \smile p(\nu) = p(\underline{\mathbb{R}}^{n+1}) = 1$, and $p(\nu) = 1$, so $p(TS^n) \equiv 1$. Since $H^*(S^n;\mathbb{Z})$ is torsion-free the congruence is an equality: $p(TS^n) = 1$, all Pontryagin classes of a sphere vanish.

2. *The complex projective plane.* Regard $T\mathbb{C}\mathrm{P}^2$ as a real rank-4 bundle. Its complexification is $(T\mathbb{C}\mathrm{P}^2)_{\mathbb{C}} \cong T' \oplus \overline{T'}$, where $T' = T^{1,0}\mathbb{C}\mathrm{P}^2$ is the holomorphic tangent bundle, a complex rank-2 bundle with total Chern class $c(T') = (1+u)^3 = 1 + 3u + 3u^2$ (from the Euler sequence, with $u \in H^2(\mathbb{C}\mathrm{P}^2;\mathbb{Z})$ the hyperplane class and $u^3 = 0$). Then

$$c\big((T\mathbb{C}\mathrm{P}^2)_{\mathbb{C}}\big) = c(T') \smile c(\overline{T'}) = (1 + 3u + 3u^2)(1 - 3u + 3u^2) = 1 - 3u^2$$

using $c(\overline{T'}) = 1 - 3u + 3u^2$ and $u^3 = 0$. Hence $c_2\big((T\mathbb{C}\mathrm{P}^2)_{\mathbb{C}}\big) = -3u^2$ and

$$p_1(T\mathbb{C}\mathrm{P}^2) = (-1)^1 c_2\big((T\mathbb{C}\mathrm{P}^2)_{\mathbb{C}}\big) = 3u^2$$

a generator of $H^4(\mathbb{C}\mathrm{P}^2;\mathbb{Z}) \cong \mathbb{Z}$ tripled. The Pontryagin number is $\langle p_1(T\mathbb{C}\mathrm{P}^2), [\mathbb{C}\mathrm{P}^2]\rangle = 3$, and the relation $\frac{1}{3}p_1[\mathbb{C}\mathrm{P}^2] = 1 = \sigma(\mathbb{C}\mathrm{P}^2)$ to the signature is the first case of the Hirzebruch signature theorem.

The Pontryagin classes exhaust the rational characteristic classes of real bundles, exactly as the Chern classes exhaust the complex ones. The precise statement is the rational cohomology of the classifying spaces, and it is proved by the same splitting philosophy used for $BU(n)$, now applied to the complexification and refined by the involution that conjugation induces on the Chern roots.

Theorem 7.6.10: Rational Cohomology of $BO(n)$ and $BSO(n)$

As graded rings,

$$H^*(BO(n);\mathbb{Q}) \cong \mathbb{Q}[p_1, \dots, p_{\lfloor n/2 \rfloor}], \qquad |p_i| = 4i$$

a polynomial ring on the universal Pontryagin classes. For the oriented classifying spaces,

$$H^*(BSO(2m+1);\mathbb{Q}) \cong \mathbb{Q}[p_1, \dots, p_m], \qquad H^*(BSO(2m);\mathbb{Q}) \cong \mathbb{Q}[p_1, \dots, p_{m-1}, e]$$

where $e \in H^{2m}(BSO(2m);\mathbb{Q})$ is the Euler class of definition 7.2.15 and $e^2 = p_m$. In every case the rational characteristic classes of real bundles are the polynomials in the Pontryagin classes, together with the Euler class in the oriented even-rank case.

***Proof*:**
All cohomology in this proof is rational. The computation is an induction on n through the oriented universal sphere bundle

$$S^{n-1} \longrightarrow S(\tilde{\gamma}_n) \xrightarrow{\rho} BSO(n)$$

of lemma 7.2.8, with Euler class $e_n \in H^n(BSO(n);\mathbb{Q})$ (definition 7.2.15). The homotopy equivalence $\tilde{\jmath}$ of that lemma identifies $H^*(S(\tilde{\gamma}_n))$ with $H^*(BSO(n-1))$, and through it ρ^* becomes the restriction $\tilde{\imath}_n^*$ along the oriented stabilization. The fiber is oriented, so the rational Gysin sequence of theorem 6.3.2 applies, reading

$$\cdots \to H^{k-n}(BSO(n)) \xrightarrow{\smile e_n} H^k(BSO(n)) \xrightarrow{\rho^*} H^k(BSO(n-1)) \xrightarrow{\int} H^{k-n+1}(BSO(n)) \to \cdots$$

with $\int$ lowering degree by $n-1$. Two pullback facts drive the induction. First, $\tilde{\imath}_n^*\tilde{\gamma}_n \cong \tilde{\gamma}_{n-1} \oplus \underline{\mathbb{R}}$ (lemma 7.2.8), so $\rho^* p_i = p_i$ by stability of the Pontryagin classes (proposition 7.6.3) and $\rho^* e_n = e(\tilde{\imath}_n^*\tilde{\gamma}_n) = 0$, the Euler class of a bundle with a nowhere-zero section. Second, for odd rank the Euler class vanishes rationally: the fiberwise negation $-1\colon \gamma_{2m+1} \to \gamma_{2m+1}$ has fiberwise determinant $(-1)^{2m+1} = -1$, so it is an orientation-reversing automorphism carrying the universal bundle to its orientation-reverse and giving $e_{2m+1} = -e_{2m+1}$, whence $e_{2m+1} = 0$ in $H^*(BSO(2m+1);\mathbb{Q})$.

Base cases. $BSO(1)$ is a point with $H^* = \mathbb{Q}$, and $BSO(2) = BU(1) = \mathbb{C}\mathrm{P}^\infty$ (as $SO(2) = U(1)$) with $H^*(BSO(2);\mathbb{Q}) = \mathbb{Q}[e_2]$, $|e_2| = 2$, the Euler class equal to c_1 (example 7.3.8). These are the cases $n = 1, 2$.

Even rank $n = 2m$. By induction $H^*(BSO(2m-1);\mathbb{Q}) = \mathbb{Q}[p_1, \ldots, p_{m-1}]$, and the fiber S^{2m-1} is odd. The target ring is generated by the classes $p_i = \rho^* p_i$, so ρ^* is surjective. By the exactness $\mathrm{im}\rho^* = \ker \int$ this forces $\int = 0$, and the Gysin sequence breaks into short exact sequences

$$0 \to H^{k-2m}(BSO(2m)) \xrightarrow{\smile e_{2m}} H^k(BSO(2m)) \xrightarrow{\rho^*} H^k(BSO(2m-1)) \to 0$$

with $\smile e_{2m}$ injective (its kernel is the image of the incoming $\int = 0$). By the module argument of theorem 7.2.5, $H^*(BSO(2m);\mathbb{Q})$ is free over $H^*(BSO(2m-1);\mathbb{Q})$ on the powers of e_{2m}, so

$$H^*(BSO(2m);\mathbb{Q}) \cong \mathbb{Q}[p_1, \ldots, p_{m-1}][e] = \mathbb{Q}[p_1, \ldots, p_{m-1}, e]$$

with $e = e_{2m}$ of degree $2m$. Finally $p_m = e^2$: as real bundles $\gamma_{2m} \otimes_{\mathbb{R}} \mathbb{C} \cong \gamma_{2m} \oplus \gamma_{2m}$, so identifying the top Chern class with the Euler class of the underlying oriented real bundle (theorem 7.4.12) and using multiplicativity of the Euler class gives $c_{2m}(\gamma_{2m} \otimes_{\mathbb{R}} \mathbb{C}) = \pm e^2$, with the sign $(-1)^m$ of definition 7.6.2 fixed precisely so that $p_m = (-1)^m c_{2m}(\gamma_{\mathbb{C}}) = e^2$.

Odd rank $n = 2m+1$. By induction $H^*(BSO(2m);\mathbb{Q}) = \mathbb{Q}[p_1, \ldots, p_{m-1}, e]$ with $|e| = 2m$ and $e^2 = p_m$, and the fiber S^{2m} is even. Here $e_{2m+1} = 0$ by the odd-rank vanishing, so $\smile e_{2m+1} = 0$ throughout, and the Gysin sequence breaks into short exact sequences

$$0 \to H^k(BSO(2m+1)) \xrightarrow{\rho^*} H^k(BSO(2m)) \xrightarrow{\int} H^{k-2m}(BSO(2m+1)) \to 0$$

Hence ρ^* is injective, realizing $H^*(BSO(2m+1);\mathbb{Q})$ as the subring $\mathrm{im}\rho^*$ of $\mathbb{Q}[p_1, \ldots, p_{m-1}, e]$, which contains $\rho^* p_i = p_i$ for $i < m$ and $\rho^* p_m = e^2$, hence contains the polynomial ring $\mathbb{Q}[p_1, \ldots, p_{m-1}, e^2]$ on m algebraically independent classes. To see there is nothing more, compare Poincaré series: the short exact sequence gives $\dim H^k(BSO(2m+1)) = \dim H^k(BSO(2m)) - \dim H^{k-2m}(BSO(2m+1))$, so $P_{2m+1} = P_{2m} - t^{2m} P_{2m+1}$, that is $P_{2m+1} = P_{2m}/(1+t^{2m})$. With $P_{2m} = (1-t^{2m})^{-1} \prod_{i=1}^{m-1} (1-t^{4i})^{-1}$ and the identity $(1-t^{2m})^{-1}(1+t^{2m})^{-1} = (1-t^{4m})^{-1}$,

$$P_{2m+1}(t) = \prod_{i=1}^{m} \frac{1}{1-t^{4i}}$$

the Poincaré series of $\mathbb{Q}[p_1, \ldots, p_m]$ with $|p_i| = 4i$. Since that polynomial ring already injects as $\mathbb{Q}[p_1, \ldots, p_{m-1}, e^2] \subseteq \mathrm{im}\rho^*$ and matches the dimension in every degree, the inclusion is an equality, $H^*(BSO(2m+1);\mathbb{Q}) = \mathbb{Q}[p_1, \ldots, p_m]$.

The unoriented case. The double cover $BSO(n) \to BO(n)$ has deck group $O(n)/SO(n) = \mathbb{Z}/2$, and since 2 is invertible in $\mathbb{Q}$ the transfer gives $H^*(BO(n);\mathbb{Q}) = H^*(BSO(n);\mathbb{Q})^{\mathbb{Z}/2}$, the invariants of an orientation-reversing element. That element fixes each p_i (pulled back from $BO(n)$)

and negates the Euler class, $e \mapsto -e$. For odd n there is no Euler class and the invariant ring is all of $\mathbb{Q}[p_1, \dots, p_m]$. For even $n = 2m$ the $e \mapsto -e$ invariants of $\mathbb{Q}[p_1, \dots, p_{m-1}, e]$ are $\mathbb{Q}[p_1, \dots, p_{m-1}, e^2] = \mathbb{Q}[p_1, \dots, p_m]$. In both cases $H^*(BO(n); \mathbb{Q}) = \mathbb{Q}[p_1, \dots, p_{\lfloor n/2 \rfloor}]$, completing the proof.

The theorem is the sense in which Pontryagin classes are indispensable: rationally they, with the Euler class, generate every characteristic class of a real bundle, just as the Chern classes generate the complex ones and the Stiefel-Whitney classes generate the mod-2 ones. The three computations $H^*(BU(n))$, $H^*(BO(n); \mathbb{Z}/2)$, and $H^*(BO(n); \mathbb{Q})$ are the three faces of the same classifying-space philosophy, one for each coefficient regime, and together they account for all characteristic classes of vector bundles. The integral cohomology of $BO(n)$, which interpolates between the mod-2 and rational answers and carries the subtle 2-torsion, is the one piece left uncomputed, here and in [33] alike.

Exercise 7.6.1

1. For bundles over a paracompact Hausdorff base, show that Pontryagin classes are stable: $p(E \oplus \underline{\mathbb{R}}^k) = p(E)$ for a trivial bundle $\underline{\mathbb{R}}^k$. Deduce that Pontryagin classes are invariants of the stable isomorphism class of E, and re-derive $p(TS^n) = 1$ from stable triviality.

2. Let $V \to X$ be a complex vector bundle of rank r over a paracompact Hausdorff base with Chern classes $c_j = c_j(V)$, and let $V_{\mathbb{R}}$ be its underlying real rank-$2r$ bundle. Deduce from corollary 7.6.8 that $p_k(V_{\mathbb{R}}) = c_k^2 - 2c_{k-1}c_{k+1} + \cdots + (-1)^k 2c_{2k}$. Specialize to a line bundle to recover $p_1(L_{\mathbb{R}}) = c_1(L)^2$.

3. Compute the total Pontryagin class $p(T\mathbb{C}\mathrm{P}^n)$ for general n from $(T\mathbb{C}\mathrm{P}^n)_{\mathbb{C}} \cong T' \oplus \overline{T'}$ and $c(T'\mathbb{C}\mathrm{P}^n) = (1+u)^{n+1}$, and verify $p(T\mathbb{C}\mathrm{P}^n) = (1+u^2)^{n+1}$ in $H^*(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) = \mathbb{Z}[u]/(u^{n+1})$. Read off $p_1(T\mathbb{C}\mathrm{P}^2) = 3u^2$ as a special case and compute $p_1(T\mathbb{C}\mathrm{P}^3)$.

Part III

Generalized Cohomology

The invariants of Part I and Part II were built by hand. Singular cohomology comes from a cochain complex, with its own proofs of homotopy invariance, of the long exact sequence of a pair and of excision. Homotopy groups come from maps out of spheres, and for them excision holds only in a range of degrees (theorem 5.7.5). In 1945 Eilenberg and Steenrod took homotopy invariance, exactness and excision as the definition of a homology or cohomology theory, and in 1962 Milnor added additivity for infinite disjoint unions. With the dimension axiom, which puts the cohomology of a point in degree zero, a natural transformation between two such theories that is an isomorphism on a point is an isomorphism on every finite CW complex, and with additivity on every CW complex (theorem 8.2.14). Without the dimension axiom the axioms admit theories whose value on a point is spread over infinitely many degrees: topological K-theory, which Atiyah and Hirzebruch built from vector bundles around 1960, and cobordism, whose groups Thom computed in 1954.

Two results of Part II already have the form that every such theory turns out to have. Ordinary cohomology is represented, $\widetilde{H}^n(X;G) \cong [X, K(G,n)]_*$ (theorem 5.5.12), and rank-n complex vector bundles over a paracompact Hausdorff space X correspond to homotopy classes of maps $X \to BU(n)$ (theorem 7.1.29). Brown's representability theorem of 1962 shows that every cohomology theory is represented in this way, by a sequence of spaces each weakly equivalent to the loop space of the next, an Ω-spectrum. Chapter 8 proves the uniqueness of ordinary cohomology and Brown's theorem, constructs the spectra of K-theory and cobordism, and computes a generalized theory from ordinary cohomology with the Atiyah-Hirzebruch spectral sequence. Chapter 9$\star$ returns to ordinary cohomology and constructs the Steenrod squares of 1947, natural operations on mod-2 cohomology.

The arguments here also assume material from two companion volumes. Section 8.1 uses the Yoneda lemma, the uniqueness of representing objects and the language of adjunctions from *Category Theory* [9], and the Milnor exact sequence of section 8.2, which the uniqueness theorem needs for infinite-dimensional complexes, uses the derived limit $\varprojlim^1$ of a tower from *Homological Algebra* [13]. Bott periodicity and Thom's computation of the unoriented cobordism ring enter section 8.5 as cited theorems. The subject continues in four companion volumes: *Vector Bundles and K-Theory* [33] develops K-theory together with Bott periodicity, *Stable Homotopy Theory* [15] the stable homotopy category of spectra, *Higher Category Theory* [12] spectra as a stable ∞-category, and *Unstable Motivic Homotopy Theory* [17] the homotopy theory of smooth algebraic varieties with the affine line in place of the interval, for which the bundle classifications of chapter 9$\star$ are the topological model.

8

Generalized Cohomology Theories and Spectra

A cohomology theory is a functor on pointed CW complexes up to homotopy, and section 8.1 fixes that category and the facts about it that every later argument uses: the wedge is the coproduct, homotopy classes into a loop space form a group, suspension and looping are adjoint, and cofiber sequences are the sequences a theory is asked to make exact. Section 8.2 states the axioms, proves the Milnor exact sequence and the uniqueness theorem for ordinary cohomology, and identifies the coefficient groups, whose nonvanishing in many degrees marks an extraordinary theory. Section 8.3 proves Brown's representability theorem, and section 8.4 assembles the representing spaces into Ω-spectra, with the Eilenberg-MacLane spectrum as the first example. Section 8.5 constructs the K-theory spectrum KU and the Thom spectra of cobordism and proves the Pontryagin-Thom isomorphism. Section 8.6 builds the Atiyah-Hirzebruch spectral sequence from the skeletal filtration, and section 8.7 surveys the theory past the axioms.

8.1 The Pointed Homotopy Category

An axiom on a cohomology theory is a condition on a functor, and a functor needs a source. That source is the pointed homotopy category hTop_* of definition 1.1.7, whose objects are pointed spaces and whose morphisms are based homotopy classes of based maps. The rest of the chapter uses four facts about it: which constructions on spaces are the categorical product and coproduct, when a set of homotopy classes carries a group structure, how suspension and looping are related, and what exactness means for a sequence of spaces. Each is a property of hTop_* alone, and none mentions a cohomology theory.

All four hold. Two properties one might expect alongside them do not. Homotopy classes of maps

into an arbitrary space form a pointed set carrying no group structure, and the wedge and the product of two circles are not homotopy equivalent, so CW_* admits no biproduct and is not an additive category. A group structure is restored by taking the target to be a loop space, and the loop space is related to the suspension by an adjunction. Suspension itself has no inverse here, and theorem 5.7.8 measures how far it is from having one.

Definition 8.1.1: Pointed Homotopy Category

Recall the pointed homotopy category hTop_* of definition 1.1.7: its objects are the pointed spaces (X, x_0), and its morphisms from (X, x_0) to (Y, y_0) form the set

$$[X, Y]_* = \{\text{based maps } X \to Y\}/\simeq$$

of based homotopy classes of based maps, with composition induced by composition of representatives. Write CW_* for the full subcategory whose objects are the pointed connected CW complexes with basepoint a 0-cell.
Each morphism set $[X, Y]_*$ is a *pointed* set, with distinguished element the class of the constant map $X \to y_0$, written 0.

Composition is well defined as in lemma 1.1.6: a based homotopy $f \simeq f'$ composed with a based map g on either side is again a based homotopy, so the class of $g \circ f$ depends only on the classes of g and f. The point of passing to homotopy classes is that a functor defined on hTop_* is by construction homotopy-invariant: the requirement $f \simeq g \implies F(f) = F(g)$, which is an axiom when F is defined on spaces and maps, is discharged by the source category when F is defined on hTop_*. Every invariant in this chapter is a functor on CW_* or on its opposite, and the restriction to CW complexes is what makes theorem 5.3.6 available: it fails for general spaces, the Warsaw circle of example 5.3.9 being the witness.

The one-point space $*$ is both initial and terminal in hTop_*: a based map $* \to Y$ must send the point to y_0 and a based map $X \to *$ is the constant map, so $[*, Y]_*$ and $[X, *]_*$ each have a single element. An object that is both initial and terminal is a zero object, and its presence is what makes the distinguished element of $[X, Y]_*$ canonical rather than a choice: the class 0 is the composite $X \to * \to Y$. With a zero object in hand, the next question is which topological constructions realise the categorical product and coproduct.

Proposition 8.1.2: The Wedge is the Coproduct and the Product is the Product

Let $\{X_\alpha\}_{\alpha\in A}$ be a family of pointed CW complexes. In hTop_*:

1. The wedge sum $\bigvee_\alpha X_\alpha$ of definition 1.2.17, together with the inclusions $\iota_\alpha\colon X_\alpha \hookrightarrow \bigvee_\alpha X_\alpha$, is the coproduct. That is, for every pointed space Y the map
$$\Big[\bigvee_\alpha X_\alpha,\ Y\Big]_* \longrightarrow \prod_\alpha [X_\alpha, Y]_*, \qquad [f] \longmapsto \big([f\circ\iota_\alpha]\big)_\alpha$$
is a bijection.
2. The product space $\prod_\alpha X_\alpha$, based at the tuple of basepoints and equipped with the projections p_α, is the product. That is, for every pointed space Z the map
$$\Big[Z,\ \prod_\alpha X_\alpha\Big]_* \longrightarrow \prod_\alpha [Z, X_\alpha]_*, \qquad [g] \longmapsto \big([p_\alpha\circ g]\big)_\alpha$$
is a bijection.

***Proof*:**
Step 1: The coproduct, at the level of maps. Write $W=\bigvee_\alpha X_\alpha$ and let $q\colon \bigsqcup_\alpha X_\alpha \to W$ be the quotient map identifying all the basepoints to a single point, which is the basepoint of W. A based map $f\colon W\to Y$ determines the family $f_\alpha = f\circ\iota_\alpha$, and each f_α is based. Conversely, given based maps $f_\alpha\colon X_\alpha\to Y$, the map $\bigsqcup_\alpha f_\alpha\colon \bigsqcup_\alpha X_\alpha\to Y$ is continuous and constant on the fibre $q^{-1}(*)$, which is the set of basepoints, since every f_α sends its basepoint to y_0. By the universal property of a quotient map [16, Universal Property Of Quotient Maps] it descends to a unique continuous $f\colon W\to Y$ with $f\circ\iota_\alpha=f_\alpha$, and f is based. The two assignments are mutually inverse on maps.

Step 2: The coproduct, at the level of homotopies. It remains to check that the correspondence respects based homotopy in both directions. If $H\colon W\times I\to Y$ is a based homotopy then each $H\circ(\iota_\alpha\times \mathrm{id}_I)$ is a based homotopy, so homotopic f give homotopic families. Conversely, let $H^\alpha\colon X_\alpha\times I\to Y$ be based homotopies. Since I is locally compact Hausdorff, $q\times\mathrm{id}_I\colon (\bigsqcup_\alpha X_\alpha)\times I\to W\times I$ is again a quotient map [16, Products of Quotient Maps]. The map $\bigsqcup_\alpha H^\alpha$ is constant on its fibres, each H^α being based and therefore sending (x^0_α,t) to y_0 for every t, so it descends to a continuous $H\colon W\times I\to Y$, which is the required based homotopy. Hence the correspondence is a bijection on homotopy classes, proving (1).

Step 3: The product. A based map $g\colon Z\to\prod_\alpha X_\alpha$ is by the universal property of the product topology the same datum as a family of maps $g_\alpha\colon Z\to X_\alpha$, and g is based exactly when every g_α is. A based homotopy $Z\times I\to\prod_\alpha X_\alpha$ is likewise the same datum as a family of based homotopies $Z\times I\to X_\alpha$, again by the universal property applied to the space $Z\times I$. No quotient is involved and no local compactness is needed. The correspondence is therefore a bijection on homotopy classes, proving (2) and completing the proof.

The proposition is what gives the additivity axiom its shape. A functor that carries coproducts to products is the categorical condition, and part (1) identifies the coproduct in hTop_* as the wedge. The axiom asking that $\widetilde{h}^n(\bigvee_\alpha X_\alpha)\to\prod_\alpha\widetilde{h}^n(X_\alpha)$ be an isomorphism is that condition and nothing more. Part (2) records that products in hTop_* are computed as products of spaces, so no separate

verification is needed on the target side.

Having both a zero object and finite products and coproducts raises the question whether they agree, as they do in an additive category, where the canonical map from a finite coproduct to the finite product is an isomorphism [13, Additive Categories: Products and Coproducts]. For pointed spaces there is such a canonical map, the inclusion of the wedge into the product, and it is not an isomorphism.

Example 8.1.3: The Wedge and the Product of Two Circles

Give the torus $T^2 = S^1 \times S^1$ its standard CW structure with one 0-cell, two 1-cells and one 2-cell. Its 1-skeleton is the wedge $S^1 \vee S^1$, and the canonical map

$$j \colon S^1 \vee S^1 \longrightarrow S^1 \times S^1$$

determined by the identity on each factor and the constant map into the other is exactly this skeletal inclusion. On fundamental groups j induces the homomorphism

$$j_* \colon \pi_1(S^1 \vee S^1) \cong \mathbb{Z} * \mathbb{Z} \longrightarrow \pi_1(S^1 \times S^1) \cong \mathbb{Z} \times \mathbb{Z}$$

from the free group on two generators a, b (computed in section 2.2) to the free abelian group of rank two, sending each generator to the corresponding generator. The commutator $aba^{-1}b^{-1}$ is a nontrivial reduced word in $\mathbb{Z} * \mathbb{Z}$ and lies in the kernel of j_*, so j_* is not injective. A based homotopy equivalence induces an isomorphism on π_1, so j is not one, and $S^1 \vee S^1$ and $S^1 \times S^1$ are not isomorphic in CW_*.

Consequently CW_* is not an additive category. It has a zero object and it has both finite products and finite coproducts, but the coproduct and the product of two objects are not identified, which is the failure of the biproduct condition.

The failure recorded in the example is the first of the two structural gaps this section locates. A cohomology theory takes values in abelian groups, so its axioms speak of a category in which morphism sets are abelian groups and the two ways of combining two objects agree. The category CW_* is not of that kind, and the morphism sets $[X, Y]_*$ are in general only pointed sets. This is not a defect in the axioms. It is the reason a theory is a functor *out of* CW_* rather than an operation *inside* it, and the reason the values are required to be groups rather than observed to be. What can be recovered is a group structure on $[X, Y]_*$ for targets Y of a restricted kind, and the restriction turns out to be exactly the condition on the spaces of a spectrum.

Proposition 8.1.4: Homotopy Classes into a Loop Space Form a Group

Let (Y, y_0) be a pointed space and ΩY its loop space (definition 5.2.14). For every pointed space X:

1. The set $[X, \Omega Y]_*$ is a group under the operation $[\alpha] + [\beta] = [\mu \circ (\alpha, \beta)]$, where $\mu \colon \Omega Y \times \Omega Y \to \Omega Y$ is concatenation of loops, with identity the class of the constant map and inverse induced by reversal of loops.
2. The operation is natural in X: for a based map $h \colon X' \to X$ the induced map $h^* \colon [X, \Omega Y]_* \to [X', \Omega Y]_*$ is a group homomorphism.
3. The group $[X, \Omega^2 Y]_*$ is abelian, where $\Omega^2 Y = \Omega(\Omega Y)$.

***Proof*:**

Step 1: Loop concatenation makes ΩY a group up to homotopy. Concatenation

$$\mu(\gamma, \delta)(t) = \begin{cases} \gamma(2t), & 0 \le t \le \frac{1}{2} \\ \delta(2t-1), & \frac{1}{2} \le t \le 1 \end{cases}$$

is continuous as a map $\Omega Y \times \Omega Y \to \Omega Y$ in the compact-open topology, and the reparametrisation homotopies constructed for the fundamental group in chapter 2 show that μ is associative up to based homotopy, that the constant loop e is a two-sided unit up to based homotopy, and that $\gamma \mapsto \bar{\gamma}$, $\bar{\gamma}(t) = \gamma(1-t)$, is a two-sided inverse up to based homotopy. These are the same homotopies as there, applied to loops in Y rather than to their classes, so nothing new is required.

Step 2: The induced operation on homotopy classes. For based maps $\alpha, \beta \colon X \to \Omega Y$ set $\alpha + \beta = \mu \circ (\alpha, \beta)$, where $(\alpha, \beta) \colon X \to \Omega Y \times \Omega Y$ is the map with components α and β. If $\alpha \simeq \alpha'$ and $\beta \simeq \beta'$ through based homotopies A and B, then $\mu \circ (A, B)$ is a based homotopy $\alpha + \beta \simeq \alpha' + \beta'$, so the operation descends to $[X, \Omega Y]_*$. Postcomposing the homotopies of Step 1 with the map (α, β, γ) transports associativity, the unit law and the inverse law from ΩY to $[X, \Omega Y]_*$: for instance the based homotopy $\mu \circ (\mu \times \mathrm{id}) \simeq \mu \circ (\mathrm{id} \times \mu)$ composed with $(\alpha, \beta, \gamma) \colon X \to (\Omega Y)^3$ gives $(\alpha + \beta) + \gamma \simeq \alpha + (\beta + \gamma)$. The identity element is the class of the constant map at the constant loop, which is the distinguished element 0 of definition 8.1.1. This proves (1).

Step 2b: Naturality. For $h \colon X' \to X$ based, $(\alpha + \beta) \circ h = \mu \circ (\alpha, \beta) \circ h = \mu \circ (\alpha \circ h, \beta \circ h) = (\alpha \circ h) + (\beta \circ h)$, an equality of maps, hence of classes. So $h^*([\alpha] + [\beta]) = h^*[\alpha] + h^*[\beta]$, proving (2).

Step 3: Commutativity in the double loop space. A map $X \to \Omega^2 Y$ is a map into the loops on ΩY, so $[X, \Omega^2 Y]_*$ carries two operations of the kind just constructed: the operation $+_1$ obtained by applying Step 2 with target $\Omega(\Omega Y)$, which concatenates in the outer loop coordinate, and the operation $+_2$ obtained by applying the functor Ω to the concatenation μ of ΩY, which concatenates in the inner coordinate. Both have the class of the constant map as identity, since concatenating a constant loop in either coordinate is the constant map. The two coordinates are independent, so concatenating in one coordinate commutes with concatenating in the other, giving the interchange law

$$(\alpha +_1 \beta) +_2 (\gamma +_1 \delta) = (\alpha +_2 \gamma) +_1 (\beta +_2 \delta)$$

for all $\alpha, \beta, \gamma, \delta \in [X, \Omega^2 Y]_*$. Writing e for the common identity and applying interchange twice,

$$\alpha +_1 \beta = (\alpha +_2 e) +_1 (e +_2 \beta) = (\alpha +_1 e) +_2 (e +_1 \beta) = \alpha +_2 \beta$$

so the two operations coincide, and

$$\alpha +_1 \beta = (e +_2 \alpha) +_1 (\beta +_2 e) = (e +_1 \beta) +_2 (\alpha +_1 e) = \beta +_2 \alpha = \beta +_1 \alpha$$

so the common operation is commutative. Hence $[X, \Omega^2 Y]_*$ is abelian, proving (3). This is the Eckmann-Hilton argument, and the case $X = S^n$ is theorem 5.1.4, completing the proof.

The proposition isolates the condition under which the morphism sets of CW_* behave like the morphism sets of an additive category: not for arbitrary targets, but for targets that are loop spaces, and abelian for targets that are double loop spaces. A theory taking values in abelian groups therefore has a chance of being represented by maps into a space only if that space is a double loop space, and the requirement that this hold in every degree at once is what the definition of an Ω-spectrum in section 8.4 imposes. The relation between the loop space of the target and the suspension of the source is the next structural fact, and it is the one on which the suspension axiom rests.

Theorem 8.1.5: The Loop-Suspension Adjunction

Let X and Y be pointed spaces, ΣX the reduced suspension of definition 5.7.1 and ΩY the loop space of definition 5.2.14. Then:

1. There is a bijection
$$\Phi \colon [\Sigma X,\, Y]_* \xrightarrow{\;\cong\;} [X,\, \Omega Y]_*$$
sending the class of $g \colon \Sigma X \to Y$ to the class of $\widehat{g} \colon X \to \Omega Y$, $\widehat{g}(x)(t) = g[x, t]$, natural in X and in Y.

2. Φ is an isomorphism of groups, where $[X, \Omega Y]_*$ carries the concatenation structure of proposition 8.1.4 and $[\Sigma X, Y]_*$ the structure induced by the pinch map $\Sigma X \to \Sigma X \vee \Sigma X$ collapsing the equatorial copy of X.

***Proof*:**
Write $\pi \colon X \times I \to \Sigma X$ for the quotient map of definition 5.7.1, which collapses $X \times \{0\}$, $X \times \{1\}$ and $\{x_0\} \times I$. Since $(x_0, 0)$ and $(x_0, 1)$ lie in $\{x_0\} \times I$, all three collapsed sets have the same image, the basepoint of ΣX.

Step 1: The correspondence on maps. Let $g \colon \Sigma X \to Y$ be based and put $G = g \circ \pi \colon X \times I \to Y$, a continuous map with $G(x, 0) = G(x, 1) = y_0$ for every x and $G(x_0, t) = y_0$ for every t. Since I is locally compact Hausdorff, the exponential law [16, The Exponential Law] says that G is continuous if and only if its adjoint

$$\widehat{G} \colon X \longrightarrow C(I, Y), \qquad \widehat{G}(x)(t) = G(x, t)$$

is continuous, where $C(I, Y)$ carries the compact-open topology. The conditions $G(x, 0) = G(x, 1) = y_0$ say that $\widehat{G}(x)$ is a loop at y_0, so $\widehat{G}$ takes values in the subspace $\Omega Y \subseteq C(I, Y)$, and the condition $G(x_0, t) = y_0$ says that $\widehat{G}(x_0)$ is the constant loop, so $\widehat{G}$ is based. Set $\widehat{g} = \widehat{G}$.

Conversely let $h \colon X \to \Omega Y$ be based and let $H \colon X \times I \to Y$, $H(x, t) = h(x)(t)$, be its adjoint,

continuous by the same exponential law. Every $h(x)$ is a loop at y_0, so H collapses $X \times \{0\}$ and $X \times \{1\}$ to y_0. The loop $h(x_0)$ is constant, so H collapses $\{x_0\} \times I$ to y_0. Thus H is constant on the fibres of π and descends, by the universal property of a quotient map [16, Universal Property Of Quotient Maps], to a unique based $\check{h}\colon \Sigma X \to Y$ with $\check{h} \circ \pi = H$. The assignments $g \mapsto \widehat{g}$ and $h \mapsto \check{h}$ are mutually inverse, since both are the exponential correspondence read through π.

Step 2: The correspondence on homotopies. A based homotopy of maps $\Sigma X \to Y$ is a map $\Sigma X \times I \to Y$, and $\pi \times \mathrm{id}_I\colon (X \times I) \times I \to \Sigma X \times I$ is a quotient map because I is locally compact Hausdorff [16, Products of Quotient Maps]. Applying Step 1 with the parameter of the homotopy carried along, a based homotopy $g \simeq g'$ corresponds to a based homotopy $\widehat{g} \simeq \widehat{g'}$, and conversely. Hence the bijection descends to homotopy classes, giving Φ.

Step 3: Naturality. For a based map $u\colon X' \to X$ one has $\widehat{g \circ \Sigma u}(x') = \widehat{g}(u(x'))$, that is $\Phi(\,(\Sigma u)^*[g]\,) = u^*\Phi([g])$. For a based map $v\colon Y \to Y'$ one has $\widehat{v \circ g}(x) = v \circ \widehat{g}(x)$ as loops, that is $\Phi(v_*[g]) = (\Omega v)_*\Phi([g])$. So Φ is natural in both variables, proving (1).

Step 4: Compatibility with the group structures. The pinch map $p\colon \Sigma X \to \Sigma X \vee \Sigma X$ collapses the equatorial copy $X \times \{\frac{1}{2}\}$, and the induced operation on $[\Sigma X, Y]_*$ sends $([g],[g'])$ to the class of the map that runs g on the image of $X \times [0, \frac{1}{2}]$ and g' on the image of $X \times [\frac{1}{2}, 1]$, reparametrised linearly in each. Under the correspondence of Step 1 this map has adjoint

$$x \longmapsto \begin{cases} \widehat{g}(x)(2t), & 0 \le t \le \frac{1}{2} \\ \widehat{g'}(x)(2t-1), & \frac{1}{2} \le t \le 1 \end{cases}$$

which is exactly $\mu \circ (\widehat{g}, \widehat{g'})$, the concatenation of proposition 8.1.4. Hence $\Phi([g] + [g']) = \Phi([g]) + \Phi([g'])$, and Φ carries the identity to the identity because the constant map corresponds to the constant loop. So Φ is a group isomorphism, proving (2) and completing the proof.

The adjunction is the mechanism behind every suspension isomorphism in this chapter. It says that raising the degree of the source by suspending and lowering the degree of the target by looping are the same operation on homotopy classes, so a graded functor built from maps into a fixed sequence of spaces will shift degree correctly precisely when consecutive spaces are related by looping. That is the content of the Ω-spectrum condition, and theorem 8.4.6 uses this theorem at exactly the point where the suspension axiom is verified. Testing the adjunction on spheres recovers a relation already in hand.

Example 8.1.6: The Adjunction on Spheres

Take $X = S^n$, so that $\Sigma S^n = S^{n+1}$ by example 5.7.2. Theorem 8.1.5 gives

$$\pi_{n+1}(Y) = [S^{n+1}, Y]_* = [\Sigma S^n, Y]_* \;\cong\; [S^n, \Omega Y]_* = \pi_n(\Omega Y)$$

an isomorphism of groups for $n \ge 1$. This is the dimension shift $\pi_{n+1}(Y) \cong \pi_n(\Omega Y)$ obtained in example 5.2.15 from the long exact sequence of the path-loop fibration, now derived without any fibration, and the two isomorphisms agree up to sign: identifying ΣS^n with $I^{n+1}/\partial I^{n+1}$ through the last coordinate, the adjunction sends the class of a cube f to the class of its adjoint $s \mapsto f(s, \cdot\,)$, while the boundary map sends it to the class of the reversed adjoint (example 5.2.18), and reversal of loops induces the inverse (proposition 8.1.4(1)). Iterating, $\pi_n(Y) \cong \pi_0(\Omega^n Y)$, and proposition 8.1.4(3) applied with $X = S^n$ and $n \ge 0$ recovers the commutativity of $\pi_{n+2}(Y)$.

Three of the four structural facts are now in place: the coproduct, the group structure on homotopy

classes into a loop space, and the adjunction relating suspension to looping. The fourth is exactness. A cohomology theory converts a sequence of spaces into an exact sequence of groups, and for this to be a condition on the theory rather than on the choice of spaces, there must be a distinguished class of sequences in CW_* that every theory is asked to convert. That class is produced by the mapping cone.

Definition 8.1.7: [Topological] Mapping Cone and Cofiber Sequence

Let $f\colon X \to Y$ be a based map of pointed CW complexes and let CX be the reduced cone of definition 5.7.1, containing X as its base $X \times \{0\}$. The *mapping cone* of f is

$$C_f = Y \cup_f CX = \big(Y \sqcup CX\big)\big/\big(x \sim f(x) \text{ for } x \in X \subseteq CX\big)$$

based at the image of the basepoint, and the *cofiber sequence* of f is the pair of maps

$$X \xrightarrow{f} Y \xrightarrow{i} C_f$$

where i is the inclusion of Y.

The mapping cone attaches to Y a cone on X along f, so that the image of X in C_f becomes contractible inside C_f without being contractible in X. For the inclusion of a subcomplex $A \subseteq X$ this recovers the quotient: the pair (X, A) is a CW pair, so CA is a contractible subcomplex of $C_i = X \cup CA$ satisfying the homotopy extension property by proposition 1.3.4, and collapsing it is a homotopy equivalence by theorem 1.3.5, giving $C_i \simeq X/A$. The construction is the homotopy-theoretic replacement for the quotient, defined for an arbitrary map rather than only for an inclusion. Its use is that maps out of C_f measure exactly the failure of maps out of Y to die on X.

Proposition 8.1.8: The Puppe Sequence and its Coexactness

Let $f\colon X \to Y$ be a based map of pointed CW complexes.

1. For every pointed space K the sequence of pointed sets
$$[C_f, K]_* \xrightarrow{i^*} [Y, K]_* \xrightarrow{f^*} [X, K]_*$$
is exact at $[Y, K]_*$, meaning that $\mathrm{im} i^* = (f^*)^{-1}(0)$.
2. Collapsing Y in the mapping cone of $i\colon Y \to C_f$ gives a based homotopy equivalence $C_i \simeq \Sigma X$, under which the cofiber sequence of i becomes
$$Y \xrightarrow{i} C_f \xrightarrow{\partial} \Sigma X$$
with ∂ the map collapsing $Y \subseteq C_f$.
3. Iterating (2) produces the *Puppe sequence*
$$X \xrightarrow{f} Y \xrightarrow{i} C_f \xrightarrow{\partial} \Sigma X \xrightarrow{\Sigma f} \Sigma Y \xrightarrow{\Sigma i} \Sigma C_f \to \cdots$$
in which every three consecutive terms are, up to based homotopy equivalence, the cofiber sequence of the first map. Consequently, applying $[-, K]_*$ yields a sequence of pointed sets that is exact at every term.

***Proof*:**

Step 1: The image of i^* is contained in the kernel of f^*. Let $[g] \in [C_f, K]_*$. The composite $i \circ f\colon X \to C_f$ sends x to the class of $f(x)$, which by the defining relation of definition 8.1.7 is the class of the base point $(x, 0)$ of the cone. The reduced cone CX is contractible, the contraction sliding each (x, s) up to the cone point, and this contraction is a based homotopy inside C_f. Hence $i \circ f$ is based nullhomotopic and $f^* i^*[g] = [g \circ i \circ f] = 0$.

Step 2: The kernel of f^* is contained in the image of i^*. Let $g\colon Y \to K$ be based with $f^*[g] = 0$, and let $H\colon X \times I \to K$ be a based nullhomotopy with $H(x, 0) = g(f(x))$ and $H(x, 1) = k_0$, the basepoint of K, and $H(x_0, t) = k_0$ for every t. The conditions $H(x, 1) = k_0$ and $H(x_0, t) = k_0$ say that H is constant on the sets collapsed by the quotient map $X \times I \to CX$ of definition 5.7.1, so H descends to a continuous $\widetilde{H}\colon CX \to K$ with $\widetilde{H}(x, 0) = g(f(x))$ on the base. The maps g on Y and $\widetilde{H}$ on CX agree on the identified copy of X, since both send x to $g(f(x))$, so they glue to a continuous based $\bar{g}\colon C_f \to K$ with $\bar{g} \circ i = g$. Therefore $[g] = i^*[\bar{g}] \in \mathrm{im} i^*$, proving (1).

Step 3: The cofiber of the inclusion is the suspension. By definition $C_i = C_f \cup_i CY$, which contains CY as a subcomplex. The pair (C_i, CY) is a CW pair, so it satisfies the homotopy extension property by proposition 1.3.4, and CY is contractible, so the collapse $C_i \to C_i/CY$ is a based homotopy equivalence by theorem 1.3.5. Collapsing CY identifies the copy of Y inside C_f with the basepoint, leaving
$$C_i/CY = C_f/Y = CX/X = \Sigma X$$
the last equality because collapsing the base $X \times \{0\}$ of the reduced cone gives the reduced suspension, by the presentation of ΣX in definition 5.7.1. Under this equivalence the inclusion

$C_f \to C_i$ becomes the collapse $\partial\colon C_f \to C_f/Y = \Sigma X$, proving (2).

Step 4: Iteration and exactness throughout. Applying (2) to i in place of f identifies the cofiber of i with ΣX. Applying it again to ∂ identifies the cofiber of ∂ with ΣY, and the resulting map $\Sigma X \to \Sigma Y$ is Σf up to composition with the self-equivalence of ΣX reversing the suspension coordinate. That self-equivalence is an isomorphism in CW_*, so it changes no image and no preimage of a distinguished element, and exactness is unaffected by it. The same remark applies at every later stage.[a] Iterating produces the displayed sequence, in which every three consecutive terms are the cofiber sequence of the first map up to based homotopy equivalence. Applying $[-, K]_*$ and invoking part (1) at each such stretch gives exactness at every term, proving (3) and completing the proof.

[a] The reversal is the reason the Puppe sequence is often written with a sign on the suspended maps. Since only exactness is used here, and exactness is invariant under composing a term with an isomorphism, the sign is suppressed throughout.

Coexactness is the property that makes a single short exactness axiom generate a long exact sequence. A theory is asked to convert one cofiber sequence into a three-term exact sequence. The Puppe sequence then gives infinitely many cofiber sequences built from the same two spaces, and the suspension isomorphism reindexes them into a single doubly infinite sequence. Proposition 8.2.3 carries this out for a reduced cohomology theory. The construction is worth grounding in a case where both the mapping cone and the resulting sequence are computable.

Example 8.1.9: The Mapping Cone of a Degree-d Self-Map

Let $d \geq 2$ and let $f\colon S^1 \to S^1$ be a map of degree d. Its mapping cone

$$C_f = S^1 \cup_f CS^1 = S^1 \cup_f e^2$$

is the space obtained by attaching a 2-cell to the circle along a loop that wraps d times, since the reduced cone on S^1 is a 2-cell attached along its boundary circle. This is the Moore space $M(\mathbb{Z}/d, 1)$: its cellular chain complex is $\mathbb{Z} \xrightarrow{d} \mathbb{Z}$ in degrees 2 and 1, giving $H_1(C_f; \mathbb{Z}) \cong \mathbb{Z}/d$ and $H_n(C_f; \mathbb{Z}) = 0$ for $n \neq 0, 1$. For $d = 2$ the cone is $\mathbb{R}\mathrm{P}^2$.

Taking $K = K(\mathbb{Z}, 1) = S^1$ (example 5.5.2) in proposition 8.1.8(1) gives the exact sequence of pointed sets

$$[C_f, S^1]_* \xrightarrow{i^*} [S^1, S^1]_* \xrightarrow{f^*} [S^1, S^1]_*$$

which by $[S^1, S^1]_* = \pi_1(S^1) \cong \mathbb{Z}$ reads $[C_f, S^1]_* \to \mathbb{Z} \xrightarrow{\cdot d} \mathbb{Z}$, the second map being multiplication by d because precomposition with a degree-d map multiplies degree by d. Multiplication by d on $\mathbb{Z}$ is injective for $d \geq 2$, so its kernel is trivial and exactness forces $\mathrm{im} i^* = 0$. Since $[C_f, S^1]_* \cong \widetilde{H}^1(C_f; \mathbb{Z})$ by theorem 5.5.12, and $\widetilde{H}^1(C_f; \mathbb{Z}) \cong \mathrm{Hom}(\mathbb{Z}/d, \mathbb{Z}) = 0$ by the universal coefficient theorem (theorem 4.1.8), the group $[C_f, S^1]_*$ is itself trivial, consistent with the computation.

The last structural fact concerns not the objects of CW_* but its functors. Several arguments in this chapter produce a natural isomorphism between two functors of the form $[-, A]_*$ and conclude that the representing spaces are homotopy equivalent. That inference is the Yoneda lemma, applied in CW_* rather than in a category of algebraic objects, and the only topological content is the identification of what an isomorphism in CW_* is.

Proposition 8.1.10: Representing Objects are Unique up to Homotopy Equivalence

Let A and B be pointed CW complexes and let

$$\Phi\colon [-,A]_* \implies [-,B]_*$$

be a natural isomorphism of contravariant functors on CW_*. Then $\epsilon := \Phi_A([\mathrm{id}_A]) \in [A,B]_*$ is represented by a based homotopy equivalence $A \to B$, and $\Phi_X = \epsilon_*$ is postcomposition with ϵ for every X.

***Proof*:**
The Yoneda lemma [9, the Yoneda lemma] applied to the category CW_*, whose morphism sets are the $[X,Y]_*$, states that natural transformations $[-,A]_* \Rightarrow [-,B]_*$ correspond bijectively to elements of $[A,B]_*$, the correspondence sending Φ to $\Phi_A([\mathrm{id}_A])$ and an element ϵ to postcomposition ϵ_*. The hypotheses are met: CW_* is a locally small category and $[-,A]_*$, $[-,B]_*$ are the contravariant hom-functors it represents. Unwinding the correspondence, naturality of Φ along a based map $u\colon X \to A$ applied to $[\mathrm{id}_A] \in [A,A]_*$ gives

$$\Phi_X\big(u^*[\mathrm{id}_A]\big) = u^*\Phi_A([\mathrm{id}_A]), \qquad \text{that is} \qquad \Phi_X([u]) = [\epsilon \circ u]$$

since $u^*[\mathrm{id}_A] = [u]$. So $\Phi_X = \epsilon_*$ for every X.

Because Φ is a natural isomorphism, the same argument applied to $\Psi = \Phi^{-1}$ produces $\delta := \Psi_B([\mathrm{id}_B]) \in [B,A]_*$ with $\Psi_X = \delta_*$. Evaluating $\Psi_A \circ \Phi_A = \mathrm{id}$ at $[\mathrm{id}_A]$ gives $[\delta \circ \epsilon] = [\mathrm{id}_A]$, and evaluating $\Phi_B \circ \Psi_B = \mathrm{id}$ at $[\mathrm{id}_B]$ gives $[\epsilon \circ \delta] = [\mathrm{id}_B]$. Hence ϵ and δ are mutually inverse in CW_*, which is to say that any representative map $A \to B$ of ϵ is a based homotopy equivalence with homotopy inverse a representative of δ, as we sought to show.

The proposition is the reason a representing space, when it exists, is determined by the functor it represents: the general statement that representing objects are unique up to canonical isomorphism [9, Uniqueness of Representing Objects] reads, in CW_*, as uniqueness up to based homotopy equivalence. On CW_* this is also uniqueness up to weak equivalence, since theorem 5.3.6 identifies the two notions for CW complexes, and it is in that form that theorem 8.3.9 states its uniqueness clause.

8.1.11 Remark: What a Homotopy Functor Can Be Asked to Preserve The four facts of this section fix what a condition on a functor $\mathsf{CW}_*^{\mathrm{op}} \to \mathsf{Ab}$ can say. One of them, proposition 8.1.4, governs the values: it says which targets make homotopy classes into a group at all. The other three identify constructions in CW_*, and each supports one condition on the functor. The wedge is the coproduct (proposition 8.1.2), so a functor may be asked to carry coproducts to products. The mapping cone produces cofiber sequences whose induced sequences of homotopy classes are exact (proposition 8.1.8), so a functor may be asked to carry a cofiber sequence to an exact sequence of groups. The suspension is related to the loop space by an adjunction (theorem 8.1.5), so a functor may be asked to carry Σ to a shift in the grading.
The three are not on the same footing. The first two are conditions on how a functor treats a construction that CW_* performs, and a functor either satisfies them or does not. The third asks the functor to convert an operation into an invertible one, and Σ is not invertible in CW_*: the suspension homomorphism of definition 5.7.3 is an isomorphism only in the range given by theorem 5.7.8, and outside that range suspending loses information, as $\pi_3(S^2) \cong \mathbb{Z}$ collapsing to $\pi_4(S^3) \cong \mathbb{Z}/2$ records. A graded functor can be required to shift degree under Σ, and singular cohomology does, but the requirement is imposed on the functor rather than inherited from an invertible operation on spaces.
This asymmetry is where the objects of section 8.4 come from. A category in which Σ is invertible, morphism sets are abelian groups, and the wedge and the product agree would carry all three conditions structurally rather than axiomatically. The category CW_* is none of these, by example 8.1.3 and the failure of Freudenthal outside the stable range. The spectra of section 8.4 are the objects of such a category, and box 8.4.13 names it.

Exercise 8.1.1

1. Show that the one-point space is a zero object of hTop_* and deduce that for any pointed spaces X and Y the constant class $0 \in [X,Y]_*$ is the unique morphism factoring through a zero object. Show also that $[X,*]_*$ and $[*,Y]_*$ are singletons, and explain why the analogous statement fails in the unpointed category hTop of definition 1.1.7.

2. Prove that $\Sigma(X \vee Y) \cong \Sigma X \vee \Sigma Y$ for pointed CW complexes X and Y, and deduce from proposition 8.1.2 and theorem 8.1.5 that $\Omega(X \times Y) \cong \Omega X \times \Omega Y$. (For the second part, compare the functors $[-, \Omega(X \times Y)]_*$ and $[-, \Omega X \times \Omega Y]_*$ and apply proposition 8.1.10.)

3. Let Y be a pointed space. Show that a natural group structure on the functor $[-,Y]_*$ determines a map $\mu \colon Y \times Y \to Y$ that is unital and associative up to based homotopy, by evaluating naturality at $X = Y \times Y$. Conclude that proposition 8.1.4 has a converse: the targets Y for which $[-,Y]_*$ is naturally a group are exactly those admitting such a μ.

4. Compute $[S^1 \vee S^1,\, S^1]_*$ and $[S^1 \times S^1,\, S^1]_*$ using $S^1 = K(\mathbb{Z},1)$ and theorem 5.5.12, and show that the map induced by the inclusion j of example 8.1.3 *is* a bijection between them. Explain why this does not contradict example 8.1.3, and name an invariant that does separate the two spaces.

5. Let $f \colon X \to Y$ be a based map of pointed CW complexes. Show that $[f] = 0$ in $[X,Y]_*$ if and only if the inclusion $i \colon Y \to C_f$ admits a based left homotopy inverse. (Both directions are proposition 8.1.8(1) applied with $K = Y$. Ask what exactness at $[Y,Y]_*$ says about the class $[\mathrm{id}_Y]$.)

6. Using proposition 8.1.8 with $K = K(\mathbb{Z}/2, 1)$, compute the image of $i^*\colon [C_f, K]_* \to [S^1, K]_*$ for $f\colon S^1 \to S^1$ of degree 2, and compare with the value of $\widetilde{H}^1(\mathbb{R}\mathrm{P}^2; \mathbb{Z}/2)$ obtained in chapter 4.

7. Show that the bijection of theorem 8.1.5 is *not* in general a bijection $[\Sigma X, Y]_* \cong [X, Y]_*$ by exhibiting X and Y with $[\Sigma X, Y]_*$ and $[X, Y]_*$ of different cardinality. Explain which step of the proof fails if ΩY is replaced by Y.

8. Prove that a based homotopy equivalence $A \to B$ induces a natural isomorphism $[-, A]_* \cong [-, B]_*$, so that the correspondence of proposition 8.1.10 is a bijection between based homotopy classes of equivalences $A \to B$ and natural isomorphisms of the represented functors. Where is connectedness of the objects of CW_* used, if anywhere?

8.2 The Eilenberg-Steenrod Axioms and Generalized Theories

The earliest list of axioms in this book, recorded in chapter 3, was a working summary: enough to recognize a homology theory when one meets it, but stated for homology only, with the category of pairs left vague and the suspension behaviour absent. The axioms deserve a proper treatment, because they carry the entire weight of the chapter. Two formulations run in parallel throughout. The *unreduced* formulation works with pairs (X, A) of CW complexes and a coboundary map raising degree. It is the natural home for the long exact sequence and for excision. The *reduced* formulation works with a single based space and replaces excision and the pair sequence by a suspension isomorphism. It is leaner and is the formulation under which representability is cleanest. The two are equivalent, and the translation between them is recorded once and used freely. The development begins with the reduced version, where the axioms are fewest.

The ambient category is CW_* of definition 8.1.1, and ΣX is the reduced suspension of definition 5.7.1, with $\Sigma S^n = S^{n+1}$ by example 5.7.2. The reduced theory assigns to each based CW complex a graded abelian group and turns the operation Σ on spaces into a degree shift on groups.

Definition 8.2.1: Reduced Cohomology Theory

A *reduced cohomology theory* is a sequence of contravariant functors $\widetilde{h}^n$, one for each integer n, from the homotopy category of based CW complexes to abelian groups, together with for each based CW complex X a natural isomorphism (the *suspension isomorphism*)

$$\sigma^n_X \colon \widetilde{h}^n(X) \xrightarrow{\cong} \widetilde{h}^{n+1}(\Sigma X)$$

natural in X, subject to the following axioms.

1. (*Homotopy*) If $f \simeq g \colon X \to Y$ are based-homotopic, then $f^* = g^* \colon \widetilde{h}^n(Y) \to \widetilde{h}^n(X)$ for every n.
2. (*Exactness*) For every based CW pair (X, A) with inclusion $i \colon A \hookrightarrow X$ and quotient map $q \colon X \to X/A$, the sequence

   $$\widetilde{h}^n(X/A) \xrightarrow{q^*} \widetilde{h}^n(X) \xrightarrow{i^*} \widetilde{h}^n(A)$$

   is exact at $\widetilde{h}^n(X)$, for every n.
3. (*Additivity*) For any indexed family $\{X_\alpha\}_\alpha$ of based CW complexes with wedge sum $\bigvee_\alpha X_\alpha$, the inclusions $X_\alpha \hookrightarrow \bigvee_\alpha X_\alpha$ induce an isomorphism

   $$\widetilde{h}^n\Big(\bigvee_\alpha X_\alpha\Big) \xrightarrow{\cong} \prod_\alpha \widetilde{h}^n(X_\alpha)$$

   for every n.

A *reduced homology theory* is the same data with all arrows reversed: covariant functors $\widetilde{h}_n$, a suspension isomorphism $\widetilde{h}_n(X) \cong \widetilde{h}_{n+1}(\Sigma X)$, exactness of $\widetilde{h}_n(A) \to \widetilde{h}_n(X) \to \widetilde{h}_n(X/A)$, and additivity with the product replaced by a direct sum.

The three axioms are the irreducible content of what it means to be a cohomology theory, stripped of any reference to chains or cocycles. Homotopy invariance says the theory sees only the homotopy type. Exactness, applied to the cofiber sequence $A \to X \to X/A$, relates the values of the theory on a subcomplex, on the whole complex and on the quotient. Iterating it produces the full long exact sequence. Additivity is the compatibility with infinite wedges that distinguishes a cohomology theory in the sense relevant to representability from the weaker notion governing only finite complexes. Without additivity the theory is determined on finite complexes but may behave erratically in the limit. The suspension isomorphism is the structural heart: it makes the graded group $\widetilde{h}^*(X)$ into a single object spread across degrees, linked rung to rung by suspension, and it is exactly this linkage that a spectrum will encode geometrically.

The axioms already determine the value of any reduced theory on every sphere, with no further data than the cohomology of S^0, which is the first concrete computation worth carrying out.

Example 8.2.2: Reduced Cohomology of Spheres from the Axioms

Let $\widetilde{h}^*$ be any reduced cohomology theory. The sphere S^m is the m-fold reduced suspension $\Sigma^m S^0$,

since $\Sigma S^k = S^{k+1}$. The suspension isomorphism, applied m times, gives

$$\widetilde{h}^n(S^m) = \widetilde{h}^n(\Sigma^m S^0) \xrightarrow{(\sigma^{-1})^m,\ \cong} \widetilde{h}^{n-m}(S^0)$$

so the cohomology of every sphere is a degree-shifted copy of the cohomology of S^0. Writing $a_k := \widetilde{h}^k(S^0)$ for the graded group $\widetilde{h}^*(S^0)$, the answer is $\widetilde{h}^n(S^m) \cong a_{n-m}$ for every $m \geq 0$ and every n. The graded group $a_* = \widetilde{h}^*(S^0)$ is the only input. Everything else is forced by suspension. For singular cohomology $\widetilde{H}^*(-;G)$, the group $a_k = \widetilde{H}^k(S^0;G)$ is G for $k = 0$ and 0 otherwise, recovering $\widetilde{H}^n(S^m;G) = G$ for $n = m$ and 0 elsewhere, the computation of chapter 4. For complex K-theory, $a_k = \widetilde{KU}^k(S^0)$ is $\mathbb{Z}$ in every even k and 0 in odd k, so $\widetilde{KU}^n(S^m)$ is $\mathbb{Z}$ whenever $n - m$ is even, illustrating how an extraordinary theory sees a sphere in infinitely many degrees.

That the short exactness in axiom (2) already forces a long exact sequence is the next thing to verify, since it is what justifies calling exactness an "axiom" in the singular. The mechanism is the iterated cofiber sequence.

Proposition 8.2.3: The Long Exact Sequence of a Cofiber Sequence

Let $\widetilde{h}^*$ be a reduced cohomology theory and (X, A) a based CW pair with inclusion $i\colon A \hookrightarrow X$ and quotient $q\colon X \to X/A$. Then there is a long exact sequence

$$\cdots \to \widetilde{h}^n(X/A) \xrightarrow{q^*} \widetilde{h}^n(X) \xrightarrow{i^*} \widetilde{h}^n(A) \xrightarrow{\delta} \widetilde{h}^{n+1}(X/A) \xrightarrow{q^*} \widetilde{h}^{n+1}(X) \to \cdots$$

natural in the pair, where the connecting map δ is σ composed with $\widetilde{h}^{n+1}$ of the collapse map $X/A \to \Sigma A$.

***Proof*:**

Step 1: The cofiber sequence of a pair. For a based CW pair (X, A), the inclusion $i\colon A \hookrightarrow X$ has mapping cone $C_i = X \cup_A CA$ (definition 8.1.7), and collapsing the contractible subcomplex CA identifies it with X/A up to based homotopy equivalence, as recorded after that definition. Proposition 8.1.8(3) then gives the *Puppe sequence*

$$A \xrightarrow{i} X \xrightarrow{q} X/A \xrightarrow{\partial} \Sigma A \xrightarrow{\Sigma i} \Sigma X \xrightarrow{\Sigma q} \Sigma(X/A) \to \cdots$$

in which each three-term stretch is, up to based homotopy equivalence, the cofiber sequence of the first map, and $\partial\colon X/A \to \Sigma A$ is the map collapsing X.

Step 2: Applying the theory. Each consecutive pair of maps in the Puppe sequence is a cofiber sequence $U \xrightarrow{f} V \to C_f$ with C_f the mapping cone, and the exactness axiom applied to the CW pair (C_f, V) (whose quotient C_f/V is the suspension ΣU) gives exactness of $\widetilde{h}^n(\Sigma U) \to \widetilde{h}^n(C_f) \to \widetilde{h}^n(V)$. Concretely, applying $\widetilde{h}^n$ to the Puppe sequence and reading it as a doubly infinite sequence yields exactness of

$$\cdots \to \widetilde{h}^n(\Sigma A) \to \widetilde{h}^n(X/A) \xrightarrow{q^*} \widetilde{h}^n(X) \xrightarrow{i^*} \widetilde{h}^n(A) \to \cdots$$

at every term, since each three consecutive terms come from a cofiber sequence to which axiom (2) applies.

Step 3: Reindexing by the suspension isomorphism. The terms $\widetilde{h}^n(\Sigma A)$ appearing in Step 2 are identified by the suspension isomorphism with $\widetilde{h}^{n-1}(A)$, and $\widetilde{h}^n(\Sigma(X/A))$ with $\widetilde{h}^{n-1}(X/A)$, naturally. Substituting these identifications converts the sequence of Step 2 into

$$\cdots \to \widetilde{h}^n(X/A) \xrightarrow{q^*} \widetilde{h}^n(X) \xrightarrow{i^*} \widetilde{h}^n(A) \xrightarrow{\delta} \widetilde{h}^{n+1}(X/A) \to \cdots$$

where the connecting map $\delta\colon \widetilde{h}^n(A) \to \widetilde{h}^{n+1}(X/A)$ is the composite of $(\partial)^*\colon \widetilde{h}^{n+1}(\Sigma A) \to \widetilde{h}^{n+1}(X/A)$ with the suspension isomorphism $\widetilde{h}^n(A) \cong \widetilde{h}^{n+1}(\Sigma A)$. Naturality of σ and of the Puppe construction in the pair makes the entire sequence natural, completing the proof.

The proposition is the reason a single short-exactness axiom suffices: the suspension isomorphism recycles it indefinitely. Each degree's worth of information feeds the next through the connecting map δ, and the resulting long exact sequence is the daily tool of the subject. The cost of the leanness of the reduced axioms is that the geometry of excision is hidden inside the homotopy equivalence "mapping cone of i is X/A," which holds for CW pairs because the cofiber is computed homotopy-theoretically. The unreduced formulation puts excision back in the foreground.

The passage from a reduced theory to an unreduced one is mechanical. Given $\widetilde{h}^*$, define for a CW pair (X, A) the group $h^n(X, A) := \widetilde{h}^n(X/A)$, with the convention $X/\emptyset = X_+ = X \sqcup \{*\}$ adjoining a disjoint basepoint, so that $h^n(X) := h^n(X, \emptyset) = \widetilde{h}^n(X_+)$ is the unreduced cohomology of X. The reduced and unreduced groups of a based space differ by the coefficients: the inclusion of the basepoint $x_0 \hookrightarrow X$ admits the constant retraction $X \to x_0$, which splits the restriction map $h^n(X) \to h^n(\mathrm{pt})$, giving $h^n(X) \cong \widetilde{h}^n(X) \oplus h^n(\mathrm{pt})$ with $\widetilde{h}^n(X) = \ker\big(h^n(X) \to h^n(\mathrm{pt})\big)$. This is the cohomological analogue of the splitting $H_0(X) \cong \widetilde{H}_0(X) \oplus \mathbb{Z}$ from definition 3.2.6. The unreduced axioms are the following.

Definition 8.2.4: Eilenberg-Steenrod Cohomology Theory

An *(unreduced) cohomology theory* on the category of CW pairs is a sequence of contravariant functors $h^n\colon (X,A)\mapsto h^n(X,A)$ to abelian groups, one for each integer n, together with a natural transformation (the *coboundary*)

$$\delta\colon h^n(A)\to h^{n+1}(X,A),\qquad h^n(A):=h^n(A,\emptyset)$$

natural in the pair, subject to the following axioms. (Here a map of pairs $f\colon (X,A)\to(Y,B)$ induces $f^*\colon h^n(Y,B)\to h^n(X,A)$.)

1. (*Homotopy*) Homotopic maps of pairs $f\simeq g\colon (X,A)\to(Y,B)$ induce $f^*=g^*$ on every h^n.
2. (*Exactness*) Each pair (X,A) with inclusions $i\colon A\hookrightarrow X$ and $j\colon (X,\emptyset)\hookrightarrow(X,A)$ induces a long exact sequence
$$\cdots\to h^n(X,A)\xrightarrow{j^*}h^n(X)\xrightarrow{i^*}h^n(A)\xrightarrow{\delta}h^{n+1}(X,A)\to\cdots$$
3. (*Excision*) If (X,A) is a CW pair and $U\subseteq A$ is a subcomplex whose closure lies in the interior of A (for CW pairs, any subcomplex U with $\overline{U}\subseteq \mathrm{int}A$, equivalently any pair admitting an excisive decomposition), then the inclusion $(X\setminus U,\,A\setminus U)\hookrightarrow(X,A)$ induces an isomorphism
$$h^n(X,A)\xrightarrow{\cong}h^n(X\setminus U,\,A\setminus U)$$
for every n.
4. (*Additivity*) For a disjoint union $X=\coprod_\alpha X_\alpha$ of CW complexes, the inclusions induce an isomorphism $h^n(X)\xrightarrow{\cong}\prod_\alpha h^n(X_\alpha)$ for every n.

A *homology theory* is the same data with all functors covariant, the coboundary replaced by a boundary $\partial\colon h_n(X,A)\to h_{n-1}(A)$ lowering degree, exactness running $\cdots\to h_n(A)\to h_n(X)\to h_n(X,A)\xrightarrow{\partial}h_{n-1}(A)\to\cdots$, the excision isomorphism in the same form, and additivity with the product replaced by a direct sum.

The first three axioms, with the dimension axiom below, are the ones isolated by Eilenberg and Steenrod and used to organize the whole of ordinary homology and cohomology. Additivity was added by Milnor. Homotopy and exactness are shared with the reduced formulation. Excision is the geometric input that makes the theory *local*: it says that cohomology of a pair does not see the part of A that can be cut away, so that $h^*(X,A)$ depends only on a neighbourhood of the part of X outside A. This is what makes Mayer-Vietoris and the cellular computation possible, and it is the axiom that singular cohomology satisfies by excision in cohomology (theorem 4.1.19), itself deduced from the excision theorem of theorem 3.4.9. Additivity, again, controls complexes with infinitely many cells. Without it the uniqueness theorem theorem 8.2.14 holds only for finite complexes. The functor $H^*(-;\mathbb{Z})\otimes\mathbb{Q}$ satisfies homotopy, exactness ($\mathbb{Q}$ being flat), excision and the dimension axiom with coefficients $\mathbb{Q}$, and its natural map to $H^*(-;\mathbb{Q})$ is an isomorphism on a point. On the discrete space $\mathbb{N}$ that map is $\mathbb{Z}^{\mathbb{N}}\otimes\mathbb{Q}\to\mathbb{Q}^{\mathbb{N}}$, which misses $(1,\frac12,\frac13,\dots)$, since an element of $\mathbb{Z}^{\mathbb{N}}\otimes\mathbb{Q}$ has a single denominator. The coboundary δ is the unreduced form of the connecting map δ

from proposition 8.2.3. The equivalence of the two formulations matches them up exactly.

That the reduced and unreduced packages carry the same information is worth stating precisely, since the proof of uniqueness and the discussion of representability both move between them without comment.

Proposition 8.2.5: Equivalence of Reduced and Unreduced Theories

The assignments

$$\widetilde{h}^* \rightsquigarrow \big(h^n(X,A) := \widetilde{h}^n(X/A)\big), \qquad h^* \rightsquigarrow \big(\widetilde{h}^n(X) := \ker\big(h^n(X) \to h^n(\mathrm{pt})\big)\big)$$

are mutually inverse bijections between reduced cohomology theories and unreduced cohomology theories, up to natural isomorphism, where the basepoint inclusion $\mathrm{pt} \hookrightarrow X$ provides the splitting $h^n(X) \cong \widetilde{h}^n(X) \oplus h^n(\mathrm{pt})$. Under this correspondence the suspension isomorphism $\widetilde{h}^n(X) \cong \widetilde{h}^{n+1}(\Sigma X)$ corresponds to the composite of the coboundary of the pair (CX, X) with excision, and the long exact sequence of proposition 8.2.3 corresponds to the exactness axiom of definition 8.2.4.

***Proof*:**

Step 1: From unreduced to reduced. Given h^*, set $\widetilde{h}^n(X) := \ker(h^n(X) \to h^n(\mathrm{pt}))$, the kernel of restriction to the basepoint. Equivalently $\widetilde{h}^n(X) = h^n(X, \mathrm{pt})$, since the long exact sequence of the pair (X, pt) together with the retraction $X \to \mathrm{pt}$ splits $h^n(X) \cong h^n(X, \mathrm{pt}) \oplus h^n(\mathrm{pt})$ and identifies $h^n(X, \mathrm{pt})$ with the kernel. Homotopy and additivity for $\widetilde{h}^*$ are inherited termwise. For exactness, the cofiber sequence $A \to X \to X/A$ of a based CW pair has $h^n(X/A, \mathrm{pt}) \cong h^n(X, A)$ by excision (collapsing A is an excision of the interior of a mapping cylinder neighbourhood, valid for CW pairs), so the pair sequence of (X, A) becomes the reduced exact sequence $\widetilde{h}^n(X/A) \to \widetilde{h}^n(X) \to \widetilde{h}^n(A)$.

Step 2: The suspension isomorphism from excision. For a based CW complex X, the reduced cone CX is contractible, so $h^n(CX) \cong h^n(\mathrm{pt})$ and the long exact sequence of the pair (CX, X) gives an isomorphism $\delta\colon \widetilde{h}^n(X) = h^n(X, \mathrm{pt}) \xrightarrow{\cong} h^{n+1}(CX, X)$. Excision applied to collapsing the base of one cone in the union $\Sigma X = CX \cup_X CX$ identifies $h^{n+1}(CX, X) \cong h^{n+1}(\Sigma X, CX) \cong \widetilde{h}^{n+1}(\Sigma X)$, the last step because the second cone CX is contractible. Composing yields the suspension isomorphism $\sigma^n_X\colon \widetilde{h}^n(X) \cong \widetilde{h}^{n+1}(\Sigma X)$, natural in X because every map in the chain is natural.

Step 3: From reduced to unreduced and back. Given $\widetilde{h}^*$, set $h^n(X, A) := \widetilde{h}^n(X/A)$ with $X/\emptyset := X_+$. The reduced long exact sequence of proposition 8.2.3 for (X_+, A_+) is the unreduced pair sequence, the wedge axiom gives unreduced additivity (since $(\coprod_\alpha X_\alpha)_+ = \bigvee_\alpha (X_\alpha)_+$), and excision holds because for a CW triad $X = A \cup B$ the collapse $X/A = (B)/(A \cap B)$ realizes the excision isomorphism as an equality of quotient spaces. Starting from h^*, passing to $\widetilde{h}^*$ by Step 1 and back recovers $h^n(X, A) = \widetilde{h}^n(X/A) = h^n(X/A, \mathrm{pt}) \cong h^n(X, A)$ by the excision identification of Step 1, and the two round trips are the identity up to natural isomorphism, as we sought to show.

With the equivalence in hand the two formulations are used interchangeably. The reduced and unreduced groups of a based space sit in the split exact sequence $0 \to \widetilde{h}^n(X) \to h^n(X) \to h^n(\mathrm{pt}) \to 0$, so the only difference is a copy of the cohomology of a point in each degree. The cohomology of a point is therefore the object that controls the bookkeeping, and it earns a name.

Definition 8.2.6: Coefficient Groups and the Dimension Axiom

Let h^* be a cohomology theory. The graded abelian group

$$h^*(\mathrm{pt}) = \{\, h^n(\mathrm{pt}) \,\}_{n\in\mathbb{Z}}$$

is the group of *coefficients* of the theory. The group $h^n(\mathrm{pt})$ is the *coefficient group in degree* n. The theory satisfies the *dimension axiom* if

$$h^n(\mathrm{pt}) = 0 \quad \text{for all } n \neq 0$$

so that the coefficients are concentrated in degree 0. A cohomology theory satisfying the dimension axiom is called an *ordinary* cohomology theory. One not required to satisfy it is called a *generalized* (or *extraordinary*) cohomology theory. The same terminology applies to homology theories, with $h_*(\mathrm{pt})$ in place of $h^*(\mathrm{pt})$.

The dimension axiom is the single hypothesis that separates the cohomology theory of the previous two chapters from everything else. For an ordinary theory the coefficients reduce to one abelian group $G = h^0(\mathrm{pt})$, and the reduced cohomology of a sphere is forced: applying the suspension isomorphism n times to $\widetilde{h}^*(S^0)$ and using $\widetilde{h}^k(S^0) = \widetilde{h}^k(\mathrm{pt} \sqcup \mathrm{pt}) = h^k(\mathrm{pt})$ gives $\widetilde{h}^m(S^n) \cong h^{m-n}(\mathrm{pt})$, which under the dimension axiom is G when $m = n$ and 0 otherwise. For a generalized theory the coefficients spread across many degrees, the cohomology of a sphere is correspondingly larger, and a sphere can carry nonzero classes in infinitely many dimensions. The non-vanishing of $h^n(\mathrm{pt})$ for $n \neq 0$ is precisely the slack that K-theory and cobordism exploit, as previewed below.

Example 8.2.7: Singular Cohomology as an Ordinary Theory

Singular cohomology with coefficients in an abelian group G, the functor $(X, A) \mapsto H^n(X, A; G)$ of chapter 4, is an ordinary cohomology theory in the sense of definition 8.2.4, and verifying each axiom is a matter of citing the relevant result from that chapter.

1. *Homotopy.* Maps of pairs that are homotopic through maps of pairs induce the same map on $H^n(-; G)$ by proposition 4.1.23.
2. *Exactness.* Theorem 4.1.17 is the required long exact sequence of the pair: the map it induces from the quotient chain map is the j^* of the axiom, restriction along $A \hookrightarrow X$ is i^*, and its δ is the connecting homomorphism.
3. *Excision.* Excision in cohomology (theorem 4.1.19) gives $H^n(X, A; G) \cong H^n(X \setminus U, A \setminus U; G)$ under the stated condition $\overline{U} \subseteq \mathrm{int} A$.
4. *Additivity.* A singular simplex $\Delta^n \to \coprod_\alpha X_\alpha$ has connected image, hence lands in a single X_α, so $C_\bullet(\coprod_\alpha X_\alpha) = \bigoplus_\alpha C_\bullet(X_\alpha)$. Applying $\mathrm{Hom}(-, G)$ turns the direct sum into the product $\prod_\alpha H^n(X_\alpha; G)$.

The coefficient groups are $H^n(\mathrm{pt}; G) = G$ for $n = 0$ and 0 for $n \neq 0$, since a point has a single 0-simplex and no higher simplices, so the dimension axiom holds and the theory is ordinary with $h^0(\mathrm{pt}) = G$. De Rham cohomology of smooth manifolds and cellular cohomology of CW complexes are two further ordinary models with $G = \mathbb{R}$ and $G = \mathbb{Z}$ respectively. Both are naturally isomorphic to singular cohomology on the spaces where all three are defined, by the comparison

theorems of chapter 4, and the uniqueness theorem below explains why such isomorphisms must exist.

The example is the calibration point: ordinary cohomology is not one theory among many but, as the next theorem shows, the only ordinary theory with a given coefficient group, up to canonical isomorphism. This is what entitles one to speak of "the" cohomology with coefficients in G regardless of which model (singular, cellular, de Rham, sheaf-theoretic with constant coefficients) is used to compute it. The uniqueness theorem makes the claim precise and proves it.

Its proof climbs the skeleta of a CW complex one at a time, and for a finite-dimensional complex the climb terminates. For an infinite-dimensional one it does not, and the theorem then needs an answer to a question the axioms do not settle by themselves: how is $\widetilde{h}^n(X)$ determined by the groups $\widetilde{h}^n(X^m)$ of the skeleta? The naive answer, that it is their limit, is very nearly right and fails by exactly one group. Settling this is the business of the next three results, which are stated here because the uniqueness theorem uses them, and which are used again for the Atiyah-Hirzebruch spectral sequence in section 8.6.

The first ingredient is a Mayer-Vietoris sequence for a reduced theory. Nothing in definition 8.2.1 mentions unions, so the sequence has to be built from the exactness axiom applied twice.

Proposition 8.2.8: Mayer-Vietoris for a Reduced Cohomology Theory

Let $\widetilde{h}^*$ be a reduced cohomology theory and let $X = A \cup B$ be a based CW complex written as the union of two subcomplexes whose intersection $C = A \cap B$ is a subcomplex containing the basepoint. Write $i_A \colon A \hookrightarrow X$, $i_B \colon B \hookrightarrow X$, $j_A \colon C \hookrightarrow A$ and $j_B \colon C \hookrightarrow B$ for the inclusions. Then there is a long exact sequence

$$\cdots \to \widetilde{h}^n(X) \xrightarrow{(i_A^*,\, i_B^*)} \widetilde{h}^n(A) \oplus \widetilde{h}^n(B) \xrightarrow{j_A^* - j_B^*} \widetilde{h}^n(C) \xrightarrow{\delta} \widetilde{h}^{n+1}(X) \to \cdots$$

natural in the triad $(X; A, B)$.

***Proof*:**

Write $\Phi = (i_A^*, i_B^*)$ and $\Psi = j_A^* - j_B^*$.

Step 1: The excision identification. Since $X = A \cup B$ is a CW triad, every cell of X not in B is a cell of A not in C, so collapsing B in X and collapsing C in A produce the same quotient: the map induced by i_A is a based homeomorphism

$$\kappa \colon A/C \xrightarrow{\cong} X/B$$

and by construction $\kappa \circ q_A = q_X \circ i_A$, where $q_A \colon A \to A/C$ and $q_X \colon X \to X/B$ are the quotient maps. Note also that $q_X \circ i_B$ is the constant map to the basepoint, so $(q_X i_B)^*$ factors through $\widetilde{h}^n(\mathrm{pt})$. That group vanishes for any reduced theory: the additivity axiom of definition 8.2.1 applied to the *empty* family of based complexes has $\bigvee_{\alpha \in \emptyset} X_\alpha = \mathrm{pt}$ on the left and the trivial group $\prod_{\alpha \in \emptyset} \widetilde{h}^n(X_\alpha) = 0$ on the right, so $\widetilde{h}^n(\mathrm{pt}) \cong 0$. Hence $(q_X i_B)^* = 0$, and that $i_A j_A = i_B j_B$, both being the inclusion $C \hookrightarrow X$.

Step 2: Two long exact sequences and a ladder. Proposition 8.2.3 applied to the based CW pairs (A, C) and (X, B) gives

$$\cdots \to \widetilde{h}^n(A/C) \xrightarrow{q_A^*} \widetilde{h}^n(A) \xrightarrow{j_A^*} \widetilde{h}^n(C) \xrightarrow{\delta_A} \widetilde{h}^{n+1}(A/C) \to \cdots$$

$$\cdots \to \widetilde{h}^n(X/B) \xrightarrow{q_X^*} \widetilde{h}^n(X) \xrightarrow{i_B^*} \widetilde{h}^n(B) \xrightarrow{\delta_X} \widetilde{h}^{n+1}(X/B) \to \cdots$$

The pair $(i_A, j_B)\colon (A, C) \to (X, B)$ is a map of based CW pairs inducing κ on quotients, so naturality of proposition 8.2.3 makes the diagram

$$\begin{array}{ccccccc} \widetilde{h}^n(X/B) & \xrightarrow{q_X^*} & \widetilde{h}^n(X) & \xrightarrow{i_B^*} & \widetilde{h}^n(B) & \xrightarrow{\delta_X} & \widetilde{h}^{n+1}(X/B) \\ \downarrow{\scriptstyle \kappa^*} & & \downarrow{\scriptstyle i_A^*} & & \downarrow{\scriptstyle j_B^*} & & \downarrow{\scriptstyle \kappa^*} \\ \widetilde{h}^n(A/C) & \xrightarrow{q_A^*} & \widetilde{h}^n(A) & \xrightarrow{j_A^*} & \widetilde{h}^n(C) & \xrightarrow{\delta_A} & \widetilde{h}^{n+1}(A/C) \end{array}$$

commute, with the outer verticals isomorphisms by Step 1. Define

$$\delta := q_X^* \circ (\kappa^*)^{-1} \circ \delta_A\colon \widetilde{h}^n(C) \to \widetilde{h}^{n+1}(X)$$

Step 3: Exactness at $\widetilde{h}^n(A) \oplus \widetilde{h}^n(B)$. That $\Psi\Phi = 0$ is the identity $i_A j_A = i_B j_B$ of Step 1. Conversely let $\Psi(a, b) = 0$, so $j_A^* a = j_B^* b =: c$. The ladder gives $\kappa^* \delta_X(b) = \delta_A(j_B^* b) = \delta_A(j_A^* a) = 0$, the last equality by exactness of the first sequence at $\widetilde{h}^n(C)$. As κ^* is injective, $\delta_X(b) = 0$, so exactness of the second sequence at $\widetilde{h}^n(B)$ yields $x \in \widetilde{h}^n(X)$ with $i_B^* x = b$. Then

$$j_A^*(i_A^* x - a) = (i_A j_A)^* x - c = (i_B j_B)^* x - c = j_B^* b - c = 0$$

so exactness of the first sequence at $\widetilde{h}^n(A)$ gives $w \in \widetilde{h}^n(A/C)$ with $q_A^* w = i_A^* x - a$. Put $x' := x - q_X^*\big((\kappa^*)^{-1} w\big)$. Using $q_X i_A = \kappa q_A$ from Step 1,

$$i_A^* x' = i_A^* x - q_A^* \kappa^* (\kappa^*)^{-1} w = i_A^* x - q_A^* w = a$$

and $i_B^* x' = i_B^* x = b$ because $(q_X i_B)^* = 0$. Hence $\Phi(x') = (a, b)$.

Step 4: Exactness at $\widetilde{h}^n(C)$. From the ladder, $(\kappa^*)^{-1} \delta_A j_B^* = \delta_X$, so

$$\delta\Psi(a, b) = q_X^* (\kappa^*)^{-1} \delta_A j_A^* a - q_X^* \delta_X b = 0 - 0 = 0$$

the first term vanishing by exactness of the first sequence at $\widetilde{h}^n(C)$ and the second by exactness of the second at $\widetilde{h}^{n+1}(X/B)$. Conversely suppose $\delta(c) = 0$ and set $u := (\kappa^*)^{-1} \delta_A(c)$. Then $q_X^* u = 0$, so $u = \delta_X(b)$ for some $b \in \widetilde{h}^n(B)$, whence $\delta_A(c) = \kappa^* \delta_X(b) = \delta_A(j_B^* b)$ and $\delta_A(c - j_B^* b) = 0$. Exactness of the first sequence at $\widetilde{h}^n(C)$ produces $a \in \widetilde{h}^n(A)$ with $j_A^* a = c - j_B^* b$, and then $\Psi(a, -b) = j_A^* a + j_B^* b = c$.

Step 5: Exactness at $\widetilde{h}^{n+1}(X)$. Both components of $\Phi\delta$ vanish: $i_B^* \delta = (q_X i_B)^*(\cdots) = 0$, and $i_A^* \delta(c) = q_A^* \kappa^* (\kappa^*)^{-1} \delta_A c = q_A^* \delta_A c = 0$ by exactness of the first sequence at $\widetilde{h}^{n+1}(A/C)$. Conversely let $\Phi(x) = 0$. From $i_B^* x = 0$ and exactness of the second sequence, $x = q_X^* u$ for some $u \in \widetilde{h}^{n+1}(X/B)$. From $i_A^* x = 0$ and $q_X i_A = \kappa q_A$, $q_A^*(\kappa^* u) = 0$, so exactness of the first sequence at $\widetilde{h}^{n+1}(A/C)$ gives $c \in \widetilde{h}^n(C)$ with $\delta_A(c) = \kappa^* u$. Then $\delta(c) = q_X^* (\kappa^*)^{-1} \kappa^* u = q_X^* u = x$.

Naturality in the triad follows from naturality of proposition 8.2.3 and of κ, completing the proof.

The second ingredient replaces an increasing union by a space built out of the stages one cylinder at a time, so that the union can be cut into two pieces and proposition 8.2.8 applied. The construction

is the mapping telescope, used informally in the proof of theorem 8.3.9 below and recorded here.

Definition 8.2.9: Mapping Telescope

Let $X_0 \xrightarrow{f_0} X_1 \xrightarrow{f_1} X_2 \to \cdots$ be a sequence of based CW complexes and based cellular maps, and let M_m denote the mapping cylinder of f_m, that is

$$M_m := \big(X_m \times I \sqcup X_{m+1}\big)/\big((x,1) \sim f_m(x)\big)$$

so that M_m contains $X_m = X_m \times \{0\}$ at its near end and X_{m+1} at its far end, and deformation retracts onto the far end by sliding the cylinder coordinate. The *mapping telescope* is

$$\mathrm{Tel}(X_\bullet) := \bigcup_{m \geq 0} M_m$$

the union taken by identifying the copy of X_{m+1} at the far end of M_m with the copy at the near end of M_{m+1}, with the basepoint ray collapsed. It is a based CW complex, and $M_m \cap M_{m+1} = X_{m+1}$.

Proposition 8.2.10: The Telescope of an Increasing Union

Let X be a based CW complex written as an increasing union $X = \bigcup_{m \geq 0} X_m$ of based subcomplexes, and let $\mathrm{Tel}(X_\bullet)$ be the telescope of the inclusions. Then the map $p\colon \mathrm{Tel}(X_\bullet) \to X$ collapsing each cylinder coordinate is a based homotopy equivalence.

***Proof*:**
Both spaces are CW complexes, so by theorem 5.3.6 it suffices to show p is a weak homotopy equivalence. Fix $q \geq 0$. A map $S^q \to X$ has compact image, hence meets only finitely many cells (lemma 1.2.4) and lands in some X_m. Likewise a homotopy $S^q \times I \to X$ lands in some X_m. Therefore $\pi_q(X) = \varinjlim_m \pi_q(X_m)$. The same argument applies to $\mathrm{Tel}(X_\bullet)$, whose finite subtelescope $\bigcup_{m \leq M} M_m$ deformation retracts onto X_{M+1} by sliding along the cylinder coordinates, so $\pi_q(\mathrm{Tel}(X_\bullet)) = \varinjlim_m \pi_q(X_m)$ as well. Under these identifications p_* is the identity of the colimit, since on the copy of X_m sitting in the telescope, p restricts to the inclusion $X_m \hookrightarrow X$. Hence p_* is an isomorphism on π_q for every q and every basepoint, the argument being the same at each since every point lies in a finite subtelescope, so p is a weak homotopy equivalence and theorem 5.3.6 makes it a homotopy equivalence.

One upgrade remains, because theorem 5.3.6 as stated concludes only that p is an *unbased* homotopy equivalence, while $\widetilde{h}^*$ is a functor on based complexes (definition 8.2.1). Its second clause gives the based form. Let M_p be the reduced mapping cylinder of the cellular map p, so that $\mathrm{Tel}(X_\bullet)$ and X both sit in M_p as subcomplexes, the retraction $r\colon M_p \to X$ is a based deformation retraction by construction, and $p = r \circ \iota$ for the inclusion $\iota\colon \mathrm{Tel}(X_\bullet) \hookrightarrow M_p$. Since r is a weak equivalence and p is one by the previous paragraph, so is ι. The map ι is the inclusion of a subcomplex, so the second clause of theorem 5.3.6 makes $\mathrm{Tel}(X_\bullet)$ a deformation retract of M_p by a homotopy performed rel $\mathrm{Tel}(X_\bullet)$, which is therefore based. Both factors of $p = r \circ \iota$ are based homotopy equivalences, hence so is p, as we sought to show.

Cutting the telescope into its even and odd cylinders and feeding the pieces to proposition 8.2.8

now produces the sequence the uniqueness theorem needs. The point to watch in the proof is that the Mayer-Vietoris map, once additivity has identified the two middle terms, turns out to be the map whose kernel and cokernel define $\varprojlim$ and $\varprojlim^1$ in [13, The Limit and $\varprojlim^1$ of a Tower], up to a sign on alternate coordinates that changes neither. Nothing is transported by analogy: it is that homomorphism, not an analogue of it.

Theorem 8.2.11: The Milnor Exact Sequence

Let $\widetilde{h}^*$ be a reduced cohomology theory and let X be a based CW complex written as an increasing union $X = \bigcup_{m\geq 0} X_m$ of based subcomplexes. For each n the groups $\widetilde{h}^n(X_m)$ form a tower under the restriction maps, and there is a short exact sequence

$$0 \to \varprojlim{}^1_m \, \widetilde{h}^{n-1}(X_m) \to \widetilde{h}^n(X) \to \varprojlim_m \widetilde{h}^n(X_m) \to 0$$

natural in the filtered complex, where $\varprojlim$ and $\varprojlim^1$ are as in [13, The Limit and $\varprojlim^1$ of a Tower]. The middle map is restriction.

The hypothesis that the X_m be subcomplexes of a single X is used only to identify $\widetilde{h}^*(X)$ with $\widetilde{h}^*(\mathrm{Tel}(X_\bullet))$. For an arbitrary sequence $X_0 \xrightarrow{f_0} X_1 \to \cdots$ of based CW complexes and based cellular maps, the same conclusion holds verbatim with $\mathrm{Tel}(X_\bullet)$ in place of X and the maps f_m^* as the transition maps of the tower.

***Proof*:**

Step 1: Replace X by the telescope. By proposition 8.2.10 the collapsing map $\mathrm{Tel}(X_\bullet) \to X$ is a based homotopy equivalence, so by the homotopy axiom it induces isomorphisms $\widetilde{h}^n(X) \cong \widetilde{h}^n(\mathrm{Tel}(X_\bullet))$ compatible with restriction to each X_m. Write $T := \mathrm{Tel}(X_\bullet)$ and work with T.

Step 2: The even-odd decomposition. Let A be the union of the cylinders M_m with m even and B the union of those with m odd. Then $A \cup B = T$, and since M_m and $M_{m'}$ meet only in the basepoint whenever $|m - m'| \geq 2$, the even cylinders are pairwise disjoint away from the basepoint, and so are the odd ones. That is what makes A and B wedges rather than disjoint unions, which is the form the additivity axiom of definition 8.2.1 requires. Each M_m deformation retracts onto the copy of X_{m+1} at its far end, so

$$A \simeq \bigvee_{m \text{ even}} X_{m+1}, \qquad B \simeq \bigvee_{m \text{ odd}} X_{m+1}$$

as based spaces, and $C := A \cap B = \bigvee_{m\geq 1} X_m$, since X_m for $m \geq 1$ lies in exactly the two cylinders M_{m-1} and M_m, one of each parity, while X_0 lies only in M_0.

Step 3: Additivity identifies the terms. The additivity axiom of definition 8.2.1 turns each wedge into a product. The odd values $m+1$ arising from A and the even ones arising from B together run over every $m \geq 1$ exactly once, so

$$\widetilde{h}^n(A) \oplus \widetilde{h}^n(B) \cong \prod_{m\geq 1} \widetilde{h}^n(X_m), \qquad \widetilde{h}^n(C) \cong \prod_{m\geq 1} \widetilde{h}^n(X_m)$$

Step 4: The Mayer-Vietoris map is the map defining $\varprojlim$ and $\varprojlim^1$. Fix $m \geq 1$ and compute the two restrictions of proposition 8.2.8 to the summand X_m of C. The subcomplex X_m sits at

the far end of M_{m-1}, and the deformation retraction of M_{m-1} onto that far end is the identity there, so restricting from the cylinder M_{m-1} is the identity of $\widetilde{h}^n(X_m)$. The same X_m sits at the near end of M_m, and the deformation retraction of M_m onto its far end X_{m+1} restricts on the near end to $f_m\colon X_m \to X_{m+1}$, so restricting from M_m is f_m^*. Under the identifications of Step 3 the Mayer-Vietoris map $j_A^* - j_B^*$ is therefore

$$\prod_{m\geq 1} \widetilde{h}^n(X_m) \to \prod_{m\geq 1} \widetilde{h}^n(X_m), \qquad (u_m)_m \longmapsto \big(u_m - f_m^*(u_{m+1})\big)_m$$

up to a sign depending on the parity of m. Which of the two cylinders M_{m-1} and M_m lies in A is decided by that parity: for m odd the identity contribution comes from A and the f_m^* contribution from B, giving $u_m - f_m^*(u_{m+1})$, while for m even the two are exchanged and the coordinate reads $f_m^*(u_{m+1}) - u_m$. So $j_A^* - j_B^*$ is $\varepsilon \circ d$ for the automorphism ε of $\prod_{m\geq 1} \widetilde{h}^n(X_m)$ negating the coordinates with m even. Post-composing with an automorphism of the target changes neither kernel nor cokernel, so $j_A^* - j_B^*$ and d have the same kernel and the same cokernel and the sign may be dropped.

The map d is precisely the one of [13, The Limit and $\varprojlim^1$ of a Tower] for the tower $\big(\widetilde{h}^n(X_m)\big)_{m\geq 1}$ with transition maps f_m^*, whose kernel is $\varprojlim_m \widetilde{h}^n(X_m)$ and whose cokernel is $\varprojlim_m^1 \widetilde{h}^n(X_m)$ by definition. Discarding the single term $m = 0$ changes neither: an element of the kernel is a coherent family, whose entry $u_0 = f_0^*(u_1)$ is determined by the rest, and the equation $u_0 - f_0^*(u_1) = b_0$ can be solved for u_0 whatever u_1 is, so no cokernel is contributed in that coordinate either.

Step 5: Read off the short exact sequence. Proposition 8.2.8 for the triad $(T; A, B)$ gives the exact sequence

$$\widetilde{h}^{n-1}(A) \oplus \widetilde{h}^{n-1}(B) \xrightarrow{d} \widetilde{h}^{n-1}(C) \xrightarrow{\delta} \widetilde{h}^n(T) \to \widetilde{h}^n(A) \oplus \widetilde{h}^n(B) \xrightarrow{d} \widetilde{h}^n(C)$$

Three exactness statements are used, one at each of the three displayed terms. Exactness at $\widetilde{h}^n(A) \oplus \widetilde{h}^n(B)$ says the image of $\widetilde{h}^n(T)$ there is $\ker d$ in degree n, so the right-hand map of the sequence below is onto its stated target. Exactness at $\widetilde{h}^{n-1}(C)$ says $\ker \delta$ is the image of d in degree $n-1$, so δ factors through an injection of $\operatorname{coker} d$ in degree $n-1$ into $\widetilde{h}^n(T)$. Exactness at $\widetilde{h}^n(T)$ itself says $\ker \Phi = \operatorname{im} \delta$, which is exactness in the middle of the spliced sequence. Together they give

$$0 \to \operatorname{coker}\big(d \text{ in degree } n-1\big) \to \widetilde{h}^n(T) \to \ker\big(d \text{ in degree } n\big) \to 0$$

which by Step 4 is the displayed sequence. The middle map is restriction because the identification of Step 3 was by restriction to the summands. Naturality follows from naturality of proposition 8.2.8 together with naturality of $\varprojlim$ and $\varprojlim^1$ in the tower, completing the proof.

Corollary 8.2.12: Mittag-Leffler Collapse of the Milnor Sequence

In the situation of theorem 8.2.11, if the tower $\big(\widetilde{h}^{n-1}(X_m)\big)_m$ satisfies the Mittag-Leffler condition, in particular if the restriction maps $\widetilde{h}^{n-1}(X_{m+1}) \to \widetilde{h}^{n-1}(X_m)$ are surjective, then restriction is an isomorphism

$$\widetilde{h}^n(X) \xrightarrow{\cong} \varprojlim_m \widetilde{h}^n(X_m)$$

***Proof*:**

Mittag-Leffler forces $\varprojlim^1_m \widetilde{h}^{n-1}(X_m) = 0$ by [13, Mittag-Leffler vanishing], and a surjective tower satisfies the condition. The left-hand term of theorem 8.2.11 vanishes, leaving the restriction map an isomorphism, as we sought to show.

The sequence is also natural in the *theory*, which is the form theorem 8.2.14 uses. By a transformation $\tau\colon \widetilde{h}^* \to \widetilde{k}^*$ of reduced theories is meant a family of homomorphisms commuting with all induced maps and with the suspension isomorphisms σ. Such a τ carries the Milnor sequence of $\widetilde{h}^*$ to that of $\widetilde{k}^*$, because every map appearing in the construction is of one of three kinds: induced by a map of spaces, hence commuting with τ by definition; a connecting map of proposition 8.2.3, which is built from σ and the Puppe collapse and so commutes with τ for the same two reasons; or an additivity identification, which is induced by the wedge inclusions. The Mayer-Vietoris connecting map $\delta = q_X^*(\kappa^*)^{-1}\delta_A$ of proposition 8.2.8 is a composite of maps of the first two kinds, so it commutes with τ as well.

The corollary is the form in which the sequence is almost always applied, and it explains why the $\varprojlim^1$ term is easy to forget: for the skeletal filtration of a CW complex and an ordinary theory, the restrictions are usually surjective and the naive answer is correct. It is not always correct, and a single example settles the matter.

Example 8.2.13: A Telescope With a Nonvanishing $\varprojlim^1$

Fix a prime p and let T be the telescope of

$$S^1 \xrightarrow{\;p\;} S^1 \xrightarrow{\;p\;} S^1 \to \cdots$$

where p denotes a degree-p map. These maps are not inclusions of subcomplexes, so it is the second form of theorem 8.2.11 that applies, computing $\widetilde{h}^*(T)$ directly. Take $\widetilde{h}^* = \widetilde{H}^*(-;\mathbb{Z})$, which is an ordinary theory satisfying additivity. A degree-p map induces multiplication by p on $\widetilde{H}^1(S^1;\mathbb{Z}) = \mathbb{Z}$ and the zero map on $\widetilde{H}^0(S^1;\mathbb{Z}) = 0$, so the two towers are

$$\big(\widetilde{H}^1(S^1)\big)_m = \big(\mathbb{Z} \xleftarrow{\;p\;} \mathbb{Z} \xleftarrow{\;p\;} \cdots\big), \qquad \big(\widetilde{H}^0(S^1)\big)_m = 0$$

The first is the tower computed in [13, A Tower With Nonvanishing $\varprojlim^1$], where its limit is 0 and its $\varprojlim^1$ is $\mathbb{Z}_p/\mathbb{Z}$. Theorem 8.2.11 in degree $n = 1$ therefore gives $\widetilde{H}^1(T;\mathbb{Z}) \cong \varprojlim_m \mathbb{Z} = 0$, and in degree $n = 2$ it gives

$$0 \to \varprojlim\nolimits^1_m \widetilde{H}^1(S^1;\mathbb{Z}) \to \widetilde{H}^2(T;\mathbb{Z}) \to \varprojlim_m \widetilde{H}^2(S^1;\mathbb{Z}) = 0 \to 0$$

so that

$$\widetilde{H}^2(T;\mathbb{Z}) \cong \mathbb{Z}_p/\mathbb{Z}$$

The telescope is a two-dimensional complex whose second cohomology is an uncountable group, all of it contributed by the $\varprojlim^1$ term. The answer can be checked independently: $H_1(T;\mathbb{Z}) = \varinjlim(\mathbb{Z} \xrightarrow{p} \mathbb{Z} \to \cdots) = \mathbb{Z}[1/p]$, Singular homology commutes with the telescope colimit, by the same compactness argument as in proposition 8.2.10, so also $H_2(T;\mathbb{Z}) = \varinjlim H_2(S^1;\mathbb{Z}) = 0$. The universal coefficient theorem (theorem 4.1.8) gives the short exact sequence $0 \to \operatorname{Ext}^1_{\mathbb{Z}}(H_1(T;\mathbb{Z}),\mathbb{Z}) \to H^2(T;\mathbb{Z}) \to \operatorname{Hom}(H_2(T;\mathbb{Z}),\mathbb{Z}) \to 0$, whose right-hand term therefore vanishes, leaving $H^2(T;\mathbb{Z}) \cong \operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}[1/p],\mathbb{Z})$, which is $\mathbb{Z}_p/\mathbb{Z}$. The two computations agree.

A natural transformation of cohomology theories is the structure that the theorem compares. Two cohomology theories h^*, k^* on CW pairs being given, a *natural transformation* $\tau\colon h^* \to k^*$ is a family of homomorphisms $\tau_{(X,A)}\colon h^n(X,A) \to k^n(X,A)$, one in each degree, commuting with all induced maps f^* and with the coboundary δ. The condition that τ commute with δ is what allows the five lemma to be applied to the ladder of long exact sequences, and it is the crux of the argument.

The motivating question is whether the value of a theory on a point determines it everywhere. For ordinary theories the answer is yes, and the proof is an induction up the skeleta of a CW complex, using the dimension axiom to compute the relative groups of each skeletal pair and the five lemma to climb from one skeleton to the next.

Theorem 8.2.14: Uniqueness of Ordinary Cohomology

Let h^* and k^* be ordinary cohomology theories on the category of CW pairs (definitions 8.2.4 and 8.2.6), and let $\tau\colon h^* \to k^*$ be a natural transformation commuting with the coboundary maps. If τ induces an isomorphism on coefficients,

$$\tau\colon h^n(\mathrm{pt}) \xrightarrow{\cong} k^n(\mathrm{pt}) \quad \text{for all } n$$

(equivalently, an isomorphism $h^0(\mathrm{pt}) \cong k^0(\mathrm{pt})$, the only possibly nonzero degree), then $\tau_{(X,A)}\colon h^n(X,A) \to k^n(X,A)$ is an isomorphism for every CW pair (X,A) and every n. For a CW pair (X,A) with X a finite complex the same conclusion holds when h^* and k^* satisfy the homotopy, exactness, excision and dimension axioms but not necessarily the additivity axiom.

***Proof*:**

Write $G := h^0(\mathrm{pt})$, so τ identifies the coefficients of the two theories. The proof reduces the general pair to a single space, climbs the skeleta of that space, and assembles the limit. Additivity enters twice: in Step 2, for a wedge of infinitely many spheres, and in Step 5, through the Milnor sequence. For a finite complex neither is needed, which gives the last clause of the theorem (Step 4).

Step 1: Reduction to absolute groups. For a CW pair (X,A) the two long exact sequences of definition 8.2.4, for h^* and for k^*, form a commutative ladder

$$\begin{array}{ccccccccc}
h^{n-1}(A) & \xrightarrow{\delta} & h^n(X,A) & \longrightarrow & h^n(X) & \longrightarrow & h^n(A) & \xrightarrow{\delta} & h^{n+1}(X,A) \\
\downarrow{\scriptstyle\tau} & & \downarrow{\scriptstyle\tau} & & \downarrow{\scriptstyle\tau} & & \downarrow{\scriptstyle\tau} & & \downarrow{\scriptstyle\tau} \\
k^{n-1}(A) & \xrightarrow{\delta} & k^n(X,A) & \longrightarrow & k^n(X) & \longrightarrow & k^n(A) & \xrightarrow{\delta} & k^{n+1}(X,A)
\end{array}$$

whose rows are exact and whose squares commute (the δ-squares by hypothesis on τ, the others by naturality). If τ is an isomorphism on the absolute groups $h^*(X) \to k^*(X)$ and $h^*(A) \to k^*(A)$ for all spaces, the five lemma forces τ to be an isomorphism on $h^n(X,A) \to k^n(X,A)$. It therefore suffices to prove τ is an isomorphism on absolute groups $h^*(X) \to k^*(X)$ for every CW complex X, and for every finite one in the last clause, where A is then finite as well.

Step 2: Spheres and wedges of spheres. On a point, $\tau\colon h^n(\mathrm{pt}) \to k^n(\mathrm{pt})$ is an isomorphism by hypothesis. For the sphere S^m, the dimension axiom gives, via the reduced theory of definition 8.2.1 and the iterated suspension isomorphism $\widetilde{h}^n(S^m) \cong \widetilde{h}^{n-m}(S^0) \cong h^{n-m}(\mathrm{pt})$, that $\widetilde{h}^n(S^m) = G$ for $n = m$ and 0 otherwise, and likewise for k. The transformation τ commutes

with the suspension isomorphism (which it does because, by proposition 8.2.5, suspension is built from δ and excision, with both of which τ commutes), so on $\widetilde{h}^*(S^m)$ it is the iterate of $\tau\colon h^*(\mathrm{pt}) \to k^*(\mathrm{pt})$, hence an isomorphism. For a wedge $\bigvee_\alpha S^m_\alpha$ of m-spheres, additivity gives $\widetilde{h}^n(\bigvee_\alpha S^m_\alpha) \cong \prod_\alpha \widetilde{h}^n(S^m_\alpha)$ compatibly under τ, so τ is an isomorphism on any wedge of equidimensional spheres. For a finite wedge the additivity axiom is not needed. For based CW complexes A and B, let i_A, i_B be the inclusions into $A \vee B$, $r_A\colon A \vee B \to A$ the retraction collapsing B, and $q\colon A \vee B \to (A \vee B)/A = B$ the quotient. Exactness gives $\ker i_A^* = \mathrm{im} q^*$, the retraction makes i_A^* split surjective, and $q \circ i_B = \mathrm{id}_B$ makes q^* split injective with $i_B^* q^* = \mathrm{id}$. Hence $(i_A^*, i_B^*)\colon \widetilde{h}^n(A \vee B) \to \widetilde{h}^n(A) \oplus \widetilde{h}^n(B)$ is an isomorphism: an element x killed by both is $x = q^* y$, and then $y = i_B^* q^* y = i_B^* x = 0$, while (a, b) is the image of $r_A^* a + q^*(b - i_B^* r_A^* a)$, since $i_A^* q^* = 0$, the map $q \circ i_A$ factoring through a point, where the reduced theory vanishes. Induction on the number of summands gives every finite wedge, compatibly under τ, from the homotopy, exactness and excision axioms alone, excision entering through the reduced exact sequence of $(A \vee B, A)$ (proposition 8.2.5).

Step 3: The skeletal induction. Let X be a CW complex and X^m its m-skeleton. The quotient X^m/X^{m-1} is a wedge of m-spheres, one for each m-cell, so by excision and proposition 8.2.5,

$$h^n(X^m, X^{m-1}) \cong \widetilde{h}^n(X^m/X^{m-1}) \cong \widetilde{h}^n\Big(\bigvee_\alpha S^m_\alpha\Big)$$

and τ is an isomorphism on these relative groups by Step 2. The claim is that τ is an isomorphism on $h^*(X^m)$ for every m, proved by induction on m. The base case $m = 0$ is a wedge of 0-spheres (a discrete set with a basepoint), handled by Step 2. For the inductive step, the ladder of Step 1 applied to the pair (X^m, X^{m-1}) reads

$$\begin{array}{ccccccc} h^n(X^m, X^{m-1}) & \longrightarrow & h^n(X^m) & \longrightarrow & h^n(X^{m-1}) & \xrightarrow{\delta} & h^{n+1}(X^m, X^{m-1}) \\ \downarrow\cong & & \downarrow\tau & & \downarrow\cong & & \downarrow\cong \\ k^n(X^m, X^{m-1}) & \longrightarrow & k^n(X^m) & \longrightarrow & k^n(X^{m-1}) & \xrightarrow{\delta} & k^{n+1}(X^m, X^{m-1}) \end{array}$$

To apply the five lemma to the term $h^n(X^m)$ one uses the five-term window

$$h^{n-1}(X^{m-1}) \xrightarrow{\delta} h^n(X^m, X^{m-1}) \to h^n(X^m) \to h^n(X^{m-1}) \xrightarrow{\delta} h^{n+1}(X^m, X^{m-1})$$

of the pair sequence, the right half of which is shown in the diagram. The two relative terms $h^n(X^m, X^{m-1})$ and $h^{n+1}(X^m, X^{m-1})$ carry isomorphisms by Step 2, and the two absolute terms $h^{n-1}(X^{m-1})$ and $h^n(X^{m-1})$ carry isomorphisms by the inductive hypothesis. With those four flanking maps isomorphisms, the five lemma forces the middle map $\tau\colon h^n(X^m) \to k^n(X^m)$ to be an isomorphism, completing the induction on m.

Step 4: Finite-dimensional and finite complexes. If X is finite-dimensional, say $\dim X = N$, then $X = X^N$ and Step 3 already gives that $\tau\colon h^n(X) \to k^n(X)$ is an isomorphism for all n. If X is moreover a finite complex, each X^m/X^{m-1} is a finite wedge of spheres, so Steps 2 and 3 use only the homotopy, exactness, excision and dimension axioms. Feeding this back into Step 1 for a pair (X, A) with X finite shows τ is an isomorphism on $h^n(X, A)$ without additivity, which is the last clause of the theorem.

Step 5: Infinite-dimensional complexes via additivity. For general X the skeleta $X^0 \subseteq X^1 \subseteq \cdots$ exhaust X, and $h^n(X)$ is related to the inverse system $\{h^n(X^m)\}_m$ by the Milnor exact sequence

$$0 \to \varprojlim{}^1_m\, h^{n-1}(X^m) \to h^n(X) \to \varprojlim_m h^n(X^m) \to 0$$

which is theorem 8.2.11, proved above from the additivity axiom by way of the mapping telescope of the skeleta. The transformation τ maps the Milnor sequence for h^* to that for k^*, compatibly, by the naturality in the theory established after that theorem (applied to the reduced theories associated with h^* and k^* under proposition 8.2.5). By Step 3, τ is an isomorphism on each $h^n(X^m) \to k^n(X^m)$, so it is an isomorphism of towers and hence induces isomorphisms on both the $\varprojlim$ terms and the $\varprojlim^1$ terms of the two sequences, these being functors of the tower. The five lemma applied to the map of Milnor short exact sequences then gives that $\tau\colon h^n(X) \to k^n(X)$ is an isomorphism. Step 1 upgrades this to arbitrary pairs, completing the proof.

The theorem is the foundation of the entire subject's right to speak of "the" ordinary cohomology with coefficients in an abelian group. Any two constructions producing ordinary cohomology theories with isomorphic coefficient groups are isomorphic as theories, provided a natural transformation between them exists carrying one coefficient group to the other. This is exactly the situation for singular, cellular, simplicial, de Rham, and constant-coefficient sheaf cohomology: each is an ordinary theory, the comparison maps between them are natural transformations commuting with coboundary, and on coefficients they all give the identity of G, so all the comparison maps are isomorphisms. The de Rham theorem (singular cohomology with $\mathbb{R}$ coefficients agrees with de Rham cohomology) and the agreement of simplicial with singular homology, proved by separate arguments in their home contexts, are both instances of one uniqueness statement. The dimension axiom is doing essential work: it is what makes $h^n(\mathrm{pt})$ a single group, which is what makes "the coefficient group" an unambiguous datum.

8.2.15 Warning: Naturality of the comparison map is not automatic The uniqueness theorem compares two theories *through a given natural transformation*. It does not assert that any two ordinary theories with isomorphic coefficients are isomorphic in the absence of a comparison map. The hypothesis that τ exist and commute with δ is genuine content. For singular versus cellular cohomology the comparison map is the one induced by the cellular-to-singular chain inclusion. For singular versus de Rham it is integration of forms over chains. Each must be exhibited and checked to commute with coboundary before uniqueness applies. What the theorem removes is the need to check the comparison is an isomorphism cell by cell: once it is the identity on coefficients, it is an isomorphism everywhere.

Dropping the dimension axiom is what produces the theories the chapter is named for. The coefficient groups $h^n(\mathrm{pt})$ of a generalized theory are no longer concentrated in a single degree, and the uniqueness theorem fails: there can be many non-isomorphic generalized theories, and a single theory carries genuine information already on a point. The two prototypes, developed later in the chapter, illustrate how far the coefficients can spread.

Definition 8.2.16: Generalized Cohomology Theory

A *generalized cohomology theory* is a cohomology theory in the sense of definition 8.2.4 (equivalently a reduced theory in the sense of definition 8.2.1) that is *not* required to satisfy the dimension axiom of definition 8.2.6. Its coefficient groups $h^n(\mathrm{pt})$ may be nonzero for infinitely many n. A generalized homology theory is defined dually. When the coefficients are concentrated in degree 0 the theory is ordinary. Otherwise it is genuinely extraordinary, and the graded group $h^*(\mathrm{pt})$ is the first invariant distinguishing it from ordinary cohomology.

The definition is a deliberate weakening: every axiom of an ordinary theory is retained except

the dimension axiom, so a generalized theory still has long exact sequences, excision, suspension isomorphisms, and additivity, and the entire formalism of the chapter (Mayer-Vietoris, the spectral sequences of chapter 6 in their generalized form, representability) applies verbatim. What is lost is the rigidity: the coefficients are now a graded group, the theory is no longer pinned down by a single abelian group, and the cohomology of a point already records nontrivial information. The two examples that follow are the reason the definition is made, and they are the subject of section 8.5.

Example 8.2.17: The Coefficients of K-theory and Cobordism

Complex topological K-theory KU^*, the generalized cohomology theory of the Ω-spectrum KU of theorem 8.5.10, whose degree-zero group on a finite CW complex X is $[X, \mathbb{Z} \times BU]$, in bijection with the virtual vector bundles of definition 8.5.1 by theorem 8.5.3, and developed further in *Vector Bundles and K-Theory* [33], has coefficient groups given by Bott periodicity:

$$KU^n(\mathrm{pt}) = \begin{cases} \mathbb{Z} & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$$

so $KU^*(\mathrm{pt}) \cong \mathbb{Z}[\beta, \beta^{-1}]$ with β the Bott class in degree -2. This is nonzero in infinitely many degrees, so KU is genuinely extraordinary: the dimension axiom fails in every even nonzero degree. Unoriented cobordism MO^*, the theory built from the Thom spectrum of definition 8.5.13, has coefficients

$$MO^*(\mathrm{pt}) \cong \mathbb{F}_2[x_i : i \geq 2,\ i \neq 2^j - 1]$$

a polynomial algebra over $\mathbb{F}_2$ on generators x_i in each degree $i \geq 2$ not of the form $2^j - 1$ (Thom's computation, carried out in section 8.5), again nonzero in infinitely many degrees. A point, which in ordinary cohomology carries a single copy of G, carries in K-theory a copy of $\mathbb{Z}$ in every even degree and in cobordism an entire polynomial ring. The non-vanishing of these coefficients in many degrees is exactly the slack the dimension axiom forbids, and it is what gives the generalized theories their additional discriminating power: a sphere S^m, which ordinary cohomology sees only in degrees 0 and m, is detected by K-theory in every degree of the same parity.

The example exhibits the dimension axiom as the precise dividing line. With it, the coefficients collapse to one group and the theory is rigid and unique. Without it, the coefficients can be an arbitrary graded ring and the resulting theories proliferate, each carrying geometry that ordinary cohomology cannot see (vector bundles for K-theory, manifolds up to cobordism for MO). The remaining sections of the chapter explain how every such theory, ordinary or generalized, is represented by a sequence of spaces and how that sequence is the object called a spectrum. K-theory and cobordism are the two extraordinary spectra whose computations close the chapter.

Exercise 8.2.1

1. Let h^* be a reduced cohomology theory in the sense of definition 8.2.1. Using only the axioms, compute $\widetilde{h}^n(S^m)$ for all $m \geq 0$ and all n in terms of the coefficient groups $h^*(\mathrm{pt}) = \widetilde{h}^*(S^0)$. Then specialize to an ordinary theory with $h^0(\mathrm{pt}) = G$ and recover the familiar answer $\widetilde{h}^n(S^m) = G$ for $n = m$ and 0 otherwise.

2. Proposition 8.2.8 gives a Mayer-Vietoris sequence for a *reduced* theory. Deduce the unreduced version from it: for an unreduced theory h^* and a CW triad $X = A \cup B$ with $C = A \cap B$ nonempty, there is a long exact sequence $\cdots \to h^n(X) \to h^n(A) \oplus h^n(B) \to h^n(C) \to h^{n+1}(X) \to \cdots$. (Use proposition 8.2.5 and note that adjoining a disjoint basepoint takes the

triad to a based triad.)

3. Verify that singular *homology* $H_*(-;G)$ satisfies the homology version of the Eilenberg-Steenrod axioms of definition 8.2.4, and identify its coefficient groups. State which axiom would fail if one used compactly supported homology (Borel-Moore homology) on non-compact spaces, and why this prevents Borel-Moore homology from being a homology theory in the unrestricted sense.

4. The reduced cohomology of a point. Show that for *any* reduced cohomology theory $\widetilde{h}^*$ one has $\widetilde{h}^n(\mathrm{pt}) = 0$ for all n (where pt is the one-point space, which as a based space is its own basepoint). Explain why this is consistent with the coefficient groups $h^n(\mathrm{pt})$ of the *unreduced* theory being nonzero, by exhibiting the relationship between the two.

5. Let $\tau\colon h^* \to k^*$ be a natural transformation of ordinary cohomology theories that is an isomorphism on coefficients. The uniqueness theorem theorem 8.2.14 concludes τ is an isomorphism on every CW pair. Show by a counterexample that the conclusion can fail for *generalized* theories: exhibit two generalized cohomology theories and a coefficient-preserving natural transformation between them that is not an isomorphism on some space. (Hint: take h^* = ordinary cohomology $H^*(-;\mathbb{Z})$ regarded as a generalized theory, $k^* = KU^*$, and the natural Chern-character-type comparison. Identify where the coefficient hypothesis fails.)

6. Prove that the suspension isomorphism of a reduced cohomology theory is compatible with the long exact sequence: for a based CW pair (X, A), the connecting map $\delta\colon \widetilde{h}^n(A) \to \widetilde{h}^{n+1}(X/A)$ of proposition 8.2.3 agrees, under the identification $\widetilde{h}^{n+1}(\Sigma A) \cong \widetilde{h}^n(A)$, with $(\partial)^*$ for the collapse $\partial\colon X/A \to \Sigma A$. Deduce that a natural transformation of theories commuting with δ automatically commutes with the suspension isomorphism.

7. Let X be a CW complex of finite dimension N, filtered by its skeleta. Show that the Milnor sequence of theorem 8.2.11 carries no information in that case: for each fixed level the images of the restriction maps are eventually constant, so $\varprojlim^1$ vanishes and restriction is an isomorphism. Which hypothesis of corollary 8.2.12 is being verified, and why does the weaker one suffice?

8. In example 8.2.13 replace the degree-p maps by degree-0 maps and recompute $\widetilde{H}^1(T;\mathbb{Z})$ and $\widetilde{H}^2(T;\mathbb{Z})$. Then replace them by degree-1 maps and recompute again. Exactly one of the three telescopes has a nonvanishing $\varprojlim^1$ term. Identify it, and say what distinguishes its tower from the other two.

9. Proposition 8.2.10 assumes every f_m is the inclusion of a subcomplex. Show the conclusion genuinely needs a hypothesis by exhibiting a sequence whose telescope is homotopy equivalent to no single X_m, and explain why this costs theorem 8.2.11 nothing.

8.3 Representable Cohomology and Brown Representability

The axioms of section 8.2 pin down what a cohomology theory *is*, but they say nothing about where it comes from. The representation theorem for ordinary cohomology (theorem 5.5.12) gave one answer in a single case: the functor $\widetilde{H}^n(-;G)$ is the set of homotopy classes of maps into a fixed space $K(G,n)$. The aim of this section is to show that this is a structural feature of every cohomology theory in the sense of section 8.2. The vehicle is Brown's representability theorem, which characterises representable functors $[-,Y]$ by two formal properties, a wedge axiom and a Mayer-Vietoris axiom,

that every reduced cohomology theory possesses degree by degree.

The argument is constructive. Given a functor F satisfying the two axioms, the representing space Y is built by attaching cells to realise larger and larger pieces of F, exactly as $K(G,n)$ was built in section 5.5 by attaching cells to kill homotopy. What replaces the homotopy-group bookkeeping there is a single element $u \in F(Y)$, the *universal element*, chosen so that pullback along it, $[X,Y] \to F(X)$, $f \mapsto f^*u$, is a bijection. Whitehead's theorem (theorem 5.3.6) certifies that the natural transformation produced is an isomorphism. The payoff, recorded at the end, is that every reduced cohomology theory is represented in each degree by a space Y_n, and the suspension isomorphism assembles the Y_n into a single homotopy-theoretic object, an Ω-spectrum, the bridge to section 8.4.

Throughout, CW_* is the category of definition 8.1.1 and the bracket $[X,Y]$ abbreviates $[X,Y]_*$. A contravariant functor $F\colon \mathsf{CW}_*^{\mathrm{op}} \to \mathbf{Set}_*$ to pointed sets is the object of study. The requirement that F factor through the homotopy category is precisely homotopy invariance, $f \simeq g \implies F(f) = F(g)$.

Before isolating the abstract axioms, the prototype must be restated in the language of the previous section. Ordinary cohomology in a fixed degree is, as a functor on CW_*, exactly $[-, K(G,n)]$.

Theorem 8.3.1: Ordinary Cohomology as a Represented Theory

Let G be an abelian group and $n \geq 1$. On the category CW_* of pointed connected CW complexes there is a natural isomorphism of functors

$$\tilde{H}^n(-;G) \;\cong\; [-,\, K(G,n)]$$

sending a class $u \in \tilde{H}^n(X;G)$ to the homotopy class of the map $f_u\colon X \to K(G,n)$ with $f_u^*(\iota_n) = u$, where $\iota_n \in H^n(K(G,n);G)$ is the fundamental class of definition 5.5.10. Under this isomorphism the coefficient group is recovered as $\tilde{H}^n(S^n;G) \cong [S^n, K(G,n)] \cong \pi_n(K(G,n)) \cong G$. Moreover, let $\varepsilon_n\colon K(G,n) \to \Omega K(G,n+1)$ be the adjoint of a based map $s_n\colon \Sigma K(G,n) \to K(G,n+1)$ with $s_n^*(\iota_{n+1}) = \sigma(\iota_n)$, where $\sigma\colon \tilde{H}^n(X;G) \xrightarrow{\cong} \tilde{H}^{n+1}(\Sigma X;G)$ is the suspension isomorphism. Then for every pointed CW complex X whose basepoint is a 0-cell, the map $(\varepsilon_n)_*\colon [X,K(G,n)] \to [X,\Omega K(G,n+1)]$, $[f] \mapsto [\varepsilon_n \circ f]$, followed by the loop-suspension adjunction $[X,\Omega K(G,n+1)] \cong [\Sigma X, K(G,n+1)]$, corresponds under representability to σ. In particular $(\varepsilon_n)_*$ is a bijection for every such X, and ε_n is a weak homotopy equivalence.

***Proof*:**

The natural bijection $T\colon [X,K(G,n)] \to \tilde{H}^n(X;G)$, $[f] \mapsto f^*(\iota_n)$, is theorem 5.5.12. Its inverse is $u \mapsto [f_u]$, which is the displayed correspondence. Naturality in X is the commuting square recorded there. It remains to verify the two identifications.

Coefficient group. Taking $X = S^n$, the bijection T gives $[S^n, K(G,n)] \cong \tilde{H}^n(S^n;G)$. By definition of the higher homotopy groups (definition 5.1.1), $[S^n, K(G,n)] = \pi_n(K(G,n))$, which is G because $K(G,n)$ is an Eilenberg-MacLane space (definition 5.5.1). On the other side, $\tilde{H}^n(S^n;G) \cong G$ by the cellular computation of the sphere. The composite isomorphism is the coefficient identification.

Suspension. Let X be a pointed CW complex whose basepoint is a 0-cell. Then ΣX is a pointed CW complex of the same kind, and theorem 5.5.12 applies to both. Write T_n for its bijection in degree n, let $\sigma_X\colon \tilde{H}^n(X;G) \to \tilde{H}^{n+1}(\Sigma X;G)$ be the suspension isomorphism of

singular cohomology, natural in X (example 8.2.7 and Step 2 of proposition 8.2.5), and let $\alpha\colon [\Sigma X, K(G,n+1)] \to [X, \Omega K(G,n+1)]$ be the bijection of theorem 8.1.5, natural in X. The composite

$$\Phi_X = \alpha \circ T_{n+1}^{-1} \circ \sigma_X \circ T_n\colon [X, K(G,n)] \longrightarrow [X, \Omega K(G,n+1)]$$

is a bijection, natural in X because each factor is: a based map $g\colon X' \to X$ acts on the middle terms through Σg. Put $K = K(G,n)$. Then $T_n[\mathrm{id}_K] = \iota_n$, so $\Phi_K[\mathrm{id}_K] = \alpha[s_n] = [\varepsilon_n]$. For a based map $f\colon X \to K$, naturality along f gives

$$\Phi_X[f] = \Phi_X\big(f^*[\mathrm{id}_K]\big) = f^*\Phi_K[\mathrm{id}_K] = [\varepsilon_n \circ f]$$

so $\Phi_X = (\varepsilon_n)_*$, which is the asserted correspondence, and $(\varepsilon_n)_*$ is a bijection. Taking $X = S^q$ for $q \geq 1$ shows that ε_n induces bijections $\pi_q(K,*) \to \pi_q(\Omega K(G,n+1),*)$ at the basepoints. Both spaces are path-connected: K by definition 5.5.1, and $\Omega K(G,n+1)$ because its path components are the based homotopy classes of loops in $K(G,n+1)$, whose fundamental group vanishes as $n+1 \geq 2$. A path from the basepoint to any other point of K gives change-of-basepoint isomorphisms on both sides that commute with $(\varepsilon_n)_*$, so ε_n is bijective on every π_q at every basepoint, and it is a weak homotopy equivalence (definition 5.3.1), completing the proof.

The theorem reframes theorem 5.5.12 as the assertion that ordinary cohomology, viewed degree by degree, is a sequence of representable functors whose representing spaces are linked by the weak homotopy equivalences $\varepsilon_n\colon K(G,n) \to \Omega K(G,n+1)$. The coefficient group is read off from the sphere, and the suspension axiom of section 8.2 becomes a formal consequence of the loop-suspension adjunction. This is the template Brown's theorem generalises: any functor behaving formally like $\tilde{H}^n(-;G)$ is $[-,Y]$ for some Y, and any cohomology theory furnishes such a functor in each degree.

Example 8.3.2: Degree-One Cohomology and the Circle

For $G = \mathbb{Z}$ and $n = 1$ the representing space is $K(\mathbb{Z},1) = S^1$ (example 5.5.2), so theorem 8.3.1 reads

$$\tilde{H}^1(X;\mathbb{Z}) \;\cong\; [X, S^1]$$

for every pointed connected CW complex X. Both sides can be computed for $X = S^1$: the left is $\tilde{H}^1(S^1;\mathbb{Z}) \cong \mathbb{Z}$, and the right is $[S^1,S^1] = \pi_1(S^1) \cong \mathbb{Z}$, the isomorphism sending a map of degree d to d times the generator. For $X = S^1 \vee S^1$ the wedge of two circles, $\tilde{H}^1(S^1 \vee S^1;\mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}$, while $[S^1 \vee S^1, S^1] \cong [S^1,S^1] \times [S^1,S^1] \cong \mathbb{Z} \times \mathbb{Z}$ because a map out of a wedge is a pair of maps out of the summands agreeing at the basepoint. The two computations match, and the matching is exactly the wedge axiom that Brown's theorem will require in the abstract. The class in $\tilde{H}^1(X;\mathbb{Z})$ corresponding to $f\colon X \to S^1$ is $f^*(\iota_1)$, the pullback of the generator $\iota_1 \in H^1(S^1;\mathbb{Z})$, which records the winding of f around the circle.

The example exhibits the two formal properties on which everything turns. A map out of a wedge is determined by its restrictions to the summands, so F applied to a wedge is the product of the values on the summands. A class on a space glued from two pieces is determined by compatible classes on the pieces. These are the wedge and Mayer-Vietoris axioms. The strength of Brown's theorem is that they suffice: no further structure on F is needed to produce a representing space.

Definition 8.3.3: Brown Functor

A *Brown functor* is a contravariant functor $F\colon \mathsf{CW}_*^{\mathrm{op}} \to \mathbf{Set}_*$ from the homotopy category of pointed connected CW complexes to pointed sets satisfying the two axioms:

1. *(Wedge axiom)* For any family $\{X_\alpha\}_{\alpha\in A}$ of pointed connected CW complexes, the inclusions $X_\alpha \hookrightarrow \bigvee_\alpha X_\alpha$ induce a bijection
$$F\Big(\bigvee_\alpha X_\alpha\Big) \xrightarrow{\cong} \prod_\alpha F(X_\alpha)$$
2. *(Mayer-Vietoris axiom)* If a CW complex $X = A \cup B$ is the union of two subcomplexes with A, B, $A \cap B$ all containing the basepoint and connected, then for any classes $a \in F(A)$ and $b \in F(B)$ whose restrictions to $A \cap B$ agree, there is a class $x \in F(X)$ restricting to a on A and to b on B.

A Brown functor is a contravariant functor that turns disjoint assembly into product and admits gluing along intersections. The wedge axiom is the infinitary version of the statement that F sends finite wedges to finite products, and it forces F of a point to be a single point (the empty product), so the distinguished basepoint of $F(X)$ is the restriction of the unique class on the basepoint. The Mayer-Vietoris axiom is existence-only: it asserts that compatible classes glue, but says nothing about uniqueness of the glued class. This asymmetry is deliberate. Uniqueness fails for general F (the kernel of the difference of restrictions can be nontrivial), and Brown's construction is arranged so that the lack of uniqueness never obstructs the building of Y. Every reduced cohomology theory provides a Brown functor in each degree, as the verification below records, so the theorem applies to all of them at once.

Example 8.3.4: Reduced Cohomology is a Brown Functor

Fix an abelian group G and $n \geq 1$, and let $F = \tilde{H}^n(-;G)$. Then F is a Brown functor. Homotopy invariance makes F a functor on CW_*, and $F(X)$ is a pointed set (an abelian group, in fact) with basepoint 0.

Wedge axiom. For a wedge $\bigvee_\alpha X_\alpha$, reduced cohomology converts the wedge into a product: $\tilde{H}^n\big(\bigvee_\alpha X_\alpha; G\big) \cong \prod_\alpha \tilde{H}^n(X_\alpha; G)$. This is the additivity (wedge) axiom for ordinary cohomology, which holds because $\bigvee_\alpha X_\alpha$ is the union of the X_α along the single basepoint, and reduced cellular cochains of a wedge are the product of the reduced cellular cochains of the summands. The induced map is restriction to the summands, exactly the map in definition 8.3.3(1).

Mayer-Vietoris axiom. For $X = A \cup B$ with A, B subcomplexes, the reduced Mayer-Vietoris sequence reads
$$\tilde{H}^n(X;G) \xrightarrow{(\,\mathrm{res}_A,\,\mathrm{res}_B\,)} \tilde{H}^n(A;G) \oplus \tilde{H}^n(B;G) \xrightarrow{(a,b)\mapsto a|_{A\cap B} - b|_{A\cap B}} \tilde{H}^n(A\cap B;G)$$
Exactness at the middle term says precisely that a pair (a, b) with equal restrictions to $A \cap B$ lies in the image of $(\mathrm{res}_A, \mathrm{res}_B)$, that is, there is $x \in \tilde{H}^n(X;G)$ restricting to a and b. This is the gluing property of definition 8.3.3(2). Hence $\tilde{H}^n(-;G)$ satisfies both axioms.

The verification is the formal core of why Brown's theorem subsumes ordinary cohomology: the two axioms are exactly the wedge and Mayer-Vietoris properties, both standard for $\tilde{H}^n$. The same verification applies verbatim to any reduced cohomology theory of section 8.2, since the wedge axiom

is its additivity axiom and the Mayer-Vietoris sequence is a formal consequence of the exactness axiom, derived from the cofiber long exact sequence. The theorem now produces, from any Brown functor, a representing space.

The construction rests on a single notion. A class $u \in F(Y)$ determines, for every X, a map of pointed sets

$$T_u \colon [X, Y] \longrightarrow F(X), \qquad [f] \longmapsto F(f)(u) = f^*u$$

by pulling u back along f. This T_u is automatically a natural transformation, by functoriality of F. The goal is to choose Y and u so that T_u is an isomorphism. The pair (Y, u) achieving this is governed by a representability criterion phrased in terms of spheres.

Definition 8.3.5: Universal Element

Let F be a Brown functor. A pair (Y, u) with Y a pointed connected CW complex and $u \in F(Y)$ is *n-universal* if the map

$$T_u \colon [S^q, Y] \longrightarrow F(S^q), \qquad [f] \longmapsto f^*u$$

is a bijection for $0 < q < n$ and a surjection for $q = n$. It is *universal* (or ∞-universal) if $T_u \colon [S^q, Y] \to F(S^q)$ is a bijection for all $q \geq 1$. The class u is then a *universal element* for F.

A universal element packages the entire functor F, tested on spheres, into a single class on a single space. The restriction to spheres is the decisive simplification: rather than control T_u on every complex at once, one controls it on the cells S^q out of which complexes are built, and the wedge and Mayer-Vietoris axioms then propagate sphere-level control to all complexes. The condition for spheres is a condition on homotopy groups, $T_u \colon \pi_q(Y) \to F(S^q)$, which the cell-attachment machinery of section 5.5 is built to adjust. The next two lemmas carry out the adjustment: the first attaches cells to enlarge $[S^q, Y]$ until T_u surjects, the second attaches cells to shrink it until T_u injects, both without disturbing the lower spheres.

Example 8.3.6: The Fundamental Class as a Universal Element

For $F = \tilde{H}^n(-; G)$, the pair $(K(G, n), \iota_n)$ is a universal element, where $\iota_n \in H^n(K(G, n); G) = F(K(G, n))$ is the fundamental class of definition 5.5.10. The map $T_{\iota_n} \colon [S^q, K(G, n)] \to \tilde{H}^n(S^q; G)$ sends $[f]$ to $f^*(\iota_n)$, which is exactly the representability bijection T of theorem 5.5.12 restricted to spheres, hence a bijection for every $q \geq 1$. Concretely, in degree $q = n$ the source is $[S^n, K(G, n)] = \pi_n(K(G, n)) = G$ and the target is $\tilde{H}^n(S^n; G) = G$, and T_{ι_n} is the identity on G under these identifications: a map $f \colon S^n \to K(G, n)$ representing $a \in G = \pi_n(K(G, n))$ pulls ι_n back to the class $a \in \tilde{H}^n(S^n; G)$, by the tautological characterisation of ι_n. For $q \neq n$ both sides vanish, so T_{ι_n} is the bijection $0 \to 0$. Thus ι_n is the universal element that Brown's construction would produce for ordinary cohomology, recovered here directly from theorem 5.5.12. The abstract construction below reconstructs $K(G, n)$ without knowing it in advance.

Lemma 8.3.7: Construction of a Universal Element

For every Brown functor F there exists a pointed connected CW complex Y and a universal element $u \in F(Y)$.

***Proof*:**
The space is built as an increasing union $Y = \bigcup_n Y_n$, each Y_n carrying a class $u_n \in F(Y_n)$ making (Y_n, u_n) n-universal, with u_{n+1} restricting to u_n. The class u is then assembled from the u_n by the wedge axiom in its mapping-telescope form.

Step 1: Realising every element of F on a wedge of spheres. Begin by arranging surjectivity of T_u in every degree on a single space. For each $q \geq 1$ and each element $a \in F(S^q)$, take a sphere S^q_a (a copy of S^q indexed by a), and form the wedge

$$Y_1 \;=\; \bigvee_{q \geq 1} \bigvee_{a \in F(S^q)} S^q_a$$

By the wedge axiom (definition 8.3.3(1)), $F(Y_1) \cong \prod_{q,a} F(S^q_a) = \prod_{q,a} F(S^q)$, so there is a unique class $u_1 \in F(Y_1)$ whose restriction to the summand S^q_a is the element $a \in F(S^q)$. Then $T_{u_1} \colon [S^q, Y_1] \to F(S^q)$ is surjective for every $q \geq 1$: the inclusion $\iota_a \colon S^q_a \hookrightarrow Y_1$ of the summand indexed by a satisfies $T_{u_1}(\iota_a) = \iota_a^* u_1 = a$ by construction. Thus (Y_1, u_1) has T_{u_1} surjective on all spheres. In particular it is 1-universal.

Step 2: From n-universal to $(n+1)$-universal by attaching cells. Suppose (Y_n, u_n) is n-universal: T_{u_n} is bijective on S^q for $0 < q < n$ and surjective for $q = n$. The aim is to enlarge Y_n to Y_{n+1}, extending u_n to u_{n+1}, so that $T_{u_{n+1}}$ becomes injective on S^n (hence bijective there) and remains surjective on S^{n+1}, while the lower spheres are undisturbed. Injectivity on S^n is arranged by gluing in cylinders that identify maps not distinguished by T_{u_n}, and surjectivity on S^{n+1} by wedging on $(n+1)$-spheres. The cylinder device replaces the difference-of-maps argument that would require $F(S^n)$ to be a group, which it need not be for a general Brown functor.

Consider first injectivity. Two maps $f_0, f_1 \colon S^n \to Y_n$ with $f_0^* u_n = f_1^* u_n$ must become homotopic after enlargement. For each such pair, glue a cylinder $S^n \times I$ to Y_n along the two ends, attaching $S^n \times \{0\}$ via f_0 and $S^n \times \{1\}$ via f_1. Performing this for every pair (f_0, f_1) with $f_0^* u_n = f_1^* u_n$ produces

$$Y_n' \;=\; Y_n \cup_{\{(f_0, f_1)\}} \Big(\bigsqcup_{(f_0, f_1)} S^n \times I \Big)$$

Each glued cylinder is built from cells of dimension $n+1$ (the cylinder $S^n \times I$ adds an $(n+1)$-cell to the spheres at its ends, which already map into Y_n), so Y_n' is obtained from Y_n by attaching cells of dimension $n+1$. The class u_n extends over Y_n': apply the Mayer-Vietoris axiom to $Y_n' = Y_n \cup C$, where C is the disjoint union of cylinders, glued along $A \cap B = \bigsqcup S^n \times \{0, 1\}$. On a cylinder $S^n \times I$ the projection to S^n followed by f_0 and the constancy of T_{u_n} on the two ends ($f_0^* u_n = f_1^* u_n$) make the restriction of u_n to the two boundary spheres equal. Hence the class $\mathrm{pr}^*(f_0^* u_n)$ on the cylinder agrees with u_n on the overlap, and the two glue to a class $u_n' \in F(Y_n')$ restricting to u_n on Y_n. In Y_n' the cylinder is a homotopy from f_0 to f_1, so $f_0 \simeq f_1$ there. Thus every pair undistinguished by T_{u_n} becomes a single homotopy class.

Next surjectivity on S^{n+1}. Wedge on a sphere S^{n+1}_b for each element $b \in F(S^{n+1})$ and use the wedge axiom to extend u_n' to a class u_{n+1} on

$$Y_{n+1} \;=\; Y_n' \vee \bigvee_{b \in F(S^{n+1})} S^{n+1}_b$$

restricting to u_n' on Y_n' and to b on each S^{n+1}_b. By construction $T_{u_{n+1}}$ is surjective on S^{n+1}.

Step 3: Checking $(n+1)$-universality. The complex Y_{n+1} is obtained from Y_n by attaching cells of dimension $n+1$ only (the cylinders of the injectivity step and the wedged $(n+1)$-spheres). For

$q < n$ the inclusion $Y_n \hookrightarrow Y_{n+1}$ induces an isomorphism on $[S^q, -]$ (cells of dimension $\geq n+1$ do not affect π_q for $q < n$, by the cellular approximation argument of lemma 5.5.3), so $T_{u_{n+1}}$ remains bijective on S^q for $0 < q < n$. On S^n, the surjectivity of T_{u_n} passes to $T_{u_{n+1}}$ because every map $S^n \to Y_n$ still maps into Y_{n+1} and u_{n+1} restricts to u_n. Injectivity is the new gain: if $f_0, f_1 \colon S^n \to Y_{n+1}$ have $f_0^* u_{n+1} = f_1^* u_{n+1}$, then by cellular approximation each is homotopic in Y_{n+1} to a map into Y_n (since S^n has no cells above dimension n and the new cells are of dimension $n+1$), and the two resulting maps into Y_n have equal u_n-pullbacks, so a cylinder was glued for exactly that pair, making them homotopic in Y_{n+1}. Hence $T_{u_{n+1}}$ is bijective on S^n. On S^{n+1} surjectivity holds by Step 2. Therefore (Y_{n+1}, u_{n+1}) is $(n+1)$-universal.

Step 4: Passing to the union. Set $Y = \bigcup_n Y_n$ with the weak topology. Any map $S^q \to Y$ has compact image, hence meets only finitely many cells (lemma 1.2.4) and lands in some Y_n with $n > q$, and similarly for homotopies, so $[S^q, Y] = \varinjlim_n [S^q, Y_n]$ and $\pi_q(Y) = \varinjlim_n \pi_q(Y_n)$, which stabilises at Y_{q+1}. The classes u_n are compatible under restriction, and the mapping telescope of the Y_n together with the wedge axiom (applied to the telescope, which is homotopy equivalent to Y) furnishes a class $u \in F(Y)$ restricting to each u_n.[a] For each q, the map $T_u \colon [S^q, Y] \to F(S^q)$ is the colimit of the bijections $T_{u_n} \colon [S^q, Y_n] \to F(S^q)$ for $n > q$ (each a bijection by n-universality), hence itself a bijection. Therefore (Y, u) is universal, completing the proof.

[a] Concretely, Y is homotopy equivalent to the mapping telescope $\mathrm{Tel}(Y_n)$, built from $\bigvee_n (Y_n)_+ \wedge I_+$ by gluing. The wedge axiom and the Mayer-Vietoris axiom together produce a class on the telescope restricting to each u_n, and the telescope's projection to Y transports it to u. The compatibility $u_{n+1}|_{Y_n} = u_n$ is what makes the gluing consistent.

The lemma is the heart of the matter. It produces a space whose homotopy groups, tested through F, exactly reproduce F on spheres. The mechanism is the cell-attachment surgery of section 5.5 run in reverse direction of control: there one killed homotopy to sculpt $K(G, n)$, here one kills and creates homotopy to make T_u a sphere-level isomorphism, with the Mayer-Vietoris axiom extending the universal class over each attachment. What remains is to promote sphere-level control to control on every CW complex, and this is where a relative refinement of the same lemma, combined with Whitehead's theorem, finishes the proof.

Lemma 8.3.8: Relative Construction over a Subcomplex

Let F be a Brown functor, let (X, A) be a CW pair with X connected, and let $u_A \in F(A)$ be a class such that (A, u_A) is universal (the map $T_{u_A} \colon [S^q, A] \to F(S^q)$ is bijective for all $q \geq 1$). Then for any class $x \in F(X)$ with $x|_A = u_A$ there is a CW complex $Y \supseteq X$, obtained by attaching cells to X away from A, and a universal element $u \in F(Y)$ with $u|_X = x$. In particular $u|_A = u_A$.

***Proof*:**
Run the construction of lemma 8.3.7 relative to X, starting from the class $x \in F(X)$ rather than from a wedge of spheres, and attaching all cells away from A. Set $Y_{(0)} = X$ with its class x. At each stage, to make T surjective on S^q wedge spheres S^q_b ($b \in F(S^q)$) onto $Y_{(k)} \setminus A$ and extend the class by the wedge axiom. To make T injective on S^q glue cylinders $S^n \times I$ as in Step 2 of lemma 8.3.7 along pairs of maps into $Y_{(k)} \setminus A$, extending the class by the Mayer-Vietoris axiom. No attachment is ever made to A, and the hypothesis that (A, u_A) is already universal means none is needed there: a map $S^q \to A$ is already detected isomorphically by T_{u_A}, and since

$x|_A = u_A$ the same holds for the part of T coming from A. The colimit Y contains X (hence A) as a subcomplex, carries a universal element u by the colimit argument of Step 4 of lemma 8.3.7, with $u|_X = x$ because x was the starting class and nothing already in X was altered, and so $u|_A = x|_A = u_A$, completing the proof.

Both surjectivity and injectivity of the natural transformation T_u on arbitrary complexes rest on the relative lemma. Surjectivity uses it to extend any class $x \in F(X)$ to a universal element on an enlargement, which then receives the universal Y. Injectivity uses it on the cylinder $X \times I$ to glue two maps into a homotopy. With these pieces assembled, Brown's theorem follows by comparing the universal Y to an arbitrary X through T_u and invoking Whitehead.

Theorem 8.3.9: Brown Representability

Let $F\colon \mathsf{CW}_*^{\mathrm{op}} \to \mathbf{Set}_*$ be a Brown functor (a homotopy-invariant contravariant functor on pointed connected CW complexes satisfying the wedge and Mayer-Vietoris axioms of definition 8.3.3). Then F is representable: there exist a pointed connected CW complex Y and a universal element $u \in F(Y)$ such that the natural transformation

$$T_u\colon [X, Y] \longrightarrow F(X), \qquad [f] \longmapsto f^*u$$

is a bijection for every pointed connected CW complex X. The representing pair (Y, u) is unique up to canonical homotopy equivalence: if (Y', u') also represents F, the class u' corresponds under T_u to a homotopy class $h\colon Y \to Y'$, and h is a homotopy equivalence.

***Proof*:**

Take (Y, u) universal, as furnished by lemma 8.3.7. The natural transformation T_u is defined for all X and is natural by functoriality of F. The content is that T_u is bijective on every connected pointed CW complex. The proof establishes surjectivity and injectivity separately, each by a gluing argument built on lemma 8.3.8.

Step 1: Surjectivity of T_u on every X. Let $x \in F(X)$ be arbitrary. Form the disjoint-then-wedge space $X \vee Y$ (joined at the common basepoint). By the wedge axiom, $F(X \vee Y) \cong F(X) \times F(Y)$, so the pair (x, u) corresponds to a class $w \in F(X \vee Y)$ restricting to x on X and to u on Y. Apply the relative construction lemma 8.3.8 to the pair $(X \vee Y,\ Y)$ with the universal class u on the subcomplex Y: it produces an enlargement $Y' \supseteq X \vee Y$ and a universal element $u' \in F(Y')$ restricting to w on $X \vee Y$, in particular to u on Y and to x on X. Now both (Y, u) and (Y', u') are universal, so T_u and $T_{u'}$ are bijective on all spheres. The inclusion $j\colon Y \hookrightarrow Y'$ satisfies $j^*u' = u'|_Y = u$, so on each sphere S^q the triangle

$$\begin{array}{ccc} [S^q, Y] & \xrightarrow{\ j_*\ } & [S^q, Y'] \\ & {\scriptstyle T_u}\searrow \quad \swarrow{\scriptstyle T_{u'}} & \\ & F(S^q) & \end{array}$$

commutes, and since T_u and $T_{u'}$ are bijections, $j_*\colon \pi_q(Y) \to \pi_q(Y')$ is a bijection for every $q \geq 1$. As Y and Y' are connected, j is a weak homotopy equivalence (definition 5.3.1), hence a homotopy equivalence by Whitehead's theorem (theorem 5.3.6). Let $r\colon Y' \to Y$ be a homotopy inverse, so $r \circ j \simeq \mathrm{id}_Y$. Consider the composite $f = r \circ \iota_X\colon X \hookrightarrow Y' \xrightarrow{r} Y$, where $\iota_X\colon X \hookrightarrow X \vee Y \hookrightarrow Y'$ is

the inclusion. Then

$$f^*u = \iota_X^* r^* u$$

and $r^*u = r^*(j^*u') = (j \circ r)^*u'$. Since $j \circ r \simeq \mathrm{id}_{Y'}$, this gives $r^*u = u'$, and hence $f^*u = \iota_X^*(r^*u) = \iota_X^* u' = u'|_X = x$. Therefore $T_u([f]) = x$, and T_u is surjective.

Step 2: Injectivity of T_u on every X. Suppose $f_0, f_1 \colon X \to Y$ satisfy $f_0^*u = f_1^*u = x$ in $F(X)$. Write $C = X \times I$ for the cylinder, a CW complex containing $X \times \{0,1\}$ as a subcomplex, with projection $\mathrm{pr}_X \colon C \to X$. The class $w := \mathrm{pr}_X^* x \in F(C)$ restricts on the two ends to $\mathrm{pr}_X^* x|_{X\times 0} = x = f_0^*u$ and likewise to $x = f_1^*u$ on $X \times 1$ (using $f_0^*u = f_1^*u = x$). Thus w is a class on the cylinder restricting to f_0^*u at one end and f_1^*u at the other.

Now build the comparison map by gluing Y to the cylinder. The maps f_0, f_1 together define $f_0 \sqcup f_1 \colon X \times \{0,1\} \to Y$ with $(f_0 \sqcup f_1)^*u = w|_{X\times\{0,1\}}$ (the two ends pull u back to f_0^*u and f_1^*u, which are the restrictions of w), so the class w on C and the class u on Y agree, after this map, on the two boundary copies of X. Form $Z := C \cup_{f_0 \sqcup f_1} Y$, gluing Y to C along $f_0 \sqcup f_1$ on the ends. By the Mayer-Vietoris axiom, w and u glue to a class $v \in F(Z)$ restricting to w on C and to u on Y. The subcomplex $Y \subseteq Z$ carries the universal u, so lemma 8.3.8 (with ambient Z, subcomplex Y, and starting class v) enlarges Z to a universal (Z', v') with $v'|_Y = u$, and as in Step 1 the inclusion $Y \hookrightarrow Z'$ is a homotopy equivalence with inverse $\rho \colon Z' \to Y$. The composite

$$H \colon X \times I = C \hookrightarrow Z \hookrightarrow Z' \xrightarrow{\rho} Y$$

is a homotopy. At the ends, $H|_{X\times 0} = \rho \circ (\text{inclusion of } X \times 0)$. The inclusion of $X \times 0$ into Z followed by $Y \hookrightarrow Z'$ agrees, by construction of the gluing, with f_0 followed by $Y \hookrightarrow Z'$, so ρ sends it back to f_0 up to homotopy (because $\rho \circ (Y \hookrightarrow Z') \simeq \mathrm{id}_Y$). Thus $H|_{X\times 0} \simeq f_0$ and likewise $H|_{X\times 1} \simeq f_1$, so H is a homotopy from f_0 to f_1. Hence $[f_0] = [f_1]$ and T_u is injective.

Step 3: Uniqueness of the representing pair. Let (Y', u') be another representing pair, so $T_{u'} \colon [-, Y'] \to F$ is a natural isomorphism. By surjectivity of T_u on Y' the class $u' \in F(Y')$ equals h^*u for some $h \colon Y' \to Y$. Symmetrically $u = k^*u'$ for some $k \colon Y \to Y'$. Then $(h \circ k)^*u = k^*h^*u = k^*u' = u = \mathrm{id}_Y^* u$, and injectivity of T_u on Y forces $h \circ k \simeq \mathrm{id}_Y$. Symmetrically $k \circ h \simeq \mathrm{id}_{Y'}$. Hence h is a homotopy equivalence, canonical in that it is the unique class with $h^*u = u'$, as we sought to show.

The theorem is the converse to the representability already known for ordinary cohomology: where theorem 8.3.1 produced the functor $\tilde{H}^n(-; G)$ from the space $K(G, n)$, Brown's theorem produces a space from any functor with the two formal axioms. The proof is purely homotopy-theoretic, using no special feature of cohomology. Three ingredients carry it: the wedge axiom, which lets one realise every element of F on a wedge of spheres; the Mayer-Vietoris axiom, which extends the universal class over every cell attachment and lets compatible classes glue; and Whitehead's theorem, which upgrades the sphere-level isomorphism T_u to a genuine homotopy equivalence and thereby converts sphere-level control into control on all complexes. The role of connectedness is to make Whitehead's theorem applicable basepoint-free. The role of the CW hypothesis is the same as everywhere in this chapter, that weak equivalences of CW complexes are equivalences.

8.3.10 Remark: What the Mayer-Vietoris Axiom Does Not Give The Mayer-Vietoris axiom of definition 8.3.3(2) asserts only that compatible classes *glue*, never that the glued class is unique. This is essential: for a genuine cohomology theory the difference of two glued classes lives in the image of a connecting map, which is generally nonzero, so uniqueness would be too strong an axiom and would exclude the very examples one wants. Brown's construction is arranged to need only existence of gluings: the universal element u is built up by extending over cells, and at no point is it claimed that the extension is unique. The price is that the representing space is determined only up to homotopy equivalence (the uniqueness clause of theorem 8.3.9), not on the nose, which is the correct level of canonicity for a homotopy functor.

A reduced cohomology theory $\tilde{h}^*$ in the sense of section 8.2 gives, in each degree n, a functor $\tilde{h}^n(-)$ that is homotopy-invariant and satisfies the wedge and Mayer-Vietoris axioms, hence a Brown functor, and Brown's theorem represents it by a space Y_n.

Corollary 8.3.11: Each Degree of a Cohomology Theory is Representable

Let $\tilde{h}^* = \{\tilde{h}^n\}_{n\in\mathbb{Z}}$ be a reduced cohomology theory (definition 8.2.1). For each n there is a pointed connected CW complex Y_n and a natural isomorphism $\tilde{h}^n(X) \cong [X, Y_n]$ for all pointed connected CW complexes X.

***Proof*:**

Fix n. The functor $\tilde{h}^n(-)$ is contravariant and homotopy-invariant by the axioms of a cohomology theory, and takes values in abelian groups, hence in pointed sets. It satisfies the wedge axiom because additivity is one of the axioms of definition 8.2.1, and the Mayer-Vietoris axiom because the exactness axiom yields the reduced Mayer-Vietoris sequence (proposition 8.2.3 spliced over a CW triad) whose middle exactness is the gluing statement, exactly as in example 8.3.4. Thus $\tilde{h}^n(-)$ is a Brown functor, and Brown's theorem (theorem 8.3.9) provides a pointed connected CW complex Y_n and a natural isomorphism $\tilde{h}^n(X) \cong [X, Y_n]$, completing the proof.

The corollary extends to an arbitrary cohomology theory what theorem 8.3.1 proved for ordinary cohomology, represented in each degree by $K(G, n)$. There the spaces are linked by the weak homotopy equivalences $\varepsilon_n \colon K(G, n) \to \Omega K(G, n+1)$, and the suspension isomorphism of an arbitrary theory links the spaces Y_n in the same way, which corollary 8.4.8 makes precise once section 8.4 has defined Ω-spectra.

8.3.12 Remark: Brown Representability and Derived Functors Brown representability (theorem 8.3.9) is the homotopy-theoretic instance of a categorical principle that the reader will meet again in homological algebra. The principle is this: a contravariant functor on a triangulated category that is homotopy-invariant, sends exact triangles to exact sequences, and carries coproducts (wedges, in the topological case) to products is representable by a single object, recovered from the functor by a universal element exactly as Y is recovered from F in the proof above. The wedge and Mayer-Vietoris axioms of definition 8.3.3 are the topological form of the coproduct and exact-triangle conditions. The stable homotopy category, glimpsed through spectra in chapter 8, is one triangulated category in which the principle operates: there ordinary cohomology is the functor represented by the Eilenberg-MacLane spectrum (definition 8.4.2), which is what corollary 8.3.11 produces degree by degree.
The derived category $D(R)$ of a ring R, constructed in [13, Existence of the Derived Category], is another triangulated category, and the same representability principle holds there. It enters as an adjoint-functor theorem: a triangulated functor out of $D(R)$ that preserves coproducts admits a right adjoint, because for each target object the maps out of its image form a contravariant functor of the kind above, hence one representable by the sought adjoint value. This is what guarantees, for a proper morphism f, the existence of the right adjoint $f^!$ underlying Grothendieck duality (see [13]). Thus the two ways the reader has met cohomology as "something derived" are one phenomenon viewed in two triangulated categories. The Ext term in the universal coefficient theorem (theorem 4.1.8), where $\mathrm{Ext}(H_{n-1}(C), G)$ is the derived functor of $\mathrm{Hom}(-, G)$, lives in $D(\mathbb{Z})$. The representability of a cohomology theory established in this section lives in the stable homotopy category. The shared mechanism is corepresentability in a triangulated category, and Brown's theorem is its first appearance in this book. The triangulated framework that makes the comparison precise is developed in [13]. The point here is only that the representability proved above and the derived functors of homological algebra are two faces of it.

Exercise 8.3.1

1. Verify that the wedge axiom of definition 8.3.3 forces $F(*) = *$ (a one-point set) for a Brown functor F, where $*$ is the one-point space. Deduce that the distinguished basepoint of $F(X)$ is $c^*(*_{F(*)})$ pulled back along the constant map $c\colon X \to *$, and explain why this makes T_u a map of *pointed* sets for any $u \in F(Y)$.

2. Let F be a Brown functor and (Y, u) a universal element. Using only the universality on spheres and the wedge axiom, show that $T_u\colon [W, Y] \to F(W)$ is a bijection for every wedge of spheres $W = \bigvee_\alpha S^{q_\alpha}$, without invoking the full Brown theorem. (This is the first nontrivial case beyond a single sphere.)

3. Show that the functor $X \mapsto \tilde{H}^n(X;\mathbb{Z}) \oplus \tilde{H}^m(X;\mathbb{Z})$ is a Brown functor for any $n, m \geq 1$, and identify its representing space in terms of Eilenberg-MacLane spaces. Generalise to a finite direct sum $\bigoplus_i \tilde{H}^{n_i}(X; G_i)$.

4. Let F be the functor sending a pointed connected CW complex X to the set of isomorphism classes of complex line bundles on X, with the basepoint the trivial bundle. Assuming the classification of line bundles by the first Chern class gives a natural isomorphism $F(X) \cong \tilde{H}^2(X;\mathbb{Z})$, identify the representing space of F and the universal element on it. (You may cite theorem 8.3.1.)

5. Explain precisely where the proof of theorem 8.3.9 uses Whitehead's theorem, and why the conclusion would fail if F were defined on *all* topological spaces rather than on CW complexes. Illustrate with the Warsaw circle (cf. the discussion following theorem 5.3.6).

6. Suppose $\tilde{h}^*$ is a reduced cohomology theory whose coefficient groups $\tilde{h}^n(S^0)$ vanish for $n \neq 0$ and equal an abelian group G for $n = 0$. Using corollary 8.3.11 and theorem 5.5.12, identify the representing spaces Y_n with Eilenberg-MacLane spaces and conclude that $\tilde{h}^*$ is ordinary cohomology with coefficients in G, compatibly with the suspension isomorphisms. (This is the representable form of the uniqueness theorem theorem 8.2.14.)

8.4 Spectra and Ω-Spectra

The representation theorem of section 5.5 produced, for each abelian group G and each degree n, a single space $K(G, n)$ that represents the functor $\tilde{H}^n(-; G)$, so that a cohomology class is the same datum as a homotopy class of maps into $K(G, n)$. The degrees are not independent. The weak homotopy equivalences $\varepsilon_n \colon K(G, n) \to \Omega K(G, n+1)$ of theorem 8.3.1 thread the representing spaces into a single sequence in which each space is, up to weak equivalence, the loop space of the next. This section abstracts that linkage. A *spectrum* is a sequence of pointed spaces tied together by structure maps relating consecutive degrees, and an *Ω-spectrum* is one in which the tie is as tight as possible: each space is, up to weak equivalence, the loop space of its successor. The payoff is twofold. An Ω-spectrum manufactures a reduced cohomology theory out of homotopy classes of maps, with the suspension isomorphism of the theory coming for free from the structure maps. Brown representability runs the construction in reverse, producing an Ω-spectrum from any cohomology theory. Ω-spectra and reduced cohomology theories are thereby equivalent data. The Eilenberg-MacLane spectrum encodes ordinary cohomology, the suspension spectrum of a point gives the sphere spectrum whose theory is stable cohomotopy, and the whole framework opens onto the stable homotopy category in which spheres of every dimension become invertible.

A spectrum is built from two pieces of data that say the same thing in adjoint forms. One either suspends E_n and maps the result into E_{n+1}, or one maps E_n into the loop space of E_{n+1}. The loop-suspension adjunction $[\Sigma X, Y]_* \cong [X, \Omega Y]_*$ makes these interchangeable. The suspension $\Sigma X = S^1 \wedge X$ used here is the reduced suspension, the smash of X with a circle, which for a CW complex agrees up to homotopy with the unreduced suspension and carries reduced cohomology up one degree by the suspension isomorphism $\tilde{H}^n(X) \cong \tilde{H}^{n+1}(\Sigma X)$ of the axioms in definition 8.2.1.

Definition 8.4.1: [Stable Homotopy] Spectrum

A *spectrum* E is a sequence $(E_n)_{n\geq 0}$ of pointed spaces together with *structure maps*

$$\sigma_n \colon \Sigma E_n \longrightarrow E_{n+1}$$

one for each $n \geq 0$, where $\Sigma X = S^1 \wedge X$ is the reduced suspension. Equivalently, by the loop-suspension adjunction $[\Sigma E_n, E_{n+1}]_* \cong [E_n, \Omega E_{n+1}]_*$, a spectrum is the same datum as a sequence (E_n) together with *adjoint structure maps*

$$\tilde{\sigma}_n \colon E_n \longrightarrow \Omega E_{n+1}$$

A *map of spectra* $f \colon E \to F$ is a sequence of pointed maps $f_n \colon E_n \to F_n$ commuting with the structure maps, that is, $f_{n+1} \circ \sigma_n^E = \sigma_n^F \circ (\Sigma f_n)$ for every n.

A spectrum packages an entire sequence of spaces into one object, with the structure maps recording how each space sits inside the loop space of the next. The two presentations are genuinely the same: passing from σ_n to $\tilde{\sigma}_n$ is the adjunction isomorphism, and a map of spectra is required to respect either presentation (the two compatibility conditions are adjoint and hence equivalent). The indexing here starts at $n \geq 0$ and the structure maps go up in degree. This is the convention adapted to cohomology, where $E^n(X)$ will be read off from the n-th space. Nothing yet requires the structure maps to be equivalences, so the definition is loose enough to admit suspension spectra, where the maps are far from equivalences, alongside the Ω-spectra introduced next, where they are equivalences. The looseness is deliberate: it is the difference between the two that distinguishes a generic spectrum from one that directly represents a cohomology theory.

The first example of a spectrum is the one already implicit in the representation theorem of section 5.5: the Eilenberg-MacLane spaces, strung together by the loop relation. It is the spectrum that represents ordinary cohomology.

Definition 8.4.2: Eilenberg-MacLane Spectrum

Let G be an abelian group. The *Eilenberg-MacLane spectrum* HG is the spectrum with n-th space

$$(HG)_n = K(G, n)$$

the Eilenberg-MacLane space of definition 5.5.1 (with $K(G, 0) = G$ the discrete group, based at 0), and adjoint structure maps

$$\tilde{\sigma}_n \colon K(G, n) \longrightarrow \Omega K(G, n+1)$$

given for $n \geq 1$ by the weak homotopy equivalence ε_n of theorem 8.3.1, and for $n = 0$ by sending $g \in G$ to a loop representing $g \in \pi_1(K(G, 1)) \cong G$, with 0 sent to the constant loop. The case $G = \mathbb{Z}$ is written $H\mathbb{Z}$.

The Eilenberg-MacLane spectrum is the prototype of the entire theory, and it is already an Ω-spectrum (the next definition makes this precise): each adjoint structure map is not an arbitrary map but a weak homotopy equivalence. This is exactly the tightness that the representation theorem of section 5.5 encoded degree by degree, now assembled into a single object. The cohomology theory this spectrum represents is ordinary cohomology with G coefficients, the loop relation that builds

it being precisely the geometric source of the suspension isomorphism in that theory. The general definition isolates this tightness as a property.

Example 8.4.3: The Structure Maps of $H\mathbb{Z}$ are Weak Equivalences

For HG with $G = \mathbb{Z}$, the adjoint structure map $\tilde{\sigma}_n \colon K(\mathbb{Z}, n) \to \Omega K(\mathbb{Z}, n+1)$ is a weak homotopy equivalence for every $n \geq 0$. For $n \geq 1$ this is theorem 8.3.1. The groups it matches are read off from the path-loop fibration $\Omega K(\mathbb{Z}, n+1) \to PK(\mathbb{Z}, n+1) \to K(\mathbb{Z}, n+1)$ of example 5.2.15, whose total space is contractible, so its long exact sequence gives

$$\pi_i\big(\Omega K(\mathbb{Z}, n+1)\big) \cong \pi_{i+1}\big(K(\mathbb{Z}, n+1)\big) \;=\; \begin{cases} \mathbb{Z} & i = n \\ 0 & i \neq n \end{cases}$$

For $n = 0$, the path components of $\Omega K(\mathbb{Z}, 1) = \Omega S^1$ are the classes in $\pi_1(S^1) \cong \mathbb{Z}$, and $\tilde{\sigma}_0$ sends $g \in \mathbb{Z}$ into the component of g, so it is a bijection on π_0. The same long exact sequence, based at any loop, gives $\pi_i(\Omega S^1) \cong \pi_{i+1}(S^1) = 0$ for $i \geq 1$, as for the discrete space $\mathbb{Z} = K(\mathbb{Z}, 0)$. The suspended forms $\sigma_n \colon \Sigma K(\mathbb{Z}, n) \to K(\mathbb{Z}, n+1)$ are the adjoints. Hence $H\mathbb{Z}$ is a spectrum whose adjoint structure maps are all weak homotopy equivalences.

Definition 8.4.4: Ω-Spectrum

An *Ω-spectrum* is a spectrum $E = (E_n, \tilde{\sigma}_n)$ in which every adjoint structure map

$$\tilde{\sigma}_n \colon E_n \longrightarrow \Omega E_{n+1}$$

is a weak homotopy equivalence (definition 5.3.1). When each E_n and each loop space ΩE_{n+1} has the homotopy type of a CW complex, this is equivalent, by Whitehead's theorem (theorem 5.3.6), to each $\tilde{\sigma}_n$ being a homotopy equivalence.

The Ω-spectrum condition says that the spectrum loses no information as it climbs in degree: the space E_n recovers, up to weak equivalence, the loop space ΩE_{n+1}, so E_n is determined by E_{n+1} and the whole spectrum is determined by its high-degree behavior. The contrast with a general spectrum is sharp. In an arbitrary spectrum the adjoint map $\tilde{\sigma}_n \colon E_n \to \Omega E_{n+1}$ can collapse or expand homotopy wildly. In an Ω-spectrum it is an isomorphism on every homotopy group. This is the precise property a spectrum needs in order to represent a cohomology theory through homotopy classes of maps, because the loop relation $E_n \simeq \Omega E_{n+1}$ is what makes the suspension isomorphism of the theory hold on the nose. The next theorem is the central construction of the section: it manufactures a reduced cohomology theory from any Ω-spectrum.

Example 8.4.5: The Sphere Spectrum is Not an Ω-Spectrum

Anticipating the suspension spectrum of definition 8.4.9, consider the sequence $E_n = S^n$ with structure maps $\sigma_n \colon \Sigma S^n = S^{n+1} \to S^{n+1}$ the identity, the homeomorphism $S^1 \wedge S^n \cong S^{n+1}$. This is a spectrum, the sphere spectrum $\mathbb{S}$. Its adjoint structure maps are

$$\tilde{\sigma}_n \colon S^n \longrightarrow \Omega S^{n+1}$$

the canonical inclusion of S^n as the constant-speed great-circle loops, which on homotopy groups is the suspension homomorphism $\pi_i(S^n) \to \pi_{i+1}(S^{n+1})$. By the Freudenthal suspension theorem this map is an isomorphism only in the stable range $i < 2n - 1$, and it fails to be an isomorphism for i large: for instance $\tilde{\sigma}_2 \colon S^2 \to \Omega S^3$ is not a weak equivalence, since $\pi_3(S^2) \cong \mathbb{Z}$ while

$\pi_3(\Omega S^3) \cong \pi_4(S^3) \cong \mathbb{Z}/2$ by example 6.3.6, and $\mathbb{Z} \ncong \mathbb{Z}/2$. Hence $\mathbb{S}$ is a spectrum but not an Ω-spectrum. The failure is exactly the instability of the homotopy of spheres below the stable range.

The example marks the boundary precisely: the sphere spectrum is the cleanest spectrum imaginable, built from spheres with identity structure maps, yet it is not an Ω-spectrum because the suspension homomorphism on spheres is not an isomorphism outside the stable range. This is not a defect to be repaired but the source of the entire subject. The stable homotopy groups, defined below as colimits over suspension, are exactly what the sphere spectrum measures, and the passage from a general spectrum to the theory it defines requires either the Ω-spectrum condition or a stabilization that achieves the same effect. With the contrast in hand, the construction of cohomology from an Ω-spectrum can be carried out in full.

The reduced cohomology theory attached to an Ω-spectrum is defined by the formula one expects from section 5.5: for $n \geq 0$ the degree-n cohomology of a pointed space X is the set of based homotopy classes of maps into E_n, and a negative degree is read off from degree 0 after suspending X. The work is to verify that this assignment satisfies the axioms of a reduced cohomology theory from definition 8.2.1: that it is a homotopy-invariant functor, that a cofibration induces an exact sequence, that the suspension isomorphism holds, and that a wedge goes to a product. The Ω-spectrum condition enters at exactly one point, the suspension isomorphism, and the group structure on the cohomology sets comes from the loop-space structure that the Ω-spectrum condition guarantees.

Theorem 8.4.6: An Ω-Spectrum Defines a Cohomology Theory

Let $E = (E_n, \tilde{\sigma}_n)$ be an Ω-spectrum. For a pointed CW complex X whose basepoint is a 0-cell and $n \geq 0$, set

$$E^n(X) \;=\; [X, E_n]_*$$

the set of based homotopy classes of maps $X \to E_n$, and for a based map $g\colon X' \to X$ let $g^*\colon E^n(X) \to E^n(X')$ be precomposition $[f] \mapsto [f \circ g]$. For $n < 0$ set

$$E^n(X) \;=\; E^0(\Sigma^{-n}X) \;=\; [\Sigma^{-n}X, E_0]_* \;\cong\; [X, \Omega^{-n}E_0]_*$$

with $g^* = (\Sigma^{-n}g)^*$, where Σ^k is the k-fold reduced suspension and the last bijection is theorem 8.1.5 applied $-n$ times. Then, for every $n \in \mathbb{Z}$:

1. Each $E^n(X)$ is an abelian group, natural in X, so $E^n(-)$ is a contravariant functor from pointed CW complexes to abelian groups.
2. *(Homotopy)* If $g \simeq g'$ as based maps then $g^* = (g')^*$.
3. *(Exactness)* For a based CW pair (X, A) with inclusion $i\colon A \hookrightarrow X$ and quotient map $q\colon X \to X/A$, the sequence

   $$E^n(X/A) \xrightarrow{q^*} E^n(X) \xrightarrow{i^*} E^n(A)$$

   is exact at $E^n(X)$.
4. *(Suspension)* There is a natural isomorphism $\Sigma\colon E^n(X) \xrightarrow{\cong} E^{n+1}(\Sigma X)$. For $n < 0$ it is the identity of $[\Sigma^{-n}X, E_0]_*$.
5. *(Wedge)* For any family (X_α) of pointed CW complexes the inclusions induce an isomorphism

   $$E^n\Big(\bigvee_\alpha X_\alpha\Big) \;\xrightarrow{\cong}\; \prod_\alpha E^n(X_\alpha)$$

Consequently the collection $(E^n)_{n\in\mathbb{Z}}$ with these structure maps is a reduced cohomology theory in the sense of definition 8.2.1.

***Proof*:**

Throughout, $[-,-]_*$ denotes based homotopy classes of based maps, the source X is a pointed CW complex whose basepoint is a 0-cell, and the loop-suspension adjunction $[\Sigma X, Y]_* \cong [X, \Omega Y]_*$ of theorem 8.1.5 is available.

Step 1: The abelian group structure on $E^n(X)$. The Ω-spectrum condition makes $\tilde{\sigma}_n\colon E_n \to \Omega E_{n+1}$ a weak homotopy equivalence, so by proposition 5.3.16 postcomposition with $\tilde{\sigma}_n$ is a bijection

$$(\tilde{\sigma}_n)_*\colon [X, E_n]_* \;\xrightarrow{\cong}\; [X, \Omega E_{n+1}]_*$$

and the right-hand side carries a group structure, natural in X, by proposition 8.1.4(1) and (2). Transporting it along the bijection $(\tilde{\sigma}_n)_*$ makes $E^n(X) = [X, E_n]_*$ a group. It is moreover abelian. Under the adjunction, $(\Omega\tilde{\sigma}_{n+1})_*\colon [X, \Omega E_{n+1}]_* \to [X, \Omega^2 E_{n+2}]_*$ is $(\tilde{\sigma}_{n+1})_*$ on $[\Sigma X, -]_*$, a bijection by proposition 5.3.16 for the CW complex ΣX, and it is a homomorphism because

$\Omega\tilde{\sigma}_{n+1}$ preserves concatenation of loops. So $E^n(X)$ is isomorphic to $[X, \Omega^2 E_{n+2}]_*$, which is abelian by proposition 8.1.4(3). Naturality in X holds because precomposition g^* commutes with postcomposition by $\tilde{\sigma}_n$, so g^* is a homomorphism. This proves (1).

Step 2: Homotopy invariance. If $g \simeq g'$ through based maps then for any $[f] \in [X, E_n]_*$ the composites $f \circ g$ and $f \circ g'$ are based homotopic (compose the homotopy $g \simeq g'$ with f), so $[f \circ g] = [f \circ g']$, that is $g^*[f] = (g')^*[f]$. Hence $g^* = (g')^*$, proving (2). In particular a based homotopy equivalence $X \simeq X'$ induces an isomorphism $E^n(X) \cong E^n(X')$, so E^n is an invariant of based homotopy type.

Step 3: Exactness. Fix n and write $K = E_n$. The pair (X, A) is a based CW pair, so proposition 8.1.8 gives the coexact Puppe sequence

$$A \xrightarrow{i} X \xrightarrow{q} X/A \xrightarrow{\partial} \Sigma A \xrightarrow{\Sigma i} \Sigma X \to \cdots$$

in which $X/A \simeq X \cup_A CA$ is the mapping cone, q the collapse, and consecutive maps are a cofibration followed by the collapse of its domain. The defining property of this sequence is that for *any* pointed space K, applying $[-, K]_*$ yields an exact sequence of pointed sets

$$[\Sigma X, K]_* \to [\Sigma A, K]_* \to [X/A, K]_* \xrightarrow{q^*} [X, K]_* \xrightarrow{i^*} [A, K]_*$$

Exactness at $[X, K]_*$ is precisely the statement to be proved: a class $[f] \in [X, K]_*$ satisfies $i^*[f] = [f \circ i] = 0$, that is $f|_A$ is null-homotopic, if and only if f extends over the cone CA, that is f factors up to homotopy through the quotient $q\colon X \to X/A$, which is the statement $[f] \in \mathrm{im} q^*$. Concretely, a based null-homotopy of $f|_A$ is a map $CA \to K$ restricting to $f|_A$ on A, and gluing it to f along A produces a map on $X \cup_A CA = X/A$ whose restriction to X is f. This is the required factorization $[f] = q^*[\bar{f}]$. Conversely if $[f] = q^*[\bar{f}]$ then $i^*[f] = i^*q^*[\bar{f}] = (q \circ i)^*[\bar{f}] = 0$ because $q \circ i$ is the constant map $A \to X/A$ (the subspace A collapses to the basepoint of X/A). Setting $K = E_n$ gives exactness of $E^n(X/A) \xrightarrow{q^*} E^n(X) \xrightarrow{i^*} E^n(A)$ as groups, the maps being homomorphisms by Step 1, proving (3).

Step 4: The suspension isomorphism. The desired isomorphism $\Sigma\colon E^n(X) \to E^{n+1}(\Sigma X)$ is the composite of three natural bijections. First, the adjoint structure map gives, by Step 1,

$$E^n(X) = [X, E_n]_* \xrightarrow[\cong]{(\tilde{\sigma}_n)_*} [X, \Omega E_{n+1}]_*$$

Second, the loop-suspension adjunction gives a natural bijection

$$[X, \Omega E_{n+1}]_* \xrightarrow{\cong} [\Sigma X, E_{n+1}]_* = E^{n+1}(\Sigma X)$$

sending a map $X \to \Omega E_{n+1}$ to its adjoint $\Sigma X \to E_{n+1}$. Both bijections are isomorphisms of groups: the first because the group structure on $E^n(X)$ was *defined* by transporting along $(\tilde{\sigma}_n)_*$, and the second by theorem 8.1.5(2), which identifies loop concatenation on the source with the pinch-map addition on $[\Sigma X, -]_*$. The composite Σ is therefore a natural isomorphism. Here, as in Step 1, the Ω-spectrum hypothesis is what makes the first arrow a bijection. This proves (4).

Step 5: The wedge axiom. Let $(X_\alpha)_{\alpha \in A}$ be pointed CW complexes with wedge $\bigvee_\alpha X_\alpha$, the quotient of $\bigsqcup_\alpha X_\alpha$ identifying all basepoints. A based map $f\colon \bigvee_\alpha X_\alpha \to E_n$ is, by the universal property of the wedge, exactly a family of based maps $f_\alpha\colon X_\alpha \to E_n$ agreeing at the common basepoint (which they do automatically, all sending it to the basepoint of E_n). That is, the

restriction maps assemble into a bijection of sets

$$\Big[\bigvee_\alpha X_\alpha, E_n\Big]_* \xrightarrow{\cong} \prod_\alpha [X_\alpha, E_n]_*$$

The same universal property applied to based homotopies (a based homotopy out of $\bigvee_\alpha X_\alpha$ is a family of based homotopies, one out of each X_α) shows the map is well defined and injective on homotopy classes, and surjective because any family of maps glues to a map on the wedge. The bijection is a group homomorphism because the group operation is computed pointwise through loop concatenation in $E_n \simeq \Omega E_{n+1}$, which commutes with restriction to each summand. Hence

$$E^n\Big(\bigvee_\alpha X_\alpha\Big) \xrightarrow{\cong} \prod_\alpha E^n(X_\alpha)$$

is an isomorphism, proving (5).

Step 6: Negative degrees. Steps 1 to 5 prove (1) to (5) for $n \geq 0$. Let $n < 0$ and $k = -n$. For X as above, $\Sigma^k X$ is a pointed CW complex whose basepoint is a 0-cell, and $E^n = E^0 \circ \Sigma^k$ on spaces and on maps. Parts (1) and (2) in degree n are therefore parts (1) and (2) in degree 0 at $\Sigma^k X$, because Σ^k carries based homotopies to based homotopies. For a based CW pair (X, A), the pair $(\Sigma^k X, \Sigma^k A)$ is a based CW pair with $\Sigma^k X/\Sigma^k A = \Sigma^k(X/A)$, both being the quotient of $X \times I^k$ that collapses $X \times \partial I^k \cup A \times I^k$, and its inclusion and quotient map are $\Sigma^k i$ and $\Sigma^k q$. So (3) in degree 0 for this pair is (3) in degree n. In the same way $\Sigma^k\big(\bigvee_\alpha X_\alpha\big) = \bigvee_\alpha \Sigma^k X_\alpha$, compatibly with the inclusions of the summands, and (5) in degree 0 gives (5) in degree n. For (4), $\Sigma^k X = \Sigma^{k-1}(\Sigma X)$, so $E^n(X) = [\Sigma^k X, E_0]_* = E^{n+1}(\Sigma X)$ as groups, and the identity is a natural isomorphism because $\Sigma^{k-1}(\Sigma g) = \Sigma^k g$.

Step 7: Assembling the cohomology theory. Parts (1)-(5) are exactly the data and axioms of a reduced cohomology theory in definition 8.2.1: a sequence of homotopy-invariant abelian-group-valued functors E^n (parts 1, 2), satisfying exactness for cofiber sequences (part 3), with natural suspension isomorphisms $E^n(X) \cong E^{n+1}(\Sigma X)$ (part 4) and the wedge-to-product property (part 5). The long exact sequence of a pair, required by that definition, is obtained by iterating the exactness of Step 3 along the Puppe sequence: applying $[-, E_n]_*$ to the full coexact sequence $A \to X \to X/A \to \Sigma A \to \Sigma X \to \cdots$ and using the suspension isomorphism to reindex $[\Sigma^k A, E_n]_* \cong E^{n-k}(A)$ converts the Puppe exactness into the long exact sequence

$$\cdots \to E^n(X/A) \to E^n(X) \to E^n(A) \to E^{n+1}(X/A) \to \cdots$$

Thus $(E^n)_n$ is a reduced cohomology theory, completing the proof.

The theorem is what makes the representable viewpoint work. It says that homotopy classes of maps into the spaces of an Ω-spectrum automatically organize themselves into a cohomology theory, with every axiom traced to a structural feature: the group law to the loop space, exactness to the Puppe sequence of a cofibration, the suspension isomorphism to the adjoint structure map, and the wedge axiom to the universal property of the wedge. The dimension axiom, conspicuously absent from the list, is what separates ordinary from generalized theories: it imposes a condition on the homotopy of the spaces E_n that the Eilenberg-MacLane spectrum satisfies and that the sphere spectrum and K-theory do not. The construction is also exhaustive: by Brown representability, every reasonable cohomology theory arises this way. That converse is recorded after the present examples.

Example 8.4.7: The Cohomology Theory of $H\mathbb{Z}$ is Ordinary Cohomology
Apply theorem 8.4.6 to the Eilenberg-MacLane spectrum $H\mathbb{Z}$ of definition 8.4.2, which is an Ω-spectrum because each $\tilde{\sigma}_n\colon K(\mathbb{Z},n)\to\Omega K(\mathbb{Z},n+1)$ is a weak homotopy equivalence (example 8.4.3). The resulting theory is

$$(H\mathbb{Z})^n(X)=[X,K(\mathbb{Z},n)]_*\ \cong\ \tilde{H}^n(X;\mathbb{Z})$$

for $n\geq 1$ the isomorphism being exactly the representation theorem $T_n\colon [X,K(\mathbb{Z},n)]_*\xrightarrow{\cong}\tilde{H}^n(X;\mathbb{Z})$, $[f]\mapsto f^*(\iota_n)$, of theorem 5.5.12. The identification is natural in X (naturality of T_n) and compatible with the suspension isomorphisms: for $n\geq 1$ the suspension isomorphism $\Sigma\colon (H\mathbb{Z})^n(X)\cong(H\mathbb{Z})^{n+1}(\Sigma X)$ built from the structure maps in Step 4 corresponds under T to the suspension isomorphism $\sigma\colon \tilde{H}^n(X;\mathbb{Z})\cong\tilde{H}^{n+1}(\Sigma X;\mathbb{Z})$ of ordinary cohomology, $T_{n+1}\circ\Sigma=\sigma\circ T_n$, because the adjoint structure map $\tilde{\sigma}_n=\varepsilon_n$ is defined so that $(\varepsilon_n)_*$ followed by the loop-suspension adjunction corresponds under T to that isomorphism (theorem 8.3.1). Theorem 5.5.12 needs $n\geq 1$, so degree 0 goes through degree 1: the composite $T_0:=\sigma^{-1}\circ T_1\circ\Sigma\colon (H\mathbb{Z})^0(X)\to\tilde{H}^0(X;\mathbb{Z})$ of three natural bijections is a natural bijection, and $T_1\circ\Sigma=\sigma\circ T_0$ holds by construction. In each degree $n<0$ both groups vanish: $(H\mathbb{Z})^n(X)=[\Sigma^{-n}X,\mathbb{Z}]_*$ is a point, since $\Sigma^{-n}X$ is path-connected and $K(\mathbb{Z},0)=\mathbb{Z}$ is discrete, and singular cohomology vanishes in negative degrees. Each T_n is additive. By Step 4 of the proof of theorem 8.4.6, Σ is a group isomorphism onto $[\Sigma X,K(\mathbb{Z},n+1)]_*$ with its pinch sum. The pinch map followed by either collapse $\Sigma X\vee\Sigma X\to\Sigma X$ is homotopic to the identity, so by the additivity axiom pulling back along a pinch sum $a+b$ gives $a^*\iota_{n+1}+b^*\iota_{n+1}$, and T_{n+1} is additive on $[\Sigma X,K(\mathbb{Z},n+1)]_*$. Hence $T_n=\sigma^{-1}\circ T_{n+1}\circ\Sigma$ is a homomorphism for every $n\geq 0$. The cohomology theory represented by $H\mathbb{Z}$ is therefore ordinary singular cohomology with integer coefficients, and replacing $\mathbb{Z}$ by G gives $\tilde{H}^n(-;G)$ from the spectrum HG. To see the values on a concrete space, take $X=S^k$ and $n\geq 0$: then

$$(H\mathbb{Z})^n(S^k)=[S^k,K(\mathbb{Z},n)]_*=\pi_k(K(\mathbb{Z},n))=\begin{cases}\mathbb{Z} & k=n\\ 0 & k\neq n\end{cases}$$

matching $\tilde{H}^n(S^k;\mathbb{Z})$, which is $\mathbb{Z}$ for $k=n$ and 0 otherwise.

The Eilenberg-MacLane spectrum $H\mathbb{Z}$ thus represents ordinary cohomology: the entire theory $\tilde{H}^*(-;\mathbb{Z})$, with all its naturality and suspension structure, is determined by the single object $H\mathbb{Z}$. The dimension axiom holds for it because the spaces $K(\mathbb{Z},n)$ have their homotopy concentrated in degree n, so the value on a sphere is concentrated in a single degree. The spectrum $H\mathbb{Z}$ links the abstract axioms of definition 8.2.4 to the concrete representing spaces. Every reduced cohomology theory arises from an Ω-spectrum in this way. The representing spaces Y_n of corollary 8.3.11 are connected, while $\tilde{h}^n(S^0)$ can be nonzero, so the spectrum is built from their loop spaces.

Corollary 8.4.8: Every Cohomology Theory is Represented by an Ω-Spectrum

Let $\tilde{h}^*$ be a reduced cohomology theory (definition 8.2.1), with suspension isomorphisms $\sigma^n\colon \tilde{h}^n(X) \xrightarrow{\cong} \tilde{h}^{n+1}(\Sigma X)$, and let Y_n be the representing spaces of corollary 8.3.11. For each n there is a based map $\epsilon_n\colon Y_n \to \Omega Y_{n+1}$ inducing isomorphisms $\pi_q(Y_n) \to \pi_q(\Omega Y_{n+1})$ at the basepoints for all $q \geq 1$, a weak homotopy equivalence onto the path component of the constant loop. The loop spaces $E_n = \Omega Y_{n+1}$, $n \geq 0$, with adjoint structure maps $\Omega\epsilon_{n+1}\colon E_n \to \Omega E_{n+1}$, form an Ω-spectrum, and $\tilde{h}^n(X) \cong [X, E_n]_*$ naturally for every pointed CW complex X whose basepoint is a 0-cell, connected or not.

***Proof*:**

Step 1: The loop relation. The suspension isomorphism and the loop-suspension adjunction combine to produce the equivalences. For every X the adjunction gives a natural bijection $[\Sigma X, Y_{n+1}] \cong [X, \Omega Y_{n+1}]$. Chaining with the representability isomorphisms of corollary 8.3.11 and the suspension isomorphism of the theory,

$$[X, Y_n] \;\cong\; \tilde{h}^n(X) \;\xrightarrow[\cong]{\sigma^n}\; \tilde{h}^{n+1}(\Sigma X) \;\cong\; [\Sigma X, Y_{n+1}] \;\cong\; [X, \Omega Y_{n+1}]$$

is a natural isomorphism $\Phi\colon [-, Y_n] \cong [-, \Omega Y_{n+1}]$ of functors on CW_*. Proposition 8.1.10 needs both spaces in CW_*, and ΩY_{n+1} need not be a CW complex, so the Yoneda argument is run at Y_n alone. Let ϵ_n represent $\Phi_{Y_n}[\mathrm{id}_{Y_n}]$. For $f\colon X \to Y_n$, naturality gives $\Phi_X[f] = f^*\Phi_{Y_n}[\mathrm{id}_{Y_n}] = [\epsilon_n \circ f]$, so $\Phi_X = (\epsilon_n)_*$, and $X = S^q$ with $q \geq 1$ shows that ϵ_n is an isomorphism on π_q at the basepoints. It need not be a bijection on π_0: the path components of ΩY_{n+1} are the elements of $\pi_1(Y_{n+1}) \cong \tilde{h}^{n+1}(S^1) \cong \tilde{h}^n(S^0)$, which can be nonzero although Y_n is connected.

Step 2: The Ω-spectrum. Let X be a pointed CW complex whose basepoint is a 0-cell. Then ΣX lies in CW_*, so the part of the display from $\tilde{h}^n(X)$ onward still applies and gives $\tilde{h}^n(X) \cong \tilde{h}^{n+1}(\Sigma X) \cong [\Sigma X, Y_{n+1}] \cong [X, E_n]_*$, natural in X. It remains to see that $\Omega\epsilon_{n+1}$ is a weak homotopy equivalence. Its effect on path components is $(\epsilon_{n+1})_*$ on π_1, a bijection. At a loop $\lambda \in \Omega Y_{n+1}$, the path-loop fibration $\Omega Y \to PY \to Y_0$ of example 5.2.15, over the path component Y_0 of the basepoint (which contains every path from it), has contractible total space, so its long exact sequence (theorem 5.2.12), based at λ in the fiber, gives $\pi_q(\Omega Y, \lambda) \cong \pi_{q+1}(Y)$ for $q \geq 1$, for any pointed space Y. A based map $Y \to Y'$ induces a map of path-loop fibrations, so these isomorphisms are natural, and under them $(\Omega\epsilon_{n+1})_*$ at λ is $(\epsilon_{n+1})_*$ on π_{q+1}, an isomorphism. Hence $\{E_n\}$ with the maps $\Omega\epsilon_{n+1}$ is an Ω-spectrum (definition 8.4.4), completing the proof.

The sequence $\{E_n\}$ with its loop maps is the universal home for the theory: every cohomology group $\tilde{h}^n(X)$ is a set of homotopy classes $[X, E_n]_*$, and the suspension axiom is encoded geometrically in the loop relation. The Eilenberg-MacLane spectrum $H\mathbb{Z}$ is the first instance, and K-theory and cobordism, treated in section 8.5, are the next two, each arising from its own Ω-spectrum.

The suspension spectrum is the most economical way to turn a single pointed space into a spectrum: suspend it repeatedly. The structure maps are then identities, since suspending the n-th space $\Sigma^n X$ produces exactly the $(n+1)$-st space $\Sigma^{n+1}X$. This is almost never an Ω-spectrum, as the sphere case in example 8.4.5 already showed, but it is the canonical bridge from unstable to stable homotopy theory and the source of the sphere spectrum.

Definition 8.4.9: Suspension Spectrum and the Sphere Spectrum

Let X be a pointed space. The *suspension spectrum* $\Sigma^\infty X$ is the spectrum with n-th space

$$(\Sigma^\infty X)_n = \Sigma^n X = \underbrace{S^1 \wedge \cdots \wedge S^1}_{n} \wedge X$$

and structure maps $\sigma_n \colon \Sigma(\Sigma^n X) = \Sigma^{n+1} X \to \Sigma^{n+1} X$ the identity. The *sphere spectrum* is

$$\mathbb{S} = \Sigma^\infty S^0$$

whose n-th space is $\Sigma^n S^0 = S^n$ (using $\Sigma S^k = S^{k+1}$ and $S^0 = \{\pm 1\}$), with identity structure maps $S^{n+1} \to S^{n+1}$.

The suspension spectrum is the left adjoint to the forgetful passage from spectra to spaces: it is the free spectrum on a pointed space, built by doing nothing but suspending. The sphere spectrum $\mathbb{S} = \Sigma^\infty S^0$ is the unit of this construction and the most basic spectrum of all. Its spaces are the spheres themselves. As example 8.4.5 recorded, $\mathbb{S}$ is not an Ω-spectrum, so the cohomology theory it defines must be extracted by a stabilization rather than read off directly from a single space, and the homotopy it carries is the stable homotopy of spheres. The cohomology theory of the sphere spectrum is stable cohomotopy, and its homotopy groups are the stable stems.

Example 8.4.10: Stable Cohomotopy and the Stable Stems

The sphere spectrum $\mathbb{S}$ defines, by the stabilized form of theorem 8.4.6, the generalized cohomology theory called *stable cohomotopy*, written $\pi_s^n(X)$ or $\mathbb{S}^n(X)$. On a finite pointed CW complex X its value is the colimit

$$\pi_s^n(X) = \mathbb{S}^n(X) = \varinjlim_k [\Sigma^k X, S^{k+n}]_*$$

the structure maps of the colimit being suspension $[\Sigma^k X, S^{k+n}]_* \to [\Sigma^{k+1} X, S^{k+1+n}]_*$. The colimit is needed precisely because $\mathbb{S}$ is not an Ω-spectrum: suspending stabilizes the unstable sets $[\Sigma^k X, S^{k+n}]_*$ into a group once k is large, by the Freudenthal suspension theorem. Dually, the *homotopy groups* of the sphere spectrum are

$$\pi_n(\mathbb{S}) = \varinjlim_k \pi_{n+k}(S^k) = \pi_n^s$$

the stable homotopy groups of spheres, the stable stems. For $n = 0$ the colimit $\varinjlim_k \pi_k(S^k) = \varinjlim_k \mathbb{Z}$ along degree-1 suspensions is $\mathbb{Z}$, so $\pi_0(\mathbb{S}) \cong \mathbb{Z}$. For $n = 1$ the colimit

$$\pi_1(\mathbb{S}) = \varinjlim_k \pi_{k+1}(S^k) = \pi_1^s \cong \mathbb{Z}/2$$

is the first stable stem, computed in example 6.3.6: $\pi_{k+1}(S^k) \cong \mathbb{Z}/2$ for all $k \geq 3$, so the colimit, already constant from $k = 3$ onward, is $\mathbb{Z}/2$, generated by the suspensions of the Hopf map η. The unstable group $\pi_3(S^2) \cong \mathbb{Z}$ at $k = 2$ lies below the stable range and collapses to $\mathbb{Z}/2$ under the first suspension, so it does not affect the colimit.

The sphere spectrum is to stable homotopy theory what the integers are to algebra: the initial object whose endomorphisms control everything else. Its homotopy groups, the stable stems π_n^s,

are the central computational problem of the subject, finite for $n \geq 1$ (a fact proved by Serre using the spectral-sequence machinery of chapter 6) and known explicitly only in a range. The contrast between $\pi_0(\mathbb{S}) \cong \mathbb{Z}$ and $\pi_1(\mathbb{S}) \cong \mathbb{Z}/2$ already exhibits the two regimes: degree zero is detected by the integer degree of a self-map of a sphere, while degree one is the 2-torsion produced by the Hopf map. The general definition of the stable homotopy groups of an arbitrary spectrum is the same colimit construction, and it is the right notion of homotopy group in the stable world.

Definition 8.4.11: Stable Homotopy Groups of a Spectrum

For a spectrum $E = (E_n, \sigma_n)$ and $n \in \mathbb{Z}$, the *n-th stable homotopy group* is the colimit

$$\pi_n(E) = \varinjlim_k \pi_{n+k}(E_k)$$

where the maps in the colimit are the composites

$$\pi_{n+k}(E_k) \xrightarrow{\Sigma} \pi_{n+k+1}(\Sigma E_k) \xrightarrow{(\sigma_k)_*} \pi_{n+k+1}(E_{k+1})$$

of the suspension homomorphism with the map induced by the structure map.

The stable homotopy groups $\pi_n(E)$ are indexed by all integers, positive and negative, reflecting that a spectrum has no bottom dimension: suspension is invertible in the stable category, so degrees may go below zero. For an Ω-spectrum the colimit stabilizes immediately, since the adjoint structure maps are equivalences and $\pi_{n+k}(E_k) \cong \pi_{n+k+1}(E_{k+1})$ already at each stage, so $\pi_n(E) \cong \pi_{n+k}(E_k)$ for any single large k. For a suspension spectrum the colimit is a genuine limit of suspensions, as in the sphere case. The two presentations of a spectrum, by structure maps or adjoints, give the same colimit because the suspension homomorphism and the loop-space dimension shift are adjoint. These groups are the homotopy-theoretic invariants of a spectrum, and a map of spectra inducing isomorphisms on all π_n is a stable equivalence, the spectrum-level analogue of a weak homotopy equivalence.

The cohomology theory of an Ω-spectrum, the theory of $H\mathbb{Z}$, and the sphere spectrum's stable cohomotopy are three instances of a single correspondence. Brown representability, established in theorem 8.3.9, asserts that every reduced cohomology theory on CW complexes is representable: in each degree n there is a space E_n with $h^n(X) \cong [X, E_n]_*$ naturally, and the suspension isomorphism of the theory upgrades these representing spaces into an Ω-spectrum. Conversely, theorem 8.4.6 produces a cohomology theory from any Ω-spectrum. The two constructions are mutually inverse up to the natural notions of equivalence, giving the correspondence that organizes the entire subject.

Theorem 8.4.12: Cohomology Theories and Ω-Spectra Correspond

There is a correspondence

$$\left\{\begin{array}{c}\text{reduced cohomology theories}\\ \text{on pointed CW complexes}\end{array}\right\} \longleftrightarrow \left\{\begin{array}{c}\Omega\text{-spectra}\\ \text{(up to weak equivalence)}\end{array}\right\}$$

given in one direction by sending an Ω-spectrum E to the theory $X \mapsto [X, E_n]_*$ of theorem 8.4.6, and in the other by sending a theory (h^n) to the Ω-spectrum whose n-th space represents h^n, which exists by Brown representability (theorem 8.3.9 and corollary 8.4.8). The two assignments are mutually inverse: a theory (h^n) is naturally isomorphic to the theory of its representing Ω-spectrum, and an Ω-spectrum is weakly equivalent to the Ω-spectrum representing its theory.

***Proof*:**

Step 1: From an Ω-spectrum to a theory and back. Let E be an Ω-spectrum and $h^n(X) = [X, E_n]_*$ the cohomology theory of theorem 8.4.6. The Ω-spectrum that corollary 8.4.8 produces from h has n-th space $E'_n = \Omega Y_{n+1}$, where $Y_{n+1} \in \mathsf{CW}_*$ represents $h^{n+1} = [-, E_{n+1}]_*$ on CW_*. The Yoneda argument of that proof, run at Y_{n+1}, gives a map $v_{n+1}\colon Y_{n+1} \to E_{n+1}$ whose postcomposition is this natural isomorphism, so v_{n+1} is an isomorphism on π_q at the basepoints for $q \geq 1$. The path-loop argument of the same proof then makes $\Omega v_{n+1}\colon E'_n \to \Omega E_{n+1}$ a weak homotopy equivalence, and $\tilde{\sigma}_n\colon E_n \to \Omega E_{n+1}$ is one by hypothesis. So E'_n and E_n are weakly equivalent through ΩE_{n+1}, compatibly with the structure maps because both sides realize the same suspension isomorphism of h^*. Hence E is weakly equivalent to the Ω-spectrum representing its own theory.

Step 2: From a theory to an Ω-spectrum and back. Let (h^n) be a reduced cohomology theory. Brown representability alone gives connected representing spaces Y_n on CW_*, which cannot represent h^n on S^0 when $h^n(S^0) \neq 0$. By corollary 8.4.8, the loop spaces $E_n = \Omega Y_{n+1}$ with the maps $\Omega\epsilon_{n+1}$ form an Ω-spectrum, with natural isomorphisms $h^n(X) \cong [X, E_n]_*$ for every pointed CW complex X whose basepoint is a 0-cell. Feeding this Ω-spectrum into theorem 8.4.6 returns, in degrees $n \geq 0$, the theory $X \mapsto [X, E_n]_* \cong h^n(X)$, naturally isomorphic to the original (h^n). The suspension isomorphisms match: under the adjunction, $(\Omega\epsilon_{n+1})_*$ on $[X, E_n]_*$ is $(\epsilon_{n+1})_*$ on $[\Sigma X, Y_{n+1}]$, which corresponds to the suspension isomorphism of h^* by the construction of ϵ_{n+1}. In a degree $n < 0$ the theory of theorem 8.4.6 is $[\Sigma^{-n}X, E_0]_* \cong h^0(\Sigma^{-n}X) \cong h^n(X)$, the last isomorphism the inverse of $-n$ suspension isomorphisms of h^*. This is natural and commutes with the suspension isomorphisms, which are identities on the left. Hence (h^n) is naturally isomorphic to the theory of its representing Ω-spectrum, completing the proof.

The correspondence is the organizing principle of stable homotopy theory: to study cohomology theories is to study Ω-spectra, and conversely. Ordinary cohomology corresponds to $H\mathbb{Z}$, K-theory corresponds to a spectrum built from $\mathbb{Z} \times BU$ by Bott periodicity (chapter 8 and [33]), and cobordism corresponds to a Thom spectrum. Each generalized theory is a spectrum, and computations in the theory become computations with the spectrum. The dictionary also explains why the dimension axiom is the dividing line: it is the condition that the representing spaces E_n be Eilenberg-MacLane spaces, with homotopy concentrated in a single degree, and dropping it admits all the extraordinary theories at once.

8.4.13 Foreshadowing: The Stable Homotopy Category The maps of spectra of definition 8.4.1 and the stable homotopy groups of definition 8.4.11 are the first ingredients of the *stable homotopy category*, the natural home for spectra. In it the objects are spectra, the morphisms are stable homotopy classes of maps (formed by inverting the maps that induce isomorphisms on all π_n), and the suspension functor Σ becomes invertible: every spectrum has a desuspension $\Sigma^{-1}E$, so degrees range over all of $\mathbb{Z}$. The sphere spectrum $\mathbb{S}$ is the unit for a symmetric monoidal smash product, making the category the stable analogue of the category of abelian groups, with $[\mathbb{S}, E] = \pi_0(E)$ playing the role of global sections. This is the setting in which the Atiyah-Hirzebruch spectral sequence of theorem 8.6.4 computes a generalized theory from ordinary cohomology, and in which the entire apparatus of generalized cohomology becomes a form of stable algebra.

Exercise 8.4.1

1. Verify that the Eilenberg-MacLane spectrum $H\mathbb{Z}$ of definition 8.4.2 is an Ω-spectrum, and compute its stable homotopy groups $\pi_n(H\mathbb{Z})$ for all $n \in \mathbb{Z}$. (Use that each $\tilde{\sigma}_n$ is a weak equivalence so the colimit stabilizes at any single stage.)

2. Let E be an Ω-spectrum and $E^n(X) = [X, E_n]_*$ its cohomology theory. Compute $E^n(S^k)$ in terms of the stable homotopy groups $\pi_*(E)$ of definition 8.4.11, and check your formula against $E = H\mathbb{Z}$ using example 8.4.7.

3. Show that for any pointed space X the suspension spectrum $\Sigma^\infty X$ has stable homotopy groups
$$\pi_n(\Sigma^\infty X) = \varinjlim_k \pi_{n+k}(\Sigma^k X)$$
the *stable homotopy groups of* X, written $\pi_n^s(X)$. Deduce $\pi_n(\mathbb{S}) = \pi_n^s(S^0) = \pi_n^s$, the stable stems, and compute $\pi_0(\mathbb{S})$.

4. Prove that the structure-map presentation and the adjoint presentation of a spectrum (definition 8.4.1) carry the same information, by writing out the bijection $[\Sigma E_n, E_{n+1}]_* \cong [E_n, \Omega E_{n+1}]_*$ explicitly and checking that a map of spectra in one presentation is a map of spectra in the other.

5. Explain why the dimension axiom holds for the theory represented by $H\mathbb{Z}$ but fails for the theory represented by the sphere spectrum $\mathbb{S}$. (Compare $H^n(\mathrm{pt})$ with the stable stems. Recall the dimension axiom concerns the value of the theory on a point, equivalently on S^0.)

6. Let E be an Ω-spectrum and let $f\colon X \to Y$ be a based map of pointed CW complexes whose cofiber is $C_f = Y \cup_f CX$. Using only the exactness axiom of theorem 8.4.6 and the suspension isomorphism, derive the long exact sequence
$$\cdots \to E^n(C_f) \to E^n(Y) \xrightarrow{f^*} E^n(X) \to E^{n+1}(C_f) \to \cdots$$

8.5 K-theory and Cobordism as Generalized Theories

The Eilenberg-Steenrod axioms of section 8.2 single out ordinary cohomology by one extra requirement, the dimension axiom: the cohomology of a point is concentrated in degree zero. Drop that axiom

and a wider world of cohomology theories opens up, each represented by an Ω-spectrum through the machinery of section 8.3 and section 8.4. This section exhibits the two theories that historically forced the generalization, and that remain its archetypes. The first is topological K-theory, built from vector bundles: $K^0(X)$ measures how far the bundles on X are from being trivial, and Bott periodicity assembles the groups $K^n(X)$ into a 2-periodic cohomology theory whose value on a point is the polynomial ring $\mathbb{Z}[\beta, \beta^{-1}]$, the loudest possible violation of the dimension axiom. The second is cobordism, built from manifolds: two closed n-manifolds are cobordant when together they bound a compact $(n+1)$-manifold, and the Pontryagin-Thom construction identifies the resulting bordism groups with the stable homotopy groups of a Thom spectrum. Both theories trade the rigidity of the dimension axiom for geometric content invisible to ordinary cohomology, and both are represented by spectra: KU for complex K-theory, MO and MU for cobordism. Vector bundles are defined in definition 5.2.8 and classified in theorem 7.1.29. The K-theory built from them, Bott periodicity included, is developed in *Vector Bundles and K-Theory* [33].

The starting point is the set of isomorphism classes of complex vector bundles over a space X. This set is a commutative monoid under direct sum (Whitney sum), but not a group: a nontrivial bundle has no additive inverse among bundles, since $E \oplus E' \cong \underline{0}$ forces both to be zero. The Grothendieck group construction repairs this, formally adjoining inverses exactly as the integers are built from the natural numbers.

Definition 8.5.1: Topological K-theory

Let X be a compact Hausdorff space. Write $\mathrm{Vect}_{\mathbb{C}}(X)$ for the commutative monoid of isomorphism classes of complex vector bundles over X under Whitney sum $\oplus$. The *(complex) K-theory group* $K^0(X)$ is the Grothendieck group of this monoid: the free abelian group on isomorphism classes $[E]$ modulo the relations $[E \oplus F] = [E] + [F]$, so that every element has the form $[E] - [F]$ for bundles E, F, with $[E] - [F] = [E'] - [F']$ if and only if $E \oplus F' \oplus G \cong E' \oplus F \oplus G$ for some bundle G. Tensor product of bundles makes $K^0(X)$ a commutative ring, with unit the class of the trivial line bundle $\underline{\mathbb{C}}$. The *rank* of a bundle E is the function $x \mapsto \dim E_x$, locally constant by definition 5.2.8(1) and additive under $\oplus$, so it extends to a homomorphism from $K^0(X)$ to the locally constant functions $X \to \mathbb{Z}$. Pullback of bundles along a map $u\colon X' \to X$ of compact Hausdorff spaces preserves $\oplus$ and $\otimes$, so it induces a ring homomorphism $u^*\colon K^0(X) \to K^0(X')$. For a basepoint $x_0 \in X$ the restriction $K^0(X) \to K^0(x_0) = \mathbb{Z}$ (the rank at x_0) is a ring homomorphism. Its kernel is the *reduced K-theory* $\tilde{K}^0(X)$, which for connected X is the subgroup of virtual bundles of rank zero. Real vector bundles give in the same way the real K-theory $KO^0(X)$ and its reduced group $\widetilde{KO}^0(X)$.

Algebraically, $K^0(X)$ is the universal target for additive invariants of bundles: any function on bundles sending $\oplus$ to $+$ factors through it. Its relation $[E] - [F] = [E'] - [F']$ involves an auxiliary bundle G about which nothing is known. Over a compact base G can be taken trivial, because every bundle is a direct summand of a trivial one: a classifying map lands in a finite Grassmannian, which embeds the bundle in a trivial bundle, and an inner product gives the complement.

Lemma 8.5.2: Bundles over a Compact Base are Summands of Trivial Bundles

Let X be a compact Hausdorff space, $\mathbb{K}$ be $\mathbb{R}$ or $\mathbb{C}$, and $E \to X$ a $\mathbb{K}$-vector bundle. Then there are a bundle $E' \to X$ and an integer $M \geq 0$ with $E \oplus E' \cong \underline{\mathbb{K}}^M$.

***Proof*:**
The sets $X_n = \{x \in X \mid \dim E_x = n\}$ are open, since the dimension is locally constant (definition 5.2.8(1)), pairwise disjoint, and cover X. By compactness only finitely many are nonempty, and each is closed, being the complement of the union of the others. So each X_n is compact Hausdorff, hence paracompact Hausdorff, a finite subcover being a locally finite refinement.

By theorem 7.1.29 there is a map $f_n\colon X_n \to \mathrm{Gr}_n(\mathbb{K}^\infty)$ with $E|_{X_n} \cong f_n^*\gamma_n$. Its image is compact, so it lies in a finite stage $\mathrm{Gr}_n(\mathbb{K}^M)$, a subcomplex carrying the subspace topology (theorem 7.1.17(1), lemma 7.1.15), and one M serves for the finitely many n. Over $\mathrm{Gr}_n(\mathbb{K}^M)$ the tautological bundle is $\{(V, v) \mid v \in V\} \subseteq \mathrm{Gr}_n(\mathbb{K}^M) \times \mathbb{K}^M$ (definition 7.1.12), so $(x, (V, v)) \mapsto (x, v)$ is a homeomorphism from $f_n^*\gamma_n$ onto

$$S_n = \left\{(x, v) \in X_n \times \mathbb{K}^M \;\middle|\; v \in f_n(x)\right\}$$

linear on fibers, with inverse $(x, v) \mapsto (x, (f_n(x), v))$. Hence S_n is a subbundle of the trivial bundle $X_n \times \mathbb{K}^M$ isomorphic to $E|_{X_n}$, and proposition 7.1.23 gives a bundle E'_n over X_n with $E|_{X_n} \oplus E'_n \cong X_n \times \mathbb{K}^M$. The sets X_n are open and disjoint, so the bundle E' equal to E'_n over each X_n is a vector bundle and the isomorphisms over the X_n assemble to $E \oplus E' \cong \underline{\mathbb{K}}^M$, completing the proof.

A rank-n bundle is the pullback of γ_n along a map into $BU(n)$ (theorem 7.1.29), and following that map by the stabilization $BU(n) \to BU(n+1)$ adds a trivial line (proposition 7.1.19(3)). A map into the colimit BU therefore records a bundle up to trivial summands, and over a compact space that is exactly the information in K^0, by the theorem below. A factor $\mathbb{Z}$ records the rank, which can be any integer once virtual bundles are allowed. For a locally constant function $r\colon X \to \mathbb{Z}_{\geq 0}$ write $\underline{\mathbb{C}}^{(r)}$ for the bundle that is $r^{-1}(j) \times \mathbb{C}^j$ over each open and closed set $r^{-1}(j)$.

Theorem 8.5.3: Representability of K^0 on Compact Spaces

Let X be a compact Hausdorff space and give $\mathbb{Z}$ the discrete topology.

1. Every map $h\colon X \to BU$ is a map into some $BU(N)$, and two maps $X \to BU(N)$ that are homotopic in BU are homotopic in some $BU(N')$.

2. A map $g = (d, h)\colon X \to \mathbb{Z} \times BU$ has locally constant d, and by (1) h is a map into some $BU(N)$ with $N \geq d$. The class

$$\Psi[g] = [h^*\gamma_N] - [\underline{\mathbb{C}}^{(N-d)}] \in K^0(X)$$

depends only on the homotopy class of g, and $\Psi\colon [X, \mathbb{Z} \times BU] \to K^0(X)$ is a bijection from unbased homotopy classes, natural in X, carrying d to the rank of $\Psi[g]$.

3. If X is connected, then for every basepoint Ψ restricts to a bijection from $[X, BU] = [X, \{0\} \times BU]$ onto $\tilde{K}^0(X)$, $[h] \mapsto [h^*\gamma_N] - N$.

4. The same holds for real bundles, with BO, KO^0 and $\widetilde{KO}^0$ in place of BU, K^0 and $\tilde{K}^0$.

***Proof*:**

Step 1: Part (1). The image of h is compact, so it lies in some $BU(N)$, which carries the subspace topology of BU (corollary 7.1.21). The same argument applied to a homotopy $X \times I \to BU$, whose domain is compact, gives the second claim.

Step 2: Ψ is well defined. A map $d\colon X \to \mathbb{Z}$ is locally constant, so it takes finitely many values by compactness and an $N \geq d$ exists. Replacing h by $i_{N+1} \circ h$ replaces $h^*\gamma_N$ by $\underline{\mathbb{C}} \oplus h^*\gamma_N$ (proposition 7.1.19(3)) and $\underline{\mathbb{C}}^{(N-d)}$ by $\underline{\mathbb{C}} \oplus \underline{\mathbb{C}}^{(N-d)}$, which leaves $\Psi[g]$ unchanged, so $\Psi[g]$ does not depend on N. A homotopy from (d,h) to (d',h') is constant in the discrete factor on each connected set $\{x\} \times I$, so $d = d'$, and by (1) it is a homotopy from h to h' in some $BU(N)$. Homotopic maps pull back isomorphic bundles (theorem 7.1.29, X being paracompact Hausdorff), so $\Psi[g] = \Psi[g']$. The rank of $\Psi[g]$ is $N - (N-d) = d$. Pullback along $u\colon X' \to X$ carries $h^*\gamma_N$ to $(h \circ u)^*\gamma_N$ and $\underline{\mathbb{C}}^{(N-d)}$ to $\underline{\mathbb{C}}^{(N-d\circ u)}$, which is naturality.

Step 3: Injectivity. Let $\Psi[(d,h)] = \Psi[(d',h')]$, with h, h' maps into one $BU(N)$, $N \geq d, d'$. Comparing ranks gives $d = d'$, and then $[h^*\gamma_N] = [h'^*\gamma_N]$. By definition 8.5.1 there is a bundle G with $h^*\gamma_N \oplus G \cong h'^*\gamma_N \oplus G$, and adding the bundle G' of lemma 8.5.2 with $G \oplus G' \cong \underline{\mathbb{C}}^M$ gives $h^*\gamma_N \oplus \underline{\mathbb{C}}^M \cong h'^*\gamma_N \oplus \underline{\mathbb{C}}^M$. By proposition 7.1.19(3) applied M times, the two sides are the pullbacks of γ_{N+M} along the composites of h and h' with the stabilization $BU(N) \to BU(N+M)$. These composites are therefore homotopic (theorem 7.1.29), so $h \simeq h'$ in BU and $(d,h) \simeq (d',h')$.

Step 4: Surjectivity. An element of $K^0(X)$ is $[E] - [F]$ for bundles E, F. By lemma 8.5.2 choose F' with $F \oplus F' \cong \underline{\mathbb{C}}^M$, so that $[E] - [F] = [E \oplus F'] - M$. Let r be the rank of $E \oplus F'$, which takes finitely many values, and $N \geq r$. The bundle $E \oplus F' \oplus \underline{\mathbb{C}}^{(N-r)}$ has rank N, so it is $h^*\gamma_N$ for some map $h\colon X \to BU(N)$ (theorem 7.1.29). With $d = r - M$, so that $N \geq d$ and $N - d = (N-r) + M$,

$$[E] - [F] = [h^*\gamma_N] - [\underline{\mathbb{C}}^{(N-r)}] - [\underline{\mathbb{C}}^M] = [h^*\gamma_N] - [\underline{\mathbb{C}}^{(N-d)}] = \Psi[(d,h)]$$

This proves (2).

Step 5: Parts (3) and (4). For connected X the function d is constant, and the rank of $\Psi[(d,h)]$ at x_0 is d. So $\Psi[(d,h)]$ lies in $\tilde{K}^0(X)$ exactly when $d = 0$, and then it is $[h^*\gamma_N] - N$. For (4) replace $\mathbb{C}$ by $\mathbb{R}$ and $BU(n)$ by $BO(n)$ throughout, using the real clauses of the cited results, completing the proof.

Compactness enters through (1) and through lemma 8.5.2, and it cannot be dropped. The CW complex BU is paracompact Hausdorff (theorem 7.1.7), so a bundle of rank N over it is pulled back from $BU(N)$. The identity of BU is homotopic to no map h into a single $BU(N)$: otherwise $c_{N+1} = h^*(c_{N+1}|_{BU(N)})$ in $H^{2N+2}(BU;\mathbb{Z})$, and c_{N+1} restricts to 0 on $BU(N)$ but is nonzero on BU (theorem 7.2.5, corollary 7.2.6). So over BU the class of the identity in $[BU, BU]$ is not the stabilized classifying map of any bundle. On infinite complexes K-theory is instead defined by maps into $\mathbb{Z} \times BU$, through the spectrum KU of theorem 8.5.10 below.

The space BU is simply connected, so a map $S^m \to BU$ needs no basepoint, and the sum of π_m, which pinches the sphere, matches the Whitney sum.

Corollary 8.5.4: Reduced K-Theory of Spheres

Every unitary group $U(N)$ and the space BU are path-connected, and BU is simply connected. For $m \geq 1$ the map

$$\pi_m(BU, b_0) \longrightarrow \tilde{K}^0(S^m), \qquad [h] \longmapsto [h^*\gamma_N] - N$$

for based maps $h\colon S^m \to BU(N)$, is an isomorphism of groups.

***Proof*:**

Step 1: The spaces. A unitary matrix is $g = Q\,\mathrm{diag}(e^{i\theta_1}, \dots, e^{i\theta_N})\,Q^{-1}$ with Q unitary, and replacing each θ_j by $t\theta_j$ joins it to 1, so $U(N)$ is path-connected. The Grassmannian $BU(N)$ is a classifying space of $U(N)$ (lemma 7.1.18(4)), so it is path-connected and $\pi_1(BU(N)) \cong \pi_0(U(N)) = 0$ (corollary 7.1.11(2)). All $BU(N)$ contain b_0, and a based loop in BU is compact, so it lies in some $BU(N)$ (corollary 7.1.21), where it is null-homotopic rel its endpoints. So BU is path-connected and simply connected.

Step 2: A bijection. A based map $S^m \to BU$ is a map into some $BU(N)$ by theorem 8.5.3(1). Forgetting basepoints is a bijection $\pi_m(BU, b_0) \to [S^m, BU]$ by proposition 5.1.15 and Step 1, and theorem 8.5.3(3) for the connected space S^m carries $[S^m, BU]$ bijectively onto $\tilde{K}^0(S^m)$. Write Ψ for these bijections.

Step 3: Additivity. Pullback along a map of compact Hausdorff spaces is a homomorphism of K-groups, and homotopic maps induce the same one, homotopic maps pulling back isomorphic bundles (theorem 7.1.29). A constant map induces 0 on $\tilde{K}^0$, as it factors through a point. Write $S^m = I^m/\partial I^m$, let ι_1, ι_2 be the inclusions of the summands of $S^m \vee S^m$, and let $P\colon S^m \to S^m \vee S^m$ send t to $\iota_1(2t_1, t_2, \dots, t_m)$ for $t_1 \leq \frac{1}{2}$ and to $\iota_2(2t_1 - 1, t_2, \dots, t_m)$ for $t_1 \geq \frac{1}{2}$. For based $f, g\colon S^m \to BU$ and $F\colon S^m \vee S^m \to BU$ equal to f on the first summand and g on the second, $F \circ P$ represents $[f] + [g]$ (definition 5.1.1). Let $c_1, c_2\colon S^m \vee S^m \to S^m$ collapse the second, respectively the first, summand to the basepoint. Then $c_1 \circ P$ represents $[\mathrm{id}] + 0 = [\mathrm{id}]$ in $\pi_m(S^m)$ (proposition 5.1.3), so $c_1 \circ P \simeq \mathrm{id}$, and likewise $c_2 \circ P \simeq \mathrm{id}$.

Restriction to the two summands is injective on $\tilde{K}^0(S^m \vee S^m)$. The wedge is a connected finite CW complex, hence compact Hausdorff (theorem 1.2.3), so by theorem 8.5.3(3) and its naturality it suffices that maps $G, G'\colon S^m \vee S^m \to BU$ with $G \circ \iota_j \simeq G' \circ \iota_j$ for $j = 1, 2$ are homotopic. The pair $(S^m \vee S^m, \{*\})$ has the homotopy extension property (proposition 1.3.4) and BU is path-connected, so G and G' may be taken based. Then $G \circ \iota_j$ and $G' \circ \iota_j$ are based and freely homotopic, hence homotopic rel the basepoint (proposition 5.1.15), and the two based homotopies agree at the wedge point and combine to $G \simeq G'$.

Let $x = \Psi[F]$ and $y = c_1^*\iota_1^*x + c_2^*\iota_2^*x$. The maps $c_2 \circ \iota_1$ and $c_1 \circ \iota_2$ are constant, so $\iota_j^*y = \iota_j^*x$ for $j = 1, 2$, and $x = y$. By naturality of Ψ,

$$\Psi[F \circ P] = P^*y = (c_1P)^*\iota_1^*x + (c_2P)^*\iota_2^*x = \iota_1^*x + \iota_2^*x = \Psi[f] + \Psi[g]$$

and Ψ is a homomorphism, completing the proof.

Example 8.5.5: K^0 of the Spheres S^2 and S^4

Take $X = S^2$. A complex vector bundle over S^2 is built by clutching: cover S^2 by two disks D_+, D_- meeting in an equatorial S^1, trivialize over each disk, and glue by a clutching function $S^1 \to U(n)$. Isomorphism classes of rank-n bundles correspond to π_0 of the space of such functions up to homotopy, that is to $\pi_1(U(n)) = \mathbb{Z}$ for $n \geq 1$ (and to $\pi_1(GL_n(\mathbb{C})) = \mathbb{Z}$ in the topological category). Thus rank-n bundles over S^2 are classified by an integer, their first Chern number. Passing to the stable range, the reduced group is

$$\tilde{K}^0(S^2) \cong \pi_1(U) \cong \mathbb{Z}$$

where $U = \varinjlim_n U(n)$ is the infinite unitary group of proposition 8.5.7. A generator is $[H] - [\underline{\mathbb{C}}]$, the class of the tautological (Hopf) line bundle H over $S^2 = \mathbb{C}\mathrm{P}^1$ minus a trivial line. The full ring is $K^0(S^2) = \mathbb{Z} \oplus \tilde{K}^0(S^2) = \mathbb{Z}[t]/(t^2)$ with $t = [H] - 1$ satisfying $t^2 = 0$. The relation $t^2 = 0$ holds because $(H \otimes H) \oplus \underline{\mathbb{C}} \cong H \oplus H$, as the clutching functions of the two bundles show [23, Example 1.13]. In $K^0(S^2)$ this reads $H^2 + 1 = 2H$, that is $t^2 = (H-1)^2 = 0$, and $K^0(S^2) \cong \mathbb{Z}[H]/(H-1)^2$ [23, Corollary 2.3]. On $\mathbb{C}\mathrm{P}^n$ the tautological line bundle obeys $(H-1)^{n+1} = 0$, as *Vector Bundles and K-Theory* [33, K-Theory of Complex Projective Space] shows.

For $X = S^4$ the same clutching analysis gives bundles classified by $\pi_3(U(n))$. By Bott periodicity of the homotopy groups of U ([28, Theorem 23.5], and *Vector Bundles and K-Theory* [33, Bott Periodicity]) $\pi_3(U) \cong \pi_1(U) \cong \mathbb{Z}$, so

$$\tilde{K}^0(S^4) \cong \pi_3(U) \cong \mathbb{Z}$$

generated by the class of a rank-2 bundle with second Chern number 1. In both cases the answer is $\mathbb{Z}$, reflecting the fact that $\tilde{K}^0(S^{2k}) \cong \mathbb{Z}$ for every $k \geq 0$ and $\tilde{K}^0(S^{2k+1}) = 0$, the 2-periodicity of K-theory seen on spheres.

By corollary 8.5.4, $\tilde{K}^0(S^m) \cong \pi_m(BU) = \pi_m(\mathbb{Z} \times BU)$ for $m \geq 1$, the homotopy groups taken at $(0, b_0)$. The reduced groups $\tilde{K}^0(S^{2k}) \cong \mathbb{Z}$ and $\tilde{K}^0(S^{2k+1}) = 0$ are therefore the homotopy groups $\pi_{2k}(\mathbb{Z} \times BU)$ and $\pi_{2k+1}(\mathbb{Z} \times BU)$. For $m \geq 1$ the clutching description of example 8.5.5 gives $\tilde{K}^0(S^m) \cong \pi_{m-1}(U)$, so $\pi_m(\mathbb{Z} \times BU) \cong \pi_{m-1}(U)$, a shift by one of the homotopy of U, whose pattern $\pi_0(U), \pi_1(U), \pi_2(U), \cdots = 0, \mathbb{Z}, 0, \mathbb{Z}, \ldots$ is $\mathbb{Z}$ in odd degrees and 0 in even. Equivalently the homotopy of $\mathbb{Z} \times BU$ itself runs $\mathbb{Z}, 0, \mathbb{Z}, 0, \ldots$ ($\mathbb{Z}$ in even degrees, 0 in odd), and that this pattern repeats with period 2 is the content of Bott's theorem, which is what turns the single group K^0 into a periodic cohomology theory.

The decisive input is Bott periodicity in its space-level form: the representing space is weakly homotopy equivalent to its own double loop space, which is the structure needed to make K^0 the degree-0 part of a 2-periodic cohomology theory. It is the composite of an equivalence $\mathbb{Z} \times BU \simeq \Omega U$ with the loops of an equivalence $U \simeq \Omega BU$, for the infinite unitary group U. The second is a delooping, proved from the frame bundles of the Grassmannians. The first is Bott's theorem proper. Bott proved it by Morse theory on the loop spaces of the unitary groups. Milnor's *Morse Theory* [28, §23, Theorem 23.3] proves the Morse-theoretic core: the inclusion of the space of minimal geodesics, the Grassmannian $\mathrm{Gr}_m(\mathbb{C}^{2m})$, into the space of paths from I to $-I$ in $SU(2m)$ is an isomorphism on π_i for $i \leq 2m$. May's *Concise Course* [25, Ch. 24, §2] outlines the space-level form, starting from Bott's map $BU \to \Omega SU$. Behrens [2, Theorem 3.1] replaces the Morse theory by an explicit quasifibration with contractible total space, and theorem 8.5.8 cites his proof. The periodicity isomorphism $\tilde{K}^0(X) \cong \tilde{K}^0(\Sigma^2 X)$ on compact spaces, which the equivalence induces, is proved by a

different route in *Vector Bundles and K-Theory* [33, Bott Periodicity].

These equivalences pass through spaces this book gives no CW structure: loop spaces, a colimit of matrix groups and a homotopy fiber. Each is therefore checked on homotopy groups at every basepoint, with lemma 5.3.2. A based weak homotopy equivalence induces one of loop spaces, and the proof uses the concatenation of loops, which carries every path component of a loop space onto the component of the constant loop.

Lemma 8.5.6: Looping Preserves Weak Equivalences

If $f\colon (Y, y_0) \to (Z, z_0)$ is a based weak homotopy equivalence, then $\Omega f\colon \Omega Y \to \Omega Z$, $\gamma \mapsto f \circ \gamma$, is a weak homotopy equivalence.

***Proof*:**
By proposition 5.2.17(4), $\Phi_0\colon \pi_{k+1}(Y, y_0) \to \pi_k(\Omega Y, c_{y_0})$ is a bijection for every $k \geq 0$, and item (2) of that proposition, applied to the map of pairs $f\colon (Y, \{y_0\}) \to (Z, \{z_0\})$, whose map of homotopy fibers is Ωf, gives $(\Omega f)_* \circ \Phi_0 = \Phi_0 \circ f_*$. So Ωf is bijective on path components and on every π_k at the constant loop. By lemma 5.3.2(2), applied to the restriction of Ωf to the path components of the constant loops, Ωf is bijective on homotopy groups at every loop in the component of c_{y_0}. Let $\omega \in \Omega Y$ be any loop, write μ for concatenation of loops and $\bar{\omega}$ for the reversed loop, and set $L(\gamma) = \mu(\bar{\omega}, \gamma)$ and $L'(\gamma) = \mu(\omega, \gamma)$. The associativity, inverse and unit homotopies of Step 1 of the proof of proposition 8.1.4 give $L \circ L' \simeq \mathrm{id}$ and $L' \circ L \simeq \mathrm{id}$, so L is a homotopy equivalence and, by lemma 5.3.2(1), bijective on every π_k. The inverse homotopy joins $L(\omega) = \mu(\bar{\omega}, \omega)$ to c_{y_0}, so Ωf is bijective on homotopy groups at $L(\omega)$. The map $L_f(\gamma) = \mu(f \circ \bar{\omega}, \gamma)$ of ΩZ is a homotopy equivalence for the same reason, and $\Omega f \circ L = L_f \circ \Omega f$, since $f \circ \mu(\bar{\omega}, \gamma) = \mu(f \circ \bar{\omega}, f \circ \gamma)$. Therefore $(\Omega f)_* = (L_f)_*^{-1} \circ (\Omega f)_* \circ L_*$ is bijective on $\pi_k(\Omega Y, \omega)$, completing the proof.

The delooping comes from the frame bundles $V_n(\mathbb{C}^\infty) \to BU(n)$ of lemma 7.1.18, principal $U(n)$-bundles with weakly contractible total spaces. Corollary 7.1.11 turns each into isomorphisms $\pi_i(BU(n)) \cong \pi_{i-1}(U(n))$ but produces no map of spaces, and the infinite unitary group, a colimit of the compact groups $U(n)$, is not locally compact, so the join construction of proposition 7.1.4 does not apply to it. The frame bundles are compatible with stabilization, their colimit is a Serre fibration over BU with fiber U, and the homotopy fiber of proposition 5.2.17 compares U with ΩBU by maps of spaces.

Proposition 8.5.7: The Infinite Unitary Group and the Loops on *BU*

Let s be the shift of proposition 7.1.19 and, for $n \geq 0$, let $\sigma_n \colon V_n(\mathbb{C}^\infty) \to V_{n+1}(\mathbb{C}^\infty)$, $\sigma_n(x_1, \dots, x_n) = (e_1, sx_1, \dots, sx_n)$, so that $q \circ \sigma_n = i_{n+1} \circ q$ for the frame bundles q of lemma 7.1.18(2). Let V_∞ be the colimit of the $V_n(\mathbb{C}^\infty)$ along the σ_n, in which a set is closed exactly when it meets every $V_n(\mathbb{C}^\infty)$ in a closed set, $q \colon V_\infty \to BU$ the induced map, $e_0 \in V_\infty$ the common image of the standard frames $(e_1, \dots, e_n)$, and $U = q^{-1}(b_0)$ the fiber over the basepoint b_0 of BU.

1. Identifying an orthonormal frame of $\mathbb{C}^n$ with the unitary matrix whose columns it lists, U is the colimit of $U(0) \to U(1) \to \cdots$ along $g \mapsto \operatorname{diag}(1, g)$, the *infinite unitary group*, with identity e_0. It is path-connected.
2. Let $F_\iota = \left\{(x, \gamma) \in U \times V_\infty^I \;\middle|\; \gamma(0) = x,\ \gamma(1) = e_0\right\}$ be the homotopy fiber of the inclusion $\iota \colon U \hookrightarrow V_\infty$ over e_0 (proposition 5.2.16), based at $\hat{e}_0 = (e_0, c_{e_0})$. Then

$$r \colon F_\iota \to U, \quad r(x, \gamma) = x, \qquad \text{and} \qquad q_\iota \colon F_\iota \to \Omega BU, \quad q_\iota(x, \gamma) = q \circ \gamma$$

   are based weak homotopy equivalences.

***Proof*:**

Step 1: The colimit V_∞. The formula for σ_n is continuous from $V_n(\mathbb{C}^M)$ to $V_{n+1}(\mathbb{C}^{M+1})$ for every M, so σ_n is continuous (definition 7.1.12). It is injective, and its image, the frames with first vector e_1, is closed, the first vector of a frame depending continuously on the frame (lemma 7.1.27(3)). On the image the inverse $(e_1, y_1, \dots, y_n) \mapsto (s^* y_1, \dots, s^* y_n)$, with $s^*(z_1, z_2, \dots) = (z_2, z_3, \dots)$ continuous on every $\mathbb{C}^M$, is continuous, so σ_n is a homeomorphism onto a closed subspace. Points of $V_n(\mathbb{C}^\infty)$ are closed, meeting each compact Hausdorff stage $V_n(\mathbb{C}^M)$ in at most one point. The arguments of Step 1 of the proof of lemma 7.1.15 and of Step 3 of the proof of lemma 7.1.18 then apply: each $V_n(\mathbb{C}^\infty)$ is a closed subspace of V_∞ carrying its own topology, and every compact subset of V_∞ lies in some $V_n(\mathbb{C}^\infty)$. The same holds for BU and the $BU(n)$ (corollary 7.1.21).

Step 2: Part (1). A point of the stage $V_m(\mathbb{C}^\infty)$ lies over b_0 exactly when it is a frame of $\mathbb{C}^m$, the basepoint of $BU(m)$ (definition 7.1.20), so $U \cap V_m(\mathbb{C}^\infty) = U(m)$, and σ_m sends the frame with columns g to the frame with columns $\operatorname{diag}(1, g)$. Points of the CW complex BU are closed, so U is closed in V_∞, and a subset of U is closed exactly when it meets every $U(m)$ in a closed set. This makes U the stated colimit. Each $U(m)$ is path-connected (corollary 8.5.4) and contains e_0, so U is path-connected.

Step 3: V_∞ is weakly contractible and q is a Serre fibration. A based map $S^k \to V_\infty$ has compact image, so by Step 1 it is a map into a stage containing the basepoint, where it is null-homotopic rel the basepoint (lemma 7.1.18(4)). Two points of V_∞ lie in a common stage, which is path-connected. So $\pi_k(V_\infty, v) = 0$ for all $k \geq 0$ and $v \in V_\infty$. Let $G \colon I^k \times I \to BU$ be a homotopy and $\tilde{g}_0 \colon I^k \to V_\infty$ a lift of $G(\,\cdot\,, 0)$. By Step 1 there is an m such that $\tilde{g}_0$ is a map into $V_m(\mathbb{C}^\infty)$ and G a map into $BU(m)$. On $V_m(\mathbb{C}^\infty)$ the map q is the frame bundle $V_m(\mathbb{C}^\infty) \to BU(m)$, a fiber bundle and so a Serre fibration (theorem 5.2.4), and a lift of G there is a lift in V_∞. So q is a Serre fibration with fiber U over b_0.

Step 4: Part (2). By proposition 5.2.17(1) for the pair (V_∞, U, e_0), $\Phi \colon \pi_n(V_\infty, U, e_0) \to \pi_{n-1}(F_\iota, \hat{e}_0)$ is a bijection for $n \geq 1$, and by item (3) of that proposition $r_* \circ \Phi = \partial$, the boundary map of the pair. For $n \geq 2$ the long exact sequence of the pair (proposition 5.1.11)

and Step 3 make $\partial\colon \pi_n(V_\infty, U, e_0) \to \pi_{n-1}(U, e_0)$ bijective. For $n = 1$ both sets are single points: $\pi_1(V_\infty, U, e_0) \cong \pi_1(BU, b_0) = 0$ by proposition 5.2.11 and corollary 8.5.4, and U is path-connected. Hence r_* is bijective on π_k at $\hat{e}_0$ for every $k \geq 0$, and F_ι is path-connected. The map q is a map of pairs $(V_\infty, U, e_0) \to (BU, \{b_0\}, b_0)$, and its map of homotopy fibers in proposition 5.2.17(2) is $(x, \gamma) \mapsto (b_0, q \circ \gamma)$, which is q_ι under the identification of item (4). Item (2) gives $(q_\iota)_* \circ \Phi = \Phi_0 \circ q_*$, where $q_*\colon \pi_n(V_\infty, U, e_0) \to \pi_n(BU, b_0)$ is the bijection of proposition 5.2.11 and $\Phi_0\colon \pi_n(BU, b_0) \to \pi_{n-1}(\Omega BU, c_{b_0})$ the bijection of item (4). So q_ι is bijective on π_k at $\hat{e}_0$ for every $k \geq 0$, and ΩBU is path-connected. By lemma 5.3.2(2), r and q_ι are weak homotopy equivalences, completing the proof.

The equivalence $\mathbb{Z} \times BU \simeq \Omega U$ is Bott's theorem proper, and Behrens proves it with an explicit quasifibration. The exponential $A \mapsto \exp(2\pi i A)$ carries the Hermitian matrices with spectrum in $[0, 1]$, a convex and hence contractible set, onto the unitary group. Its fiber over the identity consists of the matrices with eigenvalues 0 and 1 only, the orthogonal projections. A projection is a subspace, and the subspaces of $\mathbb{C}^{2N}$, stabilized in N, form $\mathbb{Z} \times BU$, the integer being the dimension minus N. Exponentiating the straight segment from a fixed projection to a projection A gives the loop in U attached to A.

Theorem 8.5.8: The Loop Space of U

For $N \geq 1$ let $P_N \in U(2N)$ be the permutation matrix with $P_N e_{N+1-j} = e_{2N+2-2j}$ and $P_N e_{N+j} = e_{2N+1-2j}$ for $1 \leq j \leq N$, and write π_W for the orthogonal projection of $\mathbb{C}^{2N}$ onto a subspace W. Every point of $\mathbb{Z} \times BU$ is $(n - N, V)$ for some $N \geq 1$ and some $V \in \mathrm{Gr}_n(\mathbb{C}^{2N}) \subseteq BU(n)$, and

$$\beta_U(n - N, V)(t) = P_N \exp\big(2\pi i\big((1-t)\pi_{\mathbb{C}^N} + t\,\pi_V\big)\big) P_N^{-1}, \qquad t \in I$$

defines a based map $\beta_U\colon \mathbb{Z} \times BU \to \Omega U$, carrying $(0, b_0)$ to the constant loop at the identity, that is a weak homotopy equivalence. Here $U(2N) \subseteq U$ as in proposition 8.5.7(1).

***Proof*:**
Step 1: Behrens's spaces. Behrens indexes his colimits by the finite-dimensional subspaces W of an inner product space with orthonormal basis $u_1, u_2, \ldots$, and the chain $W_N = \mathrm{span}(u_1, \ldots, u_N)$ is cofinal, so they may be taken along it. Identify $W_N \oplus W_N$ with $\mathbb{C}^{2N}$ by $(u_j, 0) \mapsto e_{N+1-j}$ and $(0, u_j) \mapsto e_{N+j}$. The inclusion $W_N \oplus W_N \subseteq W_{N+1} \oplus W_{N+1}$ becomes the shift $s\colon \mathbb{C}^{2N} \to \mathbb{C}^{2N+2}$, with $(u_{N+1}, 0) \mapsto e_1$ and $(0, u_{N+1}) \mapsto e_{2N+2}$. Write $\mathrm{diag}(a, X, b)$ for the matrix acting as a on e_1, as sXs^* on $s(\mathbb{C}^{2N})$ and as b on e_{2N+2}. Behrens's spaces become the following.

- His group $\widehat{U}$ is the colimit of the $U(2N)$ along $X \mapsto \mathrm{diag}(1, X, 1)$.
- His space $\widehat{E}$ is the colimit along $A \mapsto \mathrm{diag}(1, A, 0)$ of the compact convex sets $\widehat{E}_N$ of Hermitian matrices $A \in M_{2N}(\mathbb{C})$ with spectrum in $[0, 1]$, and his map is $\hat{p}\colon \widehat{E} \to \widehat{U}$, $\hat{p}(A) = \exp(2\pi i A)$.
- A Hermitian A with spectrum in $[0, 1]$ has $\exp(2\pi i A) = 1$ exactly when its eigenvalues lie in $\{0, 1\}$, that is, when $A = \pi_V$ for $V = \ker(A - 1)$ [2, Lemma 3.2]. As $\mathrm{diag}(1, \pi_V, 0) = \pi_{\mathbb{C}e_1 \oplus sV}$, the fiber $\hat{p}^{-1}(1)$ is the colimit of the spaces $\coprod_n \mathrm{Gr}_n(\mathbb{C}^{2N})$ along the stabilizations i_{n+1} of proposition 7.1.19, Behrens's $BU \times \mathbb{Z}$.

Behrens proves, by the Dold-Thom criterion, that $\hat{p}$ is a quasifibration [2, Theorem 3.1]: for every $X \in \widehat{U}$ and $A \in \hat{p}^{-1}(X)$ the map $\hat{p}_*\colon \pi_n(\widehat{E}, \hat{p}^{-1}(X), A) \to \pi_n(\widehat{U}, X)$ is bijective for every $n \geq 1$. His Lemma 3.4 rests on the representability of reduced K-theory on spheres, which is corollary 8.5.4.

Step 2: The fiber and the loops on $\widehat{U}$. Behrens contracts $\widehat{E}$ by $A \mapsto tA$, which does not commute with $A \mapsto \operatorname{diag}(1, A, 0)$. The homotopy $h_t(A) = (1-t)\pi_{\mathbb{C}^N} + tA$ on $\widehat{E}_N$ does, because $\operatorname{diag}(1, \pi_{\mathbb{C}^N}, 0) = \pi_{\mathbb{C}^{N+1}}$, and it stays in the convex set $\widehat{E}_N$. It is continuous on $\widehat{E} \times I$, which carries the colimit topology of the $\widehat{E}_N \times I$ [16, Products of Quotient Maps], and it contracts $\widehat{E}$ to the point $\pi_{\mathbb{C}^N}$ of $\hat{p}^{-1}(1)$. Let $P\widehat{U}$ be the space of paths in $\widehat{U}$ starting at 1 and $\mathrm{ev}\colon P\widehat{U} \to \widehat{U}$ the endpoint map, a Hurewicz fibration with fiber $\Omega\widehat{U}$ and contractible total space (example 5.2.15). The map $\Theta\colon \widehat{E} \to P\widehat{U}$, $\Theta(A)(t) = \hat{p}(h_t(A))$, is continuous by the exponential law [16, The Exponential Law] and satisfies $\mathrm{ev} \circ \Theta = \hat{p}$, so it is a map of pairs $(\widehat{E}, \hat{p}^{-1}(1)) \to (P\widehat{U}, \Omega\widehat{U})$, restricting on $\hat{p}^{-1}(1)$ to

$$b\colon \hat{p}^{-1}(1) \longrightarrow \Omega\widehat{U}, \qquad b(\pi_V)(t) = \exp\bigl(2\pi i((1-t)\pi_{\mathbb{C}^N} + t\pi_V)\bigr)$$

For $A \in \hat{p}^{-1}(1)$ and $n \geq 1$, $\hat{p}_* = \mathrm{ev}_* \circ \Theta_*$ on $\pi_n(\widehat{E}, \hat{p}^{-1}(1), A)$, where $\hat{p}_*$ is bijective by Theorem 3.1 of Behrens and ev_* by proposition 5.2.11, so Θ_* is bijective on these relative groups. The boundary map of a pair restricts a relative class to its bottom face, so $b_* \circ \partial = \partial \circ \Theta_*$. Both total spaces have trivial homotopy groups, so the long exact sequences of the two pairs (proposition 5.1.11) make both boundary maps bijective for $n \geq 2$ and onto the path components for $n = 1$. Hence b is bijective on π_k at every point of $\hat{p}^{-1}(1)$ for $k \geq 1$. For path components take the basepoint $A = \pi_{\mathbb{C}^N}$, which b sends to the constant loop, and at which the boundary map of the path fibration is bijective for $n = 1$ as well: up to reparametrization it is the bijection $\Phi_0\colon \pi_1(\widehat{U}) \to \pi_0(\Omega\widehat{U})$ of proposition 5.2.17(4) for $n = 1$ (example 5.2.15). Then $b_* \circ \partial = \partial \circ \Theta_*$ is onto, so b_* is onto $\pi_0(\Omega\widehat{U})$, and b_* is injective because the boundary map of $(\widehat{E}, \hat{p}^{-1}(1))$ is onto $\pi_0(\hat{p}^{-1}(1))$ while Θ_* and the boundary map of the path fibration are injective. So b is a weak homotopy equivalence.

Step 3: The fiber is $\mathbb{Z} \times BU$. Points of $\widehat{U}$ are closed, meeting each compact stage in at most one point, so $\hat{p}^{-1}(1)$ is closed in $\widehat{E}$ and carries the colimit topology of its compact stages, each homeomorphic to $\coprod_n \mathrm{Gr}_n(\mathbb{C}^{2N})$ through $V \mapsto \pi_V$, a continuous injection of a compact space into a Hausdorff one (definition 7.1.12 and the remarks after it). Let $\Lambda(\pi_V) = (n - N, V)$ for $V \in \mathrm{Gr}_n(\mathbb{C}^{2N})$. It respects the stage maps, $i_{n+1}(V)$ being the same point of BU as V and $(n+1) - (N+1) = n - N$, so $\Lambda\colon \hat{p}^{-1}(1) \to \mathbb{Z} \times BU$ is continuous. It is onto: a point (k, x) with $x \in \mathrm{Gr}_m(\mathbb{C}^M) \subseteq BU(m)$ is $\Lambda(\pi_{i^j(x)})$, where i^j is the composite of j stabilizations and $j \geq 0$ is so large that $N = m + j - k \geq 1$ and $M + j \leq 2N$, for then $i^j(x) \in \mathrm{Gr}_{m+j}(\mathbb{C}^{2N})$. This proves the first assertion. It is injective: two points with the same image, moved to a common stage N, are π_V and $\pi_{V'}$ with $V, V' \in \mathrm{Gr}_n(\mathbb{C}^{2N})$ equal in $BU(n)$, so $V = V'$. A set is closed in $\mathbb{Z} \times BU$ exactly when it meets every $\{k\} \times \mathrm{Gr}_m(\mathbb{C}^M)$ in a closed set (definition 7.1.12, definition 7.1.20, corollary 7.1.21), and there Λ^{-1} is $(k, x) \mapsto \pi_{i^j(x)}$ for one such j, continuous by proposition 7.1.19(1). So Λ is a homeomorphism, with $\Lambda(\pi_{\mathbb{C}^N}) = (0, b_0)$.

Step 4: The group is U. On $\mathbb{C}^{2N}$ the permutations satisfy $P_{N+1}s = s^2P_N$, $P_{N+1}e_1 = e_2$ and $P_{N+1}e_{2N+2} = e_1$, so $P_{N+1}\operatorname{diag}(1, X, 1)\,P_{N+1}^{-1}$ fixes e_1 and e_2 and acts as $s^2(P_NXP_N^{-1})s^{*2}$ on the remaining coordinates. That is the image of $P_NXP_N^{-1}$ under two stabilizations $g \mapsto \operatorname{diag}(1, g)$ of U. The even stages $U(2N)$ of U are cofinal, a set meeting every $U(2N)$ in a closed set meeting every $U(2N+1) \subseteq U(2N+2)$ in a closed set. Hence $\rho(X) = P_NXP_N^{-1}$ on $U(2N)$ defines a

continuous bijection $\rho\colon \widehat{U} \to U$, with continuous inverse given by conjugation by P_N^{-1} on $U(2N)$. So ρ is a homeomorphism fixing 1, and $\Omega\rho$ is a homeomorphism.

Step 5: Conclusion. The map $\beta_U = \Omega\rho \circ b \circ \Lambda^{-1}$ is given by the stated formula. It is well defined and continuous, and a weak homotopy equivalence as b is one and Λ and $\Omega\rho$ are homeomorphisms. At $(0, b_0) = \Lambda(\pi_{\mathbb{C}^N})$ the loop is constant at $\exp(2\pi i \pi_{\mathbb{C}^N}) = 1$, completing the proof.

The two equivalences compose to $\mathbb{Z} \times BU \simeq \Omega U \simeq \Omega^2 BU$, the second through the homotopy fiber F_ι of proposition 8.5.7, out of which both maps r and q_ι run. Maps out of a CW complex see only weak homotopy type (proposition 5.3.16), and $\mathbb{Z} \times BU$ is a CW complex, so this zigzag yields a single based map.

Corollary 8.5.9: The Bott Equivalence

Base $\mathbb{Z} \times BU$ at $(0, b_0)$. There is a based map $\beta\colon \mathbb{Z} \times BU \to \Omega^2(\mathbb{Z} \times BU)$ that is a weak homotopy equivalence.

***Proof*:**
A based loop in $\mathbb{Z} \times BU$, and a based loop of such loops, has connected image containing the basepoint, so it lies in the open and closed set $\{0\} \times BU$. Hence $\Omega(\mathbb{Z} \times BU) = \Omega BU$ and $\Omega^2(\mathbb{Z} \times BU) = \Omega^2 BU$. By proposition 8.5.7(2) and lemma 8.5.6, $\Omega r\colon \Omega F_\iota \to \Omega U$ and $\Omega q_\iota\colon \Omega F_\iota \to \Omega^2 BU$ are based weak homotopy equivalences. The space $\mathbb{Z} \times BU$ is a CW complex, a disjoint union of copies of BU (corollary 7.1.21), and its basepoint is a 0-cell, the point $BU(0)$. By proposition 5.3.16, $(\Omega r)_*\colon [\mathbb{Z} \times BU, \Omega F_\iota]_* \to [\mathbb{Z} \times BU, \Omega U]_*$ is a bijection, so there is a based map $\gamma\colon \mathbb{Z} \times BU \to \Omega F_\iota$ with $\Omega r \circ \gamma$ homotopic rel the basepoint to the map β_U of theorem 8.5.8. By lemma 5.3.2(1), $\Omega r \circ \gamma$ is a weak homotopy equivalence, and so is γ, its maps on homotopy groups being $(\Omega r)_*^{-1} \circ (\Omega r \circ \gamma)_*$. Put $\beta = \Omega q_\iota \circ \gamma$, a composite of based weak homotopy equivalences, completing the proof.

Theorem 8.5.10: The K-theory Spectrum KU

Base the K-theory representing space $\mathbb{Z} \times BU$ at $(0, b_0)$, and let

$$\beta \colon \mathbb{Z} \times BU \longrightarrow \Omega^2(\mathbb{Z} \times BU)$$

be the based weak homotopy equivalence of corollary 8.5.9. Set $KU_n = \mathbb{Z} \times BU$ for n even and $KU_n = \Omega(\mathbb{Z} \times BU)$ for n odd, and let the adjoint structure map $\tilde{\sigma}_n \colon KU_n \to \Omega KU_{n+1}$ be β for n even and the identity of $\Omega(\mathbb{Z} \times BU)$ for n odd. Then:

1. The collection $KU = \{KU_n\}_{n \geq 0}$ is an Ω-spectrum in the sense of definition 8.4.4.
2. The reduced cohomology theory $\widetilde{K}^*$ of KU (theorem 8.4.6) is 2-periodic: for $n \geq 0$ the functors $\widetilde{K}^n = [-, KU_n]_*$ and $\widetilde{K}^{n+2}$ are equal, with equal suspension isomorphisms. Redefining $\widetilde{K}^n := \widetilde{K}^{n+2k}$ for $n < 0$, where $n + 2k \geq 0$, with the suspension isomorphisms of degree $n + 2k$, gives a naturally isomorphic reduced cohomology theory, unchanged in degrees $n \geq 0$, in which $\widetilde{K}^n = \widetilde{K}^{n+2}$ for every $n \in \mathbb{Z}$. From here on $\widetilde{K}^*$ denotes this periodic theory.
3. The coefficient groups are

$$K^n(\mathrm{pt}) = \tilde{K}^n(S^0) \cong \begin{cases} \mathbb{Z} & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$$

In particular the dimension axiom of definition 8.2.6 fails: $K^n(\mathrm{pt}) \neq 0$ for infinitely many $n \neq 0$.

***Proof*:**

Step 1: An Ω-spectrum. By definition 8.4.4 each adjoint structure map must be a weak homotopy equivalence. For n odd, $\tilde{\sigma}_n$ is the identity of $\Omega(\mathbb{Z} \times BU) = \Omega KU_{n+1}$. For n even, $\tilde{\sigma}_n$ is $\beta \colon KU_n \to \Omega^2(\mathbb{Z} \times BU) = \Omega KU_{n+1}$, a weak homotopy equivalence by corollary 8.5.9. This proves (1).

Step 2: Periodicity. For $n \geq 0$, $KU_{n+2} = KU_n$ and $\tilde{\sigma}_{n+2} = \tilde{\sigma}_n$. The group law of $\widetilde{K}^n(X) = [X, KU_n]_*$ is transported along $(\tilde{\sigma}_n)_*$, and its suspension isomorphism is $(\tilde{\sigma}_n)_*$ followed by the loop-suspension adjunction (Steps 1 and 4 of the proof of theorem 8.4.6). So $\widetilde{K}^{n+2} = \widetilde{K}^n$ as functors to groups, with equal suspension isomorphisms. Hence the redefinition $\widetilde{K}^n := \widetilde{K}^{n+2k}$ for $n < 0$ does not depend on k. In each degree it is one of the functors $\widetilde{K}^m$, $m \geq 0$, with its suspension isomorphism, so it satisfies the axioms of definition 8.2.1, and $\widetilde{K}^n = \widetilde{K}^{n+2}$ in every degree. For $n < 0$ the theory of theorem 8.4.6 has $\widetilde{K}^0(\Sigma^{-n} X)$ in degree n, with identity suspension isomorphisms, and the $(-n)$-fold suspension isomorphism of the periodic theory, $\widetilde{K}^{n+2k}(X) \to \widetilde{K}^{2k}(\Sigma^{-n} X) = \widetilde{K}^0(\Sigma^{-n} X)$, is a natural isomorphism onto it that commutes with the suspension isomorphisms. Together with the identity in degrees $n \geq 0$ it is the isomorphism of (2). The double suspension $\widetilde{K}^0(X) \to \widetilde{K}^2(\Sigma^2 X) = \widetilde{K}^0(\Sigma^2 X)$ sends $[f]$ to the double adjoint of $\beta \circ f$, because $\tilde{\sigma}_0 = \beta$ and $\tilde{\sigma}_1$ is the identity. This proves (2).

Step 3: The coefficient groups. The coefficients of a reduced theory are the groups $\widetilde{K}^n(S^0)$, and by (2) it suffices to take $n \geq 0$. A based map $S^0 \to Y$ is a point of Y and a based homotopy is a

path, so $\widetilde{K}^n(S^0) = [S^0, KU_n]_*$ is the set $\pi_0(KU_n)$, with the group law transported along $(\tilde{\sigma}_n)_*$ from $[S^0, \Omega KU_{n+1}]_* = \pi_0(\Omega KU_{n+1})$, where it is concatenation of loops (Step 1 of the proof of theorem 8.4.6). A path component of a loop space ΩY is a based homotopy class of loops in Y, and concatenation is the product of $\pi_1(Y)$, so $\pi_0(\Omega Y) \cong \pi_1(Y)$ as groups (proposition 5.2.17(4) with $n = 1$). For n odd, $\tilde{\sigma}_n$ is the identity and $\widetilde{K}^n(S^0) \cong \pi_1(\mathbb{Z} \times BU) = \pi_1(BU) = 0$, as BU is simply connected (corollary 8.5.4). For n even, $\tilde{\sigma}_n = \beta$ and

$$\widetilde{K}^n(S^0) \cong \pi_0\big(\Omega(\Omega BU)\big) \cong \pi_1(\Omega BU, c_{b_0}) \cong \pi_2(BU, b_0)$$

the last isomorphism by proposition 5.2.17(4) with $n = 2$. It remains to see $\pi_2(BU, b_0) \cong \mathbb{Z}$. For $m \geq 2$ the sphere bundle $\rho\colon S(\gamma_m) \to BU(m)$ of lemma 7.2.4 is a Serre fibration (theorem 5.2.4) with fiber S^{2m-1}, whose π_1 and π_2 vanish (corollary 5.4.11), so its long exact sequence (theorem 5.2.12) makes ρ_* bijective on π_2. The homotopy equivalence j of that lemma satisfies $\rho \circ j = i_m$ and is bijective on π_2 (lemma 5.3.2(1)), so the stabilization $i_m\colon BU(m-1) \to BU(m)$ is bijective on π_2. A based map $S^2 \to BU$, and a based homotopy between two such, lies in some $BU(m)$ (corollary 7.1.21), so the inclusion $BU(1) \subseteq BU$ is bijective on π_2. Finally $BU(1) = \mathbb{C}\mathrm{P}^\infty$ has $\pi_2(\mathbb{C}\mathrm{P}^\infty) \cong \mathbb{Z}$ (example 7.1.6(3)). Hence, as groups,

$$K^n(\mathrm{pt}) \cong \begin{cases} \pi_2(BU) \cong \mathbb{Z} & n \text{ even} \\ \pi_1(BU) = 0 & n \text{ odd} \end{cases}$$

Because $K^n(\mathrm{pt}) = \mathbb{Z} \neq 0$ for every even n, the dimension axiom of definition 8.2.6, which requires $\tilde{K}^n(S^0) = 0$ for $n \neq 0$, fails, completing the proof.

8.5.11 Remark: The Coefficient Ring of K-theory The tensor product of bundles makes the coefficient groups of theorem 8.5.10(3) a ring. The suspension isomorphisms give $K^{-2k}(\mathrm{pt}) \cong \widetilde{K}^0(S^{2k})$, which for $k \geq 1$ is the set $[S^{2k}, \mathbb{Z} \times BU]_* = \pi_{2k}(BU)$. Its group law, transported along β from loop concatenation on $[S^{2k}, \Omega^2(\mathbb{Z} \times BU)]_*$, is the law of π_{2k}. On $[S^{2k}, \Omega Y]_*$ the sum of π_{2k}, which concatenates in the first cube coordinate, and the loop sum of proposition 8.1.4 satisfy the interchange law and have the constant map as common unit, so they coincide by the argument of Step 3 of the proof of that proposition, and β_* is a homomorphism for the sums of π_{2k} (proposition 5.1.6). By corollary 8.5.4 the group $\pi_{2k}(BU)$ is the group of rank-zero virtual bundles on S^{2k}. The reduced external product $\widetilde{K}^0(S^{2j}) \otimes \widetilde{K}^0(S^{2k}) \to \widetilde{K}^0(S^{2j} \wedge S^{2k}) = \widetilde{K}^0(S^{2j+2k})$ is the product $a * b = p_1^*(a)\, p_2^*(b)$, built from the tensor product of bundles, carried to the smash product [23, §2.2]. The *Bott class* $\beta = [H] - 1$ generates $\widetilde{K}^0(S^2) \cong K^{-2}(\mathrm{pt})$ [23, Corollary 2.3]. Multiplication by β is the Bott map $\widetilde{K}^0(X) \to \widetilde{K}^0(\Sigma^2 X)$, $\alpha \mapsto \beta * \alpha$, an isomorphism for every compact Hausdorff space X [23, Theorem 2.11], and the k-fold product $\beta^k = \beta * \cdots * \beta$ generates $\widetilde{K}^0(S^{2k}) \cong K^{-2k}(\mathrm{pt})$ [23, Corollary 2.12], so $\bigoplus_{k \geq 0} K^{-2k}(\mathrm{pt})$ is the polynomial ring $\mathbb{Z}[\beta]$. The periodic grading extends this to the Laurent ring $K^*(\mathrm{pt}) = \mathbb{Z}[\beta, \beta^{-1}]$ with $|\beta| = -2$. The same facts are developed in *Vector Bundles and K-Theory* [33, The Bott Generator and Bott Map, Bott Periodicity Isomorphism, K-Theory is a Generalized Cohomology Theory]. Products on the spectrum-level theory of infinite complexes would need a ring-spectrum structure on KU, a topic of *Stable Homotopy Theory* [15], and no argument here uses one. The class β shares its letter with the space-level map β of theorem 8.5.10, and no argument here relates the two.

The theorem is the precise sense in which Bott periodicity converts the bundle-theoretic invariant K^0, represented on compact spaces by $\mathbb{Z} \times BU$ (theorem 8.5.3), into a cohomology theory. The equivalence $\Omega^2(\mathbb{Z} \times BU) \simeq \mathbb{Z} \times BU$ is what lets the single representing space generate an entire Ω-spectrum, and it forces the coefficient ring to be periodic. The contrast with ordinary cohomology is sharp: where $H^*(\mathrm{pt}; \mathbb{Z}) = \mathbb{Z}$ sits in degree zero and nothing else, $K^*(\mathrm{pt}) = \mathbb{Z}[\beta, \beta^{-1}]$ spreads a copy of $\mathbb{Z}$ across every even degree. That spreading is the source of K-theory's computational power: classes can be moved up or down by two degrees at will, and invariants like the Chern character $K^0(X) \otimes \mathbb{Q} \cong \bigoplus_k H^{2k}(X; \mathbb{Q})$ become available, relating K-theory to ordinary cohomology through a ring isomorphism after tensoring with $\mathbb{Q}$. Real K-theory KO is built the same way from $\mathbb{Z} \times BO$, with Bott periodicity of period 8, and its coefficient ring $KO^*(\mathrm{pt})$ contains 2-torsion in degrees 1 and 2. The eightfold pattern $\mathbb{Z}, \mathbb{Z}/2, \mathbb{Z}/2, 0, \mathbb{Z}, 0, 0, 0$ is the real analogue of the twofold $\mathbb{Z}, 0$, recorded in [33, Bott Periodicity], where the real case is cited to Atiyah.

The transition to cobordism keeps the same structural goal, a generalized cohomology theory represented by a spectrum, but changes the geometry entirely. Where K-theory measures bundles, cobordism measures manifolds: the relation that two closed n-manifolds are equivalent when they jointly bound a compact $(n+1)$-manifold. The representing object is no longer a Grassmannian but a Thom space, built from the universal bundle over $BO(n)$ by collapsing its boundary at infinity. The first task is to define the Thom space of a vector bundle and to organize the Thom spaces of the universal bundles into a spectrum.

Definition 8.5.12: Thom Space

Let $\pi\colon E \to B$ be a real vector bundle of rank k over a paracompact base, equipped with a fiber metric. The *disk bundle* $D(E)$ is the subspace of vectors of length ≤ 1, and $S(E)$ is the sphere bundle of definition 7.1.24, the vectors of length $= 1$. The *Thom space* of E is the quotient

$$\mathrm{Th}(E) \;=\; D(E)/S(E)$$

the disk bundle with its boundary sphere bundle collapsed to a single point, the basepoint ∞. Equivalently, $\mathrm{Th}(E)$ is the one-point compactification of the total space E when B is compact: each fiber $\mathbb{R}^k$ is compactified to S^k and all the points at infinity are identified. The Thom space of the trivial rank-k bundle over B is the k-fold reduced suspension $\mathrm{Th}(\underline{\mathbb{R}}^k) = \Sigma^k(B_+)$, where B_+ is B with a disjoint basepoint adjoined.

In particular, letting $\gamma^k \to BO(k)$ denote the universal rank-k real vector bundle over the classifying space $BO(k)$, the associated universal Thom spaces are denoted

$$MO(k) := \mathrm{Th}(\gamma^k)$$

Let $\gamma^k \to BO(k)$ be the universal rank-k real vector bundle over the infinite Grassmannian $BO(k) = \varinjlim_N \mathrm{Gr}_k(\mathbb{R}^N)$. The Thom spaces $MO(k)$ are linked by structure maps: the inclusion $BO(k) \hookrightarrow BO(k+1)$ classifying $\gamma^k \oplus \underline{\mathbb{R}}$ induces, on Thom spaces, a map $\Sigma\, MO(k) \to MO(k+1)$ (suspension corresponds to adding a trivial line summand, and $\mathrm{Th}(\gamma^k \oplus \underline{\mathbb{R}}) = \Sigma\, \mathrm{Th}(\gamma^k)$).

Definition 8.5.13: Thom Spectrum

Given the above suspension, the resulting sequence

$$MO \;=\; \{MO(k),\ \Sigma\, MO(k) \to MO(k+1)\}_{k\geq 0}$$

is the *Thom spectrum* MO (the unoriented cobordism spectrum). Replacing $BO(k)$ by the complex Grassmannian $BU(k)$ and the universal complex bundle $\gamma^k_{\mathbb{C}}$ (of real rank $2k$) gives the *complex cobordism spectrum* MU, with $MU(k) = \mathrm{Th}(\gamma^k_{\mathbb{C}})$ and structure maps $\Sigma^2 MU(k) \to MU(k+1)$.

The Thom space is the device that turns a bundle into a single pointed space carrying the bundle's twisting in its homotopy type. Collapsing the sphere bundle to a point records exactly the data of how the fibers are glued: a trivial bundle gives a suspension (the simplest possible Thom space), and the deviation of $\mathrm{Th}(E)$ from a suspension measures the nontriviality of E. The structure maps $\Sigma\, MO(k) \to MO(k+1)$ encode that adding a trivial direction to a bundle suspends its Thom space, which is precisely the compatibility a spectrum requires. The reason the universal bundles are used is that every rank-k bundle is pulled back from γ^k, so $MO(k)$ is the universal Thom space, and its homotopy is where all cobordism information collects. Unlike KU, the spectrum MO is not an Ω-spectrum on the nose (the structure maps are not equivalences). It is a suspension-type spectrum whose homotopy groups define the theory, and one passes to an associated Ω-spectrum by the spectrification of section 8.4 when a representing space is wanted.

Example 8.5.14: The Thom Space of the Möbius Bundle

Let $L \to S^1$ be the Möbius line bundle, the nontrivial real line bundle over the circle, pulled back from the universal line bundle $\gamma^1 \to BO(1) = \mathbb{R}\mathrm{P}^\infty$ along the inclusion $S^1 = \mathbb{R}\mathrm{P}^1 \hookrightarrow \mathbb{R}\mathrm{P}^\infty$. Its disk bundle $D(L)$ is the Möbius band, a compact surface with boundary, and its sphere bundle $S(L)$ is the boundary circle (a single circle, double-covering the base, not two circles as for the trivial bundle). The Thom space is

$$\mathrm{Th}(L) = D(L)/S(L) = (\text{Möbius band})/(\text{its boundary})$$

Collapsing the boundary circle of the Möbius band to a point yields the real projective plane:

$$\mathrm{Th}(L) \cong \mathbb{R}\mathrm{P}^2$$

Indeed $\mathbb{R}\mathrm{P}^2$ is exactly a Möbius band with a disk glued along its boundary, and collapsing the boundary to a point is the same as coning it off, which fills it with a disk that is then crushed. The cell structure $\mathbb{R}\mathrm{P}^2 = e^0 \cup e^1 \cup e^2$ matches $\mathrm{Th}(L)$, whose basepoint is e^0, the collapsed boundary contributes e^1, and the open Möbius band contributes e^2.

Contrast the trivial line bundle $\underline{\mathbb{R}} \to S^1$: its disk bundle is the cylinder $S^1 \times [-1,1]$, its sphere bundle is two circles, and its Thom space is $\mathrm{Th}(\underline{\mathbb{R}}) = \Sigma(S^1_+) = \Sigma(S^1 \sqcup \mathrm{pt}) = S^2 \vee S^1$. The Thom space of the trivial bundle is a suspension, as the definition predicts, while the Thom space $\mathbb{R}\mathrm{P}^2$ of the Möbius bundle is not a suspension: its mod-2 cohomology ring $H^*(\mathbb{R}\mathrm{P}^2; \mathbb{Z}/2) = \mathbb{Z}/2[\alpha]/(\alpha^3)$ has a nontrivial cup square $\alpha^2 \neq 0$ with $|\alpha| = 1$, whereas the cup product on the reduced cohomology of any suspension ΣY vanishes identically (every reduced class on ΣY is a suspension class, and suspension classes square to zero). The nontriviality of L is visible in the Thom space as exactly this failure to be a suspension.

Example 8.5.15: The Thom Class of the Möbius Bundle

Let $L \to S^1$ be the Möbius line bundle of example 8.5.14, which is non-orientable, so it has no integral Thom class. It does have a mod-2 one, and theorem 7.3.4 predicts $H^{k+1}(L, L_0; \mathbb{Z}/2) \cong H^k(S^1; \mathbb{Z}/2)$, that is $\mathbb{Z}/2$ for $k = 0$ and $k = 1$ and zero otherwise. This is confirmed by the identification $\mathrm{Th}(L) \cong \mathbb{R}\mathrm{P}^2$ of example 8.5.14: the reduced mod-2 cohomology of $\mathbb{R}\mathrm{P}^2$ is $\mathbb{Z}/2$ in degrees 1 and 2 and zero elsewhere, matching the prediction after the shift by the rank $n = 1$. The Thom class is the generator $\alpha \in H^1(\mathbb{R}\mathrm{P}^2; \mathbb{Z}/2)$, and the Thom isomorphism is multiplication by α, which carries the degree-0 generator to α and the degree-1 generator to $\alpha^2 \neq 0$. The nonvanishing of α^2, which is what distinguished $\mathbb{R}\mathrm{P}^2$ from a suspension in example 8.5.14, is exactly the statement that the Thom isomorphism is multiplication by a class whose square is nonzero, that is, that L is nontrivial.

The two examples show the Thom-space construction at work: a nontrivial bundle produces a nontrivial space ($\mathbb{R}\mathrm{P}^2$), and the nontriviality is detected by the cup product structure that distinguishes $\mathbb{R}\mathrm{P}^2$ from a suspension. The same phenomenon at the level of the universal bundles is what makes MO a nontrivial spectrum. With the spectrum in hand, Thom's theorem identifies the homotopy groups of MO with the unoriented bordism groups of manifolds.

Closed smooth n-manifolds M_0 and M_1 are cobordant (or *bordant*) when some compact smooth $(n+1)$-manifold W has $\partial W = M_0 \sqcup M_1$ (definition 7.6.6). Cobordism is an equivalence relation. Reflexivity uses the cylinder $W = M \times [0, 1]$, symmetry is the symmetry of disjoint union, and transitivity glues two cobordisms along their common boundary component (the result is smooth after smoothing the corner). The set of cobordism classes of closed n-manifolds, denoted $\mathfrak{N}_n$, is an abelian group under disjoint union, with identity the class of the empty manifold (equivalently, of any manifold that bounds) and with every element its own inverse, since $M \sqcup M = \partial(M \times [0, 1])$ bounds. Thus $\mathfrak{N}_n$ is a vector space over $\mathbb{F}_2$. The graded group $\mathfrak{N}_* = \bigoplus_n \mathfrak{N}_n$ is a ring under Cartesian product of manifolds. Cobordant manifolds have the same Stiefel-Whitney numbers by lemma 7.6.7.

The Pontryagin-Thom construction produces an isomorphism $\mathfrak{N}_n \cong \pi_n(MO)$ directly and geometrically, by turning a manifold into a map and a map into a manifold. The construction rests on two facts: transversality, which guarantees that a generic map can be perturbed to meet a submanifold cleanly, and the collapse map, which compresses a tubular neighborhood into a Thom space.

Theorem 8.5.16: The Pontryagin-Thom Isomorphism

For each $n \geq 0$ the Pontryagin-Thom construction gives mutually inverse bijections

$$\mathfrak{N}_n \underset{\Theta}{\overset{\mathrm{PT}}{\rightleftarrows}} \varinjlim_k \pi_{n+k}(MO(k)) = \pi_n(MO)$$

defined as follows. Given a closed n-manifold M, embed it (by Whitney embedding) in $\mathbb{R}^{n+k}$ for k large, with normal bundle ν of rank k. Let $V \cong$ total space of ν be a tubular neighborhood. The *collapse map*

$$c\colon S^{n+k} = \mathbb{R}^{n+k} \cup \{\infty\} \longrightarrow V/\partial V = \mathrm{Th}(\nu)$$

sends V to $\mathrm{Th}(\nu)$ by the tubular-neighborhood identification and everything outside V to the basepoint ∞. Composing with the classifying map $\mathrm{Th}(\nu) \to MO(k)$ of the normal bundle (which classifies ν as pulled back from γ^k) gives an element $\mathrm{PT}(M) \in \pi_{n+k}(MO(k))$, stable in k. Conversely, given $f\colon S^{n+k} \to MO(k)$, make f transverse to the zero section $BO(k) \subseteq MO(k)$. The preimage $\Theta(f) = f^{-1}(BO(k))$ is a closed n-manifold whose normal bundle is $f^*\gamma^k$. The two constructions are inverse on cobordism and homotopy classes.

***Proof*:**

The Thom transversality theorem (every smooth map is homotopic, by an arbitrarily small perturbation, to one transverse to a given submanifold, with the homotopy itself takeable transverse) is used in Steps 2 and 3. The full statement and proof, resting on Sard's theorem, are standard differential topology and are given in [27, §§1-7].

Step 1: The forward map PT *is well defined.* Let $M \subseteq \mathbb{R}^{n+k}$ be a closed embedded n-manifold with tubular neighborhood V, identified with the total space of the normal bundle ν (the tubular neighborhood theorem). The one-point compactification $\mathbb{R}^{n+k} \cup \infty = S^{n+k}$ admits the collapse map c that is the identity on V (sending it to $\mathrm{Th}(\nu)$ via $V \cong D(\nu)$ and $\partial V \mapsto \infty$) and sends the complement of V to ∞. This is continuous because the complement of V is closed and maps to the single point ∞, and on $\overline{V}$ the map is the quotient $D(\nu) \to D(\nu)/S(\nu)$. The normal bundle ν of rank k is classified by a map $g\colon M \to BO(k)$ with $g^*\gamma^k \cong \nu$, inducing on Thom spaces $\mathrm{Th}(g)\colon \mathrm{Th}(\nu) \to \mathrm{Th}(\gamma^k) = MO(k)$. The composite

$$S^{n+k} \xrightarrow{c} \mathrm{Th}(\nu) \xrightarrow{\mathrm{Th}(g)} MO(k)$$

is a based map $S^{n+k} \to MO(k)$, hence an element of $\pi_{n+k}(MO(k))$. Increasing the embedding dimension from $\mathbb{R}^{n+k}$ to $\mathbb{R}^{n+k+1}$ replaces ν by $\nu \oplus \underline{\mathbb{R}}$ and the map by its suspension, matching the spectrum structure map $\Sigma MO(k) \to MO(k+1)$. Hence the class is well defined in the colimit $\pi_n(MO)$.

Independence of the embedding: any two embeddings of M in $\mathbb{R}^{n+k}$ are isotopic for k large (Whitney), and an isotopy carries the tubular neighborhood and collapse map along, giving a homotopy of the resulting maps. Independence of the choice of tubular neighborhood and classifying map is the uniqueness up to homotopy of these choices. Hence $\mathrm{PT}(M) \in \pi_n(MO)$ depends only on M.

Cobordism invariance: suppose M_0 and M_1 are cobordant via $W \subseteq \mathbb{R}^{n+k} \times [0,1]$, a compact $(n+1)$-manifold with $\partial W = M_0 \times \{0\} \sqcup M_1 \times \{1\}$, embedded so that it meets the boundary

slabs orthogonally. The normal bundle of W has rank k, and the collapse-map construction applied to W gives a map

$$S^{n+k} \times [0,1] \longrightarrow MO(k)$$

restricting on the two ends to $\mathrm{PT}(M_0)$ and $\mathrm{PT}(M_1)$. This is a homotopy, so $\mathrm{PT}(M_0) = \mathrm{PT}(M_1)$ in $\pi_n(MO)$. Therefore PT descends to a map $\mathfrak{N}_n \to \pi_n(MO)$.

Step 2: The backward map Θ is well defined. Let $f\colon S^{n+k} \to MO(k)$ be a based map. The Grassmannian $BO(k)$ sits inside $MO(k) = \mathrm{Th}(\gamma^k)$ as the zero section (the image of B under $B \to D(E) \to \mathrm{Th}(E)$), a closed submanifold of the manifold part of $MO(k)$ away from the basepoint. By the Thom transversality theorem, f is homotopic to a smooth map f' transverse to $BO(k)$. Near the part of S^{n+k} mapping into a neighborhood of $BO(k)$ the map f' is smooth, and transversality makes

$$\Theta(f) := (f')^{-1}(BO(k)) \subseteq S^{n+k}$$

a smooth closed submanifold of dimension $(n+k) - k = n$ (transversality: the preimage of a codimension-k submanifold under a transverse map is codimension k). Its normal bundle in S^{n+k} is $(f')^*$(normal bundle of $BO(k)$ in $MO(k)$) $= (f')^*\gamma^k$, since the normal bundle of the zero section B in $\mathrm{Th}(E)$ is E itself. The manifold $\Theta(f)$ lies in $\mathbb{R}^{n+k} = S^{n+k} \setminus \infty$ (the basepoint ∞ maps to the basepoint of $MO(k)$, not into $BO(k)$, so $\infty \notin \Theta(f)$), giving a closed n-manifold embedded in Euclidean space.

Independence of the choice of transverse representative: two maps f', f'' transverse to $BO(k)$ and homotopic to f are homotopic to each other, and the homotopy $H\colon S^{n+k} \times [0,1] \to MO(k)$ can itself be taken transverse to $BO(k)$ (relative transversality), so $H^{-1}(BO(k))$ is a cobordism between $\Theta(f')$ and $\Theta(f'')$. Thus $\Theta(f) \in \mathfrak{N}_n$ depends only on the homotopy class of f, giving a map $\Theta\colon \pi_n(MO) \to \mathfrak{N}_n$.

Step 3: The maps are mutually inverse. First, $\Theta \circ \mathrm{PT} = \mathrm{id}$. Start with a closed n-manifold $M \subseteq \mathbb{R}^{n+k}$, form $\mathrm{PT}(M) = \mathrm{Th}(g) \circ c$. The zero section $BO(k) \subseteq MO(k)$ pulls back under $\mathrm{Th}(g) \circ c$ to exactly M: the collapse map c sends M (the zero section of ν inside V) to the zero section of $\mathrm{Th}(\nu)$, and $\mathrm{Th}(g)$ sends the zero section of $\mathrm{Th}(\nu)$ to the zero section $BO(k)$, while everything off M either lies off the zero section or is sent to the basepoint. The composite is already transverse to $BO(k)$ along M (the collapse map is a diffeomorphism on a neighborhood of M, and g is the classifying map, so the differential is surjective onto the normal directions), so $\Theta(\mathrm{PT}(M)) = M$ as a manifold, hence as a cobordism class.

Second, $\mathrm{PT} \circ \Theta = \mathrm{id}$. Start with $f\colon S^{n+k} \to MO(k)$, taken transverse to $BO(k)$, and set $N = \Theta(f) = f^{-1}(BO(k))$, a closed n-manifold with normal bundle $\nu_N = f^*\gamma^k$. Near N the map f looks like the composite of the projection of a tubular neighborhood of N onto N followed by the bundle map $\nu_N \to \gamma^k$ classifying ν_N (transversality gives a tubular-neighborhood model in which f is fiberwise linear and isomorphic to the normal-bundle data). Away from a tubular neighborhood of N, the map f misses a neighborhood of the zero section $BO(k)$, so it can be homotoped (pushing along the radial direction of the Thom space toward the basepoint) to send the complement of the tubular neighborhood to the basepoint ∞. After this homotopy f is exactly the collapse map of the tubular neighborhood of N composed with the classifying map of ν_N, which is $\mathrm{PT}(N)$. Hence $\mathrm{PT}(\Theta(f)) = [f]$ in $\pi_n(MO)$.

Step 4: Additivity. The group operation on $\mathfrak{N}_n$ is disjoint union, and on $\pi_n(MO)$ it is the sum of homotopy classes (concatenation in the sphere). If $M = M_0 \sqcup M_1$ is a disjoint union, embed M_0 and M_1 in disjoint half-spaces of $\mathbb{R}^{n+k}$. The collapse map of the union is the based sum of the collapse maps (the two Thom-space images sit in disjoint hemispheres of S^{n+k} and the map

factors through the pinch $S^{n+k} \to S^{n+k} \vee S^{n+k}$), so $\mathrm{PT}(M_0 \sqcup M_1) = \mathrm{PT}(M_0) + \mathrm{PT}(M_1)$. Thus PT and Θ are mutually inverse group isomorphisms, completing the proof.

The Pontryagin-Thom construction is the geometric content of all of cobordism theory in a single picture: a manifold M is encoded by the way a sphere wraps around its normal-bundle Thom space, and conversely a map of spheres into a Thom space cuts out a manifold as the transverse preimage of the zero section. Transversality is what makes the dictionary two-sided. It guarantees that the cutting-out operation $f \mapsto f^{-1}(BO(k))$ produces a manifold and that homotopic maps cut out cobordant manifolds, so that homotopy classes and cobordism classes correspond exactly. The collapse map is the reverse direction, packaging the local geometry of a tubular neighborhood into a single based map. The construction adapts to other tangential structures: replacing the universal bundle γ^k over $BO(k)$ by other classifying spaces yields a Thom spectrum for each, and in each case bordism of so-structured manifolds is the homotopy of that spectrum. The oriented case is carried out in theorem 8.5.20 below. Complex structures give MU and complex cobordism, and stable framings give the sphere spectrum, recovering framed bordism. The case of framed cobordism, where the Thom spectrum is the sphere spectrum $\mathbb{S}$, recovers the original Pontryagin theorem $\Omega^{\mathrm{fr}}_n \cong \pi^s_n$ identifying framed bordism with the stable homotopy groups of spheres.

Theorem 8.5.17: Thom's Theorem

For every $n \geq 0$ there is an isomorphism of abelian groups

$$\mathfrak{N}_n \;\cong\; \pi_n(MO) \;=\; \varinjlim_k \pi_{n+k}\big(MO(k)\big)$$

between the unoriented bordism group of closed n-manifolds and the n-th stable homotopy group of the Thom spectrum MO. The isomorphism is multiplicative, identifying the bordism ring $\mathfrak{N}_*$ with the homotopy ring $\pi_*(MO)$. Moreover $\pi_*(MO)$ is a polynomial algebra over $\mathbb{F}_2$,

$$\pi_*(MO) \;\cong\; \mathfrak{N}_* \;\cong\; \mathbb{F}_2[x_i : i \geq 2,\ i \neq 2^j - 1]$$

with one polynomial generator x_i in each degree i that is not of the form $2^j - 1$. The generators in even degrees $i = 2m$ may be taken to be the classes of the real projective spaces $\mathbb{R}\mathrm{P}^{2m}$.

***Proof*:**

The isomorphism $\mathfrak{N}_n \cong \pi_n(MO)$ is the Pontryagin-Thom construction, theorem 8.5.16. This proof records the two remaining assertions and defers the homotopy-theoretic computation of $\pi_*(MO)$ to a cited source.

Step 1: Multiplicativity. The product on $\mathfrak{N}_*$ is Cartesian product of manifolds, and the product on $\pi_*(MO)$ is induced by the ring-spectrum multiplication $MO \wedge MO \to MO$ coming from the bundle maps $\gamma^j \times \gamma^k \to \gamma^{j+k}$ (Whitney sum of universal bundles, classified by $BO(j) \times BO(k) \to BO(j+k)$). Under the Pontryagin-Thom correspondence of theorem 8.5.16, a closed j-manifold and a closed k-manifold are sent to classes whose smash product is the class of their Cartesian product, because the normal bundle of $M \times N$ in $\mathbb{R}^{j+a} \times \mathbb{R}^{k+b}$ is the external Whitney sum of the normal bundles of M and N, and the collapse map of a product is the smash of the collapse maps. Hence the bijection is a ring isomorphism.

Step 2: The structure of $\pi_*(MO)$ as a polynomial algebra. This is Thom's computation, and it is the one place in the argument where the homotopy theory genuinely outruns the geometry.

The method is to compute the mod-2 cohomology $H^*(MO;\mathbb{F}_2)$ and identify it as a module over the Steenrod algebra $\mathcal{A}_2$. The Thom isomorphism $H^*(MO(k);\mathbb{F}_2) \cong H^{*-k}(BO(k);\mathbb{F}_2)$ (cup product with the Thom class $u_k \in H^k(MO(k);\mathbb{F}_2)$, the cohomological counterpart of the fiber integration in theorem 6.3.2) identifies $H^*(MO;\mathbb{F}_2)$ with $H^*(BO;\mathbb{F}_2) = \mathbb{F}_2[w_1, w_2, \dots]$, a polynomial algebra on the Stiefel-Whitney classes. The action of the Steenrod squares on the Thom class, computed by the Wu formula $\mathrm{Sq}^i(u_k) = w_i \smile u_k$, shows that $H^*(MO;\mathbb{F}_2)$ is a free module over $\mathcal{A}_2$. A free $\mathcal{A}_2$-module structure forces MO to split as a wedge of suspensions of mod-2 Eilenberg-MacLane spectra $H\mathbb{F}_2$, by a theorem of Thom and Milnor. Consequently $\pi_*(MO)$ is the polynomial algebra on generators dual to the $\mathcal{A}_2$-module generators, one in each degree i not of the form $2^j - 1$ (the degrees $2^j - 1$ are excluded because the indecomposables w_{2^j-1} are hit by Steenrod operations). The complete computation, including the Wu formula calculation and the Thom-Milnor splitting theorem, is carried out in [29, §4 and §§16-17]. Reproducing the Steenrod-algebra bookkeeping here would add several pages of mod-2 characteristic-class computation without conceptual gain.

Step 3: The even generators are projective spaces. The class $[\mathbb{R}\mathrm{P}^{2m}] \in \mathfrak{N}_{2m}$ is nonzero and indecomposable for each $m \geq 1$. It is nonzero because the Stiefel-Whitney number $w_{2m}[\mathbb{R}\mathrm{P}^{2m}]$ is odd: the total Stiefel-Whitney class of $\mathbb{R}\mathrm{P}^N$ is $(1+\alpha)^{N+1}$ in $H^*(\mathbb{R}\mathrm{P}^N;\mathbb{F}_2) = \mathbb{F}_2[\alpha]/(\alpha^{N+1})$, so for $N = 2m$ the top class is $w_{2m} = \binom{2m+1}{2m}\alpha^{2m} = (2m+1)\alpha^{2m}$, whose coefficient $2m+1$ is odd, giving $w_{2m}[\mathbb{R}\mathrm{P}^{2m}] = 1 \in \mathbb{F}_2$. Stiefel-Whitney numbers are cobordism invariants by lemma 7.6.7. It is indecomposable because the characteristic number $s_{2m}[\mathbb{R}\mathrm{P}^{2m}]$, the Newton-polynomial number that vanishes on all products $M^a \times N^b$ with $a, b > 0$, equals $2m+1 \equiv 1 \pmod 2$ for the projective space and so is nonzero. A class with a nonzero s-number cannot be a polynomial in lower generators, hence is indecomposable. The odd projective spaces $\mathbb{R}\mathrm{P}^{2m+1}$ are null-cobordant: $\mathbb{R}\mathrm{P}^{2m+1} = S^{2m+1}/\{\pm 1\}$ is the total space of an S^1-bundle over $\mathbb{C}\mathrm{P}^m$ (the circle action $\lambda \mapsto e^{i\theta}\lambda$ on S^{2m+1} descends to the antipodal-quotient), and this circle bundle is the boundary of its associated disk bundle over $\mathbb{C}\mathrm{P}^m$, so $[\mathbb{R}\mathrm{P}^{2m+1}] = 0$. The polynomial generators $x_{2m} = [\mathbb{R}\mathrm{P}^{2m}]$ therefore fill the even non-$(2^j - 1)$ degrees (every even degree avoids the odd numbers $2^j - 1$), as we sought to show.

Thom's theorem is the bridge from differential topology to stable homotopy. The left side, $\mathfrak{N}_n$, is a geometric object: equivalence classes of manifolds under the bounding relation, computable in low dimensions by hand. The right side, $\pi_n(MO)$, is a homotopy-theoretic object: a stable homotopy group of a specific spectrum, computable by the algebraic machine of Steenrod operations. The isomorphism makes each side accessible to the other's tools, and the explicit answer $\mathbb{F}_2[x_2, x_4, x_5, x_6, x_8, \dots]$ means the bordism ring is completely known: $\mathfrak{N}_0 = \mathbb{F}_2$ (a point), $\mathfrak{N}_1 = 0$ (every closed 1-manifold is a union of circles, each bounding a disk), $\mathfrak{N}_2 = \mathbb{F}_2$ generated by $\mathbb{R}\mathrm{P}^2$, $\mathfrak{N}_3 = 0$, $\mathfrak{N}_4 = (\mathbb{F}_2)^2$ generated by $\mathbb{R}\mathrm{P}^4$ and $\mathbb{R}\mathrm{P}^2 \times \mathbb{R}\mathrm{P}^2$, and so on. That a manifold's entire unoriented cobordism class is detected by its Stiefel-Whitney numbers, finitely many $\mathbb{F}_2$-valued invariants, is the geometric payoff of the computation.

Stiefel-Whitney numbers were used twice above, in Step 3 of the proof of theorem 8.5.17 to detect $[\mathbb{R}\mathrm{P}^{2m}]$ and in the discussion following it, on the strength of their invariance under cobordism (lemma 7.6.7). The converse of that invariance is a corollary of Thom's theorem, and it is the form in which the theorem applies to a manifold presented concretely: the entire cobordism class is computed by finitely many mod-2 evaluations of characteristic classes of the tangent bundle.

Corollary 8.5.18: Stiefel-Whitney Numbers Are a Complete Unoriented Cobordism Invariant

Let M be a closed smooth n-manifold and let $w_I[M] \in \mathbb{F}_2$ be its Stiefel-Whitney numbers, indexed by the partitions I of n (definition 7.6.4). Then M bounds a compact smooth $(n+1)$-manifold if and only if $w_I[M] = 0$ for every such I. Equivalently, two closed smooth n-manifolds M and N are cobordant if and only if $w_I[M] = w_I[N]$ for every such monomial.

***Proof*:**

Step 1: A manifold that bounds has all Stiefel-Whitney numbers zero. This is lemma 7.6.7.

Step 2: Vanishing Stiefel-Whitney numbers force the class to vanish. This direction is the one that uses Thom's computation, and it uses exactly the splitting recorded in Step 2 of the proof of theorem 8.5.17: the mod-2 cohomology $H^*(MO;\mathbb{F}_2)$ is a free module over the Steenrod algebra $\mathcal{A}_2$, and a Thom spectrum with that property splits as a wedge

$$MO \;\simeq\; \bigvee_j \Sigma^j H\mathbb{F}_2$$

of suspensions of the mod-2 Eilenberg-MacLane spectrum, by the theorem of Thom and Milnor, for which the source is [36]. For such a wedge the mod-2 Hurewicz map is injective: on the summand $\Sigma^j H\mathbb{F}_2$ it is the map $\pi_n(\Sigma^j H\mathbb{F}_2) \to H_n(\Sigma^j H\mathbb{F}_2;\mathbb{F}_2)$, which in the bottom degree $n = j$ is the identity $\mathbb{F}_2 \to \mathbb{F}_2$ and in every other degree has trivial source. So

$$h\colon \pi_n(MO) \longrightarrow H_n(MO;\mathbb{F}_2)$$

is injective, and a class in $\pi_n(MO)$ is determined by its image in mod-2 homology.

It remains to identify that image for $\mathrm{PT}(M)$. Represent $\mathrm{PT}(M)$ by the composite $S^{n+k} \xrightarrow{c} \mathrm{Th}(\nu) \xrightarrow{\mathrm{Th}(g)} MO(k)$ of theorem 8.5.16, with ν the normal bundle of an embedding $M \subseteq \mathbb{R}^{n+k}$ and g its classifying map. The collapse map carries the mod-2 fundamental class of S^{n+k} to the Thom-isomorphism image of $[M]$, because c is a relative homeomorphism onto $\mathrm{Th}(\nu)$ over the tubular neighbourhood. Applying $\mathrm{Th}(g)_*$ and the Thom isomorphism $H_{n+k}(MO(k);\mathbb{F}_2) \cong H_n(BO(k);\mathbb{F}_2)$ then identifies $h(\mathrm{PT}(M))$ with $g_*[M]$, an element of $H_n(BO;\mathbb{F}_2)$ once the colimit in k is taken. Pairing $g_*[M]$ against a monomial in the universal Stiefel-Whitney classes gives $\langle w_I(\nu), [M]\rangle$, a Stiefel-Whitney number of the normal bundle. Since $H^n(BO;\mathbb{F}_2)$ is the degree-n part of the polynomial algebra $\mathbb{F}_2[w_1, w_2, \dots]$ and $H_n(BO;\mathbb{F}_2)$ is its finite-dimensional dual, these pairings determine $g_*[M]$ completely. Finally $TM \oplus \nu$ is trivial, so $w(\nu) = w(TM)^{-1}$ by theorem 7.4.5, and each normal number is a universal polynomial in the tangential ones and conversely. The tangential Stiefel-Whitney numbers therefore determine $g_*[M]$, hence $h(\mathrm{PT}(M))$, hence $\mathrm{PT}(M)$, hence the cobordism class of M by theorem 8.5.16. In particular all numbers zero forces $\mathrm{PT}(M) = 0$, which says that M bounds.

Step 3: The two formulations are equivalent. A disjoint union has $T(M \sqcup N)$ restricting to TM and to TN and fundamental class $[M] + [N]$, so $w_I[M \sqcup N] = w_I[M] + w_I[N]$. Every element of $\mathfrak{N}_n$ is its own inverse, so M and N are cobordant precisely when $M \sqcup N$ bounds, which by the first formulation happens precisely when $w_I[M] + w_I[N] = 0$ for every I, that is, when $w_I[M] = w_I[N]$ in $\mathbb{F}_2$, as we sought to show.

The corollary is what makes Thom's theorem usable. Deciding whether a given closed n-manifold bounds is a question about all compact $(n+1)$-manifolds, of which there is no list. The corollary replaces that question by the evaluation of finitely many cup products of Stiefel-Whitney classes against the mod-2 fundamental class, each of which is a computation inside $H^*(M;\mathbb{F}_2)$. The two halves of the proof are not of comparable depth. The vanishing direction uses only the collar splitting $TW|_M \cong TM \oplus \underline{\mathbb{R}}$ and exactness of the sequence of the pair, while the converse is a restatement of the wedge decomposition of MO, and no geometric argument for it is known that avoids the homotopy theory.

The construction just carried out ignores orientations, and the group it computes is entirely 2-torsion for a single geometric reason: the cylinder $M \times [0,1]$ has boundary $M \sqcup M$. Requiring the manifolds and the cobordisms between them to be oriented removes that reason, since the oriented cylinder has boundary $M \sqcup (-M)$ instead. The Thom spectrum built from the oriented Grassmannians computes the group that results.

Definition 8.5.19: The Oriented Thom Spectrum and Oriented Bordism

Let $BSO(k)$ be the classifying space of the special orthogonal group, and let $\widetilde{\gamma}^{\,k} \to BSO(k)$ be the universal oriented rank-k real vector bundle, the pullback of γ^k along the double cover $BSO(k) \to BO(k)$ carrying its tautological orientation. The associated Thom spaces (definition 8.5.12) are

$$MSO(k) := \mathrm{Th}\big(\widetilde{\gamma}^{\,k}\big)$$

The inclusion $BSO(k) \hookrightarrow BSO(k+1)$ classifies $\widetilde{\gamma}^{\,k} \oplus \underline{\mathbb{R}}$, oriented by an oriented frame of $\widetilde{\gamma}^{\,k}$ followed by the positive generator of the trivial line, and induces on Thom spaces a map $\Sigma\, MSO(k) \to MSO(k+1)$. The resulting sequence

$$MSO \;=\; \{MSO(k),\ \Sigma\, MSO(k) \to MSO(k+1)\}_{k\geq 0}$$

is the *oriented Thom spectrum*, formed exactly as MO was in definition 8.5.13.

For a closed oriented smooth n-manifold N (definition 4.4.4 with $R=\mathbb{Z}$), write $-N$ for the same smooth manifold with the reversed orientation, whose fundamental class is $-[N]$ by theorem 4.4.9. Two closed oriented smooth n-manifolds M and N are *oriented-cobordant* if $M \sqcup (-N)$ is the boundary of a compact oriented smooth $(n+1)$-manifold W, the orientation induced on ∂W agreeing with the given ones. This is an equivalence relation, and the set of equivalence classes is the *oriented bordism group*

$$\Omega_n^{SO}$$

an abelian group under disjoint union whose identity is the class of the oriented manifolds that bound and in which the inverse of the class of M is the class of $-M$, since $M \sqcup (-M) = \partial(M \times [0,1])$ with the boundary orientation. The graded group $\Omega_*^{SO} = \bigoplus_{n\geq 0} \Omega_n^{SO}$ is a ring under Cartesian product with the product orientation.

The structural difference between Ω_*^{SO} and $\mathfrak{N}_*$ is already in the definition. Doubling gives only $[M]+[-M]=0$, not $2[M]=0$, so there is no reason for Ω_*^{SO} to be an $\mathbb{F}_2$-algebra, and it is not one: the class of $\mathbb{C}\mathrm{P}^2$ has infinite order in Ω_4^{SO}. The invariant that shows this is a Pontryagin number, and its cobordism invariance is Step 1 of corollary 8.5.18 run with integral coefficients. Along the boundary of a compact oriented W one has $j^*TW \cong TM \oplus \underline{\mathbb{R}}$, whose complexification is $(TM)_{\mathbb{C}} \oplus \underline{\mathbb{C}}$,

so the complex Whitney sum formula (theorem 7.4.5) with $c(\underline{\mathbb{C}}) = 1$ gives $j^*p(TW) = p(TM)$ exactly, with no 2-torsion ambiguity. When $M = \partial W$ carries the induced orientation one also has $\partial[W, M] = [M]$ integrally, so the same exactness argument makes every Pontryagin number of M vanish. Since $\langle p_1(T\mathbb{C}\mathrm{P}^2), [\mathbb{C}\mathrm{P}^2]\rangle = 3$ by example 7.6.9, and this pairing is additive under disjoint union and changes sign under orientation reversal, no nonzero multiple of $[\mathbb{C}\mathrm{P}^2]$ bounds.

The Pontryagin-Thom construction adapts to oriented manifolds, and the places where the adaptation changes the argument are the places where Ω_*^{SO} differs from $\mathfrak{N}_*$. Two of the four steps of the proof of theorem 8.5.16 are reused verbatim. The new content sits in Step 1, where the normal bundle inherits an orientation and the classifying map lifts to $BSO(k)$, in Step 2, where the transverse preimage comes out oriented, and in Step 4, where the inverse of a class is no longer the class itself.

Theorem 8.5.20: Oriented Bordism and the Spectrum MSO

Let $n \geq 0$.

1. The Pontryagin-Thom construction gives an isomorphism of abelian groups

$$\Omega_n^{SO} \;\cong\; \pi_n(MSO) \;=\; \varinjlim_k \pi_{n+k}\big(MSO(k)\big)$$

between the oriented bordism group of definition 8.5.19 and the n-th stable homotopy group of the oriented Thom spectrum. The isomorphism is multiplicative, identifying the ring Ω_*^{SO} with $\pi_*(MSO)$.

2. Rationally, the oriented bordism ring is a polynomial algebra on the even complex projective spaces,

$$\Omega_*^{SO} \otimes \mathbb{Q} \;\cong\; \mathbb{Q}\big[[\mathbb{C}\mathrm{P}^2], [\mathbb{C}\mathrm{P}^4], [\mathbb{C}\mathrm{P}^6], \dots\big]$$

with $[\mathbb{C}\mathrm{P}^{2m}]$ in degree $4m$, so that $\Omega_n^{SO} \otimes \mathbb{Q} = 0$ unless 4 divides n. Two closed oriented n-manifolds have the same class in $\Omega_n^{SO} \otimes \mathbb{Q}$ if and only if all their Pontryagin numbers agree, so the Pontryagin numbers are a complete rational invariant of oriented cobordism.

***Proof*:**
This establishes clause (1). Clause (2) is treated separately below. The four steps that follow are the four steps of the proof of theorem 8.5.16, each recording what carries over unchanged and what the orientation adds.

Step 1: The normal bundle inherits an orientation, and the forward map lands in $MSO(k)$. Let M be a closed oriented n-manifold, embedded in $\mathbb{R}^{n+k}$ for k large with normal bundle ν of rank k, so that $TM \oplus \nu \cong \underline{\mathbb{R}}^{n+k}$ over M. Here the orientation enters. The orientation of M orients each tangent space T_xM, and the standard orientation of $\mathbb{R}^{n+k}$ orients each fibre of the trivial bundle, so each fibre ν_x receives the unique orientation for which an oriented basis of T_xM followed by an oriented basis of ν_x is an oriented basis of $\mathbb{R}^{n+k}$. Both input orientations are locally constant in x, so the resulting choice is locally constant as well, and ν is an oriented rank-k bundle. Consequently its classifying map lifts across the double cover $BSO(k) \to BO(k)$ to a map $g\colon M \to BSO(k)$ with $g^*\widetilde{\gamma}^k \cong \nu$ as oriented bundles, uniquely up to homotopy once the orientation of ν is fixed, since lifting across a covering is unique given the lift of one point in each component. The collapse map is untouched: it is built from the tubular neighbourhood

alone and refers to no orientation. So the composite

$$S^{n+k} \xrightarrow{c} \mathrm{Th}(\nu) \xrightarrow{\mathrm{Th}(g)} MSO(k)$$

is a class $\mathrm{PT}(M) \in \pi_{n+k}(MSO(k))$. Raising the embedding dimension by one replaces ν by $\nu \oplus \underline{\mathbb{R}}$ with precisely the orientation used to define the structure map of MSO in definition 8.5.19, so the class is stable and defines $\mathrm{PT}(M) \in \pi_n(MSO)$. Independence of the embedding is the isotopy argument of theorem 8.5.16, unchanged, since an isotopy carries the orientation of the normal bundle along with the bundle.

Cobordism invariance needs the sign bookkeeping made explicit. Suppose M_0 and M_1 are oriented-cobordant, and realize the cobordism as a compact oriented $W \subseteq \mathbb{R}^{n+k} \times [0,1]$ meeting the two boundary slabs orthogonally, with M_0 at $t = 0$ and M_1 at $t = 1$. With the outward-normal-first convention, the orientation induced by W on its boundary agrees with the given orientation of M_1 and is opposite to that of M_0, because the outward normal points in the $-t$ direction at one end and the $+t$ direction at the other. That is exactly the statement $\partial W = (-M_0) \sqcup M_1$, which is the relation defining oriented cobordism. Orient the normal bundle of W in $\mathbb{R}^{n+k+1}$ by the rule above. At each end the two reversals compensate, and the induced orientation of ν_W restricts to the orientation of ν_{M_i} determined by the given orientation of M_i. The collapse construction applied to W therefore gives a homotopy $S^{n+k} \times [0,1] \to MSO(k)$ from $\mathrm{PT}(M_0)$ to $\mathrm{PT}(M_1)$, and PT descends to $\Omega^{SO}_n \to \pi_n(MSO)$.

Step 2: Transversality is unchanged, and the preimage comes out oriented. The Thom transversality theorem is a statement about smooth maps and smooth submanifolds and takes no account of orientations, so nothing in it needs adjusting: a based map $f\colon S^{n+k} \to MSO(k)$ is homotoped to a map f' transverse to the zero section $BSO(k) \subseteq MSO(k)$, and $\Theta(f) = (f')^{-1}(BSO(k)) \subseteq \mathbb{R}^{n+k}$ is a closed n-manifold with normal bundle $(f')^*\widetilde{\gamma}^{\,k}$, by the argument of Step 2 of theorem 8.5.16 with $BO(k)$ replaced by $BSO(k)$ throughout. What is new is the output. The bundle $\widetilde{\gamma}^{\,k}$ is oriented, so its pullback is, and reading the rule of Step 1 backwards, the orientation of the normal bundle together with the standard orientation of $\mathbb{R}^{n+k}$ orients $T\Theta(f)$. The transverse preimage is thus a closed *oriented* n-manifold, where in the unoriented setting it was only a manifold, and the orientation is given by the universal bundle rather than chosen. Independence of the transverse representative is the relative-transversality argument as before, with $H^{-1}(BSO(k))$ oriented by the same rule. The ends of that cobordism carry the orientations of $\Theta(f')$ and $\Theta(f'')$ with the boundary signs of the previous paragraph, so it is an oriented cobordism.

Step 3: The two maps are still mutually inverse. Both halves of Step 3 of the proof of theorem 8.5.16 are arguments about manifolds, tubular neighbourhoods, and homotopies, and no orientation appears anywhere in them, so they are reused word for word. What must be added is that the orientations correspond. For $\Theta \circ \mathrm{PT}$, the manifold recovered from $\mathrm{PT}(M)$ is M, and the orientation it acquires in Step 2 is the one determined by the orientation of ν together with that of $\mathbb{R}^{n+k}$. But the orientation of ν was itself defined from the orientation of M by the same rule in Step 1, and the rule is an involution, so the recovered orientation is the original one. For $\mathrm{PT} \circ \Theta$, the homotopy pushing the complement of a tubular neighbourhood to the basepoint moves no orientation data, and the classifying map it exhibits is the oriented classifying map of ν_N by the construction of Step 2.

Step 4: Additivity, and where the inverses now come from. Embedding M_0 and M_1 in disjoint half-spaces of $\mathbb{R}^{n+k}$ makes the collapse map of the disjoint union the based sum of the two collapse maps, exactly as in Step 4 of the proof of theorem 8.5.16, and the orientations of the two pieces are independent of one another. Hence $\mathrm{PT}(M_0 \sqcup M_1) = \mathrm{PT}(M_0) + \mathrm{PT}(M_1)$ and PT is a

homomorphism, and by Steps 2 and 3 it is a bijection, so it is an isomorphism $\Omega_n^{SO} \cong \pi_n(MSO)$.

The structural difference between the two theories is located here. In $\mathfrak{N}_n$ the identity $M \sqcup M = \partial(M \times [0,1])$ makes every element its own inverse, which is what makes $\mathfrak{N}_*$ an $\mathbb{F}_2$-vector space. Orienting the cylinder gives $\partial(M \times [0,1]) = M \sqcup (-M)$ instead, so the same manifold now yields only that the inverse of the class of M is the class of $-M$. On the homotopy side no separate argument is needed: PT is a homomorphism and vanishes on oriented boundaries, so

$$\mathrm{PT}(M) + \mathrm{PT}(-M) \;=\; \mathrm{PT}\big(M \sqcup (-M)\big) \;=\; 0$$

which says $\mathrm{PT}(-M) = -\mathrm{PT}(M)$: orientation reversal on the geometric side is negation in $\pi_n(MSO)$.

Multiplicativity is Step 1 of the proof of theorem 8.5.17 with BSO in place of BO. The ring-spectrum multiplication $MSO \wedge MSO \to MSO$ comes from the maps $BSO(j) \times BSO(k) \to BSO(j+k)$ classifying the Whitney sum of the two universal oriented bundles with its product orientation, the normal bundle of $M \times N$ is the external Whitney sum of the two normal bundles with the product orientation, and the collapse map of a product is the smash of the collapse maps. So the bijection is a ring isomorphism, completing the proof of clause (1).

***Proof Sketch*:**
Clause (2) is stated without proof, and the reason is the one that put the computation of $\pi_*(MO)$ outside the argument in Step 2 of the proof of theorem 8.5.17. The route to $\Omega_*^{SO} \otimes \mathbb{Q}$ runs through the rational Hurewicz isomorphism

$$\pi_*(MSO) \otimes \mathbb{Q} \;\cong\; H_*(MSO; \mathbb{Q})$$

which rests on Serre's theory of classes of abelian groups. It follows from the Hurewicz theorem modulo finite groups, which Milnor and Stasheff assemble from Serre's results and apply to Thom spaces of oriented bundles over finite complexes [29, Theorem 18.3 and Corollary 18.4].

Granting that identification, the remainder is characteristic-class bookkeeping already available here. The Thom isomorphism identifies $H_*(MSO;\mathbb{Q})$ with $H_*(BSO;\mathbb{Q})$, and theorem 7.6.10 in its stable limit gives $H^*(BSO;\mathbb{Q}) = \mathbb{Q}[p_1, p_2, \dots]$ with $|p_i| = 4i$: the Euler class present in $H^*(BSO(2m);\mathbb{Q})$ does not persist in the limit, since the proof of that theorem records that it restricts to 0 in odd rank. So $\Omega_*^{SO} \otimes \mathbb{Q}$ is a polynomial algebra on one generator in each degree $4m$, and in particular vanishes in degrees not divisible by 4. The even projective spaces give the generators: the total Pontryagin class of $T\mathbb{C}\mathrm{P}^N$ is $(1+u^2)^{N+1}$ in $H^*(\mathbb{C}\mathrm{P}^N;\mathbb{Z}) = \mathbb{Z}[u]/(u^{N+1})$, whose Pontryagin roots are $N+1$ copies of u^2, so the Newton-polynomial number in the Pontryagin classes of $\mathbb{C}\mathrm{P}^{2m}$ is $s_m[\mathbb{C}\mathrm{P}^{2m}] = 2m+1 \neq 0$ in $\mathbb{Q}$. Since s_m vanishes on decomposables, a class with nonzero s-number is indecomposable, exactly as for $\mathbb{R}\mathrm{P}^{2m}$ in Step 3 of the proof of theorem 8.5.17, and the classes $[\mathbb{C}\mathrm{P}^{2m}]$ are polynomial generators. The statement about Pontryagin numbers follows from the same identification: under the Thom isomorphism the pairings of $H^*(BSO;\mathbb{Q}) = \mathbb{Q}[p_1, p_2, \dots]$ against the Hurewicz image of a class are its Pontryagin numbers, and these pairings span the space of $\mathbb{Q}$-linear functionals on $\Omega_n^{SO} \otimes \mathbb{Q}$, so two classes agree rationally exactly when all their Pontryagin numbers agree. The full computation is Thom's [36], and [29] presents it in the characteristic-class language used here, completing the sketch.

The two clauses split the work the same way the unoriented case did. Clause (1) is geometry: an

isomorphism between a set of manifolds modulo a relation and a stable homotopy group, proved by an explicit construction in both directions. Clause (2) is homotopy theory, and it is where the answer is computed. The two coefficient rings then separate the theories. The ring $\mathfrak{N}_*$ is a polynomial algebra over $\mathbb{F}_2$ with generators in all degrees not of the form $2^j - 1$, while $\Omega^{SO}_* \otimes \mathbb{Q}$ is a polynomial algebra over $\mathbb{Q}$ with generators only in degrees divisible by 4. Clause (2) describes the rationalization alone, and whatever torsion Ω^{SO}_* carries is invisible to it, since a Pontryagin number is a homomorphism $\Omega^{SO}_n \to \mathbb{Z}$ and therefore kills every torsion class. Low degrees show the difference against $\mathfrak{N}_*$ immediately.

Example 8.5.21: Oriented Bordism in Degrees at Most Four

In degree 0 a closed oriented manifold is a finite set of points each carrying a sign, and the signed count is an isomorphism

$$\Omega^{SO}_0 \;\cong\; \mathbb{Z}$$

It is additive under disjoint union, and a signed point set bounds a compact oriented 1-manifold exactly when the count is zero, since such a 1-manifold is a union of circles and arcs and each arc contributes one point of each sign.

In degrees 1, 2 and 3 the groups vanish. Every closed oriented 1-manifold is a union of circles, each bounding a disk. Every closed oriented surface bounds a handlebody. Every closed oriented 3-manifold bounds a compact oriented 4-manifold, which is not elementary and is part of Thom's computation [36]. So

$$\Omega^{SO}_1 = \Omega^{SO}_2 = \Omega^{SO}_3 = 0$$

matching clause (2) of theorem 8.5.20, which forces these groups to be torsion, and Thom's integral computation, which shows the torsion is trivial here.

In degree 4, $\Omega^{SO}_4 \cong \mathbb{Z}$ generated by $[\mathbb{C}\mathrm{P}^2]$, again by [36]. Clause (2) gives only the rationalization $\Omega^{SO}_4 \otimes \mathbb{Q} \cong \mathbb{Q}$, and the absence of torsion is part of Thom's integral computation. The generator is pinned down by the signature. For a closed oriented $4k$-manifold M the cup-product pairing $(x, y) \mapsto \langle x \smile y, [M]\rangle$ on $H^{2k}(M;\mathbb{R})$ is symmetric, and it is nondegenerate by Poincaré duality (theorem 4.5.7) in the form corollary 4.5.6, which identifies the pairing with the Kronecker pairing against the duality isomorphism. Its *signature* $\sigma(M)$ is the number of positive eigenvalues minus the number of negative ones. It is additive under disjoint union, changes sign under orientation reversal, and vanishes on oriented boundaries [36], so it descends to a homomorphism $\sigma\colon \Omega^{SO}_{4k} \to \mathbb{Z}$. On $\mathbb{C}\mathrm{P}^2$ the group $H^2(\mathbb{C}\mathrm{P}^2;\mathbb{R})$ is spanned by the hyperplane class u with $\langle u^2, [\mathbb{C}\mathrm{P}^2]\rangle = 1$, giving a 1×1 form with entry $+1$, so $\sigma(\mathbb{C}\mathrm{P}^2) = 1$. Hence $\sigma\colon \Omega^{SO}_4 \to \mathbb{Z}$ is surjective, therefore an isomorphism, and $[\mathbb{C}\mathrm{P}^2]$ generates. The relation $\sigma(\mathbb{C}\mathrm{P}^2) = \frac{1}{3}\langle p_1(T\mathbb{C}\mathrm{P}^2), [\mathbb{C}\mathrm{P}^2]\rangle$ noted in example 7.6.9 is the first case of the Hirzebruch signature theorem, which expresses σ on Ω^{SO}_{4k} as a universal rational polynomial in the Pontryagin numbers.

Against the unoriented groups $\mathfrak{N}_0 = \mathbb{F}_2$, $\mathfrak{N}_1 = 0$, $\mathfrak{N}_2 = \mathbb{F}_2$, $\mathfrak{N}_3 = 0$, $\mathfrak{N}_4 = (\mathbb{F}_2)^2$ the pattern is different in both directions: $\Omega^{SO}_2 = 0$ while $\mathfrak{N}_2 = \mathbb{F}_2$, since the generator $\mathbb{R}\mathrm{P}^2$ of the latter is not orientable and so has no class at all in oriented bordism, and $\Omega^{SO}_4 = \mathbb{Z}$ is infinite while $\mathfrak{N}_4$ is finite.

The example records the two effects of adding an orientation to the data. It removes manifolds from the theory: the generator $\mathbb{R}\mathrm{P}^2$ of $\mathfrak{N}_2$ is non-orientable, so it represents nothing at all in Ω^{SO}_2, which is zero. It also refines the relation on those that remain, so classes of infinite order appear where before every element had order two. The invariants that come with the refinement are the Pontryagin numbers and, in dimensions divisible by four, the signature that the Hirzebruch theorem expresses in terms of them. Neither is available in $\mathfrak{N}_*$, where the coefficients are $\mathbb{F}_2$ and there is no orientation to

pair against. Complex cobordism MU, defined alongside MO in definition 8.5.13, is the next step in the same direction, replacing the orientation of the stable normal bundle by a complex structure on it.

Both theories now stand assembled as generalized cohomology theories obtained by dropping the dimension axiom. K-theory K^* is the cohomology theory represented by the Ω-spectrum KU, with coefficient ring $\mathbb{Z}[\beta, \beta^{-1}]$ periodic of period 2. Its violation of the dimension axiom is the periodicity, a copy of $\mathbb{Z}$ in every even degree. Cobordism MO^* (and MU^*) is the cohomology theory represented by the Thom spectrum MO (resp. MU), with coefficient ring $\pi_*(MO) = \mathbb{F}_2[x_i]$ the polynomial bordism ring. Its violation of the dimension axiom is the nonvanishing of $\pi_n(MO) = \mathfrak{N}_n$ in infinitely many positive degrees. These are the two prototypes: K-theory the periodic theory, built from bundles on compact spaces (theorem 8.5.3), whose coefficients are simplest, and cobordism the geometric, manifold-theoretic theory whose coefficients carry the entire classification of manifolds up to bounding. Every generalized cohomology theory is, by Brown representability (theorem 8.3.9), represented by a spectrum, and these two spectra are the templates against which the general theory of section 8.6 measures all others.

Exercise 8.5.1

1. Using the clutching description of example 8.5.5, compute $\tilde{K}^0(S^1)$ and $\tilde{K}^0(S^3)$, and confirm they vanish. Explain how this is consistent with the coefficient pattern $K^n(\mathrm{pt}) = \mathbb{Z}$ for n even and 0 for n odd from theorem 8.5.10, via the identification $\tilde{K}^0(S^m) \cong K^{-m}(\mathrm{pt})$.

2. Verify directly that the Thom space of the trivial rank-k bundle $\underline{\mathbb{R}}^k \to B$ over a compact base is the k-fold reduced suspension $\Sigma^k(B_+)$, where $B_+ = B \sqcup \{\mathrm{pt}\}$. Use this to recompute $\mathrm{Th}(\underline{\mathbb{R}} \to S^1)$ and confirm it agrees with the value $S^2 \vee S^1$ stated in example 8.5.14.

3. Show that every closed 1-manifold is null-cobordant, hence $\mathfrak{N}_1 = 0$, and confirm this against Thom's theorem (theorem 8.5.17): check that the polynomial algebra $\mathbb{F}_2[x_i : i \geq 2,\ i \neq 2^j - 1]$ has no generator in degree 1 and that there is no product of generators landing in degree 1.

4. In the Pontryagin-Thom construction (theorem 8.5.16), explain why a map $f\colon S^{n+k} \to MO(k)$ transverse to the zero section $BO(k)$ has preimage $f^{-1}(BO(k))$ of dimension exactly n, and identify its normal bundle. Then explain in one or two sentences why two homotopic transverse maps yield cobordant preimages.

5. Compute the Stiefel-Whitney number $w_2[\mathbb{RP}^2]$ and use it to show $[\mathbb{RP}^2] \neq 0$ in $\mathfrak{N}_2$, so that $\mathbb{RP}^2$ generates $\mathfrak{N}_2 \cong \mathbb{F}_2$. (You may use that the total Stiefel-Whitney class of $\mathbb{RP}^2$ is $w(\mathbb{RP}^2) = (1+\alpha)^3 = 1 + \alpha + \alpha^2$ in $H^*(\mathbb{RP}^2; \mathbb{F}_2) = \mathbb{F}_2[\alpha]/(\alpha^3)$, and that a Stiefel-Whitney number is the evaluation of a top-degree monomial on the fundamental class.)

6. Contrast the failure of the dimension axiom in K-theory and in cobordism. State the coefficient ring of each theory ($K^*(\mathrm{pt})$ and $MO^*(\mathrm{pt}) = \pi_*(MO)$), identify in which degrees each is nonzero, and explain why neither equals the ordinary-cohomology coefficient $H^*(\mathrm{pt}; \mathbb{Z}) = \mathbb{Z}$ concentrated in degree 0.

8.6 The Atiyah-Hirzebruch Spectral Sequence

The Serre spectral sequence of section 6.2 computes ordinary homology of a total space from ordinary homology of base and fiber. The preceding sections have shown that ordinary cohomology is one member of a larger family: the generalized cohomology theories of definition 8.2.16, each represented by a spectrum (definition 8.4.1, theorem 8.4.6), each satisfying every Eilenberg-Steenrod axiom (definition 8.2.4) but the dimension axiom. K-theory and cobordism, the two theories that opened this book's list of "ways of assigning algebraic information", are the prototypes. The cellular method that makes ordinary cohomology computable extends to these theories almost mechanically: the skeletal filtration of a CW complex, which fed the cellular chain complex into ordinary cohomology, feeds any generalized theory E^* into the exact-couple machine of section 6.1 and produces a spectral sequence converging to $E^*(X)$. Its second page is ordinary cohomology of X with coefficients in the coefficient ring $E^*(\mathrm{pt})$ of the theory. This is the Atiyah-Hirzebruch spectral sequence, and it is the bridge that lets every computation of ordinary cohomology in the preceding chapters be leveraged into a computation in K-theory or cobordism.

The construction parallels the Serre spectral sequence so closely that most of the work has already been done. Where the Serre construction filtered the singular chains of the total space of a fibration by the simplices of the base they cover, the present construction filters the space X by its own skeleta, and where the fiber homology $H_*(F)$ gave the coefficient grading, the coefficient groups $E^q(\mathrm{pt})$ of the generalized theory take its place. The single new ingredient is that E^* is not the ordinary cohomology that the exact-couple theorem of theorem 6.1.10 was stated for. The theorem must be re-run with E^* in place of H^*, which is legitimate because the only properties of H^* used in the construction are the Eilenberg-Steenrod axioms, and a generalized theory satisfies all of them save dimension.

Throughout, E^* is a generalized (reduced or unreduced) cohomology theory in the sense of definition 8.2.16, with coefficient groups

$$E^q(\mathrm{pt}), \qquad q \in \mathbb{Z}$$

the values of the unreduced theory on a point. These groups, taken together, form the graded *coefficient ring* $E^*(\mathrm{pt}) = \bigoplus_q E^q(\mathrm{pt})$ when E^* is multiplicative. For the construction of the spectral sequence only the additive structure is needed. Unlike ordinary cohomology, where $H^q(\mathrm{pt}) = 0$ for $q \neq 0$ by the dimension axiom, the coefficient groups of a generalized theory may be nonzero in many degrees, including negative ones, and it is exactly this spread that makes the spectral sequence carry information.

The construction begins with the filtration of the cochains of X by skeleta, exactly as in the Serre case but now applied to the generalized theory. Recall that $X^{(p)}$ denotes the p-skeleton of the CW complex X, with $X^{(p)} = \varnothing$ for $p < 0$.

Definition 8.6.1: The Skeletal Exact Couple of a Generalized Theory

Let E^* be a generalized cohomology theory and X a CW complex with skeleta $X^{(p)}$. The *skeletal exact couple* of E^* on X is the exact couple (definition 6.1.2) assembled from the long exact sequences of the pairs $(X^{(p)}, X^{(p-1)})$ in the theory E^*. Its groups are

$$D_1^{p,q} := E^{p+q}\big(X^{(p)}\big), \qquad E_1^{p,q} := E^{p+q}\big(X^{(p)},\, X^{(p-1)}\big)$$

with structure maps $i\colon D_1^{p,q} \to D_1^{p-1,q+1}$ induced by the inclusion $X^{(p-1)} \hookrightarrow X^{(p)}$, $j\colon D_1^{p,q} \to E_1^{p,q}$ the restriction to the relative group, and $k\colon E_1^{p,q} \to D_1^{p+1,q}$ the connecting homomorphism δ of the pair. The associated spectral sequence is the one generated by this exact couple in the sense of definition 6.1.6.

The exact couple records, in a single bigraded triangle, the long exact sequence of every skeletal pair at once. The auxiliary group $D_1^{p,q}$ collects the value of the theory on the p-skeleton, in total degree $p+q$. The visible group $E_1^{p,q}$ collects the value on the skeletal layer $(X^{(p)}, X^{(p-1)})$. The maps i, j, k are the inclusion, restriction, and connecting maps of the axioms. That this is genuinely an exact couple is immediate, since exactness of the triangle at its three corners is precisely the exactness of the long exact sequence of the pair $(X^{(p)}, X^{(p-1)})$ in E^*, which holds by the exactness axiom for a generalized theory. The work, exactly as for the Serre spectral sequence, is identifying the first two pages.

Example 8.6.2: The Skeletal Exact Couple for K^* on S^2

Take $X = S^2$ with its minimal CW structure: one 0-cell and one 2-cell, so $X^{(0)} = X^{(1)} = \mathrm{pt}$ and $X^{(2)} = S^2$. Let $E^* = K^*$ be complex K-theory, the theory of the spectrum KU, whose coefficient groups are $K^q(\mathrm{pt}) = \mathbb{Z}$ for q even and 0 for q odd by theorem 8.5.10(3). The exact couple of definition 8.6.1 has only two occupied filtration levels, $p = 0$ and $p = 2$. At level 0 the layer is $(X^{(0)}, X^{(-1)}) = (\mathrm{pt}, \varnothing)$, so

$$E_1^{0,q} = K^q(\mathrm{pt}) = \begin{cases} \mathbb{Z} & q \text{ even} \\ 0 & q \text{ odd} \end{cases}$$

The level $p = 1$ is empty, $E_1^{1,q} = K^{1+q}(X^{(1)}, X^{(0)}) = K^{1+q}(\mathrm{pt}, \mathrm{pt}) = 0$, there being no 1-cells. At level 2 the layer is (S^2, pt), whose reduced K-theory is $\widetilde{K}^{2+q}(S^2) = \widetilde{K}^{2+q}(\Sigma^2 S^0) \cong \widetilde{K}^q(S^0) = K^q(\mathrm{pt})$ by the double suspension isomorphism, so

$$E_1^{2,q} = \widetilde{K}^{2+q}(S^2) = \begin{cases} \mathbb{Z} & q \text{ even} \\ 0 & q \text{ odd} \end{cases}$$

The auxiliary group at level 2 is $D_1^{2,q} = K^{2+q}(S^2)$, the full K-theory of the sphere, which the spectral sequence is on its way to computing. The single first differential $d_1\colon E_1^{1,q} \to E_1^{2,q}$ has source 0 and the differential $E_1^{0,q} \to E_1^{1,q}$ has target 0, so $d_1 = 0$ identically: the cellular cochain complex of S^2 with coefficients in $K^*(\mathrm{pt})$ has no 1-cells to carry a coboundary, and the second page already equals the first.

The example is the skeletal exact couple stripped to two layers, where the absence of 1-cells forces $d_1 = 0$ and the page can be read off the two layers at once. In general the layers are not so sparse, and the first differential is the cellular coboundary of X. The next two results compute the E_1 and

E_2 pages in general, and they reduce to a single homotopy-theoretic input already used for the Serre spectral sequence: the cofiber of a skeletal pair is a wedge of spheres, on which a generalized theory is computed by the suspension axiom.

Lemma 8.6.3: The First Page is Cellular Cochains with Coefficients in $E^*(\mathrm{pt})$

Let E^* be a generalized cohomology theory and X a CW complex with p-cells indexed by a set J_p. Then for the skeletal exact couple of definition 8.6.1 there is a natural isomorphism

$$E_1^{p,q} \cong \prod_{\alpha \in J_p} E^q(\mathrm{pt}) = C^p_{\mathrm{CW}}\big(X;\, E^q(\mathrm{pt})\big)$$

the cellular p-cochains of X with coefficients in the abelian group $E^q(\mathrm{pt})$, and under this identification the differential $d_1 \colon E_1^{p,q} \to E_1^{p+1,q}$ is the cellular coboundary map δ^{CW} of X with coefficients in $E^q(\mathrm{pt})$.

***Proof*:**

Step 1: The relative group of a skeletal pair. For $p = 0$ the subspace $X^{(-1)}$ is empty, and the additivity axiom of definition 8.2.4 gives $E^q(X^{(0)}) \cong \prod_{\alpha \in J_0} E^q(\mathrm{pt})$, which is the conclusion of Step 2 for $p = 0$. For $p \geq 1$ the pair $(X^{(p)}, X^{(p-1)})$ is a good CW pair, and the quotient is a wedge of p-spheres, one for each p-cell:

$$X^{(p)}/X^{(p-1)} \cong \bigvee_{\alpha \in J_p} S^p_\alpha$$

one sphere S^p_α per characteristic map $\Phi_\alpha \colon (D^p, \partial D^p) \to (X^{(p)}, X^{(p-1)})$. By the relative-to-reduced identification for a good pair (an Eilenberg-Steenrod axiom for any generalized theory, definition 8.2.4), the relative group is the reduced group of the quotient:

$$E^{p+q}\big(X^{(p)}, X^{(p-1)}\big) \cong \widetilde{E}^{p+q}\Big(\bigvee_{\alpha} S^p_\alpha\Big)$$

Step 2: Reduced theory of a wedge of spheres. The additivity (wedge) axiom for a generalized theory turns the reduced group of a wedge into a product over the wedge summands,

$$\widetilde{E}^{p+q}\Big(\bigvee_{\alpha} S^p_\alpha\Big) \cong \prod_{\alpha \in J_p} \widetilde{E}^{p+q}(S^p)$$

and the suspension axiom, applied p times to $S^p = \Sigma^p S^0$, gives the suspension isomorphism

$$\widetilde{E}^{p+q}\big(S^p\big) = \widetilde{E}^{p+q}\big(\Sigma^p S^0\big) \cong \widetilde{E}^{q}\big(S^0\big) = E^q(\mathrm{pt})$$

the last equality because $\widetilde{E}^q(S^0) = E^q(\mathrm{pt})$ for the reduced theory on the two-point space S^0. Combining,

$$E_1^{p,q} = E^{p+q}\big(X^{(p)}, X^{(p-1)}\big) \cong \prod_{\alpha \in J_p} E^q(\mathrm{pt}) = C^p_{\mathrm{CW}}\big(X; E^q(\mathrm{pt})\big)$$

the cellular p-cochains with coefficients in $E^q(\mathrm{pt})$, free (as a product) on the p-cells. The identification is natural in X because each step (the quotient, the additivity isomorphism, the suspension isomorphism) is natural for cellular maps.

Step 3: The differential d_1 is the cellular coboundary. The differential $d_1 = j \circ k$ is the composite

$$E^{p+q}\big(X^{(p)}, X^{(p-1)}\big) \xrightarrow{\delta} E^{p+q+1}\big(X^{(p+1)}, X^{(p)}\big)$$

formed from the connecting map $k = \delta$ of the pair $(X^{(p+1)}, X^{(p)})$ followed by the restriction j through $X^{(p)}$, that is, the composite associated to the triple $(X^{(p+1)}, X^{(p)}, X^{(p-1)})$. Under the identification of Step 2 this is the connecting homomorphism of the cofiber sequence

$$\bigvee_{\beta \in J_{p+1}} S^p_\beta \longrightarrow X^{(p)}/X^{(p-1)} = \bigvee_{\alpha \in J_p} S^p_\alpha$$

induced by the attaching maps of the $(p+1)$-cells, read on $\widetilde{E}^*$. For ordinary cohomology this connecting map is, cell by cell, the degree of the composite $S^p_\beta \to X^{(p)}/X^{(p-1)} \to S^p_\alpha$ (the attaching map of e^{p+1}_β projected onto the cell e^p_α), which is the integer entry of the cellular coboundary matrix. This is the content of section 3.5 dualized to cochains. For a general theory E^*, the suspension isomorphism is natural and additive, so the connecting map acts on each coefficient copy of $E^q(\mathrm{pt})$ by multiplication by that same degree integer. Thus d_1 is the cellular coboundary δ^{CW} with the integer matrix acting on $E^q(\mathrm{pt})$-coefficients, completing the proof.

The lemma says the first page of the skeletal spectral sequence of any generalized theory is, in each row q, the cellular cochain complex of X with coefficients in the single abelian group $E^q(\mathrm{pt})$, and its differential is the ordinary cellular coboundary. The generalized theory contributes nothing new at the level of the first differential. It enters only through the coefficient groups $E^q(\mathrm{pt})$, which decorate the rows. The whole departure from ordinary cohomology is thus deferred to the higher differentials $d_2, d_3, \ldots$, which are where a generalized theory can differ from ordinary cohomology. Taking homology of each row computes the second page, and the second page is the named object of the theory.

Theorem 8.6.4: The Atiyah-Hirzebruch Spectral Sequence

Let E^* be a generalized cohomology theory and X a CW complex. Then there is a cohomological spectral sequence $\{E_r^{p,q}, d_r\}$, with differentials d_r of bidegree $(r, 1-r)$, natural in X, whose second page is ordinary cohomology of X with coefficients in the coefficient groups of E^*:

$$E_2^{p,q} \cong H^p\big(X;\, E^q(\mathrm{pt})\big)$$

If X is finite-dimensional (or, more generally, if for each n only finitely many of the groups $E_2^{p,n-p}$ are nonzero and the theory satisfies the limit axiom on X), the spectral sequence converges to the generalized cohomology of X:

$$E_2^{p,q} \implies E^{p+q}(X)$$

Convergence means that $E^n(X)$ carries a finite decreasing filtration $E^n(X) = F^0 \supseteq F^1 \supseteq \cdots \supseteq F^{N+1} = 0$, where $F^p = \ker\big(E^n(X) \to E^n(X^{(p-1)})\big)$ and $N = \dim X$, whose successive quotients are the stable terms $F^p/F^{p+1} \cong E_\infty^{p,n-p}$.

***Proof*:**
Step 1: The spectral sequence and its first page. The skeletal exact couple of definition 8.6.1 is an exact couple, since exactness of its triangle at the three corners is the exactness of the long exact sequence of each pair $(X^{(p)}, X^{(p-1)})$ in the theory E^*, which holds by the exactness axiom of definition 8.2.4. By definition 6.1.6 it generates a spectral sequence $\{E_r^{p,q}, d_r\}$, whose first page and first differential are computed by lemma 8.6.3:

$$E_1^{p,q} \cong C^p_{\mathrm{CW}}\big(X; E^q(\mathrm{pt})\big), \qquad d_1 = \delta^{\mathrm{CW}}$$

The bidegree of d_r is $(r, 1-r)$, the cohomological convention of definition 6.1.6: a connecting map of the skeletal pairs raises the filtration degree p by 1 at the first stage and by r after $r-1$ derivations, while preserving the total degree $p+q$, so the second coordinate drops by $r-1$.

Step 2: The second page. The first differential is the cellular coboundary on each row q, by lemma 8.6.3. The row $E_1^{\bullet,q} = C^\bullet_{\mathrm{CW}}(X; E^q(\mathrm{pt}))$ is therefore the cellular cochain complex of X with coefficients in the abelian group $E^q(\mathrm{pt})$, and its cohomology is, by the agreement of cellular and singular cohomology (section 3.5, dualized),

$$E_2^{p,q} = H^p\big(C^\bullet_{\mathrm{CW}}(X; E^q(\mathrm{pt})),\, \delta^{\mathrm{CW}}\big) = H^p\big(X;\, E^q(\mathrm{pt})\big)$$

the ordinary cohomology of X with coefficients in $E^q(\mathrm{pt})$. This is the claimed second page.

Step 3: Concentration in columns $0 \le p \le \dim X$. For each p, the coefficient group $E_2^{p,q} = H^p(X; E^q(\mathrm{pt}))$ vanishes when $p < 0$ (no negative skeleta, so no negative cellular cochains) and, if X is finite-dimensional of dimension N, when $p > N$ (no cells above dimension N). The rows q, by contrast, are not bounded: $E^q(\mathrm{pt})$ may be nonzero for arbitrarily large positive and negative q. The spectral sequence is therefore not first-quadrant but occupies the vertical strip $0 \le p \le N$, a finite band of columns. This is the structural difference from the Serre spectral sequence and is the one place the dimension axiom would have mattered: dropping it lets the rows extend into negative degrees.

Step 4: Convergence for finite-dimensional X. Fix a total degree n and a spot (p,q) with $p+q=n$. A differential out of (p,q) is $d_r\colon E_r^{p,q} \to E_r^{p+r,q-r+1}$. Its target lies in the strip only if $p+r \le N$, that is $r \le N-p$, so for $r > N-p$ the outgoing differential is zero. A differential into (p,q) is $d_r\colon E_r^{p-r,q+r-1} \to E_r^{p,q}$. Its source lies in the strip only if $p-r \ge 0$, that is $r \le p$, so for $r > p$ the incoming differential is zero. Setting $R(p,q) := \max(p, N-p)+1$, both differentials touching (p,q) vanish for all $r \ge R(p,q)$, and as in proposition 6.1.9 the groups $E_r^{p,q}$ stabilize to a value $E_\infty^{p,q}$ after finitely many pages. The exact couple is that of a filtration of $E^*(X)$: the auxiliary groups $D_1^{p,q} = E^{p+q}(X^{(p)})$ together with the inverse system of restrictions $E^n(X) \to E^n(X^{(p)})$ define the decreasing filtration

$$F^pE^n(X) := \ker\big(E^n(X) \to E^n(X^{(p-1)})\big)$$

the classes restricting to zero on the $(p-1)$-skeleton. For finite-dimensional X the restriction $E^n(X) \to E^n(X^{(N)}) = E^n(X)$ is the identity (as $X^{(N)} = X$), so $F^{N+1}E^n(X) = \ker(\mathrm{id}) = 0$ and $F^0E^n(X) = \ker(E^n(X) \to E^n(\varnothing)) = E^n(X)$, giving a finite filtration of length $N+1$. The general theory of the spectral sequence of an exact couple (theorem 6.1.10, transcribed from chains to the theory E^*, the transcription being valid because only the Eilenberg-Steenrod axioms were used) identifies the limit page with the associated graded of this filtration:

$$E_\infty^{p,q} \cong \frac{F^pE^{p+q}(X)}{F^{p+1}E^{p+q}(X)}$$

For infinite-dimensional X the filtration is no longer finite, and convergence becomes the statement that the restriction $E^n(X) \to \varprojlim_p E^n(X^{(p)})$ is an isomorphism. That is exactly what theorem 8.2.11 governs: it identifies the kernel and cokernel of that restriction as $\varprojlim_p{}^1 E^{n-1}(X^{(p)})$ and 0, so the restriction is an isomorphism precisely when the $\varprojlim^1$ term vanishes, which by corollary 8.2.12 holds whenever the tower $\left(E^{n-1}(X^{(p)})\right)_p$ satisfies the Mittag-Leffler condition, in particular whenever the restrictions between skeleta are surjective. Under the finite-dimensional hypothesis or this limit hypothesis, the convergence $E_2^{p,q} \Rightarrow E^{p+q}(X)$ asserted holds, completing the proof.

The theorem is the promised bridge. Its second page is built entirely from ordinary cohomology of X, every group of which the preceding chapters can compute, and its abutment is the generalized cohomology $E^*(X)$, the object of interest. A generalized theory is thereby computed in two stages: first assemble the ordinary cohomology with the coefficient groups $E^q(\mathrm{pt})$ of the theory laid out in rows, then run the differentials $d_2, d_3, \dots$ and solve the resulting extension problems. When all differentials vanish, the theory $E^*(X)$ is the associated graded of $H^*(X; E^*(\mathrm{pt}))$, so it differs from a direct sum of ordinary cohomology groups only by extensions. The differentials measure the genuine departure of E^* from ordinary cohomology, and they are concentrated in the higher pages because the first differential is always the ordinary cellular coboundary. The edge of the spectral sequence records the relationship between the generalized theory and ordinary cohomology precisely.

Proposition 8.6.5: The Edge Homomorphisms of the Atiyah-Hirzebruch Spectral Sequence

Let E^* and X be as in theorem 8.6.4, with X finite-dimensional and path-connected.

1. *(Coefficient edge.)* On the left column $p = 0$, no differential enters $E_r^{0,q}$, so the surviving term $E_\infty^{0,q}$ is a *subgroup* of $E_2^{0,q} = H^0(X; E^q(\mathrm{pt})) = E^q(\mathrm{pt})$, cut out as the intersection of the kernels of the outgoing differentials, and the composite

$$E^q(X) \twoheadrightarrow F^0E^q(X)/F^1 = E_\infty^{0,q} \hookrightarrow E_2^{0,q} = E^q(\mathrm{pt})$$

   is the restriction $E^q(X) \to E^q(\mathrm{pt})$ to a 0-cell.

2. *(Leading-term edge.)* On the bottom row $q = 0$, the surviving term $E_\infty^{n,0}$ is the subquotient $F^nE^n(X)/F^{n+1}E^n(X)$ of $E_2^{n,0} = H^n(X; E^0(\mathrm{pt}))$. If E^* is connective ($E^q(\mathrm{pt}) = 0$ for $q > 0$), then no differential enters the bottom row, so $E_\infty^{n,0}$ is a *subgroup* of $H^n(X; E^0(\mathrm{pt}))$, and there is a natural edge injection

$$E_\infty^{n,0} = F^nE^n(X)/F^{n+1}E^n(X) \hookrightarrow E_2^{n,0} = H^n\big(X; E^0(\mathrm{pt})\big)$$

***Proof*:**
Step 1: The left column receives nothing. On the left column $p = 0$, an incoming differential is $d_r \colon E_r^{-r,q+r-1} \to E_r^{0,q}$, whose source sits in column $-r < 0$. There are no negative skeleta, so $E_2^{-r,\cdot} = 0$ and no differential enters $(0, q)$. The outgoing differential $d_r \colon E_r^{0,q} \to E_r^{r,q-r+1}$ has target in column $r \geq 2 > 0$, possibly nonzero. Hence each page satisfies $E_{r+1}^{0,q} = \ker(d_r|_{E_r^{0,q}})$, a descending chain of subgroups, and $E_\infty^{0,q} = \bigcap_{r\geq 2} \ker(d_r)$ is a subgroup of $E_2^{0,q}$. For path-connected X, $H^0(X; E^q(\mathrm{pt})) = E^q(\mathrm{pt})$. By the convergence of theorem 8.6.4 the bottom

filtration step is $F^0E^q(X) = E^q(X)$ and $F^0/F^1 = E_\infty^{0,q}$. Since $F^1E^q(X) = \ker(E^q(X) \to E^q(X^{(0)})) = \ker(E^q(X) \to E^q(\mathrm{pt}))$, the quotient $E^q(X)/F^1$ is exactly the image of the restriction $E^q(X) \to E^q(\mathrm{pt})$. Thus the composite $E^q(X) \twoheadrightarrow E_\infty^{0,q} \hookrightarrow E^q(\mathrm{pt})$ is the restriction to a 0-cell, establishing (1).

Step 2: The bottom row as a subquotient. On the bottom row $q = 0$, an outgoing differential $d_r \colon E_r^{n,0} \to E_r^{n+r,1-r}$ lands in row $1 - r \leq -1 < 0$, and an incoming differential $d_r \colon E_r^{n-r,r-1} \to E_r^{n,0}$ comes from row $r - 1 \geq 1 > 0$. For a general theory both the negative rows (carrying $E^{1-r}(\mathrm{pt})$) and the positive rows are nonzero, so the bottom row both emits and receives, and $E_\infty^{n,0}$ is a genuine subquotient of $E_2^{n,0} = H^n(X; E^0(\mathrm{pt}))$, identified by convergence with $F^nE^n(X)/F^{n+1}E^n(X)$. This is the contrast with the Serre spectral sequence, whose bottom row carried permanent cycles: there the dimension axiom forced the rows below $q = 0$ to vanish, and here those rows are exactly the negative coefficient groups that the dimension axiom would have killed.

Step 3: The connective case. If $E^q(\mathrm{pt}) = 0$ for $q > 0$, then the source row $r - 1 > 0$ of every incoming differential is zero, so the bottom row receives nothing and only emits. Then $E_{r+1}^{n,0} = \ker(d_r|_{E_r^{n,0}})$ is a subgroup at every stage, and $E_\infty^{n,0} = \bigcap_{r \geq 2} \ker(d_r)$ is a subgroup of $H^n(X; E^0(\mathrm{pt}))$. By convergence $E_\infty^{n,0} = F^nE^n(X)/F^{n+1}E^n(X)$, giving the natural edge injection $F^nE^n(X)/F^{n+1}E^n(X) \hookrightarrow H^n(X; E^0(\mathrm{pt}))$, establishing (2) and completing the proof.

The coefficient edge is the device that connects the generalized theory back to its coefficient ring: every class of $E^q(X)$ restricts to a coefficient in $E^q(\mathrm{pt})$, and the spectral sequence records which coefficients are hit, namely the subgroup $E_\infty^{0,q}$ surviving all differentials out of the left column. The leading-term edge, available for connective theories, relates ordinary cohomology to the generalized theory in the bottom degree: a class in $H^n(X; E^0(\mathrm{pt}))$ either survives to contribute the top filtration quotient of $E^n(X)$ or is killed by a differential. The contrast with the Serre spectral sequence, whose bottom row consisted of permanent cycles, is the structural fingerprint of dropping the dimension axiom: the coefficient groups now populate rows away from $q = 0$, and for a non-connective theory the positive rows $q > 0$ are among them and emit differentials into the bottom row, so the bottom row is a genuine subquotient of ordinary cohomology and not a subgroup of it. With the machine assembled, the prototypical computation can be carried out in full.

The cleanest test is complex K-theory on complex projective space, where the answer is known independently from Bott periodicity (computed in [33, K-Theory of Complex Projective Space]) and the spectral sequence must reproduce it. The computation exhibits the central phenomenon of the Atiyah-Hirzebruch spectral sequence: a parity obstruction forces every differential to vanish, so the spectral sequence collapses, and the generalized theory is read off as the associated graded of ordinary cohomology with the Bott-periodic coefficients.

Example 8.6.6: $K^*(\mathbb{C}\mathrm{P}^n)$ via the Atiyah-Hirzebruch Spectral Sequence
Let $E^* = K^*$ be complex topological K-theory, with coefficient groups

$$K^q(\mathrm{pt}) = \begin{cases} \mathbb{Z} & q \text{ even} \\ 0 & q \text{ odd} \end{cases}$$

by Bott periodicity (theorem 8.5.10(3), and the coefficient ring is $\mathbb{Z}[\beta, \beta^{-1}]$ with the Bott class $\beta \in K^{-2}(\mathrm{pt})$ a unit, by box 8.5.11). Take $X = \mathbb{C}\mathrm{P}^n$, whose ordinary cohomology ring is $H^*(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) = \mathbb{Z}[x]/(x^{n+1})$ with $|x| = 2$, the truncation of the polynomial ring $H^*(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) = \mathbb{Z}[x]$ of example 6.4.5. Additively $H^p(\mathbb{C}\mathrm{P}^n; \mathbb{Z}) = \mathbb{Z}$ for p even with $0 \leq p \leq 2n$ (generated by $x^{p/2}$) and

0 otherwise.

Step 1: The second page. By theorem 8.6.4,

$$E_2^{p,q} = H^p\big(\mathbb{C}\mathrm{P}^n;\, K^q(\mathrm{pt})\big)$$

This is $\mathbb{Z}$ exactly when both p is even with $0 \le p \le 2n$ (so that $H^p(\mathbb{C}\mathrm{P}^n) = \mathbb{Z}$) and q is even (so that $K^q(\mathrm{pt}) = \mathbb{Z}$), and 0 otherwise:

$$E_2^{p,q} = \begin{cases} \mathbb{Z} & p \text{ even},\ 0 \le p \le 2n,\ q \text{ even} \\ 0 & \text{otherwise} \end{cases}$$

Every nonzero entry of E_2 sits at a spot (p, q) with both coordinates even.

Step 2: The first differential d_3 and why it vanishes. The lowest differential that could be nonzero on the K-theory page is d_3, of bidegree $(3, -2)$ (the differential d_2 has bidegree $(2, -1)$, and any source (p, q) with p, q both even maps under d_2 to $(p+2, q-1)$, where $q-1$ is odd, hence into a zero group, so $d_2 = 0$ for parity reasons). Consider $d_3 \colon E_3^{p,q} \to E_3^{p+3,q-2}$. A nonzero source requires p even, but then the target column $p+3$ is odd, so $E_2^{p+3,q-2} = H^{p+3}(\mathbb{C}\mathrm{P}^n; \cdot) = 0$ (odd-degree cohomology of $\mathbb{C}\mathrm{P}^n$ vanishes). Hence $d_3 = 0$. The same parity count rules out every higher differential: d_r has bidegree $(r, 1-r)$, sending an even-even source (p, q) to $(p+r, q+1-r)$. For the target to be nonzero one needs $p+r$ even (forcing r even, since p is even) *and* $q+1-r$ even (forcing $1-r$ even, that is r odd). No integer r is both even and odd, so for every $r \ge 2$ either the target column is odd or the target row is odd, and $E_2^{p+r,q+1-r} = 0$. Therefore

$$d_r = 0 \quad \text{for all } r \ge 2, \qquad E_2 = E_\infty$$

the spectral sequence collapses at the second page.

Step 3: Reading off $K^*(\mathbb{C}\mathrm{P}^n)$. With $E_\infty = E_2$, the group $K^m(\mathbb{C}\mathrm{P}^n)$ is, up to the extension problem, the direct sum along the diagonal $p + q = m$:

$$K^m(\mathbb{C}\mathrm{P}^n) \cong \bigoplus_{p+q=m} E_\infty^{p,q} = \bigoplus_{\substack{p+q=m \\ p,q \text{ even},\ 0 \le p \le 2n}} \mathbb{Z}$$

For $m = 0$ even: the spots are $(p, -p)$ with p even and $0 \le p \le 2n$, namely $p \in \{0, 2, 4, \dots, 2n\}$, a total of $n+1$ spots, each contributing $\mathbb{Z}$. (Here the row index $q = -p$ runs through the negative even integers, populated precisely because Bott periodicity makes $K^{-p}(\mathrm{pt}) = \mathbb{Z}$. This is where dropping the dimension axiom is felt.) Every group in the filtration is free abelian, so each successive extension $0 \to F^{p+1} \to F^p \to \mathbb{Z} \to 0$ splits, and

$$K^0(\mathbb{C}\mathrm{P}^n) \cong \mathbb{Z}^{\,n+1}$$

For $m = 1$ odd: the spots $(p, 1-p)$ on the diagonal need p even and $1-p$ even, impossible, so every term vanishes and

$$K^1(\mathbb{C}\mathrm{P}^n) = 0$$

These match the values computed in [33, K-Theory of Complex Projective Space]: $K^0(\mathbb{C}\mathrm{P}^n) \cong \mathbb{Z}[t]/(t^{n+1})$, free of rank $n+1$ on $1, t, t^2, \dots, t^n$ for $t = [H] - 1$ the reduced class of the *tautological* line bundle, and $K^1(\mathbb{C}\mathrm{P}^n) = 0$. The spectral sequence recovers the additive structure, and only that. The ring is not read off from it here, and the omission is not an economy: reading products off the abutment would need the filtration to be multiplicative in the sense of definition 6.2.13, the

induced product on E_∞ to be identified with the cup product on E_2, and an edge identification carrying $[L]-1$ to $c_1(L)$. Theorem 8.6.4 gives none of the three, and the leading-term edge of proposition 8.6.5 is gated on connectivity, which K^* fails. Both multiplicative facts are instead proved directly in [33, The Tautological Relation on $\mathbb{C}\mathrm{P}^n$, The Reduced Diagonal on the Top Cell of $\mathbb{C}\mathrm{P}^n$], the relation $t^{n+1}=0$ from an exterior-power identity among bundles over $\mathbb{C}\mathrm{P}^n$ and the generation of the top filtration step from a comparison of the reduced diagonal with the cup product. Once that ring structure is known, the $n+1$ summands above correspond to $1, t, t^2, \dots, t^n$ in the associated graded.

Consistency check on $S^2=\mathbb{C}\mathrm{P}^1$. For $n=1$ the formula gives $K^0(\mathbb{C}\mathrm{P}^1)=\mathbb{Z}^2$ and $K^1(\mathbb{C}\mathrm{P}^1)=0$, that is $K^0(S^2)=\mathbb{Z}\oplus\mathbb{Z}$ and $K^1(S^2)=0$, agreeing with the reduced computation $\widetilde{K}^0(S^2)=\mathbb{Z}$ of example 8.6.2 (the extra $\mathbb{Z}$ being $K^0(\mathrm{pt})$ from the unreduced theory). The collapse here is the same $d_3=0$ parity vanishing in its smallest instance.

The computation is the archetype of an Atiyah-Hirzebruch collapse. The coefficient ring of K-theory is concentrated in even degrees, and the ordinary cohomology of $\mathbb{C}\mathrm{P}^n$ is concentrated in even degrees, so every nonzero entry of the second page sits at an even-even spot. The differentials, which always shift the bidegree by $(r, 1-r)$ with r and $1-r$ of opposite parity, cannot connect two even-even spots, and the page collapses for parity alone. The same parity argument computes K^* of any space whose ordinary cohomology is concentrated in even degrees, which is why K-theory of flag manifolds, Grassmannians, and projective spaces is so readily accessible: the spectral sequence does no work beyond ordinary cohomology and Bott periodicity. The first nonzero differential of the K-theory spectral sequence appears only when the even-degree concentration fails, and the lowest one is d_3, which for general spaces equals a cohomology operation built from the Steenrod squares. That this d_3 is nonzero on suitable spaces is what makes K-theory a finer invariant than ordinary cohomology. Section 9.5 proves that it is the integral operation $\mathrm{Sq}^3_{\mathbb{Z}}=\beta\,\mathrm{Sq}^2\rho$, and the exercises below use that form.

The Atiyah-Hirzebruch spectral sequence returns to where this book began. The introduction listed homotopy, homology and cohomology, K-theory, and cobordism as four ways of assigning algebraic information to a space, each "creating its own theory". The closing chapter has shown that the last three of these are instances of one notion: a generalized cohomology theory (definition 8.2.16) is a functor satisfying the Eilenberg-Steenrod axioms (definition 8.2.4) except dimension, every such theory is representable by a spectrum (theorem 8.3.9, theorem 8.4.6), ordinary cohomology is the theory whose spectrum is the Eilenberg-MacLane spectrum (definition 8.4.2), and K-theory and cobordism are the theories whose spectra are the K-theory spectrum (theorem 8.5.10) and the Thom spectra (definition 8.5.13, theorem 8.5.17). The Atiyah-Hirzebruch spectral sequence is the computational expression of this unification: it computes any generalized theory E^* from ordinary cohomology with coefficients in $E^*(\mathrm{pt})$, so the ordinary cohomology developed in chapter 4 is not one theory among several but the universal first approximation to all of them. The four ways of assigning algebraic information have become one representable framework, with ordinary cohomology its E_2 page and the spectra of K-theory and cobordism its further reaches.

Exercise 8.6.1

1. Compute $K^*(S^n)$ for every $n\geq 1$ using the Atiyah-Hirzebruch spectral sequence. Write out the E_2 page (using $H^*(S^n;\mathbb{Z})$ and the Bott-periodic coefficients $K^q(\mathrm{pt})$), identify the only potentially nonzero differential, show it vanishes for parity reasons, and conclude $K^0(S^n)$ and $K^1(S^n)$. Distinguish the cases n even and n odd, and confirm the reduced values $\widetilde{K}^0(S^n)=\mathbb{Z}$

for n even and 0 for n odd.

2. Let $E^* = H^*(-;\mathbb{Z})$ be ordinary integral cohomology, regarded as a generalized theory. Show that its Atiyah-Hirzebruch spectral sequence collapses at E_2 with $E_2^{p,q} = 0$ for $q \neq 0$, and that the convergence statement of theorem 8.6.4 reduces to the identity $H^n(X;\mathbb{Z}) = H^n(X;\mathbb{Z})$. Which Eilenberg-Steenrod axiom is responsible for the concentration in the single row $q = 0$, and how does the present collapse compare with the skeletal-filtration collapse of example 6.1.11?

3. Consider the unreduced K-theory K^* of the torus $T^2 = S^1 \times S^1$. Using $H^*(T^2;\mathbb{Z}) = \mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}$ in degrees $0, 1, 2$ and the even coefficient ring $K^*(\mathrm{pt})$, write the E_2 page and determine all differentials. Compute $K^0(T^2)$ and $K^1(T^2)$ as abelian groups, and note that here the odd cohomology $H^1(T^2) = \mathbb{Z}^2$ does contribute, unlike the $\mathbb{C}\mathrm{P}^n$ case.

4. The differential d_3 of the K-theory Atiyah-Hirzebruch spectral sequence is the integral operation $\beta\,\mathrm{Sq}^2\rho\colon H^p(X;\mathbb{Z}) \to H^{p+3}(X;\mathbb{Z})$, reduction mod 2 followed by Sq^2 and the Bockstein (section 9.5). Without computing the operation, explain why on any space X with $H^*(X;\mathbb{Z})$ concentrated in even degrees this d_3 must vanish, and give two distinct such spaces (besides $\mathbb{C}\mathrm{P}^n$) where the spectral sequence consequently collapses.

5. Let E^* be a generalized theory with $E^q(\mathrm{pt}) = 0$ for all $q > 0$ and $E^0(\mathrm{pt}) = \mathbb{Z}$ (a connective theory). Show that for any finite-dimensional CW complex X the bottom row $q = 0$ of the Atiyah-Hirzebruch E_2 page is $H^p(X;\mathbb{Z})$, that no differential *enters* the bottom row, and hence that $E_\infty^{n,0}$ is a subgroup of $H^n(X;\mathbb{Z})$. Conclude using proposition 8.6.5 that there is a natural edge injection $F^nE^n(X)/F^{n+1}E^n(X) \hookrightarrow H^n(X;\mathbb{Z})$ from the top filtration quotient of $E^n(X)$. Interpret this as the statement that ordinary cohomology is the "bottom" of every connective generalized theory.

6. Explain precisely where the construction of the Atiyah-Hirzebruch spectral sequence in theorem 8.6.4 would break if the dimension axiom were imposed, and conversely identify every step that used a generalized (non-ordinary) feature of E^*. Use this to argue that the Atiyah-Hirzebruch spectral sequence is, structurally, the Serre spectral sequence's analogue with the fiber replaced by a point and ordinary cohomology replaced by E^*.

8.7 *Beyond the Axioms: Modern Directions

The chapter closed on a single identification: a cohomology theory is a spectrum (theorem 8.3.9, theorem 8.4.6), and the Atiyah-Hirzebruch spectral sequence (theorem 8.6.4) computes any theory from ordinary cohomology, which is thereby the universal first approximation to all of them. This endpoint is also a beginning. The identification of a theory with its representing spectrum remembers the homotopy type of that spectrum and little else. The structure that organizes modern stable homotopy theory (a multiplication, a hierarchy among theories, a base to work over, a universal origin, a full calculus of operations) is invisible to the bare axioms. Each direction below restores one of those layers, and in each the objects of this chapter reappear as a special case of something larger. The lesson the chapter has repeated, that ordinary cohomology is a special case of the generalized theories, applies once more to the generalized theories themselves.

The first missing layer is multiplication. The cup product made $H^\bullet(X;R)$ a graded ring (definition 4.2.7), and box 8.4.13 recorded that the stable homotopy category carries a symmetric

monoidal smash product with the sphere spectrum $\mathbb{S}$ as its unit. Combining the two, the Eilenberg-MacLane spectrum $H\mathbb{Z}$ (definition 8.4.2) is not only a spectrum but a *ring spectrum*: a multiplication $H\mathbb{Z} \wedge H\mathbb{Z} \to H\mathbb{Z}$ that induces the cup product, associative and commutative up to coherent homotopy. K-theory and the cobordism spectra MU, MO of section 8.5 carry the same structure, their products coming from tensor product of vector bundles and Cartesian product of manifolds. A spectrum whose multiplication is commutative up to coherent higher homotopy is an E_∞-ring spectrum, and the sphere $\mathbb{S}$ is the initial one, so every ring spectrum is an algebra over $\mathbb{S}$.

This is the entrance to "brave new algebra": commutative algebra carried out over the sphere spectrum in place of a commutative ring, with modules, tensor products, base change, and even Galois theory lifting from ordinary algebra to the stable category. Making the smash product strictly symmetric monoidal at the point-set level, rather than only up to homotopy, was the technical obstacle that held the theory back for decades. It was resolved in the 1990s by the model categories of Elmendorf, Kriz, Mandell, and May (*Rings, Modules, and Algebras in Stable Homotopy Theory*) and by the symmetric spectra of Hovey, Shipley, and Smith. The modern account recasts the whole framework in the language of ∞-categories, where the category of spectra is the stabilization of the category of spaces and a ring spectrum is an algebra object inside it. This is the subject of Lurie's *Higher Algebra*, and the setting for derived and spectral algebraic geometry, in which a ring spectrum plays the role a commutative ring plays in ordinary geometry.

The second layer is a hierarchy among theories. The complex cobordism spectrum MU of section 8.5 carries a universal property discovered by Quillen: its ring of coefficients $MU^\bullet(\mathrm{pt})$ is the universal ring bearing a *formal group law*, the one-variable power series $F(x, y)$ recording how the first Chern class of a tensor product of line bundles is expressed through the Chern classes of the factors. Every complex-orientable theory inherits such a law, and the two theories the reader knows sit at its simplest values: ordinary cohomology carries the additive law $F(x, y) = x + y$, and K-theory the multiplicative law $F(x, y) = x + y + \beta xy$ with β the Bott class. The geometry of formal groups, through the invariant called the *height*, then stratifies the stable homotopy category into layers (the chromatic filtration), each governed by a periodic theory, the Morava K-theories, interpolating between rational information at one extreme and mod-p ordinary cohomology at the other.

This chromatic picture is the organizing principle behind the stable homotopy groups of spheres, whose first nonzero value $\pi_1^s \cong \mathbb{Z}/2$ was computed in section 5.7: the irregular patterns in those groups resolve, layer by layer, into periodic families indexed by the heights. The foundational results are the nilpotence and periodicity theorems of Devinatz, Hopkins, and Smith, and the elliptic refinement has produced the spectrum of topological modular forms tmf, a theory as central to the subject now as K-theory was a generation ago. The standard references are Ravenel's two books, *Complex Cobordism and Stable Homotopy Groups of Spheres* and *Nilpotence and Periodicity in Stable Homotopy Theory*, named already in the closing chapter.

The third layer is a base to work over. Everything in this book takes place over topological spaces, but the axioms of definition 8.2.4 and the representability of theorem 8.3.9 make sense in any setting equipped with a workable notion of homotopy. Morel and Voevodsky gave one for algebraic geometry, replacing the unit interval by the affine line $\mathbb{A}^1$ and constructing, over any base scheme S, a motivic stable homotopy category $SH(S)$ of motivic spectra. In it algebraic K-theory and algebraic cobordism are representable exactly as their topological counterparts are here, and a distinguished ordinary theory, motivic cohomology, plays the role of $H\mathbb{Z}$ while computing the Chow groups of algebraic cycles.

The payoff was arithmetic. Voevodsky's construction of motivic cohomology and the operations acting on it resolved the Milnor conjecture relating Galois cohomology, quadratic forms, and mod-2

cohomology, work recognized with a Fields Medal. For the topologist the point is that the entire apparatus of this chapter (the axioms, the representing spectra, the Atiyah-Hirzebruch spectral sequence) has an algebro-geometric twin, and the two lie close enough that computations pass between them. The foundational reference is Morel and Voevodsky's $\mathbb{A}^1$-*homotopy theory of schemes*. The companion volume [6] develops the underlying scheme theory.

The fourth layer is a universal origin. A single algebraic variety carries several cohomology theories at once (Betti cohomology of its complex points, algebraic de Rham cohomology, ℓ-adic étale cohomology, crystalline cohomology in characteristic p), and they share so much behaviour that Grothendieck conjectured a single universal theory, the theory of *motives*, through which all of them factor. A motive is the cohomological essence of a variety stripped of the choice of which cohomology to compute. Each such Weil cohomology is then a realization functor out of the category of motives, and the agreement among the individual theories is the trace of the one motivic object standing above them. This is the exact analogue, one level up, of the chapter's own lesson that ordinary cohomology is the universal first approximation recorded on the second page of the Atiyah-Hirzebruch spectral sequence: motives are the universal cohomology of a variety, and the Weil theories are its realizations.

The theory of pure motives for smooth projective varieties rests on Grothendieck's standard conjectures, which remain open, but Voevodsky's triangulated category of mixed motives DM gives an unconditional derived-categorical home for the idea, and motivic cohomology is defined in it. A readable entry point is André's *Une introduction aux motifs*.

The fifth and most encompassing layer is a full calculus of operations. The axioms of definition 8.2.4 equip a cohomology theory with pullback f^* along every map, but a manifold or a variety supports far more: an ordinary pushforward f_*, a proper (or compactly supported) pushforward $f_!$, an exceptional pullback $f^!$, the tensor product $\otimes$, and the internal hom $\underline{\mathrm{Hom}}$. These six operations are bound together by base-change and projection formulas and by a duality exchanging $f_!$ with f_* and $f^!$ with f^*. The right adjoint $f^!$ that surfaced in box 8.3.12 as the basis of Grothendieck duality is one of the six. In this language Poincaré duality (section 4.5) becomes the statement that, for a closed oriented n-manifold mapping to a point, $f^!$ applied to the coefficients is a degree shift by n, and the Atiyah and Spanier-Whitehead dualities of stable homotopy theory are the same duality read on the sphere spectrum. The whole formalism, a cohomology theory presented not as a graded functor but as a package of six operations, originates in Grothendieck's étale cohomology (SGA 4) and its Poincaré-Verdier duality, the arithmetic counterpart taken up in [6] and [11].

Making the six operations precise and functorial across all the relevant geometry at once, with the coherence that the ∞-categorical language requires, is an active research program: the formalization through categories of correspondences in Gaitsgory and Rozenblyum's *A Study in Derived Algebraic Geometry*, and the condensed and analytic mathematics of Clausen and Scholze, in which a six-functor formalism becomes the working definition of a cohomology theory and is extended to p-adic and rigid-analytic geometry (Scholze's lecture notes on six-functor formalisms and Mann's p-adic thesis are the current expositions). This is the direction in which the very notion the chapter set out to axiomatize, "a cohomology theory", has most fully outgrown the Eilenberg-Steenrod axioms: what began as four axioms on a graded functor is now a symmetric monoidal functor equipped with six operations and a duality, and the generalized cohomology theories of definition 8.2.16 are what it induces on topological spaces.

Read together, the five directions trace a single arc. The axioms of this chapter pin a cohomology theory to the homotopy type of a spectrum. Multiplication makes that spectrum a ring, the height of a formal group sorts the rings into a tower, a base scheme carries the whole picture into algebraic geometry, motives give the universal cohomology that the individual theories realize, and the six

operations restore the calculus of pushforward and duality that a graded functor cannot see. At every step the invariants built here by hand, singular cohomology, K-theory, cobordism, reappear as the projection of a structure that remembers more. That the plainest invariant of a space, its ordinary cohomology, should sit at the foot of a hierarchy this tall is the last and largest instance of the pattern the book has followed throughout: an invariant is understood not in isolation but through the structure that produces it.

9⋆

Steenrod Squares

⋆ *Optional. This chapter is an aside; nothing later in the book depends on it.*

By theorem 5.5.12, a cohomology operation $H^n(-;G) \to H^m(-;G')$ is a homotopy class of maps $K(G,n) \to K(G',m)$, and so a single cohomology class on an Eilenberg-MacLane space. The cup square $x \mapsto x^2$ is one such operation. The Steenrod squares Sq^i extend it to a family: for each $i \geq 0$ they give a natural homomorphism $H^n(-;\mathbb{Z}/2) \to H^{n+i}(-;\mathbb{Z}/2)$, the operation Sq^n in degree n is the cup square itself, and all of them commute with suspension. Two results of earlier chapters need these operations. The Atiyah-Hirzebruch spectral sequence of section 8.6 has d_3 as the first differential that can be nonzero for K-theory, and that differential was left unidentified there. The classification of rank-two complex bundles by (c_1, c_2) in theorem 7.5.4 holds on complexes of dimension at most four, because $\pi_5(BU(2)) \cong \mathbb{Z}/2$ contributes the next obstruction, and that obstruction is built from Sq^2.

Section 9.1 constructs the squares from the spaces $K(\mathbb{Z}/2,n)$, proves their axioms and the Cartan formula, identifies Sq^1 with the Bockstein homomorphism, and defines the integral operation $\mathrm{Sq}^3_{\mathbb{Z}} = \beta\,\mathrm{Sq}^2\rho$. Section 9.2 computes the squares on $\mathbb{R}\mathrm{P}^\infty$, on $\mathbb{C}\mathrm{P}^\infty$, on $BSU(2)$ and on $K(\mathbb{Z},n)$ in its first few degrees, and proves Wu's formula for the squares of Stiefel-Whitney classes. Section 9.3 applies these computations to show that $\pi_5(S^3) \cong \mathbb{Z}/2$ and to identify the k-invariant of $BU(2)$ at that obstruction. Section 9.4 proves that $[X,-]$ is exact on a principal fibration and classifies rank-two bundles on complexes of dimension at most five and rank-three bundles on closed orientable 6-manifolds. Section 9.5 shows that the differential d_3 of the Atiyah-Hirzebruch spectral sequence for K-theory is $\mathrm{Sq}^3_{\mathbb{Z}}$.

9.1 Construction and Axioms of the Steenrod Squares

This section constructs the operations $\mathrm{Sq}^i\colon H^n(X,A;\mathbb{Z}/2)\to H^{n+i}(X,A;\mathbb{Z}/2)$ and proves the properties that the rest of the chapter computes with: naturality, additivity, $\mathrm{Sq}^0=\mathrm{id}$, vanishing above the degree, the identification of the top square with the cup square, stability, the identification of Sq^1 with the mod-2 Bockstein, and the Cartan formula. The construction is the extended-power construction. From the fundamental class of $K_n=K(\mathbb{Z}/2,n)$ it builds one class on a quotient of $S^\infty\times_{\mathbb{Z}/2}(K_n\wedge K_n)$, and the restriction of that class along a parametrized diagonal $\mathbb{R}\mathrm{P}^\infty\times K_n\to S^\infty\times_{\mathbb{Z}/2}(K_n\wedge K_n)$ carries all the squares at once as Künneth components.

The mod-2 reduction ρ and the integral Bockstein β come first, because axiom (7) names them. The section ends with the integral operation $\mathrm{Sq}^3_{\mathbb{Z}}=\beta\,\mathrm{Sq}^2\rho$, which section 9.5 identifies inside the Atiyah-Hirzebruch spectral sequence.

Definition 9.1.1: Bockstein Homomorphism

Let (X,A) be a pair of spaces, with $A=\emptyset$ allowed. The *mod-2 reduction*

$$\rho\colon H^n(X,A;\mathbb{Z})\longrightarrow H^n(X,A;\mathbb{Z}/2)$$

is the map induced on cohomology by postcomposition of cochains with the surjection $\mathbb{Z}\to\mathbb{Z}/2$. The *integral Bockstein*

$$\beta\colon H^n(X,A;\mathbb{Z}/2)\longrightarrow H^{n+1}(X,A;\mathbb{Z})$$

is the connecting homomorphism, in the sense of theorem 3.4.3, of the short exact sequence of cochain complexes obtained by applying $\mathrm{Hom}(C_\bullet(X,A),-)$ to the coefficient sequence $0\to\mathbb{Z}\xrightarrow{2}\mathbb{Z}\to\mathbb{Z}/2\to 0$. The *mod-2 Bockstein* $\widetilde{\beta}\colon H^n(X,A;\mathbb{Z}/2)\to H^{n+1}(X,A;\mathbb{Z}/2)$ is the connecting homomorphism obtained in the same way from $0\to\mathbb{Z}/2\xrightarrow{2}\mathbb{Z}/4\to\mathbb{Z}/2\to 0$.

Concretely, if a class is represented by a cocycle c with values in $\mathbb{Z}/2$, one lifts c to an integer-valued cochain $\hat{c}$, and $\delta\hat{c}$ takes even values, so $\delta\hat{c}=2b$ for a unique integral cochain b. The cochain b is a cocycle and $\beta[c]=[b]$. So β measures the failure of an integral lift of c to be a cocycle, and it vanishes on every class that has an integral cocycle lift.

Proposition 9.1.2: The Bockstein Exact Sequence

For every pair (X,A) there is a long exact sequence

$$\cdots\to H^n(X,A;\mathbb{Z})\xrightarrow{2}H^n(X,A;\mathbb{Z})\xrightarrow{\rho}H^n(X,A;\mathbb{Z}/2)\xrightarrow{\beta}H^{n+1}(X,A;\mathbb{Z})\xrightarrow{2}\cdots$$

natural in the pair, where 2 denotes multiplication by 2. Consequently $\beta\rho=0$, $2\beta=0$, and $\ker\rho=2H^n(X,A;\mathbb{Z})$. Moreover the mod-2 Bockstein satisfies $\widetilde{\beta}=\rho\beta$.

***Proof*:**

Step 1: The short exact sequence of cochain complexes. The relative chain group $C_n(X,A)=C_n(X)/C_n(A)$ is free abelian on the singular simplices of X not contained in A, as in the proof

of theorem 4.1.17. Applying $\mathrm{Hom}(C_n(X,A),-)$ to $0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$ gives

$$0 \to C^n(X,A;\mathbb{Z}) \xrightarrow{2} C^n(X,A;\mathbb{Z}) \xrightarrow{\rho} C^n(X,A;\mathbb{Z}/2) \to 0$$

in each degree. Injectivity on the left and exactness in the middle hold because a homomorphism from a free group to $\mathbb{Z}$ is divisible by 2 exactly when all its values are even, and surjectivity on the right holds because a homomorphism to $\mathbb{Z}/2$ lifts to $\mathbb{Z}$ by lifting its values on a basis. These maps commute with the coboundary, which is precomposition with ∂, so this is a short exact sequence of cochain complexes. A cochain complex is a chain complex after reindexing, so theorem 3.4.3 yields the displayed long exact sequence, with β the connecting homomorphism of definition 9.1.1.

Step 2: Naturality. A map of pairs $f\colon (X,A) \to (Y,B)$ induces $f^\sharp$ on cochains commuting with multiplication by 2 and with ρ, hence a map of short exact sequences of cochain complexes. The connecting homomorphism of theorem 3.4.3 is computed by the explicit chase recorded there: represent, lift, apply the coboundary, divide. Each step commutes with $f^\sharp$, since $f^\sharp$ of a lift of c is a lift of $f^\sharp c$ and $f^\sharp$ commutes with δ and with division by 2 on even cochains. Hence $\beta f^* = f^*\beta$, and the two inner maps are natural for the same cochain-level reason.

Step 3: The three identities. The identities $\beta\rho = 0$ and $\ker\rho = 2H^n(X,A;\mathbb{Z})$ are exactness at $H^n(X,A;\mathbb{Z}/2)$ and at the second copy of $H^n(X,A;\mathbb{Z})$. Exactness at $H^{n+1}(X,A;\mathbb{Z})$ gives $\mathrm{im}\beta = \ker(2)$, so every class in the image of β is killed by 2, which is $2\beta = 0$.

Step 4: The mod-2 Bockstein. Let c be a relative $\mathbb{Z}/2$-cocycle, let $\hat{c}$ be an integral cochain lift, and write $\delta\hat{c} = 2b$, so that $\beta[c] = [b]$. The reduction $\hat{c} \bmod 4$ is a $\mathbb{Z}/4$-cochain lifting c along $\mathbb{Z}/4 \to \mathbb{Z}/2$, and

$$\delta(\hat{c} \bmod 4) = 2b \bmod 4$$

which is the image of the $\mathbb{Z}/2$-cochain $b \bmod 2$ under the inclusion $\mathbb{Z}/2 \xrightarrow{2} \mathbb{Z}/4$. By the connecting-homomorphism chase for $0 \to \mathbb{Z}/2 \to \mathbb{Z}/4 \to \mathbb{Z}/2 \to 0$, this says $\widetilde{\beta}[c] = [b \bmod 2] = \rho[b] = \rho\beta[c]$, completing the proof.

The sequence says precisely which mod-2 classes are reductions of integral classes, namely those killed by β, and it says that an integral class dies under reduction exactly when it is divisible by 2. The Bockstein is therefore the device that lets mod-2 computations see integral torsion. The algebra this sequence starts, the derived couples and the Bockstein spectral sequence, is developed in [13, The Bockstein Exact Couple].

Example 9.1.3: The Bockstein on the Projective Plane

Let a generate $H^1(\mathbb{RP}^2;\mathbb{Z}/2)$, where $H^\bullet(\mathbb{RP}^2;\mathbb{Z}/2) = \mathbb{Z}/2[a]/(a^3)$ (example 4.2.8). The space $\mathbb{RP}^2$ is the mapping cone of a degree-2 self-map of S^1, so its homology is $\mathbb{Z},\ \mathbb{Z}/2,\ 0$ in degrees $0,1,2$ (example 8.1.9), and the universal coefficient theorem (theorem 4.1.8) gives

$$H^1(\mathbb{RP}^2;\mathbb{Z}) = \mathrm{Hom}(\mathbb{Z}/2,\mathbb{Z}) \oplus \mathrm{Ext}(\mathbb{Z},\mathbb{Z}) = 0, \qquad H^2(\mathbb{RP}^2;\mathbb{Z}) = \mathrm{Hom}(0,\mathbb{Z}) \oplus \mathrm{Ext}(\mathbb{Z}/2,\mathbb{Z}) \cong \mathbb{Z}/2$$

and $H^3(\mathbb{RP}^2;\mathbb{Z}) = 0$. In the Bockstein sequence of proposition 9.1.2,

$$0 = H^1(\mathbb{RP}^2;\mathbb{Z}) \xrightarrow{\rho} H^1(\mathbb{RP}^2;\mathbb{Z}/2) \xrightarrow{\beta} H^2(\mathbb{RP}^2;\mathbb{Z}) \xrightarrow{2} H^2(\mathbb{RP}^2;\mathbb{Z}) \xrightarrow{\rho} H^2(\mathbb{RP}^2;\mathbb{Z}/2)$$

exactness forces β injective, so βa generates $H^2(\mathbb{RP}^2;\mathbb{Z}) \cong \mathbb{Z}/2$. Multiplication by 2 kills $H^2(\mathbb{RP}^2;\mathbb{Z})$, so $\ker\rho = 2H^2(\mathbb{RP}^2;\mathbb{Z}) = 0$ and ρ is injective on $H^2(\mathbb{RP}^2;\mathbb{Z})$. Hence $\rho\beta a \neq 0$,

and since $H^2(\mathbb{R}\mathrm{P}^2;\mathbb{Z}/2) = \mathbb{Z}/2\{a^2\}$,

$$\widetilde{\beta}a = \rho\beta a = a^2$$

The class a is not the reduction of any integral class, since $\beta a \neq 0$, matching $H^1(\mathbb{R}\mathrm{P}^2;\mathbb{Z}) = 0$ directly. After axiom (7) below, this computation reads $\mathrm{Sq}^1 a = a^2$.

The construction of the squares forms products and quotients of infinite CW complexes, and the product topology on a product of two CW complexes need not agree with the topology its cells generate. Products of CW complexes are therefore given the cell topology $X \times_c Y$ of lemma 1.2.12, and singular cohomology does not distinguish the two.

Clause (5) of lemma 1.2.12 makes $L \times Y$ a CW complex in the product topology for $L = \mathbb{R}\mathrm{P}^\infty$ and any complex Y with countably many cells. The models of $K(\mathbb{Z}/2, n)$ below are built by attaching cells to kill homotopy groups, and their cells are not counted here, so the construction works in the cell topology throughout and compares cohomology through clause (2). Cross products on $X \times_c Y$ are formed as in definition 4.3.7 with its projections, which are continuous, and the identity map to $X \times Y$ carries the cross products of one space to those of the other. Iterated cell topologies are associative, since each is the final topology for the product characteristic maps.

With cell topologies understood, the smash product $X \wedge Y = (X \times_c Y)/(X \vee Y)$ of definition 1.2.18 is a CW complex for based CW complexes X and Y whose basepoints are 0-cells, by lemma 1.2.12(1) and definition 1.2.10, and this cell structure is the one used from here on: the cells of $X \wedge Y$ are a basepoint together with the products of positive-dimensional cells. Orbit spaces of free cellular actions are CW complexes as well.

Lemma 9.1.4: Quotients by Free Cellular Actions

Let Z be a CW complex and G a finite group acting on Z by homeomorphisms that carry open cells to open cells, in such a way that $g \cdot e \neq e$ for every open cell e and every $g \neq 1$. Then the orbit space Z/G is a CW complex whose cells are the images of the cells of Z, the projection $p\colon Z \to Z/G$ is open and carries each open cell homeomorphically onto its image, and the image of every G-invariant subcomplex of Z is a subcomplex of Z/G.

***Proof*:**

Step 1: Point-set properties. For open $U \subseteq Z$ the saturation $\bigcup_g gU$ is open, so p is an open map. Orbits are finite, and two disjoint finite sets in the Hausdorff space Z (theorem 1.2.3) are separated by disjoint open sets: separate each pair of points, then intersect over one index and unite over the other. Given distinct orbits Gx and Gy, choose disjoint open $U \supseteq Gx$ and $V \supseteq Gy$ and replace them by the G-invariant open sets $\bigcap_g gU$ and $\bigcap_g gV$, which still contain the orbits and remain disjoint. Their images are disjoint open neighborhoods of $p(x)$ and $p(y)$, so Z/G is Hausdorff.

Step 2: Cells. The open cells of Z partition Z and the action permutes them, so the images $p(e)$ partition Z/G. The restriction of p to an open cell e is injective: if $p(x) = p(x')$ with $x, x' \in e$, then $gx = x'$ for some g, so $ge \cap e \neq \emptyset$, so $ge = e$ by the permutation property, so $g = 1$ by freeness. An injective restriction of an open map is an open embedding, so $p|_e$ is a homeomorphism onto $p(e)$. Composing the characteristic maps of Z with p gives characteristic maps for these cells whose attaching maps land in the image of lower skeleta, and closure-finiteness is inherited from Z because $p(\bar{e}) = \overline{p(e)}$ meets only the images of the finitely many cells met by the orbit of $\bar{e}$.

Step 3: The topology. The topology of Z is final with respect to its characteristic maps, as in the proof of lemma 1.2.12, and a quotient of a final topology is final with respect to the composed maps. So the quotient topology on Z/G is the final topology for the composed characteristic maps, and since the closed image cells lie in images of finite subcomplexes, the same two-step equivalence as in lemma 1.2.12 identifies this with the weak topology of the image cell structure. The skeletal construction of definition 1.2.1 applied to the image cells therefore rebuilds Z/G. A G-invariant subcomplex Z' has closed, saturated image $p(Z')$, a union of image cells, hence a subcomplex, completing the proof.

On a smash product the cross product refines to a reduced class, and the suspension isomorphism is smashing with the circle class. Throughout, based CW complexes have 0-cell basepoints, $\pi\colon X \times_c Y \to X \wedge Y$ denotes the collapse, and coefficients are $\mathbb{Z}/2$ unless displayed.

Lemma 9.1.5: Smash Classes

Let X and Y be based CW complexes.

1. For every $m \geq 1$ the restriction $H^m(X \times_c Y) \to H^m(X \vee Y)$ is surjective, and
$$\pi^*\colon \widetilde{H}^m(X \wedge Y) \longrightarrow H^m(X \times_c Y)$$
is injective with image the kernel of that restriction.
2. For $a \in \widetilde{H}^p(X)$ and $b \in \widetilde{H}^q(Y)$ with $p+q \geq 1$ there is a unique class $a \wedge b \in \widetilde{H}^{p+q}(X \wedge Y)$ with $\pi^*(a \wedge b) = a \times b$.
3. For based maps f, g, one has $(f \wedge g)^*(a \wedge b) = f^*a \wedge g^*b$, and for the factor swap $\tau\colon X \wedge Y \to Y \wedge X$, one has $\tau^*(b \wedge a) = a \wedge b$.

***Proof*:**

Step 1: Exactness bookkeeping. The pair $(X \times_c Y,\, X \vee Y)$ is a CW pair, so the collapse identifies $H^m(X \times_c Y,\, X \vee Y)$ with $\widetilde{H}^m(X \wedge Y)$ (proposition 4.1.18(1),(3) and lemma 3.4.17), and under this identification the map to the absolute group becomes π^*. By the long exact sequence of the pair (theorem 4.1.17), clause (1) follows once the restriction $H^j(X \times_c Y) \to H^j(X \vee Y)$ is surjective for $j = m$ and $j = m-1$.

Step 2: Surjectivity of restriction to the wedge. Write $j_X\colon X \to X \vee Y$ and $j_Y\colon Y \to X \vee Y$ for the inclusions. For $j \geq 1$, let $z \in H^j(X \vee Y)$ and put $z_X = j_X^* z$ and $z_Y = j_Y^* z$. The class $p_1^* z_X + p_2^* z_Y$ on $X \times_c Y$ restricts on the axis $X \times \{y_0\}$ to z_X plus a class pulled back from the point y_0, and positive-degree classes vanish on a point, so it restricts to z_X there and likewise to z_Y on the other axis. A class on $X \vee Y$ is determined by its restrictions to X and Y (example 4.2.9), so $p_1^* z_X + p_2^* z_Y$ restricts to z. For $j = 0$, the universal coefficient theorem identifies $H^0(-;\mathbb{Z}/2)$ with the group of $\mathbb{Z}/2$-valued functions on path components, induced maps being composition. For such a function u on $X \vee Y$, the function $(u \circ j_Y) \circ p_2 + \big(u \circ j_X - u(*)\big) \circ p_1$ on $X \times_c Y$ restricts on $\{x_0\} \times Y$ to $u \circ j_Y$, because $u(j_X(x_0)) = u(*)$, and on $X \times \{y_0\}$ to $u(*) + u \circ j_X - u(*) = u \circ j_X$. So it restricts to u.

Step 3: The smash class. The cross product $a \times b = p_1^* a \smile p_2^* b$ of definition 4.3.7 restricts to zero on each axis, since one of its two factors restricts through a point and a, b are reduced. By Step 1 it lifts uniquely to $a \wedge b \in \widetilde{H}^{p+q}(X \wedge Y)$, which is clause (2). Naturality in clause

(3) holds because $\pi^*\big((f\wedge g)^*(a\wedge b)\big) = (f\times g)^*(a\times b) = f^*a\times g^*b$ by naturality of cross products, and π^* is injective. For the swap, with $t\colon X\times_c Y\to Y\times_c X$ the coordinate swap, $t^*(b\times a) = t^*p_1^*b \smile t^*p_2^*a = p_2^*b\smile p_1^*a = a\times b$, the last step by the graded commutativity of theorem 4.2.6 with the sign trivial mod 2, and injectivity of π^* again transports this to the smash classes, completing the proof.

The circle $S^1 = I/\partial I$ carries the generator $\varepsilon\in\widetilde{H}^1(S^1;\mathbb{Z}/2)$, and the reduced suspension of definition 5.7.1 is the smash with it. By lemma 1.2.12(3), $X\wedge S^1 = (X\times S^1)/(X\vee S^1)$, and $X\times I\to X\times S^1$ is a quotient map, because every characteristic map of $X\times S^1$ factors through $X\times I$. So $X\wedge S^1$ and ΣX are the same quotient of $X\times I$, and under this equality $\Sigma f = f\wedge \mathrm{id}_{S^1}$ for every based f. The suspension isomorphism $\sigma\colon \widetilde{H}^n(X;G)\to\widetilde{H}^{n+1}(\Sigma X;G)$ is the one of proposition 8.2.5, the coboundary of the cone pair followed by excision, applied to singular cohomology through example 8.2.7. Every class of positive degree is pulled back from an Eilenberg-MacLane space, so identities involving σ can be checked on fundamental classes, and models of these spaces with small skeleta make that check a computation on spheres.

Lemma 9.1.6: A Skeletal Model of $K(\mathbb{Z}/2,n)$

For each $n\geq 1$ there is a CW complex K_n of type $K(\mathbb{Z}/2,n)$ whose only 0-cell is the basepoint and whose remaining cells have dimension at least n, with n-skeleton $\Sigma^{n-1}S^1 = S^n$ and $(n+1)$-skeleton

$$M_n := \Sigma^{n-1}\mathbb{R}\mathrm{P}^2$$

a complex $S^n\cup e^{n+1}$ whose cellular boundary map is multiplication by ± 2. Writing $\iota_n\in H^n(K_n;\mathbb{Z}/2)$ for the fundamental class of definition 5.5.10 and a for the generator of $H^1(\mathbb{R}\mathrm{P}^2;\mathbb{Z}/2)$:

1. $H^j(K_n;\mathbb{Z}/2) = 0$ for $0<j<n$, and $H^n(K_n;\mathbb{Z}/2) = \mathbb{Z}/2\{\iota_n\}$.
2. For $j\leq n+1$ the restriction $H^j(K_n;\mathbb{Z}/2)\to H^j(M_n;\mathbb{Z}/2)$ is injective.
3. ι_n restricts to $\sigma^{n-1}a$ on M_n and to the generator $s_n := \sigma^{n-1}\varepsilon$ of $\widetilde{H}^n(S^n;\mathbb{Z}/2)$ on S^n.

***Proof*:**

Step 1: The Moore complex. The space $\mathbb{R}\mathrm{P}^2$ is the mapping cone of a degree-2 self-map of S^1, with cells in dimensions $0,1,2$ and a single 0-cell (example 8.1.9). Reduced suspension of a complex with 0-cell basepoint is again such a complex, with cells the basepoint and one cell of dimension $k+1$ for each positive k-cell, since $\Sigma X = X\wedge S^1$ carries the smash cell structure above. So M_n has cells in dimensions $0,n,n+1$, and its n-skeleton is Σ^{n-1} of the 1-skeleton S^1 of $\mathbb{R}\mathrm{P}^2$, namely S^n by example 5.7.2. By the suspension isomorphism with integral coefficients and the groups $H^1(\mathbb{R}\mathrm{P}^2;\mathbb{Z}) = 0$, $H^2(\mathbb{R}\mathrm{P}^2;\mathbb{Z})\cong\mathbb{Z}/2$ of example 9.1.3, the group $\widetilde{H}^j(M_n;\mathbb{Z})$ is $\mathbb{Z}/2$ for $j=n+1$ and zero otherwise. The cellular cochain complex of M_n is $\mathbb{Z}\to\mathbb{Z}$ in degrees n and $n+1$ (proposition 4.1.20), the dual of the cellular boundary map, and these groups force that boundary map to be multiplication by ± 2. Cellular homology (theorem 3.5.11) then gives

$$\widetilde{H}_j(M_n;\mathbb{Z})\cong\begin{cases}\mathbb{Z}/2 & j=n\\ 0 & j\neq n\end{cases}$$

The complex M_n has no cells in dimensions 1 through $n-1$, so it is $(n-1)$-connected, as recorded

in Step 1 of the proof of theorem 5.5.4. For $n \geq 2$ the Hurewicz theorem (theorem 5.4.9) now gives $\pi_n(M_n) \cong \mathbb{Z}/2$, and for $n = 1$ the space $M_1 = \mathbb{R}\mathrm{P}^2$ has $\pi_1 \cong \mathbb{Z}/2$ by the covering-space computation of chapter 2.

Step 2: Killing the upper homotopy. Exactly as in Steps 2 and 3 of the proof of theorem 5.5.4, attach cells of dimension $n+2$ to M_n to kill π_{n+1} by lemma 5.5.3, then cells of dimension $n+3$, and so on, and let K_n be the union with the weak topology. Each attachment preserves the homotopy groups strictly below the one killed, and every map from a sphere or a sphere times an interval lands in a finite stage by lemma 1.2.4, so K_n has type $K(\mathbb{Z}/2, n)$. Its cells beyond those of M_n have dimension at least $n+2$, so its n-skeleton and $(n+1)$-skeleton are those of M_n.

Step 3: Cohomology. The space K_n is $(n-1)$-connected, so $H_j(K_n; \mathbb{Z}) = 0$ for $0 < j < n$ and $H_n(K_n; \mathbb{Z}) \cong \pi_n(K_n) = \mathbb{Z}/2$, by theorem 5.4.9 for $n \geq 2$ and by theorem 5.4.5 for $n = 1$. The universal coefficient theorem (theorem 4.1.8) then gives $H^j(K_n; \mathbb{Z}/2) = 0$ for $0 < j < n$, using that $H_0(K_n; \mathbb{Z}) \cong \mathbb{Z}$ is free, and

$$H^n(K_n; \mathbb{Z}/2) \cong \mathrm{Hom}(\mathbb{Z}/2, \mathbb{Z}/2) \cong \mathbb{Z}/2$$

generated by ι_n, since ι_n corresponds to the identity homomorphism and is nonzero. That is clause (1).

Step 4: Restrictions. The quotient K_n/M_n is a CW complex whose positive-dimensional cells have dimension at least $n+2$, so its cellular cochain groups vanish in degrees 1 through $n+1$ and $\widetilde{H}^j(K_n/M_n; \mathbb{Z}/2) = 0$ for $j \leq n+1$ by proposition 4.1.20. By proposition 4.1.18(3) and lemma 3.4.17 this group is $H^j(K_n, M_n; \mathbb{Z}/2)$, and the long exact sequence of the pair gives clause (2). The suspension isomorphism applied $(n-1)$ times to $H^\bullet(\mathbb{R}\mathrm{P}^2; \mathbb{Z}/2)$ gives $H^n(M_n; \mathbb{Z}/2) = \mathbb{Z}/2\{\sigma^{n-1}a\}$ and $H^{n+1}(M_n; \mathbb{Z}/2) = \mathbb{Z}/2\{\sigma^{n-1}(a^2)\}$, and $\widetilde{H}^n(S^n; \mathbb{Z}/2) = \mathbb{Z}/2\{s_n\}$ likewise. The restriction of ι_n to M_n is nonzero by clause (2) with $j = n$ together with clause (1), so it equals $\sigma^{n-1}a$, the only nonzero element. Restricting further to S^n: the quotient M_n/S^n has a single cell besides the basepoint, of dimension $n+1$, so its reduced cohomology vanishes in degree n, and $H^n(M_n; \mathbb{Z}/2) \to H^n(S^n; \mathbb{Z}/2)$ is injective by the same pair sequence. So $\iota_n|_{S^n}$ is the nonzero class s_n. This proves clause (3), completing the proof.

Lemma 9.1.7: The Suspension Isomorphism is the Circle Smash

Let X be a based CW complex, $n \geq 0$, and $x \in \widetilde{H}^n(X; \mathbb{Z}/2)$. Under the equality $\Sigma X = X \wedge S^1$,

$$\sigma(x) = x \wedge \varepsilon$$

***Proof*:**
Both σ and $x \mapsto x \wedge \varepsilon$ are natural in based maps: σ by proposition 8.2.5, the smash by lemma 9.1.5(3) together with $\Sigma f = f \wedge \mathrm{id}$.

Step 1: The universal case in positive degrees. Let $n \geq 1$ and take the model K_n of lemma 9.1.6. The group $\widetilde{H}^{n+1}(\Sigma K_n; \mathbb{Z}/2) \cong \widetilde{H}^n(K_n; \mathbb{Z}/2) = \mathbb{Z}/2$ under σ, so it suffices to show both $\sigma\iota_n$ and $\iota_n \wedge \varepsilon$ are nonzero, forcing equality. The first is nonzero since σ is an isomorphism. For the second, restrict along $S^n \wedge S^1 \subseteq K_n \wedge S^1$: by lemma 9.1.5(3) and lemma 9.1.6(3) the restriction is $s_n \wedge \varepsilon$, and $\pi^*(s_n \wedge \varepsilon) = s_n \times \varepsilon$, which generates $H^{n+1}(S^n \times S^1; \mathbb{Z}/2) \cong H^n(S^n) \otimes H^1(S^1)$ by the Künneth theorem (theorem 4.3.9) with the finite complex S^n as first factor. A class with nonzero restriction is nonzero.

Step 2: Transport, positive degrees. For $x \in \widetilde{H}^n(X;\mathbb{Z}/2)$ choose a based $f\colon X \to K_n$ with $f^*\iota_n = x$ by theorem 5.5.12. Then

$$\sigma(x) = \sigma(f^*\iota_n) = (\Sigma f)^*\sigma\iota_n = (f \wedge \mathrm{id})^*(\iota_n \wedge \varepsilon) = f^*\iota_n \wedge \varepsilon = x \wedge \varepsilon$$

Step 3: Degree zero. A class $x \in \widetilde{H}^0(X;\mathbb{Z}/2)$ is a function on path components vanishing on the basepoint component, and path components of a CW complex are open, so $x = \hat{x}^*(s_0)$ for the based map $\hat{x}\colon X \to S^0$ it defines and s_0 the generator of $\widetilde{H}^0(S^0;\mathbb{Z}/2)$. By naturality it suffices to compare σs_0 and $s_0 \wedge \varepsilon$ in $\widetilde{H}^1(\Sigma S^0;\mathbb{Z}/2) = \widetilde{H}^1(S^1;\mathbb{Z}/2) = \mathbb{Z}/2$. Both are nonzero: σ is an isomorphism, and $\pi^*(s_0 \wedge \varepsilon) = s_0 \times \varepsilon \neq 0$ in $H^1(S^0 \times S^1;\mathbb{Z}/2)$, which is the cohomology of two disjoint circles with $s_0 \times \varepsilon$ the class of one of them. Hence they are equal, completing the proof.

Corollary 9.1.8: Cup Products on a Suspension Vanish

For every based CW complex X and all classes $\alpha \in \widetilde{H}^p(\Sigma X;\mathbb{Z}/2)$, $\gamma \in \widetilde{H}^q(\Sigma X;\mathbb{Z}/2)$ with $p, q \geq 1$, the product $\alpha \smile \gamma$ vanishes.

***Proof*:**
Write $\alpha = \sigma a = a \wedge \varepsilon$ and $\gamma = \sigma b = b \wedge \varepsilon$ with $a, b \in \widetilde{H}^{\geq 0}(X;\mathbb{Z}/2)$, by surjectivity of σ and lemma 9.1.7. Then, with $\pi\colon X \times_c S^1 \to X \wedge S^1$ the collapse and using that π^* is a ring homomorphism (proposition 4.2.5),

$$\pi^*(\alpha \smile \gamma) = (a \times \varepsilon) \smile (b \times \varepsilon) = p_1^*(a \smile b) \smile p_2^*(\varepsilon \smile \varepsilon)$$

after reordering the four factors by the graded commutativity of theorem 4.2.6, the signs being trivial mod 2. Since $\varepsilon^2 \in H^2(S^1;\mathbb{Z}/2) = 0$, the right side vanishes, and π^* is injective in degree $p + q \geq 2$ by lemma 9.1.5(1), so $\alpha \smile \gamma = 0$, as we sought to show.

Lemma 9.1.9: The Bockstein Commutes with Coboundaries and Suspension

For every pair (X, A) the integral Bockstein satisfies $\beta \circ \delta = \delta \circ \beta$ as maps $H^n(A;\mathbb{Z}/2) \to H^{n+2}(X, A;\mathbb{Z})$, where δ is the coboundary of theorem 4.1.17 with the respective coefficients. Consequently $\beta \circ \sigma = \sigma \circ \beta$ for the suspension isomorphism of based CW complexes. Both statements hold with β replaced by ρ, and hence by the mod-2 Bockstein $\rho\beta$.

***Proof*:**
Step 1: The cochain chase. Let $c \in C^n(A;\mathbb{Z}/2)$ be a cocycle. Choose an integral cochain lift $\tilde{c} \in C^n(A;\mathbb{Z})$ and write $\delta_{\mathrm{ch}}\tilde{c} = 2b$ with $b \in C^{n+1}(A;\mathbb{Z})$ a cocycle, so $\beta[c] = [b]$. Extend b to $\tilde{b} \in C^{n+1}(X;\mathbb{Z})$. The relative cocycle $\delta_{\mathrm{ch}}\tilde{b}$ represents $\delta\beta[c]$, by the description of the coboundary in theorem 4.1.17. For the other order, extend c to $\bar{c} \in C^n(X;\mathbb{Z}/2)$ and choose an integral lift $\hat{c} \in C^n(X;\mathbb{Z})$ of $\bar{c}$ with $\hat{c}|_A = \tilde{c}$, possible by lifting on the simplices outside A and reusing $\tilde{c}$ on those inside. The relative cocycle $\delta_{\mathrm{ch}}\bar{c}$ represents $\delta[c]$, and $u := \delta_{\mathrm{ch}}\hat{c} - 2\tilde{b}$ is an integral relative cochain lifting it, since u vanishes on chains in A by $\delta_{\mathrm{ch}}\tilde{c} = 2b$ and reduces to $\delta_{\mathrm{ch}}\bar{c}$ mod 2. Then $\delta_{\mathrm{ch}}u = -2\,\delta_{\mathrm{ch}}\tilde{b}$, so the Bockstein chase gives $\beta\,\delta[c] = [-\delta_{\mathrm{ch}}\tilde{b}] = -\delta\beta[c]$. The class $\beta\,\delta[c]$ lies in the image of β, so it is killed by 2 (proposition 9.1.2), the sign acts trivially, and $\beta\delta = \delta\beta$.

Step 2: Suspension. The suspension isomorphism of proposition 8.2.5 is a composite of the coboundary of the pair (CX, X), an excision-induced map, and restriction maps, the latter two induced by maps of pairs. The Bockstein commutes with maps of pairs by the naturality in proposition 9.1.2 and with the coboundary by Step 1, hence with σ. For ρ every step commutes at the cochain level, since reduction of coefficients commutes with δ_{ch}, with extension, and with restriction, completing the proof.

Fix once and for all the models K_n of lemma 9.1.6 with their fundamental classes ι_n. Write $L := \mathbb{R}\mathrm{P}^\infty$ with its one-cell-per-dimension structure and $H^\bullet(L; \mathbb{Z}/2) = \mathbb{Z}/2[\omega]$, where ω is the polynomial generator computed in example 5.5.9 through the identification $L = B(\mathbb{Z}/2)$ of example 5.5.6, the coefficient field there written $\mathbb{F}_2$ for the ring called $\mathbb{Z}/2$ here. Let $S^\infty = \bigcup_k S^k$ carry the antipodal involution T and the equivariant cell structure with two cells $e^k_\pm$ in each dimension, the two open hemispheres of S^k attached to S^{k-1} by the identity, so that $Te^k_\pm = e^k_\mp$, as in the construction of $\mathbb{R}\mathrm{P}^\infty$ in chapter 1. The two 0-cells are $\pm v_0$, and the basepoint of L is its 0-cell $[v_0]$. For a based CW complex X, let T also denote the swap $x \wedge y \mapsto y \wedge x$ on $X \wedge X$, a cellular homeomorphism fixing the basepoint.

Definition 9.1.10: Quadratic Construction

Let X be a based CW complex whose basepoint x_0 is a 0-cell. For $k \geq 0$ let $\Gamma_k X$ be the orbit space of $S^k \times (X \wedge X)$ under $(v, w) \mapsto (Tv, Tw)$, write $[v, w]$ for the orbit of (v, w), and let $\Gamma X := \bigcup_k \Gamma_k X$ with the weak topology. The *quadratic construction* on X is the quotient

$$\Lambda X := \Gamma X / \bar{L}, \qquad \bar{L} := \{[v, x_0 \wedge x_0] : v \in S^\infty\}$$

based at the image of $\bar{L}$. Its *parametrized diagonal* is the map

$$\nabla_X \colon L \times_c X \longrightarrow \Lambda X, \qquad \nabla_X([v], x) = [v,\, x \wedge x]$$

and a based map $f \colon X \to Y$ induces $\Lambda f \colon \Lambda X \to \Lambda Y$, $[v, x_1 \wedge x_2] \mapsto [v, f(x_1) \wedge f(x_2)]$.

Each $S^k \times (X \wedge X)$ carries the product topology, which is the cell topology by lemma 1.2.12(3) since S^k is finite. The involution permutes its product cells by $e^j_\pm \times c \mapsto e^j_\mp \times Tc$, and no cell is preserved because the hemisphere factors differ, so $\Gamma_k X$ is a CW complex by lemma 9.1.4. Each $\Gamma_k X$ is a subcomplex of $\Gamma_{k+1} X$, and the union with the weak topology is a CW complex, as in Step 3 of the proof of theorem 5.5.4. The image of $S^0 \times (X \wedge X)$ is a single copy of $X \wedge X$, via $w \mapsto [v_0, w]$, written $\Gamma_0 X = X \wedge X$: the identification pairs the two sheets, and the induced continuous bijection is open because the projection is. The basepoint of $X \wedge X$ is fixed by T, so the cells $[e^j \times *]$ form a subcomplex $\bar{L} \cong L$, and ΛX is a based CW complex by definition 1.2.10. Since $\Gamma_0 X$ meets $\bar{L}$ in one point, $X \wedge X$ is a subcomplex of ΛX, the *fiber*. The cells of ΛX other than the basepoint are the images $[e^j \times c]$ of the cells $e^j_\pm \times c$ with c a positive-dimensional cell of $X \wedge X$, and those outside the fiber have $j \geq 1$. For instance $S^n \wedge S^n$ has one cell besides the basepoint, of dimension $2n$, so ΛS^n has exactly one cell in each dimension $2n + j$, $j \geq 0$, its $2n$-skeleton being the fiber $S^n \wedge S^n$.

For each k the formula $(v, x) \mapsto [v,\, x \wedge x]$ defines a map $S^k \times X \to \Gamma_k X$: it is the composite of $\mathrm{id} \times \bar{\Delta}$ with the projection, where $\bar{\Delta} \colon X \to X \wedge X$ is the diagonal followed by the collapse, continuous because $x \mapsto (x, x)$ is continuous into the product topology and hence into $X \times_c X$ by lemma 1.2.12(2). It is invariant under $T \times \mathrm{id}$, since $[Tv, x \wedge x] = [v, T(x \wedge x)] = [v, x \wedge x]$, and the projection $S^k \times X \to \mathbb{R}\mathrm{P}^k \times X$ is an open quotient, so it descends to a continuous map $\mathbb{R}\mathrm{P}^k \times X \to \Gamma_k X$. These are compatible in k, and $L \times_c X$ has the final topology for the subcomplexes

$\mathbb{R}\mathrm{P}^k \times X$, since both test closedness on products of finite subcomplexes. Composing with the collapse to ΛX gives the continuous map ∇_X. The map Λf is continuous because $\mathrm{id} \times (f \wedge f)$ is continuous stage by stage, $f \wedge f$ being induced from $f \times f$, and it carries $\bar{L}$ into $\bar{L}$. Evaluating both sides gives $\Lambda f \circ \nabla_X = \nabla_Y \circ (\mathrm{id} \times f)$ and $\Lambda(g \circ f) = \Lambda g \circ \Lambda f$. The parametrized diagonal sends $L \times \{x_0\}$ to the basepoint and $\{[v_0]\} \times X$ into the fiber, where it is the reduced diagonal $\bar{\Delta}$.

The space ΛX is a family of copies of $X \wedge X$ indexed by the points of L, and over the loop $\mathbb{R}\mathrm{P}^1 \subseteq L$ the copies are glued by the swap T. A class on ΛX restricting to $x \wedge x$ on the fiber therefore extends $x \wedge x$ compatibly with its symmetry, and ∇_X restricts such a class to the diagonal copy of X over each point of L. Producing such a class means extending a map into K_{2N} over cells of high dimension.

Lemma 9.1.11: Extension into a Space with Vanishing Homotopy

Let (Z, A) be a CW pair and Y a path-connected space with $\pi_d(Y) = 0$ for every $d \geq 1$ such that $Z \setminus A$ has a cell of dimension $d+1$. Then every map $A \to Y$ extends to Z.

***Proof*:**

Extend inductively over $A \cup Z^{(m)}$ for increasing m, attaching all cells of a given dimension at once. A cell e^{d+1} of $Z \setminus A$ is attached along $\varphi\colon S^d \to A \cup Z^{(d)}$, and the composite of φ with the map already defined is a map $S^d \to Y$. For $d = 0$ path-connectedness gives an extension over the 1-cell by a path. For $d \geq 1$ the composite is null-homotopic: its based homotopy class lies in $\pi_d(Y) = 0$, the group at a basepoint on its image, which vanishes at one basepoint exactly when it vanishes at all of them because Y is path-connected. A null-homotopy is exactly an extension over the cell through its characteristic map. Extending over all cells of one dimension simultaneously is continuous because the enlarged skeleton carries the final topology of the attachments, and the union over m is continuous because closedness in Z is tested skeleton by skeleton, completing the proof.

Lemma 9.1.12: The Quadratic Class

Let $N \geq 1$, let W be a based CW complex whose only 0-cell is the basepoint and whose remaining cells have dimension at least N, and let $u \in \widetilde{H}^{2N}(W \wedge W; \mathbb{Z}/2)$ satisfy $T^* u = u$.

1. There is a based map $\hat{\lambda}\colon \Lambda W \to K_{2N}$ whose restriction to the fiber $W \wedge W$ represents u under theorem 5.5.12.
2. The $2N$-skeleton of ΛW is the $2N$-skeleton of $W \wedge W$, and restriction to it is injective on $\widetilde{H}^{2N}(\Lambda W; \mathbb{Z}/2)$. In particular restriction $\widetilde{H}^{2N}(\Lambda W; \mathbb{Z}/2) \to \widetilde{H}^{2N}(W \wedge W; \mathbb{Z}/2)$ is injective.

Hence $\hat{\lambda}^* \iota_{2N}$ is the unique class in $\widetilde{H}^{2N}(\Lambda W; \mathbb{Z}/2)$ restricting to u.

***Proof*:**

Step 1: Cells. Every cell of $W \wedge W$ other than the basepoint is a product of two positive-dimensional cells of W, so it has dimension at least $2N$. The cells of ΛW outside the fiber are the cells $[e^j \times c]$ with $j \geq 1$ and c such a cell, of dimension at least $2N+1$. So the $2N$-skeleton of ΛW lies in the fiber and equals the $2N$-skeleton of $W \wedge W$. The quotient of ΛW by this skeleton has no cells in dimensions 1 through $2N$, so its reduced cohomology vanishes through

degree $2N$ (proposition 4.1.20). By proposition 4.1.18(3) and lemma 3.4.17 this is the relative group, and the long exact sequence of the pair (theorem 4.1.17) makes restriction injective in degree $2N$. Restriction to the skeleton factors through restriction to the fiber, so the latter is injective too, which is (2).

Step 2: The circle stage. Let $g\colon W \wedge W \to K_{2N}$ be a based map with $g^*\iota_{2N} = u$. Since $(g \circ T)^*\iota_{2N} = T^*u = u$, theorem 5.5.12 gives a based homotopy $H\colon (W \wedge W) \times I \to K_{2N}$ with $H_0 = g$ and $H_1 = g \circ T$. Let $\gamma\colon I \to S^1$ traverse the closed hemisphere $\bar{e}^1_+$ from v_0 to $-v_0$, and let $\theta\colon I \times (W \wedge W) \to \Gamma_1 W$ be $\theta(t, w) = [\gamma(t), w]$. Every orbit meets $\bar{e}^1_+ \times (W \wedge W)$, so θ is surjective. Its only identifications are $\theta(1, w) = [-v_0, w] = [v_0, Tw] = \theta(0, Tw)$, because for $0 < t < 1$ the point $T\gamma(t)$ lies in the open lower hemisphere. The orbit projection $S^1 \times (W \wedge W) \to \Gamma_1 W$ is a closed map, since the saturation of a closed set under the finite group is a finite union of closed sets, so its restriction to the closed set $\bar{e}^1_+ \times (W \wedge W)$ is closed, and θ is a quotient map. The map $(t, w) \mapsto H(w, t)$ agrees on identified points, since $H(w, 1) = g(Tw) = H(Tw, 0)$, so it descends to a continuous $G\colon \Gamma_1 W \to K_{2N}$ with $G([v_0, w]) = g(w)$ and $G([v, *])$ the basepoint, the homotopy being based.

Step 3: Extension. The map G on $\Gamma_1 W$ and the constant map on $\bar{L}$ agree on $\Gamma_1 W \cap \bar{L}$, so together they define a continuous map on the subcomplex $\Gamma_1 W \cup \bar{L}$ of ΓW. The cells of ΓW outside this subcomplex are the cells $e^j_\pm \times c$ with $j \geq 2$ and $\dim c \geq 2N$, of dimension at least $2N + 2$, and $\pi_d(K_{2N}) = 0$ for every $d \geq 2N + 1$. By lemma 9.1.11 the map extends to ΓW. The extension is constant on $\bar{L}$, so it descends to a based map $\hat{\lambda}\colon \Lambda W \to K_{2N}$ whose restriction to the fiber is g, which is (1). The final sentence follows from (1) and (2), completing the proof.

For $n \geq 1$ the model K_n satisfies the hypotheses with $N = n$, and $u = \iota_n \wedge \iota_n$ is fixed by the swap by lemma 9.1.5(3). Write

$$\lambda_n \in \widetilde{H}^{2n}(\Lambda K_n; \mathbb{Z}/2)$$

for the unique class restricting to $\iota_n \wedge \iota_n$ on the fiber.

Lemma 9.1.13: The Expansion of the Quadratic Class

Let $n \geq 1$. There are unique classes $a_i \in H^{n+i}(K_n; \mathbb{Z}/2)$, $0 \leq i \leq n$, with

$$\nabla^*_{K_n} \lambda_n = \sum_{i=0}^{n} \omega^{n-i} \times a_i \qquad \text{in } H^{2n}(L \times_c K_n; \mathbb{Z}/2)$$

and $a_n = \iota_n \smile \iota_n$.

***Proof*:**
The space L has one cell in each dimension, so $H_k(L; \mathbb{Z}/2)$ is finitely generated for every k, and theorem 4.3.9 over the field $\mathbb{Z}/2$, transported to $L \times_c K_n$ by lemma 1.2.12(2), writes every class of $H^{2n}(L \times_c K_n; \mathbb{Z}/2)$ uniquely as $\sum_j \omega^j \times b_{2n-j}$ with $b_{2n-j} \in H^{2n-j}(K_n; \mathbb{Z}/2)$. By lemma 9.1.6(1), $b_{2n-j} = 0$ for $n < j < 2n$, and $H^0(K_n; \mathbb{Z}/2) = \mathbb{Z}/2$ because K_n is path-connected. So

$$\nabla^*_{K_n} \lambda_n = \sum_{i=0}^{n} \omega^{n-i} \times a_i + c\,\omega^{2n} \times 1, \qquad c \in \mathbb{Z}/2$$

with unique $a_i \in H^{n+i}(K_n; \mathbb{Z}/2)$. Restrict along $L \to L \times_c K_n$, $\ell \mapsto (\ell, x_0)$. The parametrized diagonal is constant on $L \times \{x_0\}$, so the left side restricts to zero. The restriction of $\omega^{n-i} \times a_i$ is

ω^{n-i} times the restriction of a_i to a point, which vanishes because a_i has positive degree, while $\omega^{2n} \times 1$ restricts to $\omega^{2n} \neq 0$ (example 5.5.9). Hence $c = 0$. Now restrict along $K_n \to L \times_c K_n$, $x \mapsto ([v_0], x)$. The parametrized diagonal restricts to $\bar{\Delta}\colon K_n \to K_n \wedge K_n$ in the fiber, so the left side restricts to $\bar{\Delta}^*(\iota_n \wedge \iota_n) = \Delta^*\pi^*(\iota_n \wedge \iota_n) = \Delta^*(\iota_n \times \iota_n) = \iota_n \smile \iota_n$, where $\Delta\colon K_n \to K_n \times_c K_n$ is the diagonal. On the right, ω^{n-i} restricts to zero on a point for $i < n$, so only a_n survives, and $a_n = \iota_n \smile \iota_n$, completing the proof.

Definition 9.1.14: Steenrod Square

For $n \geq 1$ and $i \geq 0$ put $\mathrm{Sq}^i \iota_n := a_i$ for $i \leq n$, with a_i as in lemma 9.1.13, and $\mathrm{Sq}^i \iota_n := 0$ for $i > n$, so that

$$\nabla^*_{K_n} \lambda_n = \sum_{i=0}^{n} \omega^{n-i} \times \mathrm{Sq}^i \iota_n$$

The *Steenrod square* Sq^i is the operation with these universal values:

1. for a based CW complex X, $n \geq 1$ and $x \in \widetilde{H}^n(X; \mathbb{Z}/2)$, set $\mathrm{Sq}^i x := f^* \mathrm{Sq}^i \iota_n \in \widetilde{H}^{n+i}(X; \mathbb{Z}/2)$, where $f\colon X \to K_n$ is a based map with $f^*\iota_n = x$ (theorem 5.5.12);
2. for a CW pair (X, A), $A = \emptyset$ allowed, $\mathrm{Sq}^i\colon H^n(X, A; \mathbb{Z}/2) \to H^{n+i}(X, A; \mathbb{Z}/2)$ is the operation of (1) on $\widetilde{H}^n(X/A; \mathbb{Z}/2)$ transported along the isomorphism $q^*\colon \widetilde{H}^n(X/A; \mathbb{Z}/2) \to H^n(X, A; \mathbb{Z}/2)$ induced by the collapse $q\colon (X, A) \to (X/A, *)$, which is an isomorphism by proposition 4.1.18(1),(3) and lemma 3.4.17 when $A \neq \emptyset$, and by example 4.2.9 with the splitting of proposition 8.2.5 when $A = \emptyset$, where $X/\emptyset := X_+$ is X with a disjoint basepoint;
3. in degree 0, Sq^0 is the identity and $\mathrm{Sq}^i = 0$ for $i > 0$.

The *total square* is $\mathrm{Sq} := \sum_{i \geq 0} \mathrm{Sq}^i$.

The based map f in (1) is unique up to based homotopy, so $\mathrm{Sq}^i x$ is well defined, and Sq^i is a cohomology operation in the sense of theorem 5.5.12: everything is decided on K_n, and the value on any class is a pullback of the universal value. The three clauses agree where they overlap. For a based X and $n \geq 1$, clause (2) reads $x \in \widetilde{H}^n(X) = H^n(X)$ through X_+, and the based map $r\colon X_+ \to X$ that is the identity on X and sends the added point to x_0 induces the identification $\widetilde{H}^n(X) \to \widetilde{H}^n(X_+) = H^n(X)$, so the two readings agree because r^*x is classified by $f \circ r$. For the pair $(X, \{x_0\})$ the quotient is X itself. The definition applies verbatim to a based space of CW homotopy type, where theorem 5.5.12 also holds, and naturality along a homotopy equivalence with a CW complex transfers every property proved below to such spaces.

The construction rests on one fact: $x \wedge x$ is fixed by the swap, so a map representing it is homotopic to its composite with the swap, and lemma 9.1.12 spreads that homotopy over all of L. The squares are the components of the result. Over the 0-cell of L the parametrized diagonal is the reduced diagonal, which is why the top component $\mathrm{Sq}^n x$ is the cup square x^2. The other components record the class away from that point, and the first question is whether the bottom one, the coefficient of ω^n in the universal case, is ι_n or zero, the only two elements of $H^n(K_n; \mathbb{Z}/2)$. In degree one the diagonal of L answers it.

Lemma 9.1.15: The Square Sq^0 in Degree One

$\mathrm{Sq}^0\iota_1 = \iota_1$. Consequently $\mathrm{Sq}^0 x = x$ for every based CW complex X and every $x \in \widetilde{H}^1(X;\mathbb{Z}/2)$.

***Proof*:**

Step 1: Reduction to a scalar. By lemma 9.1.6(1), $H^1(K_1;\mathbb{Z}/2) = \mathbb{Z}/2\{\iota_1\}$, so $\mathrm{Sq}^0\iota_1 = c\,\iota_1$ with $c \in \mathbb{Z}/2$, and definition 9.1.14 with lemma 9.1.13 gives

$$\nabla_{K_1}^*\lambda_1 = \omega \times c\,\iota_1 + 1 \times \iota_1^2$$

Choose a based map $g\colon L \to K_1$ with $g^*\iota_1 = \omega$ (theorem 5.5.12), and let $\Delta_L\colon L \to L \times_c L$ be the diagonal, continuous by lemma 1.2.12(2). Since $\Delta_L^*(\alpha \times \alpha') = \alpha \smile \alpha'$,

$$\big((\mathrm{id} \times g) \circ \Delta_L\big)^* \nabla_{K_1}^*\lambda_1 = \Delta_L^*\big(\omega \times c\,\omega + 1 \times \omega^2\big) = (c+1)\,\omega^2$$

Step 2: The diagonal of L is null-homotopic. By naturality of the parametrized diagonal, $\nabla_{K_1} \circ (\mathrm{id} \times g) \circ \Delta_L = \Lambda g \circ D$, where $D := \nabla_L \circ \Delta_L\colon L \to \Lambda L$ is $D([v]) = [v, [v] \wedge [v]]$. The space S^∞ is a CW complex with all homotopy groups zero (example 5.5.6), so the map to a point is a homotopy equivalence by theorem 5.3.6, and since S^∞ is path-connected there is a homotopy $\alpha\colon S^\infty \times I \to S^\infty$, $\alpha_t = \alpha(-,t)$, from the identity to the constant map at v_0. Put

$$G([v],t) = \big[v,\ [\alpha_t(v)] \wedge [\alpha_t(-v)]\big]$$

On $S^k \times I$ the formula is the composite of a map $S^k \times I \to S^k \times (L \wedge L)$, continuous because its second coordinate is continuous into $L \times L$ from a compact domain and hence into $L \times_c L$ (lemma 1.2.12(2)), with the projection to $\Gamma_k L$. These maps are compatible in k, so they define a continuous map on $S^\infty \times I$. It is invariant under $(v,t) \mapsto (-v,t)$: the second coordinate at $-v$ is the swap of the one at v, and $[-v, Tw] = [v,w]$. The projection $S^\infty \times I \to L \times I$ is a quotient map by [16, Products of Quotient Maps], so $G\colon L \times I \to \Lambda L$ is continuous. At $t = 0$ it is D, since $[-v] = [v]$ in L, and at $t = 1$ it takes the values $[v, *] \in \bar{L}$, so it is constant at the basepoint.

Step 3: Conclusion. The composite $\Lambda g \circ D$ is homotopic to a constant map, so it pulls back the positive-degree class λ_1 to zero. By Step 1, $(c+1)\omega^2 = 0$ in $H^2(L;\mathbb{Z}/2)$, and $\omega^2 \neq 0$ (example 5.5.9), so $c = 1$. For $x \in \widetilde{H}^1(X;\mathbb{Z}/2)$ classified by f, $\mathrm{Sq}^0 x = f^*\mathrm{Sq}^0\iota_1 = f^*\iota_1 = x$, as claimed.

The squares of a cross product are computed by comparing the quadratic construction on a smash product with the smash of the quadratic constructions of its factors.

Lemma 9.1.16: Squares of Cross Products

Let X and Y be CW complexes, $x \in H^p(X;\mathbb{Z}/2)$ and $y \in H^q(Y;\mathbb{Z}/2)$. For every $k \geq 0$,

$$\mathrm{Sq}^k(x \times y) = \sum_{i+j=k} \mathrm{Sq}^i x \times \mathrm{Sq}^j y \qquad \text{in } H^{p+q+k}(X \times_c Y;\mathbb{Z}/2)$$

If X and Y are based and $x \in \widetilde{H}^p(X;\mathbb{Z}/2)$, $y \in \widetilde{H}^q(Y;\mathbb{Z}/2)$ with $p+q \geq 1$, then $\mathrm{Sq}^k(x \wedge y) = \sum_{i+j=k} \mathrm{Sq}^i x \wedge \mathrm{Sq}^j y$ in $\widetilde{H}^{p+q+k}(X \wedge Y;\mathbb{Z}/2)$.

***Proof*:**

Step 1: The comparison map. Let X and Y be based CW complexes. The points of $\Lambda(X \wedge Y)$ other than the basepoint are the classes $[s, (x_1 \wedge y_1) \wedge (x_2 \wedge y_2)]$ with no coordinate at a basepoint, determined up to replacing (s, x_1, y_1, x_2, y_2) by (Ts, x_2, y_2, x_1, y_1). Define

$$v\colon \Lambda(X \wedge Y) \longrightarrow \Lambda X \wedge \Lambda Y, \qquad v\big[s, (x_1 \wedge y_1) \wedge (x_2 \wedge y_2)\big] = [s, x_1 \wedge x_2] \wedge [s, y_1 \wedge y_2]$$

and $v(*) = *$. The formula is invariant under the replacement, since $[Ts, x_2 \wedge x_1] = [s, x_1 \wedge x_2]$ in ΛX and likewise in ΛY. Applied to all of $S^\infty \times X \times Y \times X \times Y$, it sends every point with a coordinate at a basepoint of X or of Y to the basepoint, because one of its two factors then lies in $\bar{L}$. So v is induced by a function on $S^\infty \times X \times Y \times X \times Y$. A map out of a CW complex is continuous when its composites with the characteristic maps are, and the characteristic maps of $\Lambda(X \wedge Y)$ are products of characteristic maps of S^∞, X, Y, X, Y followed by the projections. Composed with v, each becomes the collapse to $\Lambda X \wedge \Lambda Y$ of a map from a compact domain into $\Lambda X \times \Lambda Y$ whose two coordinates are continuous, as composites of continuous maps from compact domains, and such a map is continuous into $\Lambda X \times_c \Lambda Y$ by lemma 1.2.12(2). Hence v is continuous. Two properties follow by evaluating at points:

(a) on the fiber, v is the shuffle $(x_1 \wedge y_1) \wedge (x_2 \wedge y_2) \mapsto (x_1 \wedge x_2) \wedge (y_1 \wedge y_2)$ onto $(X \wedge X) \wedge (Y \wedge Y) \subseteq \Lambda X \wedge \Lambda Y$;

(b) $v \circ \nabla_{X \wedge Y} \circ (\mathrm{id}_L \times \pi) = \pi' \circ (\nabla_X \times \nabla_Y) \circ e$, where $\pi'\colon \Lambda X \times_c \Lambda Y \to \Lambda X \wedge \Lambda Y$ is the collapse and $e\colon L \times_c X \times_c Y \to (L \times_c X) \times_c (L \times_c Y)$ is $e(\ell, x, y) = ((\ell, x), (\ell, y))$. The maps e and $\nabla_X \times \nabla_Y$ are continuous by lemma 1.2.12(2), being maps out of CW complexes whose coordinates are continuous.

For $\alpha, \alpha' \in H^\bullet(L)$, $b \in H^\bullet(X)$ and $b' \in H^\bullet(Y)$, the four projections of $(L \times_c X) \times_c (L \times_c Y)$ composed with e are the projections of $L \times_c X \times_c Y$ onto L, X, L, Y, so graded commutativity with signs trivial mod 2 (theorem 4.2.6) gives

$$e^*\big((\alpha \times b) \times (\alpha' \times b')\big) = (\alpha \smile \alpha') \times b \times b'$$

Step 2: The universal case. Let $m, n \geq 1$, $W := K_m \wedge K_n$ and $N := m+n$. The complex W has a single 0-cell and its other cells have dimension at least N, and a based map $\mu\colon W \to K_N$ with $\mu^* \iota_N = \iota_m \wedge \iota_n$ exists by theorem 5.5.12. The two classes $v^*(\lambda_m \wedge \lambda_n)$ and $(\Lambda\mu)^* \lambda_N$ in $\widetilde{H}^{2N}(\Lambda W)$ are equal. By lemma 9.1.12(2) it suffices to compare them on the $2N$-skeleton of $W \wedge W$, which is $Q := (S^m \wedge S^n) \wedge (S^m \wedge S^n)$ because K_m and K_n have m- and n-skeleta S^m and S^n (lemma 9.1.6). The complex Q has one cell besides the basepoint, of dimension $2N$, so $\widetilde{H}^{2N}(Q) \cong \mathbb{Z}/2$ and it

suffices that both restrictions be nonzero. On the fiber $\Lambda\mu$ is $\mu \wedge \mu$, so by lemma 9.1.5(3) the class $(\Lambda\mu)^*\lambda_N$ restricts on $W \wedge W$ to $(\iota_m \wedge \iota_n) \wedge (\iota_m \wedge \iota_n)$ and on Q to $(s_m \wedge s_n) \wedge (s_m \wedge s_n)$, using lemma 9.1.6(3). By (a) and lemma 9.1.5(3), $v^*(\lambda_m \wedge \lambda_n)$ restricts on Q to the pullback of $(s_m \wedge s_m) \wedge (s_n \wedge s_n)$ along the shuffle homeomorphism $Q \to (S^m \wedge S^m) \wedge (S^n \wedge S^n)$. If A and B are finite based complexes and a, b are nonzero reduced classes on them, then $\pi^*(a \wedge b) = a \times b \neq 0$ by theorem 4.3.9 over the field $\mathbb{Z}/2$, so $a \wedge b \neq 0$. The complexes $S^m \wedge S^n$, $S^m \wedge S^m$ and $S^n \wedge S^n$ have one cell besides the basepoint, so their smash classes $s_m \wedge s_n$, $s_m \wedge s_m$, $s_n \wedge s_n$ are nonzero, and applying the same fact once more shows that both restrictions are nonzero. The two classes are therefore equal.

Now pull both back along $\nabla_W \circ (\mathrm{id}_L \times \pi)$, with $\pi \colon K_m \times_c K_n \to W$. On one side, naturality of ∇ and definition 9.1.14 give

$$\begin{aligned}(\mathrm{id}_L \times \pi)^*\nabla_W^*(\Lambda\mu)^*\lambda_N &= (\mathrm{id}_L \times \mu\pi)^*\nabla_{K_N}^*\lambda_N \\ &= \sum_{k=0}^{N} \omega^{N-k} \times (\mu\pi)^*\mathrm{Sq}^k\iota_N = \sum_{k=0}^{N} \omega^{N-k} \times \mathrm{Sq}^k(\iota_m \times \iota_n)\end{aligned}$$

because $\mu\pi$ classifies $\pi^*(\iota_m \wedge \iota_n) = \iota_m \times \iota_n$. On the other side, (b), lemma 9.1.5(2) and the formula for e^* give

$$(\mathrm{id}_L \times \pi)^*\nabla_W^* v^*(\lambda_m \wedge \lambda_n) = e^*\big(\nabla_{K_m}^*\lambda_m \times \nabla_{K_n}^*\lambda_n\big) = \sum_{i=0}^{m}\sum_{j=0}^{n} \omega^{N-i-j} \times \big(\mathrm{Sq}^i\iota_m \times \mathrm{Sq}^j\iota_n\big)$$

By theorem 4.3.9 with the finite-type factor L, transported by lemma 1.2.12(2), the coefficient of each ω^{N-k} is unique, so $\mathrm{Sq}^k(\iota_m \times \iota_n) = \sum_{i+j=k} \mathrm{Sq}^i\iota_m \times \mathrm{Sq}^j\iota_n$ for $0 \le k \le N$. For $k > N$ both sides vanish, the left by definition 9.1.14 and the right because $\mathrm{Sq}^i\iota_m = 0$ for $i > m$ and $\mathrm{Sq}^j\iota_n = 0$ for $j > n$.

Step 3: Positive degrees. For $p, q \ge 1$ choose maps $f \colon X \to K_p$ and $g \colon Y \to K_q$ classifying x and y. Then $f \times g \colon X \times_c Y \to K_p \times_c K_q$ is continuous by lemma 1.2.12(2), $x \times y = (f \times g)^*(\iota_p \times \iota_q)$, and pulling back the universal case along $f \times g$ gives the formula.

Step 4: Degree zero. Let $p = 0$. The class x is a function on path components, and X is the disjoint union of the subcomplexes $U = x^{-1}(1)$ and $V = x^{-1}(0)$. Then $X \times_c Y$ is the disjoint union of $U \times_c Y$ and $V \times_c Y$, and its cohomology is the product of theirs (example 4.2.9). The class $x \times y$ restricts to $1 \times y$ on $U \times_c Y$ and to 0 on $V \times_c Y$, so by naturality $\mathrm{Sq}^k(x \times y)$ restricts to $1 \times \mathrm{Sq}^k y$ and to 0, which are the restrictions of $x \times \mathrm{Sq}^k y$. Since $\mathrm{Sq}^0 x = x$ and $\mathrm{Sq}^i x = 0$ for $i > 0$, the right side of the formula is also $x \times \mathrm{Sq}^k y$. The case $q = 0$ is symmetric.

Step 5: Smash products. For reduced classes with $p + q \ge 1$, naturality along π and the cross-product formula give $\pi^*\mathrm{Sq}^k(x \wedge y) = \mathrm{Sq}^k(x \times y) = \sum_{i+j=k} \mathrm{Sq}^i x \times \mathrm{Sq}^j y = \pi^*\big(\sum_{i+j=k} \mathrm{Sq}^i x \wedge \mathrm{Sq}^j y\big)$, and π^* is injective in positive degrees by lemma 9.1.5(1), completing the proof.

The following properties are all that the later sections of this chapter use of the construction. Stability comes from the circle smash, $\mathrm{Sq}^0 = \mathrm{id}$ and $\mathrm{Sq}^1 = \rho\beta$ come from stability and the small skeleta of the models K_n, and additivity comes from stability as well.

Theorem 9.1.17: The Steenrod Axioms

The Steenrod squares Sq^i, $i \geq 0$, have the following properties for all CW pairs (X, A), with $A = \emptyset$ allowed, and all degrees $n \geq 0$.

1. (*Naturality*) For every map of CW pairs $f\colon (X, A) \to (Y, B)$, $f^* \circ \mathrm{Sq}^i = \mathrm{Sq}^i \circ f^*$ on $H^n(Y, B; \mathbb{Z}/2)$.
2. (*Additivity*) $\mathrm{Sq}^i(x + y) = \mathrm{Sq}^i x + \mathrm{Sq}^i y$ for $x, y \in H^n(X, A; \mathbb{Z}/2)$.
3. $\mathrm{Sq}^0 = \mathrm{id}$.
4. $\mathrm{Sq}^i x = 0$ for $x \in H^n(X, A; \mathbb{Z}/2)$ and $i > n$.
5. $\mathrm{Sq}^n x = x \smile x$ for $x \in H^n(X, A; \mathbb{Z}/2)$, with the relative cup product of proposition 4.2.3.
6. (*Stability*) For a based CW complex X, $\mathrm{Sq}^i \circ \sigma = \sigma \circ \mathrm{Sq}^i$ on $\widetilde{H}^n(X; \mathbb{Z}/2)$, where σ is the suspension isomorphism. For a CW pair (X, A), $\mathrm{Sq}^i \circ \delta = \delta \circ \mathrm{Sq}^i$ on $H^n(A; \mathbb{Z}/2)$, where $\delta\colon H^n(A; \mathbb{Z}/2) \to H^{n+1}(X, A; \mathbb{Z}/2)$ is the coboundary of theorem 4.1.17.
7. (*Bockstein*) $\mathrm{Sq}^1 = \rho\beta\colon H^n(X, A; \mathbb{Z}/2) \to H^{n+1}(X, A; \mathbb{Z}/2)$.

***Proof*:**

In degree 0 properties (1) through (4) hold by definition 9.1.14(3), so they need proof only in positive degrees.

Step 1: Naturality, (4) and (5). For a based map $g\colon X' \to X$ and $x = f^*\iota_n$, the class g^*x is classified by $f \circ g$, so $\mathrm{Sq}^i g^* x = (fg)^* \mathrm{Sq}^i \iota_n = g^* \mathrm{Sq}^i x$. A map of CW pairs $f\colon (X, A) \to (Y, B)$ induces a based map $\bar{f}\colon X/A \to Y/B$ with $q_Y \circ f = \bar{f} \circ q_X$, so $f^* q_Y^* = q_X^* \bar{f}^*$ and (1) for pairs follows from the based case. Property (4) holds by definition 9.1.14. For (5), lemma 9.1.13 gives $\mathrm{Sq}^n \iota_n = \iota_n \smile \iota_n$, so $\mathrm{Sq}^n x = f^*(\iota_n \smile \iota_n) = x \smile x$ for $x = f^*\iota_n$, the pullback being multiplicative (proposition 4.2.5). For a relative class $x = q^* z$, the same proposition applied to the map of pairs q gives $\mathrm{Sq}^n x = q^*(z \smile z) = x \smile x$. In degree 0 a class is a function with values in $\mathbb{Z}/2$ and $x \smile x = x$, since the cup product of 0-cochains multiplies values.

Step 2: Stability under suspension. Let X be a based CW complex and $x \in \widetilde{H}^n(X; \mathbb{Z}/2)$, $n \geq 0$. By lemma 9.1.7, $\sigma x = x \wedge \varepsilon$, and by the smash form of lemma 9.1.16,

$$\mathrm{Sq}^i(x \wedge \varepsilon) = \sum_{j+l=i} \mathrm{Sq}^j x \wedge \mathrm{Sq}^l \varepsilon$$

Here $\mathrm{Sq}^0 \varepsilon = \varepsilon$ by lemma 9.1.15, $\mathrm{Sq}^1 \varepsilon = \varepsilon^2 = 0$ by (5) because $H^2(S^1; \mathbb{Z}/2) = 0$, and $\mathrm{Sq}^l \varepsilon = 0$ for $l \geq 2$ by (4). So $\mathrm{Sq}^i \sigma x = \mathrm{Sq}^i x \wedge \varepsilon = \sigma \mathrm{Sq}^i x$.

Step 3: $\mathrm{Sq}^0 = \mathrm{id}$. Let $n \geq 1$ and $s_n = \sigma^{n-1}\varepsilon \in \widetilde{H}^n(S^n; \mathbb{Z}/2)$. By Step 2 applied $n-1$ times and lemma 9.1.15, $\mathrm{Sq}^0 s_n = \sigma^{n-1} \mathrm{Sq}^0 \varepsilon = s_n$. The class ι_n restricts to s_n on the n-skeleton S^n of K_n (lemma 9.1.6(3)), so by (1) the class $\mathrm{Sq}^0 \iota_n$ restricts to $s_n \neq 0$. As $H^n(K_n; \mathbb{Z}/2) = \{0, \iota_n\}$, this forces $\mathrm{Sq}^0 \iota_n = \iota_n$, and pulling back gives $\mathrm{Sq}^0 x = x$ for every class of positive degree.

Step 4: Stability under coboundaries. Let (X, A) be a CW pair and $x \in H^n(A; \mathbb{Z}/2)$, with $A \neq \emptyset$ since otherwise there is nothing to prove. Suppose first $n \geq 1$, and let $f\colon A \to K_n$ classify x. The reduced cone CK_n of definition 5.7.1 is contractible and contains K_n as its base, so a contraction

composed with f is a homotopy from a constant map, which extends over X, to f. The homotopy extension property of (X, A) (proposition 1.3.4) then gives $F\colon X \to CK_n$ extending f, a map of CW pairs $(X, A) \to (CK_n, K_n)$ for the cell structure of CK_n as a quotient of $K_n \times I$. Naturality of δ (theorem 4.1.17) and (1) give $\delta \mathrm{Sq}^i x = F^* \delta \mathrm{Sq}^i \iota_n$ and $\mathrm{Sq}^i \delta x = F^* \mathrm{Sq}^i \delta \iota_n$, so it suffices to treat the pair (CK_n, K_n) and the class ι_n. Write $CK_n = C_+K_n$ inside $\Sigma K_n = C_-K_n \cup_{K_n} C_+K_n$, whose cones and equator are subcomplexes. By proposition 8.2.5 the suspension isomorphism is $\sigma = b^*(a^*)^{-1}\delta$, where $a\colon (C_+K_n, K_n) \to (\Sigma K_n, C_-K_n)$ and $b\colon (\Sigma K_n, *) \to (\Sigma K_n, C_-K_n)$ are inclusions of CW pairs inducing isomorphisms. By (1) the squares commute with a^* and b^*, hence with $b^*(a^*)^{-1}$, and by Step 2 with σ, so

$$b^*(a^*)^{-1}\,\mathrm{Sq}^i \delta \iota_n = \mathrm{Sq}^i \sigma \iota_n = \sigma \mathrm{Sq}^i \iota_n = b^*(a^*)^{-1}\,\delta \mathrm{Sq}^i \iota_n$$

and $b^*(a^*)^{-1}$ is injective, which gives $\mathrm{Sq}^i \delta \iota_n = \delta \mathrm{Sq}^i \iota_n$. For $n = 0$, the class x is a function on the open path components of A, so $x = h^* e$ for a map $h\colon A \to \partial I = \{0, 1\}$ and $e \in H^0(\partial I; \mathbb{Z}/2)$ the indicator of 1. As above, h extends to a map of pairs $(X, A) \to (I, \partial I)$, and it suffices to treat e. For $i = 0$ both sides equal δe by Step 3. For $i \geq 1$, $\mathrm{Sq}^i \delta e$ lies in $H^{1+i}(I, \partial I; \mathbb{Z}/2) = 0$, the pair having no relative cells above dimension one, and $\delta \mathrm{Sq}^i e = \delta 0 = 0$.

Step 5: Additivity. It suffices to treat $\widetilde{H}^n(X; \mathbb{Z}/2)$ for a based CW complex X and $n \geq 1$, since q^* is additive. First let $z_1, z_2 \in \widetilde{H}^m(\Sigma X; \mathbb{Z}/2)$, $m \geq 1$. With ΣX the quotient of $X \times I$, let $c\colon \Sigma X \to \Sigma X \vee \Sigma X$ be the pinch, $c[w, t] = j_1[w, 2t]$ for $t \leq \frac{1}{2}$ and $c[w, t] = j_2[w, 2t - 1]$ for $t \geq \frac{1}{2}$, where j_1, j_2 are the wedge inclusions, and let $p_1, p_2\colon \Sigma X \vee \Sigma X \to \Sigma X$ collapse the second and the first summand. The composites $p_1 c[w, t] = [w, \min(2t, 1)]$ and $p_2 c[w, t] = [w, \max(2t - 1, 0)]$ are homotopic to the identity through $[w, \min((1 + s)t, 1)]$ and $[w, \max((1 + s)t - s, 0)]$, $s \in I$, continuous on $\Sigma X \times I$ because $X \times I \times I \to \Sigma X \times I$ is a quotient map by [16, Products of Quotient Maps]. For $z := p_1^* z_1 + p_2^* z_2$ this gives $c^* z = z_1 + z_2$. The class $\mathrm{Sq}^i z$ restricts along j_1 to $\mathrm{Sq}^i j_1^* z = \mathrm{Sq}^i z_1$, because $p_2 j_1$ is constant and z_2 is reduced, and along j_2 to $\mathrm{Sq}^i z_2$. The class $p_1^* \mathrm{Sq}^i z_1 + p_2^* \mathrm{Sq}^i z_2$ has the same restrictions, and restriction to the summands is injective (example 4.2.9), so $\mathrm{Sq}^i z = p_1^* \mathrm{Sq}^i z_1 + p_2^* \mathrm{Sq}^i z_2$ and

$$\mathrm{Sq}^i(z_1 + z_2) = \mathrm{Sq}^i c^* z = c^* \mathrm{Sq}^i z = \mathrm{Sq}^i z_1 + \mathrm{Sq}^i z_2$$

For $x, y \in \widetilde{H}^n(X; \mathbb{Z}/2)$, Step 2 and this case give $\sigma \mathrm{Sq}^i(x + y) = \mathrm{Sq}^i(\sigma x + \sigma y) = \sigma\big(\mathrm{Sq}^i x + \mathrm{Sq}^i y\big)$, and σ is injective.

Step 6: $\mathrm{Sq}^1 = \rho\beta$. Both sides are natural for maps of CW pairs, by (1) and proposition 9.1.2, so it suffices to compare them on ι_n for $n \geq 1$ and in degree 0. In degree 0 the left side vanishes by definition 9.1.14(3). So does the right side: ρ is onto in degree 0, because a $\mathbb{Z}/2$-valued function on path components is also an integral one, and β vanishes there by exactness. For $n \geq 1$, restriction $H^{n+1}(K_n; \mathbb{Z}/2) \to H^{n+1}(M_n; \mathbb{Z}/2)$ is injective (lemma 9.1.6(2)) and carries ι_n to $\sigma^{n-1} a$ (lemma 9.1.6(3)). By Step 2 and (5), $\mathrm{Sq}^1 \sigma^{n-1} a = \sigma^{n-1} \mathrm{Sq}^1 a = \sigma^{n-1}(a^2)$. By lemma 9.1.9 and example 9.1.3, $\rho\beta\sigma^{n-1} a = \sigma^{n-1} \rho\beta a = \sigma^{n-1}(a^2)$. Hence $\mathrm{Sq}^1 \iota_n = \rho\beta \iota_n$, completing the proof.

On a class of degree n the squares run from $\mathrm{Sq}^0 x = x$ to $\mathrm{Sq}^n x = x^2$ and vanish above Sq^n. The operations strictly between are new. Stability makes all of them finer than the cup product: the squares are defined compatibly on all suspensions at once, where cup products of positive-degree classes vanish (corollary 9.1.8), so they detect maps that the ring structure cannot. Additivity was derived from stability: an operation that commutes with suspension is additive, because addition on a suspension is induced by the pinch map. Axiom (7) identifies the first square with the obstruction

to lifting a class to an integral one.

The Cartan formula expands the squares of a product. Its cross-product form is lemma 9.1.16, and the cup form follows by restricting to the diagonal.

Theorem 9.1.18: The Cartan Formula

Let X and Y be CW complexes and $k \geq 0$.

1. For homogeneous $x, y \in H^\bullet(X; \mathbb{Z}/2)$, $\mathrm{Sq}^k(x \smile y) = \sum_{i+j=k} \mathrm{Sq}^i x \smile \mathrm{Sq}^j y$.
2. For homogeneous $x \in H^\bullet(X; \mathbb{Z}/2)$ and $y \in H^\bullet(Y; \mathbb{Z}/2)$, $\mathrm{Sq}^k(x \times y) = \sum_{i+j=k} \mathrm{Sq}^i x \times \mathrm{Sq}^j y$ in $H^\bullet(X \times Y; \mathbb{Z}/2)$, computed as $H^\bullet(X \times_c Y; \mathbb{Z}/2)$ by lemma 1.2.12(2).
3. The total square $\mathrm{Sq}\colon H^\bullet(X; \mathbb{Z}/2) \to H^\bullet(X; \mathbb{Z}/2)$ is a ring homomorphism, and $\mathrm{Sq}(x \times y) = \mathrm{Sq}(x) \times \mathrm{Sq}(y)$.

***Proof*:**
Part (2) is lemma 9.1.16. For (1), the diagonal $\Delta\colon X \to X \times_c X$ is continuous by lemma 1.2.12(2) and satisfies $\Delta^*(x \times y) = x \smile y$, so naturality (theorem 9.1.17(1)) and (2) give

$$\mathrm{Sq}^k(x \smile y) = \Delta^* \mathrm{Sq}^k(x \times y) = \sum_{i+j=k} \Delta^*\left(\mathrm{Sq}^i x \times \mathrm{Sq}^j y\right) = \sum_{i+j=k} \mathrm{Sq}^i x \smile \mathrm{Sq}^j y$$

For (3), $\mathrm{Sq}(x)$ is a finite sum by theorem 9.1.17(4), Sq is additive by theorem 9.1.17(2), and $\mathrm{Sq}(1) = 1$ by definition 9.1.14(3). For homogeneous x and y, summing (1) over k gives $\mathrm{Sq}(x \smile y) = \mathrm{Sq}(x) \smile \mathrm{Sq}(y)$, and additivity extends this to all classes. The cross-product identity follows from (2) in the same way, completing the proof.

The Cartan formula makes the squares computable on any ring generated by classes whose squares are known. If $|x| = 1$, then $\mathrm{Sq}(x) = x + x^2$ by theorem 9.1.17(3), (4) and (5), so $\mathrm{Sq}(x^k) = x^k(1+x)^k$, the computation section 9.2 carries out on $\mathbb{R}\mathrm{P}^\infty$. The first values on spheres and projective planes need only the axioms.

Example 9.1.19: First Values of the Squares

1. *Spheres.* For $n \geq 1$, $\widetilde{H}^\bullet(S^n; \mathbb{Z}/2) = \mathbb{Z}/2\{s_n\}$ is concentrated in degree n. For $i > 0$ the class $\mathrm{Sq}^i s_n$ lies in $H^{n+i}(S^n; \mathbb{Z}/2) = 0$, and on $H^0(S^n; \mathbb{Z}/2)$ every Sq^i with $i > 0$ vanishes by definition 9.1.14(3). So every Sq^i with $i > 0$ vanishes on $H^\bullet(S^n; \mathbb{Z}/2)$.
2. *The real projective plane.* In $H^\bullet(\mathbb{R}\mathrm{P}^2; \mathbb{Z}/2) = \mathbb{Z}/2[a]/(a^3)$ (example 4.2.8), theorem 9.1.17(5) gives
$$\mathrm{Sq}^1 a = a^2 \neq 0$$
This agrees with theorem 9.1.17(7), since $\rho\beta a = a^2$ by example 9.1.3.
3. *The complex projective plane.* By theorem 4.5.12, $H^\bullet(\mathbb{C}\mathrm{P}^2; \mathbb{Z}) = \mathbb{Z}[u]/(u^3)$ with $|u| = 2$, torsion-free and zero in odd degrees. In the Bockstein sequence of proposition 9.1.2 multiplication by 2 on $H^{k+1}(\mathbb{C}\mathrm{P}^2; \mathbb{Z})$ is injective, so $\beta = 0$ and ρ is onto $H^k(\mathbb{C}\mathrm{P}^2; \mathbb{Z}/2)$ with kernel $2H^k(\mathbb{C}\mathrm{P}^2; \mathbb{Z})$. Reduction mod 2 commutes with cup products, since the cup product of

cochains multiplies coefficients and $\mathbb{Z} \to \mathbb{Z}/2$ is a ring homomorphism. So $x := \rho u$ generates $H^2(\mathbb{C}\mathrm{P}^2; \mathbb{Z}/2)$, the class $x^2 = \rho(u^2)$ generates $H^4(\mathbb{C}\mathrm{P}^2; \mathbb{Z}/2) \cong \mathbb{Z}/2$, and theorem 9.1.17(5) gives

$$\mathrm{Sq}^2 x = x^2 \neq 0$$

In (2) and (3) the squares are cup squares, so they repeat what the cohomology rings already record. After one suspension the cup squares vanish (corollary 9.1.8), while stability keeps the operations alive: $\mathrm{Sq}^1 \sigma a = \sigma(a^2) \neq 0$ in $\widetilde{H}^3(\Sigma\mathbb{R}\mathrm{P}^2; \mathbb{Z}/2)$ and $\mathrm{Sq}^2 \sigma x = \sigma(x^2) \neq 0$ in $\widetilde{H}^5(\Sigma\mathbb{C}\mathrm{P}^2; \mathbb{Z}/2)$. Section 9.2 uses the second to show that every suspension of the Hopf map is essential.

Composing the squares with the coefficient operations of definition 9.1.1 gives operations on integral cohomology, and one of them raises degree by three.

Definition 9.1.20: The Integral Operation $\mathrm{Sq}^3_{\mathbb{Z}}$

For a CW pair (X, A), $A = \emptyset$ allowed, and $n \geq 0$, the operation

$$\mathrm{Sq}^3_{\mathbb{Z}} := \beta\, \mathrm{Sq}^2 \rho \colon H^n(X, A; \mathbb{Z}) \longrightarrow H^{n+3}(X, A; \mathbb{Z})$$

is the composite of reduction mod 2, the second Steenrod square and the integral Bockstein.

Proposition 9.1.21: Properties of $\mathrm{Sq}^3_{\mathbb{Z}}$

The operation $\mathrm{Sq}^3_{\mathbb{Z}}$ has the following properties.

1. It is natural for maps of CW pairs and additive.
2. It is stable: $\mathrm{Sq}^3_{\mathbb{Z}} \circ \sigma = \sigma \circ \mathrm{Sq}^3_{\mathbb{Z}}$ on $\widetilde{H}^n(X; \mathbb{Z})$ for a based CW complex X, and $\mathrm{Sq}^3_{\mathbb{Z}} \circ \delta = \delta \circ \mathrm{Sq}^3_{\mathbb{Z}}$ on $H^n(A; \mathbb{Z})$ for a CW pair (X, A).
3. $2\,\mathrm{Sq}^3_{\mathbb{Z}} = 0$.
4. $\rho \circ \mathrm{Sq}^3_{\mathbb{Z}} = \mathrm{Sq}^1 \mathrm{Sq}^2 \rho$.
5. $\mathrm{Sq}^3_{\mathbb{Z}} = 0$ on $H^n(X, A; \mathbb{Z})$ for $n \leq 2$.

***Proof*:**
Step 1: Naturality, additivity and stability. The maps ρ and β are natural homomorphisms (proposition 9.1.2), and Sq^2 is natural and additive (theorem 9.1.17(1),(2)), which gives (1). For (2), ρ and β commute with σ and with δ by lemma 9.1.9, and Sq^2 does by theorem 9.1.17(6).

Step 2: The torsion and the reduction. Part (3) is $2\beta = 0$ (proposition 9.1.2). Part (4) is $\rho\beta = \mathrm{Sq}^1$ (theorem 9.1.17(7)) applied to $\mathrm{Sq}^2 \rho$.

Step 3: Low degrees. Let $z \in H^n(X, A; \mathbb{Z})$. If $n \leq 1$, then $\mathrm{Sq}^2 \rho z = 0$ by theorem 9.1.17(4), because $2 > |\rho z|$. If $n = 2$, then $\mathrm{Sq}^2 \rho z = (\rho z)^2 = \rho(z^2)$ by theorem 9.1.17(5) and the multiplicativity of ρ on relative cochains, so $\mathrm{Sq}^3_{\mathbb{Z}} z = \beta\rho(z^2) = 0$ because $\beta\rho = 0$. This proves (5), completing the proof.

The operation takes values in the 2-torsion of integral cohomology, and its reduction is $\mathrm{Sq}^1 \mathrm{Sq}^2 \rho$. It vanishes on classes of degree at most two for the two reasons of Step 3. On $K(\mathbb{Z}, n)$ with $n \geq 3$ it

gives the first integral torsion above the fundamental class, as section 9.2 computes, and section 9.5 identifies it with the differential d_3 of the Atiyah-Hirzebruch spectral sequence for complex K-theory.

Exercise 9.1.1

1. Let $n \geq 1$ and let X be a CW complex with one cell in each of the dimensions 0, n and $n+1$, whose cellular boundary map from the $(n+1)$-cell to the n-cell is multiplication by 4. Compute H^n and H^{n+1} of X with coefficients $\mathbb{Z}$ and $\mathbb{Z}/2$ from cellular cochains (proposition 4.1.20), and show that the nonzero class $y \in H^n(X;\mathbb{Z}/2)$ has $\beta y \neq 0$ but $\mathrm{Sq}^1 y = 0$. Show more generally that Sq^1 vanishes on every class in the image of ρ.

2. Deduce from theorem 9.1.17(7) and proposition 9.1.2 that $\mathrm{Sq}^1\mathrm{Sq}^1 = 0$, and from theorem 9.1.18 that $\mathrm{Sq}^1(x \smile y) = \mathrm{Sq}^1 x \smile y + x \smile \mathrm{Sq}^1 y$. Verify both identities on the class $a \times a \in H^2(\mathbb{RP}^2 \times \mathbb{RP}^2;\mathbb{Z}/2)$, where a generates $H^1(\mathbb{RP}^2;\mathbb{Z}/2)$.

3. Let $x \in H^n(X;\mathbb{Z}/2)$. Show that $\mathrm{Sq}(x^2) = \mathrm{Sq}(x)^2$, and deduce that $\mathrm{Sq}^{2i}(x^2) = (\mathrm{Sq}^i x)^2$ and $\mathrm{Sq}^{2i+1}(x^2) = 0$ for every $i \geq 0$. Use this to compute the total square on every class of $H^\bullet(\mathbb{CP}^2;\mathbb{Z}/2)$ and of $H^\bullet(S^m \times S^k;\mathbb{Z}/2)$ for $m, k \geq 1$.

4. Compute Sq^1 on $H^\bullet(\mathbb{RP}^3;\mathbb{Z}/2) = \mathbb{Z}/2[a]/(a^4)$ from the axioms and the Cartan formula. Given that $H^\bullet(\mathbb{RP}^3;\mathbb{Z})$ is $\mathbb{Z}, 0, \mathbb{Z}/2, \mathbb{Z}$ in degrees $0, 1, 2, 3$, determine from proposition 9.1.2 which classes of $H^\bullet(\mathbb{RP}^3;\mathbb{Z}/2)$ are reductions of integral classes, and check that they are exactly the classes on which Sq^1 vanishes.

5. Let $n \geq 1$, and suppose that for every based CW complex X a function $\Psi_X \colon \widetilde{H}^n(X;\mathbb{Z}/2) \to \widetilde{H}^{n+1}(\Sigma X;\mathbb{Z}/2)$ is given, natural in the sense that $\Psi_{X'} \circ g^* = (\Sigma g)^* \circ \Psi_X$ for every based map $g \colon X' \to X$. Show that Ψ is either identically zero or equal to the suspension isomorphism σ.

9.2 Computations and Wu's Formula

The axioms of section 9.1 determine the squares on the spaces the rest of this chapter computes on, and this section evaluates them: the total square on $\mathbb{RP}^\infty$ and on $\mathbb{CP}^\infty$, the cohomology of $K(\mathbb{Z}, n)$ through degree $n+3$ in both mod-2 and integral coefficients, and the cohomology of $BSU(2)$. It then proves Wu's formula for the Steenrod squares of the universal Stiefel-Whitney classes and derives the two identities for Chern classes, $\mathrm{Sq}^2 \rho c_2 = \rho c_1 \, \rho c_2 + \rho c_3$ and $\mathrm{Sq}^3_{\mathbb{Z}} c_2 = 0$, on which section 9.3, section 9.4 and section 9.5 rest.

Example 9.2.1: The Steenrod Squares on $\mathbb{RP}^\infty$
The ring is $H^*(\mathbb{RP}^\infty;\mathbb{Z}/2) = \mathbb{Z}/2[w]$ with $|w| = 1$ (example 5.5.9), and $\mathbb{RP}^\infty = K(\mathbb{Z}/2, 1)$ (example 5.5.6). On the generator, theorem 9.1.17 gives $\mathrm{Sq}^0 w = w$ by axiom (3), $\mathrm{Sq}^1 w = w^2$ by axiom (5) since $|w| = 1$, and $\mathrm{Sq}^i w = 0$ for $i > 1$ by axiom (4), so $\mathrm{Sq}(w) = w + w^2 = w(1+w)$. The total square is a ring homomorphism (theorem 9.1.18), so

$$\mathrm{Sq}(w^k) = \big(\mathrm{Sq}\, w\big)^k = w^k(1+w)^k, \qquad \mathrm{Sq}^i(w^k) = \binom{k}{i} w^{k+i}$$

with the binomial coefficient read mod 2. The inclusion $\mathbb{RP}^n \hookrightarrow \mathbb{RP}^\infty$ carries w to the degree-one generator of $H^*(\mathbb{RP}^n;\mathbb{Z}/2) = \mathbb{Z}/2[w]/(w^{n+1})$ (example 5.5.9), so by naturality the same formula holds on $\mathbb{RP}^n$, with $w^{k+i} = 0$ once $k + i > n$. Two computed instances: $\mathrm{Sq}^2(w^3) = \binom{3}{2} w^5 =$

$3w^5 = w^5$, and $\mathrm{Sq}^1(w^2) = \binom{2}{1}w^3 = 2w^3 = 0$. Restricted to $\mathbb{R}\mathrm{P}^2$, the value $\mathrm{Sq}^1 w = w^2$ recovers the computation of example 9.1.19.

Every square of every class on $\mathbb{R}\mathrm{P}^\infty$ is a binomial coefficient mod 2. Because $\mathbb{R}\mathrm{P}^\infty = K(\mathbb{Z}/2, 1)$, the example is universal for degree-one classes: a class $a \in H^1(X; \mathbb{Z}/2)$ over a CW complex is f^*w for some map f (theorem 5.5.12), so $\mathrm{Sq}^i(a^k) = \binom{k}{i}a^{k+i}$ for every such class.

Example 9.2.2: The Steenrod Squares on $\mathbb{C}\mathrm{P}^\infty$

The space $\mathbb{C}\mathrm{P}^\infty$ is a $K(\mathbb{Z}, 2)$ (example 5.5.5), with integral cohomology the polynomial ring $\mathbb{Z}[\iota_2]$ on the fundamental class (example 5.5.11). Set $x := \rho\iota_2$. The integral ring is torsion-free and vanishes in odd degrees, so in the long exact sequence of proposition 9.1.2 multiplication by 2 is injective in every degree. Consequently $\beta = 0$ out of every even degree and ρ is surjective there with kernel $2H^{2k}(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z})$, while $H^{2k+1}(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}/2) = 0$: it is preceded in the sequence by $H^{2k+1}(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) = 0$ and injected by β into the kernel of the injective multiplication by 2 on $H^{2k+2}(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z})$. Since ρ is a ring homomorphism, being induced on cochains by the ring map $\mathbb{Z} \to \mathbb{Z}/2$,

$$H^*(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}/2) = \mathbb{Z}/2[x], \qquad |x| = 2$$

On the generator, $\mathrm{Sq}^1 x \in H^3(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}/2) = 0$, $\mathrm{Sq}^2 x = x^2$ by axiom (5), and the higher squares vanish by axiom (4), so $\mathrm{Sq}(x) = x(1 + x)$. The ring-homomorphism property of the total square gives

$$\mathrm{Sq}(x^k) = x^k(1 + x)^k, \qquad \mathrm{Sq}^{2i}(x^k) = \binom{k}{i}x^{k+i}$$

and every odd square vanishes on every class: the expansion of $x^k(1+x)^k$ has terms in even degrees only, and each homogeneous class of $\mathbb{Z}/2[x]$ is 0 or a power of x. Restriction along $\mathbb{C}\mathrm{P}^n \hookrightarrow \mathbb{C}\mathrm{P}^\infty$ carries x to the generator of $H^2(\mathbb{C}\mathrm{P}^n; \mathbb{Z}/2)$, the reduction of the generator of theorem 4.5.12 by the same Bockstein argument, so the formulas hold on $\mathbb{C}\mathrm{P}^n$ with $x^{k+i} = 0$ for $k + i > n$.

Because $\mathbb{C}\mathrm{P}^\infty = K(\mathbb{Z}, 2)$, this computation is universal for reductions of integral degree-two classes: for $a \in H^2(X; \mathbb{Z})$ over a CW complex, $a = f^*\iota_2$ for some f (theorem 5.5.12), so $\mathrm{Sq}^{2i}\big((\rho a)^k\big) = \binom{k}{i}(\rho a)^{k+i}$ and all odd squares vanish on powers of ρa.

The first application is to attaching maps. Recall the Hopf map $\eta\colon S^3 \to S^2$ of example 5.2.7, and write $\eta_n := \Sigma^{n-2}\eta\colon S^{n+1} \to S^n$ for $n \geq 2$, so $\eta_2 = \eta$, using $\Sigma S^k = S^{k+1}$ (example 8.2.2). The mapping cone C_{η_2} of definition 8.1.7 is $\mathbb{C}\mathrm{P}^2$ (example 4.2.9), where the cup square already detects η. Cup products of positive-degree mod-2 classes vanish on every suspension (corollary 9.1.8), so the ring structure says nothing about η_n for $n \geq 3$. The squares commute with suspension and keep detecting the attaching map.

Proposition 9.2.3: Sq^2 Detects the Hopf Maps

For every $n \geq 2$ the groups $H^n(C_{\eta_n}; \mathbb{Z}/2)$ and $H^{n+2}(C_{\eta_n}; \mathbb{Z}/2)$ are one-dimensional over $\mathbb{Z}/2$ and

$$\mathrm{Sq}^2\colon H^n(C_{\eta_n}; \mathbb{Z}/2) \longrightarrow H^{n+2}(C_{\eta_n}; \mathbb{Z}/2)$$

is an isomorphism. Consequently η_n is essential for every $n \geq 2$.

***Proof*:**

Step 1: The case $n = 2$. Here $C_{\eta_2} = C_\eta = \mathbb{C}\mathrm{P}^2$ (example 4.2.9), whose mod-2 cohomology is $\mathbb{Z}/2[x]/(x^3)$ by example 9.2.2, so the two groups are $\mathbb{Z}/2\langle x\rangle$ and $\mathbb{Z}/2\langle x^2\rangle$. By example 9.1.19, $\mathrm{Sq}^2 x = x^2 \neq 0$.

Step 2: Induction over suspensions. Since $\eta_{n+1} = \Sigma\eta_n$, clause (3) of proposition 8.1.8 identifies the three consecutive terms $S^{n+2} \to S^{n+1} \to \Sigma C_{\eta_n}$ of the Puppe sequence of η_n, up to based homotopy equivalence, with the cofiber sequence of $\Sigma\eta_n = \eta_{n+1}$, so

$$C_{\eta_{n+1}} \simeq \Sigma C_{\eta_n}$$

The suspension isomorphism $\widetilde{H}^j(C_{\eta_n}; \mathbb{Z}/2) \cong \widetilde{H}^{j+1}(\Sigma C_{\eta_n}; \mathbb{Z}/2)$ commutes with Sq^2 by axiom (6) of theorem 9.1.17, and the homotopy equivalence transports Sq^2 isomorphically by naturality, axiom (1). Hence the two groups for η_{n+1} are again one-dimensional and Sq^2 is an isomorphism for η_{n+1} exactly when it is one for η_n. With Step 1 this proves the first claim for all $n \geq 2$ by induction.

Step 3: η_n is essential. Suppose η_n were nullhomotopic, and write $C = C_{\eta_n}$ with inclusion $i\colon S^n \to C$. Clause (1) of proposition 8.1.8, applied with $K = S^n$, gives exactness of

$$[C, S^n]_* \xrightarrow{i^*} [S^n, S^n]_* \xrightarrow{\eta_n^*} [S^{n+1}, S^n]_*$$

and $\eta_n^*[\mathrm{id}] = [\eta_n] = 0$ by assumption, so there is $r\colon C \to S^n$ with $r \circ i \simeq \mathrm{id}$. Let $s \in H^n(S^n; \mathbb{Z}/2)$ be the generator and put $u = r^*(s)$. Then $i^*u = (r \circ i)^* s = s \neq 0$, so u is the generator of $H^n(C; \mathbb{Z}/2) \cong \mathbb{Z}/2$. But

$$\mathrm{Sq}^2 u = r^*\big(\mathrm{Sq}^2 s\big) = 0$$

because $\mathrm{Sq}^2 s$ lies in $H^{n+2}(S^n; \mathbb{Z}/2) = 0$ (chapter 4). This contradicts the isomorphism of Steps 1 and 2, so η_n is not nullhomotopic, completing the proof.

For $n = 2$ the cup-product argument of example 4.2.9 reaches the same conclusion. For $n \geq 3$ that argument is unavailable on the suspension, and the proposition is the tool section 9.3 uses to produce a nonzero element of $\pi_5(S^3)$.

The Eilenberg-MacLane computations below run the Serre spectral sequence of a path-loop fibration, in which fiber classes reach the base through the transgression of definition 6.4.1. The fiber is a loop space, which need not be a CW complex, while definition 9.1.14 and the remark following it define the squares only on spaces of CW homotopy type. A CW approximation (theorem 5.3.10) carries them to every space.

Lemma 9.2.4: Squares Through a CW Approximation

1. For every map $\mu\colon W \to Y$ from a CW complex W to a space Y there are a CW pair (Z, W) and a weak homotopy equivalence $\Psi\colon Z \to Y$ with $\Psi|_W = \mu$.
2. A CW approximation $\gamma\colon X' \to X$ (theorem 5.3.10) induces isomorphisms $\gamma^*\colon H^n(X;\mathbb{Z}/2) \to H^n(X';\mathbb{Z}/2)$ for every n.
3. The operation $\mathrm{Sq}^i_X := (\gamma^*)^{-1} \circ \mathrm{Sq}^i \circ \gamma^*$ on $H^*(X;\mathbb{Z}/2)$ does not depend on the CW approximation γ, and it is the operation of definition 9.1.14 when X has the homotopy type of a CW complex.
4. $f^* \circ \mathrm{Sq}^i_Y = \mathrm{Sq}^i_X \circ f^*$ for every map $f\colon X \to Y$ of spaces, and Sq^i_X has properties (2) through (5) of theorem 9.1.17.

***Proof*:**

Step 1: Clause (1). Let M be the mapping cylinder of μ, containing $W = W \times \{0\}$ as a subspace, and let $r\colon M \to Y$ collapse the cylinder, so that $r|_W = \mu$. As in Step 1 of the proof of theorem 5.3.6, M deformation retracts onto Y along the cylinder coordinate, so r is a homotopy equivalence and hence a weak homotopy equivalence (the remark after definition 5.3.1). By proposition 5.3.11 applied to the pair (M, W) there are a CW pair (Z, W) and a weak homotopy equivalence $\zeta\colon Z \to M$ restricting to the inclusion on W. The composite $\Psi := r \circ \zeta$ induces bijections on all homotopy groups at every basepoint, because r and ζ do, and $\Psi|_W = r|_W = \mu$.

Step 2: Clause (2). The map γ is a weak homotopy equivalence, so $\gamma^*\colon H^n(X;\mathbb{Z}/2) \to H^n(X';\mathbb{Z}/2)$ is an isomorphism for every n by proposition 5.3.15.

Step 3: Clause (3). Let $\gamma'\colon X'' \to X$ be a second CW approximation. The uniqueness clause of theorem 5.3.10 gives a homotopy equivalence $h\colon X' \to X''$ with $\gamma' \circ h \simeq \gamma$, so $\gamma^* = h^*\gamma'^*$, and h^* commutes with the squares by axiom (1) of theorem 9.1.17. Hence

$$(\gamma^*)^{-1}\mathrm{Sq}^i\gamma^* = (\gamma'^*)^{-1}(h^*)^{-1}\mathrm{Sq}^i h^*\gamma'^* = (\gamma'^*)^{-1}\mathrm{Sq}^i\gamma'^*$$

If X is a CW complex, its identity map is a CW approximation, so $\mathrm{Sq}^i_X = \mathrm{Sq}^i$. If X only has the homotopy type of one, the remark after definition 9.1.14 defines $\mathrm{Sq}^i x$ through a map f into $K(\mathbb{Z}/2, n)$ classifying x, and $f \circ \gamma$ classifies $\gamma^* x$, so $\gamma^*\mathrm{Sq}^i x = \mathrm{Sq}^i\gamma^* x$ and again $\mathrm{Sq}^i_X = \mathrm{Sq}^i$.

Step 4: Clause (4). Suppose first that $f = \mu\colon W \to Y$ with W a CW complex, and take (Z, W) and Ψ from clause (1). Then Ψ is a CW approximation of Y, and $\mu = \Psi \circ j$ for the inclusion $j\colon W \to Z$ of a subcomplex. Computing Sq^i_Y through Ψ, as clause (3) allows, and using axiom (1) along j,

$$\mu^*\mathrm{Sq}^i_Y y = j^*\Psi^*\mathrm{Sq}^i_Y y = j^*\mathrm{Sq}^i\Psi^* y = \mathrm{Sq}^i j^*\Psi^* y = \mathrm{Sq}^i\mu^* y$$

For a general $f\colon X \to Y$ and a CW approximation $\gamma\colon X' \to X$, this case applied to $f \circ \gamma$ gives

$$\gamma^* f^*\mathrm{Sq}^i_Y y = \mathrm{Sq}^i(f\gamma)^* y = \mathrm{Sq}^i\gamma^* f^* y = \gamma^*\mathrm{Sq}^i_X f^* y$$

and γ^* is injective by clause (2). Finally γ^* preserves degrees, sums and cup products (proposition 4.2.5), so properties (2) through (5), which hold on X', hold for Sq^i_X, completing the proof.

From here on Sq^i on the mod-2 cohomology of any space means Sq^i_X, and $\mathrm{Sq}^3_{\mathbb{Z}} := \beta\,\mathrm{Sq}^2\rho$ on its integral cohomology, as in definition 9.1.20. The squares are then defined on the fiber of every fibration and on every paracompact base of a vector bundle, and they commute with every induced map. Through the transgression, a square of a fiber class determines a square of a base class.

Lemma 9.2.5: Kudo Transgression

Let $p\colon E \to B$ be a Serre fibration over a path-connected CW complex B, let b_0 be a 0-cell of B with path-connected fiber $F = p^{-1}(b_0)$, and suppose that $\pi_1(B)$ acts trivially on $H^*(F;\mathbb{Z}/2)$. Let $x \in H^q(F;\mathbb{Z}/2)$ be transgressive and let $y \in H^{q+1}(B;\mathbb{Z}/2)$ represent $\tau(x)$ in the zig-zag sense of theorem 6.4.3:

$$p^*(y) = \delta(x) \in H^{q+1}(E,F;\mathbb{Z}/2)$$

under the identification $H^{q+1}(B,b_0;\mathbb{Z}/2) \cong H^{q+1}(B;\mathbb{Z}/2)$ used there. Then for every $i \geq 0$ the class $\mathrm{Sq}^i x$, formed as in lemma 9.2.4, is transgressive and $\mathrm{Sq}^i y$ represents $\tau(\mathrm{Sq}^i x)$, with $p^*(\mathrm{Sq}^i y) = \delta(\mathrm{Sq}^i x)$.

***Proof*:**

Step 1: A CW pair over (E,F). Let $\gamma\colon F' \to F$ be a CW approximation (theorem 5.3.10). Clause (1) of lemma 9.2.4, applied to the composite $F' \to F \subseteq E$, gives a CW pair (Z,F') and a weak homotopy equivalence $\Psi\colon Z \to E$ with $\Psi|_{F'} = \gamma$, so Ψ is a map of pairs $(Z,F') \to (E,F)$. By clause (2) of that lemma, Ψ and γ induce isomorphisms on mod-2 cohomology in every degree. The five lemma (lemma 3.4.6), applied to the ladder formed by the long exact sequences of the two pairs, which commutes by the naturality clause of theorem 4.1.17, shows that $\Psi^*\colon H^*(E,F;\mathbb{Z}/2) \to H^*(Z,F';\mathbb{Z}/2)$ is an isomorphism. Since b_0 is a 0-cell and $\Psi(F') = \gamma(F') \subseteq F = p^{-1}(b_0)$, the composite $p \circ \Psi\colon (Z,F') \to (B,b_0)$ is a map of CW pairs.

Step 2: The squares across the zig-zag. The identification $H^{q+1}(B,b_0;\mathbb{Z}/2) \cong H^{q+1}(B;\mathbb{Z}/2)$ is induced by the map of CW pairs $(B,\emptyset) \to (B,b_0)$, so it commutes with the squares by axiom (1) of theorem 9.1.17, and y and $\mathrm{Sq}^i y$ may be read in $H^*(B,b_0;\mathbb{Z}/2)$. Write δ_Z for the coboundary of the pair (Z,F'). Then

$$\begin{aligned}\Psi^* p^*\big(\mathrm{Sq}^i y\big) &= \mathrm{Sq}^i (p\Psi)^* y = \mathrm{Sq}^i \Psi^* \delta(x) = \mathrm{Sq}^i \delta_Z(\gamma^* x)\\ &= \delta_Z\big(\mathrm{Sq}^i \gamma^* x\big) = \delta_Z\big(\gamma^* \mathrm{Sq}^i x\big) = \Psi^* \delta\big(\mathrm{Sq}^i x\big)\end{aligned}$$

by naturality of the squares along the map of CW pairs $p\Psi$ (first equality, axiom (1)), the hypothesis $p^*(y) = \delta(x)$ (second), naturality of the coboundary along Ψ, which restricts to γ on F' (third and sixth, theorem 4.1.17), stability on the CW pair (Z,F') (fourth, axiom (6)), and the definition of the squares on F through γ (fifth, lemma 9.2.4(3)). Since Ψ^* is injective, $p^*(\mathrm{Sq}^i y) = \delta(\mathrm{Sq}^i x)$ in $H^{q+i+1}(E,F;\mathbb{Z}/2)$.

Step 3: The zig-zag. In particular $\delta(\mathrm{Sq}^i x)$ lies in the image of p^*, so $\mathrm{Sq}^i x$ is transgressive by theorem 6.4.3, and by the same theorem the class $\mathrm{Sq}^i y$, whose image under p^* is $\delta(\mathrm{Sq}^i x)$, represents $\tau(\mathrm{Sq}^i x)$, as we sought to show.

The lemma turns the one computation the spectral sequence forces, the transgression of the fiber's bottom class, into a family of computations, one for each square of that class. Its first use is the cohomology of $K(\mathbb{Z},3)$.

Proposition 9.2.6: The Cohomology of $K(\mathbb{Z},3)$ in Low Degrees

For $q \leq 7$,

$$H^q(K(\mathbb{Z},3);\mathbb{Z}/2) \cong \begin{cases} \mathbb{Z}/2 & q = 0,3,5,6 \\ 0 & q = 1,2,4,7 \end{cases}$$

spanned in positive degrees by $\rho\iota_3$, by $\mathrm{Sq}^2\rho\iota_3$, and by $(\rho\iota_3)^2 = \mathrm{Sq}^3\rho\iota_3$. Integrally, $H^5(K(\mathbb{Z},3);\mathbb{Z}) = 0$ and $H^6(K(\mathbb{Z},3);\mathbb{Z}) \cong \mathbb{Z}/2$, generated by $\mathrm{Sq}^3_{\mathbb{Z}}\iota_3 = \iota_3^2$. Moreover $\mathrm{Sq}^1\mathrm{Sq}^2\rho\iota_3 = \mathrm{Sq}^3\rho\iota_3$.

***Proof*:**

Step 1: The fibration and its second page. Let B be a CW complex of type $K(\mathbb{Z},3)$ (theorem 5.5.4), based at a 0-cell b_0, and take the path-loop fibration $\Omega B \to PB \to B$ of example 5.2.15, with contractible total space, as in the proof of proposition 6.3.5. Its fiber $F := \Omega B$ is path-connected with $\pi_k(F) \cong \pi_{k+1}(B)$ for every $k \geq 0$ (example 5.2.15). So for a CW approximation $\gamma\colon F' \to F$ (theorem 5.3.10) the CW complex F' has type $K(\mathbb{Z},2)$ and is homotopy equivalent to $\mathbb{C}\mathrm{P}^\infty$ by theorem 5.5.4 and example 5.5.5. The induced maps on mod-2 cohomology are ring isomorphisms commuting with the squares (lemma 9.2.4), so example 9.2.2 gives $H^*(\Omega B;\mathbb{Z}/2) = \mathbb{Z}/2[x]$ with $|x| = 2$. The base is simply connected, so the action of $\pi_1(B)$ is trivial and theorem 6.2.8 applies with $\mathbb{Z}/2$ coefficients: $E_2^{p,q} = H^p\big(B;H^q(\Omega B;\mathbb{Z}/2)\big)$, which is $H^p(B;\mathbb{Z}/2)\otimes H^q(\Omega B;\mathbb{Z}/2)$ because every coefficient group in sight is 0 or $\mathbb{Z}/2$. The sequence is multiplicative (theorem 6.2.15), with page-two product the cup products of base and fiber, and every class on the base row is a cycle for every differential, the targets lying in negative rows. The fiber cohomology vanishes in odd degrees, so every d_2 vanishes and $E_2 = E_3$. Contractibility of PB forces $E_\infty = 0$ away from $(0,0)$.

Step 2: Degrees up to 3. The entries $(1,0)$ and $(2,0)$ admit no nonzero differentials in or out, so $H^1(B;\mathbb{Z}/2) = H^2(B;\mathbb{Z}/2) = 0$. The class x at $(0,2)$ has exactly one possible differential, d_3 into $(3,0)$, later ones landing in negative rows. Both x and $E_3^{3,0} = H^3(B;\mathbb{Z}/2)$ must die, so $d_3\colon E_3^{0,2} \to E_3^{3,0}$ is an isomorphism and $H^3(B;\mathbb{Z}/2) = \mathbb{Z}/2\cdot d_3(x)$. By proposition 6.3.5, $H_3(B;\mathbb{Z}) = \mathbb{Z}$ and $H_4(B;\mathbb{Z}) = 0$, so the universal coefficient theorem (theorem 4.1.8, the singular chain complex being free) gives $H^3(B;\mathbb{Z}) = \mathbb{Z}\cdot\iota_3$, the fundamental class corresponding to the identity (definition 5.5.10), and $H^4(B;\mathbb{Z}) = 0$. In the Bockstein sequence (proposition 9.1.2) the map β out of $H^3(B;\mathbb{Z}/2)$ lands in $H^4(B;\mathbb{Z}) = 0$, so ρ is surjective and $\rho\iota_3 \neq 0$. Hence $d_3(x) = \rho\iota_3$, the unique nonzero class. Since no earlier differential lands in $(3,0)$, theorem 6.4.3 further gives $p^*(\rho\iota_3) = \delta(x)$: the class $\rho\iota_3$ represents $\tau(x)$ in the zig-zag sense.

Step 3: Degrees 4 *and* 5. The only candidate differential into $(4,0)$ is d_3 from $(1,2) = H^1(B;\mathbb{Z}/2)\otimes\mathbb{Z}/2 = 0$, so $H^4(B;\mathbb{Z}/2) = E_\infty^{4,0} = 0$. Next, $x^2 = \mathrm{Sq}^2 x$ by property (5) of theorem 9.1.17, which holds on $H^*(\Omega B;\mathbb{Z}/2)$ by lemma 9.2.4(4). The base is a simply connected CW complex based at the 0-cell b_0, so lemma 9.2.5 applies: x^2 is transgressive and $\mathrm{Sq}^2\rho\iota_3$ represents $\tau(x^2)$. Consistently, the Leibniz rule gives $d_3(x^2) = 2x\,d_3(x) = 0$ directly. No earlier differential reaches $(5,0)$, the only candidate being d_3 from $(2,2) = 0$, so $d_5(x^2) = \mathrm{Sq}^2\rho\iota_3$ holds exactly in $E_5^{5,0} = H^5(B;\mathbb{Z}/2)$. The entry $(0,4)$ is hit by nothing, its lower differentials vanish (d_2 and d_4 land in the zero rows 3 and 1, and $d_3(x^2) = 0$), and both $(0,4)$ and $(5,0)$ must die, so d_5 is an isomorphism:

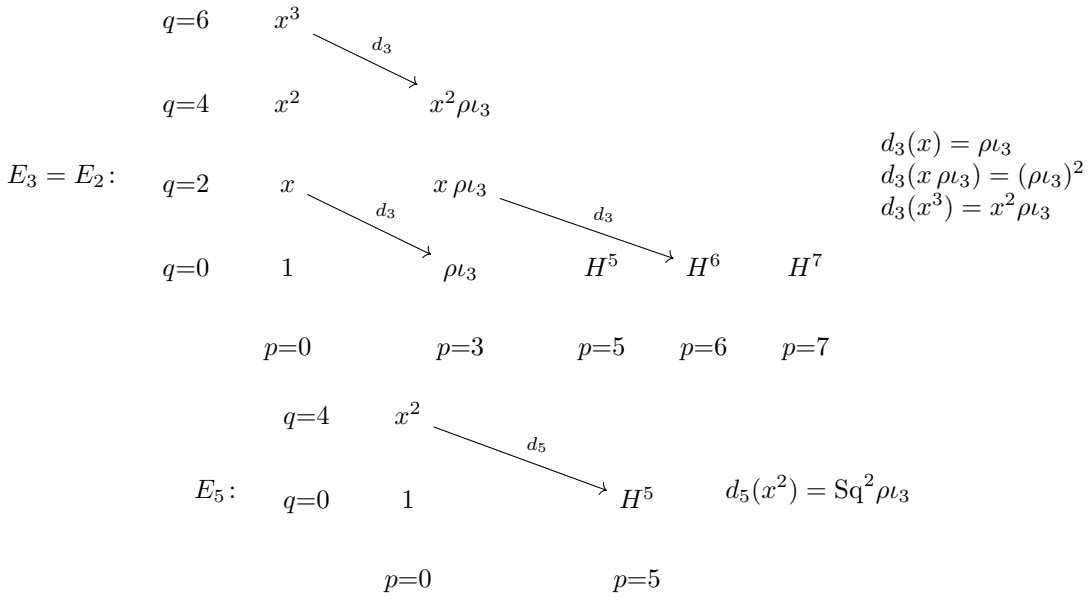
$$H^5(B;\mathbb{Z}/2) = \mathbb{Z}/2\cdot\mathrm{Sq}^2\rho\iota_3$$

Step 4: Degree 6. The entry $(3,2)$ is $\mathbb{Z}/2 \cdot (x\,\rho\iota_3)$, the product of the fiber class x with the base class $\rho\iota_3$. By the Leibniz rule, $d_3(x\,\rho\iota_3) = d_3(x)\,\rho\iota_3 = (\rho\iota_3)^2$ in $E_3^{6,0} = H^6(B;\mathbb{Z}/2)$, no earlier differential having touched $(6,0)$. The entry $(3,2)$ receives nothing, since $d_3(x^2) = 0$, and its only outgoing differential is this d_3, so injectivity is forced: $(\rho\iota_3)^2 \neq 0$. After this differential, $E_4^{6,0} = H^6(B;\mathbb{Z}/2)/\langle(\rho\iota_3)^2\rangle$ can no longer be hit, the candidates d_5 from $(1,4) = 0$ and d_7 from column -1 being empty, so it must already vanish:

$$H^6(B;\mathbb{Z}/2) = \mathbb{Z}/2 \cdot (\rho\iota_3)^2$$

and $(\rho\iota_3)^2 = \mathrm{Sq}^3\rho\iota_3$ by axiom (5).

Step 5: Degree 7. Nothing can reach $(7,0)$: d_3 from $(4,2) = H^4(B;\mathbb{Z}/2) \otimes \mathbb{Z}/2 = 0$ by Step 3, d_5 from $(2,4) = 0$, and d_7 from $(0,6) = \mathbb{Z}/2 \cdot x^3$ vanishes because x^3 dies on page 3: by the Leibniz rule $d_3(x^3) = 3x^2\,d_3(x) = x^2\rho\iota_3$, the generator of $E_3^{3,4} = H^3(B;\mathbb{Z}/2) \otimes \mathbb{Z}/2 \cdot x^2$, so $E_4^{0,6} = 0$. Hence $H^7(B;\mathbb{Z}/2) = E_\infty^{7,0} = 0$. The two relevant pages, with the differentials as arrows:



Step 6: The integral groups. From proposition 6.3.5, $H_4(B;\mathbb{Z}) = 0$ and $H_5(B;\mathbb{Z}) = \mathbb{Z}/2$. The universal coefficient sequence for $H^5(B;\mathbb{Z})$ has $\mathrm{Ext}(H_4,\mathbb{Z}) = 0$ on the left and $\mathrm{Hom}(H_5,\mathbb{Z}) = \mathrm{Hom}(\mathbb{Z}/2,\mathbb{Z}) = 0$ on the right, so $H^5(B;\mathbb{Z}) = 0$. For degree 6, take $\mathbb{Z}/2$ coefficients first: the sequence

$$0 \to \mathrm{Ext}(H_5,\mathbb{Z}/2) \to H^6(B;\mathbb{Z}/2) \to \mathrm{Hom}(H_6,\mathbb{Z}/2) \to 0$$

has $\mathrm{Ext}(\mathbb{Z}/2,\mathbb{Z}/2) = \mathbb{Z}/2$ injecting into $H^6(B;\mathbb{Z}/2) = \mathbb{Z}/2$ of Step 4, so $\mathrm{Hom}(H_6,\mathbb{Z}/2) = 0$. Hence $H_6 = 2H_6$, and any homomorphism $H_6 \to \mathbb{Z}$ has image equal to twice itself, which forces the image, a subgroup of $\mathbb{Z}$, to vanish: $\mathrm{Hom}(H_6,\mathbb{Z}) = 0$. The integral sequence then reads $0 \to \mathrm{Ext}(H_5,\mathbb{Z}) \to H^6(B;\mathbb{Z}) \to 0$, so $H^6(B;\mathbb{Z}) \cong \mathrm{Ext}(\mathbb{Z}/2,\mathbb{Z}) \cong \mathbb{Z}/2$. For the generators: in proposition 9.1.2 the map β on $H^5(B;\mathbb{Z}/2)$ is injective, its predecessor ρ coming from $H^5(B;\mathbb{Z}) = 0$, so $\mathrm{Sq}^3_{\mathbb{Z}}\iota_3 = \beta\,\mathrm{Sq}^2\rho\iota_3 \neq 0$ generates $H^6(B;\mathbb{Z})$. The square ι_3^2 is the same element: ρ is a ring homomorphism, so $\rho(\iota_3^2) = (\rho\iota_3)^2 \neq 0$ by Step 4, whence $\iota_3^2 \neq 0$, and a group of order two has a single nonzero element. Up to the sign of the chosen generator $b = \pm\iota_3$, which cancels in the square, this class is the class b^2 produced in Step 5 of the proof of proposition 6.3.5.

Step 7: The composition identity. By axiom (7) and the above,

$$\mathrm{Sq}^1\mathrm{Sq}^2\rho\iota_3 = \rho\beta\,\mathrm{Sq}^2\rho\iota_3 = \rho\big(\mathrm{Sq}^3_{\mathbb{Z}}\iota_3\big) = \rho(\iota_3^2) = (\rho\iota_3)^2 = \mathrm{Sq}^3\rho\iota_3$$

completing the proof.

As a consistency check, the mod-2 groups match proposition 6.3.5 under the universal coefficient theorem: $H^4(K(\mathbb{Z},3);\mathbb{Z}/2)$ sits between $\mathrm{Ext}(H_3,\mathbb{Z}/2) = 0$, the group $H_3 = \mathbb{Z}$ being free, and $\mathrm{Hom}(H_4,\mathbb{Z}/2) = 0$, so it vanishes, and $H^5(K(\mathbb{Z},3);\mathbb{Z}/2)$ sits between $\mathrm{Ext}(H_4,\mathbb{Z}/2) = 0$ and $\mathrm{Hom}(H_5,\mathbb{Z}/2) = \mathrm{Hom}(\mathbb{Z}/2,\mathbb{Z}/2) = \mathbb{Z}/2$, giving $\mathbb{Z}/2$, in agreement with Step 3.

Looping once more raises the computation to every $K(\mathbb{Z},n)$, where the window above the fundamental class stabilizes.

Proposition 9.2.7: The Cohomology of $K(\mathbb{Z},n)$ Through Degree $n+3$

For every $n \geq 3$,

$$H^{n+1}(K(\mathbb{Z},n);\mathbb{Z}/2) = 0,$$
$$H^{n+2}(K(\mathbb{Z},n);\mathbb{Z}/2) = \mathbb{Z}/2\cdot\mathrm{Sq}^2\rho\iota_n,$$
$$H^{n+3}(K(\mathbb{Z},n);\mathbb{Z}/2) = \mathbb{Z}/2\cdot\mathrm{Sq}^3\rho\iota_n$$

while integrally $H^{n+2}(K(\mathbb{Z},n);\mathbb{Z}) = 0$ and $H^{n+3}(K(\mathbb{Z},n);\mathbb{Z}) \cong \mathbb{Z}/2$, generated by $\mathrm{Sq}^3_{\mathbb{Z}}\iota_n$. For $n = 1,2$ one has instead $H^{n+3}(K(\mathbb{Z},n);\mathbb{Z}) = 0$. The case $n = 4$ computes $H^*(K(\mathbb{Z},4))$ through degree 7.

***Proof*:**
The mod-2 statement is proved by induction on n. The base case $n = 3$ is proposition 9.2.6, with $\mathrm{Sq}^3\rho\iota_3 = (\rho\iota_3)^2$.

Step 1: Setup and fiber input. For $n \geq 4$ let B be a CW complex of type $K(\mathbb{Z},n)$, based at a 0-cell, and take the path-loop fibration $\Omega B \to PB \to B$ as before. Its fiber $F := \Omega B$ is path-connected with $\pi_k(F) \cong \pi_{k+1}(B)$ for every $k \geq 0$ (example 5.2.15). So for a CW approximation $\gamma\colon F' \to F$ (theorem 5.3.10) the CW complex F' has type $K(\mathbb{Z},n-1)$, and γ^* is a ring isomorphism on mod-2 cohomology commuting with the squares (lemma 9.2.4). Write $t \in H^{n-1}(F;\mathbb{Z}/2)$ for the class with $\gamma^*t = \rho\iota_{n-1}$, where ι_{n-1} is the fundamental class of F'. The fiber cohomology in degrees $q \leq n+2$, computed on F' and transported by γ^*:

- $H^0(F;\mathbb{Z}/2) = \mathbb{Z}/2$, and $H^q(F;\mathbb{Z}/2) = 0$ for $0 < q < n-1$: the CW complex F' is $(n-2)$-connected, so its low homology vanishes (theorem 5.4.9) and the universal coefficient sequence vanishes with it.
- $H^{n-1}(F;\mathbb{Z}/2) = \mathbb{Z}/2\cdot t$.
- $H^n(F;\mathbb{Z}/2) = 0$: the universal coefficient sequence of F' in degree n sits between $\mathrm{Ext}(H_{n-1}(F'),\mathbb{Z}/2) = 0$, the group $H_{n-1}(F') = \mathbb{Z}$ being free, and $\mathrm{Hom}(H_n(F'),\mathbb{Z}/2) = 0$, with $H_n(F') = 0$ by proposition 6.3.5.
- $H^{n+1}(F;\mathbb{Z}/2) = \mathbb{Z}/2\cdot\mathrm{Sq}^2t$ and $H^{n+2}(F;\mathbb{Z}/2) = \mathbb{Z}/2\cdot\mathrm{Sq}^3t$, by the inductive hypothesis applied to F'.

For $n = 4$ these are the groups of proposition 9.2.6, with $\mathrm{Sq}^3 t = t^2$. The base is simply connected, theorem 6.2.8 and theorem 6.2.15 apply with $\mathbb{Z}/2$ coefficients, $E_2^{p,q} = H^p(B;\mathbb{Z}/2) \otimes H^q(F;\mathbb{Z}/2)$ in the range where the coefficients are 0 or $\mathbb{Z}/2$, and E_∞ vanishes away from the origin.

Step 2: The base below degree n. For $0 < p < n$ the entry $(p, 0)$ supports no differential and receives none: an incoming d_r needs fiber degree $r-1$ with nonzero cohomology, so $r - 1 \geq n - 1$ and the source column $p - r < 0$ is empty. Hence $H^p(B;\mathbb{Z}/2) = 0$ for $0 < p < n$.

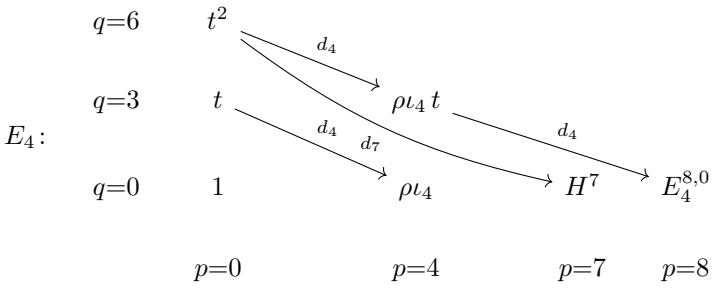
Step 3: The transgression of t. For $2 \leq r \leq n-1$ the differential d_r sends $(0, n-1)$ into fiber degrees between 1 and $n-2$, which vanish, so t is transgressive, and $(n, 0)$ receives only d_n from $(0, n-1)$. Both must die, so $d_n \colon \mathbb{Z}/2 \cdot t \to H^n(B;\mathbb{Z}/2)$ is an isomorphism. As in Step 2 of the previous proof, proposition 6.3.5, definition 5.5.10 and the universal coefficient theorem give $H^n(B;\mathbb{Z}) = \mathbb{Z} \cdot \iota_n$ and $H^{n+1}(B;\mathbb{Z}) = 0$, so ρ is surjective onto $H^n(B;\mathbb{Z}/2)$ by proposition 9.1.2 and $d_n(t) = \rho\iota_n \neq 0$. Since nothing earlier lands in $(n, 0)$, theorem 6.4.3 gives $p^*(\rho\iota_n) = \delta(t)$.

Step 4: Degree $n+1$. A differential d_r into $(n+1, 0)$ starts at $(n+1-r, r-1)$, a column that is negative for $r > n+1$, and the fiber cohomology vanishes in degrees 1 through $n-2$. So only d_n from $(1, n-1) = H^1(B;\mathbb{Z}/2) \otimes \mathbb{Z}/2 \cdot t = 0$ and d_{n+1} from $(0, n) = H^n(F;\mathbb{Z}/2) = 0$ can map into it, and nothing leaves it. Hence $H^{n+1}(B;\mathbb{Z}/2) = E_\infty^{n+1,0} = 0$.

Step 5: Degree $n+2$. Lemma 9.2.5 applies as before, the base being a simply connected CW complex based at a 0-cell: $\mathrm{Sq}^2 t$ is transgressive and $\mathrm{Sq}^2 \rho\iota_n$ represents its transgression. Into $(n+2, 0)$ only d_n from $(2, n-1) = H^2(B;\mathbb{Z}/2) \otimes \mathbb{Z}/2 = 0$ and the transgression d_{n+2} from $(0, n+1)$ can map, so $d_{n+2}(\mathrm{Sq}^2 t) = \mathrm{Sq}^2 \rho\iota_n$ holds exactly in $E_{n+2}^{n+2,0} = H^{n+2}(B;\mathbb{Z}/2)$. The class $\mathrm{Sq}^2 t$ is hit by nothing, and its differentials below the transgression land in $(3, n-1)$, zero because $3 < n$, or in zero fiber rows. Both ends must die, so d_{n+2} is an isomorphism:

$$H^{n+2}(B;\mathbb{Z}/2) = \mathbb{Z}/2 \cdot \mathrm{Sq}^2 \rho\iota_n$$

Step 6: Degree $n+3$. Off the axes, an entry of total degree $n+3$ has fiber degree $q \in \{n-1, n+1, n+2\}$ and base degree $p = 4, 2, 1$ respectively. The entries with $p = 1, 2$ vanish, and for $n \geq 5$ so does the one with $p = 4 < n$. So for $n \geq 5$ the only entries bearing on $H^{n+3}(B;\mathbb{Z}/2)$ are $(n+3, 0)$ and the source $(0, n+2)$ of its transgression, of total degree $n+2$. For $n = 4$ the entry $(4, 3) = \mathbb{Z}/2 \cdot (\rho\iota_4\, t)$ is present as well, and the page is



$$d_4(t) = \rho\iota_4$$
$$d_4(t^2) = 2\,\rho\iota_4\, t = 0$$
$$d_4(\rho\iota_4\, t) = (\rho\iota_4)^2$$
$$d_7(t^2) = \mathrm{Sq}^3 \rho\iota_4$$

By lemma 9.2.5, $\mathrm{Sq}^3 t$ is transgressive with representative $\mathrm{Sq}^3 \rho\iota_n$. For $n = 4$ the survival is visible directly. Here $\mathrm{Sq}^3 t = t^2$, and on page 4, where t transgresses, the Leibniz rule gives $d_4(t^2) = d_4(t)\, t + (-1)^3 t\, d_4(t) = 2\,\rho\iota_4\, t = 0$, the sign immaterial mod 2. Pages 2 and 3 send $(0, 6)$ into the columns 2 and 3, zero by Step 2, and pages 5 and 6 send it into $(5, 2)$ and $(6, 1)$, where $H^2(F;\mathbb{Z}/2) = H^1(F;\mathbb{Z}/2) = 0$ by proposition 9.2.6. The companion entry $(4, 3)$ interferes with nothing in total degree at most 7: its differential $d_4(\rho\iota_4\, t) = \rho\iota_4 \cdot d_4(t) = (\rho\iota_4)^2$ lands in

$(8,0)$, total degree 8, and the only candidate map into $(4,3)$, the d_4 from $(0,6)$, vanishes as computed above. Into $(n+3,0)$, every differential except the transgression has vanishing source: d_n from $(3,n-1)$ with $H^3(B;\mathbb{Z}/2)=0$, d_{n+2} from $(1,n+1)$ with $H^1(B;\mathbb{Z}/2)=0$, and for $n=4$ additionally d_5 from $(2,4)$ and d_6 from $(1,5)$, with $H^2(B;\mathbb{Z}/2)=H^1(B;\mathbb{Z}/2)=0$. So the transgression $d_{n+3}(\mathrm{Sq}^3 t)=\mathrm{Sq}^3\rho\iota_n$ holds exactly, it is the only differential touching $(0,n+2)$ or $(n+3,0)$, and both ends must die:

$$H^{n+3}(B;\mathbb{Z}/2)=\mathbb{Z}/2\cdot\mathrm{Sq}^3\rho\iota_n$$

Step 7: The integral groups. For $n=3$ the integral statement is proposition 9.2.6. For $n\geq 4$, proposition 6.3.5 gives $H_{n+1}(B;\mathbb{Z})=0$ and $H_{n+2}(B;\mathbb{Z})=\mathbb{Z}/2$. The universal coefficient sequence for $H^{n+2}(B;\mathbb{Z})$ has $\mathrm{Ext}(H_{n+1},\mathbb{Z})=0$ and $\mathrm{Hom}(H_{n+2},\mathbb{Z})=\mathrm{Hom}(\mathbb{Z}/2,\mathbb{Z})=0$, so $H^{n+2}(B;\mathbb{Z})=0$. With $\mathbb{Z}/2$ coefficients in degree $n+3$, the subgroup $\mathrm{Ext}(H_{n+2},\mathbb{Z}/2)=\mathbb{Z}/2$ injects into $H^{n+3}(B;\mathbb{Z}/2)=\mathbb{Z}/2$, so $\mathrm{Hom}(H_{n+3},\mathbb{Z}/2)=0$, whence $H_{n+3}=2H_{n+3}$ and, as in the previous proof, $\mathrm{Hom}(H_{n+3},\mathbb{Z})=0$. The integral sequence then gives $H^{n+3}(B;\mathbb{Z})\cong\mathrm{Ext}(H_{n+2},\mathbb{Z})\cong\mathbb{Z}/2$. In proposition 9.1.2, β is injective on $H^{n+2}(B;\mathbb{Z}/2)$, its predecessor ρ coming from $H^{n+2}(B;\mathbb{Z})=0$, so $\mathrm{Sq}^3_{\mathbb{Z}}\iota_n=\beta\,\mathrm{Sq}^2\rho\iota_n\neq 0$ generates $H^{n+3}(B;\mathbb{Z})$.

Step 8: The cases $n=1,2$. Any CW complex of type $K(\mathbb{Z},1)$ is homotopy equivalent to S^1 (example 5.5.2, theorem 5.5.4), and $H^4(S^1;\mathbb{Z})=0$ (example 5.5.9). Any one of type $K(\mathbb{Z},2)$ is homotopy equivalent to $\mathbb{C}\mathrm{P}^\infty$ (example 5.5.5), whose integral cohomology vanishes in odd degrees (example 5.5.11), so $H^5(K(\mathbb{Z},2);\mathbb{Z})=0$, completing the proof.

For $n\geq 3$ the window above the fundamental class no longer depends on n: one zero group, then $\mathrm{Sq}^2\rho\iota_n$, then $\mathrm{Sq}^3\rho\iota_n$, with the first integral torsion above degree n appearing in degree $n+3$ as the value of the integral operation. Section 9.5 reads this list as the classification of the natural operations $H^p(-;\mathbb{Z})\to H^{p+3}(-;\mathbb{Z})$ in the stable range.

The integral operation of definition 9.1.20 was built to lift Sq^3 across the reduction ρ, and the universal computation proves that it does.

Corollary 9.2.8: The Integral Operation Lifts Sq^3

For every space X and every $n\geq 0$,

$$\rho\circ\mathrm{Sq}^3_{\mathbb{Z}} \;=\; \mathrm{Sq}^3\circ\rho\;:\; H^n(X;\mathbb{Z})\longrightarrow H^{n+3}(X;\mathbb{Z}/2)$$

***Proof*:**

Step 1: Reduction to CW complexes. Let $\gamma\colon X'\to X$ be a CW approximation (theorem 5.3.10). The map γ^* commutes with ρ and β (proposition 9.1.2) and with Sq^2 and Sq^3 (lemma 9.2.4(4)), hence with $\mathrm{Sq}^3_{\mathbb{Z}}$, and it is injective on $H^{n+3}(X;\mathbb{Z}/2)$ (lemma 9.2.4(2)). So the identity on X' gives it on X, and X may be taken to be a CW complex.

Step 2: Degrees at most 2. For $n=0$ both sides vanish: for $|a|=0$, axiom (4) gives $\mathrm{Sq}^2\rho a=0$, hence $\mathrm{Sq}^3_{\mathbb{Z}}a=\beta\,\mathrm{Sq}^2\rho a=0$, and $\mathrm{Sq}^3\rho a=0$. For $n=1,2$, the left side vanishes by proposition 9.1.21 and the right side by axiom (4), since $3>|\rho a|$.

Step 3: Degrees at least 3. Let $n\geq 3$. The cohomology of a disjoint union is the product of the cohomologies of the pieces (example 4.2.9), and both operations act componentwise by naturality, so assume X connected and based. Given $a\in H^n(X;\mathbb{Z})$, representability (theorem 5.5.12)

produces $f\colon X \to K(\mathbb{Z}, n)$, the target the CW model of theorem 5.5.4, with $f^*\iota_n = a$. Naturality of ρ, of $\mathrm{Sq}^3_{\mathbb{Z}}$ (proposition 9.1.21) and of Sq^3 (axiom (1)) reduces the identity to the universal class ι_n. There, $\mathrm{Sq}^3_{\mathbb{Z}}\iota_n$ generates $H^{n+3}(K(\mathbb{Z}, n); \mathbb{Z}) \cong \mathbb{Z}/2$ by proposition 9.2.7, and ρ is injective on this group, its kernel $2H^{n+3}(K(\mathbb{Z}, n); \mathbb{Z})$ vanishing (proposition 9.1.2). So $\rho\,\mathrm{Sq}^3_{\mathbb{Z}}\iota_n$ is a nonzero element of $H^{n+3}(K(\mathbb{Z}, n); \mathbb{Z}/2) = \mathbb{Z}/2 \cdot \mathrm{Sq}^3\rho\iota_n$ and therefore equals $\mathrm{Sq}^3\rho\iota_n$, completing the proof.

On the reduction of an integral class, Sq^3 therefore lands back in the image of ρ, and the corollary names the integral class it comes from. This identity is what section 9.5 evaluates against K-theory.

The last space on the section's list is the classifying space of $SU(2)$, whose associated rank-two bundle has $c_1 = 0$. Its model below is the quotient of the infinite sphere by $SU(2)$ acting as the unit quaternions, and it rests on identifying $SU(2)$ with the three-sphere.

Lemma 9.2.9: $SU(2)$ Is the Three-Sphere

The group $SU(2)$ consists exactly of the matrices

$$\begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad a, b \in \mathbb{C}, \quad |a|^2 + |b|^2 = 1$$

and the map $A \mapsto Ae_1$ sending a matrix to its first column is a homeomorphism $SU(2) \to S^3 \subseteq \mathbb{C}^2$.

***Proof*:**
A unitary matrix has orthonormal columns. If the first column of $A \in SU(2)$ is (a, b), a unit vector, then the second column is a unit vector orthogonal to it, hence of the form $\lambda(-\bar{b}, \bar{a})$ with $|\lambda| = 1$: the vector $(-\bar{b}, \bar{a})$ is a unit vector with $\langle(-\bar{b}, \bar{a}), (a, b)\rangle = -\bar{b}\bar{a} + \bar{a}\bar{b} = 0$, and the orthogonal complement of a unit vector in $\mathbb{C}^2$ is a complex line. The determinant is then $a(\lambda\bar{a}) - (-\lambda\bar{b})b = \lambda(|a|^2 + |b|^2) = \lambda$, so $\det A = 1$ exactly when $\lambda = 1$, which proves the description of $SU(2)$. The assignment $A \mapsto Ae_1 = (a, b)$ is therefore a bijection onto S^3. It is continuous as the restriction of a linear map, and its inverse, sending (a, b) to the displayed matrix, is continuous as well, completing the proof.

In particular $SU(2)$ is compact, path-connected and simply connected, since $\pi_1(S^3) = 0$ (corollary 5.4.11). The computation of $H^*(BSU(2))$ requires a classifying space that is a CW complex, because the Gysin sequence and the Serre spectral sequence are both applied over it. The Milnor model of proposition 7.1.4 is a CW complex only for discrete groups, so the proof below constructs a CW model by hand: the infinite quaternionic projective space $\mathbb{H}\mathrm{P}^\infty$.

Theorem 9.2.10: The Cohomology of $BSU(2)$

There is a universal principal $SU(2)$-bundle whose base is a CW complex of finite type, the quaternionic projective space $BSU(2) := \mathbb{H}\mathrm{P}^\infty$. Let V be the rank-two complex vector bundle associated to it through the standard action of $SU(2)$ on $\mathbb{C}^2$, and write $c_2 := c_2(V)$. Then

$$H^*(BSU(2);\mathbb{Z}) = \mathbb{Z}[c_2], \qquad H^*(BSU(2);\mathbb{Z}/2) = \mathbb{Z}/2[\rho c_2], \qquad |c_2| = 4$$

the operations Sq^1, Sq^2 and Sq^3 vanish identically on $H^*(BSU(2);\mathbb{Z}/2)$, and

$$\mathrm{Sq}^4(\rho c_2)^k = k\,(\rho c_2)^{k+1}$$

***Proof*:**

Step 1: Matrix quaternions. Let $\mathbb{H} \subseteq M_2(\mathbb{C})$ be the set of matrices $q = \begin{pmatrix} a & -\bar b \\ b & \bar a \end{pmatrix}$ with $a, b \in \mathbb{C}$. It is a real subspace, closed under multiplication: the product of the displayed matrix with the one determined by (c, d) is the one determined by $(e, f) = (ac - \bar b d,\ bc + \bar a d)$, as matrix multiplication shows entry by entry. The determinant $\det q = |a|^2 + |b|^2$ is multiplicative, positive for $q \neq 0$, and the inverse $\det(q)^{-1}\operatorname{adj}(q)$ of a nonzero element again lies in $\mathbb{H}$, the adjugate being the matrix determined by $(\bar a, -b)$. So $\mathbb{H}$ is a real division algebra with multiplicative norm $|q| := \det(q)^{1/2}$, and by lemma 9.2.9 its unit sphere is exactly $SU(2)$.

Step 2: The bundle $S^\infty \to \mathbb{H}\mathrm{P}^\infty$. Let $\mathbb{H}^\infty = \bigcup_n \mathbb{H}^n$ be the space of columns with finitely many nonzero quaternion entries and $|v|^2 := \sum_j \det(v_j)$. Identifying each entry with its first column identifies $\mathbb{H}^n$ with $\mathbb{C}^{2n}$ preserving norms, so the unit sphere of $\mathbb{H}^\infty$ is the sphere $S^\infty = \bigcup_n S^{4n-1}$ of example 5.5.5. The group $SU(2)$ acts on S^∞ on the right by coordinatewise multiplication, through isometries since the norm is multiplicative, and freely: if $vg = v$ and v_j is a nonzero entry, then $v_j g = v_j$ and $g = v_j^{-1}(v_j g) = 1$. Let $\mathbb{H}\mathrm{P}^\infty := S^\infty/SU(2)$ with the quotient topology and projection p, and let $\mathbb{H}\mathrm{P}^n := p(S^{4n+3})$, the classes of vectors supported in the first $n+1$ coordinates.

The quotient is a principal $SU(2)$-bundle. Over $U_j := \{[v] : v_j \neq 0\}$ the assignment $s_j([v]) := v \cdot (v_j^{-1}|v_j|)$ is well defined, since replacing v by vg replaces $v_j^{-1}|v_j|$ by $g^{-1}v_j^{-1}|v_j|$, and it is continuous: p is an open map, the saturation of an open set being the union of its images under the homeomorphisms $v \mapsto vg$, and $s_j \circ p$ is the continuous map $v \mapsto v \cdot (v_j^{-1}|v_j|)$ on the open set $p^{-1}(U_j)$. Then

$$U_j \times SU(2) \longrightarrow p^{-1}(U_j), \qquad ([v], g) \longmapsto s_j([v]) \cdot g$$

is a homeomorphism, with inverse $v \mapsto \big([v], |v_j|^{-1}v_j\big)$, and it is equivariant for the right translation action on the $SU(2)$ factor.

The total space S^∞ is contractible. Let σ shift the coordinates of $\mathbb{H}^\infty$ one step. The linear path $(1-t)v + t\sigma(v)$ never vanishes for $v \in S^\infty$: at $t = 1$ it is $\sigma(v) \neq 0$, and for $t < 1$ the equation $(1-t)v = -t\sigma(v)$ forces $v_1 = 0$, then inductively $v_{j+1} = 0$ from $(\sigma v)_{j+1} = v_j$, so $v = 0$, a contradiction. Normalizing gives a homotopy from the identity of S^∞ to σ. Likewise $(1-t)\sigma(v) + te_1$ never vanishes, its first coordinate being $t \cdot 1$ for $t > 0$ and the vector being $\sigma(v) \neq 0$ at $t = 0$, so normalizing contracts σ to the constant map at e_1. Hence p is a principal $SU(2)$-bundle with contractible total space: a universal bundle in the sense of definition 7.1.3, and $\mathbb{H}\mathrm{P}^\infty$ is a classifying space $BSU(2)$.

Step 3: The CW structure. Each $\mathbb{H}\mathrm{P}^n$ is Hausdorff: two distinct orbits in S^{4n+3} are disjoint compact sets, $SU(2)$ being compact, and since right translations are isometries of $\mathbb{H}^{n+1}$, the open $\varepsilon/2$-neighborhoods of the two orbits, ε their distance, are saturated and disjoint, so their images separate the two points. Define $\Phi_n\colon D^{4n} \to \mathbb{H}\mathrm{P}^n$ on the unit disk of $\mathbb{H}^n$ by $\Phi_n(u) := \left[u : (1-|u|^2)^{1/2}\right]$, the last homogeneous coordinate a nonnegative real multiple of $1 \in \mathbb{H}$. It is surjective: an orbit with last coordinate zero is Φ_n of a boundary point, and one with $v_{n+1} \neq 0$ contains the representative $v \cdot (v_{n+1}^{-1}|v_{n+1}|)$, whose last coordinate is $|v_{n+1}| \cdot 1$. It is injective on the open disk: if $[u : r] = [u' : r']$ with $r, r' > 0$, the group element g carrying one representative to the other satisfies $r' \cdot 1 = r \cdot g$, so g is a positive real multiple of 1 of norm one, hence $g = 1$. On the boundary, Φ_n restricts to the orbit projection $S^{4n-1} \to \mathbb{H}\mathrm{P}^{n-1}$. As a continuous surjection from a compact space onto a Hausdorff space, Φ_n is closed, so the induced continuous bijection from the adjunction space $\mathbb{H}\mathrm{P}^{n-1} \cup_p D^{4n}$ onto $\mathbb{H}\mathrm{P}^n$ is a homeomorphism. Inductively, $\mathbb{H}\mathrm{P}^n$ is a CW complex with one cell in each of the dimensions $0, 4, \ldots, 4n$, containing $\mathbb{H}\mathrm{P}^{n-1}$ as a subcomplex. Finally, the quotient topology on $\mathbb{H}\mathrm{P}^\infty$ is the weak topology of this filtration: a set U is open exactly when $p^{-1}(U)$ meets every S^{4n+3} in an open set, and since $p^{-1}(U)$ is saturated, $p^{-1}(U) \cap S^{4n+3}$ is the preimage of $U \cap \mathbb{H}\mathrm{P}^n$ under the restricted projection, itself a quotient map as a closed continuous surjection. So $Q := \mathbb{H}\mathrm{P}^\infty$ is a CW complex of finite type, with one cell in each dimension divisible by 4, and it is paracompact Hausdorff by theorem 7.1.7.

Step 4: The integral ring, by Gysin. The bundle p is a fiber bundle with fiber $SU(2) \cong S^3$ (lemma 9.2.9), hence a Serre fibration (theorem 5.2.4). The base Q is path-connected, as the image of the path-connected S^∞, and simply connected: in the tail $\pi_1(S^\infty) \to \pi_1(Q) \to \pi_0(SU(2))$ of the long exact sequence (theorem 5.2.12), the set $\pi_0(SU(2))$ is a single point, so all of $\pi_1(Q)$ is the kernel of the final map, which equals the image of $\pi_1(S^\infty) = 0$. The simply connected base makes the local system $H^3(S^3;\mathbb{Z})$ trivial, so theorem 6.3.2 applies with $k = 3$, and $\widetilde{H}^*(S^\infty;\mathbb{Z}) = 0$ by contractibility. For $j = 1, 2, 3$ the segment

$$H^{j-4}(Q;\mathbb{Z}) \xrightarrow{\smile e} H^j(Q;\mathbb{Z}) \xrightarrow{p^*} H^j(S^\infty;\mathbb{Z})$$

has vanishing outer terms, so $H^j(Q;\mathbb{Z}) = 0$. For $j \geq 4$, the segment $H^{j-1}(S^\infty) \to H^{j-4}(Q) \xrightarrow{\smile e} H^j(Q) \to H^j(S^\infty)$ has vanishing outer terms, so cup product with the Euler class $e \in H^4(Q;\mathbb{Z})$ is an isomorphism $H^{j-4}(Q;\mathbb{Z}) \cong H^j(Q;\mathbb{Z})$. Starting from $H^0(Q;\mathbb{Z}) = \mathbb{Z}$,

$$H^*(Q;\mathbb{Z}) = \mathbb{Z}[e], \qquad |e| = 4$$

Step 5: The generator is c_2 of the associated bundle. Let $V := S^\infty \times_{SU(2)} \mathbb{C}^2$, the quotient of $S^\infty \times \mathbb{C}^2$ by $(vg, u) \sim (v, gu)$. The charts of Step 2 trivialize it, so V is a rank-two complex vector bundle over the paracompact Hausdorff base Q, with Chern classes defined by definition 7.2.7 as pullbacks of the universal classes of theorem 7.2.5. Immediately $c_1(V) \in H^2(Q;\mathbb{Z}) = 0$.

Let $T := \{\mathrm{diag}(\lambda, \bar{\lambda}) : |\lambda| = 1\} \subseteq SU(2)$. Under the identification of each quaternion coordinate with its first column, right multiplication by $\mathrm{diag}(\lambda, \bar{\lambda})$ multiplies the column by the complex scalar λ, so S^∞/T is the quotient of S^∞ by the unit complex scalars: the space $\mathbb{C}\mathrm{P}^\infty$ of example 5.5.5. Write $\pi\colon \mathbb{C}\mathrm{P}^\infty \to Q$ for the induced projection. It is a fiber bundle with fiber $SU(2)/T$: the equivariant charts of Step 2 descend to homeomorphisms over each U_j. The map $gT \mapsto [ge_1]$ is a continuous bijection from the compact coset space $SU(2)/T$ onto $\mathbb{C}\mathrm{P}^1$, hence a homeomorphism: it is surjective because $SU(2)$ carries e_1 to every unit vector (lemma 9.2.9), and injective because an element of $SU(2)$ whose first column is proportional to e_1 has, by the matrix description, first column $(\lambda, 0)$ and so lies in T. With $\mathbb{C}\mathrm{P}^1 = S^2$ (example 5.2.7), π is

an S^2-bundle over the simply connected Q, its local system is trivial, and its Euler class lies in $H^3(Q;\mathbb{Z}) = 0$ by Step 4. The Gysin sequence of π therefore breaks into short exact sequences

$$0 \to H^m(Q;\mathbb{Z}) \xrightarrow{\pi^*} H^m(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \longrightarrow H^{m-2}(Q;\mathbb{Z}) \to 0$$

and at $m = 4$, where $H^2(Q;\mathbb{Z}) = 0$, the map $\pi^*\colon H^4(Q;\mathbb{Z}) \to H^4(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z})$ is an isomorphism.

The pullback π^*V is the bundle with structure group restricted to T: the map $S^\infty \times_T \mathbb{C}^2 \to \pi^*V$, $[v,u] \mapsto ([v]_T, [v,u])$, is well defined, continuous, fiberwise linear and bijective, every class of V over $[v]$ having a unique representative of the form $[v,u]$, and over each chart it is the identity of $U \times \mathbb{C}^2$, so it is an isomorphism of bundles. The circle T acts on $\mathbb{C}^2$ diagonally with the characters λ and $\bar\lambda$, so

$$\pi^*V \cong L_+ \oplus L_-, \qquad L_\pm := S^\infty \times_T \mathbb{C} \text{ with } \lambda \text{ acting as } \lambda^{\pm 1}$$

The bundle L_+ is the tautological line bundle $\gamma^1_{\mathbb{C}} \to \mathbb{C}\mathrm{P}^\infty$: the assignment $[v,u] \mapsto ([v], uv)$ is well defined, both $(v\lambda, u)$ and $(v, \lambda u)$ mapping to $([v], \lambda uv)$, and it is a fiberwise linear isomorphism onto the lines. Hence $c_1(L_+) = c_1(\gamma^1_{\mathbb{C}}) = -u$, where u denotes the positive generator of $H^2(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z})$ (proposition 7.4.8). The pairing $[v,s] \otimes [v,s'] \mapsto ss'$ is a well-defined isomorphism of $L_+ \otimes L_-$ with the trivial line bundle, the two characters cancelling, and the trivial bundle has $c_1 = 0$, being pulled back from a point (proposition 7.4.1). Additivity of c_1 on tensor products (proposition 7.4.7) then gives $c_1(L_-) = -c_1(L_+) = u$. The Whitney sum formula (theorem 7.4.5), with line bundles carrying no higher Chern classes (definition 7.2.7), gives

$$\pi^* c_2(V) = c_2(\pi^*V) = c_1(L_+)\, c_1(L_-) = -u^2$$

using naturality (proposition 7.4.1) for the first equality. Since u^2 generates $H^4(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z})$ (example 5.5.11) and π^* is an isomorphism in degree 4, the class $c_2 = c_2(V)$ generates $H^4(Q;\mathbb{Z}) = \mathbb{Z}\cdot e$. So $c_2 = \pm e$ and

$$H^*(BSU(2);\mathbb{Z}) = \mathbb{Z}[c_2]$$

Step 6: Mod 2 *and the squares.* The integral ring is torsion-free, so multiplication by 2 is injective in every degree and the sequence of proposition 9.1.2 shows, exactly as in example 9.2.2, that ρ is surjective in each degree with kernel twice the integral group and that the mod-2 cohomology vanishes in degrees where the integral cohomology of the next degree is torsion-free and the current one vanishes. Since ρ is a ring homomorphism,

$$H^*(BSU(2);\mathbb{Z}/2) = \mathbb{Z}/2[\rho c_2]$$

concentrated in degrees divisible by 4. The operations Sq^1, Sq^2, Sq^3 raise degree by 1, 2, 3 and therefore vanish identically. On the generator, $\mathrm{Sq}^4(\rho c_2) = (\rho c_2)^2$ by axiom (5), the intermediate squares vanish by the degree argument, and higher ones by axiom (4), so $\mathrm{Sq}(\rho c_2) = \rho c_2\,(1+\rho c_2)$. The total square is a ring homomorphism, so $\mathrm{Sq}\big((\rho c_2)^k\big) = (\rho c_2)^k(1+\rho c_2)^k$, whose degree-$(4k+4)$ component is

$$\mathrm{Sq}^4(\rho c_2)^k = \binom{k}{1}(\rho c_2)^{k+1} = k\,(\rho c_2)^{k+1}$$

completing the proof.

By corollary 7.1.11(1), any classifying space of $SU(2)$ is joined to $\mathbb{H}\mathrm{P}^\infty$ by a zigzag of weak homotopy

equivalences, which induce ring isomorphisms on integral and mod-2 cohomology (proposition 5.3.15), and one that is a CW complex is homotopy equivalent to $\mathbb{H}\mathrm{P}^\infty$. So the cohomology ring does not depend on the model, and on CW models neither do the squares. The space $\mathbb{H}\mathrm{P}^\infty$ stands to $SU(2)$ as $\mathbb{C}\mathrm{P}^\infty = BU(1)$ stands to the circle (example 5.5.5): one cell every four dimensions, a polynomial ring on one generator. The vanishing of Sq^2 on ρc_2 reappears below as a consistency check against Wu's formula.

Wu's formula computes every Steenrod square of every Stiefel-Whitney class. Splitting the universal bundle into line bundles reduces it to a computation with symmetric polynomials, the route of [29, Problem 8-A].

Theorem 9.2.11: Wu's Formula

In $H^*(BO(n);\mathbb{Z}/2)$, for all $0 \le k \le m \le n$,

$$\mathrm{Sq}^k w_m \;=\; \sum_{t=0}^{k} \binom{m-k+t-1}{t}\, w_{k-t}\, w_{m+t}$$

where the binomial coefficients are read mod 2 with the conventions $\binom{-1}{0} = 1$, $w_0 = 1$, and $w_j = 0$ for $j > n$. Consequently the same identity holds for the Stiefel-Whitney classes of every real vector bundle over a paracompact Hausdorff base.[a]

[a] The identity $\mathrm{Sq}^i u = w_i \smile u$ for the Thom class u, cited under the name "Wu formula" in the proof of theorem 8.5.17, is Thom's formula, a separate statement that this chapter does not prove.

***Proof*:**

Step 1: The universal line bundles. Write $P := (\mathbb{R}\mathrm{P}^\infty)^n$, with $\mathbb{R}\mathrm{P}^\infty = BO(1)$ (theorem 7.2.10) and projections $p_i\colon P \to \mathbb{R}\mathrm{P}^\infty$. Each factor has one cell in each dimension (example 5.5.9), so lemma 1.2.12(1) and (5), applied one factor at a time, show that P with the product topology is a CW complex with countably many cells, hence paracompact Hausdorff (theorem 7.1.7). Each factor is of finite type, so with $a \in H^1(BO(1);\mathbb{Z}/2)$ the generator of theorem 7.2.10 for $n = 1$, the Künneth theorem with field coefficients (theorem 4.3.9) gives

$$H^*(P;\mathbb{Z}/2) = \mathbb{Z}/2[a_1,\dots,a_n], \qquad a_i := p_i^* a, \quad |a_i| = 1$$

The line bundle $\ell_i := p_i^*\gamma^1$ is classified by p_i, so $w(\ell_i) = 1 + a_i$ (definition 7.2.13). Let $\mu\colon P \to BO(n)$ classify $\ell_1 \oplus \cdots \oplus \ell_n$ (theorem 7.1.29). By definition 7.2.13 and the Whitney sum formula (theorem 7.4.5),

$$\mu^* w = w(\ell_1 \oplus \cdots \oplus \ell_n) = \prod_{i=1}^{n} (1 + a_i)$$

so $\mu^* w_j = \sigma_j = \sigma_j(a_1,\dots,a_n)$, the j-th elementary symmetric polynomial, for every $j \ge 0$. For $j > n$ both w_j and σ_j are zero.

Step 2: The map μ^* is injective. Let $\kappa\colon B' \to BO(n)$ be a CW approximation (theorem 5.3.10). The bundle $\kappa^*\gamma^n$ has the paracompact Hausdorff base B' (theorem 7.1.7), so the splitting principle (theorem 7.4.3) gives a map $\pi\colon F \to B'$ from the flag bundle $F := F(\kappa^*\gamma^n)$ with π^* injective on mod-2 cohomology and $\pi^*\kappa^*\gamma^n \cong L_1 \oplus \cdots \oplus L_n$. The space F is paracompact Hausdorff (corollary 7.4.4), so each L_i is classified by a map $g_i\colon F \to BO(1)$, and $g := (g_1,\dots,g_n)\colon F \to P$,

continuous because P carries the product topology, satisfies $g^*\ell_i \cong g_i^*\gamma^1 \cong L_i$. So $\mu\circ g$ and $\kappa\circ\pi$ both classify $L_1\oplus\cdots\oplus L_n$, and theorem 7.1.29 gives $\mu\circ g\simeq\kappa\circ\pi$, hence $g^*\mu^* = \pi^*\kappa^*$. The map κ^* is an isomorphism on mod-2 cohomology (lemma 9.2.4(2)) and π^* is injective, so μ^* is injective. By naturality of the squares (lemma 9.2.4(4)) it therefore suffices to prove

$$\mathrm{Sq}^k\sigma_m \;=\; \sum_{t=0}^{k}\binom{m-k+t-1}{t}\,\sigma_{k-t}\,\sigma_{m+t} \qquad \text{in } \mathbb{Z}/2[a_1,\dots,a_n]$$

Step 3: The left side. Each a_i has degree one, so $\mathrm{Sq}(a_i) = a_i(1+a_i)$ by axioms (3), (5) and (4). For $S\subseteq\{1,\dots,n\}$ write $a_S := \prod_{i\in S} a_i$. The total square is a ring homomorphism (theorem 9.1.18), so

$$\mathrm{Sq}(a_S) = a_S\prod_{i\in S}(1+a_i), \qquad \mathrm{Sq}^k(a_S) = \sum_{T\subseteq S,\ |T|=k} a_T a_S$$

and summing over the m-element sets,

$$\mathrm{Sq}^k\sigma_m = \sum_{|S|=m}\ \sum_{T\subseteq S,\ |T|=k} a_T a_S$$

Every monomial here has exactly k variables squared and $m-k$ to the first power, and for disjoint D,E with $|D|=k$, $|E|=m-k$ the monomial $a_D^2 a_E$ occurs exactly once, from the unique pair $(S,T)=(D\cup E,\,D)$.

Step 4: The right side, coefficient by coefficient. A product $\sigma_{k-t}\sigma_{m+t} = \sum a_U a_W$ over $|U|=k-t$, $|W|=m+t$ is a sum of monomials $a_{U\cap W}^2\,a_{U\triangle W}$. Fix disjoint D,E with $d:=|D|$, $e:=|E|$ and $2d+e=m+k$. The pairs (U,W) with $U\cap W=D$ and $U\cup W=D\cup E$ are $U=D\cup A$, $W=D\cup(E\setminus A)$ for $A\subseteq E$ with $|A|=k-t-d$, so there are $\binom{e}{k-t-d}$ of them, and the coefficient of $a_D^2a_E$ on the right side is

$$c(d) = \sum_{t=0}^{k}\binom{m-k+t-1}{t}\binom{e}{k-d-t} \pmod 2$$

the coefficient of s^{k-d} in the product of the power series $\sum_j\binom{m-k+j-1}{j}s^j$ and $(1+s)^e$ over $\mathbb{Z}/2$.

Step 5: The binomial identity. For $r\ge 0$ the series $\sum_j\binom{r+j-1}{j}s^j\in\mathbb{Z}[[s]]$ equals $(1+s+s^2+\cdots)^r$: the coefficient of s^j in the r-fold product counts the tuples $(j_1,\dots,j_r)$ of nonnegative integers with sum j, of which there are $\binom{r+j-1}{j}$, and for $r=0$ the convention $\binom{-1}{0}=1$ gives the constant series 1. Since $(1-s)(1+s+s^2+\cdots)=1$ and $-1=1$ in $\mathbb{Z}/2$, reduction mod 2 gives $\sum_j\binom{r+j-1}{j}s^j=(1+s)^{-r}$ in $\mathbb{Z}/2[[s]]$, where $1+s$ is a unit. With $r=m-k$ and $e=m+k-2d$,

$$c(d) = \big[s^{k-d}\big](1+s)^{-(m-k)}(1+s)^{m+k-2d} = \big[s^{k-d}\big](1+s)^{2(k-d)} = \binom{2(k-d)}{k-d}$$

Writing $j=k-d$, Pascal's rule and the symmetry $\binom{2j-1}{j}=\binom{2j-1}{j-1}$ give $\binom{2j}{j}=2\binom{2j-1}{j-1}\equiv 0 \bmod 2$ for $j\ge 1$, while $j=0$ gives 1. So the right side has coefficient 1 exactly on the monomial types with $d=k$ and coefficient 0 otherwise, matching the left side. This proves the identity in $\mathbb{Z}/2[a_1,\dots,a_n]$, and injectivity of μ^* proves it in $H^*(BO(n);\mathbb{Z}/2)$.

Step 6: Bundles. For a real bundle E of rank r over a paracompact Hausdorff base, the classes $w_j(E)$ are the pullbacks of the universal classes under a classifying map into $BO(r)$, with $w_j(E) = 0$ for $j > r$ (definition 7.2.13). Applying the classifying map's pullback to the universal identity and using naturality of the squares along it (lemma 9.2.4(4)) gives the identity for E, completing the proof.

Three instances, each computed from the coefficients and used below. For $k = m = 1$ only $t = 0$ contributes, with $\binom{-1}{0} = 1$, so $\mathrm{Sq}^1 w_1 = w_1^2$, as axiom (5) requires of a degree-one class. The same collapse for general $k = m$ gives $\mathrm{Sq}^m w_m = w_m^2$. For $k = 1$, $m = 2$: the coefficients are $\binom{0}{0} = 1$ at $t = 0$ and $\binom{1}{1} = 1$ at $t = 1$, so

$$\mathrm{Sq}^1 w_2 = w_1 w_2 + w_3$$

For $k = 2$, $m = 4$: the coefficients at $t = 0, 1, 2$ are $\binom{1}{0} = 1$, $\binom{2}{1} = 2 \equiv 0$ and $\binom{3}{2} = 3 \equiv 1$, so

$$\mathrm{Sq}^2 w_4 = w_2 w_4 + w_6$$

Through the comparison of Chern and Stiefel-Whitney classes, this last instance becomes a constraint on Chern classes mod 2.

Corollary 9.2.12: Sq^2 on the Chern Classes

Let E be a complex vector bundle over a paracompact Hausdorff base. Then

$$\mathrm{Sq}^2 \rho c_1(E) = \big(\rho c_1(E)\big)^2, \qquad \mathrm{Sq}^2 \rho c_2(E) = \rho c_1(E)\, \rho c_2(E) + \rho c_3(E)$$

equivalently $\rho c_3(E) = \mathrm{Sq}^2 \rho c_2(E) + \rho\big(c_1(E) c_2(E)\big)$ in H^6 of the base. If E has rank at most 2, then $c_3(E) = 0$ and $\mathrm{Sq}^2 \rho c_2(E) = \rho c_1(E)\, \rho c_2(E)$.

***Proof*:**
By proposition 7.4.14, whose ρ_2 is the reduction ρ of this chapter, the underlying real bundle $E_{\mathbb{R}}$ has $w_{2i}(E_{\mathbb{R}}) = \rho c_i(E)$ and vanishing odd-degree classes. Wu's formula for $E_{\mathbb{R}}$ with $k = m = 2$ reads $\mathrm{Sq}^2 w_2 = w_2^2$, which is the first identity, and is also property (5) of theorem 9.1.17, which lemma 9.2.4(4) extends to every base, applied to the degree-two class w_2. The instance $\mathrm{Sq}^2 w_4 = w_2 w_4 + w_6$ computed above gives

$$\mathrm{Sq}^2 \rho c_2(E) = \mathrm{Sq}^2 w_4(E_{\mathbb{R}}) = w_2(E_{\mathbb{R}}) w_4(E_{\mathbb{R}}) + w_6(E_{\mathbb{R}}) = \rho c_1(E)\, \rho c_2(E) + \rho c_3(E)$$

Rearranging over $\mathbb{Z}/2$ and using that ρ is a ring homomorphism, so that $\rho c_1\, \rho c_2 = \rho(c_1 c_2)$, gives the equivalent form. For rank at most 2, $c_3(E) = 0$ by definition 7.2.7, completing the proof.

On the bundle V over $BSU(2)$ the corollary is consistent with theorem 9.2.10: there $c_1(V) = 0$ and $c_3(V) = 0$, so it predicts $\mathrm{Sq}^2 \rho c_2 = 0$, which holds because $H^6(BSU(2); \mathbb{Z}/2) = 0$. The identity binds the mod-2 reduction of c_3 to the lower two Chern classes, and section 9.4 turns it into the condition cutting out the image of the Chern-class map in rank three.

Corollary 9.2.13: $\mathrm{Sq}^3_{\mathbb{Z}}$ Vanishes on c_2

For every complex vector bundle E over a paracompact Hausdorff base X,

$$\mathrm{Sq}^3_{\mathbb{Z}}\, c_2(E) = 0 \in H^7(X;\mathbb{Z})$$

***Proof*:**
By definition 9.1.20, corollary 9.2.12, and the ring-homomorphism property of ρ,

$$\mathrm{Sq}^3_{\mathbb{Z}} c_2(E) = \beta\,\mathrm{Sq}^2 \rho c_2(E) = \beta\big(\rho c_1(E)\,\rho c_2(E) + \rho c_3(E)\big) = \beta\rho\big(c_1(E)c_2(E) + c_3(E)\big) = 0$$

since $\beta\rho = 0$ (proposition 9.1.2), completing the proof.

The vanishing is a genuine condition: on $K(\mathbb{Z},4)$ the universal class has $\mathrm{Sq}^3_{\mathbb{Z}}\iota_4 \neq 0$ (proposition 9.2.7), so ι_4 is not the second Chern class of any bundle over $K(\mathbb{Z},4)$. Section 9.5 builds its identification of the Atiyah-Hirzebruch differential on exactly this constraint.

Exercise 9.2.1

1. Let a and b be the pullbacks of the degree-one generators under the two projections of $\mathbb{RP}^n \times \mathbb{RP}^m$, so that $H^*(\mathbb{RP}^n \times \mathbb{RP}^m;\mathbb{Z}/2) = \mathbb{Z}/2[a,b]/(a^{n+1}, b^{m+1})$ by theorem 4.3.9. Show that $\mathrm{Sq}(a^j b^k) = a^j(1+a)^j\, b^k(1+b)^k$, and compute $\mathrm{Sq}^2(a^2 b)$ in $H^*(\mathbb{RP}^4 \times \mathbb{RP}^4;\mathbb{Z}/2)$.

2. Derive from theorem 9.2.11 the identity $\mathrm{Sq}^1 w_m = w_1 w_m + (m-1)\,w_{m+1}$, coefficients mod 2, and verify that for $m=1$ it reduces to axiom (5) of theorem 9.1.17 and for $m=2$ to the instance computed in the text.

3. Show that in $H^*(\Sigma X;R)$, for X a based CW complex, every cup product of two positive-degree classes vanishes: cover ΣX by contractible open neighborhoods U_+ and U_- of its two cones, lift the two classes to $H^*(\Sigma X, U_+;R)$ and $H^*(\Sigma X, U_-;R)$, and use the relative cup product of proposition 4.2.3(5). Deduce that $H^*(\Sigma\mathbb{CP}^2;\mathbb{Z}/2)$ and $H^*(S^3 \vee S^5;\mathbb{Z}/2)$ are isomorphic as rings, and that $\Sigma\mathbb{CP}^2$ and $S^3 \vee S^5$ are nonetheless not homotopy equivalent.

4. Show that the operations Sq^{2i} with $0 < i < k$ all vanish on $x^k \in H^{2k}(\mathbb{CP}^\infty;\mathbb{Z}/2)$ exactly when k is a power of 2.

5. Let X be a CW complex with cells only in even dimensions, so that its integral cohomology vanishes in odd degrees (proposition 4.1.20). Show that $\mathrm{Sq}^3_{\mathbb{Z}} = 0$ on $H^*(X;\mathbb{Z})$, and deduce from corollary 9.2.8 and proposition 9.1.2 that $\mathrm{Sq}^3 = 0$ on $H^*(X;\mathbb{Z}/2)$.

6. For which $k \geq 1$ is $\mathrm{Sq}^4 \colon H^{4k}(BSU(2);\mathbb{Z}/2) \to H^{4k+4}(BSU(2);\mathbb{Z}/2)$ an isomorphism? Justify the answer with theorem 9.2.10.

9.3 Applications in Homotopy Theory

This section computes two invariants that lie one degree beyond the range of section 7.5. The first is the group $\pi_5(S^3)$. It is cyclic of order two, generated by the composite $\eta_3 \circ \eta_4$ of two suspended Hopf maps: the operation Sq^2 shows that the composite is not null-homotopic, and the sphere bundle

$S^5 \to BU(2) \to BU(3)$, together with the coefficient groups of K-theory, bounds the order of $\pi_5(S^3)$ by two. The same computation gives $\pi_6(BU(2)) \cong \mathbb{Z}/2$. The second invariant is the k-invariant of $BU(2)$ that attaches $\pi_5(BU(2)) \cong \mathbb{Z}/2$ to $K(\mathbb{Z},2) \times K(\mathbb{Z},4)$, and Wu's formula in rank two identifies it as $\mathrm{Sq}^2\rho\iota_4 + \rho\iota_2\,\rho\iota_4$.

Throughout, $\eta_n = \Sigma^{n-2}\eta\colon S^{n+1} \to S^n$ for $n \geq 2$ are the suspended Hopf maps of section 9.2, and C_f is the mapping cone of definition 8.1.7. A based map $S^m \to S^n$ with $m \geq n$ is cellular for the cell structures $e^0 \cup e^m$ and $e^0 \cup e^n$, and so is a based map into $S^3 = e^0 \cup e^3$ from a CW complex whose 2-skeleton is the basepoint. The mapping cones below are therefore CW complexes: C_f has the cells of the target of f together with one $(k+1)$-cell for each k-cell of the source other than its basepoint.

A null-homotopy of $\eta_3 \circ \eta_4$ would build a complex with cells in dimensions 3, 5 and 7, whose 5-skeleton is a copy of C_{η_3} and whose quotient by the bottom sphere is the suspension of C_{η_4}. By proposition 9.2.3, applied to each of these, $\mathrm{Sq}^2\mathrm{Sq}^2$ would be nonzero on the bottom class. That class is the reduction of an integral class, and every such class of degree three has $\mathrm{Sq}^2\mathrm{Sq}^2$ pulled back from $H^7(K(\mathbb{Z},3);\mathbb{Z}/2)$, which vanishes by proposition 9.2.6.

Lemma 9.3.1: The Composite $\eta_3 \circ \eta_4$ Is Not Null-Homotopic

The composite $\eta_3 \circ \eta_4\colon S^5 \to S^3$ represents a nonzero element of $\pi_5(S^3)$.

***Proof*:**

Suppose that $\eta_3 \circ \eta_4$ is based null-homotopic. Write $A = C_{\eta_4} = S^4 \cup_{\eta_4} CS^5$, with $a\colon S^4 \to A$ the inclusion. In positive degrees reduced and unreduced cohomology agree (proposition 4.1.14), and the reduced cohomology of a sphere S^m with coefficients in G is G in degree m and zero in every other degree, by corollary 3.4.15 and the universal coefficient theorem (theorem 4.1.8).

Step 1: The extension. A based null-homotopy of $\eta_3 \circ \eta_4$ descends to a map $CS^5 \to S^3$ that agrees with $\eta_3 \circ \eta_4$ on the base S^5, and gluing it to η_3 on S^4 gives a based map $\varphi\colon A \to S^3$ with $\varphi \circ a = \eta_3$. This is the construction of Step 2 of the proof of proposition 8.1.8. Let $i\colon S^3 \to C_\varphi$ be the inclusion.

Step 2: Exact sequences. Let $f\colon X \to Y$ be a based map of pointed CW complexes, G an abelian group and $k \geq 1$. By proposition 8.1.8(3), the Puppe sequence $X \to Y \to C_f \to \Sigma X \to \Sigma Y$ becomes exact under $[-, K(G,k)]_*$. By theorem 5.5.12 this functor is $\widetilde{H}^k(-;G)$ on pointed CW complexes, with the pullbacks as induced maps, and these are homomorphisms, so exactness of pointed sets is exactness of groups. This gives an exact sequence

$$\widetilde{H}^k(\Sigma Y;G) \longrightarrow \widetilde{H}^k(\Sigma X;G) \xrightarrow{\partial^*} \widetilde{H}^k(C_f;G) \xrightarrow{i^*} \widetilde{H}^k(Y;G) \xrightarrow{f^*} \widetilde{H}^k(X;G)$$

For $f = \eta_4$ the terms $\widetilde{H}^3(\Sigma S^5;\mathbb{Z}) = \widetilde{H}^3(S^6;\mathbb{Z})$ and $\widetilde{H}^3(S^4;\mathbb{Z})$ vanish (example 5.7.2), so $\widetilde{H}^3(A;\mathbb{Z}) = 0$. For $f = \varphi$ the sequence reads

$$\widetilde{H}^k(S^4;G) \longrightarrow \widetilde{H}^k(\Sigma A;G) \xrightarrow{\partial^*} \widetilde{H}^k(C_\varphi;G) \xrightarrow{i^*} \widetilde{H}^k(S^3;G) \xrightarrow{\varphi^*} \widetilde{H}^k(A;G)$$

With $k = 3$ and $G = \mathbb{Z}$ the last group vanishes, so $i^*\colon H^3(C_\varphi;\mathbb{Z}) \to H^3(S^3;\mathbb{Z})$ is onto. With $k = 5$ and $G = \mathbb{Z}/2$ the groups $\widetilde{H}^5(S^4;\mathbb{Z}/2)$ and $\widetilde{H}^5(S^3;\mathbb{Z}/2)$ vanish, so ∂^* is an isomorphism in degree 5. With $k = 7$ and $G = \mathbb{Z}/2$ the group $\widetilde{H}^7(S^4;\mathbb{Z}/2)$ vanishes, so ∂^* is injective in degree 7.

Step 3: The first square. By Step 2 there is $\tilde{x} \in H^3(C_\varphi;\mathbb{Z})$ with $i^*\tilde{x}$ a generator of $H^3(S^3;\mathbb{Z}) \cong \mathbb{Z}$. Put $x = \rho\tilde{x}$. Since $H^4(S^3;\mathbb{Z}) = 0$, proposition 9.1.2 makes $\rho\colon H^3(S^3;\mathbb{Z}) \to H^3(S^3;\mathbb{Z}/2)$ onto, so it carries the generator to the nonzero element and $i^*x = \rho\, i^*\tilde{x} \neq 0$. The inclusion of S^3 and the composite of the cone $Ca\colon CS^4 \to CA$ with the quotient map to C_φ respect the identifications $s \sim \eta_3(s)$, $s \in S^4$, that define C_{η_3}, because $a(s)$ is identified with $\varphi(a(s)) = \eta_3(s)$ in C_φ. They therefore induce a map $u\colon C_{\eta_3} \to C_\varphi$ with $u \circ i' = i$, where $i'\colon S^3 \to C_{\eta_3}$ is the inclusion. Then $i'^*(u^*x) = i^*x \neq 0$, so u^*x is the nonzero element of $H^3(C_{\eta_3};\mathbb{Z}/2) \cong \mathbb{Z}/2$, and $\mathrm{Sq}^2u^*x \neq 0$ by proposition 9.2.3 with $n = 3$. By naturality (theorem 9.1.17(1)), $u^*\mathrm{Sq}^2x = \mathrm{Sq}^2u^*x \neq 0$, hence $\mathrm{Sq}^2x \neq 0$.

Step 4: The second square. By Step 2 with $k = 5$ there is a nonzero $z \in \widetilde{H}^5(\Sigma A;\mathbb{Z}/2)$ with $\partial^*z = \mathrm{Sq}^2x$. The suspension isomorphism $\widetilde{H}^4(A;\mathbb{Z}/2) \cong \widetilde{H}^5(\Sigma A;\mathbb{Z}/2)$ carries some nonzero w to z, and $\mathrm{Sq}^2w \neq 0$ by proposition 9.2.3 with $n = 4$. By stability (theorem 9.1.17(6)), Sq^2z is the suspension of Sq^2w, so $\mathrm{Sq}^2z \neq 0$. By naturality and the injectivity of ∂^* in degree 7,

$$\mathrm{Sq}^2\mathrm{Sq}^2x = \mathrm{Sq}^2\partial^*z = \partial^*\mathrm{Sq}^2z \neq 0 \quad \text{in } H^7(C_\varphi;\mathbb{Z}/2)$$

Step 5: The contradiction. The space C_φ is a connected CW complex, so theorem 5.5.12 gives a map $F\colon C_\varphi \to K(\mathbb{Z},3)$ with $F^*\iota_3 = \tilde{x}$. The reduction ρ and the squares are natural, so

$$\mathrm{Sq}^2\mathrm{Sq}^2x = \mathrm{Sq}^2\mathrm{Sq}^2\rho F^*\iota_3 = F^*\big(\mathrm{Sq}^2\mathrm{Sq}^2\rho\iota_3\big)$$

and $\mathrm{Sq}^2\mathrm{Sq}^2\rho\iota_3$ lies in $H^7(K(\mathbb{Z},3);\mathbb{Z}/2)$, which is zero by proposition 9.2.6. This contradicts Step 4, so $\eta_3 \circ \eta_4$ is not null-homotopic, completing the proof.

The class x restricts on C_{η_3} to the class on which Sq^2 is nonzero, and Sq^2x comes from the suspension of C_{η_4}, where Sq^2 is again nonzero. A null-homotopy of $\eta_3 \circ \eta_4$ places both nonzero squares on the single complex C_φ, and the composite $\mathrm{Sq}^2\mathrm{Sq}^2$ is then nonzero on the reduction of an integral class of degree three. The computation $H^7(K(\mathbb{Z},3);\mathbb{Z}/2) = 0$ of proposition 9.2.6 excludes this.

The upper bound on $\pi_5(S^3)$ comes from the classifying space $BU(2)$ through the isomorphism $\pi_6(BU(2)) \cong \pi_5(U(2))$ of corollary 7.1.11. As a space, the group $U(2)$ is the product of $SU(2)$ and the circle of determinants.

Lemma 9.3.2: The Unitary Group $U(2)$ as a Product of Spaces

The map

$$\Phi\colon SU(2) \times U(1) \longrightarrow U(2), \qquad \Phi(A,z) = A\begin{pmatrix} 1 & 0 \\ 0 & z \end{pmatrix}$$

is a homeomorphism. Consequently $U(2)$ is homeomorphic to $S^3 \times S^1$, and $\pi_k(U(2)) \cong \pi_k(S^3)$ for every $k \geq 2$.

***Proof*:**
Write $D(z)$ for the diagonal matrix with entries 1 and z, a unitary matrix with determinant z when $|z| = 1$. For $A \in SU(2)$ the product $AD(z)$ is unitary, so Φ is a well-defined continuous map. Define $\Psi\colon U(2) \to SU(2) \times U(1)$ by $\Psi(B) = \big(BD((\det B)^{-1}), \det B\big)$. The matrix $BD((\det B)^{-1})$ is unitary with determinant $\det B \cdot (\det B)^{-1} = 1$, and Ψ is continuous, since the determinant is a polynomial in the entries and inversion on $U(1)$ is complex conjugation.

The composites are

$$\Phi\Psi(B) = BD((\det B)^{-1})D(\det B) = B, \qquad \Psi\Phi(A,z) = \big(AD(z)D(z^{-1}), \det A \cdot z\big) = (A,z)$$

using $\det A = 1$. So Φ is a homeomorphism. By lemma 9.2.9 and $U(1) = S^1$, the space $U(2)$ is homeomorphic to $S^3 \times S^1$, by a homeomorphism carrying the identity matrix to a point $(s_0, 1)$. A based map into a product is a pair of based maps, and likewise for based homotopies, so $\pi_k(S^3 \times S^1) \cong \pi_k(S^3) \times \pi_k(S^1)$, and $\pi_k(S^1) = 0$ for $k \geq 2$ by example 5.5.2. Hence $\pi_k(U(2)) \cong \pi_k(S^3)$ for every $k \geq 2$, completing the proof.

The order of $\pi_6(BU(2))$ is bounded through the sphere bundle $S^5 \to S(\gamma_3) \to BU(3)$ of lemma 7.2.4, whose total space is $BU(2)$ up to homotopy. Its base has the homotopy groups of BU in degrees five and six, and there the coefficient groups of K-theory in theorem 8.5.10 give $\pi_5(BU) = 0$ and $\pi_6(BU) \cong \mathbb{Z}$.

Theorem 9.3.3: The Group $\pi_5(S^3)$

The group $\pi_5(S^3)$ is cyclic of order two, generated by $\eta_3 \circ \eta_4$.

***Proof*:**
For $n \geq 2$ let $\rho_n \colon S(\gamma_n) \to BU(n)$ be the sphere bundle of lemma 7.2.4, a fiber bundle with fiber S^{2n-1}, hence a Serre fibration by theorem 5.2.4. The homotopy equivalence of that lemma from $BU(n-1)$ to $S(\gamma_n)$ induces isomorphisms on all homotopy groups, and through it the long exact sequence of ρ_n (theorem 5.2.12) relates the homotopy groups of S^{2n-1}, $BU(n-1)$ and $BU(n)$. The map ρ_n is onto, its fibers being nonempty. Since $BU(2)$ is path-connected (proposition 7.5.1), every $BU(n)$ with $n \geq 2$ is path-connected.

Step 1: The homotopy of $BU(3)$ in degrees five and six. In the space BU of definition 7.1.20, every cell outside the subcomplex $BU(3)$ has dimension at least 8 (corollary 7.1.21), so the 7-skeleton of BU lies in $BU(3)$. Let $k \leq 6$, and base maps at the basepoint of BU, a 0-cell. By cellular approximation, the theorem imported in the proof of theorem 5.3.6, a based map $S^k \to BU$ is homotopic rel the basepoint to a cellular map, which lands in $BU(3)$. Two based maps $S^k \to BU(3)$ may likewise be taken cellular within $BU(3)$, and a based homotopy $S^k \times I \to BU$ between them is then homotopic, rel the subcomplex $S^k \times \partial I \cup \{s_0\} \times I$ on which it is cellular, to a cellular map, which lands in the $(k+1)$-skeleton and so in $BU(3)$. Hence $\pi_k(BU(3)) \to \pi_k(BU)$ is an isomorphism for $k \leq 6$.

Write $\widetilde{K}^m(X) = [X, KU_m]_*$ for the reduced theory of theorem 8.5.10(2), so that $KU_0 = KU_6 = \mathbb{Z} \times BU$. For $1 \leq k \leq 6$ a based map $S^k \to \mathbb{Z} \times BU$ lands in the path component of the basepoint, a copy of the path-connected space BU, and likewise for based homotopies, so $\pi_k(BU) \cong \widetilde{K}^0(S^k) = \widetilde{K}^6(S^k)$. Since $S^k = \Sigma^k S^0$ (example 5.7.2), applying the suspension isomorphism of theorem 8.4.6 k times gives $\widetilde{K}^6(S^k) \cong \widetilde{K}^{6-k}(S^0)$, which by theorem 8.5.10(3) is $\mathbb{Z}$ when k is even and 0 when k is odd. Therefore $\pi_5(BU(3)) = 0$ and $\pi_6(BU(3)) \cong \mathbb{Z}$.

Step 2: The sphere bundle over $BU(3)$. By lemma 7.2.4 with $n = 3$, the map ρ_3 is a Serre fibration with fiber S^5, and its long exact sequence, with $\pi_k(S(\gamma_3))$ identified with $\pi_k(BU(2))$ as above, contains the segment

$$\pi_6(S^5) \longrightarrow \pi_6(BU(2)) \xrightarrow{(\rho_3)_*} \pi_6(BU(3)) \xrightarrow{\partial} \pi_5(S^5) \xrightarrow{j_*} \pi_5(BU(2)) \longrightarrow \pi_5(BU(3))$$

with j the inclusion of the fiber. Here $\pi_5(BU(3)) = 0$ and $\pi_6(BU(3)) \cong \mathbb{Z}$ by Step 1, $\pi_5(S^5) \cong \mathbb{Z}$ by corollary 5.4.11, $\pi_5(BU(2)) \cong \mathbb{Z}/2$ by proposition 7.5.1, and $\pi_6(S^5) \cong \mathbb{Z}/2$ by corollary 6.3.7. Exactness at $\pi_5(BU(2))$ makes j_* onto, so its kernel is the subgroup $2\mathbb{Z}$ of index two. Exactness at $\pi_5(S^5)$ makes $2\mathbb{Z}$ the image of ∂, and a homomorphism $\mathbb{Z} \to \mathbb{Z}$ with nonzero image is injective, so ∂ is injective. Exactness at $\pi_6(BU(3))$ then makes $(\rho_3)_*$ zero, and exactness at $\pi_6(BU(2))$ makes $\pi_6(S^5) \to \pi_6(BU(2))$ onto. Hence $\pi_6(BU(2))$ has at most two elements.

Step 3: From $BU(2)$ to S^3. By corollary 7.1.11 with $G = U(2)$, $\pi_6(BU(2)) \cong \pi_5(U(2))$, and $\pi_5(U(2)) \cong \pi_5(S^3)$ by lemma 9.3.2. So $\pi_5(S^3)$ has at most two elements.

Step 4: Conclusion. By lemma 9.3.1 the class of $\eta_3 \circ \eta_4$ is a nonzero element of $\pi_5(S^3)$. A group with at most two elements containing a nonzero element is cyclic of order two, generated by that element, completing the proof.

The two bounds come from different sources. The lower bound is a computation with Steenrod squares on a three-cell complex. The upper bound enters through the sphere bundle over $BU(3)$, using the stable groups $\pi_5(BU) = 0$ and $\pi_6(BU) \cong \mathbb{Z}$ given by Bott periodicity and the first stable stem $\pi_6(S^5) \cong \mathbb{Z}/2$. Step 2 also shows that the boundary map $\partial\colon \pi_6(BU(3)) \to \pi_5(S^5)$ is injective with image $2\mathbb{Z}$, so it is multiplication by ± 2 under any identification of both groups with $\mathbb{Z}$.

The theorem is consistent with theorem 5.7.8. The sphere S^3 is 2-connected, so the suspension $\sigma\colon \pi_5(S^3) \to \pi_6(S^4)$ is onto, the degree 5 being the borderline value $2n-1$ for $n = 3$. It carries the class of $\eta_3 \circ \eta_4$ to the class of $\Sigma\eta_3 \circ \Sigma\eta_4 = \eta_4 \circ \eta_5$, so $\pi_6(S^4)$ is generated by $\eta_4 \circ \eta_5$ and has at most two elements.

Step 3 of the proof identifies $\pi_6(BU(2))$ with $\pi_5(S^3)$, and the same identification holds for $BSU(2)$.

Corollary 9.3.4: The Sixth Homotopy Groups of $BU(2)$ and $BSU(2)$

There are isomorphisms

$$\pi_6(BU(2)) \cong \pi_5(U(2)) \cong \pi_5(S^3) \cong \mathbb{Z}/2, \qquad \pi_6(BSU(2)) \cong \pi_5(SU(2)) \cong \pi_5(S^3) \cong \mathbb{Z}/2$$

***Proof*:**
The first chain is Step 3 of the proof of theorem 9.3.3 followed by the theorem itself. For the second, corollary 7.1.11 with $G = SU(2)$ gives $\pi_6(BSU(2)) \cong \pi_5(SU(2))$, the homeomorphism $SU(2) \cong S^3$ of lemma 9.2.9 gives $\pi_5(SU(2)) \cong \pi_5(S^3)$, and theorem 9.3.3 gives $\pi_5(S^3) \cong \mathbb{Z}/2$, completing the proof.

Example 9.3.5: Rank-Two Bundles on S^6

The sphere $S^6 = e^0 \cup e^6$ is a CW complex, hence paracompact Hausdorff (theorem 7.1.7), and $BU(2)$ is simply connected (proposition 7.5.1). By theorem 7.1.29, proposition 5.1.15 and corollary 9.3.4,

$$\mathrm{Vect}^2_{\mathbb{C}}(S^6) \cong [S^6, BU(2)] \cong \pi_6(BU(2)) \cong \mathbb{Z}/2$$

so S^6 carries exactly two rank-two complex bundles up to isomorphism, the trivial bundle, classified by the constant map, and one other. Since $H^2(S^6;\mathbb{Z}) = H^4(S^6;\mathbb{Z}) = 0$, both have $c_1 = c_2 = 0$, and the Chern classes do not distinguish them. The same happens on S^5, where $\mathrm{Vect}^2_{\mathbb{C}}(S^5) \cong \pi_5(BU(2)) \cong \mathbb{Z}/2$ (the remark after theorem 7.5.4). On S^4, by contrast, rank-two bundles are classified by c_2 (example 7.5.5).

By proposition 7.5.3, the map $\kappa = (\kappa_1, \kappa_2)\colon BU(2) \to K(\mathbb{Z},2) \times K(\mathbb{Z},4)$ representing (c_1, c_2) is 5-connected, and its target has no homotopy above degree four. These are the two conditions placed on the fourth stage of a Postnikov tower in definition 5.6.2. The group $\pi_5(BU(2)) \cong \mathbb{Z}/2$ enters the tower through the k-invariant $k_5 \in H^6(X_4; \pi_5(BU(2)))$ of definition 5.6.11,[1] and comparing X_4 with $K(\mathbb{Z},2) \times K(\mathbb{Z},4)$ along κ places k_5 in the mod-2 cohomology of the product of Eilenberg-MacLane spaces, which in degree six has dimension three.

Fix $K(\mathbb{Z},2) = \mathbb{CP}^\infty$ (example 5.5.5), with ι_2 the generator of example 5.5.11, fix a CW complex $K(\mathbb{Z},4)$ (theorem 5.5.4), and write $T = K(\mathbb{Z},2) \times K(\mathbb{Z},4)$. A class on one factor is written with the same symbol as its pullback to T along the projection, so that $\rho\iota_2\,\rho\iota_4$ is the cross product $\rho\iota_2 \times \rho\iota_4$ of definition 4.3.7.

Proposition 9.3.6: The Cohomology of $K(\mathbb{Z},2) \times K(\mathbb{Z},4)$ in Degrees Five and Six

The group $H^5(T; \mathbb{Z}/2)$ is zero, and $H^6(T; \mathbb{Z}/2)$ is a $\mathbb{Z}/2$-vector space with basis

$$(\rho\iota_2)^3, \qquad \rho\iota_2\,\rho\iota_4, \qquad \mathrm{Sq}^2\rho\iota_4$$

***Proof*:**

Step 1: The factor $\mathbb{CP}^\infty$. By example 5.5.11, $H^*(\mathbb{CP}^\infty; \mathbb{Z}) = \mathbb{Z}[\iota_2]$, which is torsion-free and zero in odd degrees. For every m, proposition 9.1.2 gives an exact sequence

$$H^m(\mathbb{CP}^\infty;\mathbb{Z}) \xrightarrow{\rho} H^m(\mathbb{CP}^\infty;\mathbb{Z}/2) \xrightarrow{\beta} H^{m+1}(\mathbb{CP}^\infty;\mathbb{Z}) \xrightarrow{2} H^{m+1}(\mathbb{CP}^\infty;\mathbb{Z})$$

Multiplication by 2 is injective on the torsion-free group $H^{m+1}(\mathbb{CP}^\infty;\mathbb{Z})$, so $\beta = 0$ and ρ is onto, with kernel $2H^m(\mathbb{CP}^\infty;\mathbb{Z})$ by the same proposition. The reduction ρ is a ring homomorphism, being induced by the ring homomorphism $\mathbb{Z} \to \mathbb{Z}/2$ on cochains, so $H^*(\mathbb{CP}^\infty;\mathbb{Z}/2) = \mathbb{Z}/2[\rho\iota_2]$, which is $\mathbb{Z}/2$ in each even degree and zero in each odd degree. The complex $\mathbb{CP}^\infty$ has one cell in each even dimension (example 5.5.5), so its cellular chain complex shows that $H_k(\mathbb{CP}^\infty;\mathbb{Z}/2)$ is finite for every k, and $\mathbb{CP}^\infty$ is of finite type over $\mathbb{Z}/2$ in the sense of theorem 4.3.9.

Step 2: The factor $K(\mathbb{Z},4)$ through degree six. The space $K(\mathbb{Z},4)$ is 3-connected, so theorem 5.4.9 gives $\widetilde{H}_i(K(\mathbb{Z},4)) = 0$ for $i \le 3$ and $H_4(K(\mathbb{Z},4)) \cong \mathbb{Z}$, and $H_5(K(\mathbb{Z},4)) = 0$ by proposition 6.3.5. The universal coefficient theorem (theorem 4.1.8) then gives $H^i(K(\mathbb{Z},4);\mathbb{Z}/2) = 0$ for $1 \le i \le 3$, since both $\mathrm{Hom}(H_i,\mathbb{Z}/2)$ and $\mathrm{Ext}(H_{i-1},\mathbb{Z}/2)$ vanish there, and $H^5(K(\mathbb{Z},4);\mathbb{Z}) = 0$, since $\mathrm{Hom}(H_5,\mathbb{Z}) = 0$ and $\mathrm{Ext}(H_4,\mathbb{Z}) = \mathrm{Ext}(\mathbb{Z},\mathbb{Z}) = 0$. With $H^5(K(\mathbb{Z},4);\mathbb{Z}) = 0$, proposition 9.1.2 makes $\rho\colon H^4(K(\mathbb{Z},4);\mathbb{Z}) \to H^4(K(\mathbb{Z},4);\mathbb{Z}/2)$ onto with kernel $2H^4(K(\mathbb{Z},4);\mathbb{Z})$. The group $H^4(K(\mathbb{Z},4);\mathbb{Z}) \cong \mathbb{Z}$ is generated by ι_4 (definition 5.5.10), so $H^4(K(\mathbb{Z},4);\mathbb{Z}/2) = \mathbb{Z}/2 \cdot \rho\iota_4$. By proposition 9.2.7 with $n = 4$, $H^5(K(\mathbb{Z},4);\mathbb{Z}/2) = 0$ and $H^6(K(\mathbb{Z},4);\mathbb{Z}/2) = \mathbb{Z}/2 \cdot \mathrm{Sq}^2\rho\iota_4$.

Step 3: The Künneth formula. By theorem 4.3.9, with the field $\mathbb{Z}/2$ and the finite-type factor $\mathbb{CP}^\infty$, the cross product is an isomorphism

$$\bigoplus_{k+\ell=m} H^k(\mathbb{CP}^\infty;\mathbb{Z}/2) \otimes H^\ell(K(\mathbb{Z},4);\mathbb{Z}/2) \xrightarrow{\cong} H^m(T;\mathbb{Z}/2)$$

By Steps 1 and 2, for $\ell \le 6$ the first factor is nonzero only for even k and the second only for $\ell \in \{0,4,6\}$. In total degree 5 an even k forces an odd ℓ, so every summand vanishes and

[1] The index follows definition 5.6.11, where $k_n \in H^{n+1}(X_{n-1};\pi_n)$ classifies the passage from X_{n-1} to X_n. Antieau and Elmanto's primer on unstable motivic homotopy theory indexes the same class k_4, after the section X_4 on which it is defined.

$H^5(T;\mathbb{Z}/2)=0$. In total degree 6 the pairs with both factors nonzero are $(k,\ell)=(6,0)$, $(2,4)$ and $(0,6)$, contributing the basis vectors $(\rho\iota_2)^3\times 1$, $\rho\iota_2\times\rho\iota_4$ and $1\times \mathrm{Sq}^2\rho\iota_4$. In the notation fixed above these are $(\rho\iota_2)^3$, $\rho\iota_2\,\rho\iota_4$ and the pullback of $\mathrm{Sq}^2\rho\iota_4$ along the second projection, which is Sq^2 of the pullback of $\rho\iota_4$ by naturality (theorem 9.1.17(1)), as claimed.

By theorem 5.5.12, a pair $(a,b)\in H^2(X;\mathbb{Z})\times H^4(X;\mathbb{Z})$ on a connected CW complex X is the pullback of (ι_2,ι_4) along a map $X\to T$, unique up to homotopy, so every class in $H^6(T;\mathbb{Z}/2)$ defines a natural operation $(a,b)\mapsto u(a,b)\in H^6(X;\mathbb{Z}/2)$. The basis of the proposition gives three of them, $(\rho a)^3$, $\rho a\,\rho b$ and $\mathrm{Sq}^2\rho b$, and the operation defined by any other class is a sum of these. The first two are cup products, and only the third uses a Steenrod square.

The fourth section X_4 of the Postnikov tower of $BU(2)$ in theorem 5.6.4 is $BU(2)$ with cells of dimension at least six attached, and κ extends over these cells to a map into T that is an isomorphism on homotopy groups. A map that is an isomorphism on homotopy groups through some degree is an isomorphism on cohomology below that degree, and for simply connected spaces this follows from the relative Hurewicz theorem applied to a mapping cylinder.

Lemma 9.3.7: Cohomology Isomorphisms from Homotopy Isomorphisms

Let $n\geq 1$, let $\lambda\colon X\to Y$ be a map of path-connected spaces with X simply connected, and let $x_0\in X$. Suppose that $\lambda_*\colon \pi_k(X,x_0)\to\pi_k(Y,\lambda(x_0))$ is an isomorphism for $1\leq k<n$ and is surjective for $k=n$. Then for every abelian group G the map $\lambda^*\colon H^k(Y;G)\to H^k(X;G)$ is an isomorphism for $k<n$ and is injective for $k=n$. In particular this holds for an n-connected map (definition 5.3.13) of simply connected spaces, and a map of simply connected spaces inducing isomorphisms on all homotopy groups at one basepoint induces isomorphisms on cohomology in every degree.

***Proof*:**

Step 1: The mapping cylinder. Let $M=(X\times I\sqcup Y)/\big((x,1)\sim\lambda(x)\big)$, with $i\colon X\to M$, $i(x)=[x,0]$, the inclusion $e\colon Y\to M$, and the retraction $r\colon M\to Y$, $r[x,t]=\lambda(x)$ and $r(y)=y$. The set $X\times\{0\}$ is closed in $X\times I\sqcup Y$ and is a union of equivalence classes, so i is a homeomorphism onto a closed subspace [16, Subspaces And Quotient Maps], and X is identified with $i(X)$. The homotopy $([x,t],s)\mapsto[x,(1-s)t+s]$, fixed on Y, deforms id_M to $e\circ r$ (example 1.1.13), and $r\circ e=\mathrm{id}_Y$, so r is a homotopy equivalence. Moreover $r\circ i=\lambda$, and $(x,s)\mapsto[x,s]$ is a homotopy from i to $e\circ\lambda$. The space M is path-connected, each point $[x,t]$ being joined to $\lambda(x)\in Y$ by the path $s\mapsto[x,t+(1-t)s]$.

Step 2: The homotopy of the pair (M,X). The group $\pi_1(Y,\lambda(x_0))$ is the image of $\pi_1(X,x_0)=0$, so Y is simply connected, and M is simply connected by theorem 2.1.45, since r is a homotopy equivalence. For the simply connected spaces X, Y, M and every $k\geq 1$, proposition 5.1.15 identifies π_k at any basepoint with the set $[S^k,-]$ of free homotopy classes, compatibly with induced maps. On free homotopy classes $i_\#=e_\#\circ\lambda_\#$, by the homotopy $i\simeq e\circ\lambda$ of Step 1, and $e_\#$ is a bijection with inverse $r_\#$, since $r\circ e=\mathrm{id}_Y$ and $e\circ r\simeq\mathrm{id}_M$. Hence $i_*\colon\pi_k(X,x_0)\to\pi_k(M,x_0)$ is bijective for $k<n$ and surjective for $k=n$. Let $2\leq k\leq n$, and consider the segment

$$\pi_k(X,x_0)\xrightarrow{i_*}\pi_k(M,x_0)\longrightarrow\pi_k(M,X,x_0)\xrightarrow{\partial}\pi_{k-1}(X,x_0)\xrightarrow{i_*}\pi_{k-1}(M,x_0)$$

of the exact sequence of the pair (proposition 5.1.11). Surjectivity of the first i_* makes the second map zero, and injectivity of the last i_* (trivial when $k=2$, since $\pi_1(X,x_0)=0$) makes ∂

zero, so $\pi_k(M, X, x_0) = 0$. In degree one, $\pi_1(M, x_0) = 0$ and $\pi_0(X)$ is a point, so exactness of $\pi_1(M, x_0) \to \pi_1(M, X, x_0) \to \pi_0(X)$ as pointed sets gives $\pi_1(M, X, x_0) = 0$.

Step 3: Homology and cohomology of the pair. The pair (M, X) consists of path-connected spaces with X nonempty and simply connected, and $\pi_k(M, X) = 0$ for $k < n + 1$, so theorem 5.4.10, applied with $n + 1 \geq 2$ in place of n, gives $H_k(M, X) = 0$ for $k \leq n$. The relative singular chain complex $C_\bullet(M)/C_\bullet(X)$ is free on the singular simplices of M not contained in X, so theorem 4.1.8 applies to it. For $k \leq n$ both $\mathrm{Ext}(H_{k-1}(M, X), G)$ and $\mathrm{Hom}(H_k(M, X), G)$ vanish, hence $H^k(M, X; G) = 0$.

Step 4: The comparison. In the exact sequence

$$H^k(M, X; G) \longrightarrow H^k(M; G) \xrightarrow{i^*} H^k(X; G) \longrightarrow H^{k+1}(M, X; G)$$

of theorem 4.1.17, both relative groups vanish for $k < n$, and the first vanishes for $k = n$. So i^* is an isomorphism for $k < n$ and is injective for $k = n$. The map r^* is an isomorphism by proposition 4.1.23, since r is a homotopy equivalence, and $\lambda^* = i^* r^*$. An n-connected map satisfies the hypothesis at every basepoint, and a map inducing isomorphisms on all homotopy groups at x_0 satisfies it for every n, completing the proof.

The lemma requires no cell structure on Y, and below Y is the product T of two CW complexes with the product topology. For the inclusion $f_n \colon BU(2) \to X_n$ of theorem 5.6.4, an isomorphism on π_k for $k \leq n$ into a space with no homotopy above degree n, the lemma with $n + 1$ in place of n gives $H^k(X_n; G) \cong H^k(BU(2); G)$ for $k \leq n$.

Theorem 9.3.8: The k-Invariant of $BU(2)$ at π_5

Let $\{X_n, f_n, p_n\}$ be the Postnikov tower of $BU(2)$ constructed in theorem 5.6.4, let $\kappa \colon BU(2) \to T$ be the map of proposition 7.5.3 representing (c_1, c_2), and put

$$\theta = \mathrm{Sq}^2 \rho\iota_4 + \rho\iota_2 \, \rho\iota_4 \in H^6(T; \mathbb{Z}/2)$$

1. The map κ extends to a map $\lambda \colon X_4 \to T$ with $\lambda \circ f_4 = \kappa$, which induces isomorphisms on all homotopy groups at every basepoint in $BU(2)$ and isomorphisms $\lambda^* \colon H^m(T; G) \to H^m(X_4; G)$ for every m and every abelian group G.
2. The kernel of $\kappa^* \colon H^6(T; \mathbb{Z}/2) \to H^6(BU(2); \mathbb{Z}/2)$ is $\{0, \theta\}$.
3. The k-invariant $k_5 \in H^6(X_4; \pi_5(BU(2))) = H^6(X_4; \mathbb{Z}/2)$ of definition 5.6.11 is $k_5 = \lambda^* \theta$.

***Proof*:**

1. By theorem 5.6.4, X_4 is a CW complex obtained from $BU(2)$ by attaching cells of dimension at least 6, and f_4 is the inclusion. The homotopy groups of T are the products of those of its factors, so $\pi_k(T) = 0$ for every $k \geq 5$ at every basepoint (definition 5.5.1). Let a cell of dimension $d \geq 6$ be attached along $\psi \colon S^{d-1} \to X_4$, and suppose a map extending κ has been defined on the cells of lower dimension. Its composite with ψ is a map $S^{d-1} \to T$ with $d - 1 \geq 5$, hence null-homotopic, so the map extends over the cell. Proceeding by increasing dimension, exactly as in Step 3 of the proof of theorem 5.6.4, extends κ to $\lambda \colon X_4 \to T$ with $\lambda \circ f_4 = \kappa$. Fix $b \in BU(2)$. For $1 \leq k \leq 4$ the map $(f_4)_* \colon \pi_k(BU(2), b) \to \pi_k(X_4, b)$ is an

isomorphism by definition 5.6.2(2), and so is $\kappa_* = \lambda_* \circ (f_4)_*$ by proposition 7.5.3, hence λ_* is an isomorphism. For $k \geq 5$ both $\pi_k(X_4, b)$ (definition 5.6.2(3)) and $\pi_k(T)$ vanish. The space X_4 is path-connected and simply connected, its π_1 being that of $BU(2)$, and T is path-connected, so lemma 9.3.7 applies for every n and λ^* is an isomorphism in every degree.

2. By theorem 7.2.5, $H^*(BU(2);\mathbb{Z}) = \mathbb{Z}[c_1, c_2]$, torsion-free and zero in odd degrees. As in Step 1 of the proof of proposition 9.3.6, proposition 9.1.2 makes ρ onto $H^m(BU(2);\mathbb{Z}/2)$ with kernel $2H^m(BU(2);\mathbb{Z})$ for every m. Hence $H^5(BU(2);\mathbb{Z}/2) = 0$, and $H^6(BU(2);\mathbb{Z}/2)$ has basis $(\rho c_1)^3$, $\rho c_1\,\rho c_2$, the reductions of the basis c_1^3, c_1c_2 of $H^6(BU(2);\mathbb{Z})$. By definition 7.2.7 the generators c_i are the Chern classes of the universal bundle γ_2 over $BU(2)$, a CW complex and so paracompact Hausdorff (theorem 7.1.7), and $c_3(\gamma_2) = 0$ because γ_2 has rank two. So corollary 9.2.12 gives $\mathrm{Sq}^2\rho c_2 = \rho c_1\,\rho c_2$. Since $\kappa_1^*\iota_2 = c_1$ and $\kappa_2^*\iota_4 = c_2$, and κ^* commutes with ρ, with cup products and with Sq^2 (theorem 9.1.17(1)),

$$\kappa^*(\rho\iota_2)^3 = (\rho c_1)^3, \qquad \kappa^*(\rho\iota_2\,\rho\iota_4) = \rho c_1\,\rho c_2, \qquad \kappa^*\mathrm{Sq}^2\rho\iota_4 = \mathrm{Sq}^2\rho c_2 = \rho c_1\,\rho c_2$$

By proposition 9.3.6 a general element of $H^6(T;\mathbb{Z}/2)$ is $u = e_1(\rho\iota_2)^3 + e_2\,\rho\iota_2\,\rho\iota_4 + e_3\,\mathrm{Sq}^2\rho\iota_4$ with $e_1, e_2, e_3 \in \mathbb{Z}/2$, and then $\kappa^*u = e_1(\rho c_1)^3 + (e_2 + e_3)\,\rho c_1\,\rho c_2$. This vanishes exactly when $e_1 = 0$ and $e_2 = e_3$, so the kernel of κ^* is $\{0, \theta\}$.

3. Identify $\pi_5(BU(2))$ with $\mathbb{Z}/2$ (proposition 7.5.1), write $K = K(\mathbb{Z}/2, 6)$ with fundamental class ι_6, and let $*$ be its basepoint. By (1) there is a unique $\theta' \in H^6(T;\mathbb{Z}/2)$ with $\lambda^*\theta' = k_5$, and it remains to show $\theta' = \theta$.

   The class θ' lies in the kernel of κ^*. By (5.2) in the proof of proposition 5.6.10, $p_5^*k_5 = 0$, and $f_4 \simeq p_5 \circ f_5$ by definition 5.6.2(1). Hence

   $$\kappa^*\theta' = f_4^*\lambda^*\theta' = f_4^*k_5 = f_5^*p_5^*k_5 = 0$$

   and $\theta' \in \{0, \theta\}$ by (2).

   The class k_5 is nonzero. The space $BU(2)$ is a CW complex through its Schubert cells (theorem 7.1.17) and is simply connected (proposition 7.5.1), so proposition 5.6.10 gives a based map $g\colon X_4 \to K$ with $g^*\iota_6 = k_5$ and a weak homotopy equivalence $\Psi\colon X_5 \to g^*PK$ over X_4, where

   $$PK = \{\gamma\colon I \to K \,|\, \gamma(0) = *\}, \qquad g^*PK = \{(x, \gamma) \in X_4 \times PK \,|\, \gamma(1) = g(x)\}$$

   Suppose $k_5 = 0$. Then g is based null-homotopic by the based bijection of theorem 5.5.12: there is $H\colon X_4 \times I \to K$ with $H(x, 0) = g(x)$, $H(x, 1) = *$ and $H(x_0, t) = *$ for the basepoint x_0 of X_4. Let $\gamma_x(t) = H(x, t)$, a path from $g(x)$ to $*$. The formula $r(x, \gamma) = \gamma \cdot \gamma_x$, the path γ followed by γ_x, defines a map $r\colon g^*PK \to \Omega K$ into the loops at $*$, continuous by the exponential law [16, The Exponential Law]. The fiber over x_0 is $\{x_0\} \times \Omega K$, with inclusion $\epsilon(\omega) = (x_0, \omega)$, and $r(\epsilon(\omega))$ is ω followed by the constant path, which the reparametrization homotopy, continuous in ω, deforms to ω. So $r \circ \epsilon \simeq \mathrm{id}_{\Omega K}$, and ϵ^*r^* is the identity of $H^*(\Omega K;\mathbb{Z}/2)$.

   By example 8.1.6, $\pi_0(\Omega K)$ corresponds to $\pi_1(K) = 0$ and $\pi_k(\Omega K) \cong \pi_{k+1}(K)$ for $k \geq 1$, so ΩK is path-connected and 4-connected with $\pi_5(\Omega K) \cong \mathbb{Z}/2$. By theorem 5.4.9, $\widetilde{H}_i(\Omega K) = 0$ for $i \leq 4$ and $H_5(\Omega K) \cong \mathbb{Z}/2$, and the universal coefficient theorem gives $H^5(\Omega K;\mathbb{Z}/2) \cong \mathrm{Hom}(\mathbb{Z}/2, \mathbb{Z}/2) \cong \mathbb{Z}/2$. Since ϵ^*r^* is the identity, r^* is injective, so $H^5(g^*PK;\mathbb{Z}/2) \neq 0$, and $H^5(X_5;\mathbb{Z}/2) \neq 0$ because Ψ^* is an isomorphism (proposition 5.3.15).

On the other hand, $f_5\colon BU(2) \to X_5$ is an isomorphism on π_k for $k \leq 5$ by definition 5.6.2(2), and is onto $\pi_6(X_5) = 0$ by definition 5.6.2(3). By lemma 9.3.7 with $n = 6$, $f_5^*\colon H^5(X_5;\mathbb{Z}/2) \to H^5(BU(2);\mathbb{Z}/2)$ is an isomorphism, and its target is zero by (2). This contradiction shows $k_5 \neq 0$. Hence $\theta' \neq 0$, so $\theta' = \theta$ and $k_5 = \lambda^*\theta$, completing the proof.

Part (1) identifies the fourth Postnikov section of $BU(2)$ with T through λ, and part (3) identifies the k-invariant at π_5 with θ. Part (2) identifies θ as the only nonzero class of $H^6(T;\mathbb{Z}/2)$ that pulls back to zero on $BU(2)$. For a rank-two bundle E over a CW complex, classified by $g\colon X \to BU(2)$, the class $(\kappa \circ g)^*\theta$ is $\mathrm{Sq}^2\rho c_2(E) + \rho c_1(E)\,\rho c_2(E)$, and its vanishing is the rank-two case of corollary 9.2.12. Section 9.4 shows that, for a map from a CW complex to X_4, the pullback of k_5 is the complete obstruction to lifting it to X_5, and uses this to classify rank-two bundles on complexes of dimension five.

The two terms of θ can be checked separately. Along $j\colon K(\mathbb{Z},4) \to T$, $y \mapsto (*, y)$, the class $\rho\iota_2$ pulls back to zero, since its pullback factors through the cohomology of a point, so $j^*\theta = \mathrm{Sq}^2\rho\iota_4$. If a rank-two bundle E over a connected CW complex has $c_1(E) = 0$, the first component of the map classifying $(c_1(E), c_2(E))$ is null-homotopic, so that map is homotopic to one factoring through j, and part (2) gives $\mathrm{Sq}^2\rho c_2(E) = 0$. For the bundle underlying the universal $SU(2)$-bundle over $BSU(2)$, whose first Chern class lies in $H^2(BSU(2);\mathbb{Z}) = 0$, this agrees with theorem 9.2.10, by which Sq^2 vanishes on $H^4(BSU(2);\mathbb{Z}/2)$.

Example 9.3.9: The Class θ on $\mathbb{C}\mathrm{P}^3$

Let $h \in H^2(\mathbb{C}\mathrm{P}^3;\mathbb{Z})$ be the generator of theorem 4.5.12, so that $H^*(\mathbb{C}\mathrm{P}^3;\mathbb{Z}) = \mathbb{Z}[h]/(h^4)$, and put $x = \rho h$. By theorem 5.5.12 there is a map $f = (f_1, f_2)\colon \mathbb{C}\mathrm{P}^3 \to T$ with $f_1^*\iota_2 = h$ and $f_2^*\iota_4 = h^2$. Then

$$f^*\theta = \mathrm{Sq}^2(x^2) + x \cdot x^2 = \binom{2}{1}x^3 + x^3 = x^3$$

where $\mathrm{Sq}^2(x^2) = \binom{2}{1}x^3$ is example 9.2.2 restricted to $\mathbb{C}\mathrm{P}^3$. The group $H^7(\mathbb{C}\mathrm{P}^3;\mathbb{Z})$ vanishes, so proposition 9.1.2 makes $\rho\colon H^6(\mathbb{C}\mathrm{P}^3;\mathbb{Z}) \to H^6(\mathbb{C}\mathrm{P}^3;\mathbb{Z}/2)$ onto with kernel $2H^6(\mathbb{C}\mathrm{P}^3;\mathbb{Z}) = 2\mathbb{Z}h^3$, and $x^3 = \rho(h^3)$ generates $H^6(\mathbb{C}\mathrm{P}^3;\mathbb{Z}/2) \cong \mathbb{Z}/2$. Hence $f^*\theta \neq 0$.

Consequently no rank-two complex bundle E on $\mathbb{C}\mathrm{P}^3$ has $c_1(E) = h$ and $c_2(E) = h^2$. Such a bundle would be classified by a map $g\colon \mathbb{C}\mathrm{P}^3 \to BU(2)$ (theorem 7.1.29, valid since the CW complex $\mathbb{C}\mathrm{P}^3$ is paracompact Hausdorff by theorem 7.1.7), the two components of $\kappa \circ g$ would pull ι_2 and ι_4 back to $c_1(E) = h$ and $c_2(E) = h^2$, so $\kappa \circ g \simeq f$ by theorem 5.5.12, and then $f^*\theta = g^*\kappa^*\theta = 0$ by theorem 9.3.8(2), a contradiction. Over the four-dimensional complex $\mathbb{C}\mathrm{P}^2$, every pair of classes is (c_1, c_2) of a rank-two bundle by theorem 7.5.4. The obstruction on $\mathbb{C}\mathrm{P}^3$ lies in $H^6(\mathbb{C}\mathrm{P}^3;\mathbb{Z}/2)$, the degree of the cell that $\mathbb{C}\mathrm{P}^3$ adds to $\mathbb{C}\mathrm{P}^2$.

Exercise 9.3.1

1. Adapt the proof of lemma 9.3.1 to show that the composite $\eta \circ \eta_3\colon S^4 \to S^2$ is not null-homotopic, using $K(\mathbb{Z},2) = \mathbb{C}\mathrm{P}^\infty$ and example 9.2.2 in place of $K(\mathbb{Z},3)$ and proposition 9.2.6. Then use the Hopf fibration of example 5.2.7 and corollary 6.3.7 to show that $\pi_4(S^2) \cong \mathbb{Z}/2$, generated by $\eta \circ \eta_3$.
2. Show that $\pi_5(S^2) \cong \mathbb{Z}/2$, generated by $\eta \circ \eta_3 \circ \eta_4$. Show also that the mod-2 cohomology ring of the mapping cone $C_{\eta\circ\eta_3\circ\eta_4}$, together with its Steenrod squares, is isomorphic to that of $S^2 \vee S^6$, so that no square and no cup product on the mapping cone distinguishes this

composite from the constant map, whose mapping cone is $S^2 \vee S^6$.

3. Let $h \in H^2(\mathbb{C}\mathrm{P}^3;\mathbb{Z})$ be a generator, let $m,n \in \mathbb{Z}$, and let $f\colon \mathbb{C}\mathrm{P}^3 \to T$ satisfy $f^*\iota_2 = mh$ and $f^*\iota_4 = nh^2$. Show that $f^*\theta = mn\,(\rho h)^3$, and deduce that if a rank-two complex bundle E on $\mathbb{C}\mathrm{P}^3$ has $c_1(E) = mh$ and $c_2(E) = nh^2$, then mn is even. Check this condition directly for $E = L \oplus L'$, where L and L' are line bundles with $c_1(L) = ah$ and $c_1(L') = bh$ (such line bundles exist by proposition 7.4.7, and theorem 7.4.5 computes $c(E)$).

4. Show that $H^7(T;\mathbb{Z}/2)$ is one-dimensional, spanned by $\mathrm{Sq}^3\rho\iota_4$, and that $\mathrm{Sq}^1\theta = \mathrm{Sq}^3\rho\iota_4$. Conclude that θ is not the mod-2 reduction of any class in $H^6(T;\mathbb{Z})$.

5. Let $c\colon BSU(2) \to K(\mathbb{Z},4)$ represent the class c_2 of theorem 9.2.10. Show that c is 5-connected. Following the proof of theorem 9.3.8, show that c extends over the fourth Postnikov section of $BSU(2)$ to a map λ' inducing isomorphisms on cohomology, and that the fifth k-invariant of $BSU(2)$ is $\lambda'^*\mathrm{Sq}^2\rho\iota_4$.

9.4 Fiber Sequences and Bundles in Dimensions Five and Six

Theorem 7.5.4 classifies rank-two complex bundles over a CW complex of dimension at most four by the pair (c_1,c_2), and the bound is sharp: $\mathrm{Vect}^2_{\mathbb{C}}(S^5) \cong \pi_5(BU(2)) \cong \mathbb{Z}/2$ by theorem 7.1.29, proposition 5.1.15 and proposition 7.5.1, while $H^2(S^5;\mathbb{Z})$ and $H^4(S^5;\mathbb{Z})$ vanish. Theorem 9.3.8 identified the class responsible. The fifth Postnikov stage of $BU(2)$ is a principal fibration over the fourth stage X_4, the extension $\lambda\colon X_4 \to K(\mathbb{Z},2)\times K(\mathbb{Z},4)$ of (c_1,c_2) is an isomorphism on homotopy groups, and the k-invariant is $k_5 = \lambda^*\theta$ with $\theta = \mathrm{Sq}^2\rho\iota_4 + \rho\iota_2\,\rho\iota_4$, the sum of two of the three basis elements of proposition 9.3.6. This section converts that computation into classifications. Theorem 9.4.3 makes $[X,-]$ exact along a principal fibration: it says which maps into the base lift to the total space, it makes the group $H^n(X;A)$ act transitively on the set of lifts, and it computes the stabilizer of a lift. Applied to k_5 it classifies rank-two bundles over five-complexes (theorem 9.4.5). Applied to the sixth Postnikov stage of $BU(3)$ it classifies rank-three bundles over six-complexes (theorem 9.4.12) and over closed six-manifolds (corollary 9.4.13). Box 9.4.14 records the two published statements these results correct.

Two computations recur throughout. A loop of maps $X \to B$ based at f is a map $X \times S^1 \to B$ restricting to f, so the indeterminacy of a lift lives in the cohomology of $X \times S^1$, and the first lemma fixes coordinates there. The acting group of the lifting theorem is the set of homotopy classes $[X,\Omega K(A,n+1)]$, and the second lemma identifies it with $H^n(X;A)$. Throughout, $S^1 = I/\partial I$ with quotient map $q\colon I \to S^1$ and basepoint $1 := q(0) = q(1)$, and paths are composed by $*$ in the order written, as in definition 2.1.4.

Lemma 9.4.1: Cohomology of the Product with a Circle

Let X be a CW complex, A an abelian group, and $m \geq 2$. Write $i\colon X = X \times \{1\} \hookrightarrow X \times S^1$ for the inclusion and $\mathrm{pr}\colon X \times S^1 \to X$ for the projection.

1. There is an injective homomorphism $\varphi\colon H^{m-1}(X;A) \to H^m(X \times S^1;A)$ with image $\ker i^*$, and
$$H^m(X \times S^1;A) = \mathrm{pr}^* H^m(X;A) \oplus \varphi\big(H^{m-1}(X;A)\big)$$
Accordingly every $u \in H^m(X \times S^1;A)$ is written $u = \mathrm{pr}^* u_0 + u_1\sigma$, where $u_1\sigma := \varphi(u_1)$, and $u_1 \in H^{m-1}(X;A)$ is the *σ-coefficient* of u.
2. The decomposition is natural: for a map $h\colon X' \to X$ of CW complexes, $(h \times \mathrm{id})^* \circ \varphi = \varphi \circ h^*$, and for a homomorphism $A \to A'$ of coefficient groups, φ commutes with the induced coefficient maps.
3. Let Y be a space, let $F, F'\colon X \times S^1 \to Y$ satisfy $F|_{X\times\{1\}} = F'|_{X\times\{1\}}$, and let $F * F'$ denote the circle-wise concatenation, $(F * F')(x, q(t)) = F(x, q(2t))$ for $t \leq \frac{1}{2}$ and $F'(x, q(2t-1))$ for $t \geq \frac{1}{2}$. Then for every $y \in H^m(Y;A)$ the σ-coefficient of $(F * F')^* y$ is the sum of the σ-coefficients of $F^* y$ and of $F'^* y$.
4. For $A = \mathbb{Z}/2$ and X connected, $\varphi(x) = \mathrm{pr}^* x \smile s$, where $s \in H^1(X \times S^1;\mathbb{Z}/2)$ is the pullback of the generator of $H^1(S^1;\mathbb{Z}/2)$ along the projection to S^1.

***Proof*:**

Step 1: A comparison map to the reduced suspension. Give $X_+ = X \sqcup \{+\}$ the basepoint $+$ and let $\Sigma(X_+)$ be its reduced suspension, the quotient of $X_+ \times I$ collapsing $X_+ \times \{0\}$, $X_+ \times \{1\}$ and $\{+\} \times I$ to the basepoint e (definition 5.7.1), with quotient map Q. Define
$$\psi\colon X \times S^1 \longrightarrow \Sigma(X_+), \qquad \psi\big(x, q(t)\big) = Q(x,t)$$
This is well defined, since $Q(x,0) = Q(x,1) = \mathrm{e}$, and it sends $X \times \{1\}$ to e. It is continuous: let $V \subseteq \Sigma(X_+)$ be open and $(x_0, z_0) \in \psi^{-1}(V)$. If $z_0 = q(t_0)$ with $0 < t_0 < 1$, then $Q^{-1}(V)$ is open in $X_+ \times I$ and contains (x_0, t_0), so it contains a basic neighborhood $U \times J$ with $J \subseteq (0,1)$, and $U \times q(J)$ is an open neighborhood of (x_0, z_0) inside $\psi^{-1}(V)$. If $z_0 = 1$, then $V \ni \mathrm{e}$, so the open set $Q^{-1}(V)$ contains $(x_0, 0)$ and $(x_0, 1)$, hence contains $U_0 \times [0, \delta)$ and $U_1 \times (1-\delta, 1]$ for some $\delta > 0$ and open $U_0, U_1 \ni x_0$. Then $(U_0 \cap U_1) \times q\big([0,\delta) \cup (1-\delta, 1]\big)$ is an open neighborhood of $(x_0, 1)$ contained in $\psi^{-1}(V)$.

Step 2: Two Mayer-Vietoris sequences and the definition of φ. Refine the CW structures. Give S^1 the structure with 0-cells $1 = q(0)$ and $w := q(\frac{1}{2})$ and two 1-cells, and give $X \times S^1$ the product structure, a CW complex because the factor S^1 is finite (definition 1.2.8). Set $J_+ := q([0, \frac{1}{2}])$ and $J_- := q([\frac{1}{2}, 1])$, so that $X \times J_+$ and $X \times J_-$ are subcomplexes covering $X \times S^1$ with intersection $X \times \{1\} \sqcup X \times \{w\}$. On the suspension side, $\Sigma(X_+) = C_- \cup C_+$ is the union of the two reduced cones, subcomplexes meeting in the equatorial copy of X_+ (definition 5.7.1), and ψ carries $X \times J_+$ into the image of $X_+ \times [0, \frac{1}{2}]$ and $X \times J_-$ into the image of $X_+ \times [\frac{1}{2}, 1]$, and carries $X \times \{w\}$ to the equatorial copy of X and $X \times \{1\}$ to e.

Reduced singular cohomology is a reduced cohomology theory (example 8.2.7 together with proposition 8.2.5), so both covers have Mayer-Vietoris sequences (proposition 8.2.8, available for these covers since CW pairs are good pairs, lemma 3.4.17), natural for maps respecting the

covers. For the suspension, both cones are contractible, so in every degree the connecting map

$$\delta_\Sigma \colon \widetilde{H}^{m-1}\big(X_+; A\big) \xrightarrow{\cong} \widetilde{H}^{m}\big(\Sigma(X_+); A\big)$$

is an isomorphism, where the source is the reduced cohomology of the equatorial copy. The restriction $\widetilde{H}^{m-1}(X_+; A) \to H^{m-1}(X; A)$ to the summand X is an isomorphism, since $X_+ = X \sqcup \{+\}$ with basepoint $+$. Define

$$\varphi := \psi^* \circ \delta_\Sigma \circ r^{-1} \colon H^{m-1}(X; A) \longrightarrow H^m(X \times S^1; A)$$

where r is that restriction isomorphism, and where ψ^* is pullback on reduced cohomology followed by the identification $\widetilde{H}^m = H^m$ in degrees $m \geq 1$ (proposition 4.1.14(1)).

Step 3: Image and injectivity. Write the Mayer-Vietoris sequence of the cover $X \times S^1 = (X \times J_+) \cup (X \times J_-)$, based at $(x_0, 1)$ for a 0-cell x_0 when $X \neq \varnothing$. Each arc $J_\pm$ deformation retracts to the endpoint 1, sliding along the arc, so the inclusions $X \times \{1\} \hookrightarrow X \times J_\pm$ are homotopy equivalences and $H^k(X \times J_\pm; A) \cong H^k(X; A)$ compatibly with restriction to either boundary copy of X. In degrees $k \geq 1$ reduced and unreduced cohomology agree (proposition 4.1.14(1)), and the sequence reads

$$H^{m-1}(X; A)^{\oplus 2} \xrightarrow{\partial} H^{m-1}(X; A)^{\oplus 2} \xrightarrow{\delta} H^m(X \times S^1; A) \longrightarrow H^m(X; A)^{\oplus 2} \xrightarrow{\partial} H^m(X; A)^{\oplus 2}$$

where the middle target of ∂ is the cohomology of the two intersection copies $X \times \{1\}$ and $X \times \{w\}$ and $\partial(u, v) = (u - v, u - v)$ under the identifications above. The kernel of ∂ in degree m is the diagonal, which is the image of pr^*, and the cokernel of ∂ in degree $m - 1$ maps isomorphically to $H^{m-1}(X; A)$ by $(u, v) \mapsto v - u$. Exactness therefore gives a short exact sequence

$$0 \longrightarrow H^{m-1}(X; A) \xrightarrow{\bar\delta} H^m(X \times S^1; A) \longrightarrow \mathrm{pr}^* H^m(X; A) \longrightarrow 0$$

split by pr^*, where $\bar\delta(x) := \delta(0, x)$ for the class $(0, x)$ carried by the copy $X \times \{w\}$. The image of $\bar\delta$ is the set of classes restricting to zero on both pieces $X \times J_\pm$. A class u with $i^* u = 0$ restricts to zero on each piece, because each piece deformation retracts to $X \times \{1\}$, and conversely a class vanishing on the pieces satisfies $i^* u = 0$ because $X \times \{1\}$ lies in a piece. So $\mathrm{im}\bar\delta = \ker i^*$, and $H^m(X \times S^1; A) = \mathrm{pr}^* H^m(X; A) \oplus \ker i^*$.

It remains to compare $\bar\delta$ with φ. The map ψ respects the two covers, so by naturality of the Mayer-Vietoris sequence the square

$$\begin{array}{ccc} \widetilde{H}^{m-1}(X_+; A) & \xrightarrow{\delta_\Sigma} & \widetilde{H}^{m}(\Sigma(X_+); A) \\ \downarrow{\scriptstyle (\psi|)^*} & & \downarrow{\scriptstyle \psi^*} \\ \widetilde{H}^{m-1}\big(X \times \{1\} \sqcup X \times \{w\}; A\big) & \xrightarrow{\delta} & \widetilde{H}^{m}(X \times S^1; A) \end{array}$$

commutes. The restriction of ψ to the intersection sends $X \times \{1\}$ to the basepoint e and $X \times \{w\}$ to the equatorial copy of X by the identity, so $(\psi|)^*$ carries a class $\tilde x$ with $r(\tilde x) = x$ to the pair $(0, x)$. Hence $\varphi(x) = \psi^* \delta_\Sigma r^{-1}(x) = \delta(0, x) = \bar\delta(x)$, and part (1) follows, injectivity included.

Step 4: Naturality. For $h \colon X' \to X$, the map $h \times \mathrm{id}$ respects the covers and $\Sigma(h_+) \circ \psi' = \psi \circ (h \times \mathrm{id})$, where h_+ extends h by the basepoint, so part (2) for maps follows from naturality of the Mayer-Vietoris connecting map and of r. A coefficient homomorphism $A \to A'$ induces a map of singular cochain complexes commuting with every map in sight, so it commutes with δ_Σ, r and ψ^*, which is part (2) for coefficients.

Step 5: Concatenation. Let $f_1 := F|_{X\times\{1\}} = F'|_{X\times\{1\}}$ and let D be the CW complex obtained from two copies of $X \times S^1$ by gluing the subcomplex $X \times \{1\}$ of the first to the same subcomplex of the second. The two copies are closed in D, and F and F' agree on the gluing locus, so they paste to a continuous map $F \cup F' \colon D \to Y$. Define $c \colon X \times S^1 \to D$ by $c(x, q(t)) = (x, q(2t))$ in the first copy for $t \le \frac{1}{2}$ and $c(x, q(t)) = (x, q(2t-1))$ in the second copy for $t \ge \frac{1}{2}$. On the closed overlap $t = \frac{1}{2}$ both formulas give glued points of $X \times \{1\}$, and at $t \in \{0, 1\}$ both land in the glued $X \times \{1\}$ as well, so c is continuous by pasting, and $(F \cup F') \circ c = F * F'$.

Set $v := (F \cup F')^* y \in H^m(D; A)$ and let $j_1, j_2 \colon X \times S^1 \to D$ be the two copies, so $j_1^* v = F^* y$ and $j_2^* v = F'^* y$. Write $\pi_D \colon D \to X$ for the common projection and $u_X := f_1^* y \in H^m(X; A)$, the common restriction of $F^* y$ and $F'^* y$ to $X \times \{1\}$. Then $v' := v - \pi_D^* u_X$ restricts on the two copies to $\varphi(d_1)$ and $\varphi(d_2)$, where d_1, d_2 are the σ-coefficients of $F^* y$ and $F'^* y$, and $c^* \pi_D^* u_X = \mathrm{pr}^* u_X$, so the claim reduces to showing that the σ-coefficient of $c^* v'$ is $d_1 + d_2$.

The Mayer-Vietoris sequence of the closed cover of D by its two copies (again proposition 8.2.8 and lemma 3.4.17) shows, exactly as in Step 3, that the restriction (j_1^*, j_2^*) is injective in degree m: the map preceding the connecting homomorphism is onto, since pr^* provides preimages. Let $r_1 \colon D \to X \times S^1$ collapse the second copy by $(x, z) \mapsto (x, 1)$ and fix the first, and let r_2 do the opposite. Both are continuous by pasting. The class $r_1^* \varphi(d_1) + r_2^* \varphi(d_2)$ has the same two restrictions as v', using $r_1 \circ j_1 = \mathrm{id}$, $r_1 \circ j_2 = i \circ \mathrm{pr}$ and $i^* \varphi = 0$, so

$$v' = r_1^* \varphi(d_1) + r_2^* \varphi(d_2)$$

The composite $r_1 \circ c \colon X \times S^1 \to X \times S^1$ is $\mathrm{id}_X \times c_1$, where c_1 traverses the lower half-circle at double speed and is constant on the upper half. The maps c_1 and id_{S^1} are homotopic through maps fixing 1, by the reparametrization $h_s(q(t)) = q\big(\min(2t/(1+s), 1)\big)$, so $\mathrm{id}_X \times c_1 \simeq \mathrm{id}$ and $c^* r_1^* \varphi(d_1) = \varphi(d_1)$. The composite $r_2 \circ c = \mathrm{id}_X \times c_2$ with c_2 constant on the lower half is handled by the mirror reparametrization, giving $c^* r_2^* \varphi(d_2) = \varphi(d_2)$. Hence $c^* v' = \varphi(d_1) + \varphi(d_2)$, whose σ-coefficient is $d_1 + d_2$, as required for part (3).

Step 6: The cup-product form for $\mathbb{Z}/2$. Both φ and $x \mapsto \mathrm{pr}^* x \smile s$ are natural in X, the second because $(h \times \mathrm{id})^* s = s$ and pullback is a ring homomorphism. Fix $m \ge 2$ and let $K := K_{m-1}$ be the model of $K(\mathbb{Z}/2, m-1)$ in lemma 9.1.6, with fundamental class $\iota \in H^{m-1}(K; \mathbb{Z}/2)$ (definition 5.5.10). Every class $x \in H^{m-1}(X; \mathbb{Z}/2)$ on a connected CW complex X is $g^* \iota$ for a map $g \colon X \to K$ (theorem 5.5.12, free classes), and naturality gives $\varphi(g^* \iota) = (g \times \mathrm{id})^* \varphi(\iota)$ and $\mathrm{pr}^*(g^* \iota) \smile s = (g \times \mathrm{id})^*(\mathrm{pr}^* \iota \smile s)$. So it is enough to prove $\varphi(\iota) = \mathrm{pr}^* \iota \smile s$ in $H^m(K \times S^1; \mathbb{Z}/2)$.

Both classes lie in $\ker i^*$: for the second, $i^*(\mathrm{pr}^* \iota \smile s) = \iota \smile i^* s = 0$, since s restricts to $K \times \{1\}$ through $H^1(\mathrm{pt}; \mathbb{Z}/2) = 0$. By part (1), $\ker i^* \cong H^{m-1}(K; \mathbb{Z}/2)$ through φ, and $H^{m-1}(K; \mathbb{Z}/2) = \{0, \iota\}$ by lemma 9.1.6(1). So $\ker i^*$ has exactly two elements, 0 and $\varphi(\iota)$, and the claim follows once $\mathrm{pr}^* \iota \smile s \neq 0$.

The $(m-1)$-skeleton of K is S^{m-1}, and ι restricts there to the generator a of $H^{m-1}(S^{m-1}; \mathbb{Z}/2)$ (lemma 9.1.6(3)). On $S^{m-1} \times S^1$, the Künneth theorem with field coefficients and the finite-type factor S^{m-1} (theorem 4.3.9) shows $a \times s_{S^1} \neq 0$ in degree m, and $a \times s_{S^1}$ is the restriction of $\mathrm{pr}^* \iota \smile s$ along $S^{m-1} \times S^1 \hookrightarrow K \times S^1$ by definition 4.3.7. Hence $\mathrm{pr}^* \iota \smile s \neq 0$, so $\varphi(\iota) = \mathrm{pr}^* \iota \smile s$, completing the proof.

The lemma gives a name to the two coordinates of a class on $X \times S^1$: the part pulled back from X, and the σ-coefficient, which records what happens around the circle. The group that acts on lifts is a group of homotopy classes of maps into the loop space of an Eilenberg-MacLane complex, and its identification with cohomology runs through the loop-suspension adjunction and the comparison

map ψ of the proof above.

Lemma 9.4.2: Maps into the Loops on an Eilenberg-MacLane Space

Let A be an abelian group, $n \geq 1$, and $K := K(A, n+1)$ the Eilenberg-MacLane complex of theorem 5.5.4, with basepoint $*$ and fundamental class ι_{n+1}. Let X be a connected CW complex. For a map $W\colon X \to \Omega K$ define the adjoint $\widehat{W}\colon X \times S^1 \to K$ by $\widehat{W}(x, q(t)) = W(x)(t)$.

1. $\widehat{W}$ is continuous, sends $X \times \{1\}$ to $*$, and its class $\widehat{W}^*\iota_{n+1}$ lies in $\ker i^*$, so
$$\Theta[W] := \varphi^{-1}\big(\widehat{W}^*\iota_{n+1}\big) \in H^n(X; A)$$
is defined, with φ as in lemma 9.4.1, and depends only on the free homotopy class of W.
2. The set $[X, \Omega K]$ of free homotopy classes is a group under pointwise concatenation of loops, and $\Theta\colon [X, \Omega K] \to H^n(X; A)$ is a group isomorphism, natural in X.

***Proof*:**

Step 1: Continuity and well-definedness. The evaluation $(x, t) \mapsto W(x)(t)$ is continuous as a map $X \times I \to K$, since I is locally compact Hausdorff and the compact-open topology makes evaluation jointly continuous in such a variable, as in the lifting formulas of example 5.2.15. Continuity of $\widehat{W}$ at points $(x, q(t))$ with $0 < t < 1$ follows because q restricts to a homeomorphism of $(0, 1)$ onto its image. At a point $(x_0, 1)$, given an open $V \ni * = \widehat{W}(x_0, 1)$, continuity of the evaluation at $(x_0, 0)$ and at $(x_0, 1)$ produces $\delta > 0$ and a neighborhood $U \ni x_0$ with $W(x)(t) \in V$ for $x \in U$ and $t \in [0, \delta) \cup (1-\delta, 1]$, and $U \times q\big([0, \delta) \cup (1-\delta, 1]\big)$ is the required neighborhood. A free homotopy $W \simeq W'$ has an adjoint homotopy $\widehat{W} \simeq \widehat{W'}$ by the same two-case argument applied with $X \times I$ in place of X, so $\widehat{W}^*\iota_{n+1}$, and with it $\Theta[W]$, depends only on the class of W. Finally $\widehat{W}(x, 1) = W(x)(0) = *$, so $i^*\widehat{W}^*\iota_{n+1}$ is pulled back along a constant map and vanishes, and $\widehat{W}^*\iota_{n+1} \in \ker i^* = \mathrm{im}\varphi$.

Step 2: The group structure and additivity of Θ. Pointwise concatenation descends to classes: homotopies concatenate. Associativity and the unit and inverse laws hold up to homotopy by the usual uniform reparametrizations, as for a single loop space (proposition 8.1.4), so $[X, \Omega K]$ is a group, with unit the constant map and inverse induced by loop reversal. For $V, W\colon X \to \Omega K$, the adjoint of the pointwise concatenation $V * W$ is the circle-wise concatenation $\widehat{V} * \widehat{W}$ of lemma 9.4.1(3), and both adjoints restrict to the constant map on $X \times \{1\}$, so that clause gives
$$\Theta[V * W] = \Theta[V] + \Theta[W]$$
Naturality in X holds because $\widehat{W \circ h} = \widehat{W} \circ (h \times \mathrm{id})$ and φ is natural.

Step 3: Bijectivity. A map $X \to \Omega K$ is the same thing as a based map $X_+ \to \Omega K$, and free homotopies are based homotopies of X_+, so $[X, \Omega K] = [X_+, \Omega K]_*$ as sets. The loop-suspension adjunction (theorem 8.1.5) is a bijection $[\Sigma(X_+), K]_* \cong [X_+, \Omega K]_*$, and theorem 5.5.12 identifies $[\Sigma(X_+), K]_*$ with $\widetilde{H}^{n+1}(\Sigma(X_+); A)$ by pulling back ι_{n+1}. Here $\Sigma(X_+)$ is a connected based CW complex. Under these bijections, a class $[W]$ corresponds to a based map $g\colon \Sigma(X_+) \to K$ with $\widehat{g} \simeq W$, where $\widehat{g}(x)(t) = g\big(Q(x, t)\big)$ in the notation of the proof of lemma 9.4.1, and then
$$\widehat{W}^*\iota_{n+1} = \big(g \circ \psi\big)^*\iota_{n+1} = \psi^*\big(g^*\iota_{n+1}\big)$$

since the adjoint of $\widehat{g}$ in the sense of part (1) is exactly $g \circ \psi$. By the proof of lemma 9.4.1, $\varphi = \psi^* \circ \delta_\Sigma \circ r^{-1}$ with δ_Σ and r isomorphisms, and $\mathrm{im}\psi^* \supseteq \mathrm{im}\varphi = \ker i^*$ with $\psi^* \circ \delta_\Sigma$ injective, so ψ^* is injective on $\widetilde{H}^{n+1}(\Sigma(X_+); A)$ with image $\ker i^*$. Therefore

$$\Theta[W] = \varphi^{-1}\psi^*\big(g^*\iota_{n+1}\big) = r\,\delta_\Sigma^{-1}\big(g^*\iota_{n+1}\big)$$

is the composite of four bijections applied to $[W]$, so Θ is a bijection, completing the proof.

A principal fibration is the homotopy-theoretic form of a short exact sequence, and the theorem below is the exactness statement for the functor $[X, -]$ applied to it. The lifting criterion in part (1) is the analogue of exactness at the base. Parts (2) and (3) refine exactness at the total space: the fiber of p_* over a liftable class is a single orbit of a cohomology group, and the stabilizer is computed by a cohomology class on $X \times S^1$. Part (3) measures the failure of lifts to be classified by a quotient of $H^n(X; A)$ chosen independently of the map being lifted, and it is exactly the term the published rank-two statement of box 9.4.14 dropped.

Theorem 9.4.3: Lifting Along a Principal Fibration

Let A be an abelian group and $n \geq 1$. Let B be a simply connected CW complex, let $k\colon B \to K(A, n+1)$ be a map to the Eilenberg-MacLane complex $K := K(A, n+1)$ of theorem 5.5.4, and let $p\colon E \to B$ be the homotopy fiber of k:

$$E = \big\{(b, \gamma) \in B \times K^I \;\big|\; \gamma(0) = *,\ \gamma(1) = k(b)\big\}, \qquad p(b, \gamma) = b$$

For a map $f\colon X \to B$ write $f^*k := (k \circ f)^*\iota_{n+1} \in H^{n+1}(X; A)$. Let X be a connected CW complex.

1. A map $f\colon X \to B$ admits a lift $g\colon X \to E$ with $p \circ g = f$ if and only if $f^*k = 0$.
2. The group $H^n(X; A) \cong [X, K(A, n)]$ of theorem 5.5.12 acts on $[X, E]$, through the isomorphism Θ of lemma 9.4.2 and concatenation of paths, so that $p_*\big(d \cdot [g]\big) = p_*[g]$ for all d and $[g]$, and the orbits of the action are exactly the nonempty fibers of $p_*\colon [X, E] \to [X, B]$.
3. Let $f\colon X \to B$ be a map that lifts, with lift g. Write $\pi_1\big(\mathrm{Map}(X, B), f\big)$ for the group of homotopy classes rel $X \times \{1\}$ of maps $F\colon X \times S^1 \to B$ with $F|_{X\times\{1\}} = f$, under circle-wise concatenation. The stabilizer of $[g]$ is the image of the homomorphism

$$\Delta_f\colon \pi_1\big(\mathrm{Map}(X, B), f\big) \longrightarrow H^n(X; A)$$

sending a loop F to the σ-coefficient of $F^*k \in H^{n+1}(X \times S^1; A) = H^{n+1}(X; A) \oplus H^n(X; A)\sigma$ (lemma 9.4.1).

***Proof*:**
Throughout, PK is the path space of K at $*$ and ΩK its loop space (definition 5.2.14). Paths in PK start at $*$, so E is the pullback of the endpoint fibration $PK \to K$ of example 5.2.15 along k. By proposition 5.2.6, p is a Serre fibration whose fiber over b is the space of paths from $*$ to $k(b)$. Concatenation defines a map

$$a\colon \Omega K \times E \longrightarrow E, \qquad a\big(\omega, (b, \gamma)\big) = (b, \omega * \gamma)$$

over B. It is continuous: the evaluation $(\omega, \gamma, t) \mapsto (\omega * \gamma)(t)$ is continuous by pasting over the closed pieces $t \leq \frac{1}{2}$ and $t \geq \frac{1}{2}$ together with joint continuity of evaluation on the locally compact factor I, and passing to the adjoint, which is always continuous into the compact-open topology, gives continuity of a.

1. If $g = (f, \gamma)$ is a lift, then $H(x,t) := \gamma(x)(t)$ is continuous by the evaluation argument above and is a homotopy from the constant map at $*$ to $k \circ f$, so $[k \circ f] = 0$ in $[X, K]$ and $f^*k = 0$ by the free-class form of theorem 5.5.12, X being connected. Conversely, if $f^*k = 0$ then $k \circ f$ is freely nullhomotopic: there is $H \colon X \times I \to K$ with H_0 constant at some point c and $H_1 = k \circ f$. The space K is path-connected, so prepending a path from $*$ to c, constant in x, arranges H_0 constant at $*$. Then $g(x) := \big(f(x), H(x, -)\big)$ is a lift of f, using the adjoint direction of the exponential correspondence for continuity.

2. For $d \in H^n(X; A)$ choose $W \colon X \to \Omega K$ with $\Theta[W] = d$ and set $d \cdot [g] := \big[a \circ (W, g)\big]$, the class of $x \mapsto \big(p(g(x)),\, W(x) * \gamma_g(x)\big)$ where γ_g is the path coordinate of g. This is well defined: Θ is a bijection, so two choices of W are freely homotopic, and homotopies in either slot of a give homotopies of the result. The constant map represents 0 and acts by const $* \gamma \simeq \gamma$, a uniform reparametrization, and for d, d' with representatives W, W' the maps $(W * W') * \gamma$ and $W * (W' * \gamma)$ differ by a uniform reparametrization, while $\Theta[W * W'] = d + d'$ by lemma 9.4.2. So the formula is an action of the group $H^n(X; A)$. The projection is untouched: $p \circ (W \cdot g) = p \circ g$ on the nose, so each orbit lies in one fiber of p_*.

   For transitivity, let $[g_0], [g_1]$ lie in the fiber over the same class, put $f := p \circ g_0$, and let $H \colon X \times I \to B$ be a homotopy from $p \circ g_1$ to f. The path coordinate $\gamma \colon X \to PK$ of g_1 lifts $k \circ H$ at time 0 through the endpoint fibration $PK \to K$, which has the homotopy lifting property with respect to every space (example 5.2.15). So there is $\Gamma \colon X \times I \to PK$ with $\Gamma(x, 0) = \gamma(x)$ and $\Gamma(x, u)(1) = k(H(x, u))$. Then $(x, u) \mapsto \big(H(x, u), \Gamma(x, u)\big)$ is a homotopy in E from g_1 to a lift of f, and replacing g_1 by its end gives $p \circ g_1 = f$ exactly. Write $g_j = (f, \gamma_j)$ and set $W(x) := \gamma_1(x) * \overline{\gamma_0(x)}$, a loop at $*$, continuous by the same pasting and adjoint arguments. Then

$$W(x) * \gamma_0(x) = \gamma_1(x) * \overline{\gamma_0(x)} * \gamma_0(x) \simeq \gamma_1(x)$$

   by the uniform path-cancellation homotopy, rel endpoints in every slot, so $W \cdot g_0 \simeq g_1$ through maps to E over f. Hence $[g_1] = \Theta[W] \cdot [g_0]$, and the orbit is the whole fiber, as claimed.

3. The set $\pi_1\big(\mathrm{Map}(X, B), f\big)$ is a group: circle-wise concatenation of two loops at f is again a loop at f, continuity is the pasting argument of lemma 9.4.1(3), and the group laws hold up to homotopy rel $X \times \{1\}$ by the usual reparametrizations, constant on the X coordinate. The map Δ_f is well defined, since maps homotopic rel $X \times \{1\}$ have equal pullbacks, and it is a homomorphism by lemma 9.4.1(3) applied to $Y = K$ and $y = \iota_{n+1}$, the two loops agreeing with f on $X \times \{1\}$.

   Fix the lift $g = (f, \gamma)$. For a loop F at f write $\tilde{F} \colon X \times I \to B$ for $\tilde{F}(x, t) = F(x, q(t))$ and

$$\lambda(x) := \big(t \mapsto k\big(\tilde{F}(x, t)\big)\big), \qquad \omega_F(x) := \gamma(x) * \lambda(x) * \overline{\gamma(x)}$$

   so $\lambda(x)$ is a loop at $k(f(x))$ and $\omega_F(x)$ a loop at $*$, and $\omega_F \colon X \to \Omega K$ is continuous.

Step 1: $\Delta_f(F) = \Theta[\omega_F]$. For $u \in [0,1]$ let $c_u(x)$ be the path $\gamma(x)$ traversed from time u to time 1, a path from $\gamma(x)(u)$ to $k(f(x))$, and set $\ell_u(x) := c_u(x) * \lambda(x) * \overline{c_u(x)}$, a loop at $\gamma(x)(u)$. The adjoints of the family ℓ_u assemble to a continuous map $X \times S^1 \times I \to K$ by the pasting and basepoint arguments already used, since $\ell_u(x)(0) = \ell_u(x)(1) = \gamma(x)(u)$. At $u = 0$ the loop is $\omega_F(x)$, whose adjoint is $\widehat{\omega_F}$. At $u = 1$ the conjugating path is constant and the adjoint of ℓ_1 is homotopic to $k \circ F$ by a uniform reparametrization. Hence $k \circ F \simeq \widehat{\omega_F}$, so

$$F^*k = \widehat{\omega_F}^{\,*}\iota_{n+1} = \varphi\big(\Theta[\omega_F]\big)$$

by the definition of Θ in lemma 9.4.2. The right side has vanishing pr^*-component, so the σ-coefficient of F^*k is $\Theta[\omega_F]$, which is the claim. In particular $f^*k = i^*F^*k = 0$, consistent with the existence of g.

Step 2: The image of Δ_f *stabilizes* $[g]$. Let F be a loop at f and let $G(x,u) := \big(\tilde{F}(x,u),\ \gamma(x) * \lambda(x)|_{[0,u]}\big)$, where $\lambda(x)|_{[0,u]}$ is the initial segment, reparametrized to $[0,1]$. This is a homotopy of maps into E. At $u = 0$ it is $(f, \gamma * \mathrm{const}) \simeq g$, and at $u = 1$ it is $(f, \gamma * \lambda)$. Also

$$\omega_F \cdot g = \big(f,\ (\gamma * \lambda * \overline{\gamma}) * \gamma\big) \simeq (f, \gamma * \lambda)$$

by path cancellation. Chaining the three homotopies gives $\omega_F \cdot g \simeq g$, so $\Delta_f(F) = \Theta[\omega_F]$ stabilizes $[g]$ by Step 1.

Step 3: The stabilizer lies in the image of Δ_f. Suppose $d \cdot [g] = [g]$, with representative $W := W_d$, so there is a homotopy $G\colon X \times I \to E$ from g to $(f, W * \gamma)$. Let $\tilde{F} := p \circ G$, a homotopy from f to f, and F the corresponding loop at f. Write $\Gamma(x,u)$ for the path coordinate of $G(x,u)$ and $\Phi_x(u,t) := \Gamma(x,u)(t)$, a map of the square, continuous in all three variables. Along $t = 0$ its edge is the constant path at $*$, and along $t = 1$ it is the loop $\lambda(x)$, in the notation of Step 1 for this F. Along $u = 0$ it is the path $\gamma(x)$, and along $u = 1$ it is $W(x) * \gamma(x)$. The two edge routes of a square from one corner to the opposite corner are homotopic rel endpoints, by composing Φ_x with the linear homotopy between the routes inside the convex square I^2, uniformly in x. Reading the routes from the corner $(0,0)$ to $(1,1)$ gives, rel endpoints and uniformly in x,

$$\mathrm{const}_* * \big(W(x) * \gamma(x)\big) \ \simeq\ \gamma(x) * \lambda(x)$$

Concatenating both sides with $\overline{\gamma(x)}$ and cancelling, $W(x) \simeq \gamma(x) * \lambda(x) * \overline{\gamma(x)} = \omega_F(x)$, through loops at $*$ and uniformly in x. So $[W] = [\omega_F]$ in $[X, \Omega K]$ and $d = \Theta[W] = \Theta[\omega_F] = \Delta_f(F)$ by Step 1. Together with Step 2 this identifies the stabilizer of $[g]$ with the image of Δ_f, completing the proof.

The theorem reduces a lifting problem with values in a space to two computations with values in cohomology. Existence of a lift is the vanishing of one class, f^*k. The set of lifts, when nonempty, is the quotient of $H^n(X;A)$ by the subgroup $\mathrm{im}\Delta_f$, and that subgroup is computed by pulling the single class k back along loops at f and reading off σ-coefficients. When k is given by a formula in cohomology operations, as it will be for k_5, the stabilizer becomes a formula too.

Example 9.4.4: The Path Fibration over an Eilenberg-MacLane Space
Take $B = K(A, n+1)$ and $k = \mathrm{id}$, so that $E = PK$ is the path space and p the endpoint fibration of example 5.2.15. The space PK is contractible: $H(\gamma, u)(t) := \gamma\big((1-u)t\big)$ contracts it to the constant path. Part (1) of theorem 9.4.3 says a map $f\colon X \to K$ lifts to PK exactly

when $f^*\iota_{n+1} = 0$, that is, exactly when f is nullhomotopic, which matches the contraction: a lift g of f gives the nullhomotopy $(x,t) \mapsto \gamma_g(x)(t)$ of $k \circ f = f$. Since $[X, PK]$ has one element, part (2) says p_* has exactly one nonempty fiber, the class of nullhomotopic maps, and that the single class is a full orbit of $H^n(X;A)$. Part (3) then forces the stabilizer to be everything, and a direct computation gives $\mathrm{im}\Delta_f = H^n(X;A)$: for f constant and any $d \in H^n(X;A)$, the loop $F := \widehat{W}$ adjoint to a representative W of d restricts to the constant map on $X \times \{1\}$ and has $\Delta_f(F) = \Theta[W] = d$ by the definition of Θ. The two extremes are both realized: for the path fibration every lift class is stabilized by everything, while for a product fibration $B \times K(A,n) \to B$, where k is constant, every loop at f has $F^*k = 0$ and the lifts form a full torsor of $H^n(X;A)$.

The applications feed Postnikov data into theorem 9.4.3. The Postnikov stages that serve as bases are CW complexes, but the Chern classes live on $BU(n)$ and on products of Eilenberg-MacLane spaces, which are compared with the stages through highly connected maps and weak homotopy equivalences. Cohomology classes pass down a highly connected map of simply connected spaces by lemma 9.3.7, so that a Chern class on $BU(3)$ is, in the degrees the map controls, the pullback of a unique class on a Postnikov stage. Homotopy classes of maps out of a CW complex pass along a weak homotopy equivalence (proposition 5.3.16), so that a CW approximation can stand in for its target.

In rank two the base of the principal fibration is the fourth Postnikov stage X_4 of $BU(2)$, the k-invariant is the class $k_5 = \lambda^*\theta$ of theorem 9.3.8, pulled back from $K(\mathbb{Z},2) \times K(\mathbb{Z},4)$, and the acting group on a five-complex is $H^5(X;\mathbb{Z}/2)$. The stabilizer formula of theorem 9.4.3(3) becomes, through the Cartan formula, an explicit subgroup of $H^5(X;\mathbb{Z}/2)$ depending on the mod 2 reductions of the Chern classes.

Theorem 9.4.5: Rank-Two Bundles on Five-Complexes

Let X be a connected CW complex of dimension at most 5.

1. Every pair $(c_1, c_2) \in H^2(X;\mathbb{Z}) \times H^4(X;\mathbb{Z})$ is the pair of Chern classes of a rank-two complex vector bundle over X.
2. For each pair (c_1, c_2), the isomorphism classes of rank-two complex bundles E over X with $c_1(E) = c_1$ and $c_2(E) = c_2$ are in bijection with $H^5(X;\mathbb{Z}/2)/I(c_1,c_2)$, where
$$I(c_1,c_2) = \left\{ \mathrm{Sq}^2\rho b + \rho c_1\, \rho b + \rho a\, \rho c_2 \;\middle|\; a \in H^1(X;\mathbb{Z}),\ b \in H^3(X;\mathbb{Z}) \right\}$$
3. In particular, if $\rho c_1 = \rho c_2 = 0$, then $I(c_1,c_2) = \mathrm{Sq}^2\rho\big(H^3(X;\mathbb{Z})\big)$.

***Proof*:**

Step 1: The principal model. Let $T = K(\mathbb{Z},2) \times K(\mathbb{Z},4)$ with its classes ι_2, ι_4, the class $\theta = \mathrm{Sq}^2\rho\iota_4 + \rho\iota_2\,\rho\iota_4 \in H^6(T;\mathbb{Z}/2)$, the Postnikov tower $\{X_n, f_n, p_n\}$ of $BU(2)$ constructed in theorem 5.6.4, the map $\kappa = (\kappa_1,\kappa_2)\colon BU(2) \to T$ representing (c_1,c_2) and its extension $\lambda\colon X_4 \to T$ be as in theorem 9.3.8. For every connected CW complex W, a map or homotopy into T is a pair of maps or homotopies into the factors, so
$$[W,T] \xrightarrow{\cong} H^2(W;\mathbb{Z}) \times H^4(W;\mathbb{Z}), \qquad [u] \longmapsto \big(u^*\iota_2,\, u^*\iota_4\big)$$
is a bijection by theorem 5.5.12.

The space $BU(2)$ is a connected CW complex through its Schubert cells (theorem 7.1.17), simply

connected by proposition 7.5.1, hence simple (definition 5.1.13, the trivial fundamental group acting trivially). Each stage X_n is a CW complex obtained from $BU(2)$ by attaching cells (theorem 5.6.4), so it is path-connected, and $\pi_1(X_n) \cong \pi_1(BU(2)) = 0$ by definition 5.6.2(2). In particular X_4 is a simply connected CW complex. Identify $\pi_5(BU(2))$ with $\mathbb{Z}/2$ (proposition 7.5.1) and write $K := K(\mathbb{Z}/2, 6)$, with fundamental class ι_6. By proposition 5.6.10 and definition 5.6.11 there are a based map $k_5 \colon X_4 \to K$ whose class $k_5^*\iota_6$ is the k-invariant k_5, and a weak homotopy equivalence over X_4 from X_5 to the pullback k_5^*PK, whose points are pairs (x, γ) with γ a path from the basepoint to $k_5(x)$. By theorem 9.3.8(3), $k_5 = \lambda^*\theta$.

Apply theorem 9.4.3 with $B = X_4$, $A = \mathbb{Z}/2$, $n = 5$ and $k = k_5$, and let $p \colon E \to X_4$ be the homotopy fiber of k_5 in the sense of that theorem. It is the pullback k_5^*PK, so there is a weak homotopy equivalence $e \colon X_5 \to E$ with $p \circ e = p_5$. For every map f from a CW complex to X_4,

$$f^*k_5 = (k_5 \circ f)^*\iota_6 = f^*\lambda^*\theta = (\lambda \circ f)^*\theta$$

Finally, the section $f_5 \colon BU(2) \to X_5$ induces isomorphisms on π_i for $i \leq 5$, and $\pi_6(X_5) = 0$ (definition 5.6.2), so f_5 is 6-connected (definition 5.3.13).

Step 2: The chain of bijections. Let X be as in the statement. The space X is paracompact Hausdorff (theorem 7.1.7, theorem 1.2.3), so theorem 7.1.29 gives $\mathrm{Vect}^2_{\mathbb{C}}(X) \cong [X, BU(2)]$ by $[h] \mapsto h^*\gamma_2$. Composition with f_5 is a bijection $[X, BU(2)] \to [X, X_5]$ by lemma 5.3.14 with $n = 6$, since f_5 is 6-connected, source and target are simply connected, and $\dim X \leq 5 = n - 1$. Composition with the weak homotopy equivalence e is a bijection $[X, X_5] \to [X, E]$ by proposition 5.3.16. Together these give

$$\mathrm{Vect}^2_{\mathbb{C}}(X) \;\cong\; [X, E], \qquad h^*\gamma_2 \longmapsto [e \circ f_5 \circ h]$$

On the base side, T is path-connected and simply connected by example 5.1.7 and definition 5.5.1, and so is X_4 by Step 1. For such spaces proposition 5.1.15 identifies π_k at every basepoint with the set $[S^k, -]$ of free homotopy classes, compatibly with λ. Since λ_* is bijective on π_k at the basepoints in $BU(2)$ (theorem 9.3.8(1)), it is bijective at every basepoint, and it is bijective on path components, both spaces being path-connected. So λ is a weak homotopy equivalence (definition 5.3.1), and by proposition 5.3.16 and Step 1,

$$[X, X_4] \xrightarrow{\lambda_*} [X, T] \xrightarrow{\cong} H^2(X; \mathbb{Z}) \times H^4(X; \mathbb{Z})$$

is a bijection. Under these bijections $p_* \colon [X, E] \to [X, X_4]$ computes the Chern classes. For a bundle with classifying map h,

$$\lambda \circ p \circ e \circ f_5 \circ h = \lambda \circ p_5 \circ f_5 \circ h \simeq \lambda \circ f_4 \circ h = \kappa \circ h$$

by the tower triangle (definition 5.6.2(1)) and $\lambda \circ f_4 = \kappa$, so the pair of coordinates is $\big(h^*\kappa_1^*\iota_2,\, h^*\kappa_2^*\iota_4\big) = \big(c_1(h^*\gamma_2), c_2(h^*\gamma_2)\big)$ by definition 7.2.7.

Step 3: Existence. Fix (c_1, c_2) and let $f \colon X \to X_4$ represent the corresponding class, so that $(\lambda \circ f)^*\iota_2 = c_1$ and $(\lambda \circ f)^*\iota_4 = c_2$. By Step 1 the lifting obstruction of theorem 9.4.3(1) is

$$f^*k_5 = (\lambda \circ f)^*\theta = \mathrm{Sq}^2\rho\big((\lambda \circ f)^*\iota_4\big) + \rho\big((\lambda \circ f)^*\iota_2\big)\,\rho\big((\lambda \circ f)^*\iota_4\big) = \mathrm{Sq}^2\rho c_2 + \rho c_1\,\rho c_2$$

by naturality of the squares (theorem 9.1.17(1)), of ρ and of the cup product. This class lies in $H^6(X; \mathbb{Z}/2)$, which vanishes because the cellular cochain groups of X vanish above degree 5 (proposition 4.1.20). So f lifts to E, the fiber of p_* over $[f]$ is nonempty, and part (1) follows through Step 2.

Step 4: Orbit and stabilizer. By theorem 9.4.3(2) the fiber over $[f]$ is a single orbit of $H^5(X;\mathbb{Z}/2)$, and by part (3) of that theorem the stabilizer of the class of a lift of f is $\mathrm{im}\Delta_f$. For an abelian group action the stabilizer is constant along an orbit, and $d \mapsto d \cdot \xi$ descends to a bijection $H^5(X;\mathbb{Z}/2)/\mathrm{im}\Delta_f \cong$ orbit for any point ξ of it. So part (2) reduces to the equality $\mathrm{im}\Delta_f = I(c_1, c_2)$.

Step 5: The image of Δ_f. Let $F\colon X \times S^1 \to X_4$ be a loop at f. By Step 1, $\Delta_f(F)$ is the σ-coefficient of $F^*k_5 = (\lambda \circ F)^*\theta$. Set $\alpha := (\lambda \circ F)^*\iota_2$ and $\beta := (\lambda \circ F)^*\iota_4$ and decompose, by lemma 9.4.1(1),

$$\alpha = \mathrm{pr}^*c_1 + \varphi(a), \qquad \beta = \mathrm{pr}^*c_2 + \varphi(b), \qquad a \in H^1(X;\mathbb{Z}),\ b \in H^3(X;\mathbb{Z})$$

where the pr^*-components are c_1, c_2 because $i^*\alpha = (\lambda \circ f)^*\iota_2$ and $i^*\beta = (\lambda \circ f)^*\iota_4$. Reducing mod 2 and applying parts (2) and (4) of lemma 9.4.1,

$$\rho\alpha = \mathrm{pr}^*\rho c_1 + \mathrm{pr}^*(\rho a) \smile s, \qquad \rho\beta = \mathrm{pr}^*\rho c_2 + \mathrm{pr}^*(\rho b) \smile s$$

Then $(\lambda \circ F)^*\theta = \mathrm{Sq}^2(\rho\beta) + \rho\alpha \smile \rho\beta$. The Cartan formula (theorem 9.1.18) gives

$$\mathrm{Sq}^2\big(\mathrm{pr}^*(\rho b) \smile s\big) = \mathrm{pr}^*(\mathrm{Sq}^2\rho b) \smile s + \mathrm{pr}^*(\mathrm{Sq}^1\rho b) \smile \mathrm{Sq}^1 s + \mathrm{pr}^*(\rho b) \smile \mathrm{Sq}^2 s$$

Here $\mathrm{Sq}^1 s = s \smile s$ by theorem 9.1.17(5) in degree one, and $s \smile s$ is pulled back from $H^2(S^1;\mathbb{Z}/2) = 0$, while $\mathrm{Sq}^2 s = 0$ by theorem 9.1.17(4). So

$$\mathrm{Sq}^2(\rho\beta) = \mathrm{pr}^*\big(\mathrm{Sq}^2\rho c_2\big) + \mathrm{pr}^*\big(\mathrm{Sq}^2\rho b\big) \smile s$$

For the product, expand and use $s \smile s = 0$ together with graded commutativity (theorem 4.2.6), whose signs are $+1$ in the degrees involved:

$$\rho\alpha \smile \rho\beta = \mathrm{pr}^*\big(\rho c_1\, \rho c_2\big) + \mathrm{pr}^*\big(\rho c_1\, \rho b + \rho a\, \rho c_2\big) \smile s$$

Adding, $(\lambda \circ F)^*\theta = \mathrm{pr}^*\big(f^*k_5\big) + \mathrm{pr}^*\big(\mathrm{Sq}^2\rho b + \rho c_1\, \rho b + \rho a\, \rho c_2\big) \smile s$, so

$$\Delta_f(F) = \mathrm{Sq}^2\rho b + \rho c_1\, \rho b + \rho a\, \rho c_2 \in I(c_1, c_2)$$

Conversely, let $a \in H^1(X;\mathbb{Z})$ and $b \in H^3(X;\mathbb{Z})$ be given. The product $X \times S^1$ is a connected CW complex (definition 1.2.8, the factor S^1 finite), so by Step 1 there is a map $G\colon X \times S^1 \to T$ with $G^*\iota_2 = \mathrm{pr}^*c_1 + \varphi(a)$ and $G^*\iota_4 = \mathrm{pr}^*c_2 + \varphi(b)$. Its restriction to $X \times \{1\}$ has classes (c_1, c_2), since $i^*\mathrm{pr}^* = \mathrm{id}$ and $i^*\varphi = 0$, so it is homotopic to $\lambda \circ f$ by Step 1. By proposition 5.3.16 there is $F''\colon X \times S^1 \to X_4$ with $\lambda \circ F'' \simeq G$. Restricting, $\lambda \circ (F''|_{X\times\{1\}}) \simeq \lambda \circ f$, so $F''|_{X\times\{1\}} \simeq f$ by the injectivity half of proposition 5.3.16. The pair $(X \times S^1, X \times \{1\})$ has the homotopy extension property (proposition 1.3.4), so F'' is homotopic to a loop F at f. Then $\lambda \circ F \simeq G$, and $\Delta_f(F)$ is the σ-coefficient of $(\lambda \circ F)^*\theta = G^*\theta$. By the computation above, now with α, β the classes of G, it is $\mathrm{Sq}^2\rho b + \rho c_1\, \rho b + \rho a\, \rho c_2$. So $\mathrm{im}\Delta_f = I(c_1, c_2)$, proving part (2).

Step 6: The special case. If $\rho c_1 = \rho c_2 = 0$, the two cup terms of $I(c_1, c_2)$ vanish and $I(c_1, c_2) = \big\{\mathrm{Sq}^2\rho b \;\big|\; b \in H^3(X;\mathbb{Z})\big\}$, proving part (3) and completing the proof.

The group $H^5(X;\mathbb{Z}/2)$ is the universal count: it is the set of lifts over a fixed (c_1, c_2) before any identifications. The subgroup $I(c_1, c_2)$ is the monodromy, the identifications produced by moving a map into the fourth Postnikov stage X_4 around loops of maps, whose classes λ reads off in

$K(\mathbb{Z},2)\times K(\mathbb{Z},4)$, and the theorem says those are the only identifications. The dependence of $I(c_1,c_2)$ on the Chern classes themselves, through the terms $\rho c_1\,\rho b$ and $\rho a\,\rho c_2$, is the feature that resists any count by a quotient of $H^5(X;\mathbb{Z}/2)$ fixed in advance.

Two checks calibrate the statement. On $X=S^5$ the groups H^1, H^2, H^3 and H^4 all vanish, so the only pair is $(0,0)$ and $I(0,0)=0$, and the theorem counts $H^5(S^5;\mathbb{Z}/2)\cong\mathbb{Z}/2$, two bundles: the count $\mathrm{Vect}^2_{\mathbb{C}}(S^5)\cong\pi_5(BU(2))\cong\mathbb{Z}/2$ of proposition 7.5.1 and proposition 5.1.15, recovered with its two elements now labelled by $H^5(S^5;\mathbb{Z}/2)$. The second check is the space on which the published description of the lifts fails.

Example 9.4.6: Rank-Two Bundles on $S^1\times S^4$

Let $X=S^1\times S^4$, a connected CW complex of dimension 5 with the product cell structure (definition 1.2.8). Write $s_1\in H^1(X;\mathbb{Z})$ and $s_4\in H^4(X;\mathbb{Z})$ for the pullbacks of generators along the projections. The Künneth theorem (theorem 4.3.9, the factor S^1 of finite type, all groups of S^1 free) gives

$$H^*(X;\mathbb{Z}):\quad H^0=\mathbb{Z},\quad H^1=\mathbb{Z}s_1,\quad H^4=\mathbb{Z}s_4,\quad H^5=\mathbb{Z}\,s_1s_4$$

with $H^2=H^3=0$, and likewise $H^5(X;\mathbb{Z}/2)\cong\mathbb{Z}/2$ generated by $\rho(s_1s_4)=\rho s_1\,\rho s_4$. Since $H^2(X;\mathbb{Z})=0$, every rank-two bundle has $c_1=0$, and theorem 9.4.5(1) realizes every $c_2=\mu s_4$. Take $(c_1,c_2)=(0,s_4)$, realized for instance by pulling back a bundle on S^4 with second Chern number one (example 7.5.5) along the projection. The indeterminacy subgroup is

$$I(0,s_4)=\left\{\mathrm{Sq}^2\rho b+\rho a\,\rho s_4\;\middle|\;a\in H^1(X;\mathbb{Z}),\ b\in H^3(X;\mathbb{Z})\right\}=\left\{\rho a\,\rho s_4\;\middle|\;a\in H^1(X;\mathbb{Z})\right\}$$

since $H^3(X;\mathbb{Z})=0$. Taking $a=s_1$ gives the generator $\rho s_1\,\rho s_4$ of $H^5(X;\mathbb{Z}/2)$, so $I(0,s_4)=H^5(X;\mathbb{Z}/2)$ and the quotient is trivial: there is exactly *one* rank-two bundle on $S^1\times S^4$ with $(c_1,c_2)=(0,s_4)$. A count of the lifts by $H^5(X;\mathbb{Z}/2)/\mathrm{Sq}^2\rho H^3(X;\mathbb{Z})\cong\mathbb{Z}/2$ would predict two. The term $\rho a\,\rho c_2$, present exactly because $\rho c_2\neq0$, collapses the two lifts into one isomorphism class. For even multiples the collapse disappears: for $(c_1,c_2)=(0,2\mu' s_4)$ the reduction ρc_2 vanishes, $I=0$, and there are exactly two bundles.

The rank-three classification runs along the same track one stage higher: the acting group becomes $H^6(X;\mathbb{Z})$, through a fiber $K(\mathbb{Z},6)$, and the stabilizer computation is replaced by a computation of how c_3 varies along an orbit. Pinning down that variation requires explicit bundles on $\mathbb{C}\mathrm{P}^3$, assembled from the tautological line bundle by tensor powers and duals. Duals of bundles of higher rank have not yet appeared in this book, so the two constructions get definitions first.

Definition 9.4.7: Dual Bundle

Let $E\to X$ be a complex vector bundle. The *dual bundle* $E^*\to X$ has fiber $(E^*)_x=\mathrm{Hom}_{\mathbb{C}}(E_x,\mathbb{C})$, topologized so that a local trivialization $E|_U\cong U\times\mathbb{C}^n$ induces the trivialization $E^*|_U\cong U\times(\mathbb{C}^n)^*$ by dualizing fiberwise. The transition functions of E^* are the inverse transposes of those of E.

Dualizing is compatible with the constructions already in play: there are natural isomorphisms $(E\oplus F)^*\cong E^*\oplus F^*$, given fiberwise by restriction of functionals to the summands, and $(\underline{\mathbb{C}}^n)^*\cong\underline{\mathbb{C}}^n$, given by the dual of the standard basis, both checked on the dualized trivializations. For a line bundle L the evaluation pairing trivializes $L\otimes L^*$, and proposition 7.4.7(1) records the consequence $c_1(L^*)=-c_1(L)$.

Definition 9.4.8: The Line Bundles $\mathcal{O}(k)$ on Projective Space

On $\mathbb{C}\mathrm{P}^m$ let $\mathcal{O}(-1) := \gamma^1$ be the tautological line bundle of example 7.4.10 and $\mathcal{O}(1) := \mathcal{O}(-1)^*$. For $k \geq 1$ set $\mathcal{O}(k) := \mathcal{O}(1)^{\otimes k}$ and $\mathcal{O}(-k) := \mathcal{O}(-1)^{\otimes k}$, the tensor powers in the sense of proposition 7.4.7, and set $\mathcal{O} := \mathcal{O}(0) := \underline{\mathbb{C}}$.

Write $h := c_1(\mathcal{O}(1)) \in H^2(\mathbb{C}\mathrm{P}^m;\mathbb{Z})$. By example 7.4.10 and proposition 7.4.7(1), $h = -c_1(\gamma^1)$ is a generator, and $H^*(\mathbb{C}\mathrm{P}^m;\mathbb{Z}) = \mathbb{Z}[h]/(h^{m+1})$ by theorem 4.5.12. Additivity of c_1 on tensor products gives $c_1(\mathcal{O}(k)) = kh$ for every $k \in \mathbb{Z}$, and since c_1 is an isomorphism from line bundles over the CW complex $\mathbb{C}\mathrm{P}^m$ to H^2 (proposition 7.4.7(2)), the bundle $\mathcal{O}(k)$ is the unique line bundle with first Chern class kh, and $\mathcal{O}(j) \otimes \mathcal{O}(k) \cong \mathcal{O}(j+k)$ and $\mathcal{O}(k)^* \cong \mathcal{O}(-k)$.

Example 9.4.9: Chern Classes of Rank-Three Bundles on $\mathbb{C}\mathrm{P}^3$

Computations on $\mathbb{C}\mathrm{P}^3$, with $H^*(\mathbb{C}\mathrm{P}^3;\mathbb{Z}) = \mathbb{Z}[h]/(h^4)$ as above. All three feed the results below: the first two are the test cases for the parity constraint on c_3, and the third produces the bundle that calibrates the orbit computation of lemma 9.4.11.

1. *A sum of hyperplane bundles.* For $E = \mathcal{O}(1)^{\oplus 3}$, example 7.4.6 gives $c(E) = (1+h)^3 = 1 + 3h + 3h^2 + h^3$, so
$$c_1c_2 = 9h^3, \qquad c_3 = h^3$$
both *odd* multiples of the generator h^3. This is consistent with corollary 9.2.12: here $\rho c_2 = \rho(3h^2) = (\rho h)^2$, and the Cartan formula (theorem 9.1.18) with the axioms of theorem 9.1.17 gives
$$\mathrm{Sq}^2\big((\rho h)^2\big) = \mathrm{Sq}^2\rho h \smile \rho h + \mathrm{Sq}^1\rho h \smile \mathrm{Sq}^1\rho h + \rho h \smile \mathrm{Sq}^2\rho h = 2(\rho h)^3 + \big(\mathrm{Sq}^1\rho h\big)^2 = 0$$
since $\mathrm{Sq}^2\rho h = (\rho h)^2$ in the top degree of ρh, the ambient coefficients are $\mathbb{Z}/2$, and $\mathrm{Sq}^1\rho h = \rho\beta\rho h = 0$ because $\mathrm{Sq}^1 = \rho\beta$ (theorem 9.1.17(7)) and $\beta\rho = 0$ (proposition 9.1.2). So $\mathrm{Sq}^2\rho c_2 + \rho(c_1c_2) = 0 + (\rho h)^3 = \rho c_3$, the identity the corollary asserts for every bundle.

2. *The tangent bundle.* Let T be the complex tangent bundle of $\mathbb{C}\mathrm{P}^3$, as in the $\mathbb{C}\mathrm{P}^2$ computation of example 7.6.9. The total Chern class is $c(T) = (1+h)^4$, proved in [29, Theorem 14.10] through the isomorphism $T \oplus \underline{\mathbb{C}} \cong \mathrm{Hom}_{\mathbb{C}}(\gamma^1, \underline{\mathbb{C}}^4) \cong \mathcal{O}(1)^{\oplus 4}$, the complex analogue of the real splitting recalled in example 7.4.20, together with the Whitney sum formula (theorem 7.4.5). So $c(T) = 1 + 4h + 6h^2 + 4h^3$, and
$$c_1c_2 = 24h^3, \qquad c_3 = 4h^3$$
both *even*. Correspondingly $w_2(\mathbb{C}\mathrm{P}^3) = \rho c_1(T) = \rho(4h) = 0$ by proposition 7.4.14, whose coefficient map ρ_2 is this chapter's ρ, and the constraint of corollary 9.2.12 is the vacuous $0 = 0$: $\rho c_2 = \rho(6h^2) = 0$ and $\rho(c_1c_2) = 0$ and $\rho c_3 = 0$.

3. *A rank-three bundle with total class* $1 + 2h^3$. First the arithmetic, in $\mathbb{Z}[h]/(h^4)$. The inverse of $(1+2h)^3 = 1 + 6h + 12h^2 + 8h^3$ is $1 - 6h + 24h^2 - 80h^3$, as multiplying out confirms: the coefficients of h, h^2, h^3 in the product are $6-6 = 0$, $12-36+24 = 0$ and $8-72+144-80 = 0$. Also $(1+h)^3(1+3h) = 1 + 6h + 12h^2 + 10h^3$. Multiplying,
$$(1+h)^3(1+3h)(1+2h)^{-3} = 1 + (6-6)h + (12-36+24)h^2 + (10-72+144-80)h^3 = 1 + 2h^3$$

Now the realization. Inside $\mathbb{C}\mathrm{P}^3 \times (\mathbb{C}^4 \otimes \mathbb{C}^4) = \underline{\mathbb{C}}^{16}$ sits the subspace $\{(\ell, z) \mid z \in \ell \otimes \ell\}$. Over an open set carrying a nowhere-zero local section ς of $\mathcal{O}(-1)$, the assignment $(\ell, \lambda) \mapsto (\ell, \lambda\,\varsigma(\ell) \otimes \varsigma(\ell))$ is a fiberwise-linear homeomorphism from the trivial line bundle, so the subspace is a line subbundle, and the same trivializations exhibit it as a model of $\mathcal{O}(-1) \otimes \mathcal{O}(-1) = \mathcal{O}(-2)$. By proposition 7.1.23, over the compact Hausdorff and hence paracompact base $\mathbb{C}\mathrm{P}^3$,

$$\underline{\mathbb{C}}^{16} \cong \mathcal{O}(-2) \oplus Q$$

for a rank-15 bundle Q. Dualizing and using $\mathcal{O}(-2)^* \cong \mathcal{O}(2)$ and $(\underline{\mathbb{C}}^{16})^* \cong \underline{\mathbb{C}}^{16}$ gives $\underline{\mathbb{C}}^{16} \cong \mathcal{O}(2) \oplus Q^*$, so by theorem 7.4.5 and the invertibility of total classes (definition 7.2.7), $c(Q^*) = (1 + 2h)^{-1}$. The rank-49 bundle

$$F := \mathcal{O}(1)^{\oplus 3} \oplus \mathcal{O}(3) \oplus (Q^*)^{\oplus 3}$$

then has $c(F) = (1+h)^3(1+3h)(1+2h)^{-3} = 1+2h^3$, so $c_1(F) = c_2(F) = 0$ and $c_3(F) = 2h^3$.

Finally, descend to rank three. Write $i_{n+1}\colon BU(n) \to BU(n+1)$ for the stabilization of proposition 7.1.19. By lemma 7.2.4 it is a homotopy equivalence $BU(n) \to S(\gamma_{n+1})$ followed by the sphere bundle $S(\gamma_{n+1}) \to BU(n+1)$, a fibration with fiber S^{2n+1}. The long exact sequence of that fibration (theorem 5.2.12) shows that i_{n+1} induces isomorphisms on π_k for $k \le 2n$ and a surjection in degree $2n+1$, so i_{n+1} is $(2n+1)$-connected, and also, starting from proposition 7.5.1, that every $BU(n)$ with $n \ge 2$ is simply connected. The composite $\jmath\colon BU(3) \to BU(49)$ of the stabilizations is therefore 7-connected: the first map is 7-connected and the later ones are isomorphisms on π_k for $k \le 7$. By lemma 5.3.14, composition with $\jmath$ is a bijection $[\mathbb{C}\mathrm{P}^3, BU(3)] \to [\mathbb{C}\mathrm{P}^3, BU(49)]$, since $\dim \mathbb{C}\mathrm{P}^3 = 6$. Let $h_F\colon \mathbb{C}\mathrm{P}^3 \to BU(49)$ classify F (theorem 7.1.29) and let $h_3\colon \mathbb{C}\mathrm{P}^3 \to BU(3)$ satisfy $\jmath \circ h_3 \simeq h_F$. The generators of theorem 7.2.5 satisfy $\jmath^* c_i = c_i$ for $i \le 3$, so the rank-three bundle $F_3 := h_3^* \gamma_3$ has

$$c_i(F_3) = h_3^* c_i = h_3^* \jmath^* c_i = h_F^* c_i = c_i(F) \qquad (i = 1, 2, 3)$$

that is, $\big(c_1(F_3), c_2(F_3), c_3(F_3)\big) = (0, 0, 2h^3)$.

The rank-three analogue of the model built in theorem 9.4.5 replaces $BU(2)$ by $BU(3)$ and the fiber $K(\mathbb{Z}/2, 5)$ by $K(\mathbb{Z}, 6)$. The k-invariant of the relevant stage will never be computed, and it does not need to be: on a six-complex the lifting obstruction lives in a vanishing group. What the classification does need is the effect of the $H^6(X; \mathbb{Z})$-action on the third Chern class, and that effect is governed by a single integer slope m.

Lemma 9.4.10: The Sixth Postnikov Stage of $BU(3)$

1. $\pi_5(BU(3)) = 0$ and $\pi_6(BU(3)) \cong \mathbb{Z}$, and $BU(3)$ is simply connected.
2. There are a simply connected CW complex B_5, a map $\kappa_6 \colon B_5 \to K(\mathbb{Z}, 7)$ with homotopy fiber $p \colon E_6 \to B_5$ as in theorem 9.4.3, and classes $v_2 \in H^2(B_5; \mathbb{Z})$, $v_4 \in H^4(B_5; \mathbb{Z})$, $\bar{c}_3 \in H^6(E_6; \mathbb{Z})$, such that for every connected CW complex X with $\dim X \leq 6$ there is a bijection $\Phi_X \colon \mathrm{Vect}^3_{\mathbb{C}}(X) \to [X, E_6]$ with
$$c_1(E) = f^* v_2, \qquad c_2(E) = f^* v_4, \qquad c_3(E) = \Phi_X(E)^* \bar{c}_3, \qquad \text{where } f := p \circ \Phi_X(E)$$
3. For every connected CW complex W, the map $[W, B_5] \to H^2(W; \mathbb{Z}) \times H^4(W; \mathbb{Z})$, $[u] \mapsto (u^* v_2, u^* v_4)$, is a bijection.
4. There is an integer m with the following property: for every connected CW complex X with $\dim X \leq 6$, every $g \in [X, E_6]$ and every $t \in H^6(X; \mathbb{Z})$,
$$(t \cdot g)^* \bar{c}_3 = g^* \bar{c}_3 + m\, t$$
where $\cdot$ is the action of $H^6(X; \mathbb{Z})$ on $[X, E_6]$ from theorem 9.4.3(2).

***Proof*:**

Step 1: The homotopy groups of $BU(3)$. By Step 1 of the proof of theorem 9.3.3, $\pi_5(BU(3)) = 0$ and $\pi_6(BU(3)) \cong \mathbb{Z}$, and $BU(3)$ is simply connected by example 9.4.9(3). This proves part (1).

Step 2: A CW model and its tower. Let $w \colon Y \to BU(3)$ be a CW approximation (theorem 5.3.10), with Y connected since $BU(3)$ is path-connected. Then $\pi_1(Y) \cong \pi_1(BU(3)) = 0$, so Y is a simply connected, hence simple, connected CW complex. Set $c_i^Y := w^* c_i$ for $i = 1, 2, 3$, with $c_i \in H^{2i}(BU(3); \mathbb{Z})$ the generators of theorem 7.2.5. For every CW complex W, proposition 5.3.16 and theorem 7.1.29 give bijections
$$\mathrm{Vect}^3_{\mathbb{C}}(W) \;\cong\; [W, BU(3)] \;\cong\; [W, Y]$$
under which a bundle of class $[u] \in [W, Y]$ has $c_i = u^* c_i^Y$.

Let $\{Y_n\}$ be the Postnikov tower of Y constructed in theorem 5.6.4, with CW stages, sections $f_n \colon Y \to Y_n$ and tower maps p_n. By part (1), $\pi_5(Y) = 0$ and $\pi_6(Y) \cong \mathbb{Z}$. Fix such an isomorphism. By definition 5.6.2, f_n is an isomorphism on π_i for $i \leq n$ while $\pi_i(Y_n) = 0$ for $i > n$, so f_5 is 6-connected and f_6 is 7-connected (definition 5.3.13: the surjectivity in the top degree is onto a zero group), and Y_5, Y_6 are simply connected. Set $B_5 := Y_5$. By proposition 5.6.5, proposition 5.6.10 and definition 5.6.11 there are a map $\kappa_6 \colon B_5 \to K(\mathbb{Z}, 7)$ and a weak homotopy equivalence over B_5 from Y_6 to $\kappa_6^* PK(\mathbb{Z}, 7)$. Let $p \colon E_6 \to B_5$ be the homotopy fiber of κ_6 in the sense of theorem 9.4.3, which is the pullback $\kappa_6^* PK(\mathbb{Z}, 7)$. So there is a weak homotopy equivalence $\bar{e} \colon Y_6 \to E_6$ over B_5. The composite $\varrho := \bar{e} \circ f_6 \colon Y \to E_6$ is 7-connected, as a 7-connected map followed by a weak homotopy equivalence. The space E_6 is path-connected and simply connected, since Y_6 is and $\bar{e}$ is a bijection on path components and an isomorphism on π_1 at every basepoint (definition 5.3.1).

By lemma 9.3.7 applied to f_5 (with $n = 6$) and to ϱ (with $n = 7$), there are unique classes v_2, v_4 on B_5 with $f_5^* v_i = c_i^Y$ and a unique class $\bar{c}_3 \in H^6(E_6; \mathbb{Z})$ with $\varrho^* \bar{c}_3 = c_3^Y$. For X with

$\dim X \le 6$, composition with ϱ is a bijection $[X,Y] \to [X,E_6]$ (lemma 5.3.14, $n=7$), and Φ_X is defined as the composite of the three bijections. For a bundle E of class $[u] \in [X,Y]$, the third Chern class is $u^*c_3^Y = u^*\varrho^*\bar{c}_3 = \Phi_X(E)^*\bar{c}_3$, and $p \circ \varrho = p \circ \bar{e} \circ f_6 \simeq p'_6 \circ f_6 \simeq f_5$ with p'_6 the tower map, since $\bar{e}$ is over B_5 and the tower triangle commutes (definition 5.6.2(1)). So $f = p \circ \Phi_X(E) \simeq f_5 \circ u$ and $f^*v_i = u^*c_i^Y = c_i(E)$ for $i = 1,2$. This proves part (2).

Step 3: The base. Let $v\colon B_5 \to K(\mathbb{Z},2) \times K(\mathbb{Z},4)$ have components representing v_2 and v_4 (theorem 5.5.12 on the connected CW complex B_5). The stabilization $i_3\colon BU(2) \to BU(3)$ is 5-connected (example 9.4.9(3)), so $\pi_3(BU(3)) \cong \pi_3(BU(2)) = 0$ and $\pi_2(BU(3)) \cong \pi_4(BU(3)) \cong \mathbb{Z}$ by proposition 7.5.1. With part (1), the weak equivalence w, the isomorphisms $(f_5)_*$ on π_i for $i \le 5$ and the vanishing of $\pi_i(B_5)$ for $i > 5$ (definition 5.6.2), the space B_5 has homotopy only in degrees 2 and 4, where both groups are $\mathbb{Z}$, and so does the target of v (example 5.1.7). To see v_* is an isomorphism there, compare with $BU(2)$. By proposition 5.3.16 (the source $BU(2)$ being a CW complex, section 7.5) there is $\tilde{\imath}\colon BU(2) \to Y$ with $w \circ \tilde{\imath} \simeq i_3$. Then $\tilde{\imath}^*c_i^Y = i_3^*c_i = c_i$ on $BU(2)$ for $i = 1,2$ (theorem 7.2.5), so the composite $v \circ f_5 \circ \tilde{\imath}$ has component classes (c_1, c_2) and is homotopic to the map κ of proposition 7.5.3, by the representability bijection on $BU(2)$. On π_2 and π_4 the map $\tilde{\imath}_* = w_*^{-1} \circ (i_3)_*$ is an isomorphism, i_3 being 5-connected and w a weak equivalence, and $(f_5)_*$ is an isomorphism there as well, while κ_* is an isomorphism because κ is 5-connected (proposition 7.5.3). From $v \circ f_5 \circ \tilde{\imath} \simeq \kappa$ it follows that v_* is an isomorphism on π_2 and π_4. Hence v is a weak homotopy equivalence, and for every connected CW complex W,

$$[W, B_5] \xrightarrow{v_*} \big[W, K(\mathbb{Z},2) \times K(\mathbb{Z},4)\big] \xrightarrow{\cong} H^2(W;\mathbb{Z}) \times H^4(W;\mathbb{Z})$$

is a bijection by proposition 5.3.16 and the product description in Step 1 of the proof of theorem 9.4.5, and it sends $[u]$ to (u^*v_2, u^*v_4). This proves part (3).

Step 4: The slope. Let $K' := K(\mathbb{Z},6)$ be the model produced by Step 1 of the proof of theorem 5.5.4 for $G = \mathbb{Z}$ and $n = 6$: it begins with S^6, whose sixth homotopy group is already $\mathbb{Z}$ (corollary 5.4.11), and attaches only cells of dimension 8 and higher. Let $s\colon S^6 \hookrightarrow K'$ be the inclusion of that sphere. Every stage of the construction attaches cells of dimension at least 8, hence leaves π_6 unchanged and only quotients π_7 and above (lemma 5.5.3), and a map or homotopy from a sphere lands in a finite stage (lemma 1.2.4), so s_* is an isomorphism on π_k for $k \le 6$ while $\pi_7(K') = 0$: the map s is 7-connected. The class $\sigma_6 := s^*\iota_6$ generates $H^6(S^6;\mathbb{Z})$: as in Step 2 of the proof of proposition 7.5.3, $\langle s^*\iota_6, [S^6]\rangle = \langle \iota_6, s_*[S^6]\rangle = \langle \iota_6, h[s]\rangle = \pm 1$, using the definition of the Hurewicz map (definition 5.4.2), the Hurewicz isomorphism for the 5-connected K' (theorem 5.4.9), the evaluation isomorphism of definition 5.5.10, and the evaluation isomorphism on S^6 from theorem 4.1.8 and corollary 3.4.15.

Choose $V\colon S^6 \to \Omega K(\mathbb{Z},7)$ with $\Theta[V] = \sigma_6$ (lemma 9.4.2). Since $\Omega K(\mathbb{Z},7)$ is path-connected, the homotopy extension property of (S^6, z_0) lets V be chosen with $V(z_0)$ the constant loop, z_0 the basepoint. Define

$$\mathcal{A}\colon S^6 \times E_6 \longrightarrow E_6, \qquad \mathcal{A}\big(z,(b,\gamma)\big) = \big(b,\, V(z) * \gamma\big)$$

continuous by the pasting and adjoint arguments used for the action map in the proof of theorem 9.4.3. The factor S^6 has finitely generated homology and free cohomology, so the Tor terms of the Künneth theorem (theorem 4.3.9) vanish. That theorem and the vanishing of $H^k(S^6;\mathbb{Z})$ for $0 < k < 6$ give

$$H^6(S^6 \times E_6;\mathbb{Z}) = \big(1 \times H^6(E_6;\mathbb{Z})\big) \oplus \big(\sigma_6 \times H^0(E_6;\mathbb{Z})\big)$$

with $H^0(E_6;\mathbb{Z}) = \mathbb{Z} \cdot 1$ for the path-connected E_6. Write $\mathcal{A}^*\bar{c}_3 = 1 \times u' + m\,(\sigma_6 \times 1)$ with $u' \in H^6(E_6;\mathbb{Z})$ and $m \in \mathbb{Z}$. Restricting to $\{z_0\} \times E_6$, where $\mathcal{A}$ is concatenation with the constant

loop and hence homotopic to the identity by a uniform reparametrization, kills the second summand and gives $u' = \bar{c}_3$.

Now let X, g and t be as in part (4). By representability and lemma 5.3.14 applied to the 7-connected s (both spaces simply connected, $\dim X \leq 6$), there is $t' \colon X \to S^6$ with $t = t'^*\sigma_6$. The map $W := V \circ t'$ has $\Theta[W] = t'^*\Theta[V] = t$ by naturality of Θ, so it computes the action, and $W \cdot g = \mathcal{A} \circ (t', g)$ for any representative g. Pulling back, with $\times$ as in definition 4.3.7,

$$(t \cdot g)^*\bar{c}_3 = (t', g)^*\big(1 \times \bar{c}_3\big) + m\,(t', g)^*\big(\sigma_6 \times 1\big) = g^*\bar{c}_3 + m\,t'^*\sigma_6 = g^*\bar{c}_3 + m\,t$$

which is part (4), completing the proof.

The slope m is the whole remaining question: the model says rank-three bundles over a fixed (c_1, c_2) form one orbit of $H^6(X;\mathbb{Z})$ on which c_3 moves in increments of m. The two computations of example 9.4.9 now pin m from opposite sides, Wu's identity forcing m even and the bundle F_3 forcing m to divide 2.

Lemma 9.4.11: The Third Chern Class on $\pi_6(BU(3))$

$\pi_6(BU(3)) \cong \mathbb{Z}$, the integer m of lemma 9.4.10 is ± 2, and the map

$$\pi_6(BU(3)) \longrightarrow H^6(S^6;\mathbb{Z}), \qquad [g] \longmapsto c_3\big(g^*\gamma_3\big)$$

is injective with image $2H^6(S^6;\mathbb{Z})$. Under an isomorphism $\pi_6(BU(3)) \cong \mathbb{Z}$ and the identification $H^6(S^6;\mathbb{Z}) = \mathbb{Z}\sigma_6$ it is multiplication by ± 2.

***Proof*:**

Step 1: One orbit on the six-sphere. Take $X = S^6$ in lemma 9.4.10. By part (3), $[S^6, B_5] \cong H^2(S^6;\mathbb{Z}) \times H^4(S^6;\mathbb{Z}) = 0$ has a single element, so by theorem 9.4.3(2) the whole set $[S^6, E_6]$ is one nonempty fiber of p_*, a single orbit of $H^6(S^6;\mathbb{Z})$. The stabilizer is trivial: a loop F at a lifting representative f_0 is a map $S^6 \times S^1 \to B_5$, and by part (3) again together with lemma 9.4.1,

$$[S^6 \times S^1, B_5] \cong H^2(S^6 \times S^1;\mathbb{Z}) \times H^4(S^6 \times S^1;\mathbb{Z}) = 0$$

since $H^k(S^6 \times S^1;\mathbb{Z}) \cong H^k(S^6;\mathbb{Z}) \oplus H^{k-1}(S^6;\mathbb{Z}) = 0$ for $k = 2, 4$. So F is freely nullhomotopic, $F^*[\kappa_6] = 0$, and $\Delta_{f_0}(F) = 0$. By theorem 9.4.3(3) the stabilizer vanishes, and $t \mapsto t \cdot g_0$ is a bijection $H^6(S^6;\mathbb{Z}) \to [S^6, E_6]$ for any g_0. Take $g_0 := \Phi_{S^6}(\underline{\mathbb{C}}^3)$, the class of the trivial bundle, which has $g_0^*\bar{c}_3 = c_3(\underline{\mathbb{C}}^3) = 0$, the trivial bundle being pulled back along a constant map. By lemma 9.4.10(2) and (4), the bijection

$$\mathrm{Vect}^3_{\mathbb{C}}(S^6) \;\cong\; [S^6, E_6] \;\cong\; H^6(S^6;\mathbb{Z}) \cong \mathbb{Z}$$

matches $E =$ the bundle of class $t \cdot g_0$ with $c_3(E) = m\,t$, so the values of c_3 on rank-three bundles over S^6 are exactly $m\,H^6(S^6;\mathbb{Z})$.

Step 2: m is even. For any rank-three bundle E over S^6, corollary 9.2.12 gives $\rho c_3(E) = \mathrm{Sq}^2\rho c_2(E) + \rho\big(c_1(E)c_2(E)\big) = 0$, since $c_1(E)$ and $c_2(E)$ live in vanishing groups. By proposition 9.1.2, $\ker\rho = 2H^6(S^6;\mathbb{Z})$, so every value of c_3 is even and $m\mathbb{Z} \subseteq 2\mathbb{Z}$.

Step 3: m divides 2. On $\mathbb{C}\mathrm{P}^3$, the bundle F_3 of example 9.4.9(3) and the trivial bundle $\underline{\mathbb{C}}^3$ both have $(c_1, c_2) = (0, 0)$, so by lemma 9.4.10(2),(3) their classes in $[\mathbb{C}\mathrm{P}^3, E_6]$ lie over the same

element of $[\mathbb{C}\mathrm{P}^3, B_5]$ and, by theorem 9.4.3(2), differ by the action: $\Phi(F_3) = t \cdot \Phi(\underline{\mathbb{C}}^3)$ for some $t \in H^6(\mathbb{C}\mathrm{P}^3;\mathbb{Z}) = \mathbb{Z}h^3$. Applying lemma 9.4.10(4),

$$2h^3 = c_3(F_3) = c_3(\underline{\mathbb{C}}^3) + m\,t = m\,t$$

so m divides 2 in $\mathbb{Z}$. With Step 2, $m = \pm 2$.

Step 4: Conclusion. $BU(3)$ is simply connected, so $\pi_6(BU(3)) \cong [S^6, BU(3)]$ by proposition 5.1.15, and $[S^6, BU(3)] \cong \mathrm{Vect}^3_{\mathbb{C}}(S^6)$ by theorem 7.1.29, compatibly with $[g] \mapsto c_3(g^*\gamma_3)$. By Steps 1 to 3 the map $[g] \mapsto c_3(g^*\gamma_3)$ becomes $t \mapsto m\,t = \pm 2t$ on $H^6(S^6;\mathbb{Z})$, so it is injective with image $2H^6(S^6;\mathbb{Z})$. It is also a homomorphism. For a based map $g\colon S^6 \to BU(3)$, definition 7.2.7, the definition of the Hurewicz homomorphism h (definition 5.4.2) and the naturality of the evaluation pairing give

$$\langle c_3(g^*\gamma_3), [S^6]\rangle = \langle g^*c_3, [S^6]\rangle = \langle c_3, h[g]\rangle$$

as in Step 2 of the proof of proposition 7.5.3. The right side is additive in $[g]$ because h is a homomorphism (proposition 5.4.3), and evaluation on $[S^6]$ is an isomorphism $H^6(S^6;\mathbb{Z}) \to \mathbb{Z}$ by theorem 4.1.8 and corollary 3.4.15. So $[g] \mapsto c_3(g^*\gamma_3)$ is an injective homomorphism from $\pi_6(BU(3)) \cong \mathbb{Z}$ (lemma 9.4.10(1)) to $H^6(S^6;\mathbb{Z}) = \mathbb{Z}\sigma_6$ with image $2\mathbb{Z}\sigma_6$. It carries a generator to a generator of the image, $\pm 2\sigma_6$, and is therefore multiplication by ± 2, completing the proof.

On the six-sphere the lemma reads: a rank-three bundle is determined by its third Chern class, and the classes that occur are exactly the even multiples of the generator. Both halves repeat below for an arbitrary six-complex with torsion-free H^6, with Wu's identity as the general form of the parity constraint.

Theorem 9.4.12: Rank-Three Bundles on Six-Complexes

Let X be a CW complex of dimension at most 6 with $H^6(X;\mathbb{Z})$ torsion-free. Then

$$\mathrm{Vect}^3_{\mathbb{C}}(X) \longrightarrow H^2(X;\mathbb{Z}) \times H^4(X;\mathbb{Z}) \times H^6(X;\mathbb{Z}), \qquad E \longmapsto \big(c_1(E), c_2(E), c_3(E)\big)$$

is injective, and its image is exactly the set of triples (a, b, c) satisfying

$$\rho c = \mathrm{Sq}^2 \rho b + \rho a\, \rho b \quad \text{in } H^6(X;\mathbb{Z}/2)$$

***Proof*:**

Step 1: Reduction to connected X. As in Step 1 of the proof of theorem 7.5.4, the path components X_α are subcomplexes and X is their topological disjoint union, bundles over X are families of bundles over the X_α, and restriction identifies $H^m(X;\mathbb{Z})$ with $\prod_\alpha H^m(X_\alpha;\mathbb{Z})$ compatibly with Chern classes (proposition 7.4.1), with ρ and with the squares (theorem 9.1.17(1)). Each $H^6(X_\alpha;\mathbb{Z})$ is torsion-free, being isomorphic to a subgroup of the product via the splitting of the restriction, and the stated condition on a triple holds exactly when it holds on every component. The map of the theorem is the product of the maps for the components, so assume X connected.

Step 2: Necessity. The complex X is paracompact Hausdorff (theorem 7.1.7, theorem 1.2.3), so for every rank-three bundle E corollary 9.2.12 gives $\rho c_3(E) = \mathrm{Sq}^2 \rho c_2(E) + \rho\big(c_1(E)c_2(E)\big)$, which is the stated condition for the triple of E.

Step 3: Every admissible triple occurs. Let (a, b, c) satisfy the condition. By lemma 9.4.10(3) there is $f\colon X \to B_5$ with $(f^*v_2, f^*v_4) = (a, b)$. The lifting obstruction of theorem 9.4.3(1) is $f^*[\kappa_6] \in H^7(X;\mathbb{Z})$, which vanishes because the cellular cochains of X vanish above degree 6 (proposition 4.1.20). So f lifts, say to $g \in [X, E_6]$, and the bundle E_0 with $\Phi_X(E_0) = g$ has $(c_1, c_2)(E_0) = (a, b)$ by lemma 9.4.10(2). Write $c_0 := c_3(E_0) = g^*\bar{c}_3$. By Step 2, $\rho c_0 = \mathrm{Sq}^2\rho b + \rho a\,\rho b = \rho c$, so $c - c_0 \in \ker\rho = 2H^6(X;\mathbb{Z})$ (proposition 9.1.2), say $c - c_0 = 2u$. By lemma 9.4.11 the slope is $m = \pm 2$, so $t := (m/2)\,u \in H^6(X;\mathbb{Z})$ satisfies $m\,t = 2u$, and the bundle E with $\Phi_X(E) = t \cdot g$ has, by lemma 9.4.10(2) and (4),

$$\big(c_1(E), c_2(E)\big) = (a, b), \qquad c_3(E) = g^*\bar{c}_3 + m\,t = c_0 + 2u = c$$

using that the action does not move the projection to B_5 (theorem 9.4.3(2)).

Step 4: Injectivity. Let E, E' have the same triple, with classes $g := \Phi_X(E)$ and $g' := \Phi_X(E')$. Equality of (c_1, c_2) and the injectivity in lemma 9.4.10(3) force $p_*g = p_*g'$ in $[X, B_5]$, so $g' = t \cdot g$ for some $t \in H^6(X;\mathbb{Z})$ by theorem 9.4.3(2). Equality of c_3 and lemma 9.4.10(4) give $m\,t = 0$, so $2t = 0$, so $t = 0$ since $H^6(X;\mathbb{Z})$ is torsion-free, and $g' = 0 \cdot g = g$. Since Φ_X is a bijection, $E \cong E'$, completing the proof.

The torsion-freeness is doing one job: it converts $2t = 0$ into $t = 0$, so that the increment-two action of $H^6(X;\mathbb{Z})$ on the lifts is free and c_3 separates the points of each orbit. The image condition is Wu's identity and nothing else: within one orbit the values of c_3 sweep out a full coset of $2H^6(X;\mathbb{Z})$, and the identity picks out which coset.

Closed six-manifolds are the setting of the published statement this corrects, and there the theorem specializes to a parity rule.

Corollary 9.4.13: Rank-Three Bundles on Closed Six-Manifolds

Let X be a closed connected orientable smooth 6-manifold, with fundamental class $[X] \in H_6(X;\mathbb{Z})$ and $[X]_2 \in H_6(X;\mathbb{Z}/2)$ its mod 2 reduction. Then $H^6(X;\mathbb{Z}) \cong \mathbb{Z}$, generated by the class s_X with $\langle s_X, [X]\rangle = 1$, the map (c_1, c_2, c_3) of theorem 9.4.12 is injective on $\mathrm{Vect}^3_{\mathbb{C}}(X)$, and its image is the set of triples $(a, b, \lambda s_X)$ with

$$\lambda \equiv \big\langle \mathrm{Sq}^2\rho b + \rho a\,\rho b,\ [X]_2 \big\rangle \pmod 2$$

***Proof*:**

Step 1: CW type. The manifold X carries a smooth function with no degenerate critical points [28, Corollary 6.7]. Such critical points are isolated [28, Corollary 2.3], so the compact X has finitely many, each of index at most 6. Every sublevel set is closed in X, hence compact, so X has the homotopy type of a CW complex $\bar{X}$ with one cell of dimension μ for each critical point of index μ [28, Theorem 3.5], a finite complex of dimension at most 6. Let $\chi\colon X \to \bar{X}$ be a homotopy equivalence. Both spaces are paracompact Hausdorff, X being compact and $\bar{X}$ a CW complex, so theorem 7.1.29 and composition with χ give a bijection $\mathrm{Vect}^3_{\mathbb{C}}(\bar{X}) \cong \mathrm{Vect}^3_{\mathbb{C}}(X)$ under which Chern classes correspond through the ring isomorphism χ^* (proposition 7.4.1), and the condition $\rho c = \mathrm{Sq}^2\rho b + \rho a\,\rho b$ transfers along χ^* by naturality of ρ and of the squares (theorem 9.1.17(1)). So theorem 9.4.12 applies to X once its hypothesis on H^6 is checked.

Step 2: The top cohomology. By theorem 4.4.7 and definition 4.4.8, $H_6(X;\mathbb{Z}) \cong \mathbb{Z}$ generated by $[X]$, and Poincaré duality (theorem 4.5.7) gives $H^6(X;\mathbb{Z}) \cong H_0(X;\mathbb{Z}) \cong \mathbb{Z}$. The

homology of X is isomorphic to that of the finite complex $\bar{X}$, hence finitely generated, being computed from a finite cellular chain complex (theorem 3.5.11), so $\mathrm{Ext}(H_5(X),\mathbb{Z})$ is a finite group, and the universal coefficient sequence (theorem 4.1.8) presents $H^6(X;\mathbb{Z}) \cong \mathbb{Z}$ as $\mathrm{Hom}(H_6(X),\mathbb{Z}) \oplus \mathrm{Ext}(H_5(X),\mathbb{Z})$. A finite direct summand of $\mathbb{Z}$ vanishes, so evaluation $H^6(X;\mathbb{Z}) \to \mathrm{Hom}(H_6(X),\mathbb{Z}) = \mathrm{Hom}(\mathbb{Z}[X],\mathbb{Z})$ is an isomorphism. Define s_X as the generator evaluating to 1 on $[X]$. In particular $H^6(X;\mathbb{Z})$ is torsion-free and theorem 9.4.12 applies, giving injectivity and the image condition $\rho c = \mathrm{Sq}^2\rho b + \rho a\,\rho b$.

Step 3: The parity form. Since X has the homotopy type of a complex of dimension at most 6, $H^7(X;\mathbb{Z}) = 0$ (proposition 4.1.20), so in the Bockstein sequence (proposition 9.1.2) the map β out of $H^6(X;\mathbb{Z}/2)$ is zero and $\rho\colon H^6(X;\mathbb{Z}) \to H^6(X;\mathbb{Z}/2)$ is onto with kernel $2H^6(X;\mathbb{Z})$: thus $H^6(X;\mathbb{Z}/2) \cong \mathbb{Z}/2$, generated by ρs_X. At the level of cochains and chains, evaluating the mod 2 reduction of an integral cocycle on the mod 2 reduction of an integral cycle is the mod 2 reduction of the integer evaluation, so $\langle \rho s_X, [X]_2\rangle = 1$ and pairing with $[X]_2$ is an isomorphism $H^6(X;\mathbb{Z}/2) \to \mathbb{Z}/2$. For $c = \lambda s_X$ the condition $\rho c = \mathrm{Sq}^2\rho b + \rho a\,\rho b$ becomes, after pairing with $[X]_2$, exactly the stated congruence, completing the proof.

The parity of c_3 on a closed six-manifold is not a constant of the manifold: it is a function of (c_1, c_2). The rule λ even is correct precisely over the pairs with $\mathrm{Sq}^2\rho b + \rho a\,\rho b = 0$, for instance over $(0,0)$, where it recovers the image $2H^6(S^6;\mathbb{Z})$ computed on the sphere in lemma 9.4.11. On $\mathbb{CP}^3$ with $(a,b) = (3h, 3h^2)$, the pair of the bundle $\mathcal{O}(1)^{\oplus 3}$, the class $\mathrm{Sq}^2\rho b + \rho a\,\rho b = (\rho h)^3$ pairs to 1 against $[\mathbb{CP}^3]_2$, by the computation of example 9.4.9(1) and the duality of theorem 4.5.12, so there the realized values of c_3 are the *odd* multiples of h^3.

9.4.14 Remark: Two Published Statements and Their Corrections The classifications of this section serve as the topological template for the motivic classification of vector bundles, and the standard reference there is the primer of Antieau and Elmanto, *A primer for unstable motivic homotopy theory*, arXiv:1605.00929. Two of its statements differ from the results above. Its Proposition 2.30 parametrizes the rank-two bundles on a five-complex with fixed (c_1, c_2) by $H^5(X;\mathbb{Z}/2)/\mathrm{Sq}^2\rho H^3(X;\mathbb{Z})$, with no dependence on the pair. Theorem 9.4.5 shows the correct indeterminacy is $I(c_1, c_2)$, which agrees with that quotient where $\rho c_1 = \rho c_2 = 0$ and is larger in general: on $S^1 \times S^4$ over $(0, s_4)$ there is one bundle where the published count gives two (example 9.4.6). Its Proposition 2.36 asserts that on a closed orientable 6-manifold the image of (c_1, c_2, c_3) consists of the triples with c_3 an even multiple of the generator. Corollary 9.4.13 shows the parity of c_3 is $\langle \mathrm{Sq}^2\rho b + \rho a\,\rho b, [X]_2\rangle$, and $\mathcal{O}(1)^{\oplus 3}$ on $\mathbb{CP}^3$, with $c_3 = h^3$, realizes an odd value (example 9.4.9). In comparing indexings, note that the class written k_4 in the primer is the k-invariant this book writes k_5, following definition 5.6.11.

Exercise 9.4.1

1. Let $X = S^2 \times S^3$, with s_2, s_3 the pullbacks of generators along the projections. Show that every rank-two complex bundle on X has $c_2 = 0$, and that the number of isomorphism classes of rank-two bundles with $c_1 = \lambda s_2$ is 2 when λ is even and 1 when λ is odd.

2. Let $X = S^1 \times S^4$. Show that the rank-two complex bundles on X with $c_2 = \mu s_4$ number 2 when μ is even and 1 when μ is odd, extending example 9.4.6.

3. Use $\pi_5(BU(3)) = 0$ (lemma 9.4.10(1)) to show that every rank-three complex bundle on S^5 is trivial. Conclude that the two rank-two bundles on S^5 become isomorphic after a single

trivial line is added to each.

4. Determine which triples $(\alpha h, \beta h^2, \lambda h^3)$ arise as (c_1, c_2, c_3) of a rank-three complex bundle on $\mathbb{C}\mathrm{P}^3$, and show each realized triple is realized by exactly one bundle. Verify the answer on $\mathcal{O}(1)^{\oplus 3}$.

5. Determine which triples $(\alpha s_2, \beta s_4, \lambda\, s_2 s_4)$ arise as (c_1, c_2, c_3) of a rank-three complex bundle on $S^2 \times S^4$, and show each realized triple is realized by exactly one bundle.

6. Compute the integral cohomology of $\mathbb{R}\mathrm{P}^5$ and show that $\mathbb{R}\mathrm{P}^5$ carries exactly eight isomorphism classes of rank-two complex bundles, two for each of the four possible pairs (c_1, c_2).

9.5 Steenrod Squares in the Atiyah-Hirzebruch Spectral Sequence

The Atiyah-Hirzebruch spectral sequence of complex K-theory collapses on every space whose integral cohomology is concentrated in even degrees, and example 8.6.6 computed $K^*(\mathbb{C}\mathrm{P}^n)$ from exactly that collapse. On a space with odd-degree cohomology the first differential that can act is d_3, and the book has so far not identified it. This section computes it: for every finite-dimensional CW complex, $d_3 = \mathrm{Sq}^3_{\mathbb{Z}} = \beta\,\mathrm{Sq}^2\rho$, the integral operation of definition 9.1.20. Three inputs combine. The differential is a natural, additive, suspension-compatible operation on integral cohomology, which takes a based form of the spectral sequence and its invariance under suspension (lemma 9.5.2, lemma 9.5.3, lemma 9.5.4). The computation of $H^{p+3}(K(\mathbb{Z}, p); \mathbb{Z})$ in proposition 9.2.7 leaves only 0 and $\mathrm{Sq}^3_{\mathbb{Z}}$ as candidates (proposition 9.5.5). A three-cell complex built from the suspended Hopf map eliminates the candidate 0 (example 9.5.7): a K-theory class on it is represented by a map into a Grassmannian $BU(N)$ and so has a second Chern class (lemma 9.5.6), and $\mathrm{Sq}^3_{\mathbb{Z}}$ annihilates second Chern classes (corollary 9.2.13).

Throughout, $\widetilde{K}^*$ is the reduced K-theory of theorem 8.5.10: the reduced cohomology theory $\widetilde{K}^n(X) = [X, KU_n]_*$ of the Ω-spectrum KU, produced by theorem 8.4.6, with coefficient groups $\widetilde{K}^q(S^0) = K^q(\mathrm{pt})$ equal to $\mathbb{Z}$ for q even and 0 for q odd (theorem 8.5.10, example 8.2.17). In negative degrees $\widetilde{K}^*$ is the periodic theory of theorem 8.5.10(2), so $\widetilde{K}^n = \widetilde{K}^{n+2}$, with equal suspension isomorphisms, in every degree. This literal 2-periodicity is used repeatedly below. The unreduced theory K^* of section 8.6 is the one associated to $\widetilde{K}^*$ by proposition 8.2.5: its relative groups are $K^n(X, A) = \widetilde{K}^n(X/A)$, and on a based space the basepoint retraction splits $K^n(X) \cong \widetilde{K}^n(X) \oplus K^n(\mathrm{pt})$, with $\widetilde{K}^n(X) = \ker\big(K^n(X) \to K^n(\mathrm{pt})\big)$. Every K-theory group in this section is a group of homotopy classes into the spaces KU_n, and vector bundles enter only through the Grassmannians $BU(N)$ inside $KU_0 = \mathbb{Z} \times BU$ (lemma 9.5.6). Ordinary cohomology enters the same framework: singular cohomology is an unreduced theory by example 8.2.7, so proposition 8.2.5 equips reduced singular cohomology $\widetilde{H}^*(-; G)$ with a suspension isomorphism, written σ, and with the long exact sequences of proposition 8.2.3. Finally, β always denotes the integral Bockstein of definition 9.1.1. The periodicity class of theorem 8.5.10 is never written by that letter here.

Two computations with pinch maps are needed several times: the pullback along a pinch sum of maps is the sum of the pullbacks, and reversing a suspension coordinate changes the sign of every class. Both follow from the wedge axiom, so they hold in every reduced theory at once.

Lemma 9.5.1: Pinch Sums and Reversals in a Reduced Theory

Let $\widetilde{h}^*$ be a reduced cohomology theory (definition 8.2.1). Let W and Z be pointed CW complexes, let $P\colon \Sigma W \to \Sigma W \vee \Sigma W$ be the pinch map collapsing the equatorial copy of W, and for based maps $f, g\colon \Sigma W \to Z$ write $f + g := \nabla \circ (f \vee g) \circ P$, where $\nabla\colon Z \vee Z \to Z$ is the fold map.

1. If P_0 is a one-point based CW complex, then $\widetilde{h}^n(P_0) = 0$ for every n, and every based map that factors through P_0 induces the zero map on $\widetilde{h}^n$.
2. $(f+g)^* = f^* + g^*\colon \widetilde{h}^n(Z) \to \widetilde{h}^n(\Sigma W)$.
3. The reversal $r\colon \Sigma W \to \Sigma W$, $r[w,t] = [w, 1-t]$, satisfies $r^* = -\mathrm{id}$ on $\widetilde{h}^n(\Sigma W)$, and the transposition of the two suspension coordinates $\tau\colon \Sigma^2 W \to \Sigma^2 W$, $\tau[w,s,t] = [w,t,s]$, satisfies $\tau^* = -\mathrm{id}$ on $\widetilde{h}^n(\Sigma^2 W)$.

Here $[w,t]$ denotes the image of (w,t) in $\Sigma W = (W \times I)/(W \times \partial I \cup \{w_0\} \times I)$ of definition 5.7.1, and $[w,s,t] = [[w,s],t]$ in $\Sigma^2 W = \Sigma(\Sigma W)$, with s the inner and t the outer suspension coordinate.

***Proof*:**

1. The wedge $P_0 \vee P_0$ is again P_0, with both inclusions equal to the identity. The additivity axiom of definition 8.2.1 states that restriction along the two inclusions is an isomorphism $\widetilde{h}^n(P_0) \to \widetilde{h}^n(P_0) \times \widetilde{h}^n(P_0)$. Here that map is the diagonal $x \mapsto (x,x)$. A diagonal map of a group onto the full product forces the group to have a single element, since (x,y) with $x \neq y$ lies in the image of no element. Hence $\widetilde{h}^n(P_0) = 0$. A based map factoring through P_0 induces a map on $\widetilde{h}^n$ factoring through $\widetilde{h}^n(P_0) = 0$.

2. Write $\iota_1, \iota_2\colon \Sigma W \to \Sigma W \vee \Sigma W$ for the two inclusions and $q_1, q_2\colon \Sigma W \vee \Sigma W \to \Sigma W$ for the maps that are the identity on the named summand and constant on the other. The wedge axiom makes (ι_1^*, ι_2^*) an isomorphism onto $\widetilde{h}^n(\Sigma W) \times \widetilde{h}^n(\Sigma W)$. Since $q_j \circ \iota_j = \mathrm{id}$ and $q_j \circ \iota_k$ is constant for $k \neq j$, part (1) gives $\iota_j^* q_j^* = \mathrm{id}$ and $\iota_k^* q_j^* = 0$, so the element $q_1^* a + q_2^* b$ has coordinates (a,b). By injectivity of the wedge isomorphism, every element of $\widetilde{h}^n(\Sigma W \vee \Sigma W)$ is $q_1^* a + q_2^* b$ for unique a and b. Next, $q_1 \circ P$ sends $[w,t]$ to $[w,2t]$ for $t \leq \frac{1}{2}$ and to the basepoint for $t \geq \frac{1}{2}$, and

$$H_s[w,t] := [w, \min(1, (2-s)t)]$$

is a based homotopy from $q_1 \circ P$ to the identity. It is well defined on the quotient because the loci $t \in \{0,1\}$ and $w = w_0$ are sent to the basepoint for every s. Symmetrically $q_2 \circ P \simeq \mathrm{id}$. Now take $z \in \widetilde{h}^n(Z)$. Restriction along the two inclusions of $Z \vee Z$ sends $\nabla^* z$ to (z,z), so $(f \vee g)^* \nabla^* z$ has coordinates (f^*z, g^*z) and equals $q_1^* f^* z + q_2^* g^* z$. Applying the homomorphism P^*,

$$(f+g)^* z = (q_1 P)^* f^* z + (q_2 P)^* g^* z = f^* z + g^* z$$

3. The sum $\mathrm{id} + r$ runs up the suspension coordinate and back down: $(\mathrm{id} + r)[w,t]$ is $[w,2t]$

for $t \le \frac{1}{2}$ and $[w, 2-2t]$ for $t \ge \frac{1}{2}$. The formula

$$G_s[w,t] := \begin{cases} [w,\ 2t(1-s)] & t \le \frac{1}{2} \\ [w,\ (2-2t)(1-s)] & t \ge \frac{1}{2} \end{cases}$$

is continuous, matches at $t = \frac{1}{2}$, respects the collapsed loci, and is a based homotopy from $\mathrm{id} + r$ to the constant map. By part (1) and part (2), $0 = (\mathrm{id} + r)^* = \mathrm{id} + r^*$, so $r^* = -\mathrm{id}$.

For the transposition, write $\Sigma^2 W = (W \times I^2)/\big(W \times \partial I^2 \cup \{w_0\} \times I^2\big)$ with coordinates $[w, s, t]$. Let $R\colon I^2 \to I^2$ be the quarter turn $R(s,t) = (t, 1-s)$, a homeomorphism of the square preserving its boundary, and let $\varrho[w,s,t] := [w, R(s,t)]$. Conjugating the Euclidean rotations about the center of the square by the radial homeomorphism $(I^2, \partial I^2) \cong (D^2, \partial D^2)$ yields a continuous family $R_\theta\colon I^2 \to I^2$, $0 \le \theta \le \frac{\pi}{2}$, of boundary-preserving homeomorphisms with $R_0 = \mathrm{id}$ and $R_{\pi/2} = R$. The conjugation returns R itself at $\theta = \frac{\pi}{2}$ because the quarter turn is a symmetry of the square and therefore commutes with the radial rescaling. The family $[w, (s,t)] \mapsto [w, R_\theta(s,t)]$ is then a based homotopy $\varrho \simeq \mathrm{id}$ on $\Sigma^2 W$. Writing $\mathrm{fl}[w,s,t] := [w, 1-s, t]$, a direct evaluation gives $\varrho \circ \mathrm{fl} = \tau$, so $\tau \simeq \mathrm{fl}$. Finally $\mathrm{fl} = \Sigma r'$ for the reversal r' of ΣW, and naturality of the suspension isomorphism σ of the theory with respect to r' gives

$$(\Sigma r')^* = \sigma \circ (r')^* \circ \sigma^{-1} = \sigma(-\mathrm{id})\sigma^{-1} = -\mathrm{id}$$

using the first half of this part for r'. Hence $\tau^* = -\mathrm{id}$, completing the proof.

The first two parts say that the group structure of a reduced theory is computed by the pinch map whenever the argument is a suspension. The third is the reason every comparison of suspended spectral sequences below carries a global sign and nothing worse.

The spectral sequence of theorem 8.6.4 is built from the unreduced theory, and its column $p = 0$ carries the coefficients of the theory. Suspension is a based construction, so comparing the spectral sequences of X and ΣX takes a based variant, in which the skeletal exact couple is formed in the reduced theory. The next lemma records that variant and its relation to the unreduced spectral sequence.

Lemma 9.5.2: The Reduced Atiyah-Hirzebruch Spectral Sequence

Let $\widetilde{h}^*$ be a reduced cohomology theory, h^* the associated unreduced theory of proposition 8.2.5, and X a based CW complex whose basepoint x_0 is a 0-cell.

1. The groups $\widetilde{D}_1^{p,q} = \widetilde{h}^{p+q}\big(X^{(p)}\big)$ and $\widetilde{E}_1^{p,q} = \widetilde{h}^{p+q}\big(X^{(p)}/X^{(p-1)}\big)$, with the convention $X^{(-1)} := \{x_0\}$ and with the restriction, quotient and connecting maps of the long exact sequences of proposition 8.2.3 for the skeletal pairs $(X^{(p)}, X^{(p-1)})$, form an exact couple. Its spectral sequence $\widetilde{E}_r^{p,q}(X)$, the *reduced Atiyah-Hirzebruch spectral sequence* of $\widetilde{h}^*$ on X, is natural in based cellular maps, and
$$\widetilde{E}_2^{p,q} \cong \widetilde{H}^p\big(X;\, \widetilde{h}^q(S^0)\big)$$
2. For every $r \geq 1$ and every $p \geq 1$ there is an isomorphism $\widetilde{E}_r^{p,q} \cong E_r^{p,q}$ with the spectral sequence of theorem 8.6.4 for h^* on X, compatible with every differential whose source and target both lie in columns ≥ 1. In column 0 there is a natural surjection $E_r^{0,q} \to \widetilde{E}_r^{0,q}$, compatible with the differentials leaving column 0.
3. If X is finite-dimensional of dimension N, the reduced spectral sequence converges: with $F^p := \ker\big(\widetilde{h}^n(X) \to \widetilde{h}^n(X^{(p-1)})\big)$ for $p \geq 1$, the chain $\widetilde{h}^n(X) \supseteq F^1 \supseteq \cdots \supseteq F^{N+1} = 0$ is a finite filtration with
$$F^p/F^{p+1} \cong \widetilde{E}_\infty^{p,\,n-p} \quad (p \geq 1), \qquad \widetilde{h}^n(X)/F^1 \cong \widetilde{E}_\infty^{0,n}$$
and these identifications are natural in based cellular maps.

***Proof*:**

1. Each skeletal pair is a based CW pair containing x_0, so proposition 8.2.3 applies to it, and exactness of the couple at its three corners is exactness of those sequences. By proposition 8.2.5, the relative groups of the unreduced theory are $h^{p+q}(X^{(p)}, X^{(p-1)}) = \widetilde{h}^{p+q}(X^{(p)}/X^{(p-1)})$, the same groups as $\widetilde{E}_1^{p,q}$, and the unreduced long exact sequence of each skeletal pair is obtained from the reduced one through the natural splittings $h^n(Y) \cong \widetilde{h}^n(Y) \oplus h^n(\mathrm{pt})$ of that proposition. The restrictions between skeleta respect the splittings and are the identity between the $h^n(\mathrm{pt})$ summands, because the basepoint retractions commute with the skeletal inclusions. The map from the relative group lands in the reduced summand, and the unreduced coboundary is the connecting map of proposition 8.2.3 composed with the projection to the reduced summand. Consequently the first differential of the reduced couple agrees with that of the unreduced couple in columns $p \geq 1$, where the E_1-terms coincide, and lemma 8.6.3 identifies the row $\widetilde{E}_1^{\bullet,q}$ there with the cellular cochain complex $C^\bullet_{\mathrm{CW}}(X; \widetilde{h}^q(S^0))$ under the cellular coboundary. In column 0, the convention $X^{(-1)} = \{x_0\}$ gives $\widetilde{E}_1^{0,q} = \widetilde{h}^q\big(X^{(0)}/\{x_0\}\big) = \widetilde{h}^q\big(X^{(0)}\big)$. The skeleton $X^{(0)}$ is the wedge of one copy of S^0 for each 0-cell other than x_0, so the wedge axiom gives $\widetilde{E}_1^{0,q} \cong \prod_{\alpha \neq x_0} \widetilde{h}^q(S^0)$, the kernel of the restriction of cellular 0-cochains to the basepoint cell.

   The cellular retraction $X \to \{x_0\}$ induces a map of cochain complexes splitting the restriction $C^\bullet_{\mathrm{CW}}(X; A) \to C^\bullet_{\mathrm{CW}}(\{x_0\}; A)$, so the cellular cochain complex of X is the direct

sum, as a complex, of the kernel subcomplex and a copy of the complex of the point. The row complex $(\widetilde{E}_1^{\bullet,q}, d_1)$ is that kernel subcomplex. Taking cohomology of the direct sum,

$$\widetilde{E}_2^{p,q} \cong \ker\Big(H^p\big(X;\widetilde{h}^q(S^0)\big) \to H^p\big(\mathrm{pt};\widetilde{h}^q(S^0)\big)\Big) = \widetilde{H}^p\big(X;\widetilde{h}^q(S^0)\big)$$

using proposition 4.1.20 to identify cellular with singular cohomology and proposition 4.1.14 to identify the kernel with reduced cohomology. For $p \geq 1$ the kernel is all of H^p. Naturality holds because a based cellular map preserves skeleta and quotients and proposition 8.2.3 is natural.

2. By the description in (1), the projections $h^{p+q}(X^{(p)}) \to \widetilde{h}^{p+q}(X^{(p)})$, the identity on the E_1-terms in columns $p \geq 1$, and the projection $E_1^{0,q} = h^q(X^{(0)}) \to \widetilde{h}^q(X^{(0)}) = \widetilde{E}_1^{0,q}$ in column 0 constitute a morphism from the unreduced skeletal couple to the reduced one: each square commutes by that description. On second pages the induced map is the surjection $H^p \to \widetilde{H}^p$, an isomorphism for $p \geq 1$. Suppose inductively that on page r the induced maps are isomorphisms in columns ≥ 1 and a surjection in column 0, compatibly with d_r. Kernels of d_r then correspond in columns ≥ 1, since the targets lie in columns $\geq 1 + r$, where the map is an isomorphism. Images of d_r landing in a column ≥ 1 also correspond: a source in a column ≥ 1 is matched by the isomorphism, and a source in column 0 is matched because the column-0 map is surjective and the d_r-squares commute. Hence the maps on page $r + 1$, being induced on subquotients, are again isomorphisms in columns ≥ 1 and a surjection in column 0.

3. Apply the convergence clause of theorem 8.6.4 to the unreduced theory h^* on the finite-dimensional complex X: the filtration $F^p h^n(X) = \ker\big(h^n(X) \to h^n(X^{(p-1)})\big)$ is finite with quotients $E_\infty^{p,n-p}$. For $p \geq 1$ the subgroup $F^p h^n(X)$ lies inside the reduced summand, because a class vanishing on $X^{(p-1)}$ vanishes at the basepoint, and under the splittings of (1) it equals $F^p = \ker\big(\widetilde{h}^n(X) \to \widetilde{h}^n(X^{(p-1)})\big)$: a reduced class restricts into the reduced summand of $h^n(X^{(p-1)})$, so its restriction vanishes in h^n exactly when it vanishes in $\widetilde{h}^n$. In both spectral sequences every differential touching a fixed spot vanishes once r exceeds $\max(p, N - p) + 1$, the pages being confined to the strip $0 \leq p \leq N$, so each spot stabilizes after finitely many pages, and by (2) the stable values agree in columns $p \geq 1$. This gives $F^p/F^{p+1} \cong \widetilde{E}_\infty^{p,n-p}$ for $p \geq 1$. For the top quotient, the classes in column 0 pulled back along the cellular retraction $c\colon X \to \{x_0\}$ are permanent cycles: the spectral sequence of a point carries no nonzero differential, and naturality applied to c makes every d_r vanish on them. Hence $E_\infty^{0,n} \cong \widetilde{E}_\infty^{0,n} \oplus h^n(\mathrm{pt})$ compatibly with the splitting $h^n(X)/F^1 \cong \big(\widetilde{h}^n(X)/F^1\big) \oplus h^n(\mathrm{pt})$, and the reduced top quotient is $\widetilde{E}_\infty^{0,n}$. Naturality of every identification follows from naturality of the filtration, a kernel of restrictions, and of the spectral sequence, completing the proof.

The reduced spectral sequence differs from the unreduced one only in the column $p = 0$, which no longer carries the coefficient ring. What is gained is a spectral sequence of based complexes, and based complexes can be suspended. The next lemma is what the based form is for: suspension shifts the reduced spectral sequence one column to the right and changes the sign of every differential, and nothing else.

Lemma 9.5.3: Suspension Invariance of the Atiyah-Hirzebruch Spectral Sequence

Let $\widetilde{h}^*$ be a reduced cohomology theory, equivalently a generalized cohomology theory in reduced form (definition 8.2.16), and let X be a based CW complex whose basepoint is a 0-cell. Then the suspension isomorphisms of $\widetilde{h}^*$ induce, for every $r \geq 1$, isomorphisms

$$\Phi\colon \widetilde{E}_r^{p,q}(X) \xrightarrow{\cong} \widetilde{E}_r^{p+1,q}(\Sigma X)$$

of the reduced Atiyah-Hirzebruch spectral sequences of lemma 9.5.2, satisfying $d_r \circ \Phi = -\Phi \circ d_r$ for every r. In particular the pages correspond and the differentials commute with Φ up to sign.

***Proof*:**

Step 1: The suspended skeleta. Present $\Sigma X = (X \times I)/(X \times \partial I \cup \{x_0\} \times I)$ as in definition 5.7.1. The product $X \times I$ carries the product cell structure of definition 1.2.8, which is a CW structure in the product topology by lemma 1.2.12(3), the factor I being finite. In it $X \times \partial I \cup \{x_0\} \times I$ is a subcomplex, and the quotient inherits the cell structure whose cells are the basepoint together with one $(n+1)$-cell $e \times (0,1)$ for each n-cell e of X other than the basepoint, with characteristic maps the composites $D^{n+1} \cong D^n \times I \to X \times I \to \Sigma X$. The basepoint of ΣX is a 0-cell, and the skeleta are

$$(\Sigma X)^{(p+1)} = \Sigma\big(X^{(p)}\big)$$

for every $p \geq -1$, both sides being the basepoint when $p = -1$ under the convention $X^{(-1)} = \{x_0\}$ of lemma 9.5.2. Moreover, for every $p \geq 0$, $\Sigma\big(X^{(p)}/X^{(p-1)}\big) = (\Sigma X)^{(p+1)}/(\Sigma X)^{(p)}$ as an equality of quotient spaces: both sides are the quotient of $X^{(p)} \times I$ by the union of $X^{(p-1)} \times I$, $X^{(p)} \times \partial I$ and $\{x_0\} \times I$.

Step 2: The terms, restrictions and quotient maps. Define Φ on the couple by the suspension isomorphism of the theory:

$$\sigma\colon \widetilde{h}^{p+q}\big(X^{(p)}\big) \longrightarrow \widetilde{h}^{p+q+1}\big(\Sigma X^{(p)}\big) = \widetilde{D}_1^{p+1,q}(\Sigma X), \qquad \sigma\colon \widetilde{E}_1^{p,q}(X) \longrightarrow \widetilde{E}_1^{p+1,q}(\Sigma X)$$

the second map using Step 1 to identify the suspended quotient with the skeletal quotient of ΣX. The restriction maps of the two couples are induced by the skeletal inclusions and their suspensions, and the quotient maps by the collapses and their suspensions. The suspension of an inclusion is the inclusion of the suspensions, and the suspension of a collapse is the collapse of the suspensions, so naturality of σ makes the corresponding squares commute.

Step 3: The connecting squares anticommute. By proposition 8.2.3, the connecting map of a based CW pair (Y, A) is $\delta = \partial^* \circ \sigma$, where $\partial\colon Y/A \to \Sigma A$ is the composite, in the based homotopy category, of the inverse of the collapse equivalence $c\colon C_i \to Y/A$ (which collapses the cone CA inside the mapping cone $C_i = Y \cup_A CA$, as in the proof of that proposition) with the collapse $\partial'\colon C_i \to C_i/Y = \Sigma A$. The claim is that for the suspended pair, under the identification $\Sigma(Y/A) = \Sigma Y/\Sigma A$ of Step 1,

$$\partial_\Sigma = \tau \circ \Sigma\partial\colon\ \Sigma(Y/A) \longrightarrow \Sigma^2 A$$

with τ the transposition of the two suspension coordinates of $\Sigma^2 A$. To see this, identify $C_{\Sigma i} = \Sigma Y \cup_{\Sigma A} C(\Sigma A)$ with $\Sigma C_i = \Sigma Y \cup \Sigma(CA)$ by the homeomorphism κ that is the identity

on ΣY and sends the cone point $\langle [a,s],t\rangle$, with cone coordinate t over the suspension point $[a,s]$, to the suspension point $[\langle a,t\rangle,s]$, with suspension coordinate s over the cone point $\langle a,t\rangle$. The gluing loci match, since at $t=0$ both sides give the point of ΣY over $(i(a),s)$, and the collapsed loci correspond. Under κ the two cone collapses agree, $\Sigma c\circ\kappa = c_\Sigma$: each map collapses exactly the cone part and is the suspended quotient map on ΣY. For the collapses of the ambient space, write the identification $C_i/Y = \Sigma A$ so that the cone coordinate becomes the suspension coordinate. Evaluating both maps on a point $\langle [a,s],t\rangle$ then gives

$$\partial'_\Sigma\langle [a,s],t\rangle = [a,s,t], \qquad \big(\Sigma\partial'\circ\kappa\big)\langle [a,s],t\rangle = [a,t,s]$$

so $\partial'_\Sigma = \tau\circ\Sigma\partial'\circ\kappa$ as an equality of maps. Composing with the inverses of the collapse equivalences and using $\Sigma(c^{-1}) = (\Sigma c)^{-1}$ in the homotopy category yields $\partial_\Sigma = \tau\circ\Sigma\partial$ as claimed. Now let $(Y,A) = (X^{(p+1)}, X^{(p)})$ and compute, for $x\in\widetilde{h}^{p+q}(X^{(p)})$,

$$\delta_\Sigma(\sigma x) = \partial^*_\Sigma\,\sigma\sigma x = (\tau\circ\Sigma\partial)^*\,\sigma\sigma x = (\Sigma\partial)^*\,\tau^*\,\sigma\sigma x = -(\Sigma\partial)^*\,\sigma\sigma x = -\sigma\big(\partial^*\sigma x\big) = -\sigma\,\delta(x)$$

where the fourth equality is lemma 9.5.1(3) applied to $W = X^{(p)}$ and the fifth is naturality of σ with respect to ∂. So the Φ-squares for the connecting maps anticommute, with the same sign at every (p,q).

Step 4: Pages and differentials. Steps 2 and 3 say that Φ is an isomorphism from the couple of X onto the couple of ΣX with its connecting map negated. Negating the connecting map of an exact couple changes no group of any derived couple and negates each derived connecting map, the derived maps being induced by the original ones. Each differential of the generated spectral sequence involves the connecting map exactly once, so every d_r is negated and no page changes. Hence Φ induces isomorphisms of all pages with $d_r\circ\Phi = -\Phi\circ d_r$, completing the proof.

With the based spectral sequence and its suspension invariance available, the differential d_3 for K-theory can be isolated as an operation. The coefficient groups of KU vanish in odd rows, which makes d_2 vanish and places d_3 between full cohomology groups.

Lemma 9.5.4: The Differential d_3 for KU is a Natural Operation

Let X be a CW complex, and consider the Atiyah-Hirzebruch spectral sequence of theorem 8.6.4 for complex K-theory on X, together with, for based X with basepoint a 0-cell, the reduced spectral sequence of lemma 9.5.2 for $\widetilde{K}^*$.

1. $d_2 = 0$, so $E_3 = E_2$, in the reduced spectral sequence as well.
2. For even q the differential $d_3\colon E_3^{p,q} = H^p(X;\mathbb{Z}) \to E_3^{p+3,q-2} = H^{p+3}(X;\mathbb{Z})$ is one and the same homomorphism for every even q. It defines an operation
$$\theta_p\colon H^p(X;\mathbb{Z}) \longrightarrow H^{p+3}(X;\mathbb{Z})$$
which is additive and natural for every map of CW complexes. For based X and $p\geq 1$ the differential d_3 of the reduced spectral sequence equals θ_p under $\widetilde{H}^p = H^p$.
3. Between the reduced spectral sequences of X and ΣX, the isomorphisms Φ of lemma 9.5.3 satisfy $d_3\circ\Phi = -\Phi\circ d_3$.

***Proof*:**

1. The second page is $E_2^{p,q} = H^p(X; K^q(\mathrm{pt}))$, which vanishes for odd q by theorem 8.5.10(3). The differential d_2 has bidegree $(2,-1)$: a nonzero source needs q even, and then the target row $q-1$ is odd. So $d_2 = 0$, and the reduced page $\widetilde{H}^p(X; K^q(\mathrm{pt}))$ vanishes in the same rows, so the same parity count applies to it.

2. The spaces KU_n and the structure maps of the spectrum KU depend only on the parity of n (theorem 8.5.10), so the functors $\widetilde{K}^n$ and $\widetilde{K}^{n+2}$ coincide, together with their suspension isomorphisms, for $n \geq 0$ and, by the convention fixed at the start of this section, for $n < 0$. Hence they share the connecting maps of proposition 8.2.3 and the splittings of proposition 8.2.5 built from them. The skeletal exact couples in rows q and $q+2$ are therefore identical, and so are all pages and differentials: $E_r^{p,q} = E_r^{p,q+2}$ with equal d_r. In particular the maps $d_3 \colon H^p(X;\mathbb{Z}) \to H^{p+3}(X;\mathbb{Z})$ in the even rows are one homomorphism θ_p. Differentials in the spectral sequence of an exact couple of abelian groups are homomorphisms, so θ_p is additive. The skeletal exact couple is functorial for cellular maps, which preserve skeleta, so $\theta_p \varphi^* = \varphi^* \theta_p$ for every cellular map $\varphi \colon Y \to X$. An arbitrary map φ of CW complexes is homotopic to a cellular map φ' by the cellular approximation theorem, and homotopic maps induce the same homomorphism on cohomology, so $\theta_p \varphi^* = \theta_p \varphi'^* = \varphi'^* \theta_p = \varphi^* \theta_p$, because θ_p is defined on the groups $H^p(-;\mathbb{Z})$ themselves. The final sentence is lemma 9.5.2(2), the agreement of the two spectral sequences in columns ≥ 1: both the source column p and the target column $p+3$ lie in that range.

3. This is lemma 9.5.3 for the theory $\widetilde{K}^*$ at $r = 3$, completing the proof.

The operation θ_p is exactly the kind of object this chapter classifies: a natural, additive map $H^p(-;\mathbb{Z}) \to H^{p+3}(-;\mathbb{Z})$. Its value on the fundamental class of $K(\mathbb{Z},p)$ determines it, and proposition 9.2.7 computed the group in which that value lives.

Proposition 9.5.5: The Differential d_3 is a Multiple of $\mathrm{Sq}^3_{\mathbb{Z}}$

There is a single $\lambda \in \mathbb{Z}/2$ such that, for every CW complex X and every $p \geq 0$,

$$\theta_p = \lambda \,\mathrm{Sq}^3_{\mathbb{Z}} \colon H^p(X;\mathbb{Z}) \longrightarrow H^{p+3}(X;\mathbb{Z})$$

***Proof*:**

Step 1: Reduction to connected complexes. The cellular cochain complex of a disjoint union of CW complexes is the product of the cellular cochain complexes, so restriction gives an injection $H^{p+3}(X;\mathbb{Z}) \to \prod_\alpha H^{p+3}(X_\alpha;\mathbb{Z})$ over the path components (proposition 4.1.20). Both θ_p and $\mathrm{Sq}^3_{\mathbb{Z}}$ commute with restriction to each component, by naturality (lemma 9.5.4, proposition 9.1.21), so it is enough to prove the identity on connected X.

Step 2: Degrees $p \leq 2$. Let X be connected. For $p = 0$, choose a point $j \colon \mathrm{pt} \to X$ and let $c \colon X \to \mathrm{pt}$, so $c \circ j = \mathrm{id}$. Then $H_0(j)$ is a split injection $\mathbb{Z} \to \mathbb{Z}$ (proposition 3.2.5, proposition 3.2.4), hence an isomorphism, and $H_0(c)$ is its inverse. By naturality of the universal coefficient isomorphism of theorem 4.1.8, $c^* \colon H^0(\mathrm{pt};\mathbb{Z}) \to H^0(X;\mathbb{Z})$ is an isomorphism. Every

$x \in H^0(X;\mathbb{Z})$ is therefore c^*x', and

$$\theta_0(x) = c^*\theta_0(x') \in c^*H^3(\mathrm{pt};\mathbb{Z}) = 0$$

For $p \in \{1,2\}$: fix a CW model K of $K(\mathbb{Z},p)$ (theorem 5.5.4). Every $x \in H^p(X;\mathbb{Z})$ is $f^*\iota_p$ for a map $f\colon X \to K$ (theorem 5.5.12, with reduced and unreduced cohomology agreeing in degrees ≥ 1 by proposition 4.1.14 and with free homotopy classes sufficing on connected X). The pages and differentials of theorem 8.6.4 exist for every CW complex, finite-dimensional or not, and θ_p commutes with f^* for the arbitrary map f by lemma 9.5.4(2), so

$$\theta_p(x) = \theta_p(f^*\iota_p) = f^*\big(\theta_p(\iota_p)\big) \in f^*H^{p+3}(K;\mathbb{Z}) = 0$$

by the vanishing $H^{p+3}(K(\mathbb{Z},p);\mathbb{Z}) = 0$ for $p \leq 2$ (proposition 9.2.7). Since $\mathrm{Sq}^3_{\mathbb{Z}}$ vanishes on $H^p(-;\mathbb{Z})$ for $p \leq 2$ as well (proposition 9.1.21), the identity holds in these degrees for either value of λ.

Step 3: A multiplier in each degree $p \geq 3$. With K as above, $H^{p+3}(K(\mathbb{Z},p);\mathbb{Z}) \cong \mathbb{Z}/2$ generated by $\mathrm{Sq}^3_{\mathbb{Z}}\iota_p$ (proposition 9.2.7), so $\theta_p(\iota_p) = \lambda_p\,\mathrm{Sq}^3_{\mathbb{Z}}\iota_p$ for a unique $\lambda_p \in \mathbb{Z}/2$. For connected X and $x = f^*\iota_p$ as in Step 2, naturality of θ_p and of $\mathrm{Sq}^3_{\mathbb{Z}}$ for the arbitrary map f (lemma 9.5.4(2), proposition 9.1.21(1)) gives

$$\theta_p(x) = f^*\theta_p(\iota_p) = \lambda_p\, f^*\mathrm{Sq}^3_{\mathbb{Z}}\iota_p = \lambda_p\,\mathrm{Sq}^3_{\mathbb{Z}}(x)$$

so $\theta_p = \lambda_p\,\mathrm{Sq}^3_{\mathbb{Z}}$ in degree p, on every CW complex by Step 1.

Step 4: The multipliers agree. Fix $p \geq 3$, let K be a CW model of $K(\mathbb{Z},p)$ based at a 0-cell, and write $\iota := \iota_p \in \widetilde{H}^p(K;\mathbb{Z})$. By definition 5.5.10, $H^p(K;\mathbb{Z}) \cong \mathrm{Hom}(\mathbb{Z},\mathbb{Z}) \cong \mathbb{Z}$ with generator ι, and $\mathrm{Sq}^3_{\mathbb{Z}}\iota$ generates $H^{p+3}(K;\mathbb{Z}) \cong \mathbb{Z}/2$. Consider the isomorphism $\Phi\colon \widetilde{E}_3^{p,q}(K) \to \widetilde{E}_3^{p+1,q}(\Sigma K)$ of lemma 9.5.3 for $\widetilde{K}^*$, in an even row q. Its source is $\widetilde{H}^p(K;\mathbb{Z}) \cong \mathbb{Z}$ with generator ι, and its target $\widetilde{H}^{p+1}(\Sigma K;\mathbb{Z})$ is identified by the suspension isomorphism σ of ordinary reduced cohomology with $\widetilde{H}^p(K;\mathbb{Z})$, so it is infinite cyclic with generator $\sigma\iota$. An isomorphism carries generators to generators, so

$$\Phi(\iota) = \pm\,\sigma\iota$$

Likewise $\widetilde{H}^{p+4}(\Sigma K;\mathbb{Z}) \cong \widetilde{H}^{p+3}(K;\mathbb{Z}) \cong \mathbb{Z}/2$, a group with a unique nonzero element. Now evaluate θ_{p+1} on $\Phi(\iota) \in \widetilde{H}^{p+1}(\Sigma K;\mathbb{Z})$ in two ways, reading every reduced d_3 in columns ≥ 1 as θ by lemma 9.5.4(2). First, by lemma 9.5.4(3) and Step 3,

$$\theta_{p+1}\big(\Phi\iota\big) = -\Phi\big(\theta_p\iota\big) = -\lambda_p\,\Phi\big(\mathrm{Sq}^3_{\mathbb{Z}}\iota\big)$$

and $\Phi(\mathrm{Sq}^3_{\mathbb{Z}}\iota)$ is nonzero because Φ is injective. Being nonzero in a group of order two, it is the unique nonzero element, and the sign in front of it changes nothing. Second, by Step 3 in degree $p+1$ applied on ΣK, by additivity, and by the stability of $\mathrm{Sq}^3_{\mathbb{Z}}$ (proposition 9.1.21, the sign again being immaterial because $2\,\mathrm{Sq}^3_{\mathbb{Z}} = 0$),

$$\theta_{p+1}\big(\Phi\iota\big) = \pm\,\theta_{p+1}(\sigma\iota) = \pm\,\lambda_{p+1}\,\mathrm{Sq}^3_{\mathbb{Z}}(\sigma\iota) = \lambda_{p+1}\,\sigma\big(\mathrm{Sq}^3_{\mathbb{Z}}\iota\big)$$

and $\sigma(\mathrm{Sq}^3_{\mathbb{Z}}\iota)$ is nonzero since σ is an isomorphism, so it is that same unique nonzero element. Comparing the two evaluations gives $\lambda_{p+1} = \lambda_p$. Setting $\lambda := \lambda_3$ yields $\theta_p = \lambda\,\mathrm{Sq}^3_{\mathbb{Z}}$ for every $p \geq 3$, and Step 2 extends the identity to $p \leq 2$, completing the proof.

What remains is to decide whether λ is 0 or 1, and a single space with a single nonvanishing $\mathrm{Sq}^3_{\mathbb{Z}}$ decides it, provided the spectral sequence can be shown not to collapse there. The deciding space is a three-cell complex: a 4-sphere, a 6-cell attached along the suspended Hopf map, which makes Sq^2 act, and a 7-cell attached along a doubled map, which makes the Bockstein act. K-theory detects the bottom cell of this complex through second Chern classes, and corollary 9.2.13 turns that detection into the non-collapse.

A class in $\widetilde{K}^0$ of a finite complex is a based homotopy class of maps into $\mathbb{Z} \times BU$. Compactness pushes such a map into a single Grassmannian $BU(N)$, and the bundle it pulls back there has a second Chern class that depends only on the class.

Lemma 9.5.6: Second Chern Classes of Reduced K-Theory Classes

Give $BU(n) = \mathrm{Gr}_n(\mathbb{C}^\infty)$ its tautological bundle γ_n, whose fiber over a plane V is V, and the basepoint $\mathbb{C}^n = \mathrm{span}(e_1, \dots, e_n)$. Let BU be the colimit of the $BU(n)$ along the stabilizations $i_{n+1}(V) = \mathbb{C}e_1 \oplus s(V)$ of definition 7.1.20, in which these basepoints agree, and base $KU_0 = \mathbb{Z} \times BU$ of theorem 8.5.10 at that common basepoint in $\{0\} \times BU$. Let Y be a connected based finite CW complex.

1. Every class $v \in \widetilde{K}^0(Y) = [Y, KU_0]_*$ is the class of a based map $h\colon Y \to BU(N) \subseteq \{0\} \times BU$ for some N.
2. For v represented by h as in (1), the class $c_2(v) := c_2(h^*\gamma_N) \in H^4(Y;\mathbb{Z})$ depends only on v.
3. On $Y = S^4$ the map $c_2\colon \widetilde{K}^0(S^4) \to H^4(S^4;\mathbb{Z})$ is an isomorphism of infinite cyclic groups.

***Proof*:**

Step 1: The model. Each i_{n+1} is a based closed embedding (proposition 7.1.19(1)). Each $BU(n)$ carries the subspace topology of BU, and every compact subset of BU lies in some $BU(N)$ (corollary 7.1.21). So a map from a compact space into BU is a continuous map into some $BU(N)$.

Step 2: Part (1). The complex Y is compact, as a finite union of images of disks, and Hausdorff (theorem 1.2.3). A based map $Y \to \mathbb{Z} \times BU$ has connected image containing the basepoint, and the sets $\{k\} \times BU$ are open and closed, so the map lands in $\{0\} \times BU = BU$ and then, by Step 1, in some $BU(N)$. This proves (1).

Step 3: Stability of the Chern classes. Let $h\colon C \to BU(n)$ be a map from a compact Hausdorff space. Pulling back the isomorphism of proposition 7.1.19(3) along h gives an isomorphism of vector bundles $\underline{\mathbb{C}} \oplus h^*\gamma_n \to (i_{n+1} \circ h)^*\gamma_{n+1}$. A trivial bundle is pulled back from a point, whose cohomology vanishes in positive degrees, so its total Chern class is 1 by proposition 7.4.1. The Whitney formula (theorem 7.4.5, the base C being paracompact Hausdorff) then gives $c\big((i_{n+1} \circ h)^*\gamma_{n+1}\big) = c(h^*\gamma_n)$. By induction, the total Chern class of $h^*\gamma_n$ does not change when h is followed by the inclusion $BU(n) \subseteq BU(m)$ for any $m \geq n$.

Step 4: Part (2). Let $h\colon Y \to BU(N)$ and $h'\colon Y \to BU(N')$ be based maps representing the same class v. A based homotopy between them in KU_0 is a map from the compact connected space $Y \times I$, so by the argument of Step 2 it is a homotopy $H\colon Y \times I \to BU(M)$, for some $M \geq N, N'$, from h to h' regarded as maps into $BU(M)$. Homotopic maps pull back isomorphic

bundles (theorem 7.1.29, the compact Hausdorff space Y being paracompact), so Step 3 gives

$$c_2(h^*\gamma_N) = c_2(H_0^*\gamma_M) = c_2(H_1^*\gamma_M) = c_2(h'^*\gamma_{N'})$$

Step 5: The two groups. By literal 2-periodicity $\widetilde{K}^0(S^4) = \widetilde{K}^4(S^4)$, and four suspension isomorphisms (theorem 8.4.6(4), with $S^4 = \Sigma^4 S^0$ by example 5.7.2) identify this group with $\widetilde{K}^0(S^0) = K^0(\mathrm{pt}) \cong \mathbb{Z}$ (theorem 8.5.10(3)). Also $H^4(S^4;\mathbb{Z}) = \widetilde{H}^4(S^4;\mathbb{Z}) \cong \mathbb{Z}$ (proposition 4.1.14, example 8.2.2).

Step 6: Additivity on $\widetilde{K}^0(S^4)$. Let $v_1, v_2 \in \widetilde{K}^0(S^4)$. By (1) they are represented by based maps $h_1, h_2 \colon S^4 \to BU(N)$ with a common N, since following a map into $BU(N_k)$ by the inclusion into a larger $BU(N)$ does not change the map into BU. Write $S^4 = \Sigma S^3$, let $P \colon S^4 \to S^4 \vee S^4$ be the pinch map and ι_1, ι_2 the two wedge inclusions, and let $F \colon S^4 \vee S^4 \to BU(N)$ be h_1 on the first summand and h_2 on the second. The composite $\nabla \circ (\iota_1 \vee \iota_2)$ is the identity of $S^4 \vee S^4$, so the pinch sum $\iota_1 + \iota_2$ of lemma 9.5.1 is P, and part (2) of that lemma, with $W = S^3$ and $Z = S^4 \vee S^4$, gives $P^* = \iota_1^* + \iota_2^*$ in every reduced cohomology theory. In $\widetilde{K}^0$, applied to the class of F, this says that the based map $F \circ P \colon S^4 \to BU(N)$ represents $[F \circ \iota_1] + [F \circ \iota_2] = v_1 + v_2$. In reduced singular cohomology, applied to $c_2(F^*\gamma_N) \in H^4(S^4 \vee S^4;\mathbb{Z}) = \widetilde{H}^4(S^4 \vee S^4;\mathbb{Z})$ and combined with proposition 7.4.1, it gives

$$c_2(v_1+v_2) = P^* c_2(F^*\gamma_N) = \iota_1^* c_2(F^*\gamma_N) + \iota_2^* c_2(F^*\gamma_N) = c_2(h_1^*\gamma_N) + c_2(h_2^*\gamma_N) = c_2(v_1) + c_2(v_2)$$

Step 7: Surjectivity, and part (3). The sphere S^4 is a CW complex of dimension four, so by theorem 7.5.4 it carries a rank-two bundle E_0 with $c_1(E_0) = 0$ and $c_2(E_0)$ a generator of $H^4(S^4;\mathbb{Z})$. By theorem 7.1.29, $E_0 \cong g^*\gamma_2$ for a map $g \colon S^4 \to BU(2)$. The space $BU(2)$ is path-connected (proposition 7.5.1), so a path from the image of the basepoint s_0 to $\mathbb{C}^2$, extended to a homotopy of g by the homotopy extension property of the CW pair $(S^4, \{s_0\})$ (proposition 1.3.4), deforms g to a based map g' with $g'^*\gamma_2 \cong E_0$ by the same theorem. The class of g' therefore has c_2 equal to the generator $c_2(E_0)$. By Steps 5 and 6, c_2 is a homomorphism between infinite cyclic groups whose image contains a generator, so it is onto. Its kernel is then a subgroup of $\mathbb{Z}$ with quotient $\mathbb{Z}$, and a nonzero subgroup of $\mathbb{Z}$ has finite quotient, so the kernel is zero. This proves (3), completing the proof.

Part (3) compares a spectrum computation with a bundle computation. The group $\widetilde{K}^0(S^4) \cong \mathbb{Z}$ comes only from the coefficients of KU and periodicity, and the second Chern class matches it with $H^4(S^4;\mathbb{Z})$: a based map $S^4 \to BU(N)$ represents a generator of $\widetilde{K}^0(S^4)$ exactly when the bundle it pulls back has a generator as its second Chern class.

Example 9.5.7: The Complex $S^4 \cup_{\eta_4} e^6 \cup_2 e^7$ Detects d_3

Write $\eta_4 = \Sigma^2\eta \colon S^5 \to S^4$ for the twice-suspended Hopf map, as in proposition 9.2.3 (the Hopf map $\eta \colon S^3 \to S^2$ is example 5.2.7), and let $2\iota \colon S^5 \to S^5$ denote the pinch sum $\mathrm{id} + \mathrm{id}$ in the notation of lemma 9.5.1, formed on $S^5 = \Sigma S^4$ (example 5.7.2). This example constructs a based 7-dimensional finite CW complex X with

$$\widetilde{H}^*(X;\mathbb{Z}) \colon \quad \mathbb{Z} \text{ in degree } 4, \qquad \mathbb{Z}/2 \text{ in degree } 7, \qquad 0 \text{ otherwise}$$

and establishes three facts: that $\mathrm{Sq}^3_{\mathbb{Z}}$ is nonzero on $H^4(X;\mathbb{Z})$, that the generator of $H^4(X;\mathbb{Z})$ does not survive to E_∞ in the Atiyah-Hirzebruch spectral sequence for $\widetilde{K}^*$, and that $\widetilde{K}^0(X) \cong \mathbb{Z}$ and $\widetilde{K}^1(X) = 0$.

Step 1: The complex. Composition with η_4 commutes with fold maps, so $\eta_4 \circ 2\iota = \eta_4 \circ \nabla \circ (\mathrm{id} \vee \mathrm{id}) \circ P = \nabla \circ (\eta_4 \vee \eta_4) \circ P = \eta_4 + \eta_4$. The pinch sum of based maps $S^5 \to S^4$ represents the group operation of $\pi_5(S^4)$ (definition 5.7.3), and $\pi_5(S^4) \cong \mathbb{Z}/2$ (corollary 6.3.7) is a group in which every element added to itself gives the identity. Hence $\eta_4 \circ 2\iota$ is based-nullhomotopic. A based nullhomotopy is a map $H\colon CS^5 \to S^4$ restricting to $\eta_4 \circ 2\iota$ on the base of the cone, and gluing H to η_4 defines

$$g\colon M := C_{2\iota} = S^5 \cup_{2\iota} CS^5 \longrightarrow S^4, \qquad g|_{S^5} = \eta_4$$

on the mapping cone M of definition 8.1.7: a cone-base point lying over $s \in S^5$ is identified in M with $2\iota(s)$, and H sends it to $\eta_4(2\iota(s))$, so the two prescriptions agree where they meet. Set

$$X := C_g = S^4 \cup_g CM$$

Give S^4 and S^5 their minimal structures $e^0 \cup e^4$ and $e^0 \cup e^5$. For a based finite CW complex B with basepoint b_0 a 0-cell, the reduced cone CB is the quotient of the product complex $B \times I$ (definition 1.2.8, a CW complex in the product topology by lemma 1.2.12(3)) by its subcomplex $B \times \{1\} \cup \{b_0\} \times I$. By definition 1.2.10 it is a CW complex containing the base $B = B \times \{0\}$ as a subcomplex, with one $(k+1)$-cell $e \times (0,1)$ for each k-cell $e \neq b_0$ of B. For a based cellular map $f\colon B \to T$ of finite complexes, the mapping cone $C_f = T \cup_f CB$ is T with the cells of $CB \setminus B$ attached in order of dimension. The characteristic map of the cell $e \times (0,1)$, for a k-cell e, is its characteristic map in CB followed by the quotient map to C_f, and its boundary lands in the k-skeleton of C_f: the part of the boundary lying in B lies in $B^{(k)}$, which f carries into $T^{(k)}$, and the rest lies in the cone on $B^{(k-1)}$. The map 2ι is cellular, since below dimension five both skeleta of S^5 are the basepoint, so M has one cell in each of the dimensions $0, 5, 6$. Then g is cellular too, since below dimension five M is the basepoint and from dimension four on the skeleta of S^4 are all of S^4, so X has one cell in each of the dimensions $0, 4, 6, 7$: the 6-cell is the cone on the 5-cell of M and the 7-cell is the cone on its 6-cell. In particular, using the equality of maps $g|_{S^5} = \eta_4$,

$$X^{(3)} = \{x_0\}, \qquad X^{(4)} = X^{(5)} = S^4, \qquad X^{(6)} = S^4 \cup_{\eta_4} CS^5 = C_{\eta_4}, \qquad X^{(7)} = X$$

Step 2: Ordinary cohomology. All computations run along the Puppe sequences of 2ι and of g (proposition 8.1.8(3)), applied to $[-, K(\mathbb{Z}, n)]_*$ and $[-, K(\mathbb{Z}/2, n)]_*$ and transported to reduced cohomology by the natural bijection of theorem 5.5.12. The resulting sequences of reduced cohomology groups are exact, every map being an induced homomorphism whose kernel is the preimage of 0. The values on spheres are $\widetilde{H}^n(S^m; G) = G$ for $n = m$ and 0 otherwise (example 8.2.2). The maps appearing in the Puppe sequence of 2ι are the suspensions $\Sigma^j(2\iota)$, and $\Sigma^j(2\iota) \simeq \mathrm{id} + \mathrm{id}$ on S^{5+j}, because the suspension homomorphism of definition 5.7.3 carries pinch sums to pinch sums and the suspension of the identity is the identity. By lemma 9.5.1(2), each of these maps induces multiplication by 2 on every reduced cohomology group of the sphere. With coefficients $\mathbb{Z}$, exactness of

$$\widetilde{H}^n(S^6; \mathbb{Z}) \xrightarrow{2} \widetilde{H}^n(S^6; \mathbb{Z}) \longrightarrow \widetilde{H}^n(M; \mathbb{Z}) \longrightarrow \widetilde{H}^n(S^5; \mathbb{Z}) \xrightarrow{2} \widetilde{H}^n(S^5; \mathbb{Z})$$

at the two middle spots gives two values. For $n = 5$, the group $\widetilde{H}^5(M; \mathbb{Z})$ injects into $\ker(2\colon \mathbb{Z} \to \mathbb{Z}) = 0$, its left neighbor vanishing, so $\widetilde{H}^5(M; \mathbb{Z}) = 0$. For $n = 6$, the map from $\widetilde{H}^6(S^6; \mathbb{Z})$ is onto with kernel $2\mathbb{Z}$, so $\widetilde{H}^6(M; \mathbb{Z}) \cong \mathbb{Z}/2$. All other degrees vanish. With coefficients $\mathbb{Z}/2$ the maps labelled 2 are zero, and the same exactness gives $\widetilde{H}^5(M; \mathbb{Z}/2) \cong \widetilde{H}^6(M; \mathbb{Z}/2) \cong \mathbb{Z}/2$. One suspension later in the same Puppe sequence, the identical pattern computes ΣM: integrally $\mathbb{Z}/2$ in degree 7 and nothing else, and mod 2 a copy of $\mathbb{Z}/2$ in each of the degrees 6 and 7. Finally, with either

coefficient group, the Puppe sequence of g, namely $M \to S^4 \to X \to \Sigma M \to S^5 \to \cdots$, gives in degree 4 an isomorphism $i^*\colon \widetilde{H}^4(X;G) \to \widetilde{H}^4(S^4;G)$ along $i\colon S^4 \hookrightarrow X$, the neighbors $\widetilde{H}^4(\Sigma M;G)$ and $\widetilde{H}^4(M;G)$ vanishing. In degree $n \neq 4$ the group $\widetilde{H}^n(X;G)$ sits between $\widetilde{H}^n(\Sigma M;G)$ and $\widetilde{H}^n(S^4;G) = 0$, so it is a quotient of $\widetilde{H}^n(\Sigma M;G)$. That quotient vanishes for $n \notin \{6,7\}$, since $\widetilde{H}^n(\Sigma M;G) = 0$ there. For $n \in \{6,7\}$ the term $\widetilde{H}^n(S^5;G)$ before $\widetilde{H}^n(\Sigma M;G)$ also vanishes, so $\widetilde{H}^n(\Sigma M;G) \cong \widetilde{H}^n(X;G)$. Assembling, and fixing a generator x_4 of $H^4(X;\mathbb{Z}) \cong \mathbb{Z}$:

$$\widetilde{H}^*(X;\mathbb{Z}) = \begin{cases} \mathbb{Z}\langle x_4\rangle & *=4 \\ \mathbb{Z}/2 & *=7 \\ 0 & \text{else} \end{cases} \qquad \widetilde{H}^*(X;\mathbb{Z}/2) = \begin{cases} \mathbb{Z}/2 & *=4,6,7 \\ 0 & \text{else} \end{cases}$$

Step 3: $\mathrm{Sq}^3_{\mathbb{Z}} x_4 \neq 0$. Let $w\colon C_{\eta_4} = X^{(6)} \hookrightarrow X$ be the skeletal inclusion. The restriction w^*x_4 generates $H^4(C_{\eta_4};\mathbb{Z}) \cong \mathbb{Z}$: the Puppe sequence of η_4 identifies $H^4(C_{\eta_4};\mathbb{Z}) \cong H^4(S^4;\mathbb{Z})$ by restriction, the composite of the two restrictions is the isomorphism i^* of Step 2, and an isomorphism carries generators to generators. By proposition 9.2.3, $\mathrm{Sq}^2\colon H^4(C_{\eta_4};\mathbb{Z}/2) \to H^6(C_{\eta_4};\mathbb{Z}/2)$ is an isomorphism of groups of order two. Since $\ker\rho = 2H^4(C_{\eta_4};\mathbb{Z})$ (proposition 9.1.2) and w^*x_4 generates $H^4(C_{\eta_4};\mathbb{Z}) \cong \mathbb{Z}$, the reduction $\rho(w^*x_4)$ is the nonzero element of $H^4(C_{\eta_4};\mathbb{Z}/2)$. Naturality of Sq^2 (theorem 9.1.17(1)) and of ρ (proposition 9.1.2) then gives

$$w^*\big(\mathrm{Sq}^2\rho\, x_4\big) = \mathrm{Sq}^2\rho\,(w^*x_4) \neq 0$$

whence $\mathrm{Sq}^2\rho\, x_4 \neq 0$ in $H^6(X;\mathbb{Z}/2) \cong \mathbb{Z}/2$. In the Bockstein sequence of proposition 9.1.2 for X,

$$H^6(X;\mathbb{Z}) \xrightarrow{\ \rho\ } H^6(X;\mathbb{Z}/2) \xrightarrow{\ \beta\ } H^7(X;\mathbb{Z}) \xrightarrow{\ 2\ } H^7(X;\mathbb{Z})$$

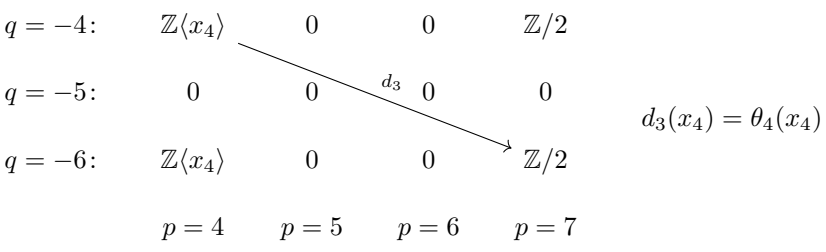
the group $H^6(X;\mathbb{Z})$ vanishes by Step 2, so β is injective on $H^6(X;\mathbb{Z}/2)$ and

$$\mathrm{Sq}^3_{\mathbb{Z}} x_4 = \beta\,\mathrm{Sq}^2\rho\, x_4 \neq 0 \in H^7(X;\mathbb{Z}) \cong \mathbb{Z}/2$$

Step 4: The class x_4 does not survive. Consider the reduced Atiyah-Hirzebruch spectral sequence of $\widetilde{K}^*$ on X (lemma 9.5.2). The complex X is 7-dimensional, so the convergence clause applies. By Step 2 the second page is concentrated in the columns $p = 4$ and $p = 7$ and the even rows:

$$\widetilde{E}_2^{4,q} \cong \mathbb{Z}, \qquad \widetilde{E}_2^{7,q} \cong \mathbb{Z}/2 \qquad (q \text{ even})$$

and every other spot vanishes. No differential ever enters the column $p = 4$, the columns $4 - r$ being empty for every $r \geq 2$, and the only differential that can leave either column is d_3, of bidegree $(3,-2)$, from column 4 into column 7: every other candidate lands in an empty column, in an odd row, or outside the strip $0 \leq p \leq 7$. The populated part of the third page around total degrees 0 and 1, with the one possible differential drawn, is



$$d_3(x_4) = \theta_4(x_4)$$

the identification of d_3 with θ_4 being lemma 9.5.4. Suppose x_4 survives, that is, $d_3 x_4 = 0$. Then $x_4 \in \widetilde{E}_\infty^{4,-4}$, and the convergence identification of lemma 9.5.2(3) produces a class $u \in F^4\widetilde{K}^0(X)$ whose class in $F^4/F^5 \cong \widetilde{E}_\infty^{4,-4}$ is x_4. Here $F^4 = \ker\big(\widetilde{K}^0(X) \to \widetilde{K}^0(X^{(3)})\big) = \widetilde{K}^0(X)$, since $X^{(3)}$ is the basepoint and $\widetilde{K}^0$ of a point vanishes (lemma 9.5.1(1)). The skeletal inclusion $S^4 = X^{(4)} \hookrightarrow X$ is based cellular, and the naturality clauses of lemma 9.5.2 give a commutative comparison between X and S^4. For S^4 the reduced spectral sequence has the single column $p = 4$, every differential vanishes, and the filtration reads $F^4\widetilde{K}^0(S^4) = \widetilde{K}^0(S^4)$ and $F^5\widetilde{K}^0(S^4) = 0$, so the convergence identification is a group isomorphism

$$\widetilde{K}^0(S^4) \xrightarrow{\cong} \widetilde{E}_\infty^{4,-4}(S^4) = \widetilde{H}^4(S^4;\mathbb{Z})$$

Under the comparison, the restriction $u|_{S^4}$ corresponds to the restriction of x_4, which generates $\widetilde{H}^4(S^4;\mathbb{Z})$ by Step 2. Hence $u|_{S^4}$ is a generator of $\widetilde{K}^0(S^4) \cong \mathbb{Z}$.

Represent u by a based map $h\colon X \to BU(N)$ as in lemma 9.5.6(1), which applies because the finite complex X has a single 0-cell and is therefore connected, and set $E := h^*\gamma_N$, a complex vector bundle over X. The based map $h \circ i\colon S^4 \to BU(N)$ represents $i^*u = u|_{S^4}$, so lemma 9.5.6(2) and proposition 7.4.1 give

$$i^*c_2(E) = c_2\big((h \circ i)^*\gamma_N\big) = c_2\big(u|_{S^4}\big)$$

The isomorphism c_2 of lemma 9.5.6(3) carries the generator $u|_{S^4}$ to a generator of $H^4(S^4;\mathbb{Z})$, and i^* is an isomorphism carrying x_4 to a generator (Step 2), so $c_2(E) = \pm x_4$. The base X is compact and Hausdorff (theorem 1.2.3), hence paracompact, so corollary 9.2.13 applies to E. With the additivity of $\mathrm{Sq}^3_{\mathbb{Z}}$ (proposition 9.1.21) it gives

$$\mathrm{Sq}^3_{\mathbb{Z}} x_4 = \pm\,\mathrm{Sq}^3_{\mathbb{Z}} c_2(E) = 0$$

contradicting Step 3. Therefore x_4 does not survive, and $d_3 x_4 \neq 0$.

Step 5: The K-theory of X. Since $d_3 x_4$ generates $\widetilde{E}_3^{7,-6} \cong \mathbb{Z}/2$, the fourth page has $\widetilde{E}_4^{4,q} = \ker(d_3) = 2\mathbb{Z} \cong \mathbb{Z}$ and $\widetilde{E}_4^{7,q} = 0$ in even rows, and no later differential can act, so $\widetilde{E}_4 = \widetilde{E}_\infty$. Reading the diagonals: in total degree 0 the only contribution is $\widetilde{E}_\infty^{4,-4} \cong \mathbb{Z}$, and the filtration of lemma 9.5.2(3) has this as its only nonzero quotient, so $\widetilde{K}^0(X) \cong \mathbb{Z}$. In total degree 1 every spot vanishes, the odd rows by parity and $\widetilde{E}_\infty^{7,-6} = 0$, so $\widetilde{K}^1(X) = 0$.

The Puppe sequence of g confirms this and ties the vanishing of $\widetilde{K}^1(X)$ to the non-collapse. The values $\widetilde{K}^n(S^m)$ are those of example 8.2.2 with the coefficients of theorem 8.5.10, and the Puppe sequence of 2ι applied to $\widetilde{K}^* = [-, KU_*]_*$, with $(\Sigma^j 2\iota)^*$ acting as multiplication by 2 exactly as in Step 2, gives by the same exactness pattern

$$\widetilde{K}^0(M) \cong \mathbb{Z}/2, \quad \widetilde{K}^1(M) = 0, \qquad \widetilde{K}^0(\Sigma M) = 0, \quad \widetilde{K}^1(\Sigma M) \cong \mathbb{Z}/2$$

the last two from the first two by the suspension isomorphism of theorem 8.4.6 and the literal 2-periodicity of $\widetilde{K}^*$. The Puppe sequence of g then reads, around total degree 1,

$$\widetilde{K}^1(S^5) \longrightarrow \widetilde{K}^1(\Sigma M) \longrightarrow \widetilde{K}^1(X) \longrightarrow \widetilde{K}^1(S^4) = 0$$

so $\widetilde{K}^1(X)$ is a quotient of $\widetilde{K}^1(\Sigma M) \cong \mathbb{Z}/2$ and vanishes exactly when the first map, which the suspension isomorphism conjugates to $g^*\colon \widetilde{K}^0(S^4) \to \widetilde{K}^0(M)$, is onto. Around total degree 0 it reads

$$0 = \widetilde{K}^0(\Sigma M) \longrightarrow \widetilde{K}^0(X) \longrightarrow \widetilde{K}^0(S^4) \xrightarrow{g^*} \widetilde{K}^0(M)$$

The spectral-sequence computation above forces $g^* \neq 0$: were $g^* = 0$, exactness would give $\widetilde{K}^1(X) \cong \mathbb{Z}/2$, contradicting $\widetilde{K}^1(X) = 0$. With g^* onto, exactness returns $\widetilde{K}^0(X) = \ker(g^*\colon \mathbb{Z} \to \mathbb{Z}/2) = 2\mathbb{Z} \cong \mathbb{Z}$ and $\widetilde{K}^1(X) = 0$. The two computations agree, and they agree precisely because $d_3 \neq 0$ on X.

The example pins the multiplier, and the identification of the differential follows.

Theorem 9.5.8: The Differential d_3 Equals $\mathrm{Sq}^3_{\mathbb{Z}}$ for KU

In the Atiyah-Hirzebruch spectral sequence of complex K-theory on a finite-dimensional CW complex X (theorem 8.6.4, and equally in the reduced spectral sequence of lemma 9.5.2 for based X), the third differential is the integral Steenrod operation of definition 9.1.20: for every p and every even q,

$$d_3 \;=\; \mathrm{Sq}^3_{\mathbb{Z}} \;=\; \beta\,\mathrm{Sq}^2\rho\colon\; E_3^{p,q} = H^p(X;\mathbb{Z}) \longrightarrow E_3^{p+3,q-2} = H^{p+3}(X;\mathbb{Z})$$

***Proof*:**

By lemma 9.5.4 and proposition 9.5.5, $d_3 = \theta_p = \lambda\,\mathrm{Sq}^3_{\mathbb{Z}}$ with a single $\lambda \in \mathbb{Z}/2$ for all p, q and X, in the unreduced and the reduced spectral sequences alike. On the complex of example 9.5.7, $d_3x_4 \neq 0$ while $\mathrm{Sq}^3_{\mathbb{Z}}x_4$ generates a group of order two, and the identity $d_3x_4 = \lambda\,\mathrm{Sq}^3_{\mathbb{Z}}x_4$ then forces $\lambda = 1$, completing the proof.

The theorem identifies the operation that section 8.6 left unnamed: the first differential beyond the cellular one that complex K-theory carries is the composite of reduction mod 2, the second Steenrod square, and the integral Bockstein. It also explains the collapse computations of that section from one principle. On $\mathbb{C}\mathrm{P}^n$, and on every space whose integral cohomology is concentrated in even degrees, $\mathrm{Sq}^3_{\mathbb{Z}}$ shifts degree by three and therefore vanishes identically, so $d_3 = 0$. This is the operation-level reading of the parity argument of example 8.6.6. The differential is also a torsion phenomenon: $2\,\mathrm{Sq}^3_{\mathbb{Z}} = 0$ (proposition 9.1.21), so d_3 annihilates every doubled class and vanishes once 2 is inverted.

Running the identification backward converts the differential into an obstruction: a cohomology class that comes from K-theory must be annihilated by $\mathrm{Sq}^3_{\mathbb{Z}}$.

Corollary 9.5.9: Classes Surviving to K-theory are Annihilated by $\mathrm{Sq}^3_{\mathbb{Z}}$

Let X be a finite-dimensional CW complex and $x \in H^p(X;\mathbb{Z})$. If x survives to E_∞ in the Atiyah-Hirzebruch spectral sequence of complex K-theory, and in particular if p is even and x is a representative of the class in $E_\infty^{p,-p} \cong F^pK^0(X)/F^{p+1}K^0(X)$ of some $u \in F^pK^0(X)$, then

$$\mathrm{Sq}^3_{\mathbb{Z}}\,x = 0$$

***Proof*:**

The differentials d_2 vanish (lemma 9.5.4), so $E_3^{p,q} = E_2^{p,q} = H^p(X;\mathbb{Z})$ in even rows. A class surviving to E_∞ is in particular a d_3-cycle on the third page, and $d_3 = \mathrm{Sq}^3_{\mathbb{Z}}$ by theorem 9.5.8, so $\mathrm{Sq}^3_{\mathbb{Z}}x = 0$. For the second clause, the convergence of theorem 8.6.4 identifies $F^pK^0(X)/F^{p+1}K^0(X)$ with $E_\infty^{p,-p}$, a quotient of the subgroup of permanent cycles in $E_2^{p,-p}$, which is $H^p(X;\mathbb{Z})$ because the row $-p$ is even. A representative x of the class of u is such a

permanent cycle, and the first clause applies, completing the proof.

On $\mathbb{C}\mathrm{P}^n$ the corollary is vacuous: the cohomology is even, every class survives, and $\mathrm{Sq}^3_{\mathbb{Z}}$ already vanishes for degree reasons, consistently with the collapse of example 8.6.6. Its content appears on spaces with torsion. The complex X of example 9.5.7 carries the integral class x_4 with $\mathrm{Sq}^3_{\mathbb{Z}} x_4 \neq 0$, and the corollary says that no K-theory class on X has x_4 as the representative of its leading filtration quotient, which is what Step 4 of that example verified at the level of Chern classes. Since $\mathrm{Sq}^3_{\mathbb{Z}}$ kills $2x$ for every x, the obstruction takes values in 2-torsion, invisible to rational cohomology, and this is the first point in the book at which K-theory and ordinary cohomology are separated by an odd-degree operation.

Exercise 9.5.1

1. Let X be a finite CW complex all of whose cells have even dimension. Show that $H^*(X;\mathbb{Z})$ is free abelian and concentrated in even degrees, that every differential of the Atiyah-Hirzebruch spectral sequence for K^* vanishes, and that
$$K^0(X) \cong \bigoplus_k H^{2k}(X;\mathbb{Z}), \qquad K^1(X) = 0$$
Conclude that corollary 9.5.9 is vacuous on such complexes.

2. Let $C_{\eta_4} = S^4 \cup_{\eta_4} e^6$ be the 6-skeleton of the complex of example 9.5.7. Compute $\widetilde{H}^*(C_{\eta_4};\mathbb{Z})$, show that $\mathrm{Sq}^2\rho$ is nonzero on $H^4(C_{\eta_4};\mathbb{Z})$ while $\mathrm{Sq}^3_{\mathbb{Z}}$ vanishes there, and compute $\widetilde{K}^0(C_{\eta_4}) \cong \mathbb{Z}^2$ and $\widetilde{K}^1(C_{\eta_4}) = 0$. Identify the group in which $\mathrm{Sq}^3_{\mathbb{Z}}$ of a generator of H^4 lands, and explain how its value on the two complexes separates this computation from that of example 9.5.7.

3. Let X be the complex of example 9.5.7. Compute $\widetilde{K}^0(\Sigma X)$ and $\widetilde{K}^1(\Sigma X)$ twice: once from the suspension isomorphism and periodicity of $\widetilde{K}^*$ together with the values computed there, and once by running the reduced Atiyah-Hirzebruch spectral sequence of ΣX with theorem 9.5.8, using the stability of $\mathrm{Sq}^3_{\mathbb{Z}}$ (proposition 9.1.21) to evaluate d_3 on the suspended generator.

4. Deduce from $d_3 \circ d_3 = 0$, proposition 9.5.5 and theorem 9.5.8 that
$$\mathrm{Sq}^3_{\mathbb{Z}} \circ \mathrm{Sq}^3_{\mathbb{Z}} = 0 \colon H^p(X;\mathbb{Z}) \longrightarrow H^{p+6}(X;\mathbb{Z})$$
for every CW complex X and every p. (The identity $\theta_p = \lambda\,\mathrm{Sq}^3_{\mathbb{Z}}$ of the proposition holds on every CW complex, finite-dimensional or not, and theorem 9.5.8 forces $\lambda = 1$.)

5. For $n \geq 2$ let $Y_n = S^n \cup_{2\iota} e^{n+1}$ be the mapping cone of a pinch sum $\mathrm{id} + \mathrm{id}$ on S^n, as in Step 1 of example 9.5.7. Compute $\widetilde{H}^*(Y_n;\mathbb{Z})$, show that the Atiyah-Hirzebruch spectral sequence of $\widetilde{K}^*$ on Y_n has no room for a nonzero differential, and conclude
$$\widetilde{K}^0(Y_n) \cong \begin{cases} \mathbb{Z}/2 & n \text{ odd} \\ 0 & n \text{ even} \end{cases} \qquad \widetilde{K}^1(Y_n) \cong \begin{cases} 0 & n \text{ odd} \\ \mathbb{Z}/2 & n \text{ even} \end{cases}$$
Check the case $n = 5$ against the groups $\widetilde{K}^*(M)$ computed in example 9.5.7.

10

What Next?

This book has carried the reader from the fundamental group and covering spaces through singular and cellular homology, cohomology rings and Poincaré duality, the higher homotopy groups, spectral sequences, characteristic classes, and a first encounter with stable homotopy theory through Brown representability and spectra. Each of these is the entrance to a subject far larger than any one chapter can hold. This closing chapter is a map of where to go next, pairing every direction with the companion EYNTKA volume where one exists and with the standard texts a working topologist keeps within reach.

Consolidating the foundations. The closest companions for a second pass over this material are Hatcher's *Algebraic Topology*, the source whose organisation this book most often follows; May's *A Concise Course in Algebraic Topology*, terser and more categorical; tom Dieck's *Algebraic Topology*; and, as a classical reference, Spanier's *Algebraic Topology*. Bredon's *Topology and Geometry* threads the subject together with manifold theory from the outset.

Homotopy theory and the stable category. chapter 5 and chapter 8 only opened the door to homotopy theory and spectra. A systematic treatment of the stable category is the natural sequel: Adams's *Stable Homotopy and Generalised Homology*, Switzer's *Algebraic Topology: Homotopy and Homology*, and the modern *Foundations of Stable Homotopy Theory* of Barnes and Roitzheim. The abstract homotopy theory beneath these (model categories, and past them ∞-categories) is the companion volume *Higher Category Theory* [12], and is built in Hovey's *Model Categories*, Riehl's *Categorical Homotopy Theory*, Goerss and Jardine's *Simplicial Homotopy Theory*, and Lurie's *Higher Topos Theory*. The computational frontier, the stable homotopy groups of spheres whose first nonzero value $\pi_1^s \cong \mathbb{Z}/2$ was identified in section 5.7, is the subject of Ravenel's *Complex Cobordism and Stable Homotopy Groups of Spheres* and *Nilpotence and Periodicity in Stable Homotopy Theory*.

Vector bundles, K-theory, and characteristic classes. chapter 7 computed the cohomology of BO and BU and built the Stiefel-Whitney, Chern, and Euler classes. The reference that develops them in full, together with the Pontryagin classes and the Thom isomorphism, is Milnor and Stasheff's *Characteristic Classes*. The K-theory those invariants feed is taken up in the companion volume

Vector Bundles and K-Theory ([33]), covering topological K^0 and K^1, Bott periodicity, and the splitting principle. Its algebraic and operator-theoretic siblings are *Algebraic K-Theory* ([7]: K_0, K_1, K_2, and Quillen's higher K-theory) and *C*-Algebras and K-Theory* ([8]). Atiyah's *K-theory* remains the classic short account.

The differentiable and geometric side. Much of the homotopy-invariant machinery of this book has a smooth-manifold counterpart that is frequently more computable. de Rham cohomology, Morse theory, and the geometry of bundles appear in Bott and Tu's *Differential Forms in Algebraic Topology*, in Milnor's *Morse Theory* and *Topology from the Differentiable Viewpoint* ([27]), and in the companion *Differential Topology* ([32]).

Homological algebra and spectral sequences. The algebra behind chapter 3, chapter 4, and chapter 6, the derived functors of the universal coefficient and Künneth theorems and the machinery of the Serre spectral sequence, is developed for its own sake in the companion *Homological Algebra* ([13]) and in Weibel's *An Introduction to Homological Algebra*. Spectral sequences as a practical craft are the subject of McCleary's *A User's Guide to Spectral Sequences*. The categorical language used throughout is laid out in *Category Theory* ([9]).

Cohomology in algebra and arithmetic. The cohomological methods of this book reappear across the series in new settings: sheaf cohomology and the cohomology of schemes in *Algebraic Geometry* ([6]), group and Galois cohomology with its arithmetic duality theorems in *Group and Galois Cohomology* ([11]), and Hodge theory on Kähler manifolds in the Hodge-theory volumes. Milne's *Étale Cohomology* is the standard bridge from the topological theory to its arithmetic counterpart.

A closing suggestion. The surest way to consolidate this material is to compute with it. Re-deriving, without looking, the groups and rings tabulated in appendix A, and pushing one genuinely nontrivial example through each major tool (a long exact sequence, a spectral sequence, a Gysin sequence, an obstruction class), fixes the machinery in a way no second reading can. From there, every direction above is open.

A

Common Structures and Related Algebraic Structure

This appendix collects, for quick reference, common spaces together with their fundamental groups and (co)homology groups. Every entry can be derived by the methods of the preceding chapters. The tables are meant as a check on those computations and as a map of the small zoo of spaces that recur throughout the subject.

A.1 Δ-Complex and Cell Complexes

The torus T^2, the Klein bottle K, and the projective plane $\mathbb{R}\mathrm{P}^2$ all arise as quotients of a single square with its four edges identified in pairs. The three identification patterns, and the Δ-complex structures they induce, are shown below. Reading the boundary word of each square off its identifications gives the standard presentations $\langle a, b \mid aba^{-1}b^{-1}\rangle$, $\langle a, b \mid abab^{-1}\rangle$, and $\langle a \mid a^2\rangle$ of chapter 2, and the induced cellular chain complexes compute the homology recorded in appendix A.3.

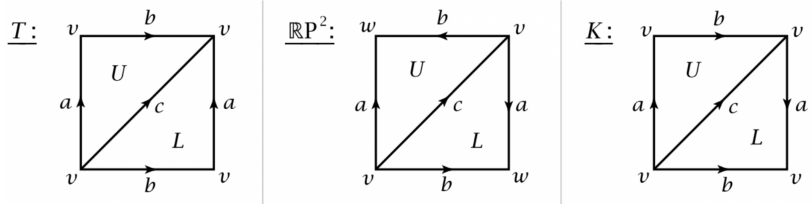


Figure A.1: Edge identifications and Δ-complex structures on the square presenting T^2, K, and $\mathbb{R}\mathrm{P}^2$.

A.2 Fundamental Groups and Covering Spaces

The fundamental groups of the spaces met most often are collected below. Each may be computed by the van Kampen and covering-space methods of chapter 2. The abelianisation of each group recovers the first homology H_1 of the corresponding row in the table of appendix A.3.

Space	Fundamental group π_1
S^1	$\mathbb{Z}$
S^n, $n \geq 2$	0
$T^2 = S^1 \times S^1$	$\mathbb{Z}^2$
$T^n = (S^1)^n$	$\mathbb{Z}^n$
$\bigvee_{i=1}^{k} S^1$	free group F_k of rank k
$\bigvee_{i=1}^{k} S^n$, $n \geq 2$	0
M_g (orientable surface, genus g)	$\left\langle a_1, b_1, \ldots, a_g, b_g \;\middle\vert\; \prod_{i=1}^{g} [a_i, b_i] \right\rangle$
N^g (non-orientable, g cross-caps)	$\left\langle a_1, \ldots, a_g \;\middle\vert\; a_1^2 a_2^2 \cdots a_g^2 \right\rangle$
K (Klein bottle $= N_2$)	$\langle a, b \vert abab^{-1} \rangle$
$\mathbb{R}\mathrm{P}^1 = S^1$	$\mathbb{Z}$
$\mathbb{R}\mathrm{P}^n$, $n \geq 2$	$\mathbb{Z}/2\mathbb{Z}$
$\mathbb{C}\mathrm{P}^n$	0
$L(p;q)$ (lens space)	$\mathbb{Z}/p\mathbb{Z}$

A.3 Homologies and Cohomologies

Note that $H_0(X) \cong \tilde{H}_0(X) \oplus \mathbb{Z}$. Hence, $\tilde{H}_0(X)$ shall always be omitted, since we are only working with path-connected spaces. To calculate the cohomology groups, if $T_n \subseteq H_n$ is the torsion part:

$$H^n(C;\mathbb{Z}) \cong (H_n/T_n) \oplus T_{n-1}$$

i.e., the torsion part "moves up" while the torsion-free part stays put.

Shape	Homology	Cohomology with $\mathbb{Z}$
$S^1 \simeq M$[1]	$\tilde{H}_n(S^1) = \begin{cases} \mathbb{Z} & n = 1 \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(S^1;G) = \begin{cases} G & n = 1 \\ 0 & \text{otherwise} \end{cases}$
S^k	$\tilde{H}_n(S^k) = \begin{cases} \mathbb{Z} & n = k \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(S^k;G) = \begin{cases} G & n = k \\ 0 & \text{otherwise} \end{cases}$
$S^1 \times S^1 = T^2$	$\tilde{H}_n(T^2) = \begin{cases} \mathbb{Z}^2 & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(T^2;\mathbb{Z}) = \begin{cases} \mathbb{Z}^2 & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & \text{otherwise} \end{cases}$
M_g	$\tilde{H}_n(M_g) = \begin{cases} \mathbb{Z}^{2g} & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(M_g;\mathbb{Z}) = \begin{cases} \mathbb{Z}^{2g} & n = 1 \\ \mathbb{Z} & n = 2 \\ 0 & \text{otherwise} \end{cases}$

[1] Möbius Band

K	$\tilde{H}_n(K) = \begin{cases} \mathbb{Z} \oplus \mathbb{Z}_2 & n = 1 \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(K;\mathbb{Z}) = \begin{cases} \mathbb{Z} & n = 1 \\ \mathbb{Z}_2 & n = 2 \\ 0 & \text{otherwise} \end{cases}$
$\mathbb{R}\mathrm{P}^2$	$\tilde{H}_n(\mathbb{R}\mathrm{P}^2) = \begin{cases} \mathbb{Z}_2 & n = 1 \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(\mathbb{R}\mathrm{P}^2;\mathbb{Z}) = \begin{cases} 0 & n = 1 \\ \mathbb{Z}/2\mathbb{Z} & n = 2 \\ 0 & \text{otherwise} \end{cases}$
$\mathbb{R}\mathrm{P}^k$, k odd	$\tilde{H}_n(\mathbb{R}\mathrm{P}^k) = \begin{cases} \mathbb{Z}_2 & n \text{ odd},\ 1 \le n \le k-2 \\ \mathbb{Z} & n = k \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(\mathbb{R}\mathrm{P}^k;\mathbb{Z}) = \begin{cases} \mathbb{Z}_2 & n \text{ even},\ 2 \le n \le k-1 \\ \mathbb{Z} & n = k \\ 0 & \text{otherwise} \end{cases}$
$\mathbb{R}\mathrm{P}^k$, k even	$\tilde{H}_n(\mathbb{R}\mathrm{P}^k) = \begin{cases} \mathbb{Z}_2 & n \text{ odd},\ 1 \le n \le k-1 \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(\mathbb{R}\mathrm{P}^k;\mathbb{Z}) = \begin{cases} \mathbb{Z}_2 & n \text{ even},\ 2 \le n \le k \\ 0 & \text{otherwise} \end{cases}$
$\mathbb{C}\mathrm{P}^2$	$\tilde{H}_n(\mathbb{C}\mathrm{P}^2) = \begin{cases} \mathbb{Z} & n = 2 \\ \mathbb{Z} & n = 4 \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(\mathbb{C}\mathrm{P}^2;\mathbb{Z}) = \begin{cases} \mathbb{Z} & n = 2 \\ \mathbb{Z} & n = 4 \\ 0 & \text{otherwise} \end{cases}$
$\mathbb{C}\mathrm{P}^k$	$\tilde{H}_n(\mathbb{C}\mathrm{P}^k) = \begin{cases} \mathbb{Z} & n \text{ even},\ 2 \le n \le 2k \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(\mathbb{C}\mathrm{P}^k;\mathbb{Z}) = \begin{cases} \mathbb{Z} & n \text{ even},\ 2 \le n \le 2k \\ 0 & \text{otherwise} \end{cases}$
N^g	$\tilde{H}_n(N^g) = \begin{cases} \mathbb{Z}^{g-1} \oplus \mathbb{Z}_2 & n = 1 \\ 0 & \text{otherwise} \end{cases}$	$\tilde{H}^n(N^g;\mathbb{Z}) = \begin{cases} \mathbb{Z}^{g-1} & n = 1 \\ \mathbb{Z}_2 & n = 2 \\ 0 & \text{otherwise} \end{cases}$

A.3.1 Homology and Cohomology with $\mathbb{Z}/2$ Coefficients

Over the field $\mathbb{Z}/2$ the universal coefficient theorem identifies homology and cohomology degree by degree, $\tilde{H}_n(X;\mathbb{Z}/2) \cong \tilde{H}^n(X;\mathbb{Z}/2)$, so a single column records both. Against the integral table, the $\mathbb{Z}/2$ coefficients see every cell and split all 2-torsion off into extra dimensions, which is why $\mathbb{R}\mathrm{P}^k$ acquires a generator in every degree up to k and the Betti numbers of a non-orientable surface grow.

Shape	$\tilde{H}_n(\,\cdot\,;\mathbb{Z}/2) \cong \tilde{H}^n(\,\cdot\,;\mathbb{Z}/2)$
S^k	$\mathbb{Z}/2$ for $n = k$
T^2	$(\mathbb{Z}/2)^2$ for $n = 1$; $\quad \mathbb{Z}/2$ for $n = 2$
T^k	$(\mathbb{Z}/2)^{\binom{k}{n}}$ for $1 \le n \le k$
K (Klein bottle)	$(\mathbb{Z}/2)^2$ for $n = 1$; $\quad \mathbb{Z}/2$ for $n = 2$
$\mathbb{R}\mathrm{P}^k$	$\mathbb{Z}/2$ for $1 \le n \le k$
$\mathbb{C}\mathrm{P}^k$	$\mathbb{Z}/2$ for n even, $2 \le n \le 2k$
N^g (g cross-caps)	$(\mathbb{Z}/2)^g$ for $n = 1$; $\quad \mathbb{Z}/2$ for $n = 2$

A.3.2 Cohomology Rings

Note that

$$\tilde{H}^\bullet\left(\bigvee_\alpha X_\alpha; R\right) \cong \prod_\alpha \tilde{H}^\bullet\left(X_\alpha; R\right)$$

Giving us some flexibility for finding cohomology rings:

Shape	Ring
S^k	$H^\bullet(S^k;\mathbb{Z}) \cong \mathbb{Z}[\alpha]/(\alpha^2),\ \lvert\alpha\rvert = k$ $H^\bullet(S^k;\mathbb{Z}_2) \cong \mathbb{Z}_2[\alpha]/(\alpha^2),\ \lvert\alpha\rvert = k$
T^2	$H^\bullet(T^2;R) \cong \Lambda_R[\alpha,\beta],\ \lvert\alpha\rvert = \lvert\beta\rvert = 1$
T^k	$H^\bullet(T^k;R) \cong \Lambda_R[\alpha_1,\alpha_2,\ldots,\alpha_k],\ \lvert\alpha_i\rvert = 1$
$\mathbb{R}\mathrm{P}^2$	$H^\bullet(\mathbb{R}\mathrm{P}^2;\mathbb{Z}_2) \cong \mathbb{Z}_2[\alpha]/(\alpha^3)$ $H^\bullet(\mathbb{R}\mathrm{P}^2;\mathbb{Z}) \cong \mathbb{Z}[\alpha]/(2\alpha,\alpha^2)$
$\mathbb{R}\mathrm{P}^k$	$H^\bullet(\mathbb{R}\mathrm{P}^k;\mathbb{Z}_2) \cong \mathbb{Z}_2[\alpha]/(\alpha^{k+1})$
$\mathbb{R}\mathrm{P}^\infty$	$H^\bullet(\mathbb{R}\mathrm{P}^\infty;\mathbb{Z}_2) \cong \mathbb{Z}_2[\alpha],\ \lvert\alpha\rvert = 1$ $H^\bullet(\mathbb{R}\mathrm{P}^\infty;\mathbb{Z}) \cong \mathbb{Z}[\alpha]/(2\alpha),\ \lvert\alpha\rvert = 2$
$\mathbb{R}\mathrm{P}^{2k}$	$H^\bullet(\mathbb{R}\mathrm{P}^{2k};\mathbb{Z}) \cong \mathbb{Z}[\alpha]/(2\alpha,\alpha^{k+1}),\ \lvert\alpha\rvert = 2$
$\mathbb{R}\mathrm{P}^{2k+1}$	$H^\bullet(\mathbb{R}\mathrm{P}^{2k+1};\mathbb{Z}) \cong \mathbb{Z}[\alpha,\beta]/(2\alpha,\alpha^{k+1},\beta^2,\alpha\beta)$, $\lvert\alpha\rvert = 2,\ \lvert\beta\rvert = 2k+1$
$\mathbb{C}\mathrm{P}^2$	$H^\bullet(\mathbb{C}\mathrm{P}^2;\mathbb{Z}) \cong \mathbb{Z}[\alpha]/(\alpha^3),\ \lvert\alpha\rvert = 2$
$\mathbb{C}\mathrm{P}^k$	$H^\bullet(\mathbb{C}\mathrm{P}^k;\mathbb{Z}) \cong \mathbb{Z}[\alpha]/(\alpha^{k+1}),\ \lvert\alpha\rvert = 2$
$\mathbb{C}\mathrm{P}^\infty$	$H^\bullet(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z}) \cong \mathbb{Z}[\alpha],\ \lvert\alpha\rvert = 2$

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